IHP

SG13G2

SG13G2 DC CV NMOS LV

Rev. 1.2 (190924)

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1 Documentation

1.1 Included Effects in Simulation Model

General	Date, Version	24.09.2019
	Processversion	SG13G2
	Simulator	ADS 2019, Spectre 18.1
	Simulation model	PSP 103.6
	Warning Boundaries	On

Statistic	Corner	x
	Process variations for Monte Carlo Simulation	e
	Mismatch	x
Transistor	DC behavior	x
	Drain diode	x
	Overlap cap.	x
	Junction cap.	x
	1/f Noise	x_1
	HF / Thermal Noise	-
	Substrate resistance (RF)	x_2
	Gate resistance (RF)	x_2
Temperature	-40°C, 25°C, 70°C, 125°C	x

- x measured and modelled
- x_1 parameters taken from SG13
- x_2 measurements are done from a different wafer run than DC model which leads to slight offsets
- e estimated and modelled
- o not included
- - not applicable

1.2 Time range measurement

The verification of the time range measurements is done by the standard IHP verification procedure.

2 Setup

2.1 Measurement Conditions

2.1.1 Measurement Conditions for Transistors

	Sweep	Start[V]	Stop[V]	No. Points	List or f(Vth)
idvg	vg	-500m	1.35	38	
	vb	0	-1.2	5	
	vd	List		3	0.05, 0.6, 1.2
	vs	0		1	
idvd	vd	0	1.35	28	
	vg	Lin f(Vth)	1.35	5	Ref. Setup: idvg Idref: 100.0n Offset: 100.0m Resolution: 1.000m
	vb	0		1	
	vs	0		1	
idvd_vbmin	vd	0	1.35	28	
	vg	Lin f(Vth)	1.35	5	Ref. Setup: idvg Idref: 100.0n Offset: 100.0m Resolution: 1.000m
	vb	-1.2		1	
	vs	0		1	
idvd_vth	vd	0	1.35	28	
	vg	List f(Vth)		3	Ref. Setup: idvg Idref: 100.0n Offset: -200.0m Resolution: 1.000m Steps: 200.0m 200.0m
	vb	0		1	
	vs	0		1	
idvd_vth_vbmin	vd	0	1.35	28	
	vg	List		3	Ref. Setup: idvg

		f(Vth)			Idref: 100.0n Offset: -200.0m Resolution: 1.000m Steps: 200.0m 200.0m
	vb	-1.2		1	
	vs	0		1	
RF idvg	vg	-500m	1.35	38	
	vd	List		3	0.05, 0.6, 1.2
RF idvd	vd	0	1.35	28	
	vg	0	1.35	5	
RF sdut	freq	100MEG	47.6G	31	
	vd	0	1.35	10	
	vg	0	1.35	10	
spad	freq	100MEG	47.6G	31	
	Port1	0		1	
	Port2	0		1	

2.1.2 Measurement Conditions for Capacitance

	Sweep	Start[V]	Step[V]	Stop[V]	High Node	Low Node
Junction Bulk-Drain	1 VHI	-5	200m	200m	B	D
Junction Bulk-Source	1 VHI	-1	20m	300m	B	S
Oxide	1 VHI	1.35	-50m	-1.35	B	G
Overlap	1 VHI	-1.35	50m	1.35	G	D

2.2 Devices

2.2.1 Setup of DC Test Devices

Name	W[um]	L[um]	AD[um²]	AS[um²]	PD[um]	PS[um]	NRD	NRS	NF	Comment	Size Category
W10u0_L10u0_S541_5	10	10	3.4	3.4	20.7	20.7	0	0	1		Large
W0u15_L10u0_S547_3	150m	10	101m	101m	1.34	1.34	0	0	1		Narrow
W10u0_L0u13_T356_S380_2	10	130m	0	0	0	0	0	0	1		Short
W10u0_L0u13_S540_2	10	130m	3.4	3.4	20.7	20.7	0	0	1		Short
W10u0_L0u13_S380_2	10	130m	0	0	0	0	0	0	1		Short
W0u15_L0u13_S546_3	150m	130m	101m	101m	1.34	1.34	0	0	1		Small
W10u0_L0u18_S540_5	10	180m	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L0u5_S541_1	10	500m	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L1u2_S541_2	10	1.2	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L2u0_S541_3	10	2	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L5u0_S541_4	10	5	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L0u15_S540_4	10	150m	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L0u14_S540_3	10	140m	3.4	3.4	20.7	20.7	0	0	1		L Scale
W0u3_L10u0_S545_2	300m	10	102m	102m	1.28	1.28	0	0	1		W Scale
W0u18_L10u0_S546_1	180m	10	103m	103m	1.34	1.34	0	0	1		W Scale
W05u0_L10u0_S542_3	5	10	1.7	1.7	10.7	10.7	0	0	1		W Scale
W02u0_L10u0_S543_1	2	10	680	680	4.6	4.6	0	0	1		W

			m	m	8	8					Scale
W0u6_L10u0_S544_1	600 m	10	204 m	204 m	1.8 8	1.8 8	0	0	1		W Scale
W0u3_L0u13_S544_2	300 m	130 m	102 m	102 m	1.2 8	1.2 8	0	0	1		LW Scale
W0u15_L0u5_S547_2	150 m	500 m	101 m	101 m	1.3 4	1.3 4	0	0	1		LW Scale
W05u0_L5u0_S542_2	5	5	1.7	1.7	10. 7	10. 7	0	0	1		LW Scale
W0u6_L0u18_S543_3	600 m	180 m	204 m	204 m	1.8 8	1.8 8	0	0	1		LW Scale
W0u3_L0u14_S544_3	300 m	140 m	102 m	102 m	1.2 8	1.2 8	0	0	1		LW Scale
W0u18_L0u14_S545_4	180 m	140 m	103 m	103 m	1.3 4	1.3 4	0	0	1		LW Scale
W0u18_L0u13_S545_3	180 m	130 m	103 m	103 m	1.3 4	1.3 4	0	0	1		LW Scale
W0u3_L0u15_S544_4	300 m	150 m	102 m	102 m	1.2 8	1.2 8	0	0	1		LW Scale
W02u0_L2u0_S542_5	2	2	680 m	680 m	4.6 8	4.6 8	0	0	1		LW Scale
W0u15_L0u15_S546_5	150 m	150 m	101 m	101 m	1.3 4	1.3 4	0	0	1		LW Scale
W0u6_L0u5_S543_4	600 m	500 m	204 m	204 m	1.8 8	1.8 8	0	0	1		LW Scale
W0u18_L0u15_S545_5	180 m	150 m	103 m	103 m	1.3 4	1.3 4	0	0	1		LW Scale
W0u3_L0u5_S545_1	300 m	500 m	102 m	102 m	1.2 8	1.2 8	0	0	1		LW Scale
W0u6_L0u13_S543_2	600 m	130 m	204 m	204 m	1.8 8	1.8 8	0	0	1		LW Scale
W0u15_L0u18_S547_1	150 m	180 m	101 m	101 m	1.3 4	1.3 4	0	0	1		LW Scale
W0u6_L1u2_S543_5	600 m	1.2	204 m	204 m	1.8 8	1.8 8	0	0	1		LW Scale
W0u15_L0u14_S546_4	150 m	140 m	101 m	101 m	1.3 4	1.3 4	0	0	1		LW Scale
W02u0_L0u13_S542_4	2	130 m	680 m	680 m	4.6 8	4.6 8	0	0	1		LW Scale
W0u3_L0u18_S544_5	300 m	180 m	102 m	102 m	1.2 8	1.2 8	0	0	1		LW Scale

W05u0_L0u13_S542_1	5	130 m	1.7	1.7	10.7	10.7	0	0	1		LW Scale
W0u15_L0u12_S546_2	150 m	120 m	101 m	101 m	1.34	1.34	0	0	1		Additional
W10u0_L0u12_S540_1	10	120 m	3.4	3.4	20.7	20.7	0	0	1		Additional

2.2.2 Setup of CV Test Devices

Name	W[um]	L[um]	NF	AD[um²]	AS[um²]	PD[um]	PS[um]	Comment	Size Category
Junction Capacitance Bulk - Drain									
CJBDarea_S350B_1			1	37.5K		850		B-High, S/D-Low	BD Area
CJBDperim_S350B_2			1	28.6K		102K		B-High, S/D-Low	BD Perim
Junction Capacitance Bulk - Source									
CJBSarea_S350B_1			1	0		0		B-High, S/D-Low	BS Area
CJBSperim_S350B_2			1	0		0		B-High, S/D-Low	BS Perim
Oxide Capacitance									
Cox_S346K_3	160	250	1					B-High, G-Low	Oxide
Overlap Capacitance									
Cgds1_S350B_11	8K	130 m	800	1.52K	1.52K	8.32K	8.32K	G-High, S/D-	Overlap GDS

								Low, B+Sub -GND	
Cgdsb1_S350B_11	8K	130 m	80 0	1.52 K	1.52 K	8.32 K	8.32 K	G-High, S/D+B -Low	Overlap GDSB
Cgds2_S350B_12	6.6 K	250 m	66 0	1.26 K	1.26 K	6.87 K	6.87 K	G-High, S/D-Low, B+Sub -GND	Overlap 1 GDS
Cgdsb2_S350B_12	6.6 K	250 m	66 0	1.26 K	1.26 K	6.87 K	6.87 K	G-High, S/D+B -Low	Overlap 1 GDSB
Cgds3_S350B_13	400	10	40	79.8	79.8	436	436	G-High, S/D-Low, B+Sub -GND	Overlap 2 GDS
Cgdsb3_S350B_13	400	10	40	79.8	79.8	436	436	G-High, S/D+B -Low	Overlap 2 GDSB

3 Corner Devices

3.1 Target Values vs. Simulated Values

3.1.1 W = 10.00u, L = 130.0n, NF = 1

	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	550m	550m	0	500m	498m	0.38	440m	436m	0.99
Ron*W	-	-	-	650m	656m	0.93	-	-	-
Ion/W	380	382	-0.6	480	479	0.18	580	581	-0.1
log(Ioff/W)	-11.3	-11.4	2.26	-10.1	-10.2	2.25	-9.3	-9.43	2.24

3.1.2 W = 10.00u, L = 10.00u, NF = 1

	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	230m	230m	0	200m	200m	0	170m	171m	-0.64
Ion/W	16.7	16.6	0.39	18.1	18	0.39	19.5	19.5	0

3.1.3 W = 150.0n, L = 130.0n, NF = 1

	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	620m	618m	0.41	530m	529m	0.19	440m	436m	0.85
Ion/W	420	420	0	550	548	0.37	680	683	-0.4

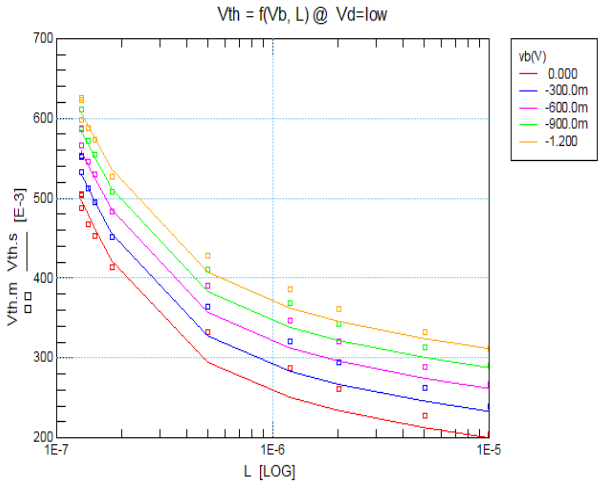
3.1.4 W = 150.0n, L = 10.00u, NF = 1

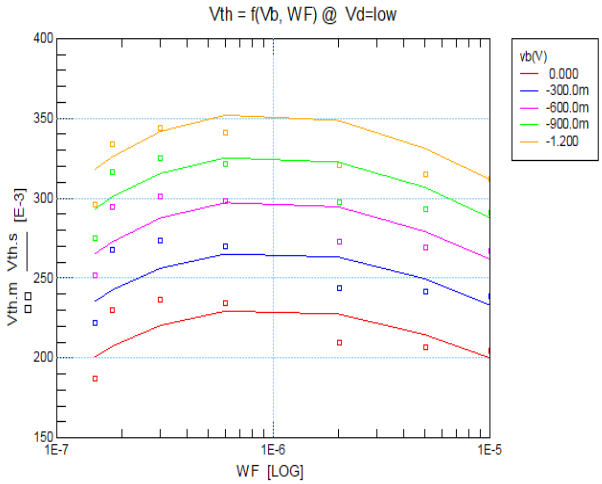
	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	270m	270m	0	200m	201m	-0.51	130m	131m	-1.01
Ion/W	13	13.1	-0.52	15.5	15.5	0.26	18	18	-0.12

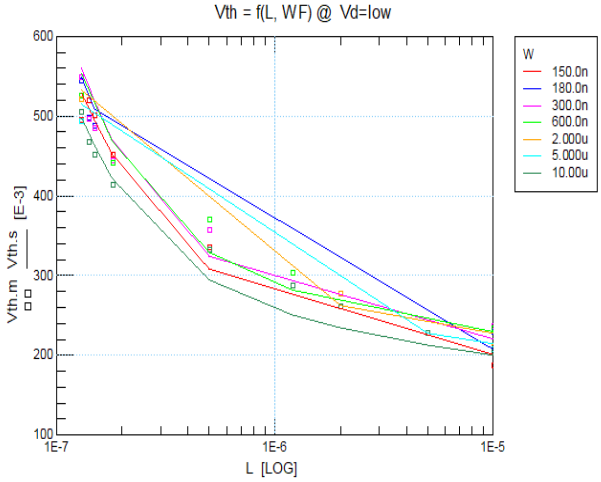
4 DC Transistors

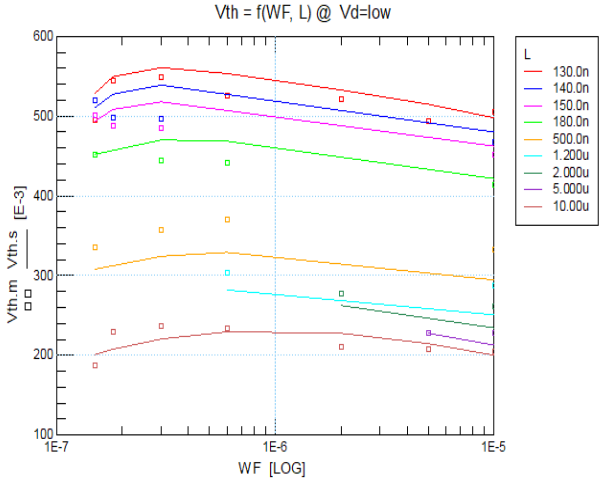
4.1 Summarized Results

4.1.1 Threshold Voltage Linear Region

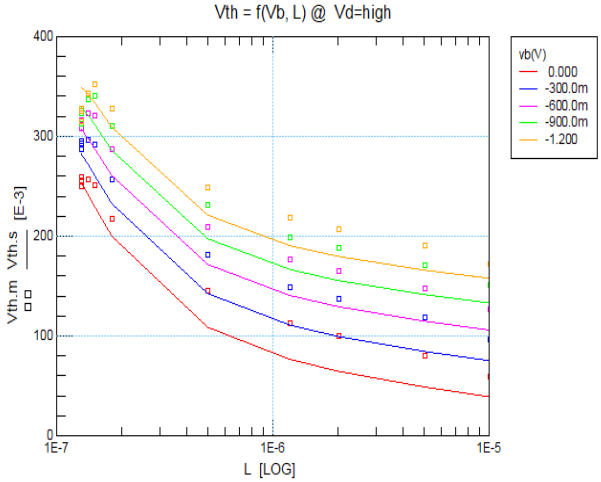


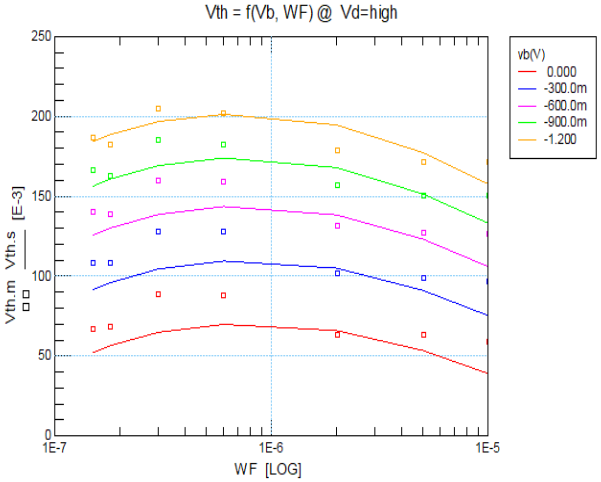


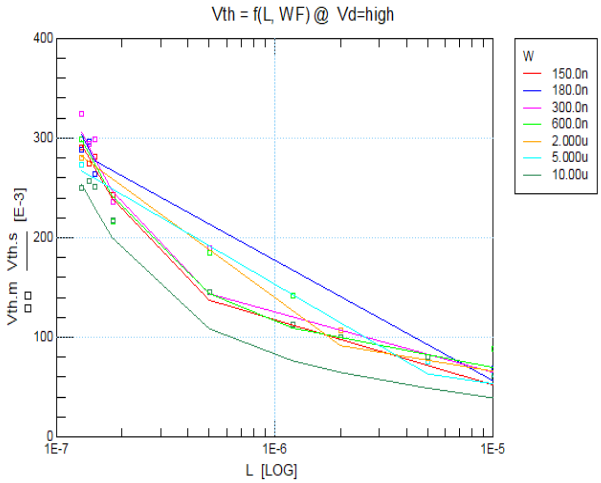


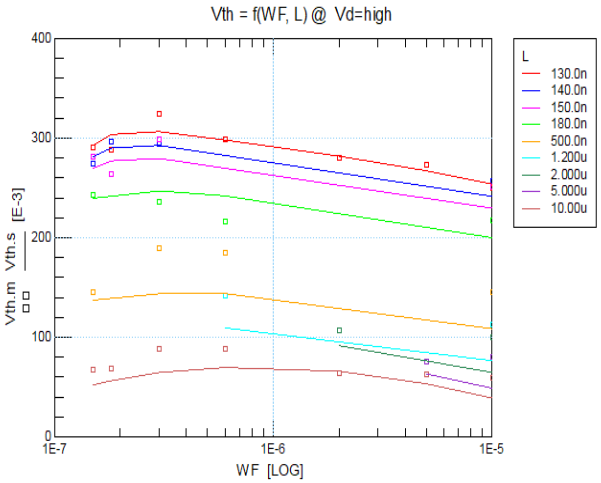


4.1.2 Threshold Voltage in Saturation

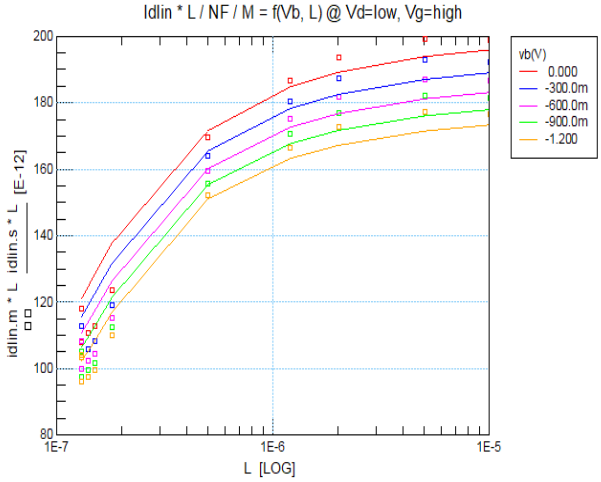


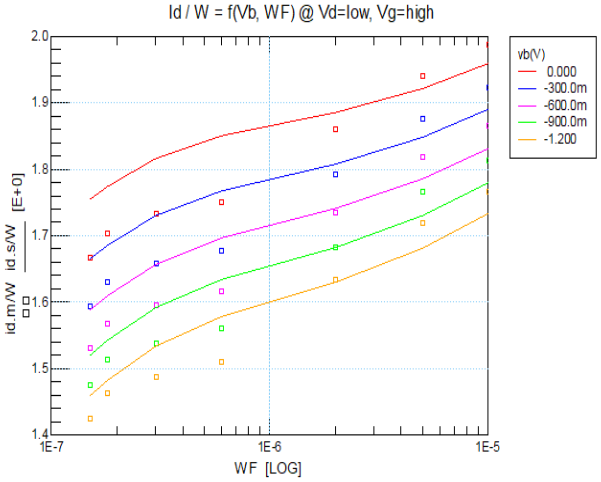


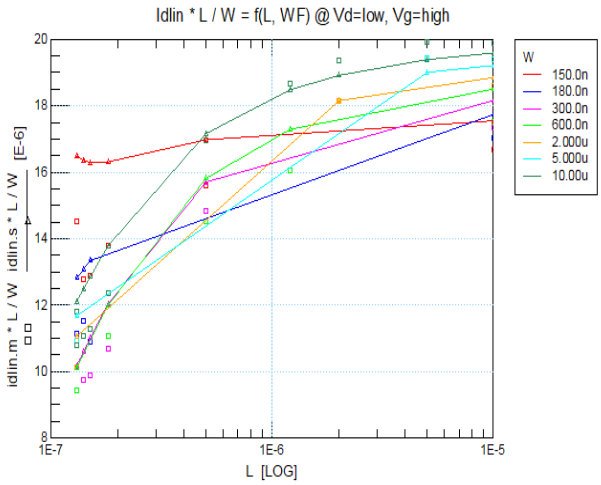


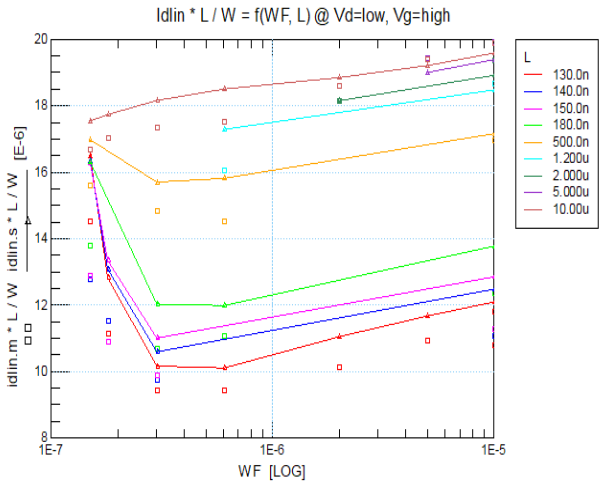


4.1.3 Maximum Drain Current in Linear Region

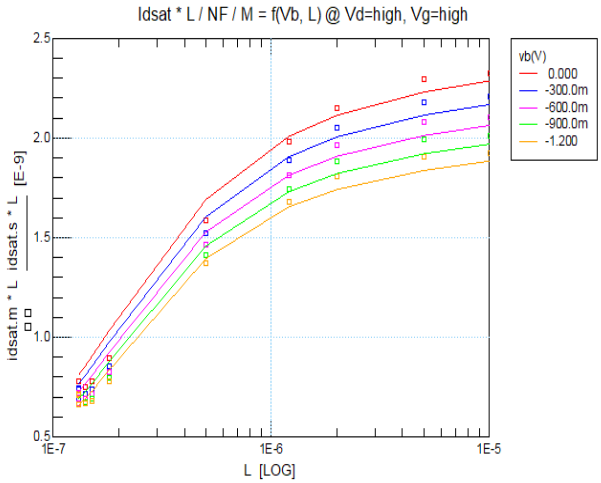


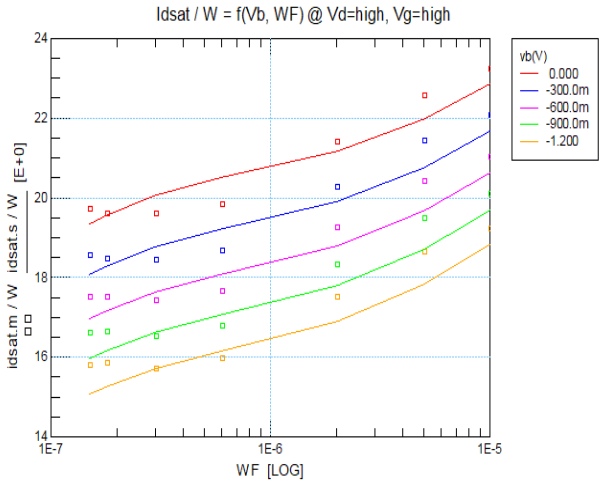


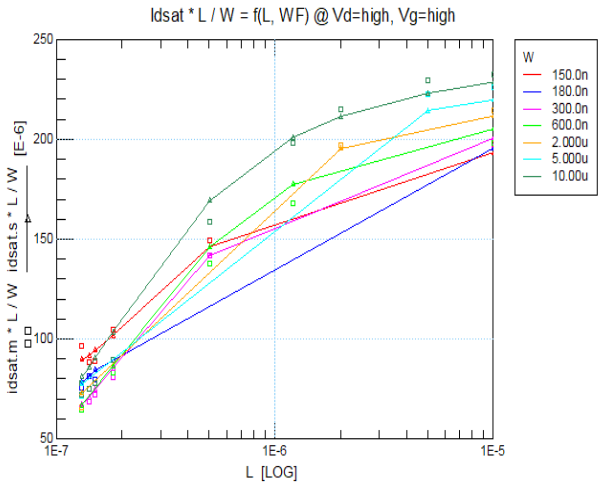


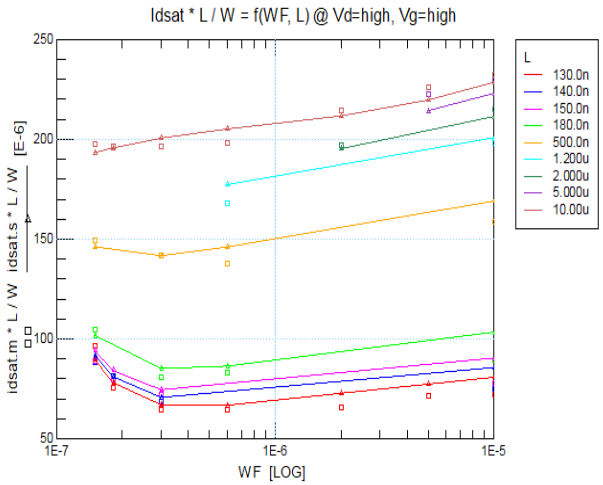


4.1.4 Maximum Drain Current in Saturation Region



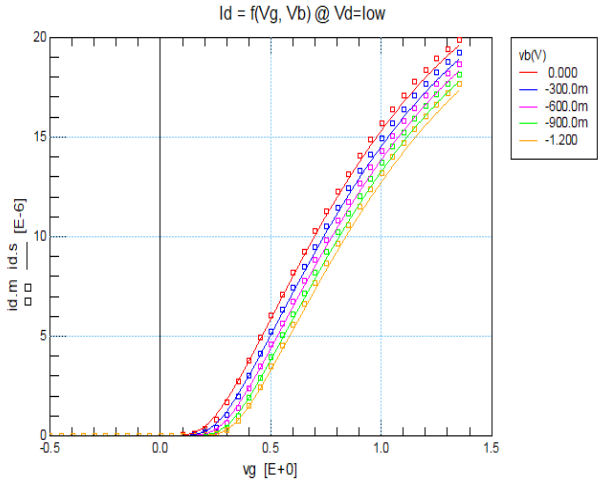


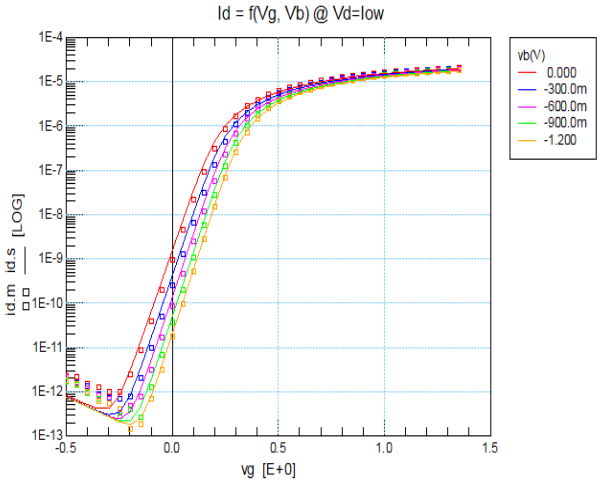


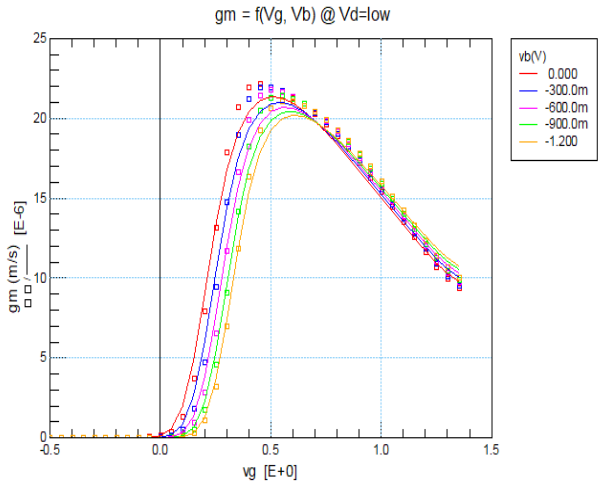


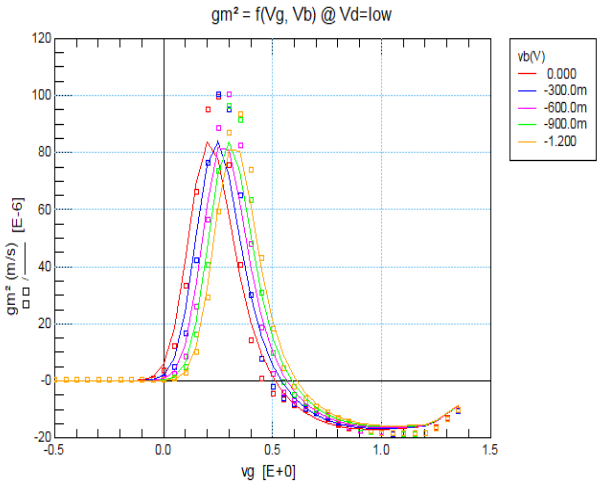
4.2 W10u0_L10u0_S541_5

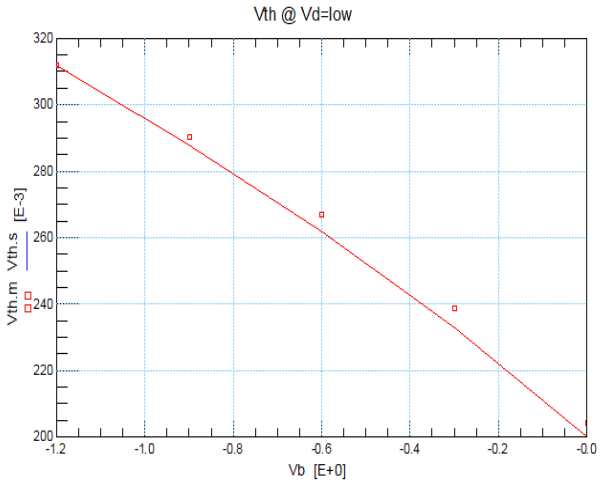
idvg, Vd = 0.05V, T = 27°C

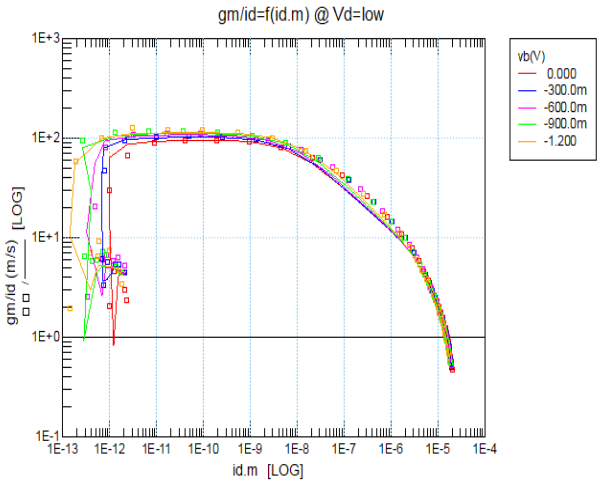


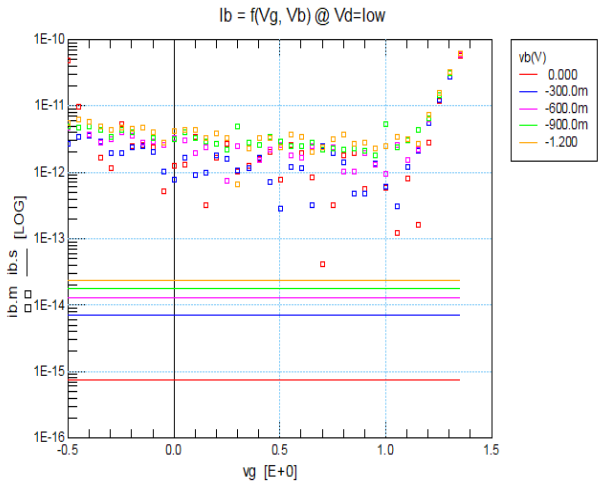


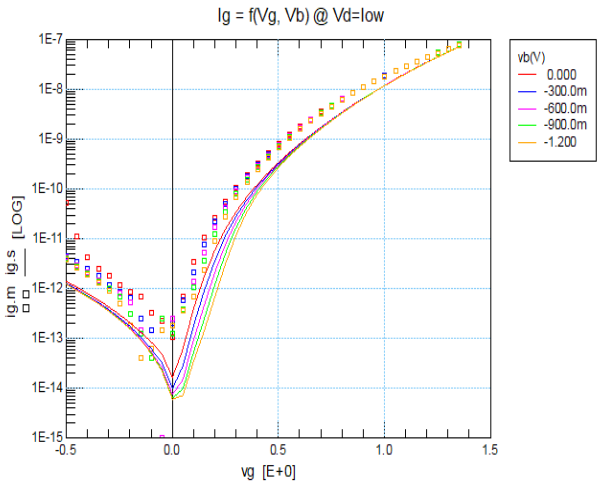




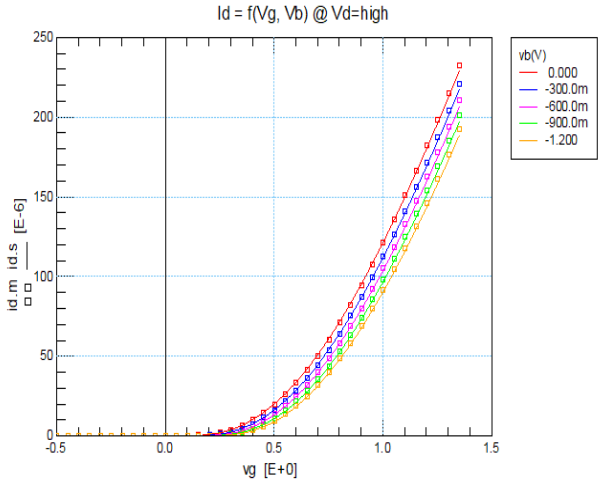


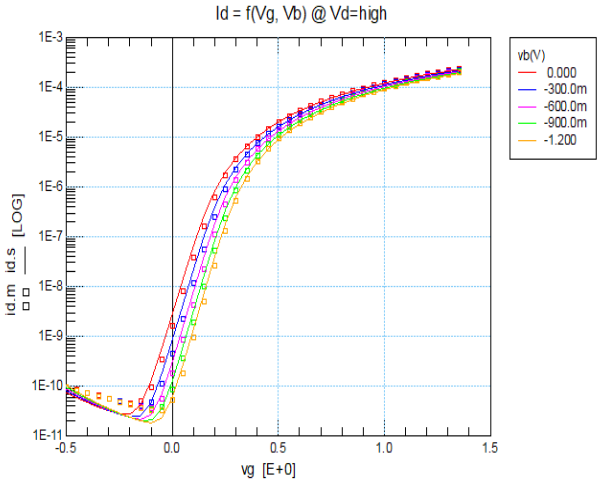


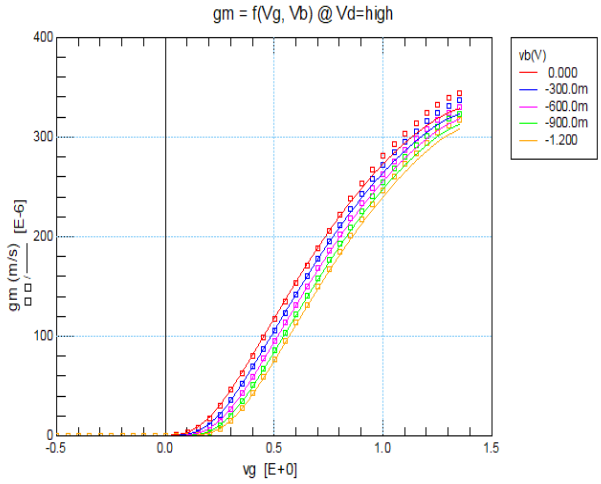


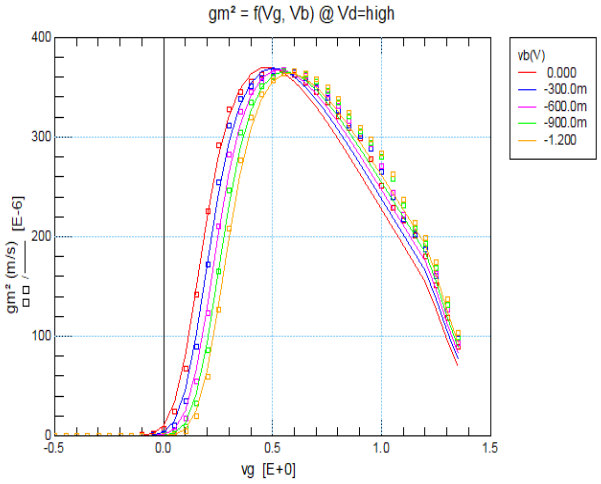


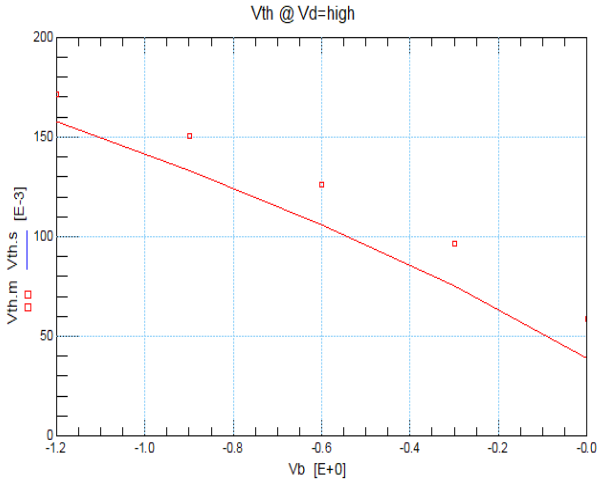
idvg, Vd = 1.2V, T = 27°C

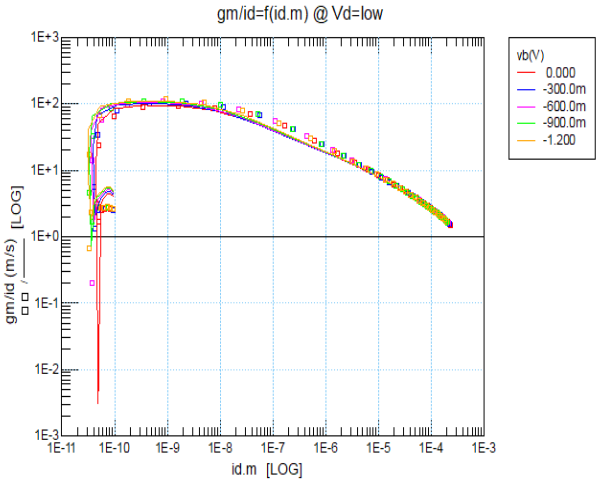


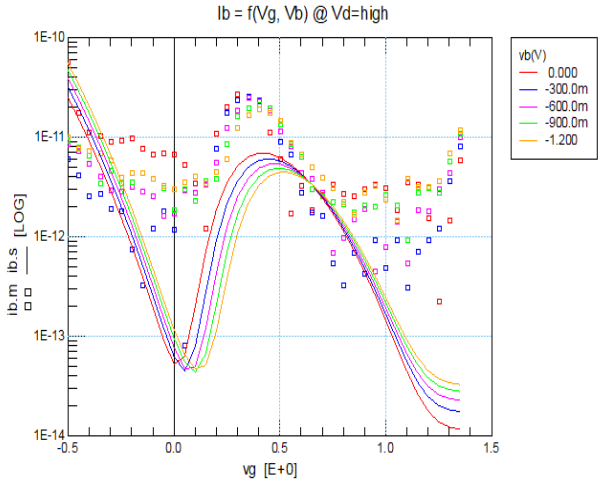


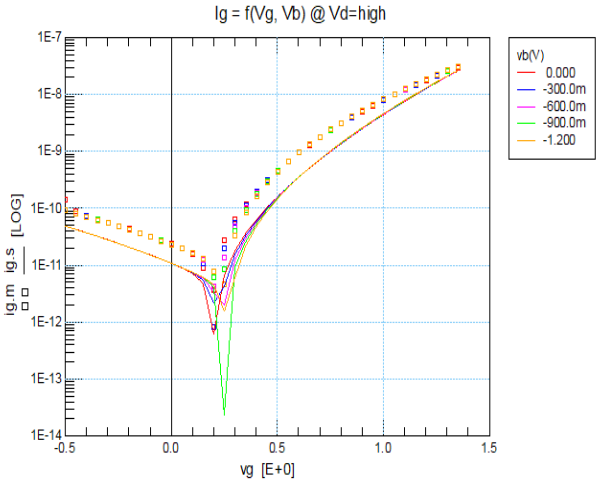




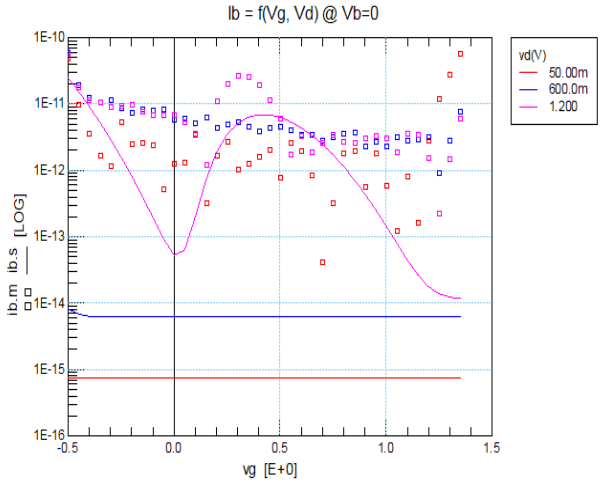


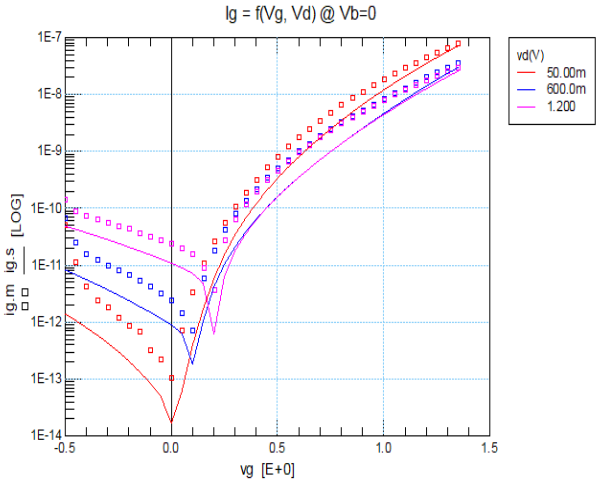




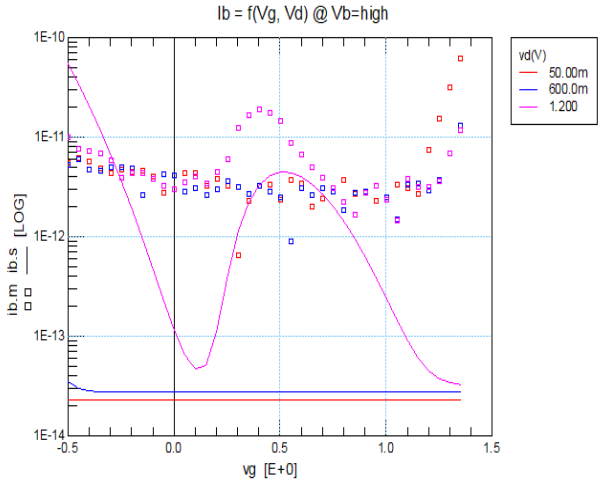


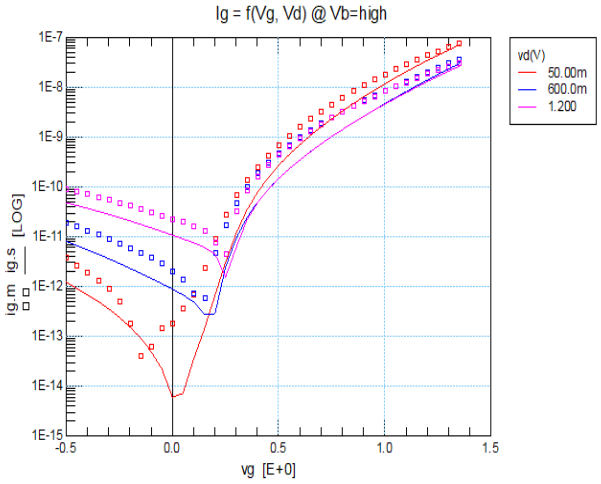
idvg, Ib,Ig, Vb = 0V, T = 27°C



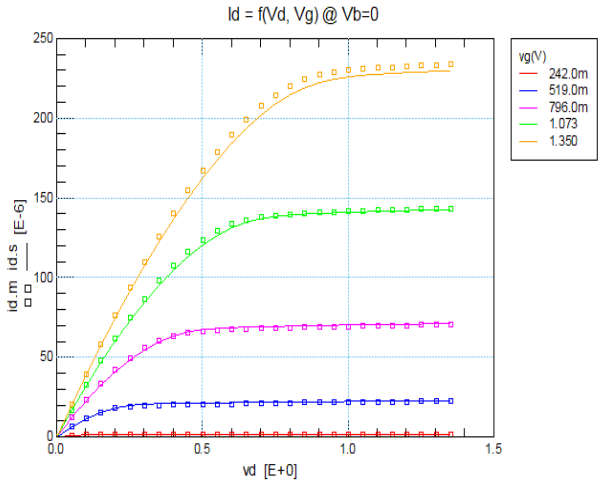


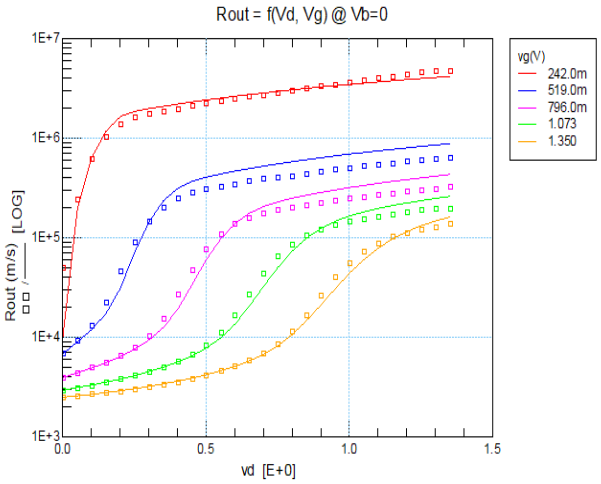
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

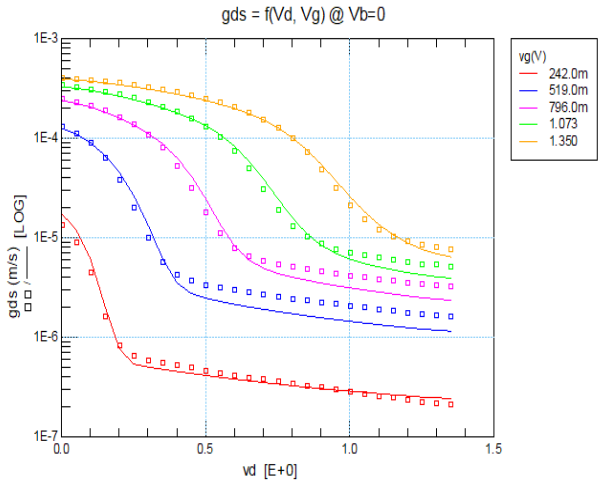




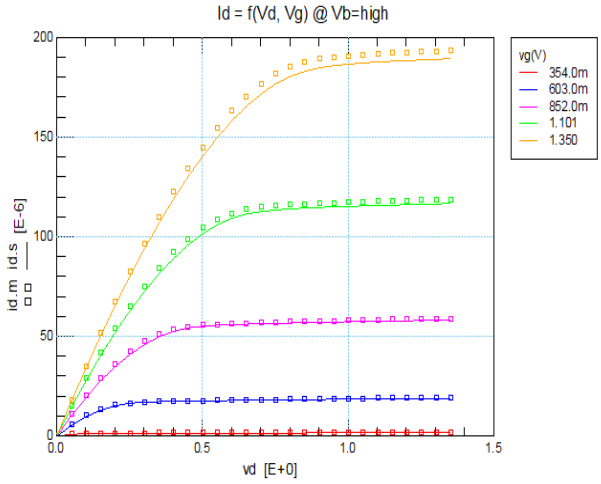
idvd, Vb = 0V, T = 27°C

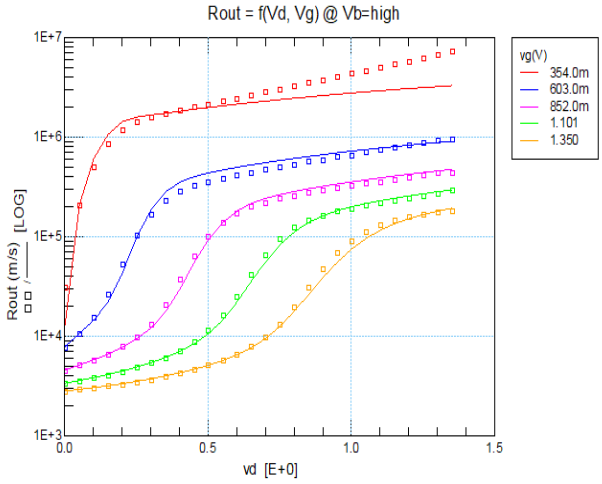


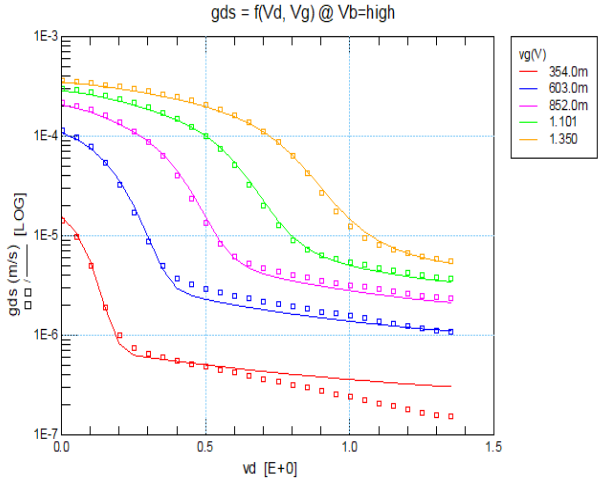




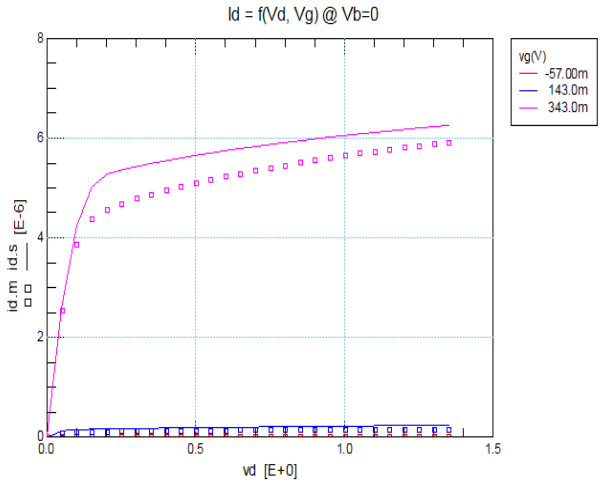
idvd_vbmin, Vb = -1.2V, T = 27°C

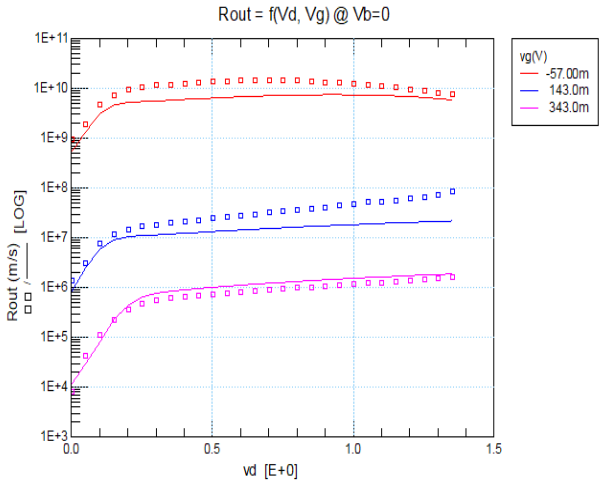


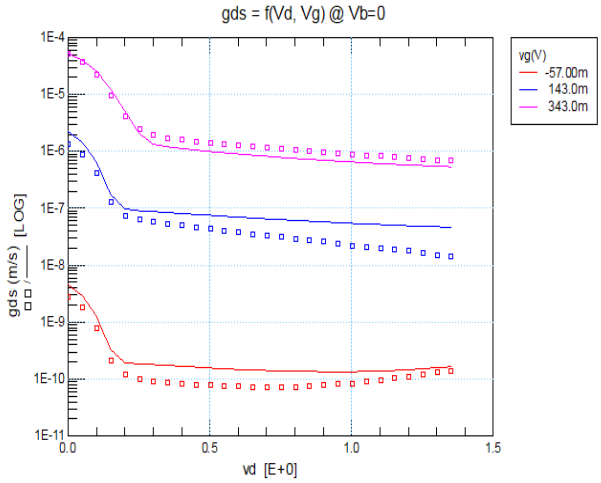




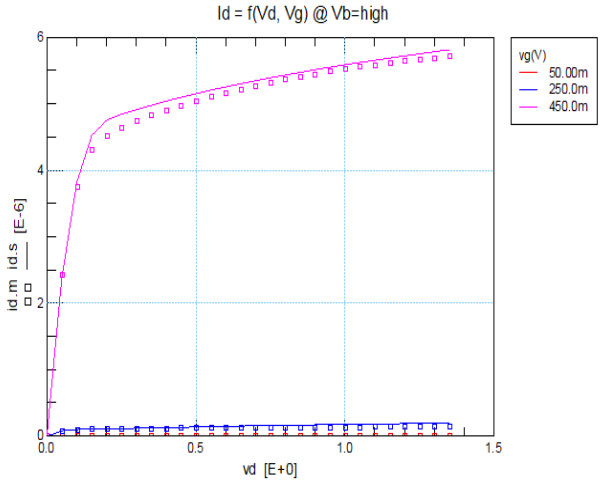
idvd_vth, Vb = 0V, T = 27°C

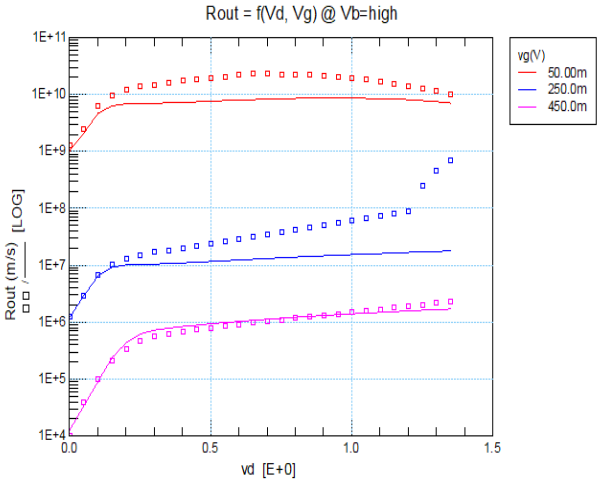


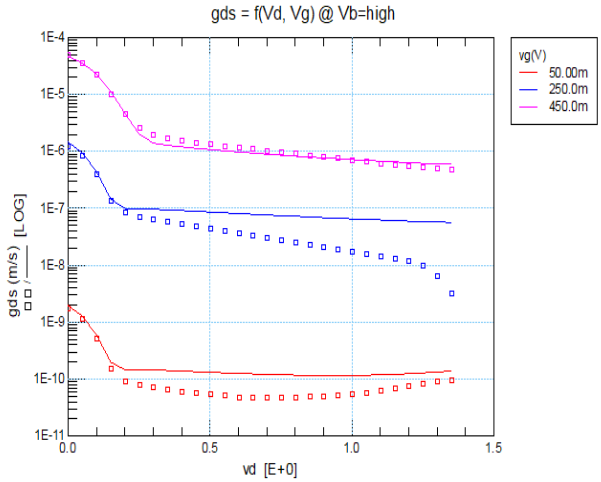




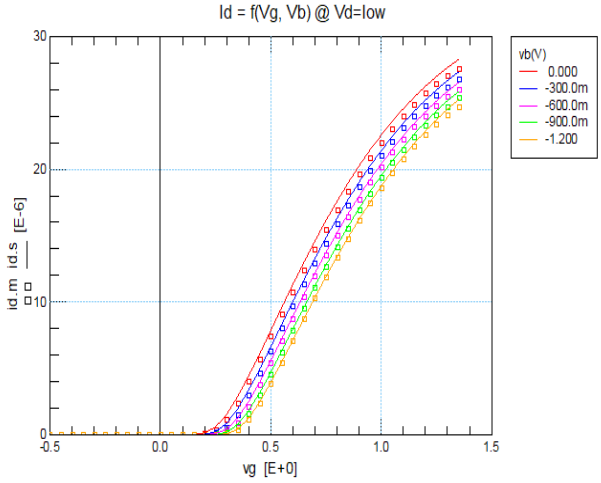
idvd_vth_vbmin, Vb = -1.2V, T = 27°C

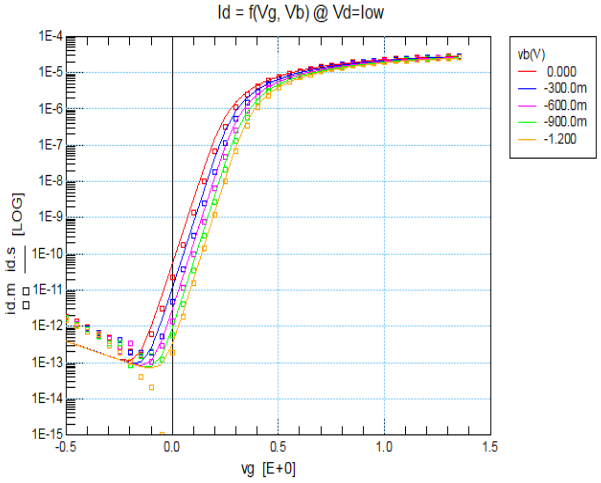


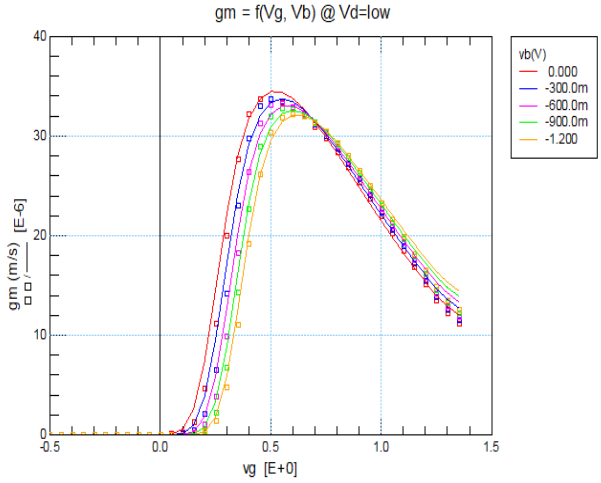


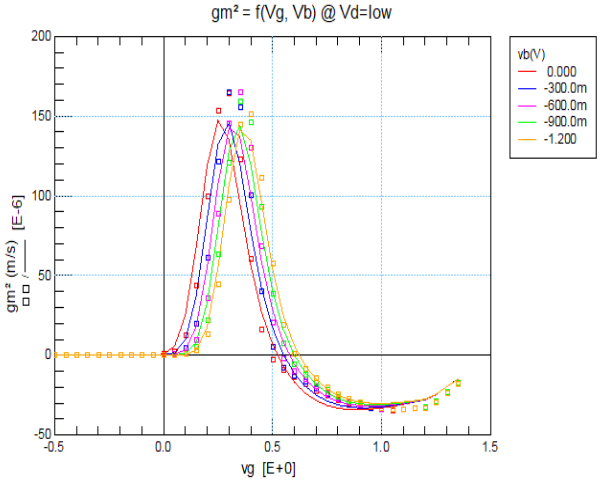


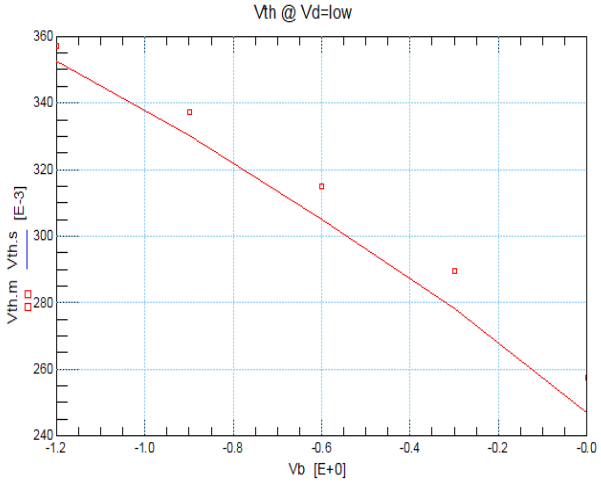
idvg, Vd = 0.05V, T = -40°C

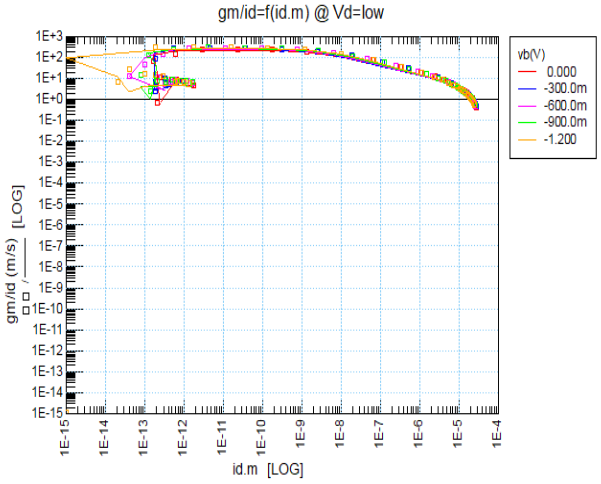


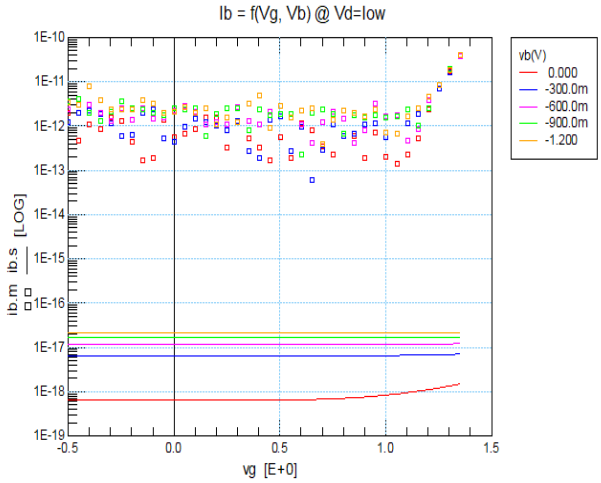


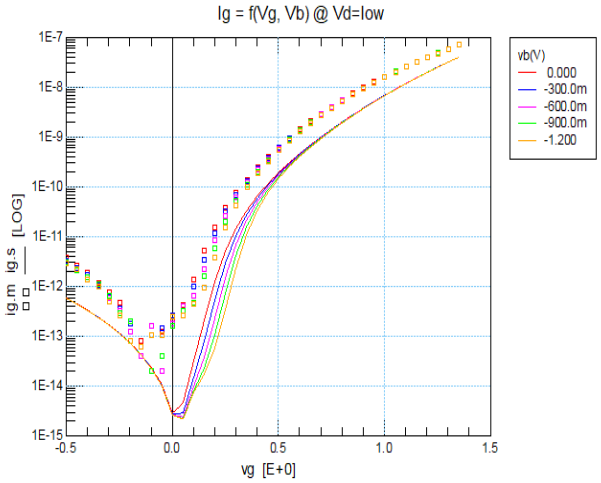




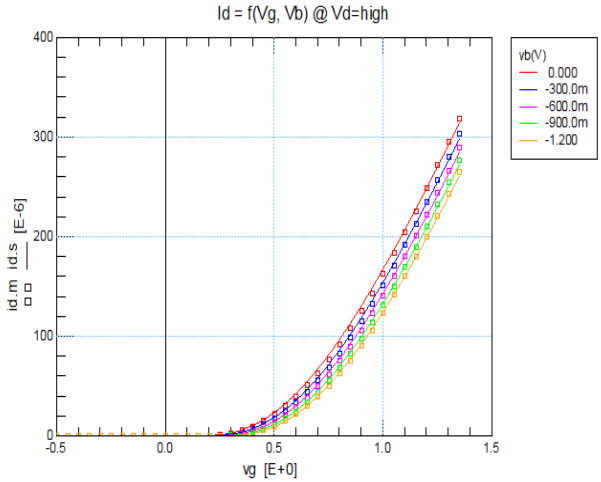


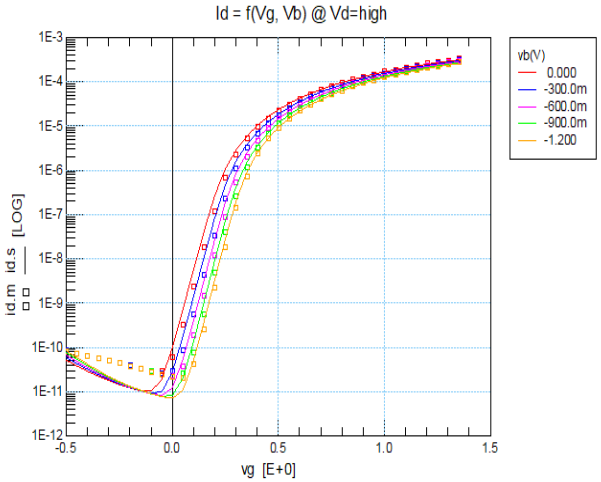


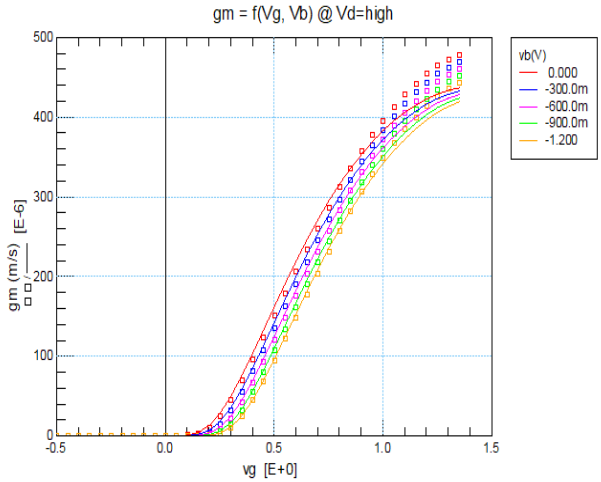


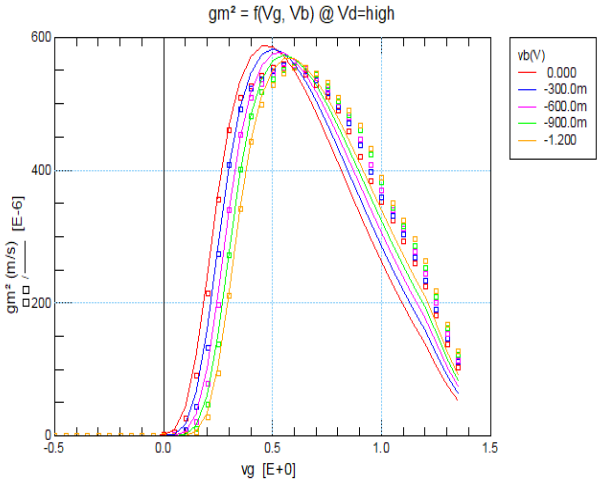


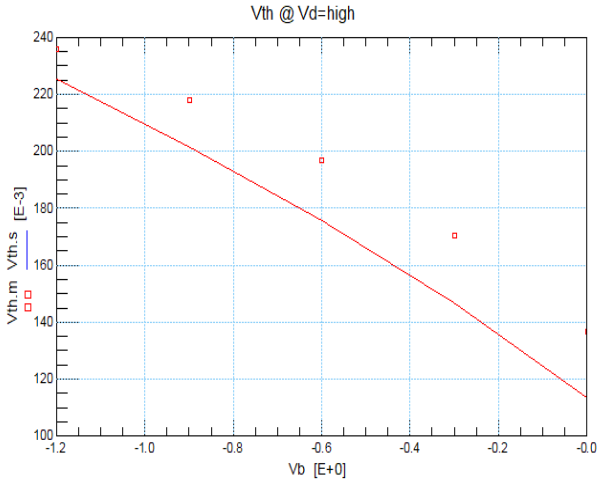
idvg, Vd = 1.2V, T = -40°C

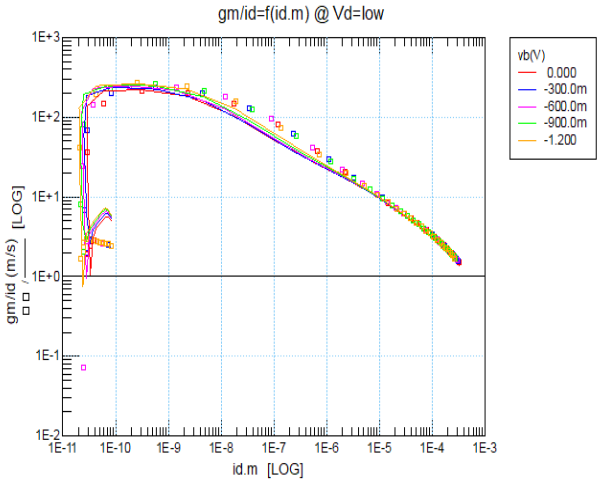


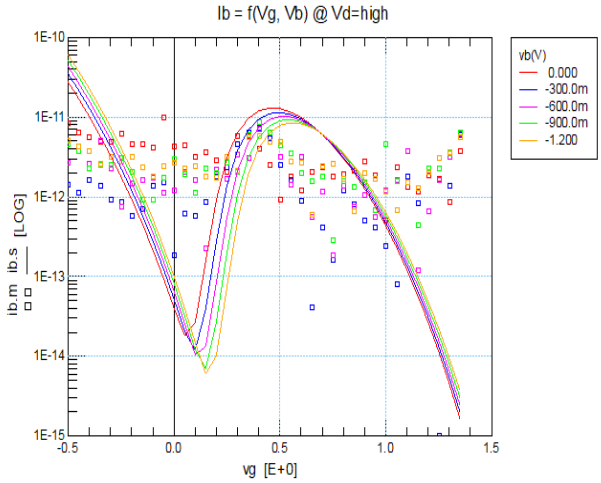


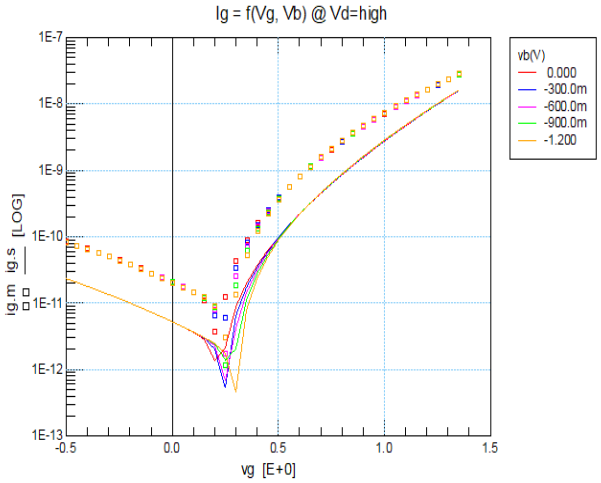




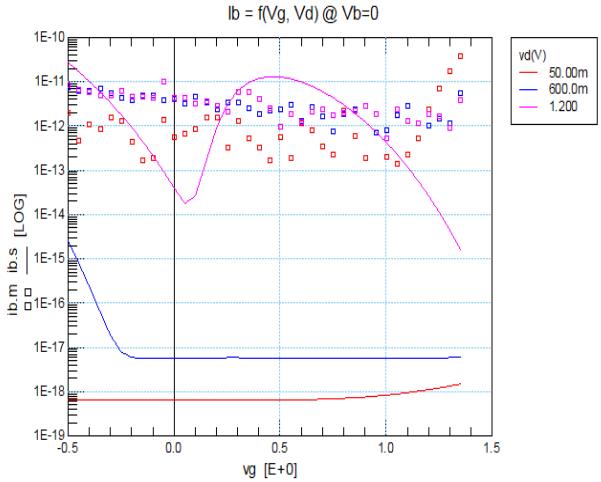


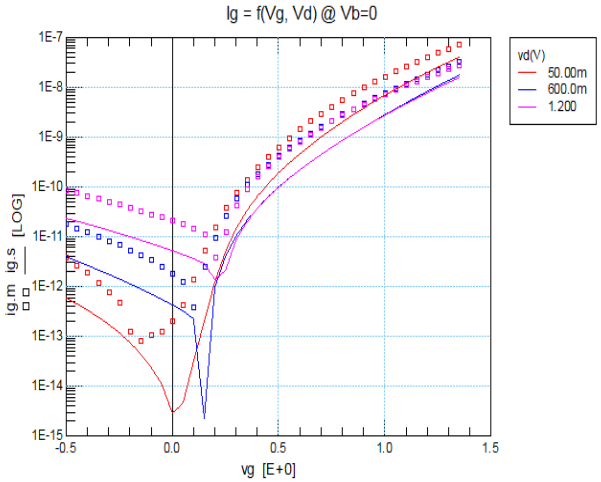




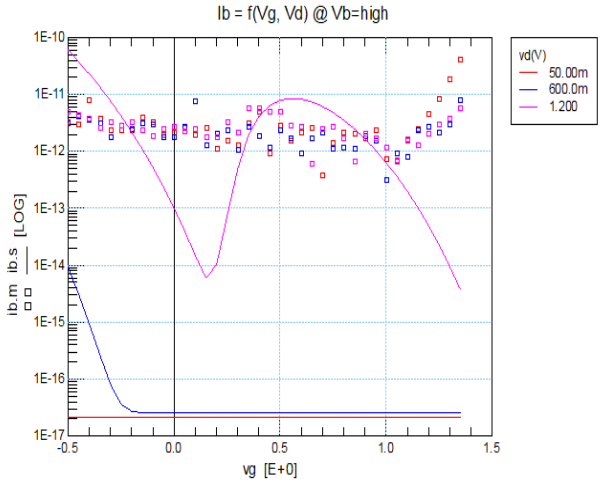


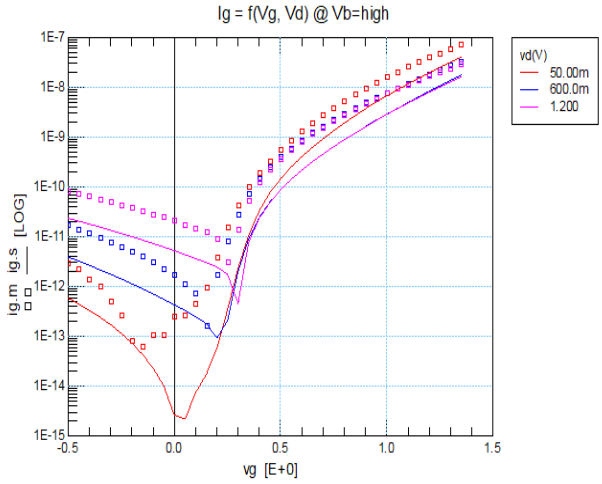
idvg, Ib,Ig, Vb = 0V, T = -40°C



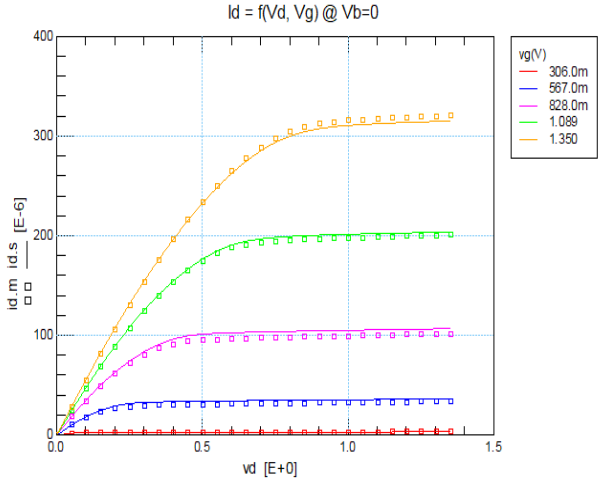


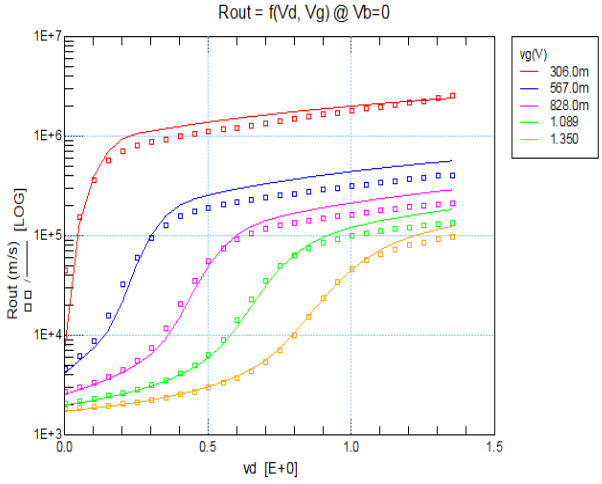
idvg, Ib,Ig, Vb = -1.2V, T = -40°C

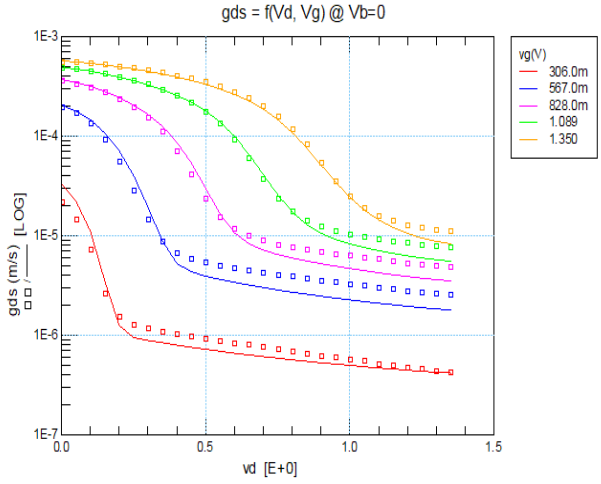




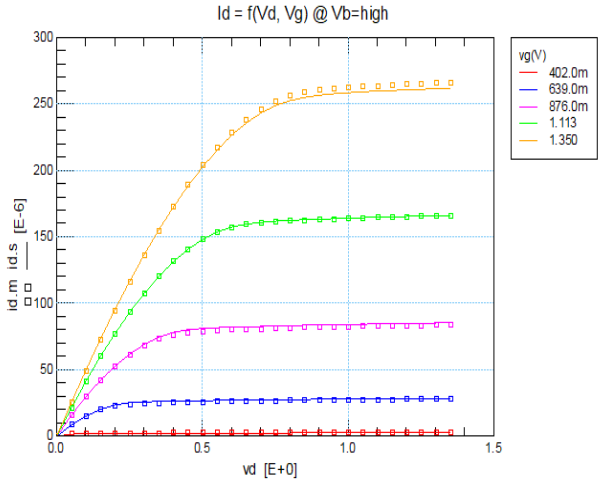
idvd, Vb = 0V, T = -40°C

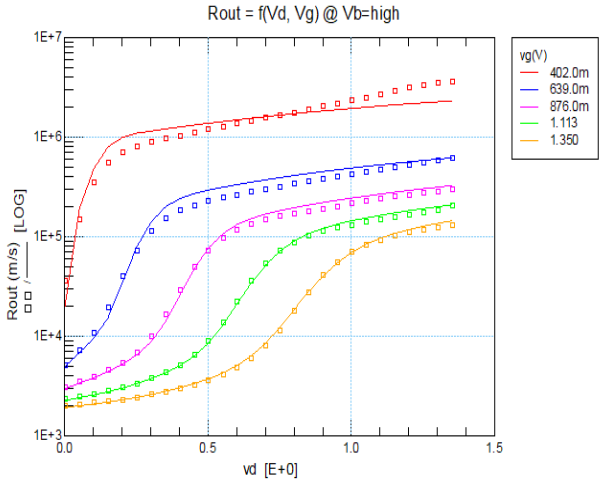


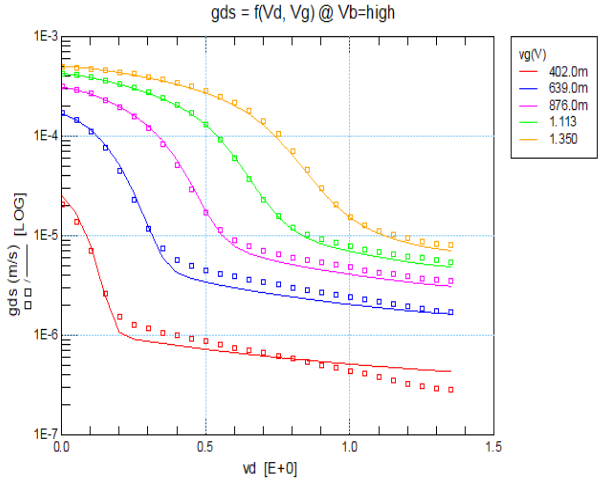




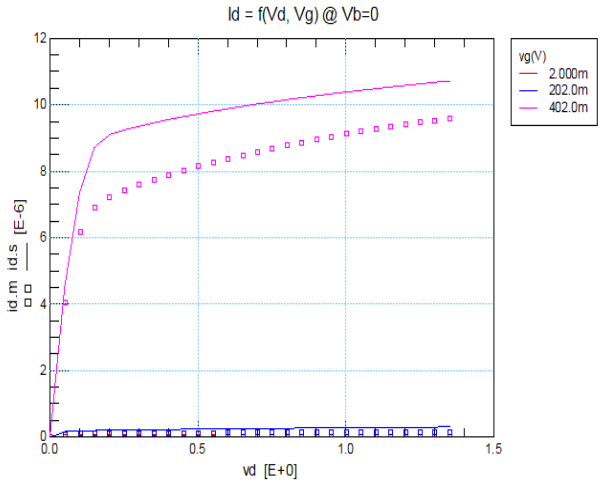
idvd_vbmin, Vb = -1.2V, T = -40°C

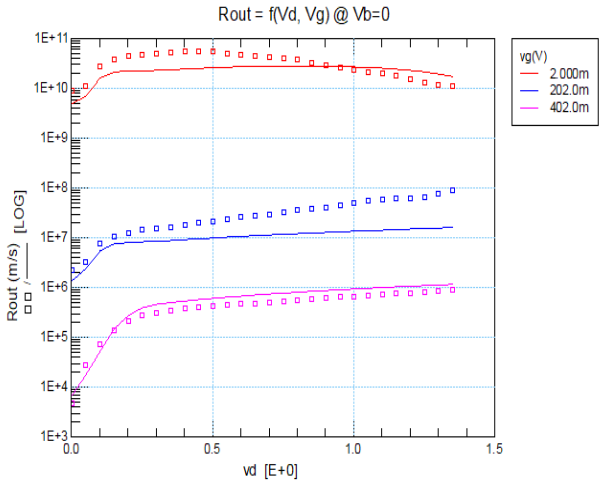


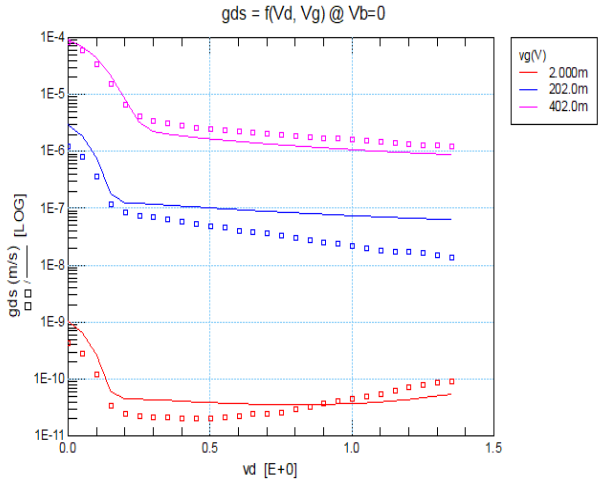




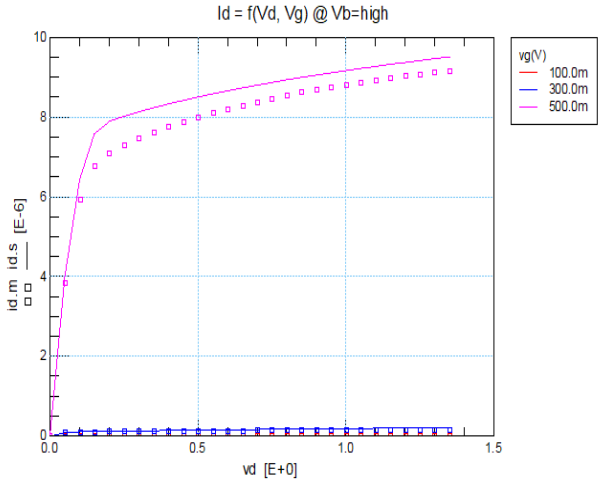
idvd_vth, Vb = 0V, T = -40°C

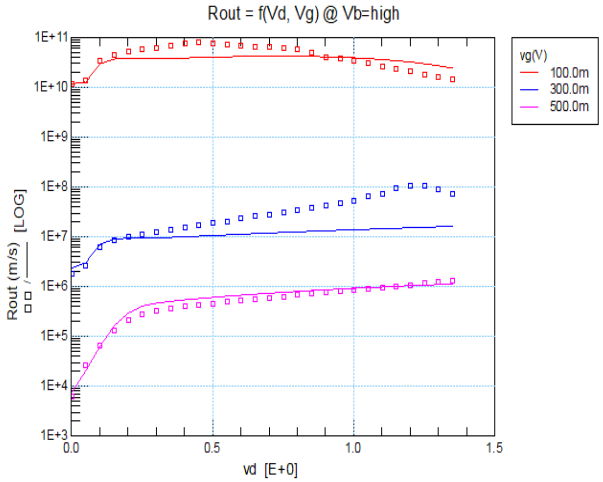


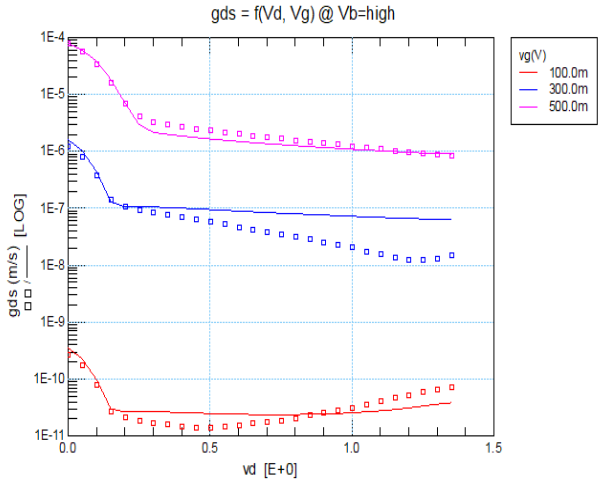




idvd_vth_vbmin, Vb = -1.2V, T = -40°C

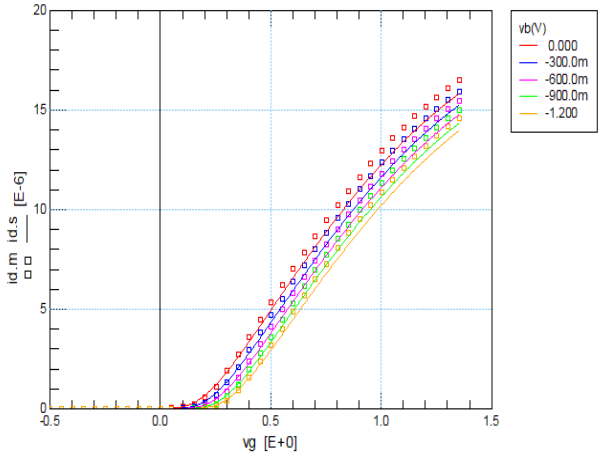




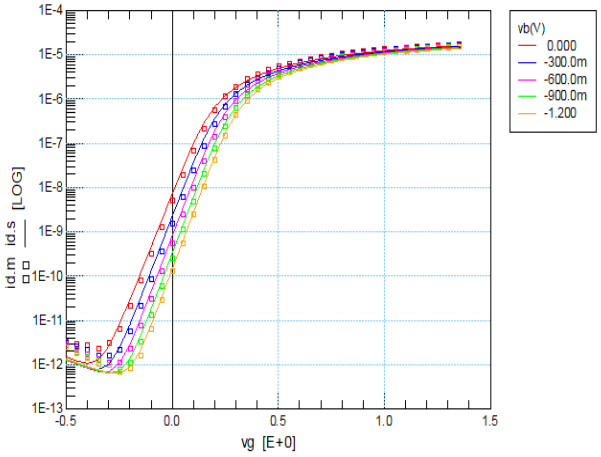


idvg, Vd = 0.05V, T = 70°C

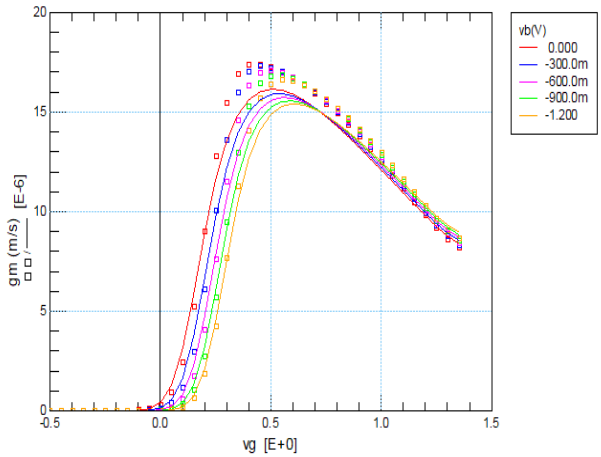
Id = f(Vg, Vb) @ Vd=low



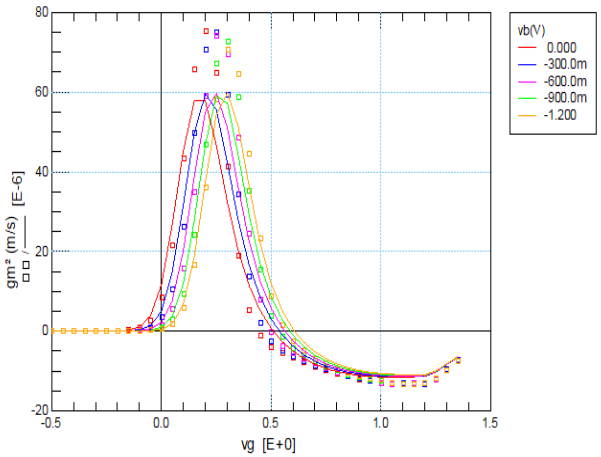
Id = f(Vg, Vb) @ Vd=low



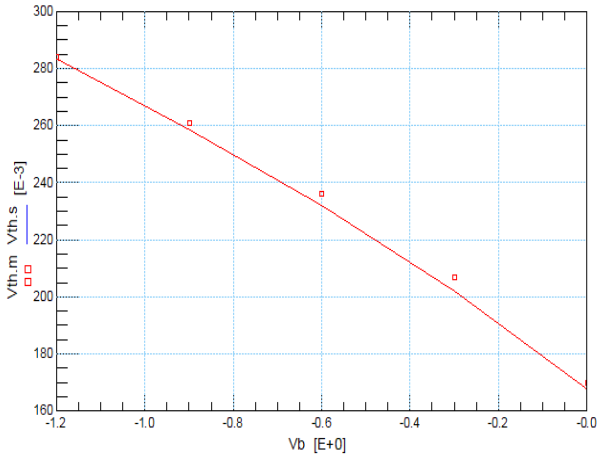
gm = f(Vg, Vb) @ Vd=low



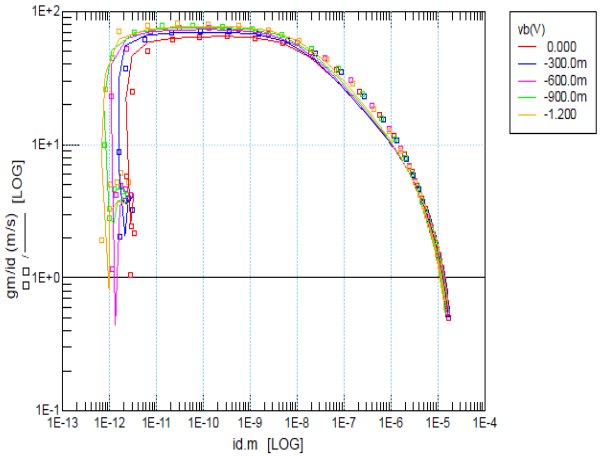
gm² = f(Vg, Vb) @ Vd=low

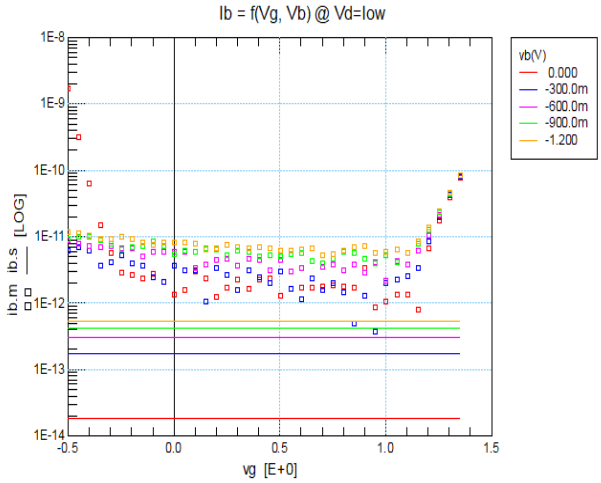


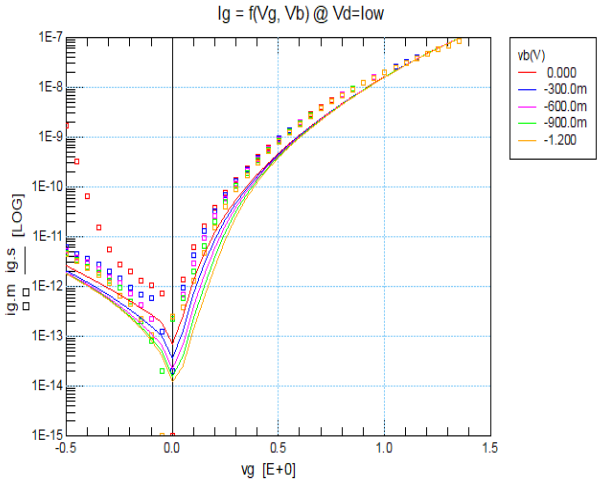
Vth @ Vd=low



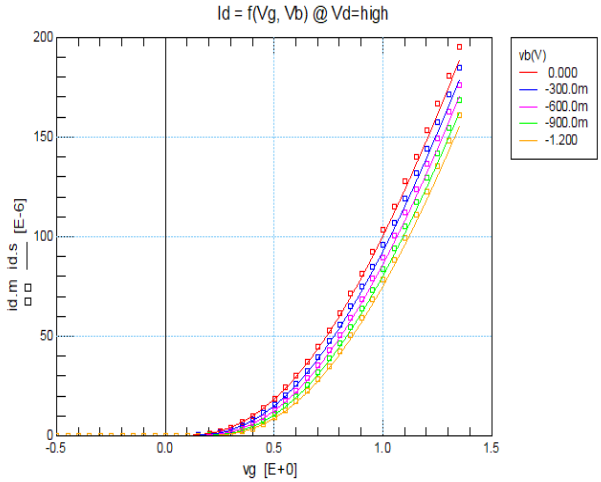
gm/id=f(id.m) @ Vd=low

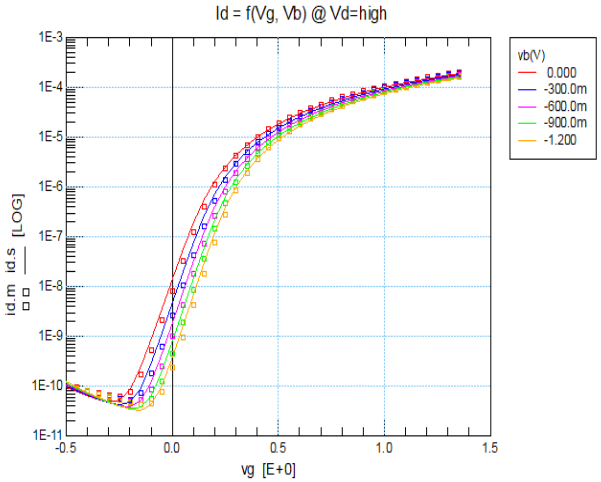


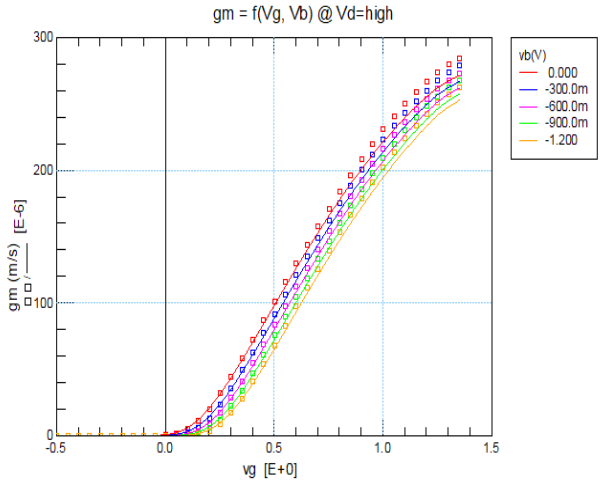


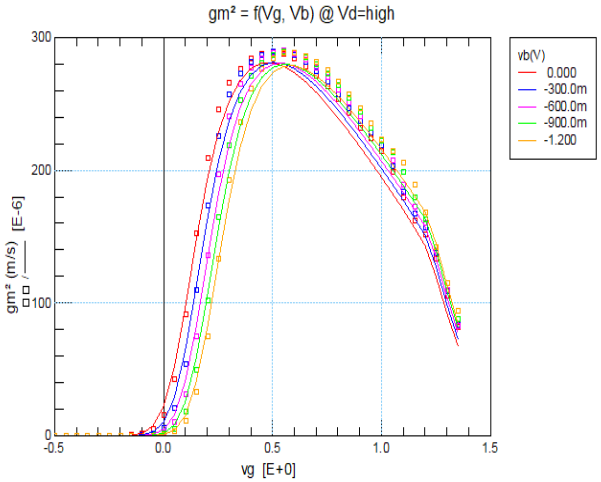


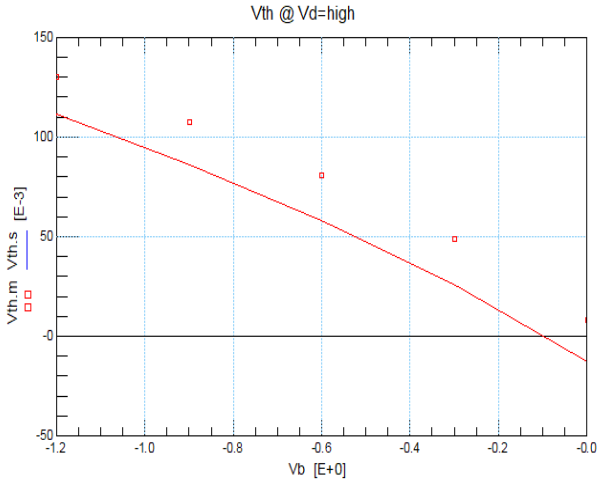
idvg, Vd = 1.2V, T = 70°C

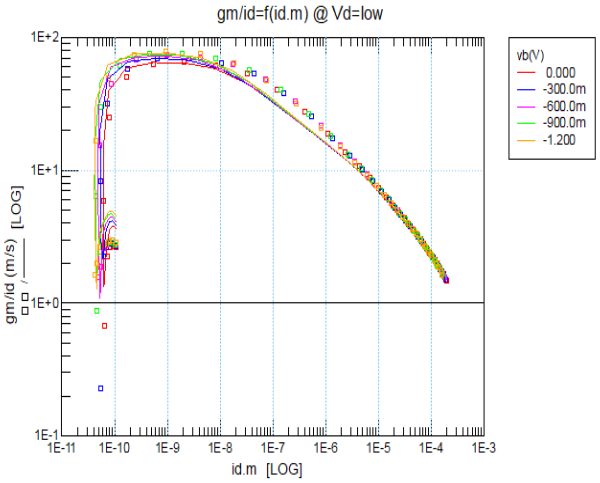


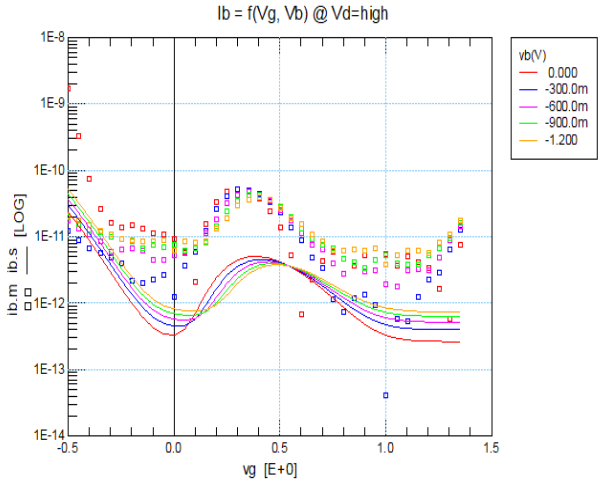


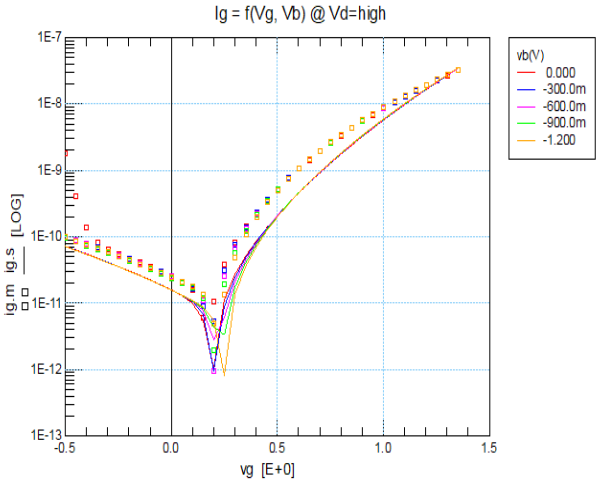




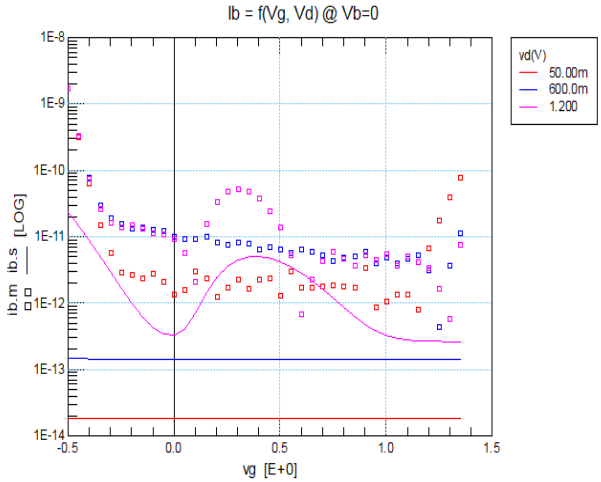


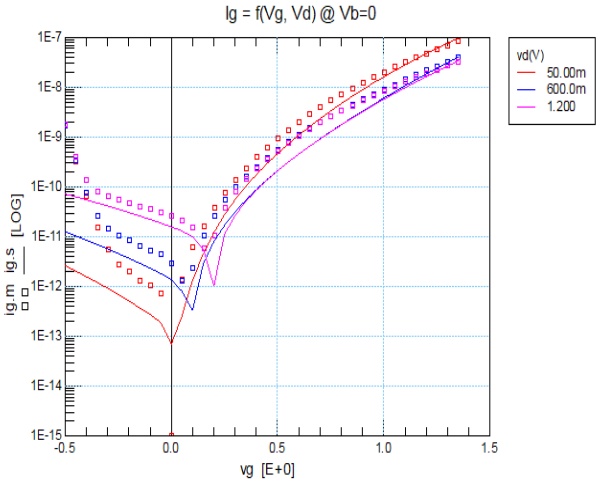




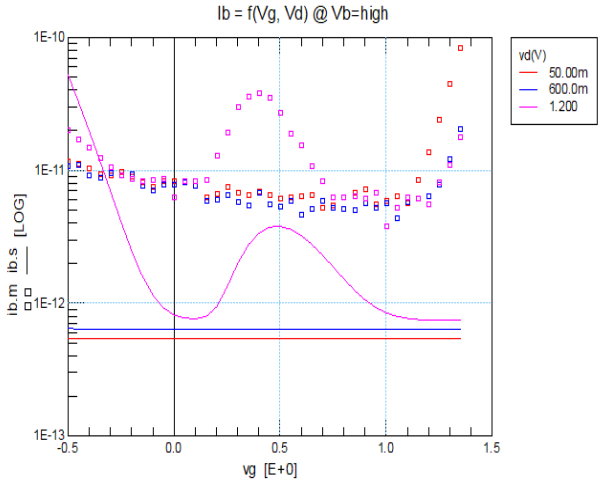


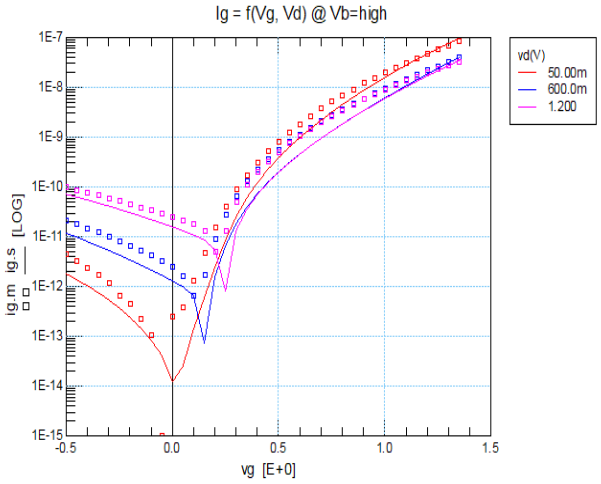
idvg, Ib,Ig, Vb = 0V, T = 70°C



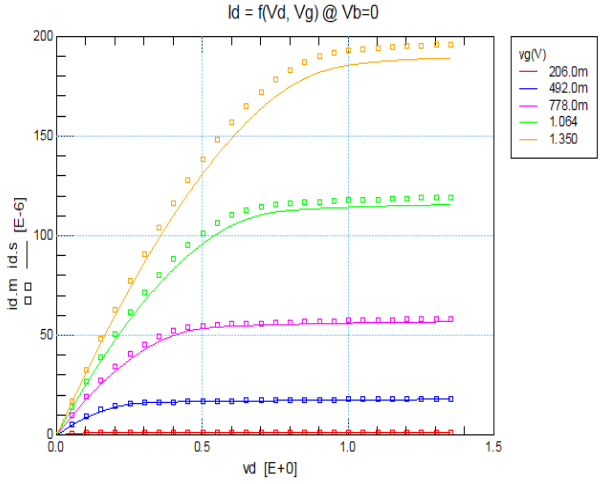


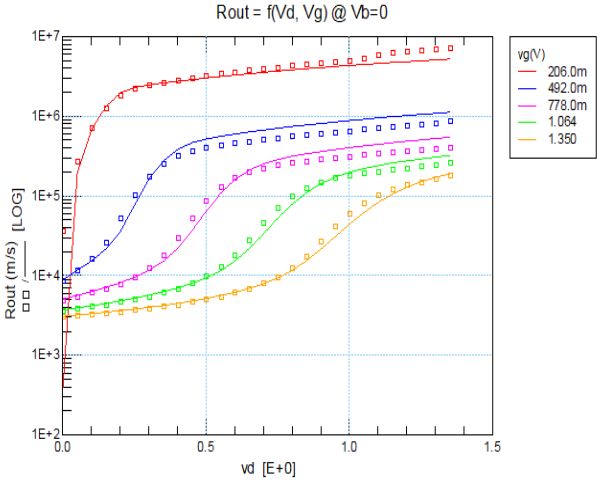
idvg, Ib,Ig, Vb = -1.2V, T = 70°C

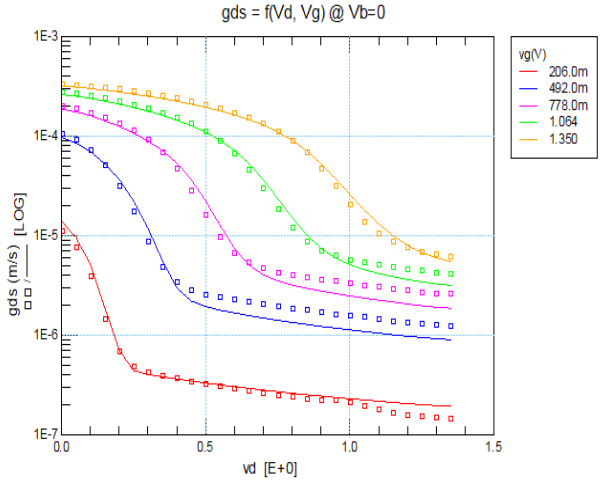




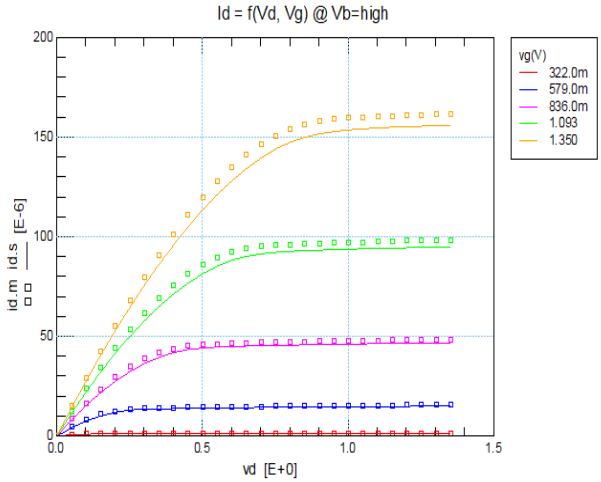
idvd, Vb = 0V, T = 70°C

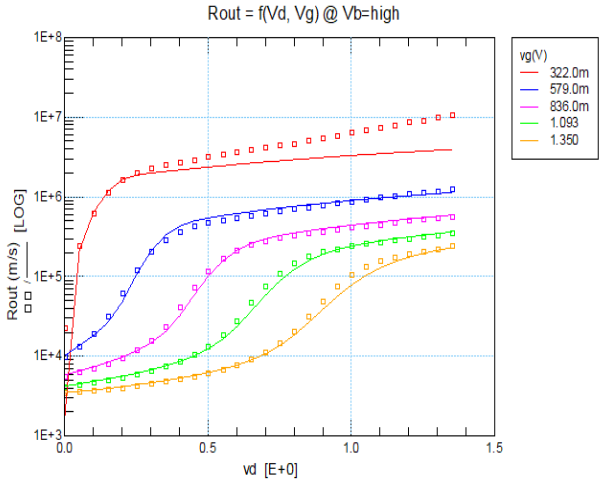


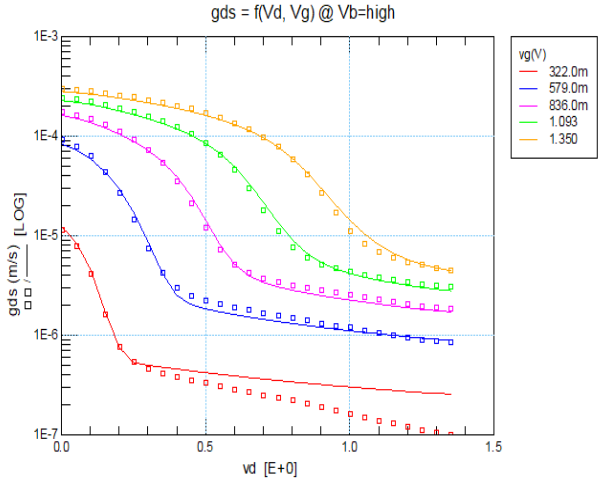




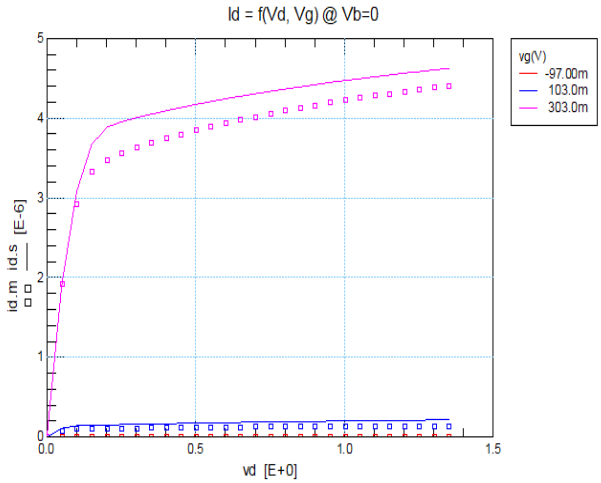
idvd_vbmin, Vb = -1.2V, T = 70°C

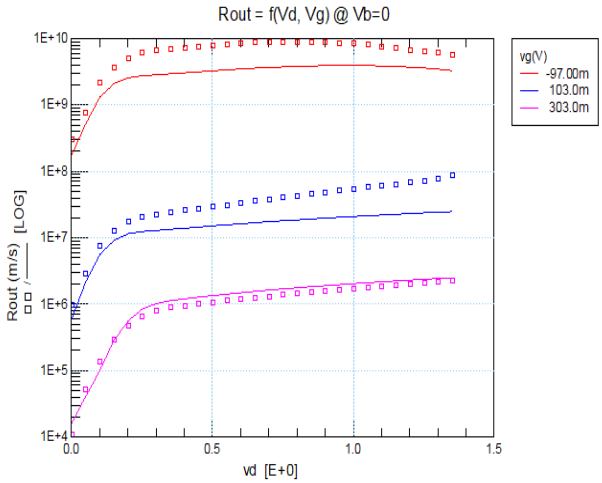


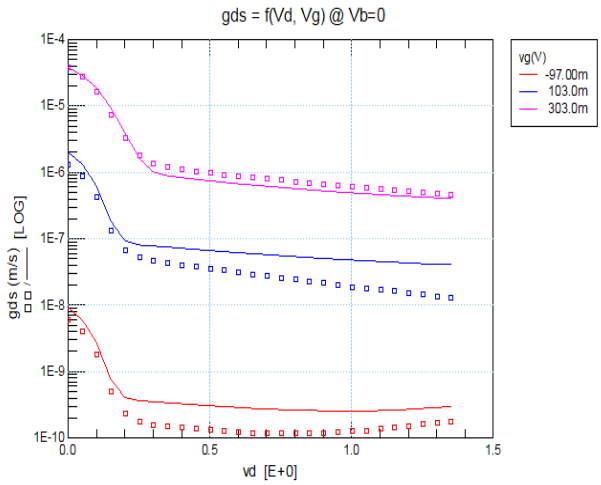




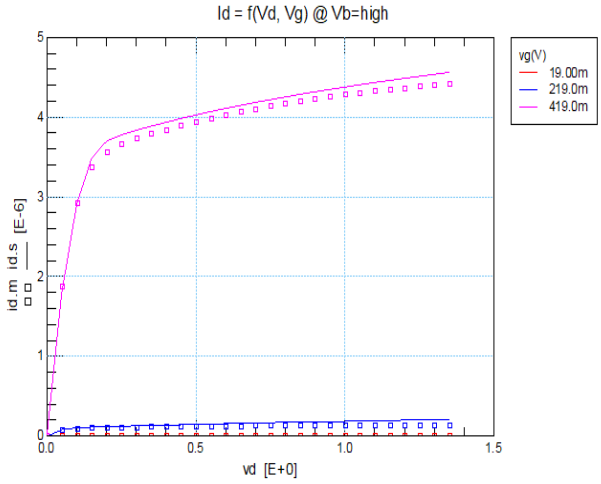
idvd_vth, Vb = 0V, T = 70°C

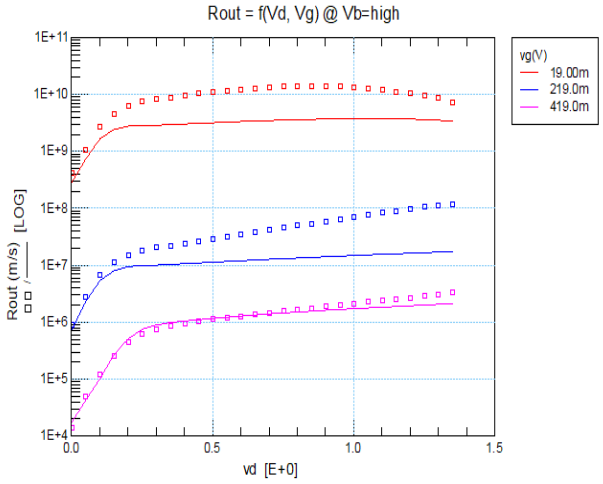


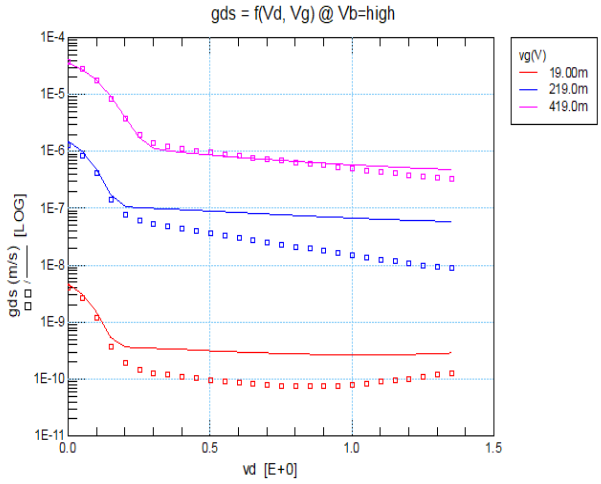




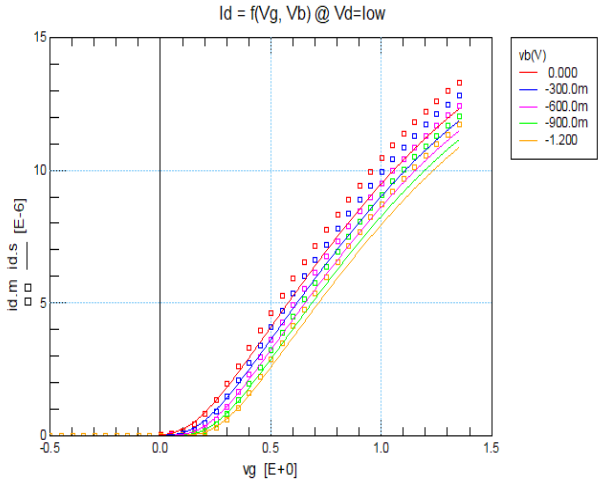
idvd_vth_vbmin, Vb = -1.2V, T = 70°C

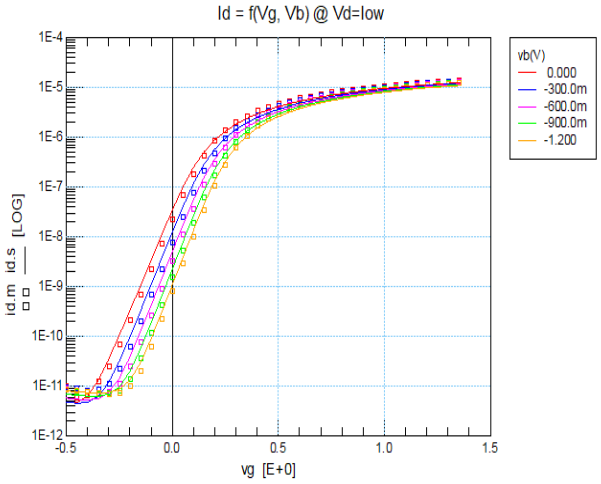


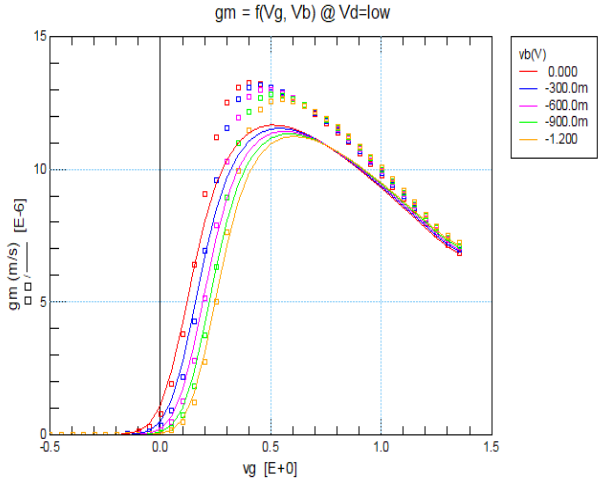


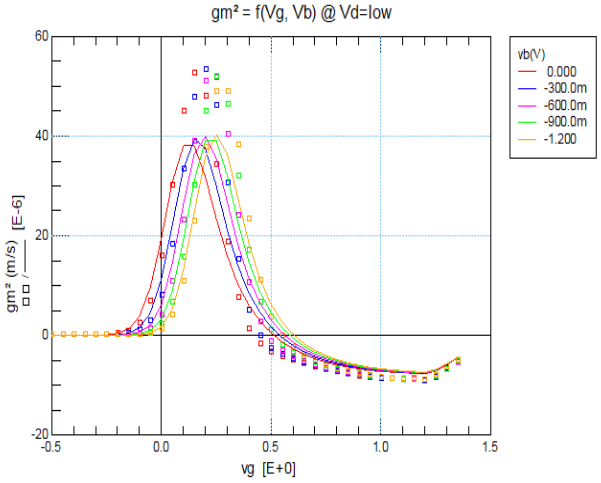


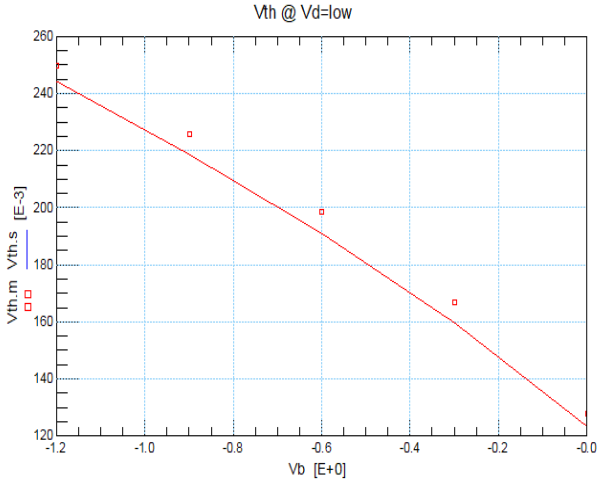
idvg, Vd = 0.05V, T = 125°C

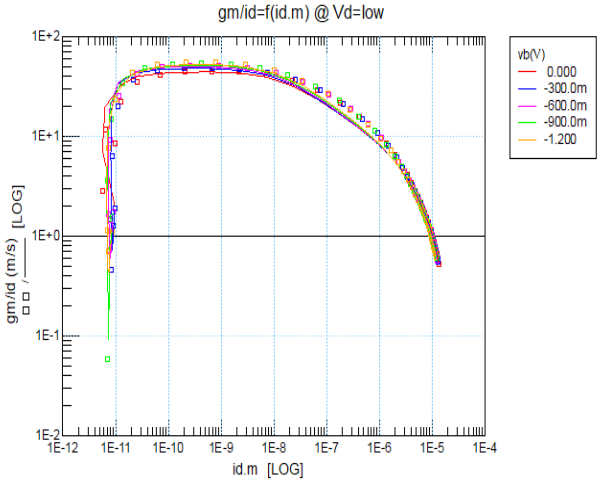


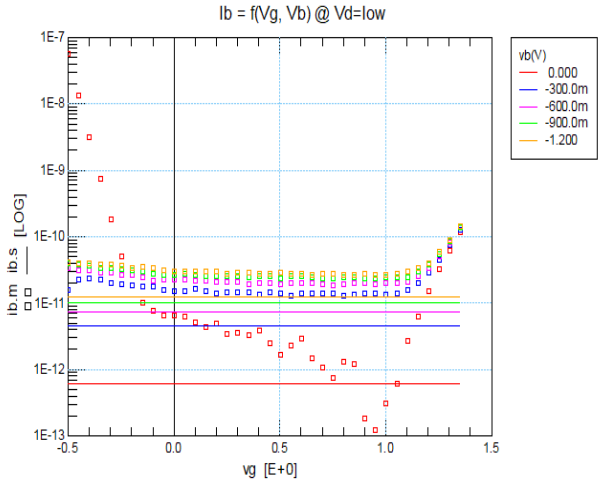


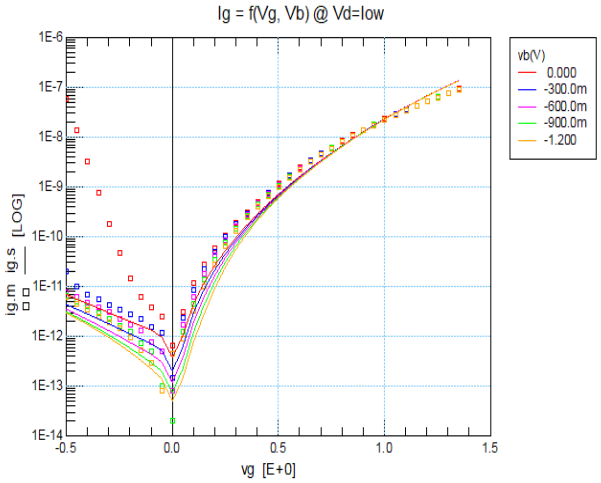




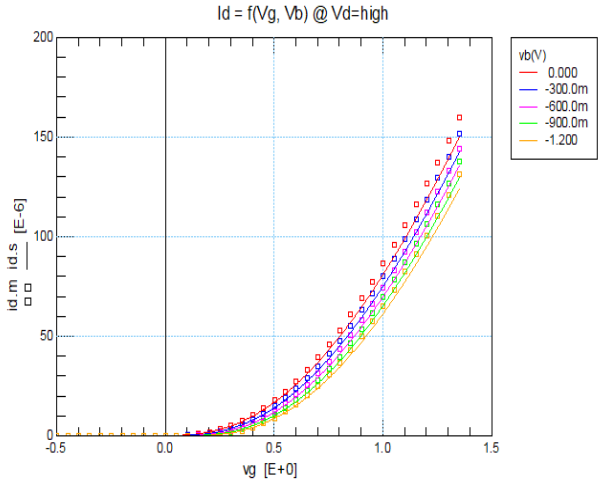


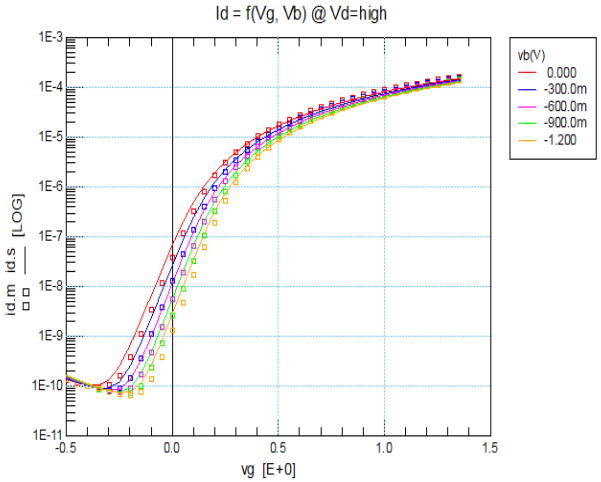


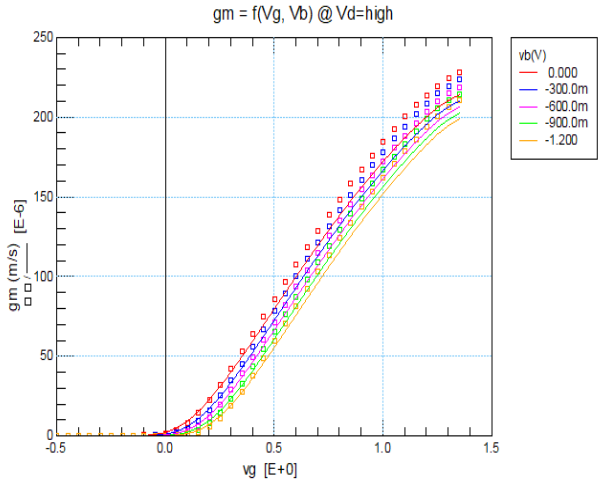


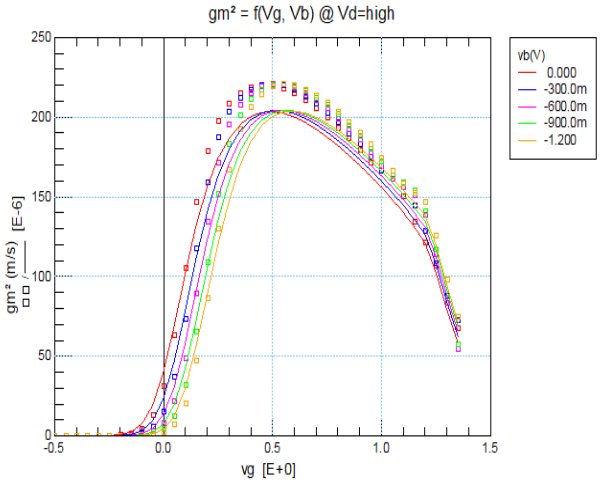


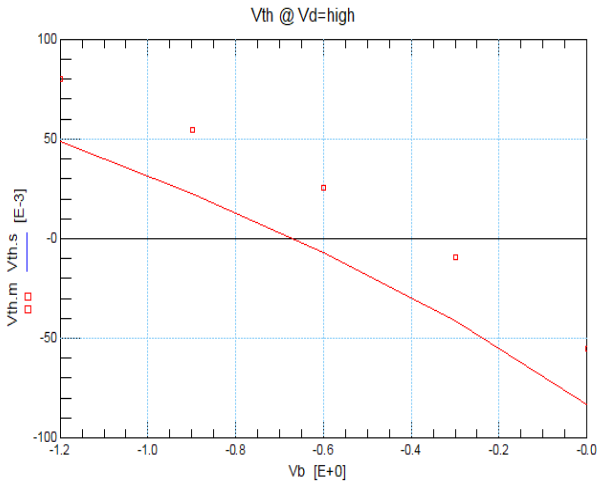
idvg, Vd = 1.2V, T = 125°C

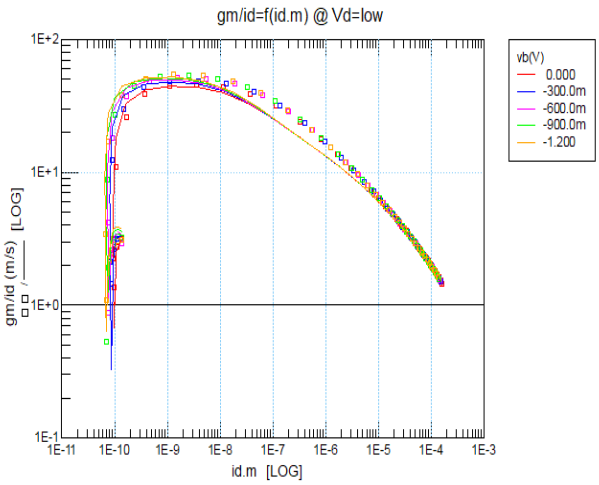


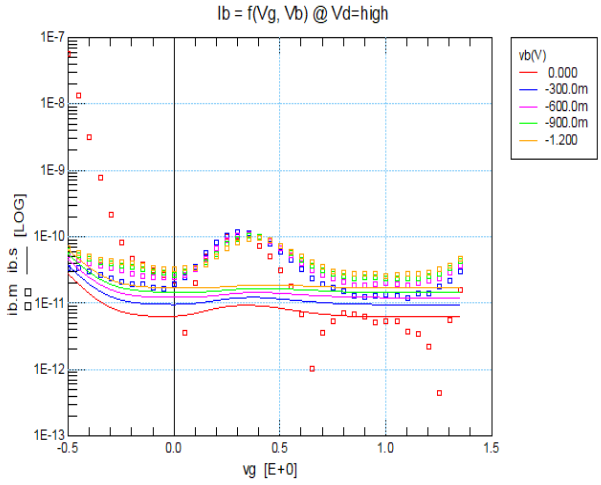


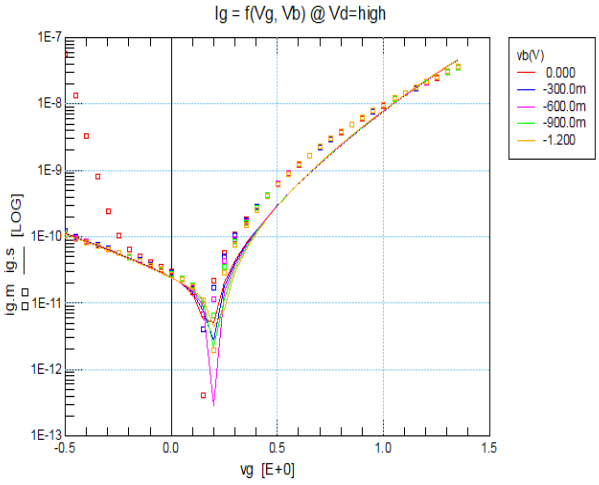




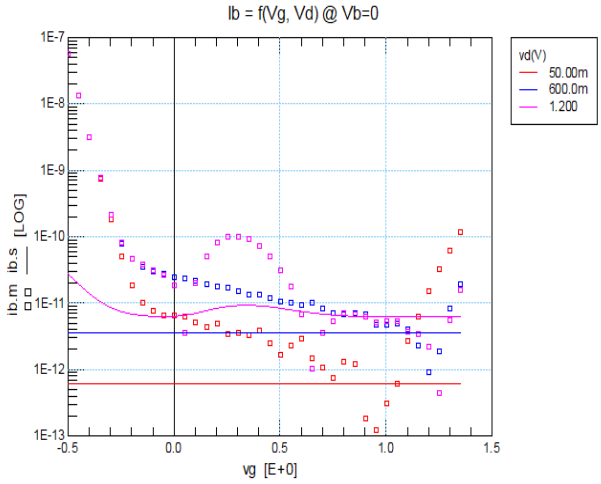


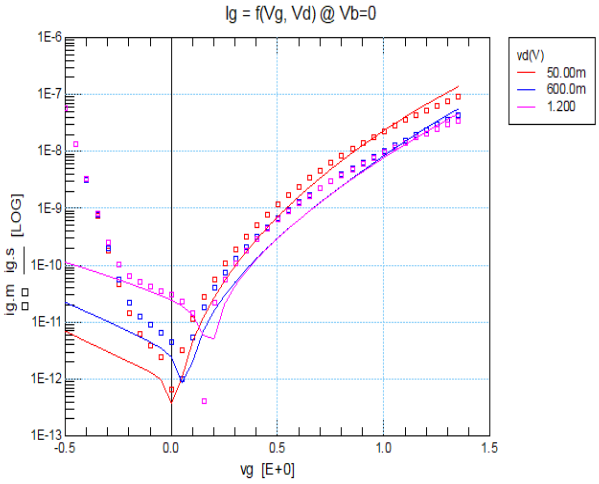




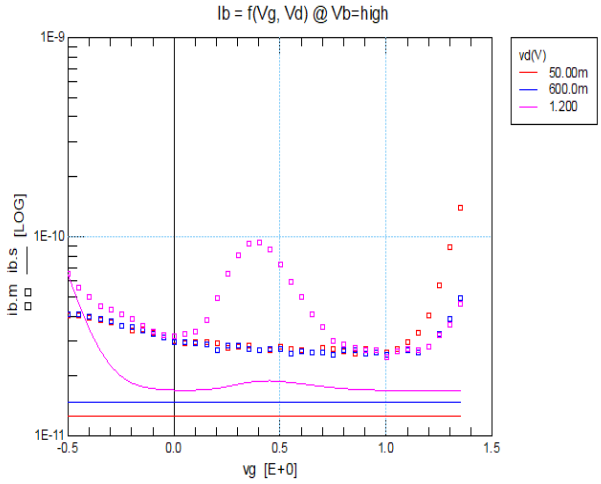


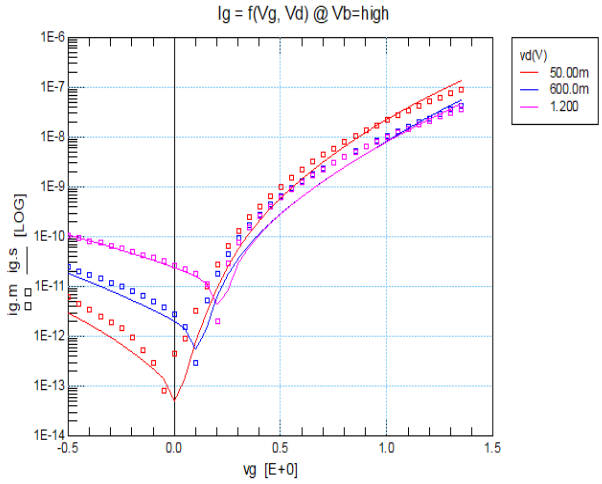
idvg, Ib,Ig, Vb = 0V, T = 125°C



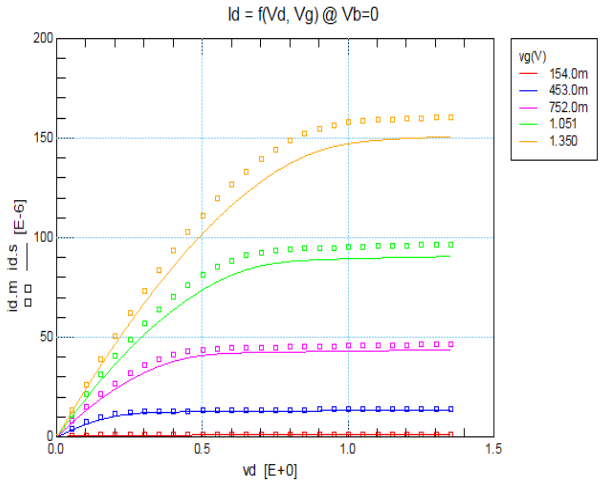


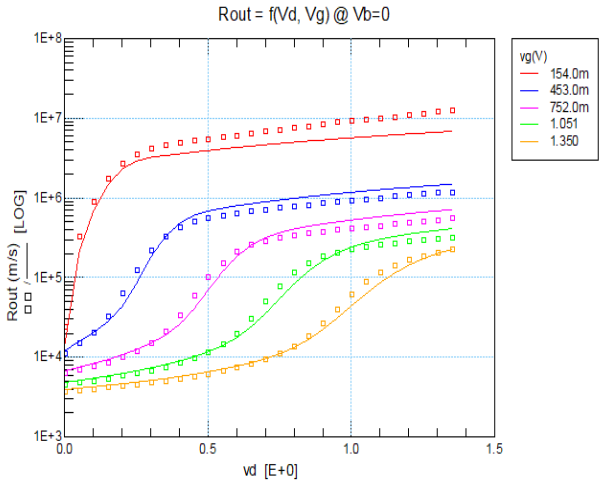
idvg, Ib,Ig, Vb = -1.2V, T = 125°C

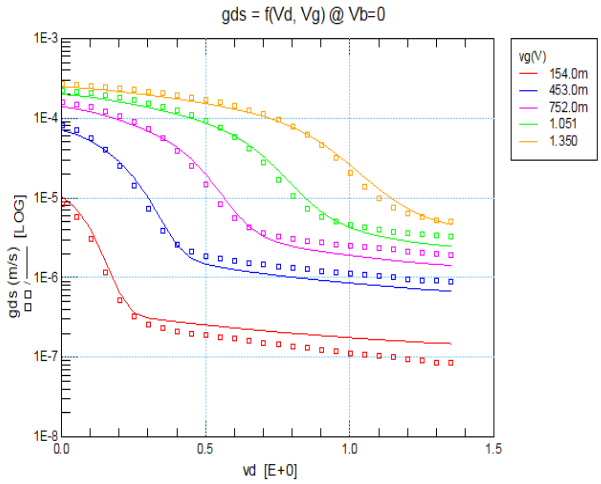




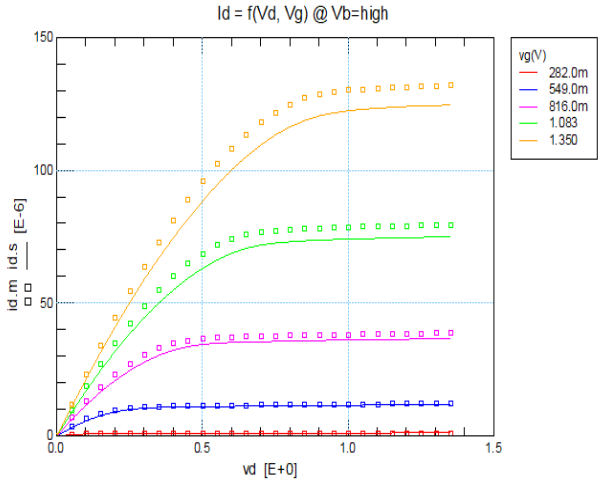
idvd, Vb = 0V, T = 125°C

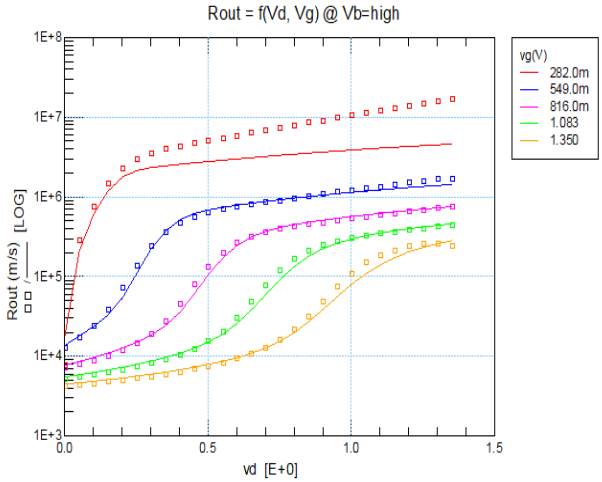


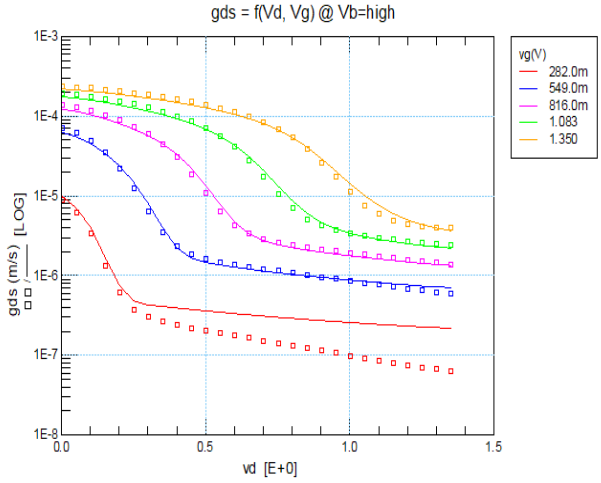




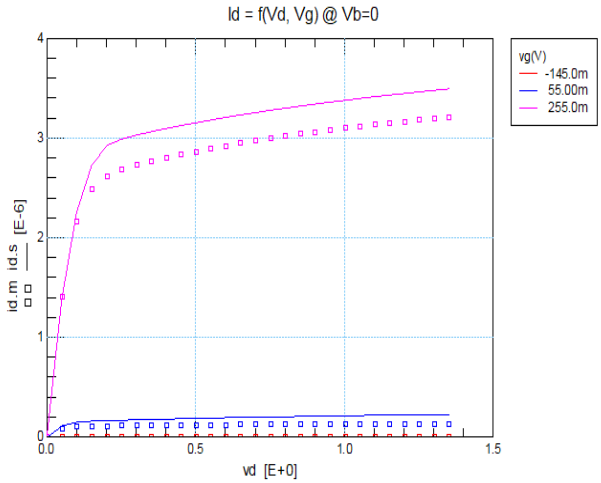
idvd_vbmin, Vb = -1.2V, T = 125°C

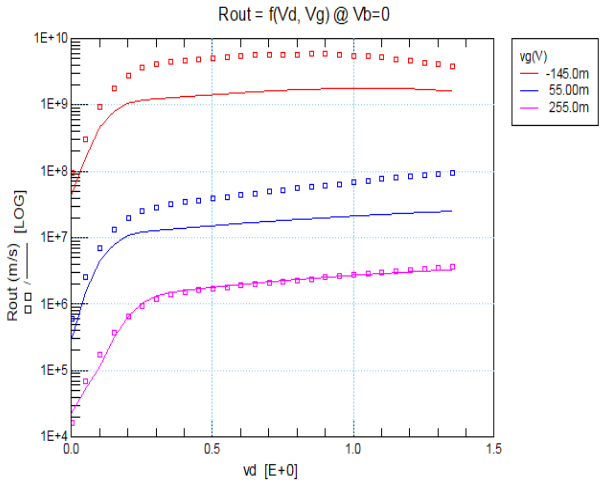


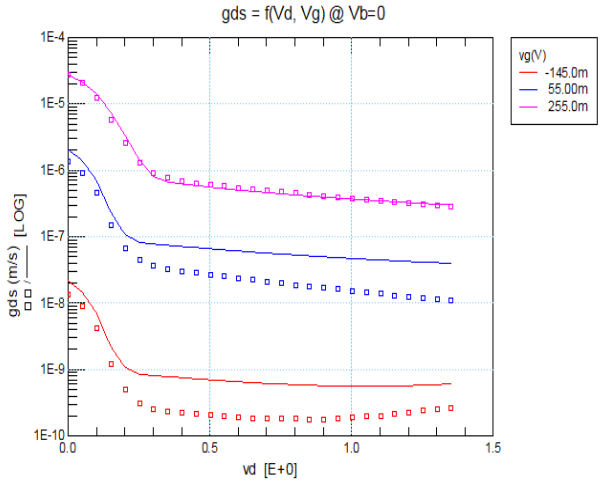




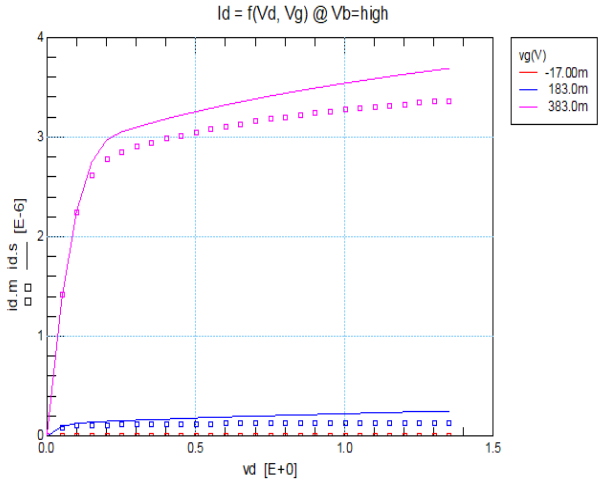
idvd_vth, Vb = 0V, T = 125°C

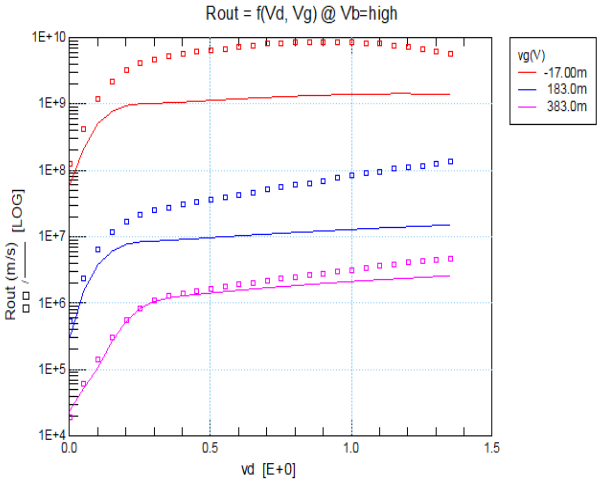


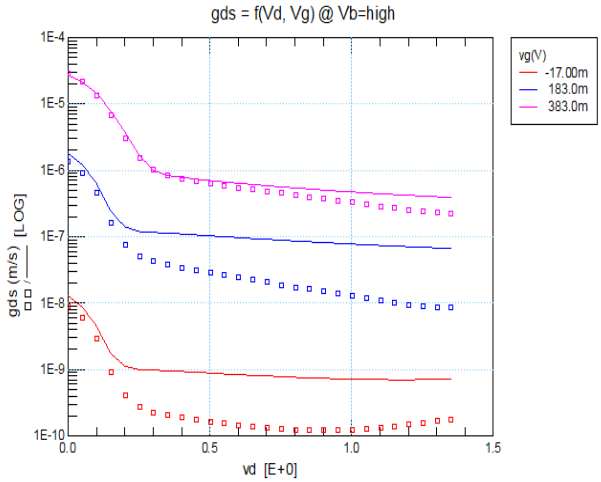




idvd_vth_vbmin, Vb = -1.2V, T = 125°C

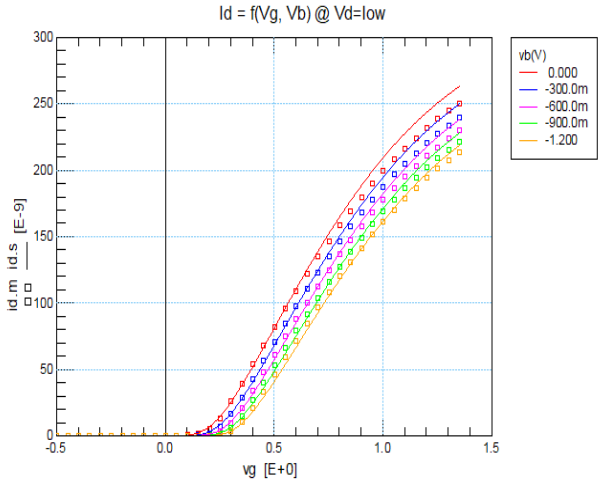


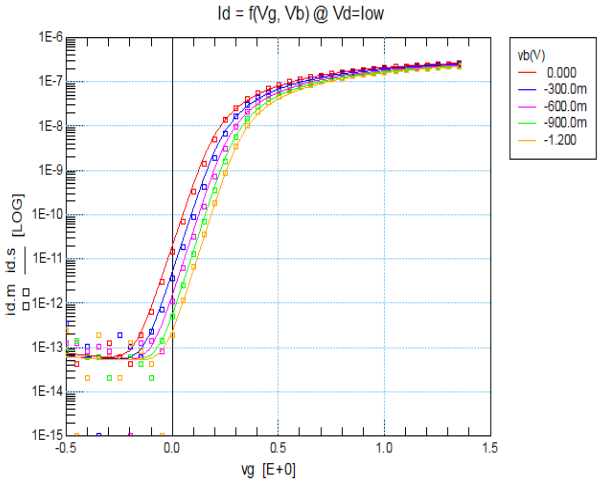


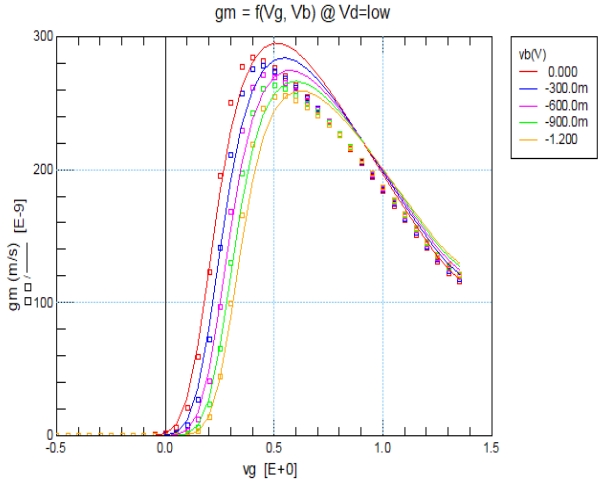


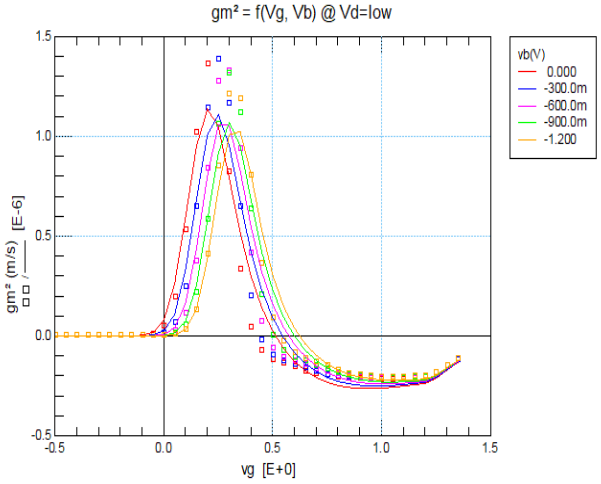
4.3 W0u15_L10u0_S547_3

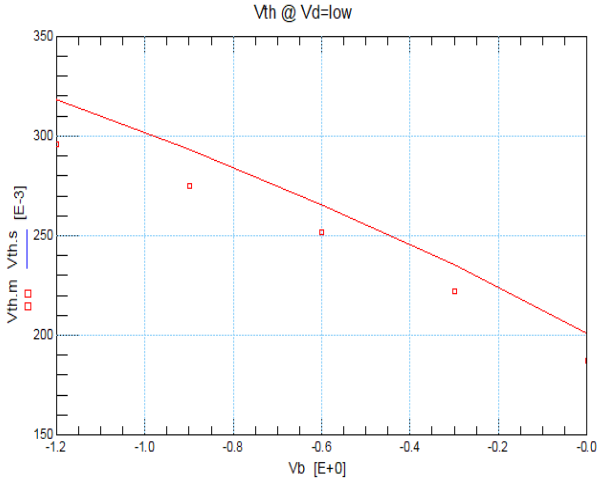
idvg, Vd = 0.05V, T = 27°C

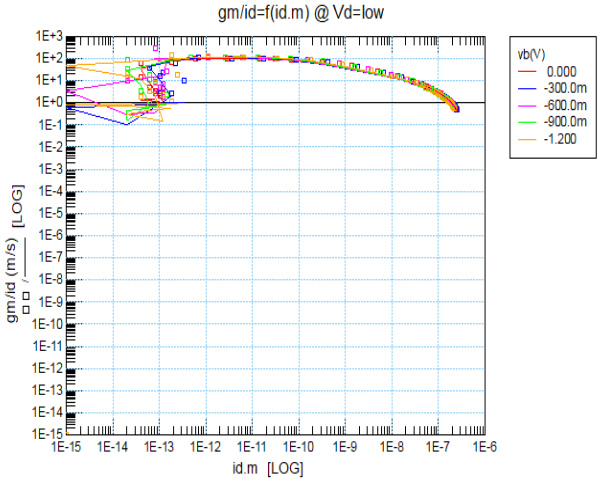


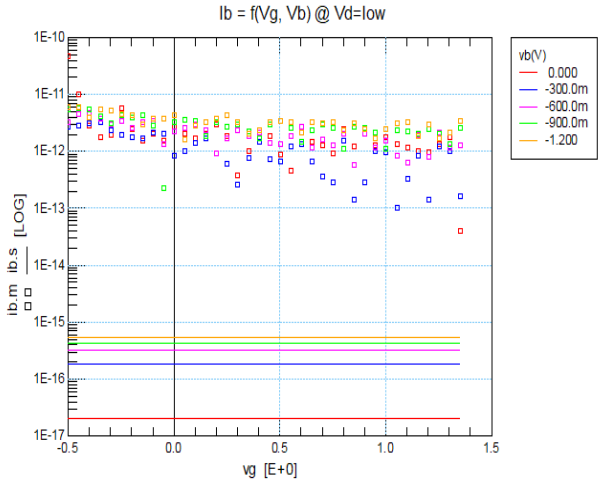


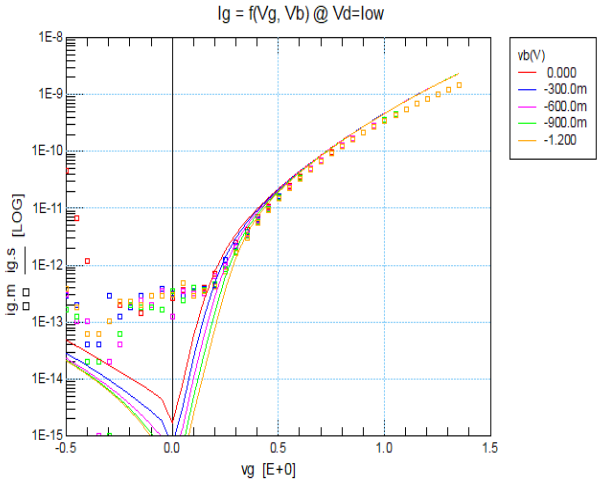




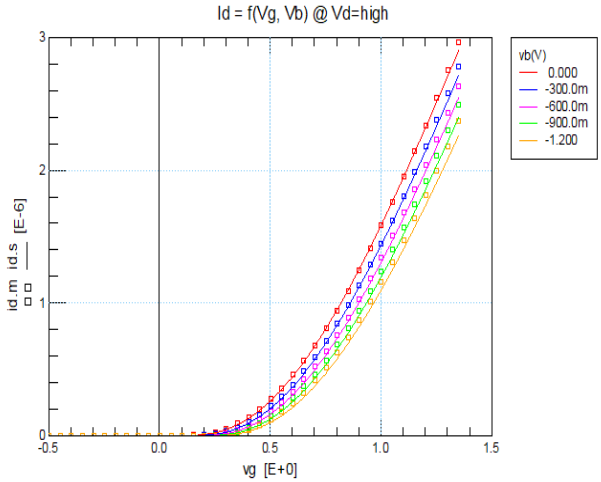


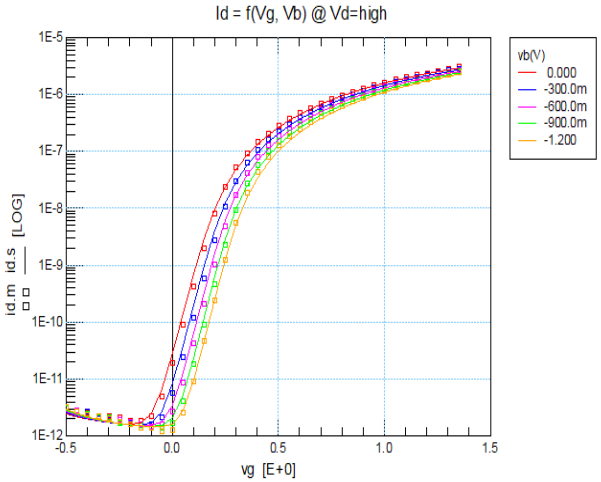


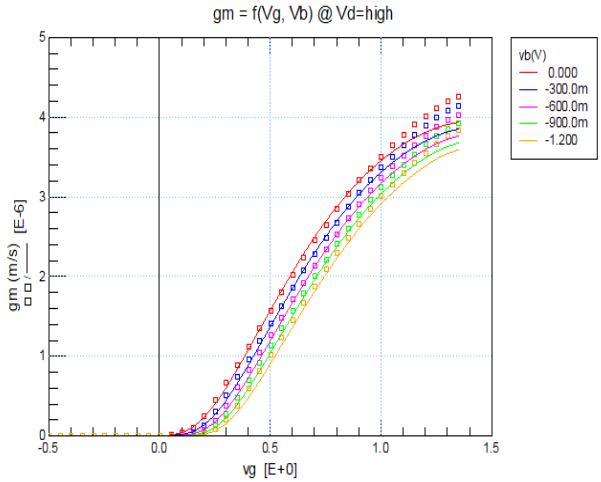


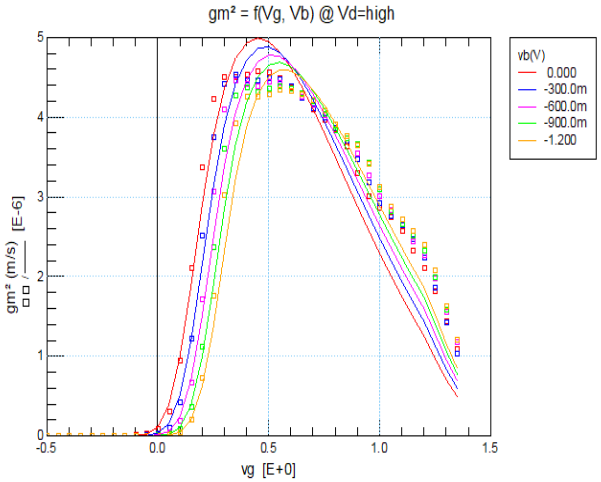


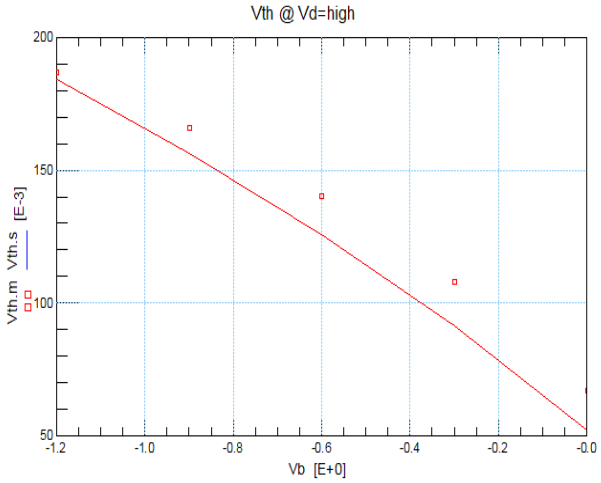
idvg, Vd = 1.2V, T = 27°C

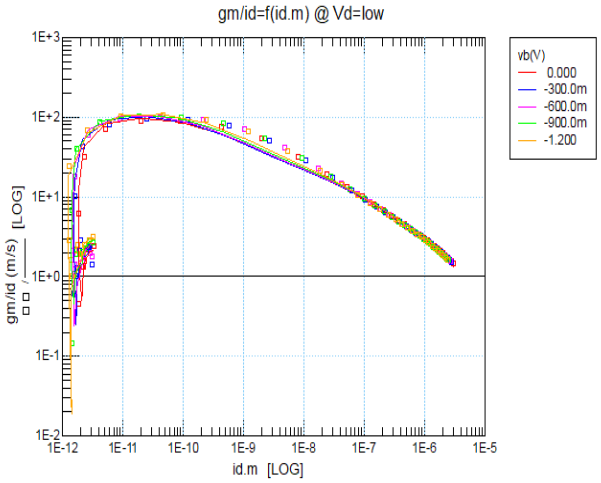


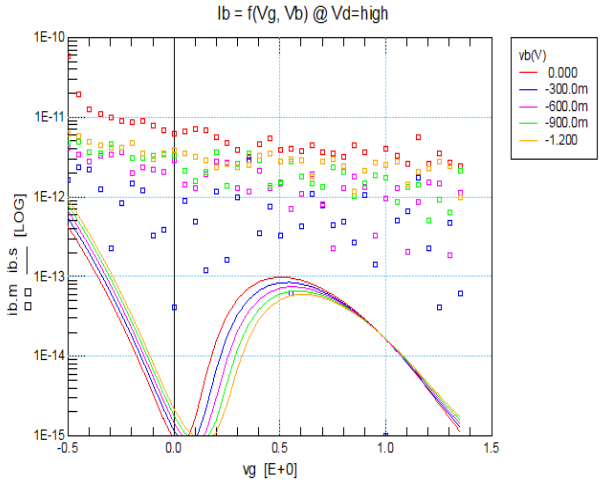


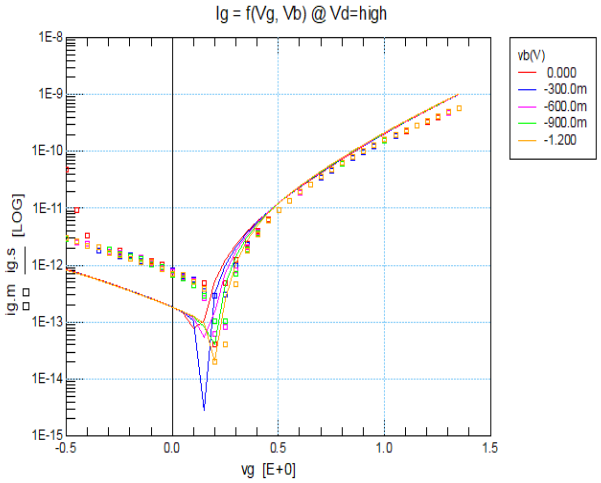




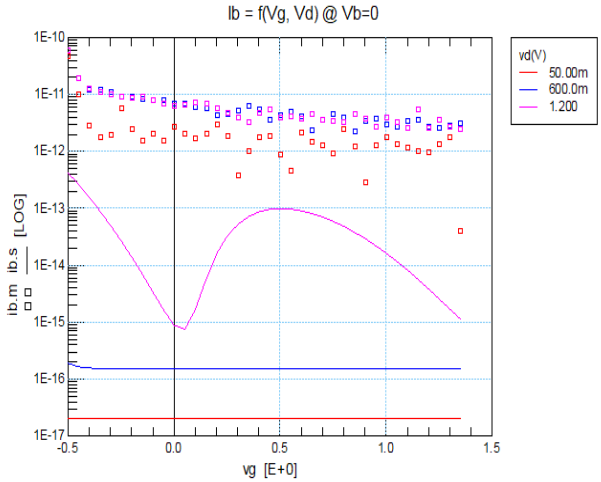


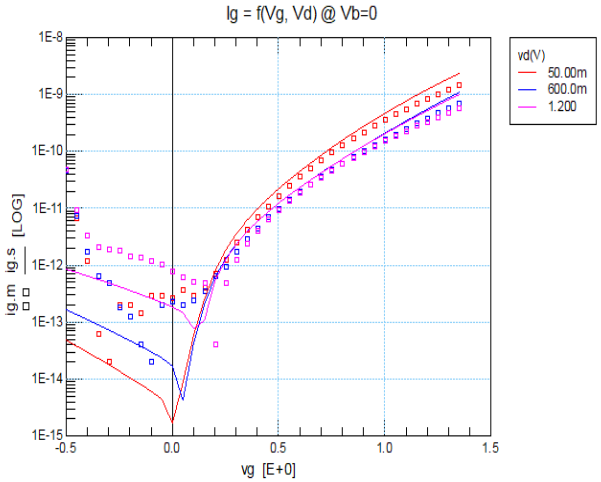




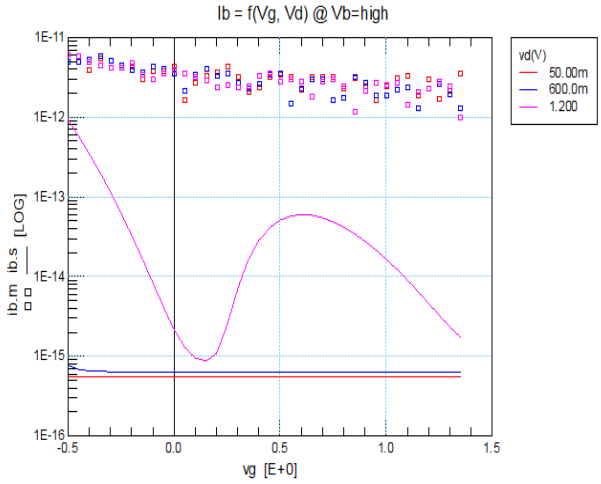


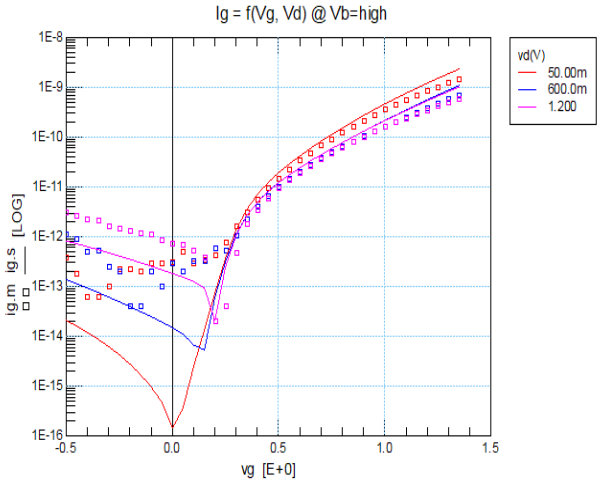
idvg, Ib,Ig, Vb = 0V, T = 27°C



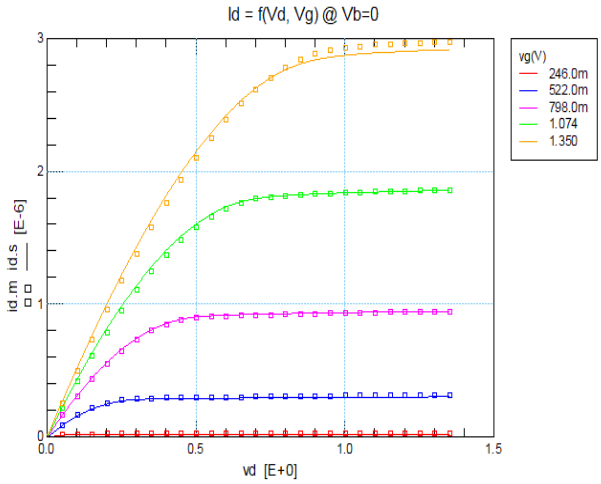


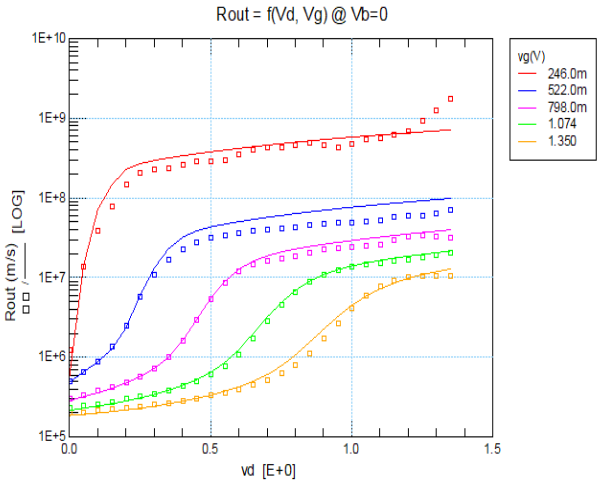
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

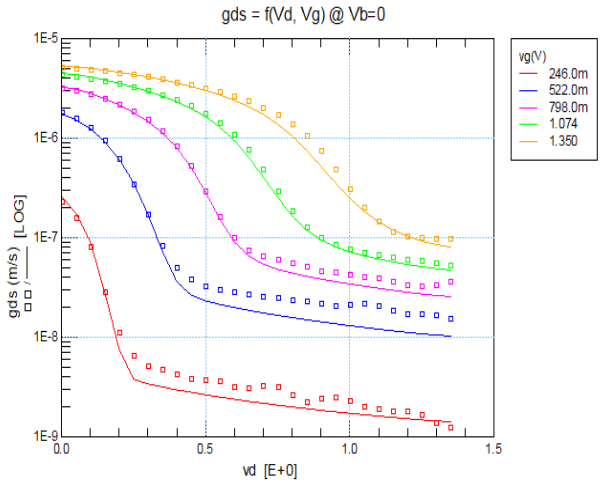




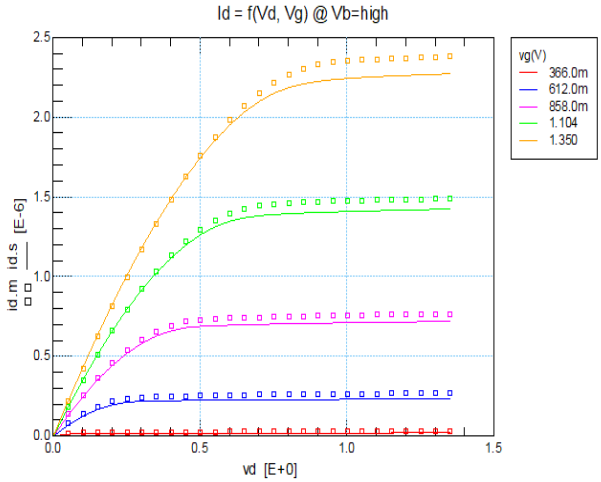
idvd, Vb = 0V, T = 27°C

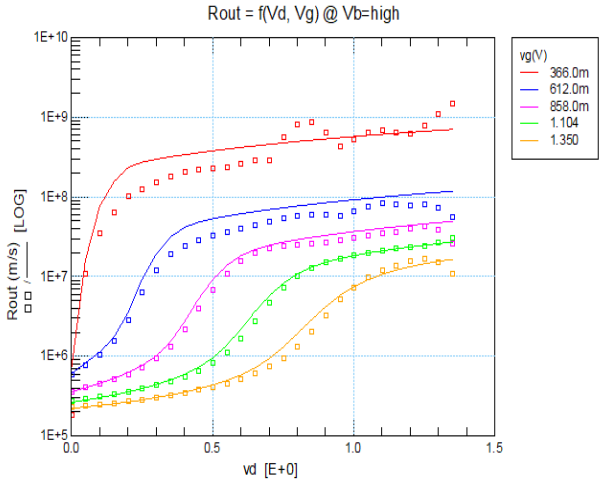


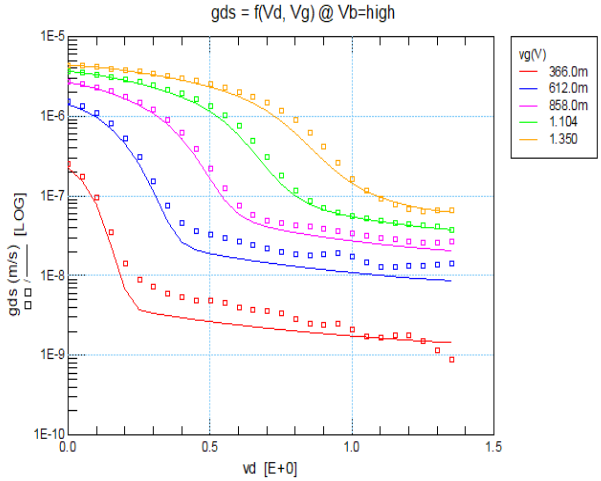




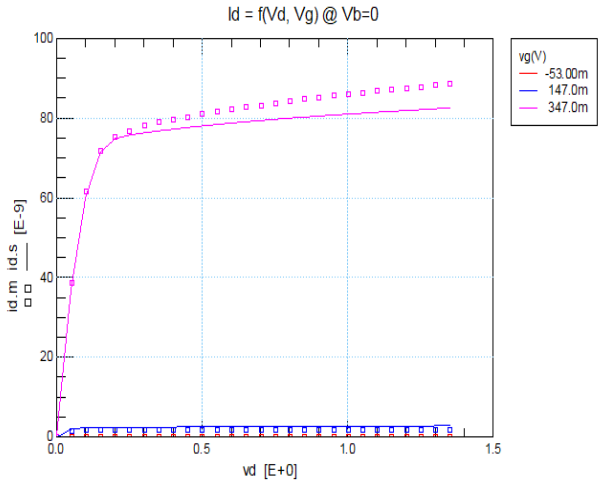
idvd_vbmin, Vb = -1.2V, T = 27°C

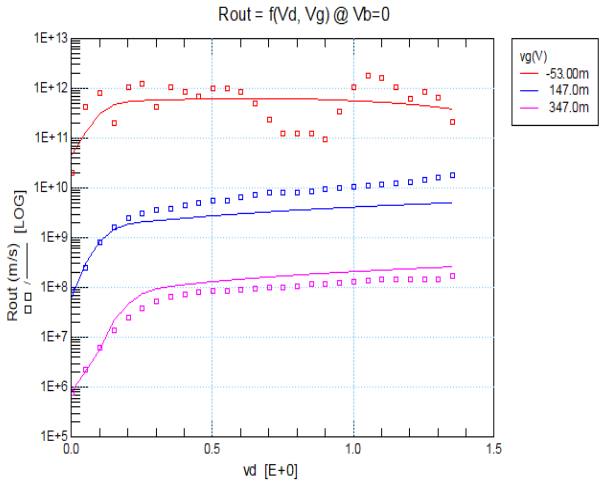


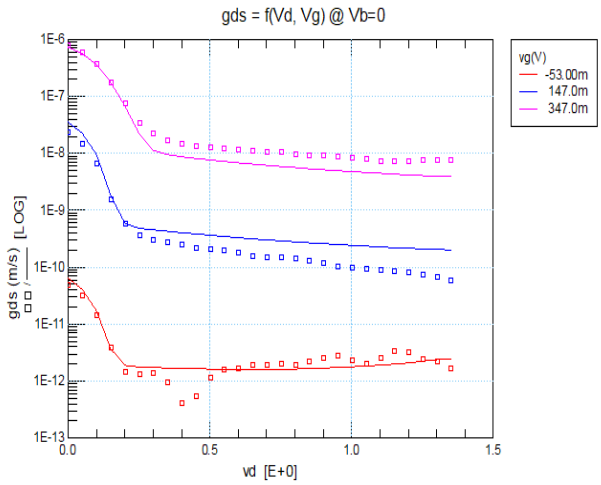




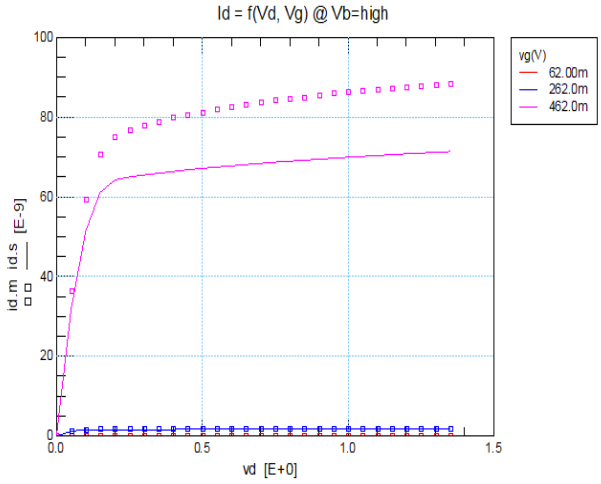
idvd_vth, Vb = 0V, T = 27°C

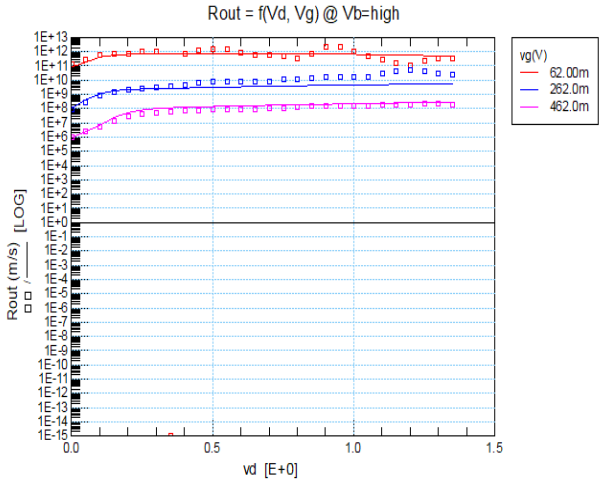


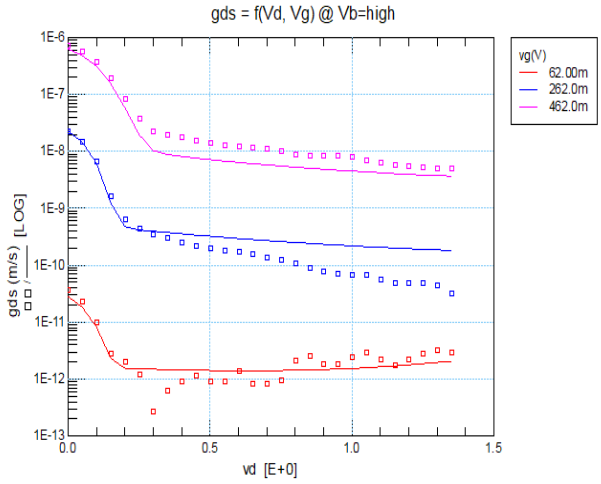




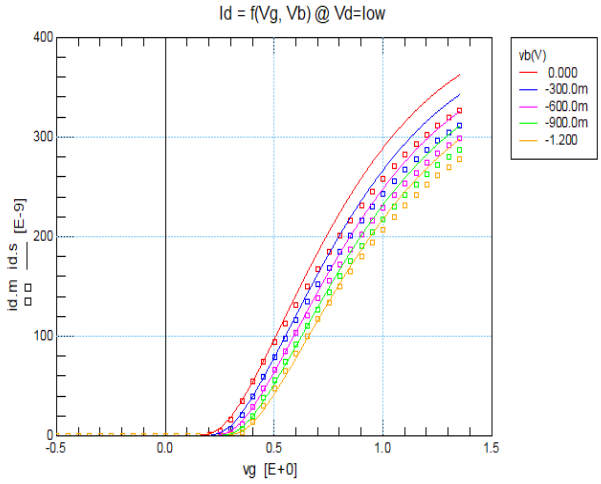
idvd_vth_vbmin, Vb = -1.2V, T = 27°C

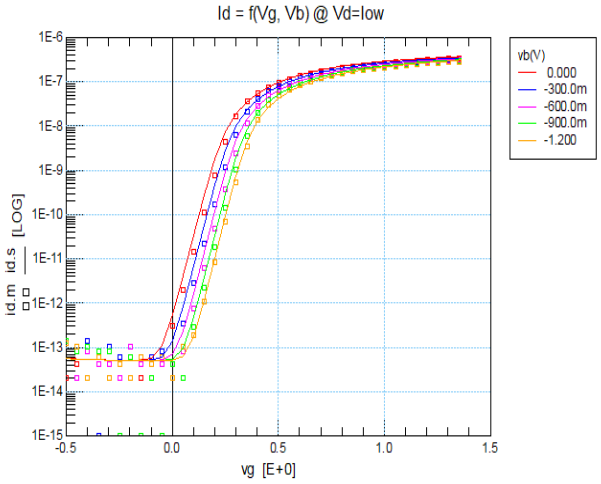


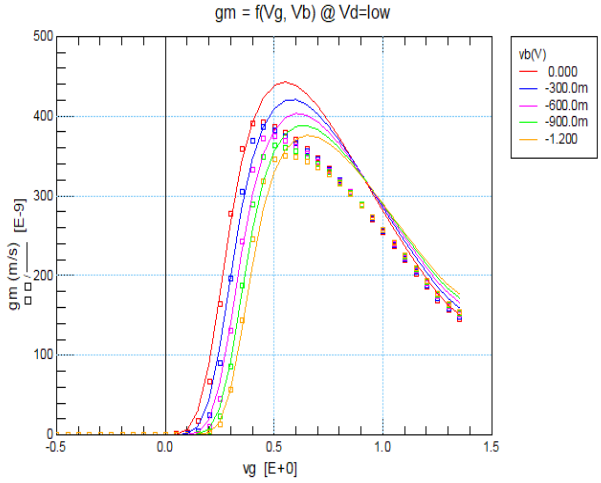


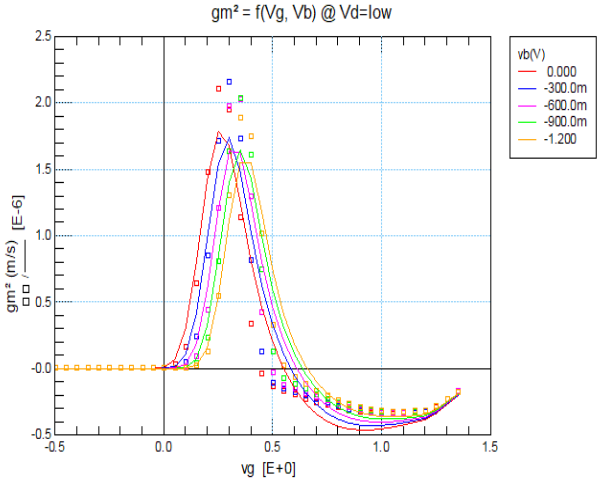


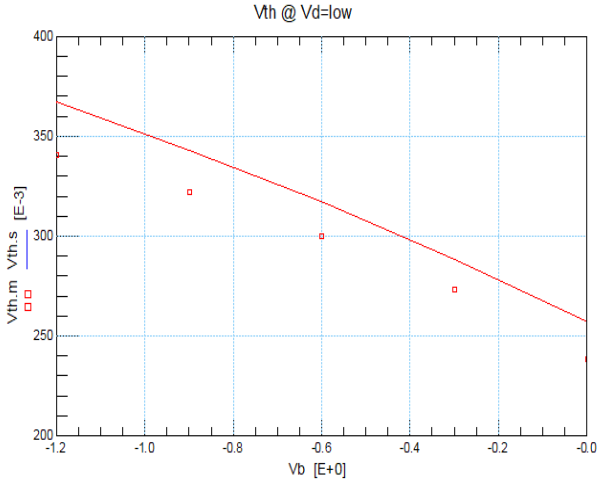
idvg, Vd = 0.05V, T = -40°C

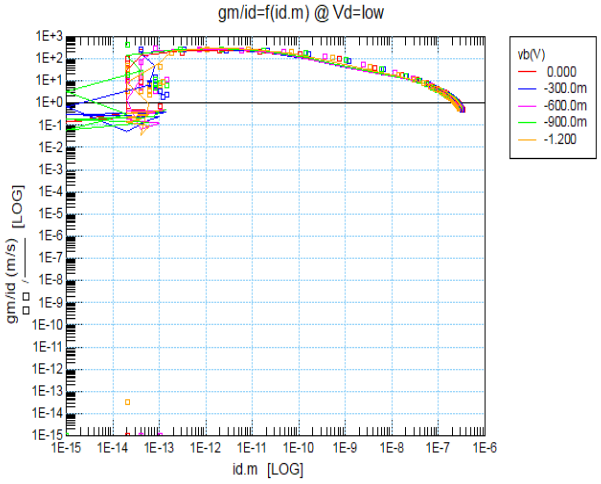


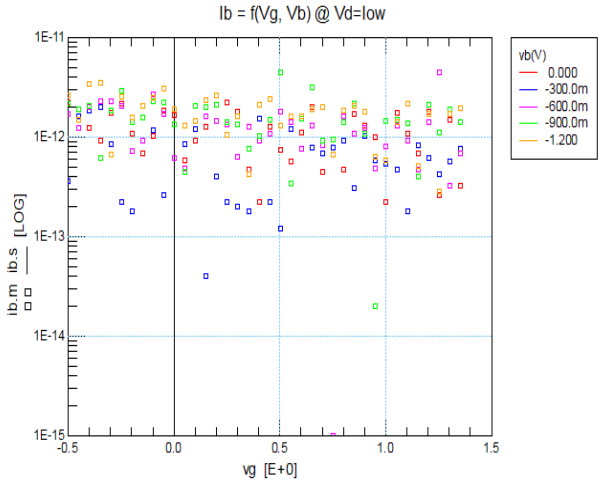


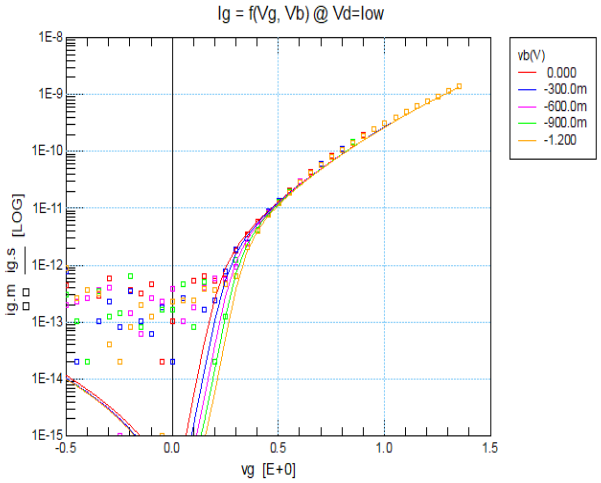




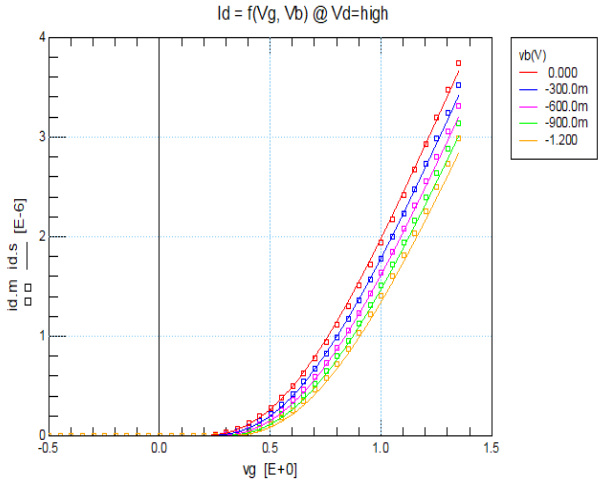


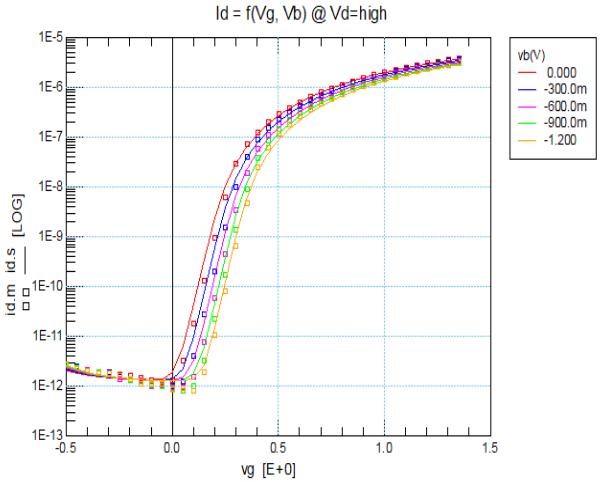


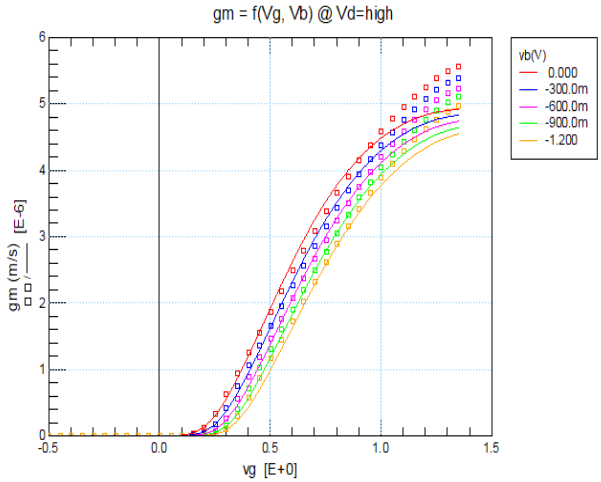


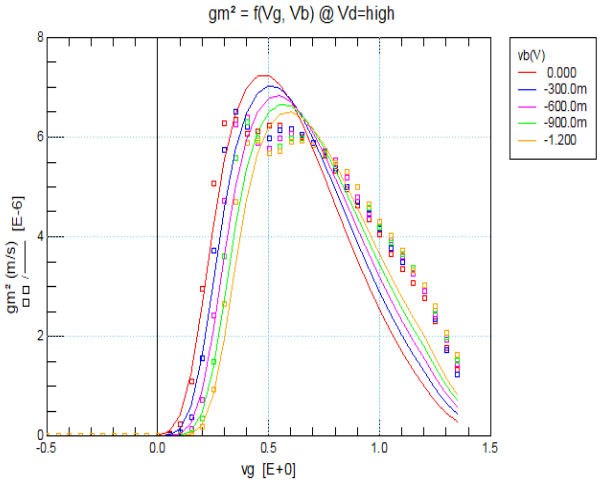


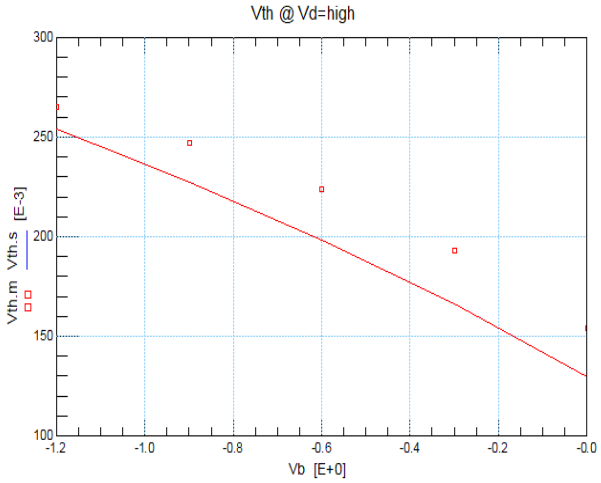
idvg, Vd = 1.2V, T = -40°C

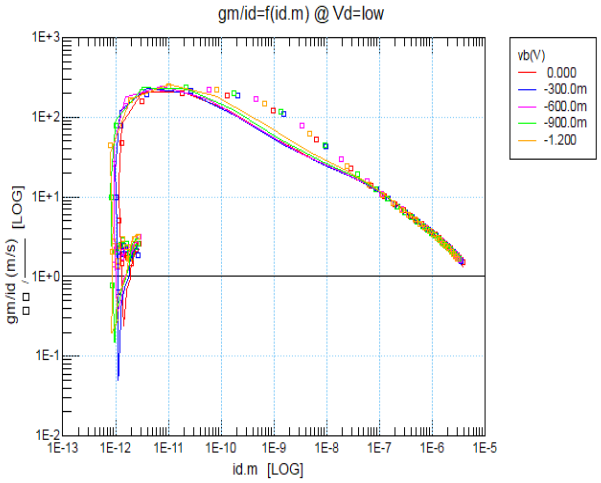


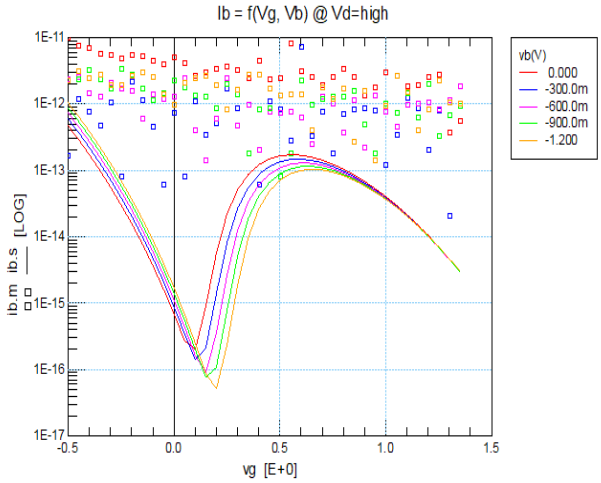


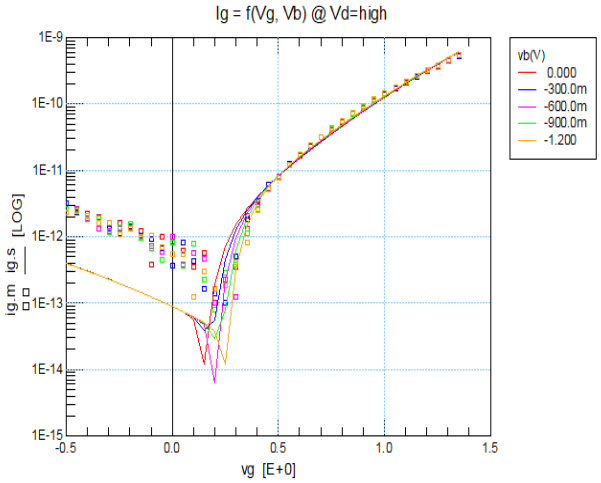




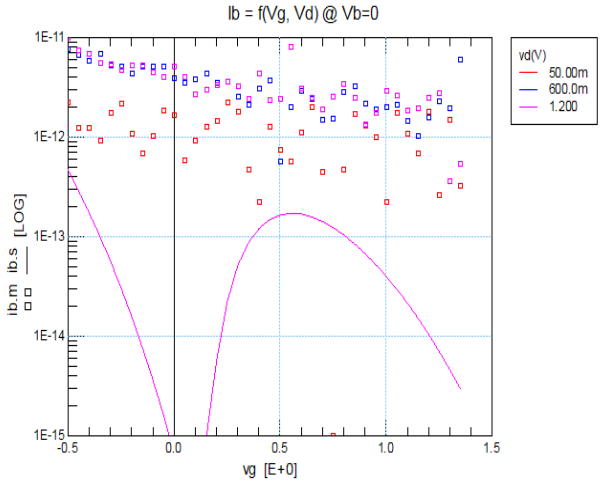


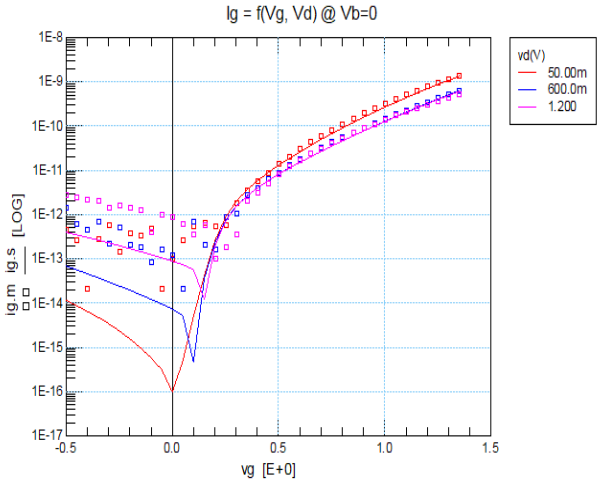




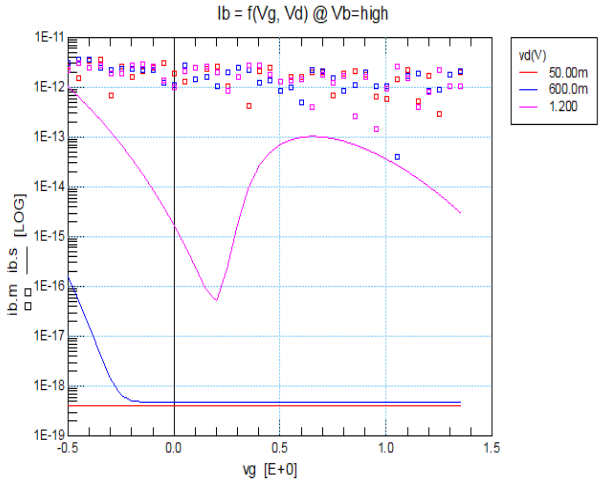


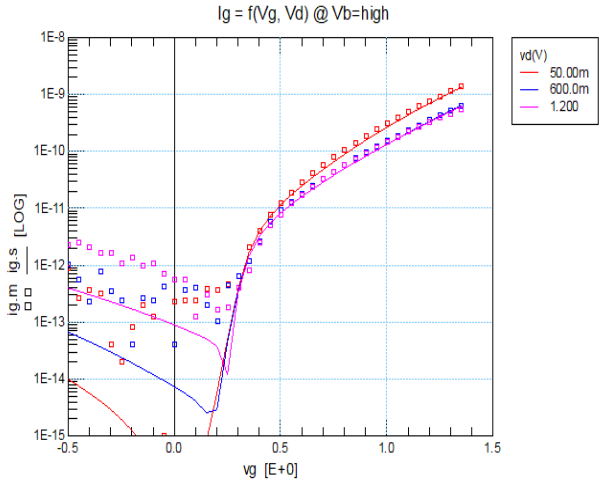
idvg, Ib,Ig, Vb = 0V, T = -40°C



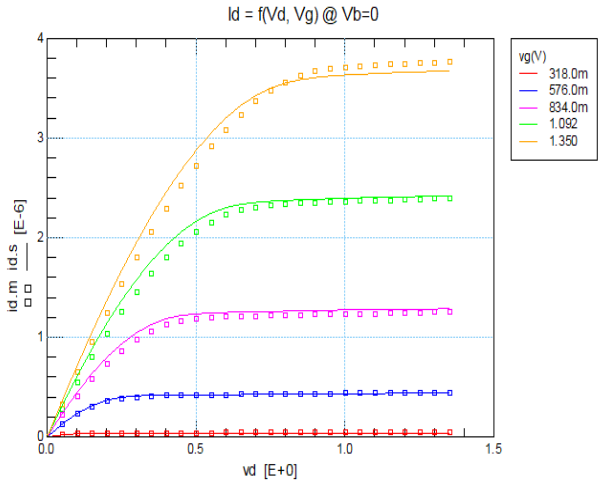


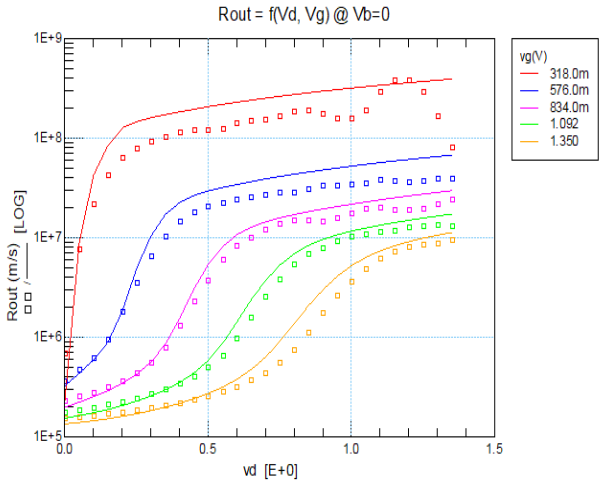
idvg, Ib,Ig, Vb = -1.2V, T = -40°C

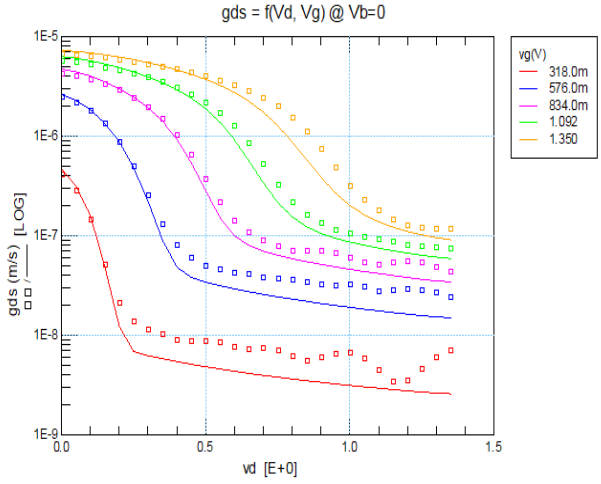




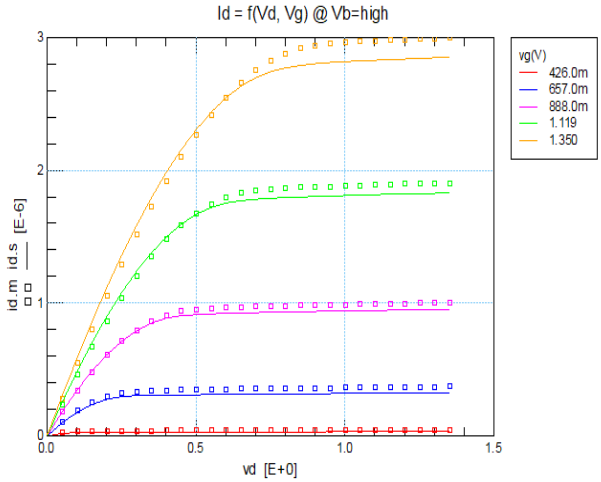
idvd, Vb = 0V, T = -40°C

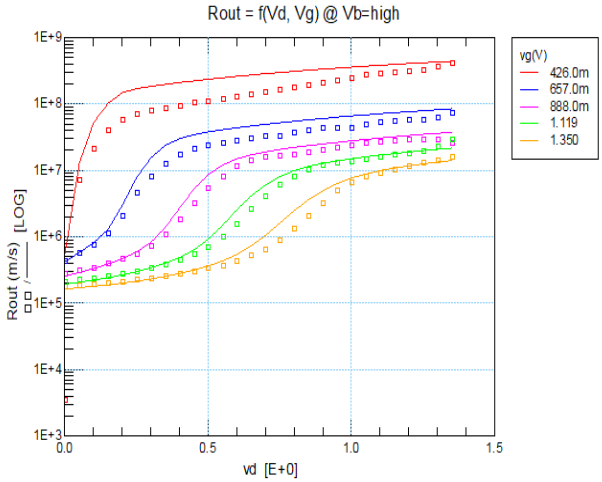


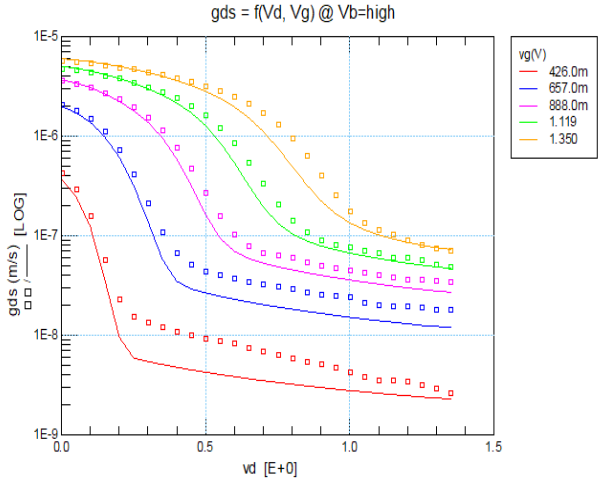




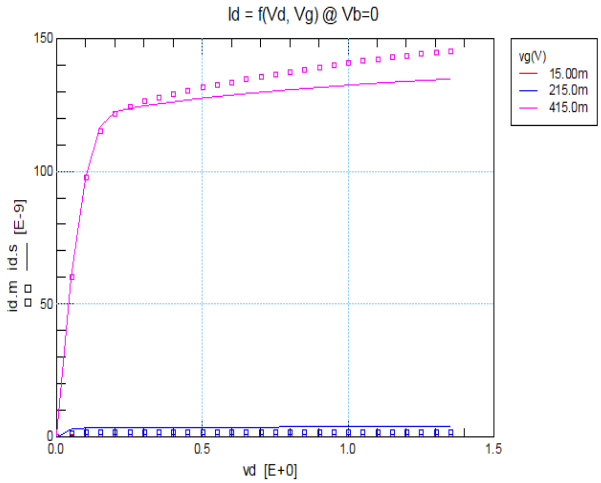
idvd_vbmin, Vb = -1.2V, T = -40°C

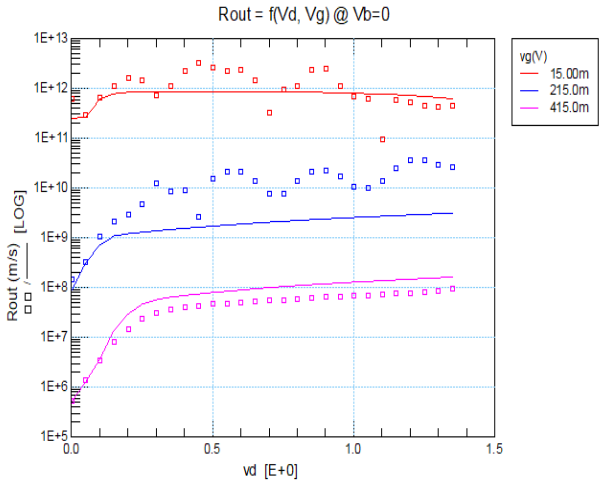


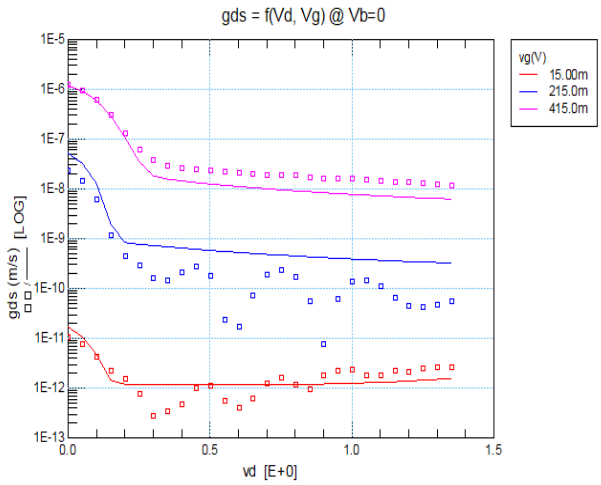




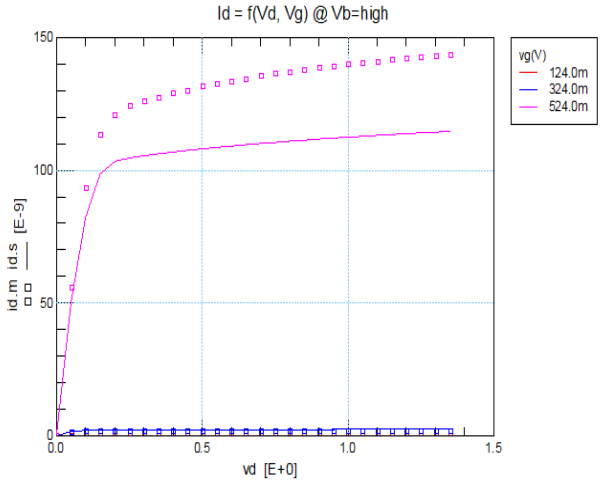
idvd_vth, Vb = 0V, T = -40°C

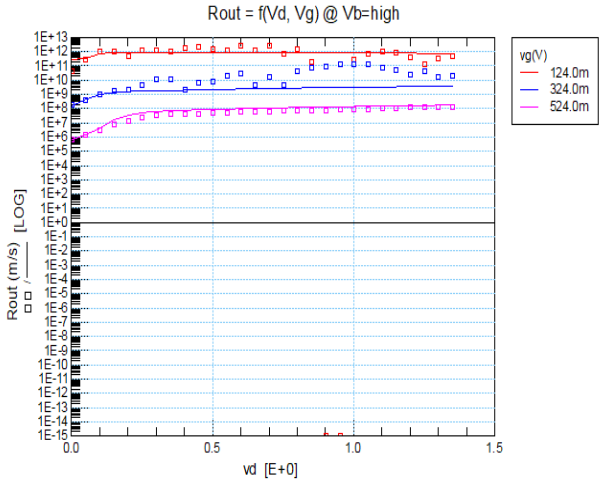


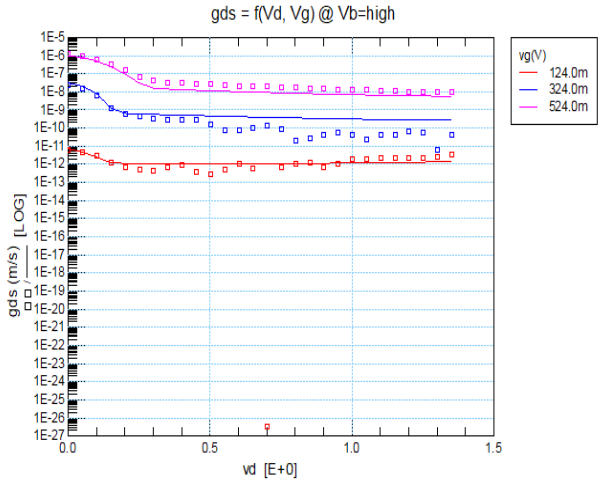




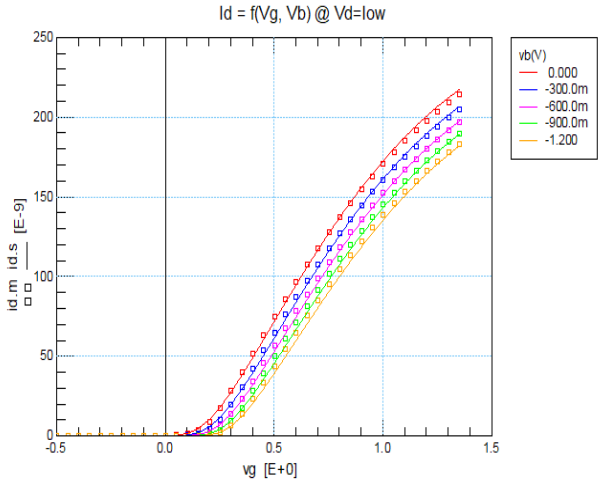
idvd_vth_vbmin, Vb = -1.2V, T = -40°C

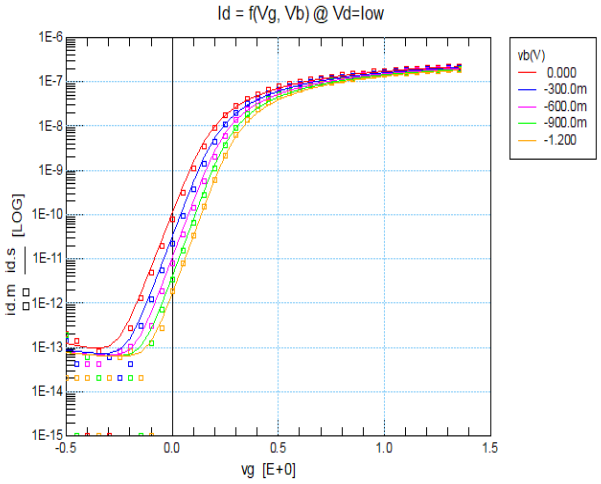


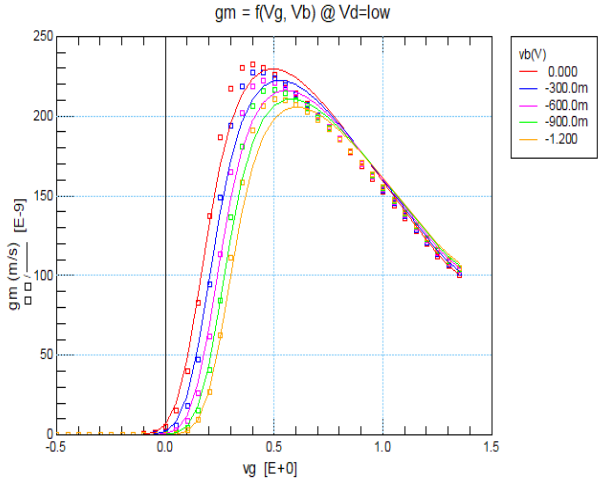


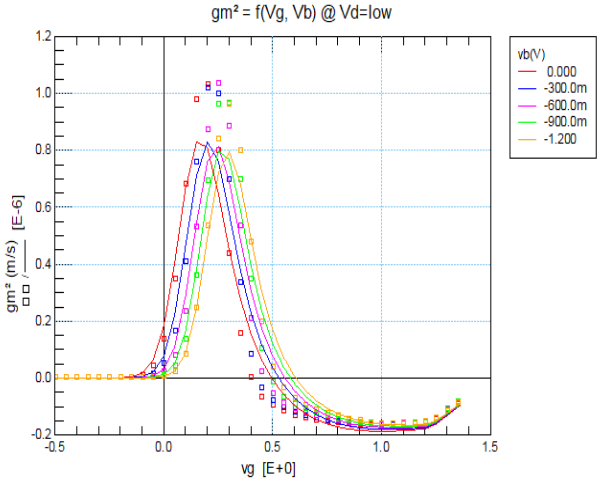


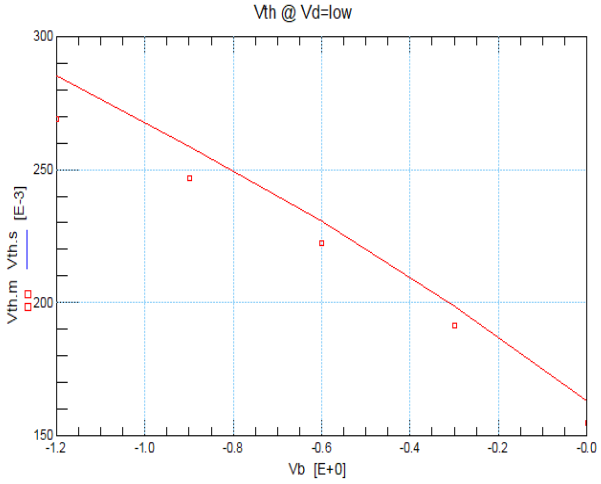
idvg, Vd = 0.05V, T = 70°C

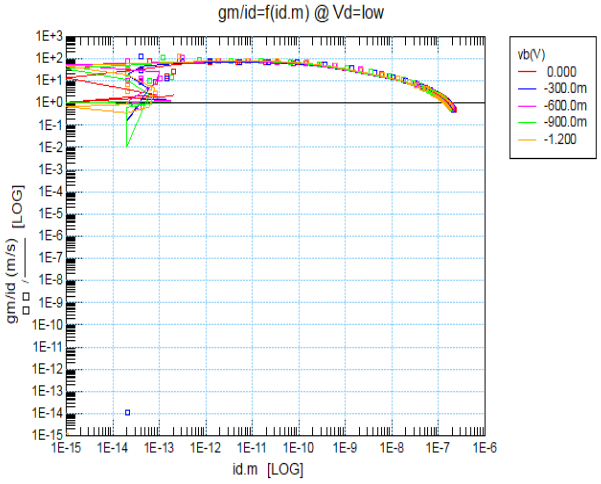


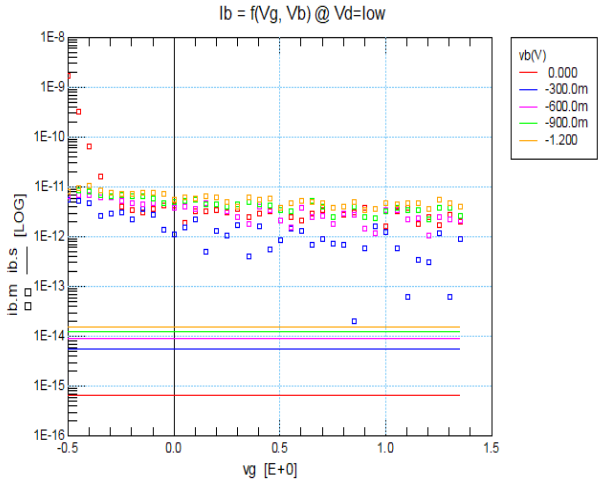


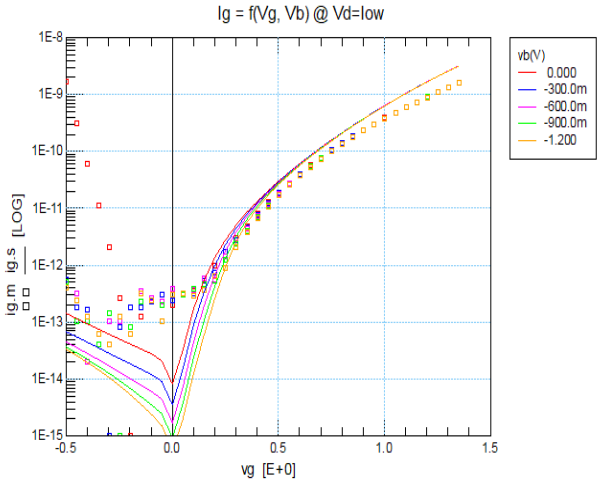




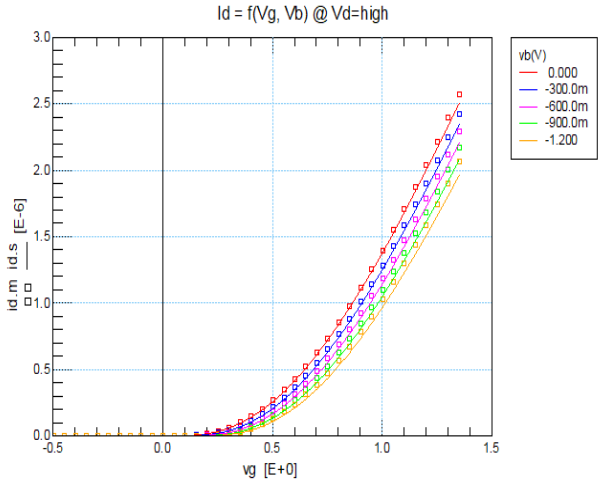


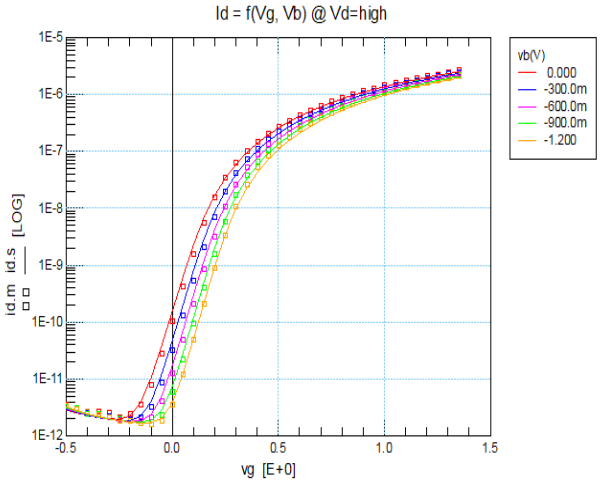


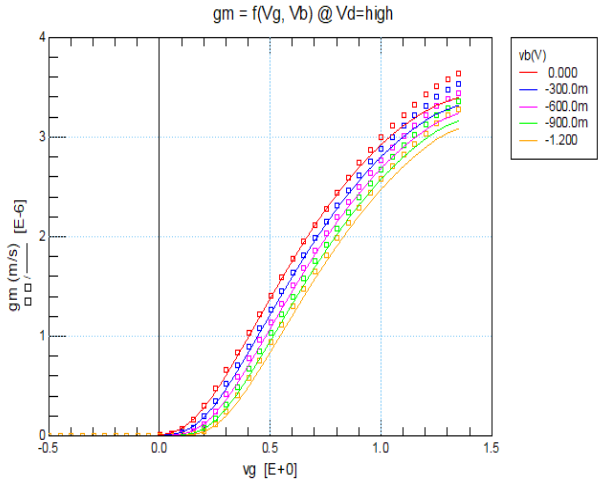


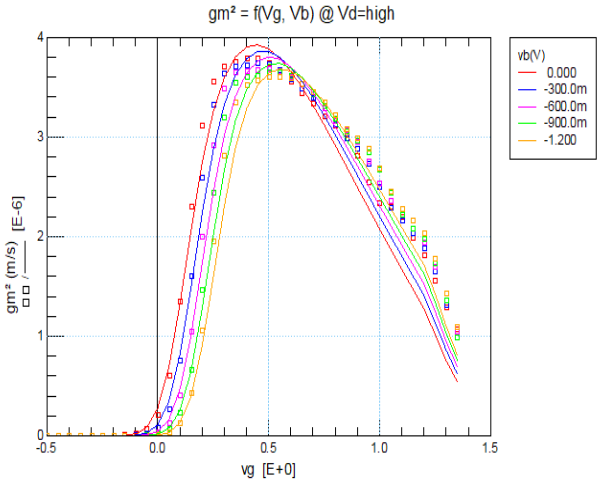


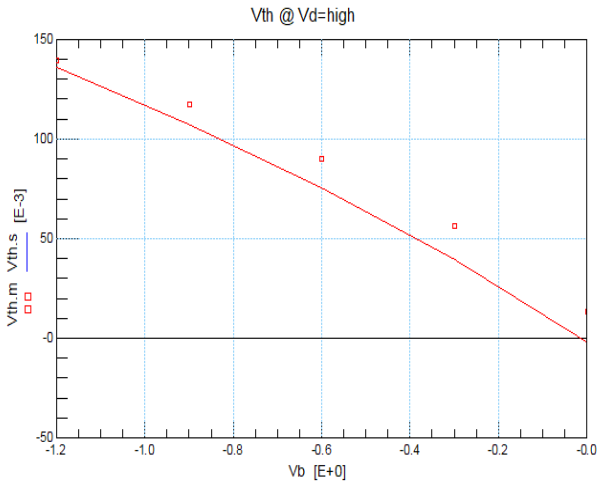
idvg, Vd = 1.2V, T = 70°C

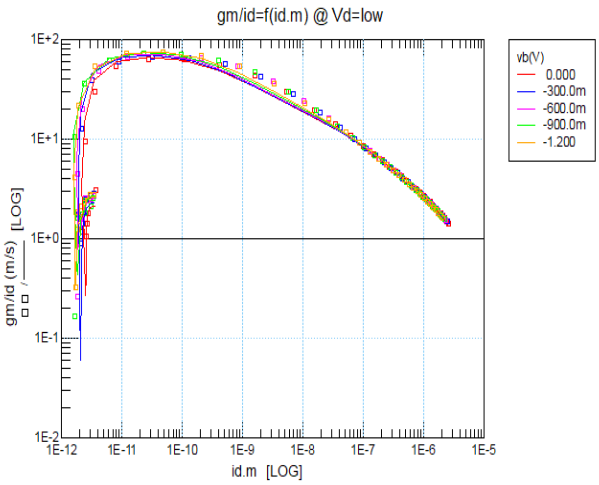


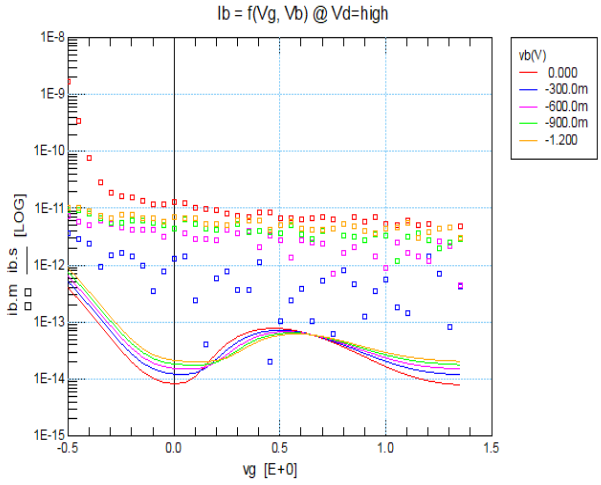


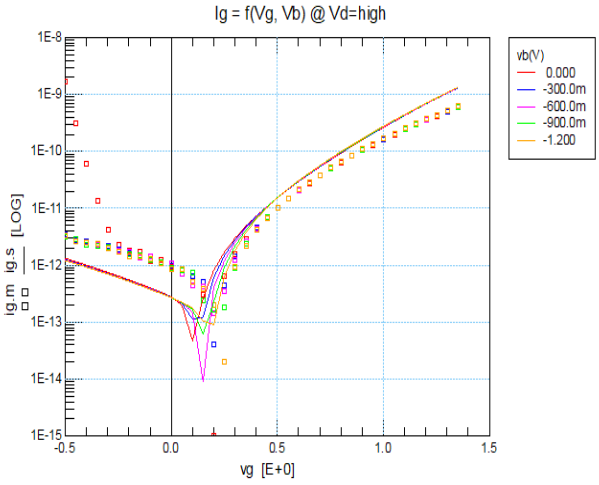




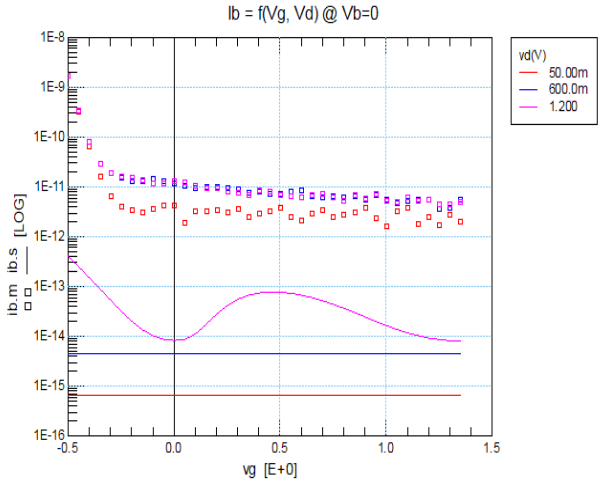


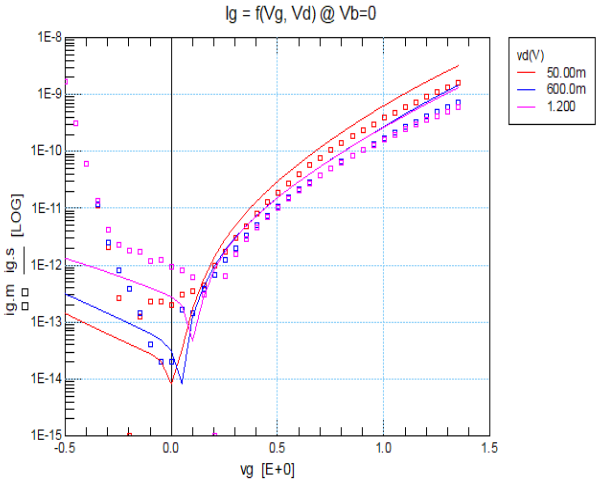




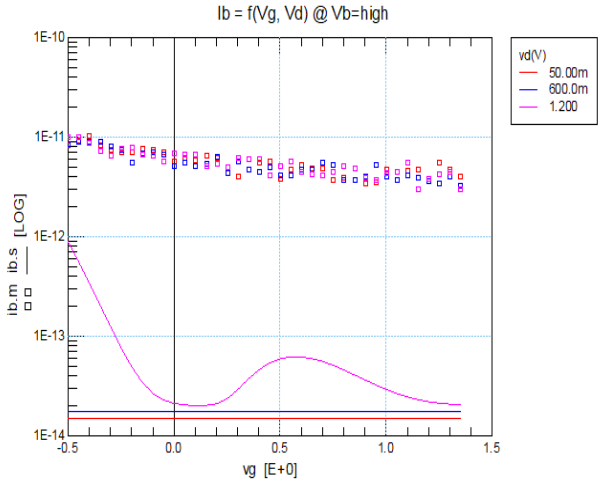


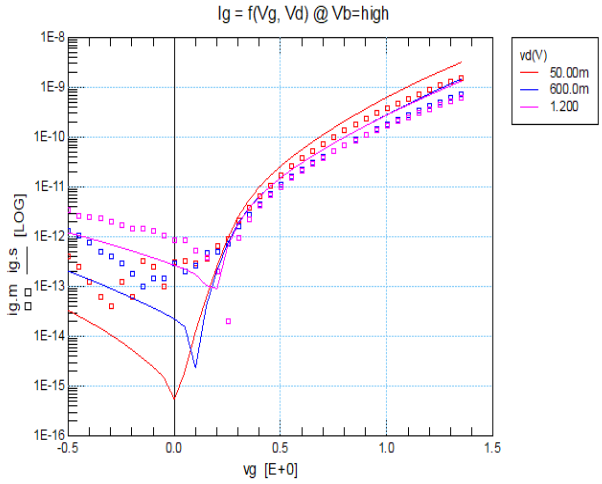
idvg, Ib,Ig, Vb = 0V, T = 70°C



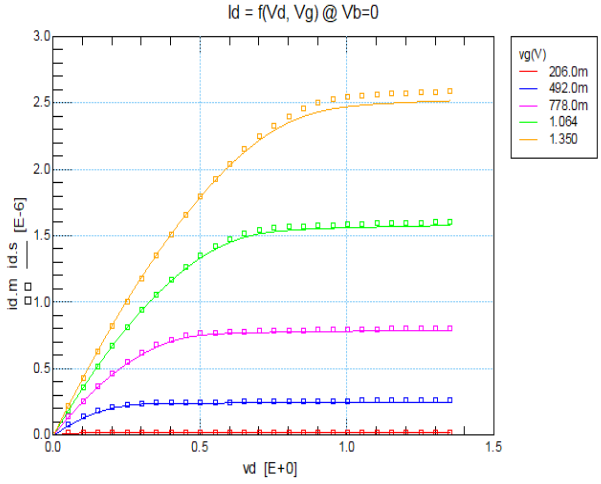


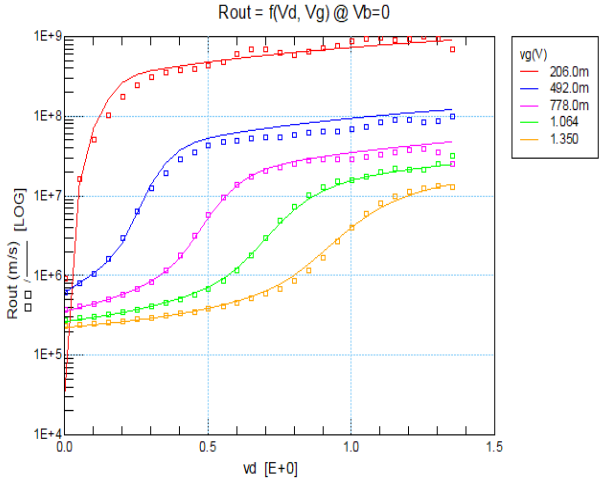
idvg, Ib,Ig, Vb = -1.2V, T = 70°C

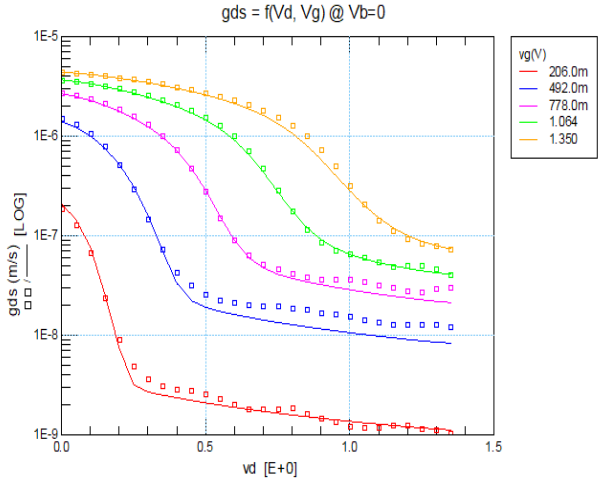




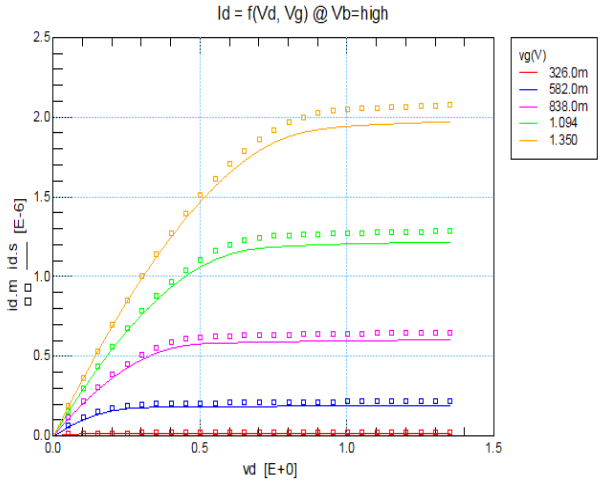
idvd, Vb = 0V, T = 70°C

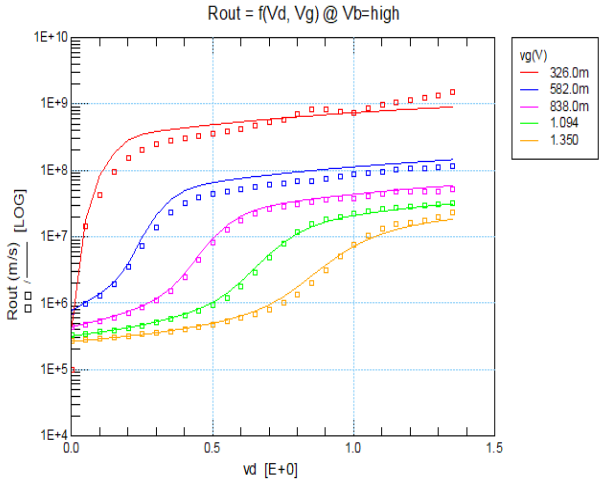


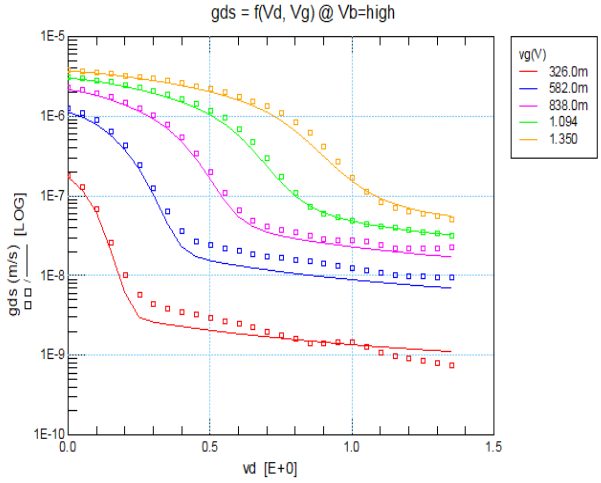




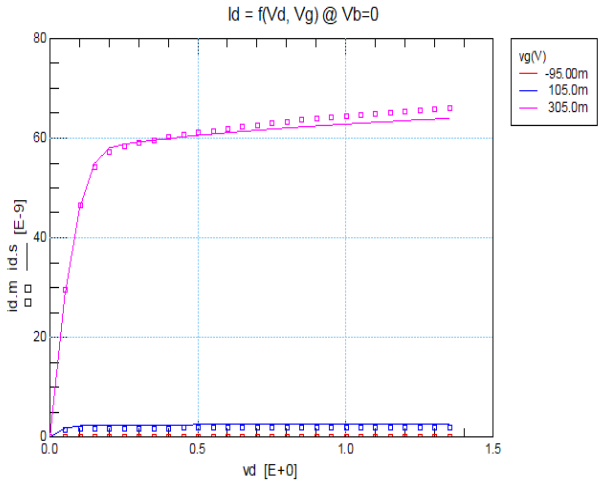
idvd_vbmin, Vb = -1.2V, T = 70°C

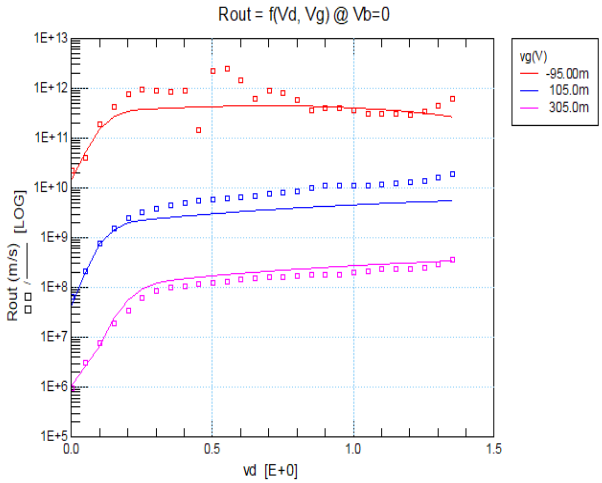


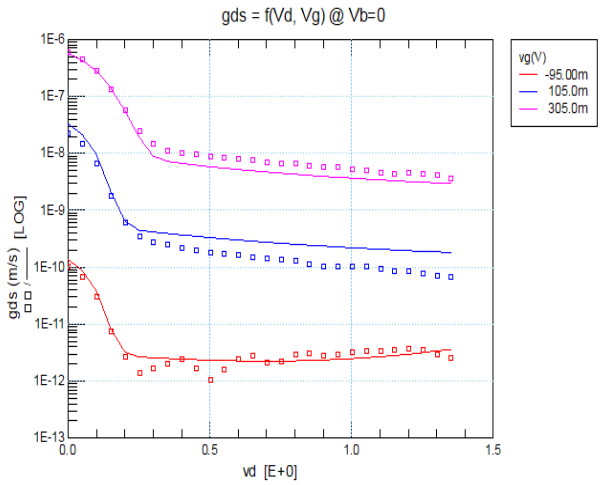




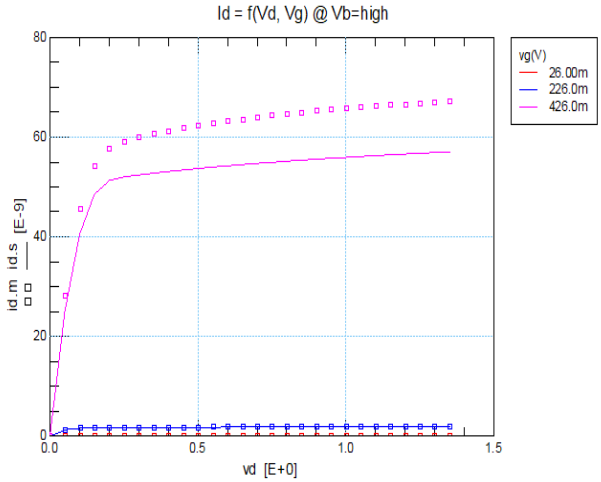
idvd_vth, Vb = 0V, T = 70°C

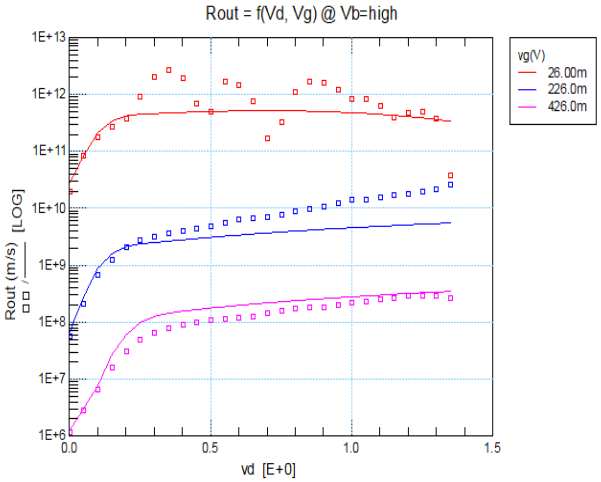


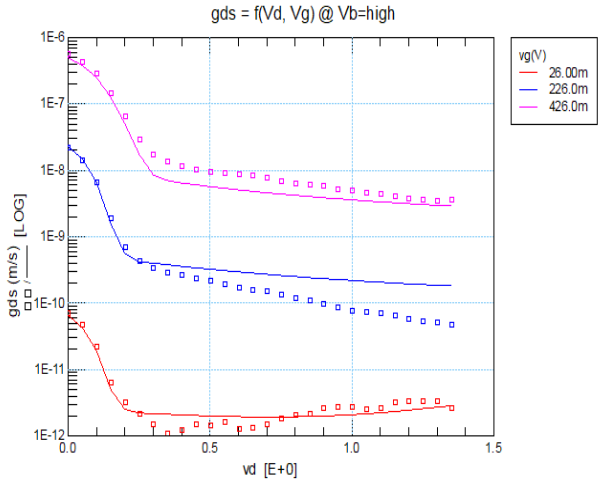




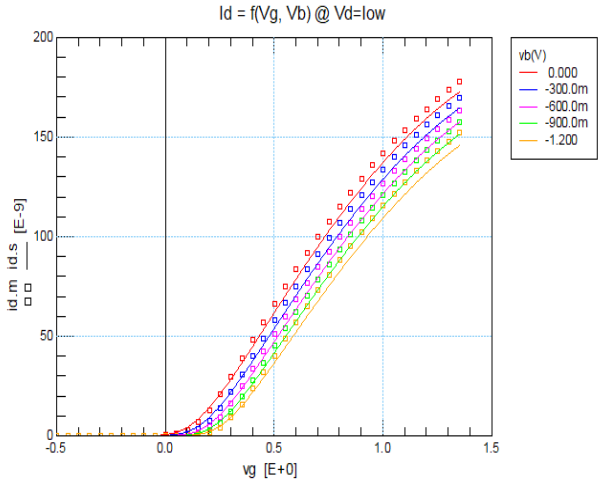
idvd_vth_vbmin, Vb = -1.2V, T = 70°C

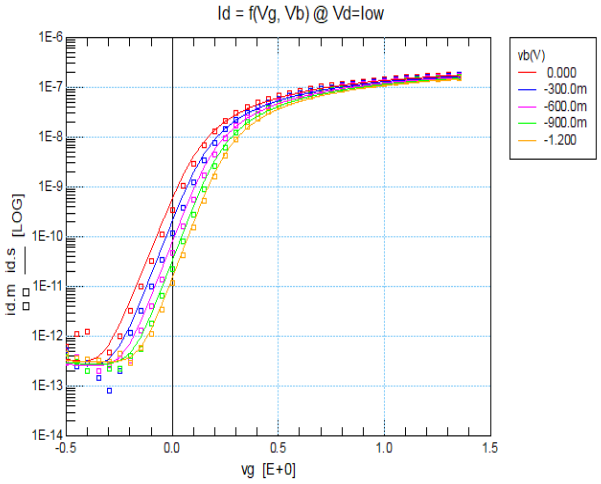


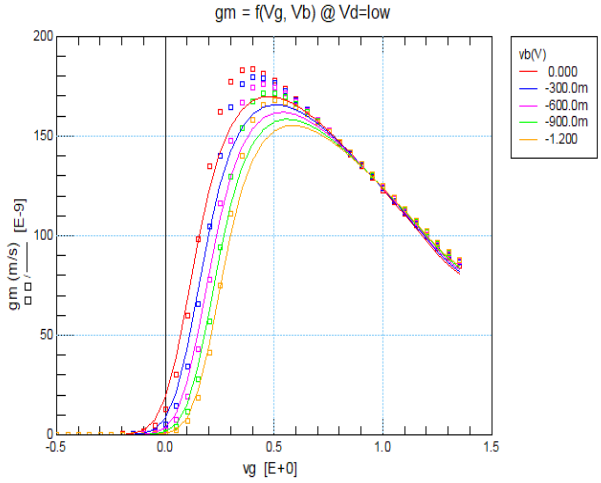


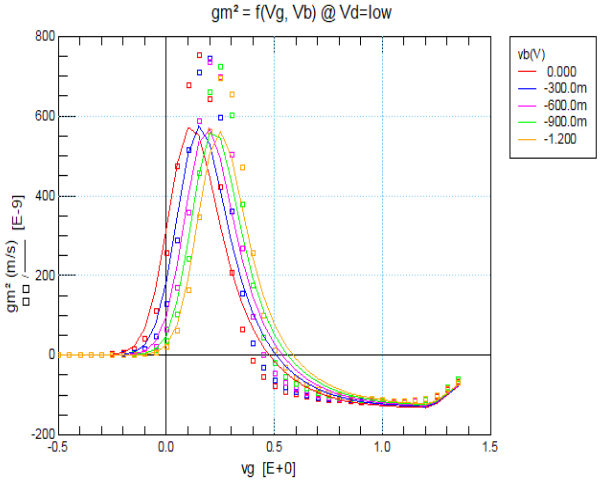


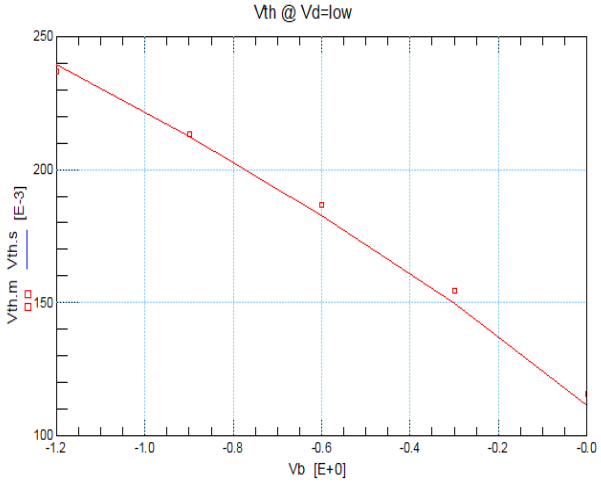
idvg, Vd = 0.05V, T = 125°C

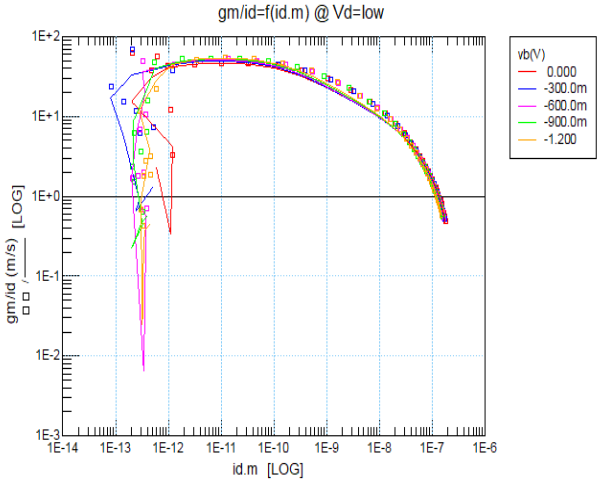


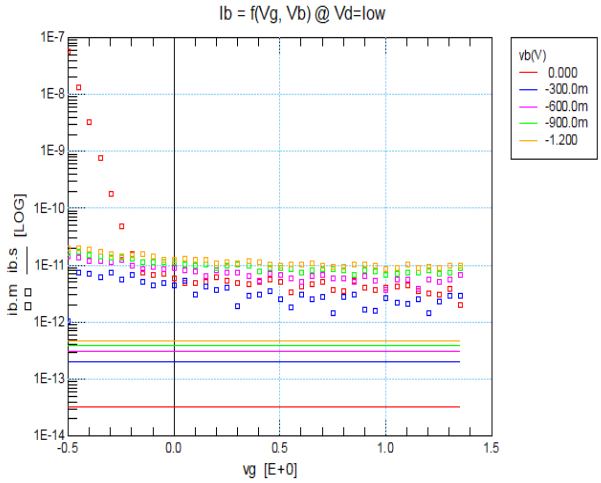


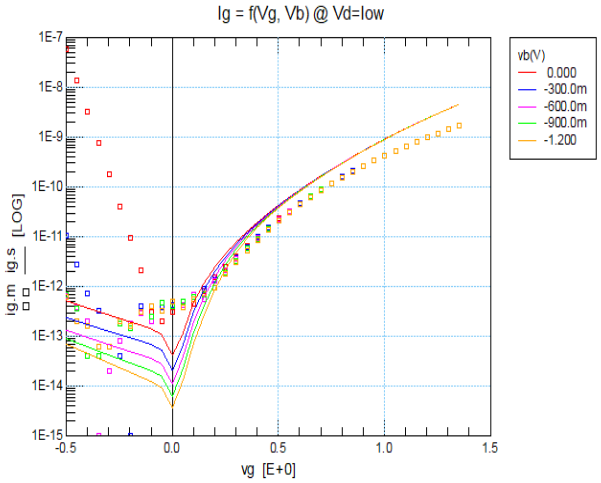




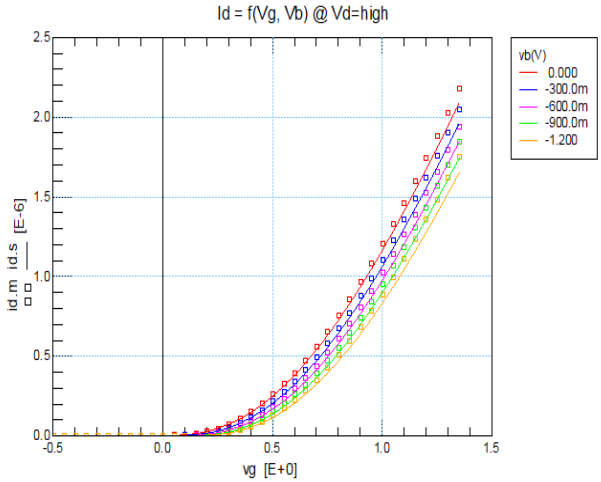


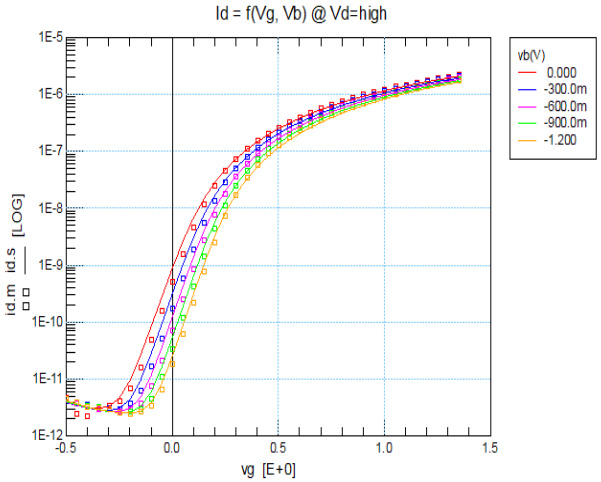


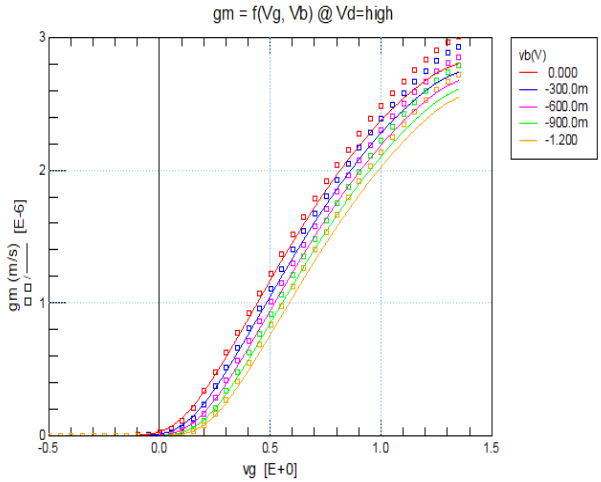


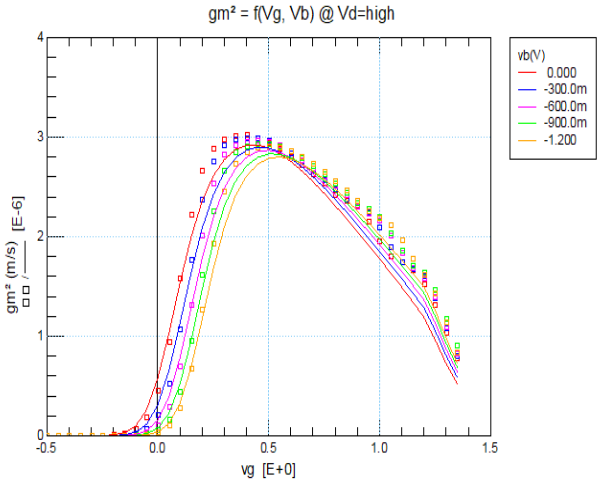


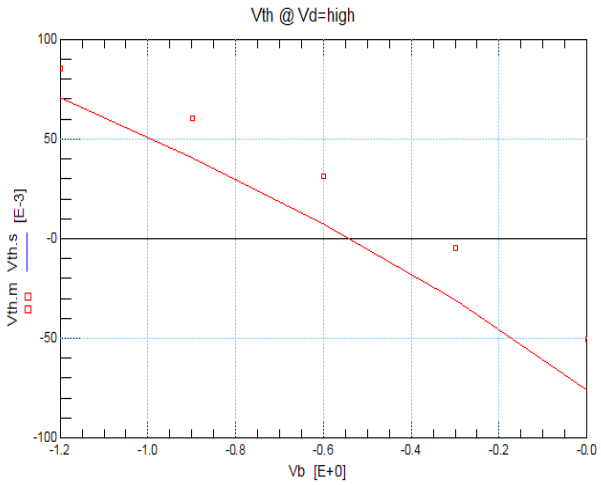
idvg, Vd = 1.2V, T = 125°C

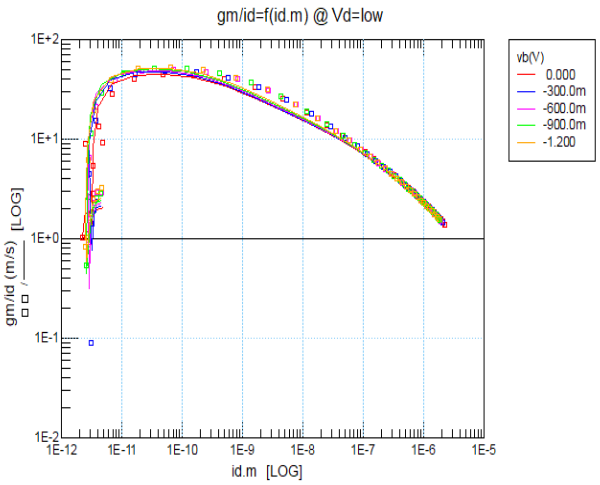


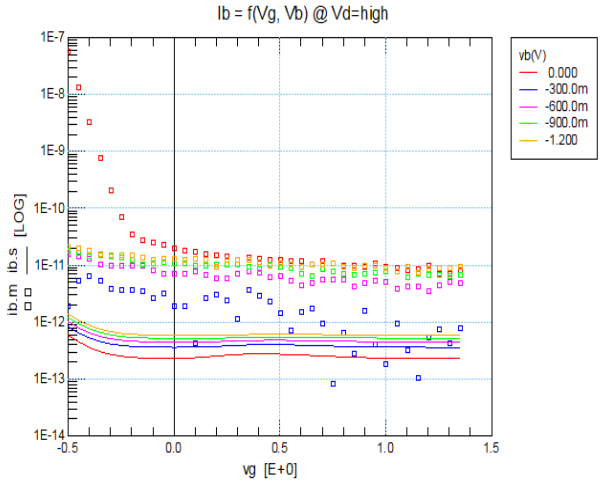


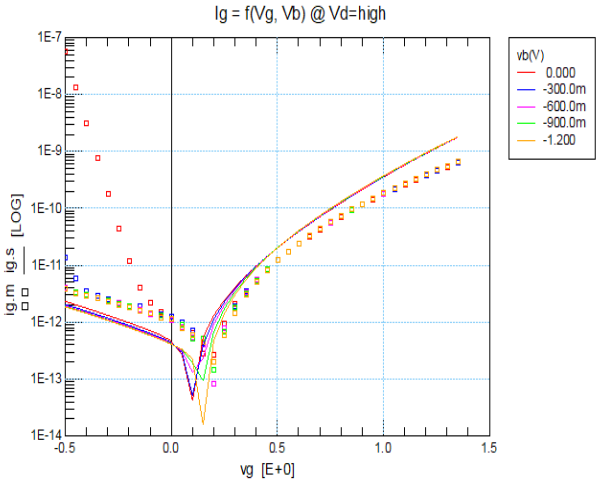




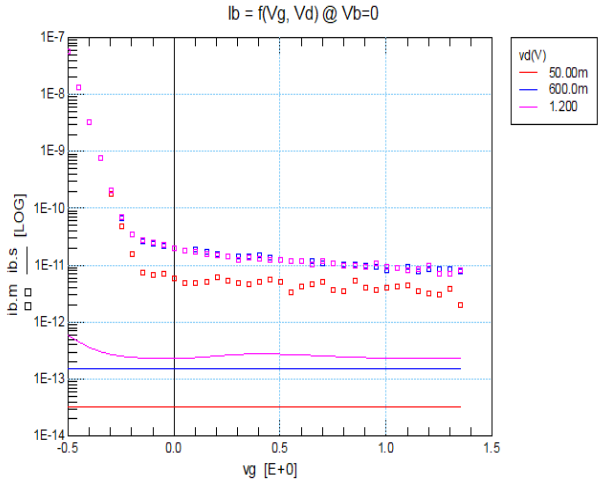


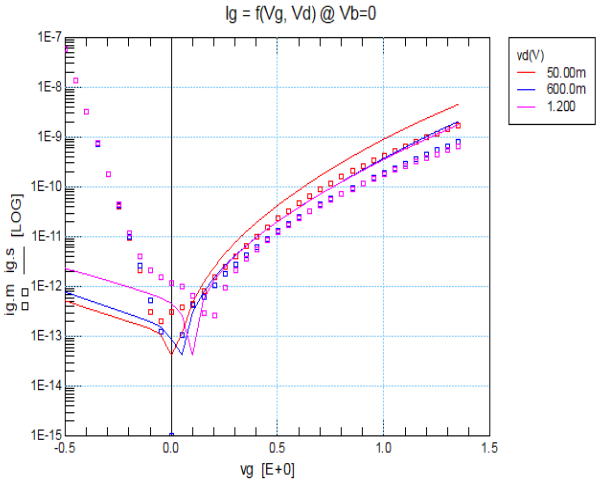




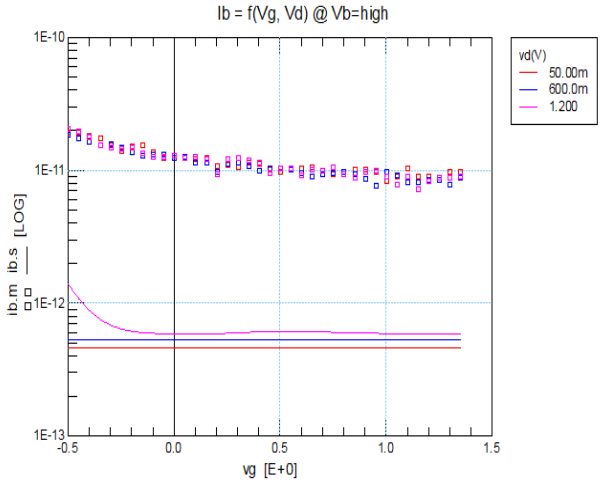


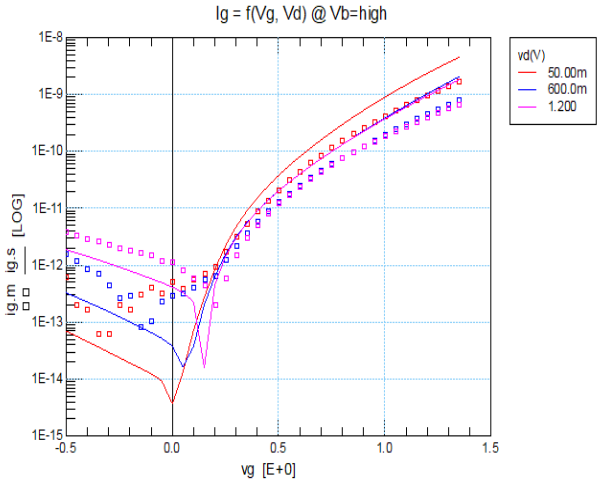
idvg, Ib,Ig, Vb = 0V, T = 125°C



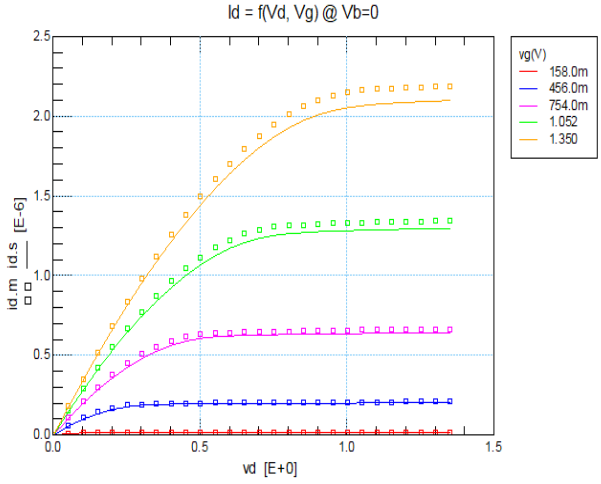


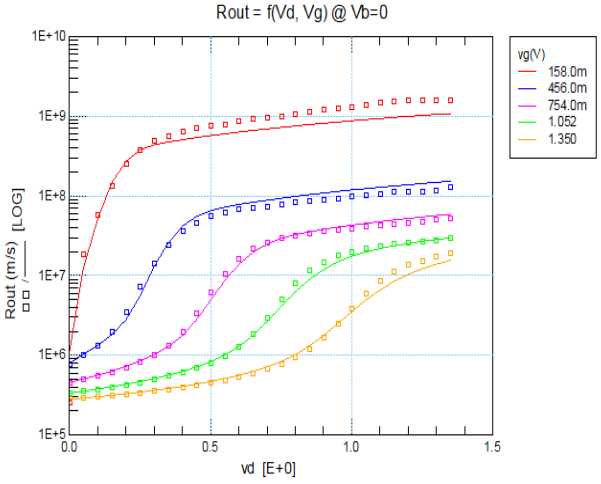
idvg, Ib,Ig, Vb = -1.2V, T = 125°C

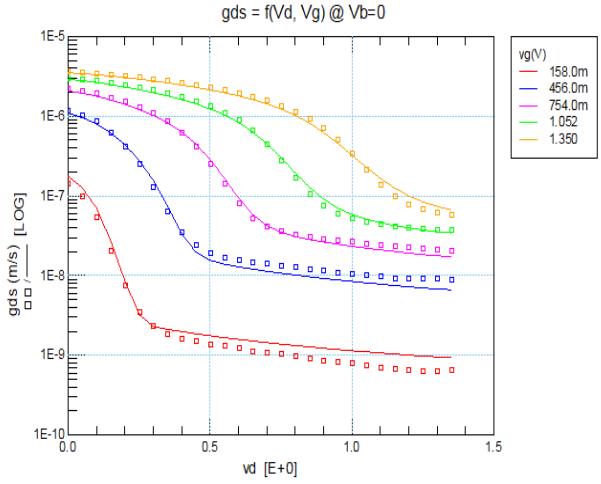




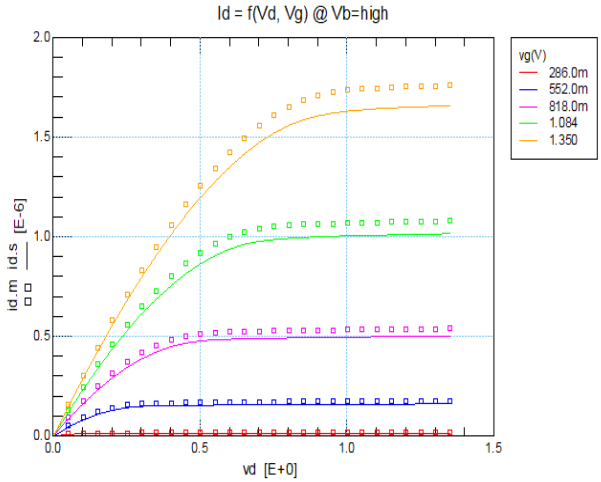
idvd, Vb = 0V, T = 125°C

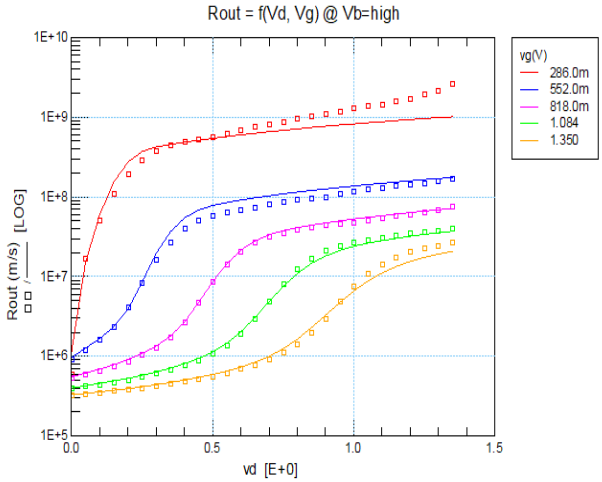


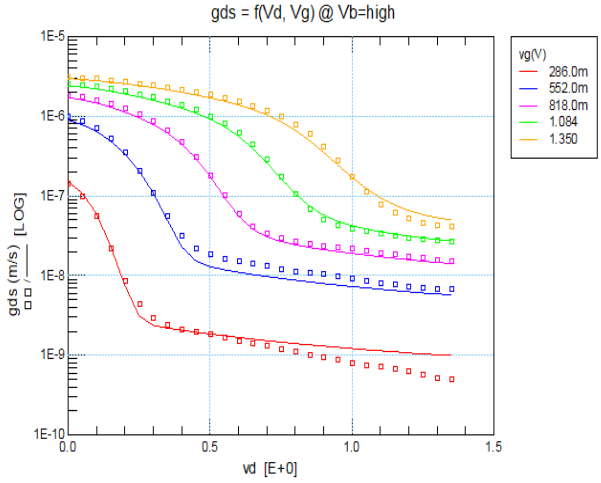




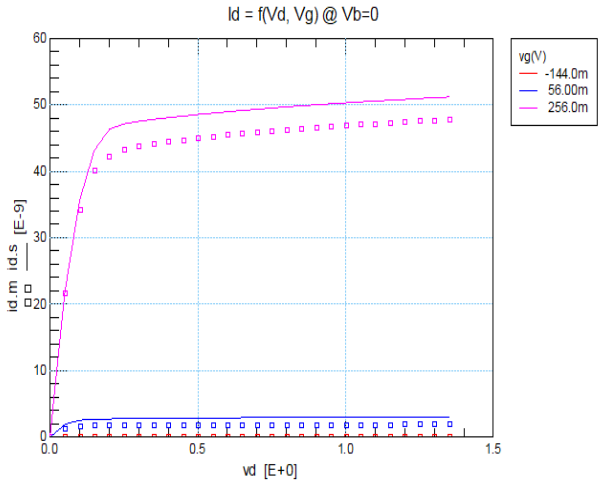
idvd_vbmin, Vb = -1.2V, T = 125°C

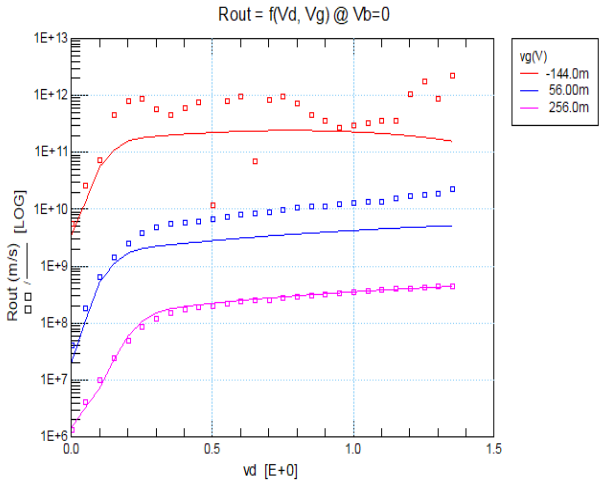


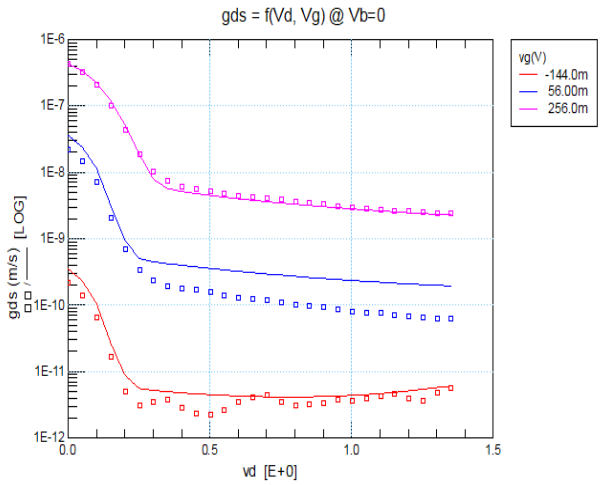




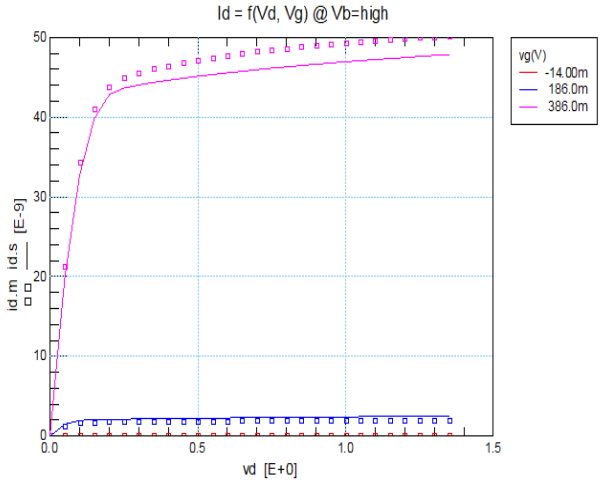
idvd_vth, Vb = 0V, T = 125°C

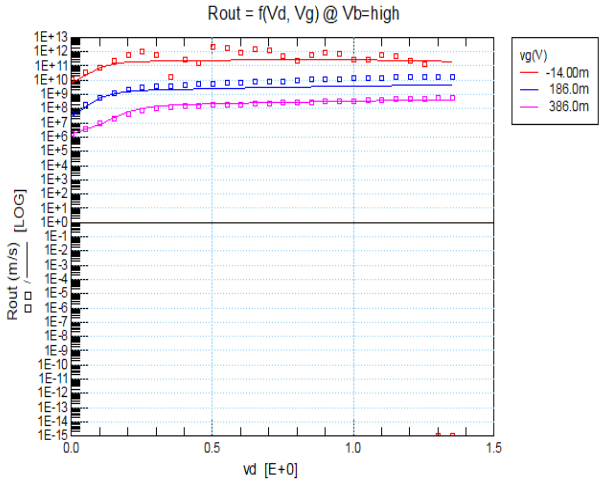


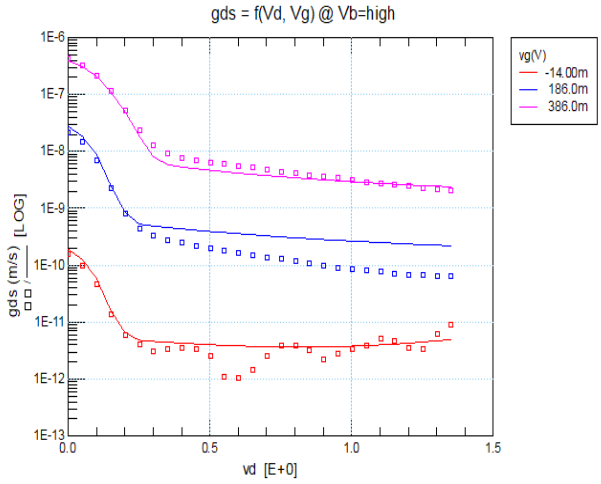




idvd_vth_vbmin, Vb = -1.2V, T = 125°C

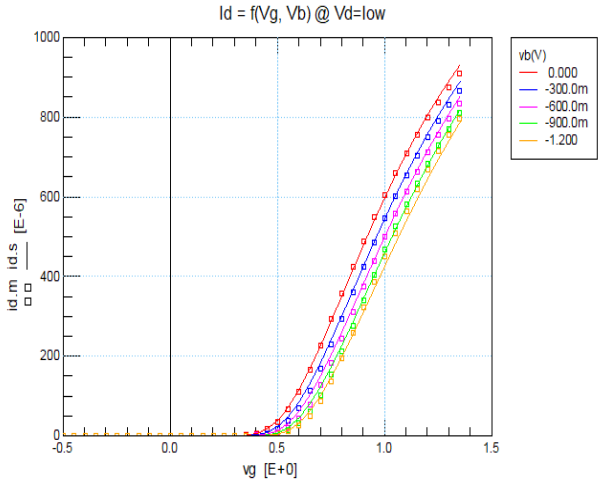


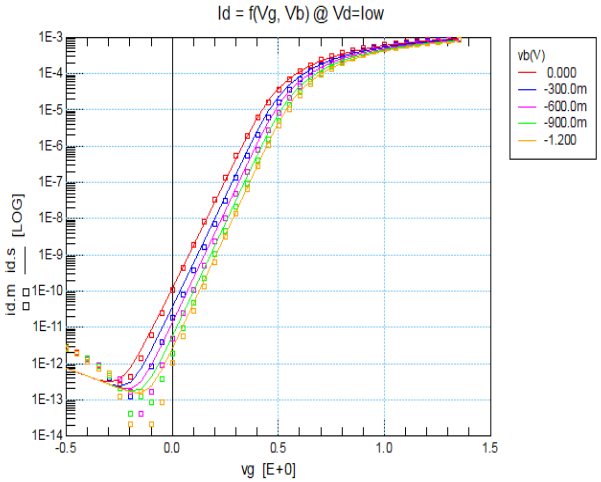


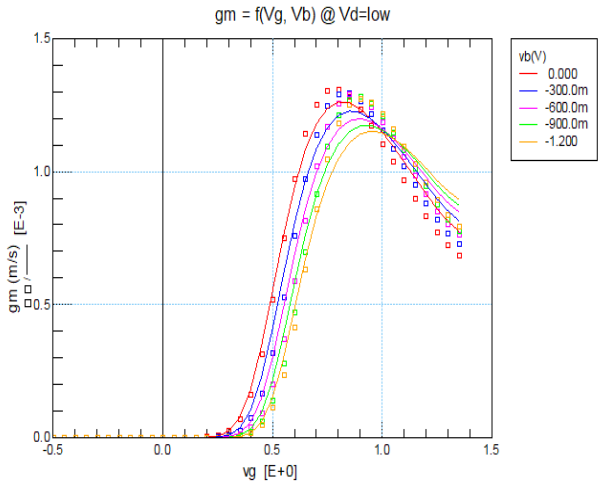


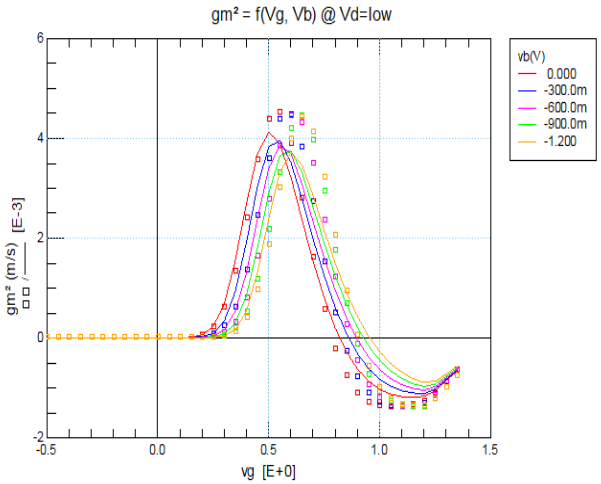
4.4 W10u0_L0u13_T356_S380_2

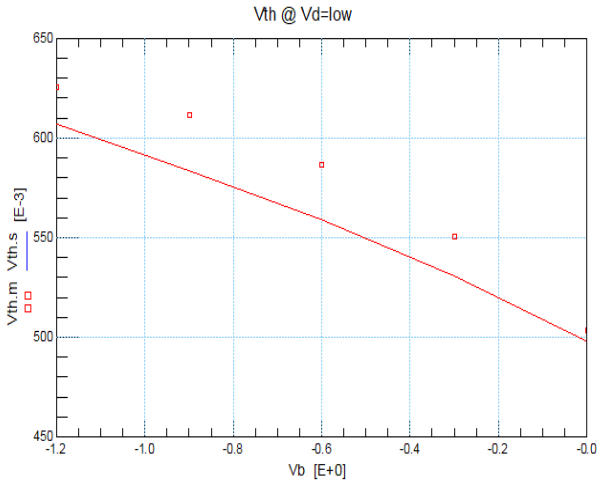
idvg, Vd = 0.05V, T = 27°C

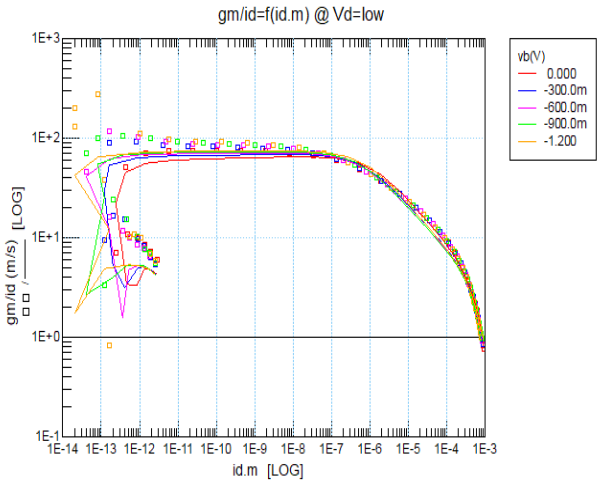


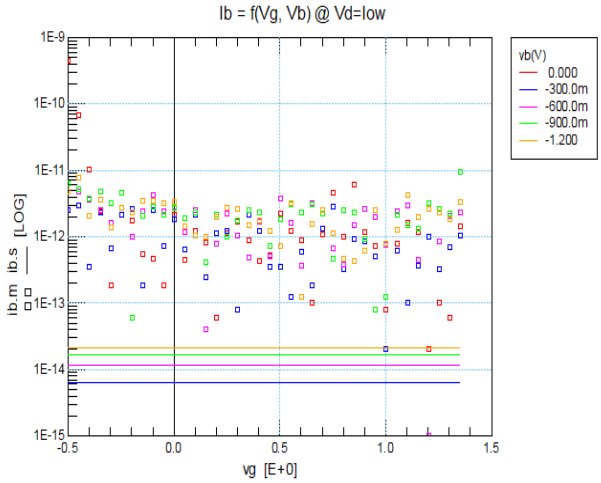


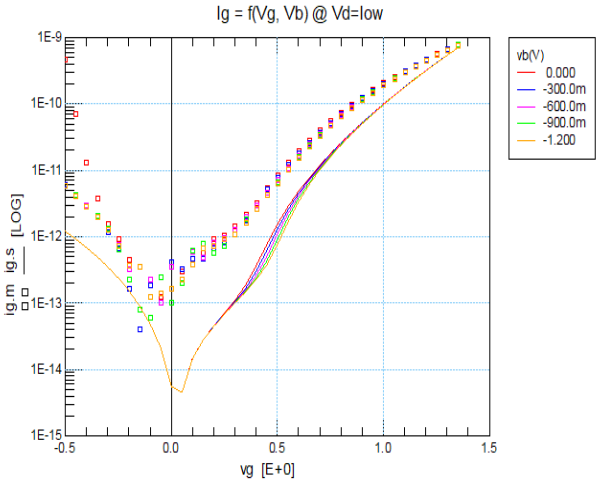




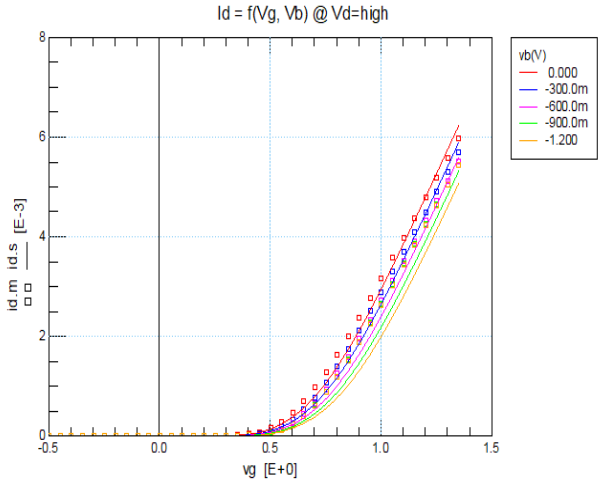


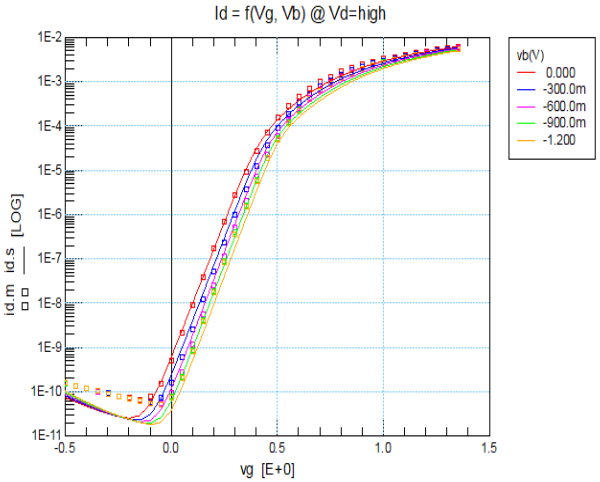


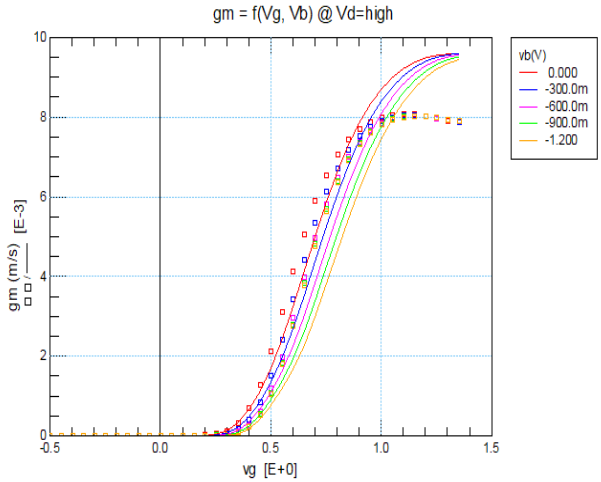


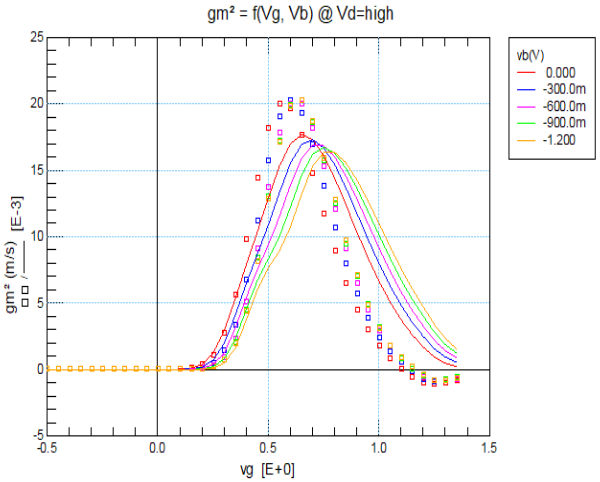


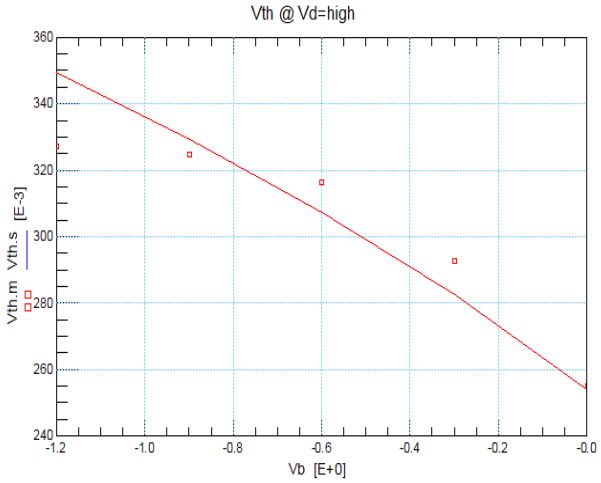
idvg, Vd = 1.2V, T = 27°C

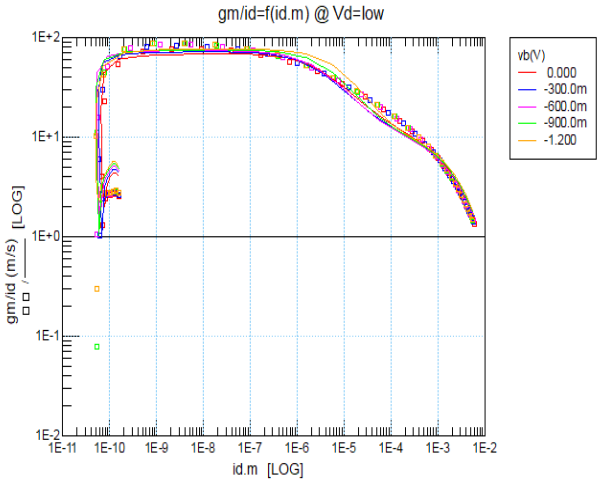


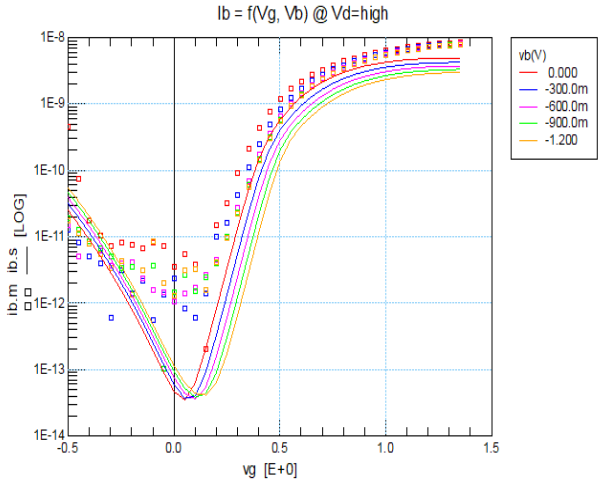


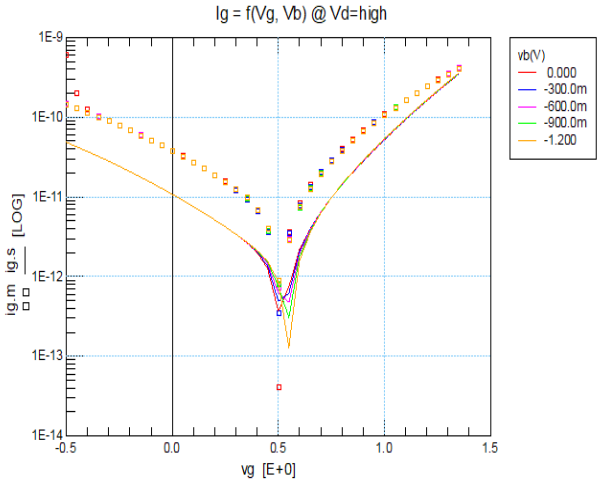




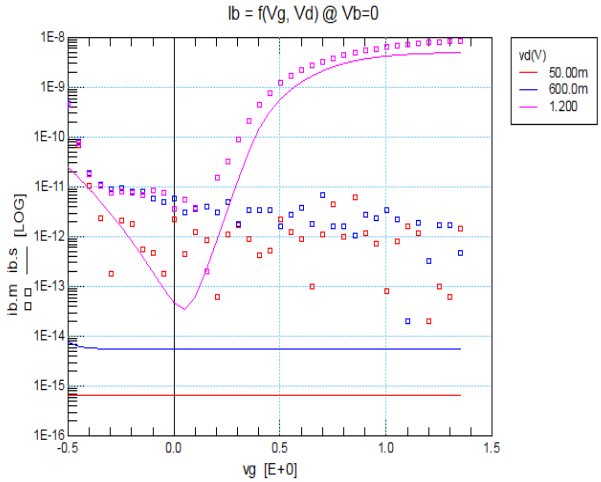


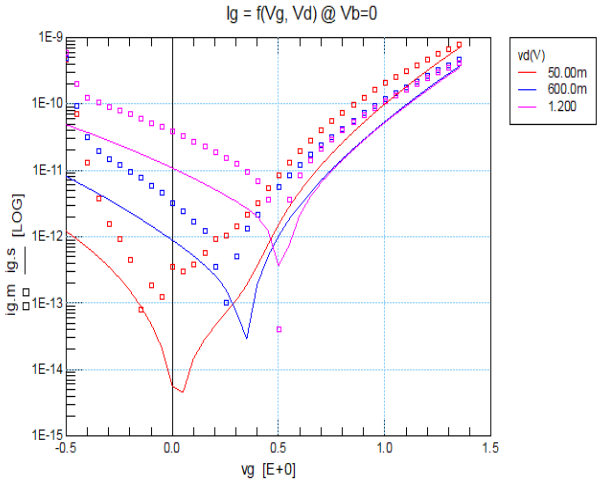




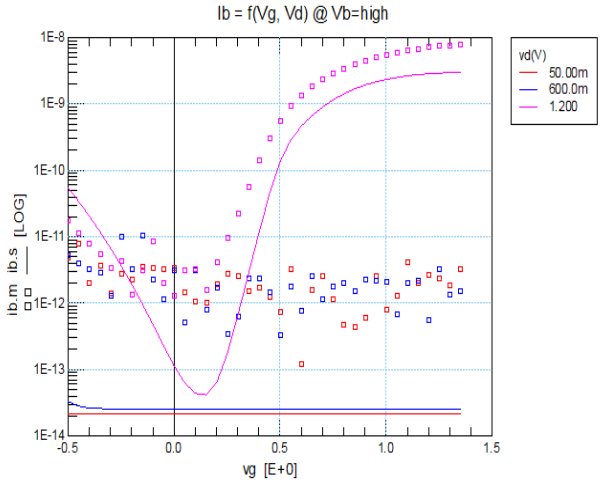


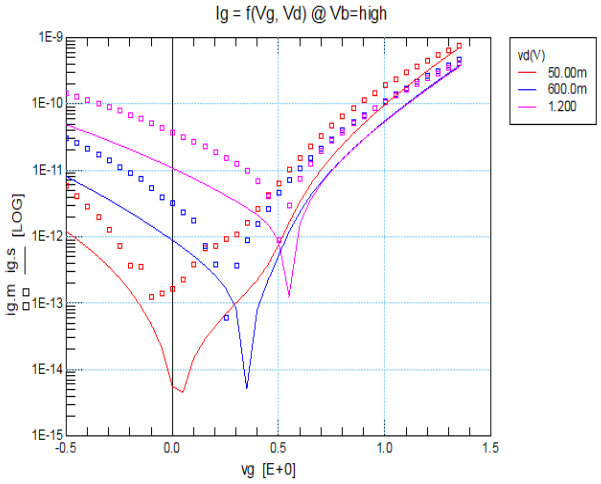
idvg, Ib,Ig, Vb = 0V, T = 27°C



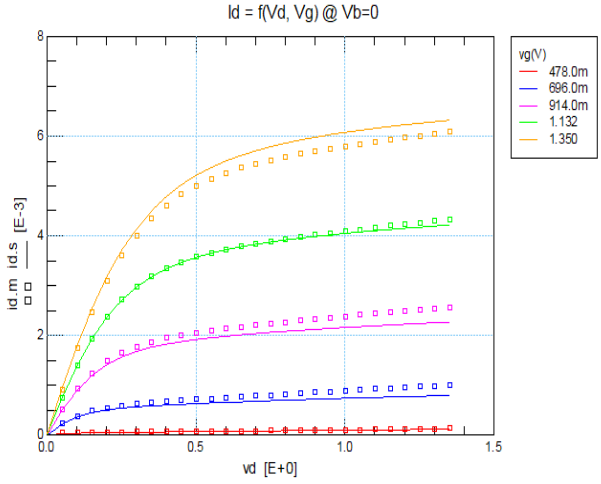


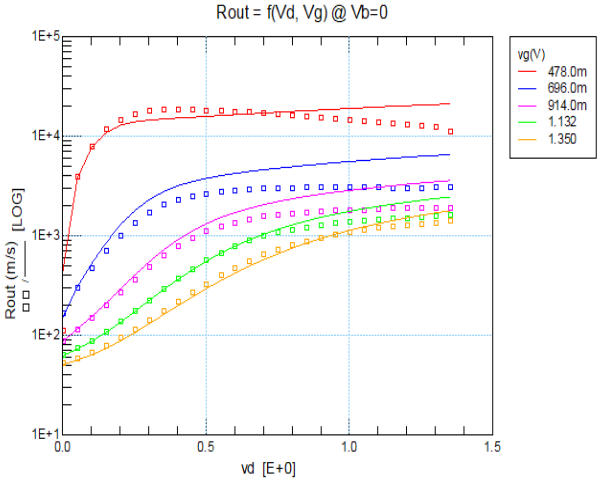
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

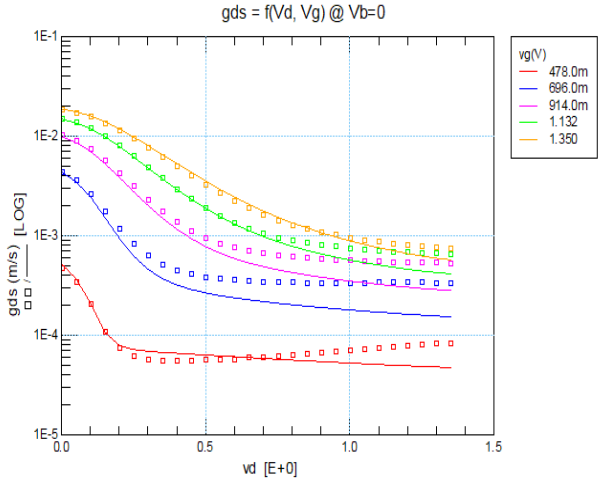




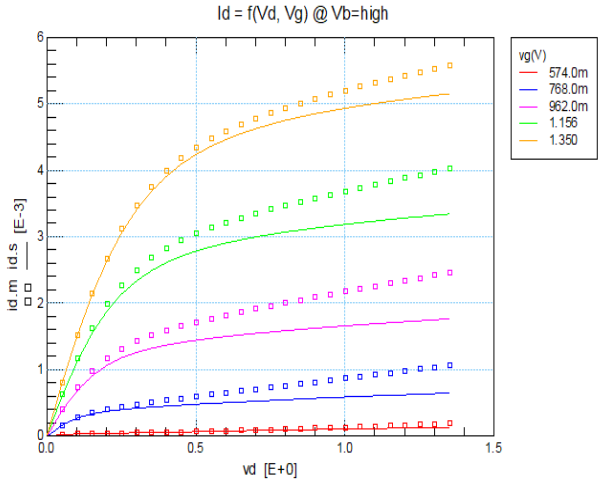
idvd, Vb = 0V, T = 27°C

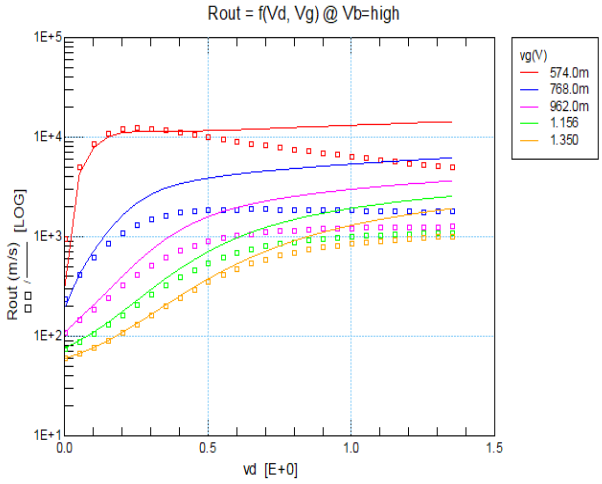


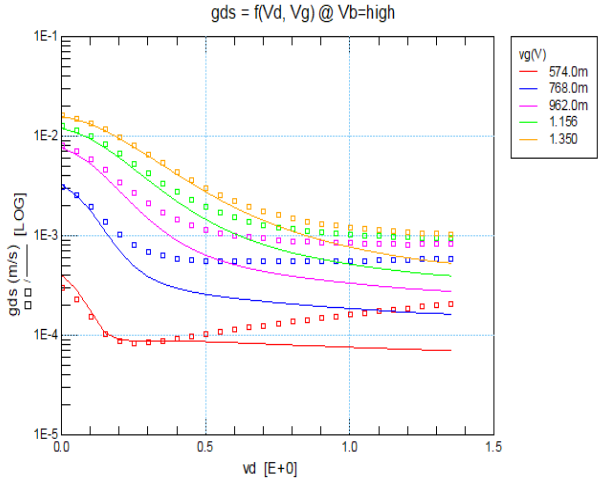




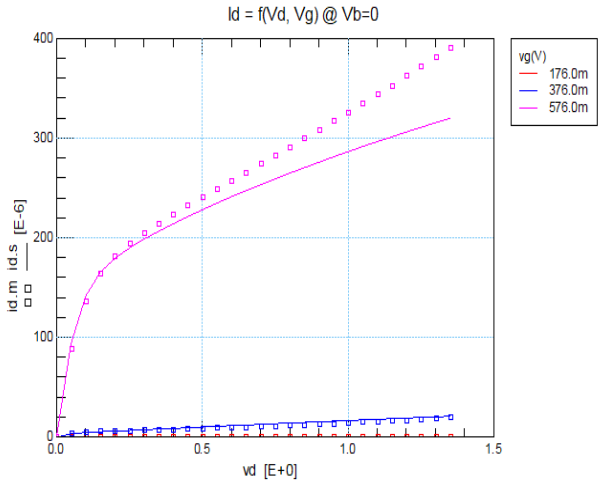
idvd_vbmin, Vb = -1.2V, T = 27°C

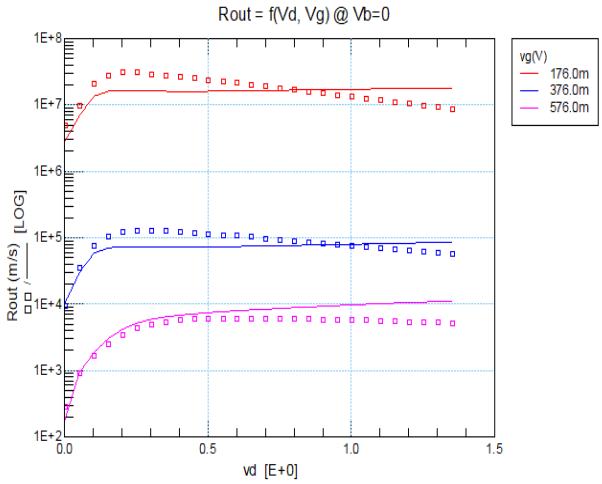


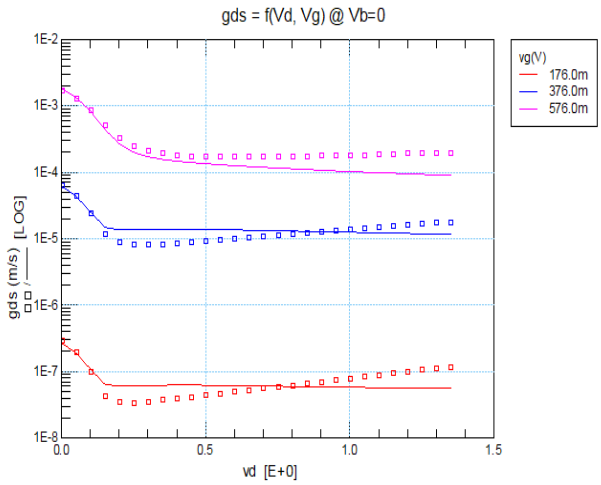




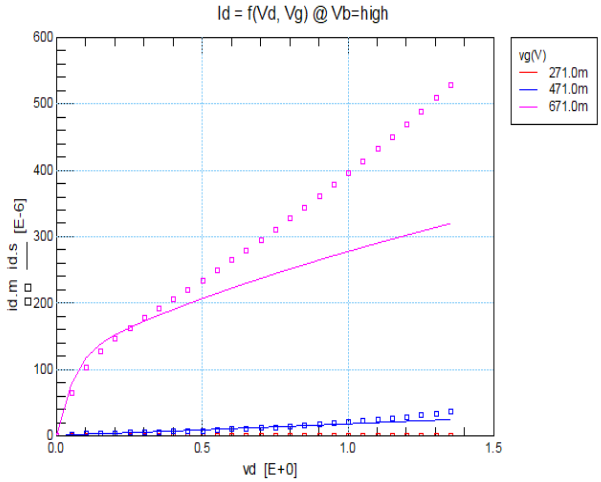
idvd_vth, Vb = 0V, T = 27°C

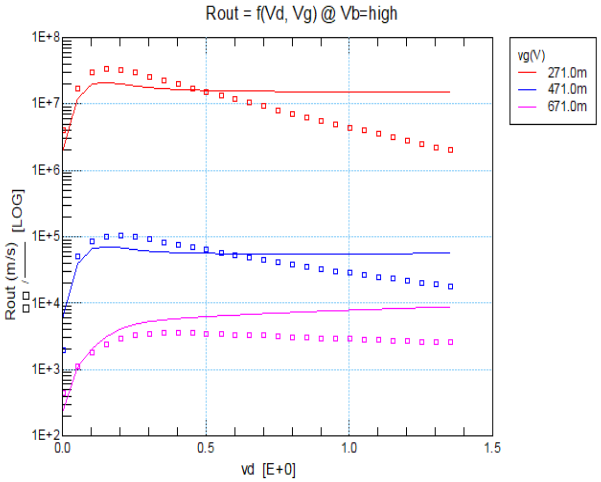


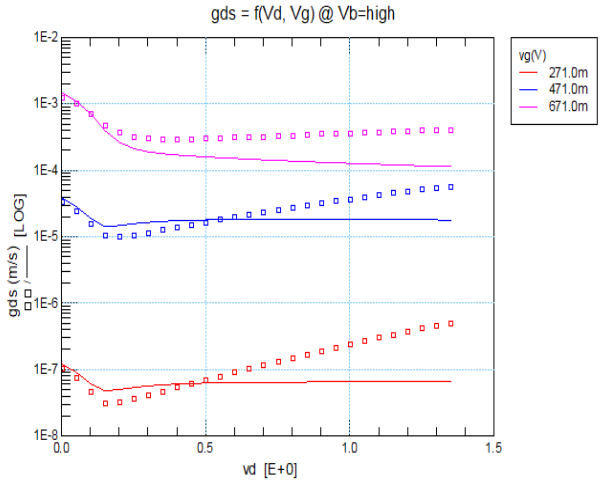




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

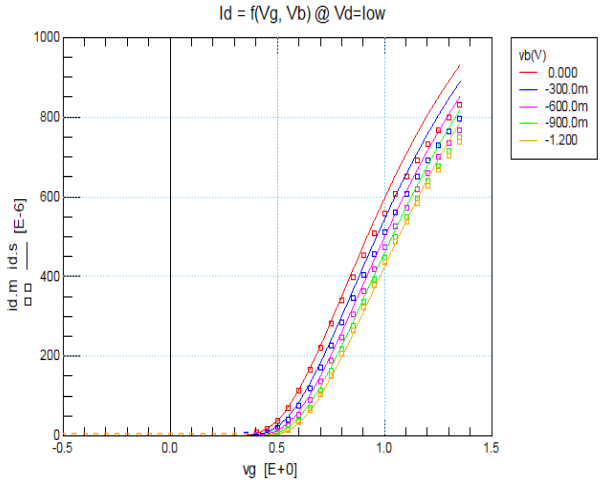


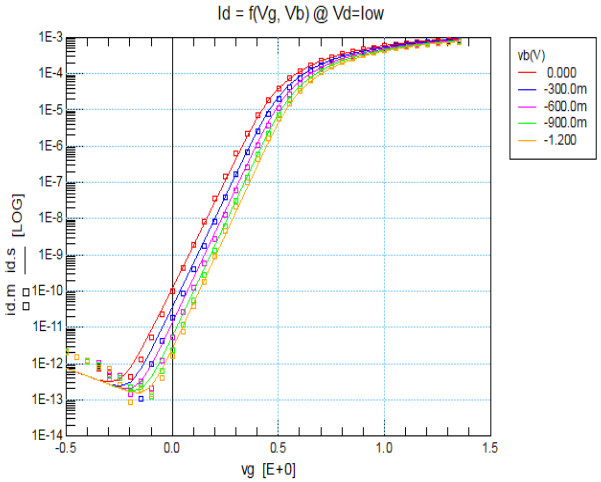


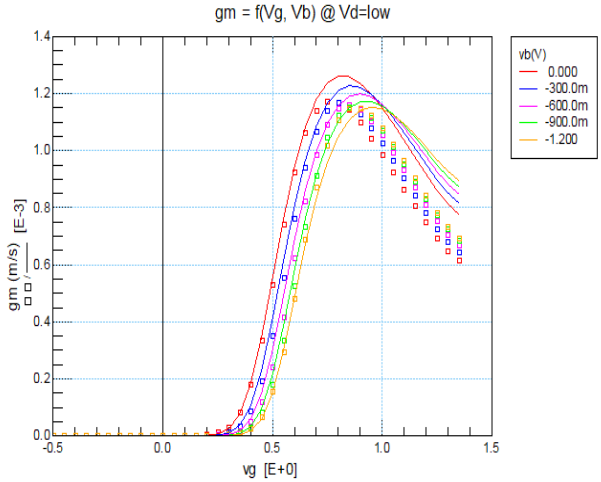


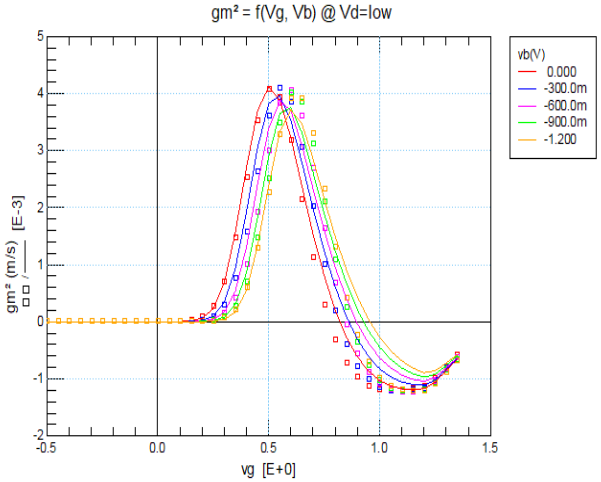
4.5 W10u0_L0u13_S540_2

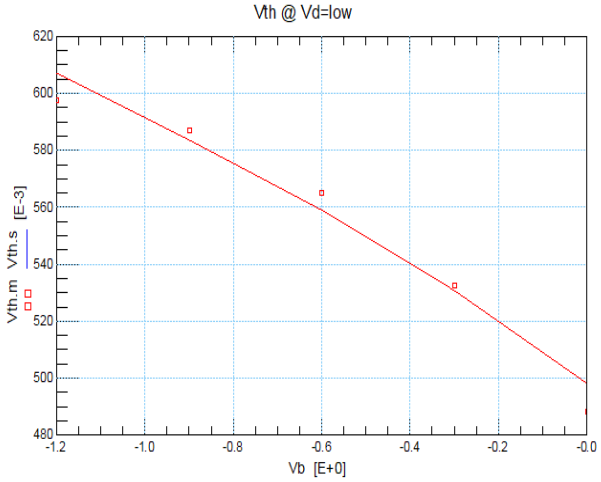
idvg, Vd = 0.05V, T = 27°C

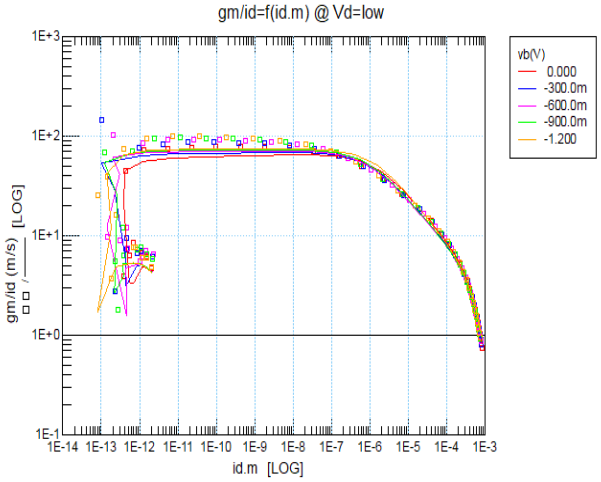


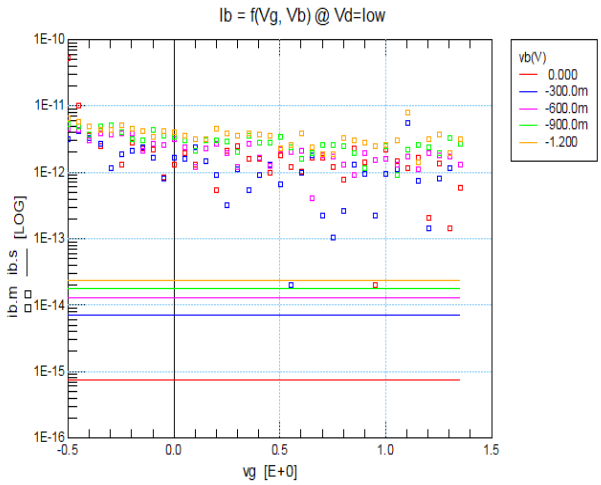


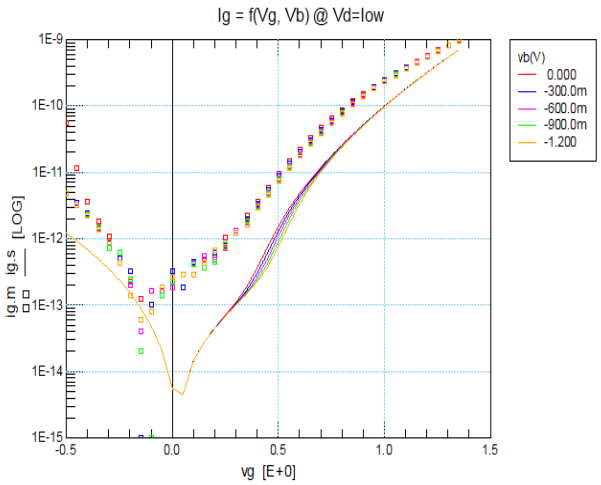




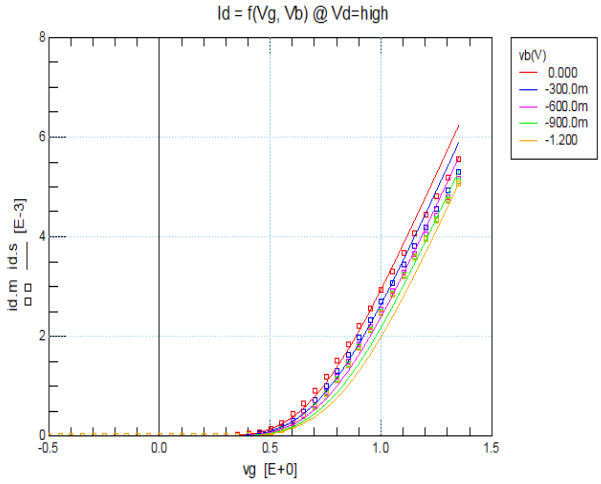


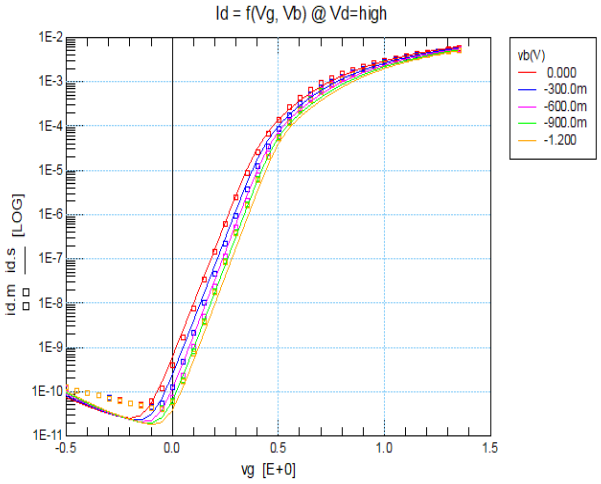


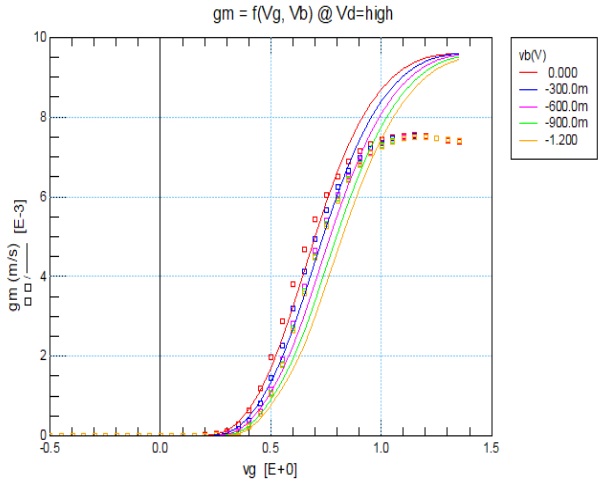


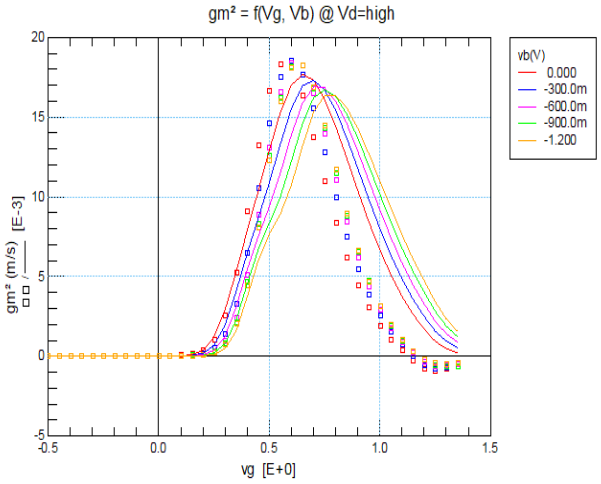


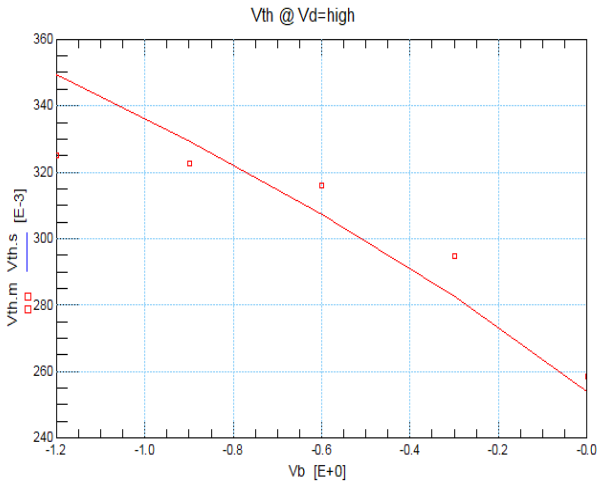
idvg, Vd = 1.2V, T = 27°C

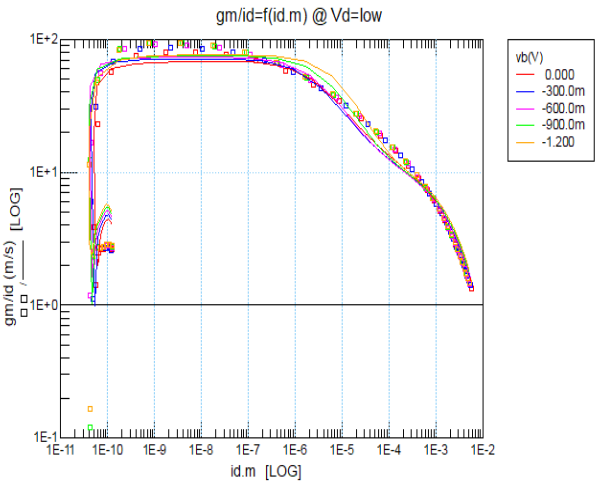


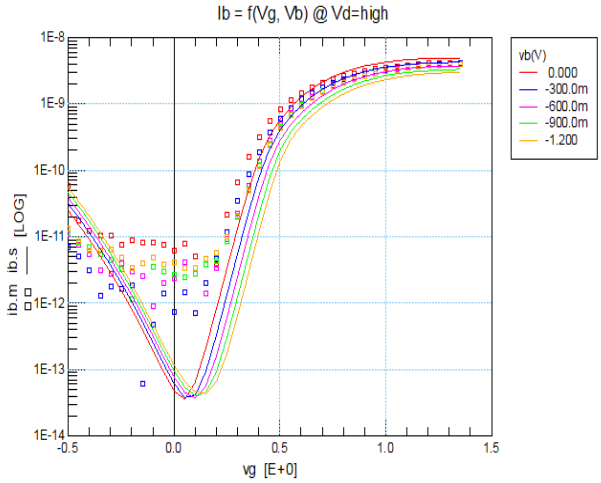


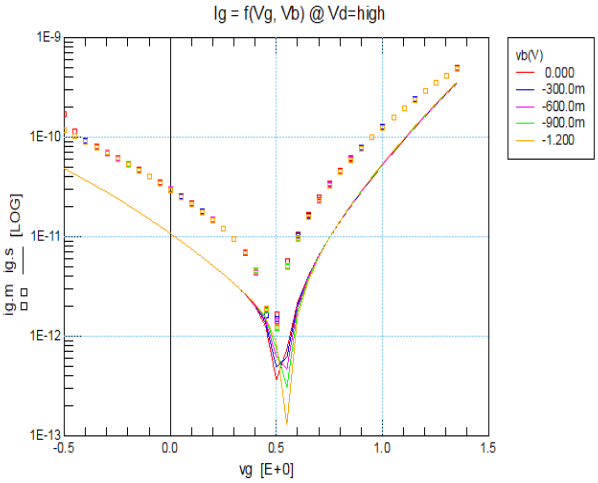




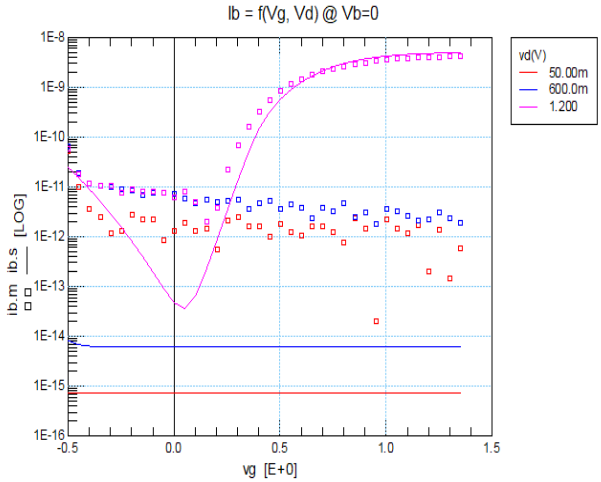


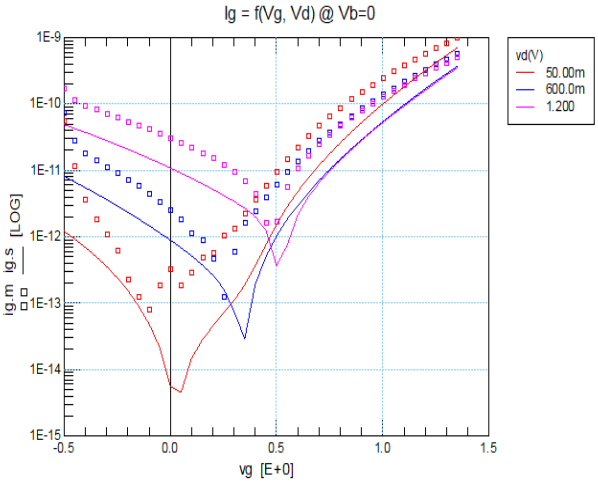




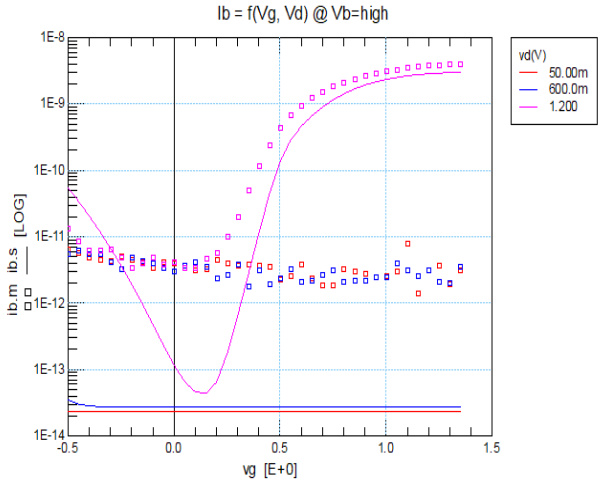


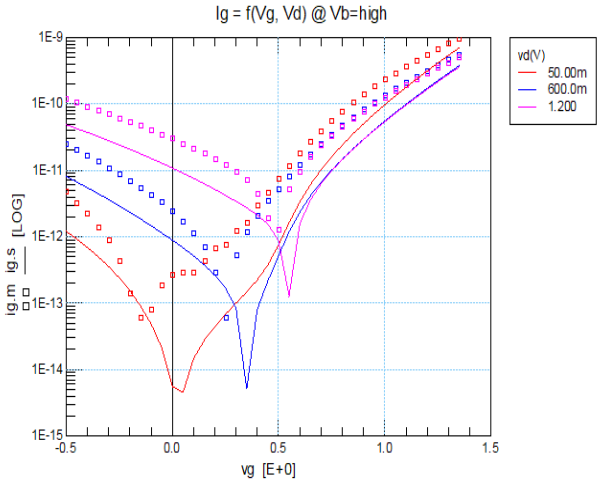
idvg, Ib,Ig, Vb = 0V, T = 27°C



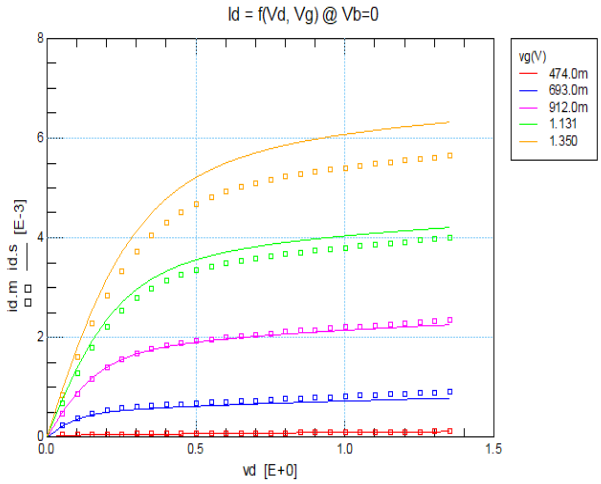


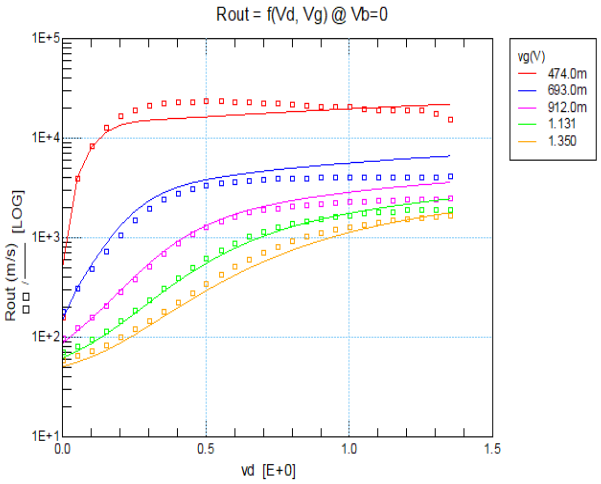
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

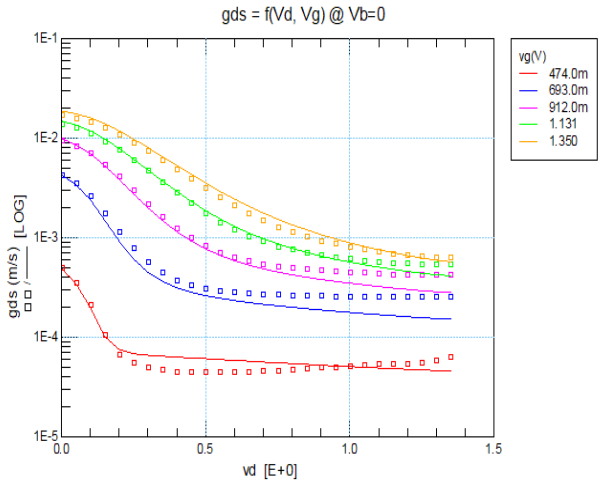




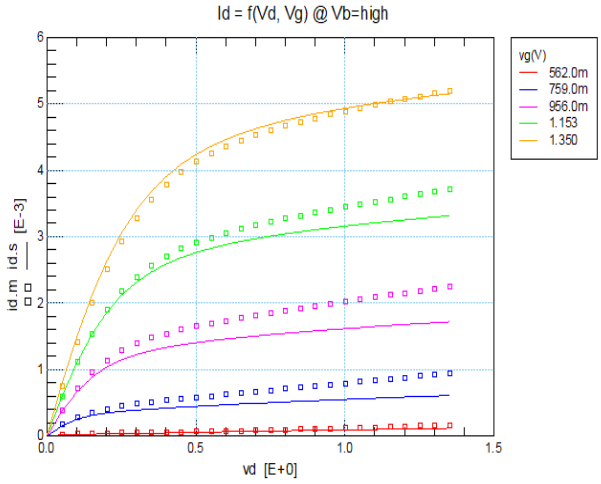
idvd, Vb = 0V, T = 27°C

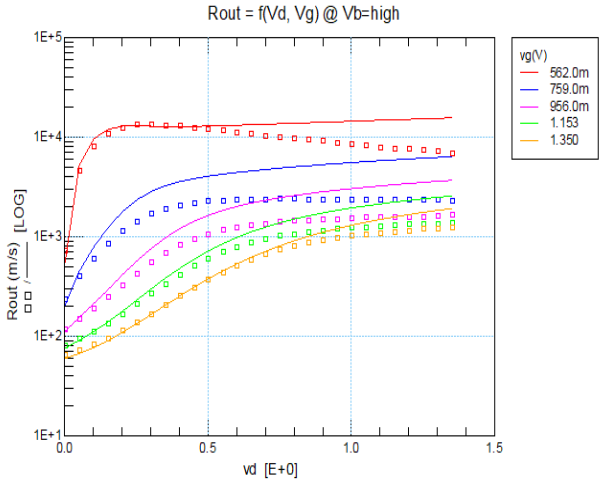


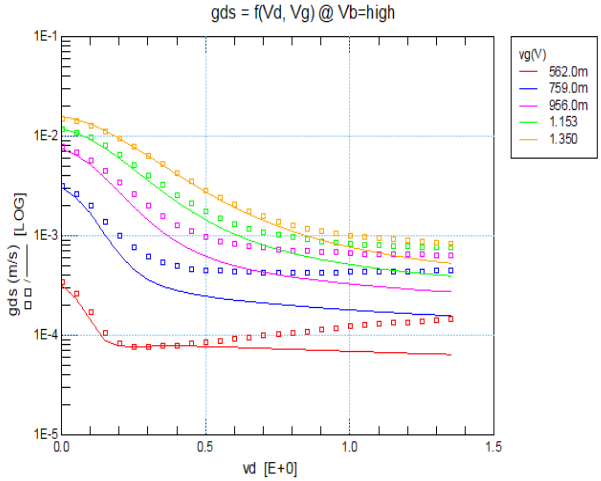




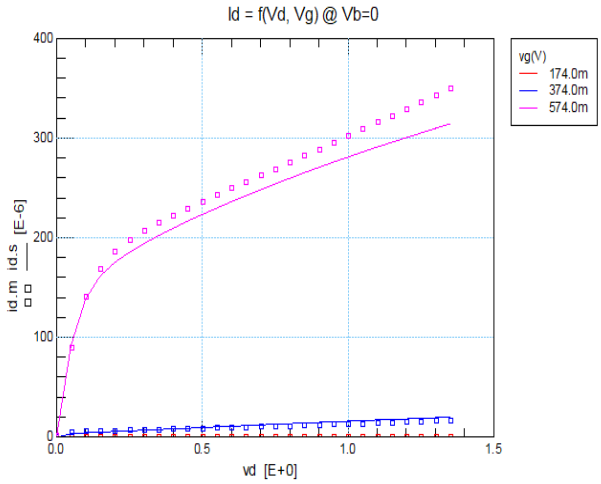
idvd_vbmin, Vb = -1.2V, T = 27°C

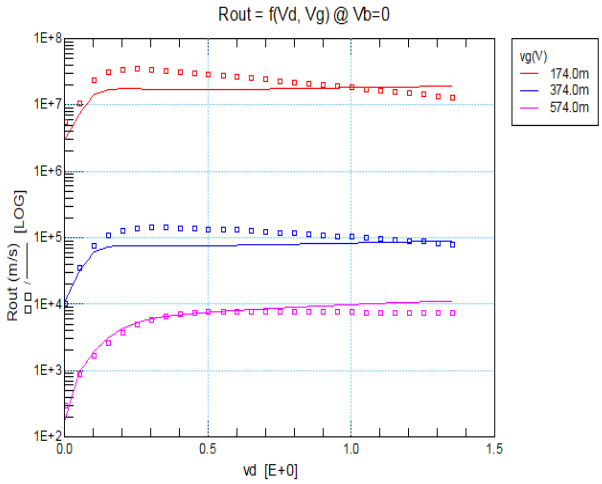


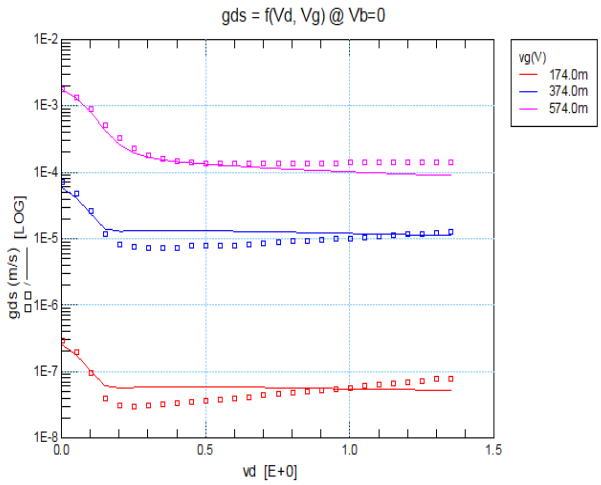




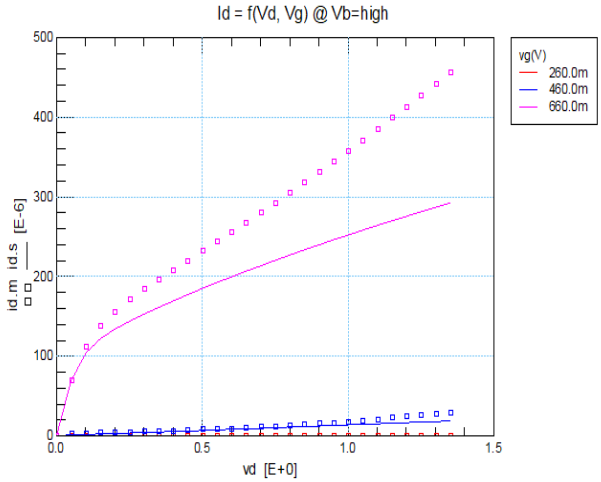
idvd_vth, Vb = 0V, T = 27°C

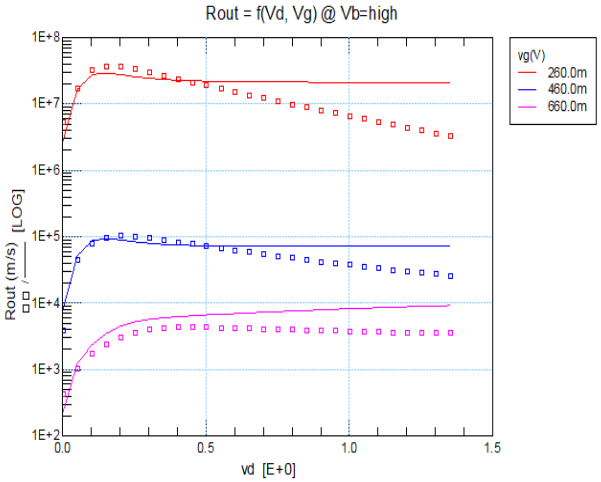


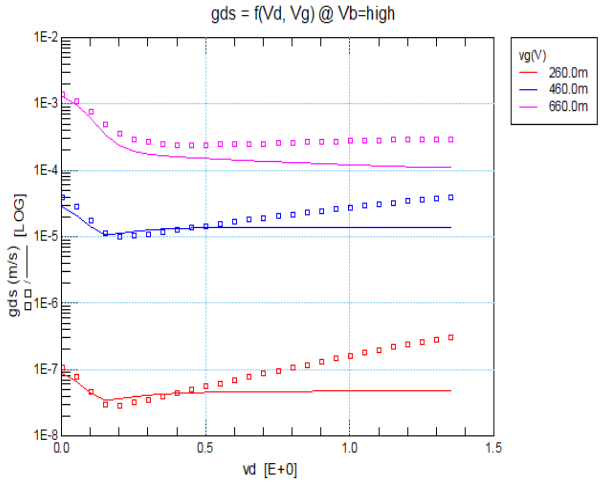




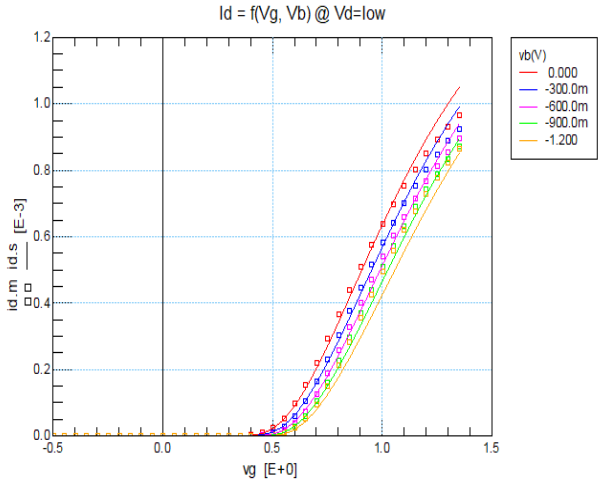
idvd_vth_vbmin, Vb = -1.2V, T = 27°C

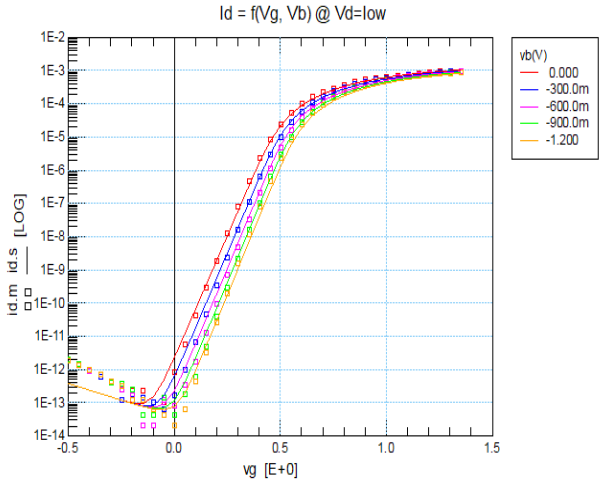


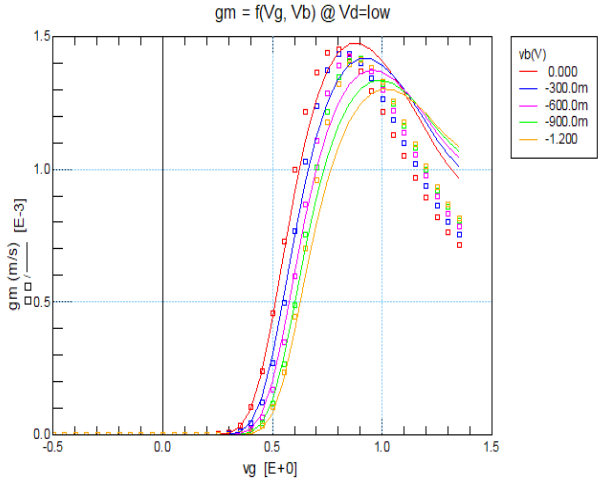


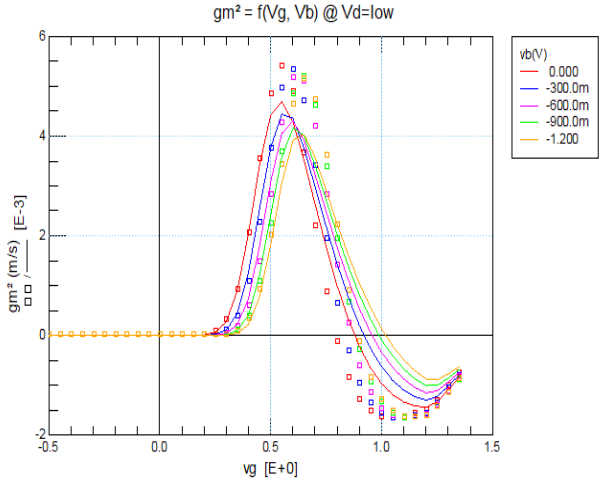


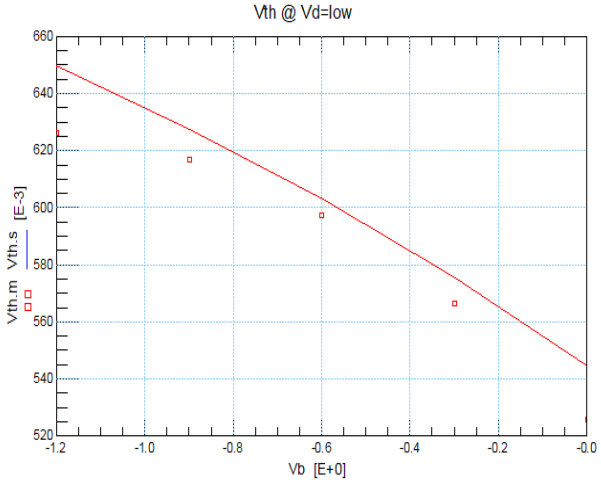
idvg, Vd = 0.05V, T = -40°C

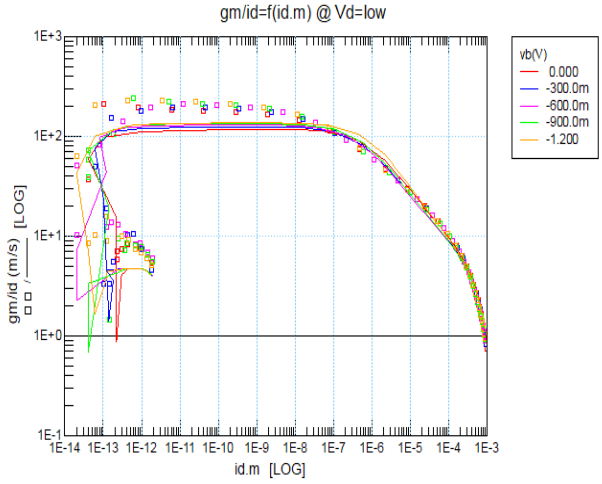


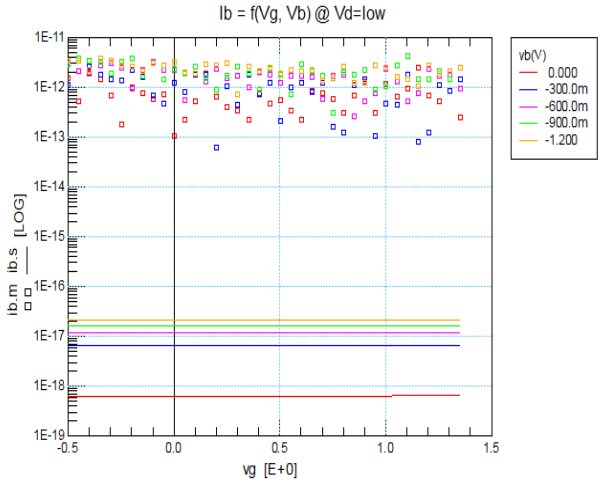


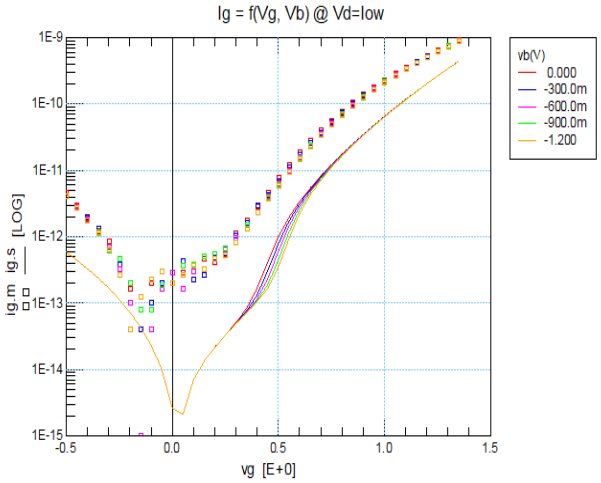




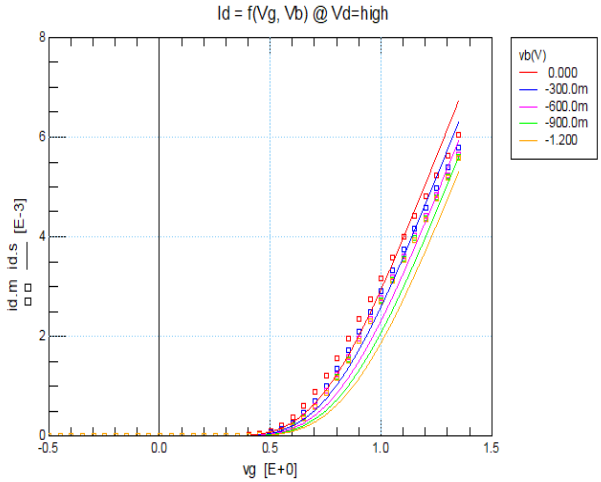


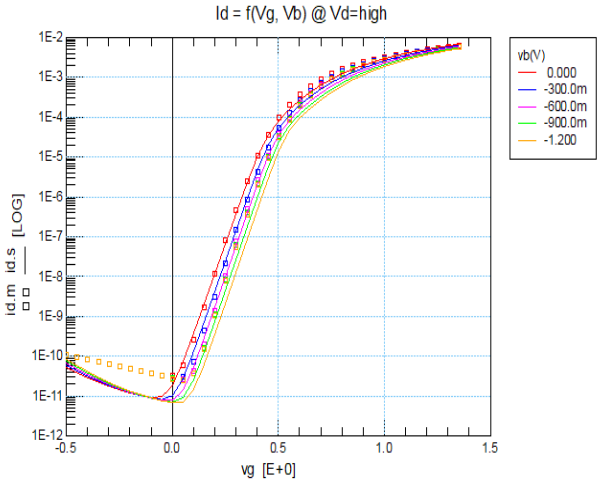


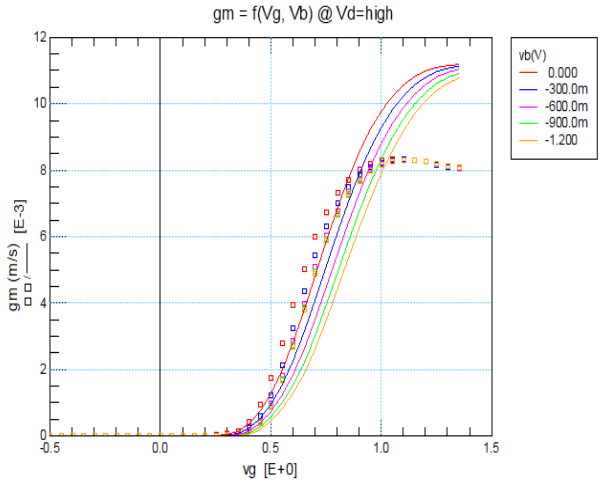


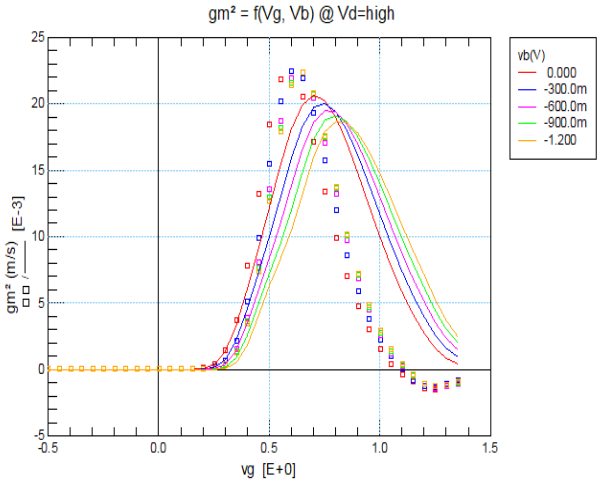


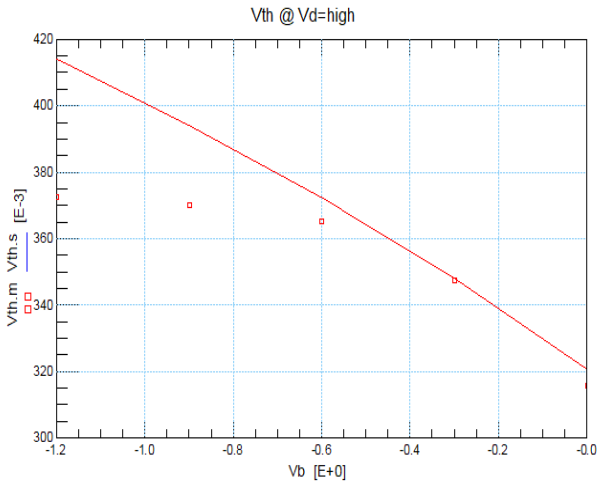
idvg, Vd = 1.2V, T = -40°C

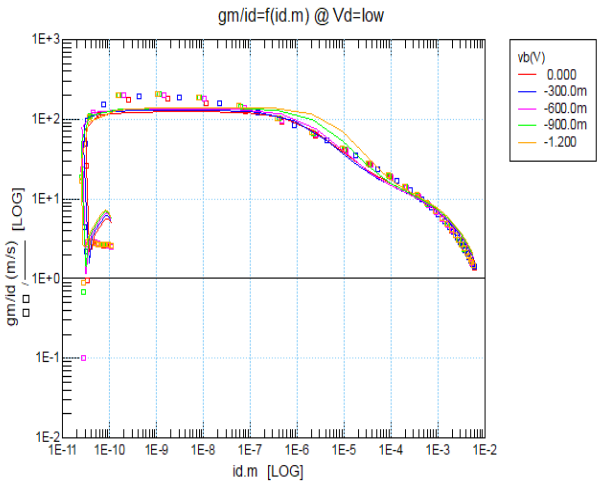


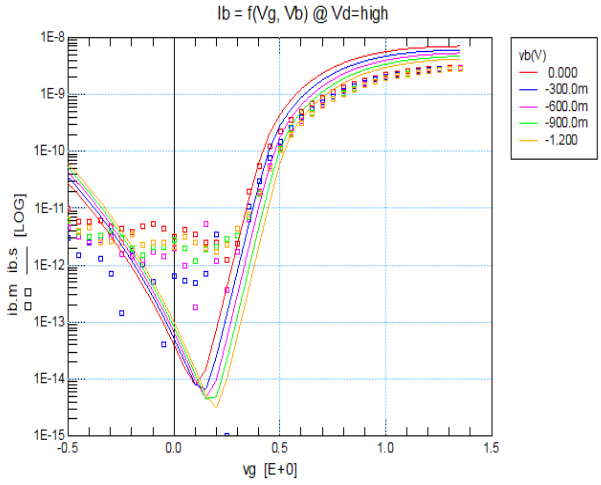


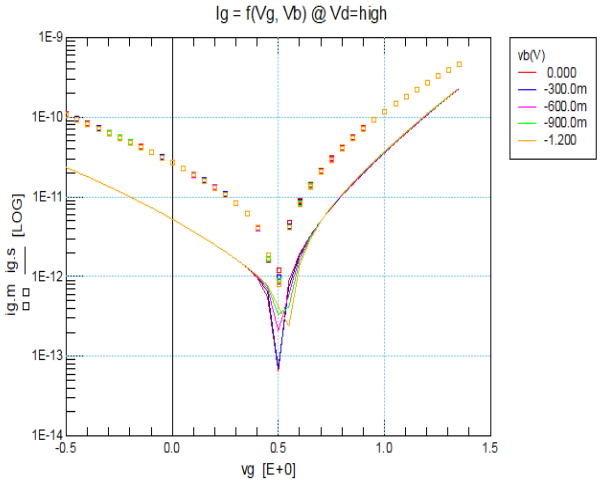




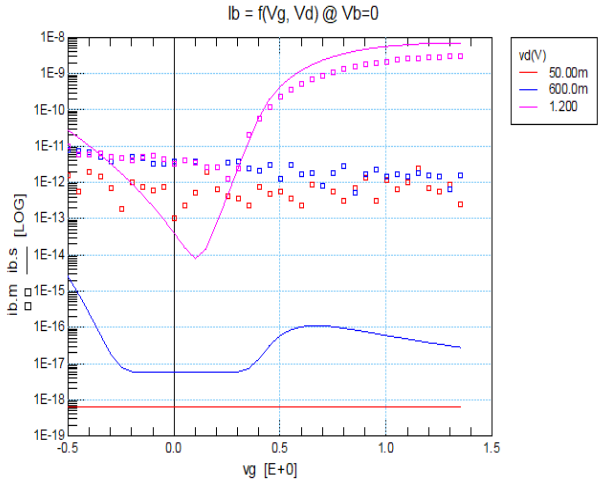


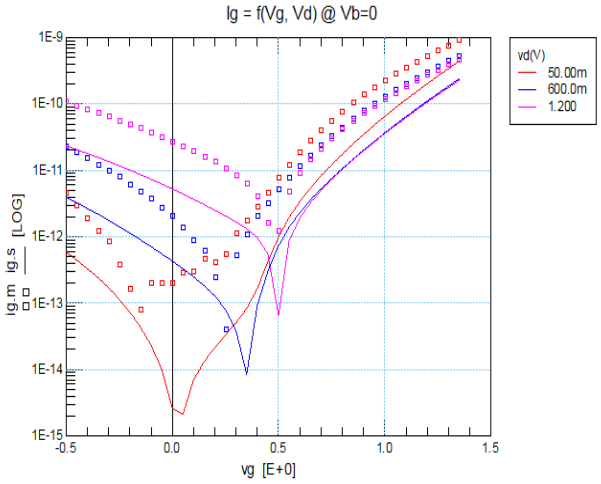




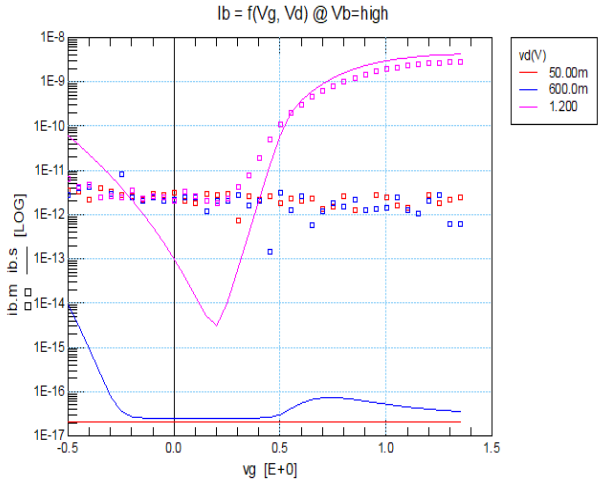


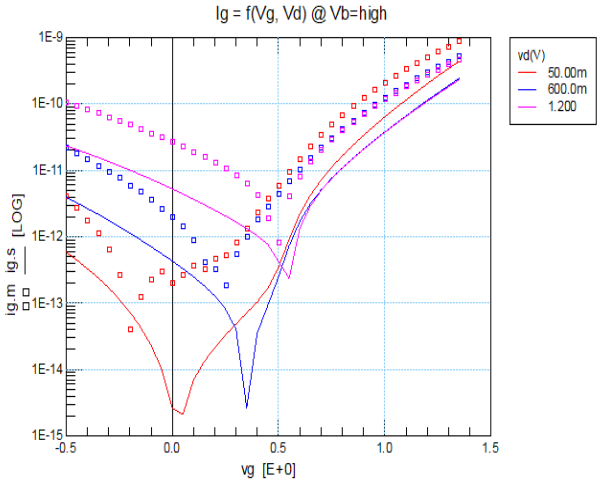
idvg, Ib,Ig, Vb = 0V, T = -40°C



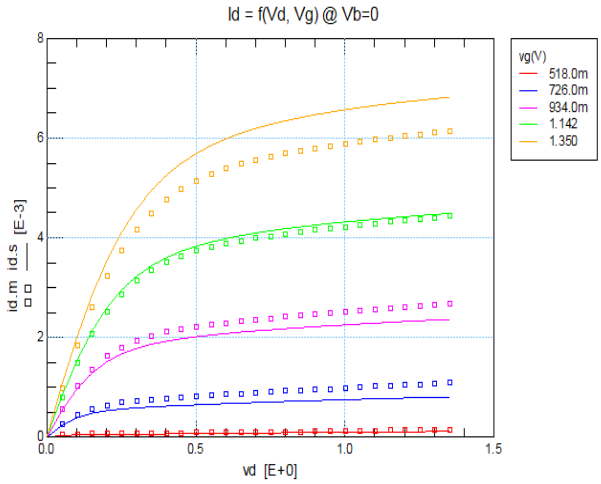


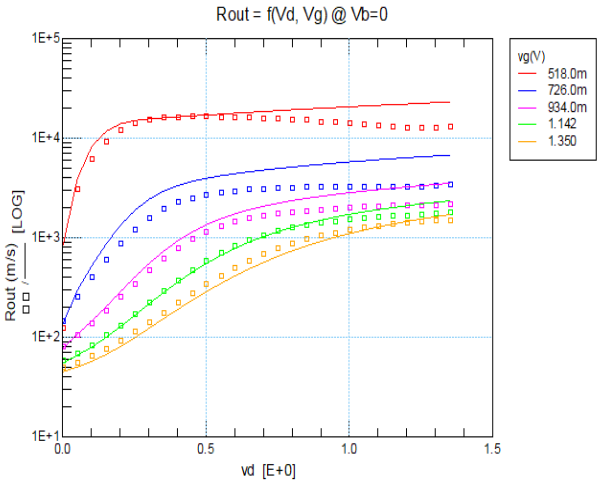
idvg, Ib,Ig, Vb = -1.2V, T = -40°C

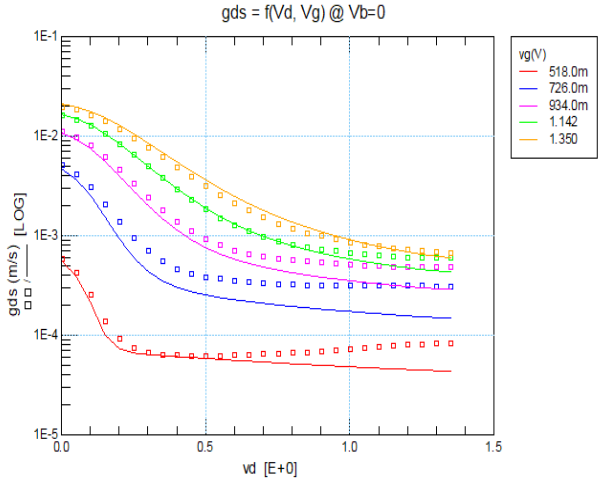




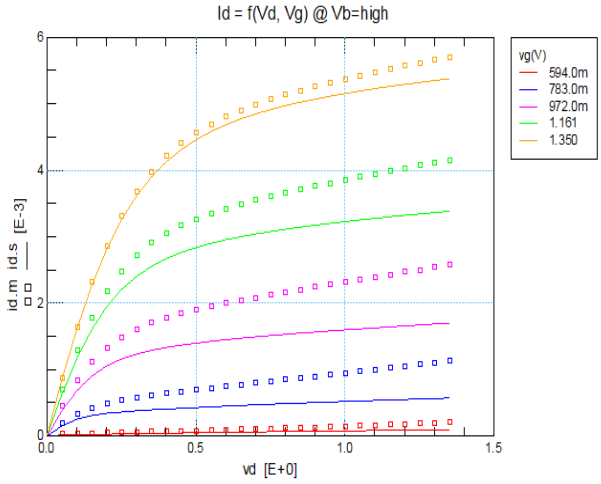
idvd, Vb = 0V, T = -40°C

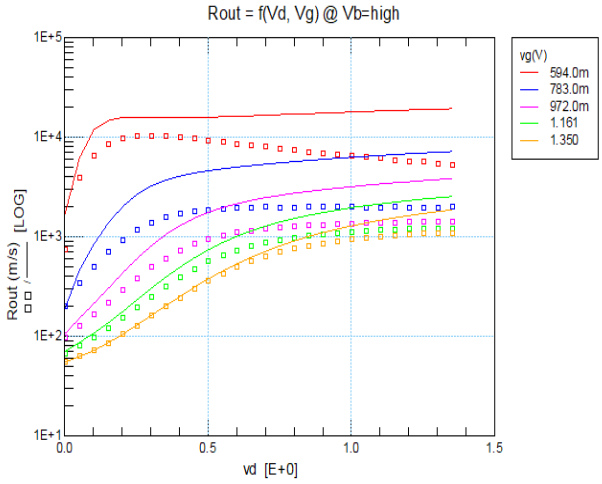


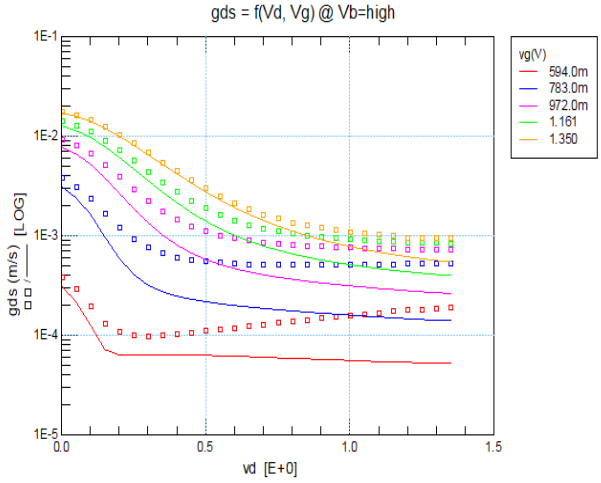




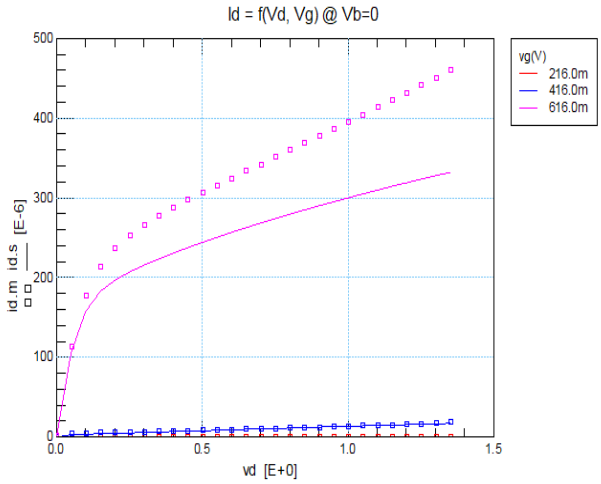
idvd_vbmin, Vb = -1.2V, T = -40°C

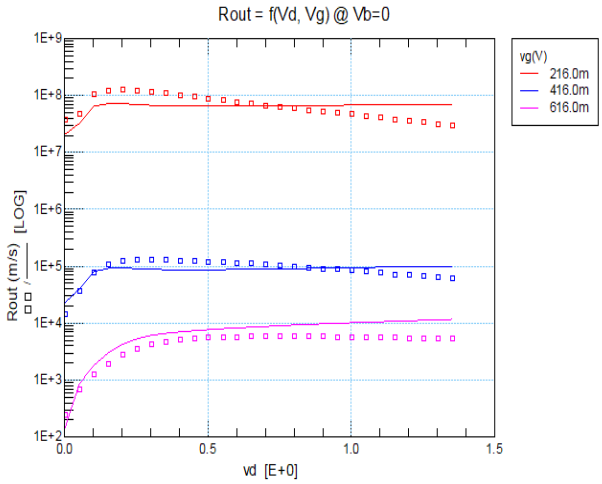


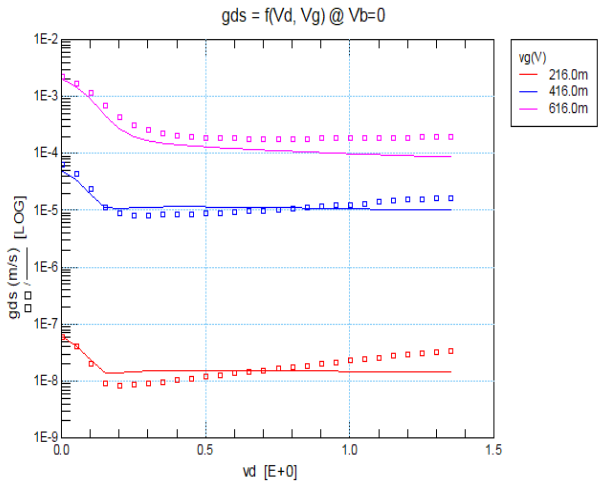




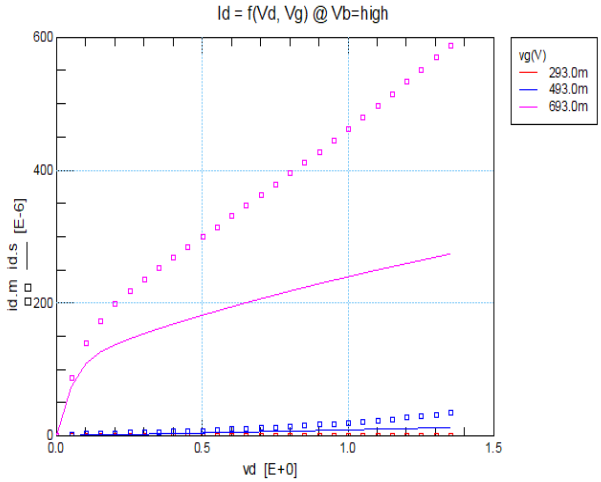
idvd_vth, Vb = 0V, T = -40°C

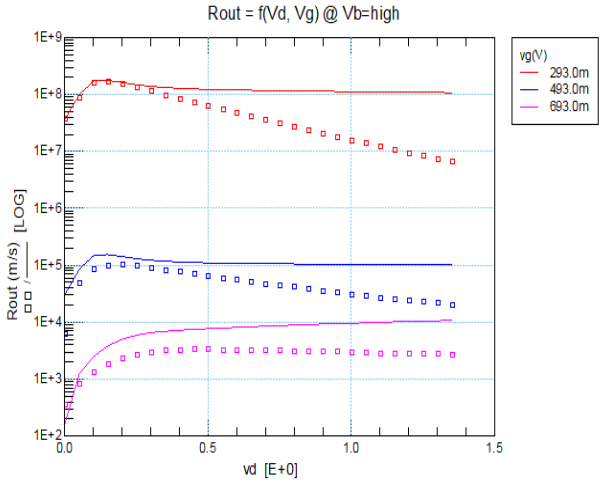


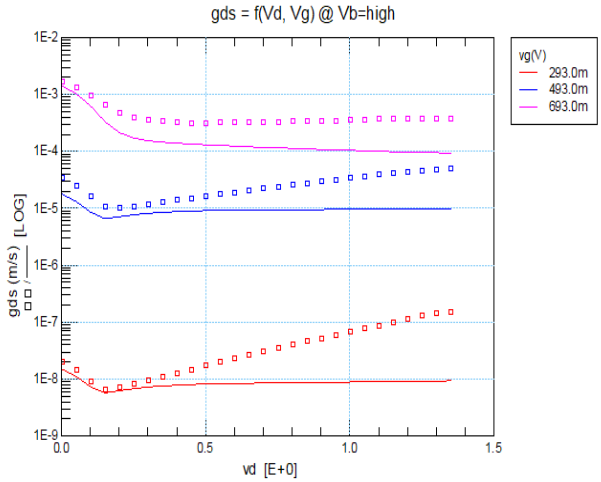




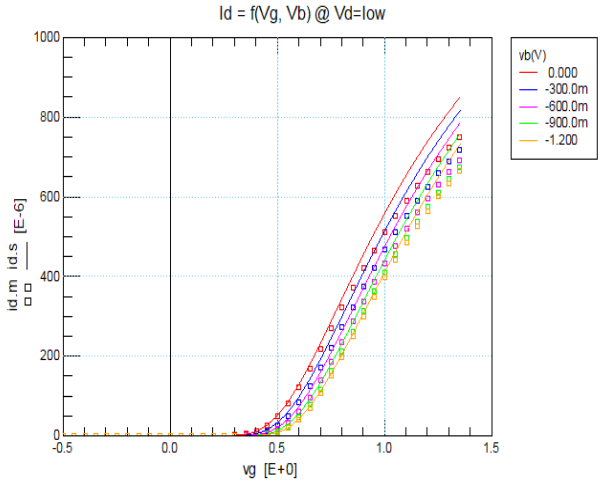
idvd_vth_vbmin, Vb = -1.2V, T = -40°C

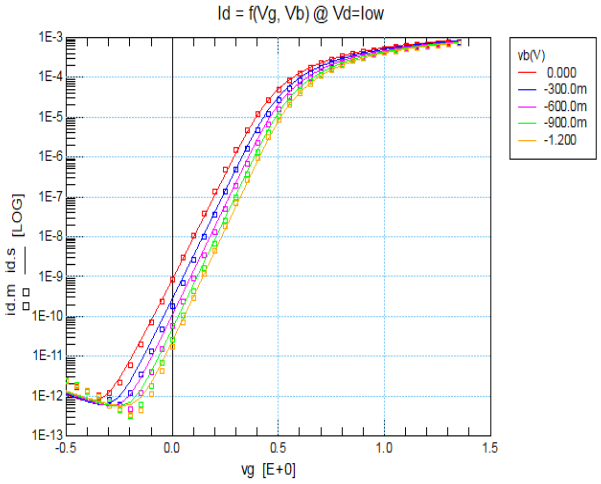


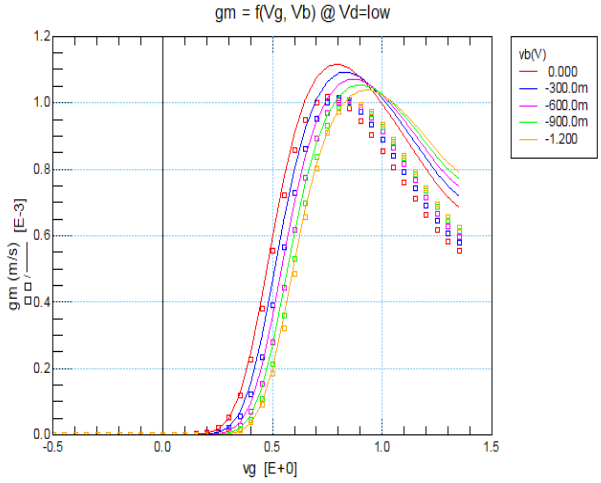


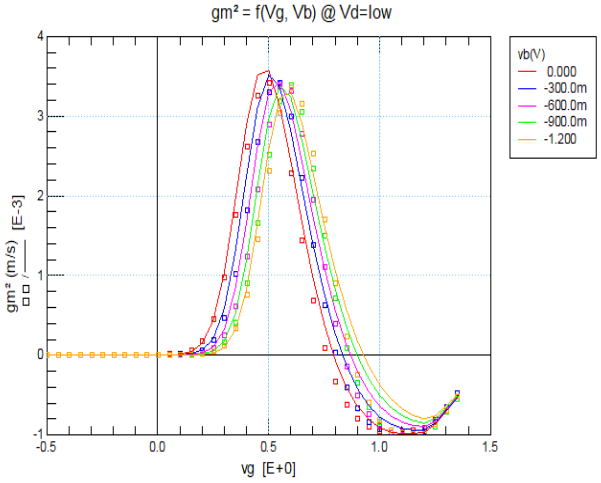


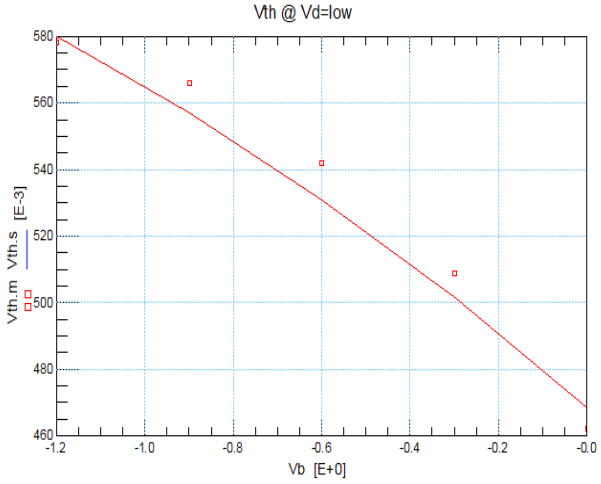
idvg, Vd = 0.05V, T = 70°C

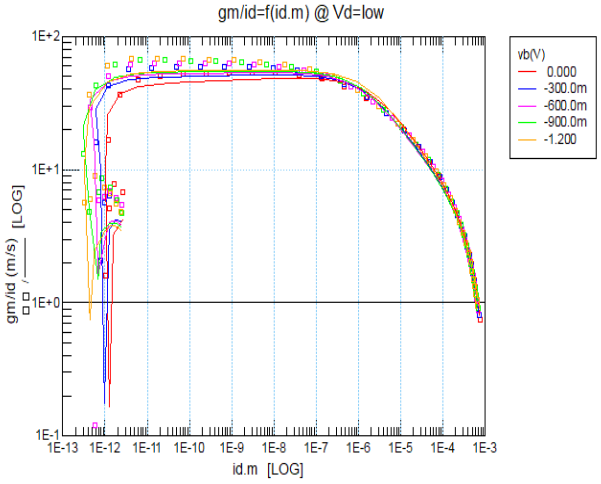


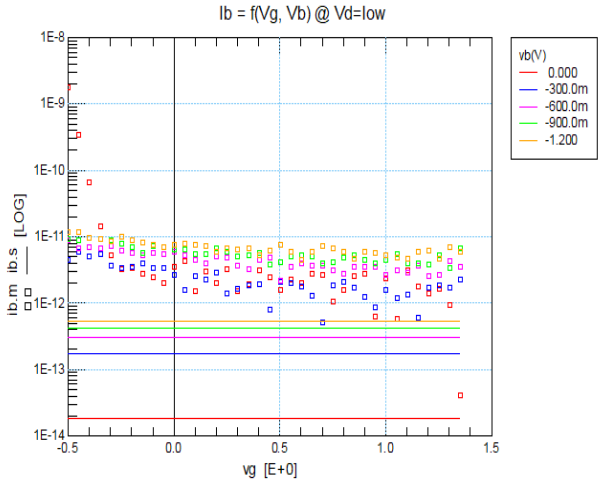


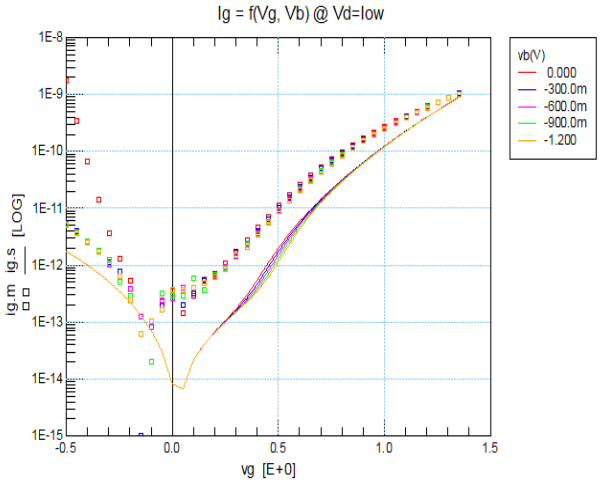




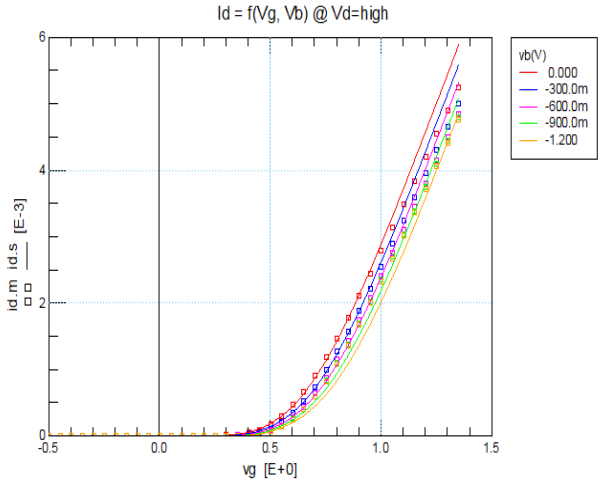


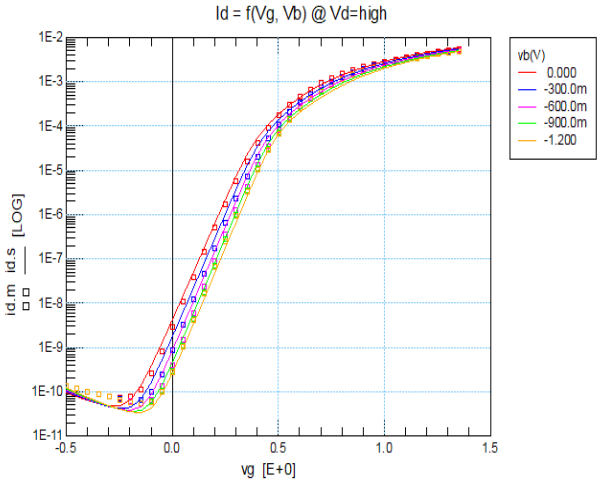


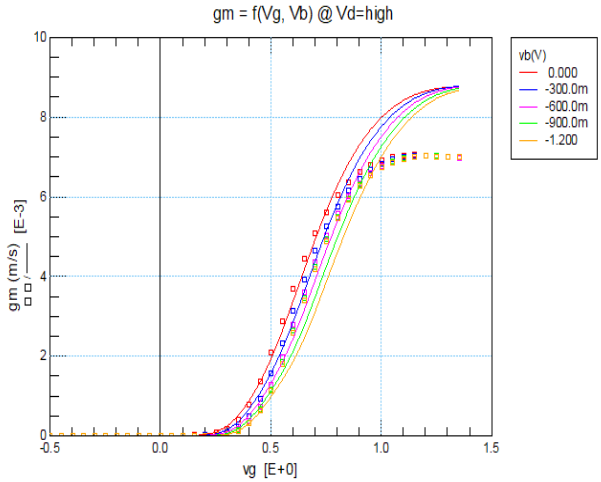


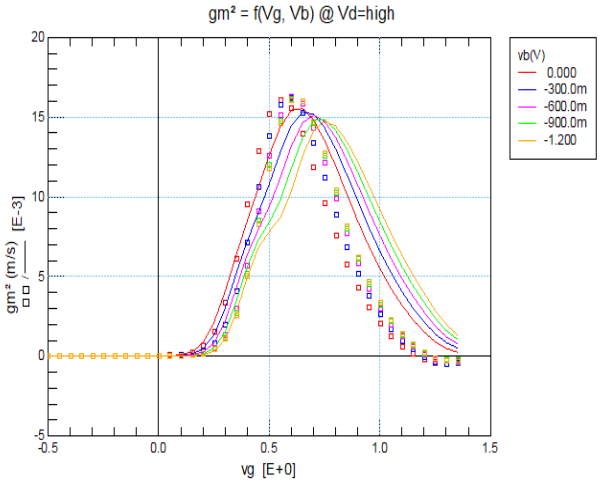


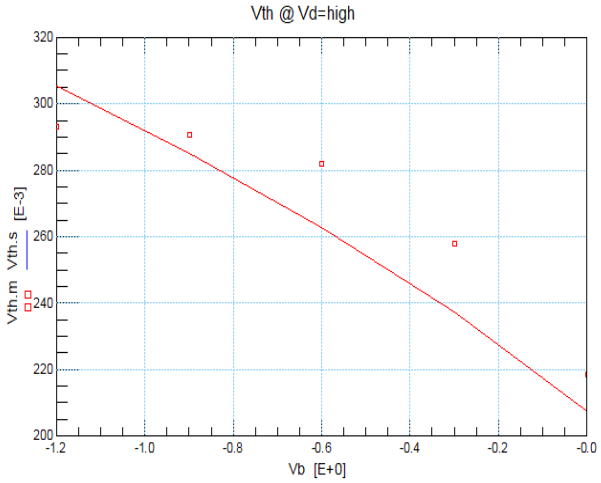
idvg, Vd = 1.2V, T = 70°C

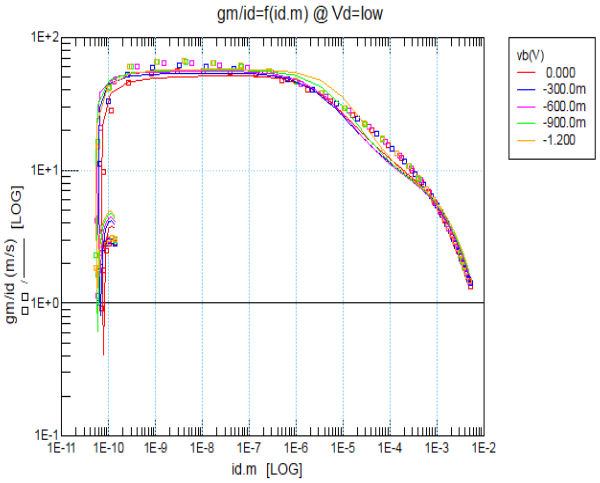


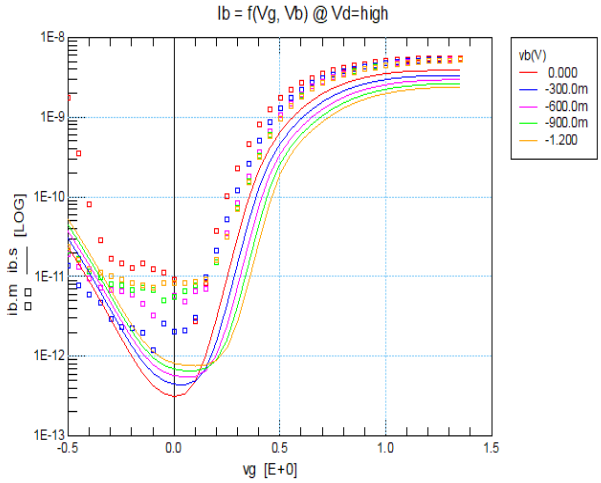


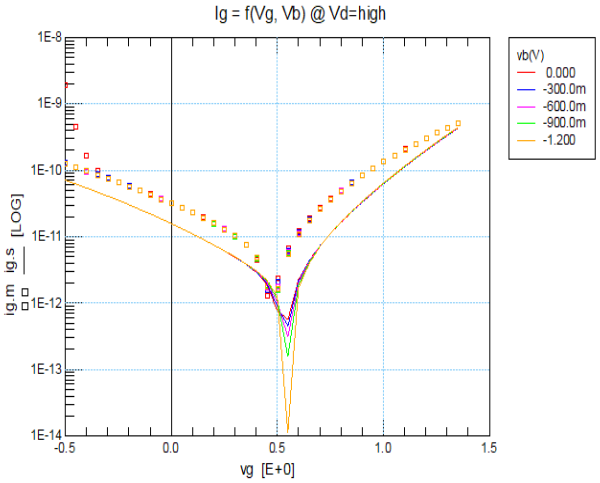




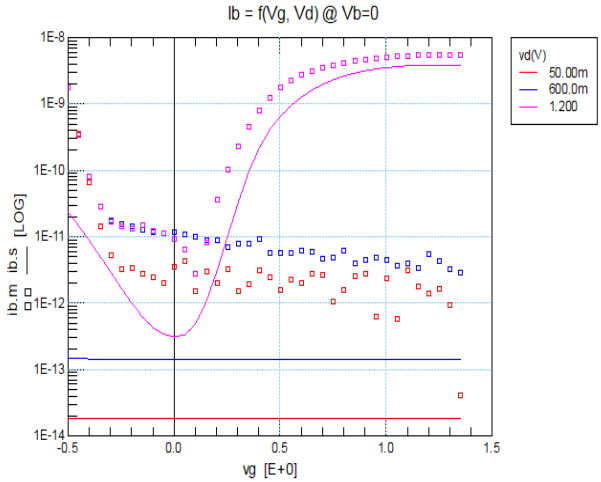


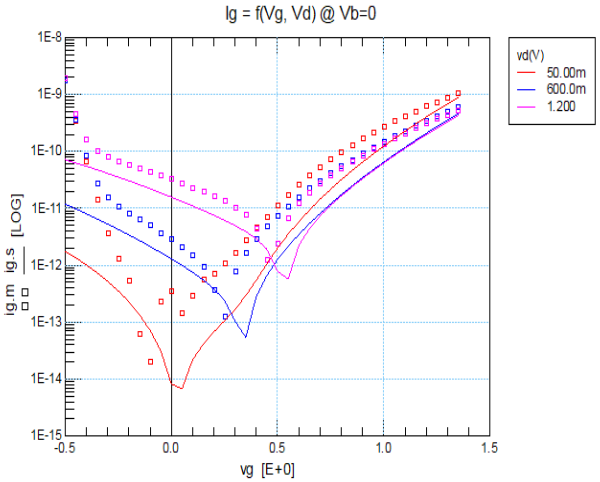




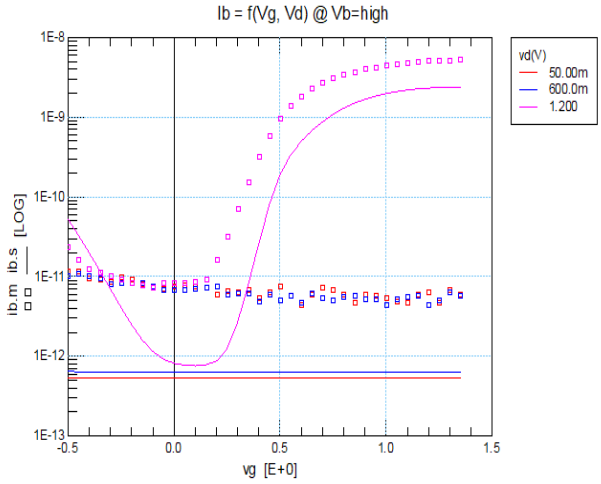


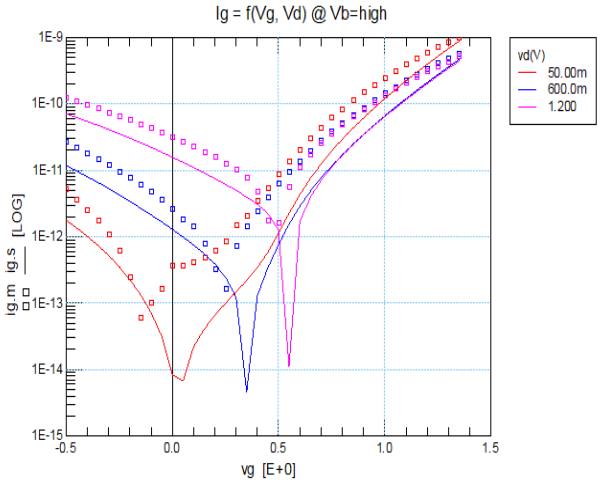
idvg, Ib,Ig, Vb = 0V, T = 70°C



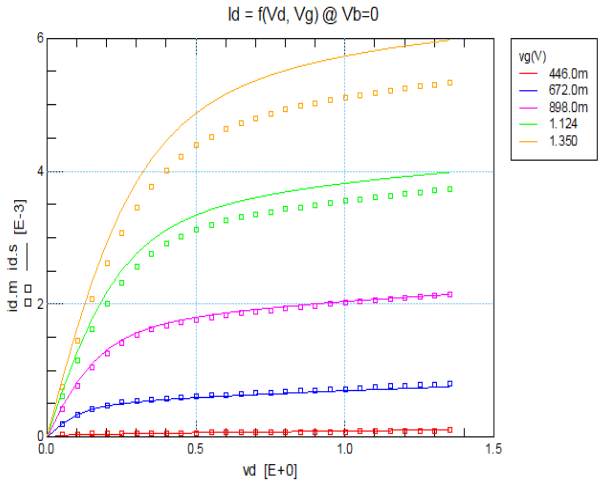


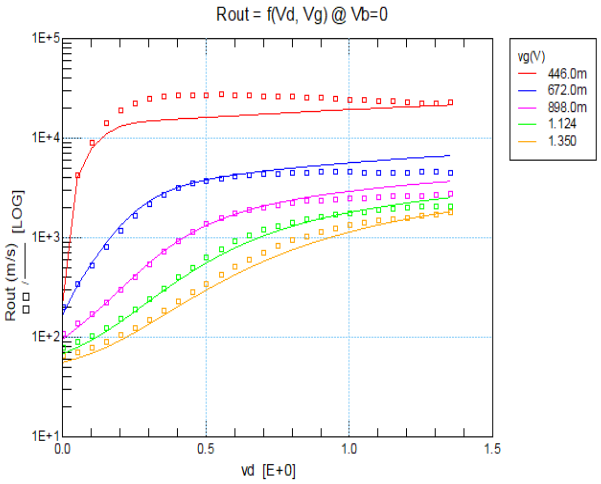
idvg, Ib,Ig, Vb = -1.2V, T = 70°C

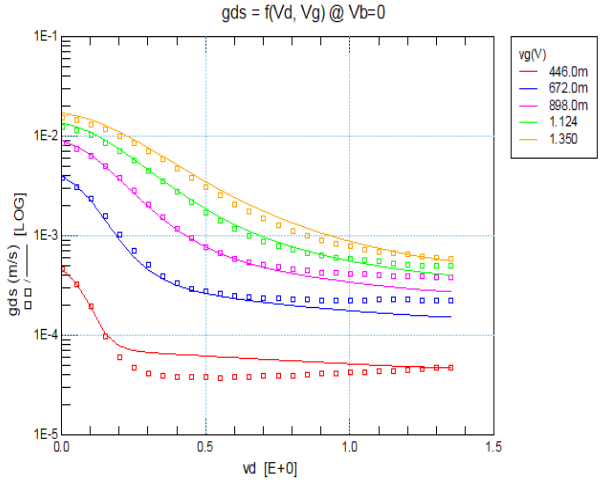




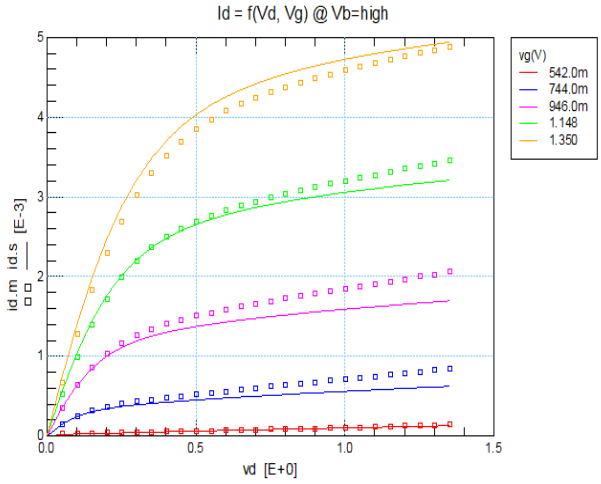
idvd, Vb = 0V, T = 70°C

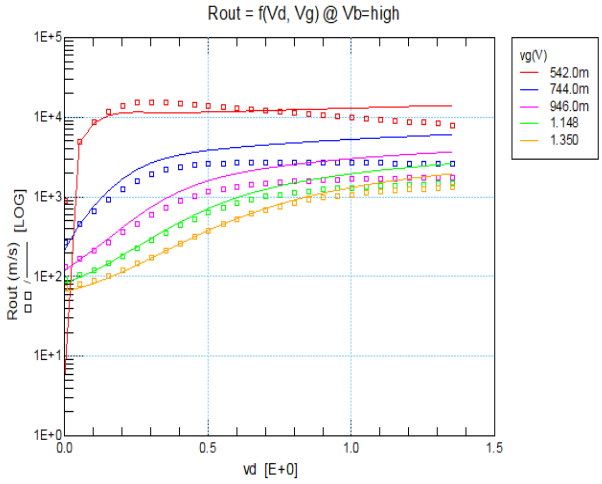


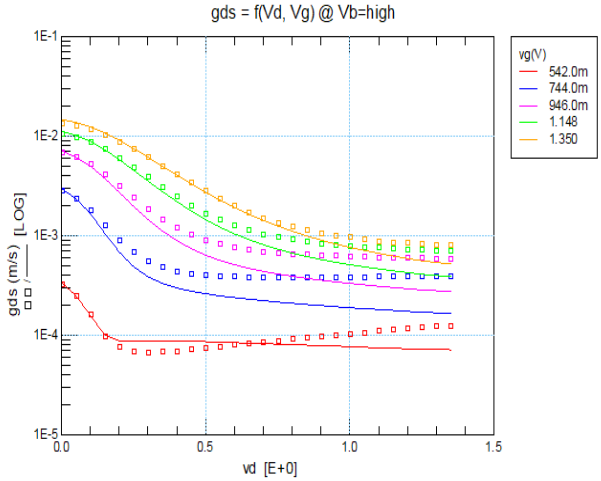




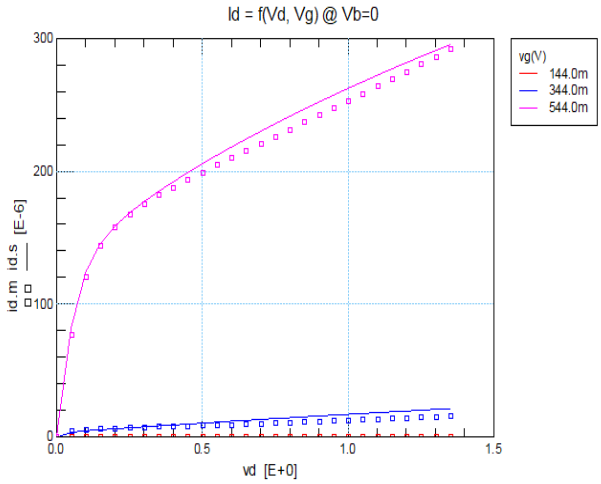
idvd_vbmin, Vb = -1.2V, T = 70°C

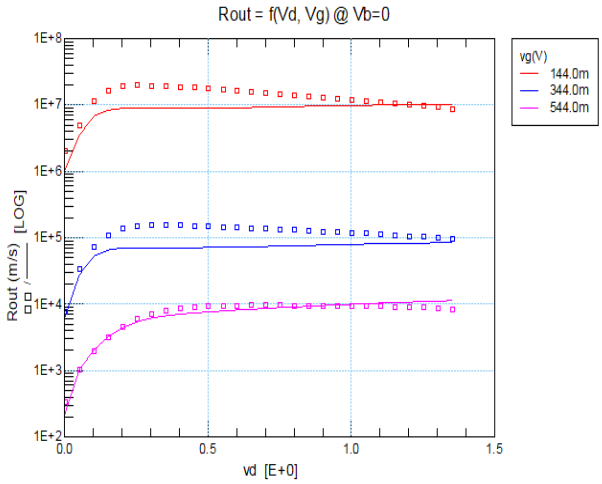


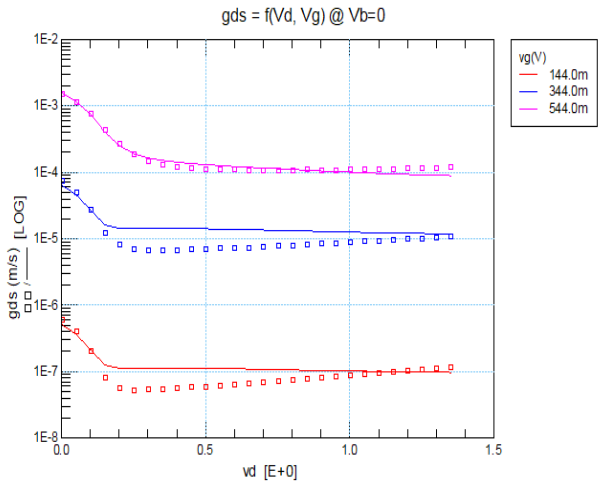




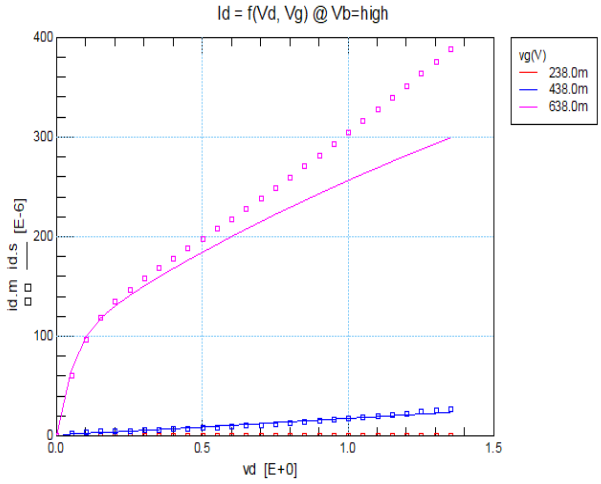
idvd_vth, Vb = 0V, T = 70°C

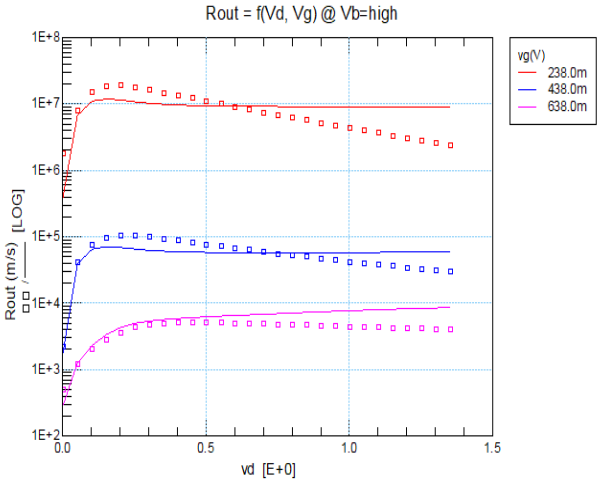


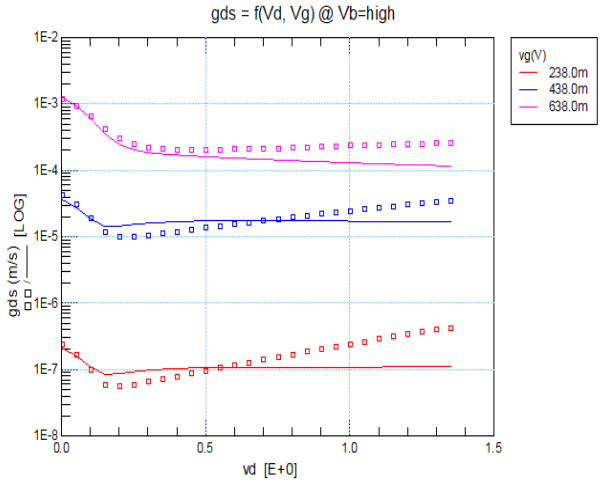




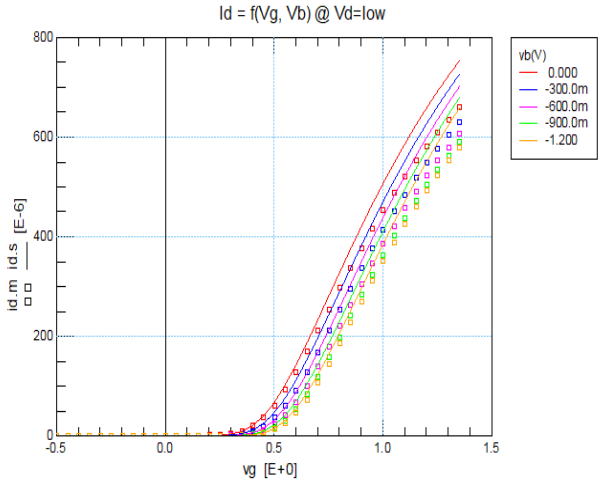
idvd_vth_vbmin, Vb = -1.2V, T = 70°C

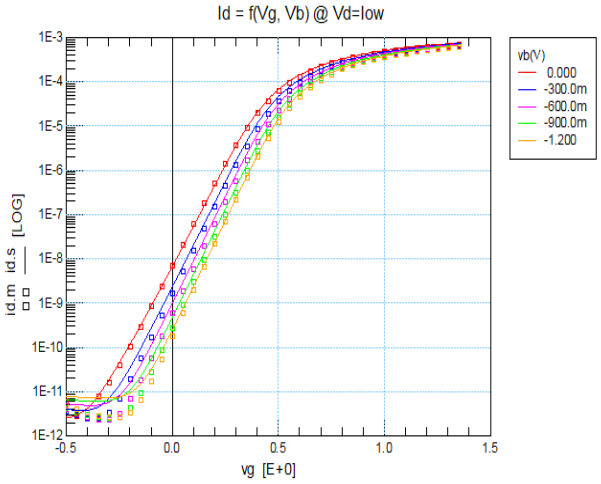


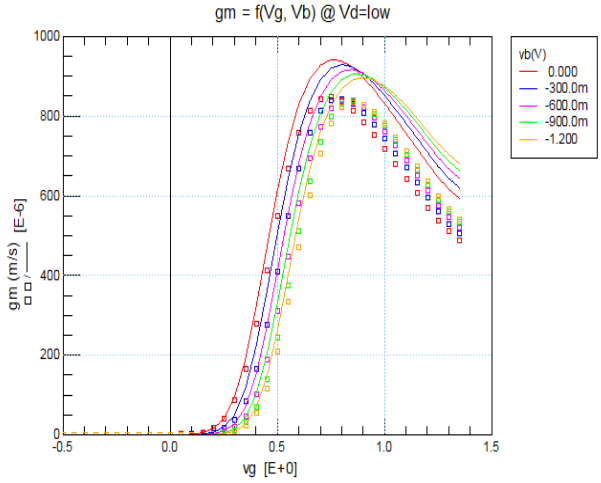


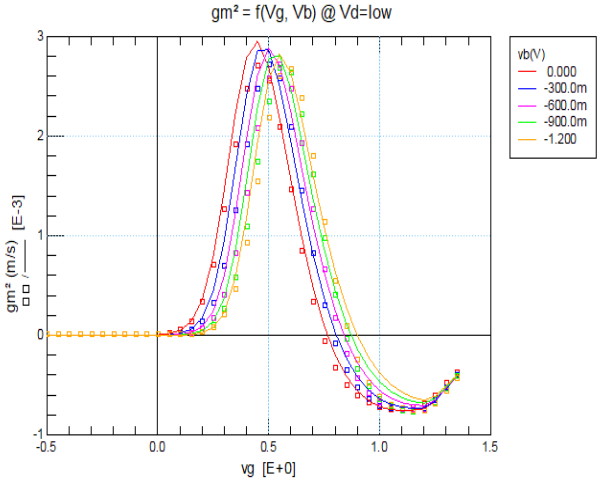


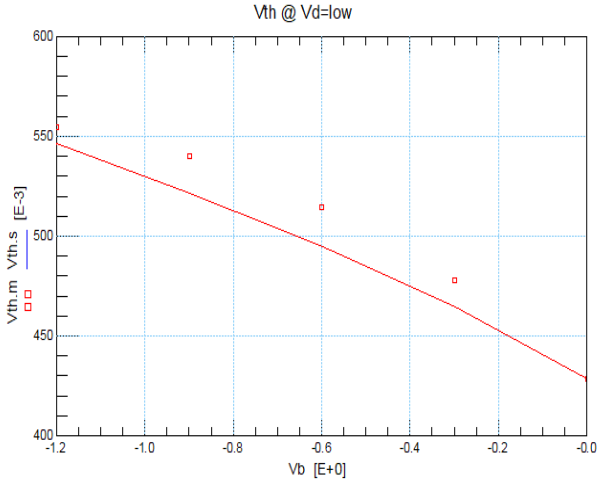
idvg, Vd = 0.05V, T = 125°C

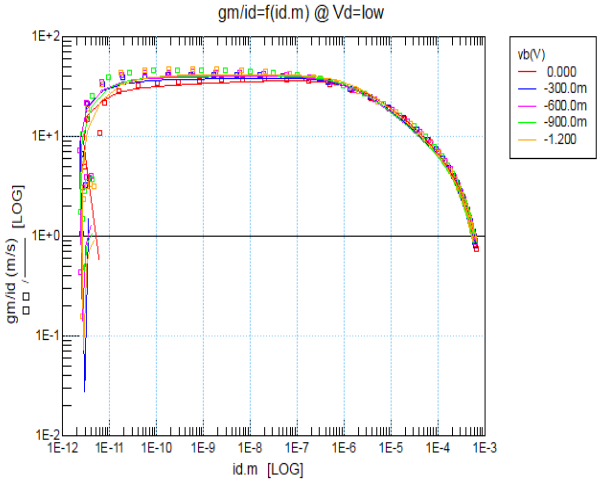


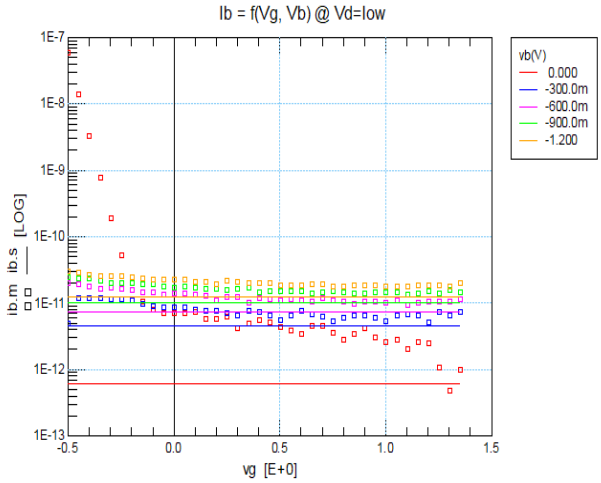


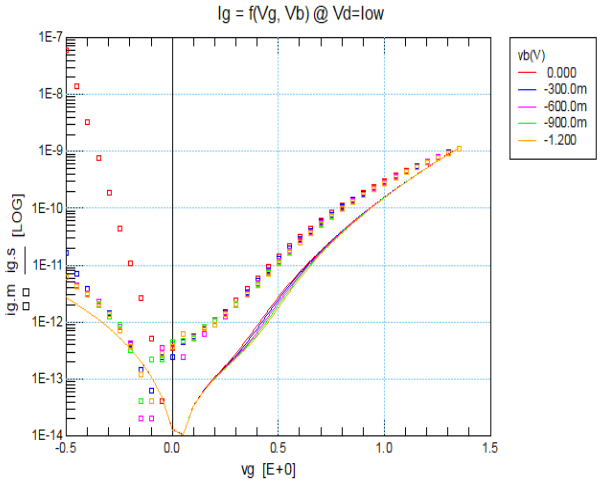




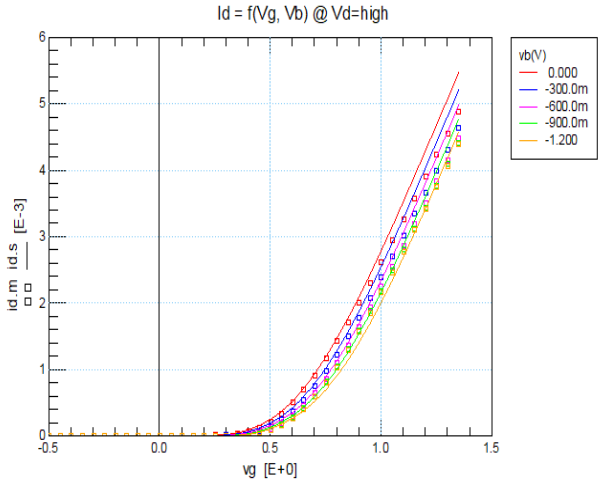


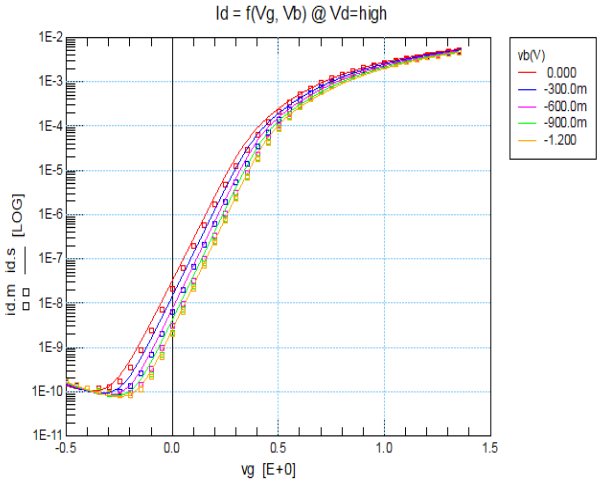


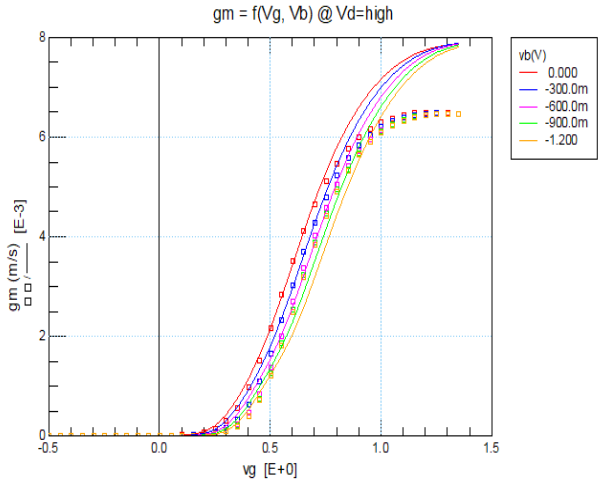


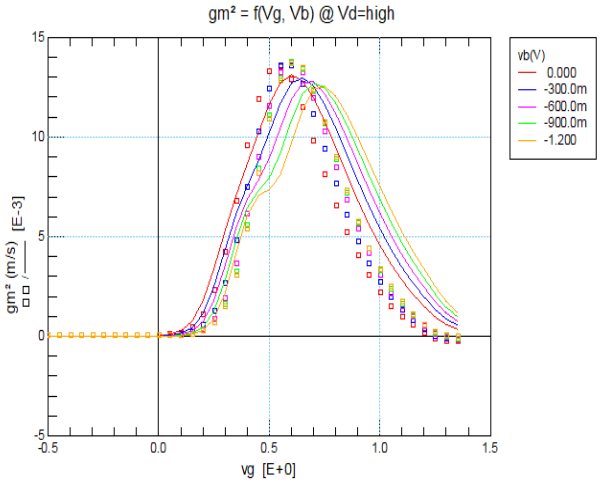


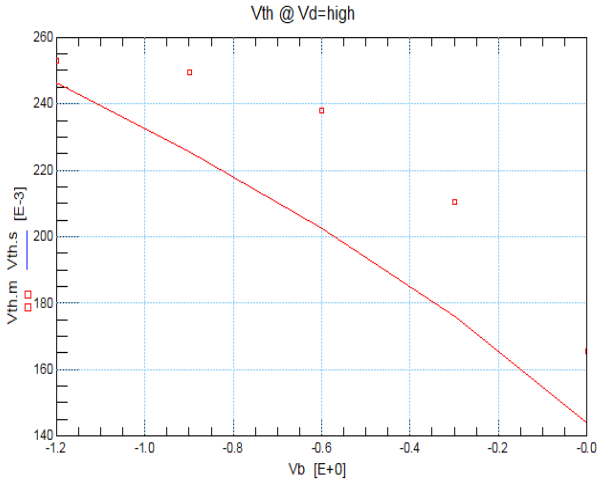
idvg, Vd = 1.2V, T = 125°C

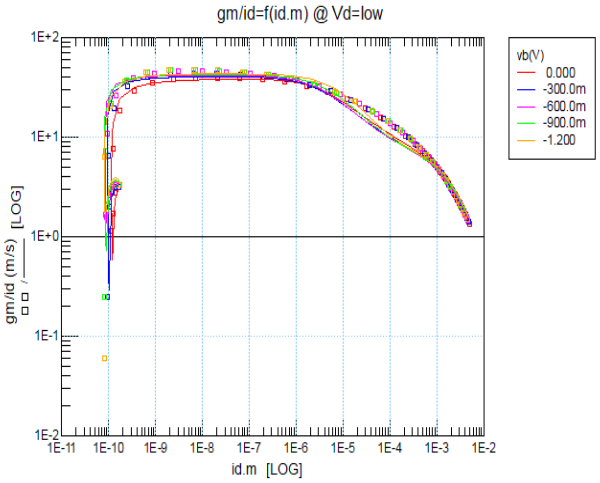


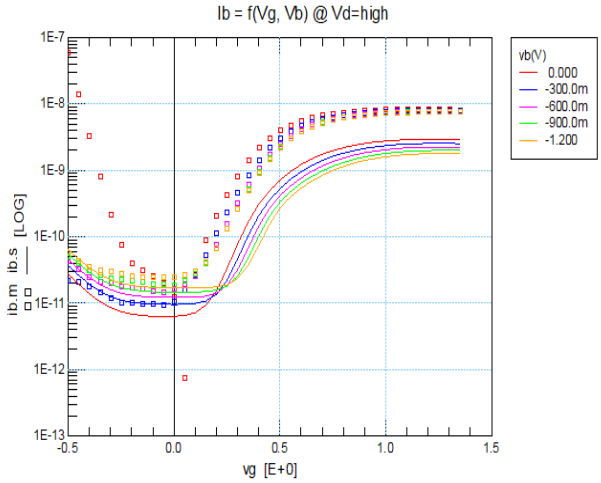


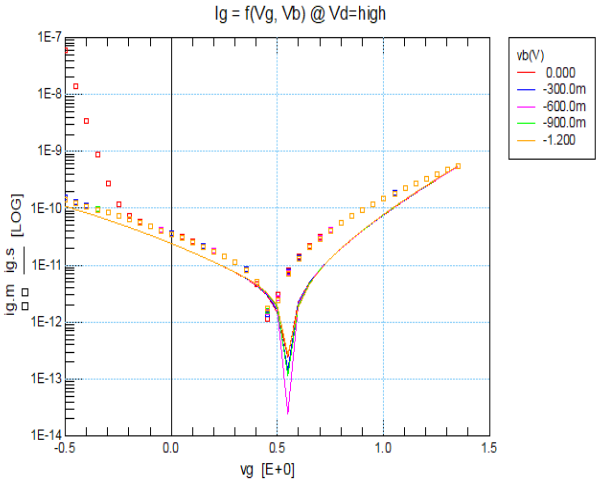




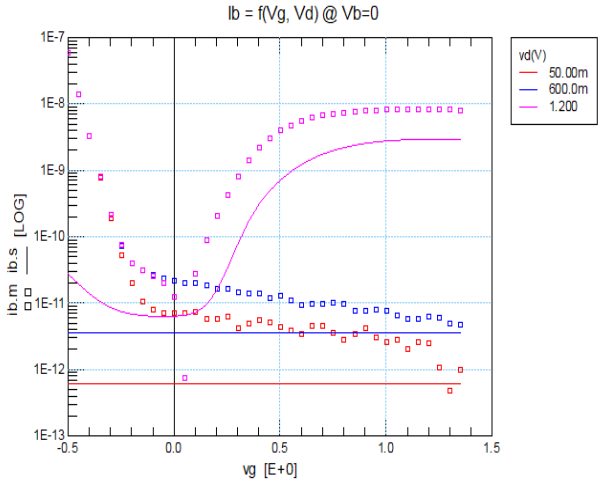


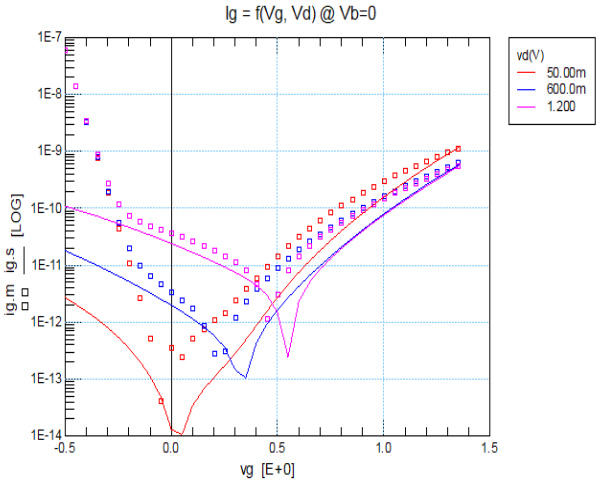




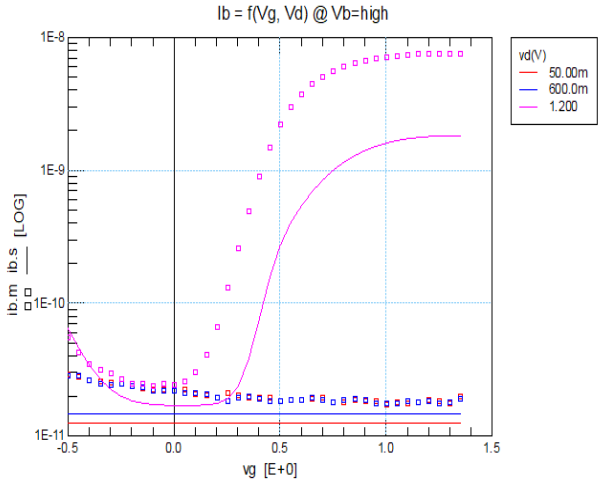


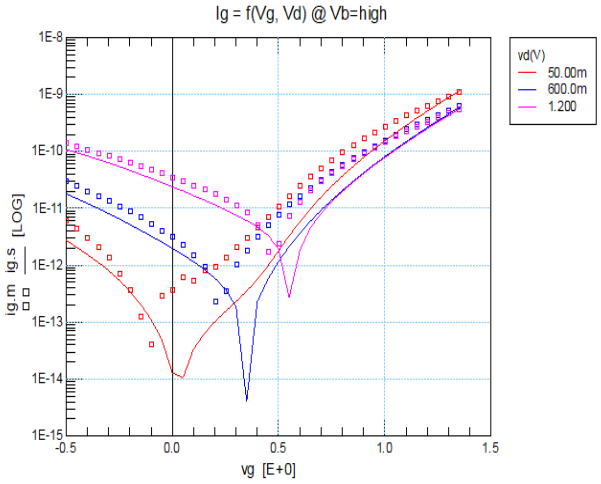
idvg, Ib,Ig, Vb = 0V, T = 125°C



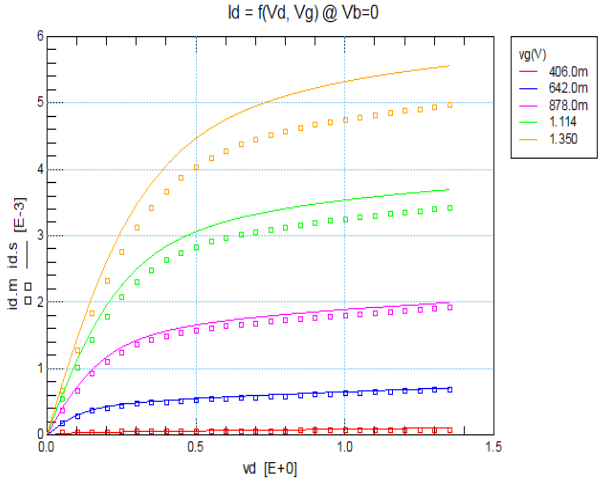


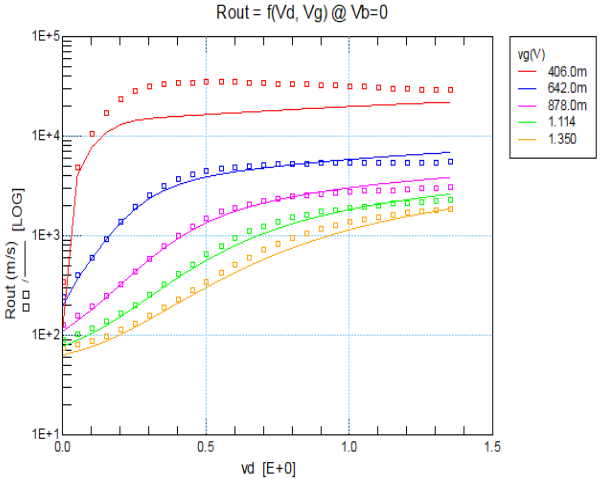
idvg, Ib,Ig, Vb = -1.2V, T = 125°C

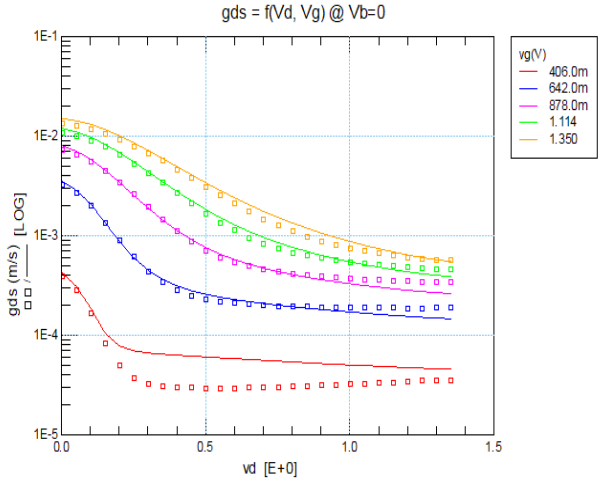




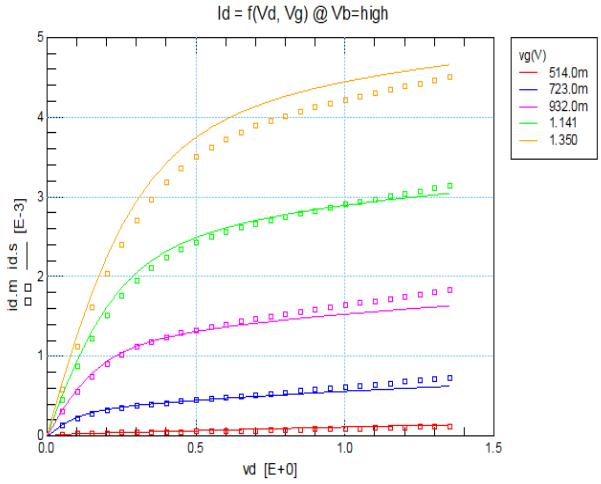
idvd, Vb = 0V, T = 125°C

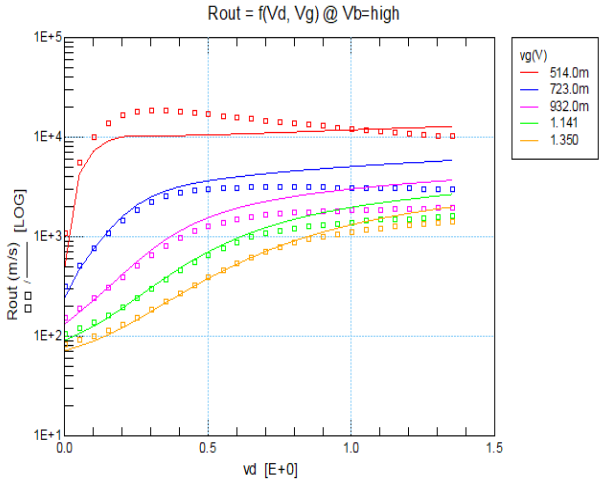


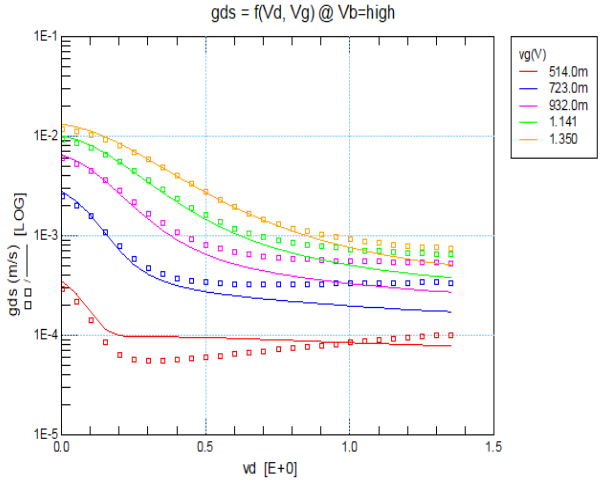




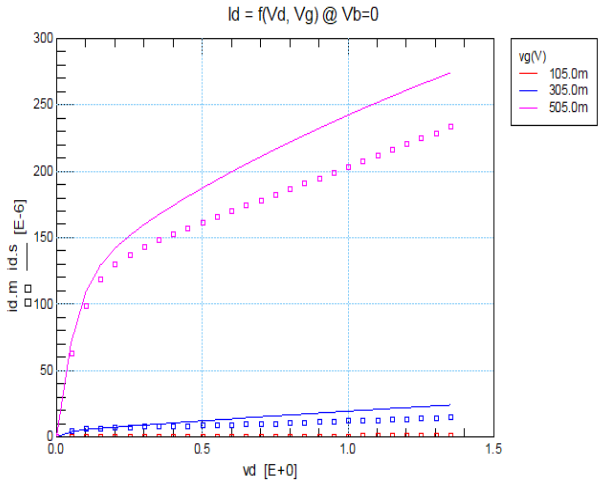
idvd_vbmin, Vb = -1.2V, T = 125°C

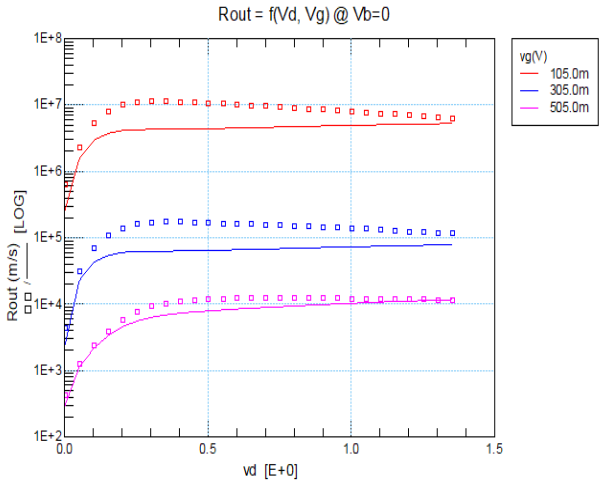


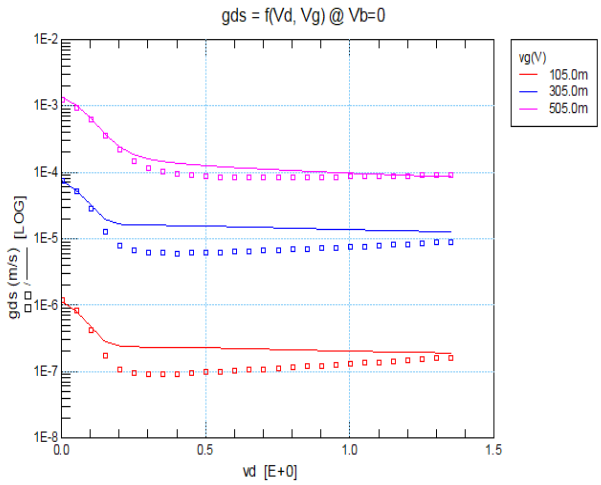




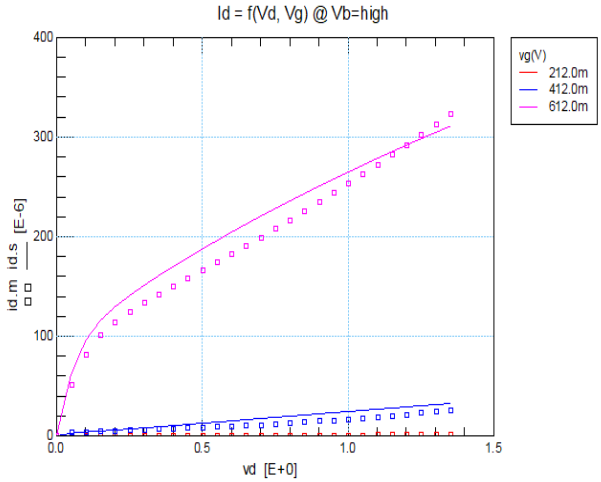
idvd_vth, Vb = 0V, T = 125°C

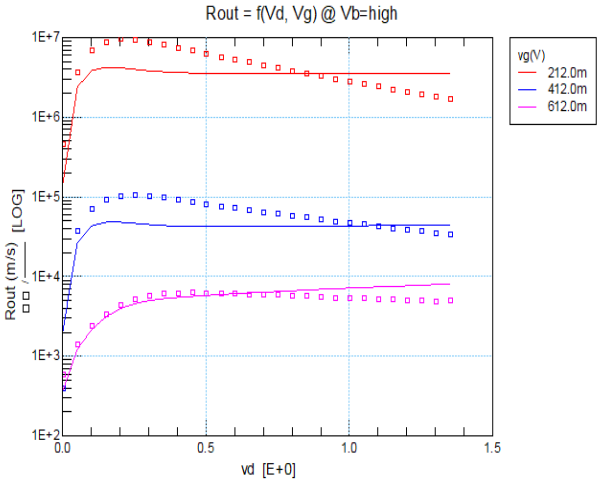


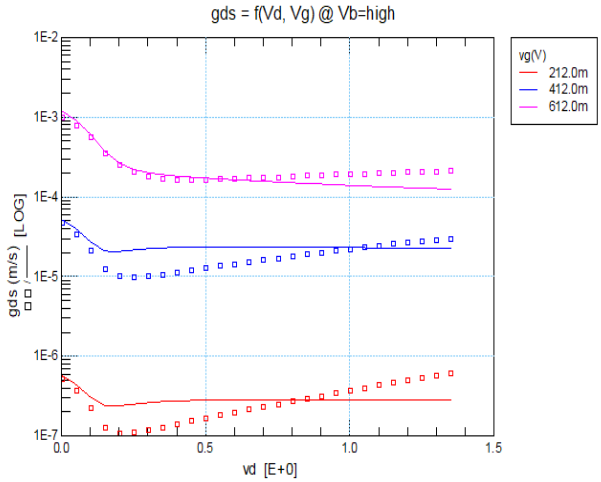




idvd_vth_vbmin, Vb = -1.2V, T = 125°C

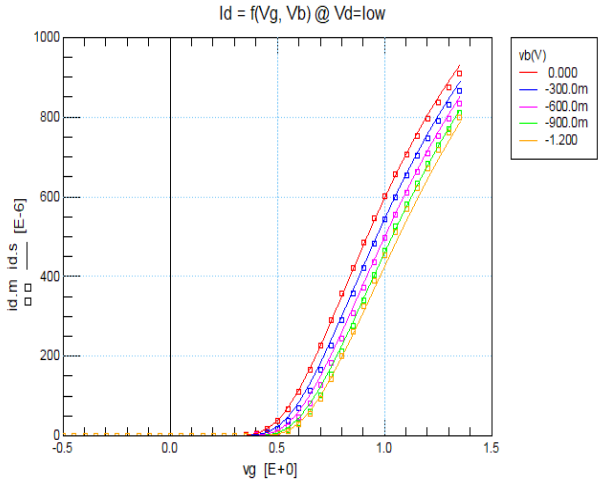


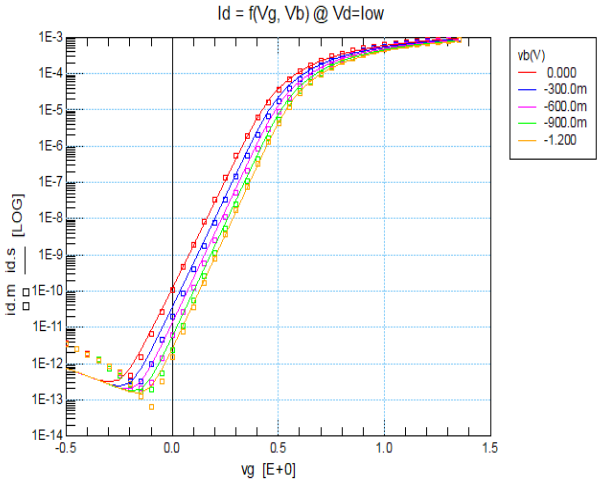


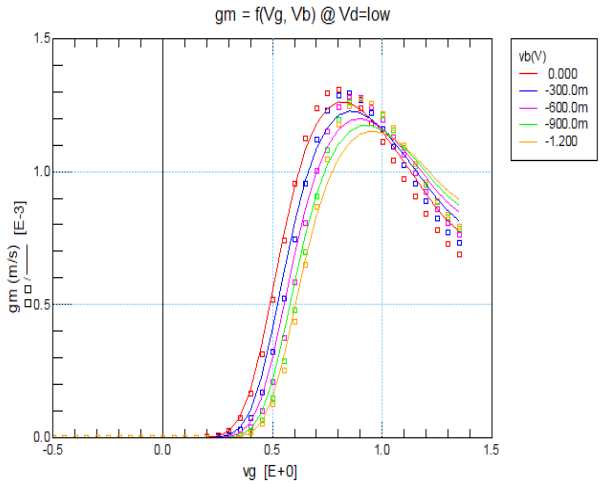


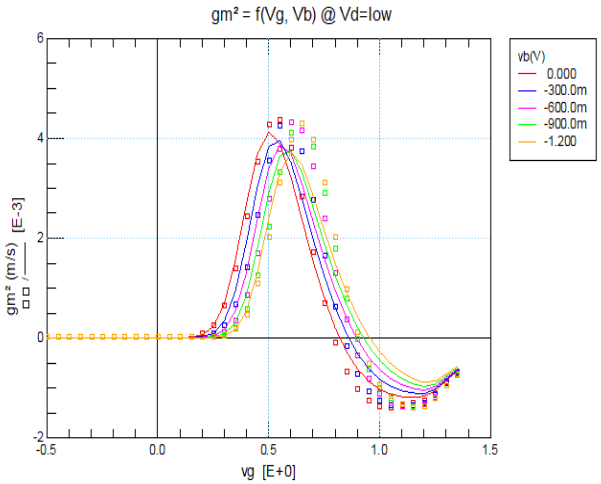
4.6 W10u0_L0u13_S380_2

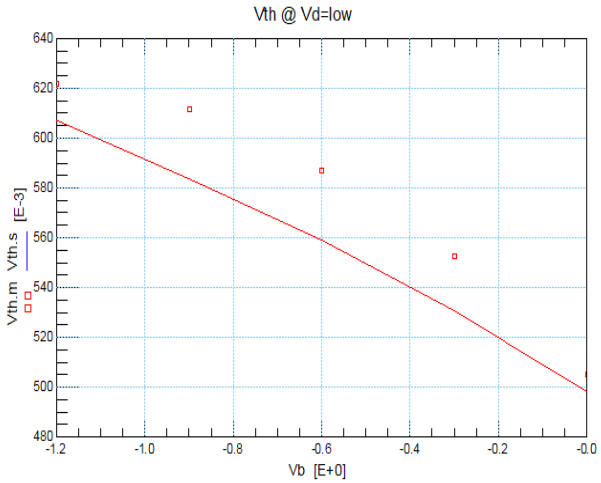
idvg, Vd = 0.05V, T = 27°C

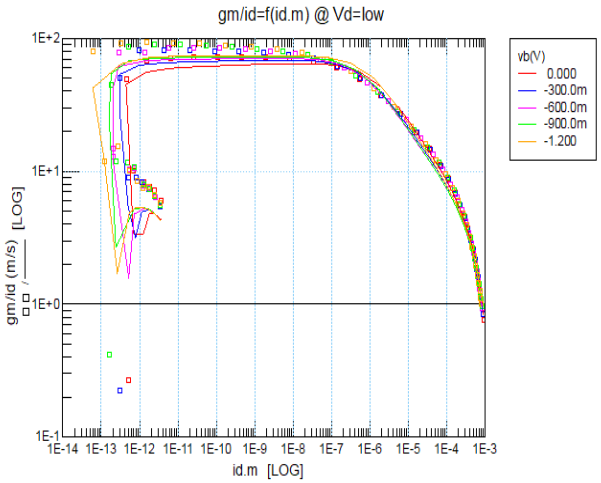


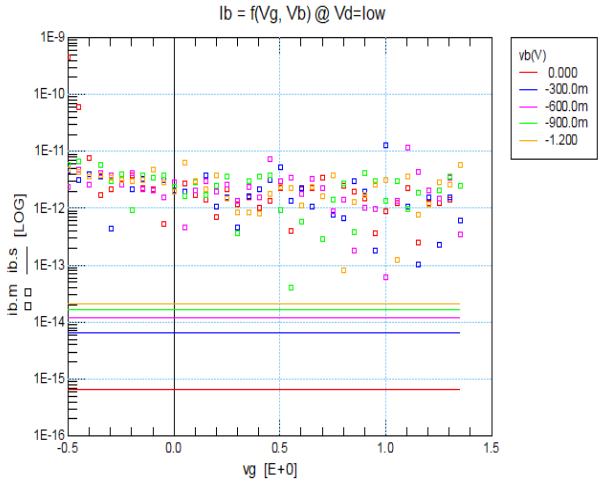


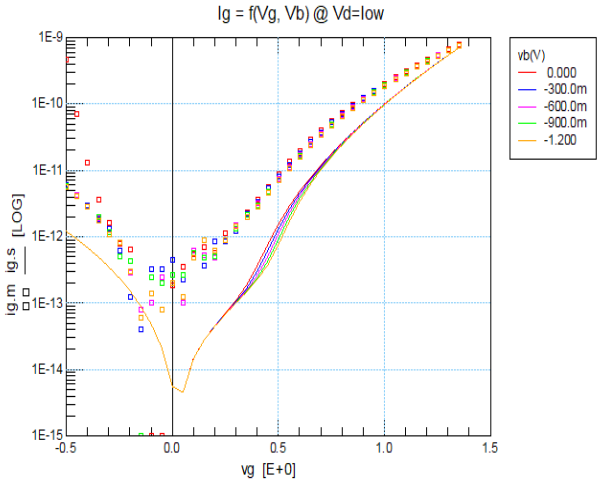




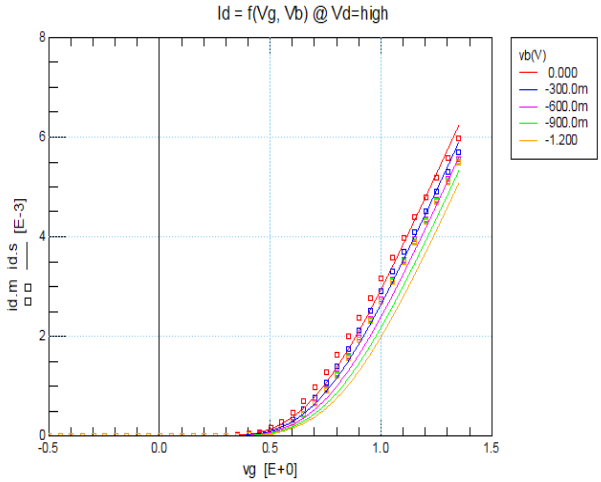


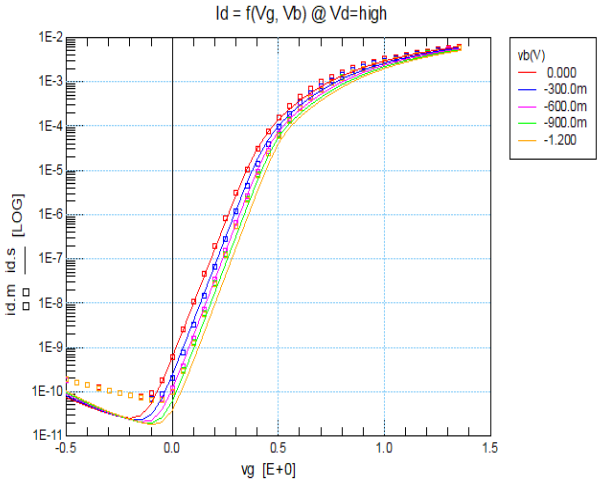


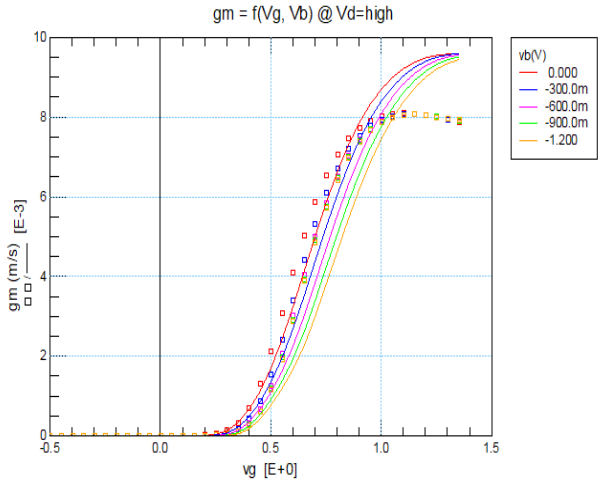


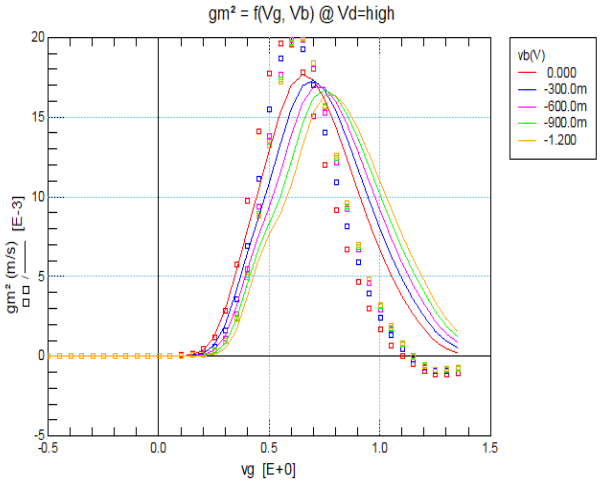


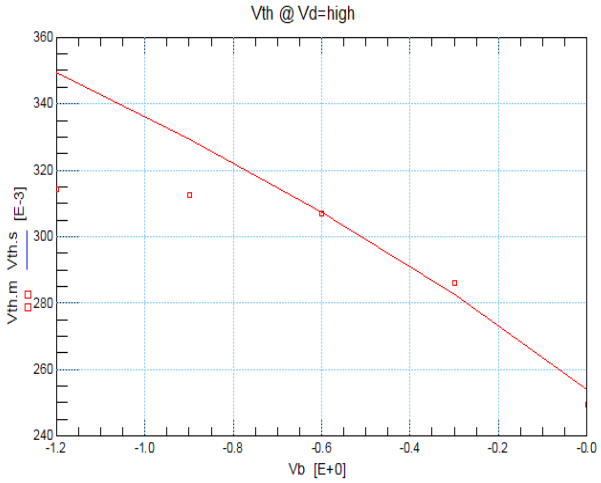
idvg, Vd = 1.2V, T = 27°C

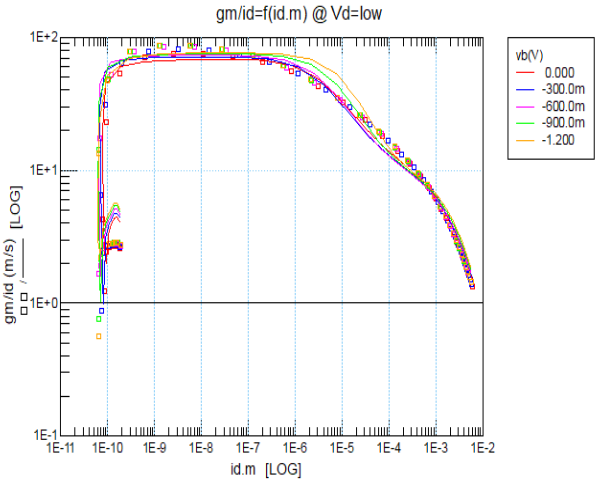


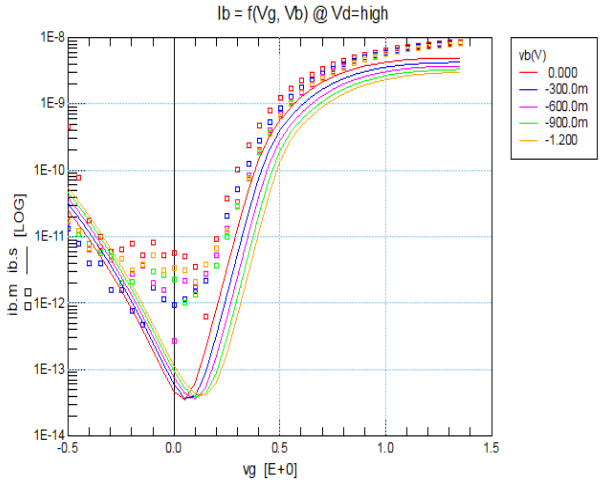


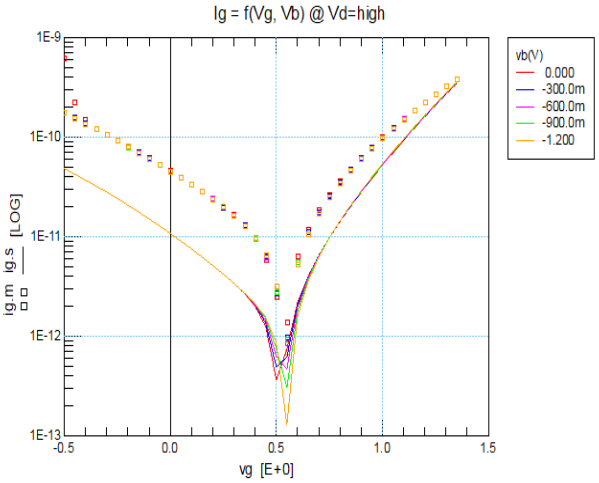




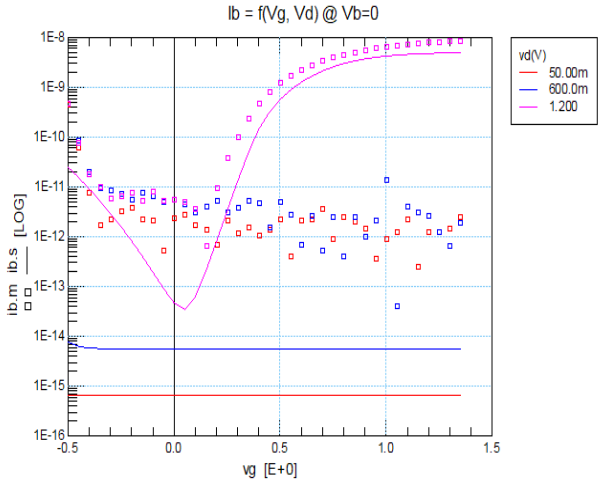


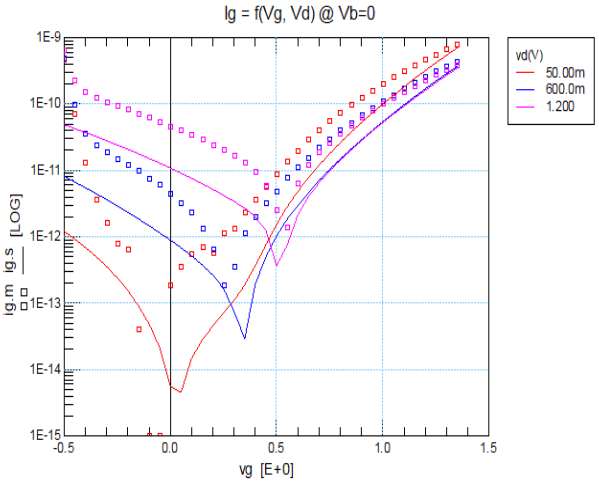




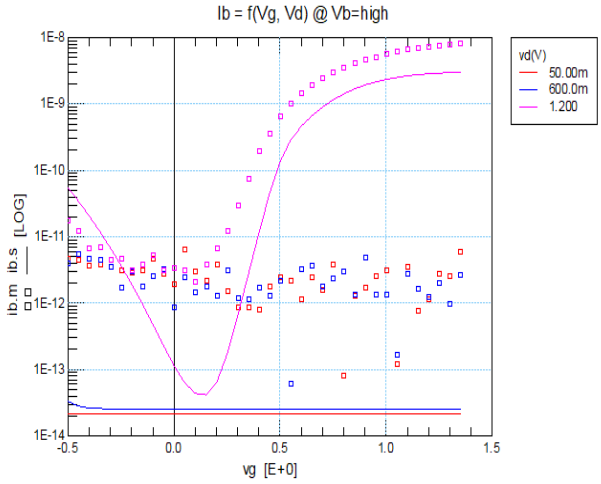


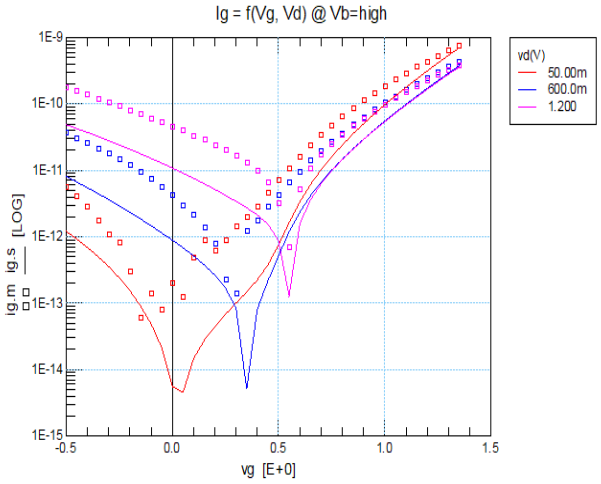
idvg, Ib,Ig, Vb = 0V, T = 27°C



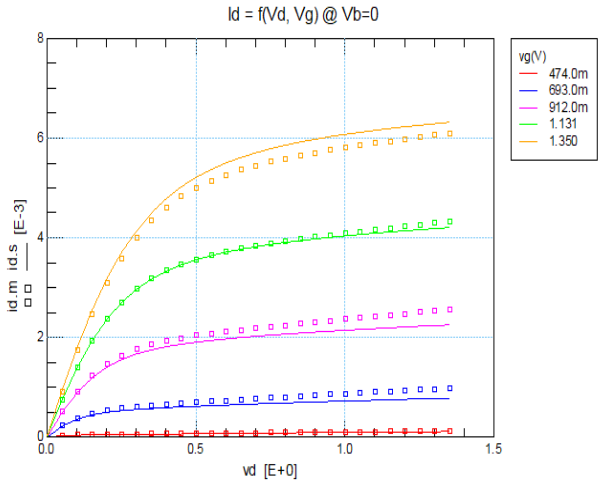


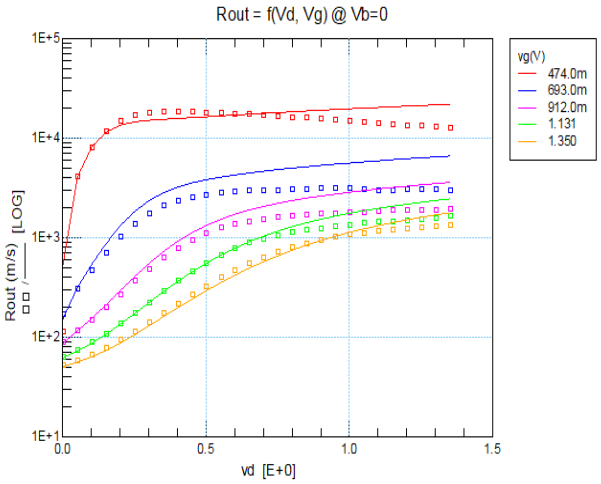
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

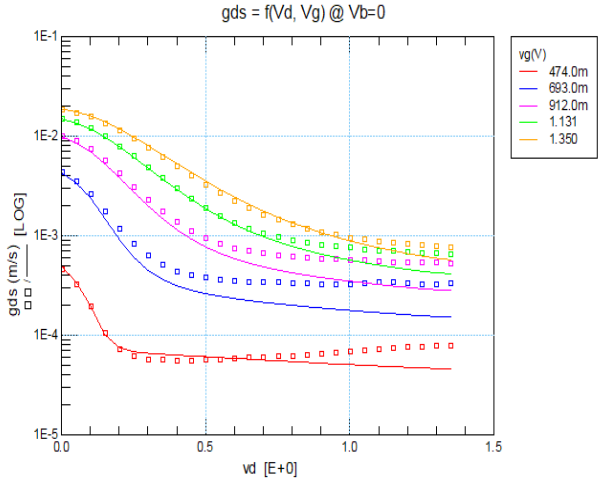




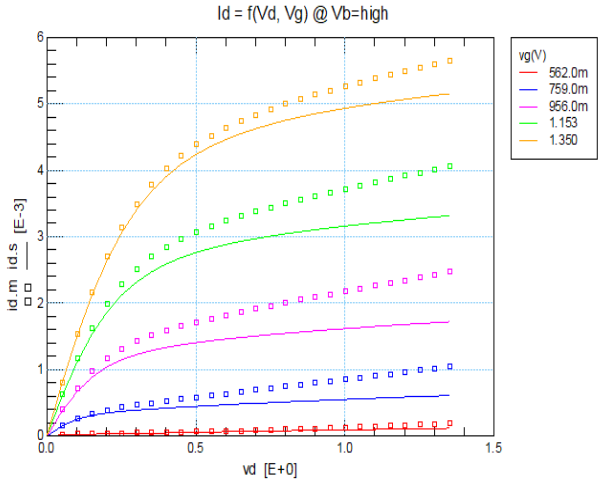
idvd, Vb = 0V, T = 27°C

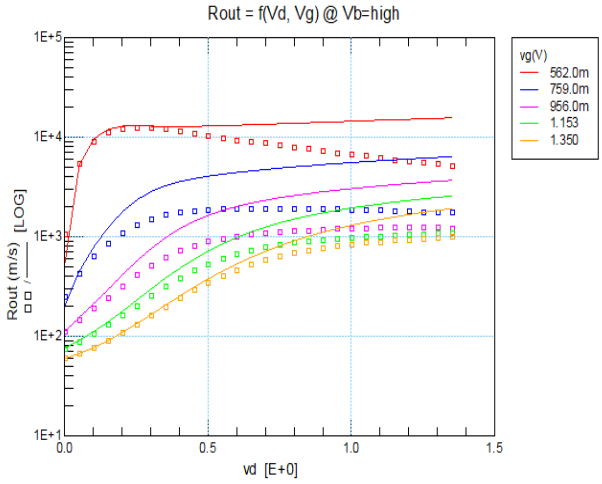


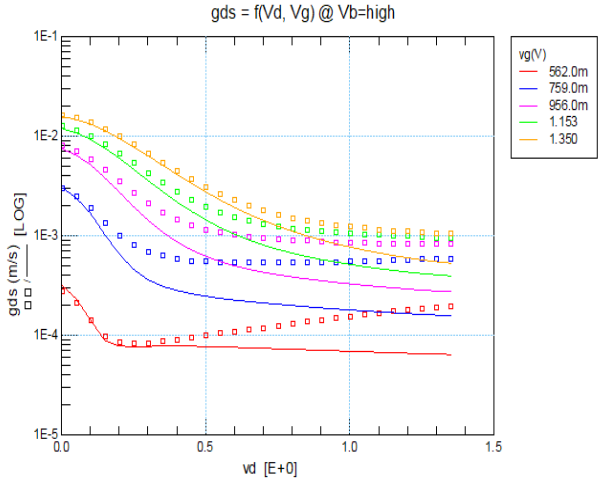




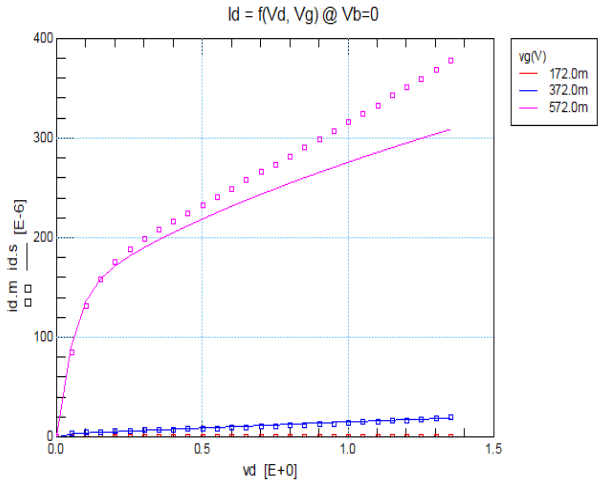
idvd_vbmin, Vb = -1.2V, T = 27°C

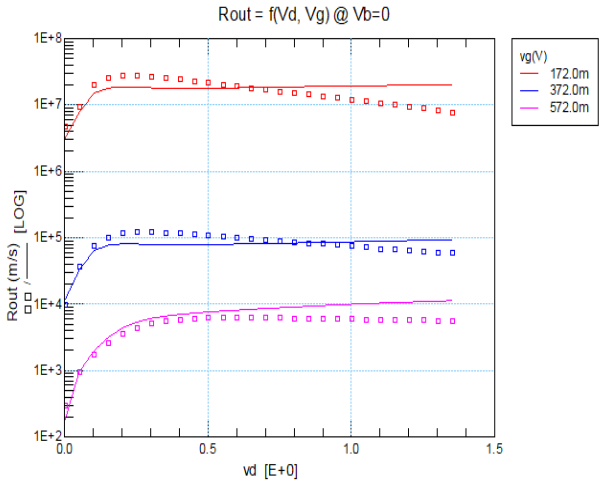


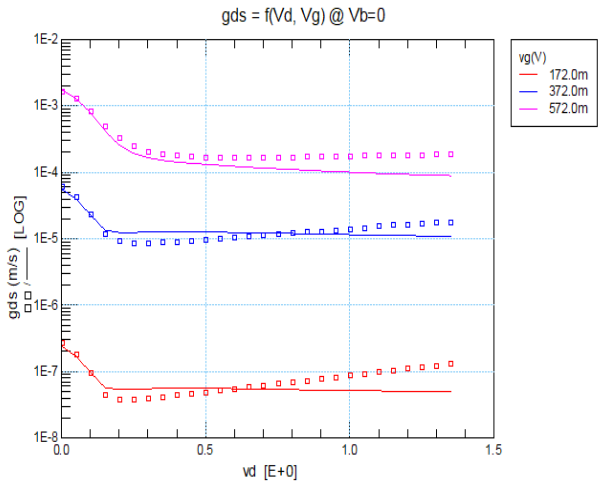




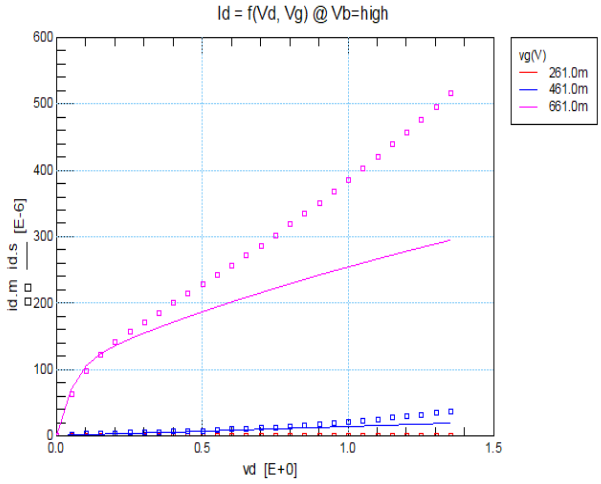
idvd_vth, Vb = 0V, T = 27°C

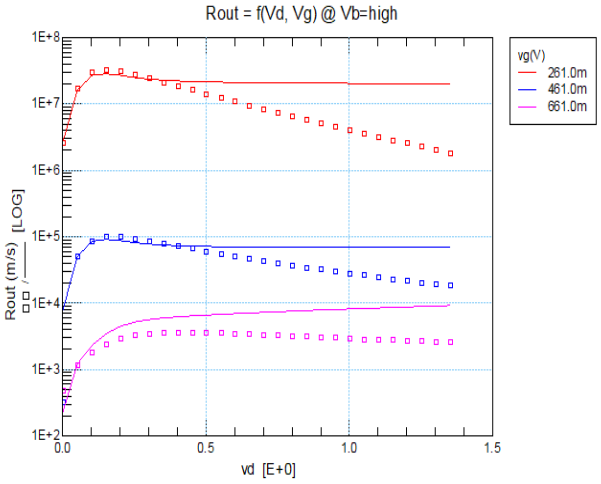


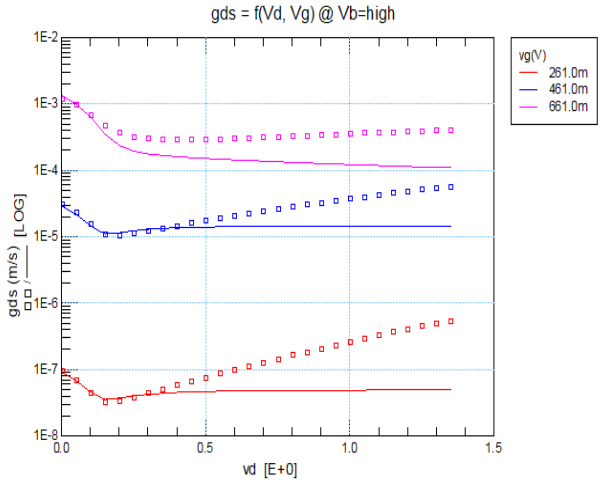




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

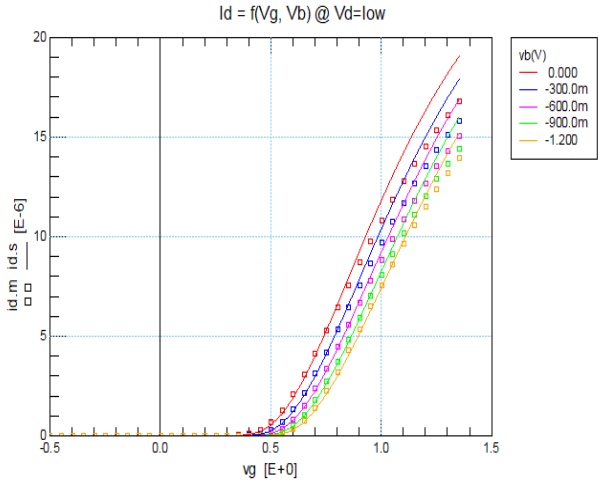


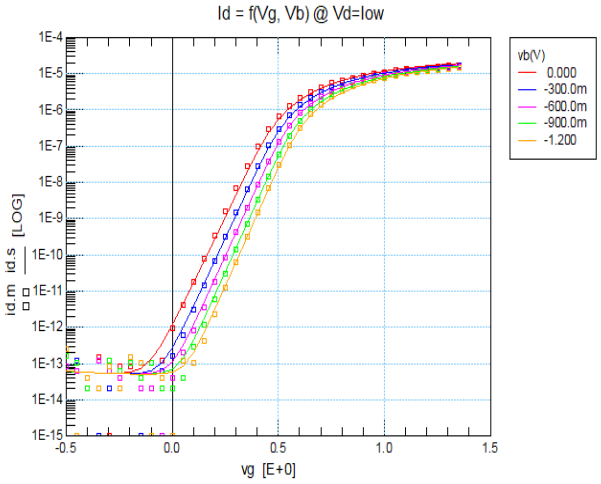


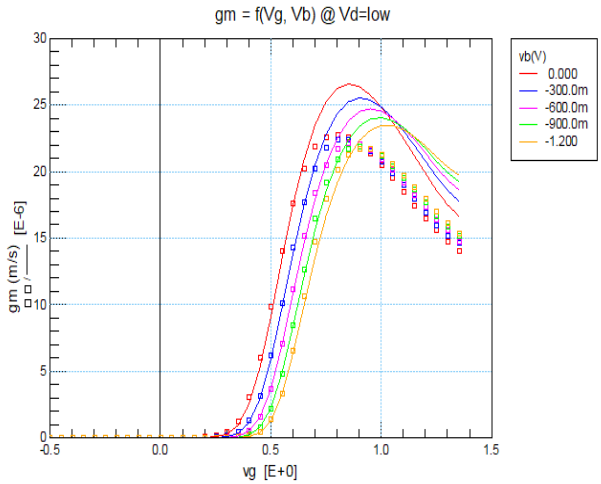


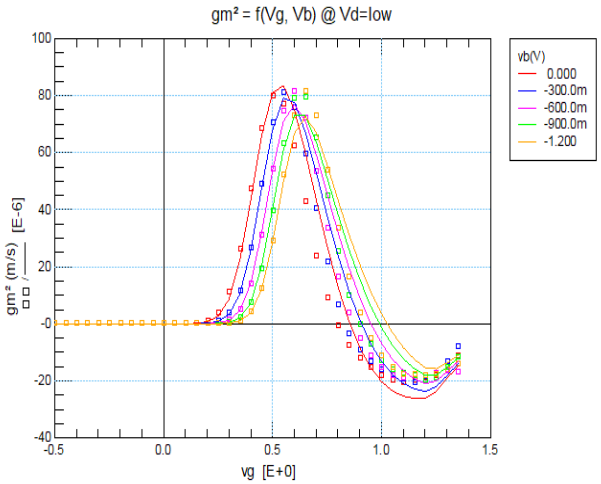
4.7 W0u15_L0u13_S546_3

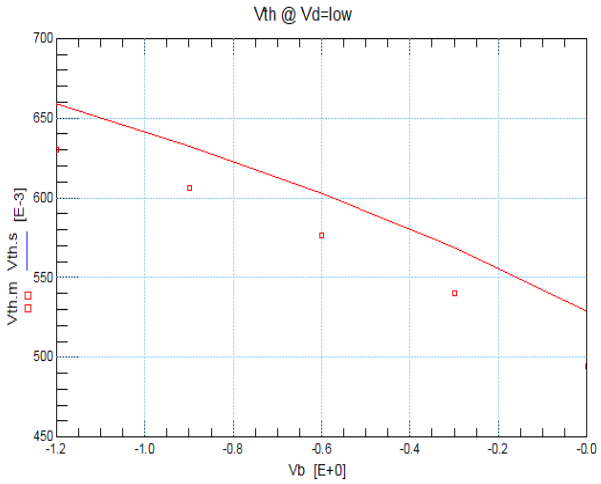
idvg, Vd = 0.05V, T = 27°C

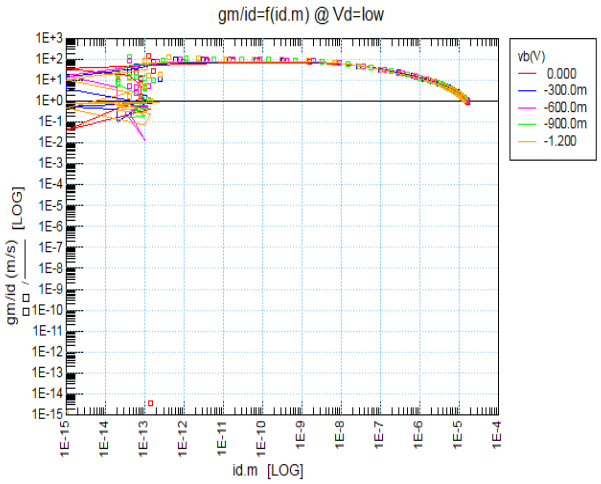


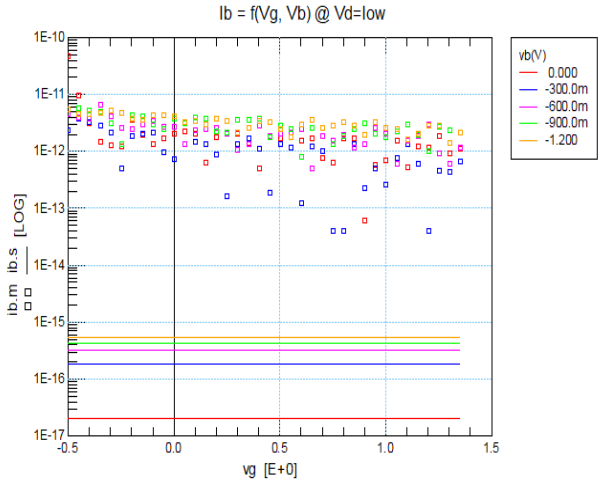


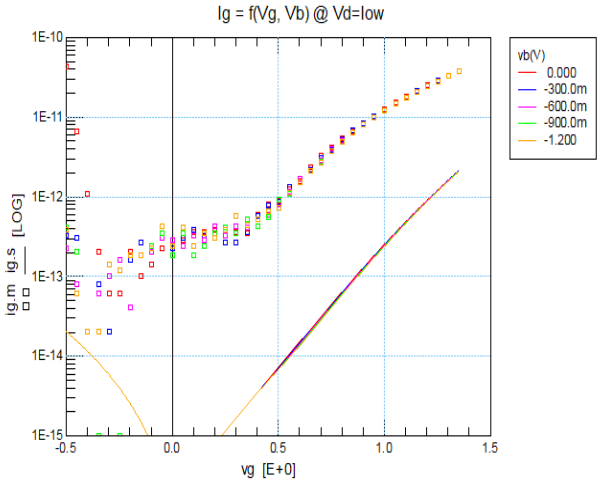




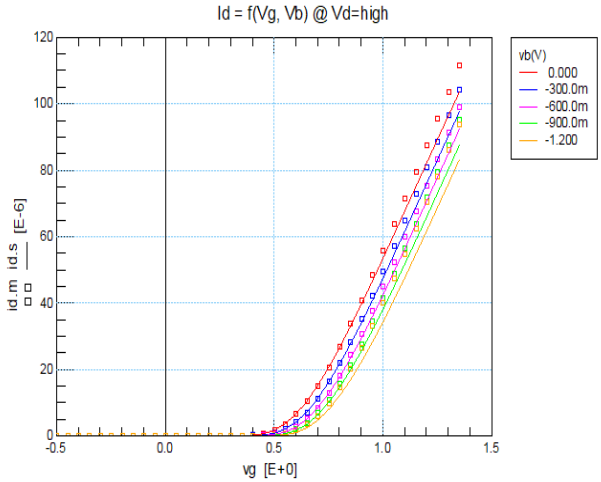


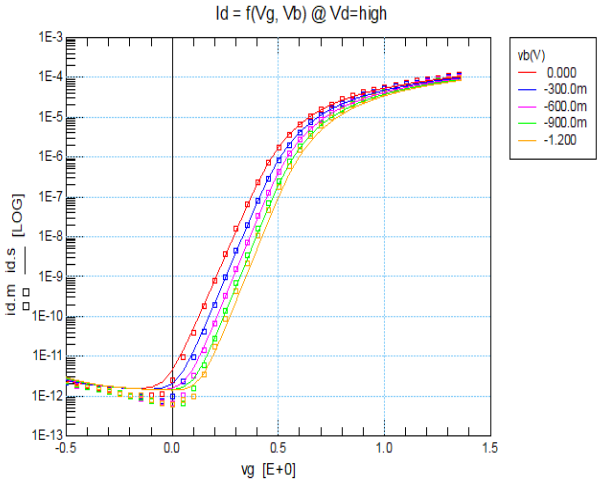


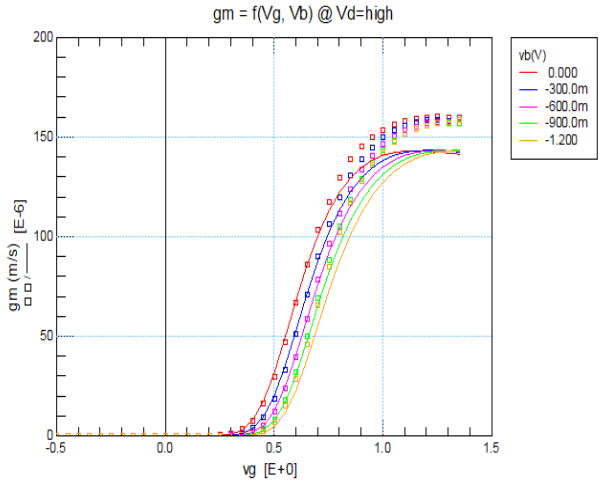


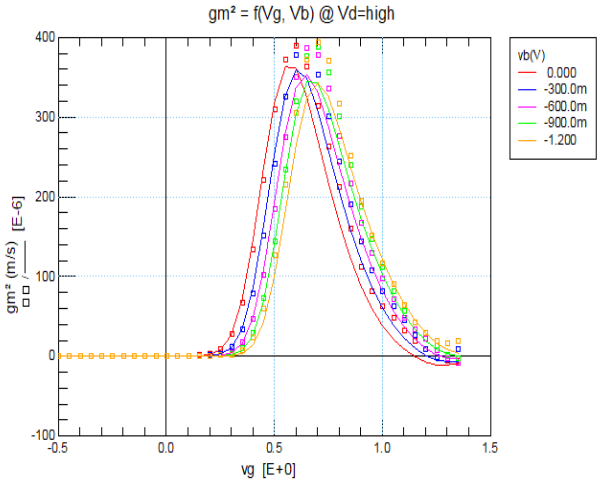


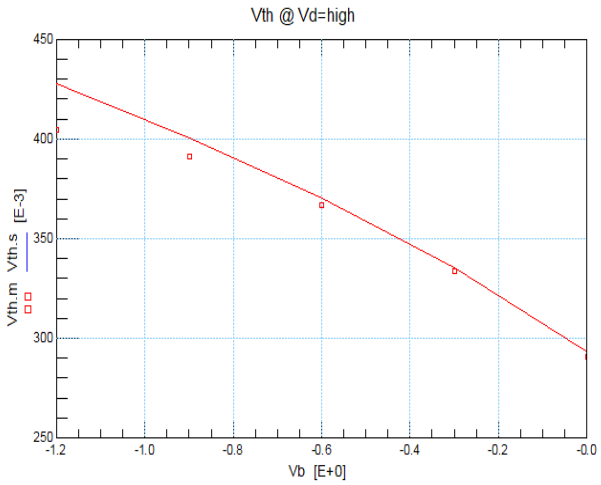
idvg, Vd = 1.2V, T = 27°C

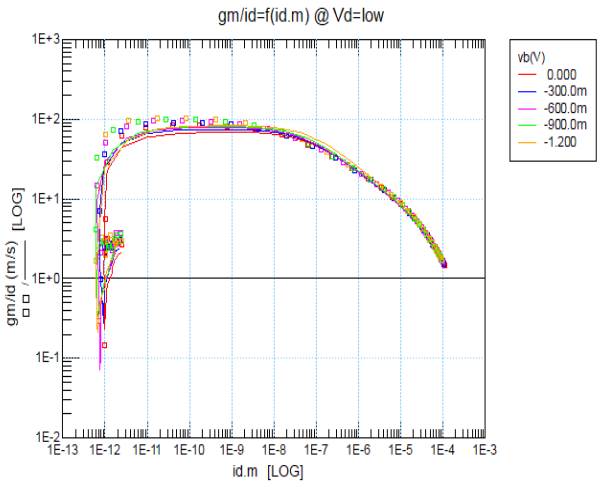


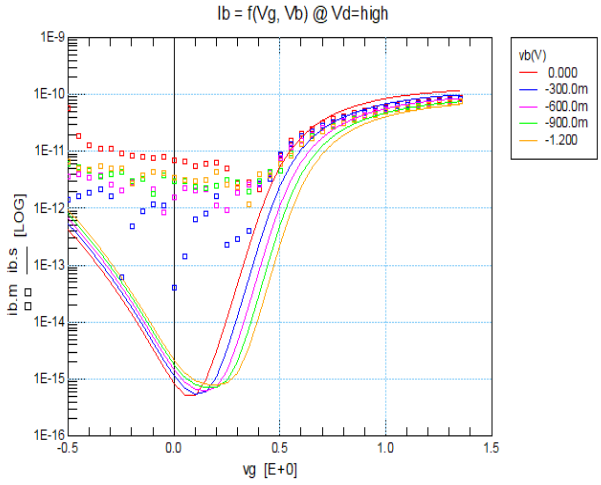


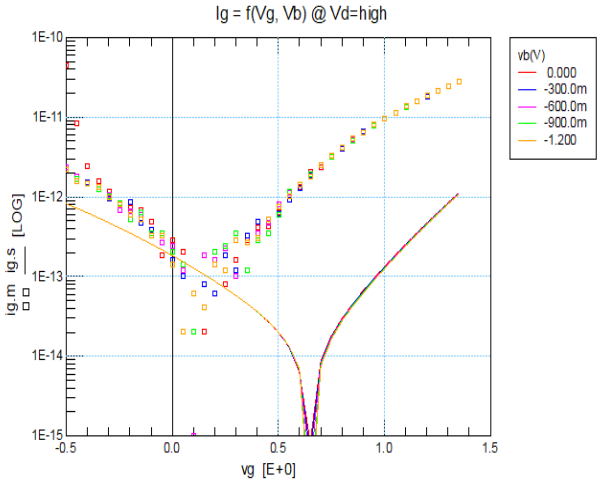




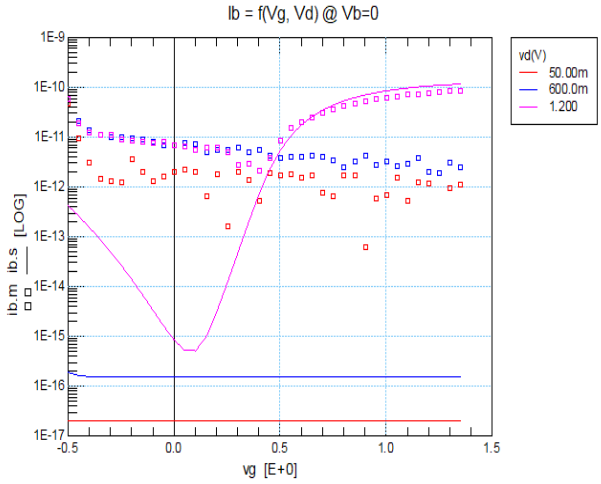


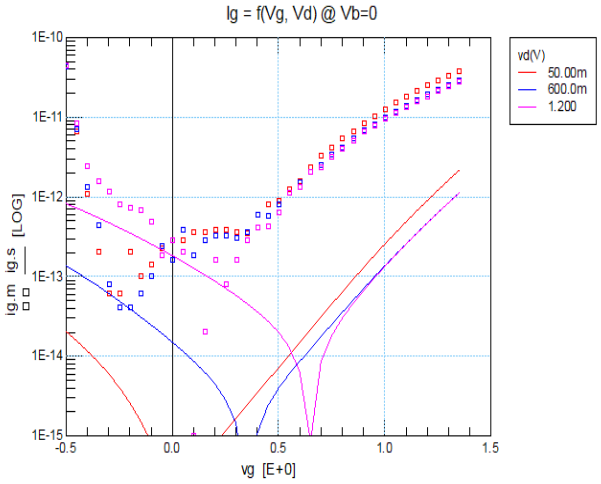




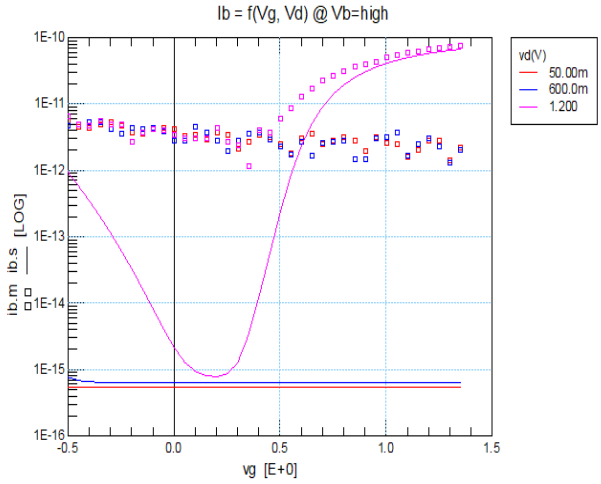


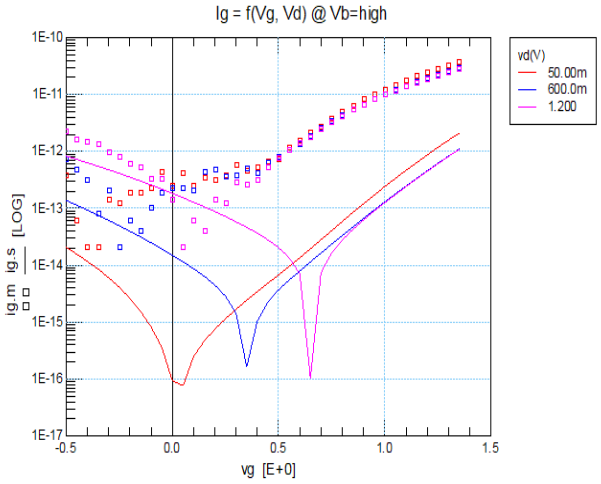
idvg, Ib,Ig, Vb = 0V, T = 27°C



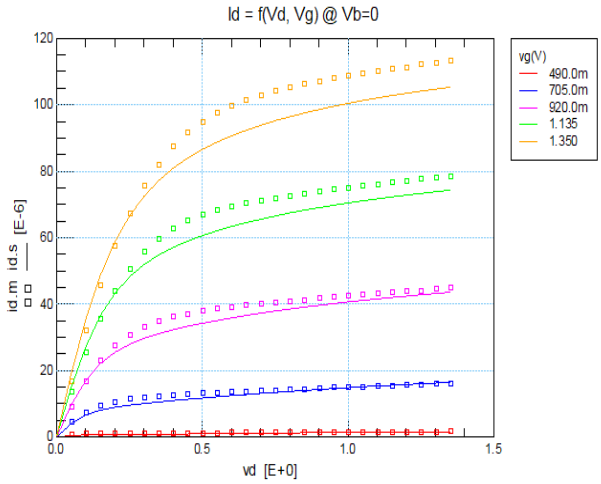


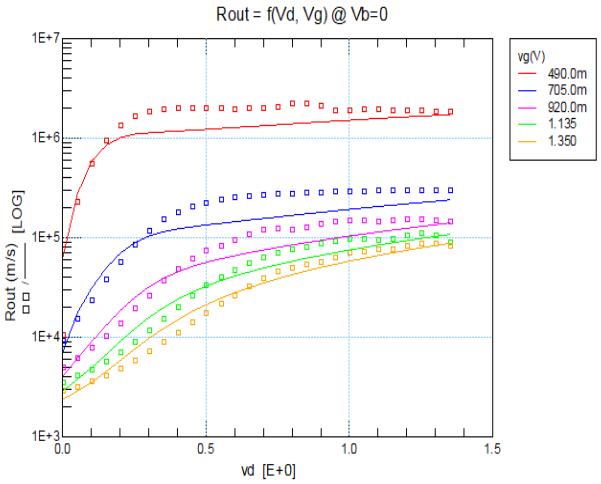
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

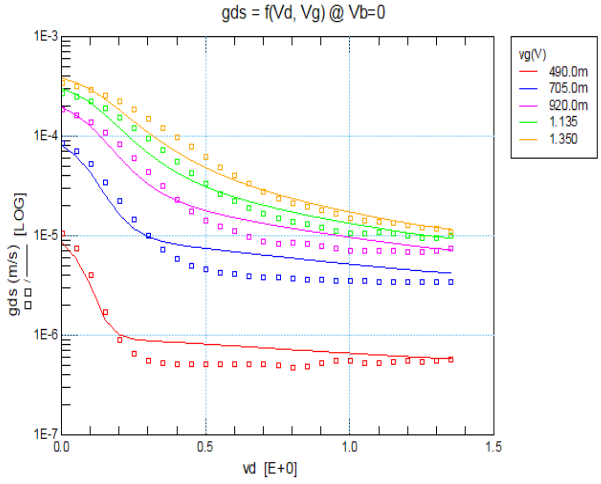




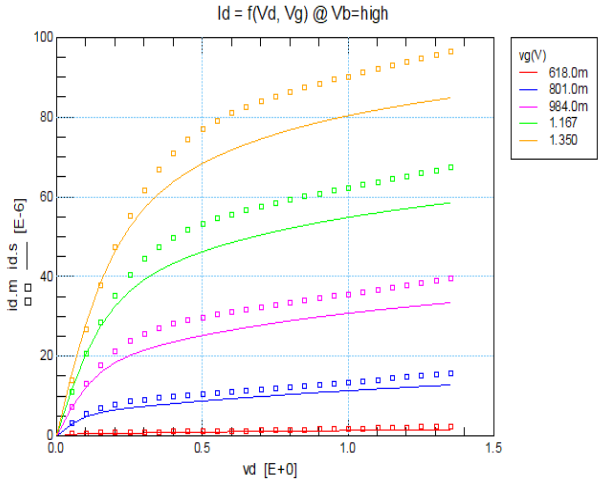
idvd, Vb = 0V, T = 27°C

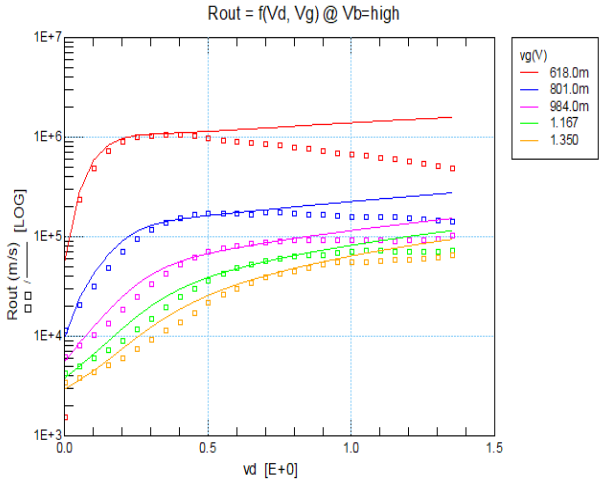


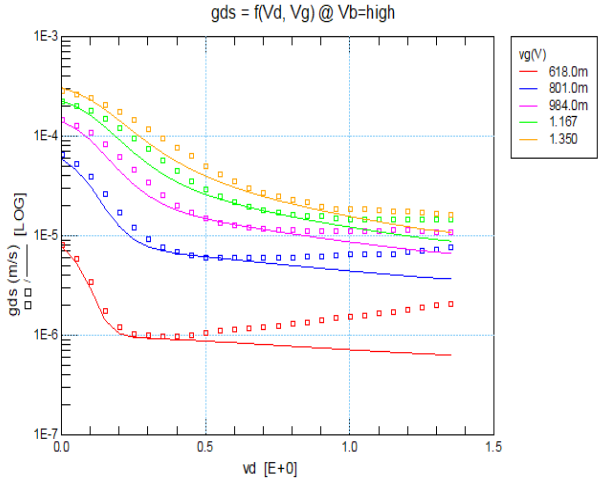




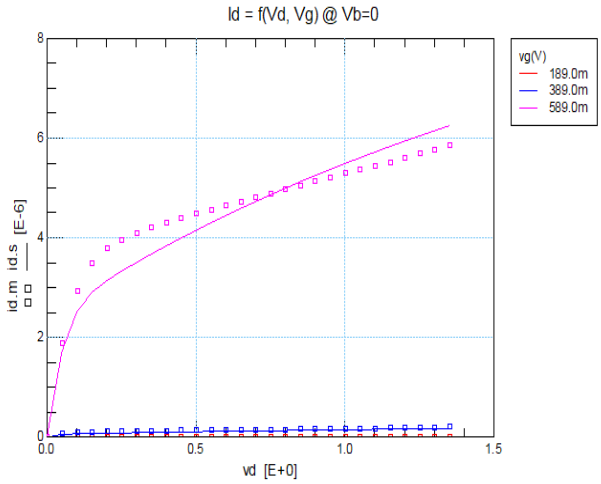
idvd_vbmin, Vb = -1.2V, T = 27°C

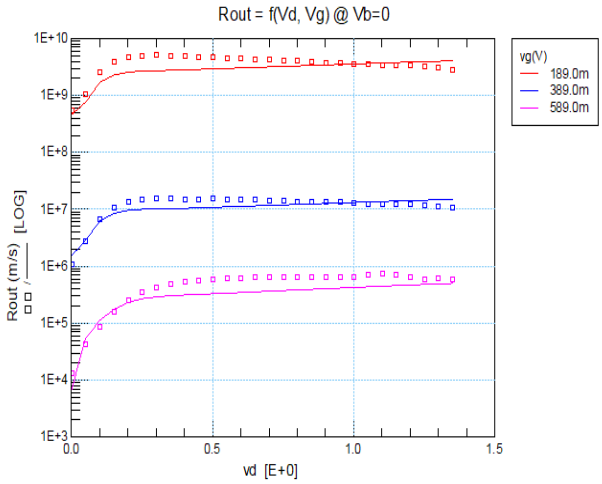


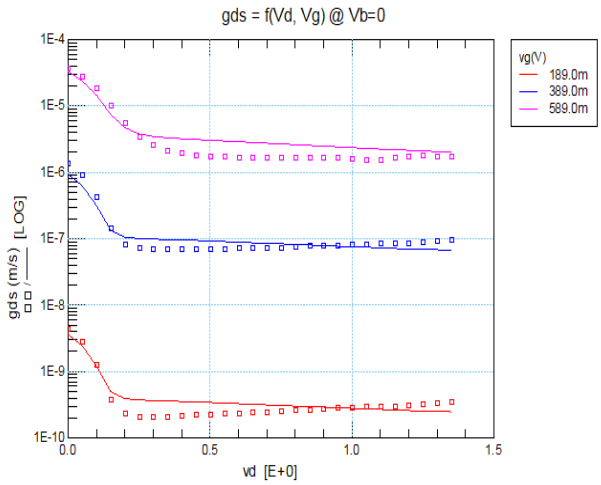




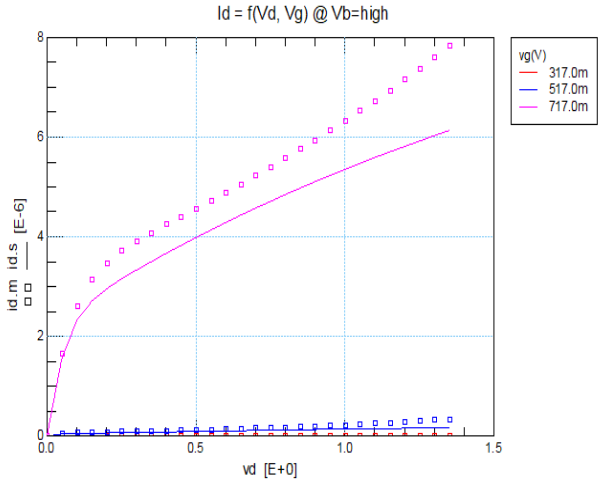
idvd_vth, Vb = 0V, T = 27°C

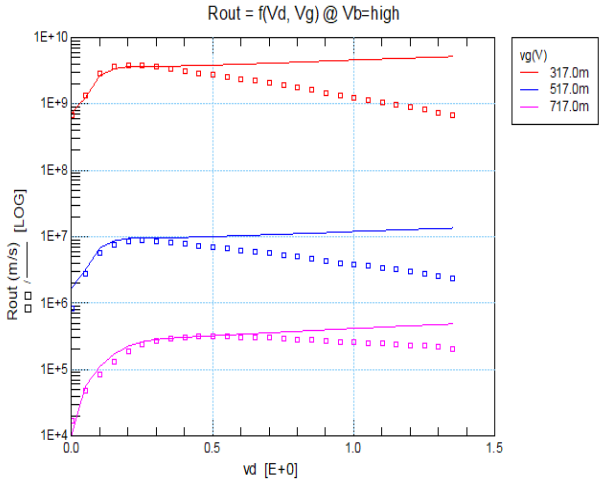


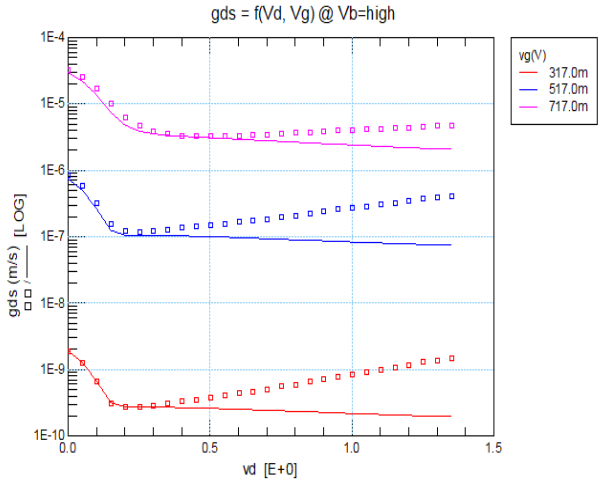




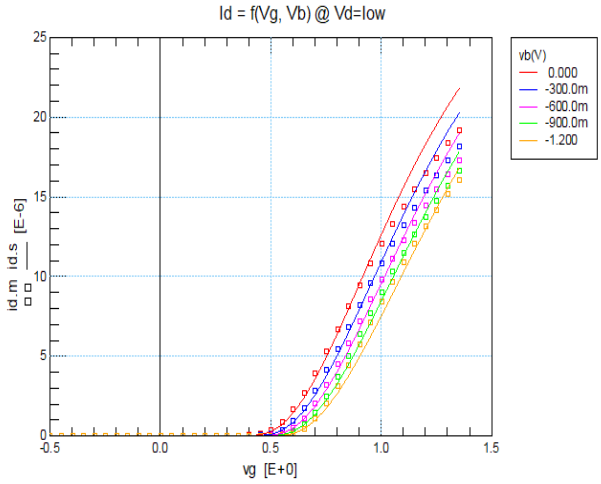
idvd_vth_vbmin, Vb = -1.2V, T = 27°C

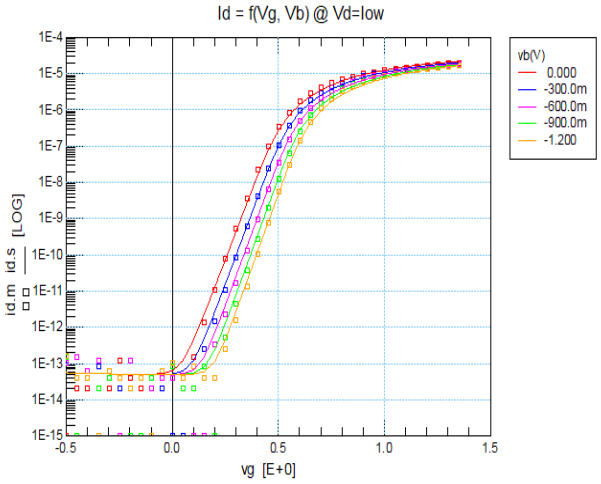


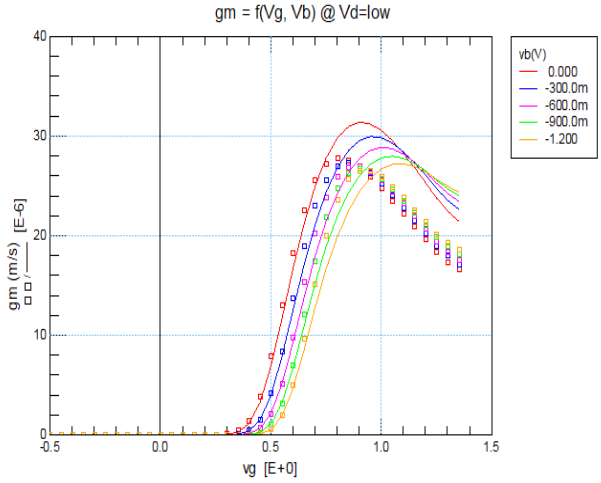


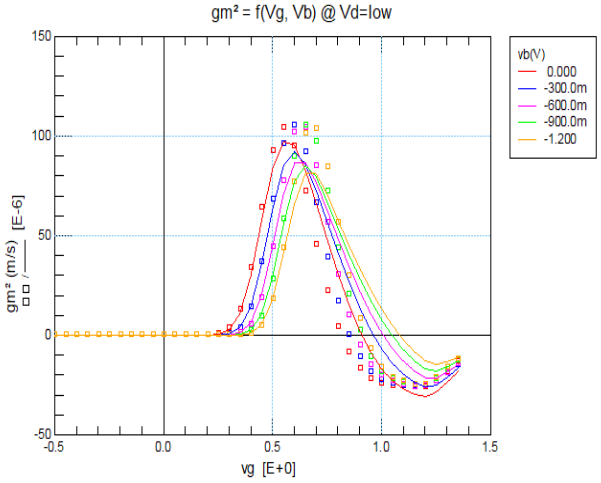


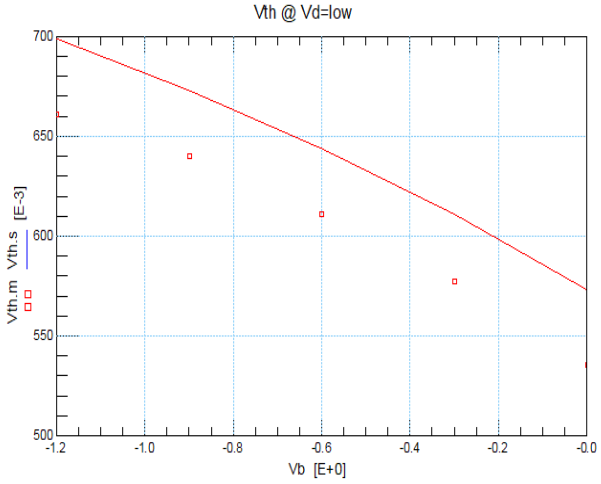
idvg, Vd = 0.05V, T = -40°C

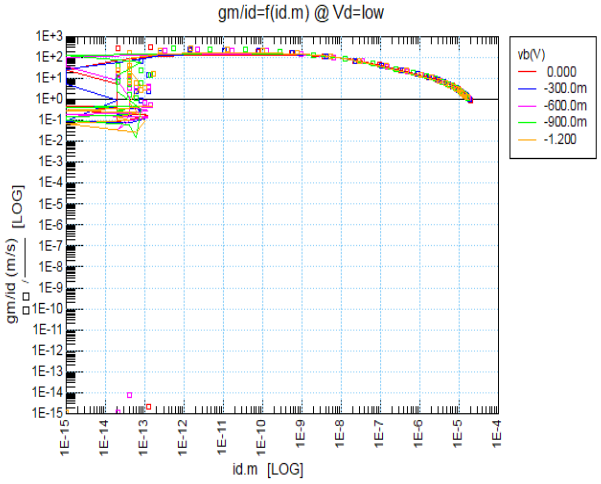


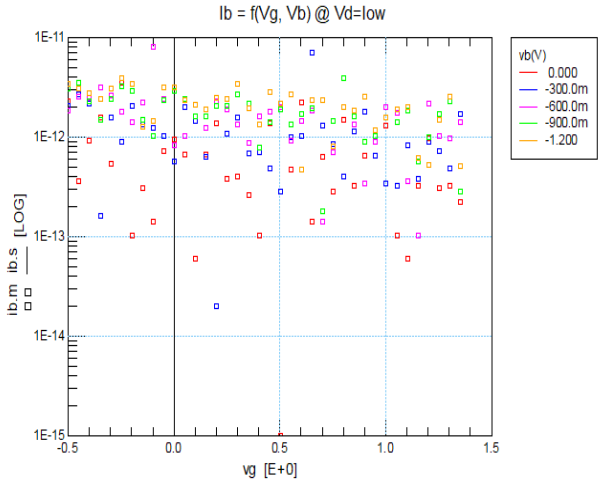


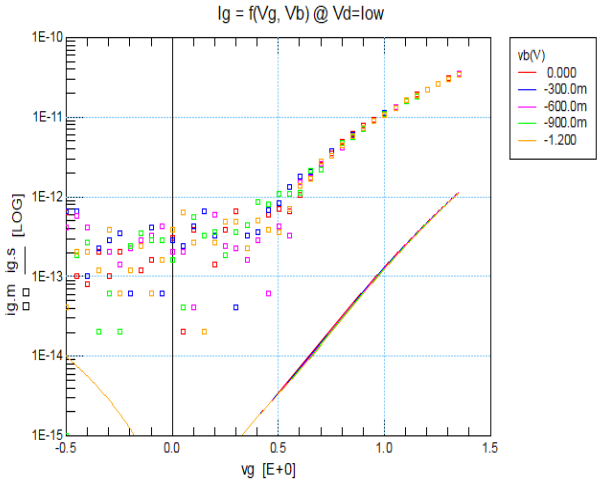




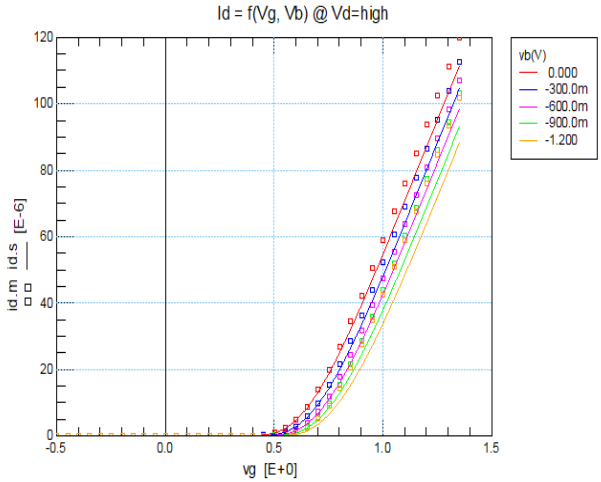


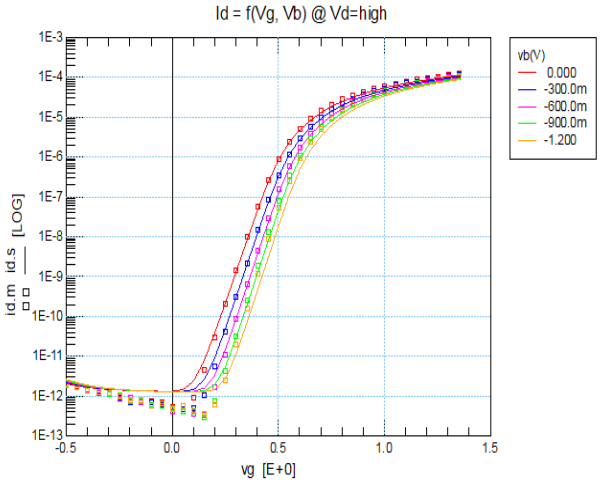


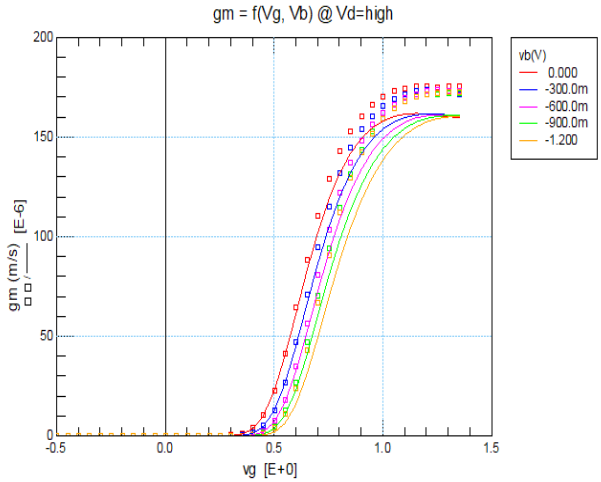


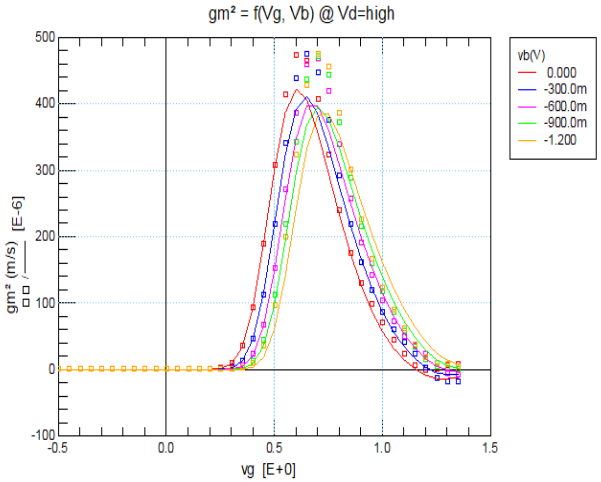


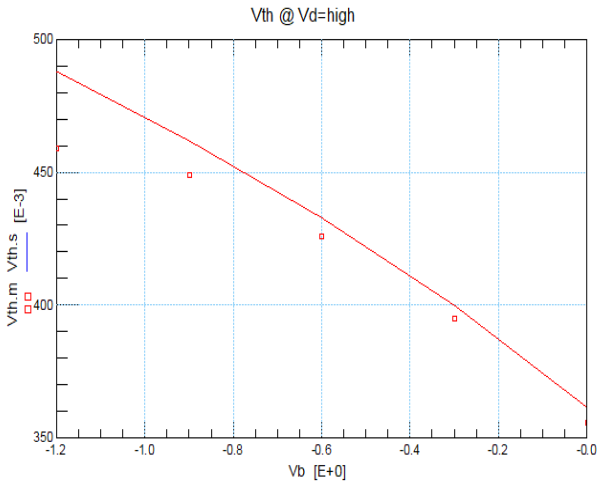
idvg, Vd = 1.2V, T = -40°C

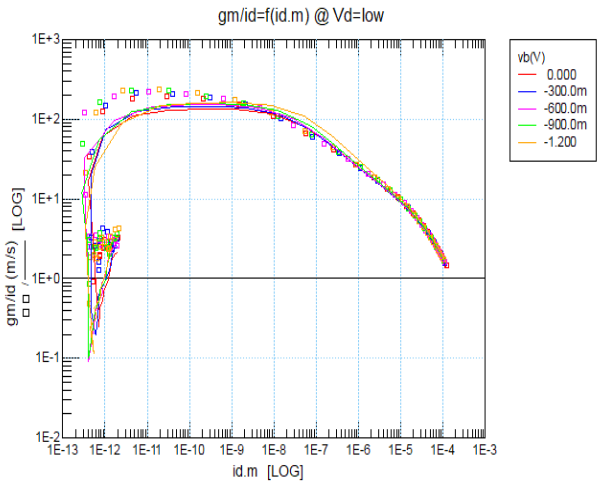


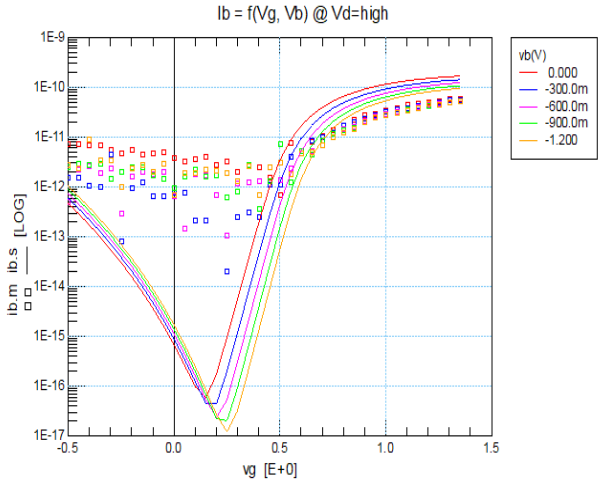


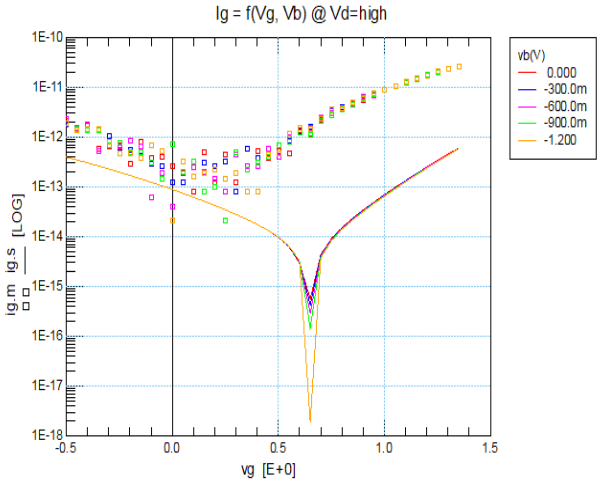




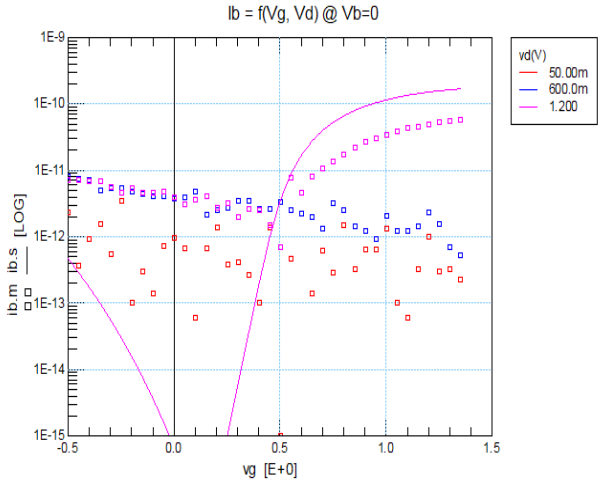


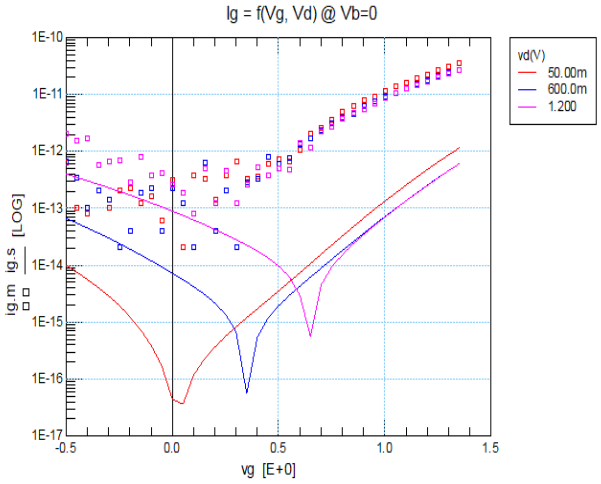




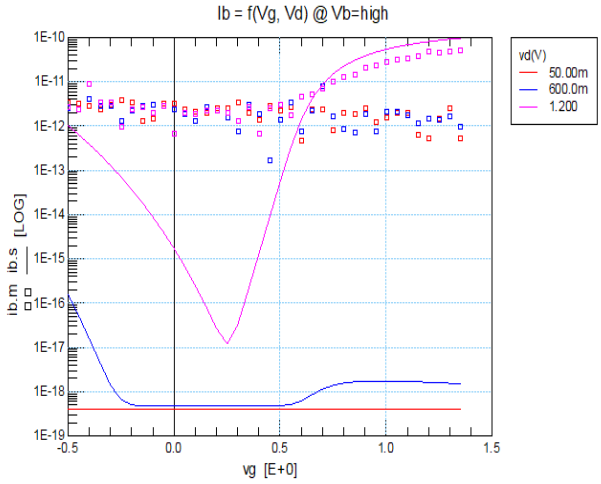


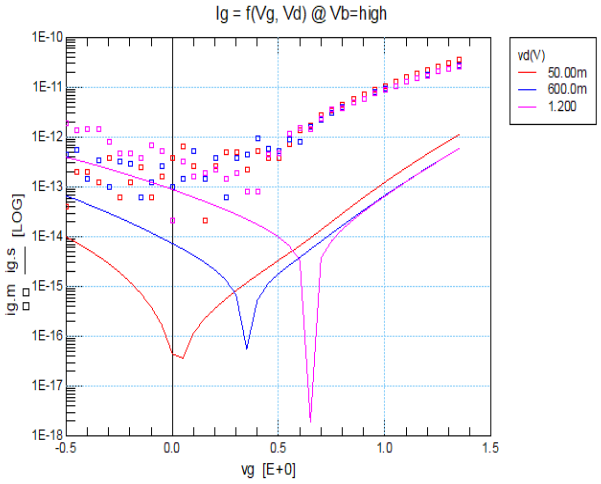
idvg, Ib,Ig, Vb = 0V, T = -40°C



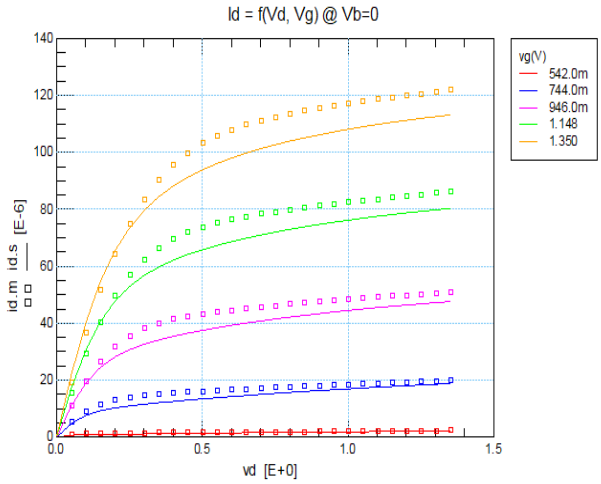


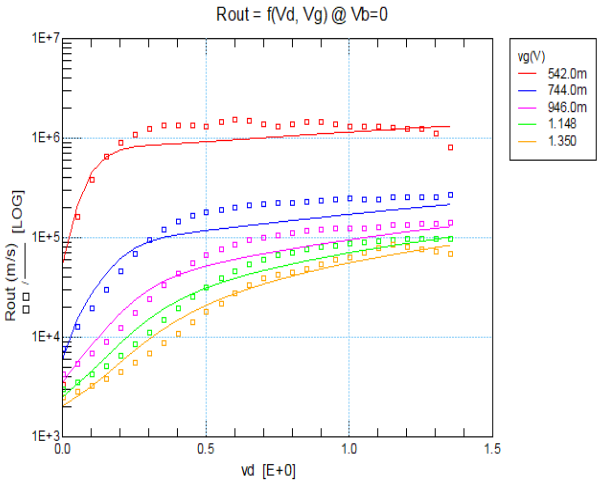
idvg, Ib,Ig, Vb = -1.2V, T = -40°C

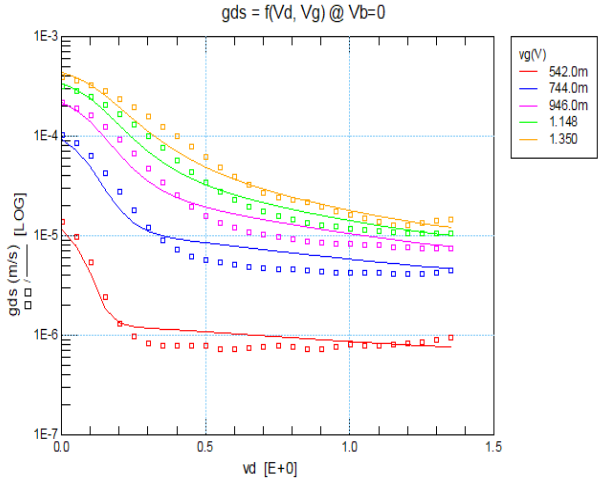




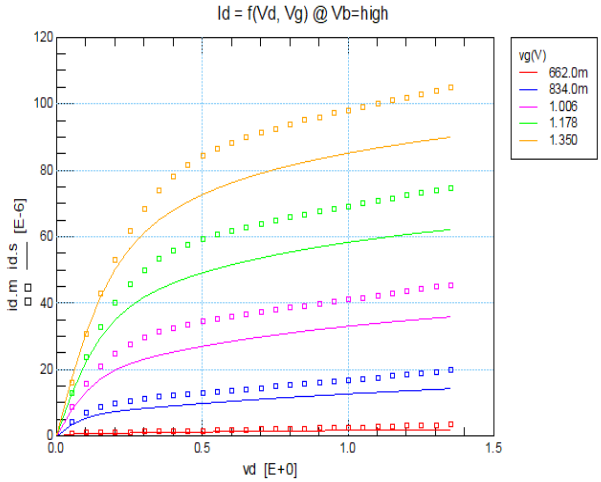
idvd, Vb = 0V, T = -40°C

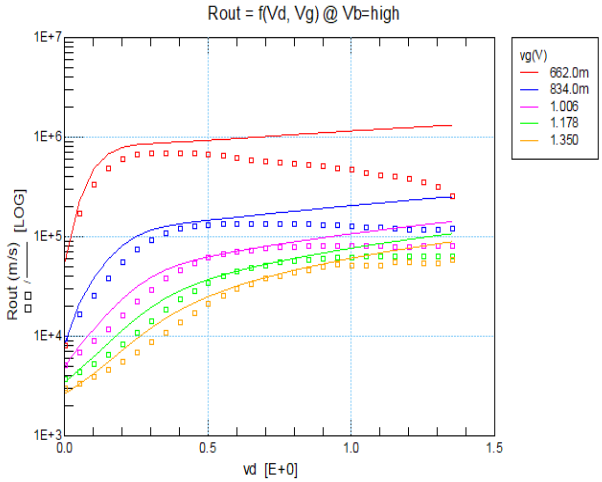


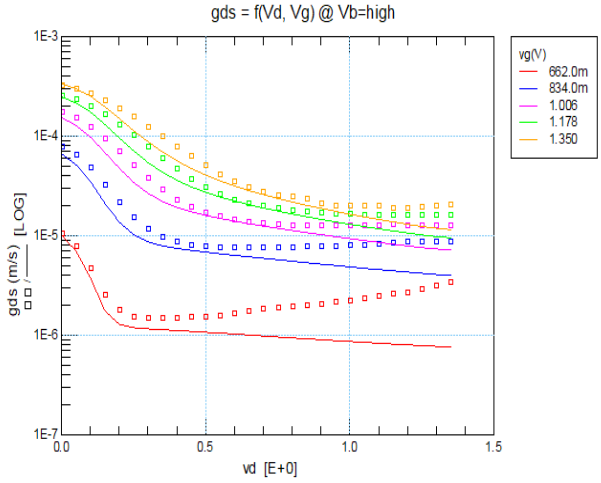




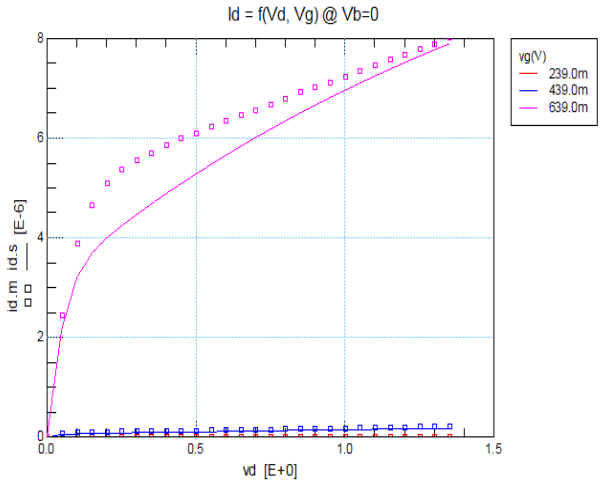
idvd_vbmin, Vb = -1.2V, T = -40°C

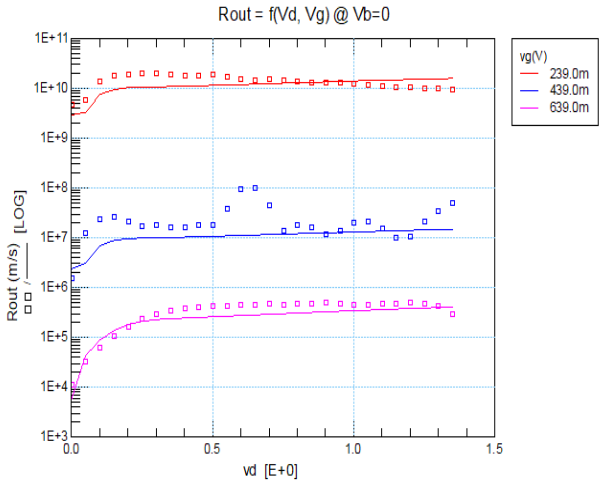


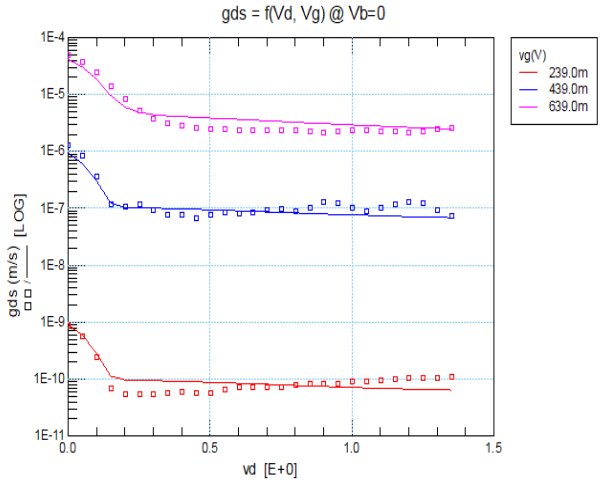




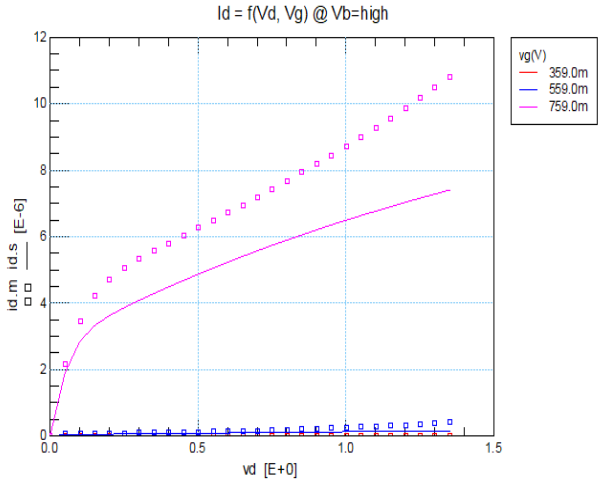
idvd_vth, Vb = 0V, T = -40°C

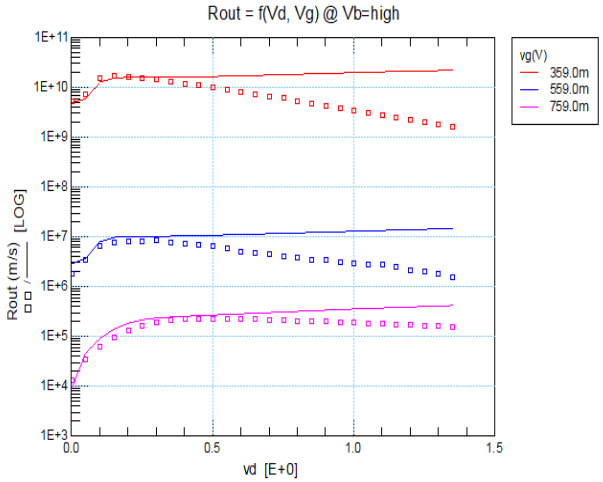


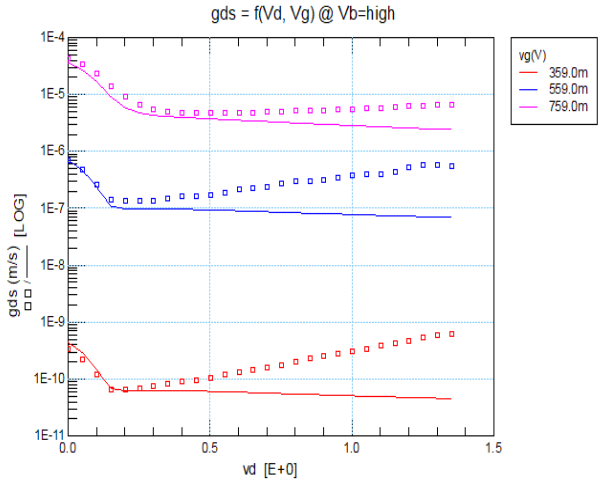




idvd_vth_vbmin, Vb = -1.2V, T = -40°C

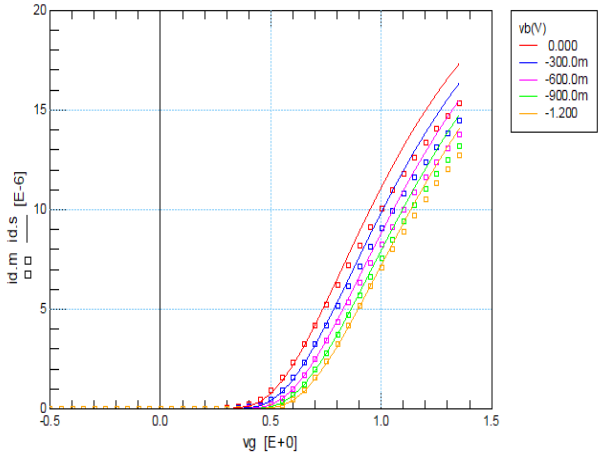




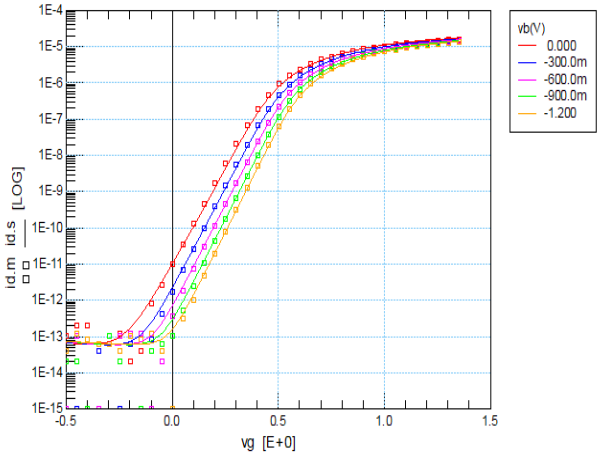


idvg, Vd = 0.05V, T = 70°C

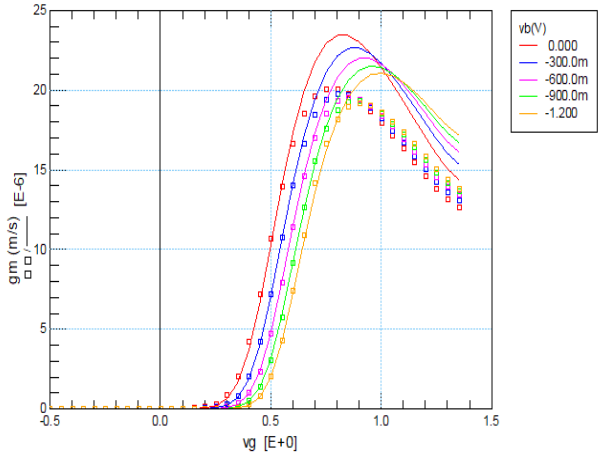
Id = f(Vg, Vb) @ Vd=low



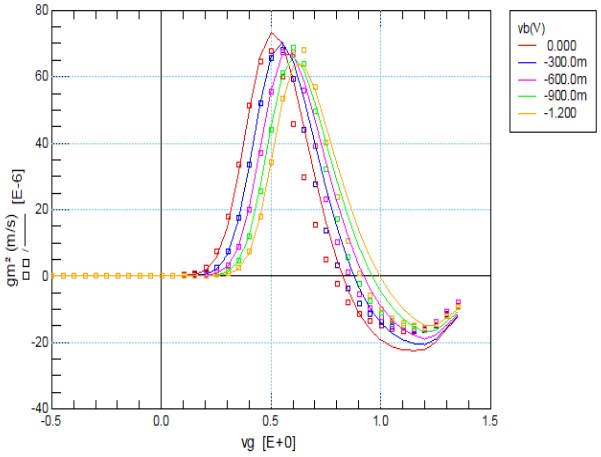
Id = f(Vg, Vb) @ Vd=low



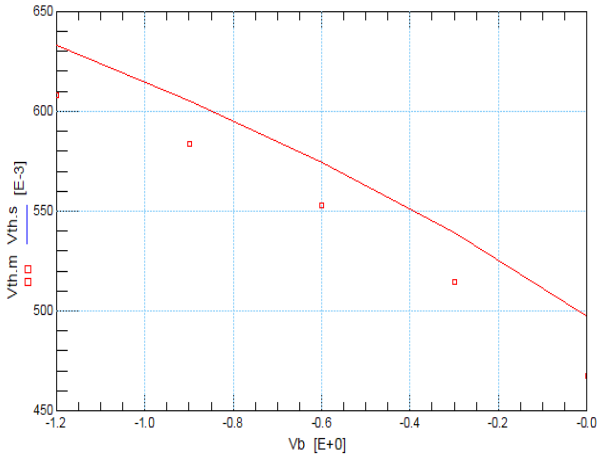
gm = f(Vg, Vb) @ Vd=low



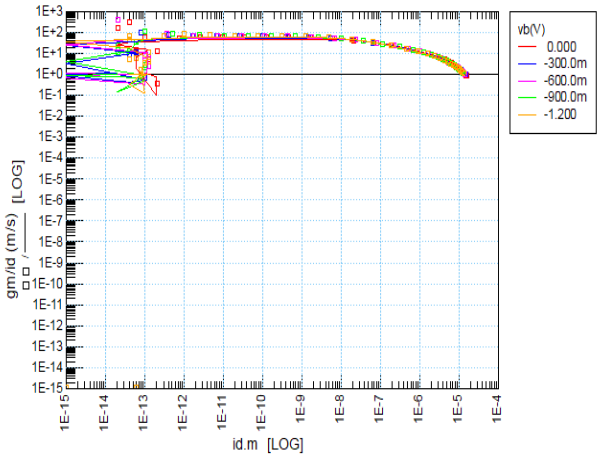
gm² = f(Vg, Vb) @ Vd=low

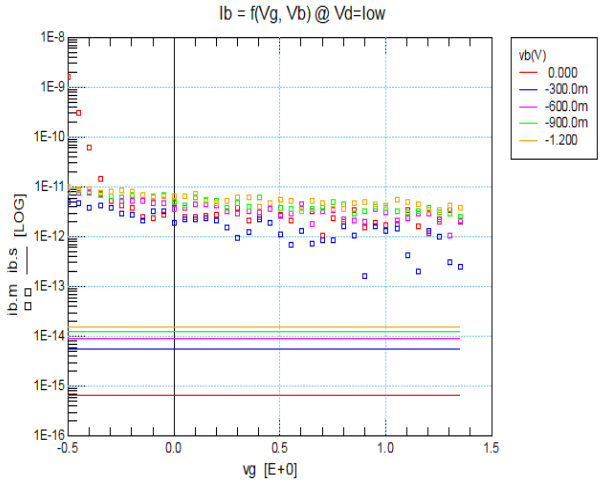


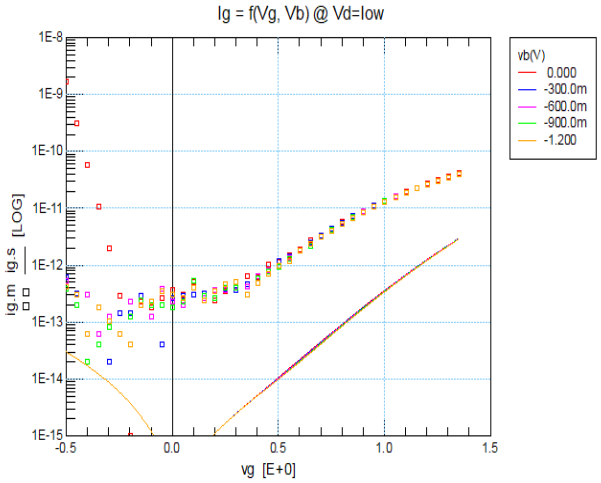
Vth @ Vd=low



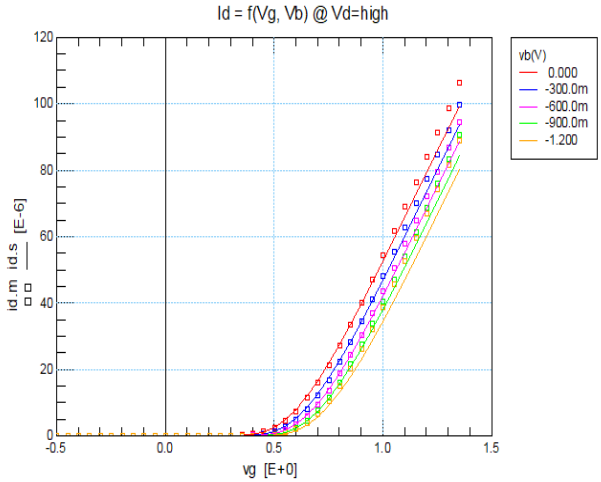
gm/id=f(id.m) @ Vd=low

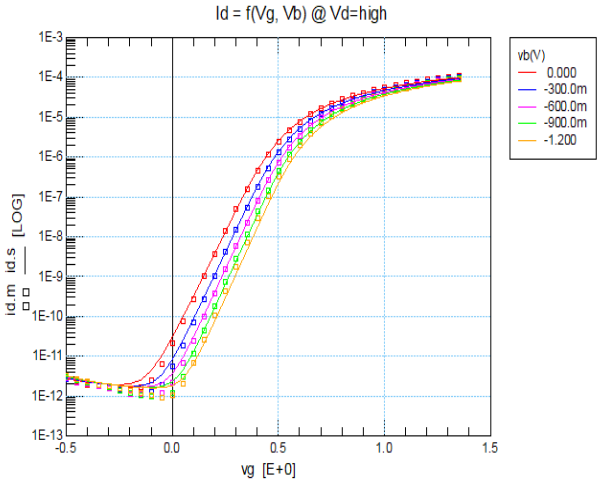


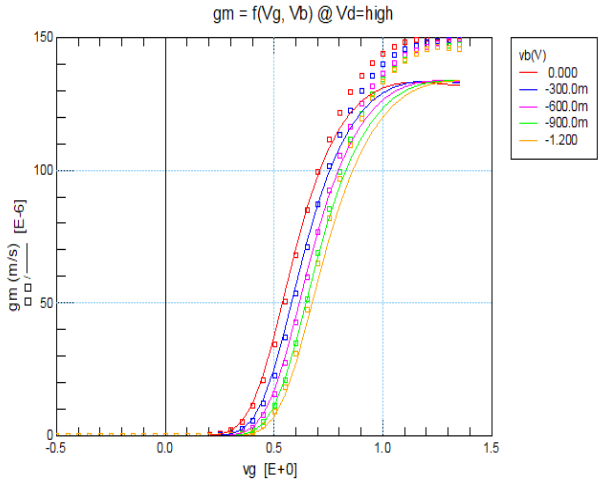


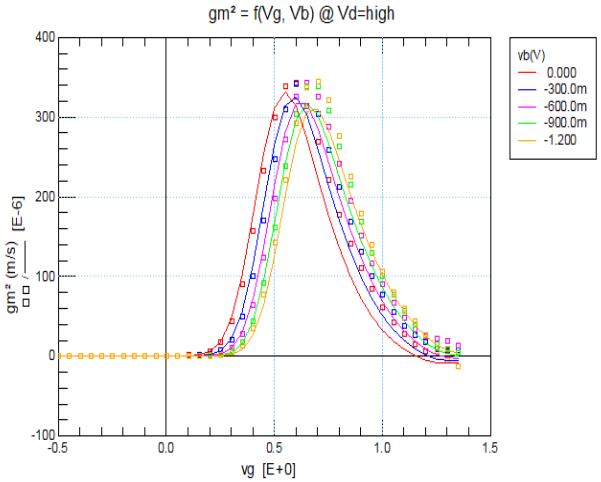


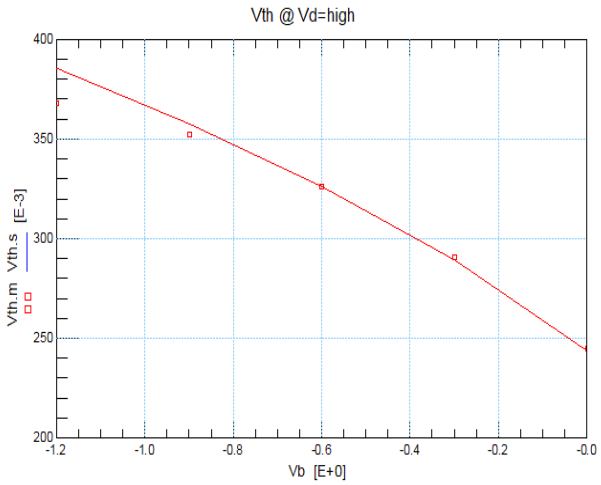
idvg, Vd = 1.2V, T = 70°C

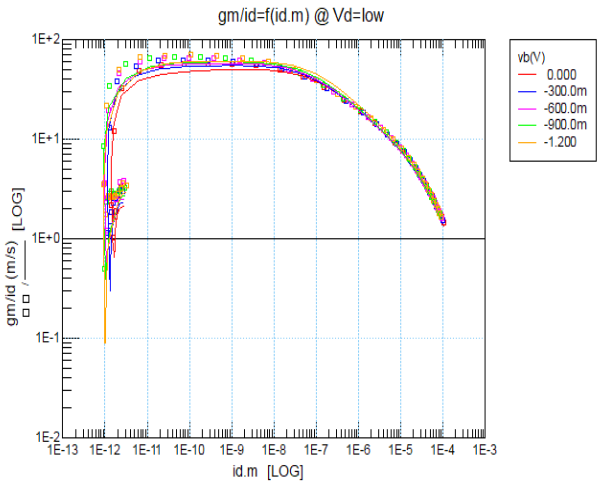


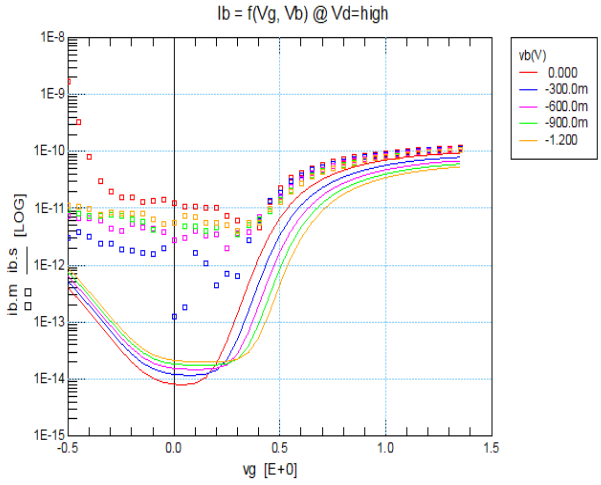


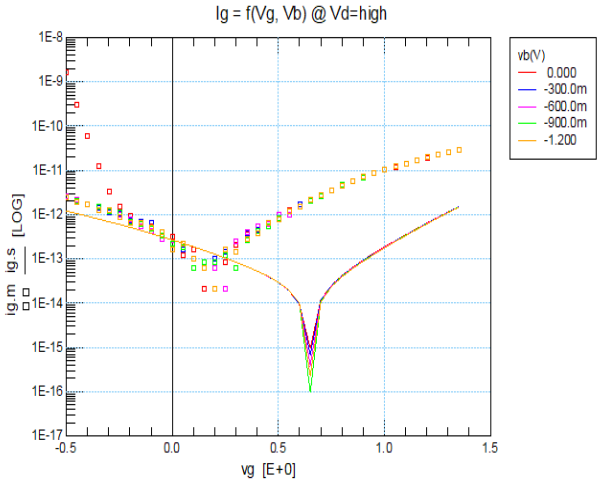




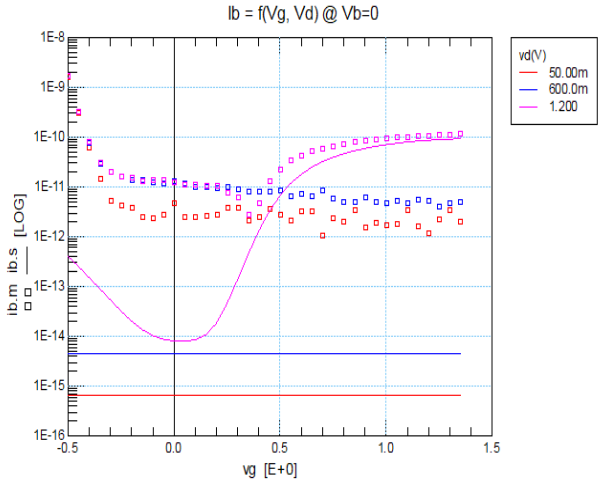


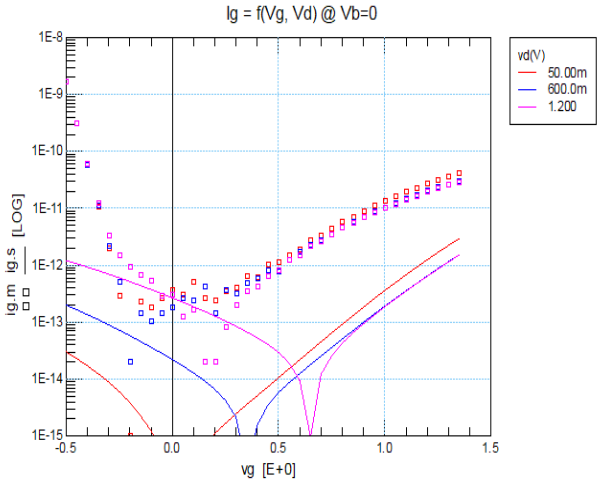




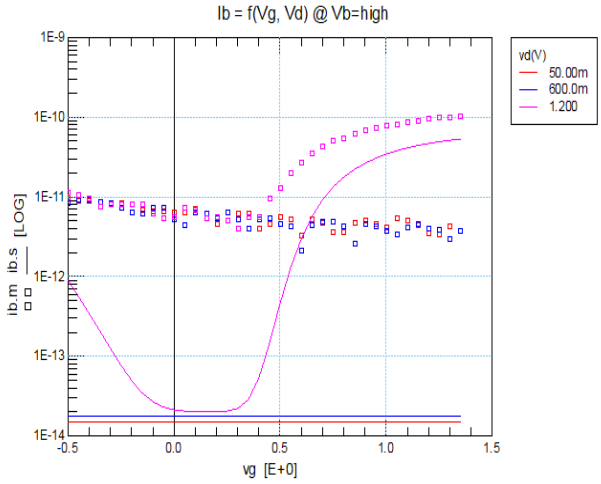


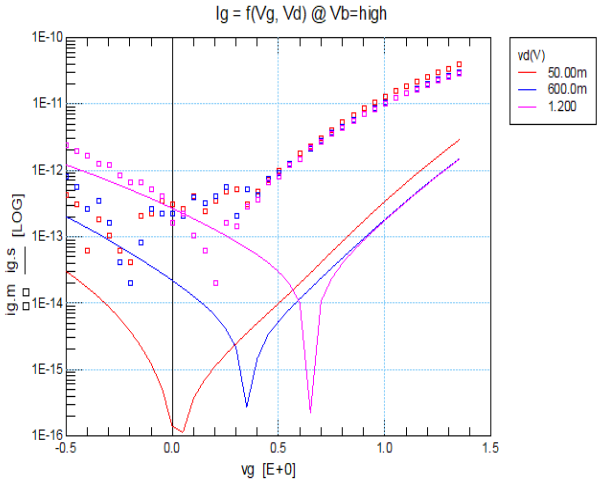
idvg, Ib,Ig, Vb = 0V, T = 70°C



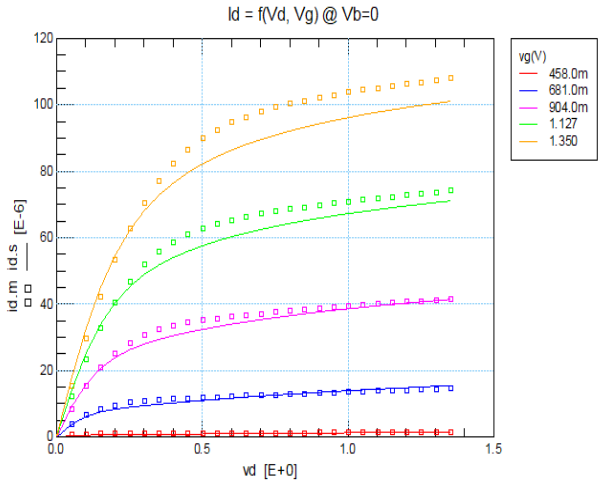


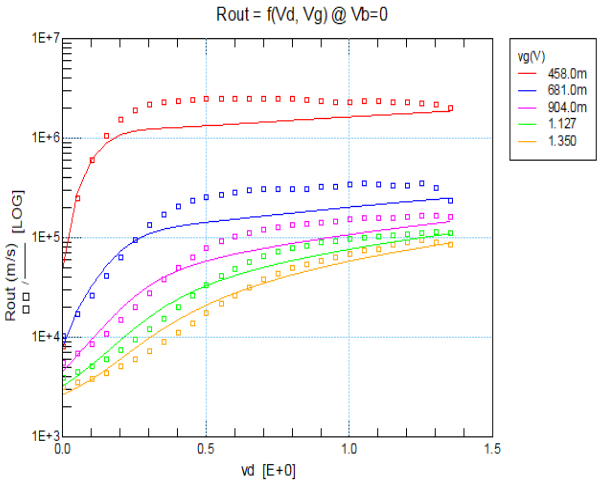
idvg, Ib,Ig, Vb = -1.2V, T = 70°C

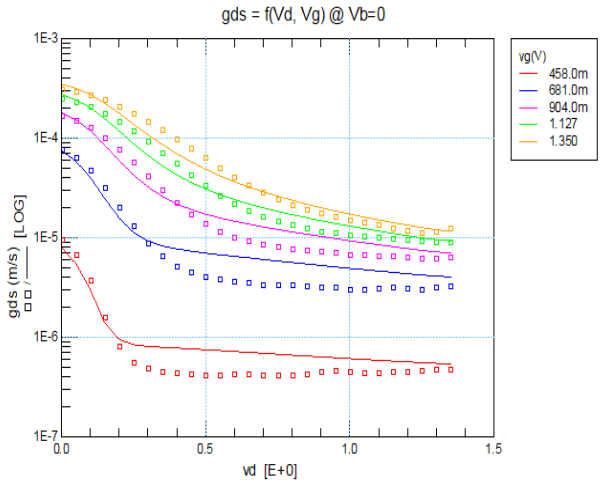




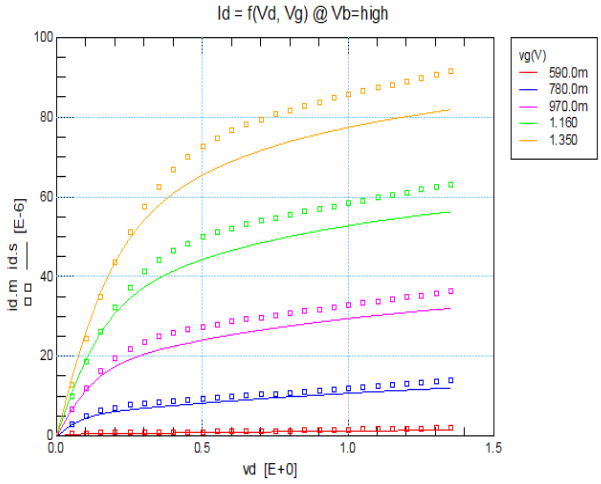
idvd, Vb = 0V, T = 70°C

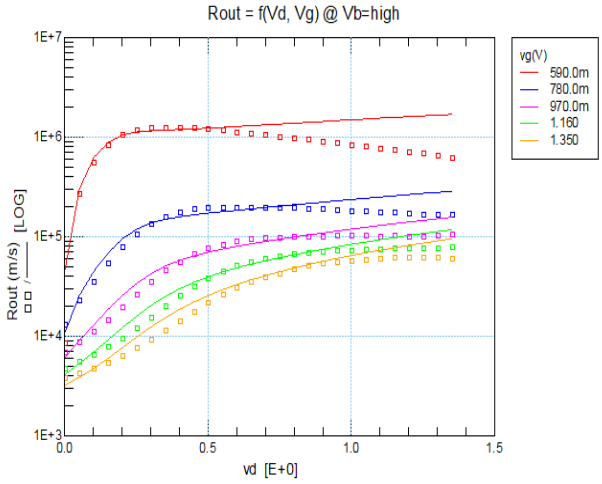


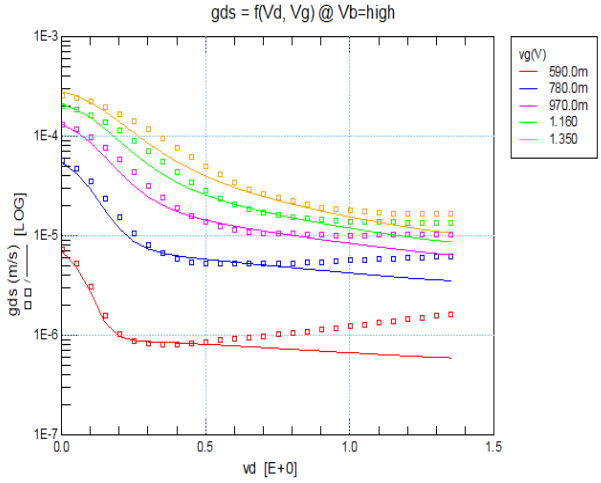




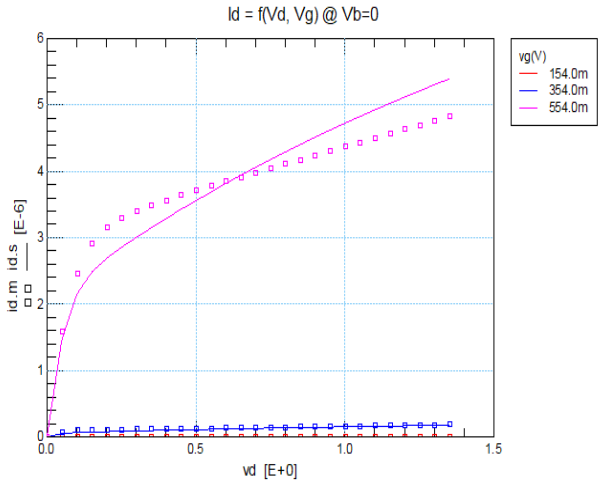
idvd_vbmin, Vb = -1.2V, T = 70°C

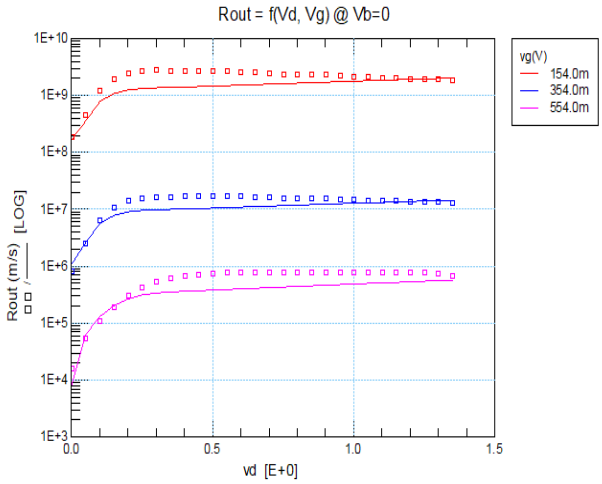


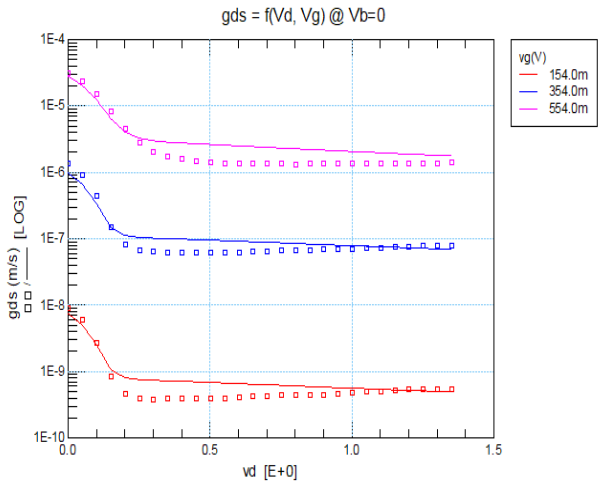




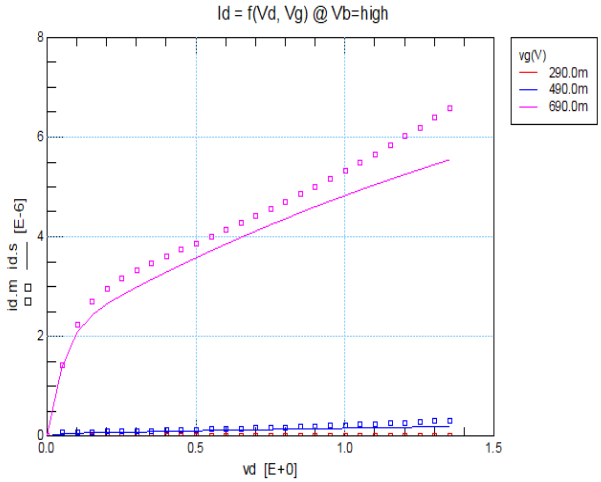
idvd_vth, Vb = 0V, T = 70°C

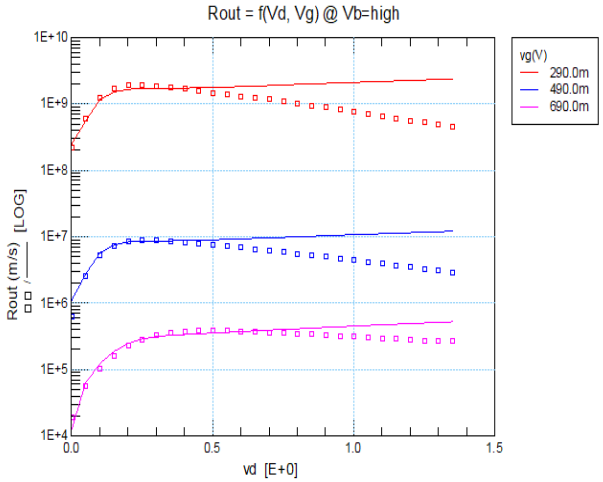


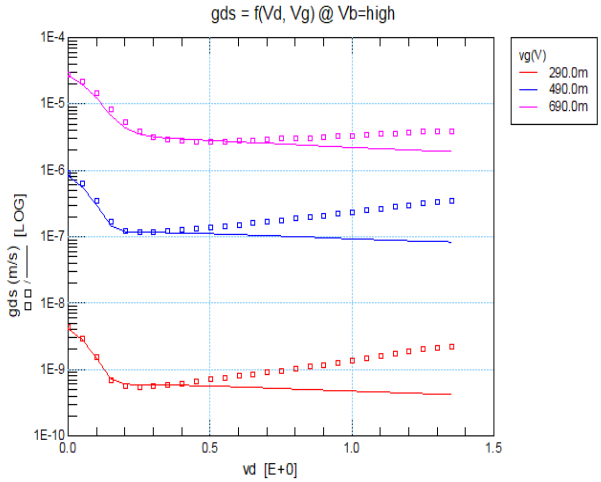




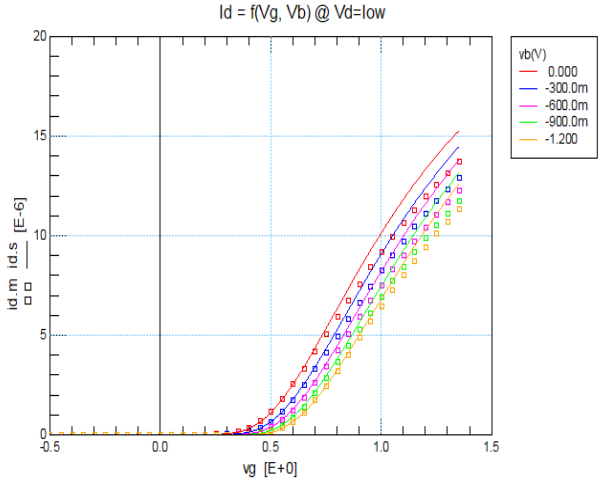
idvd_vth_vbmin, Vb = -1.2V, T = 70°C

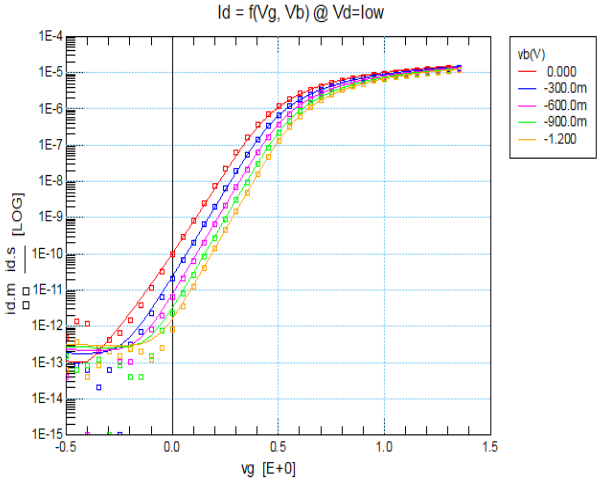


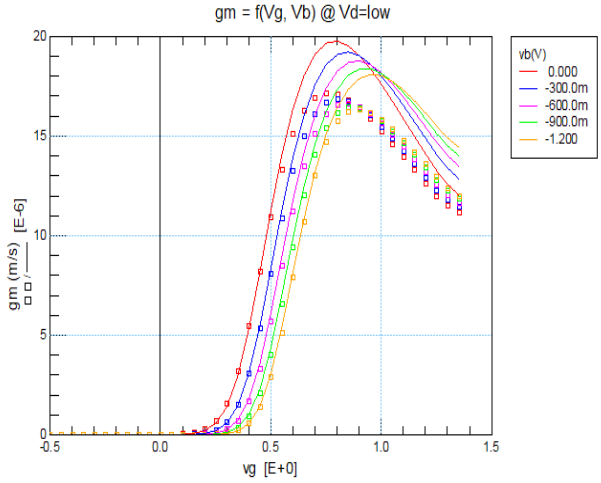


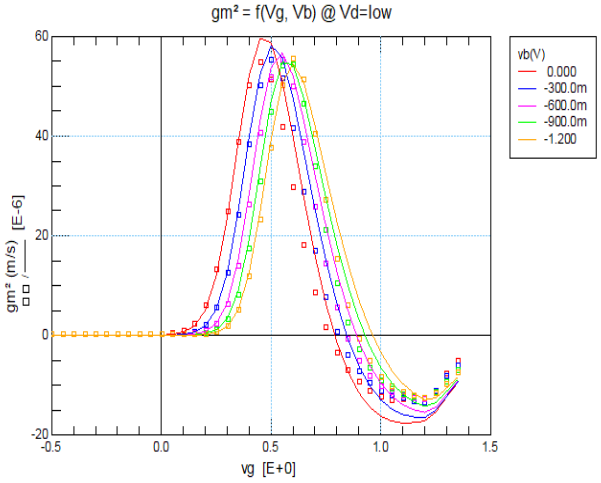


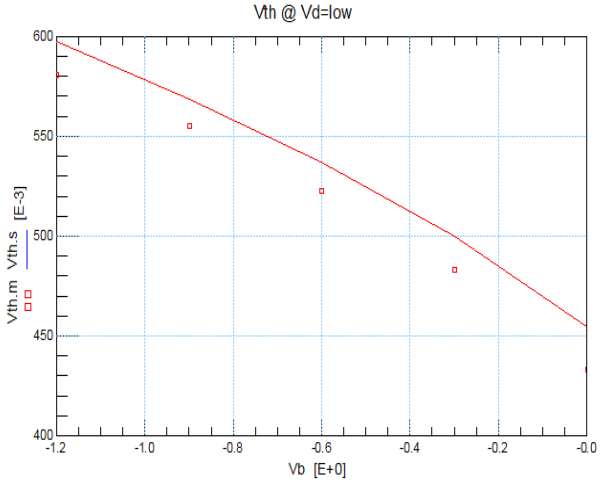
idvg, Vd = 0.05V, T = 125°C

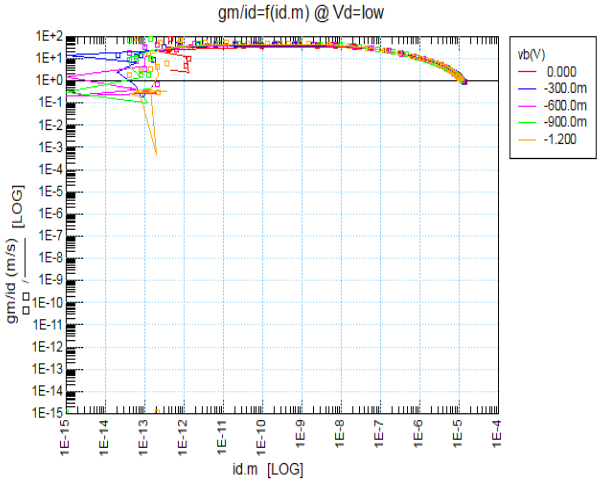


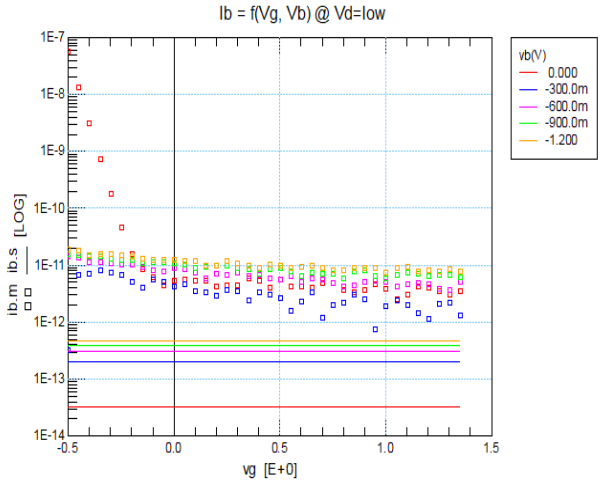


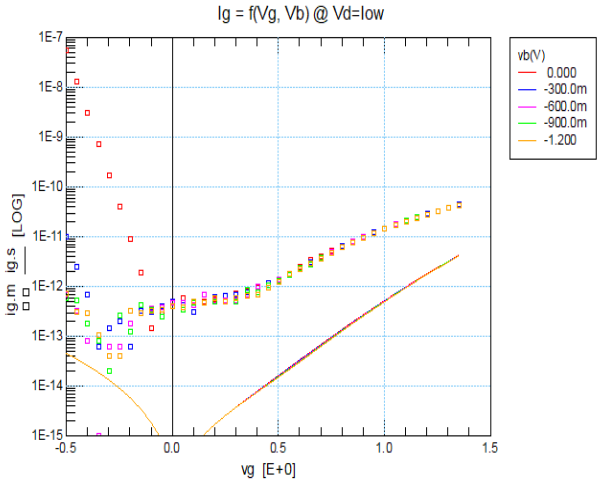




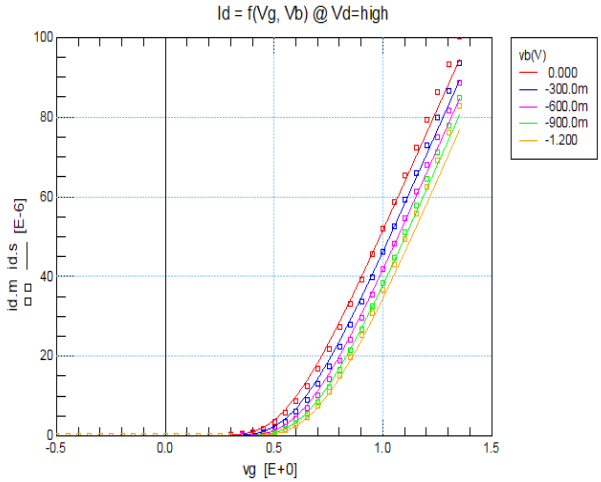


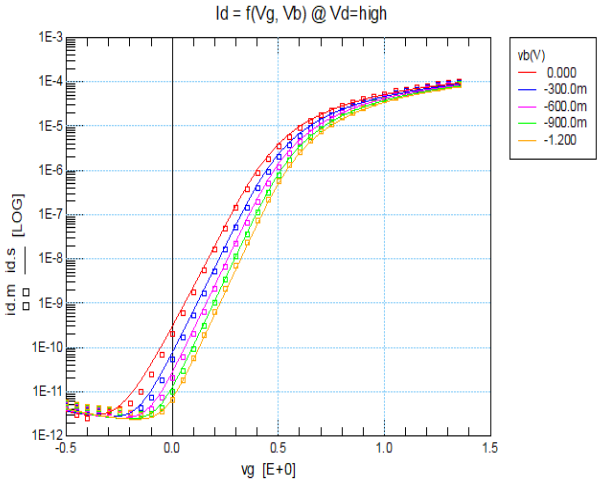


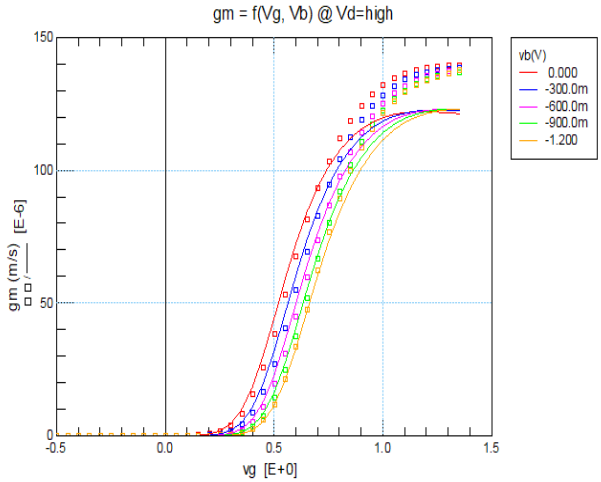


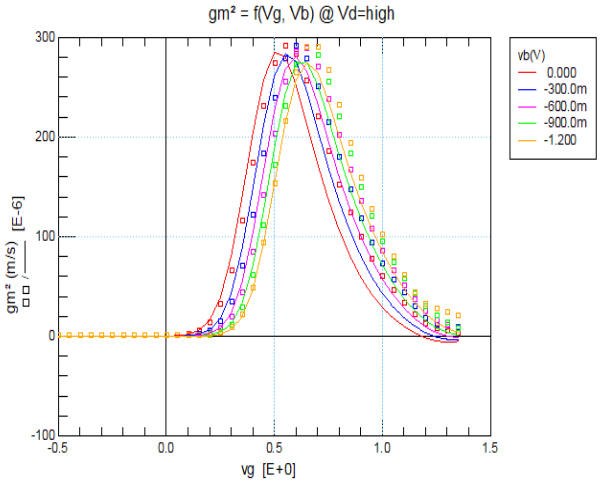


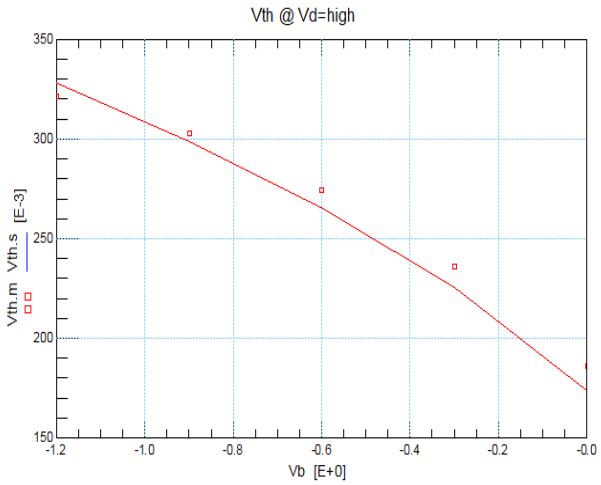
idvg, Vd = 1.2V, T = 125°C

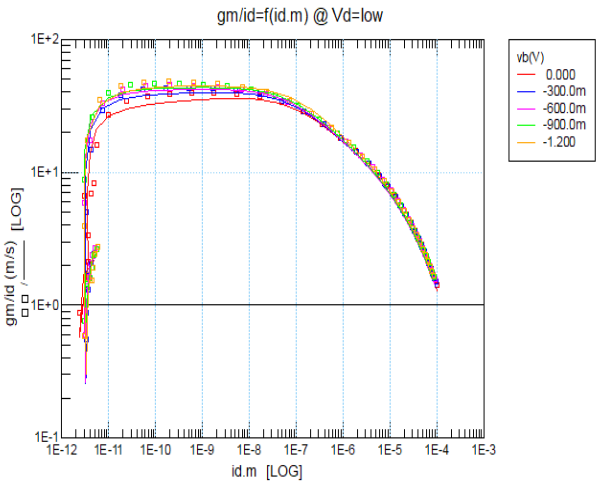


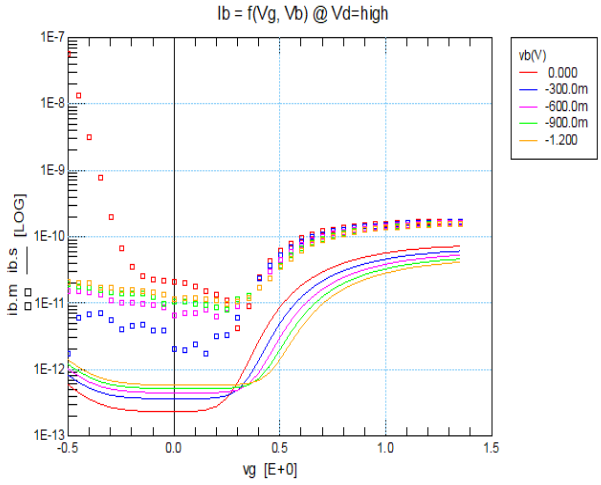


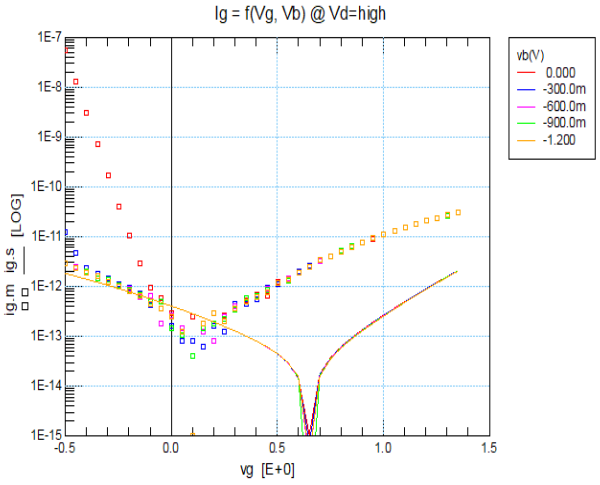




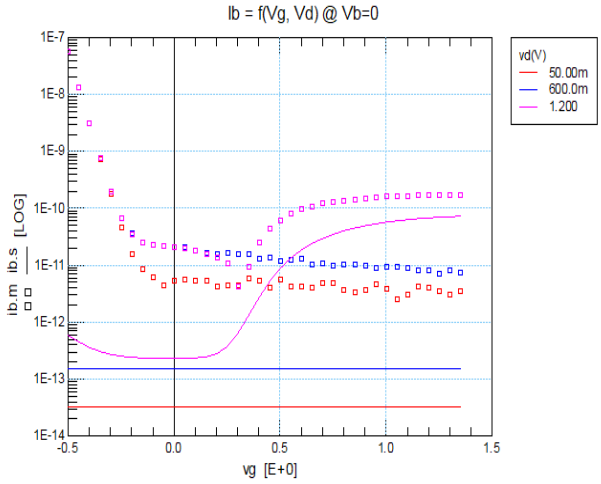


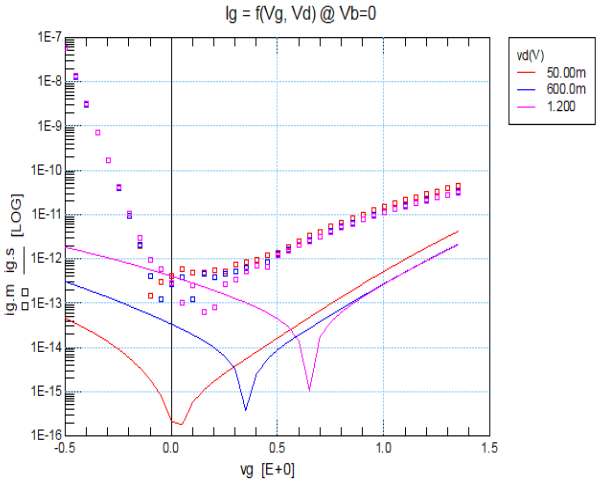




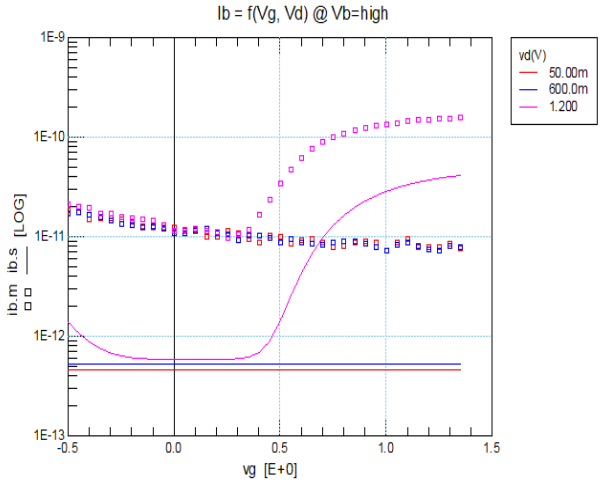


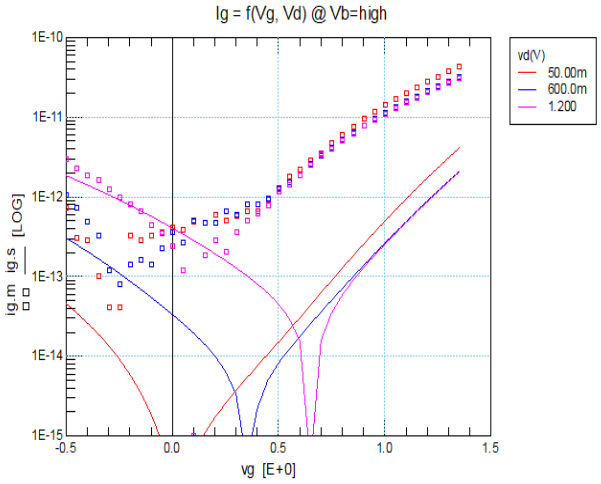
idvg, Ib,Ig, Vb = 0V, T = 125°C



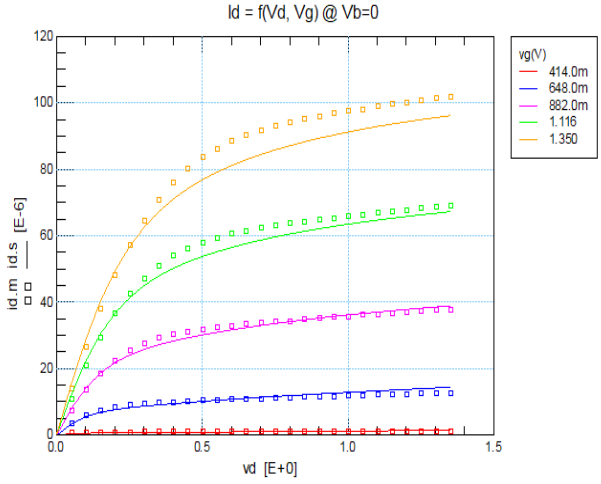


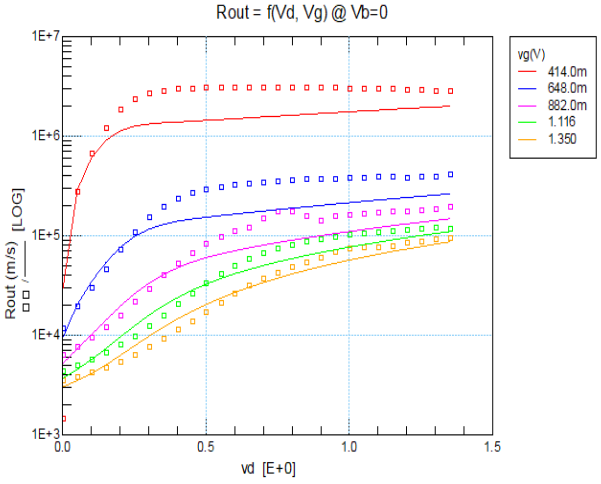
idvg, Ib,Ig, Vb = -1.2V, T = 125°C

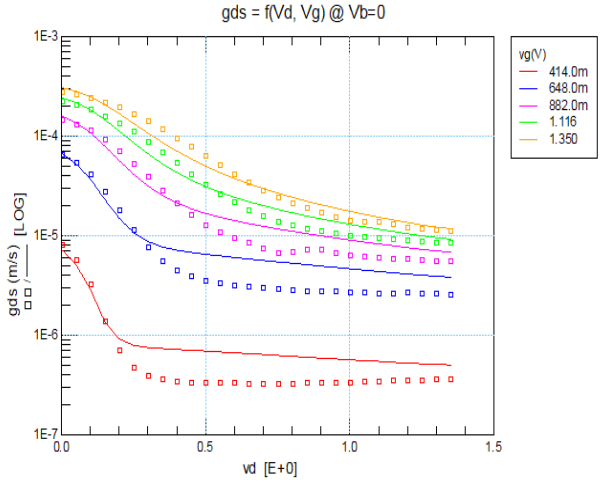




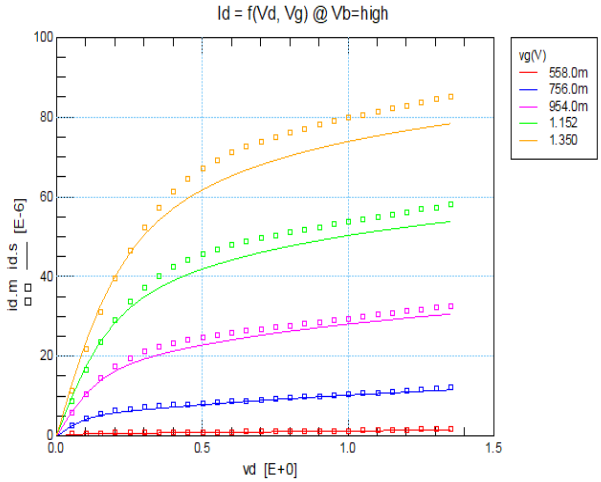
idvd, Vb = 0V, T = 125°C

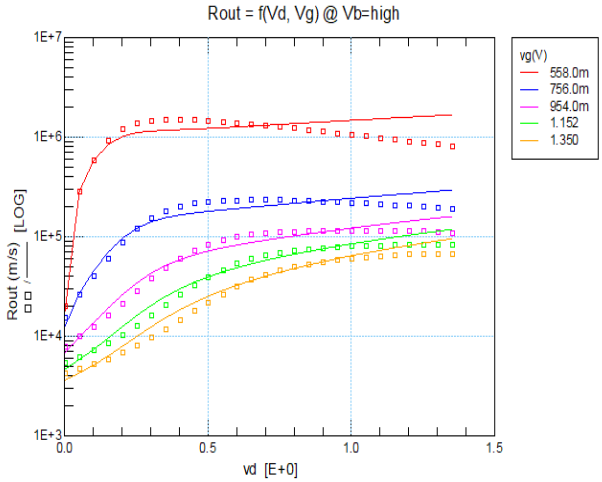


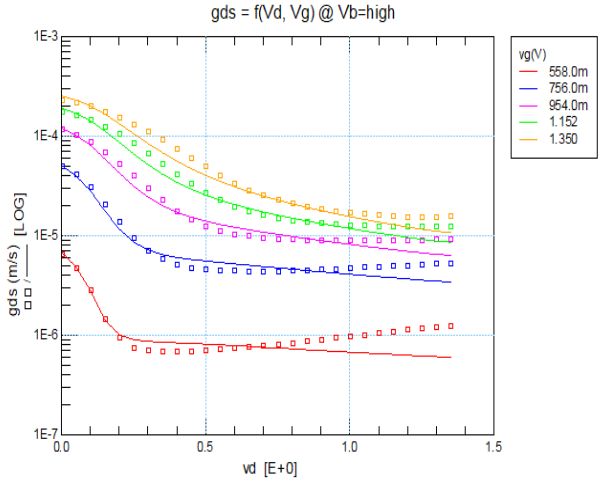




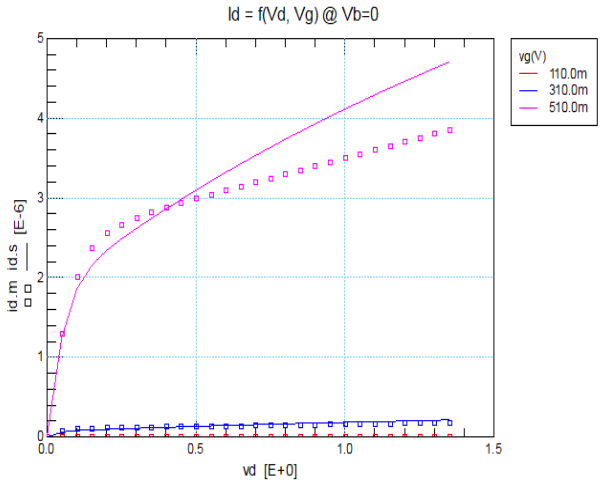
idvd_vbmin, Vb = -1.2V, T = 125°C

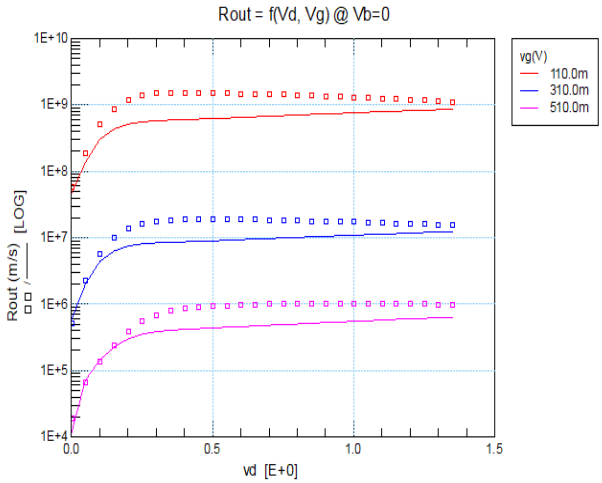


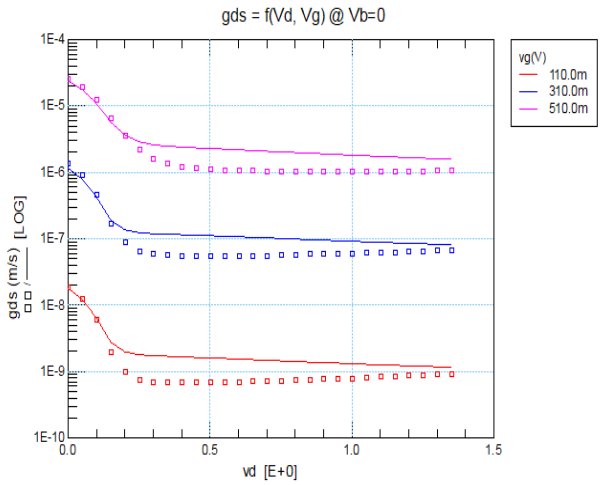




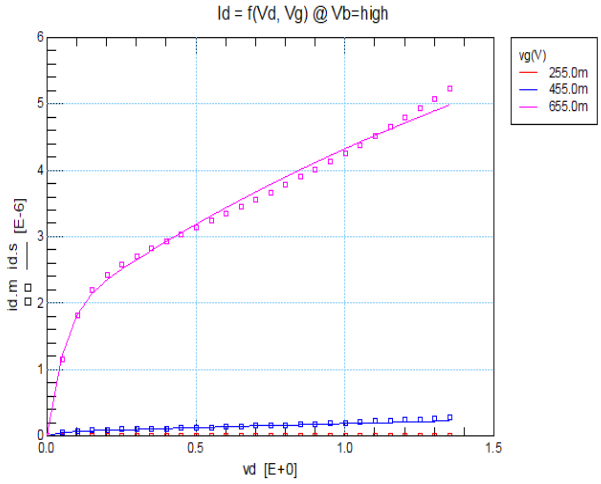
idvd_vth, Vb = 0V, T = 125°C

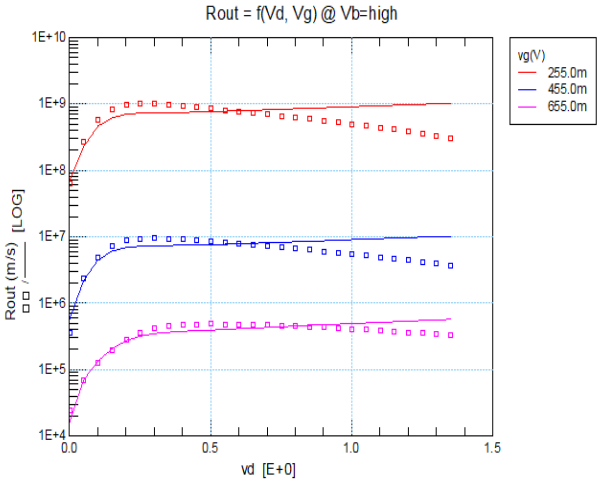


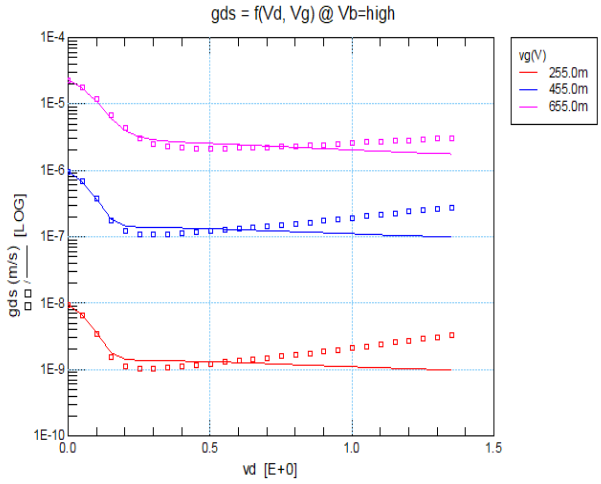




idvd_vth_vbmin, Vb = -1.2V, T = 125°C

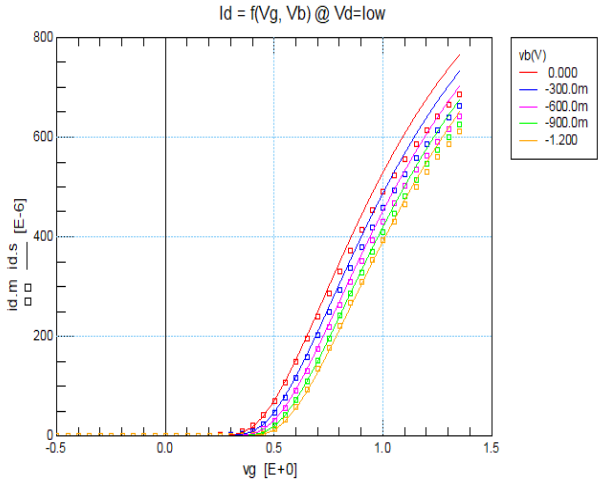


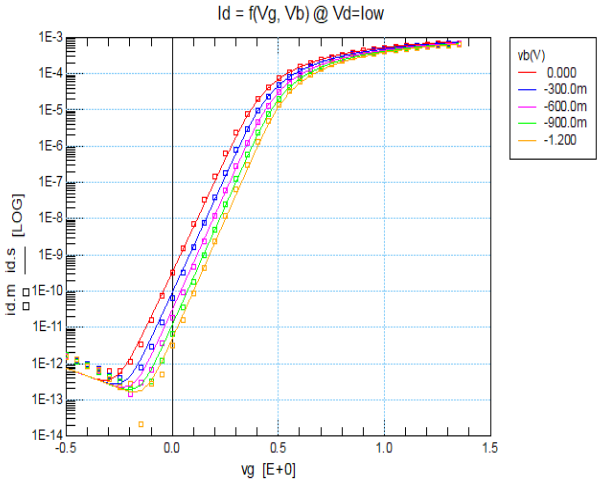


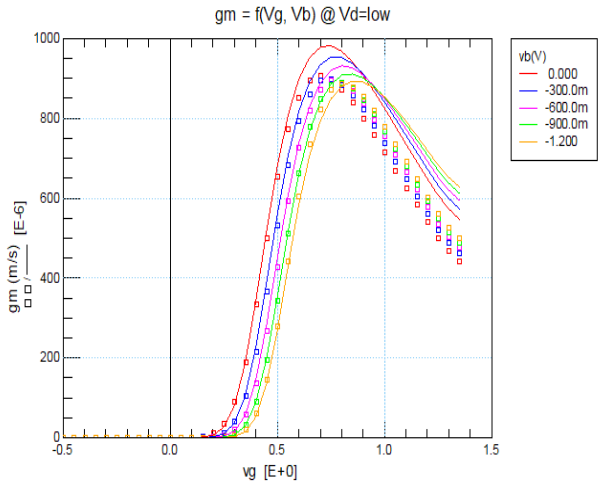


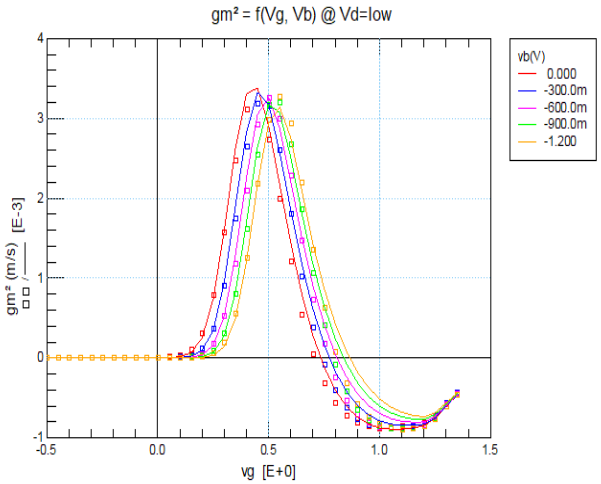
4.8 W10u0_L0u18_S540_5

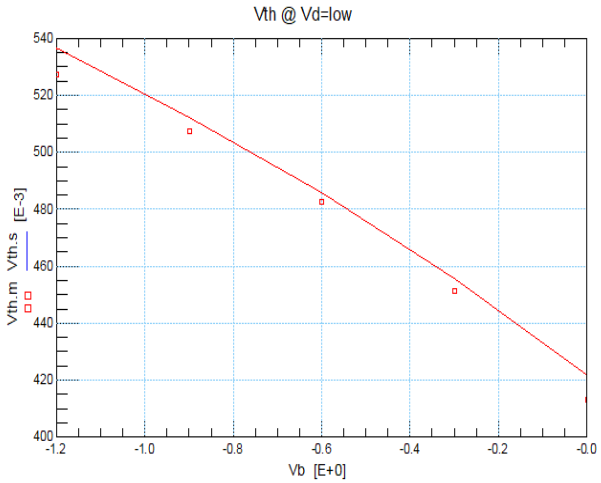
idvg, Vd = 0.05V, T = 27°C

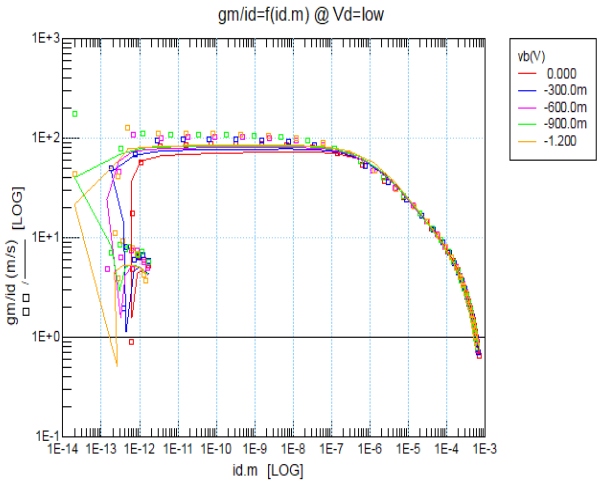


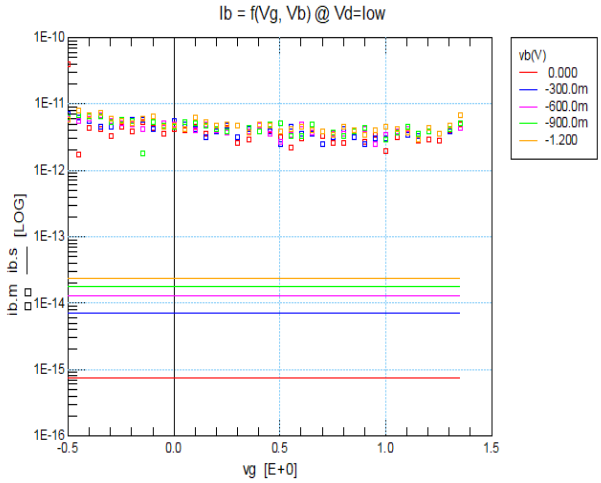


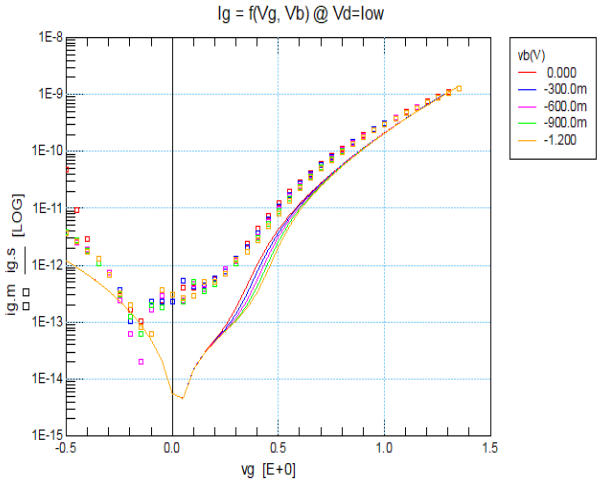




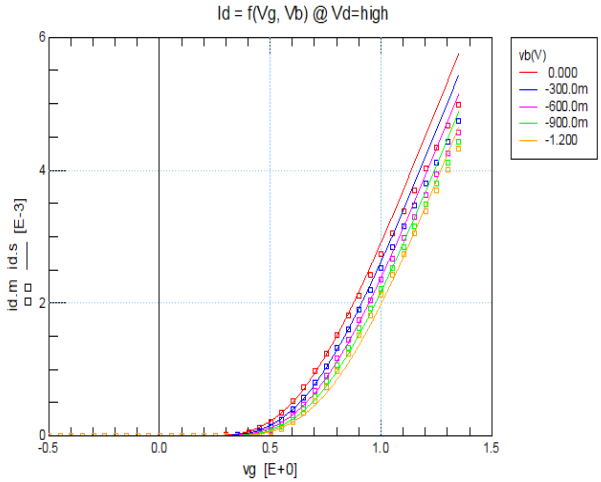


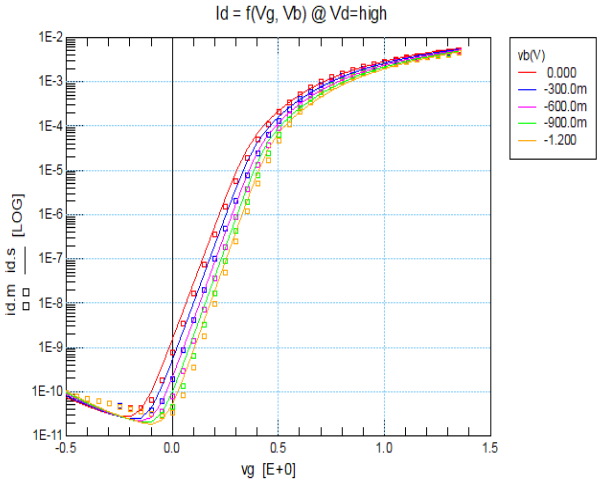


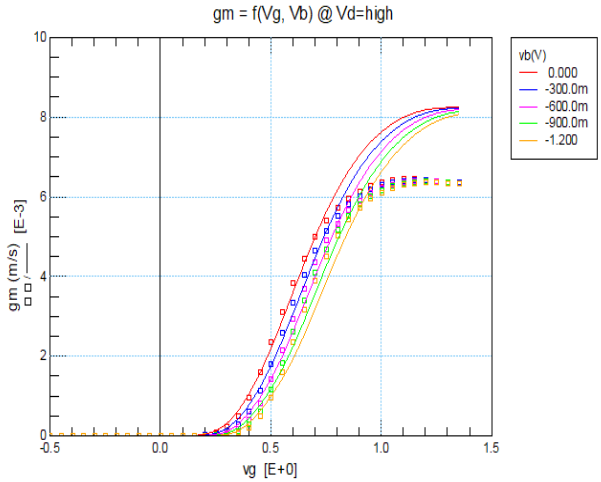


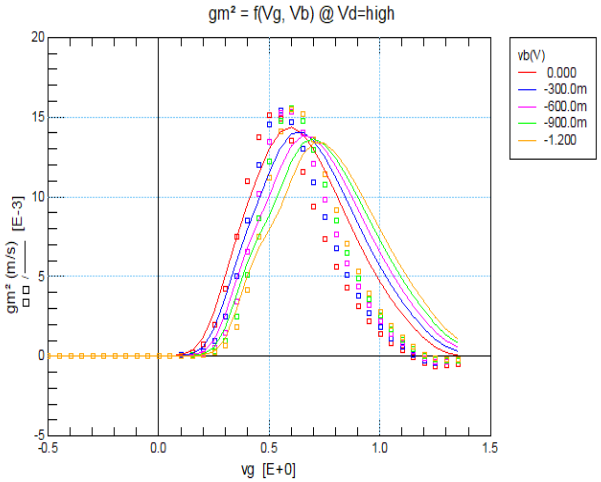


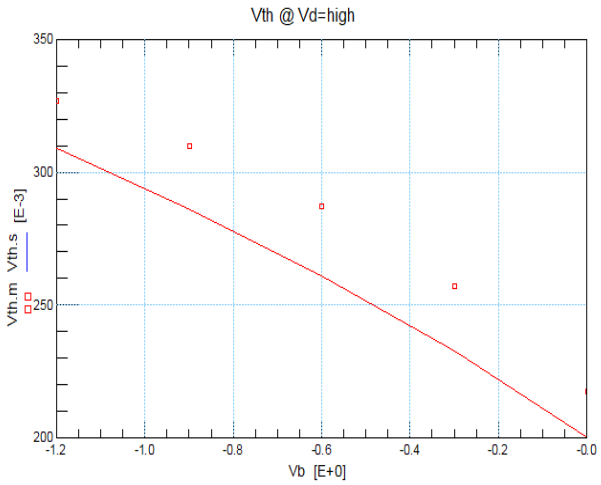
idvg, Vd = 1.2V, T = 27°C

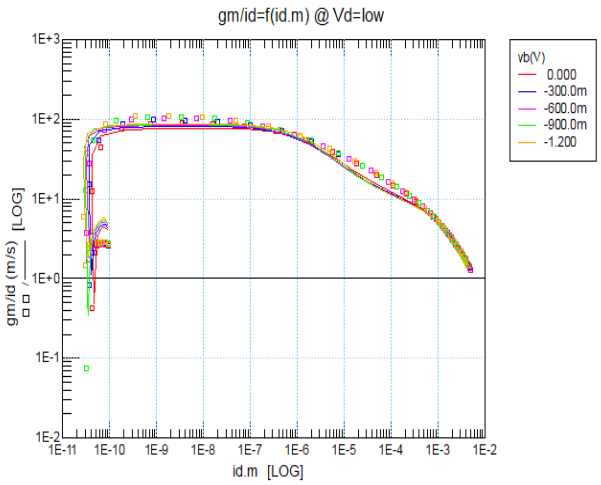


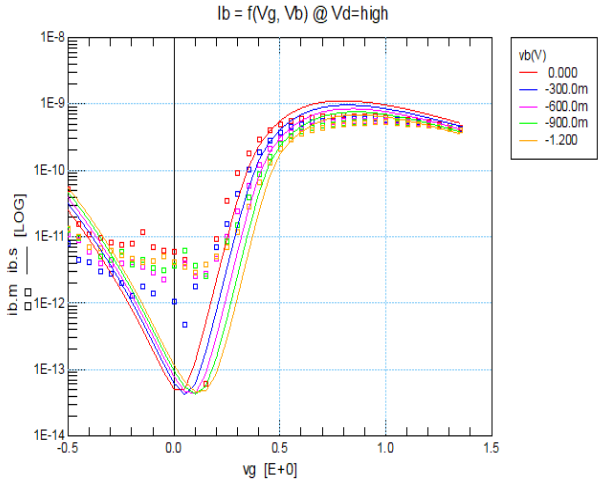


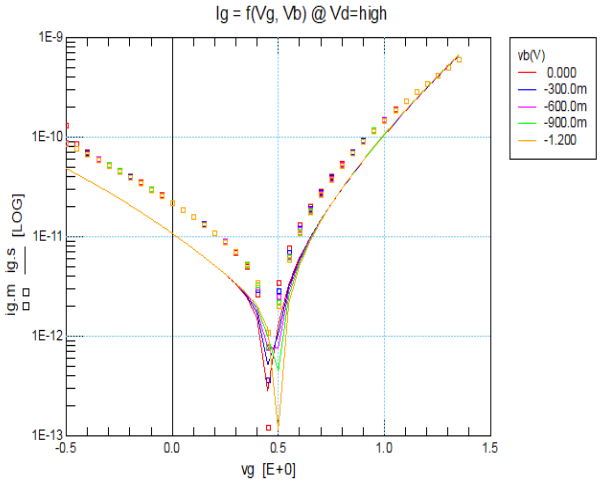




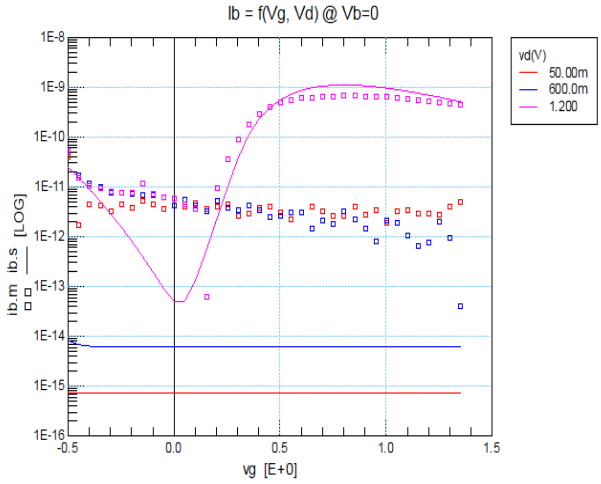


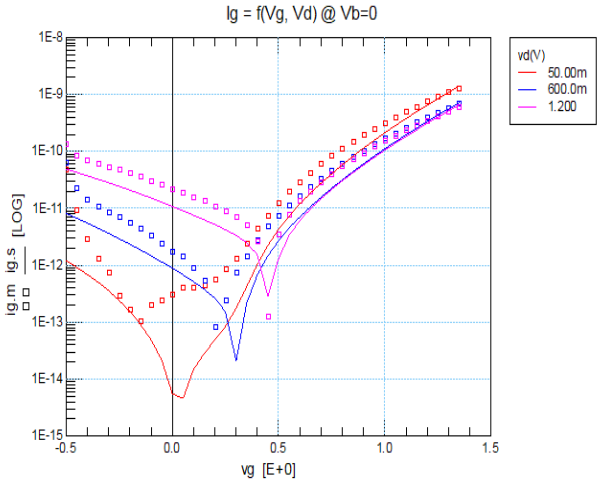




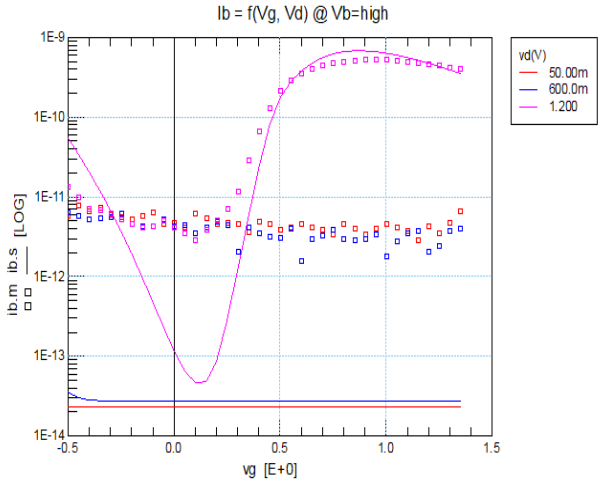


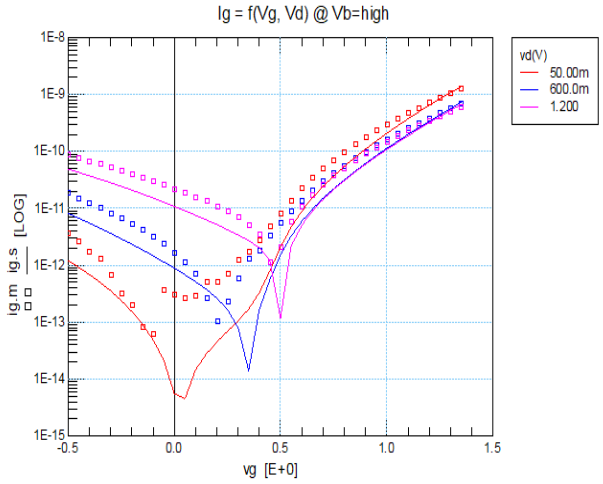
idvg, Ib,Ig, Vb = 0V, T = 27°C



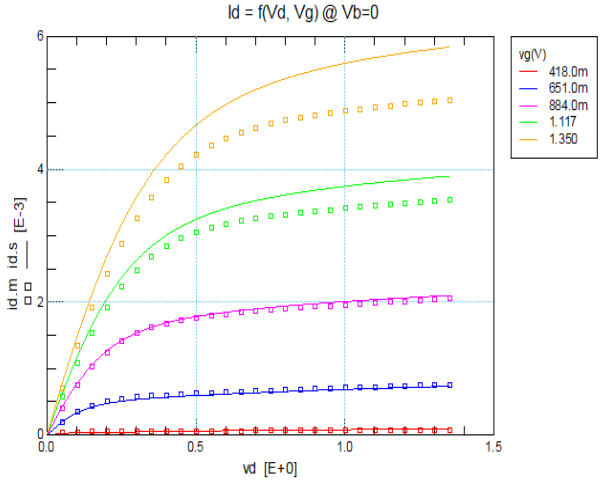


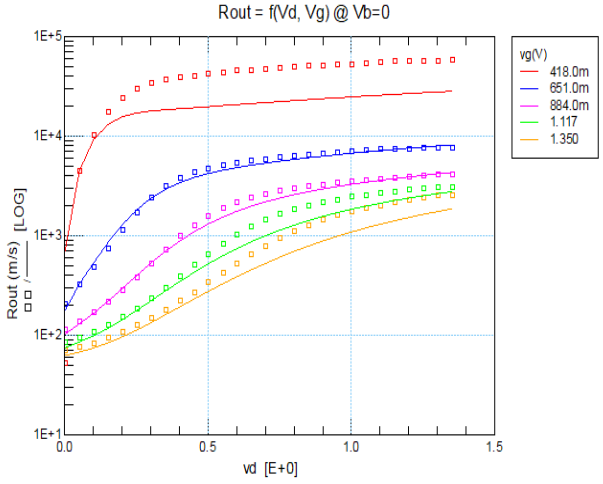
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

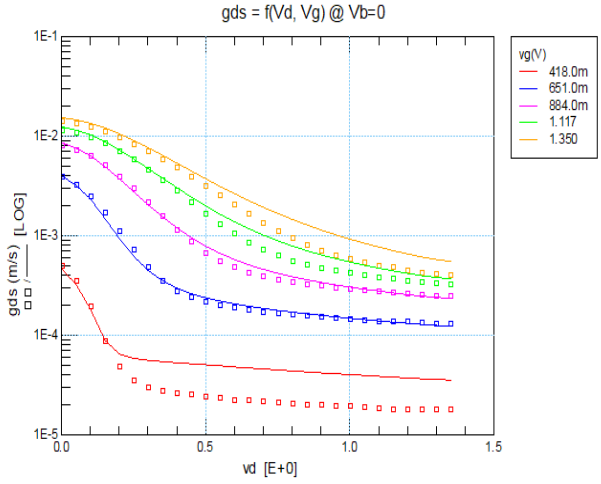




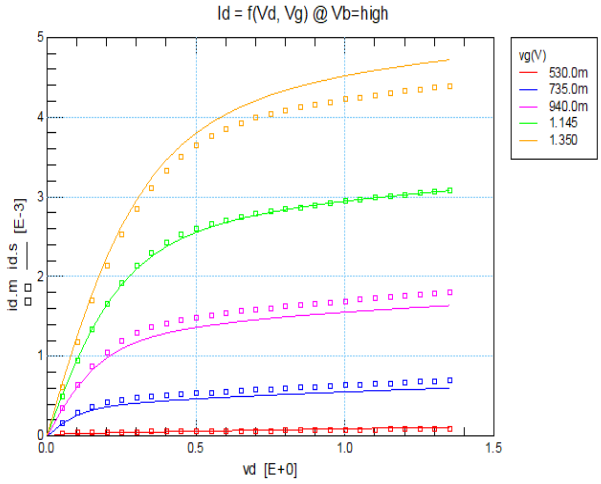
idvd, Vb = 0V, T = 27°C

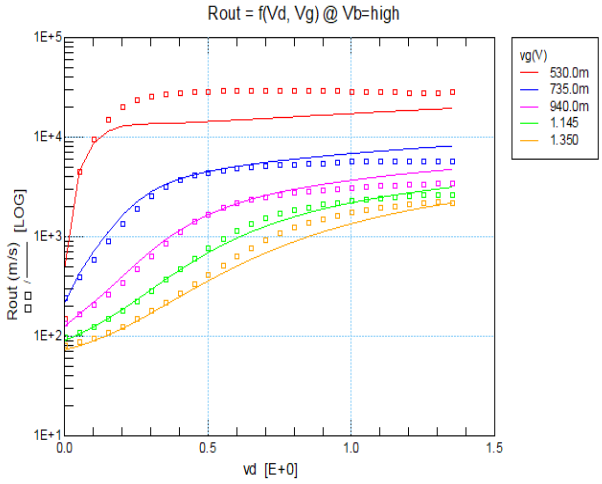


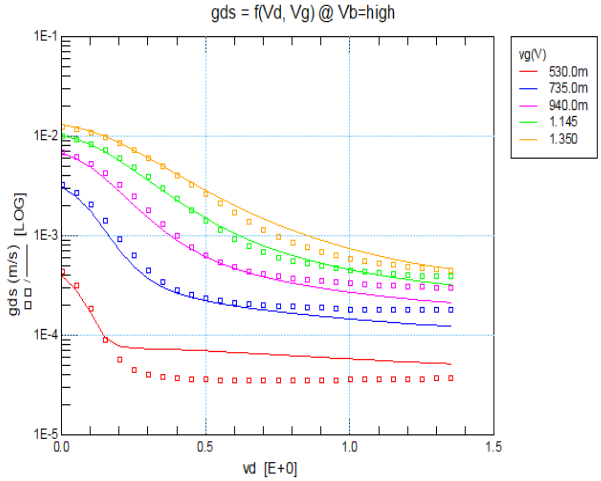




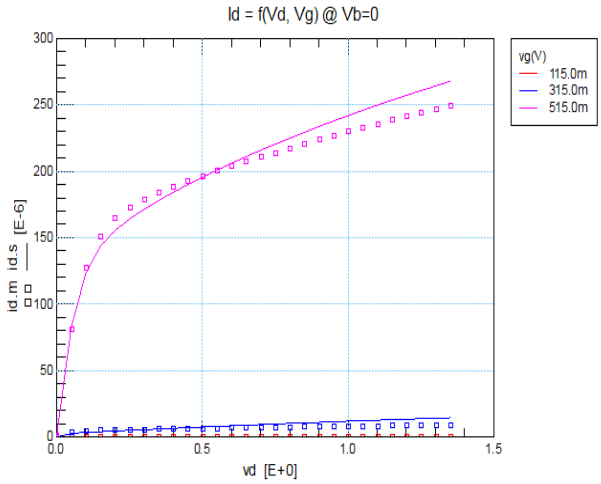
idvd_vbmin, Vb = -1.2V, T = 27°C

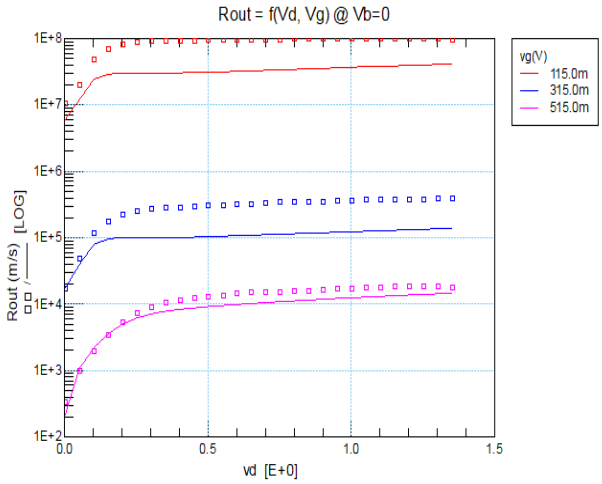


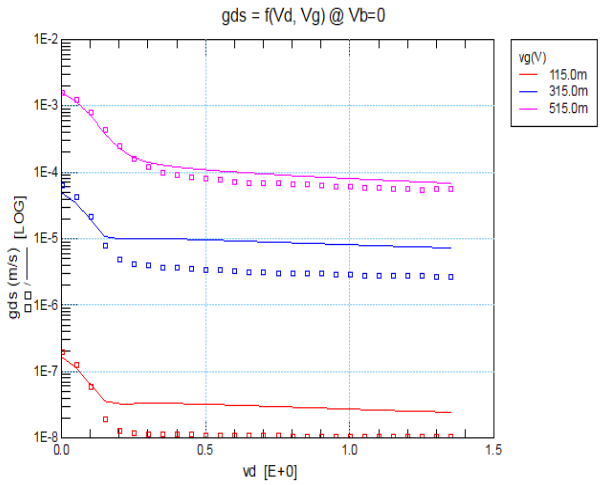




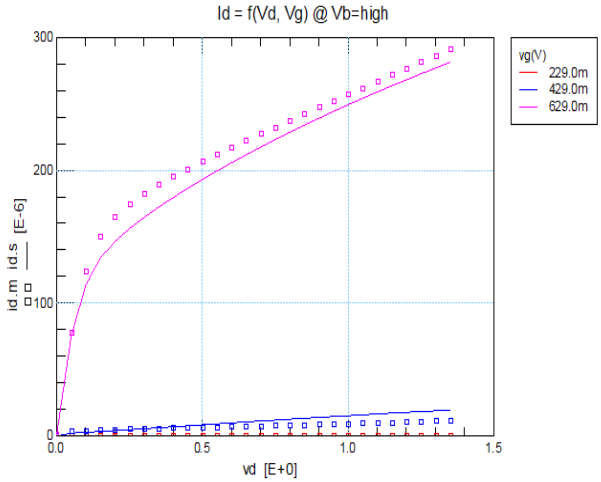
idvd_vth, Vb = 0V, T = 27°C

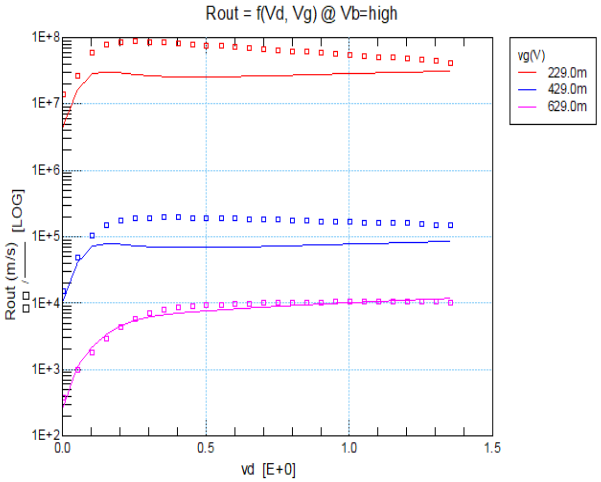


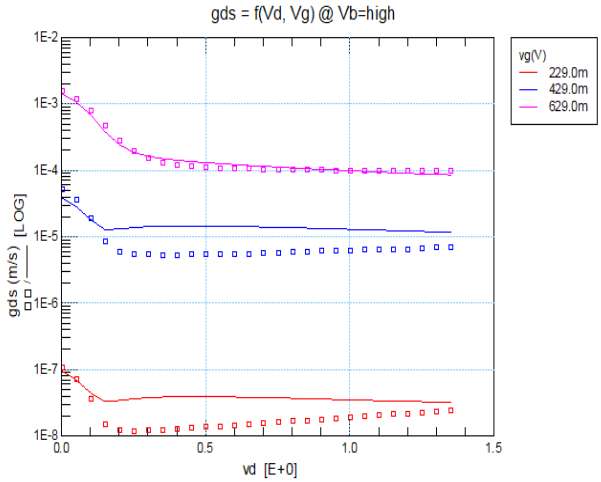




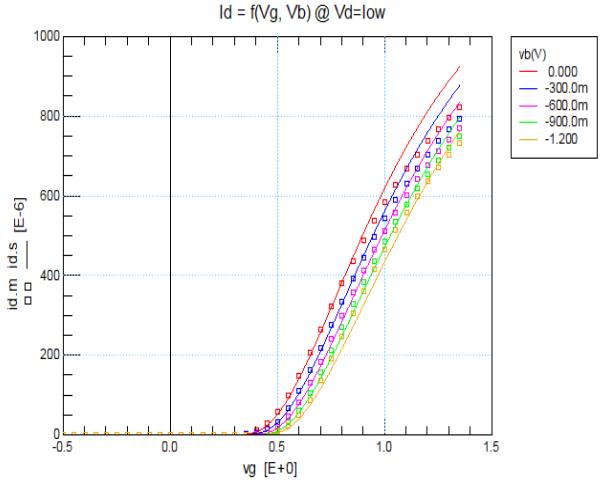
idvd_vth_vbmin, Vb = -1.2V, T = 27°C

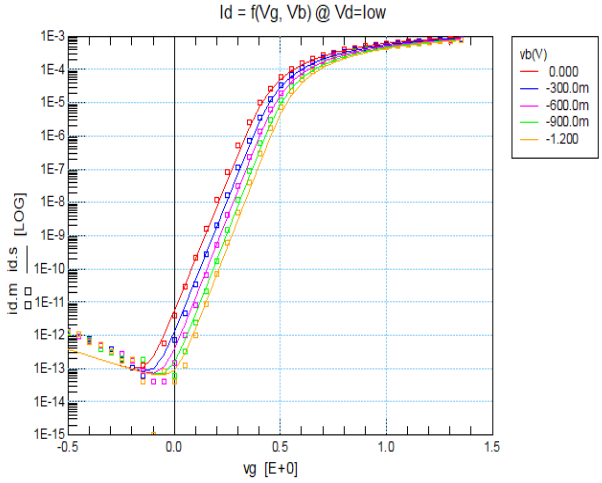


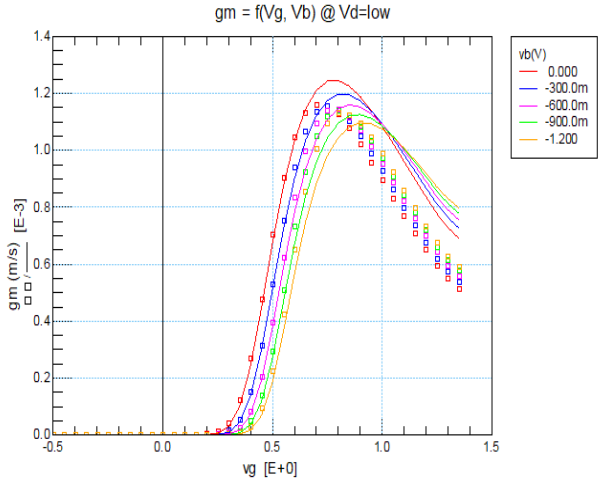


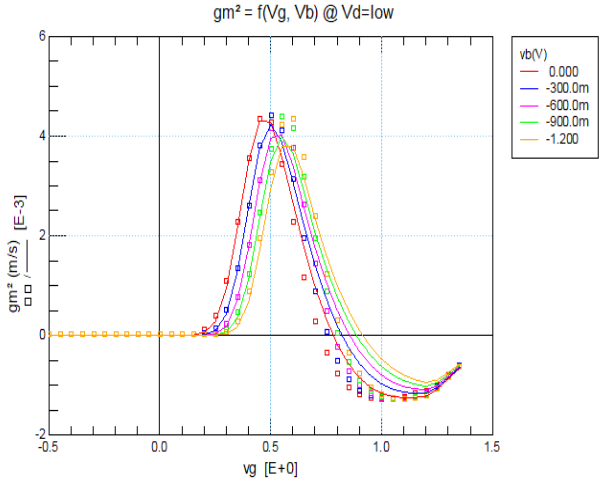


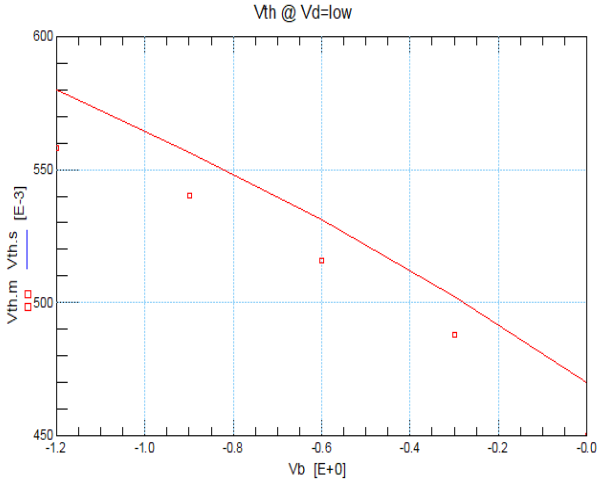
idvg, Vd = 0.05V, T = -40°C

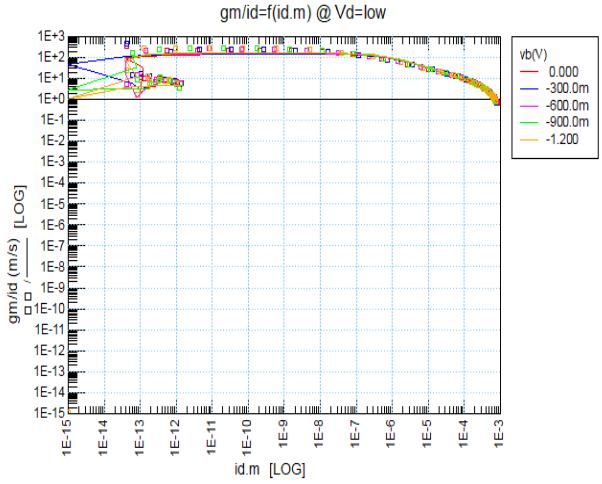


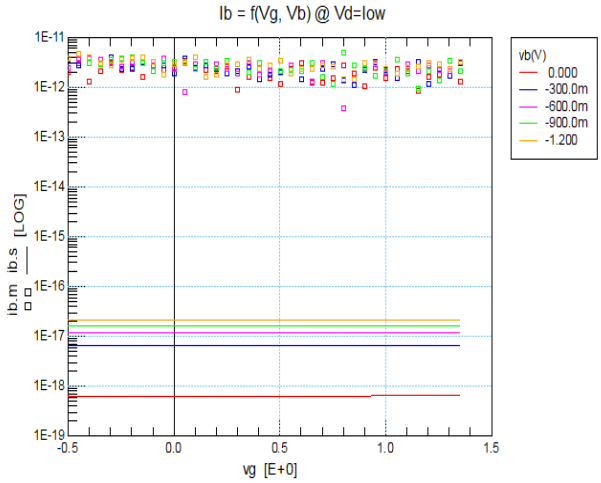


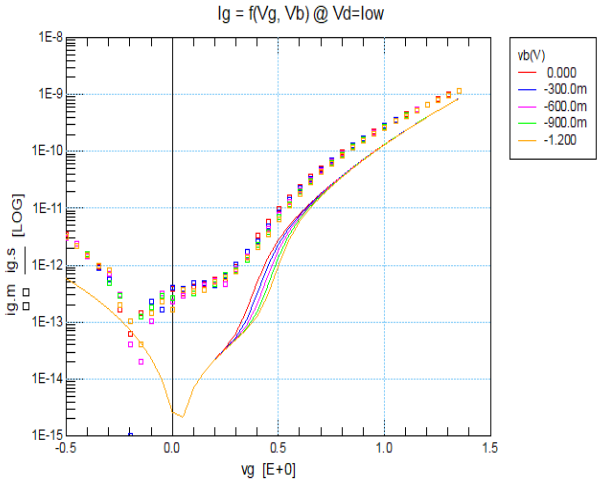




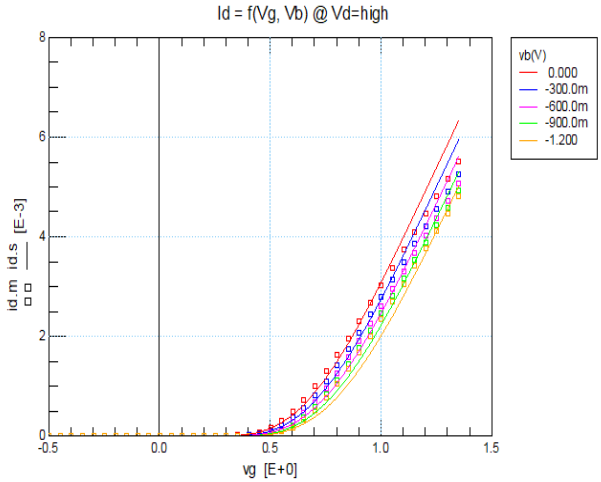


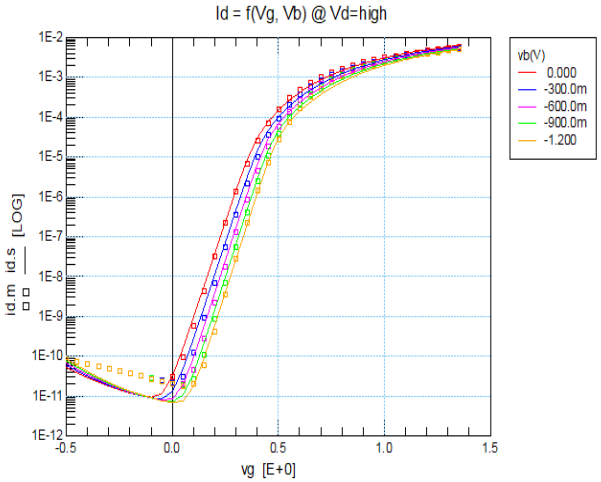


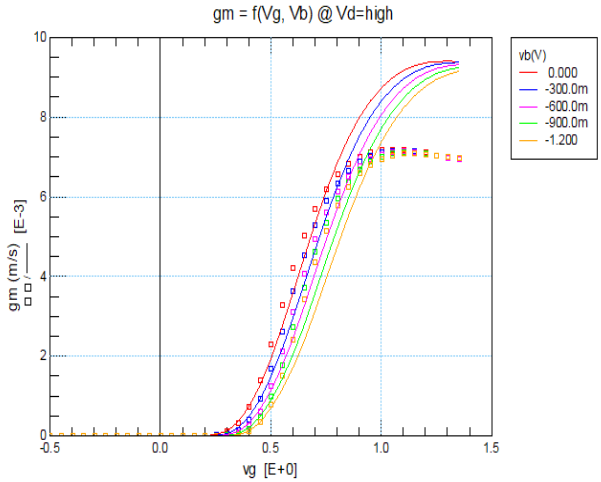


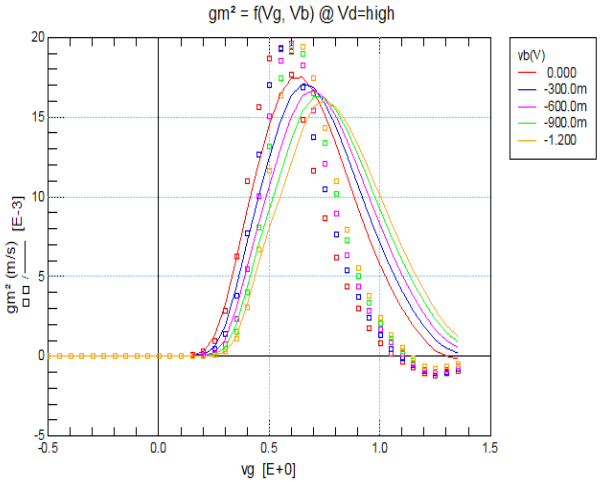


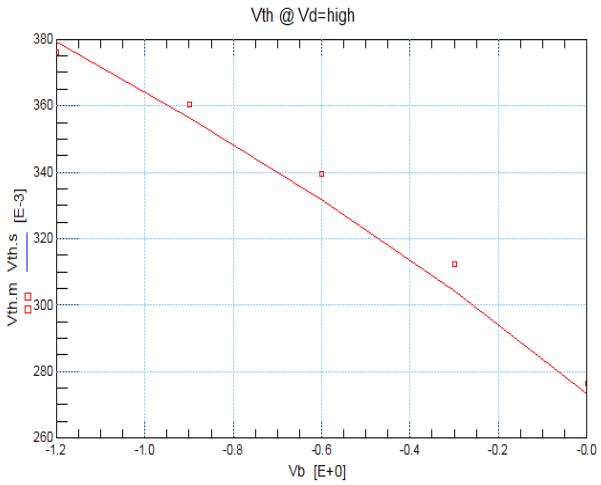
idvg, Vd = 1.2V, T = -40°C

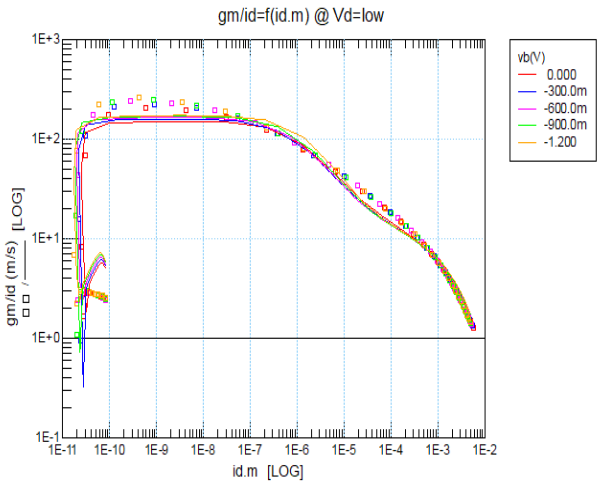


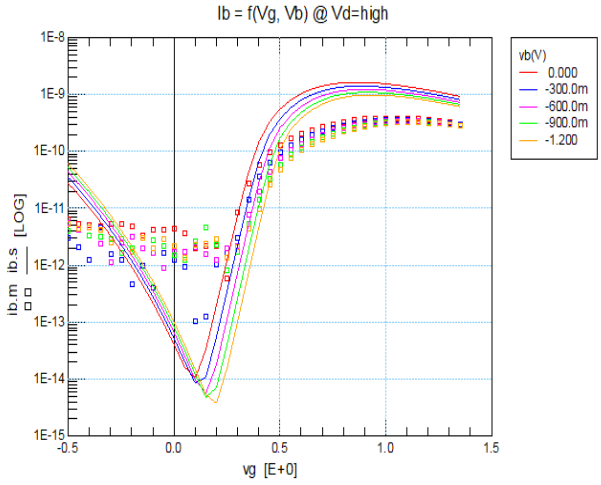


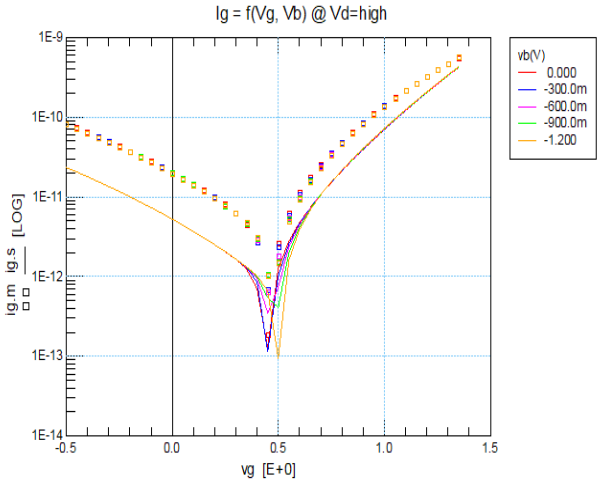




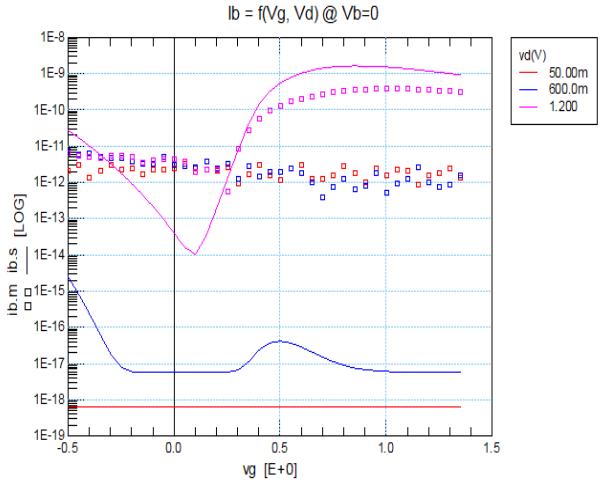


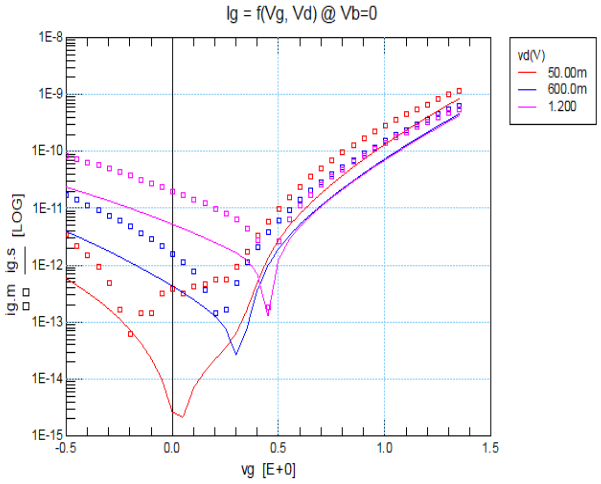




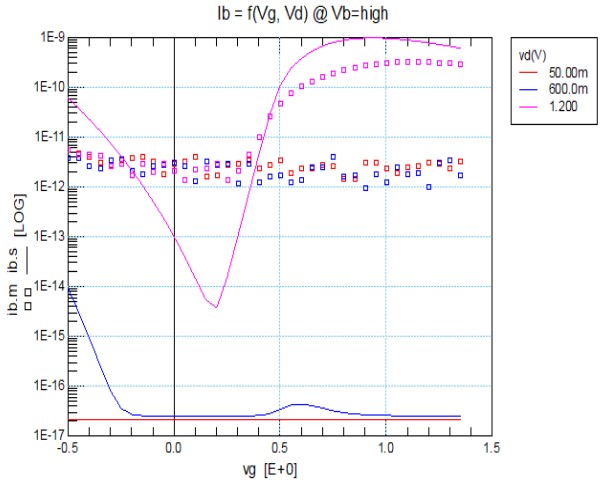


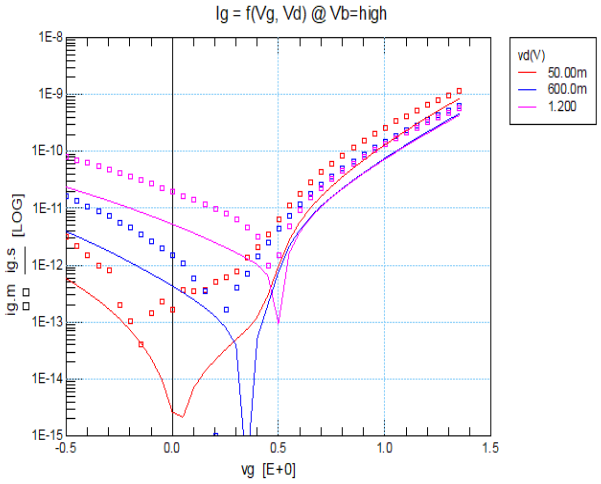
idvg, Ib,Ig, Vb = 0V, T = -40°C



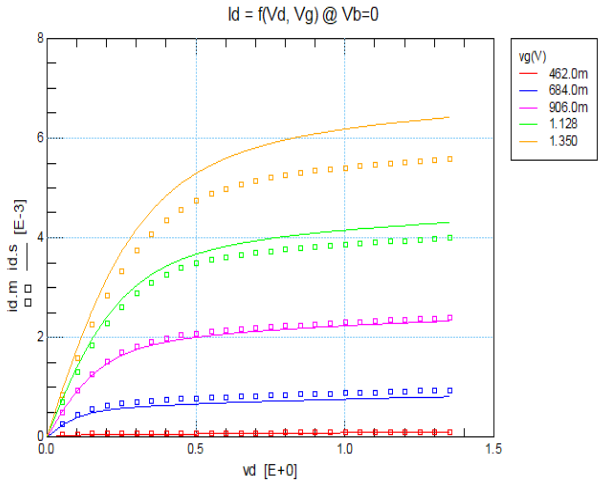


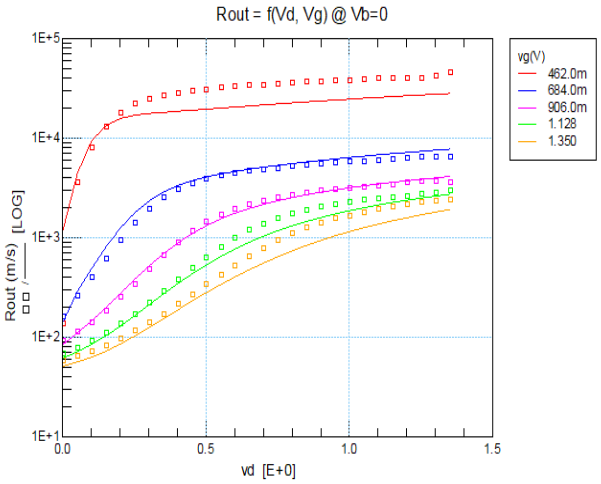
idvg, Ib,Ig, Vb = -1.2V, T = -40°C

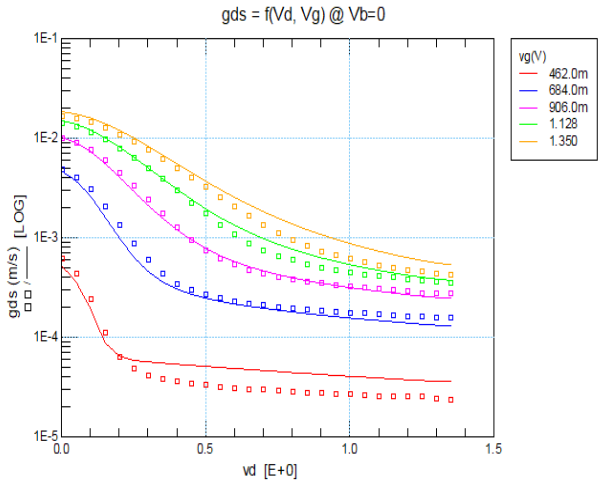




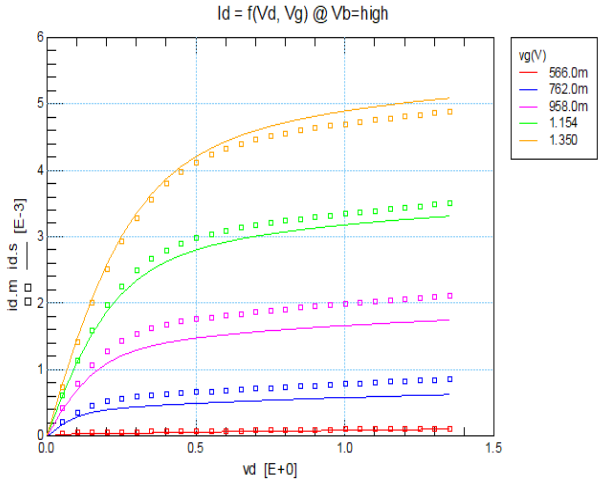
idvd, Vb = 0V, T = -40°C

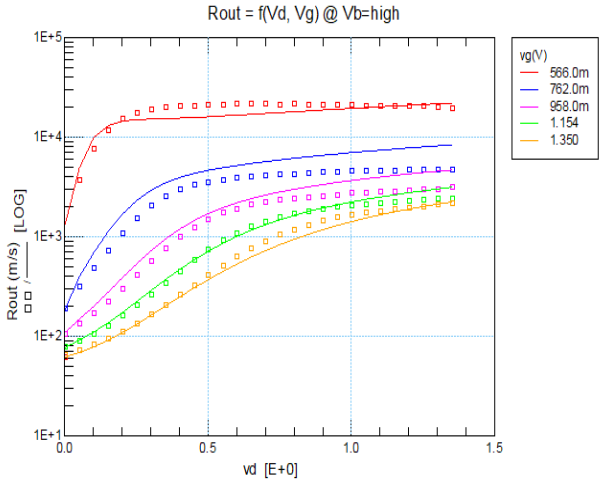


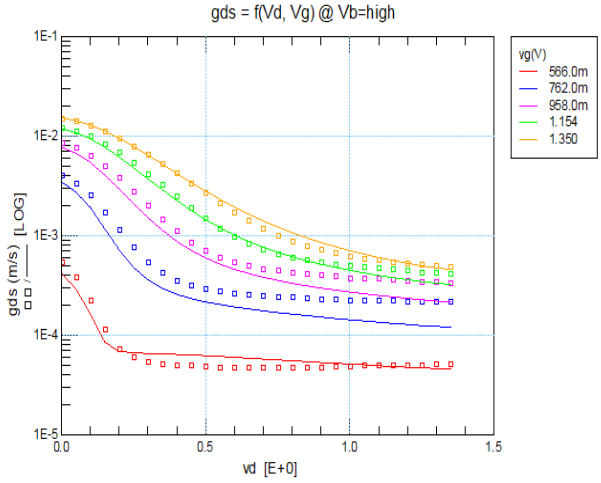




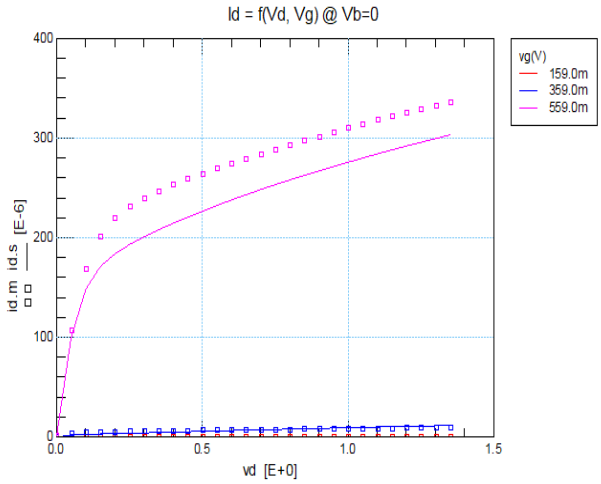
idvd_vbmin, Vb = -1.2V, T = -40°C

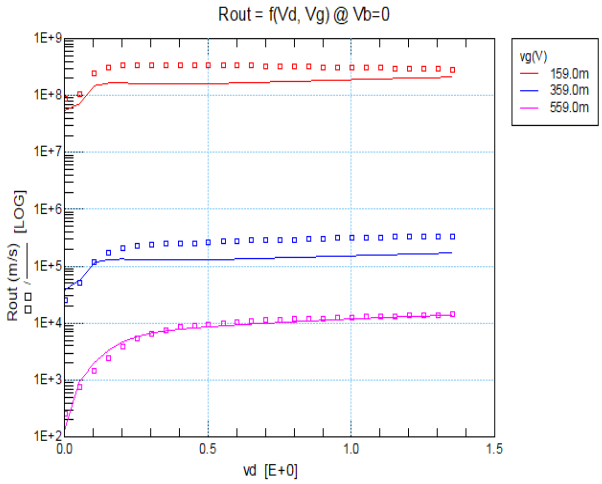


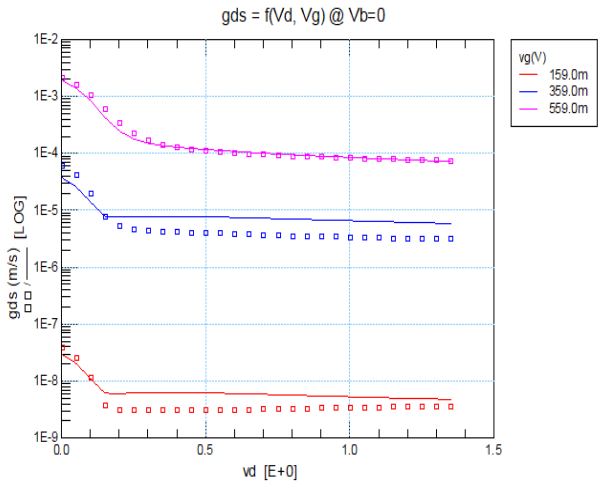




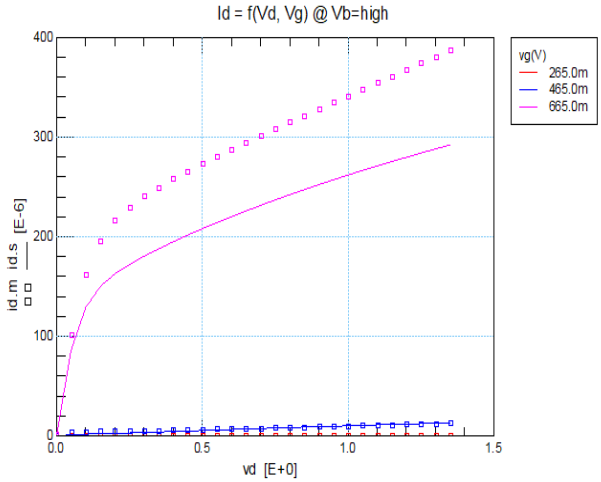
idvd_vth, Vb = 0V, T = -40°C

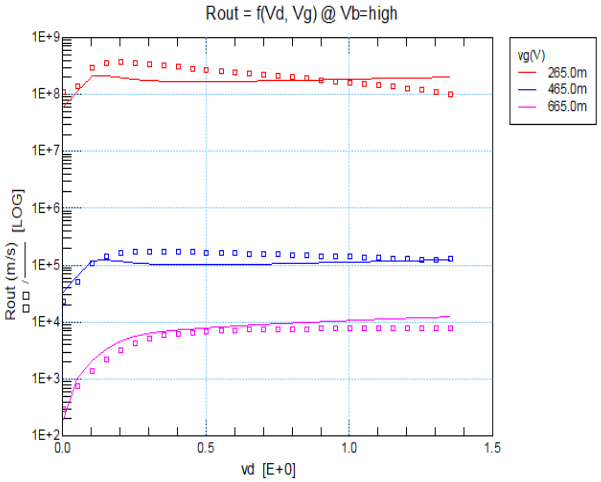


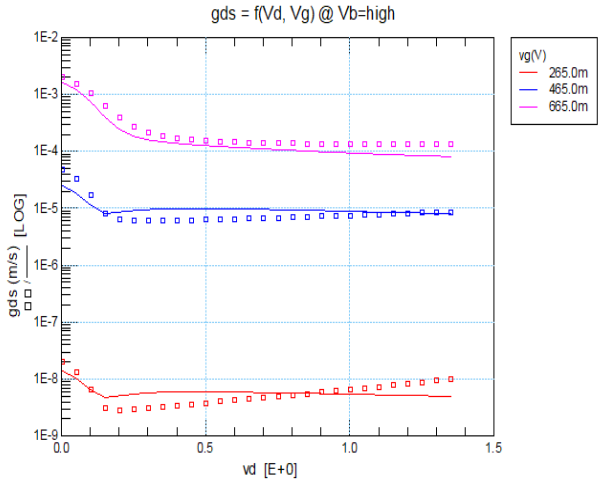




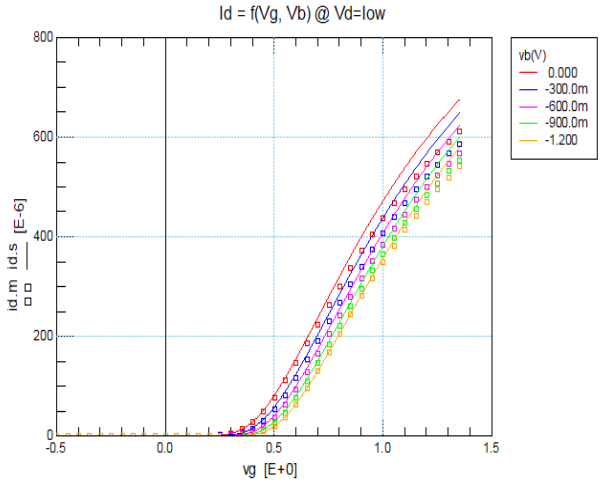
idvd_vth_vbmin, Vb = -1.2V, T = -40°C

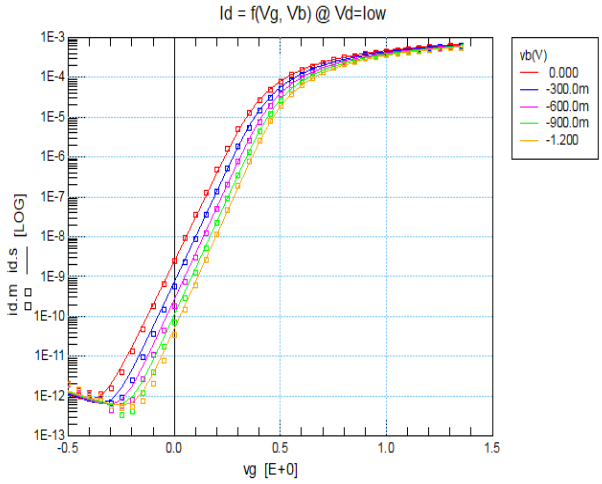


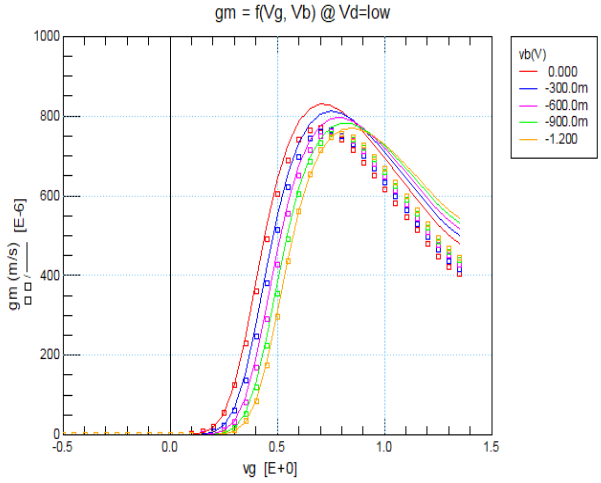


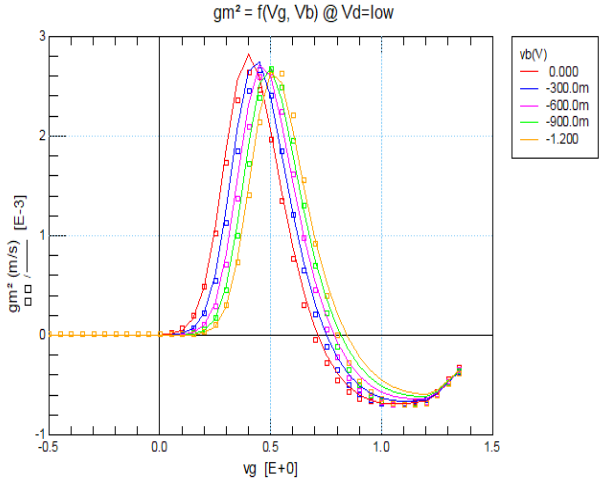


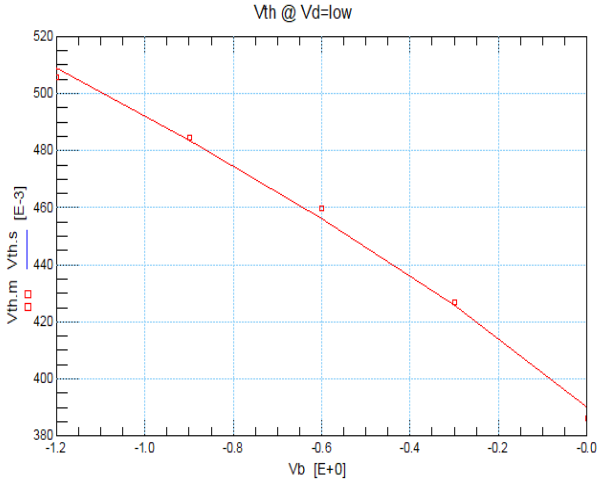
idvg, Vd = 0.05V, T = 70°C

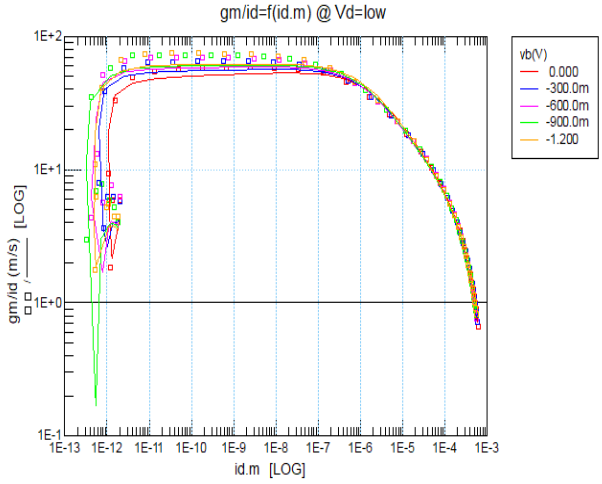


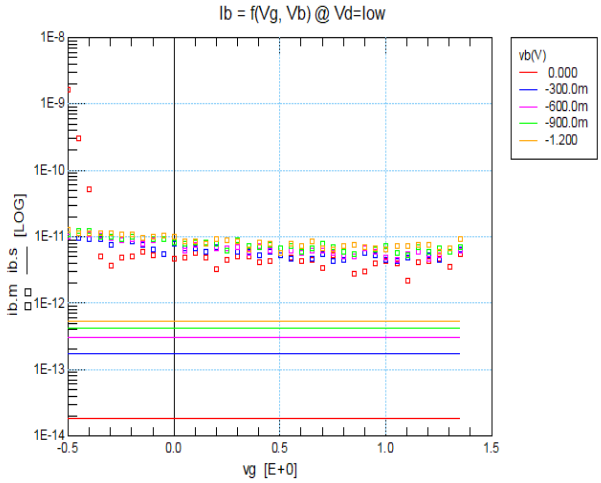


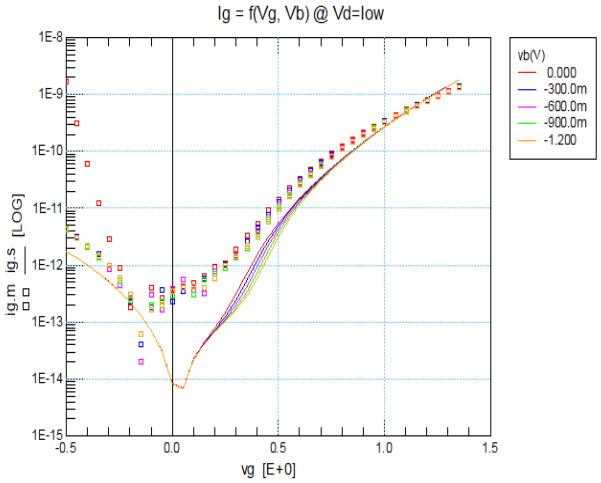




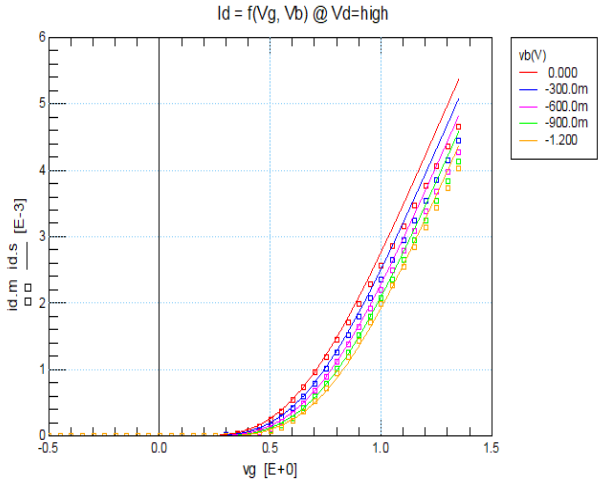


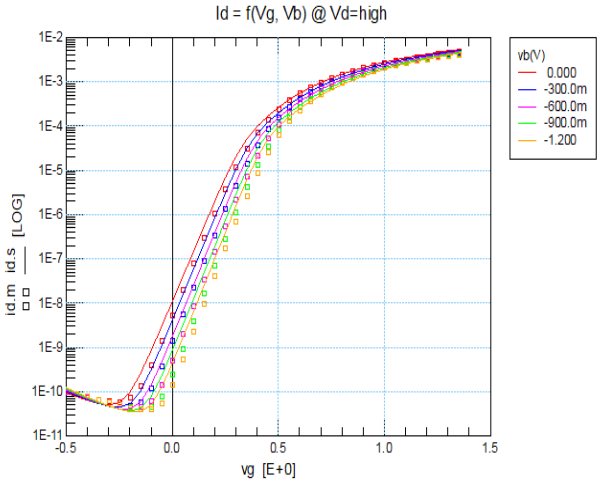


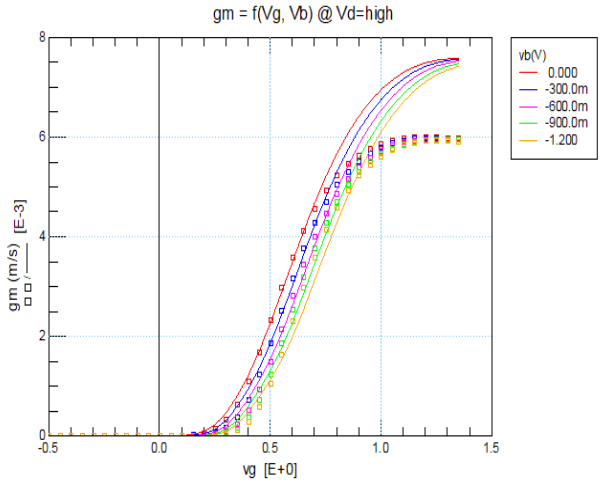


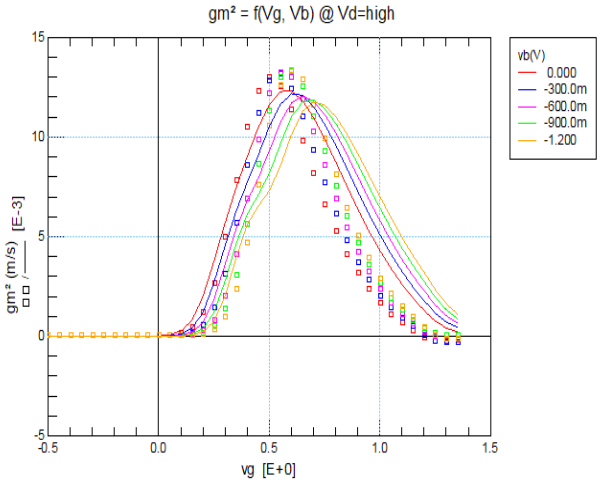


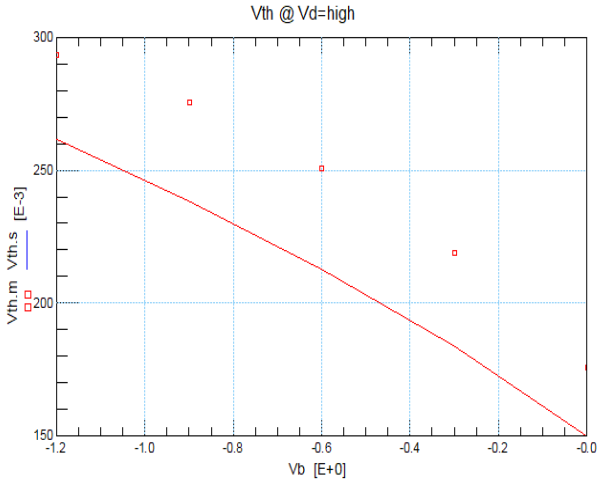
idvg, Vd = 1.2V, T = 70°C

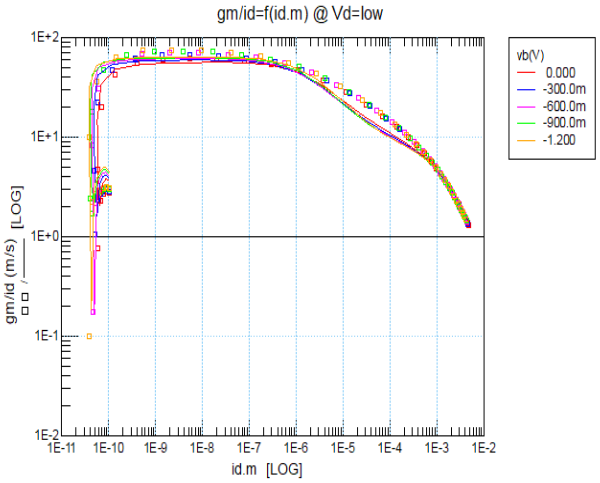


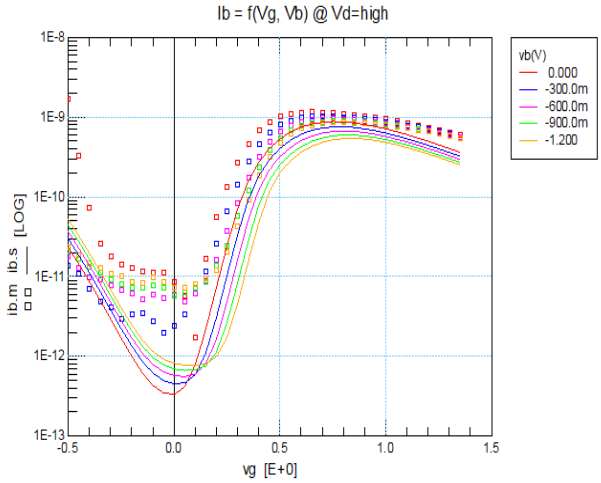


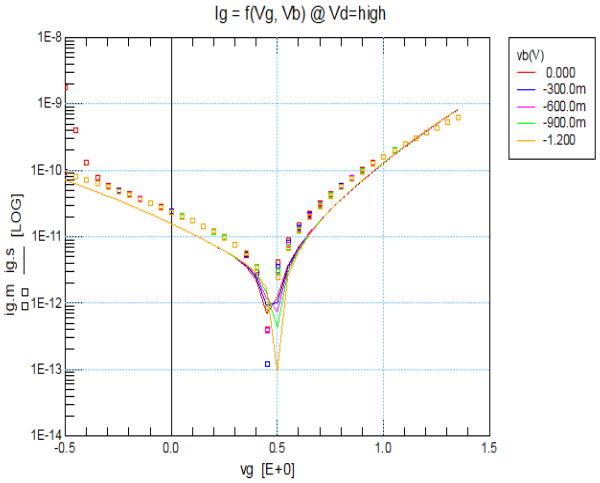




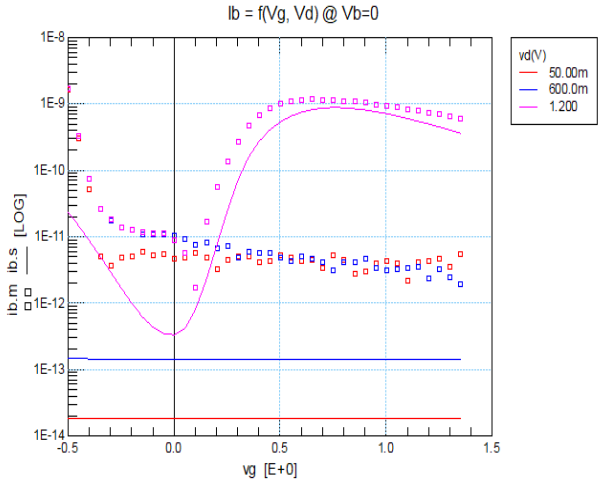


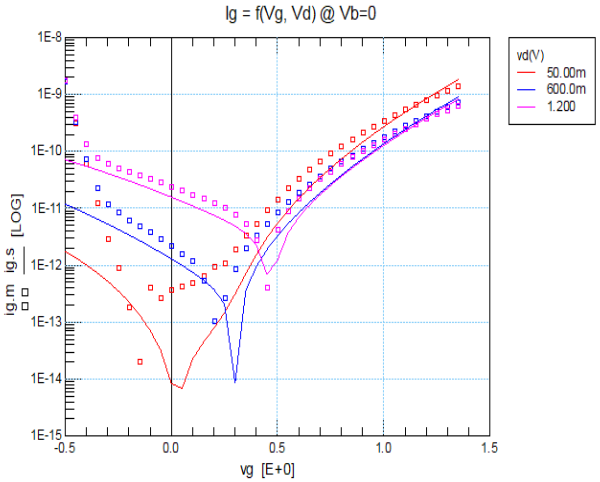




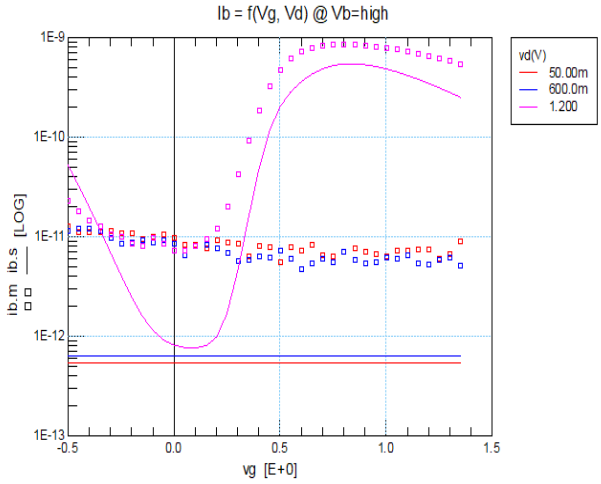


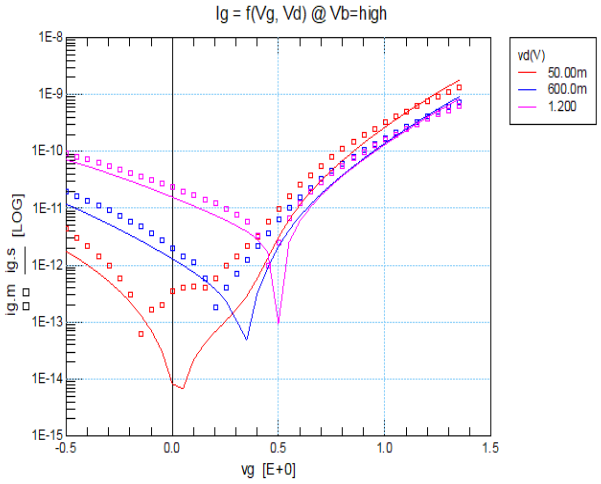
idvg, Ib,Ig, Vb = 0V, T = 70°C



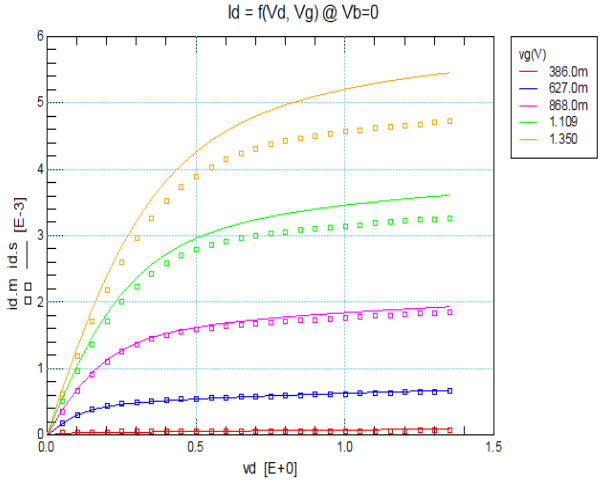


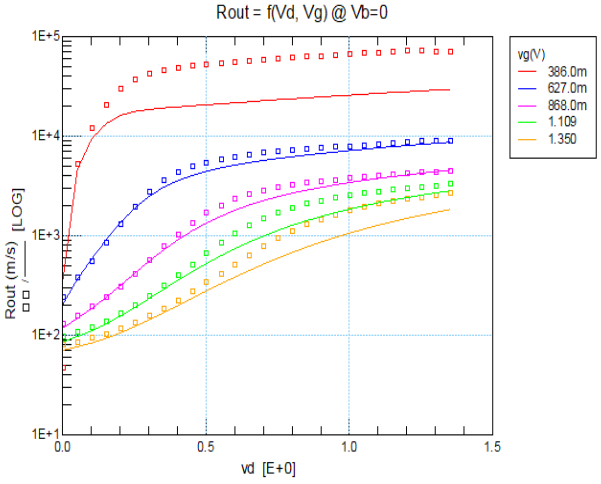
idvg, Ib,Ig, Vb = -1.2V, T = 70°C

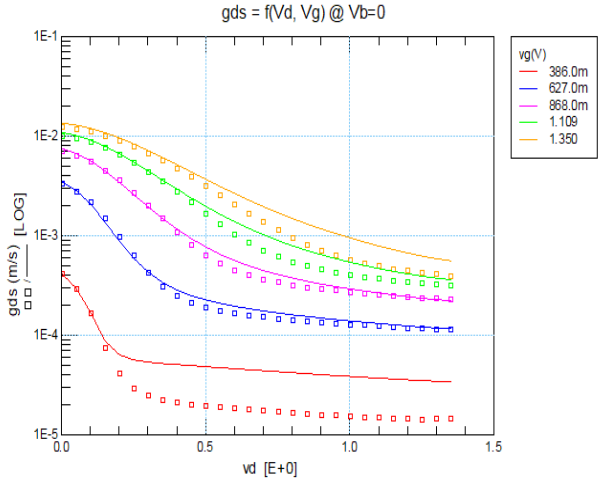




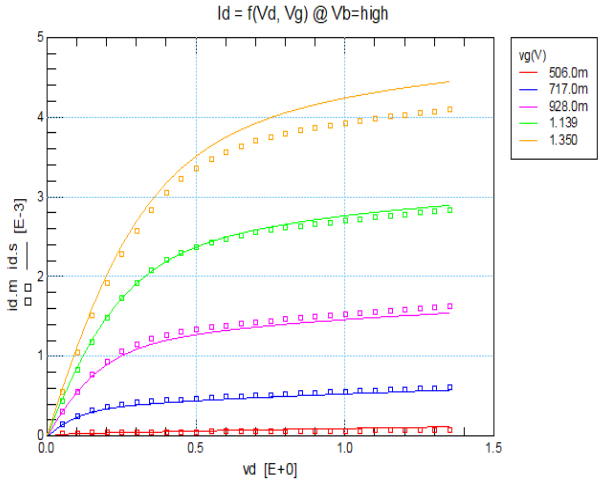
idvd, Vb = 0V, T = 70°C

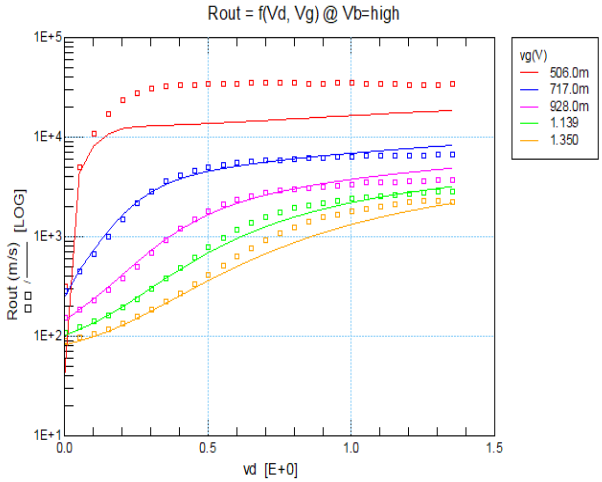


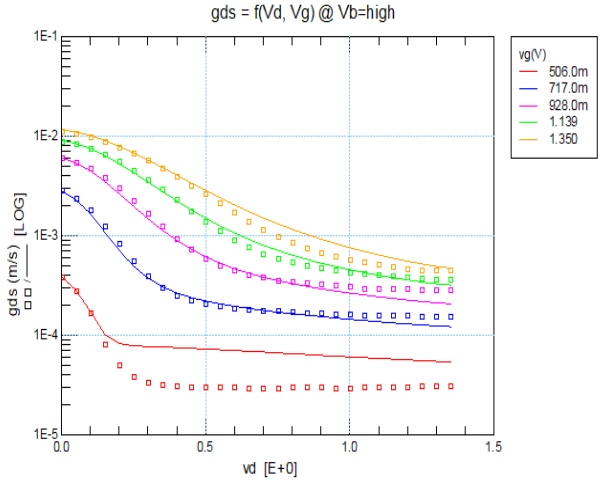




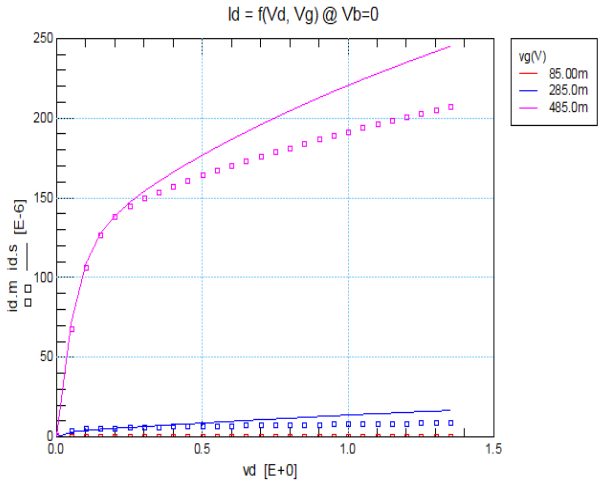
idvd_vbmin, Vb = -1.2V, T = 70°C

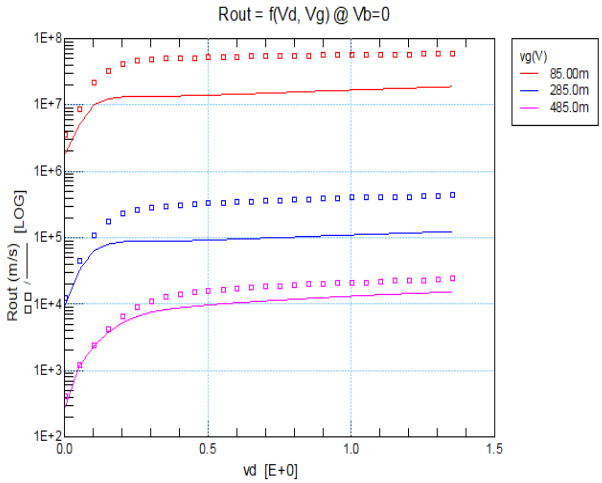


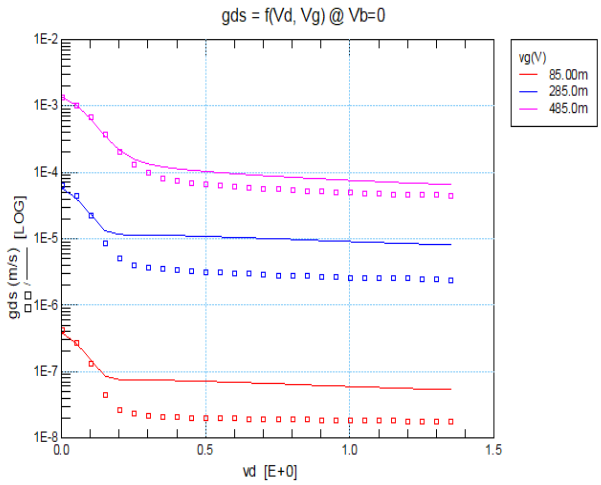




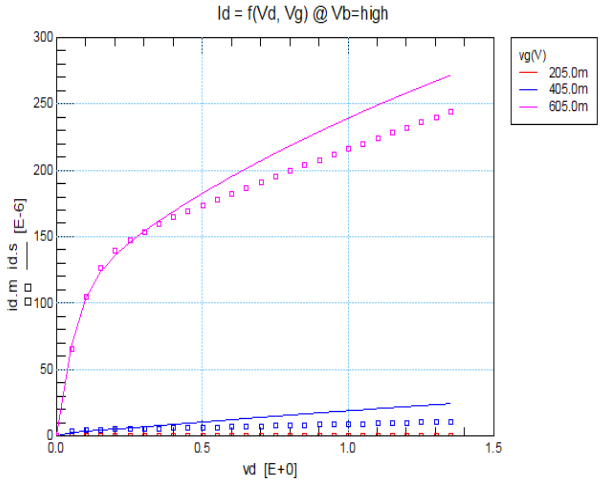
idvd_vth, Vb = 0V, T = 70°C

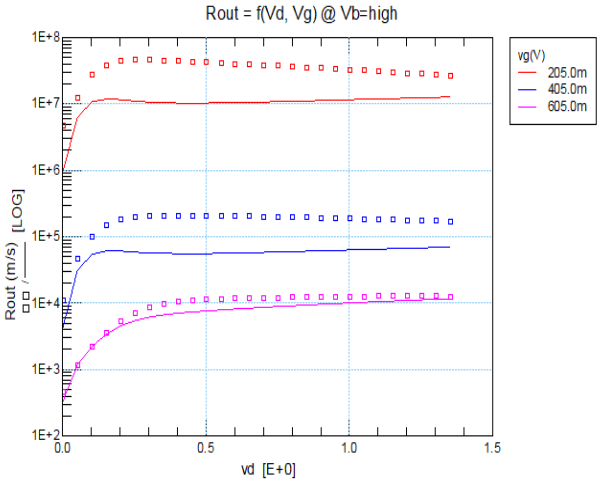


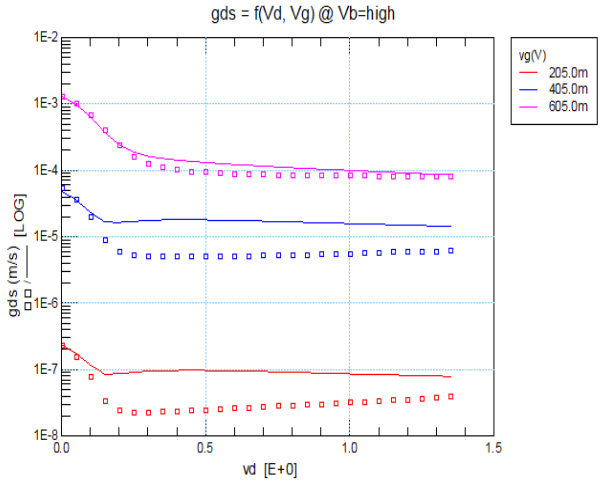




idvd_vth_vbmin, Vb = -1.2V, T = 70°C

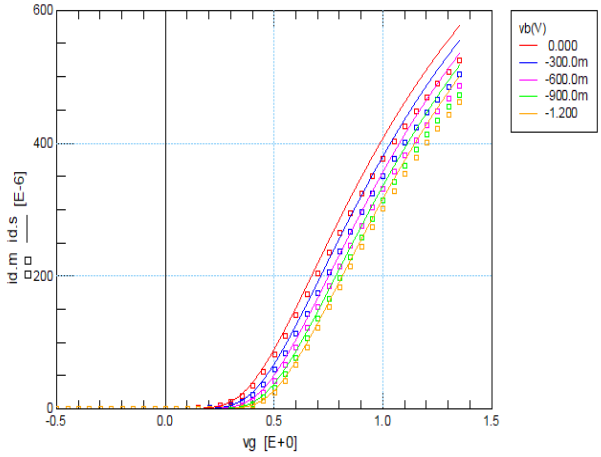




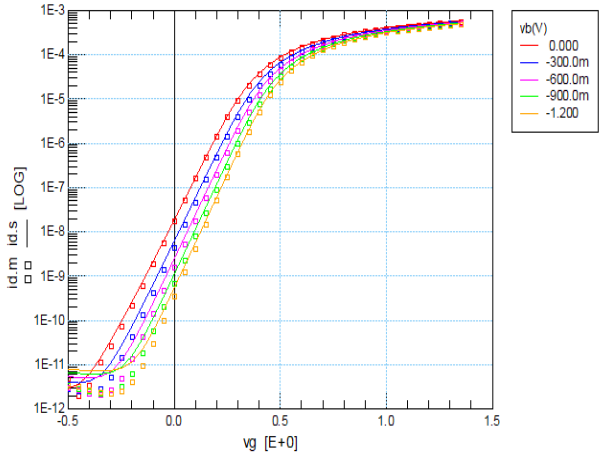


idvg, Vd = 0.05V, T = 125°C

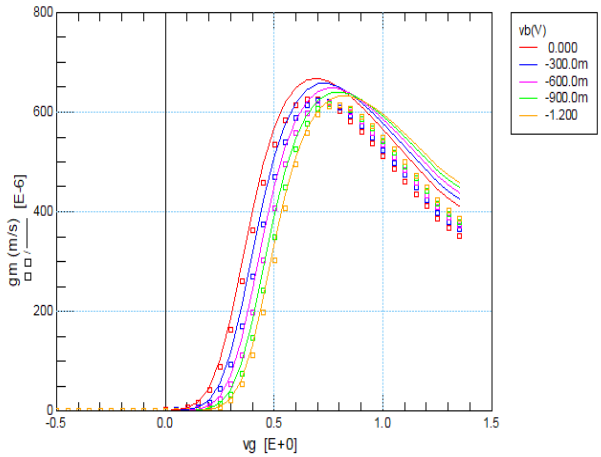
Id = f(Vg, Vb) @ Vd=low



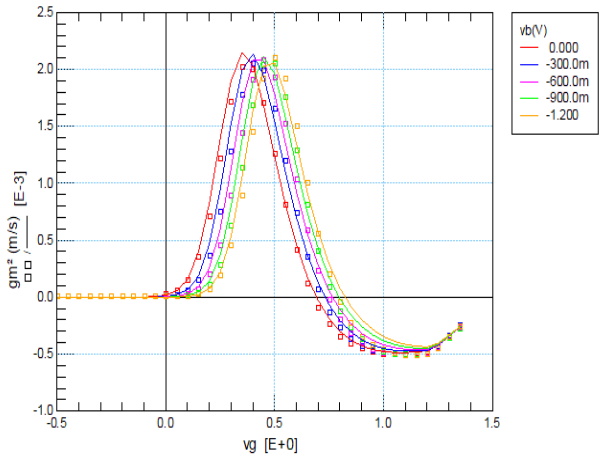
Id = f(Vg, Vb) @ Vd=low



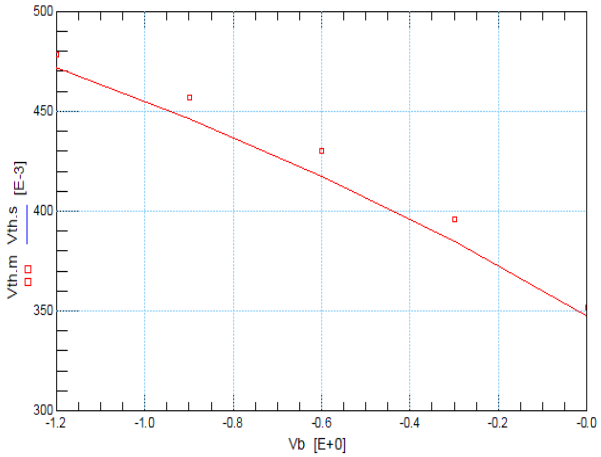
gm = f(Vg, Vb) @ Vd=low



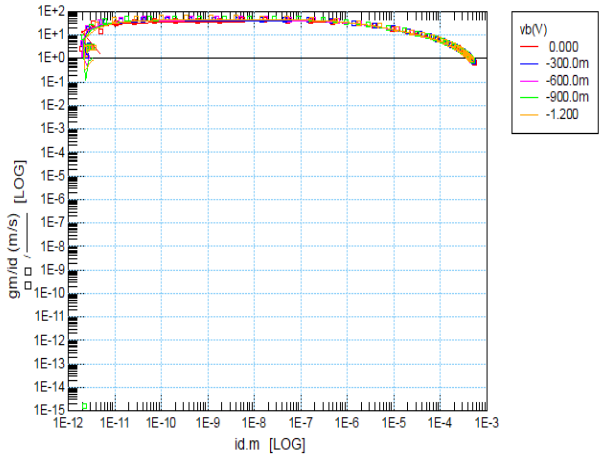
gm² = f(Vg, Vb) @ Vd=low

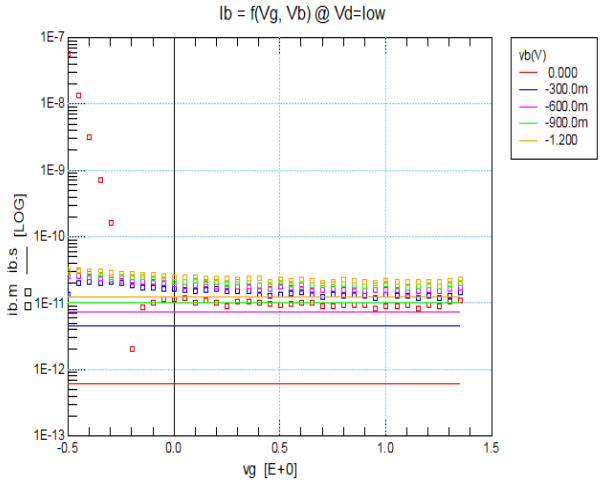


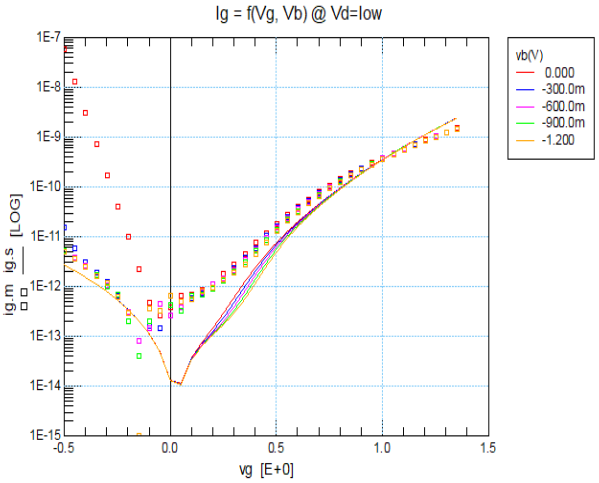
Vth @ Vd=low



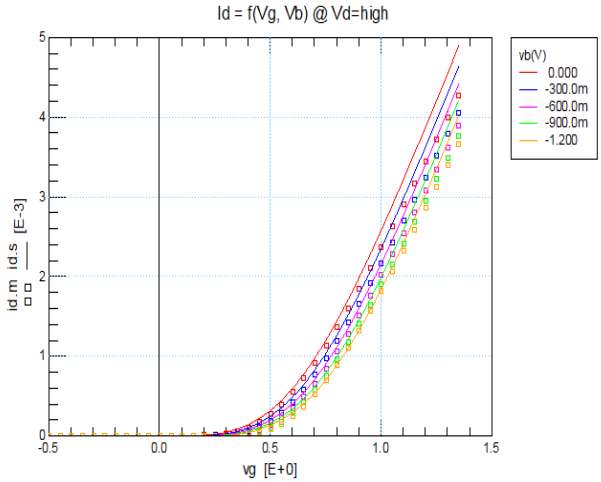
gm/id=f(id.m) @ Vd=low

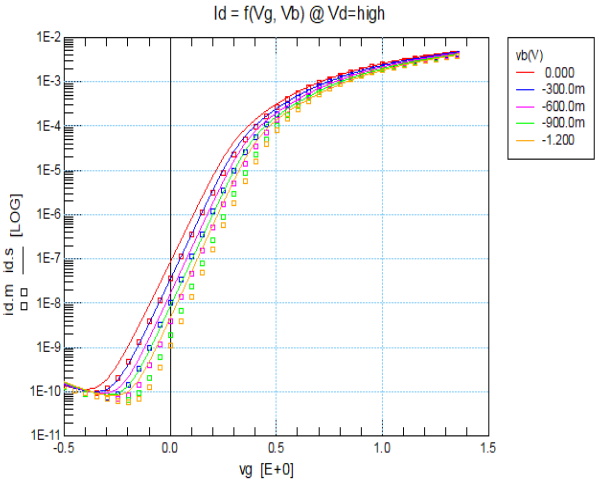


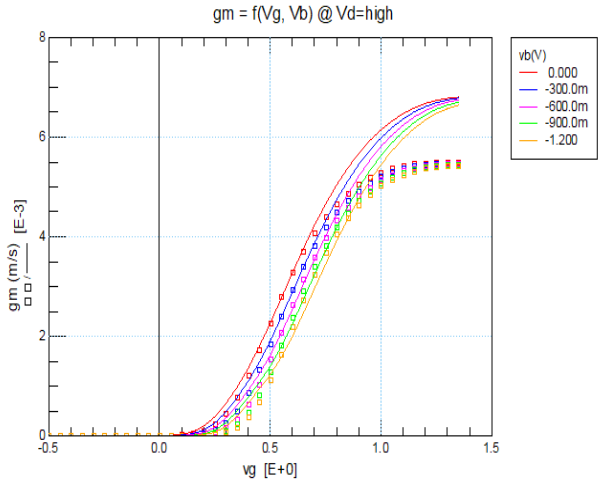


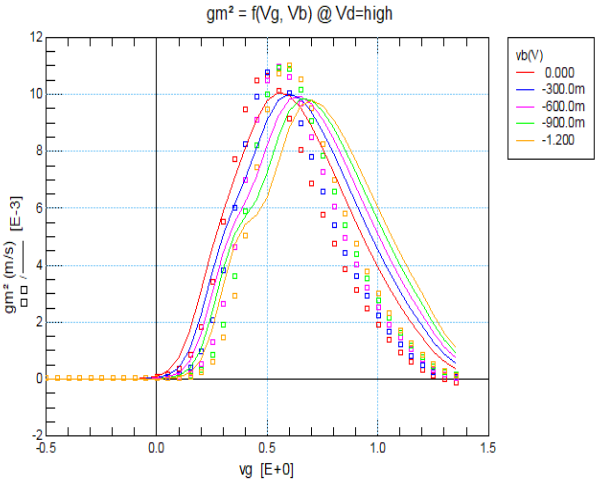


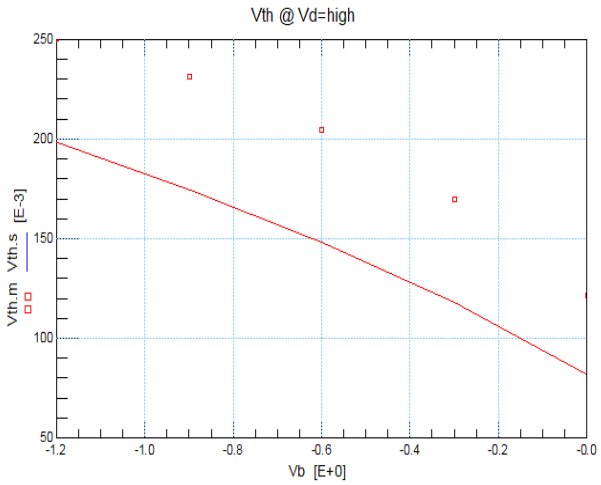
idvg, Vd = 1.2V, T = 125°C

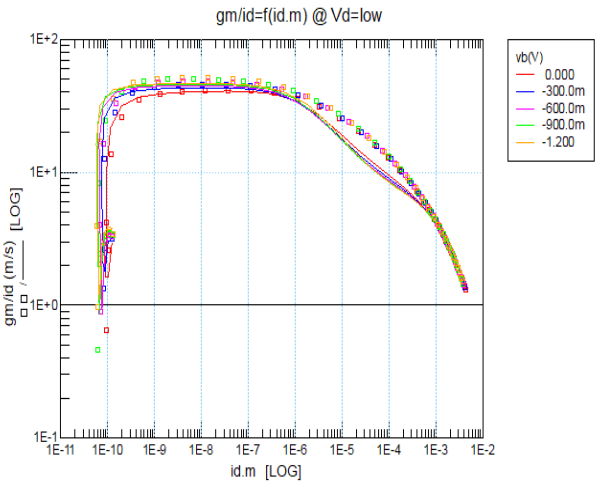


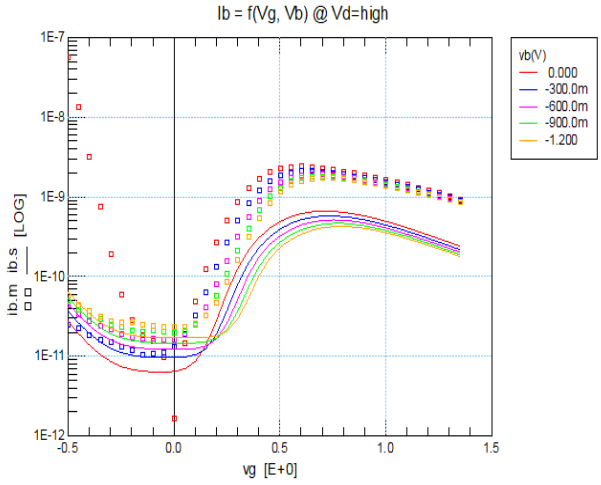


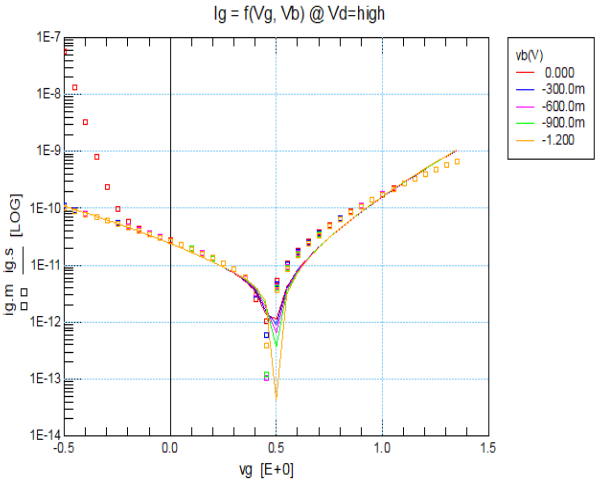




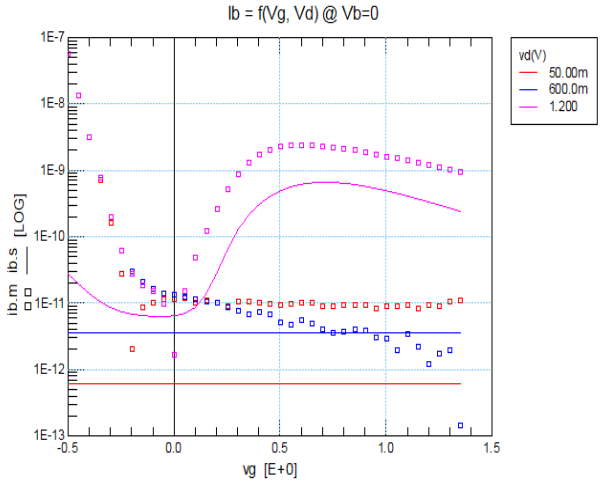


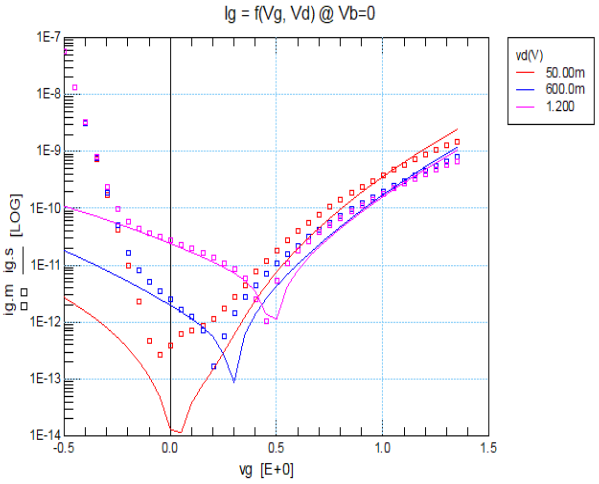




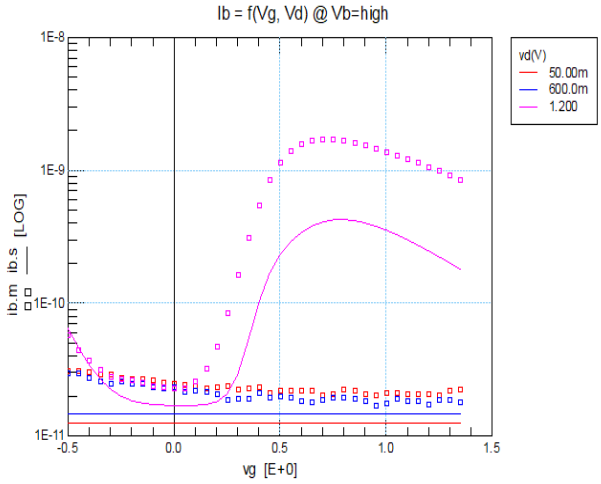


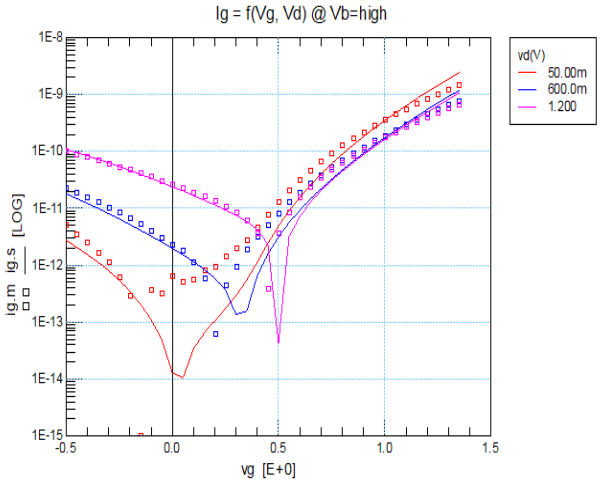
idvg, Ib,Ig, Vb = 0V, T = 125°C



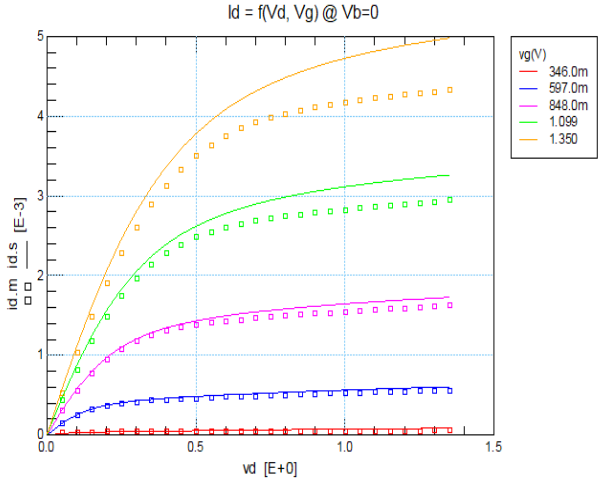


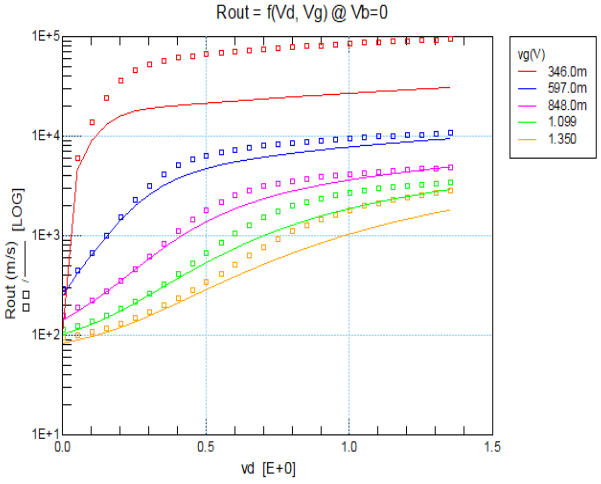
idvg, Ib,Ig, Vb = -1.2V, T = 125°C

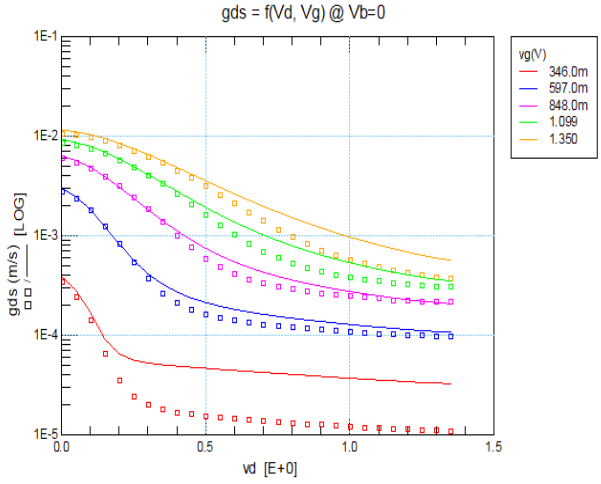




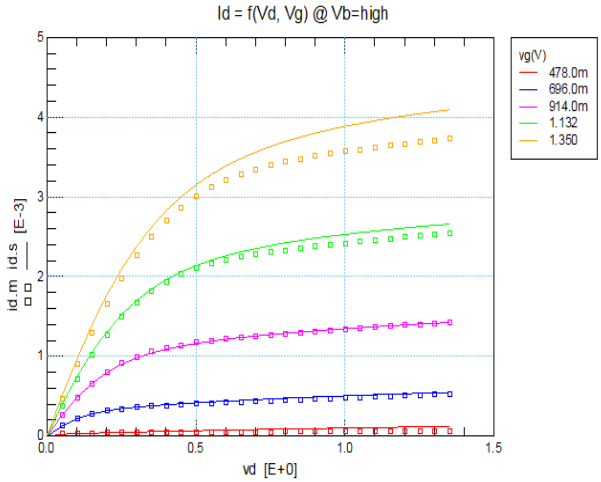
idvd, Vb = 0V, T = 125°C

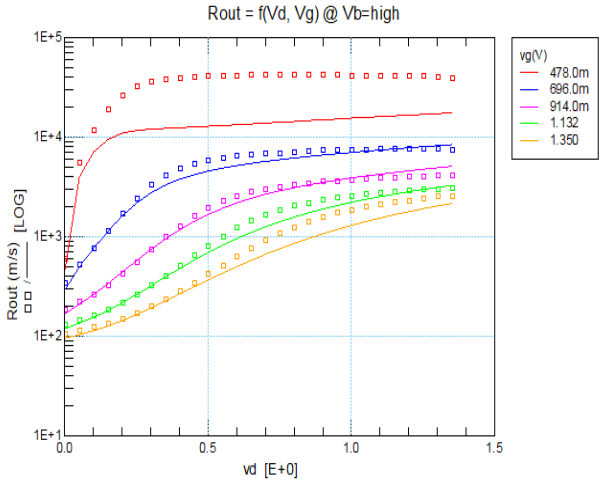


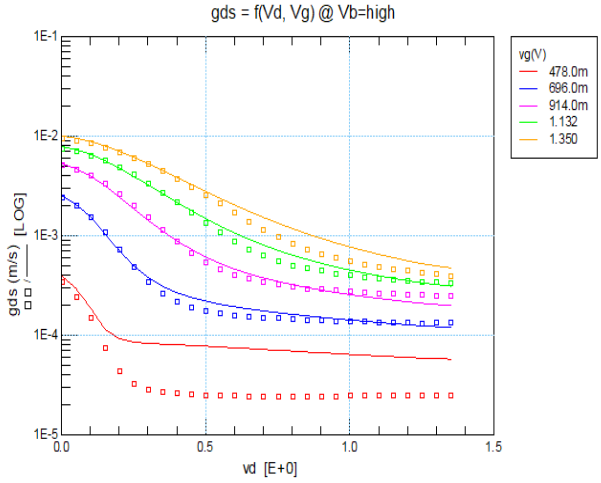




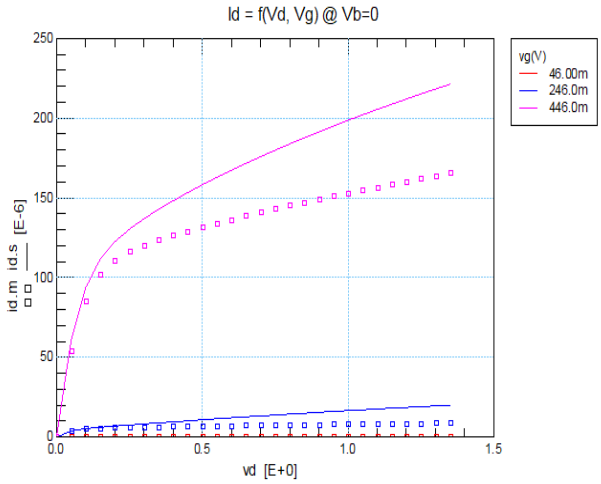
idvd_vbmin, Vb = -1.2V, T = 125°C

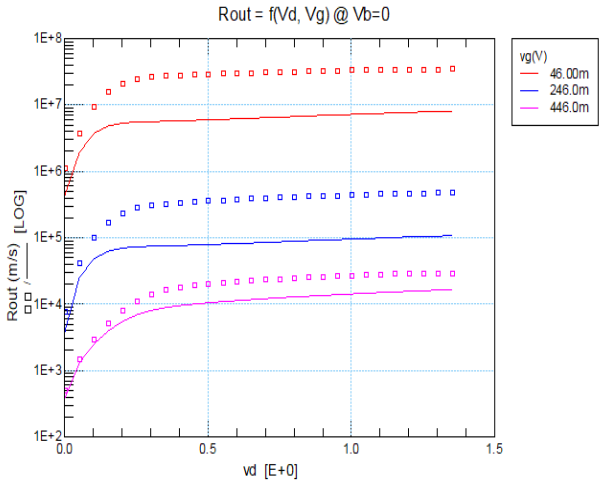


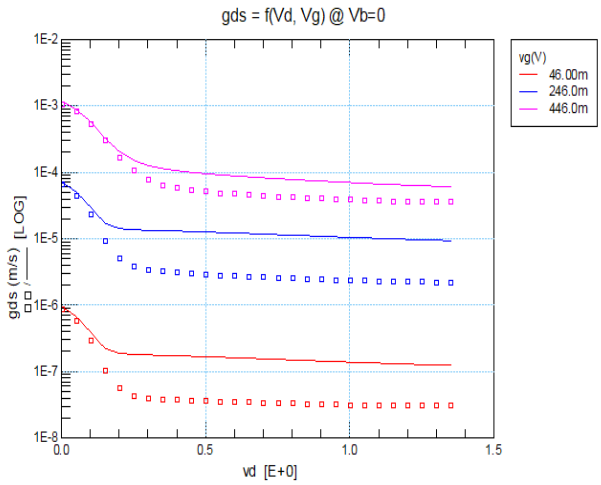




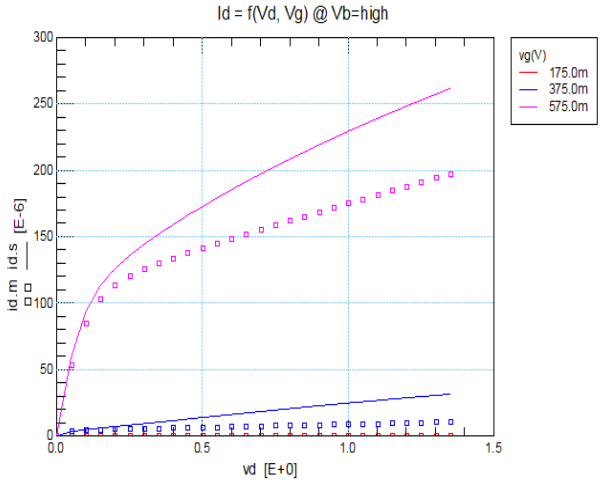
idvd_vth, Vb = 0V, T = 125°C

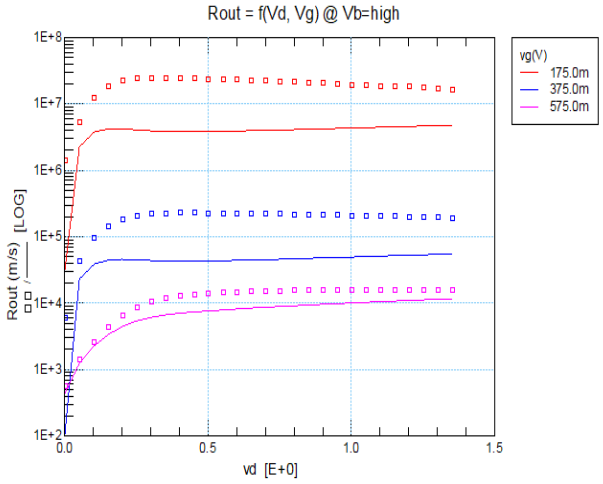


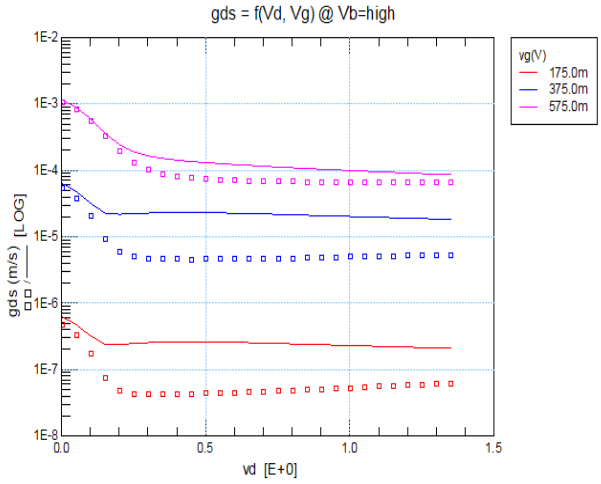




idvd_vth_vbmin, Vb = -1.2V, T = 125°C

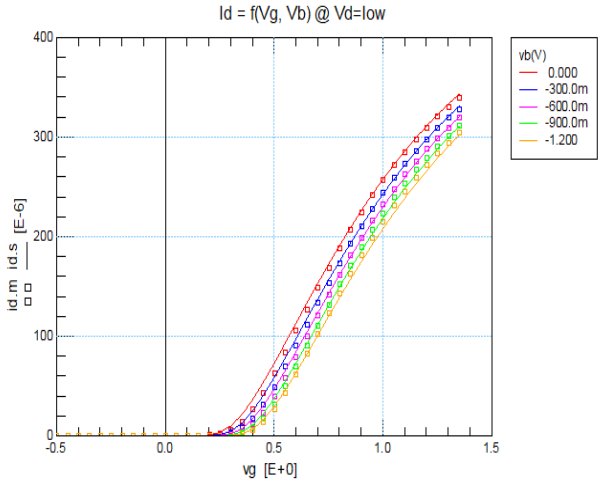


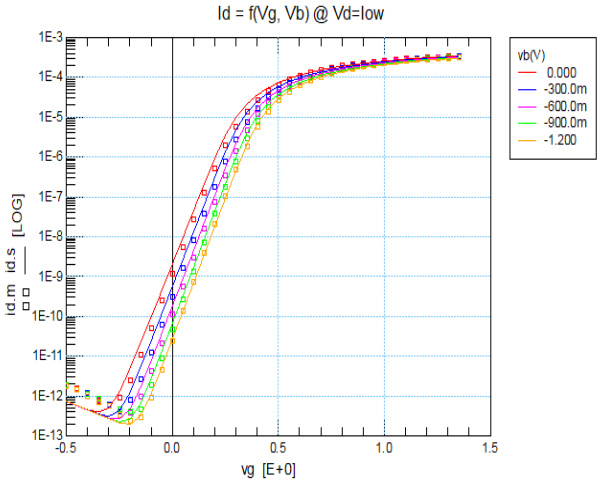


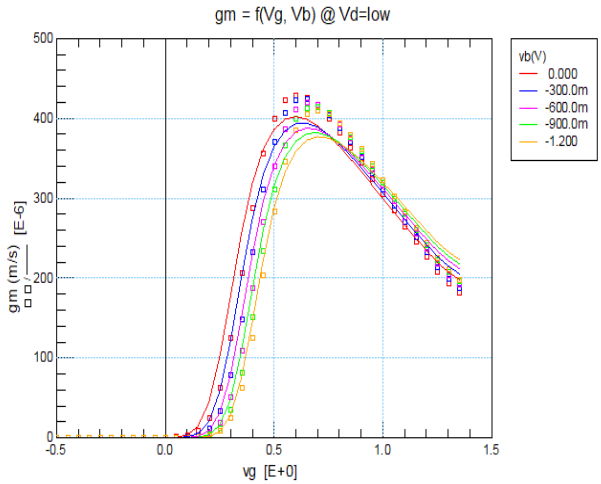


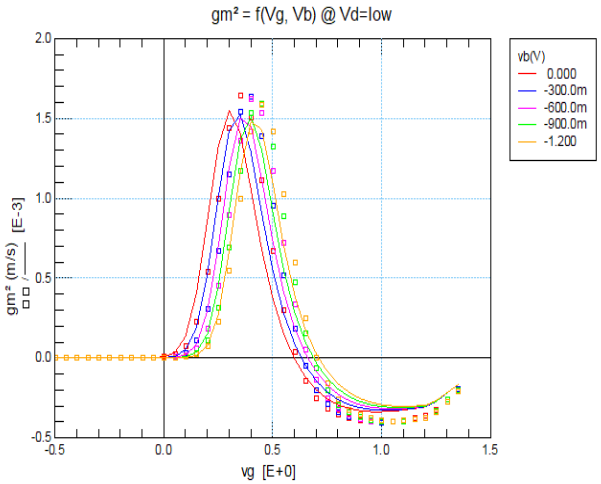
4.9 W10u0_L0u5_S541_1

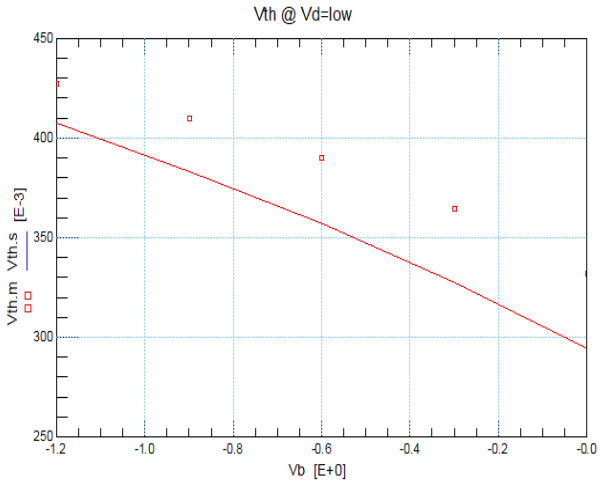
idvg, Vd = 0.05V, T = 27°C

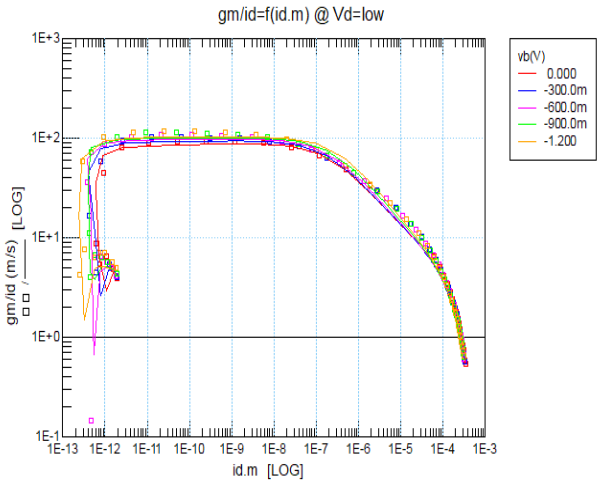


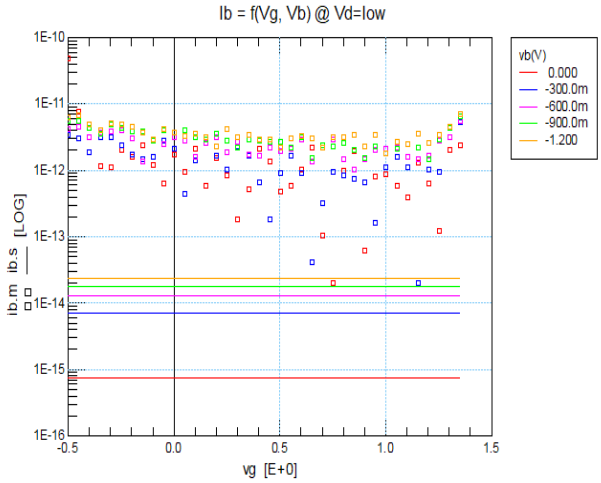


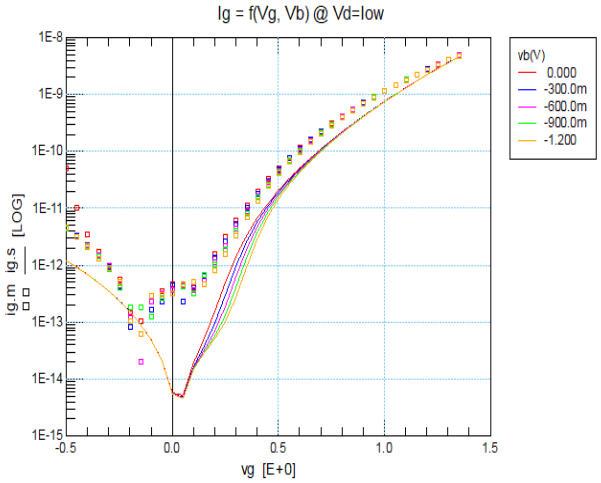




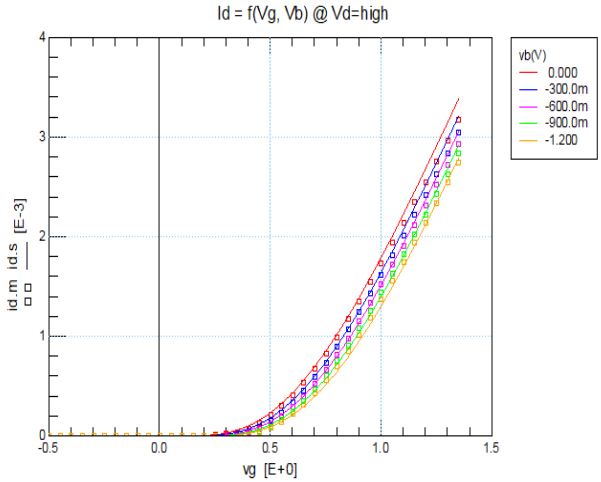


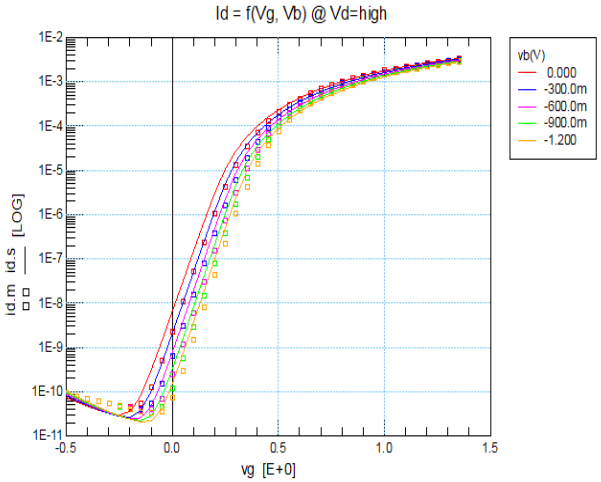


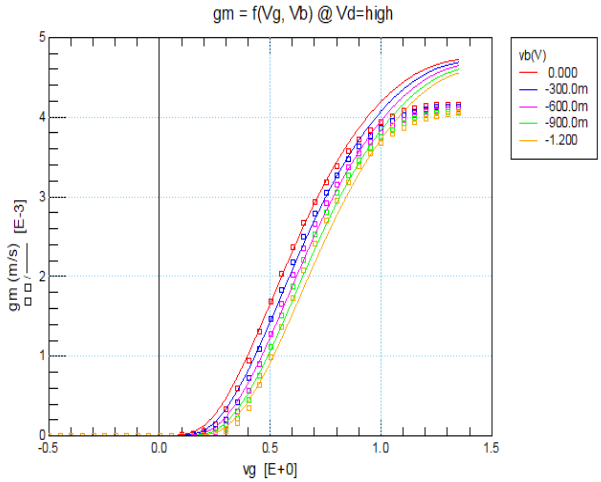


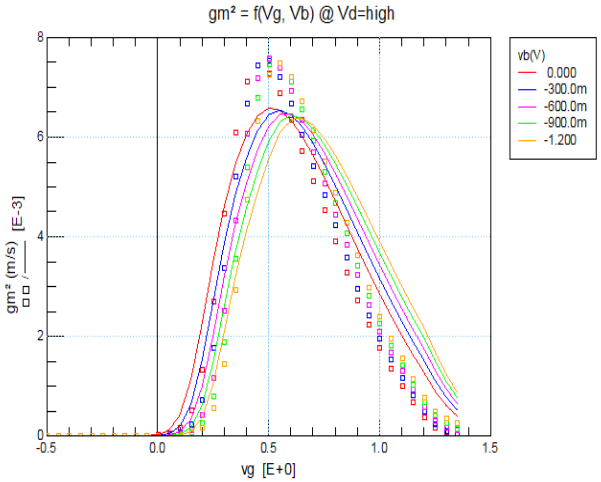


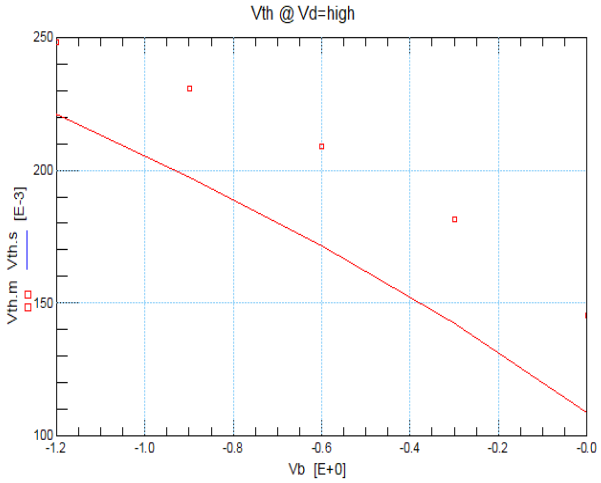
idvg, Vd = 1.2V, T = 27°C

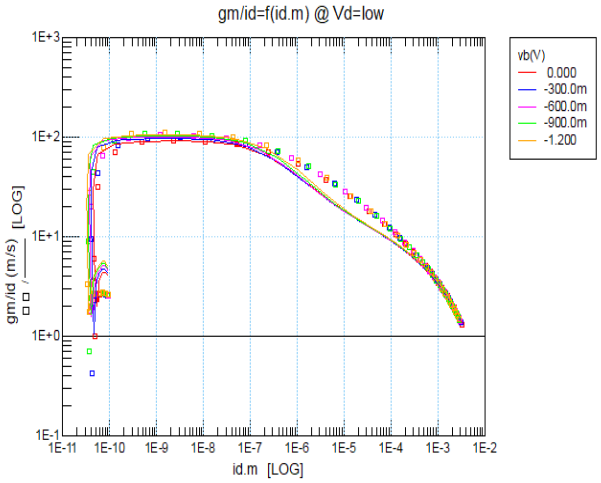


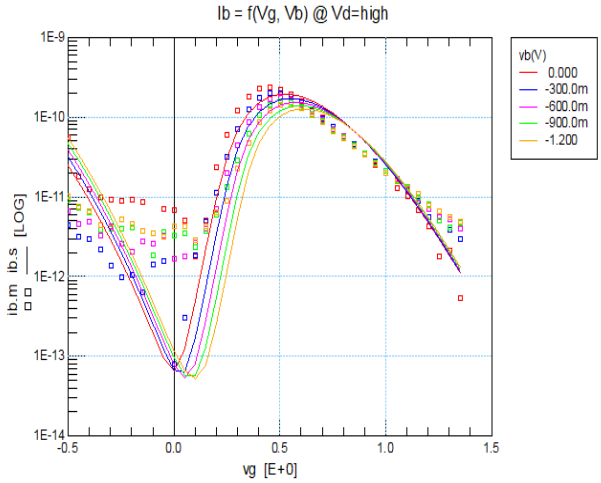


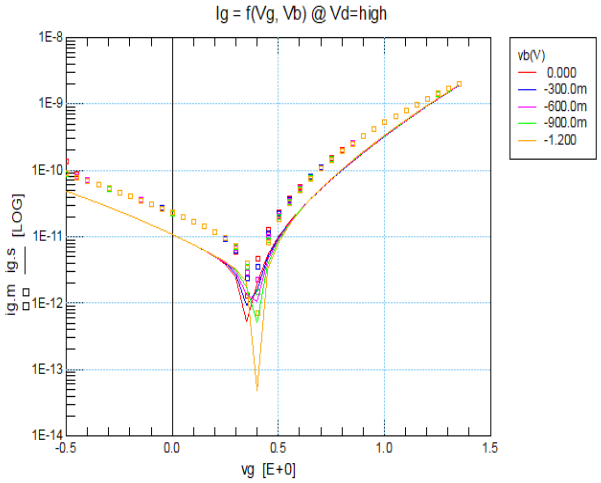




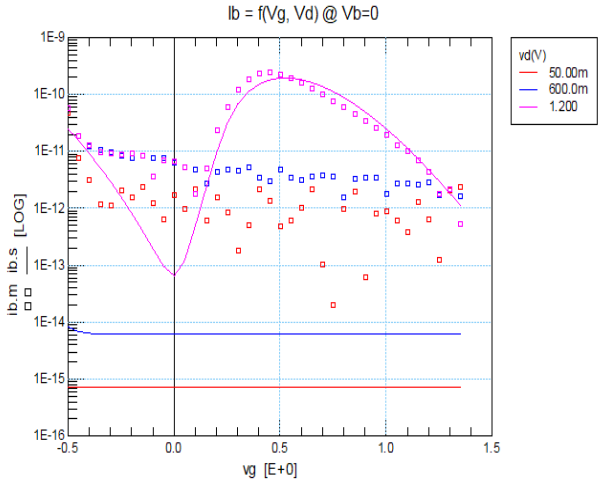


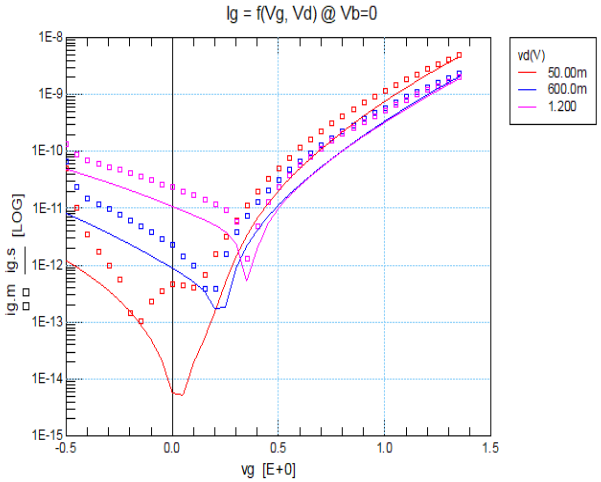




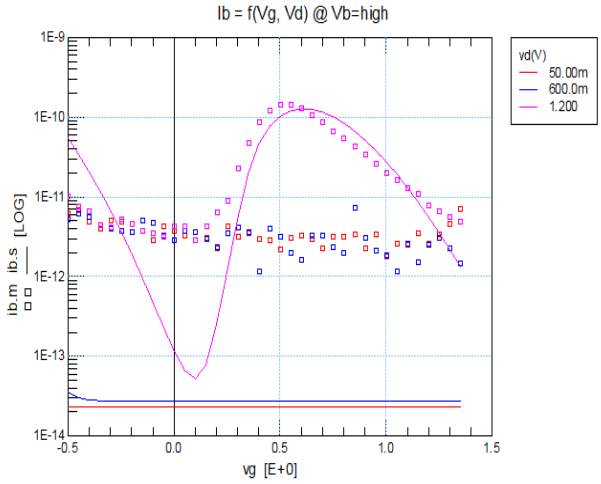


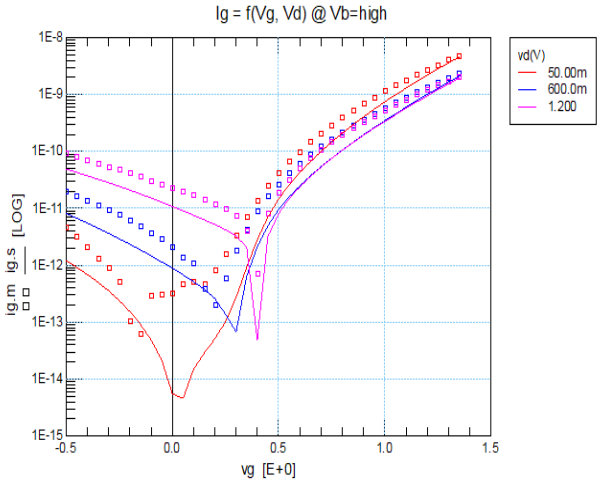
idvg, Ib,Ig, Vb = 0V, T = 27°C



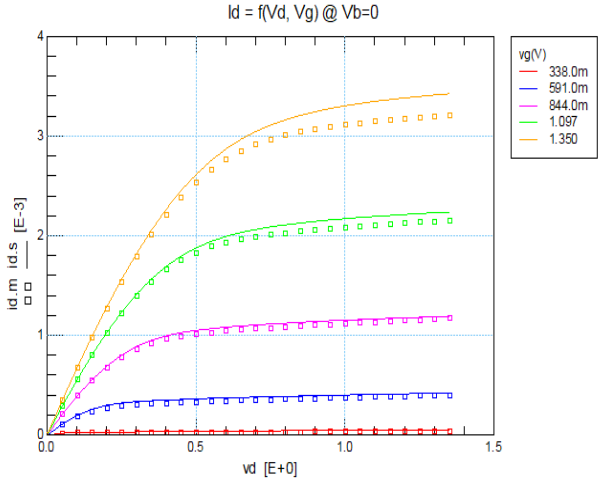


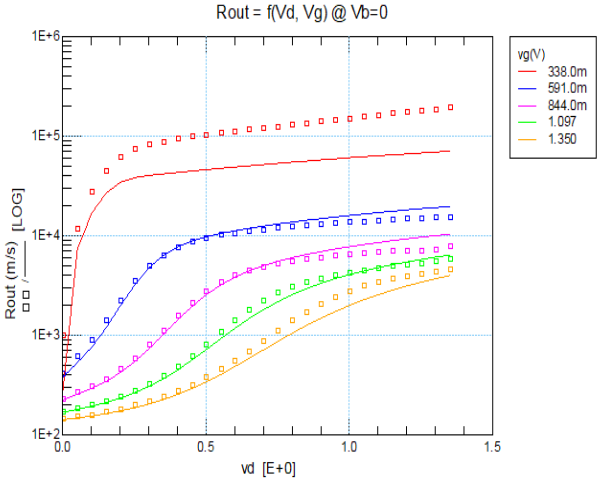
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

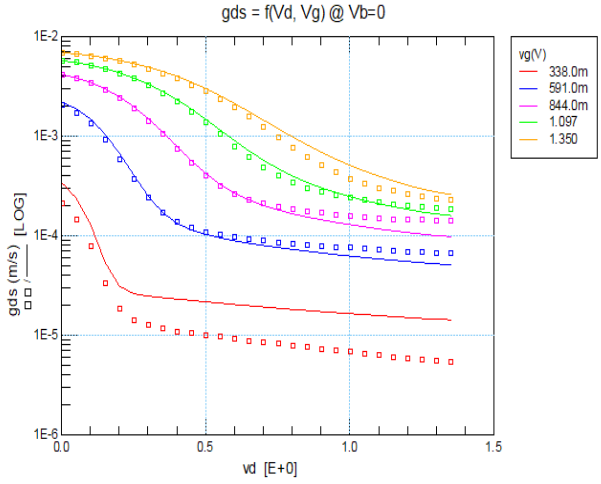




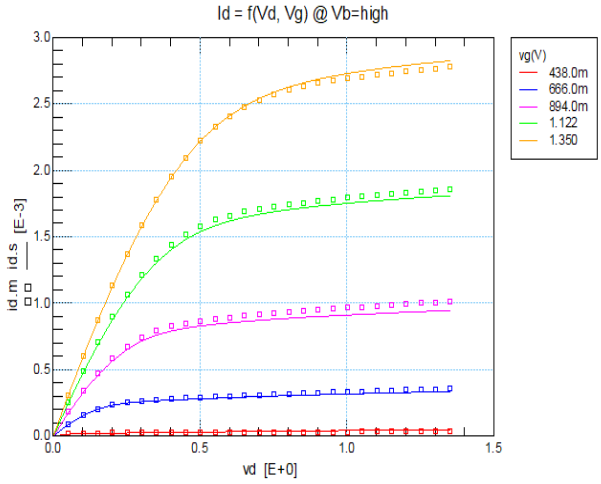
idvd, Vb = 0V, T = 27°C

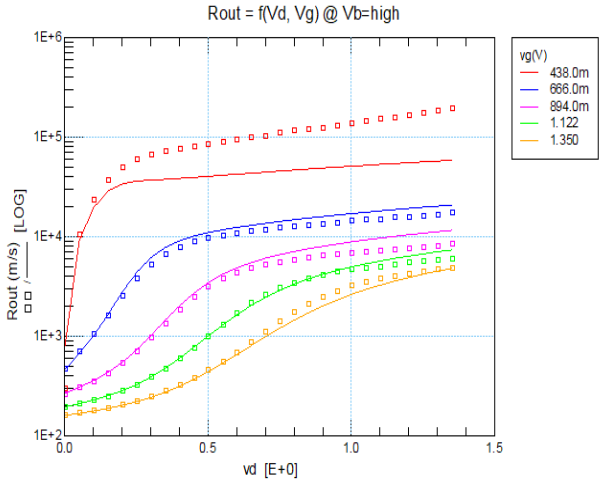


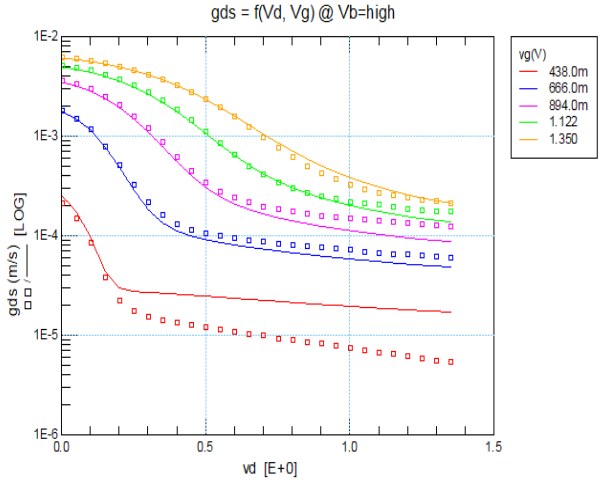




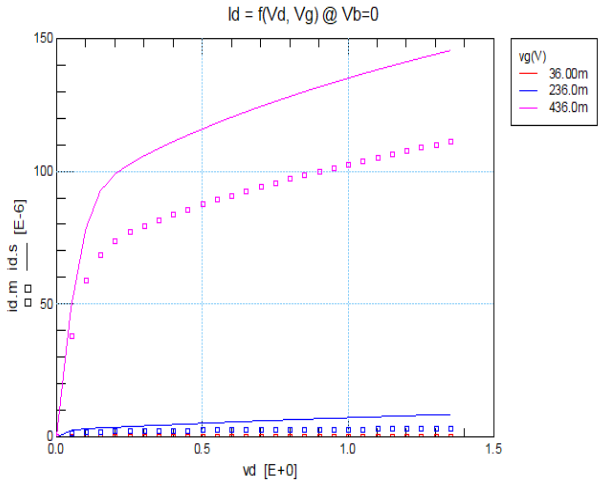
idvd_vbmin, Vb = -1.2V, T = 27°C

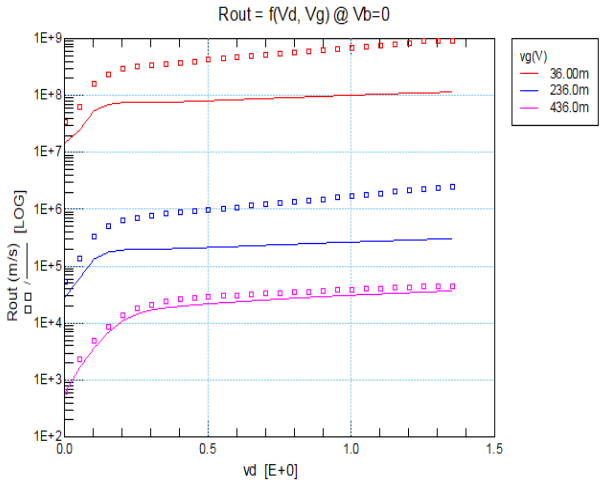


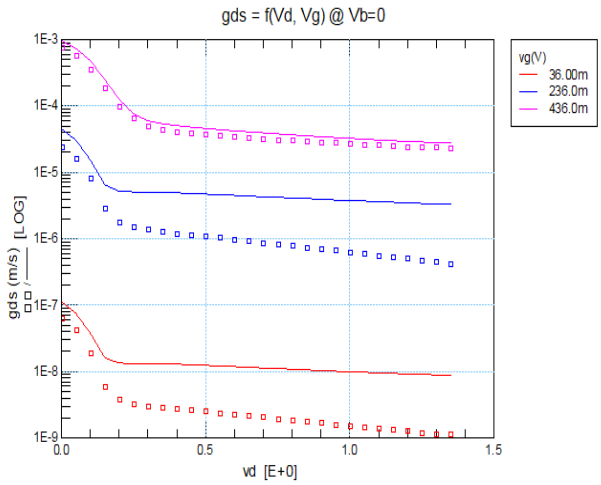




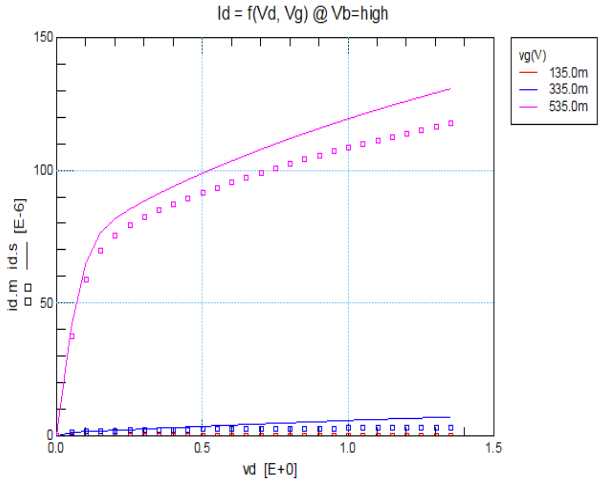
idvd_vth, Vb = 0V, T = 27°C

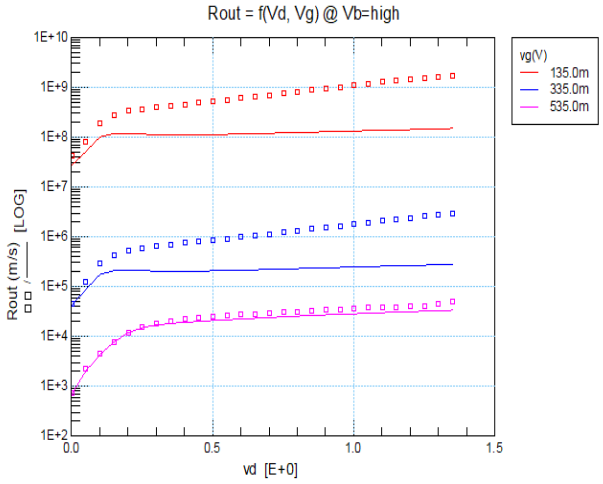


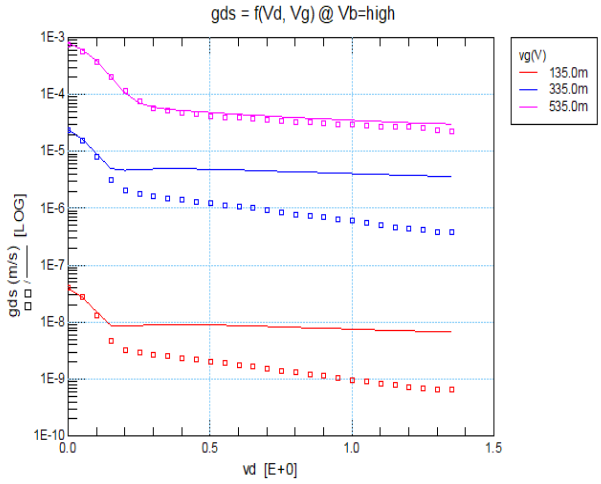




idvd_vth_vbmin, Vb = -1.2V, T = 27°C



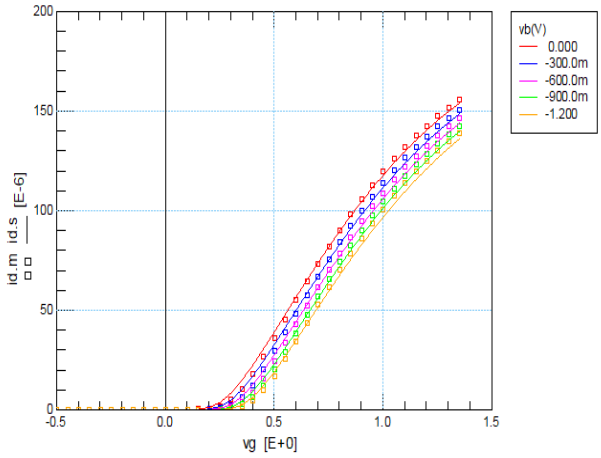




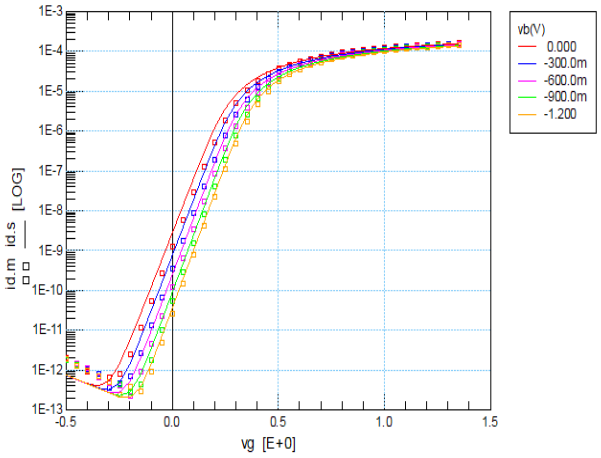
4.10W10u0_L1u2_S541_2

idvg, Vd = 0.05V, T = 27°C

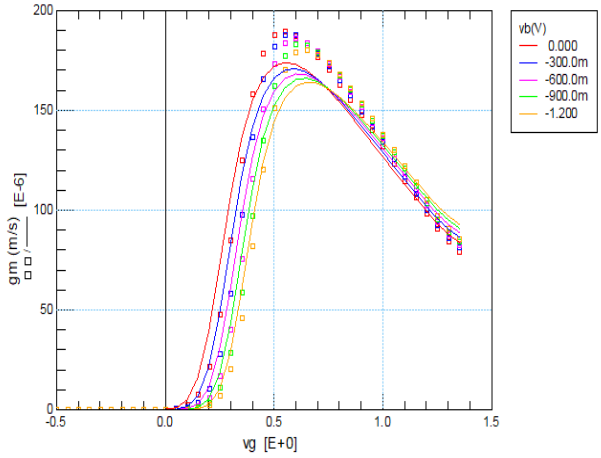
Id = f(Vg, Vb) @ Vd=low



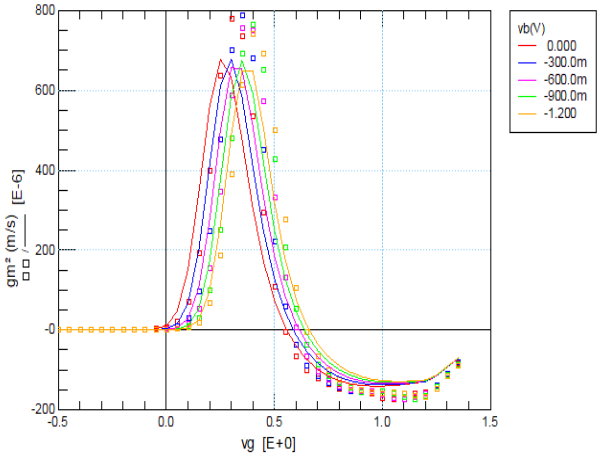
Id = f(Vg, Vb) @ Vd=low



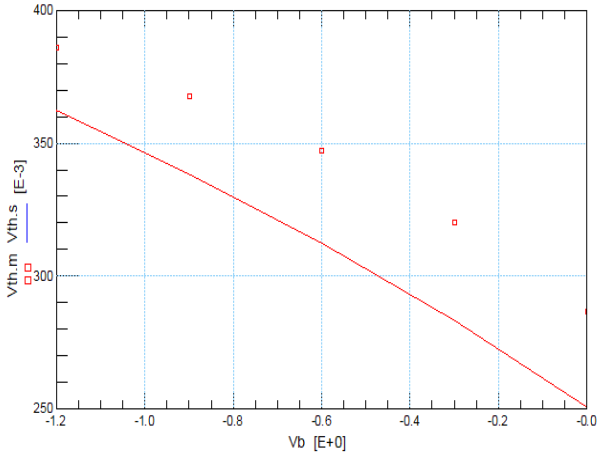
gm = f(Vg, Vb) @ Vd=low



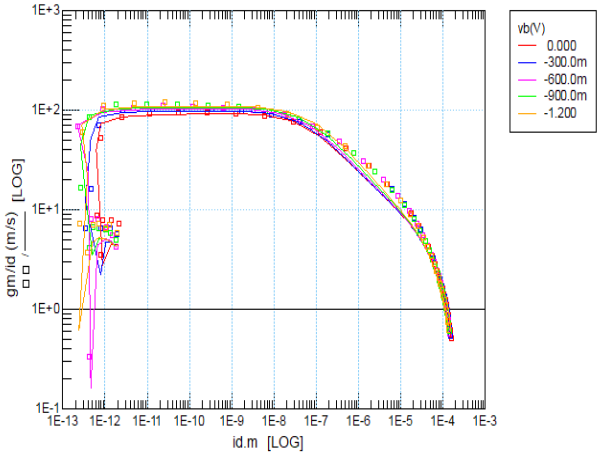
gm² = f(Vg, Vb) @ Vd=low

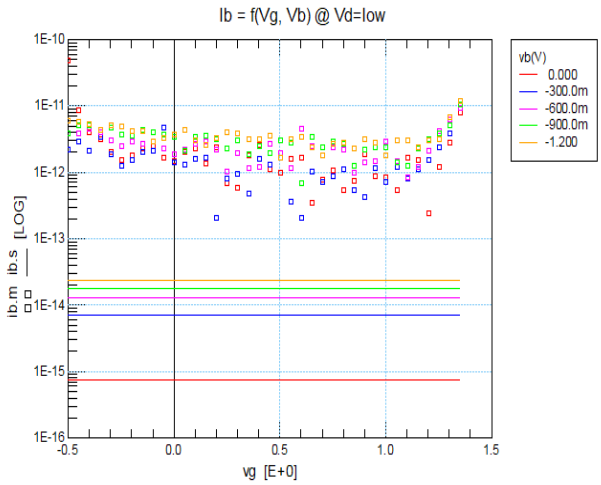


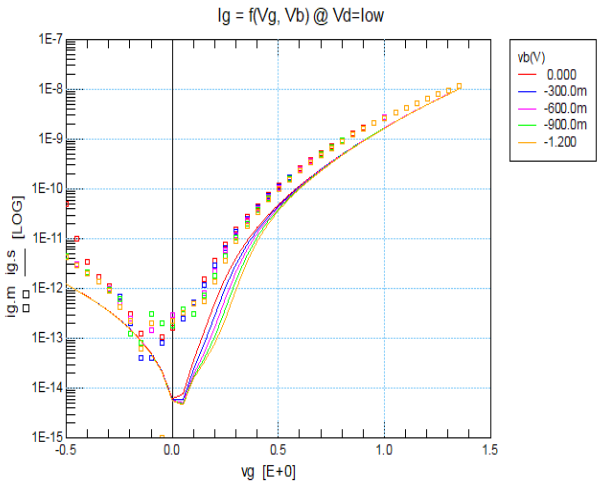
Vth @ Vd=low



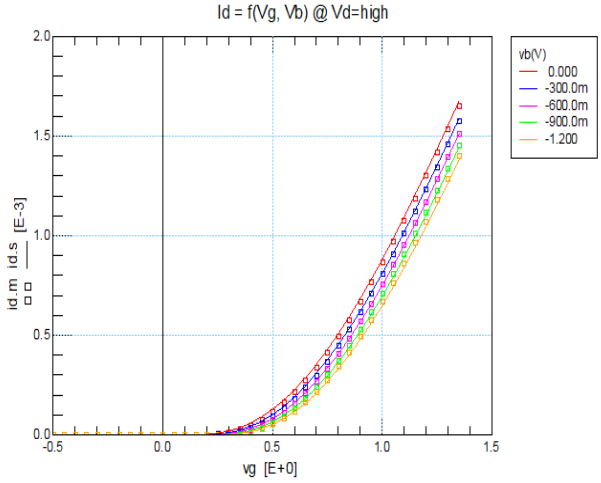
gm/id=f(id.m) @ Vd=low

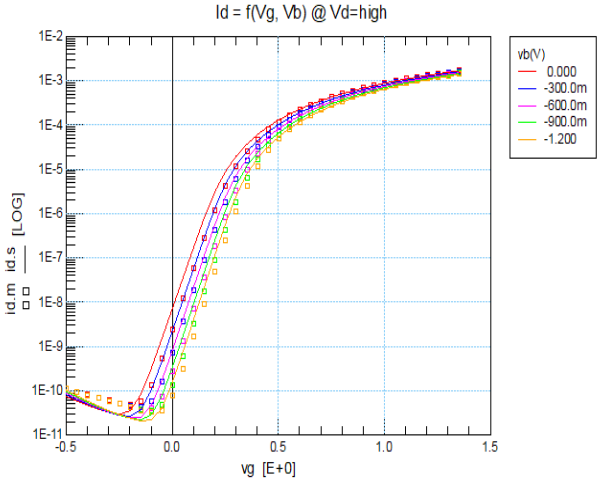


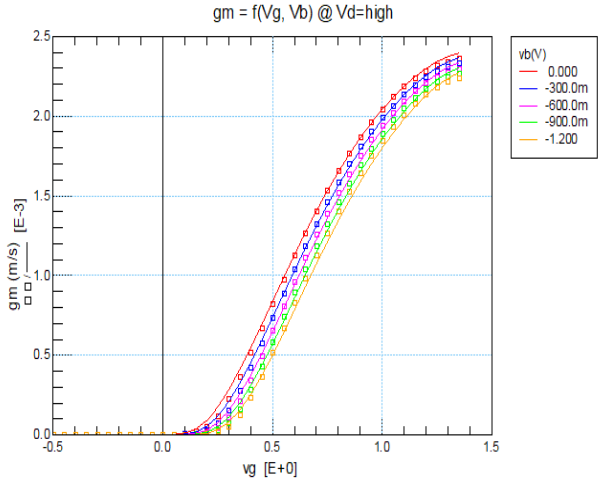


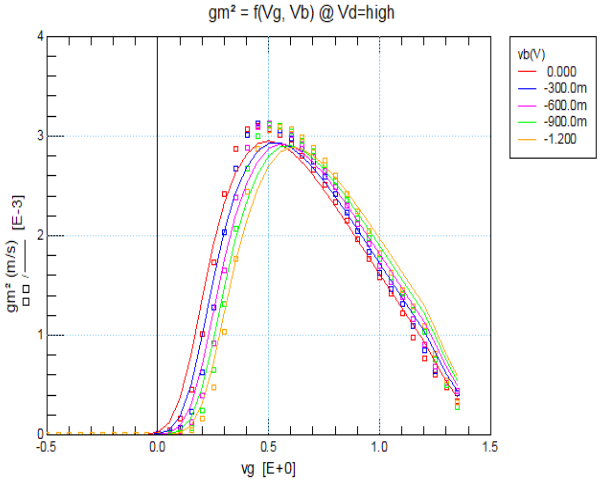


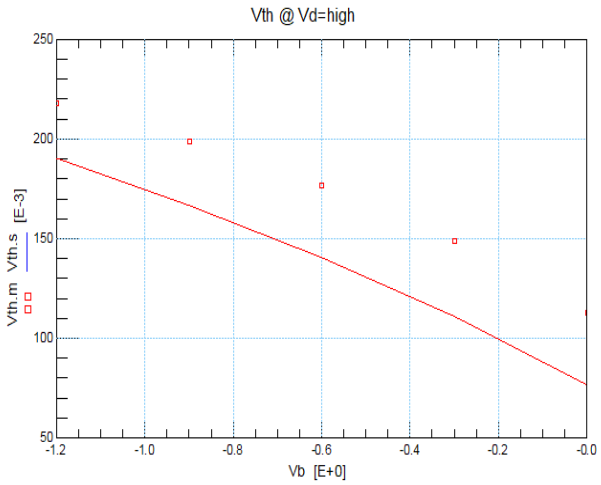
idvg, Vd = 1.2V, T = 27°C

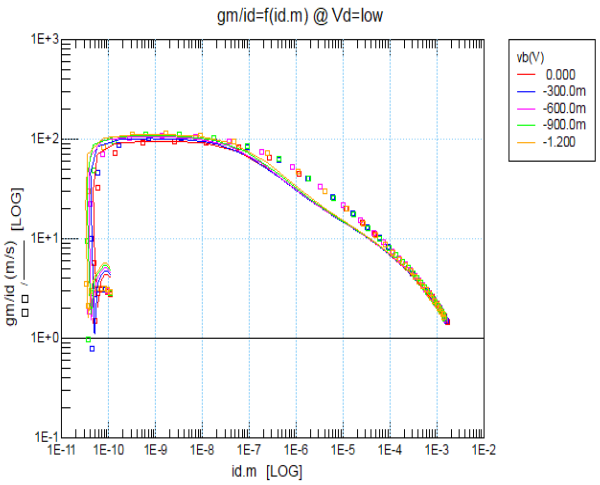


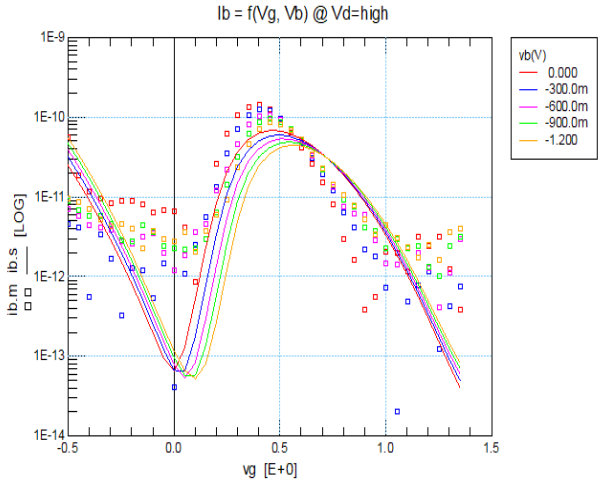


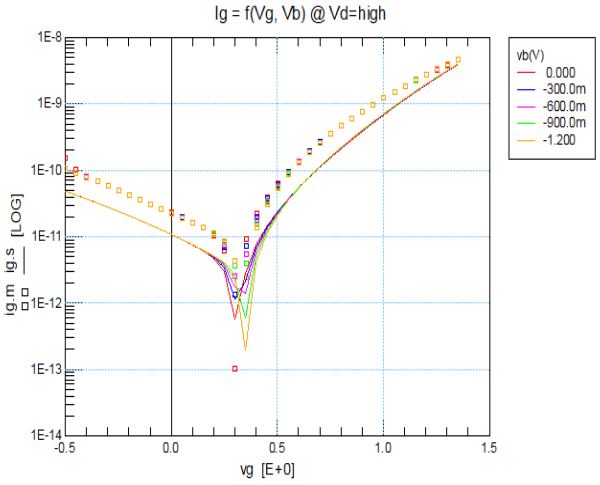




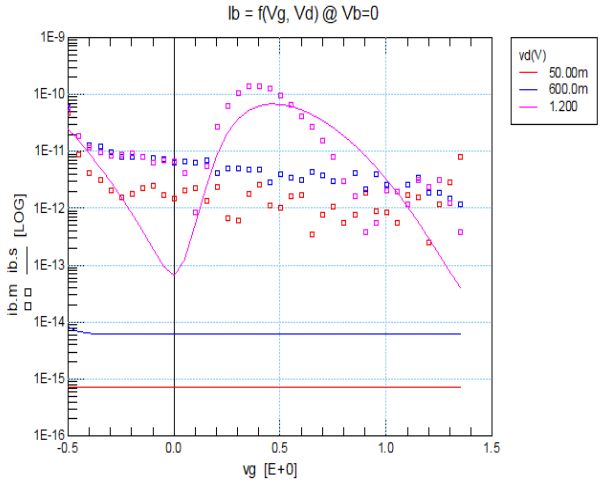


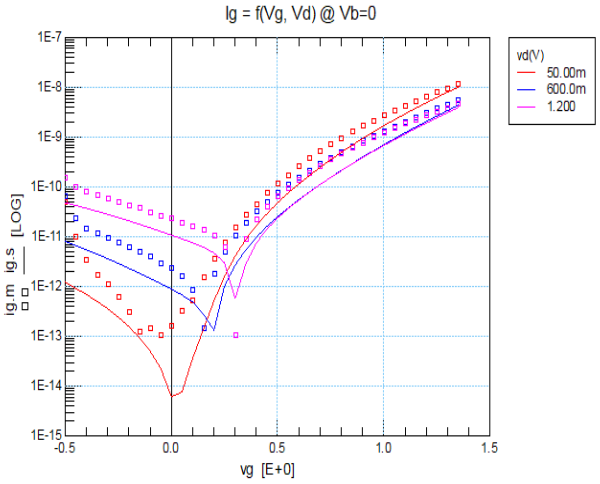




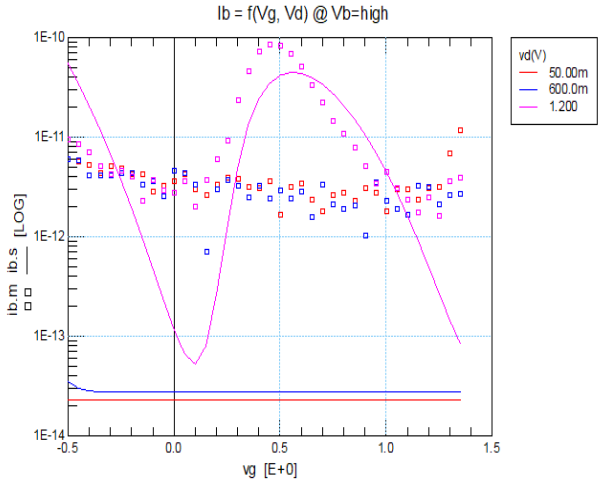


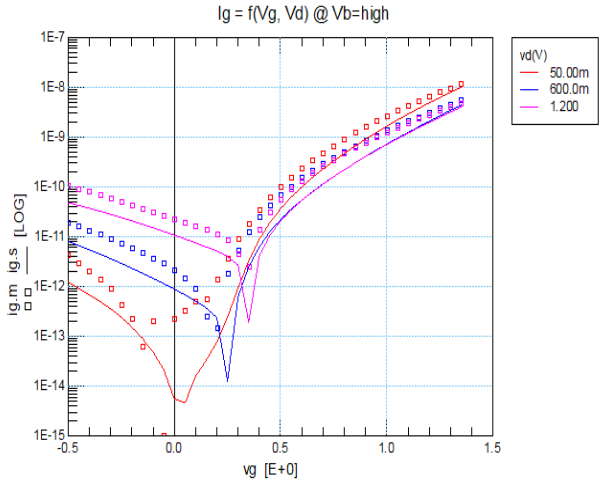
idvg, Ib,Ig, Vb = 0V, T = 27°C



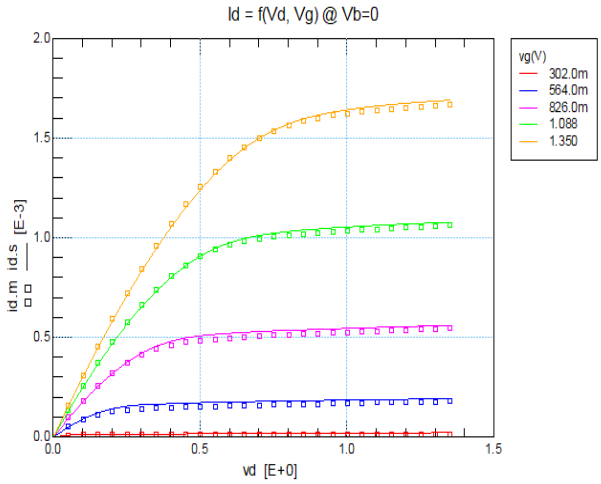


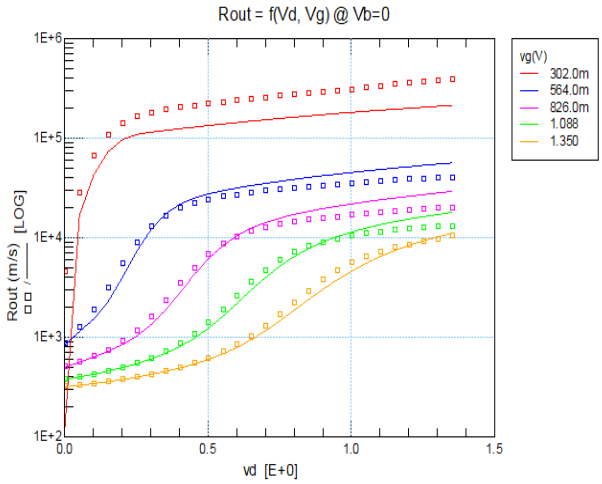
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

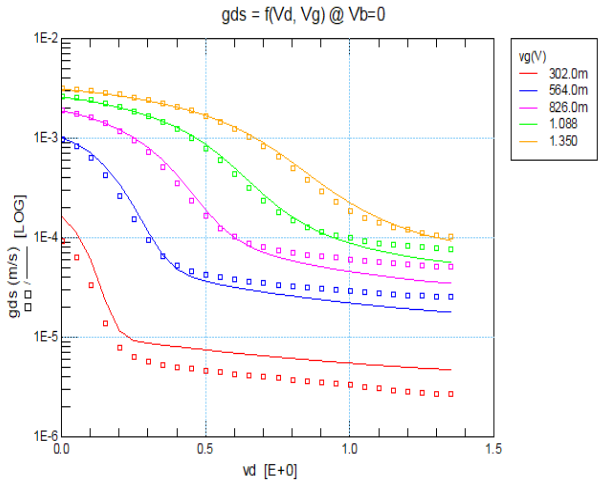




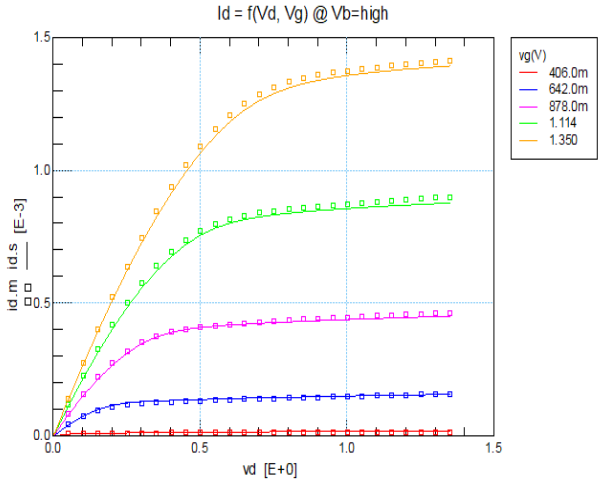
idvd, Vb = 0V, T = 27°C

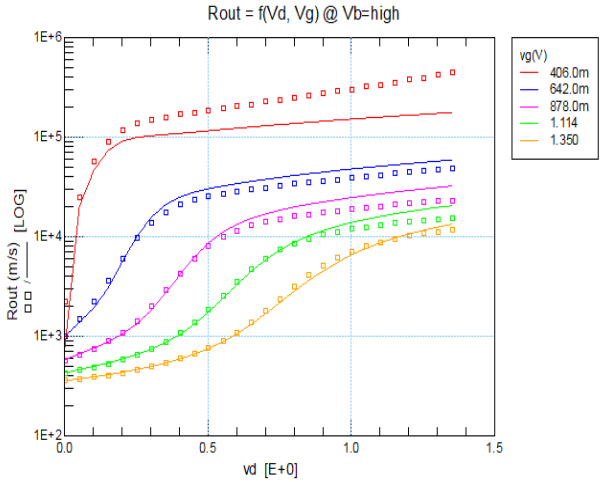


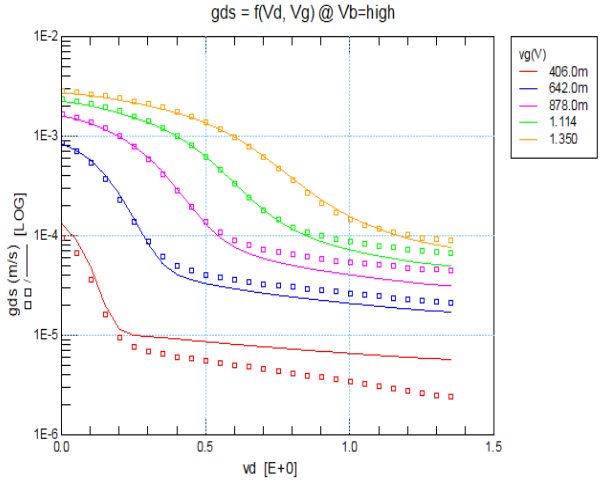




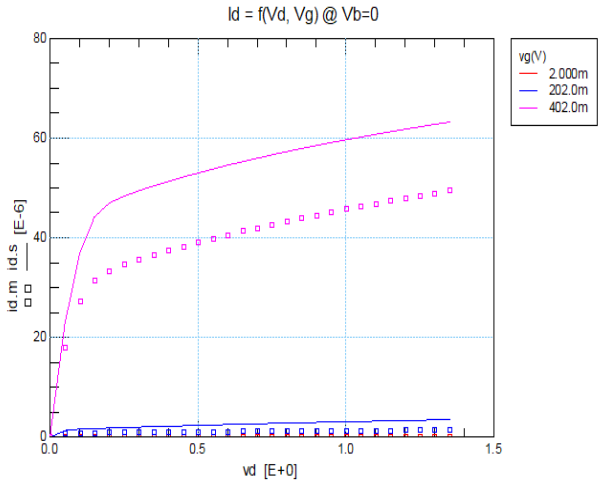
idvd_vbmin, Vb = -1.2V, T = 27°C

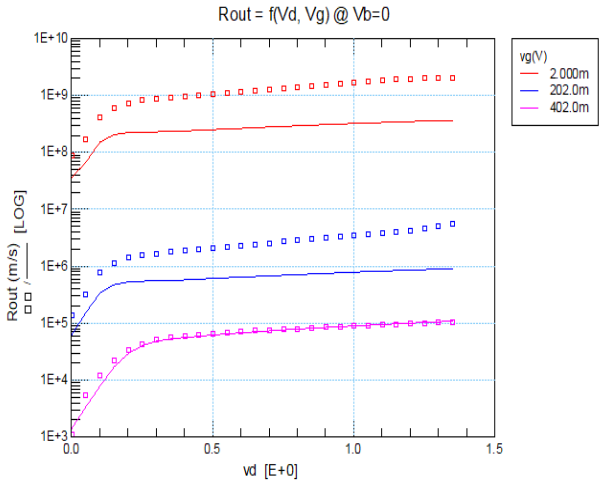


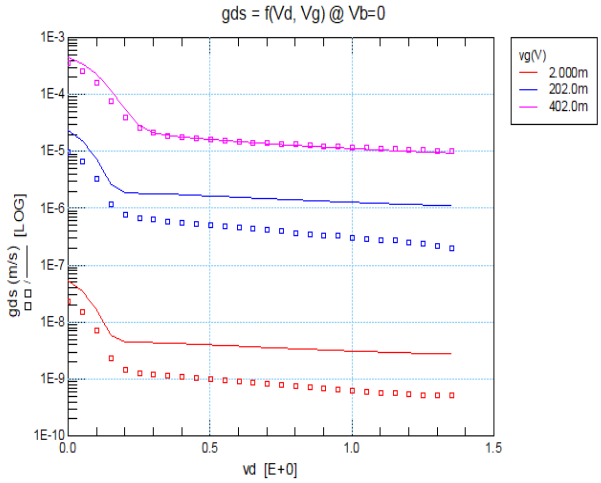




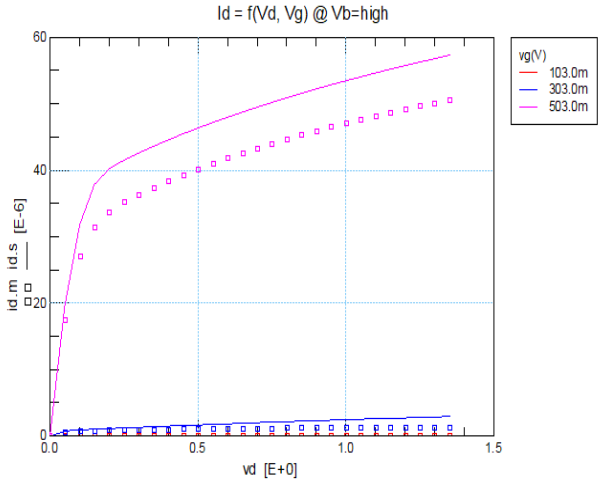
idvd_vth, Vb = 0V, T = 27°C

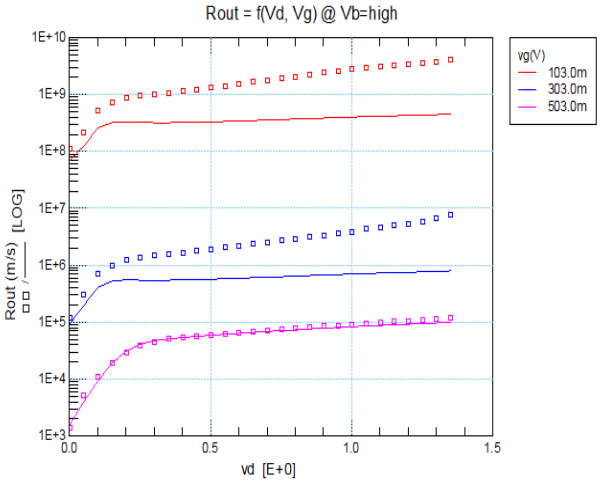


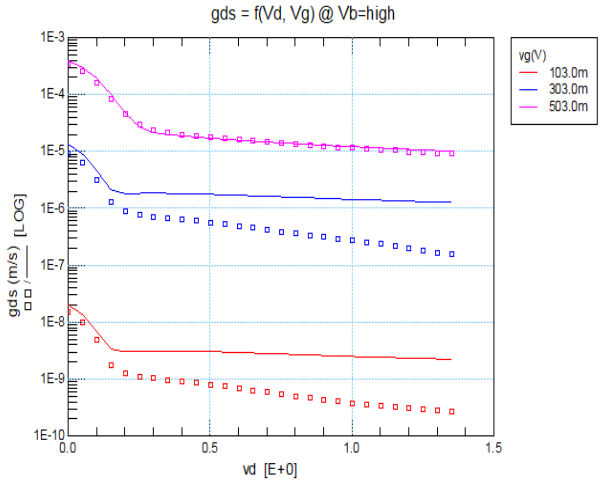




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

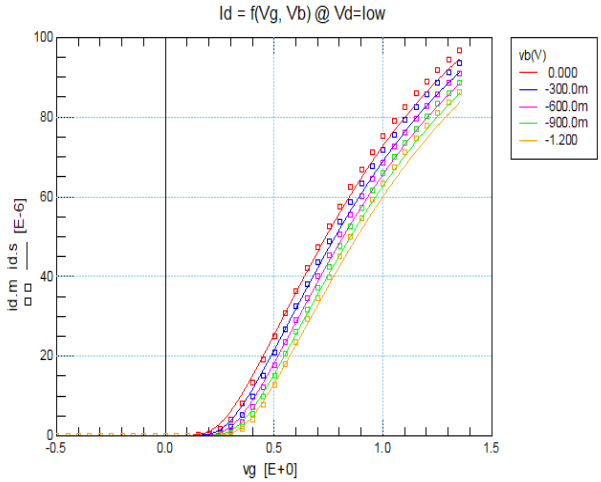


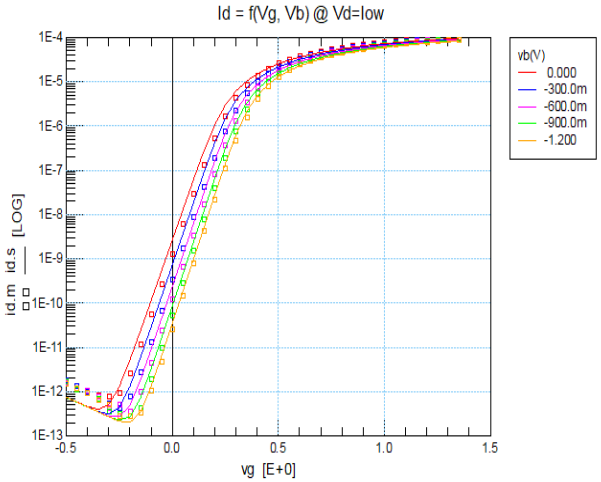


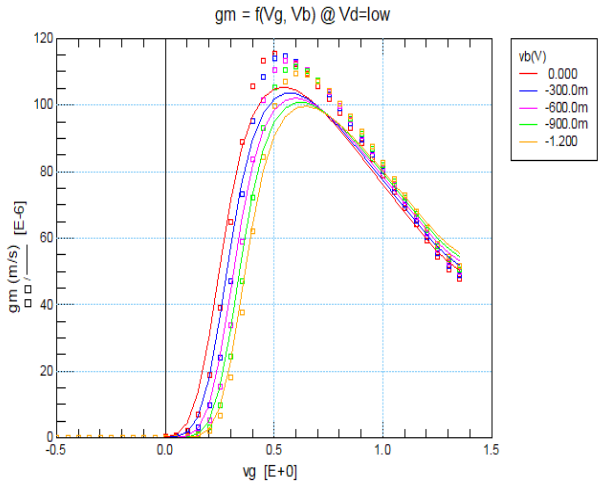


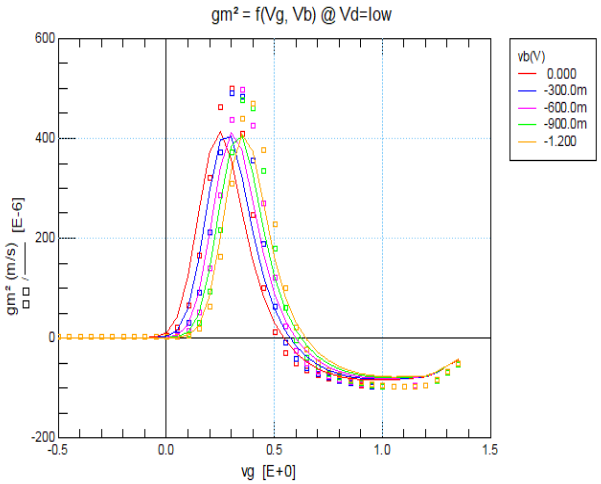
4.11 W10u0_L2u0_S541_3

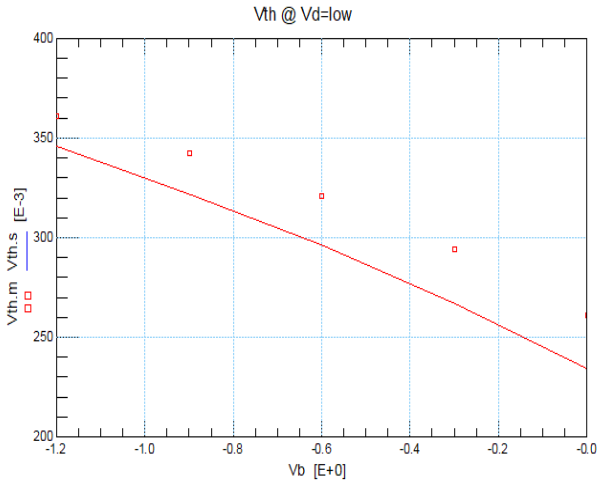
idvg, Vd = 0.05V, T = 27°C

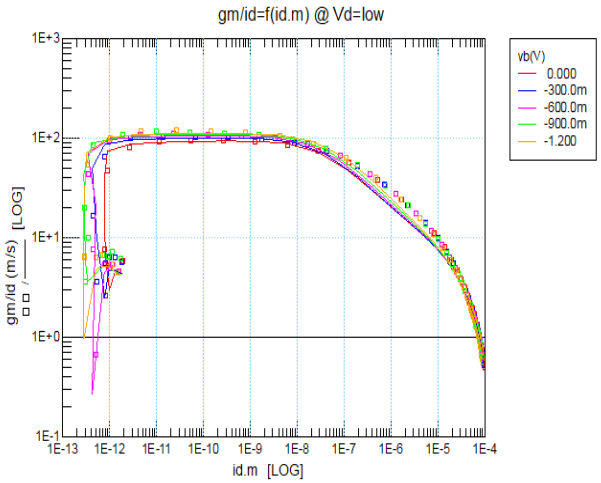


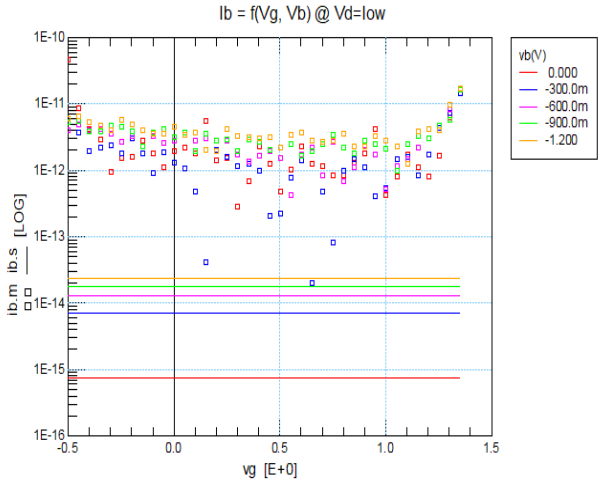


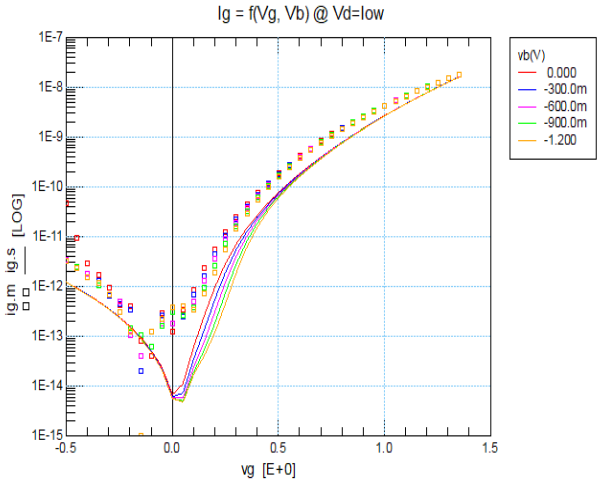




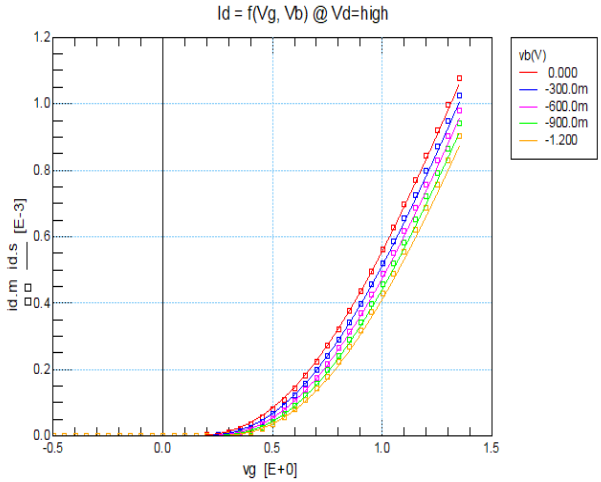


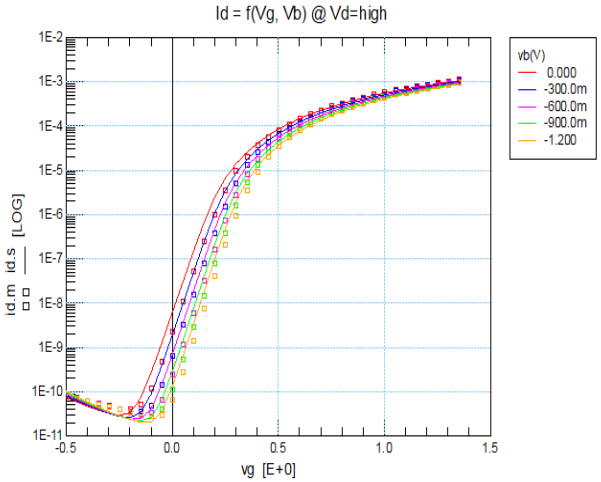


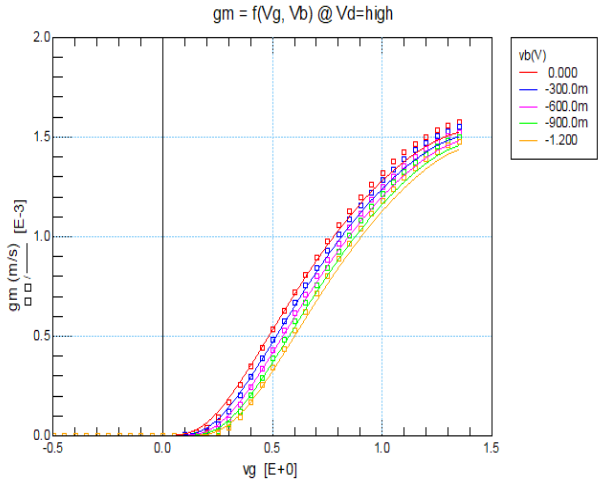


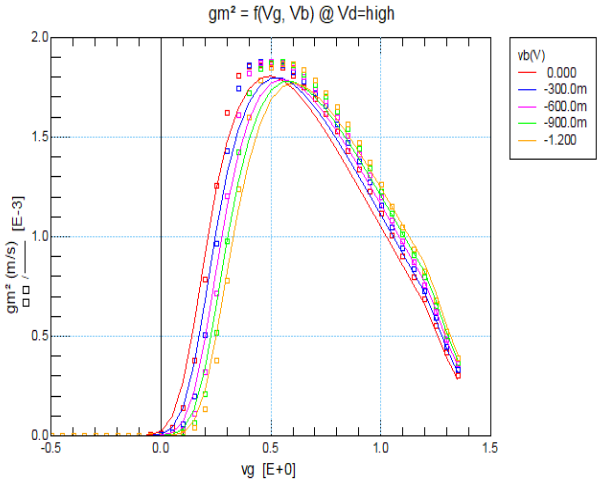


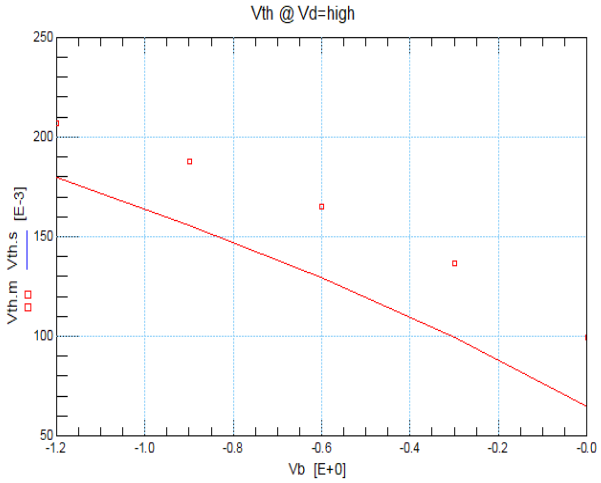
idvg, Vd = 1.2V, T = 27°C

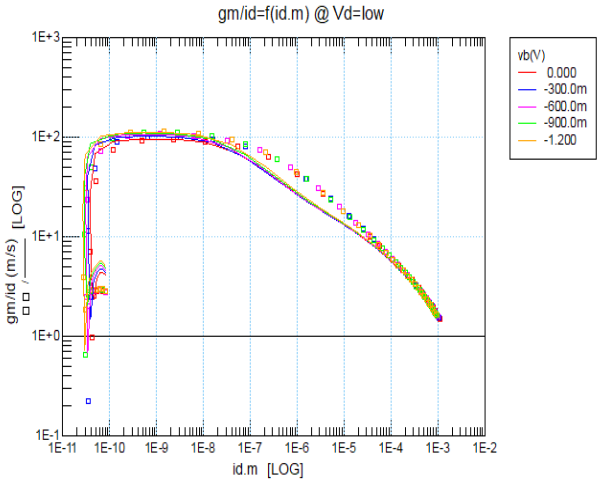


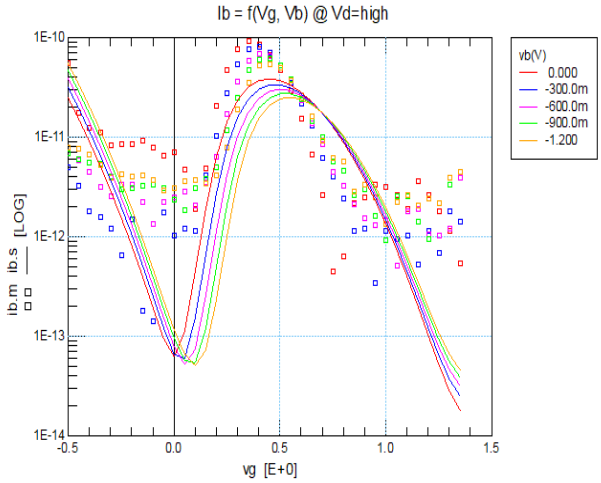


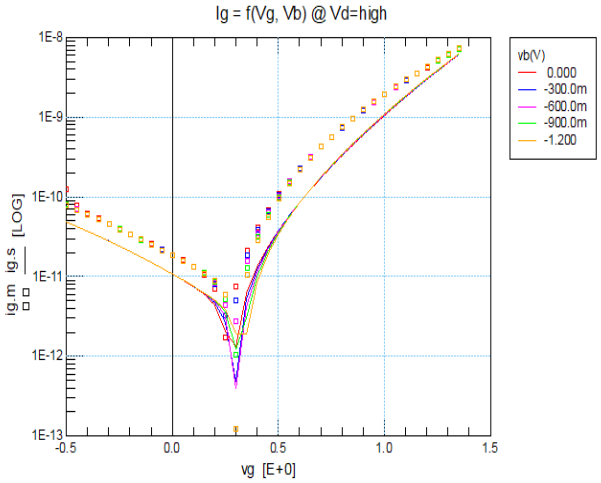




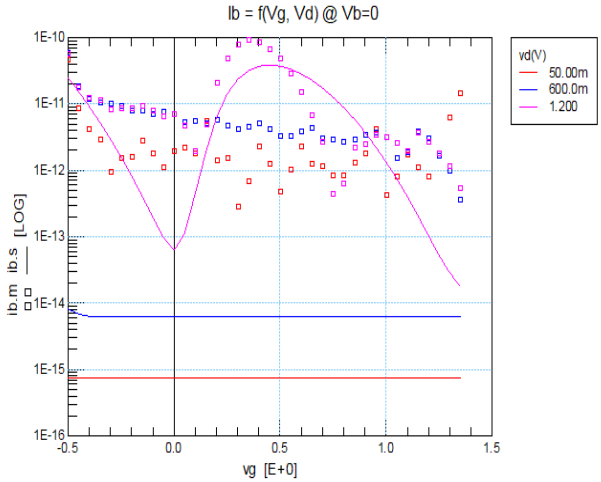


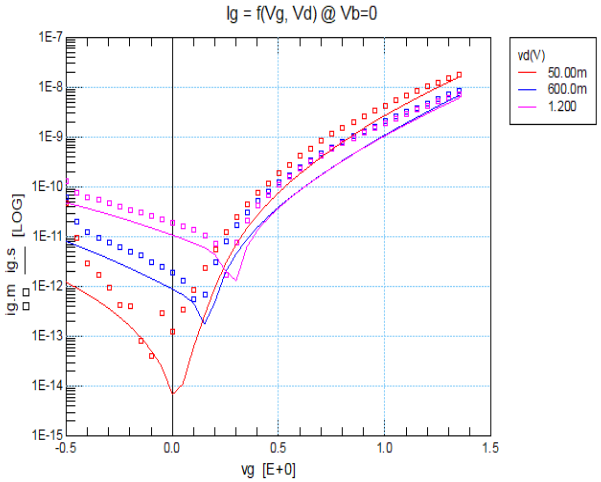




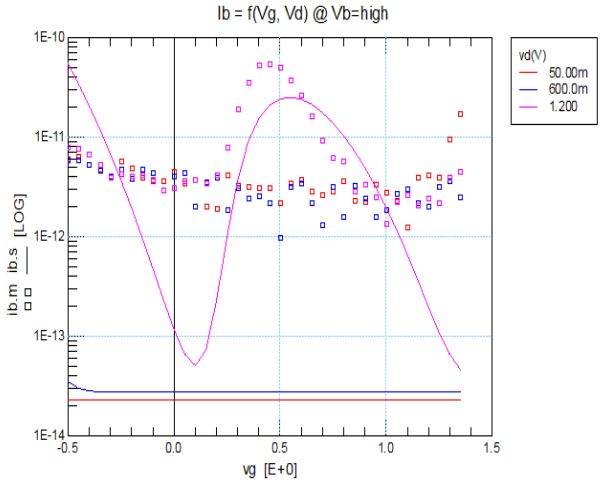


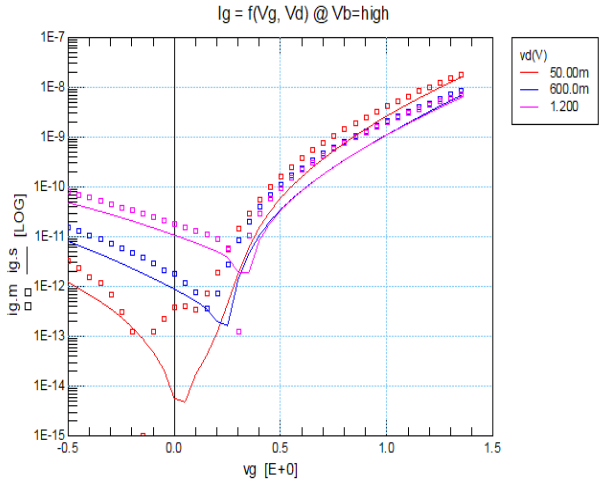
idvg, Ib,Ig, Vb = 0V, T = 27°C



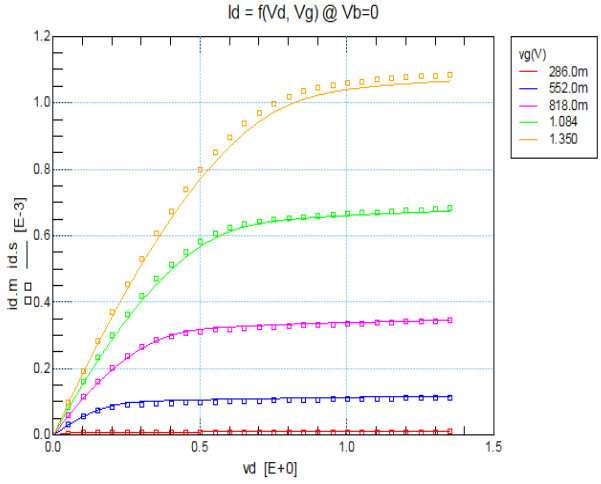


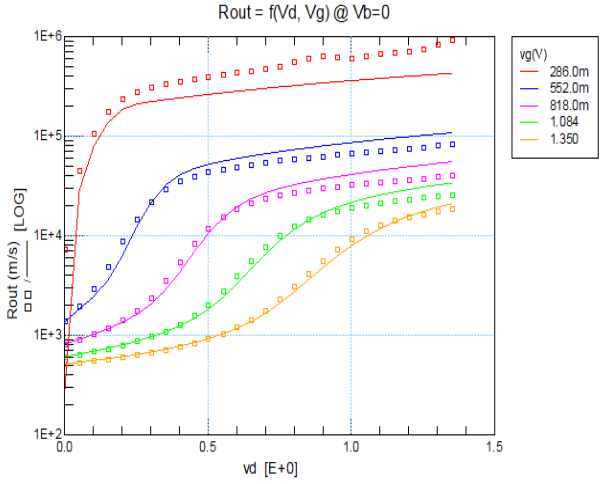
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

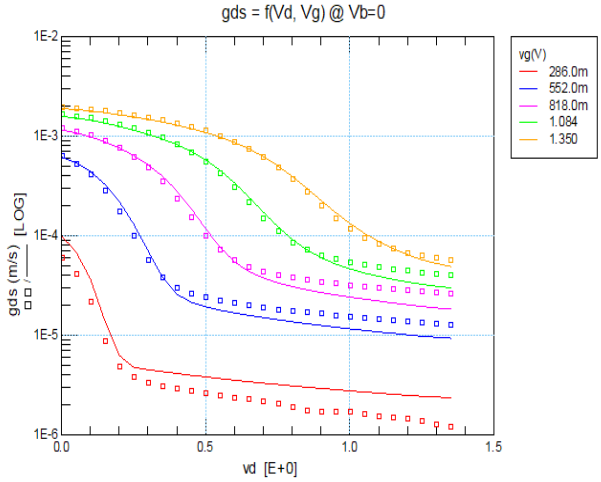




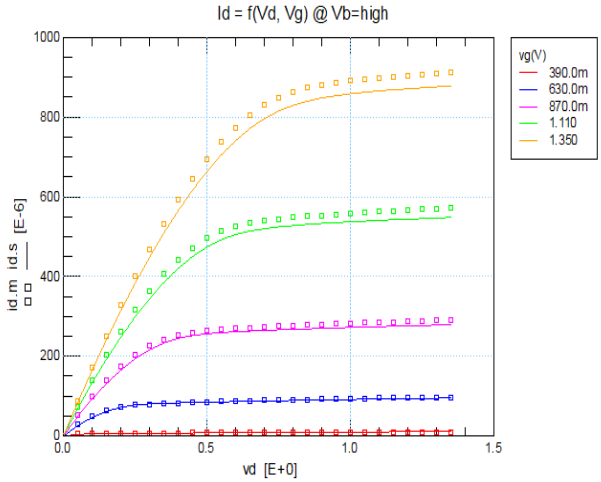
idvd, Vb = 0V, T = 27°C

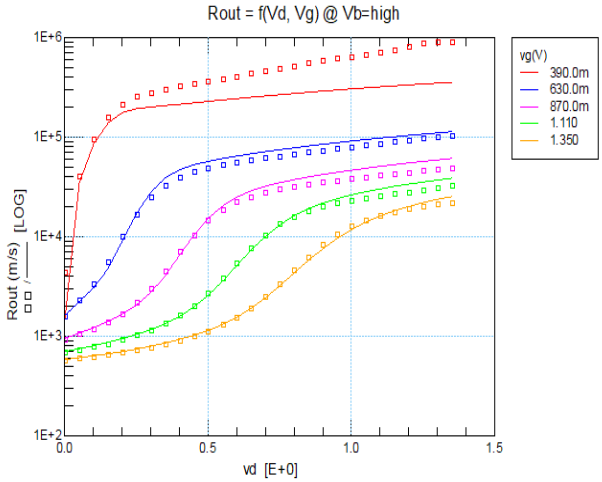


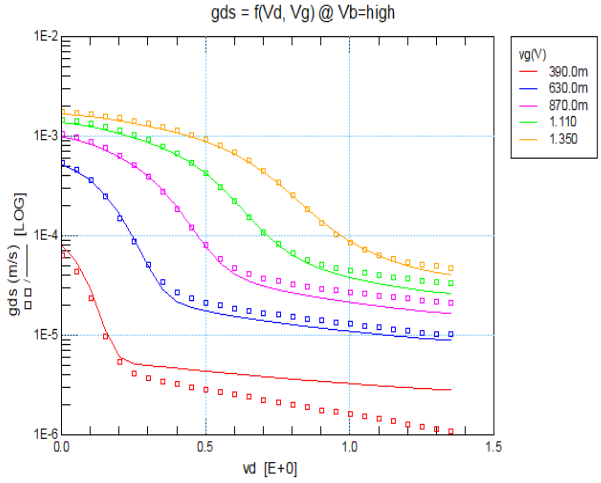




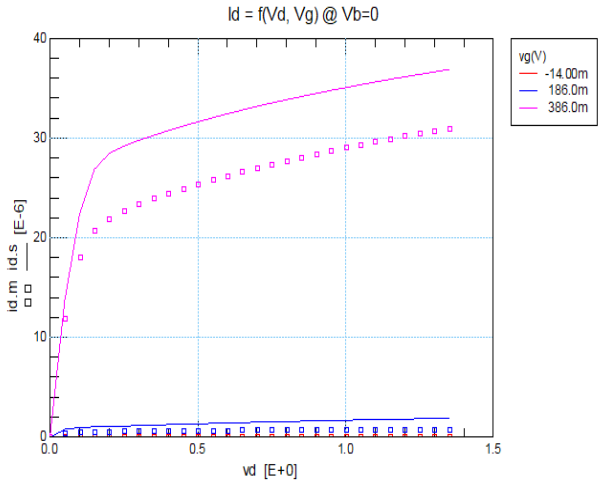
idvd_vbmin, Vb = -1.2V, T = 27°C

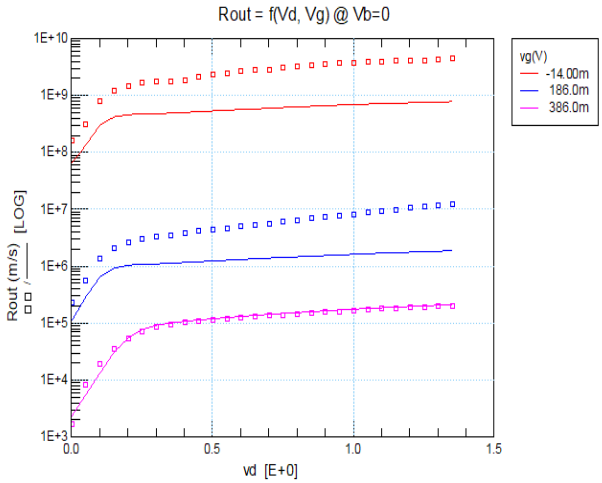


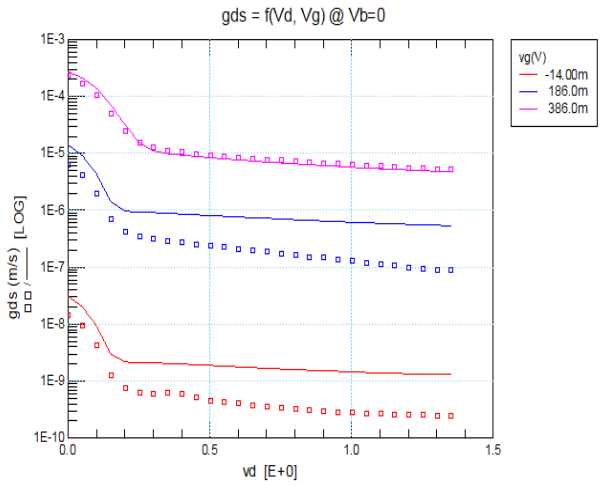




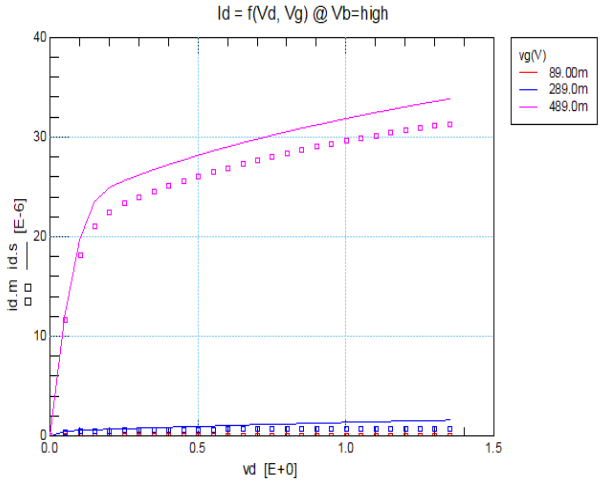
idvd_vth, Vb = 0V, T = 27°C

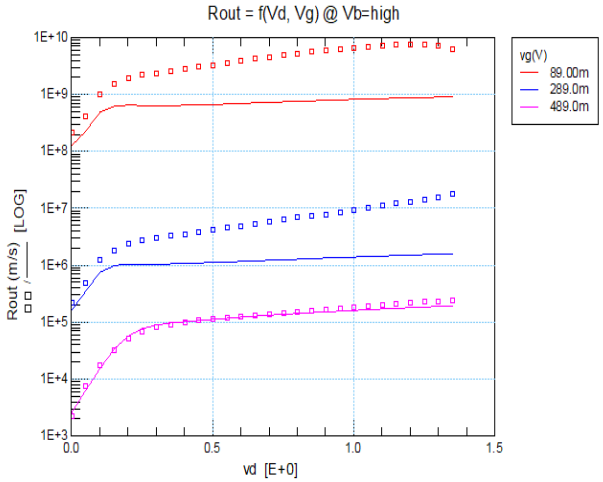


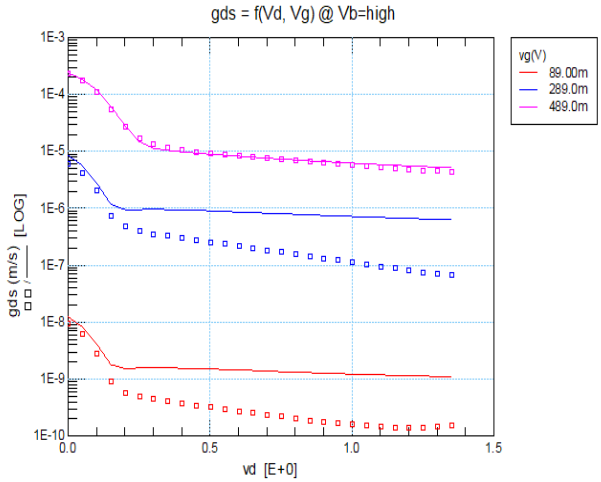




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

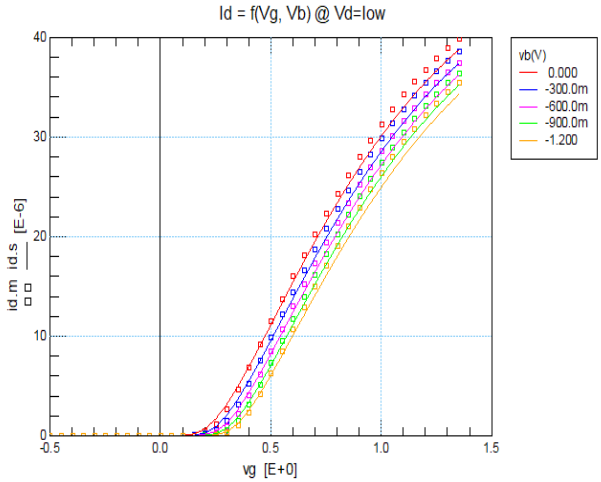


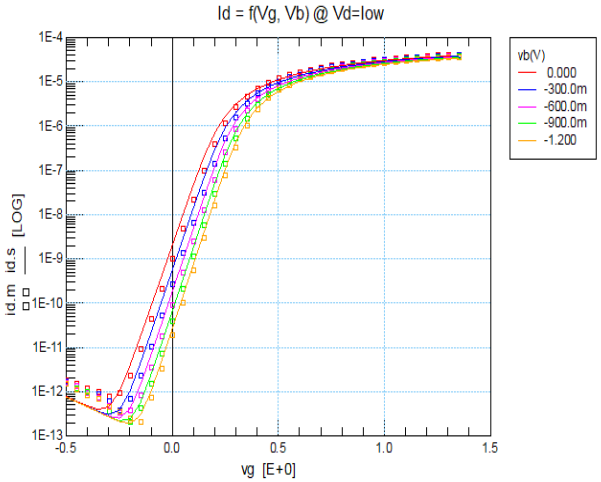


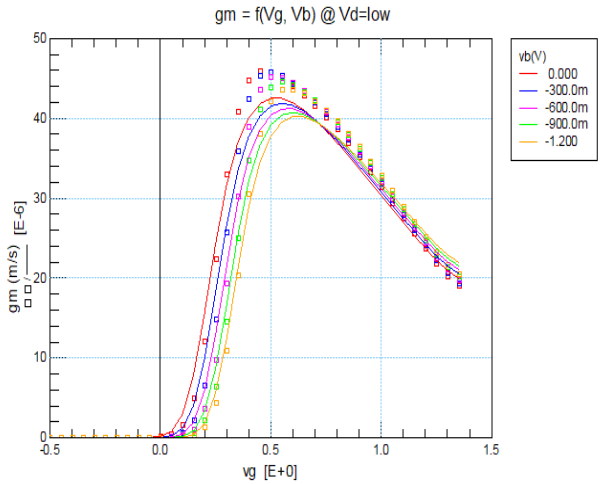


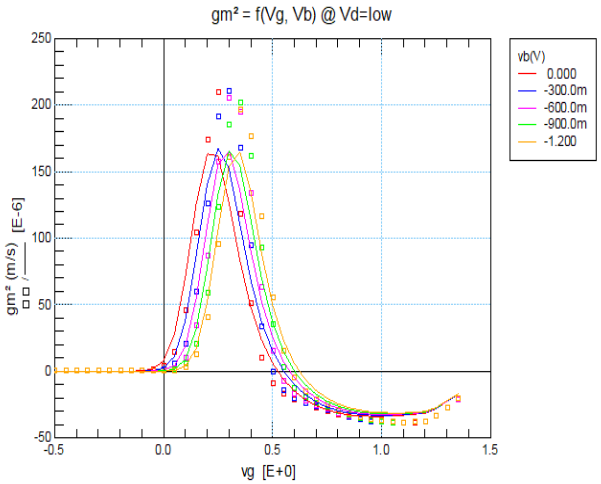
4.12 W10u0_L5u0_S541_4

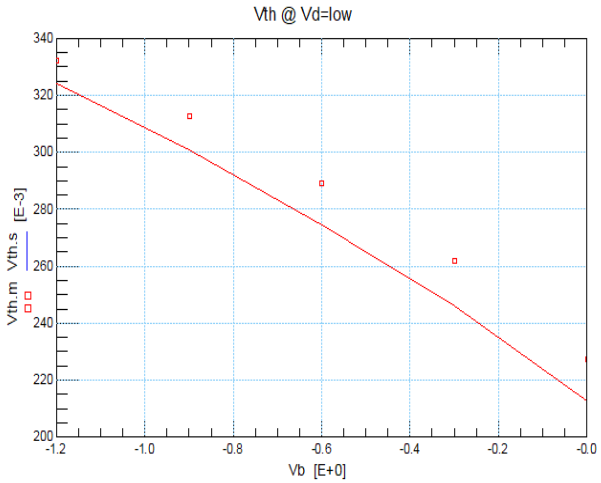
idvg, Vd = 0.05V, T = 27°C

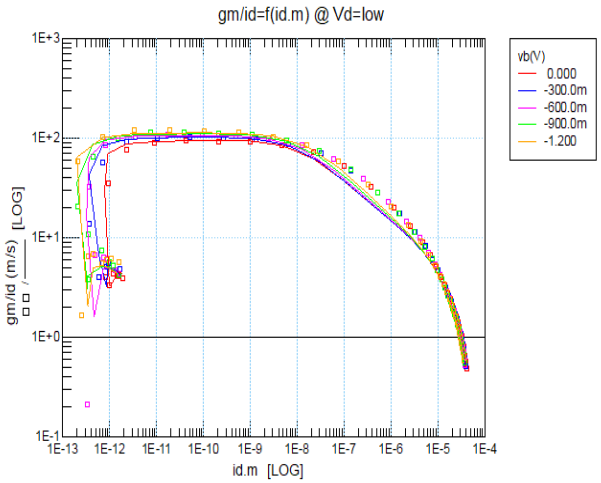


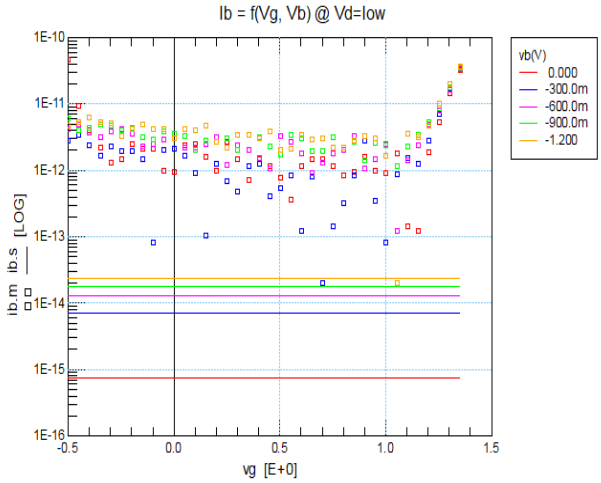


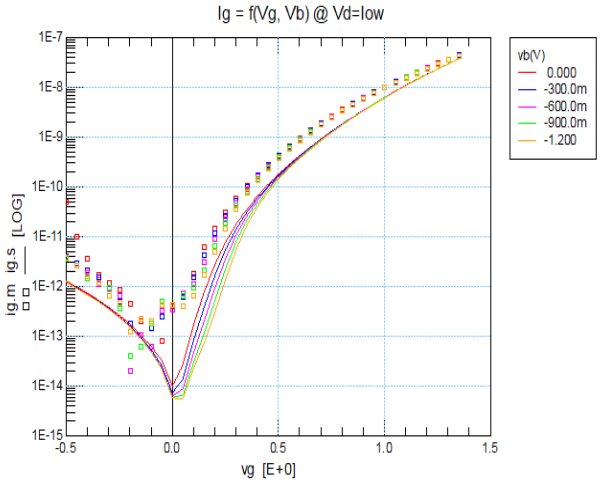




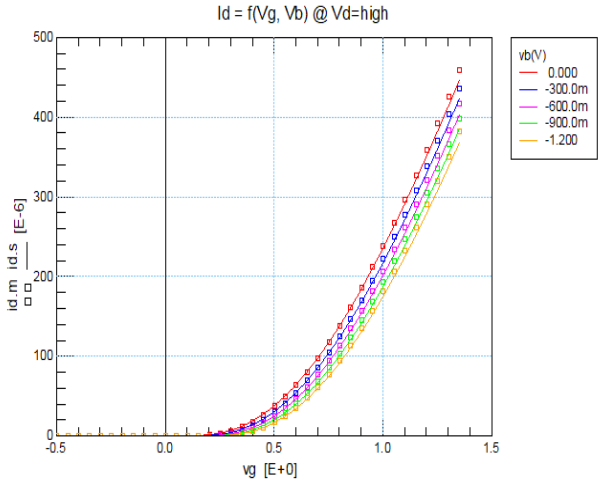


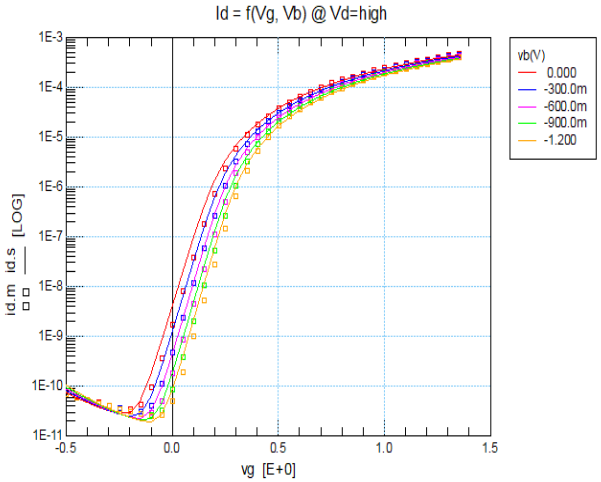


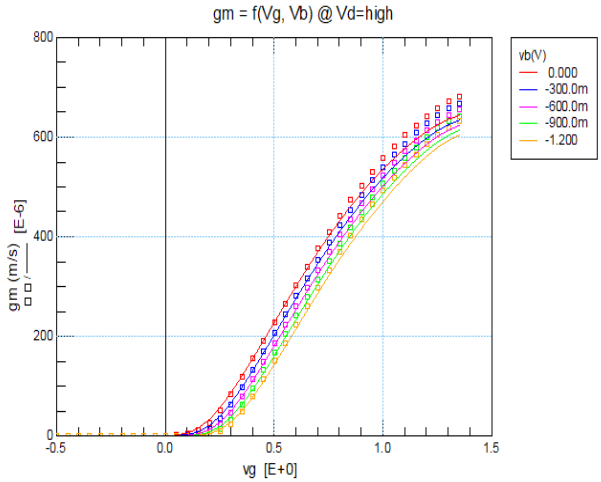


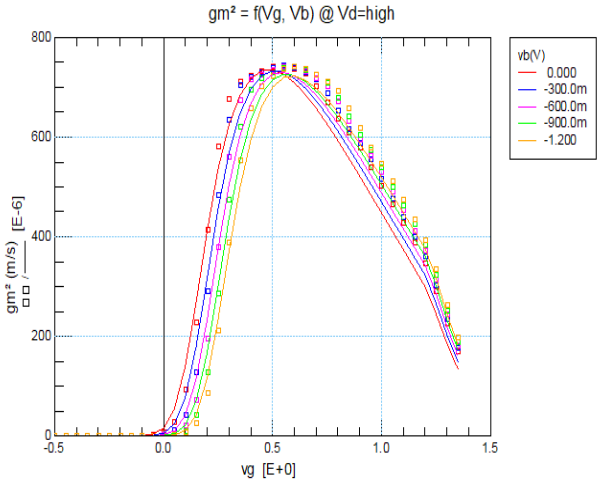


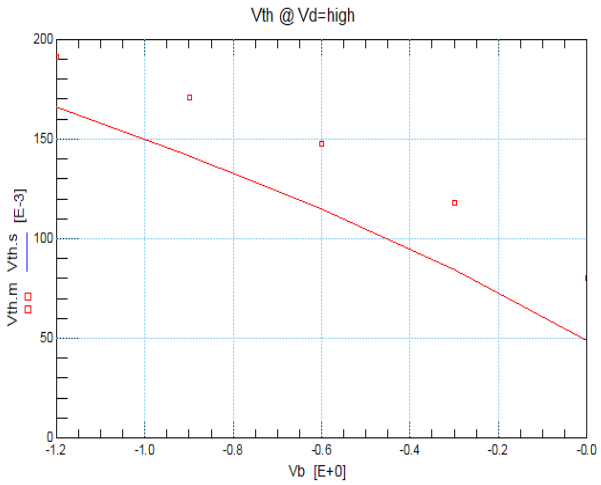
idvg, Vd = 1.2V, T = 27°C

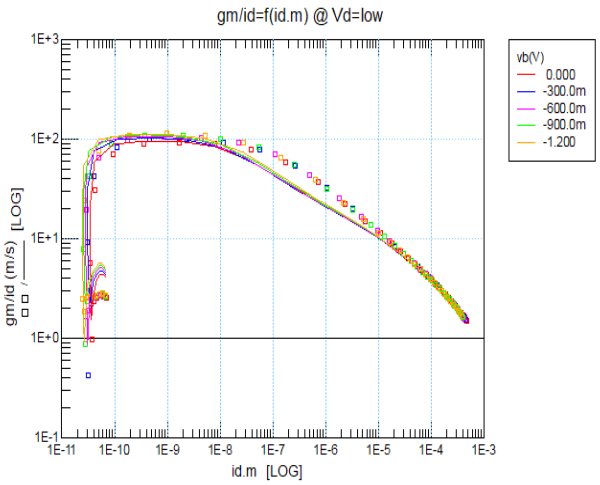


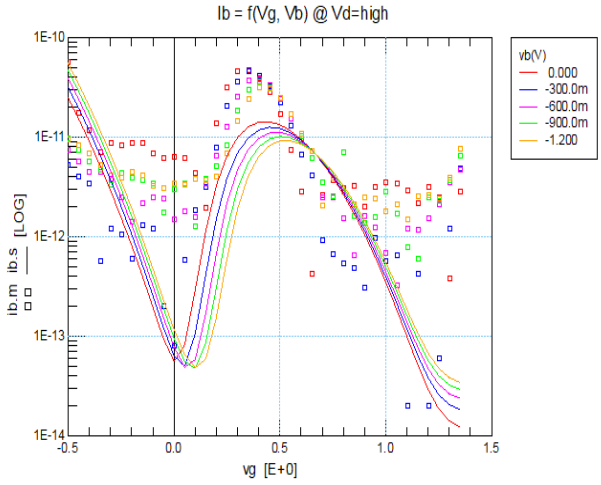


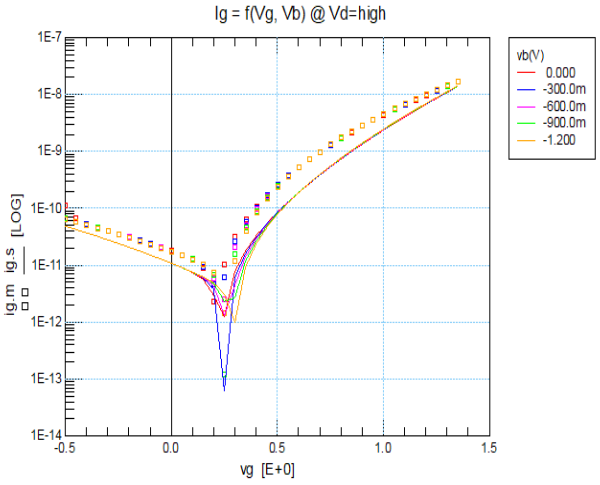




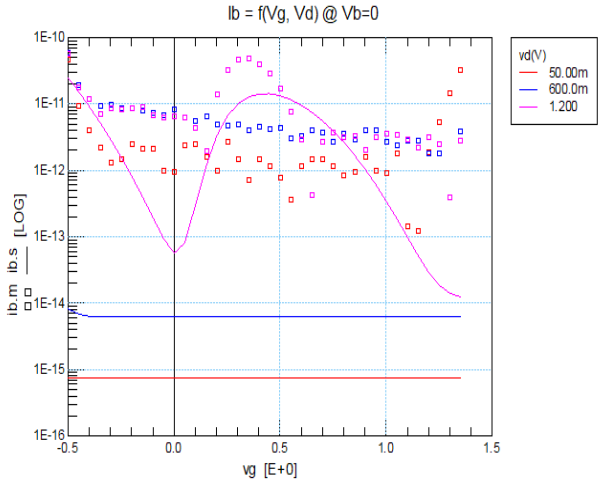


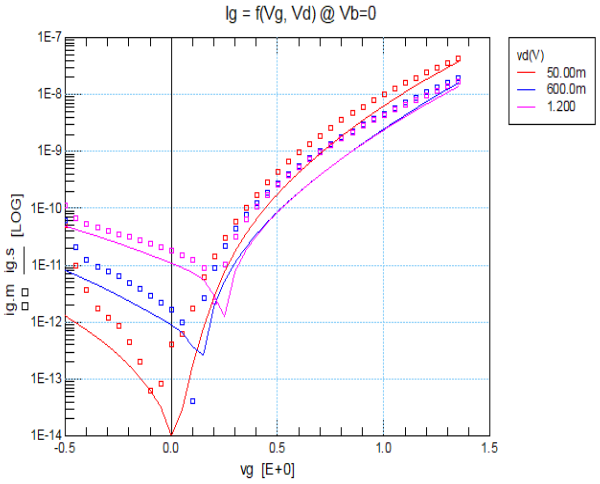




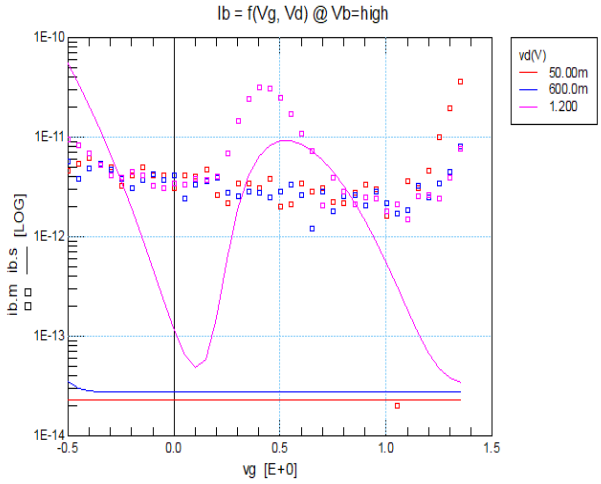


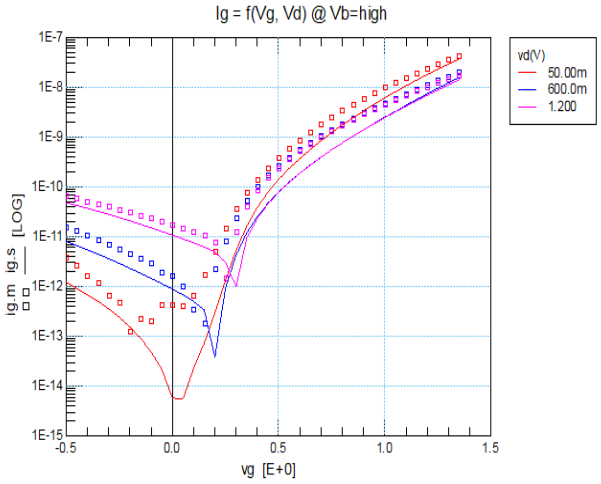
idvg, Ib,Ig, Vb = 0V, T = 27°C



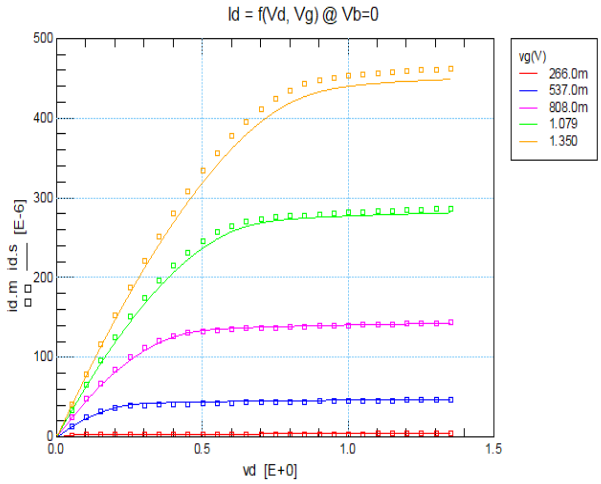


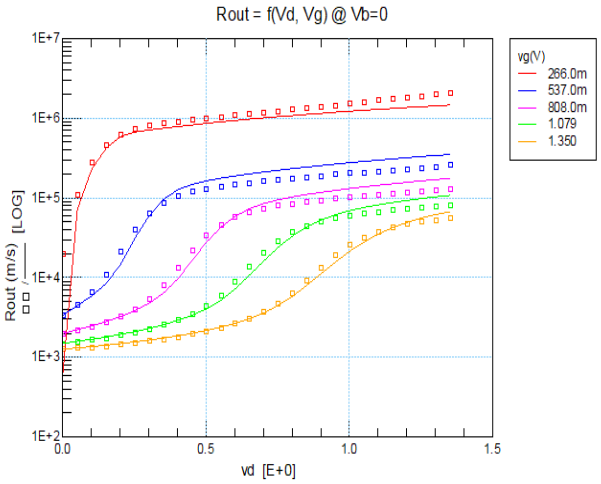
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

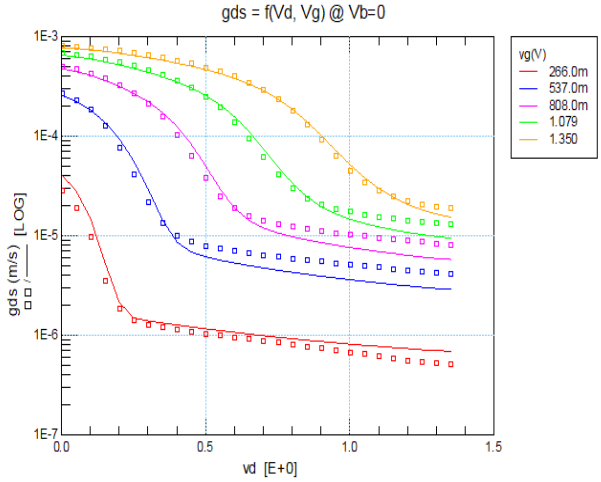




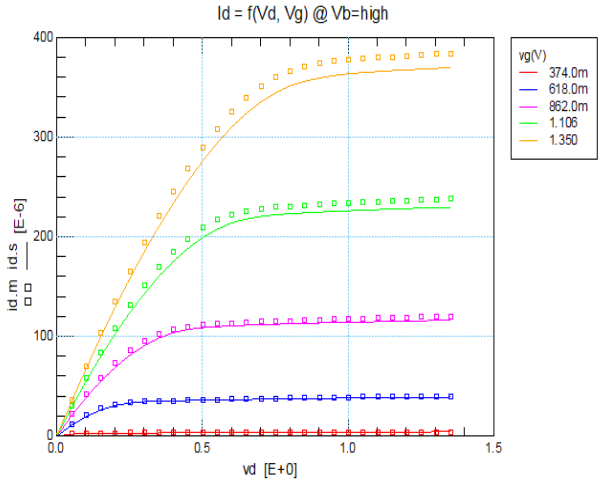
idvd, Vb = 0V, T = 27°C

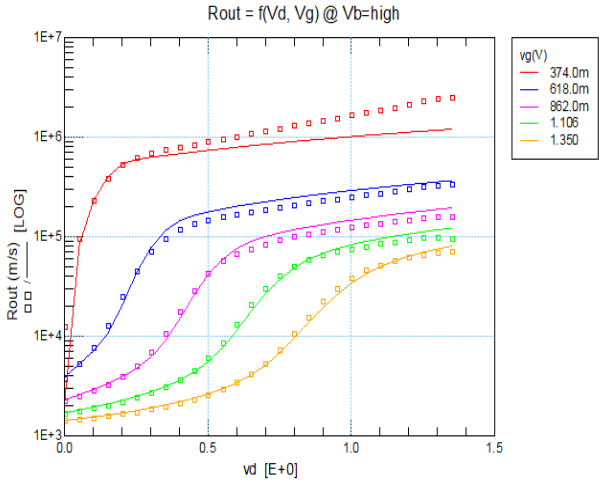


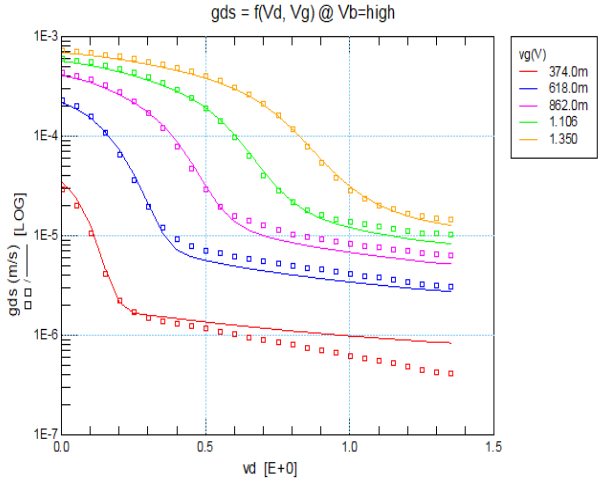




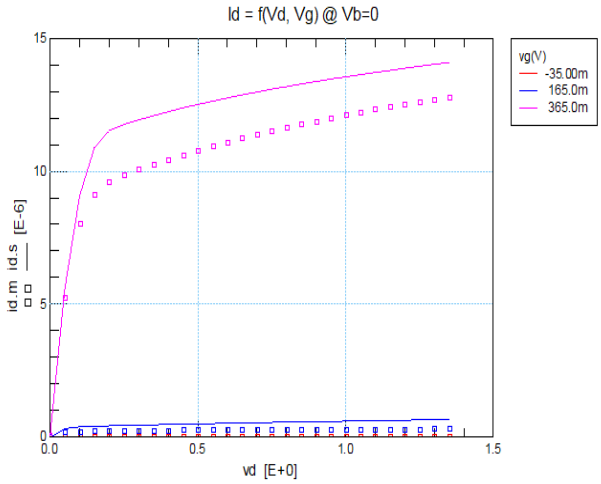
idvd_vbmin, Vb = -1.2V, T = 27°C

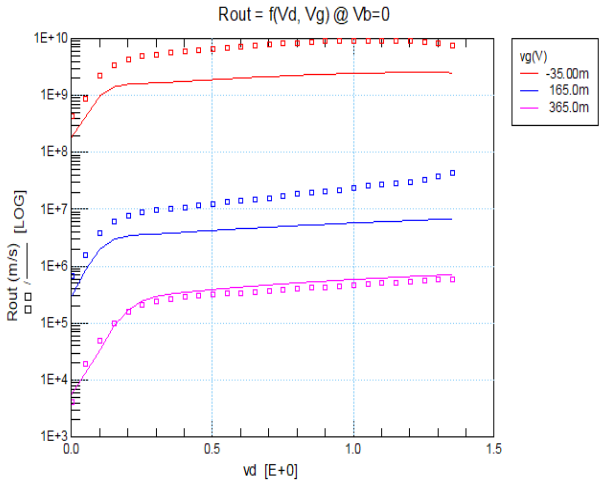


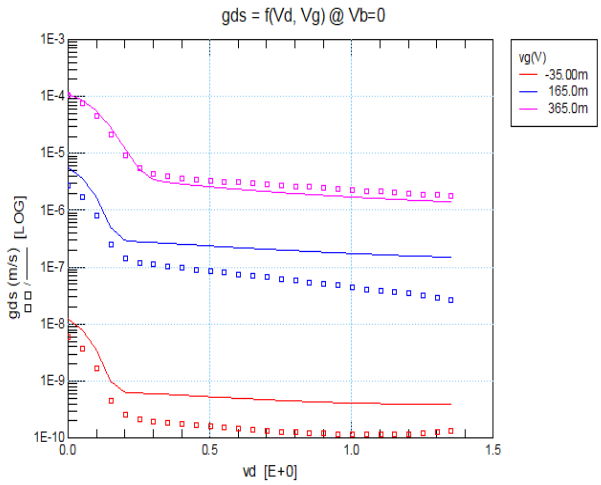




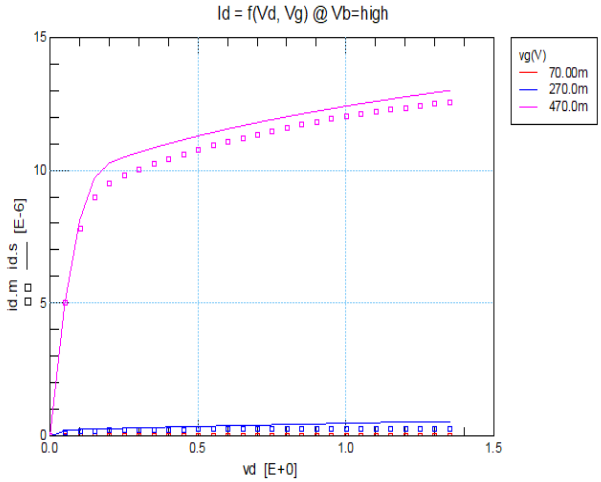
idvd_vth, Vb = 0V, T = 27°C

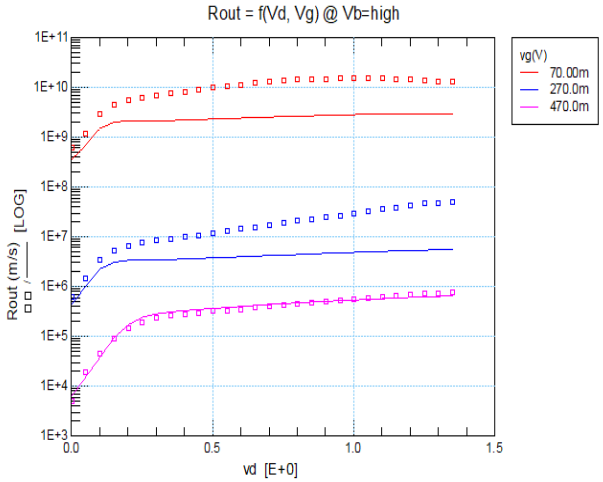


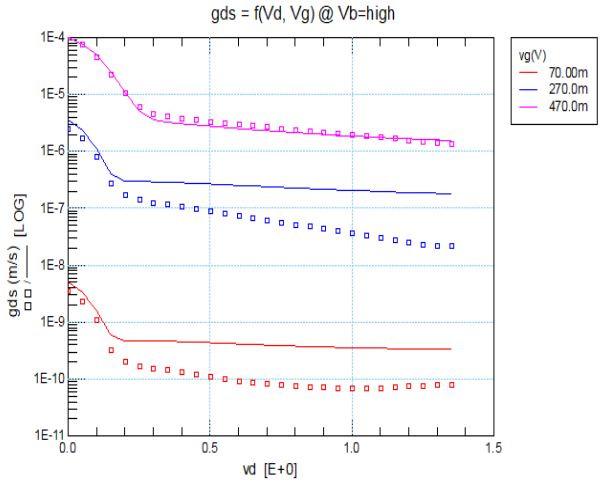




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

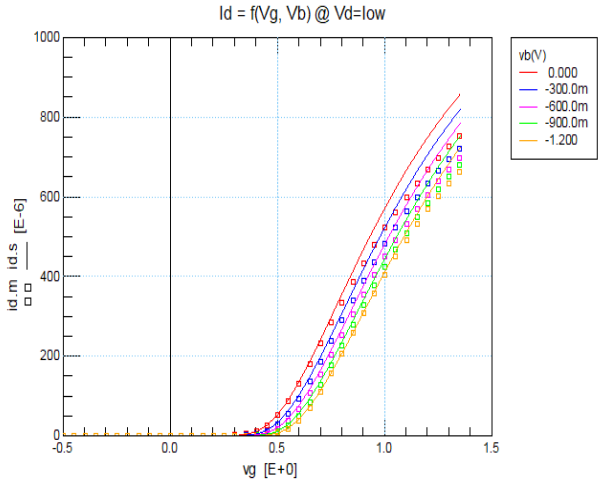


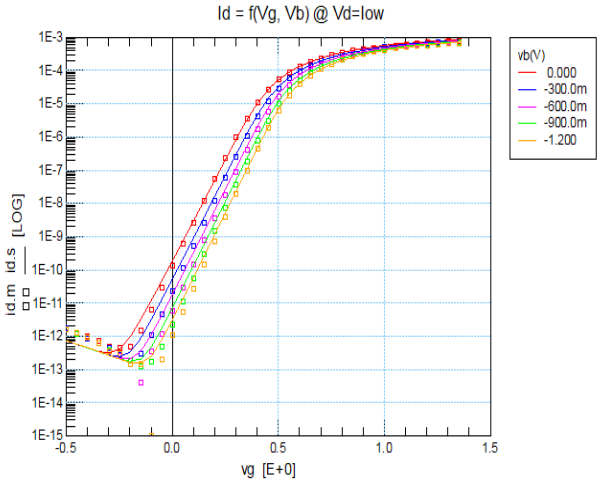


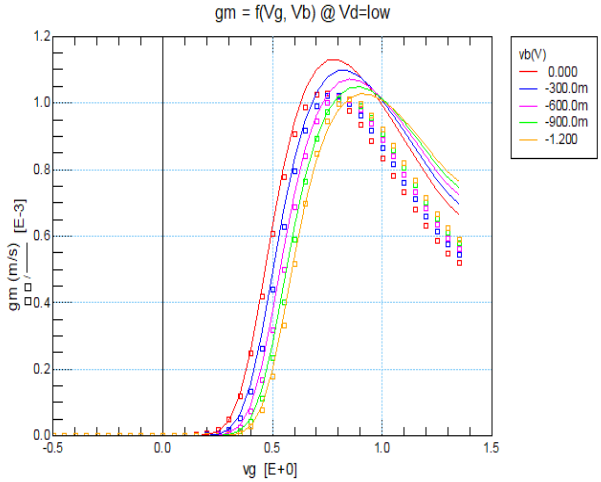


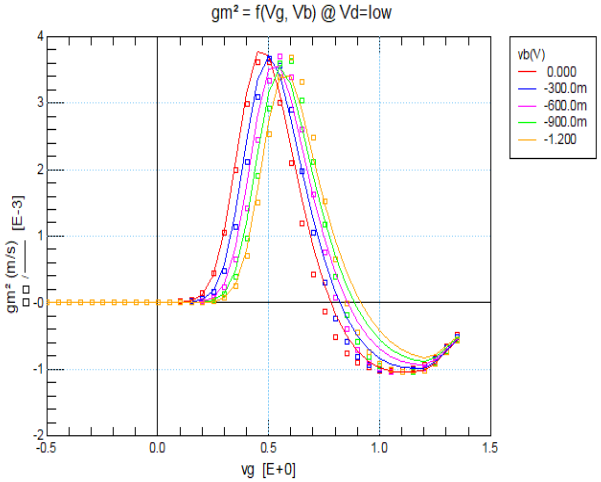
4.13 W10u0_L0u15_S540_4

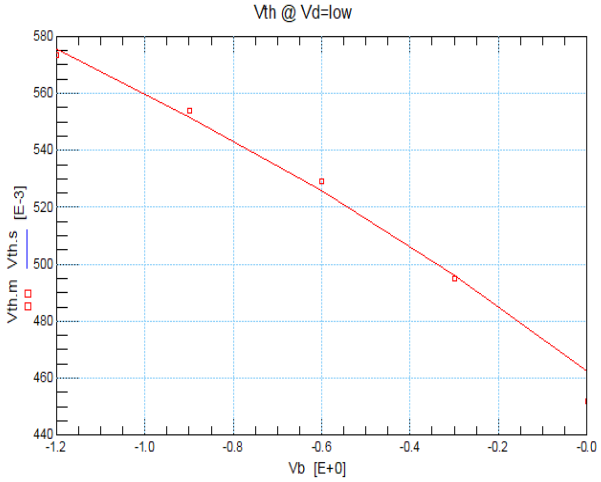
idvg, Vd = 0.05V, T = 27°C

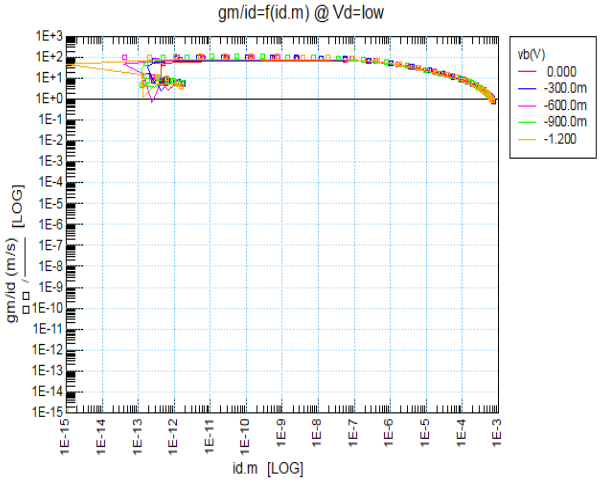


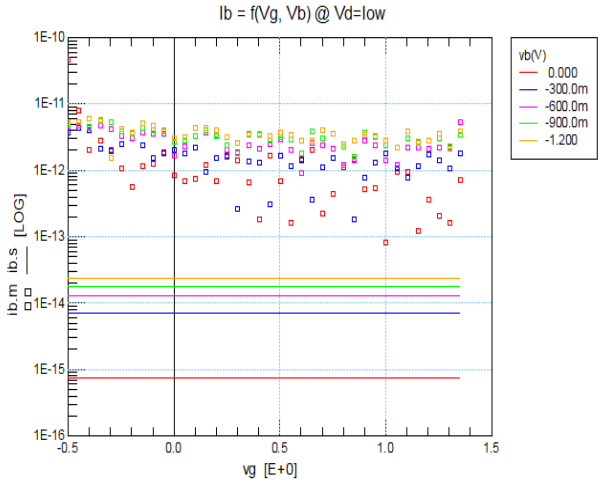


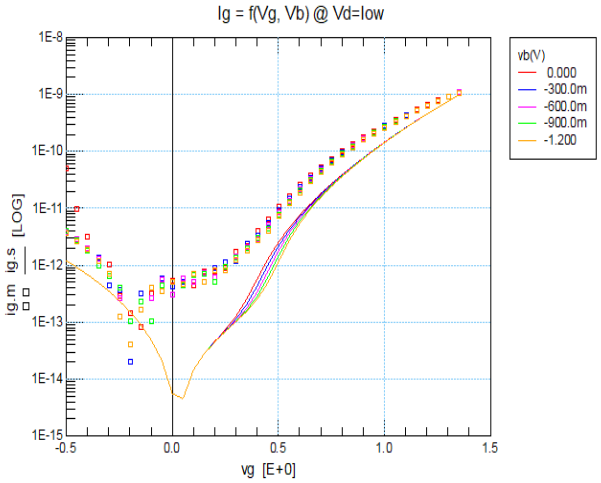




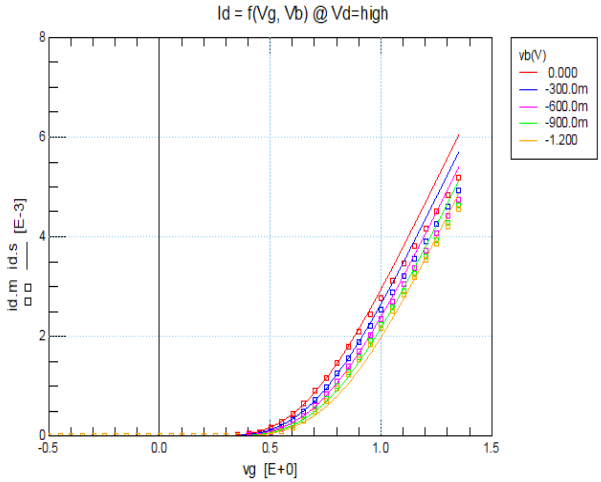


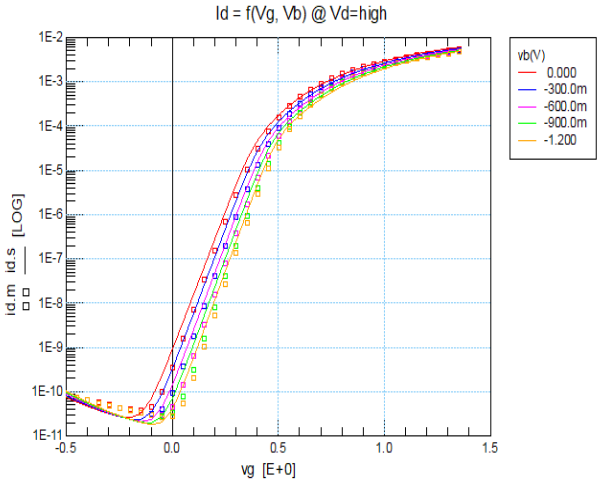


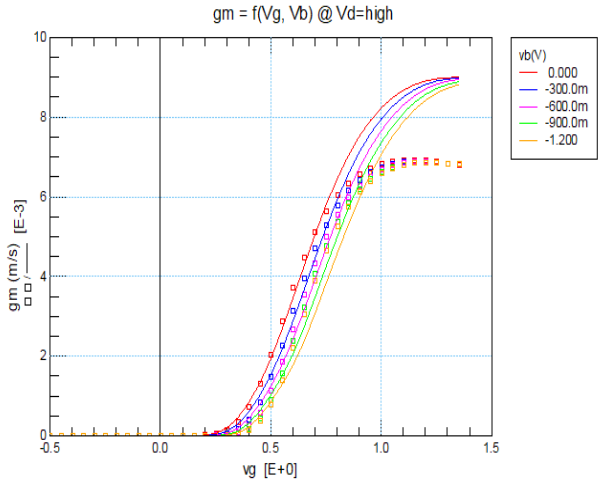


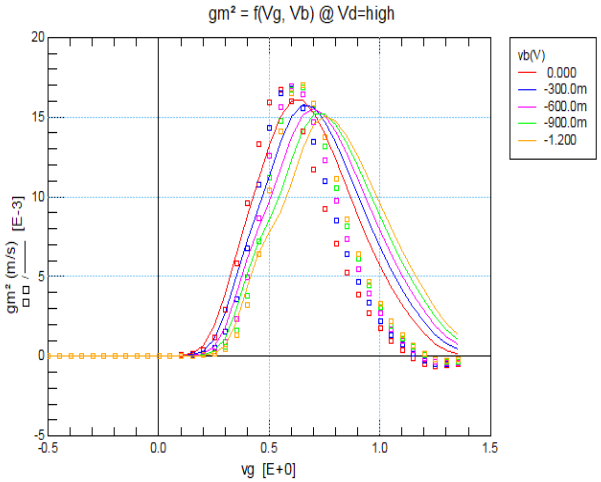


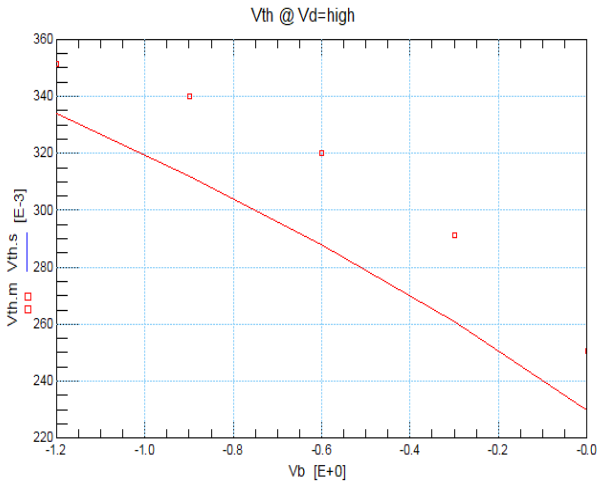
idvg, Vd = 1.2V, T = 27°C

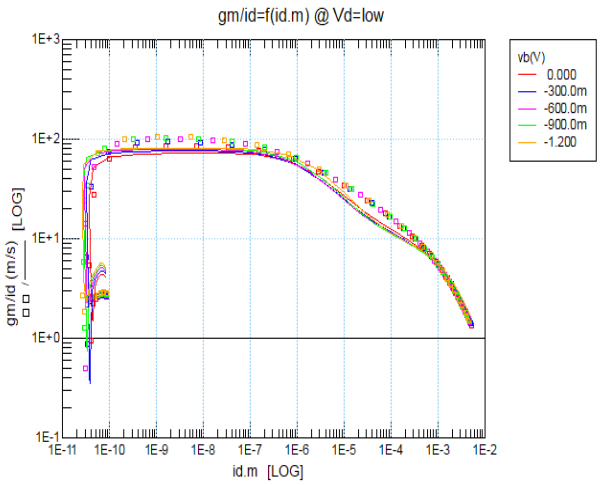


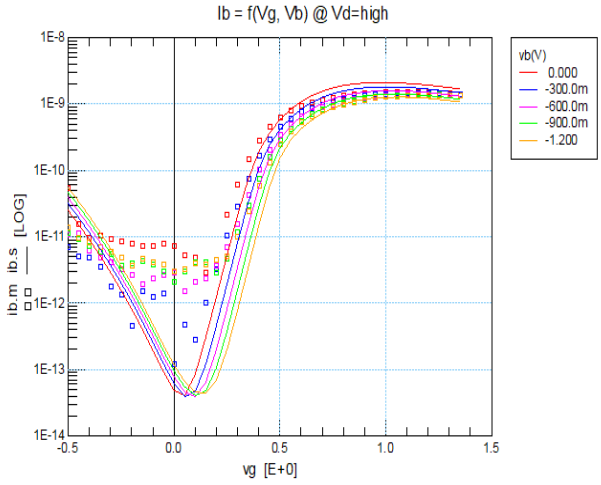


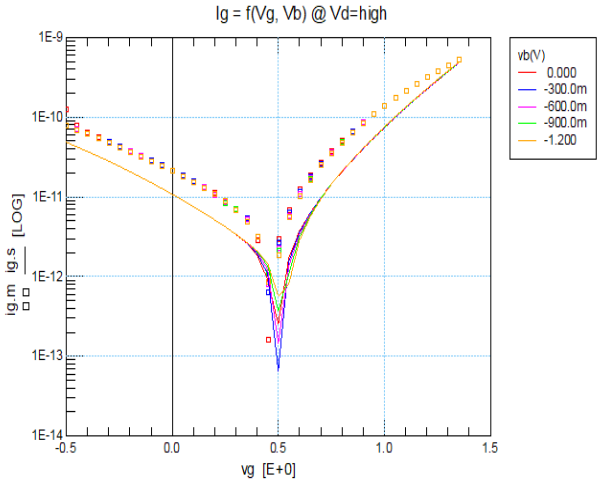




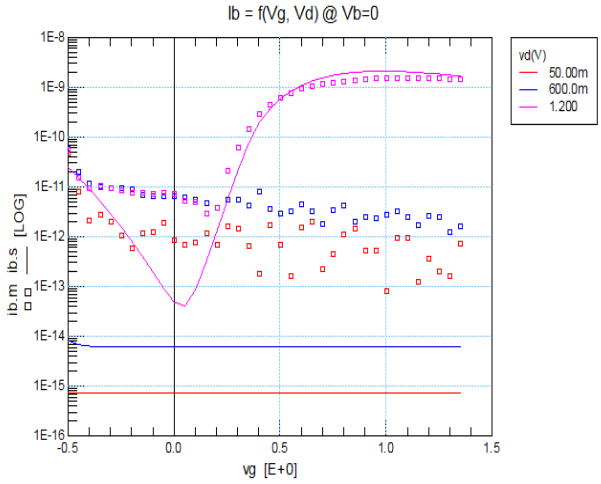


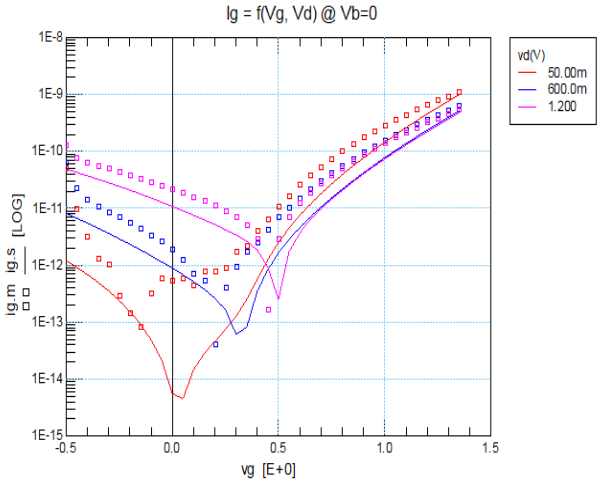




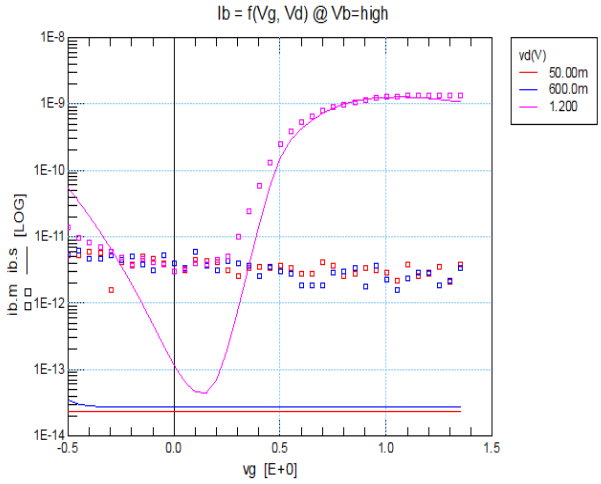


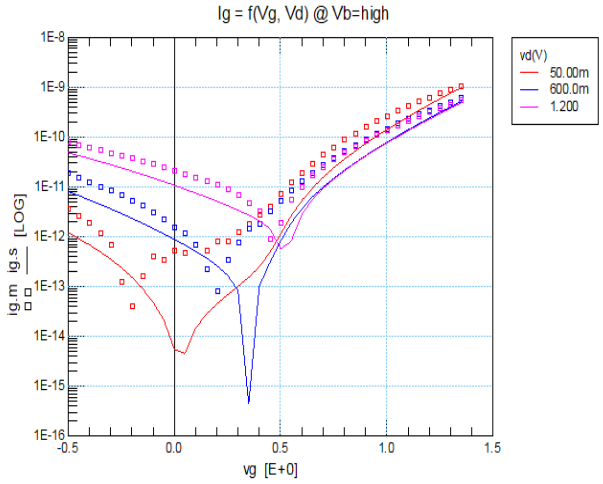
idvg, Ib,Ig, Vb = 0V, T = 27°C



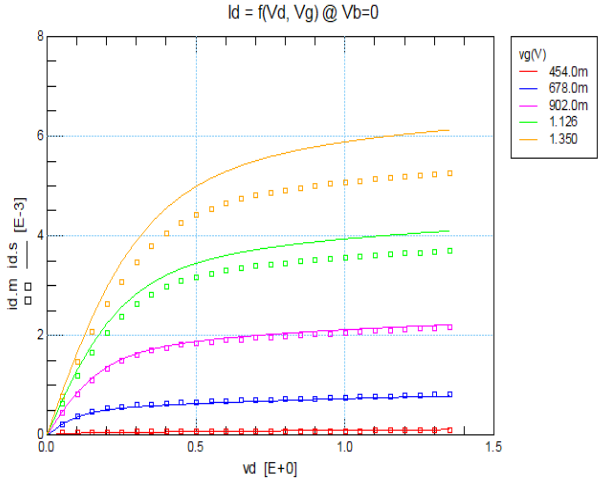


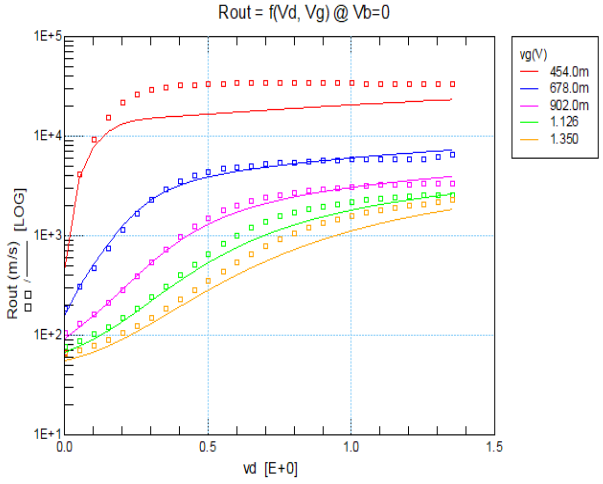
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

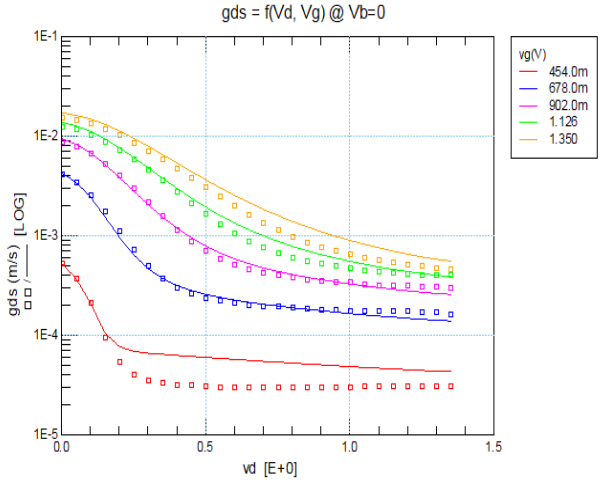




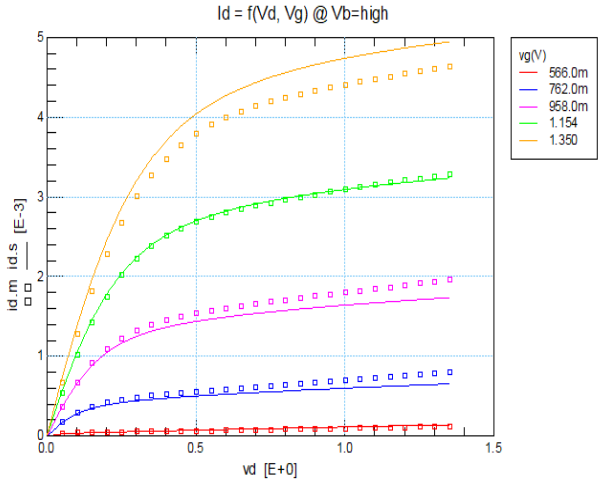
idvd, Vb = 0V, T = 27°C

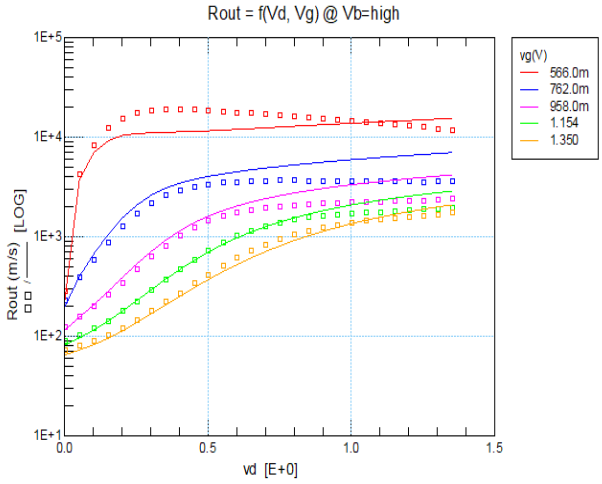


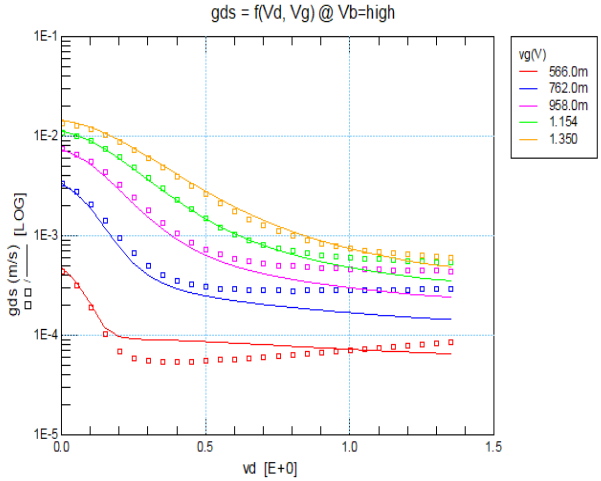




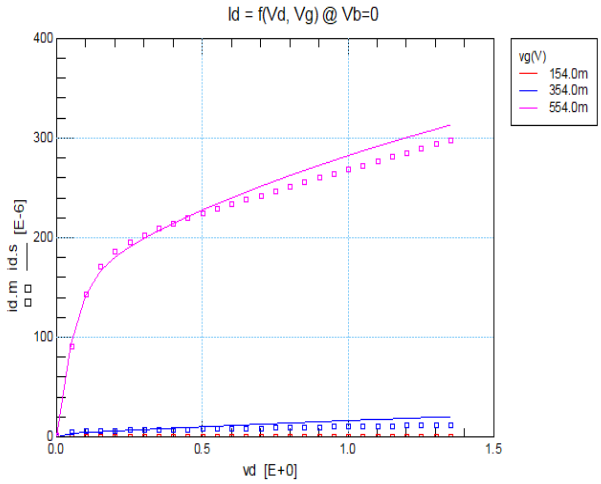
idvd_vbmin, Vb = -1.2V, T = 27°C

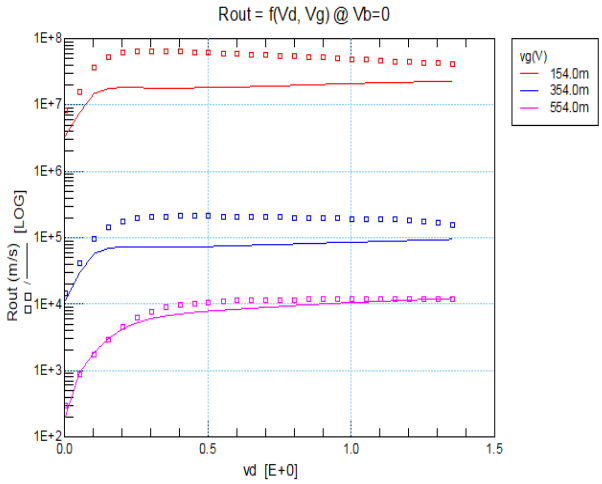


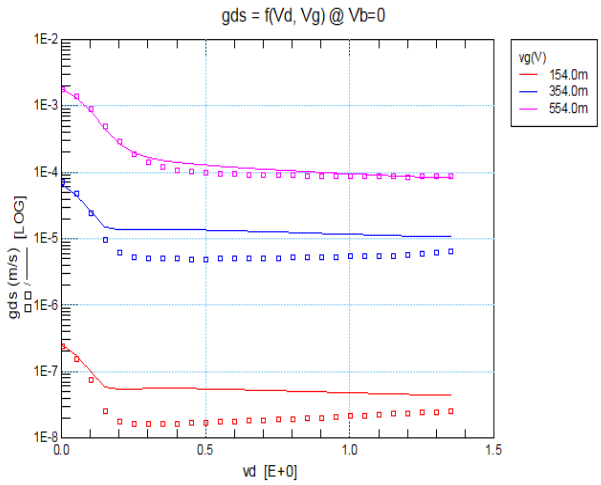




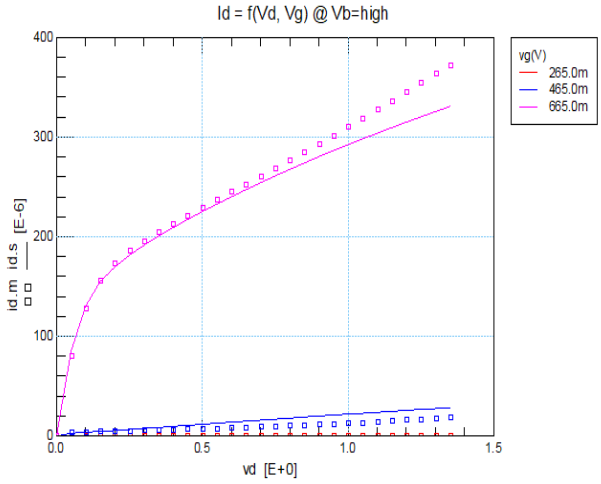
idvd_vth, Vb = 0V, T = 27°C

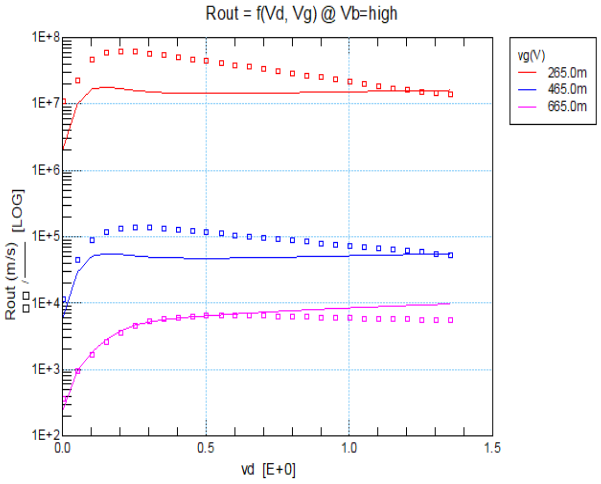


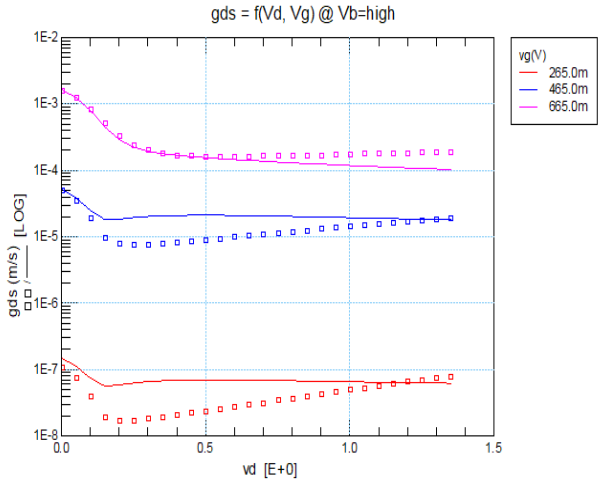




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

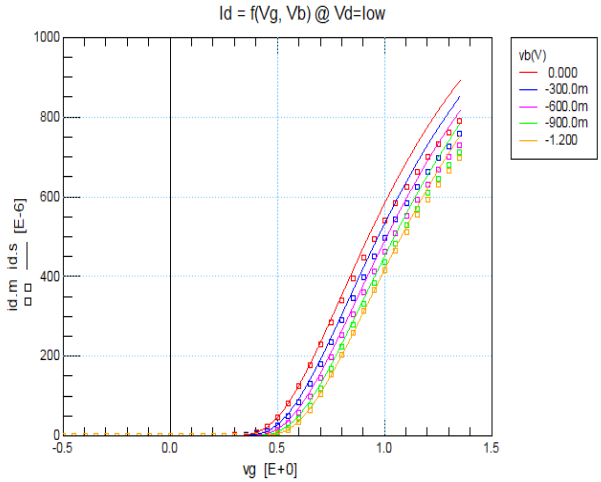


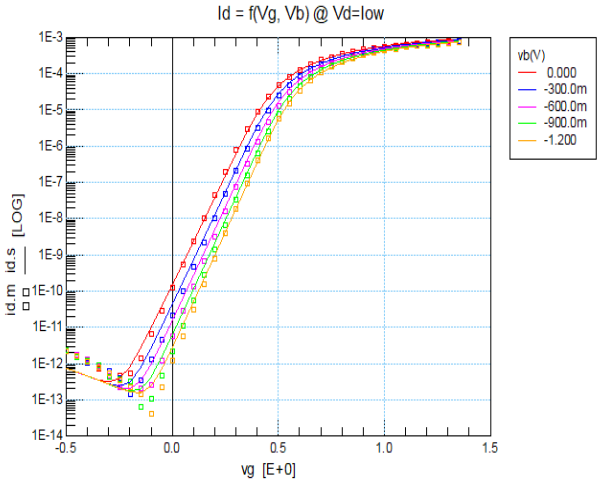


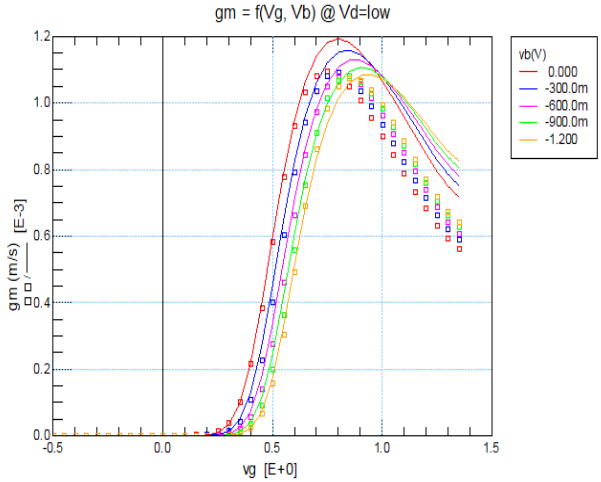


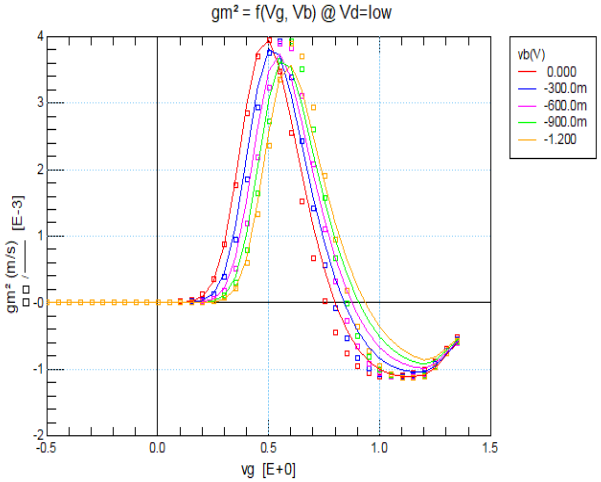
4.14 W10u0_L0u14_S540_3

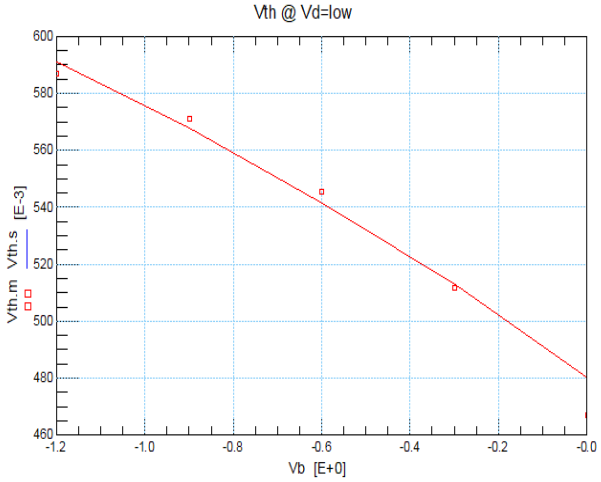
idvg, Vd = 0.05V, T = 27°C

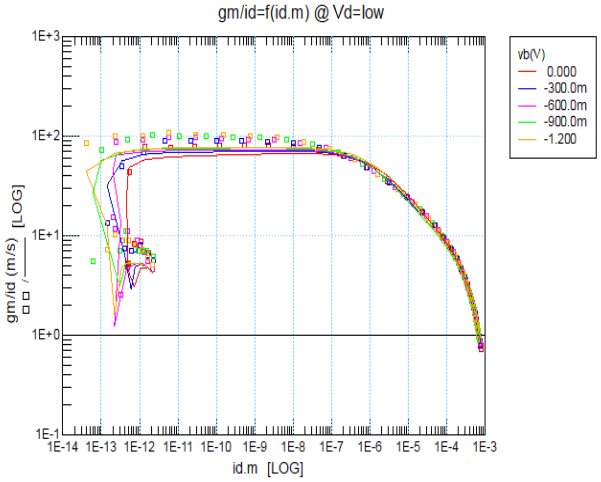


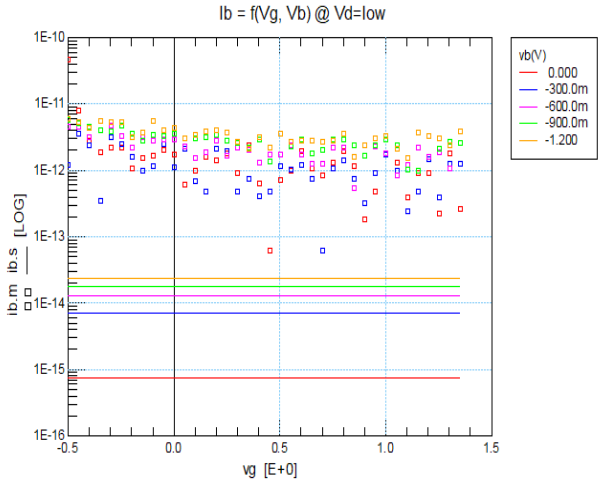


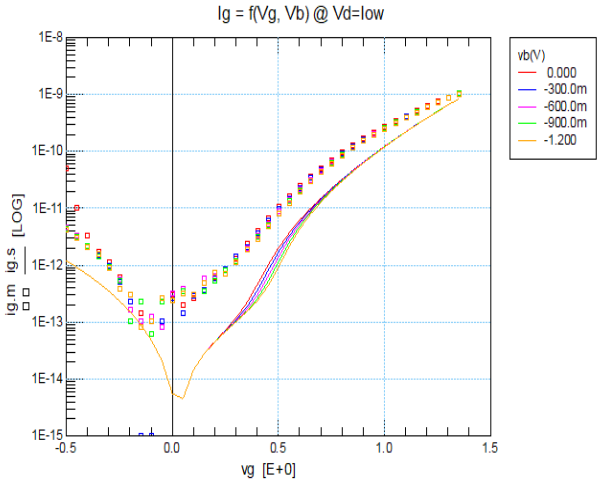




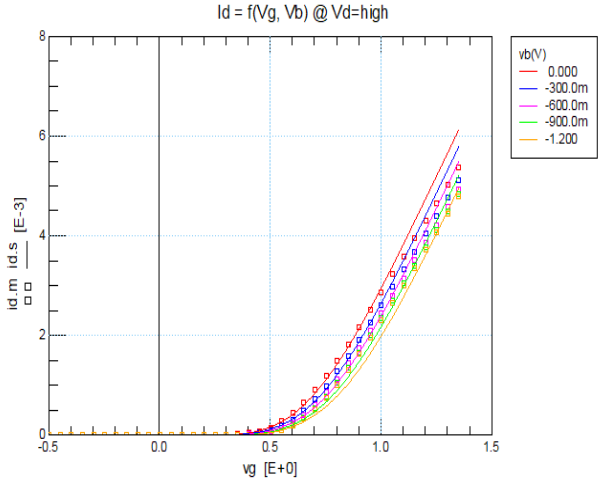


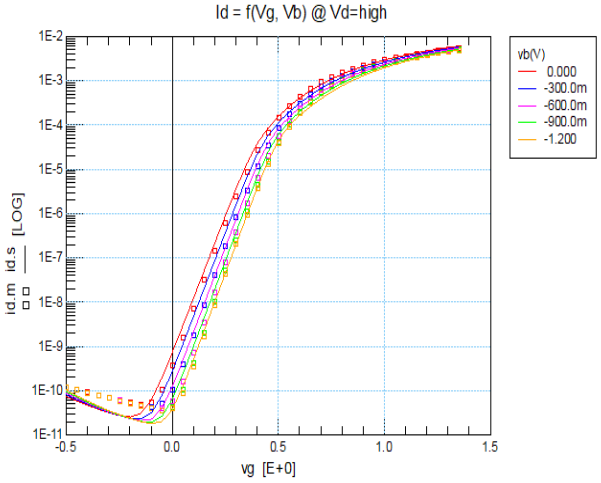


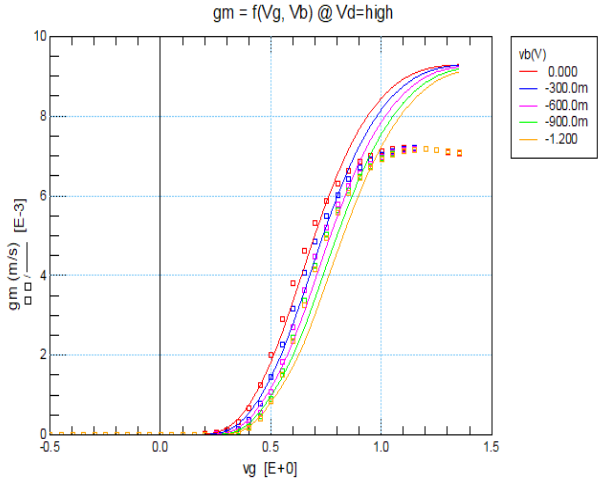


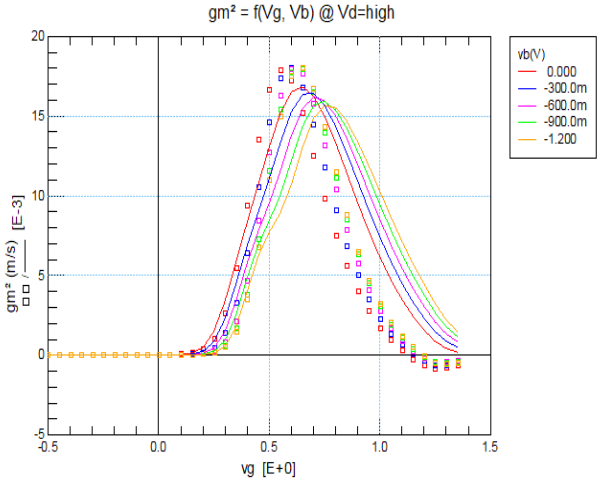


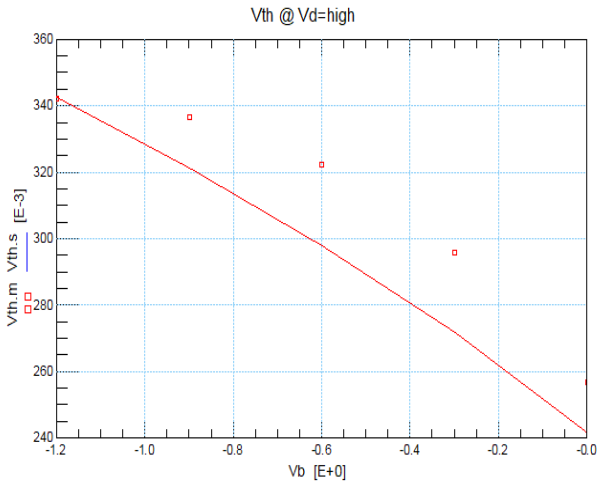
idvg, Vd = 1.2V, T = 27°C

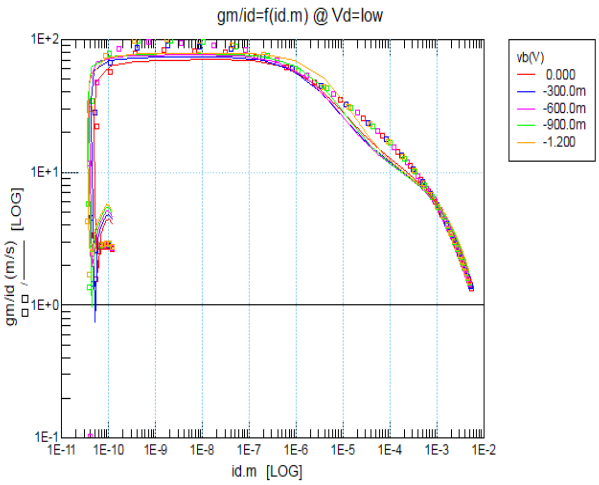


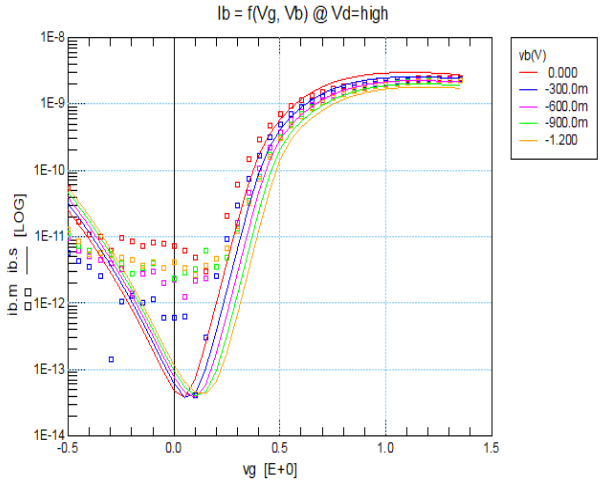


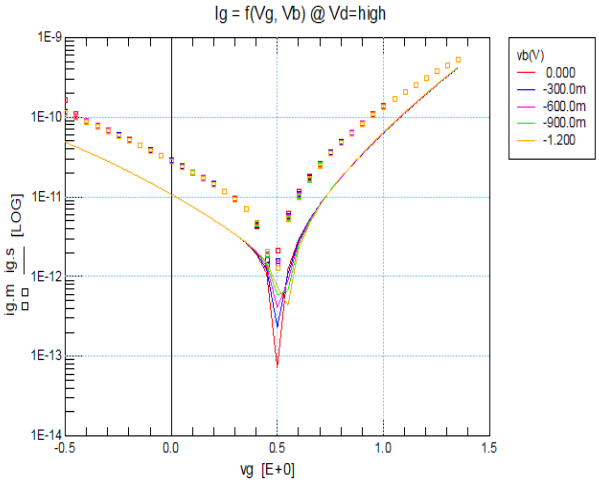




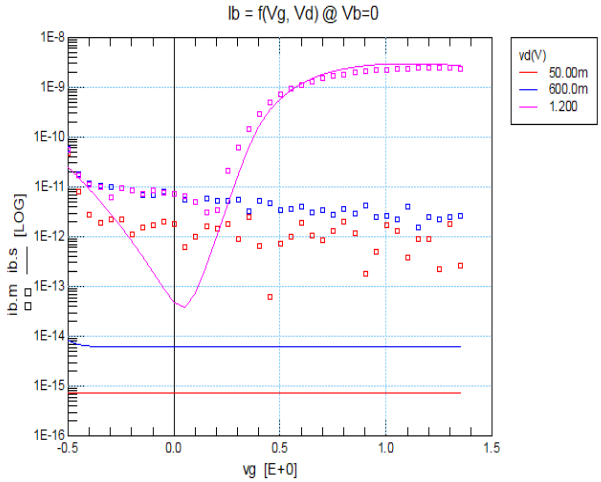


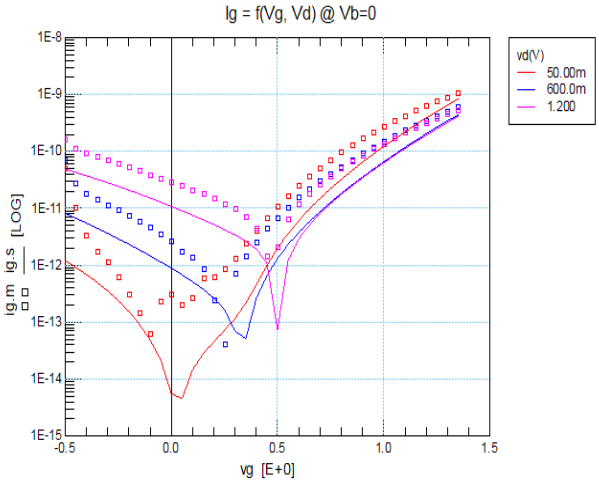




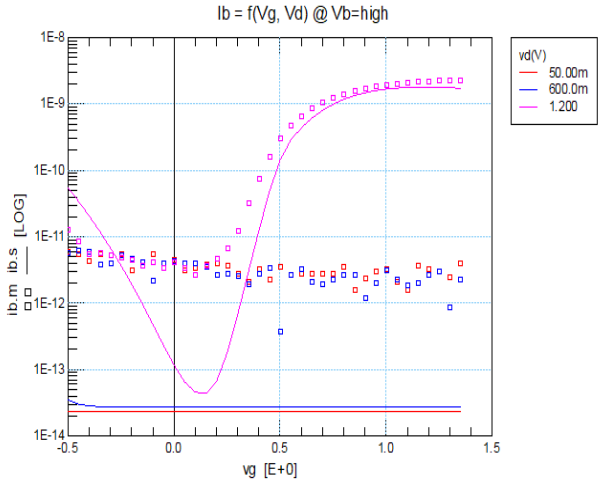


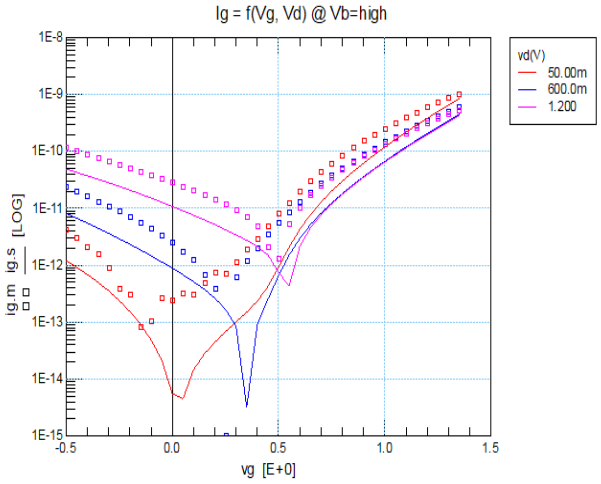
idvg, Ib,Ig, Vb = 0V, T = 27°C



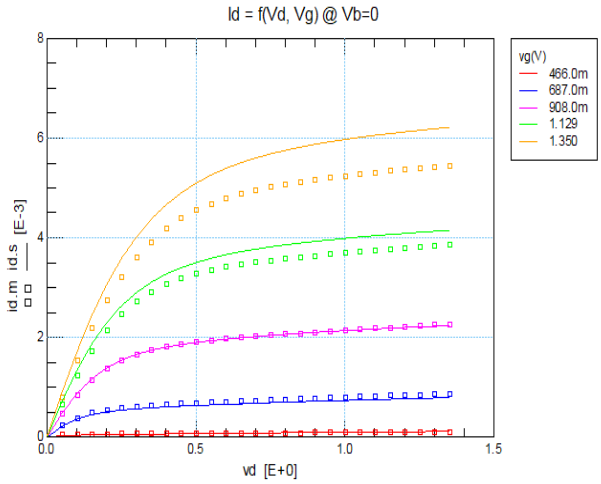


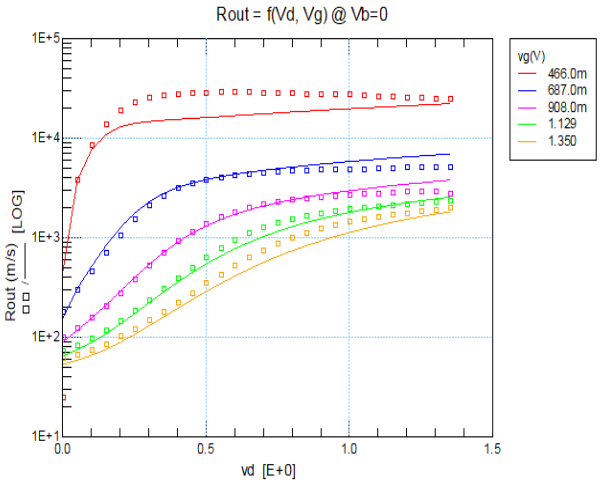
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

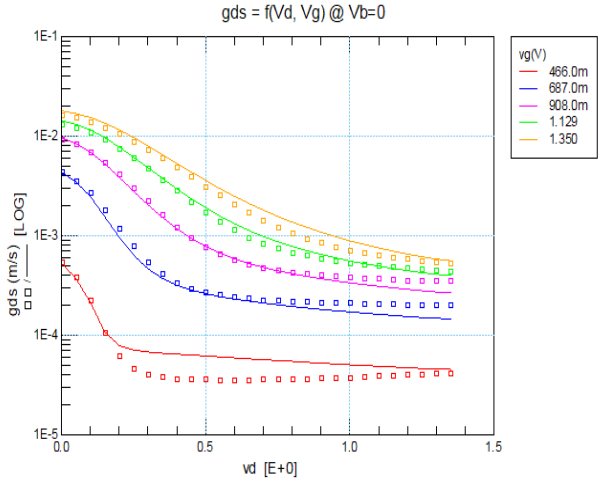




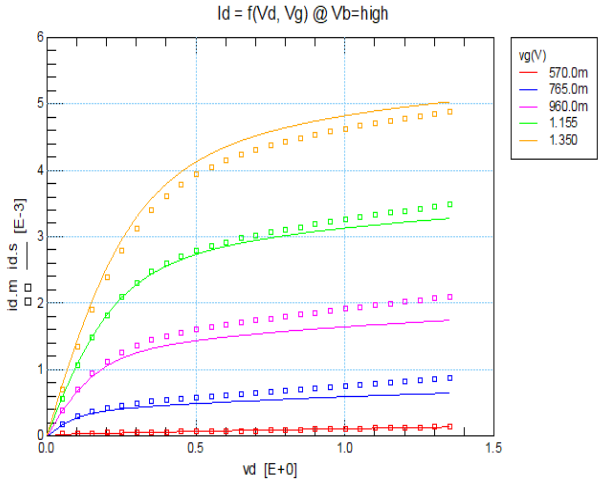
idvd, Vb = 0V, T = 27°C

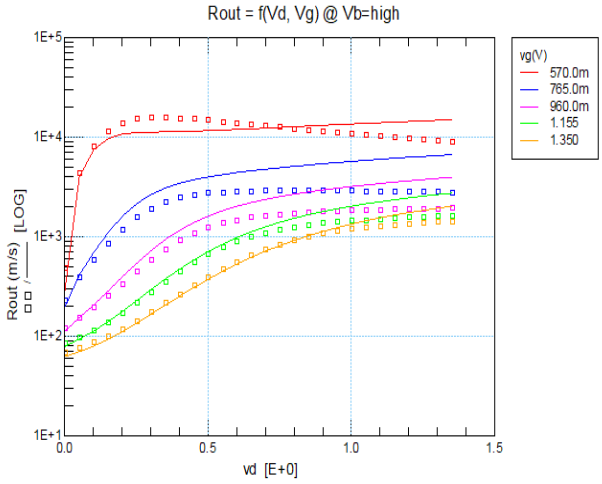


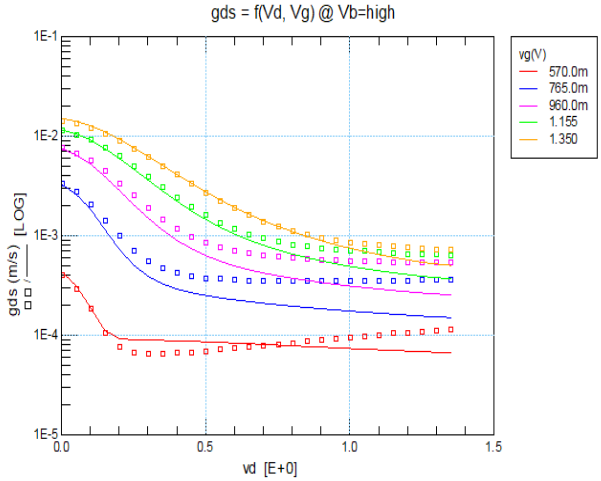




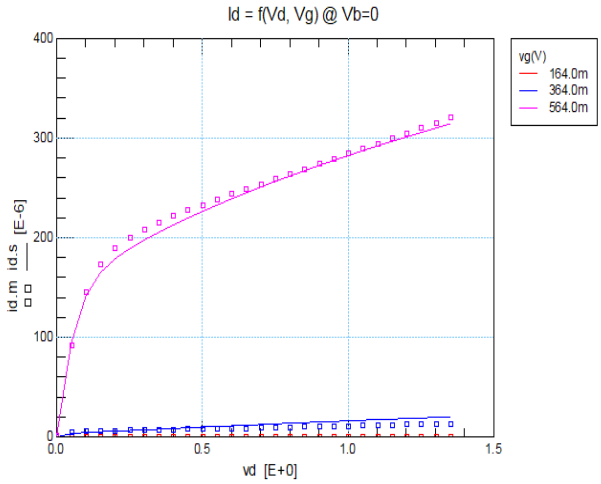
idvd_vbmin, Vb = -1.2V, T = 27°C

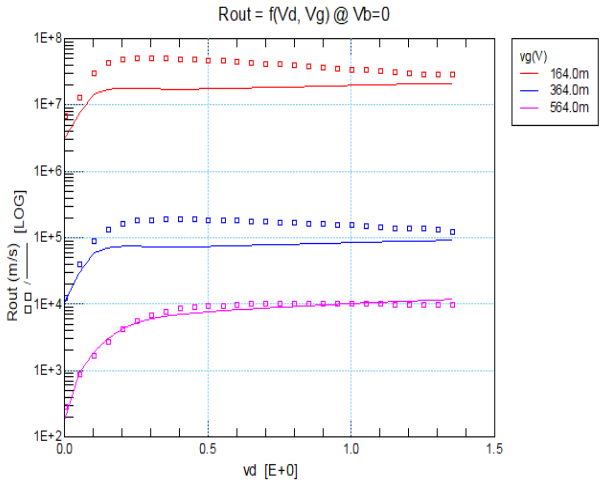


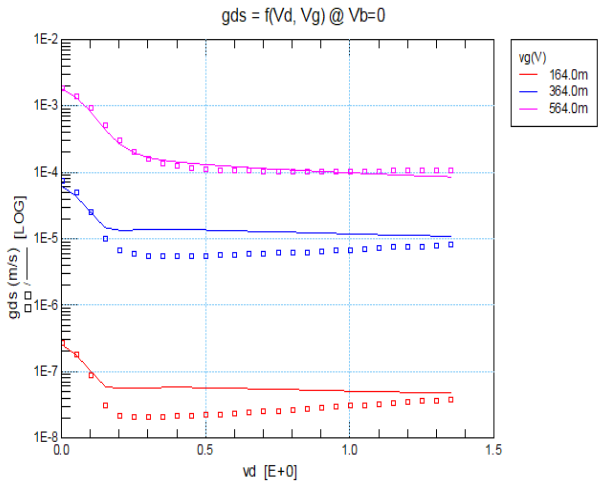




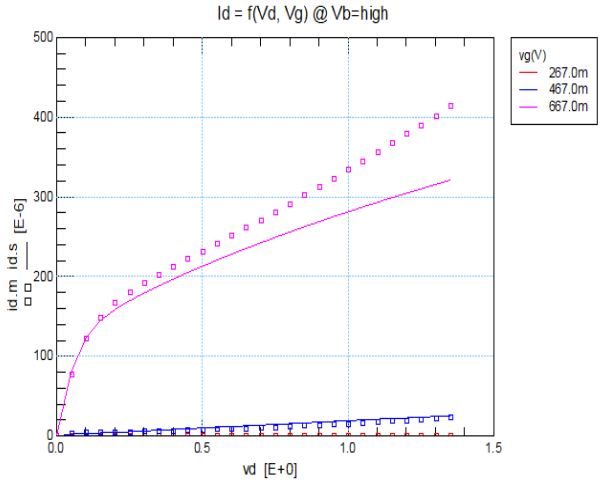
idvd_vth, Vb = 0V, T = 27°C

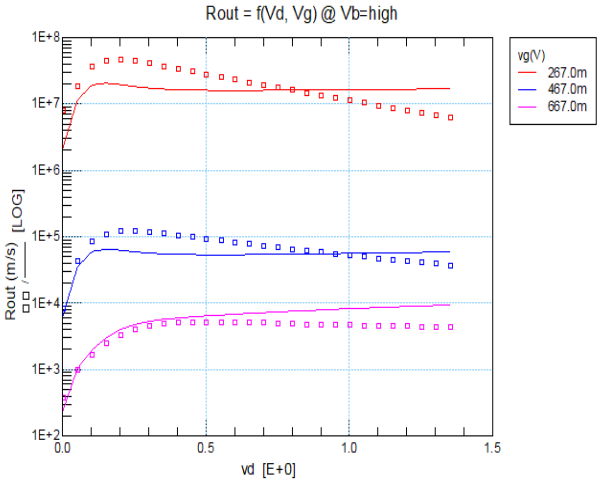


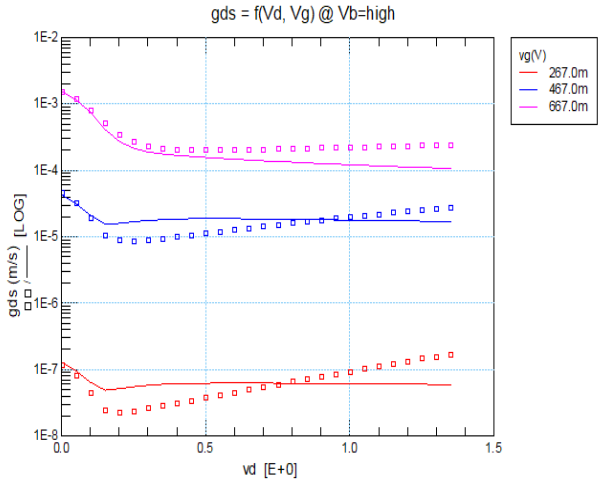




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

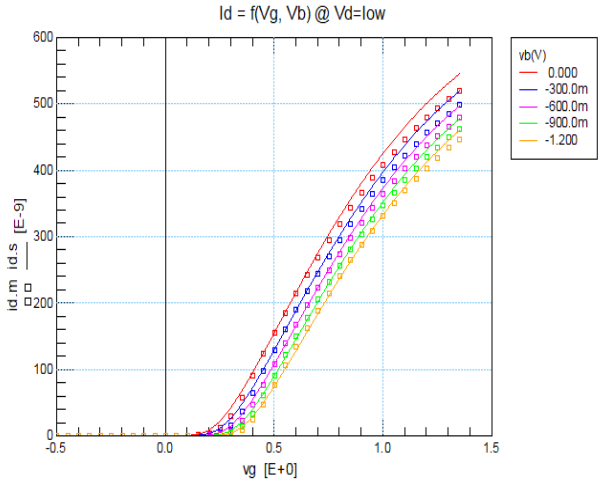


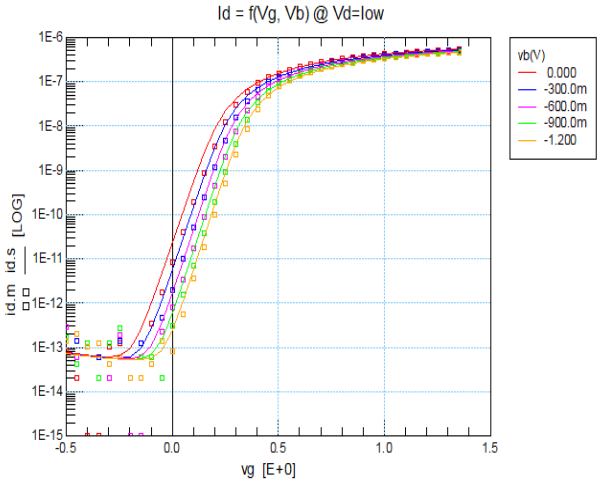


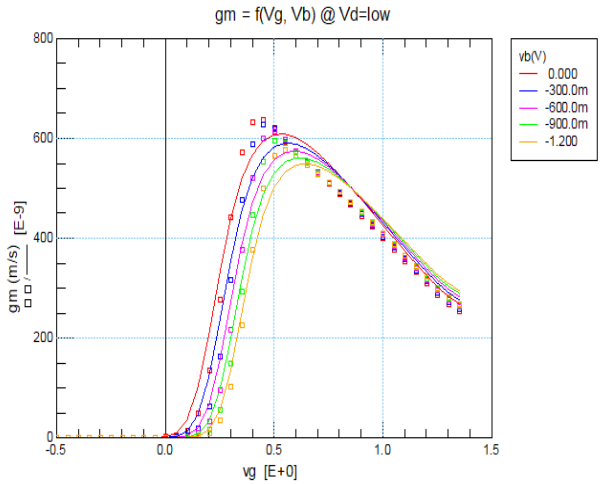


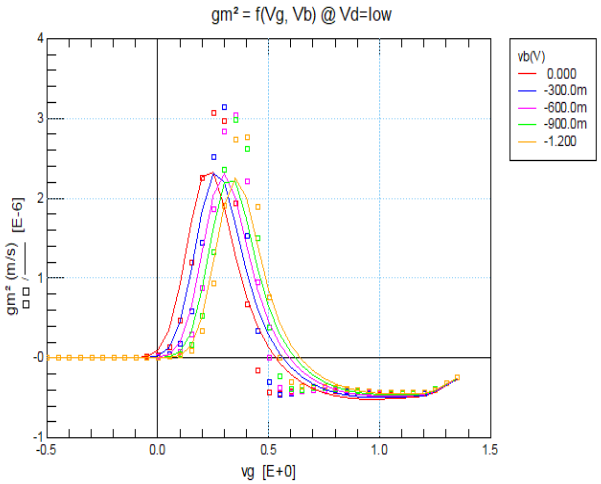
4.15W0u3_L10u0_S545_2

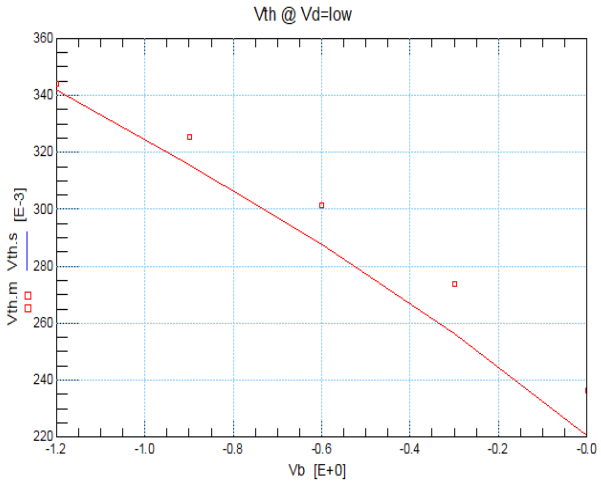
idvg, Vd = 0.05V, T = 27°C

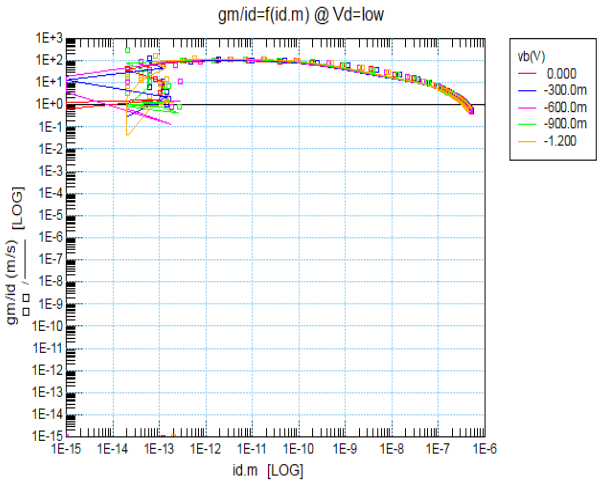


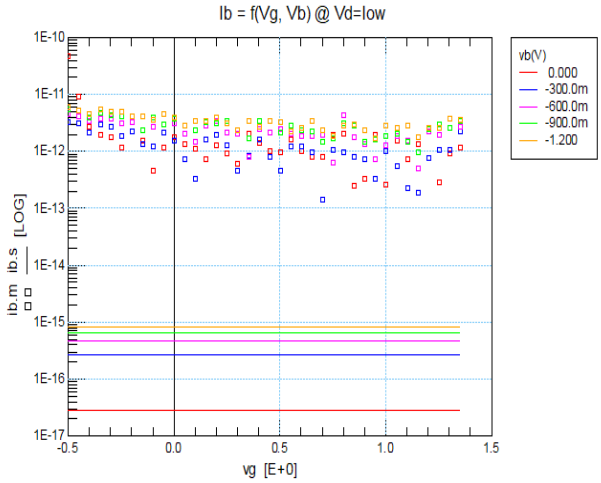


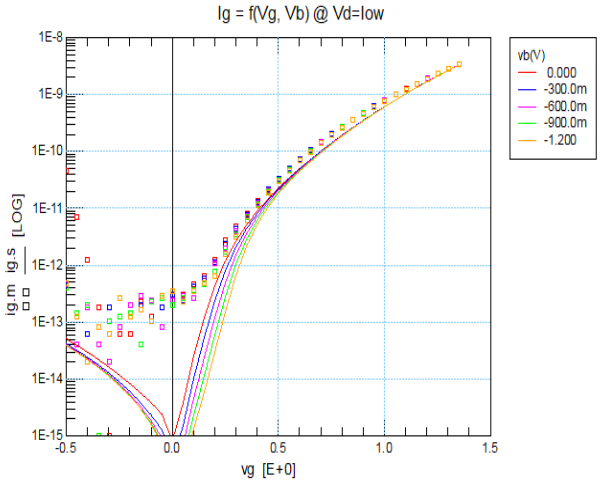




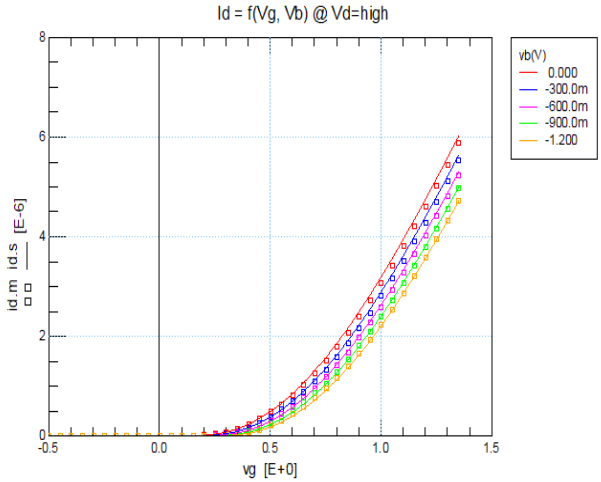


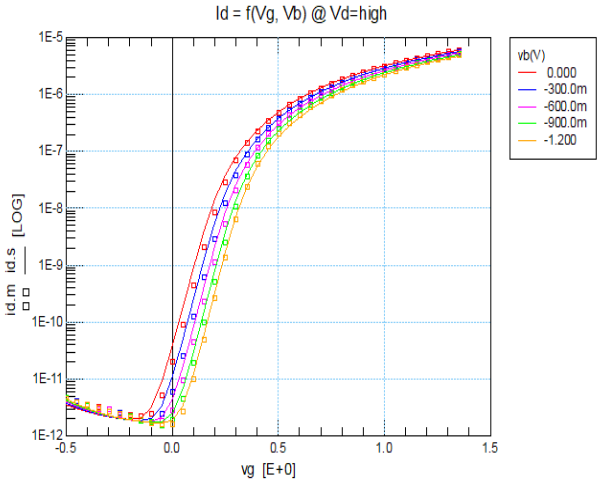


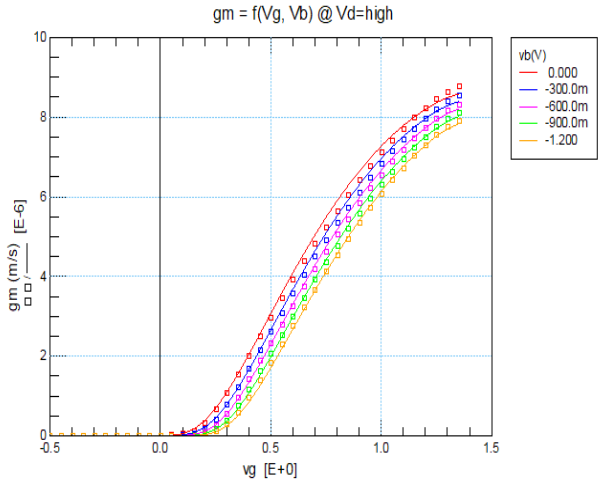


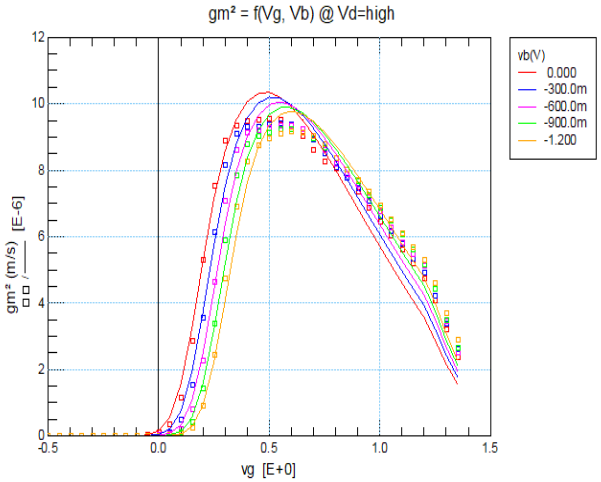


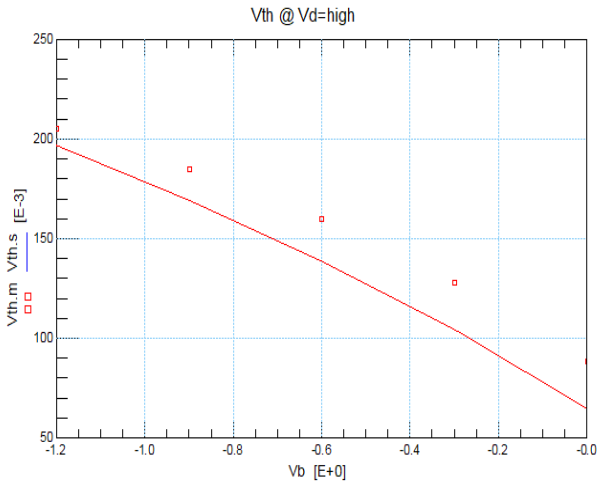
idvg, Vd = 1.2V, T = 27°C

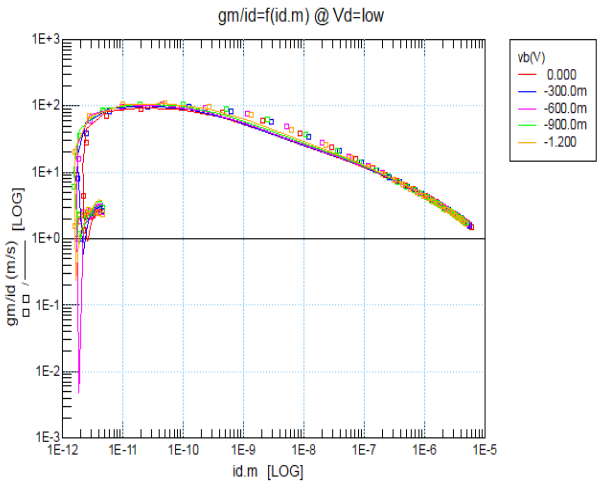


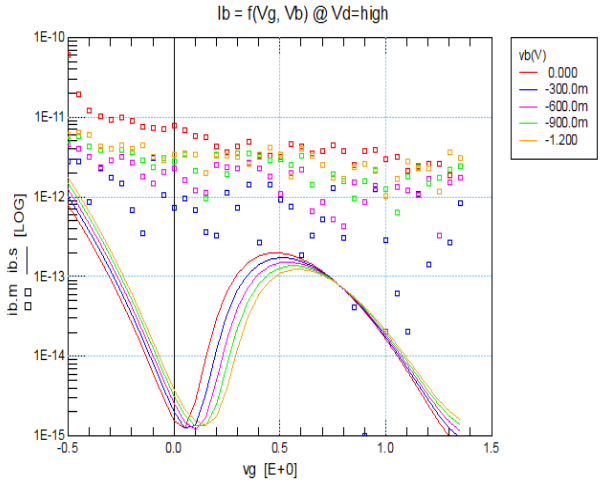


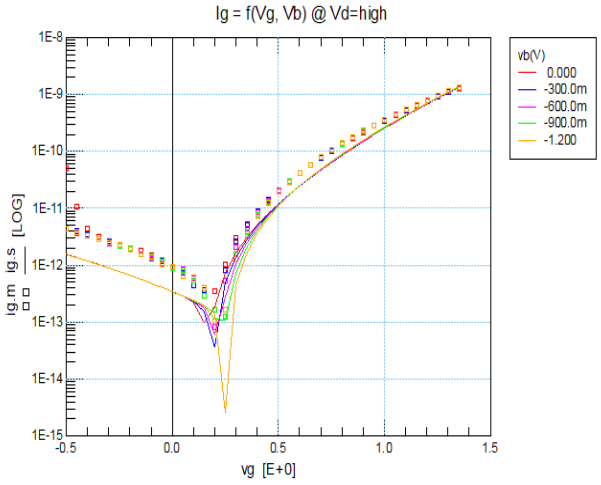




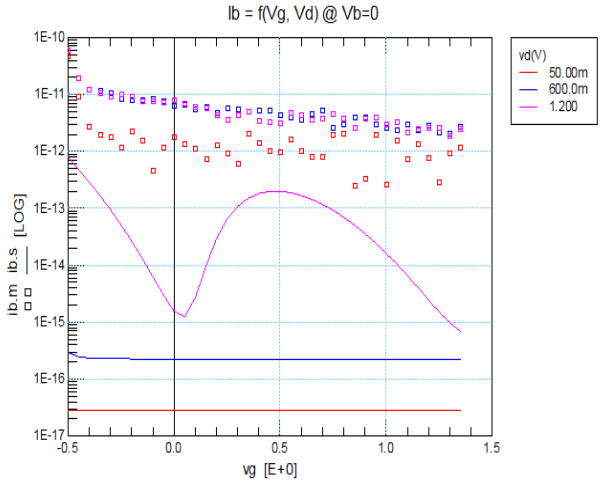


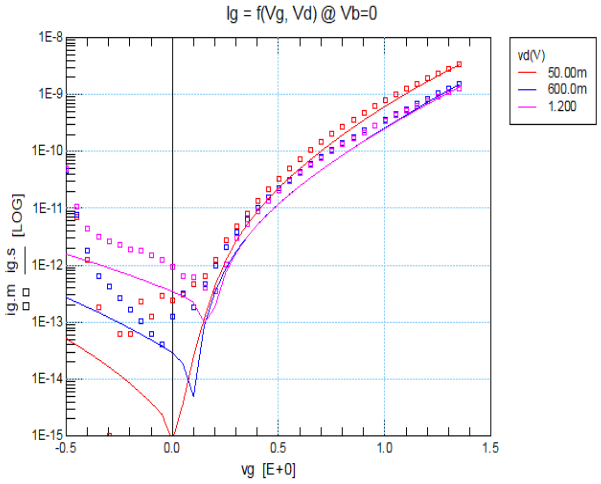




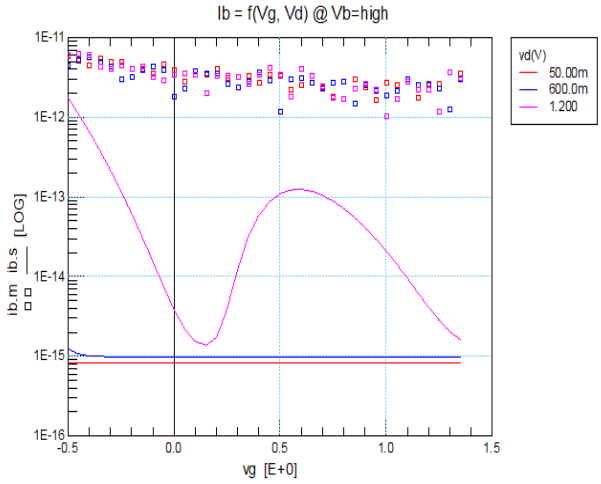


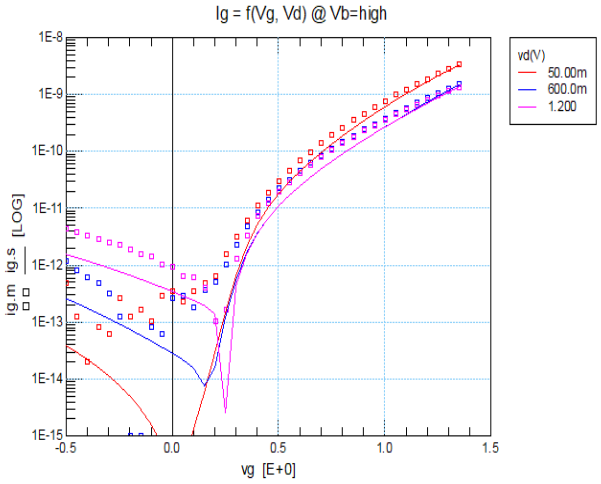
idvg, Ib,Ig, Vb = 0V, T = 27°C



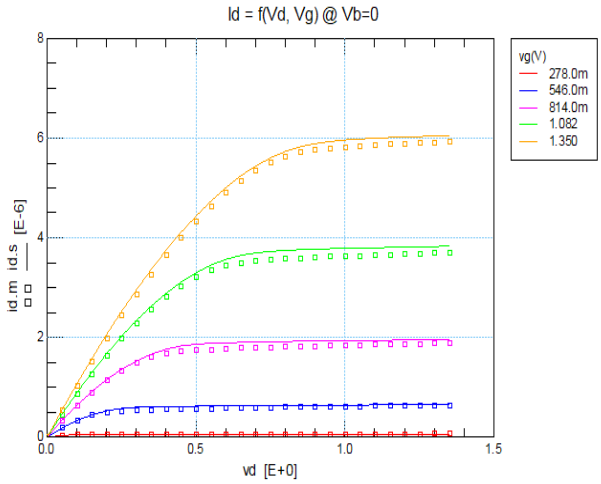


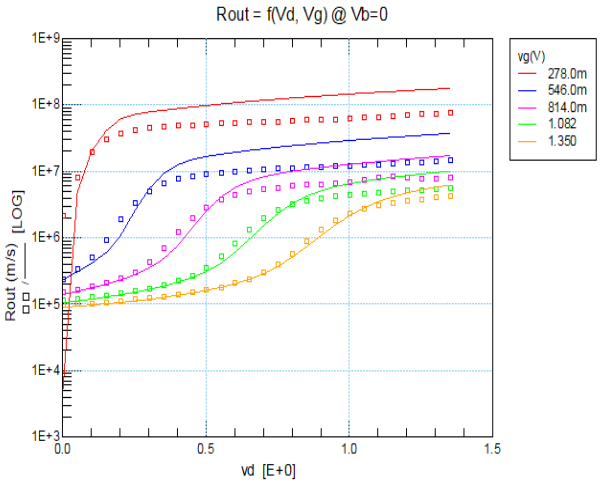
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

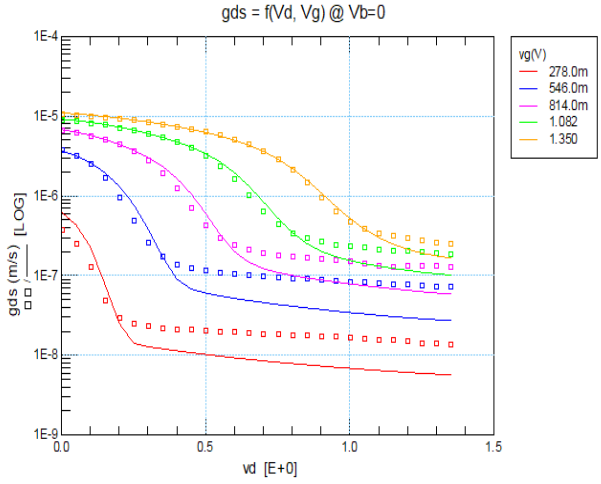




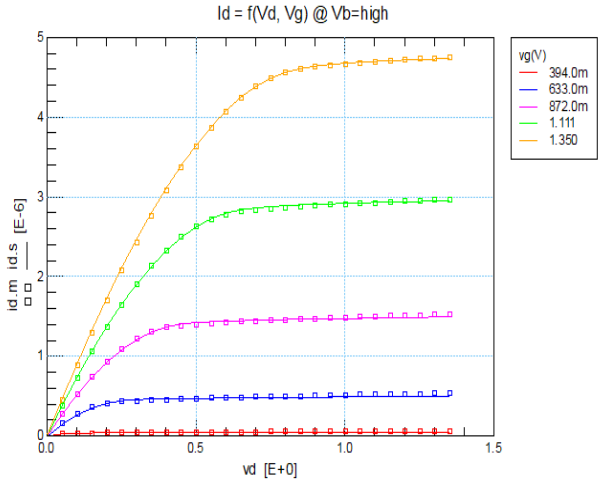
idvd, Vb = 0V, T = 27°C

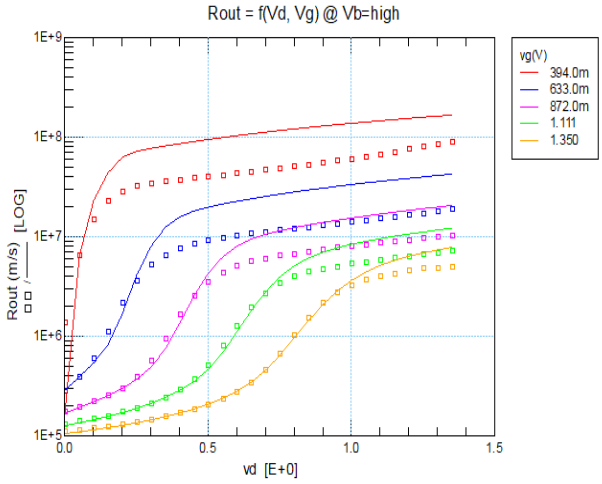


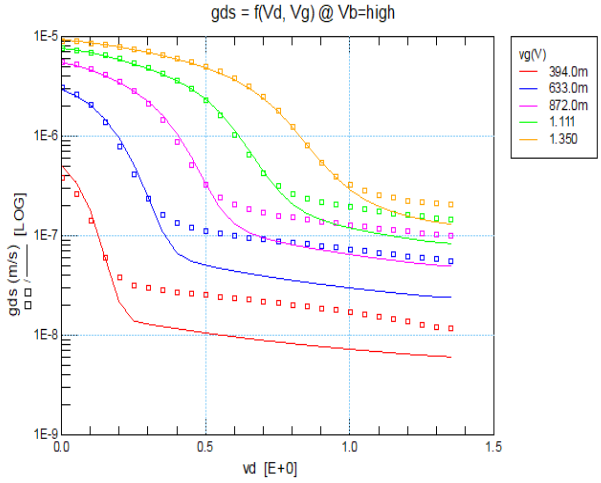




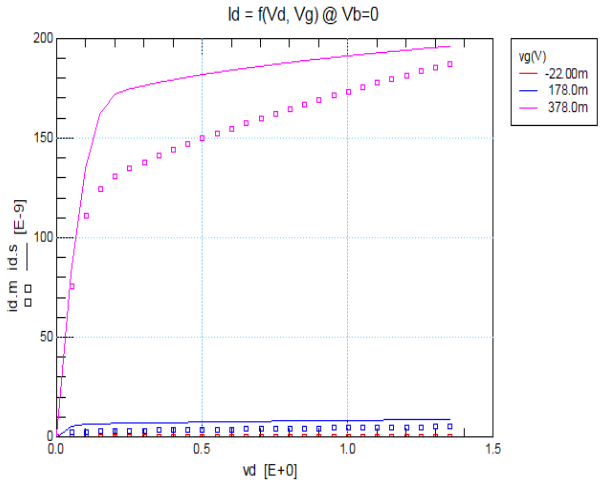
idvd_vbmin, Vb = -1.2V, T = 27°C

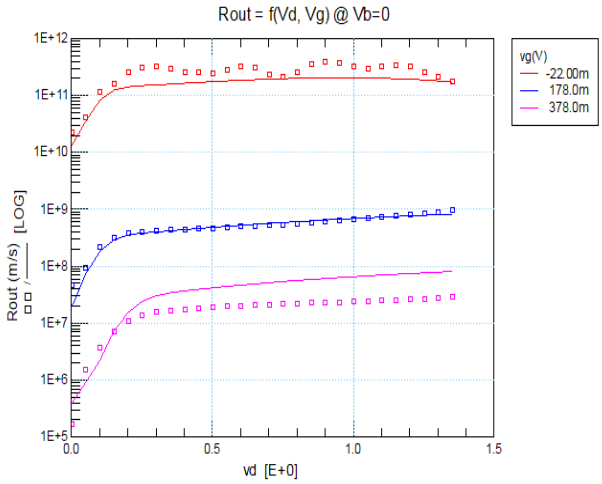


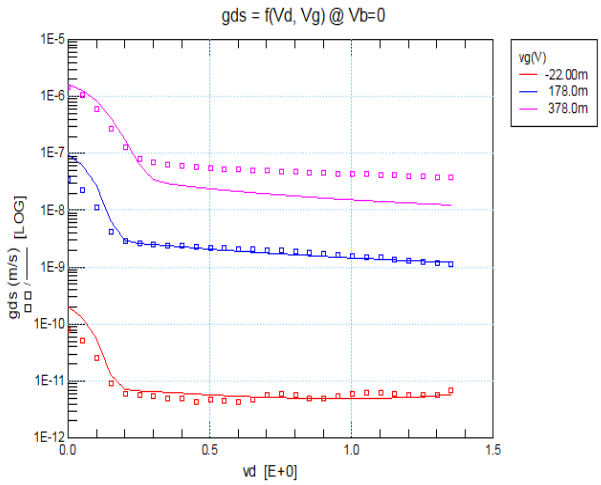




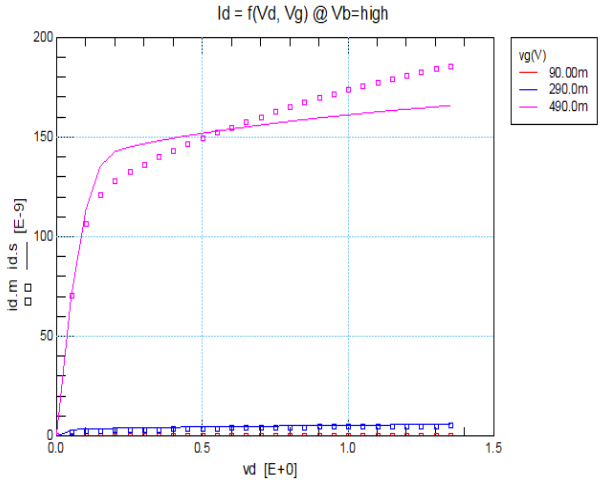
idvd_vth, Vb = 0V, T = 27°C

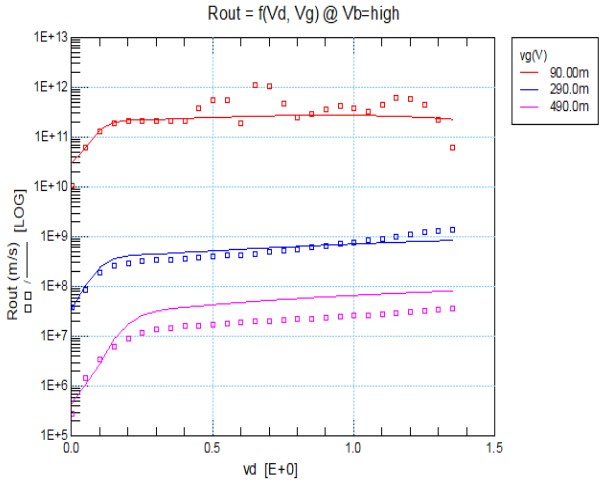


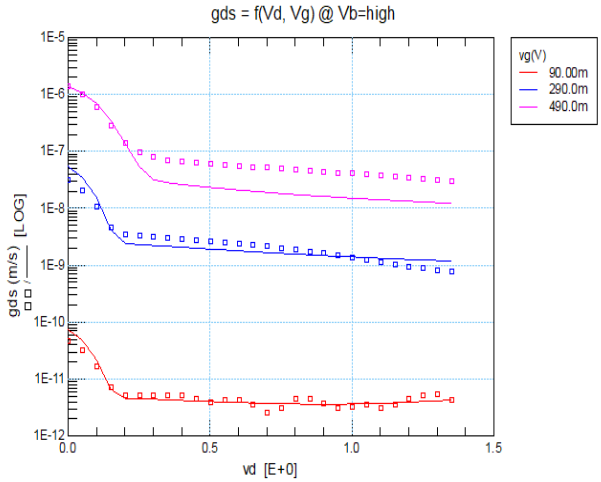




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

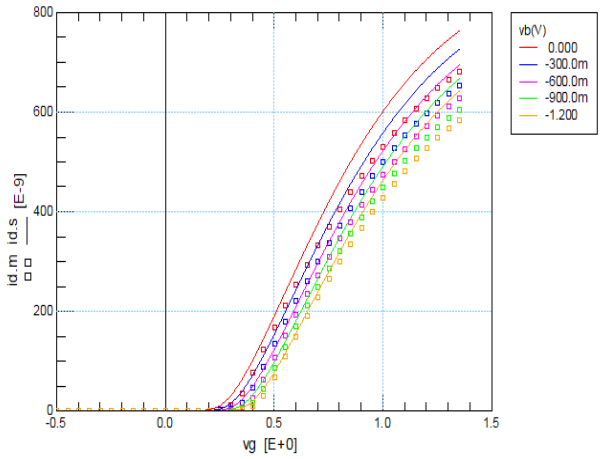




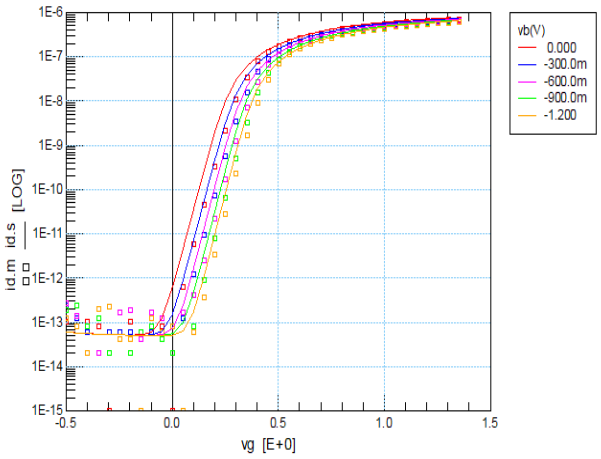


idvg, Vd = 0.05V, T = -40°C

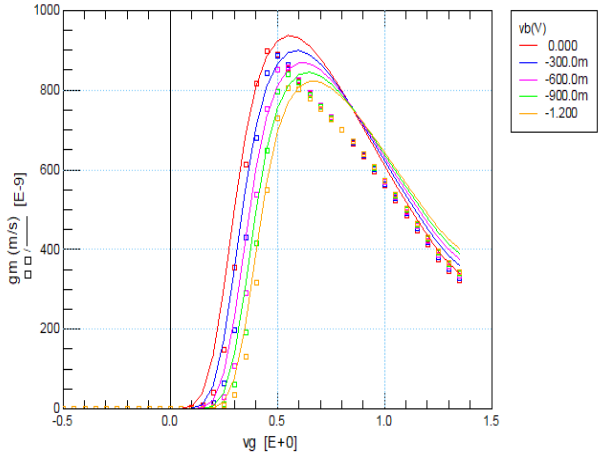
Id = f(Vg, Vb) @ Vd=low



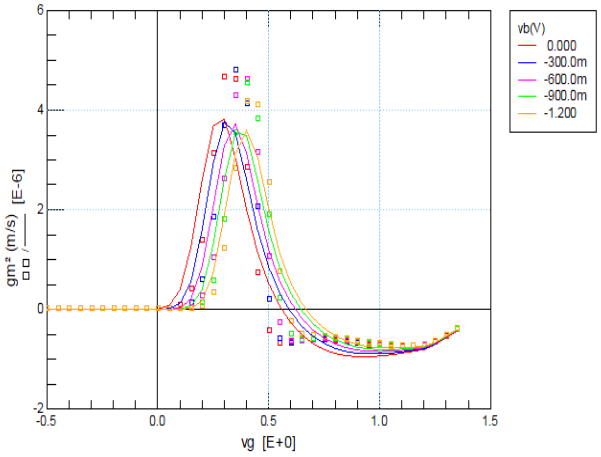
Id = f(Vg, Vb) @ Vd=low



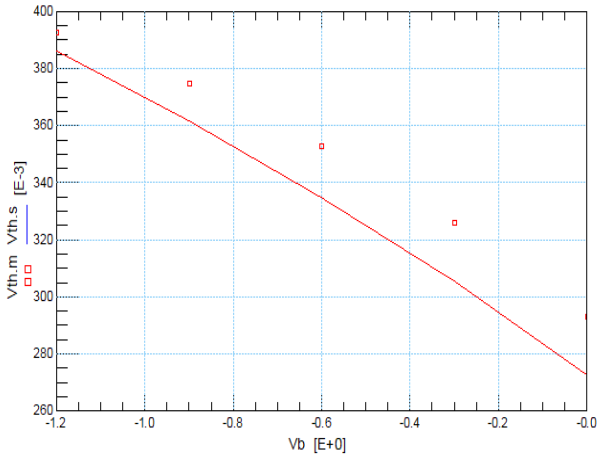
gm = f(Vg, Vb) @ Vd=low



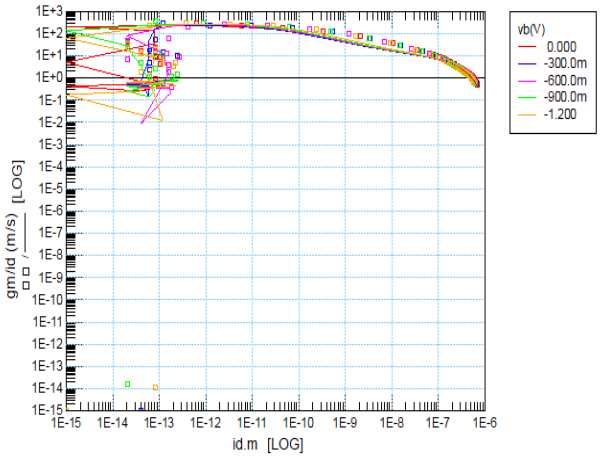
gm² = f(Vg, Vb) @ Vd=low

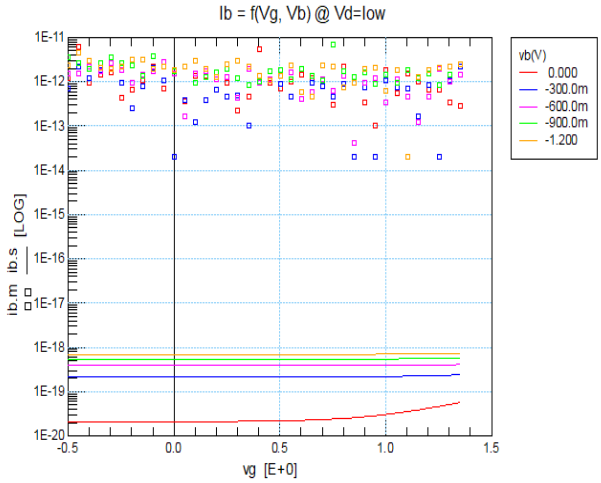


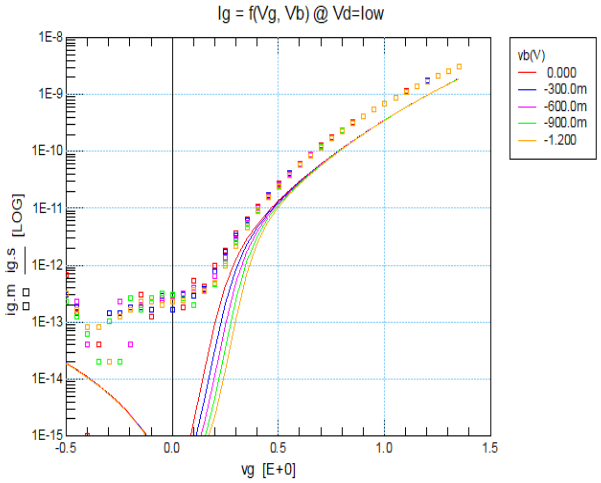
Vth @ Vd=low



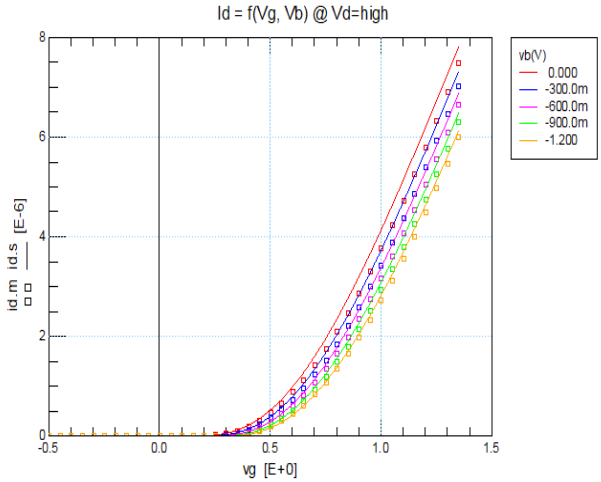
gm/id=f(id.m) @ Vd=low

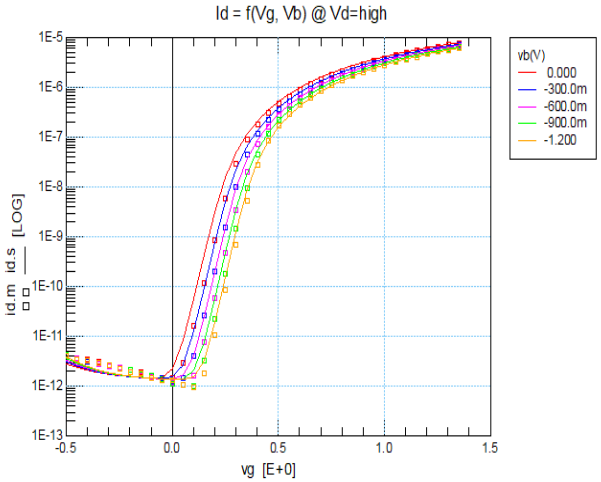


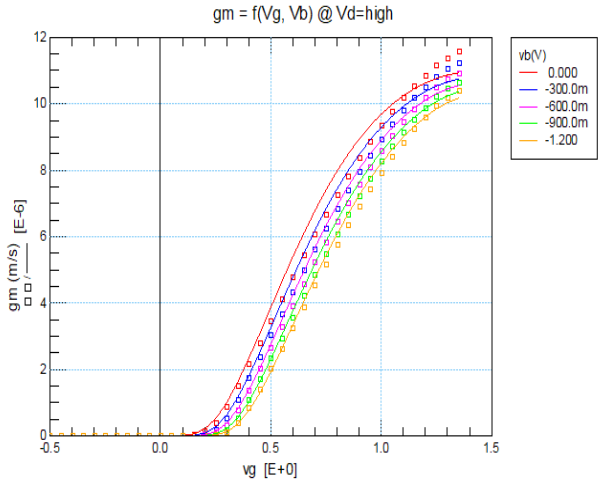


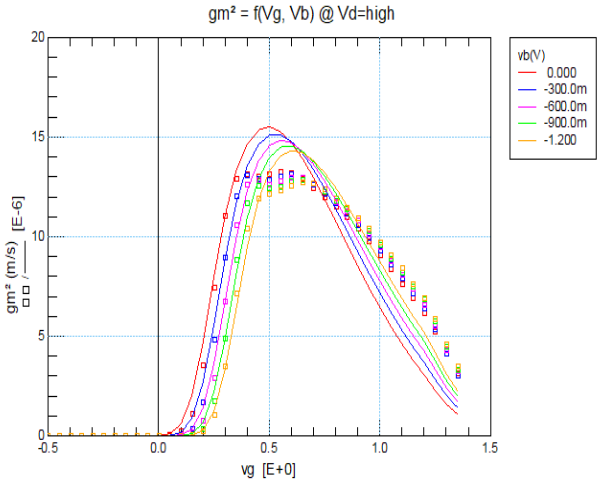


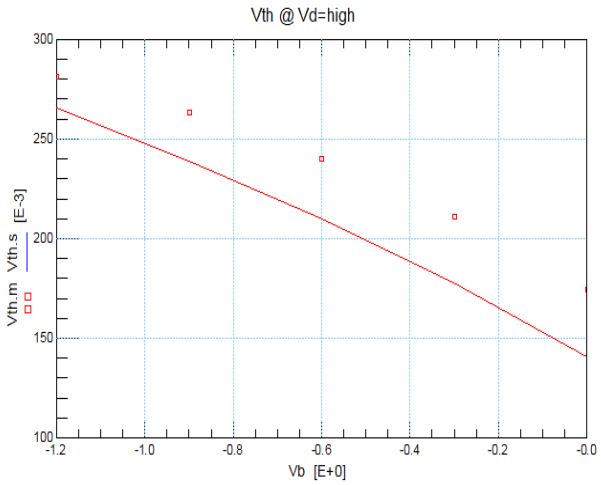
idvg, Vd = 1.2V, T = -40°C

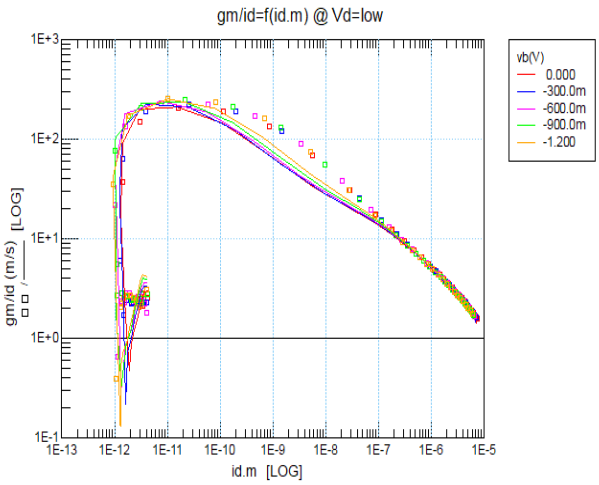


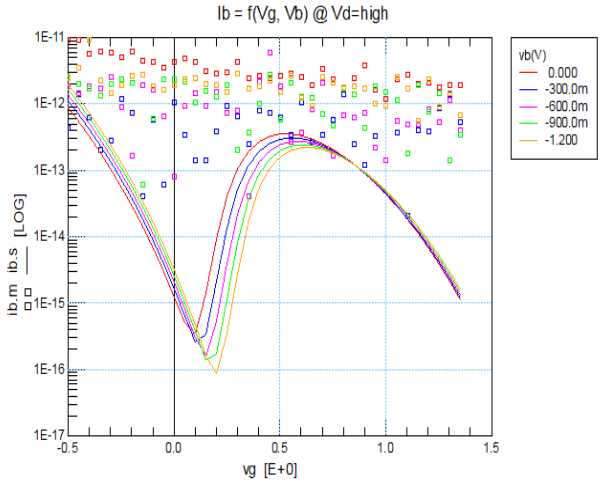


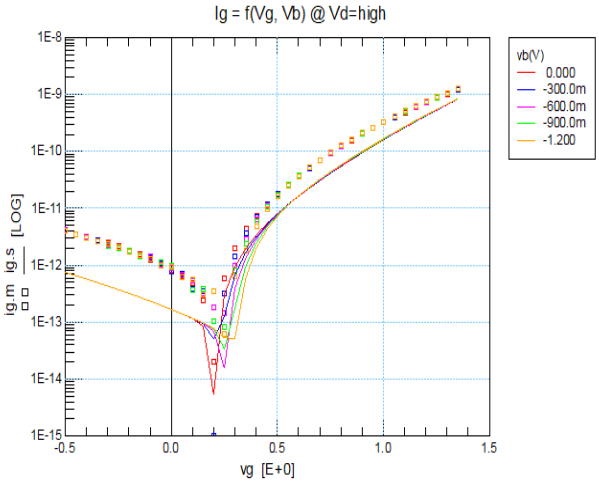




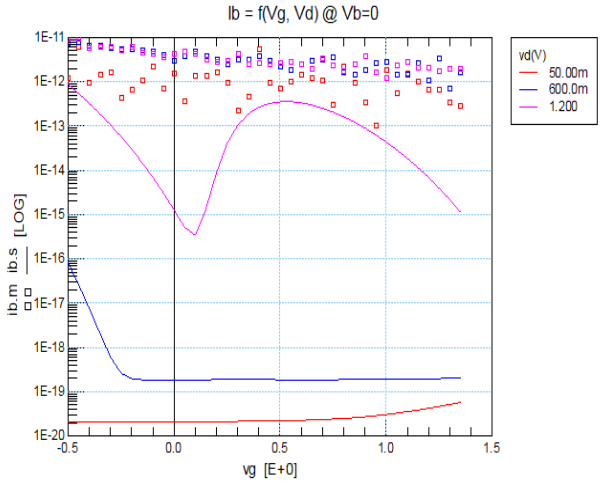


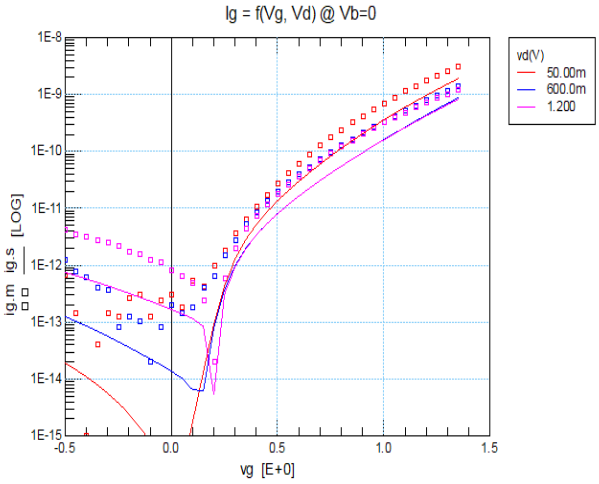




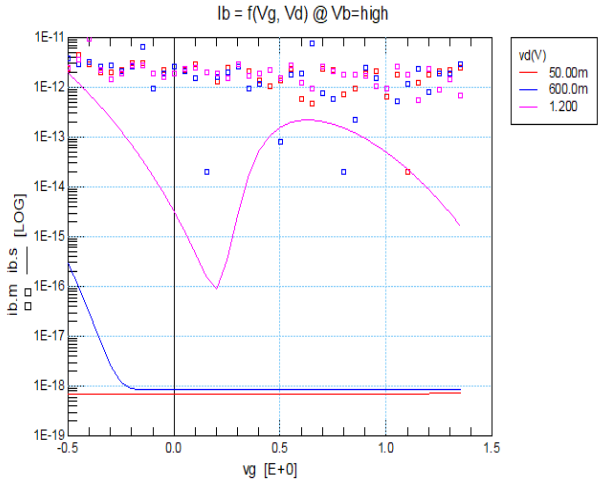


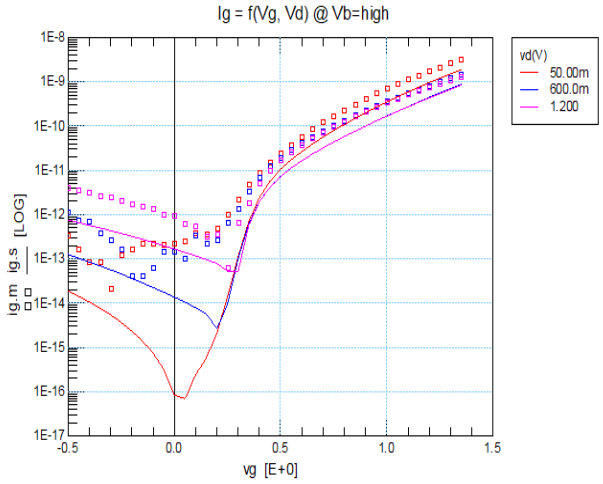
idvg, Ib,Ig, Vb = 0V, T = -40°C



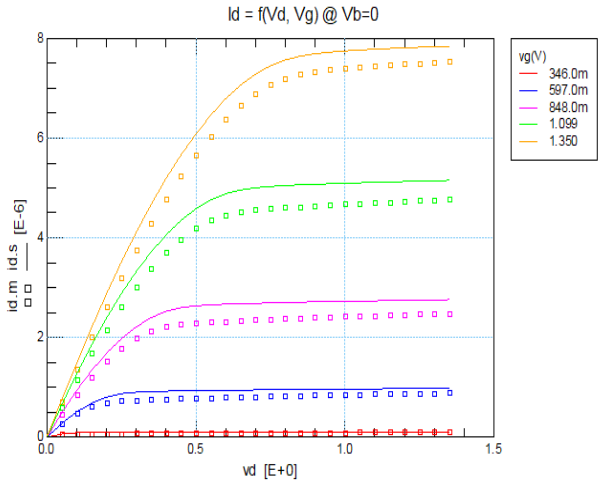


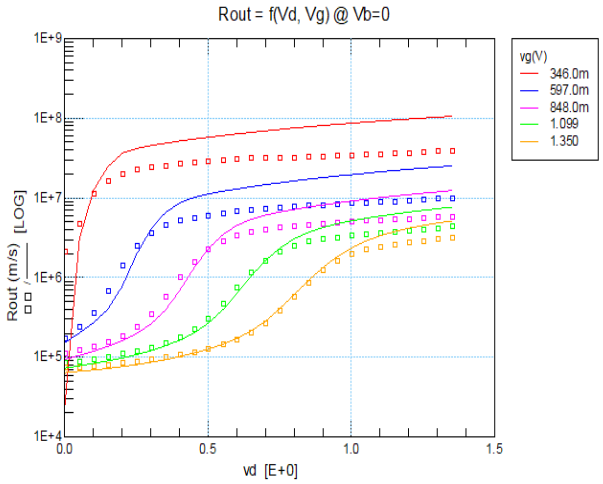
idvg, Ib,Ig, Vb = -1.2V, T = -40°C

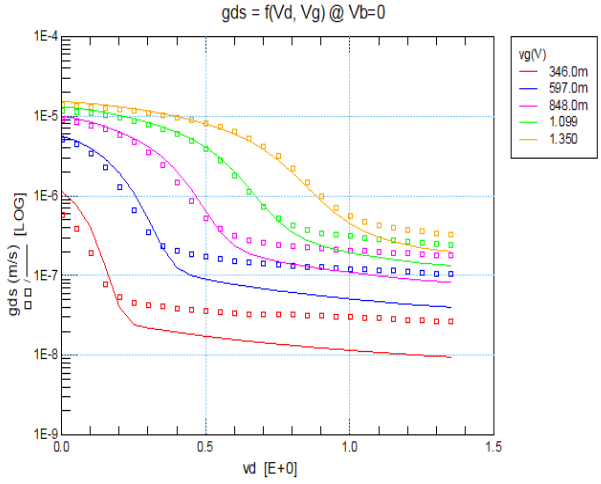




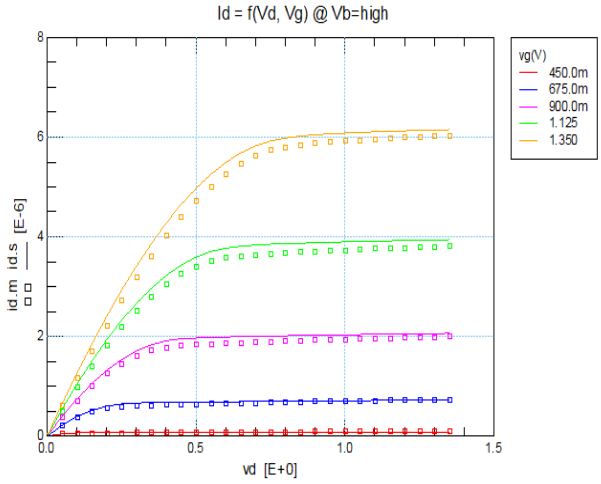
idvd, Vb = 0V, T = -40°C

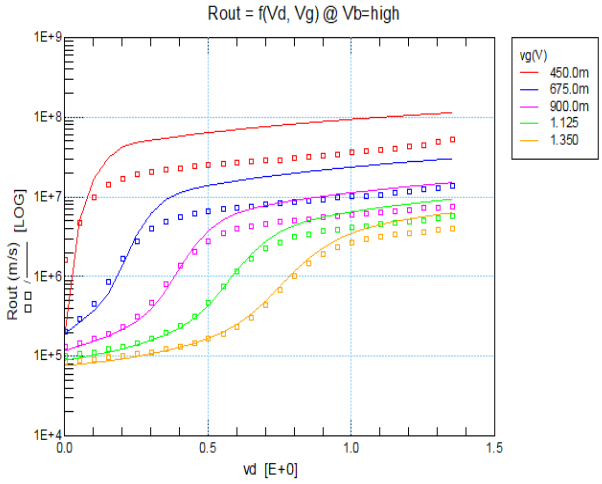


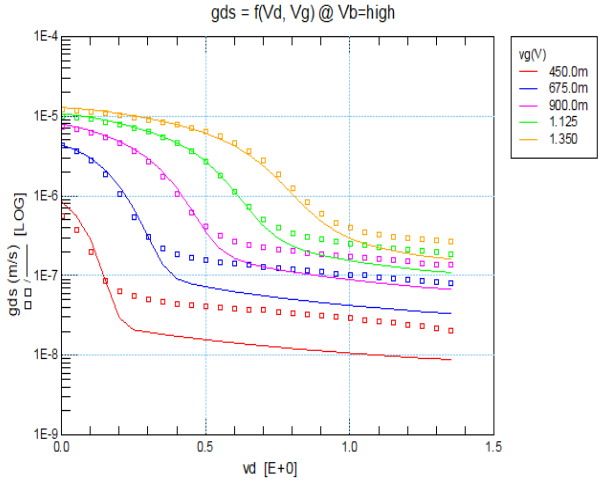




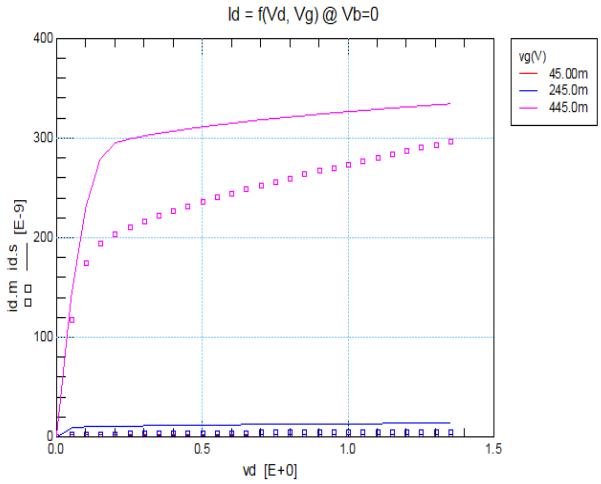
idvd_vbmin, Vb = -1.2V, T = -40°C

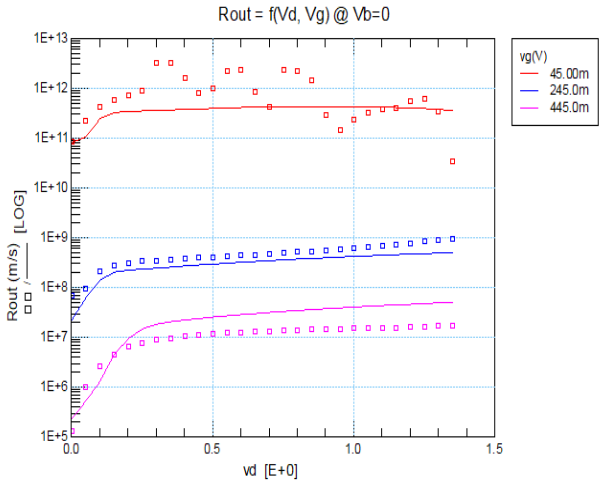


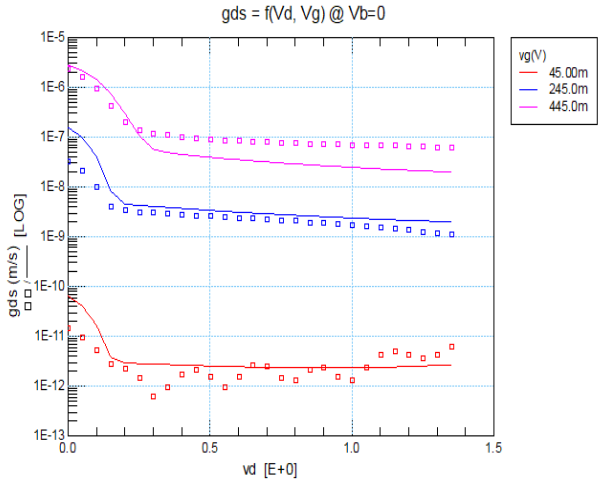




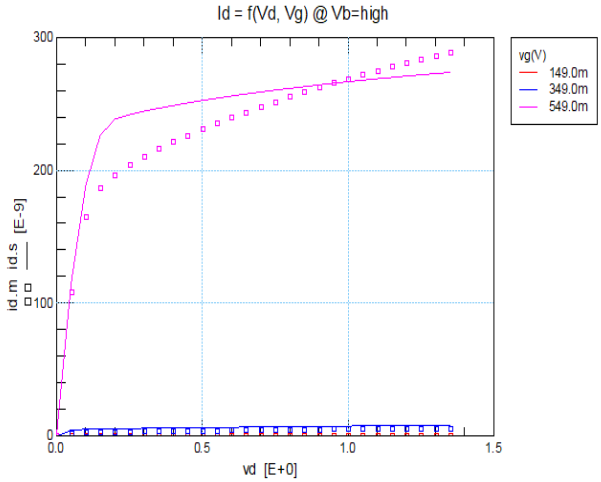
idvd_vth, Vb = 0V, T = -40°C

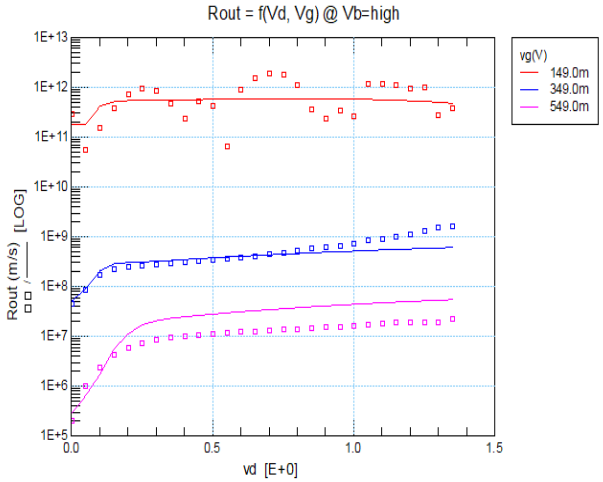


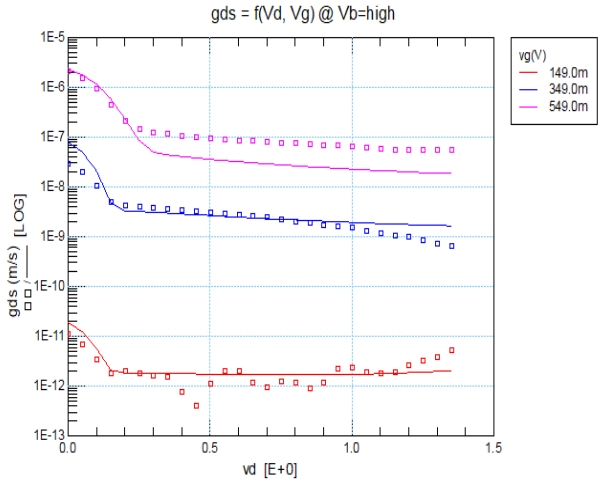




idvd_vth_vbmin, Vb = -1.2V, T = -40°C

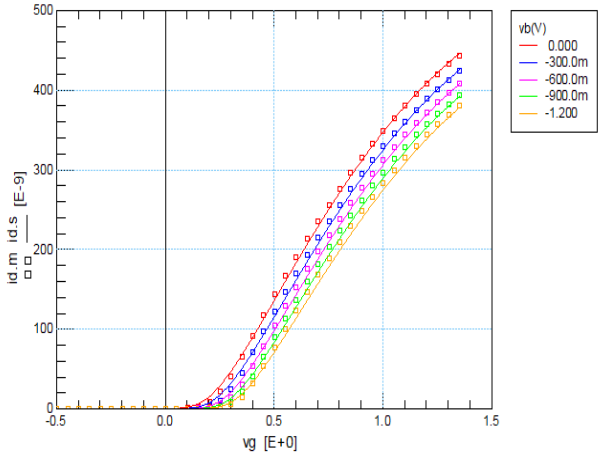




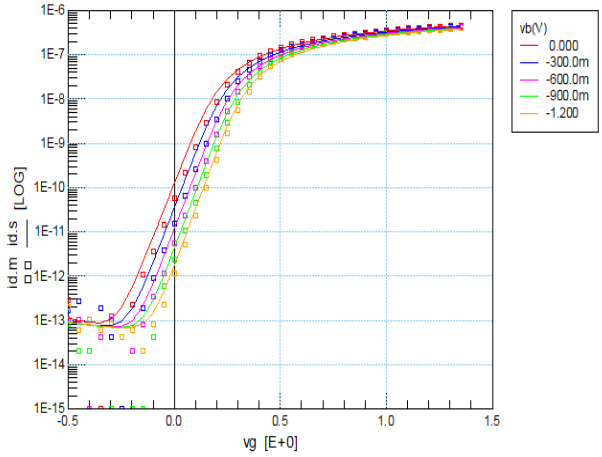


idvg, Vd = 0.05V, T = 70°C

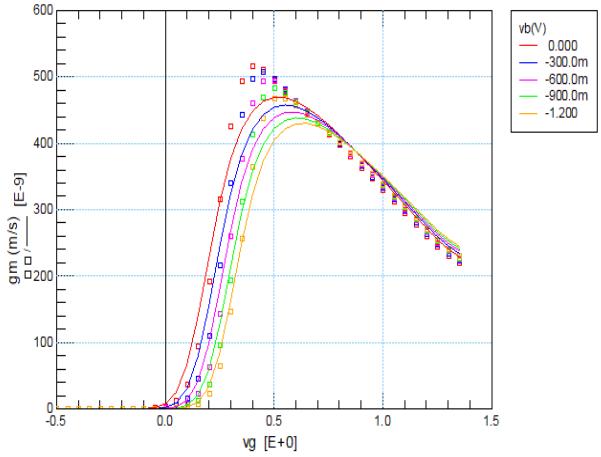
Id = f(Vg, Vb) @ Vd=low



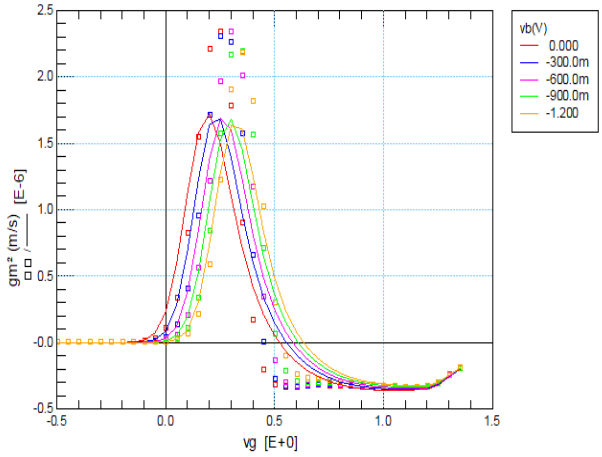
Id = f(Vg, Vb) @ Vd=low



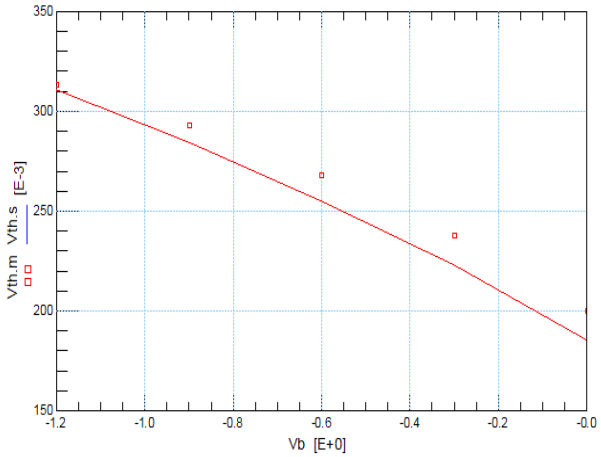
gm = f(Vg, Vb) @ Vd=low



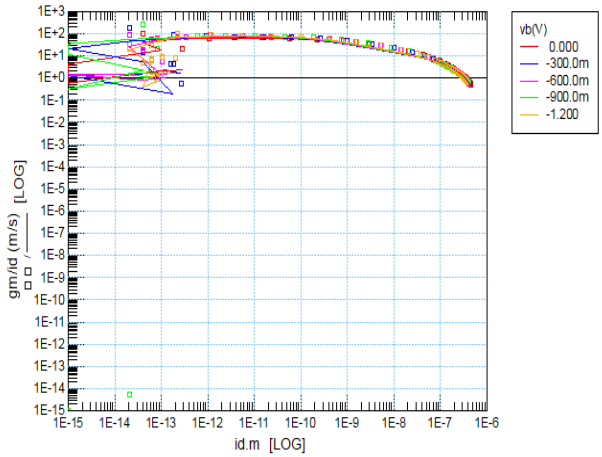
gm² = f(Vg, Vb) @ Vd=low

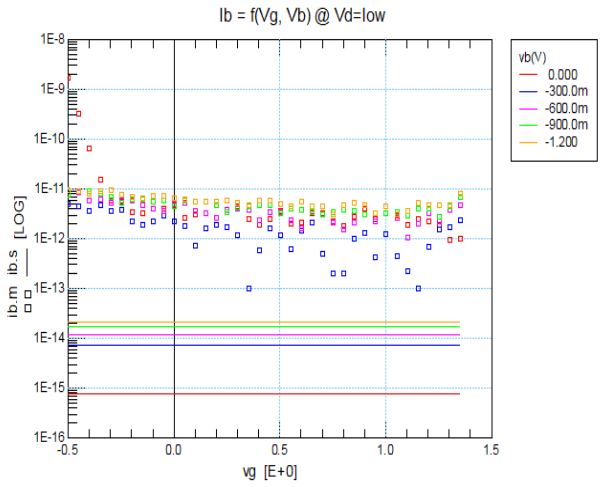


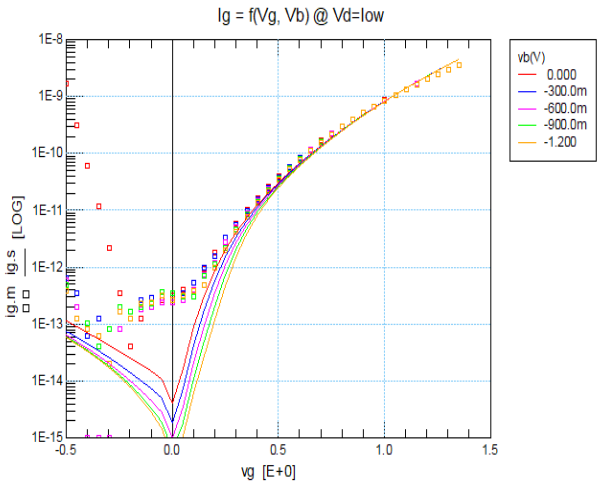
Vth @ Vd=low



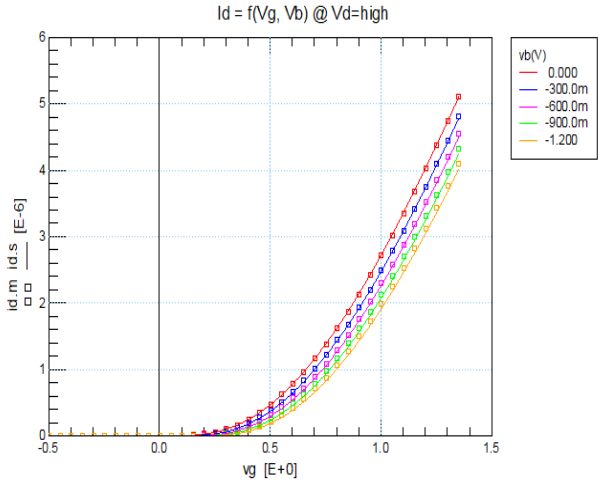
gm/id=f(id.m) @ Vd=low

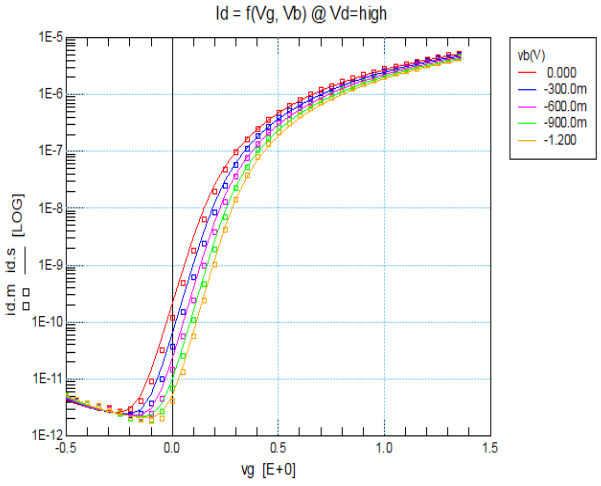


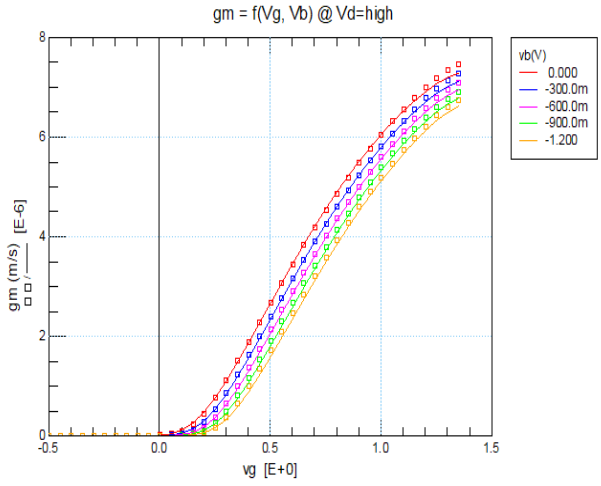


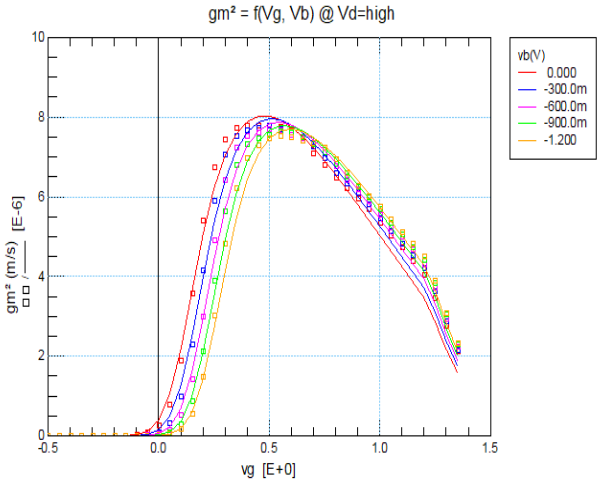


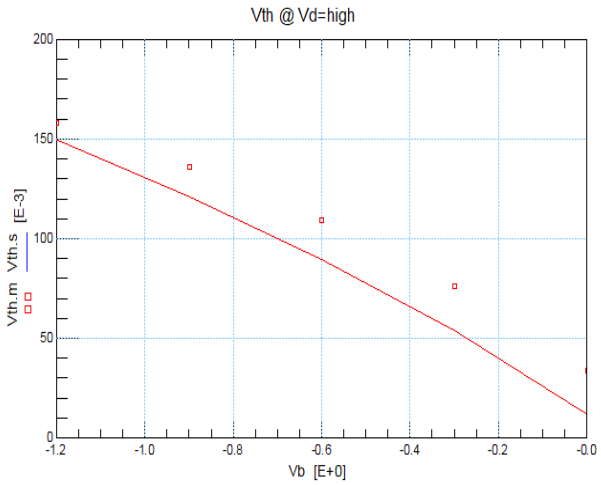
idvg, Vd = 1.2V, T = 70°C

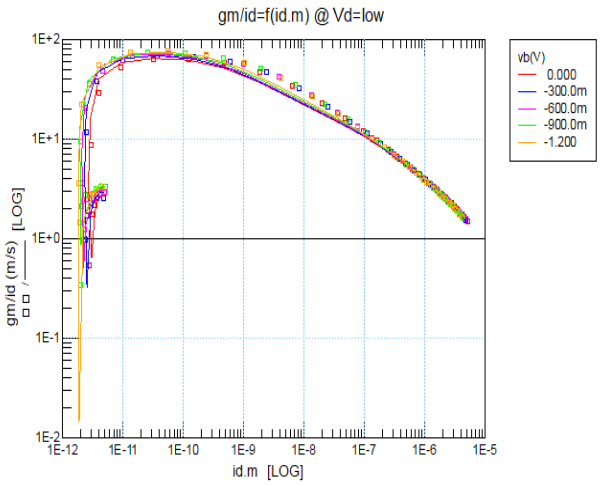


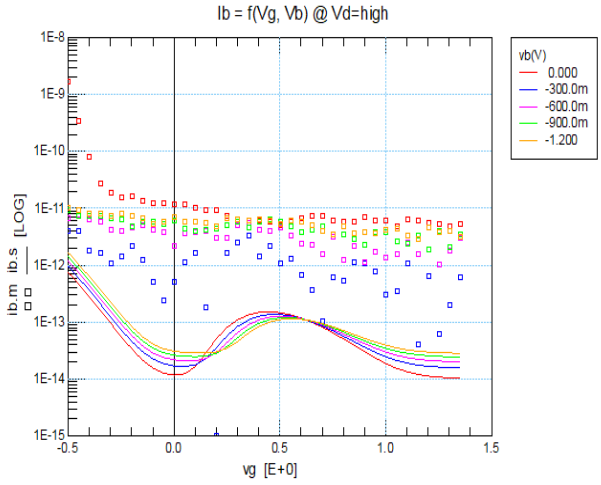


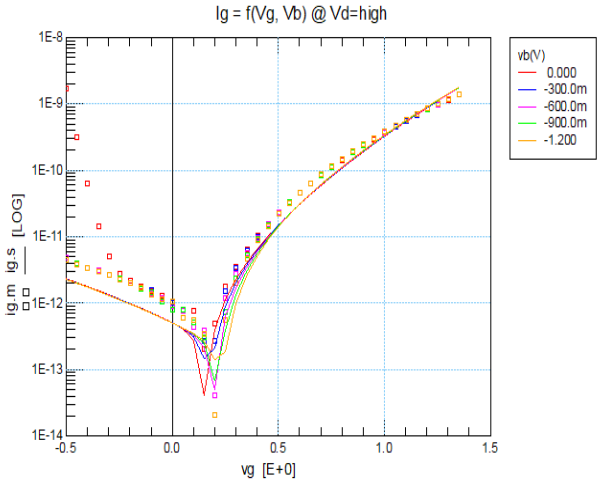




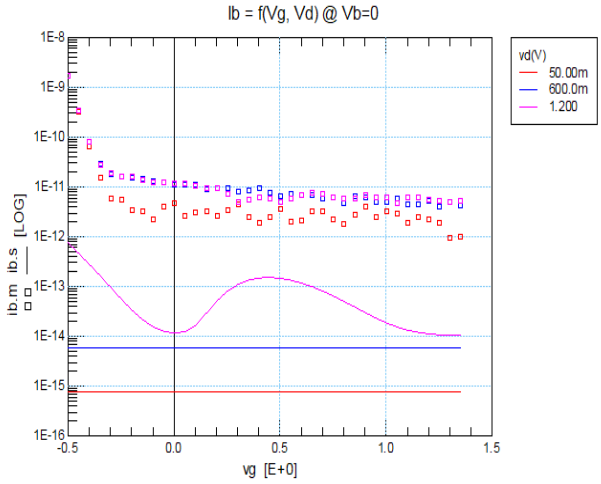


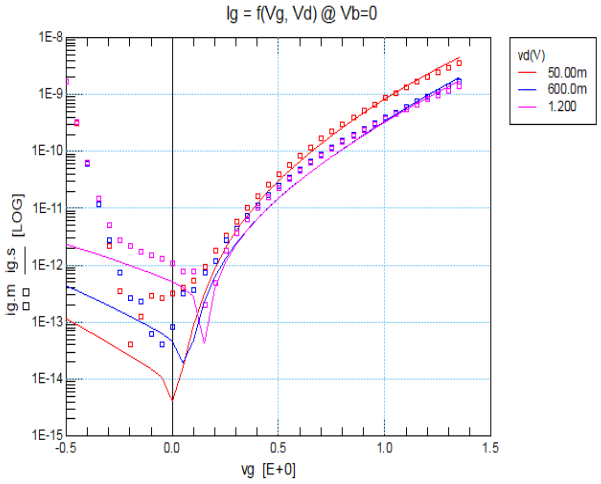




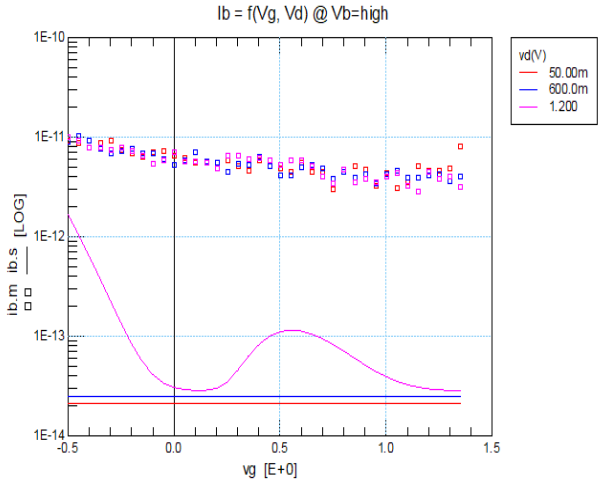


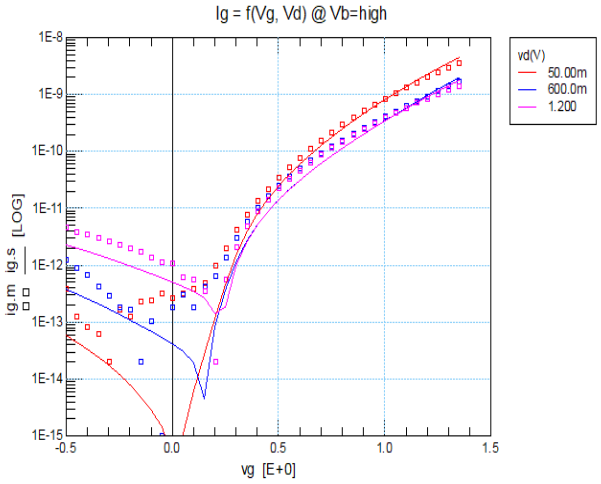
idvg, Ib,Ig, Vb = 0V, T = 70°C



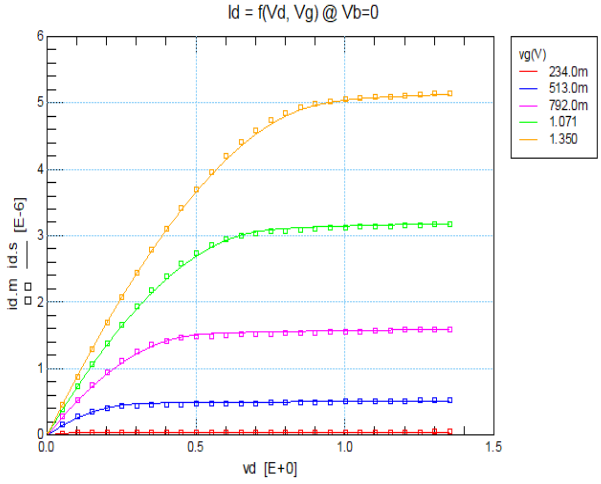


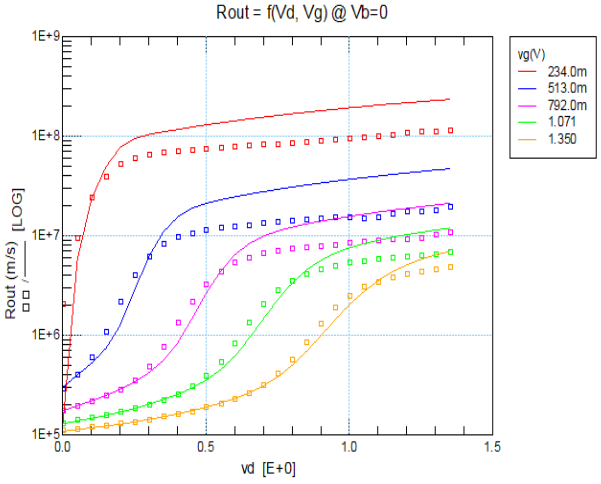
idvg, Ib,Ig, Vb = -1.2V, T = 70°C

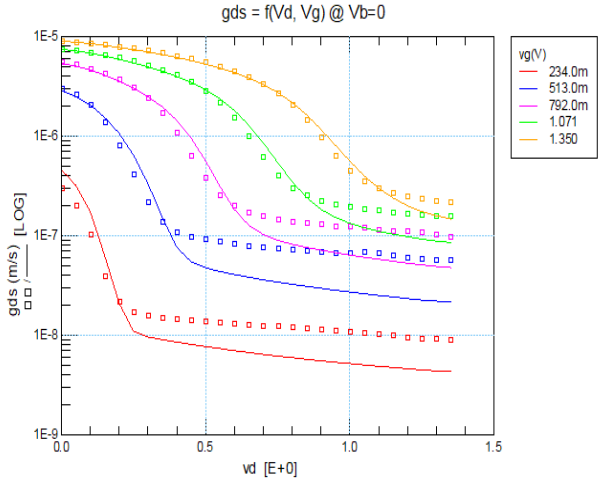




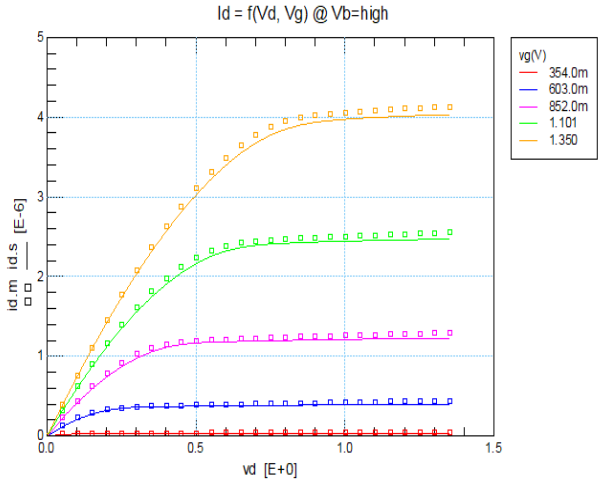
idvd, Vb = 0V, T = 70°C

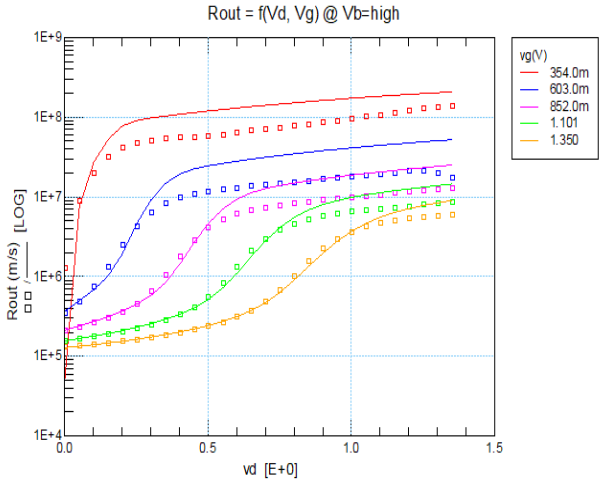


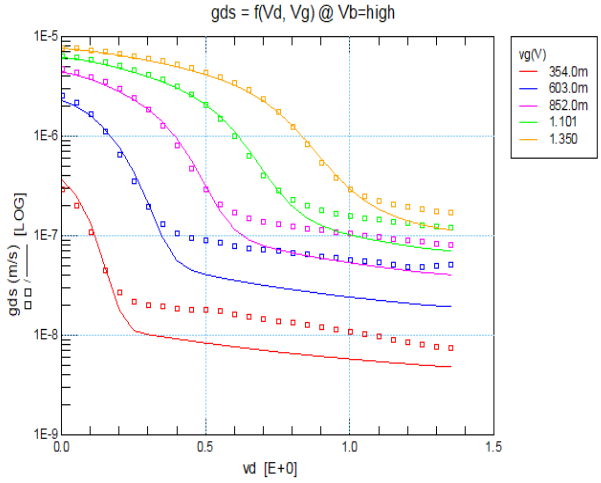




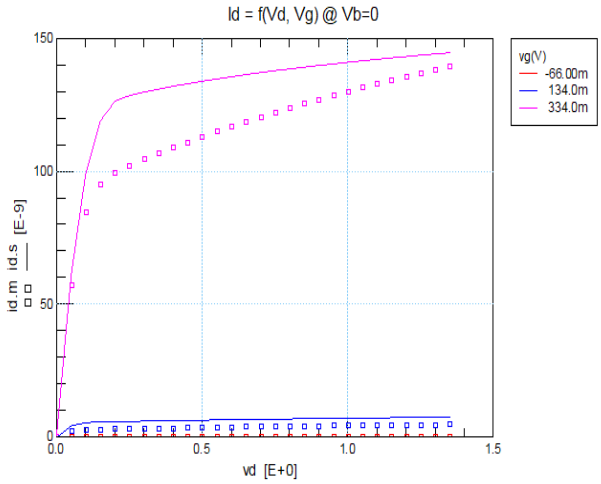
idvd_vbmin, Vb = -1.2V, T = 70°C

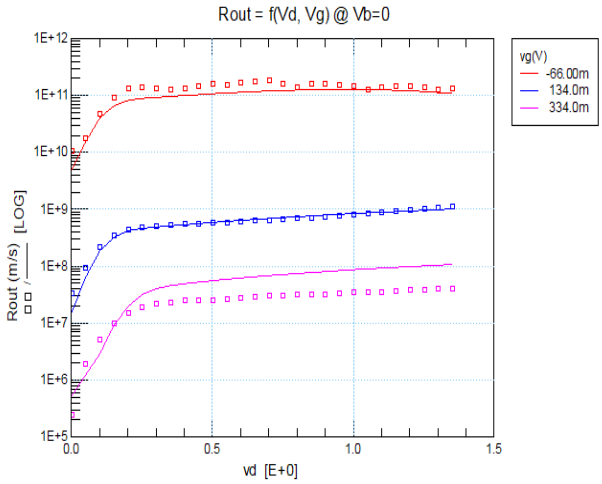


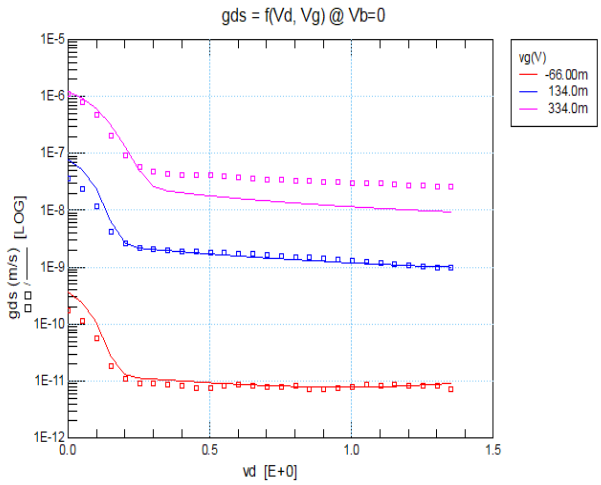




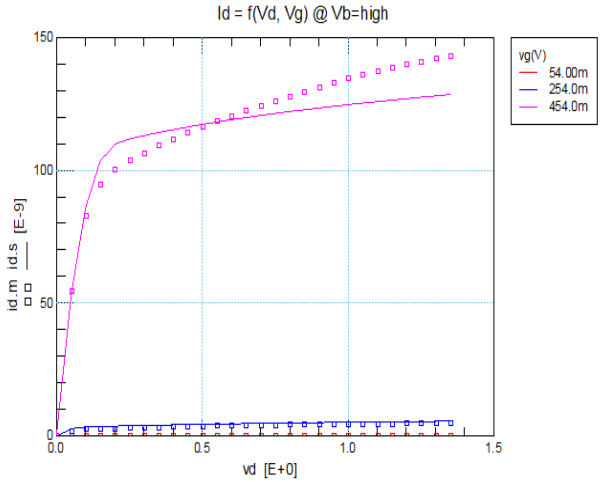
idvd_vth, Vb = 0V, T = 70°C

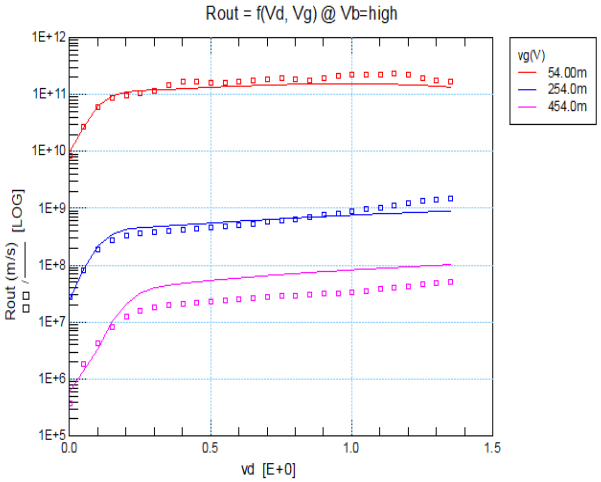


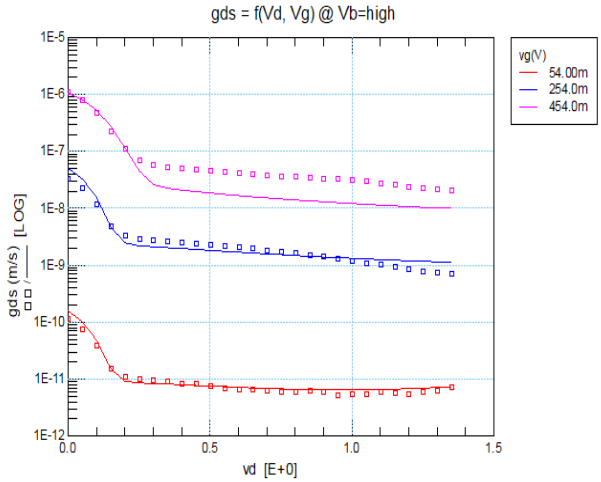




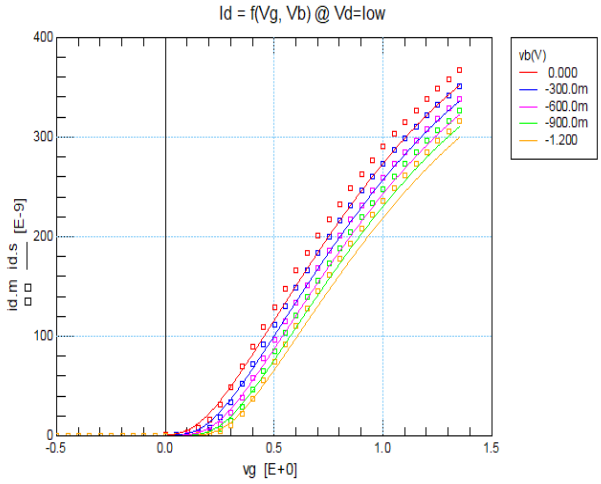
idvd_vth_vbmin, Vb = -1.2V, T = 70°C

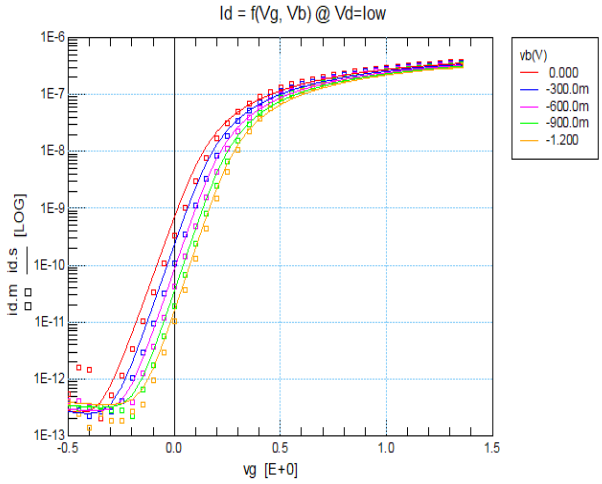


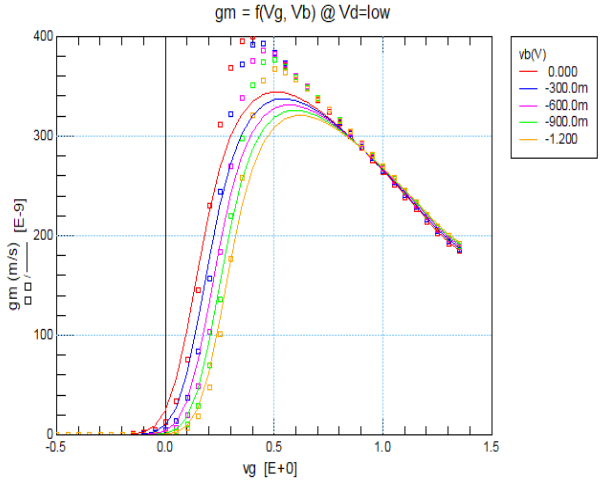


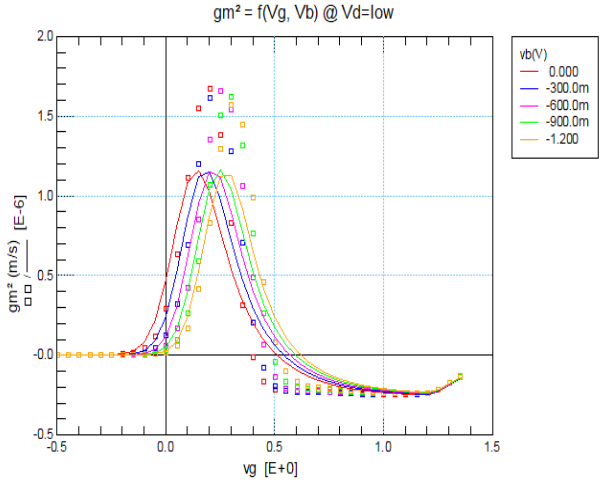


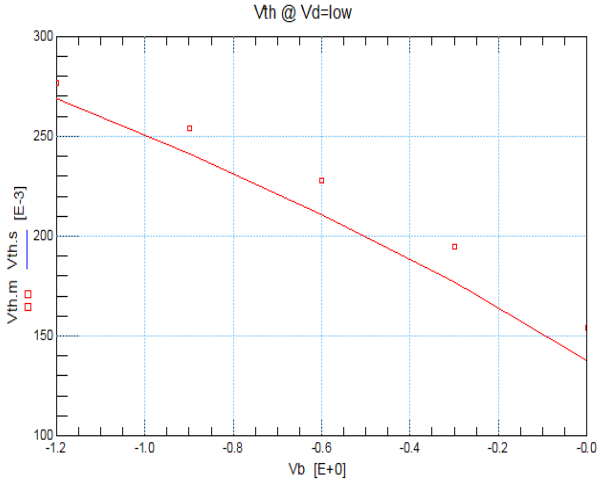
idvg, Vd = 0.05V, T = 125°C

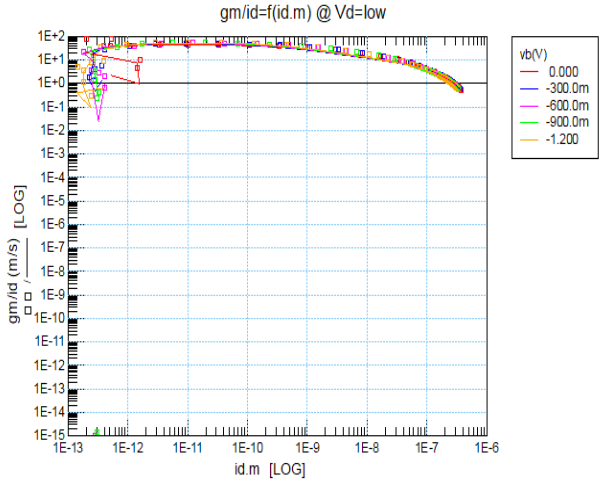


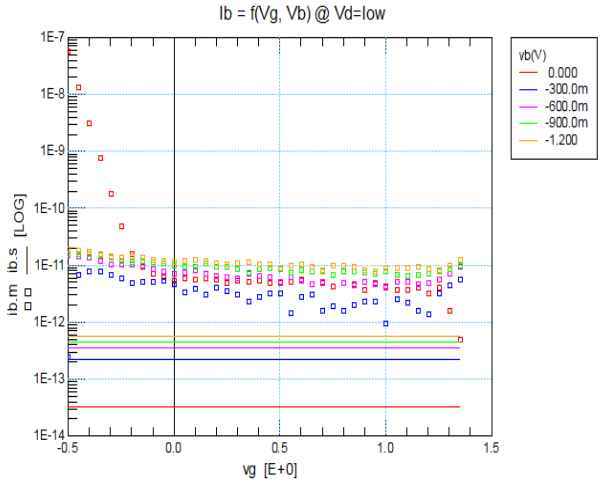


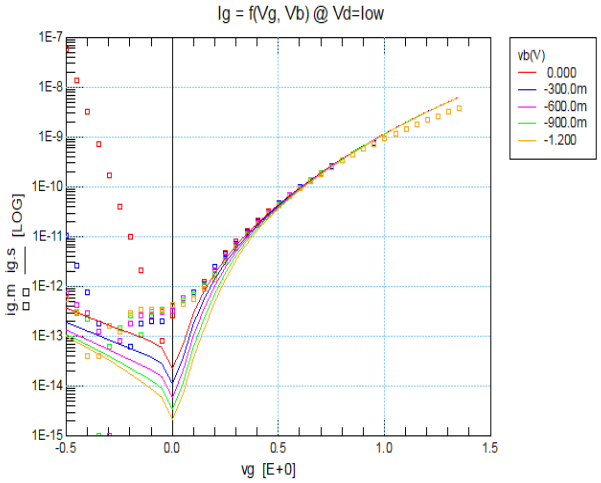




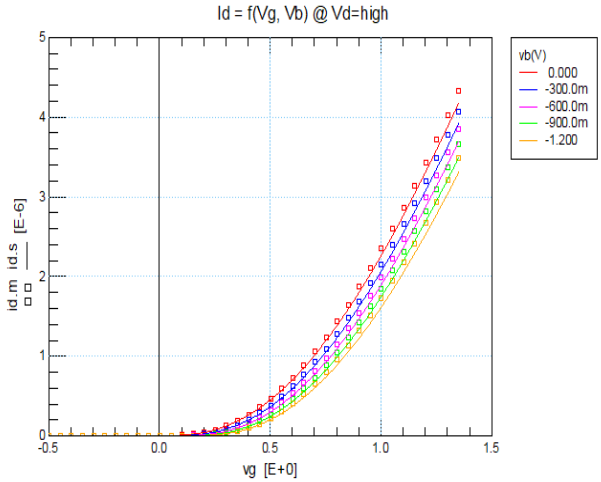


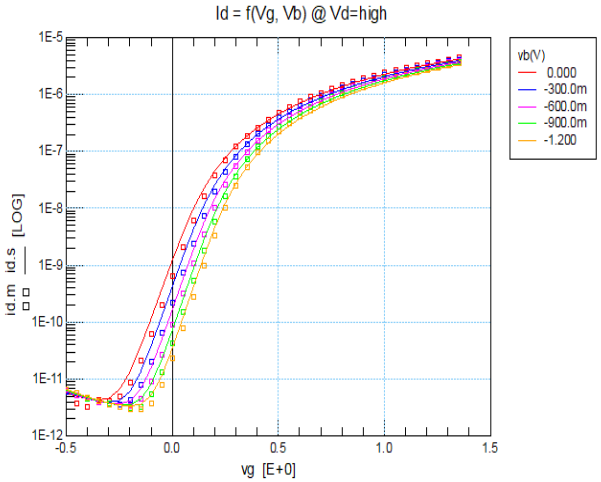


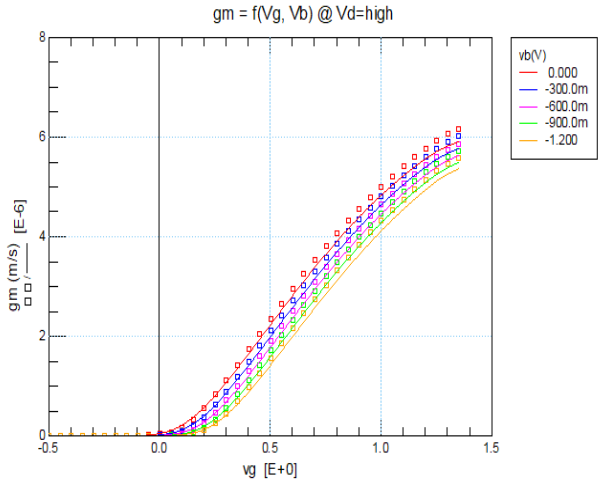


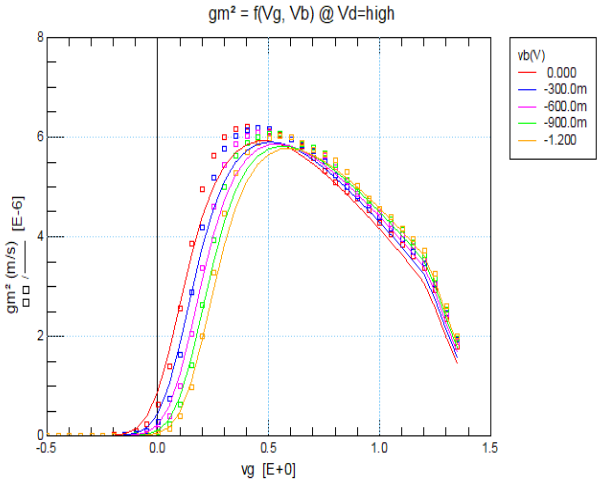


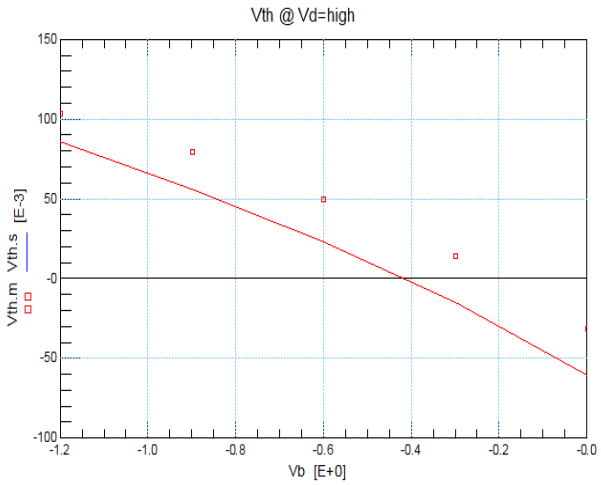
idvg, Vd = 1.2V, T = 125°C

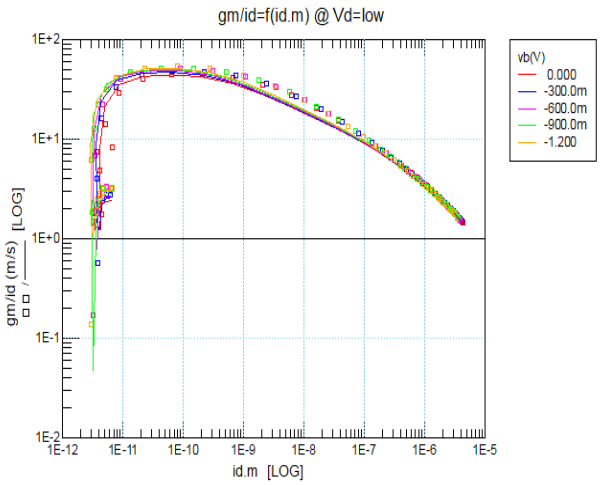


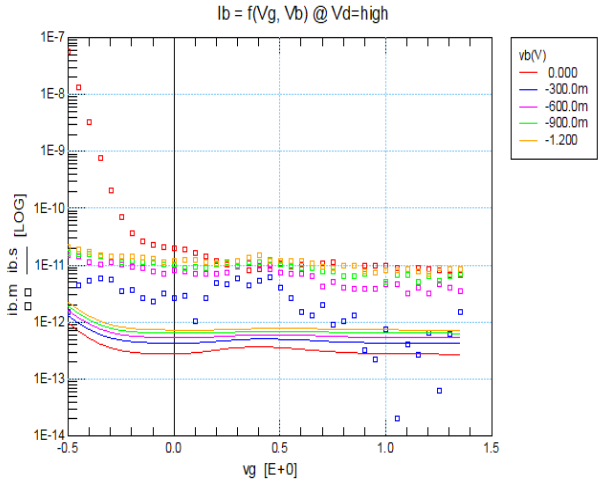


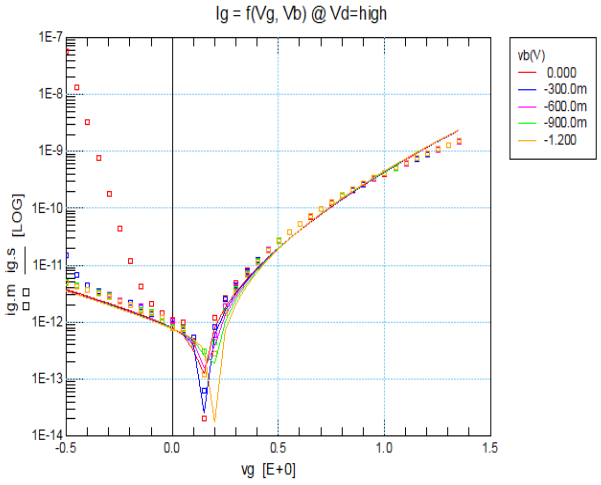




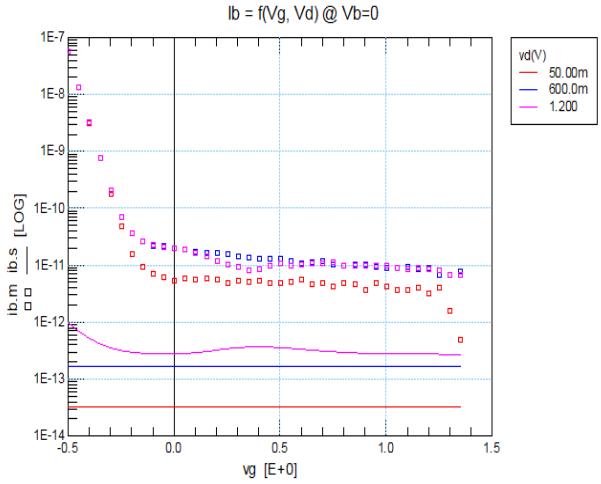


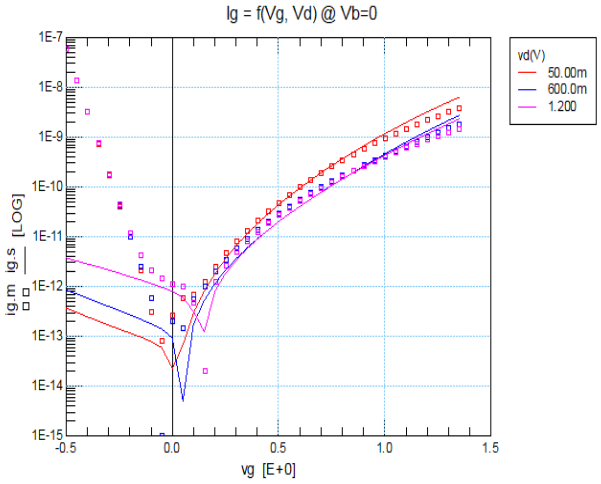




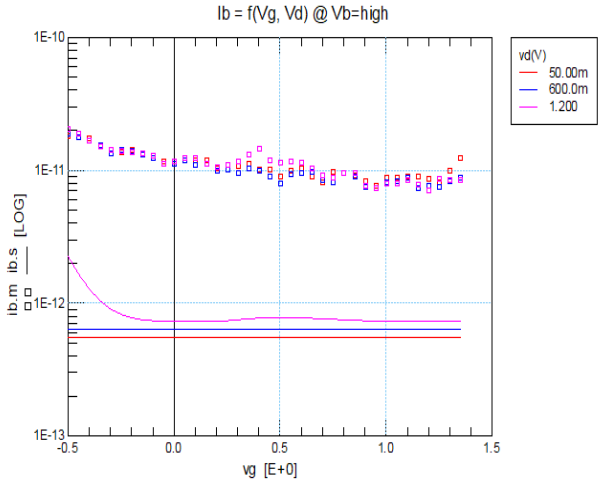


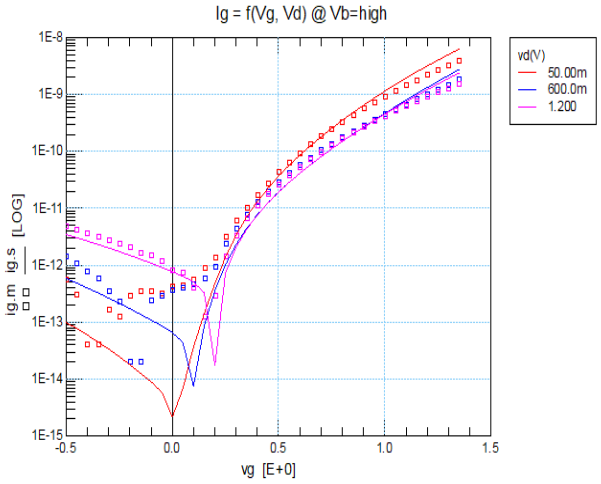
idvg, Ib,Ig, Vb = 0V, T = 125°C



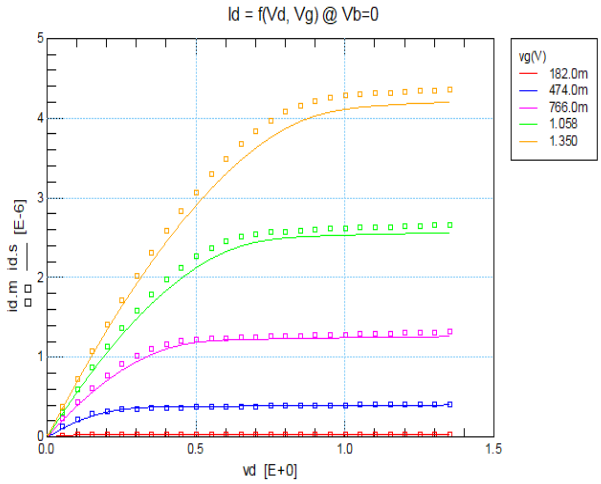


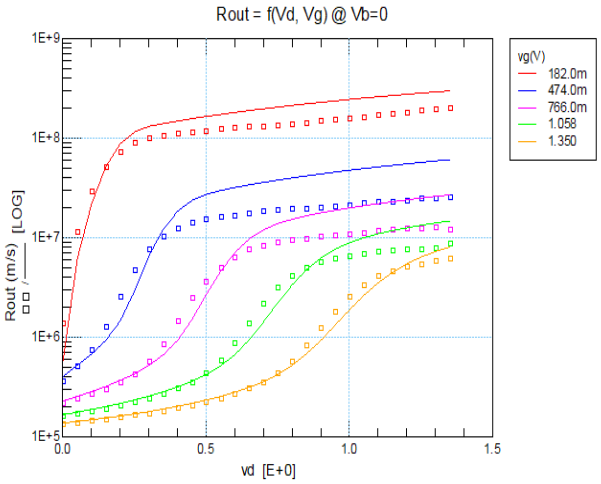
idvg, Ib,Ig, Vb = -1.2V, T = 125°C

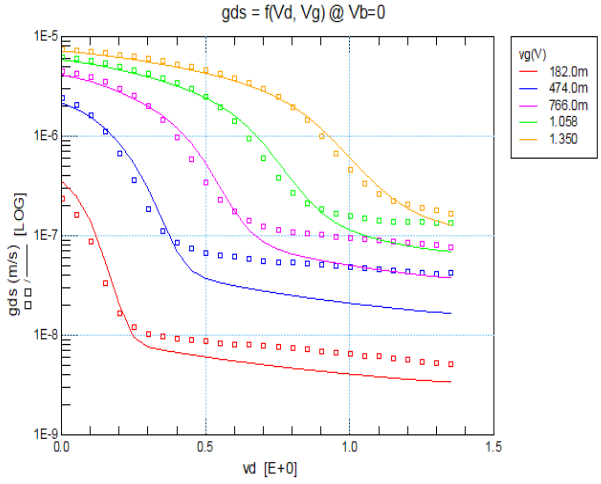




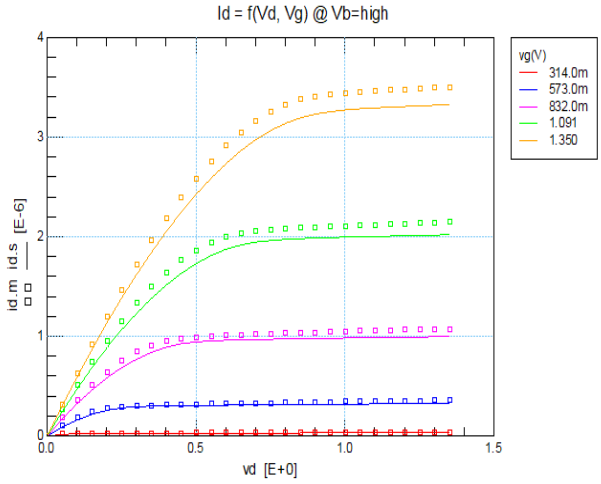
idvd, Vb = 0V, T = 125°C

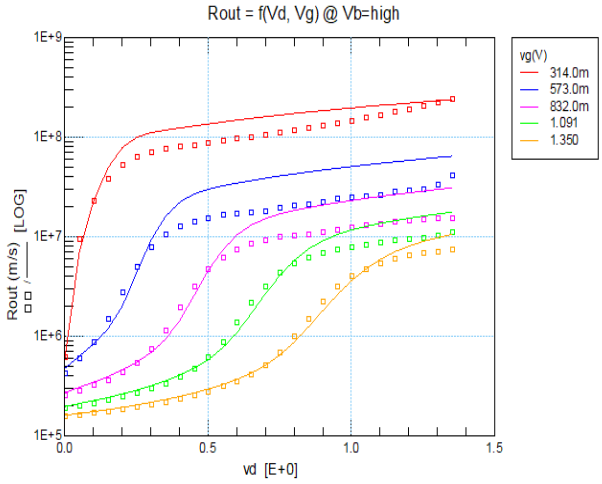


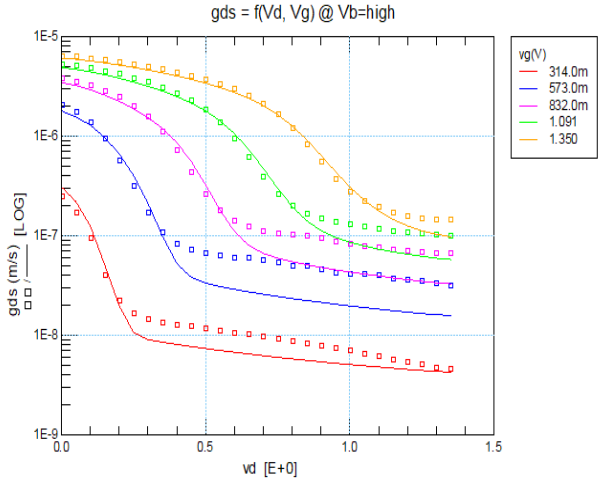




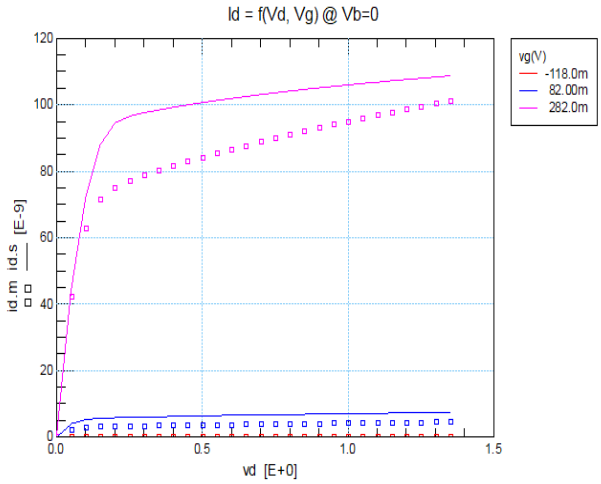
idvd_vbmin, Vb = -1.2V, T = 125°C

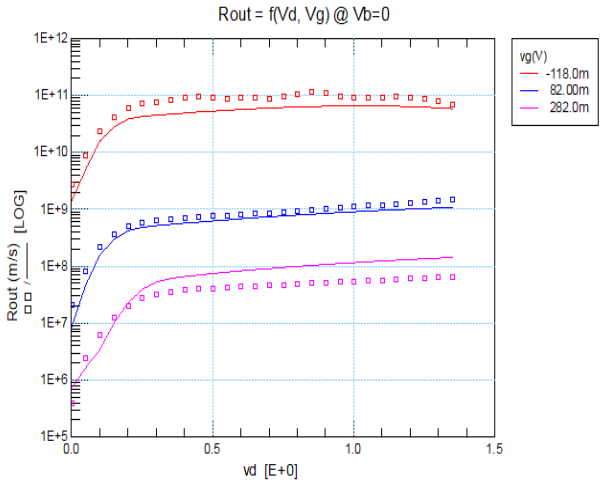


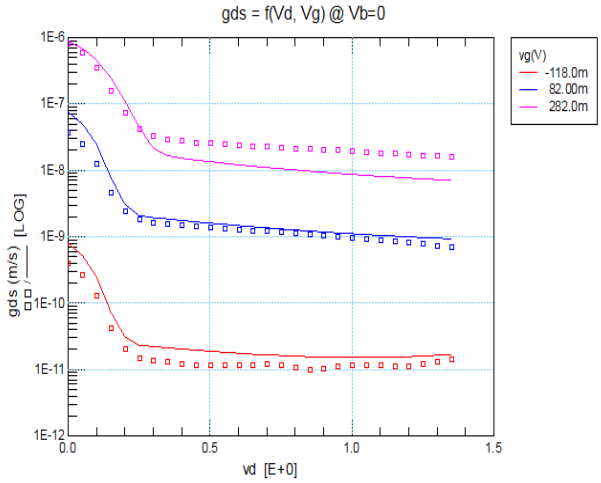




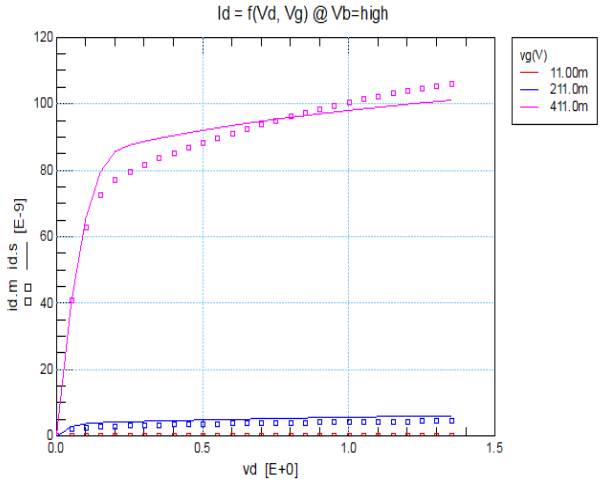
idvd_vth, Vb = 0V, T = 125°C

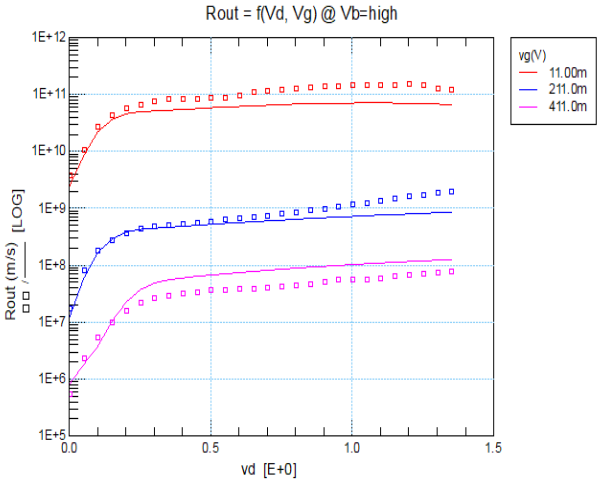


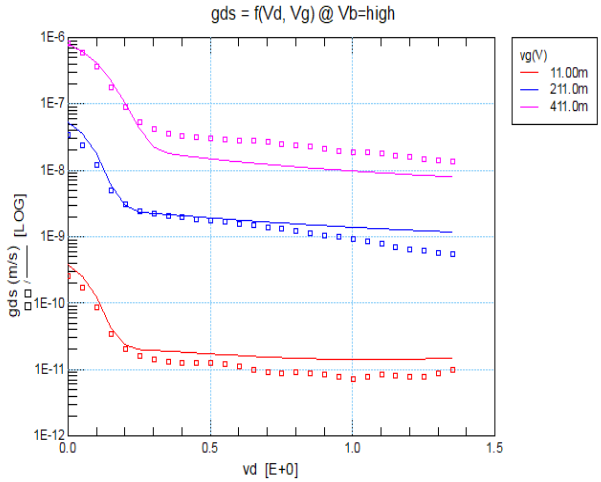




idvd_vth_vbmin, Vb = -1.2V, T = 125°C

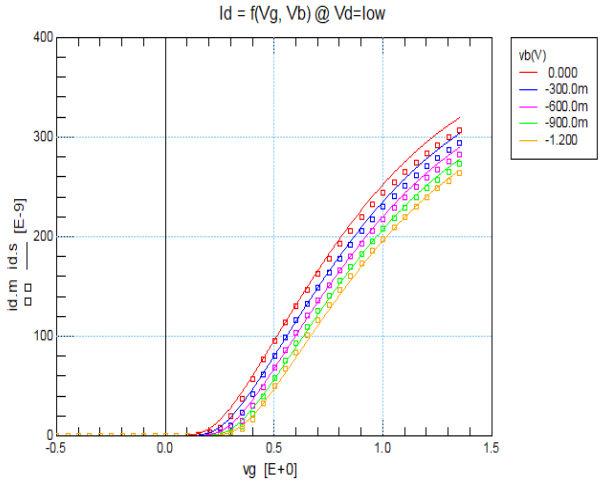


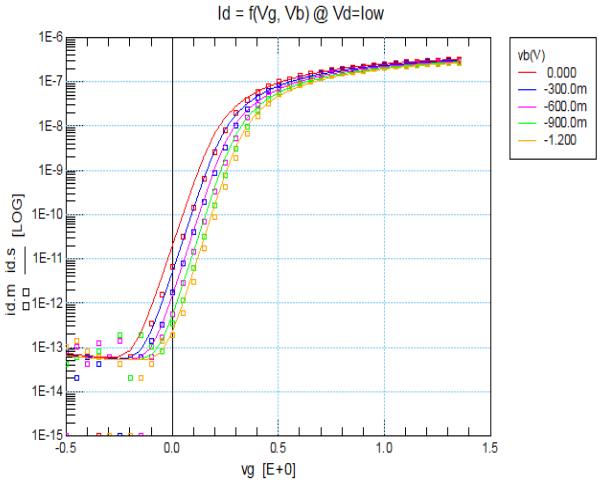


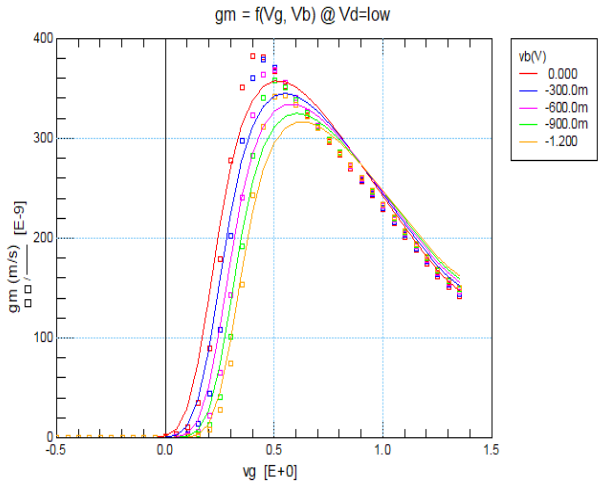


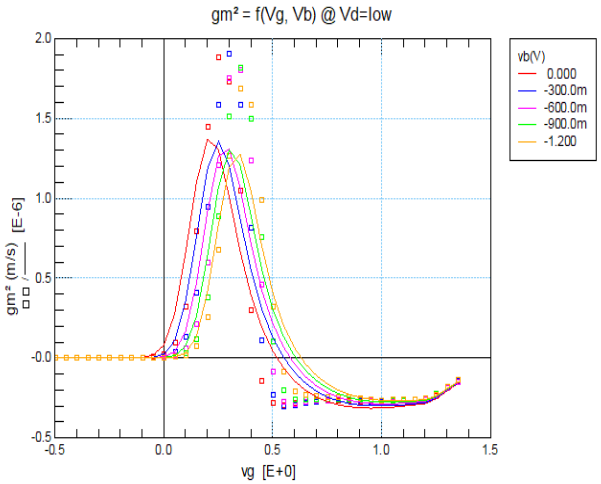
4.16W0u18_L10u0_S546_1

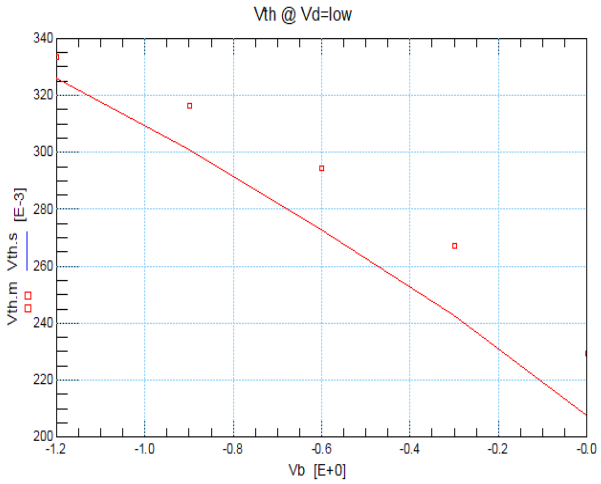
idvg, Vd = 0.05V, T = 27°C

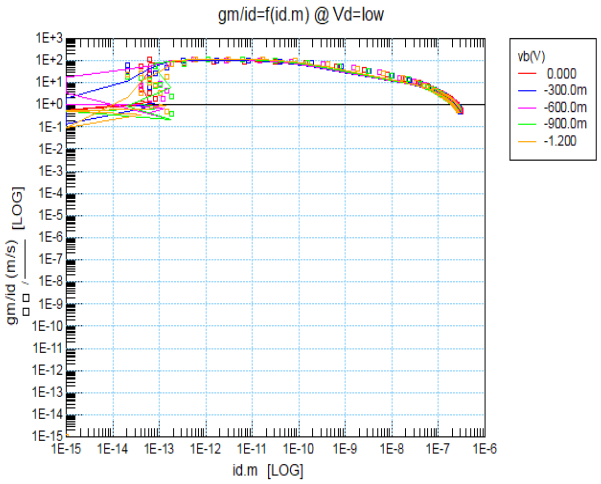


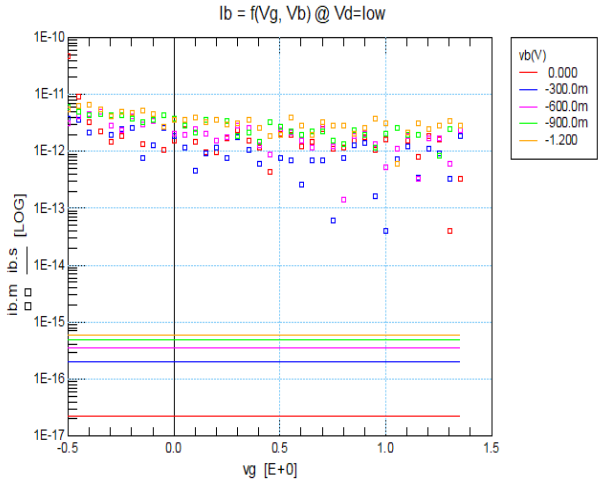


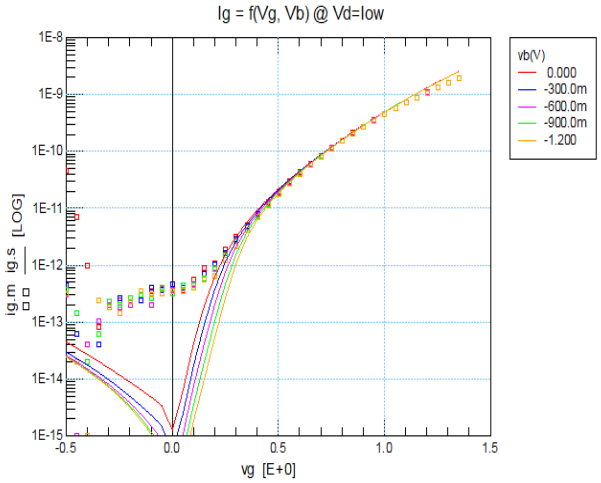




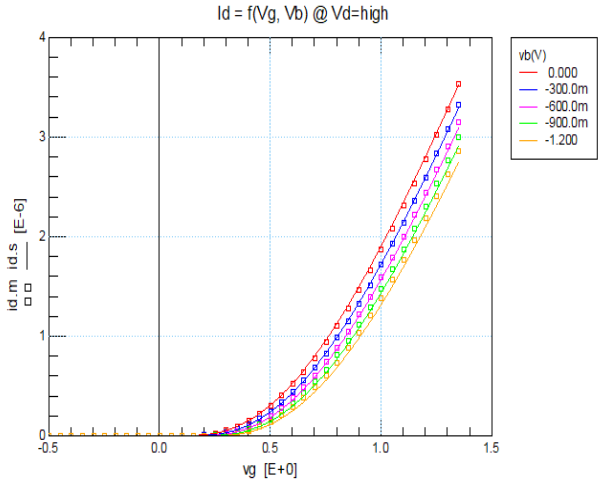


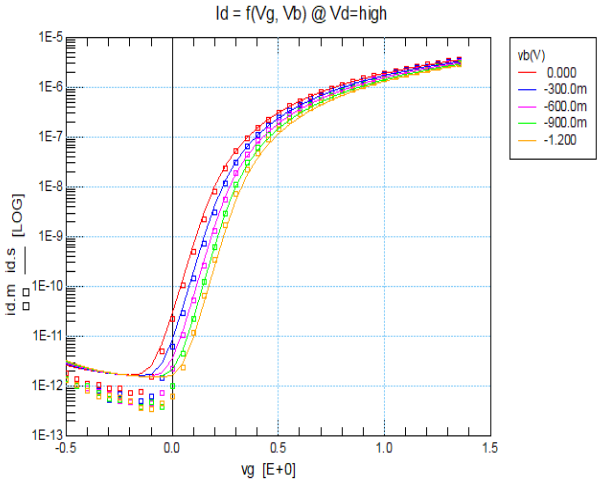


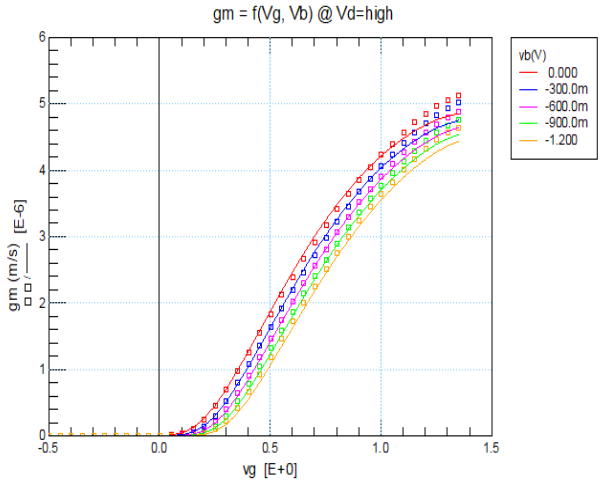


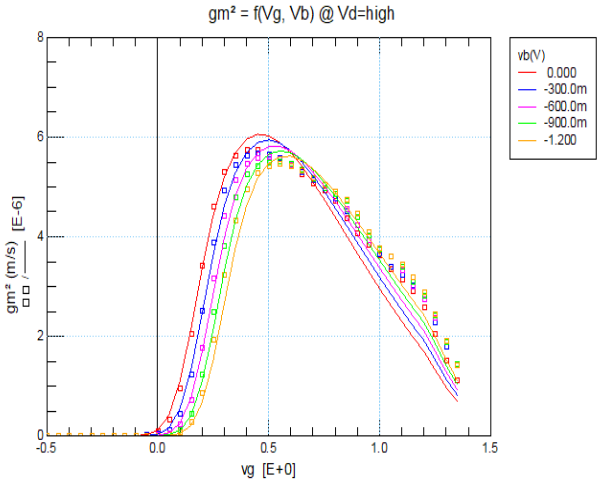


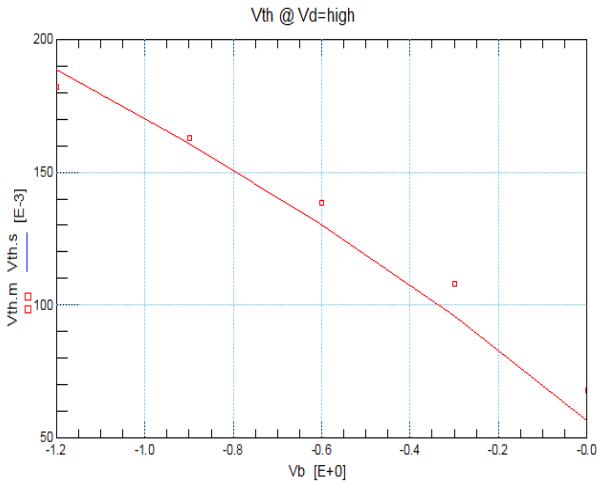
idvg, Vd = 1.2V, T = 27°C

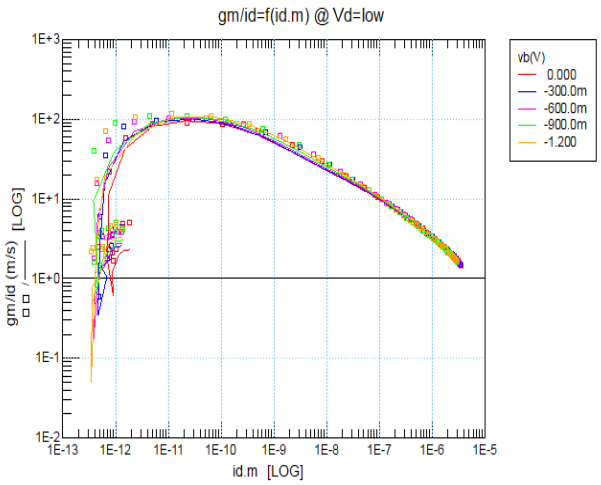


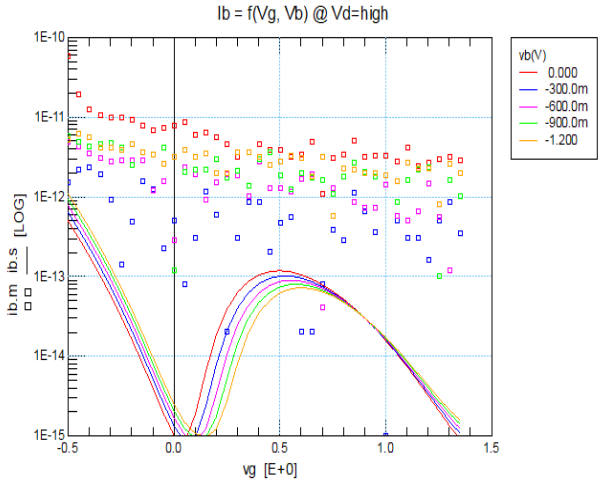


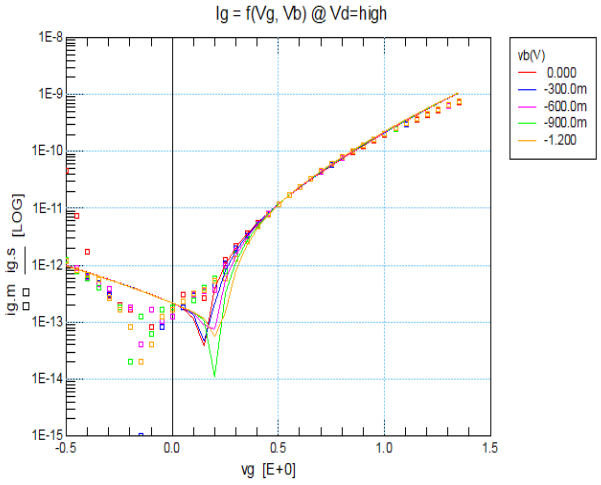




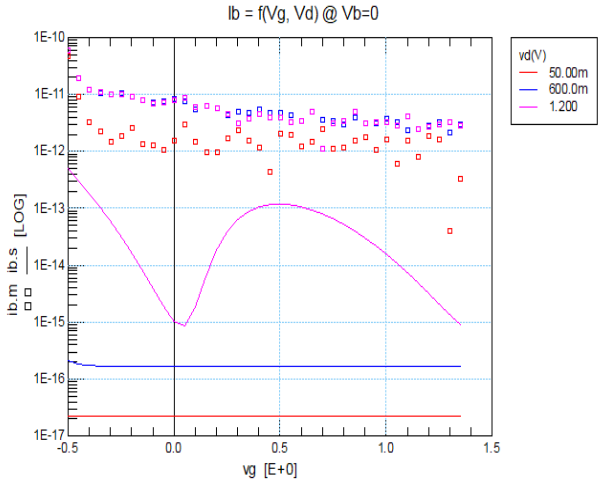


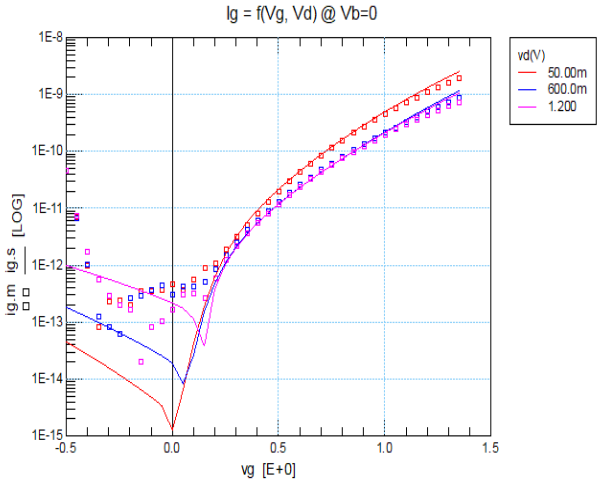




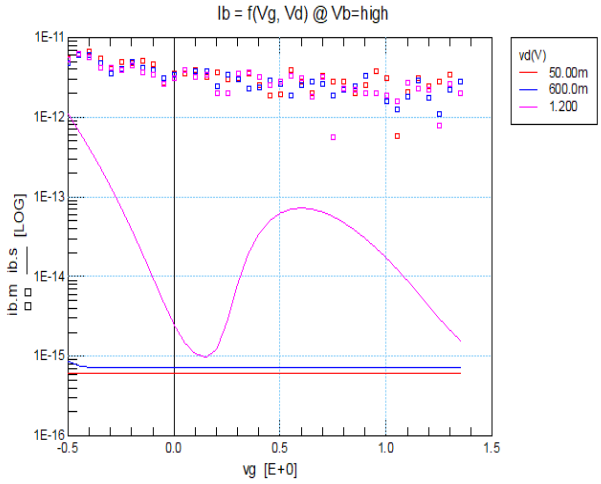


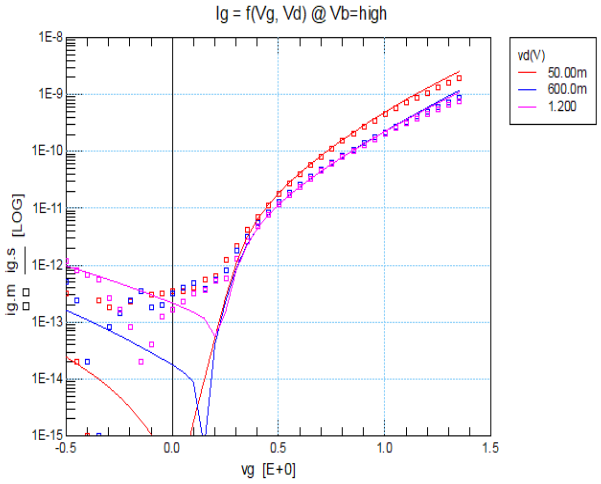
idvg, Ib,Ig, Vb = 0V, T = 27°C



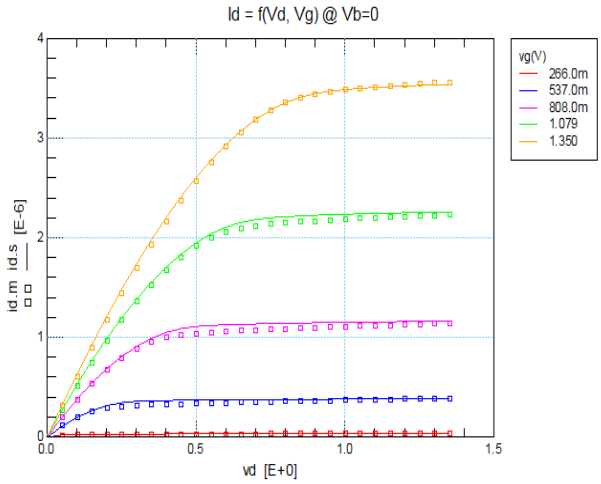


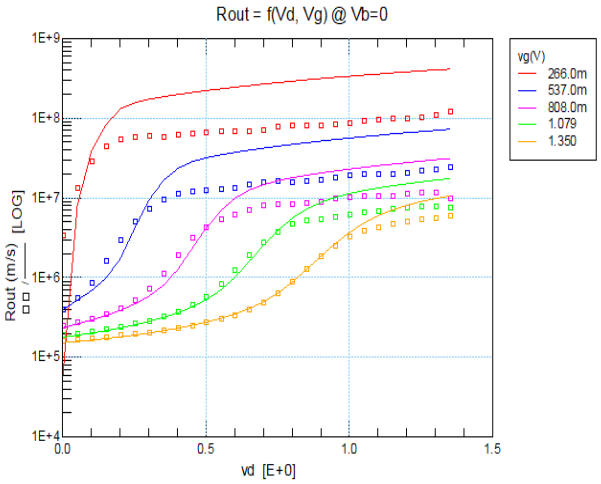
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

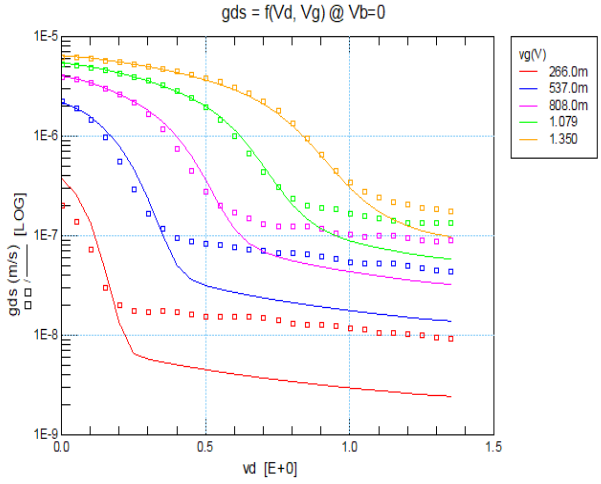




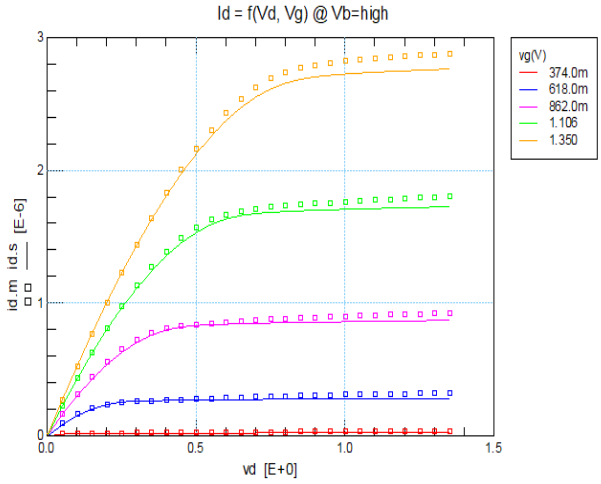
idvd, Vb = 0V, T = 27°C

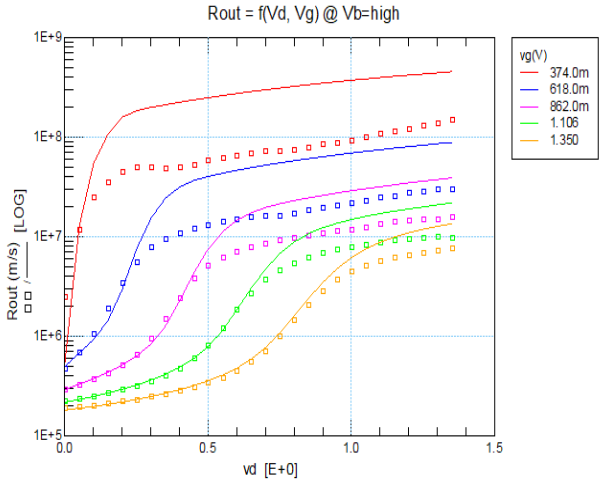


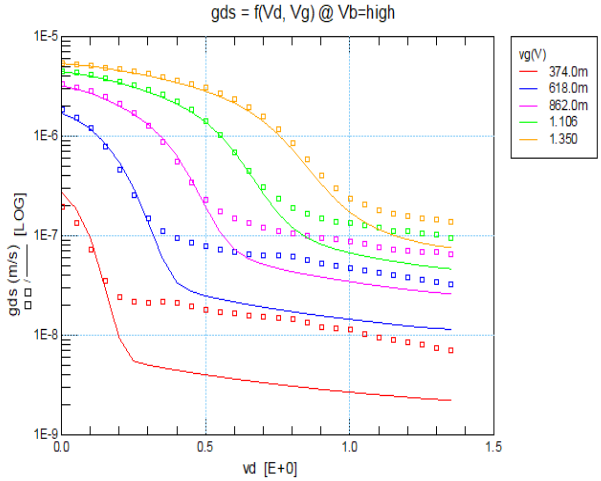




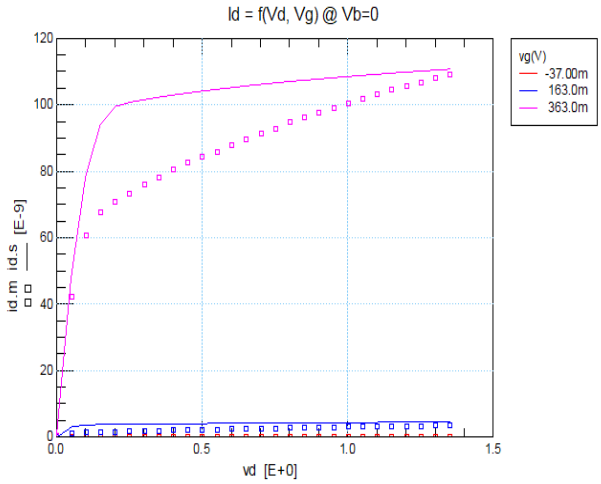
idvd_vbmin, Vb = -1.2V, T = 27°C

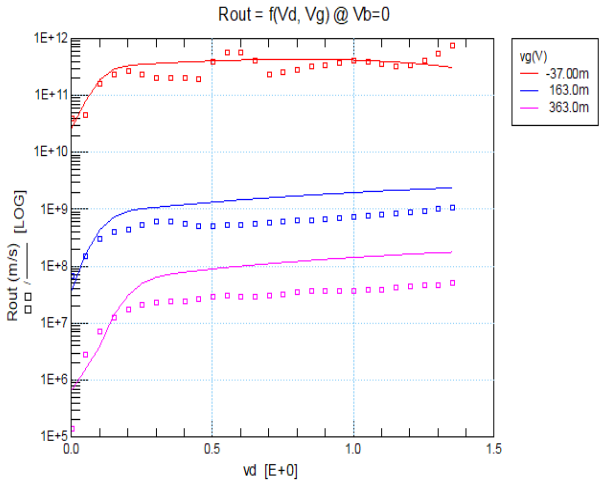


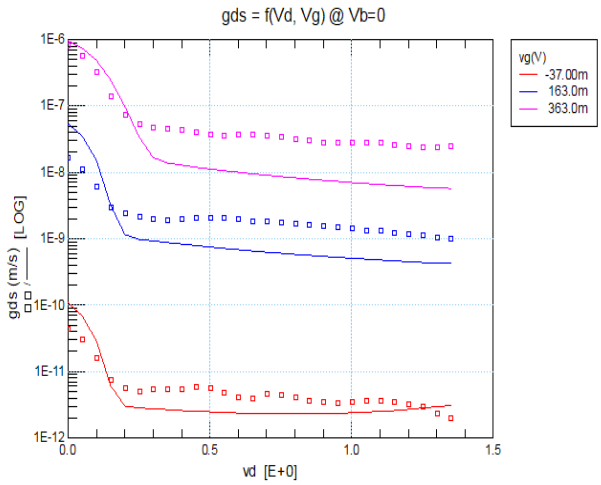




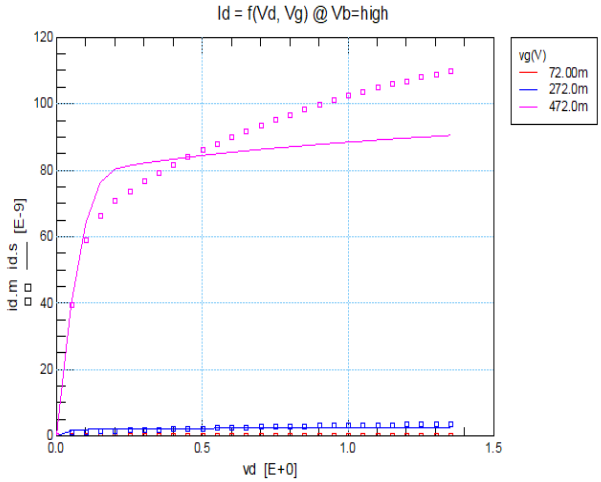
idvd_vth, Vb = 0V, T = 27°C

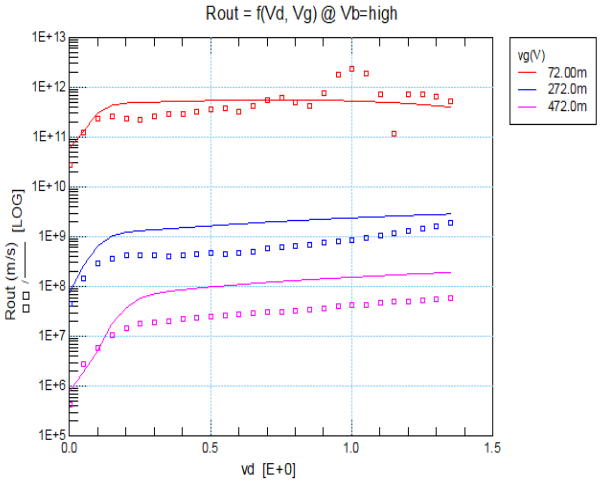


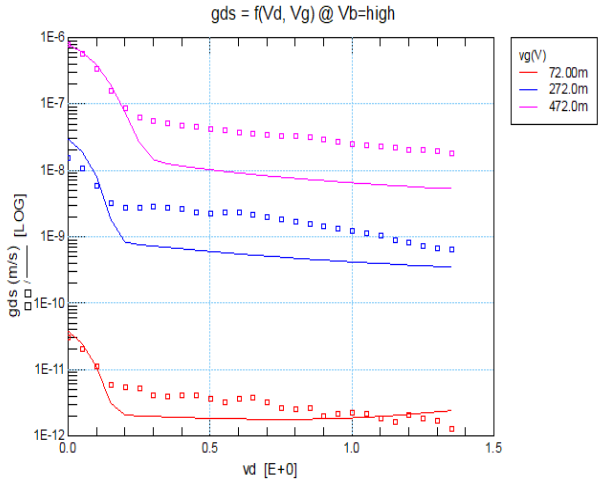




idvd_vth_vbmin, Vb = -1.2V, T = 27°C



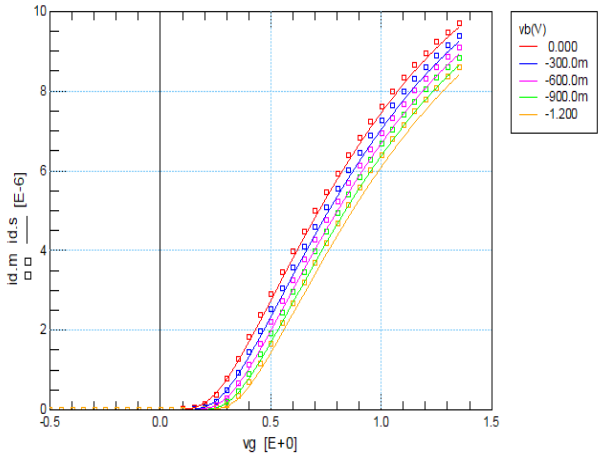




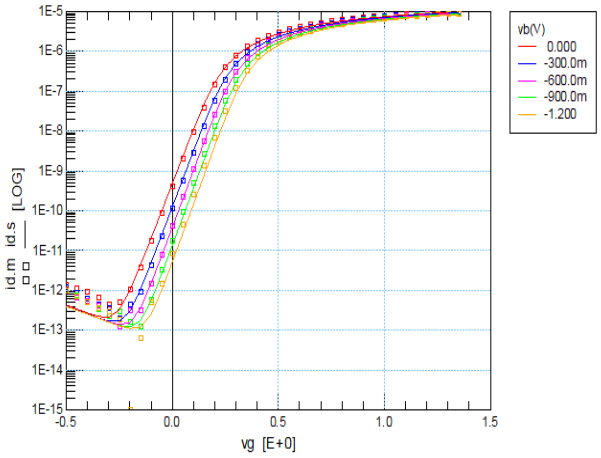
4.17W05u0_L10u0_S542_3

idvg, Vd = 0.05V, T = 27°C

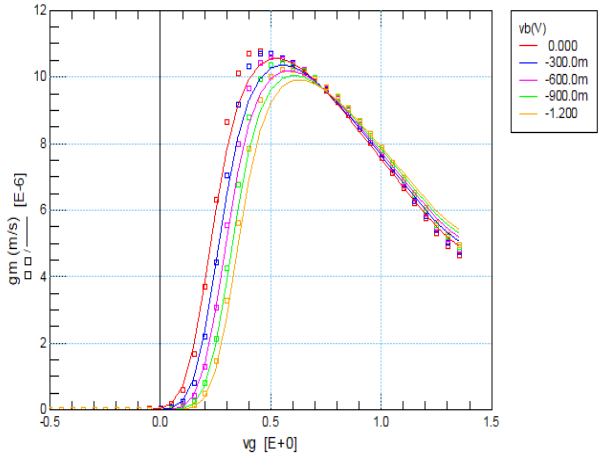
Id = f(Vg, Vb) @ Vd=low



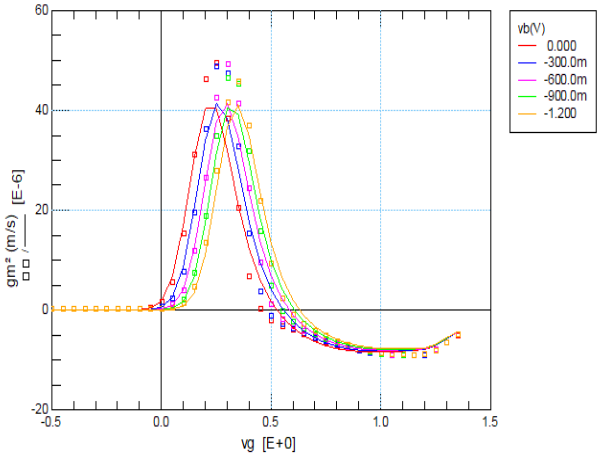
Id = f(Vg, Vb) @ Vd=low



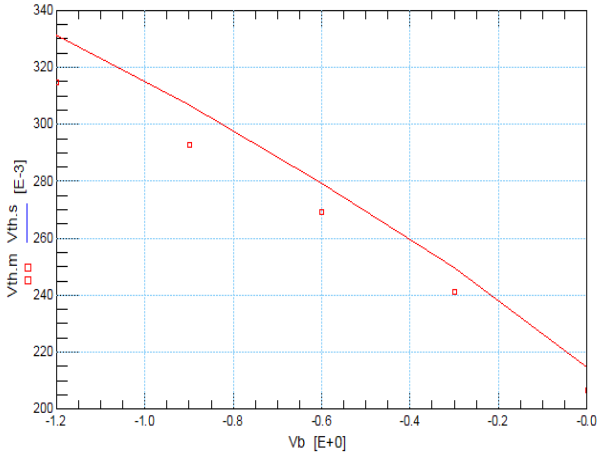
gm = f(Vg, Vb) @ Vd=low



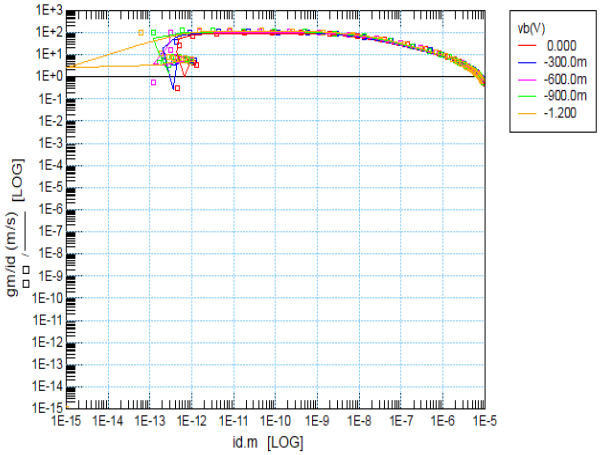
gm² = f(Vg, Vb) @ Vd=low

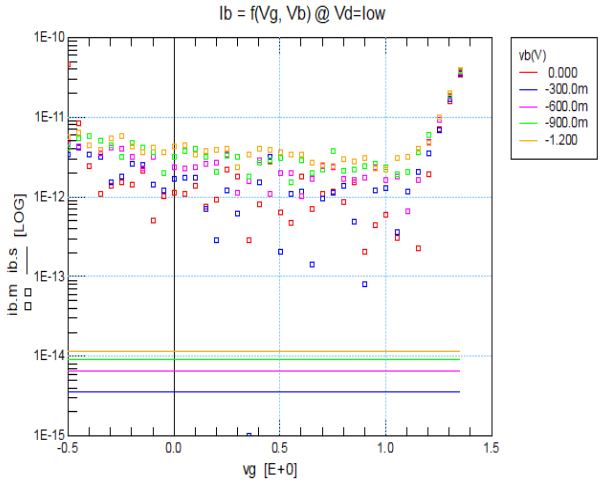


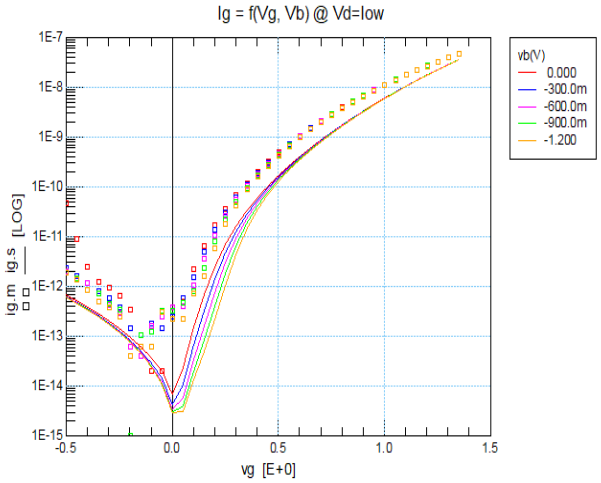
Vth @ Vd=low



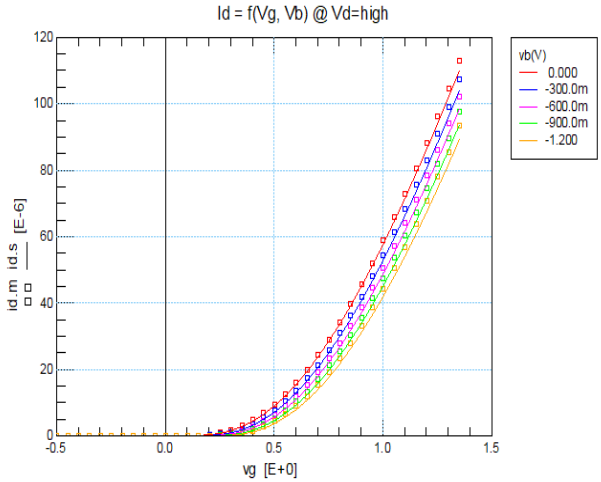
gm/id=f(id.m) @ Vd=low

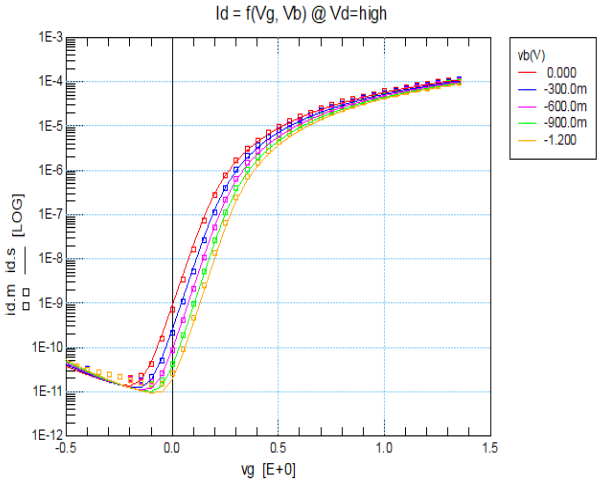


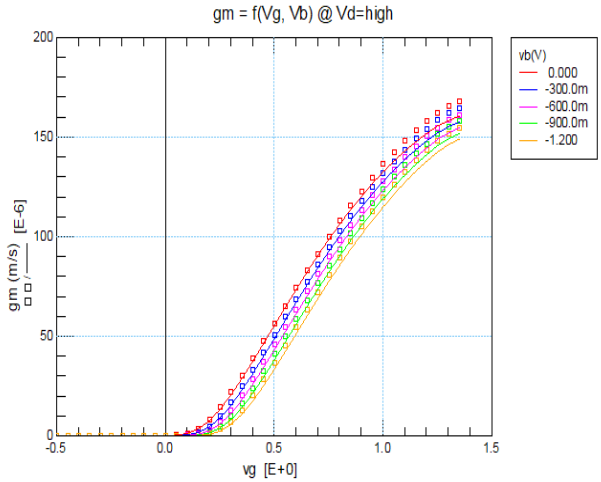


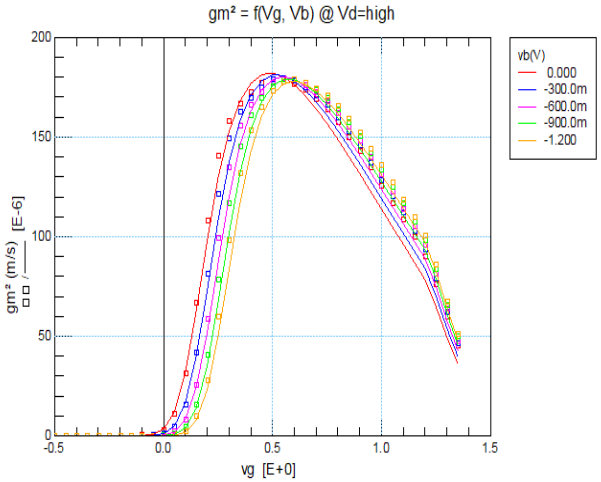


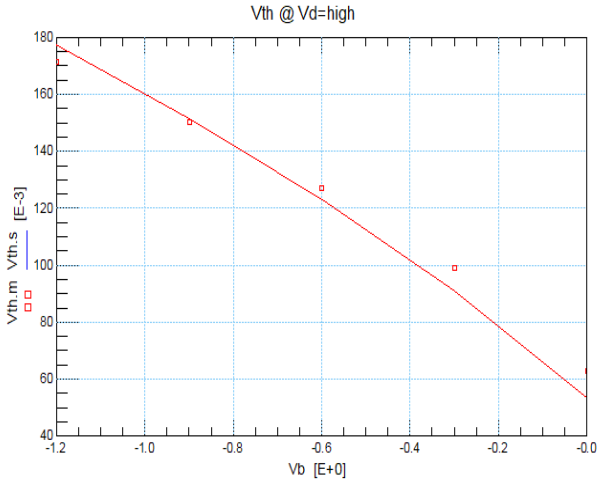
idvg, Vd = 1.2V, T = 27°C

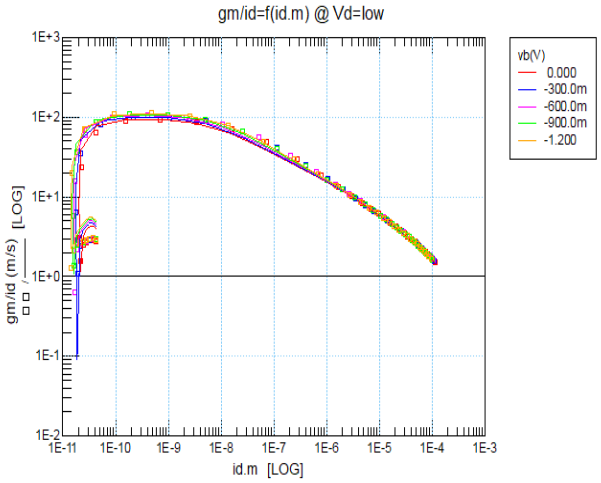


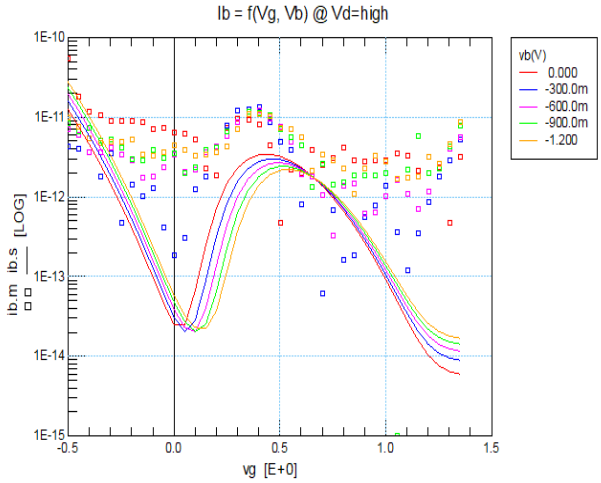


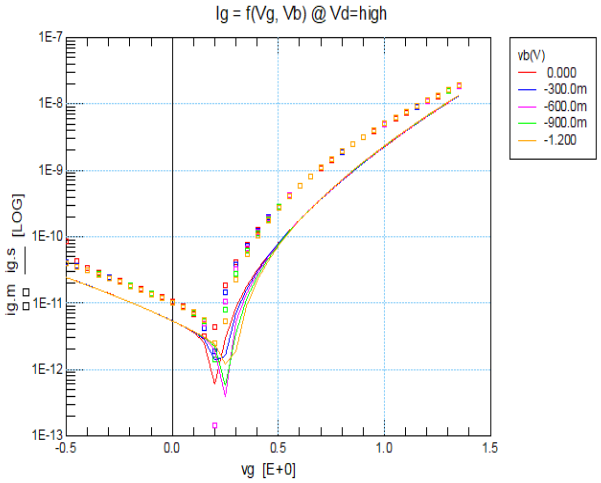




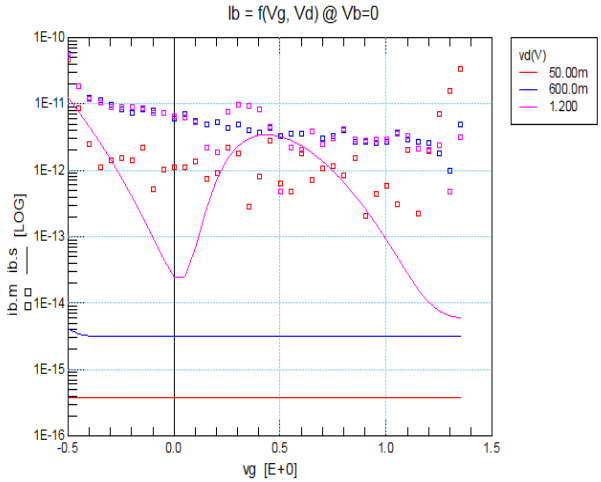


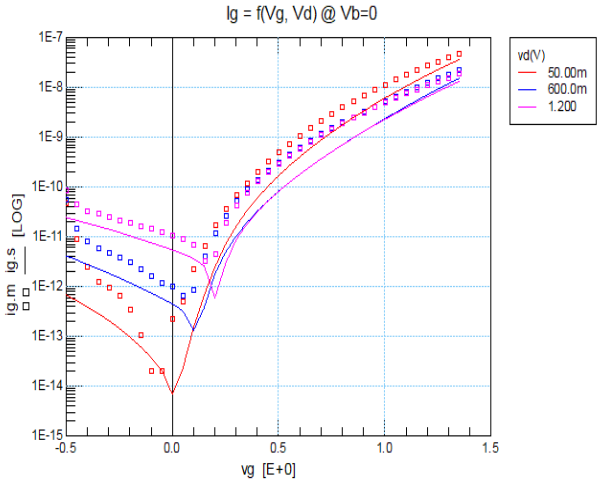




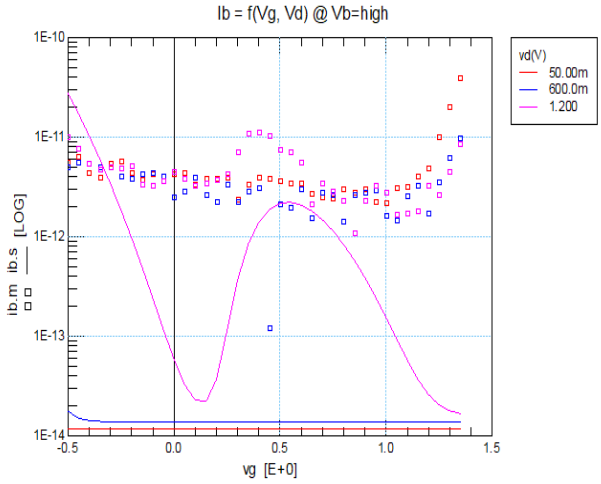


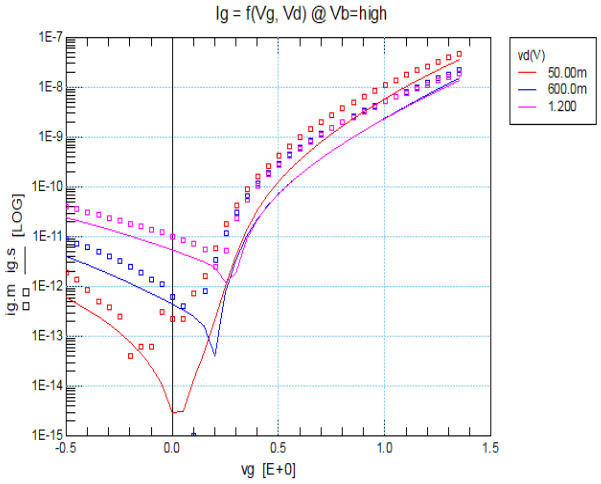
idvg, Ib,Ig, Vb = 0V, T = 27°C



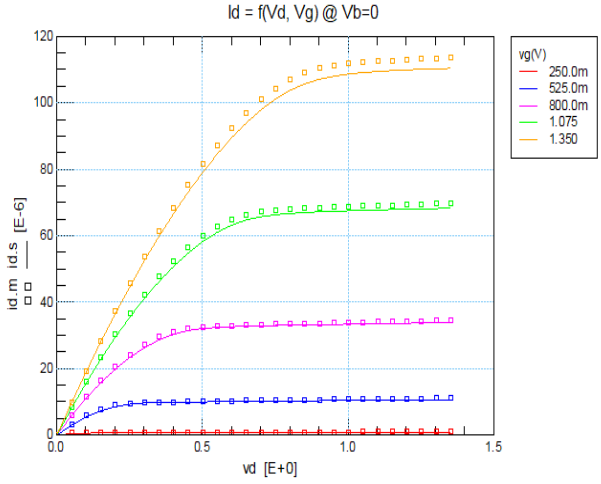


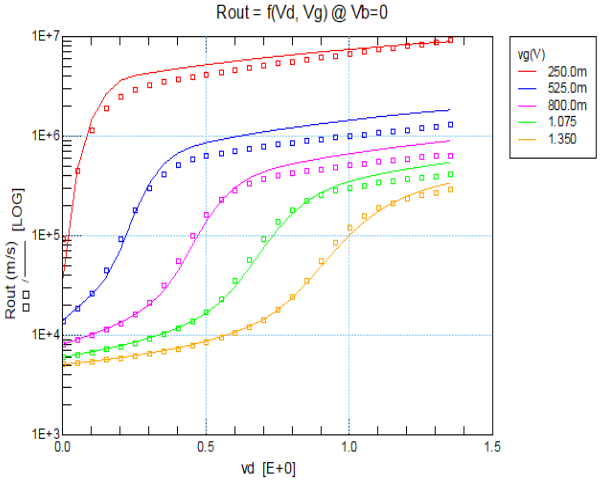
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

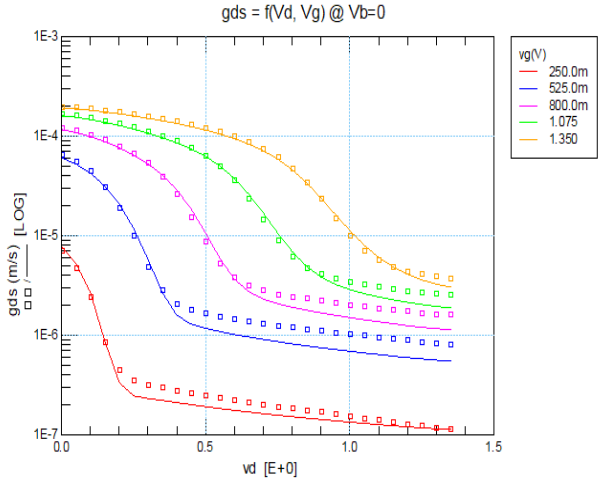




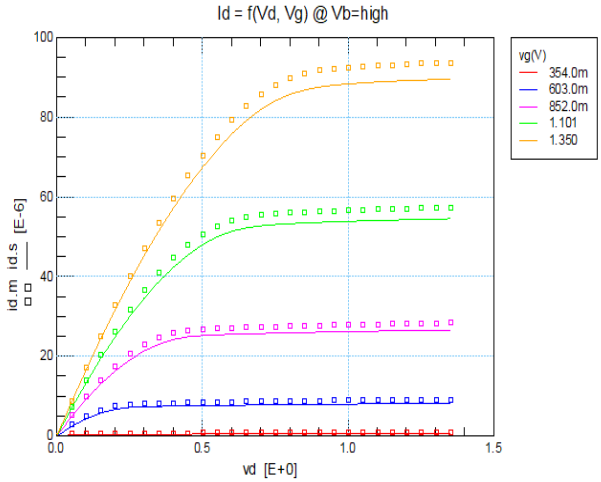
idvd, Vb = 0V, T = 27°C

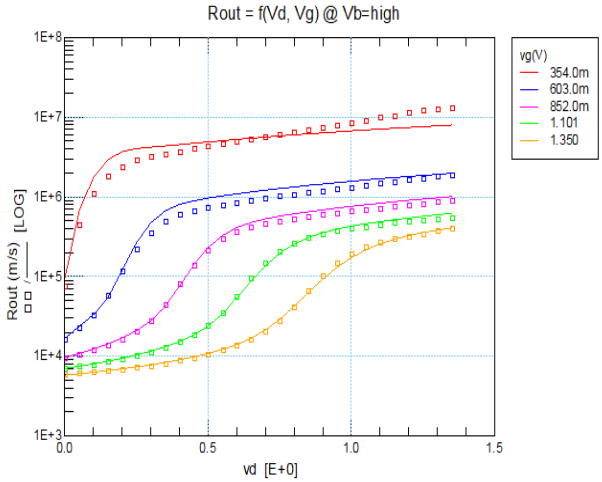


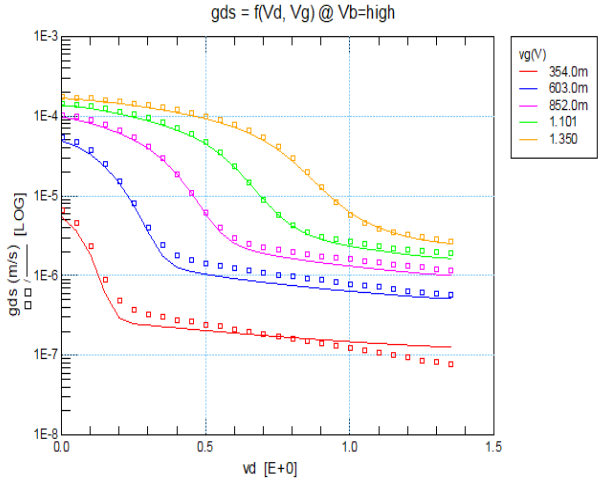




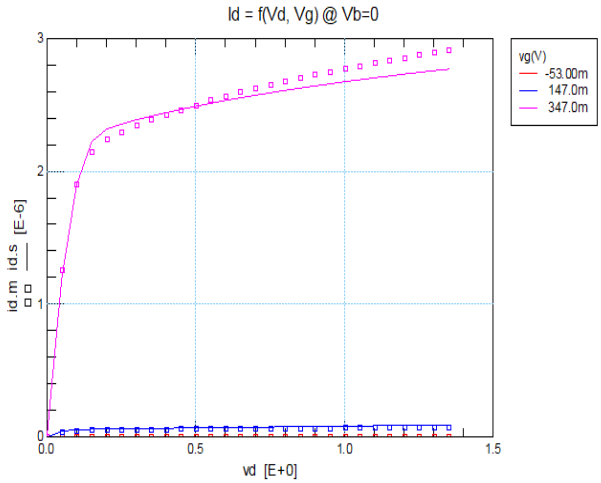
idvd_vbmin, Vb = -1.2V, T = 27°C

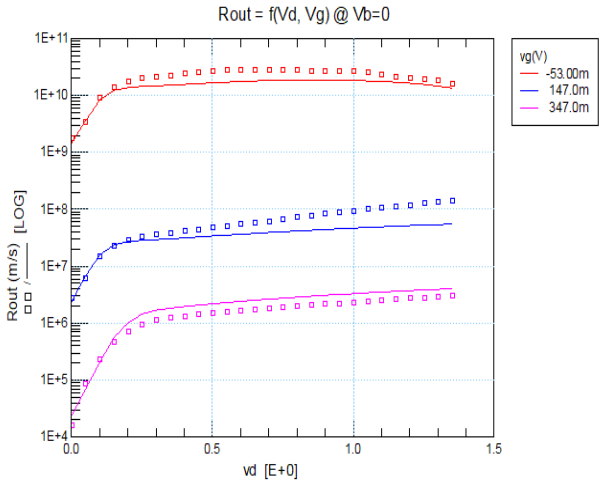


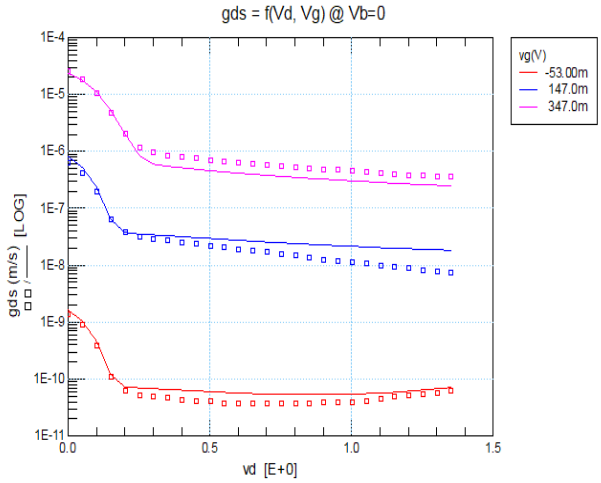




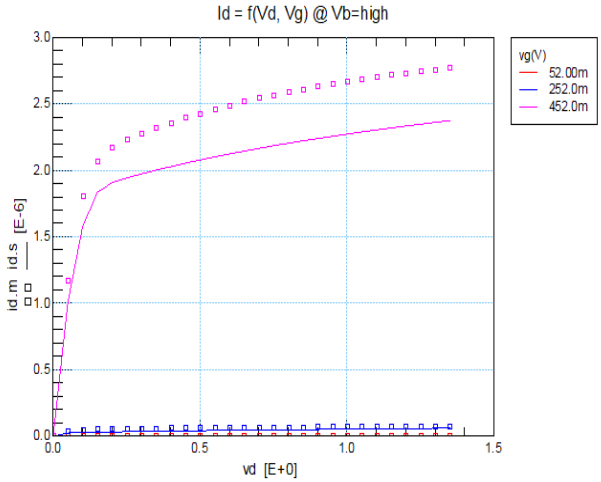
idvd_vth, Vb = 0V, T = 27°C

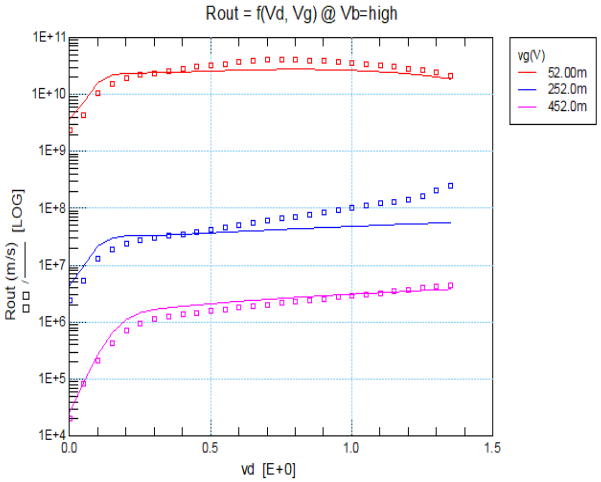


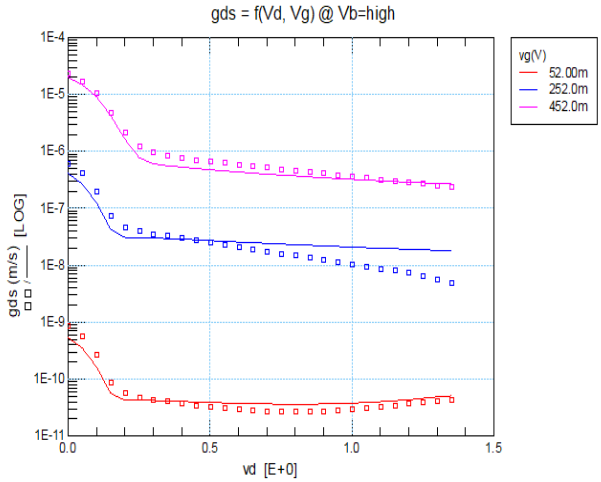




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

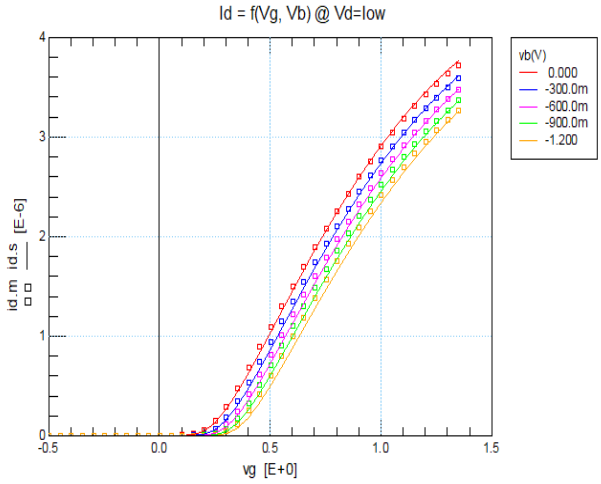


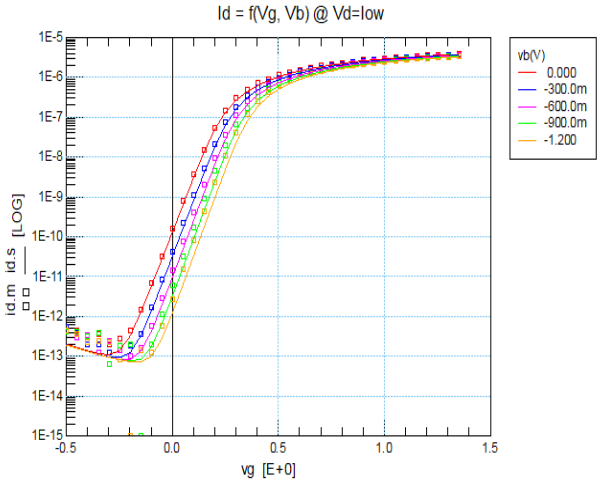


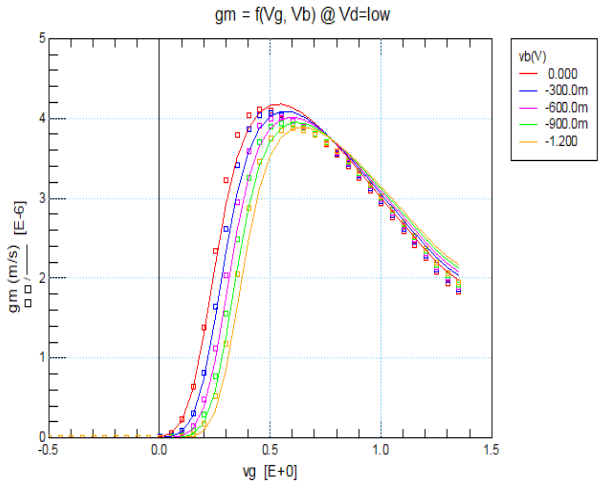


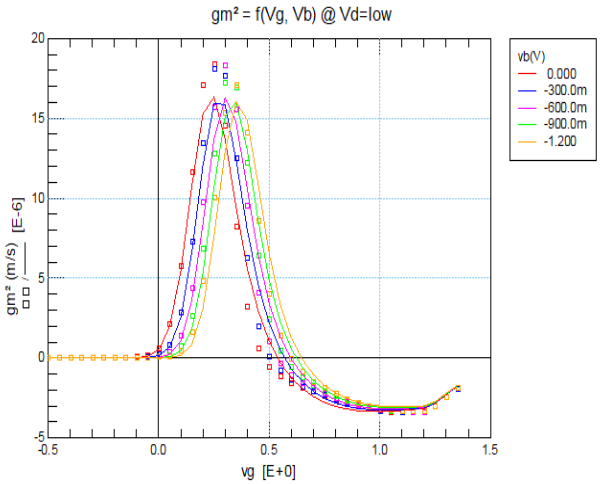
4.18 W02u0_L10u0_S543_1

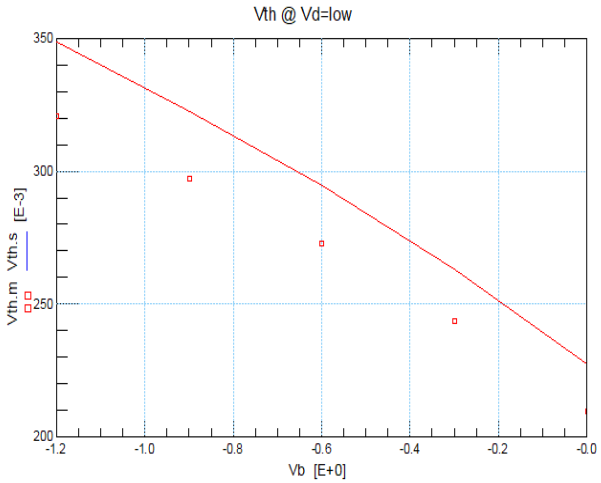
idvg, Vd = 0.05V, T = 27°C

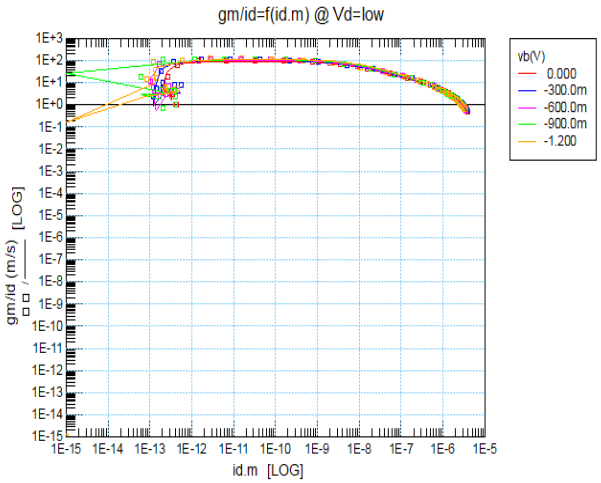


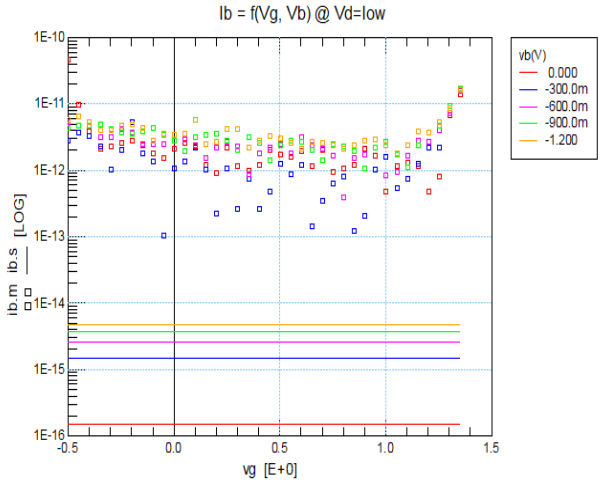


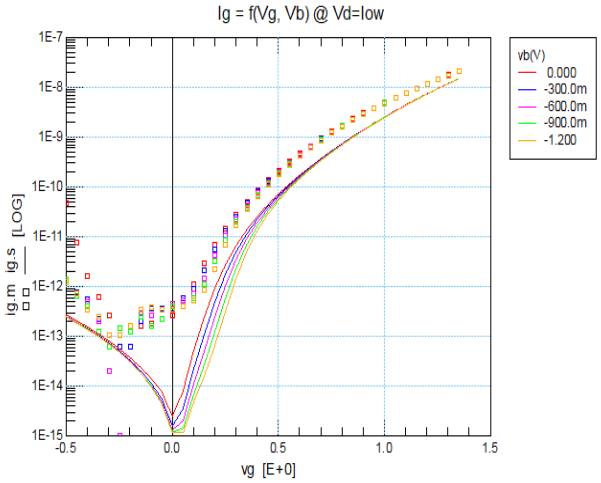




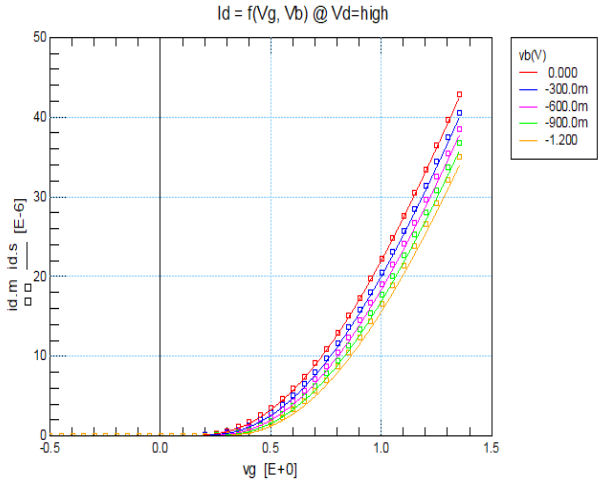


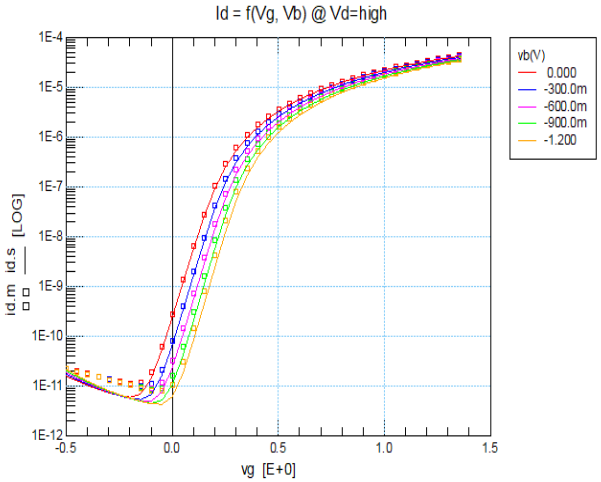


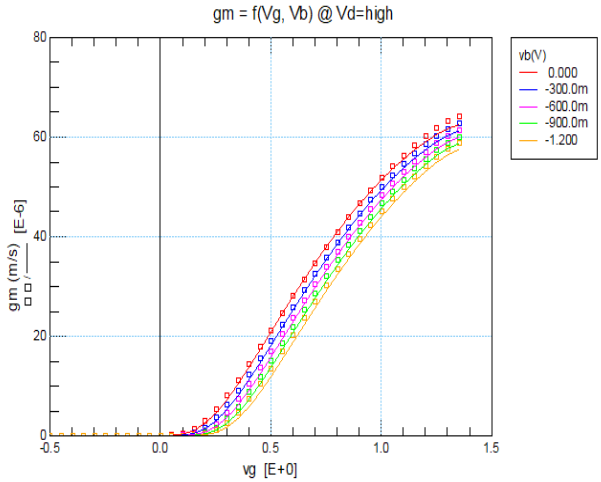


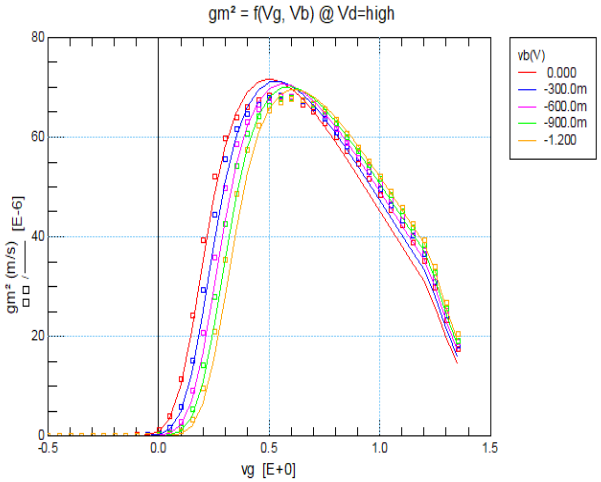


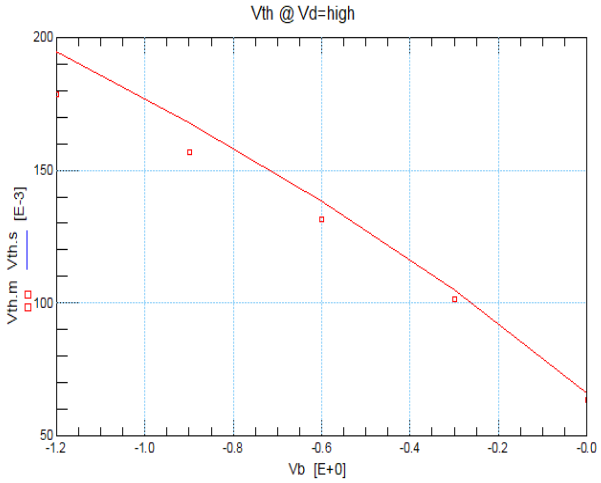
idvg, Vd = 1.2V, T = 27°C

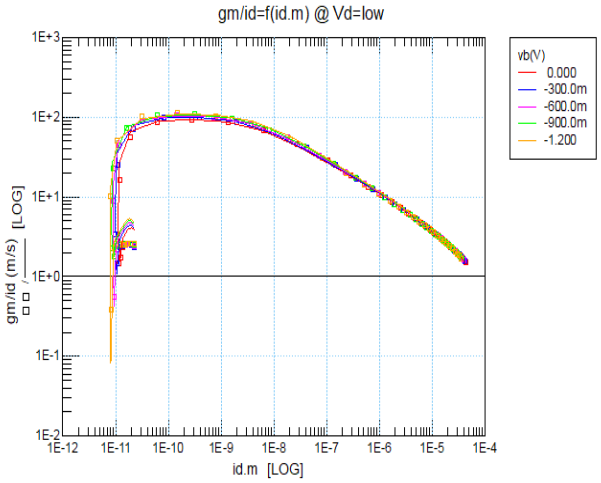


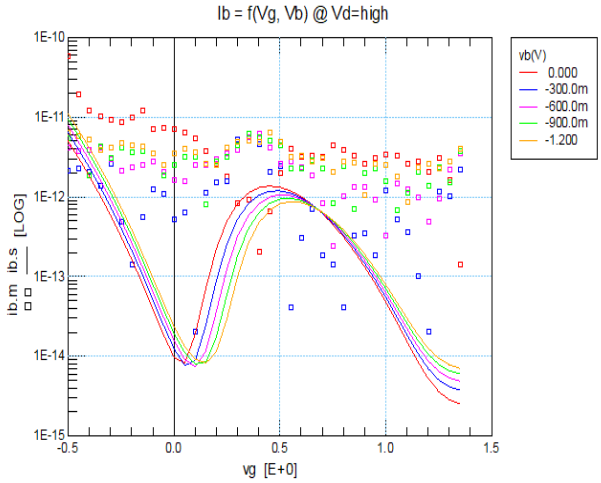


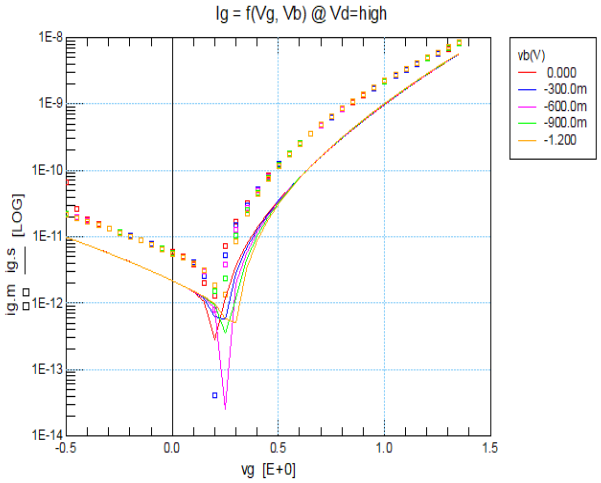




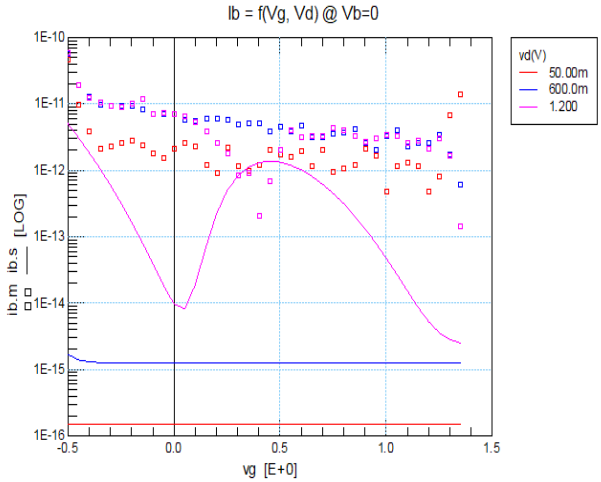


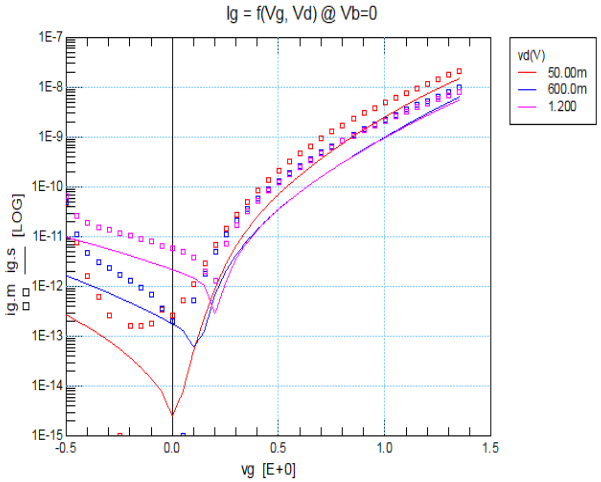




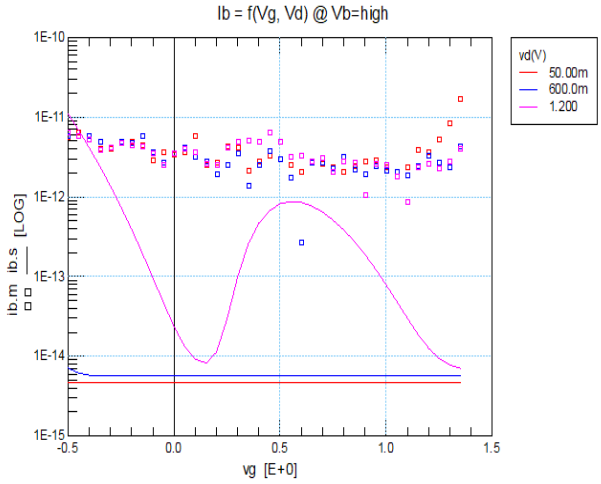


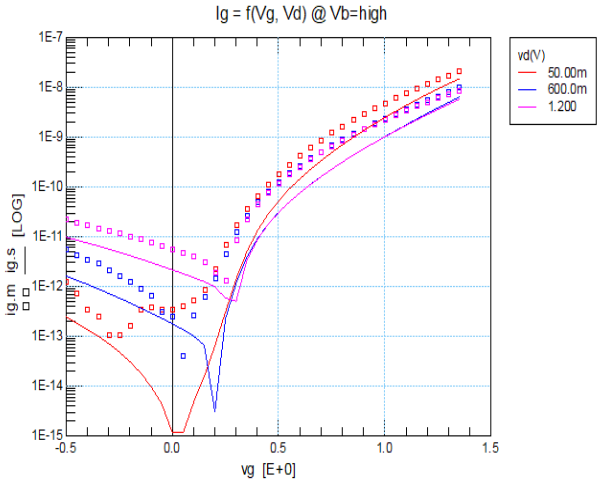
idvg, Ib,Ig, Vb = 0V, T = 27°C



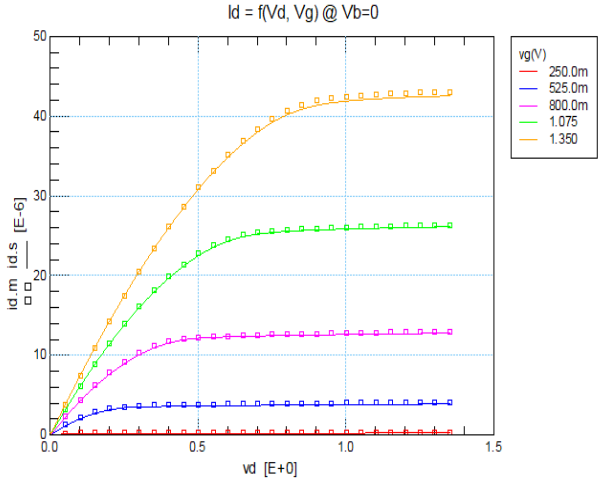


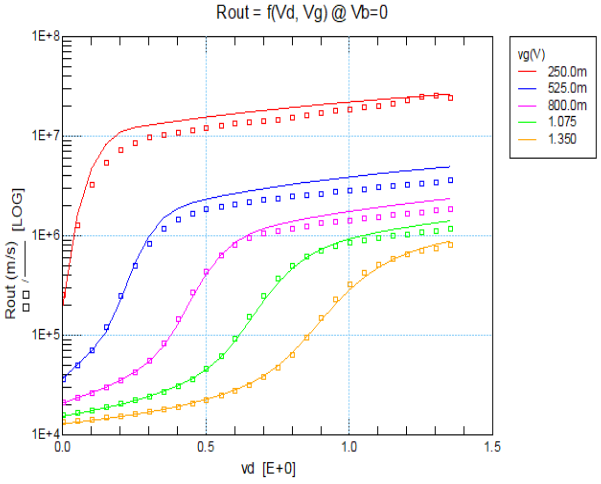
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

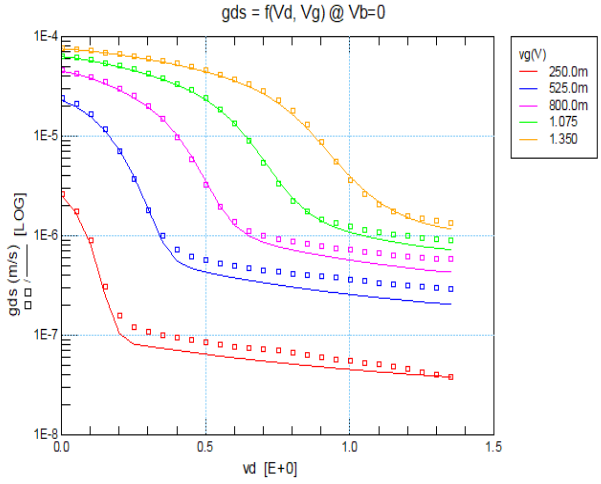




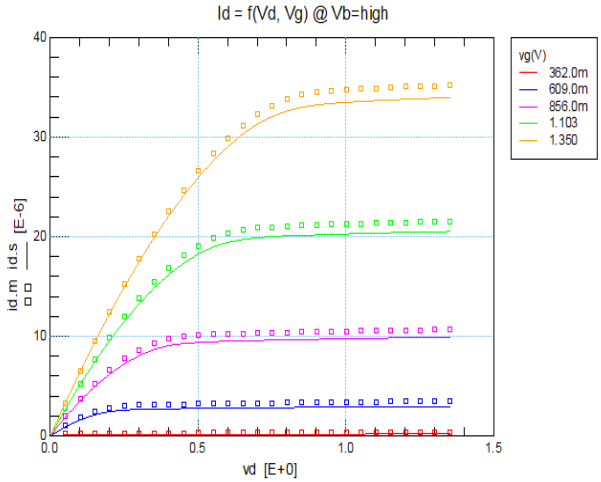
idvd, Vb = 0V, T = 27°C

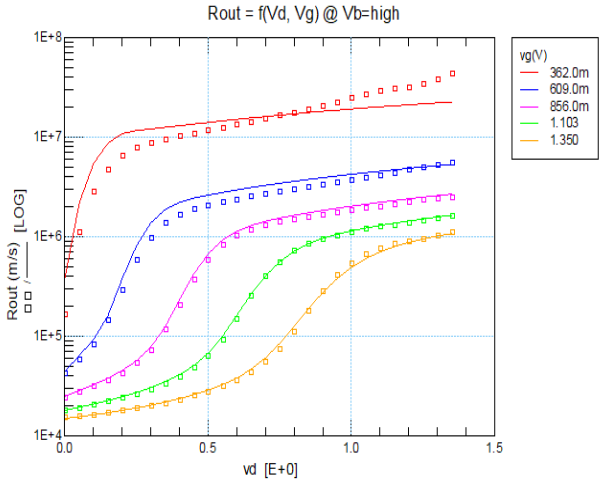


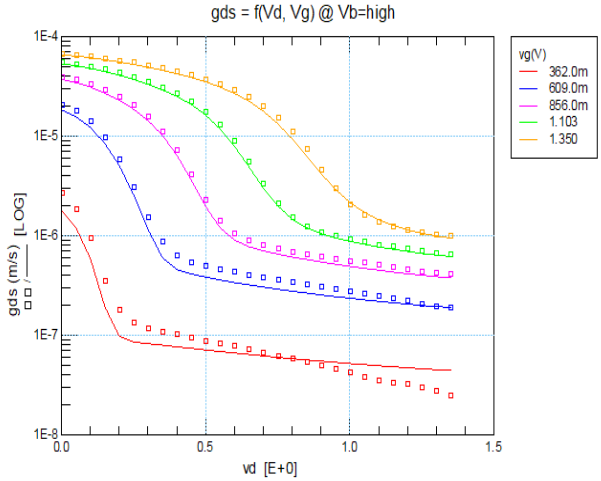




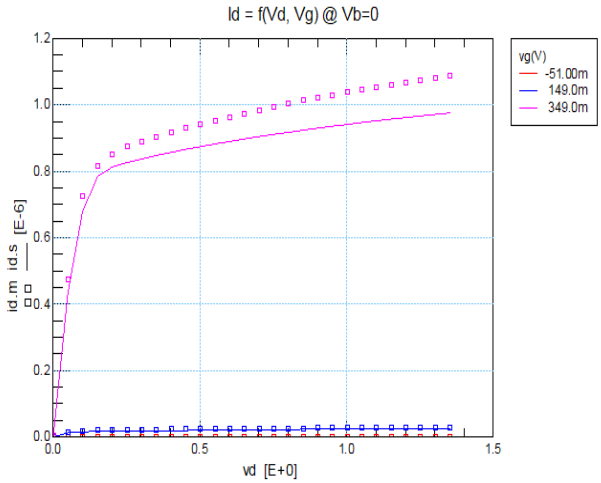
idvd_vbmin, Vb = -1.2V, T = 27°C

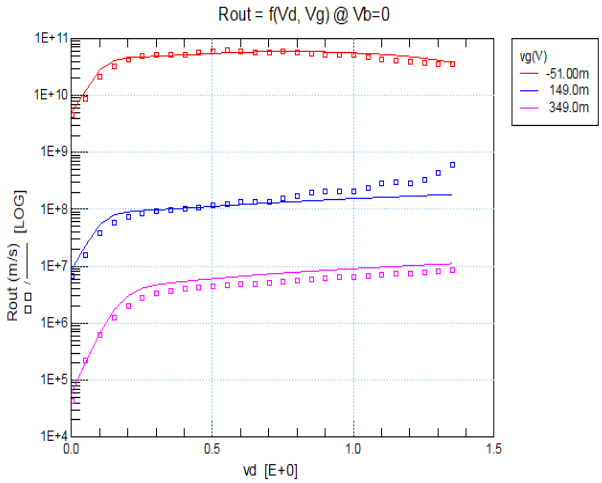


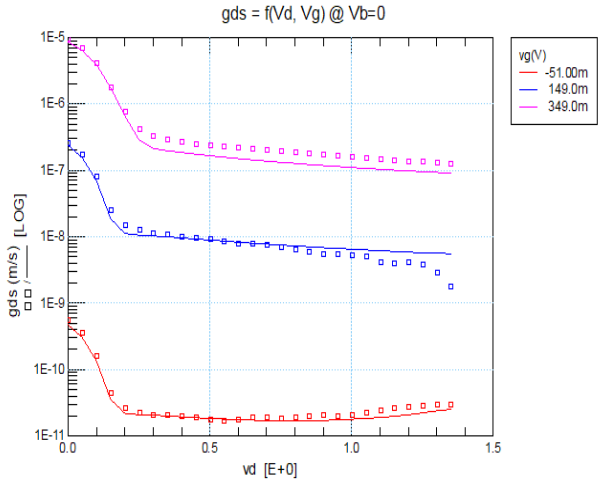




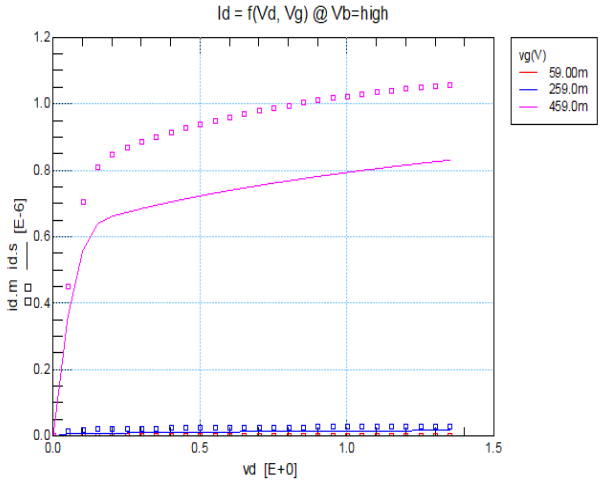
idvd_vth, Vb = 0V, T = 27°C

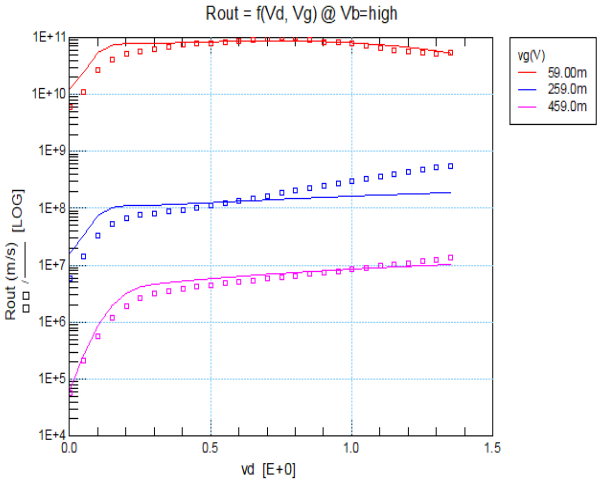


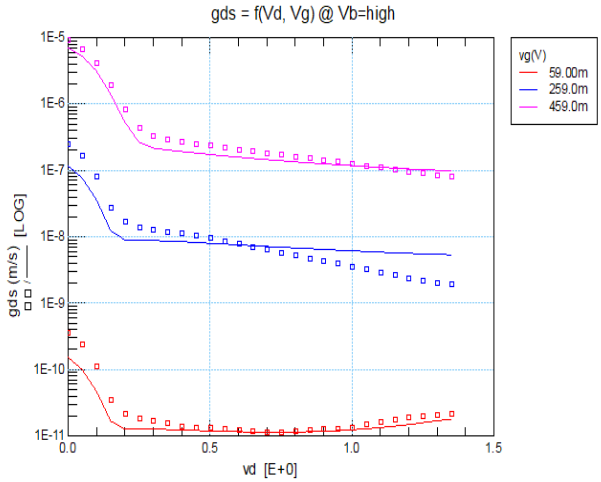




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

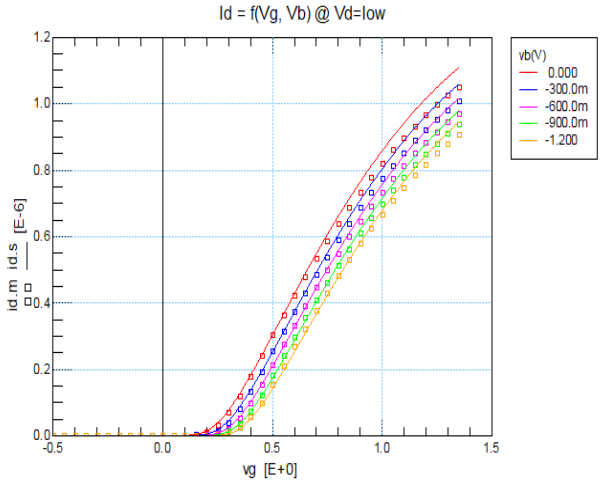


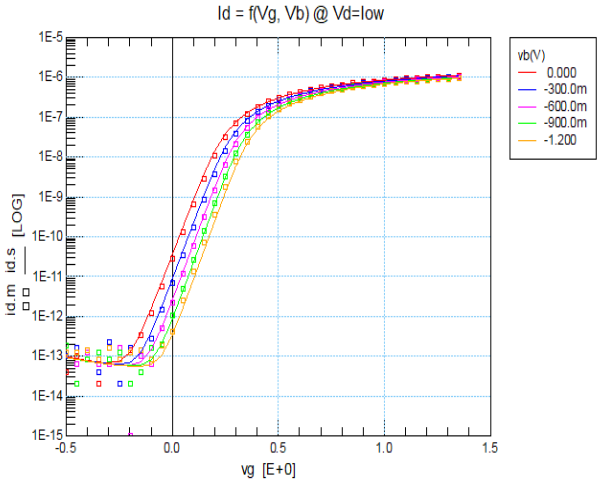


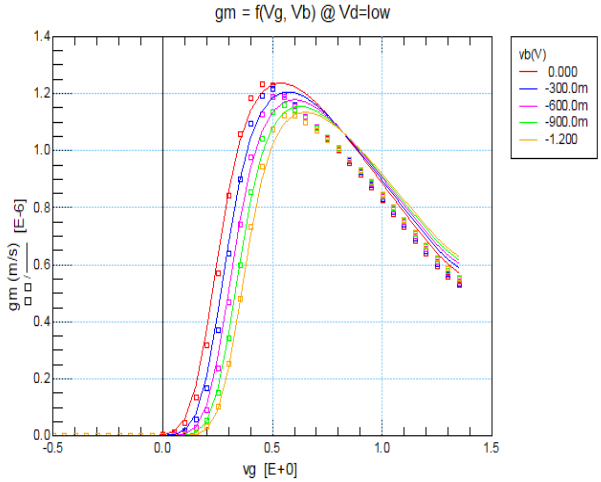


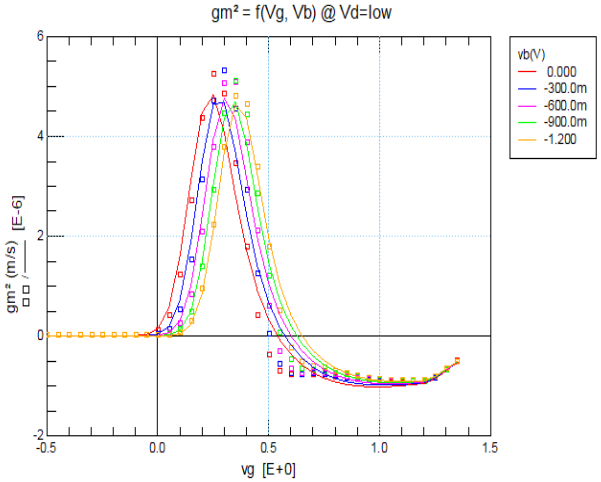
4.19W0u6_L10u0_S544_1

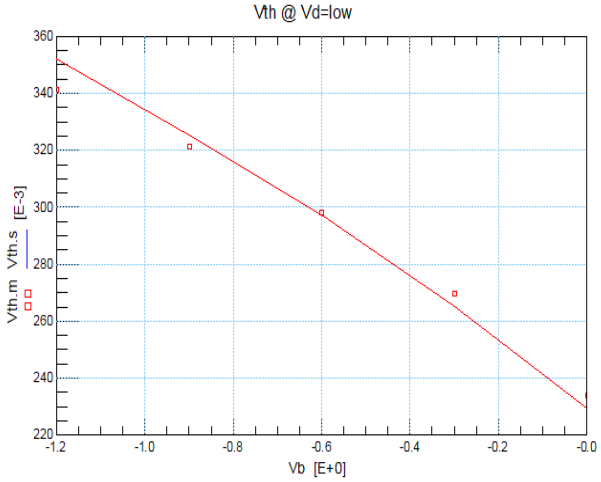
idvg, Vd = 0.05V, T = 27°C

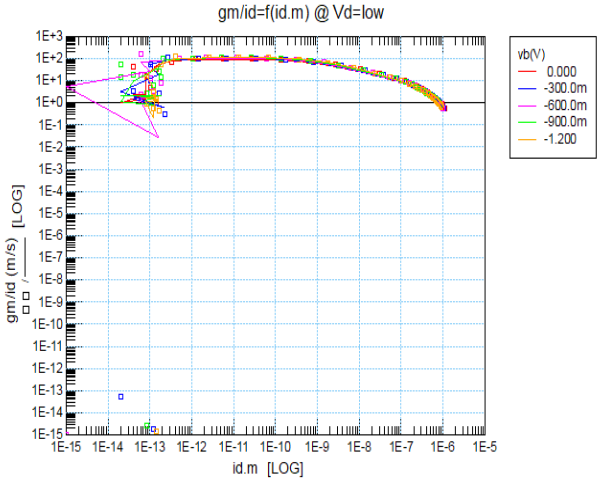


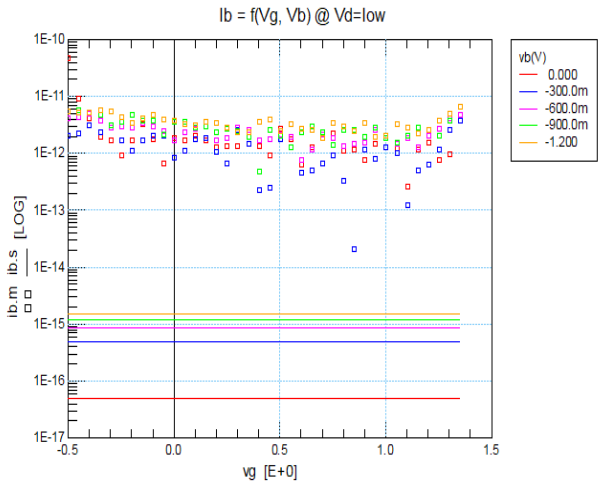


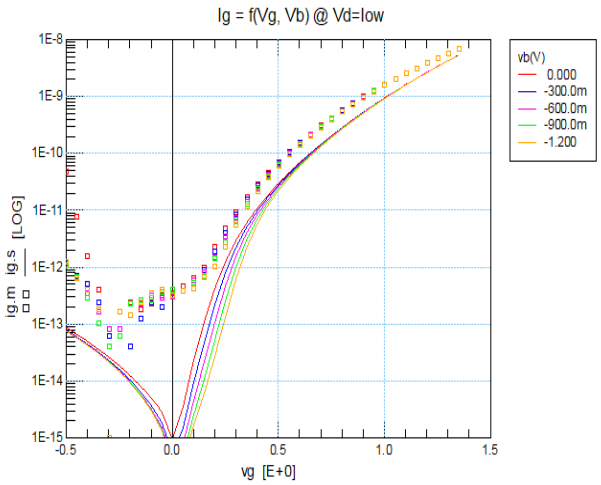




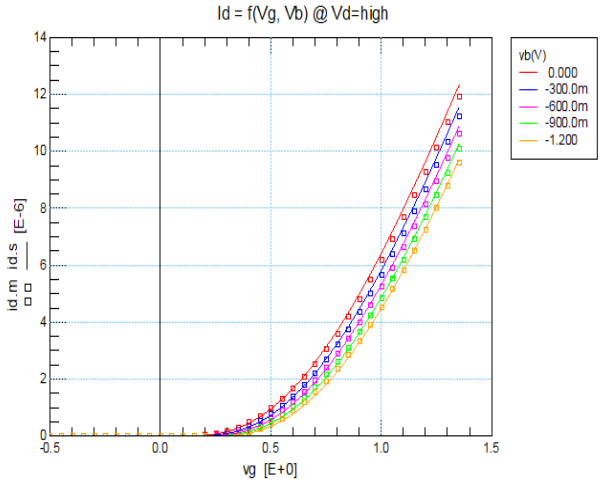


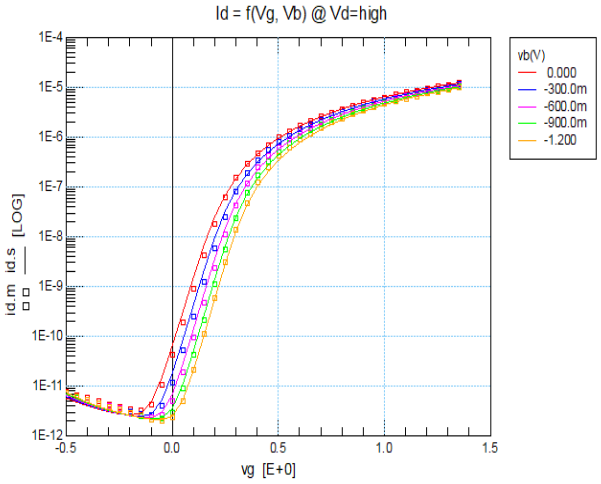


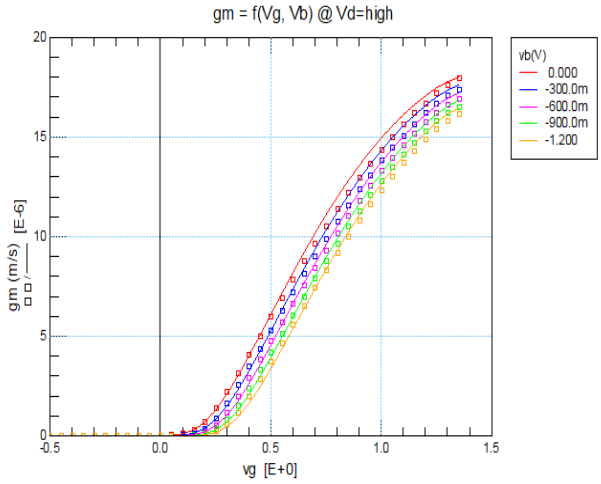


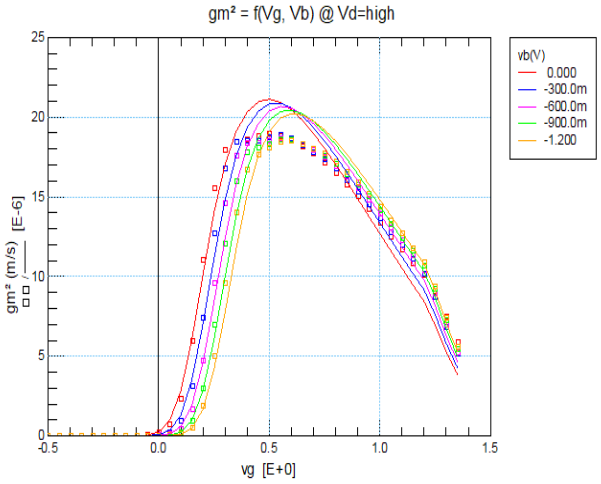


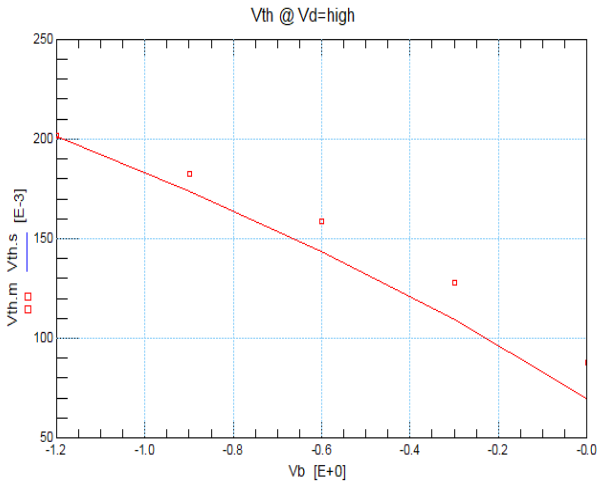
idvg, Vd = 1.2V, T = 27°C

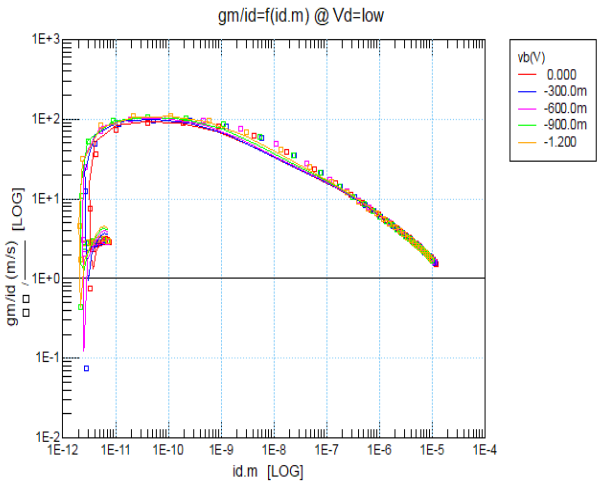


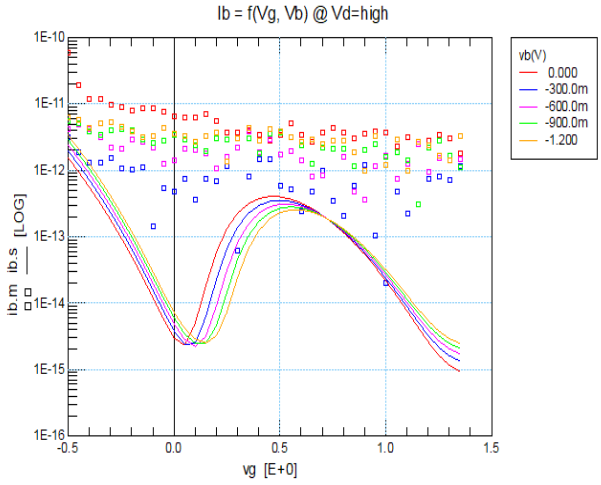


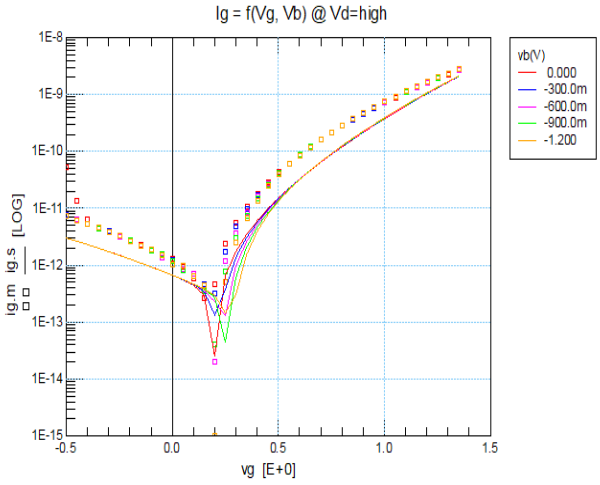




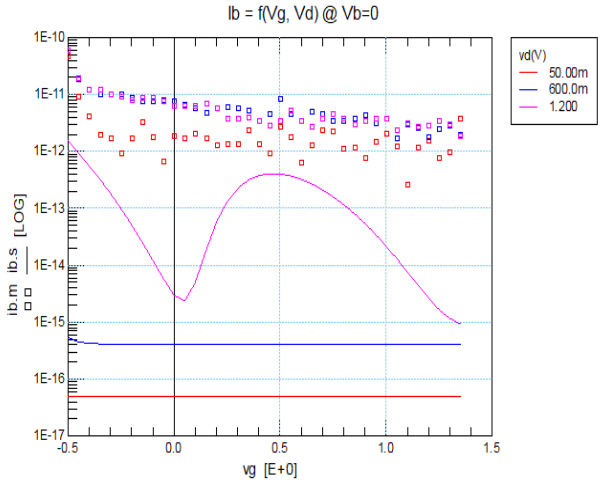


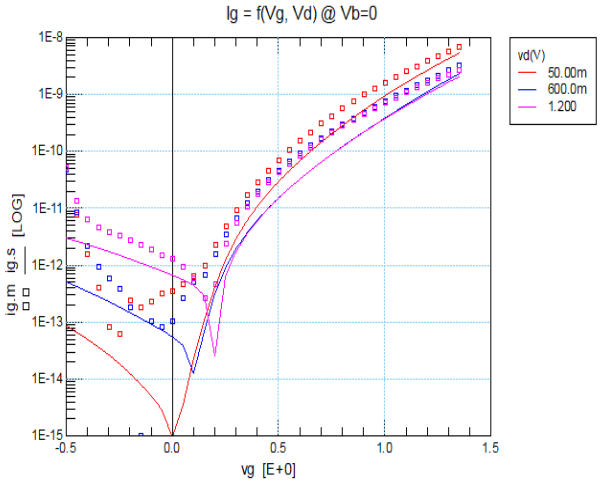




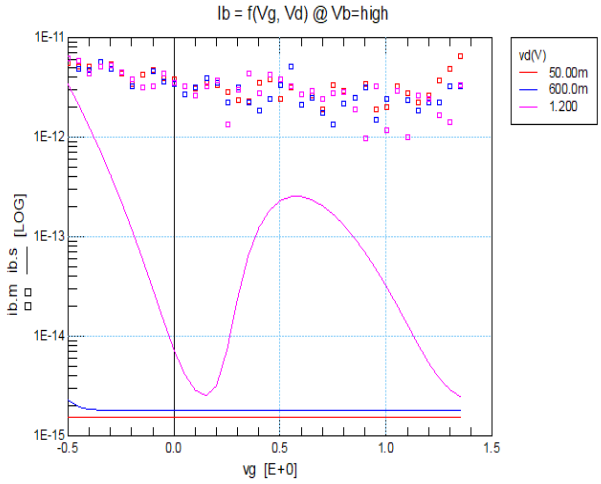


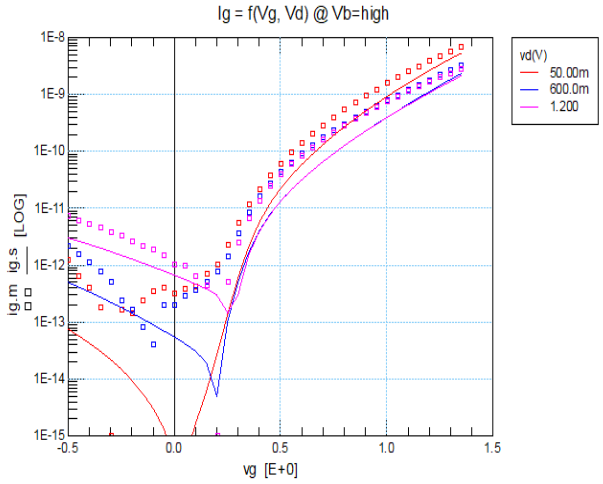
idvg, Ib,Ig, Vb = 0V, T = 27°C



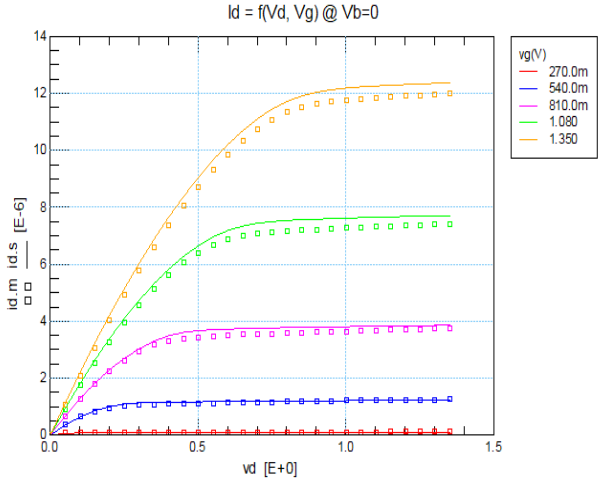


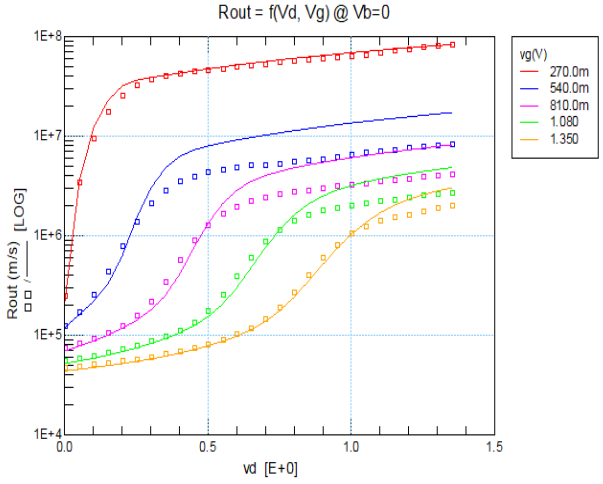
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

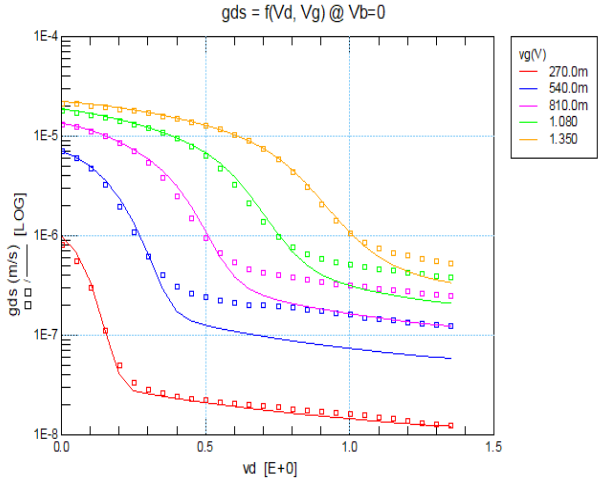




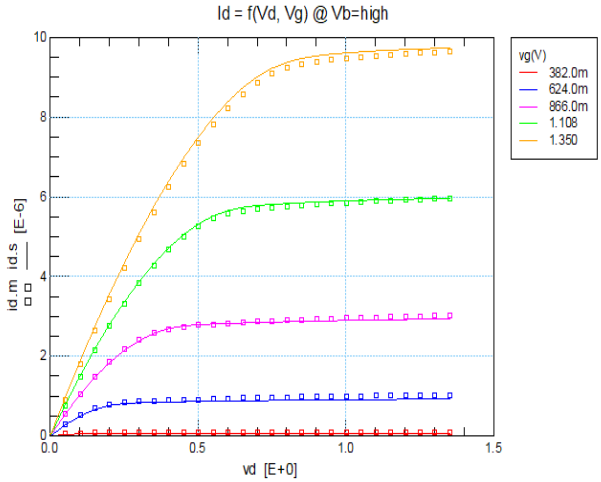
idvd, Vb = 0V, T = 27°C

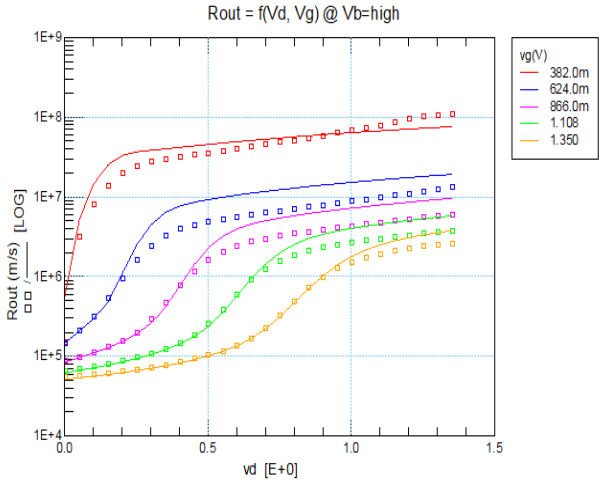


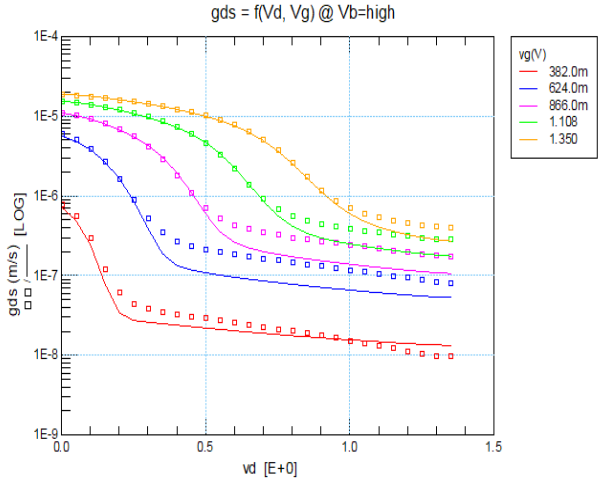




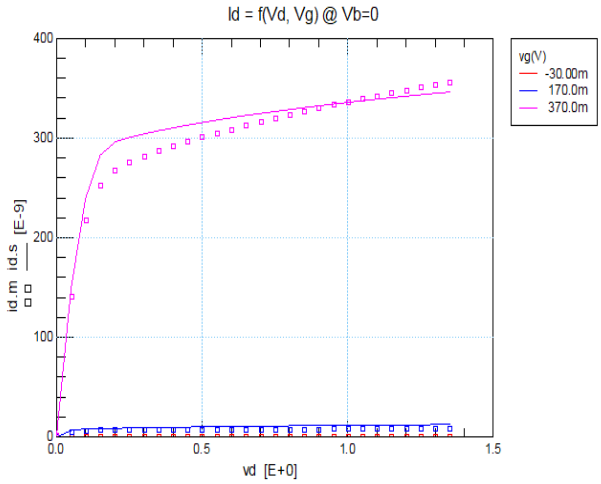
idvd_vbmin, Vb = -1.2V, T = 27°C

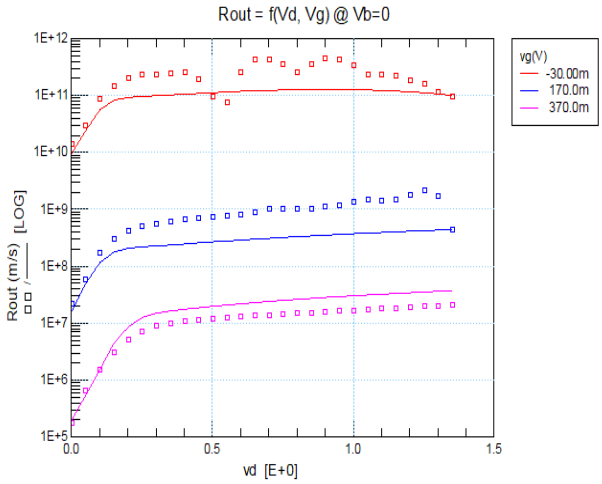


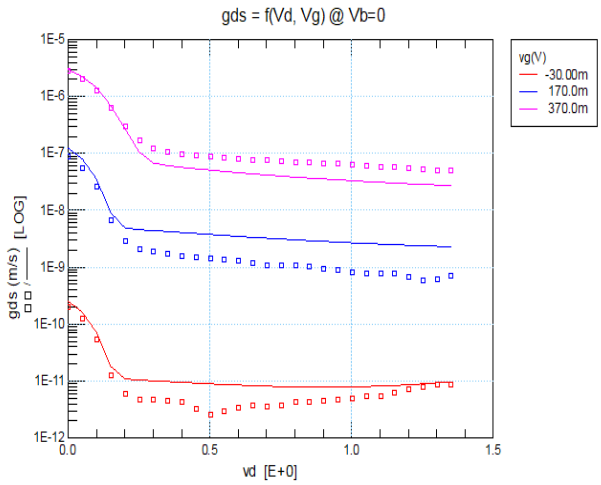




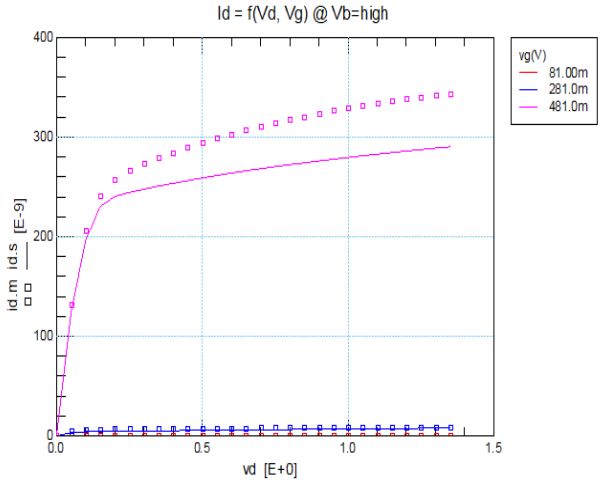
idvd_vth, Vb = 0V, T = 27°C

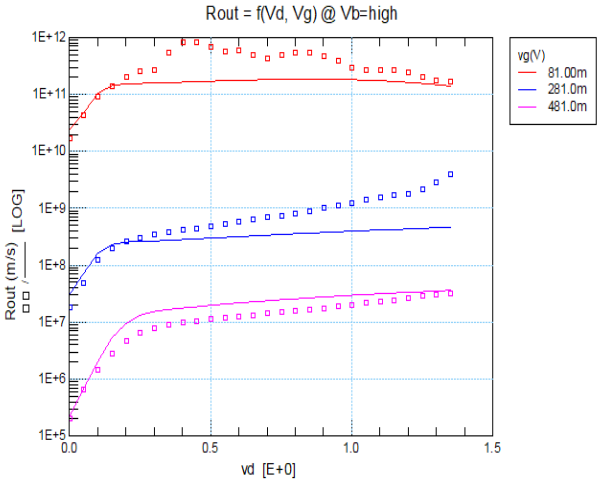


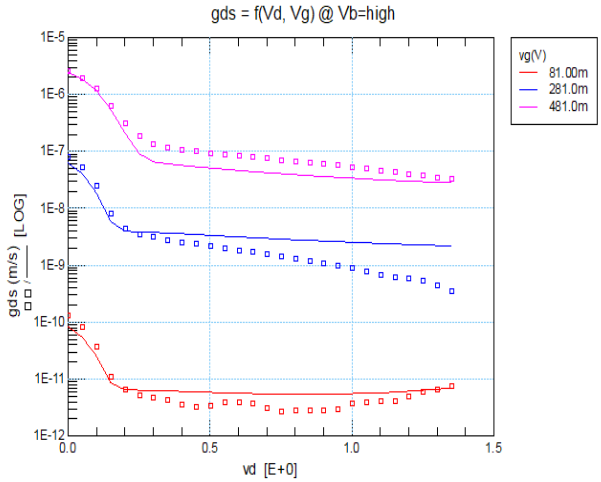




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

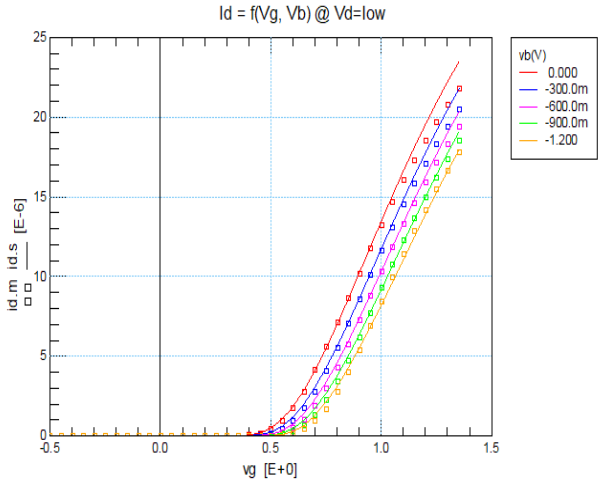


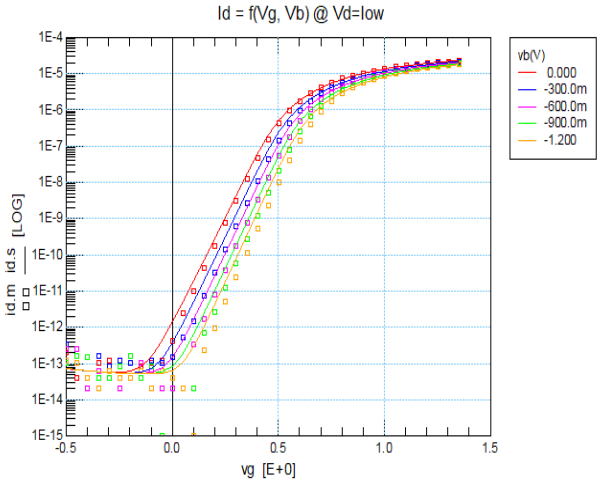


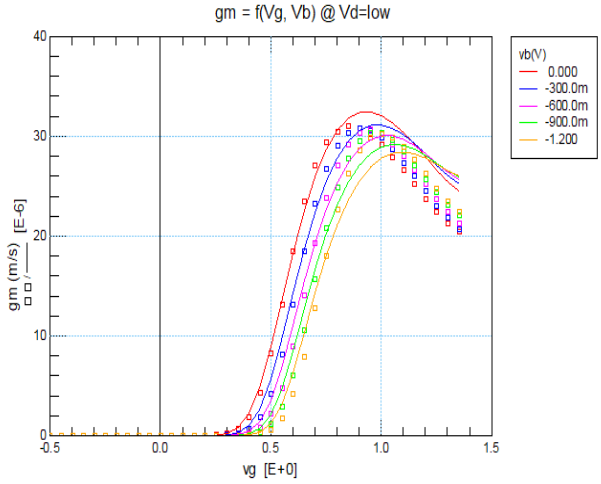


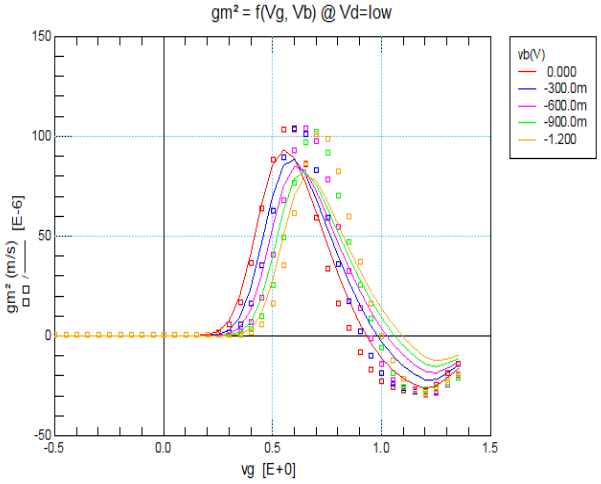
4.20W0u3_L0u13_S544_2

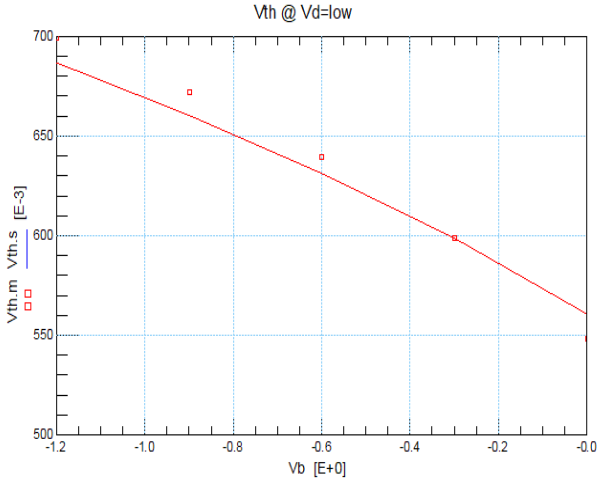
idvg, Vd = 0.05V, T = 27°C

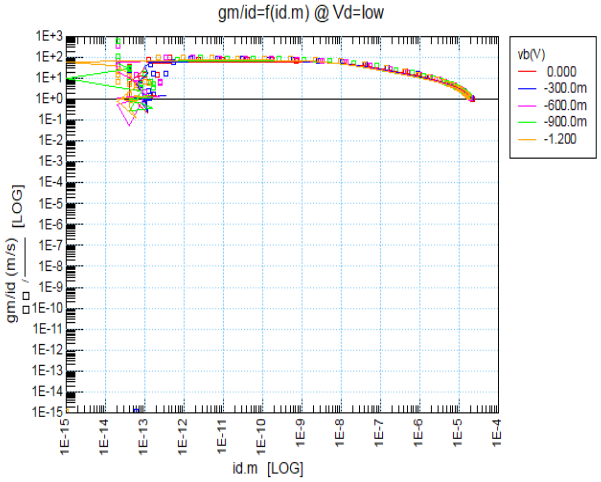


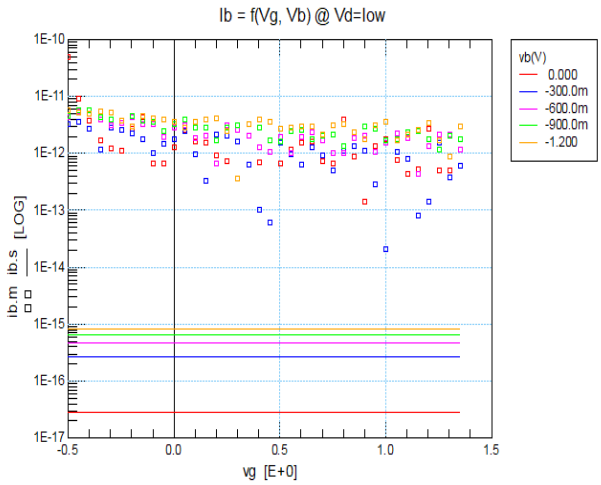


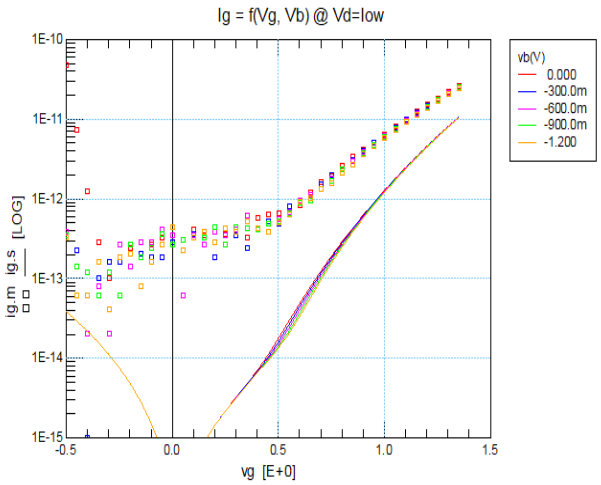




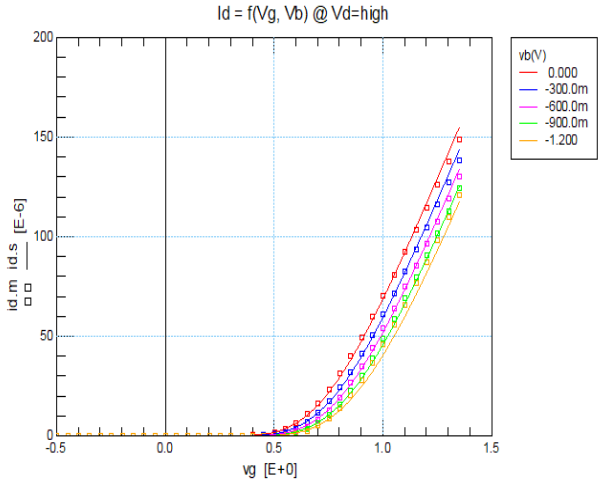


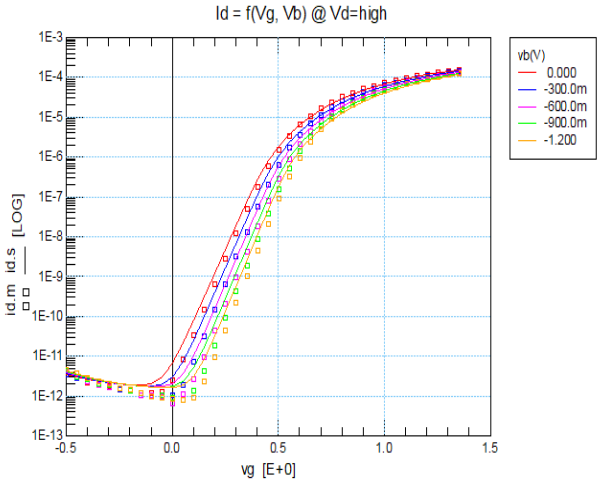


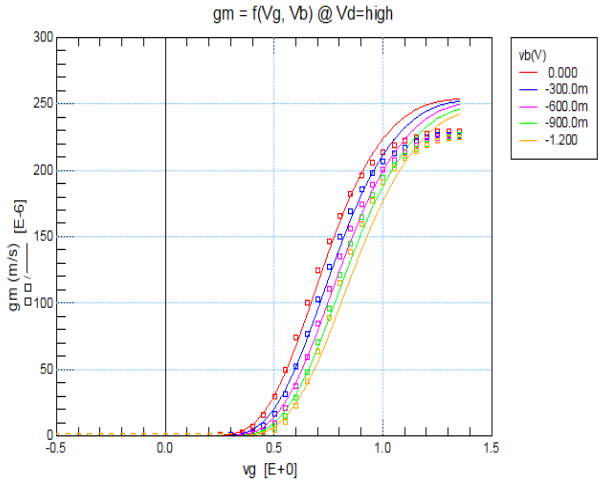


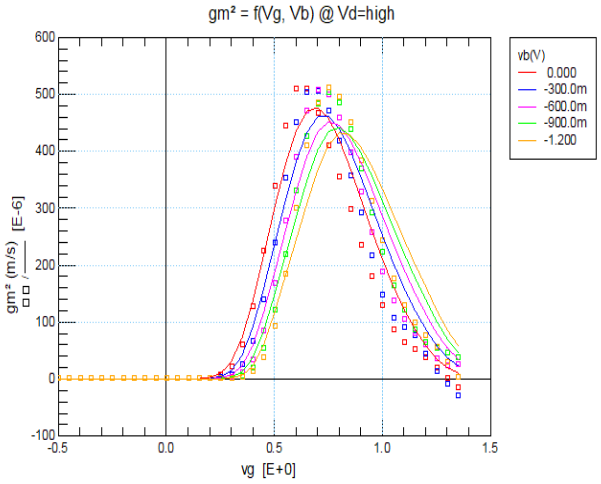


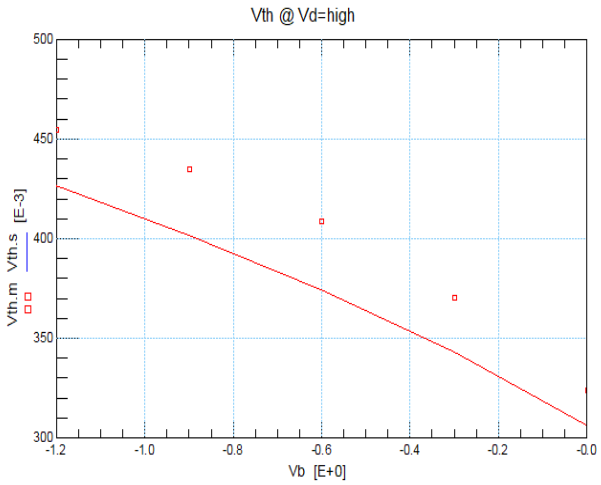
idvg, Vd = 1.2V, T = 27°C

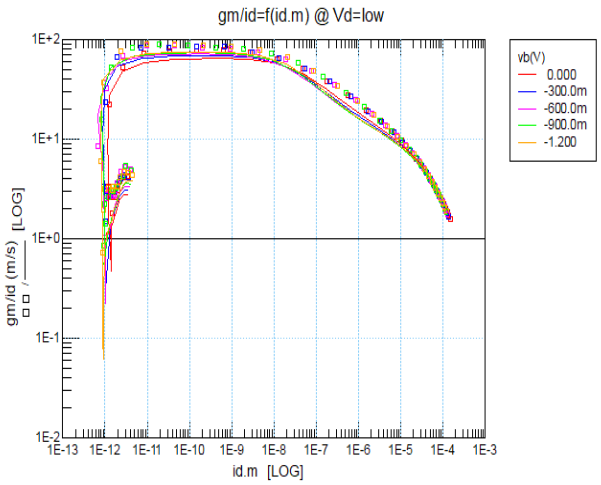


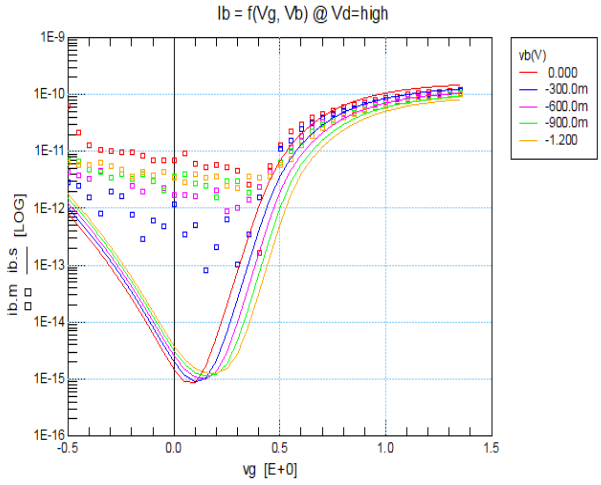


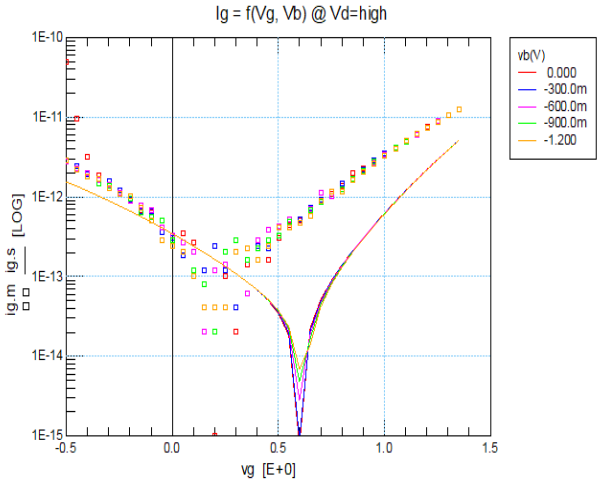




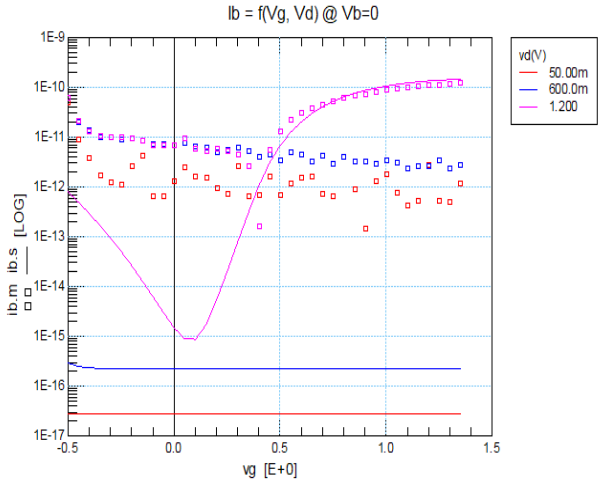


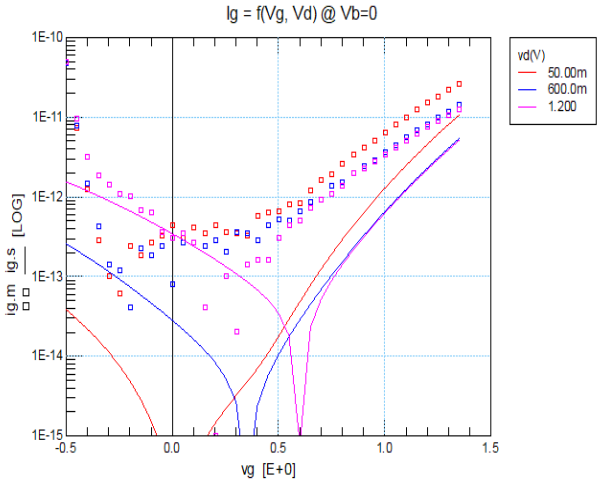




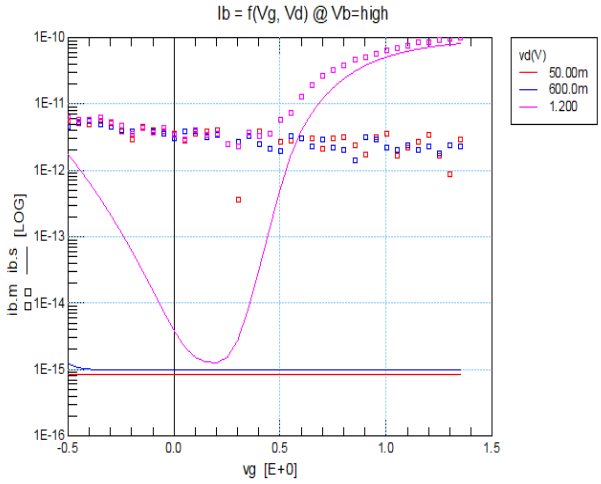


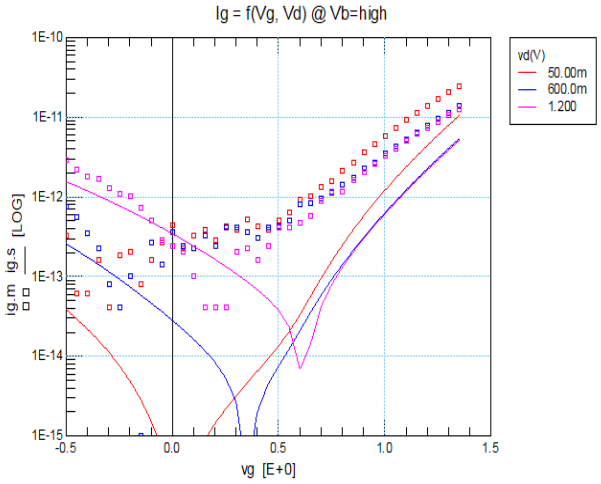
idvg, Ib,Ig, Vb = 0V, T = 27°C



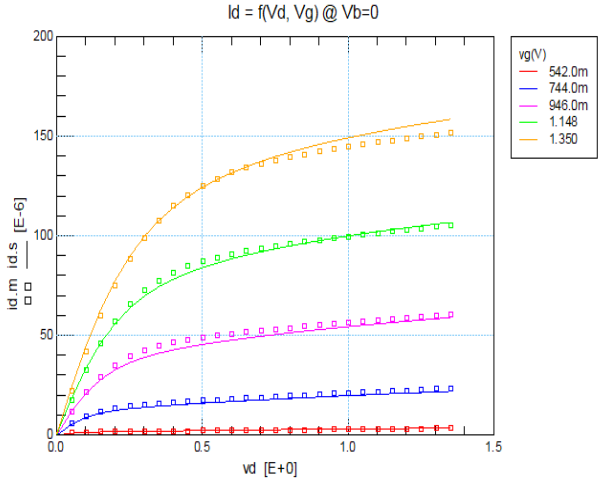


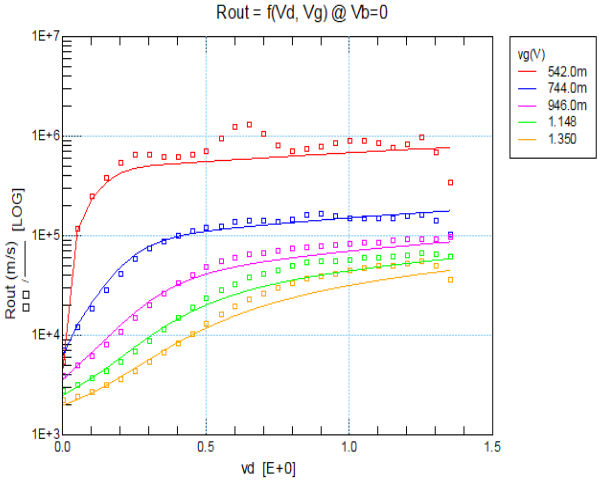
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

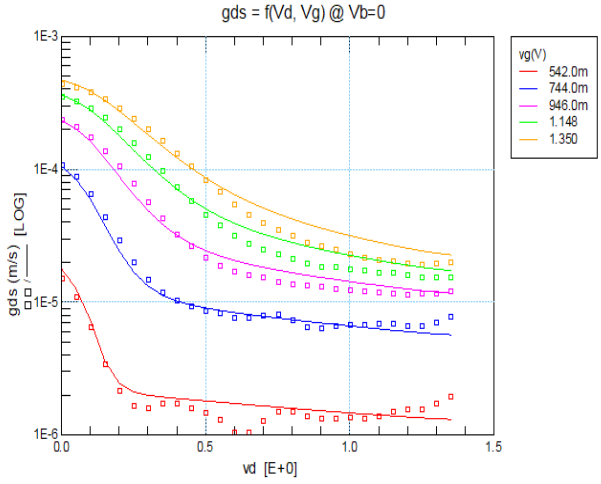




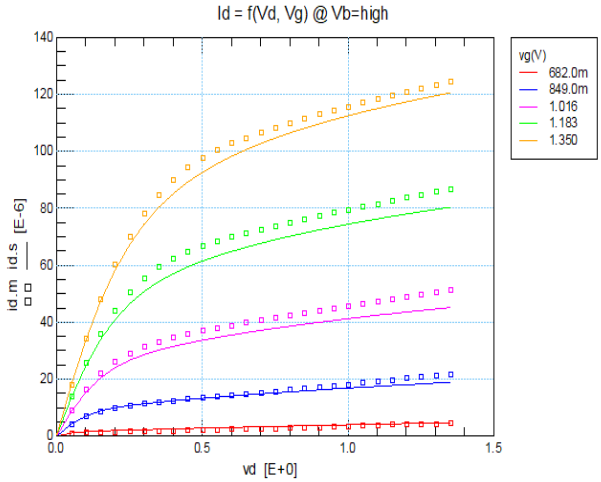
idvd, Vb = 0V, T = 27°C

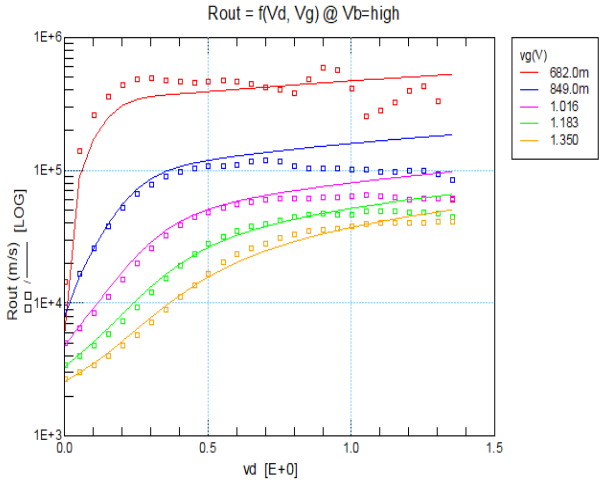


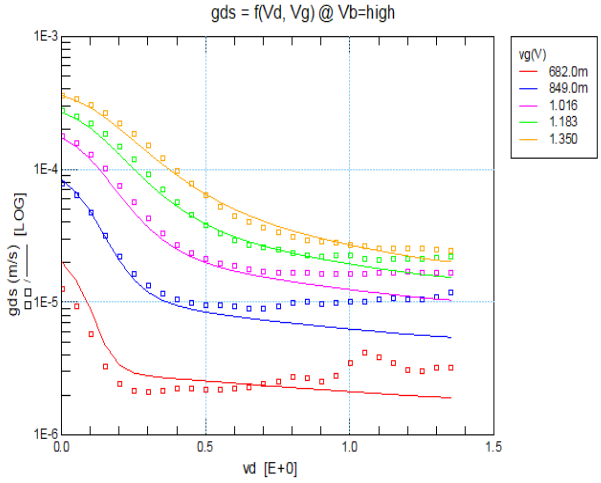




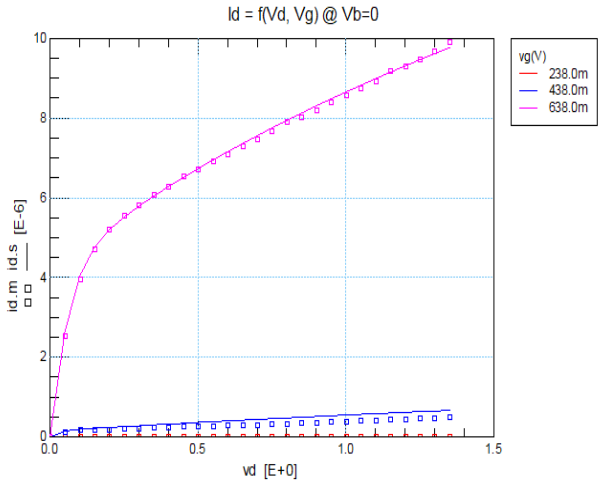
idvd_vbmin, Vb = -1.2V, T = 27°C

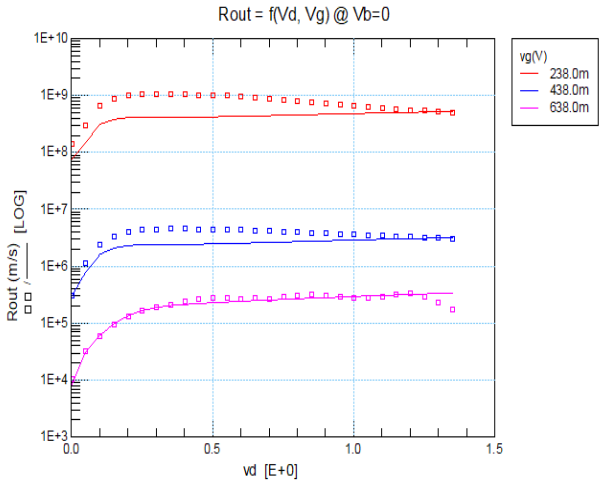


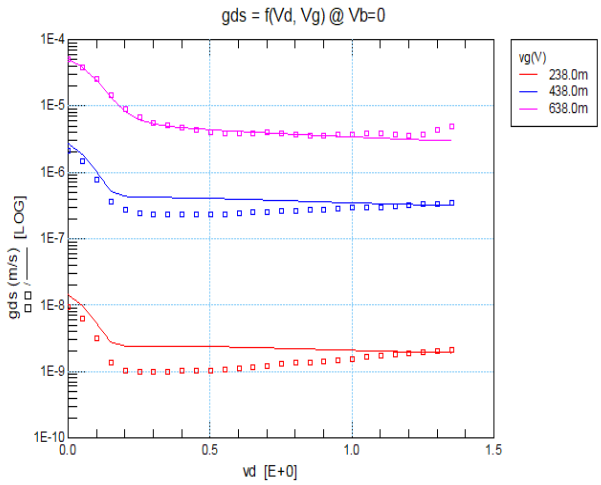




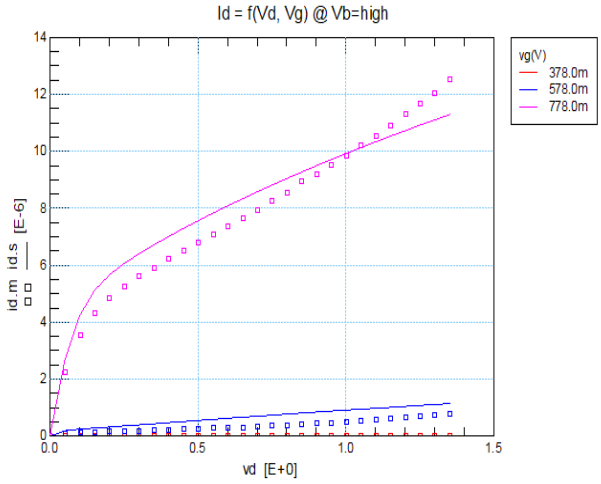
idvd_vth, Vb = 0V, T = 27°C

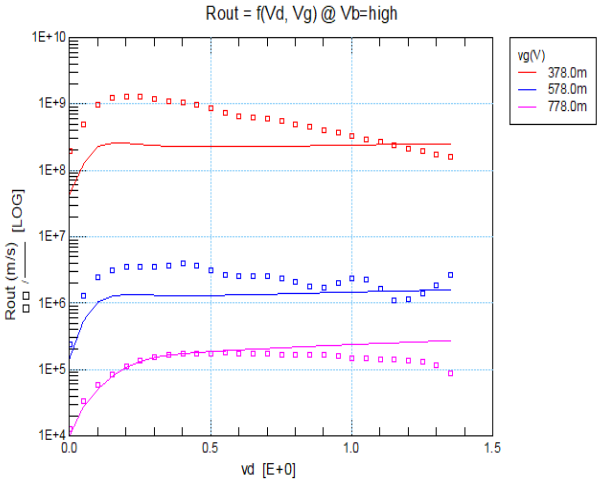


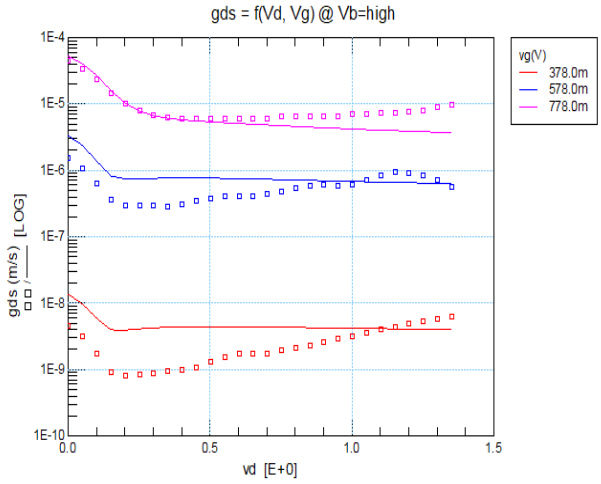




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

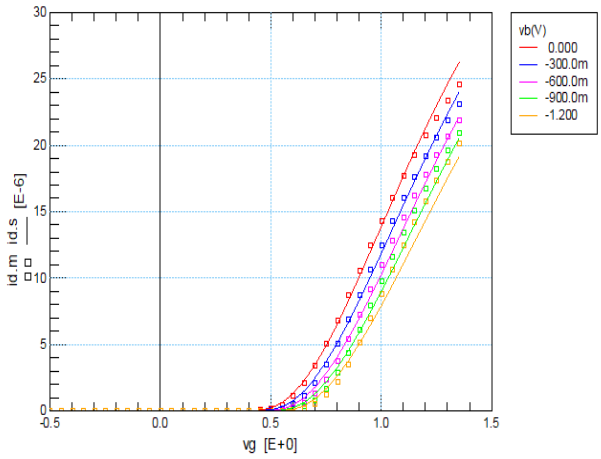




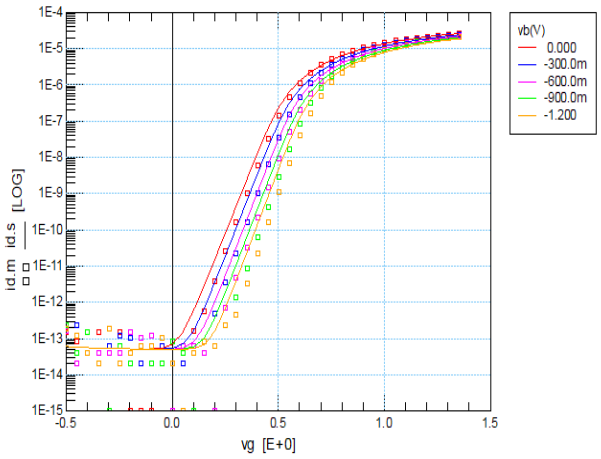


idvg, Vd = 0.05V, T = -40°C

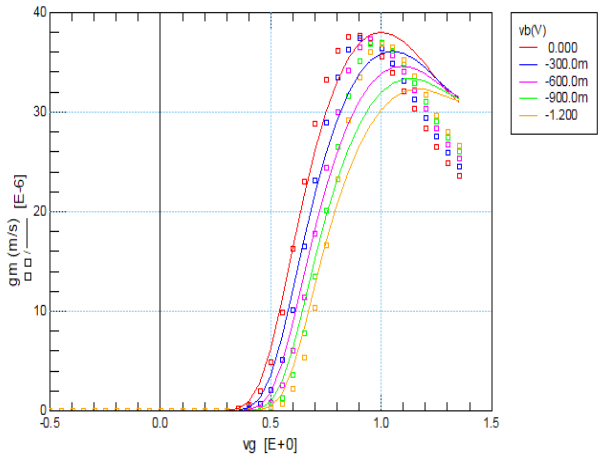
Id = f(Vg, Vb) @ Vd=low



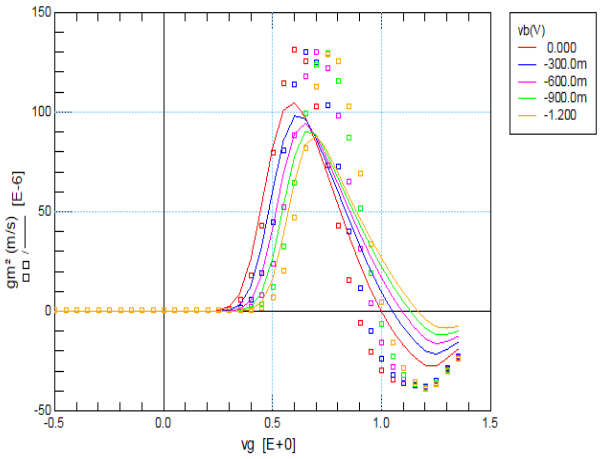
Id = f(Vg, Vb) @ Vd=low



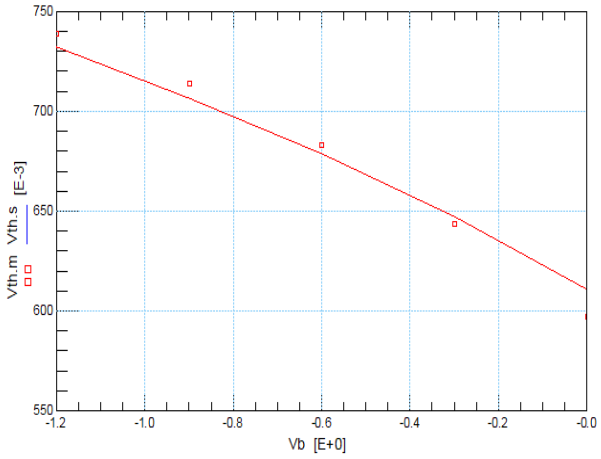
gm = f(Vg, Vb) @ Vd=low



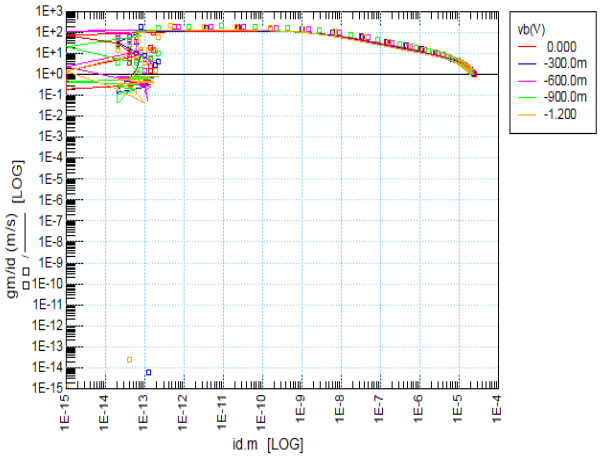
gm² = f(Vg, Vb) @ Vd=low

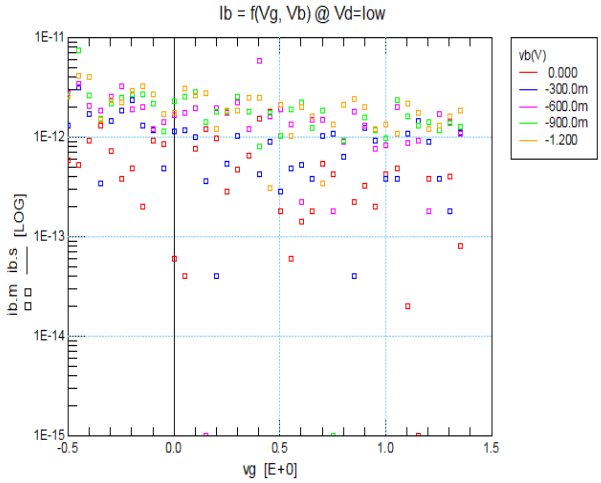


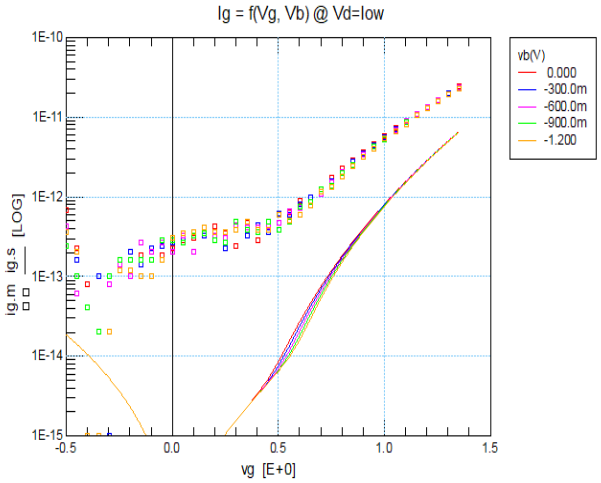
Vth @ Vd=low



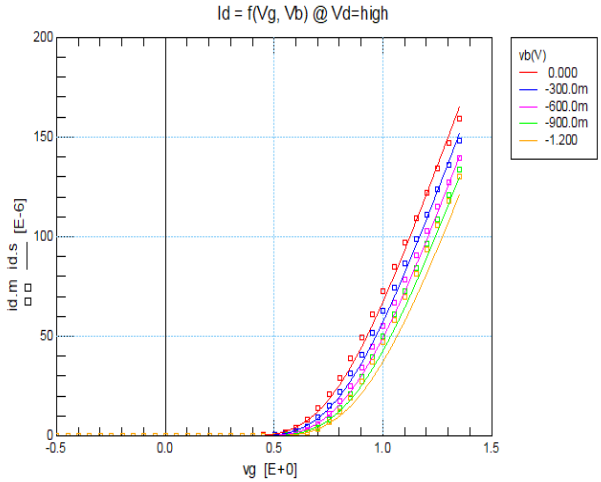
gm/id=f(id.m) @ Vd=low

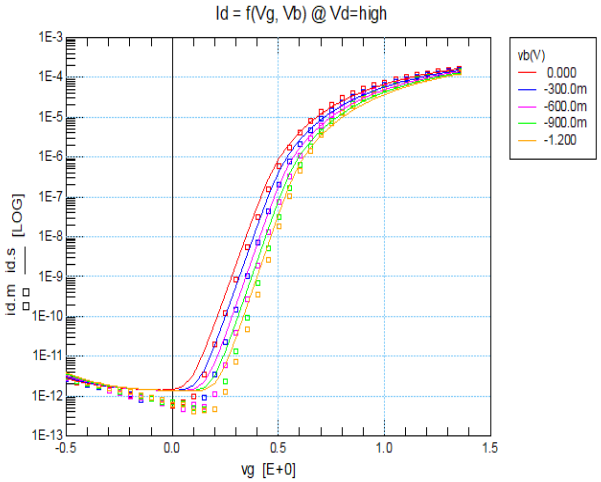


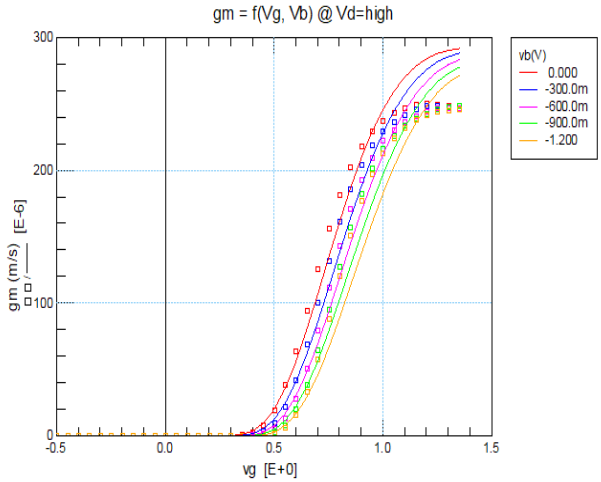


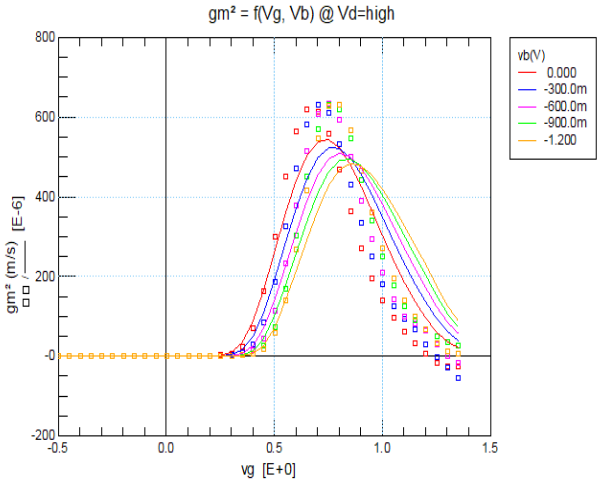


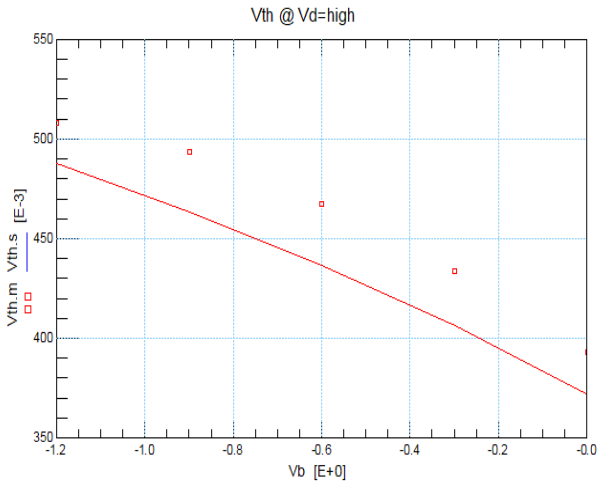
idvg, Vd = 1.2V, T = -40°C

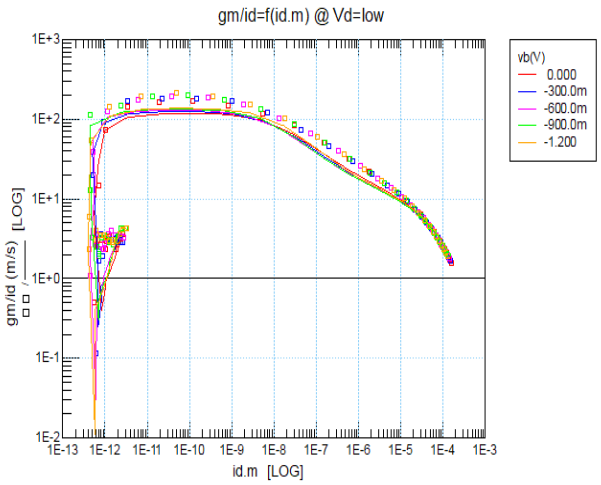


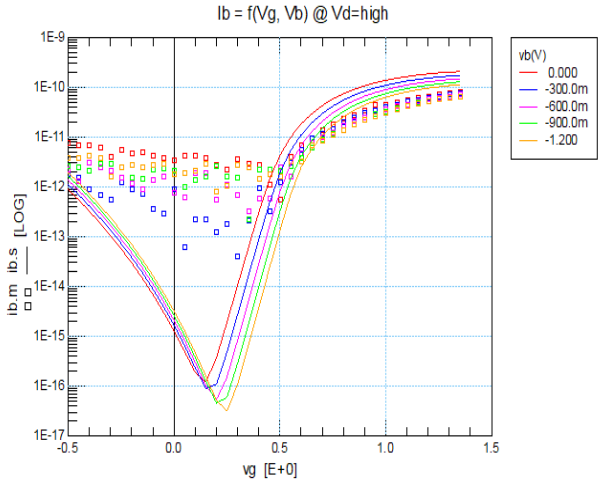


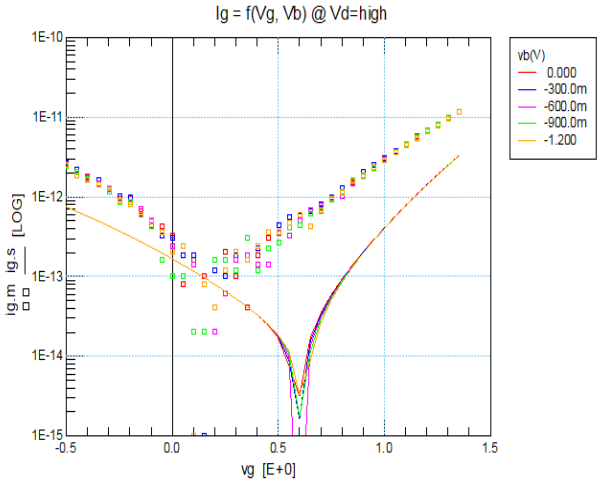




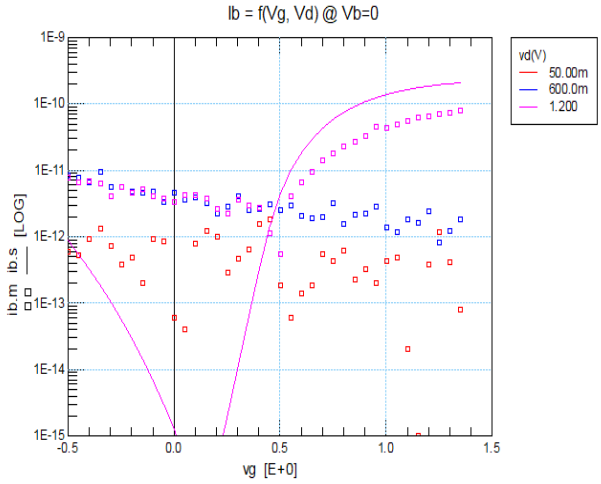


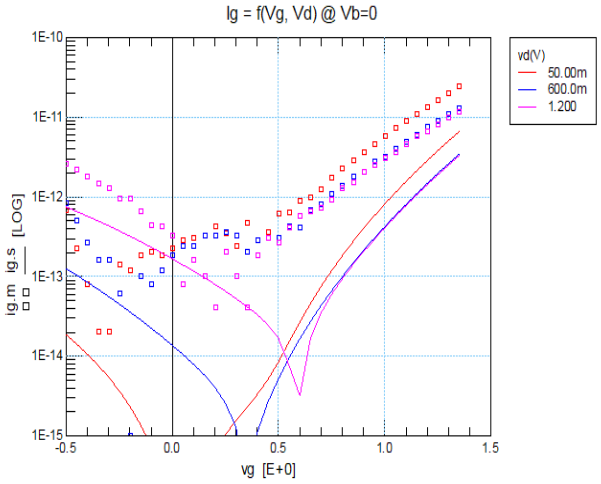




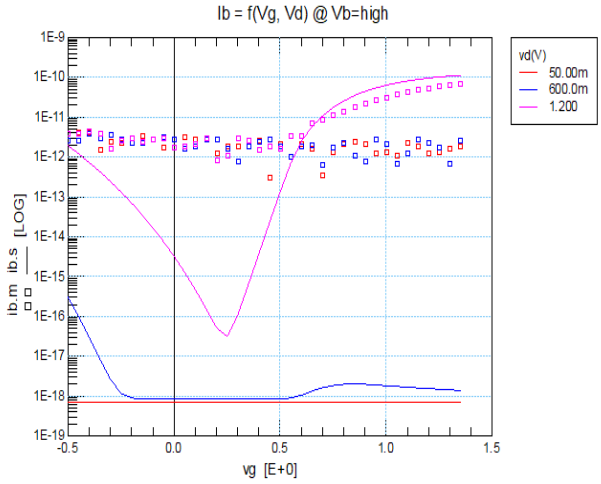


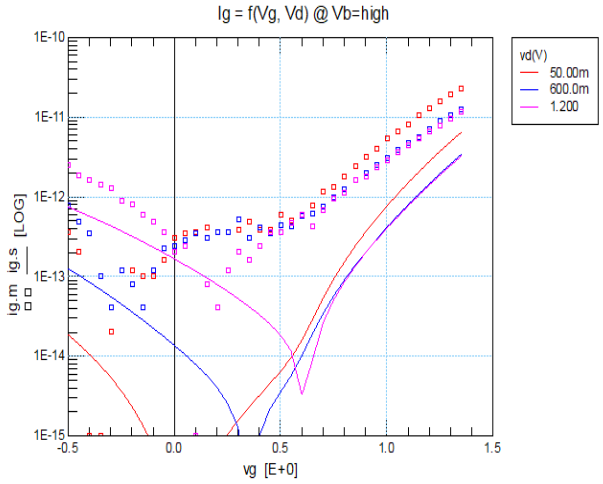
idvg, Ib,Ig, Vb = 0V, T = -40°C



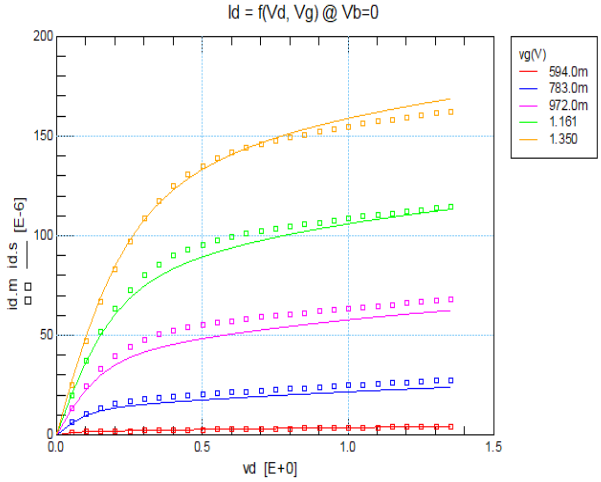


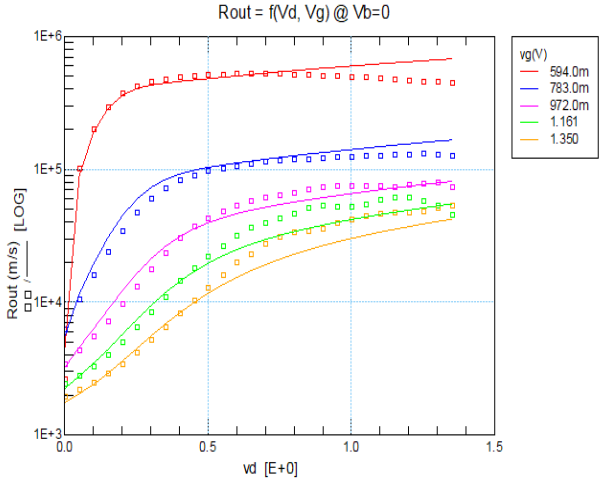
idvg, Ib,Ig, Vb = -1.2V, T = -40°C

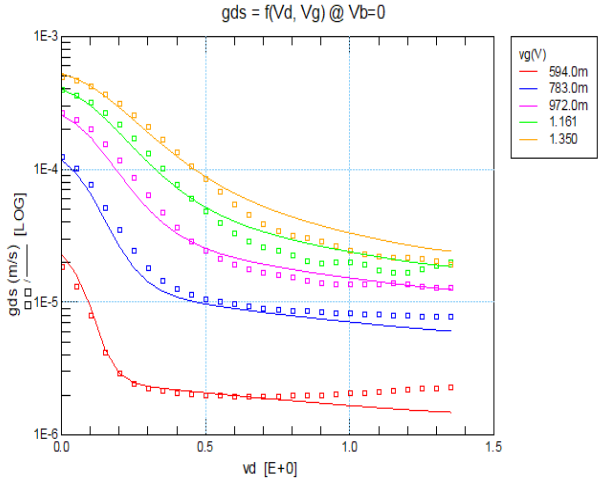




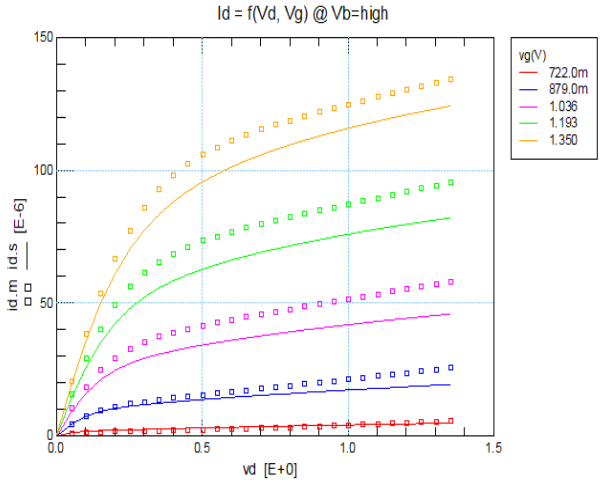
idvd, Vb = 0V, T = -40°C

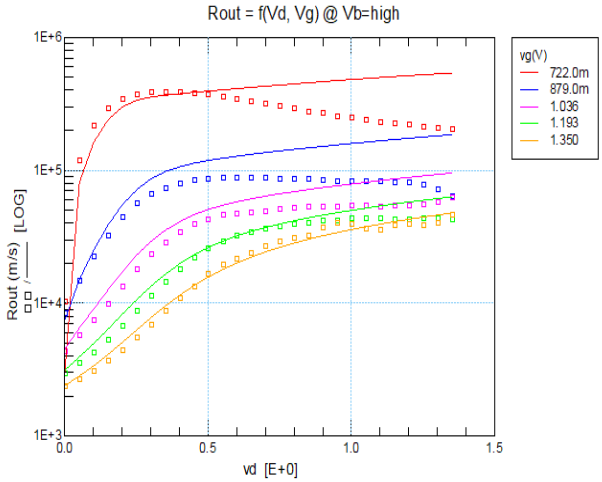


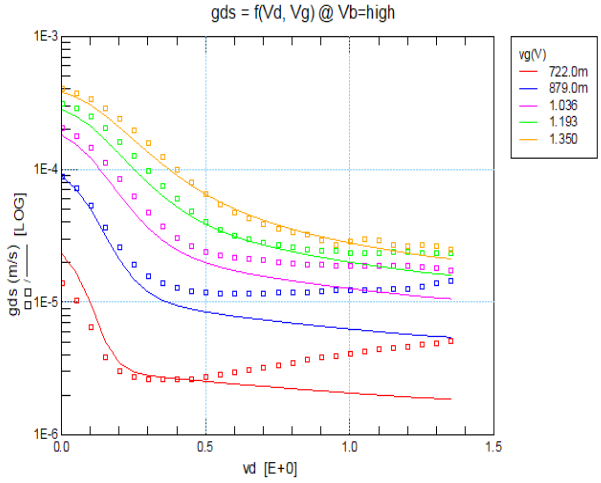




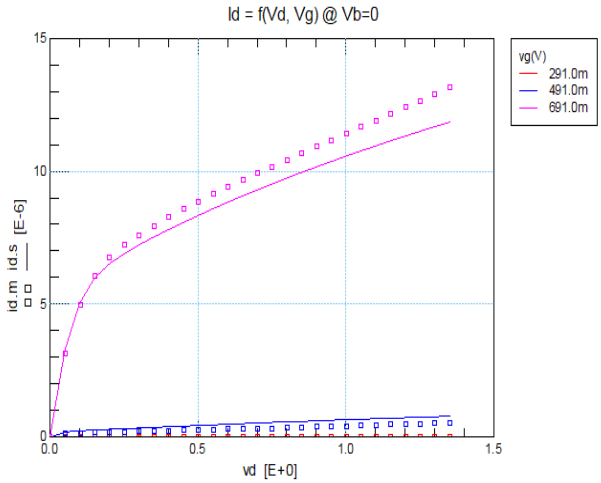
idvd_vbmin, Vb = -1.2V, T = -40°C

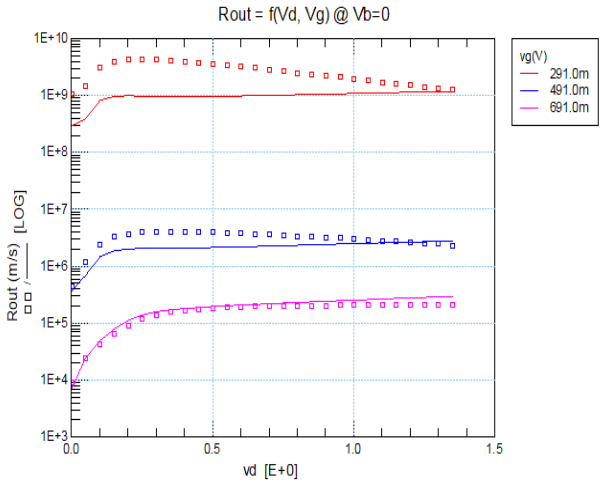


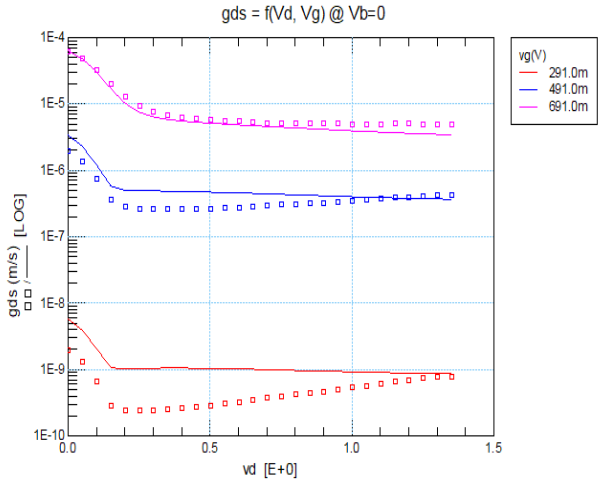




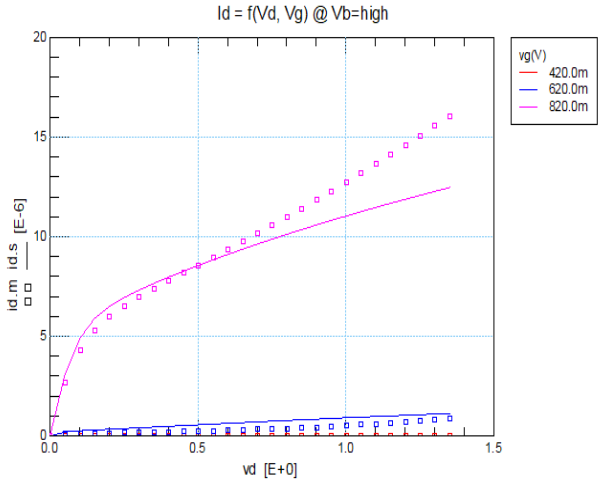
idvd_vth, Vb = 0V, T = -40°C

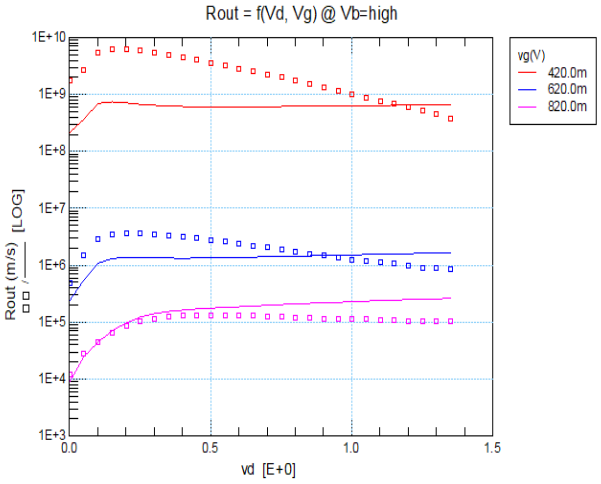


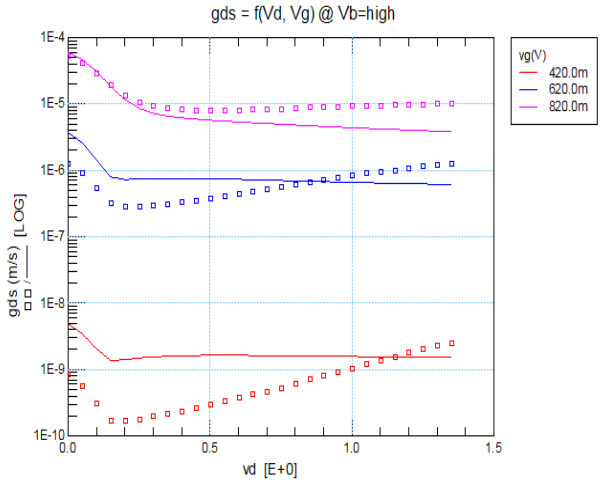




idvd_vth_vbmin, Vb = -1.2V, T = -40°C

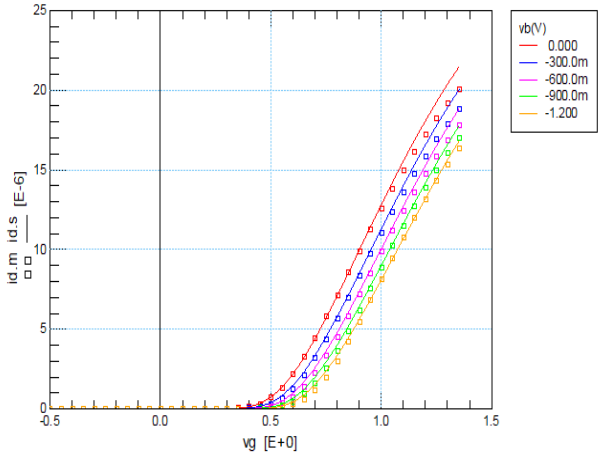




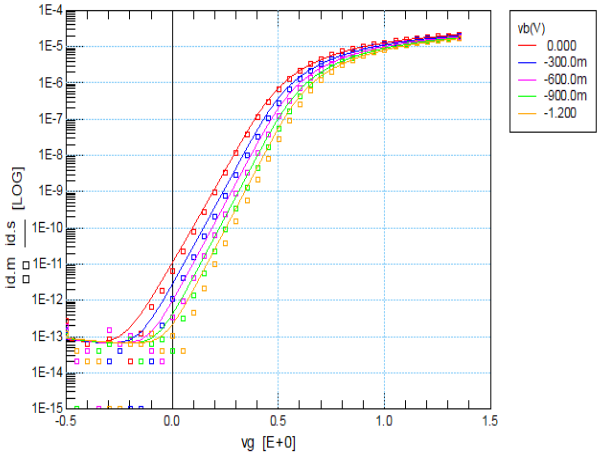


idvg, Vd = 0.05V, T = 70°C

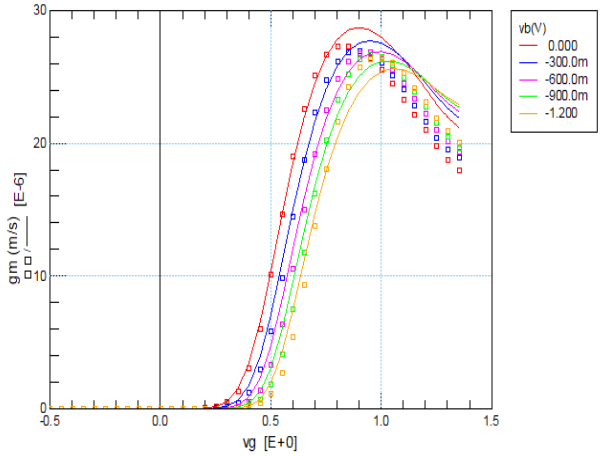
Id = f(Vg, Vb) @ Vd=low



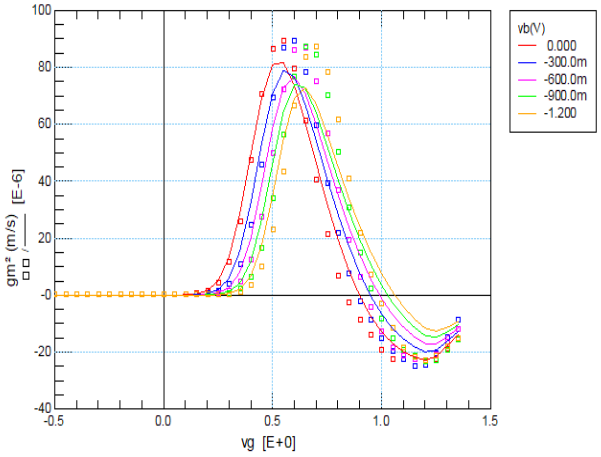
Id = f(Vg, Vb) @ Vd=low



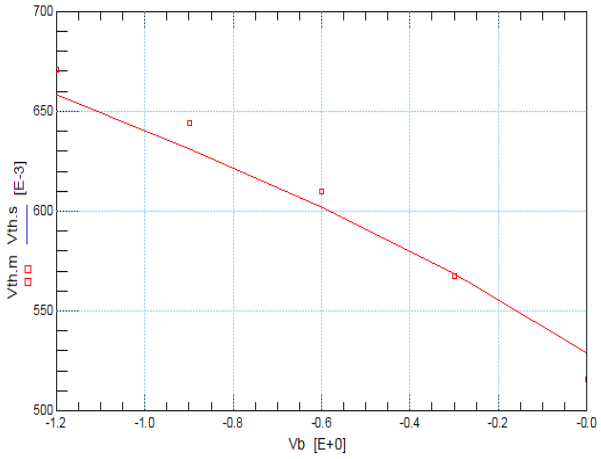
gm = f(Vg, Vb) @ Vd=low



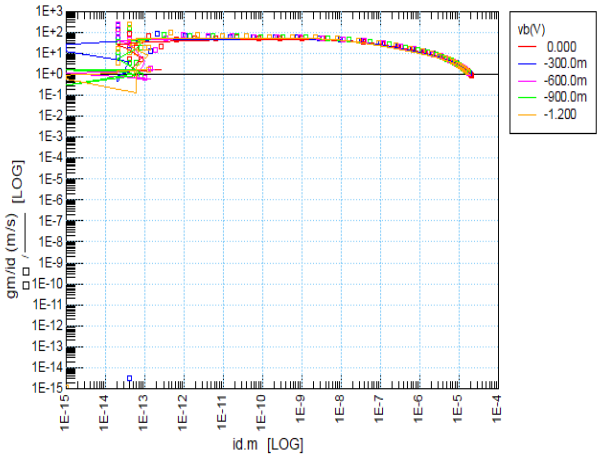
gm² = f(Vg, Vb) @ Vd=low

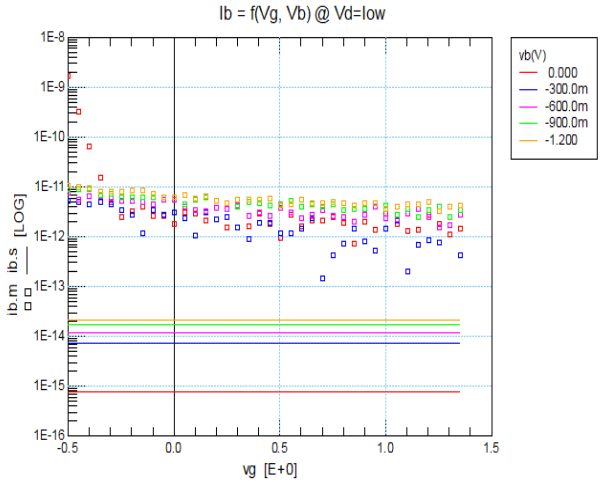


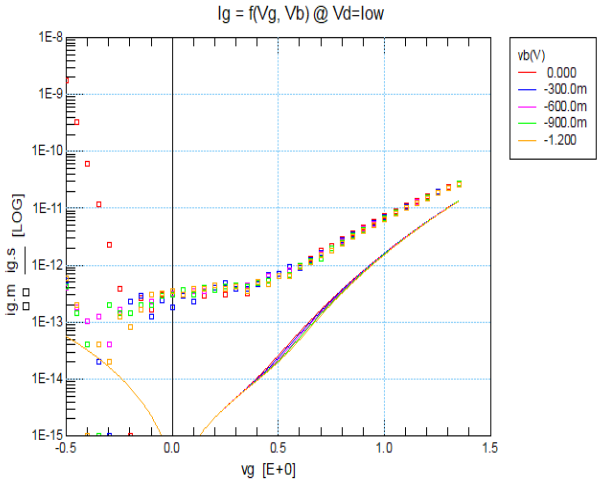
Vth @ Vd=low



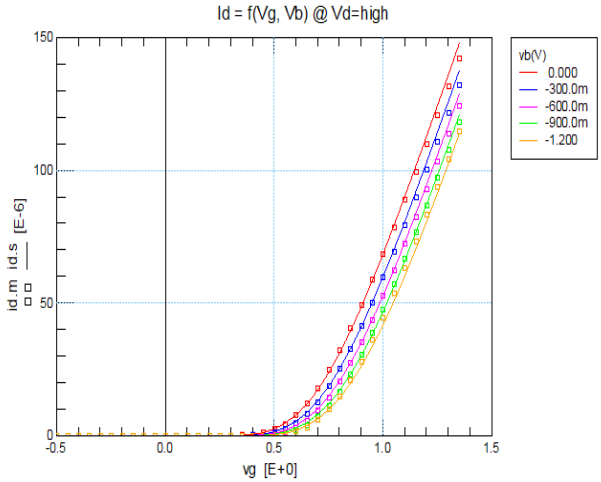
gm/id=f(id.m) @ Vd=low

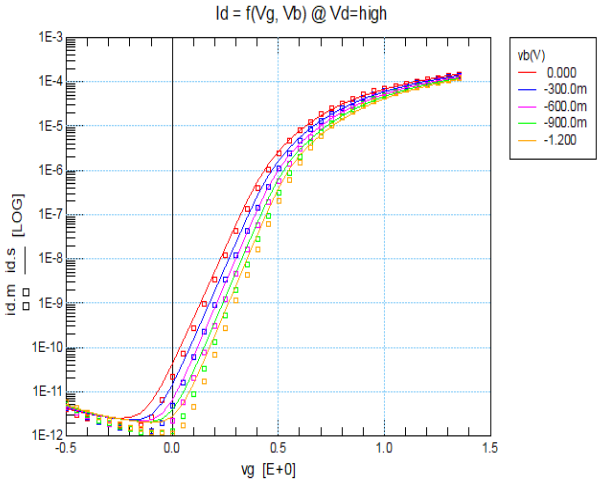


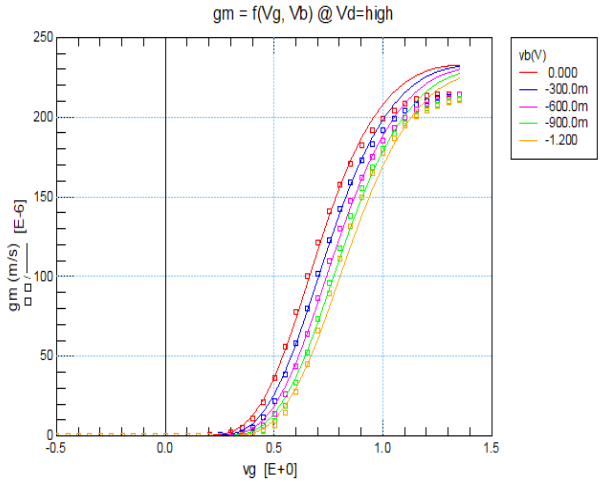


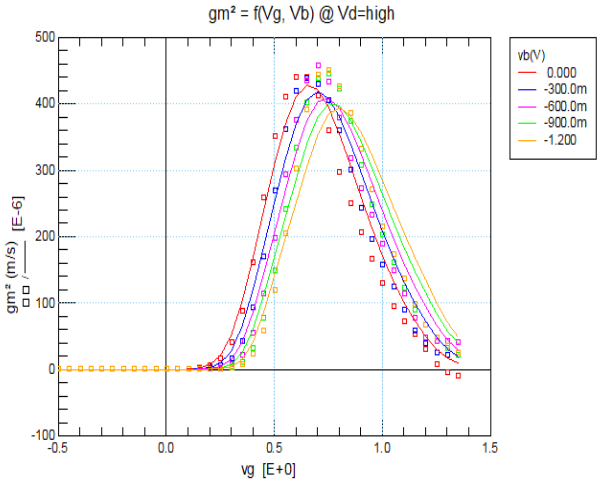


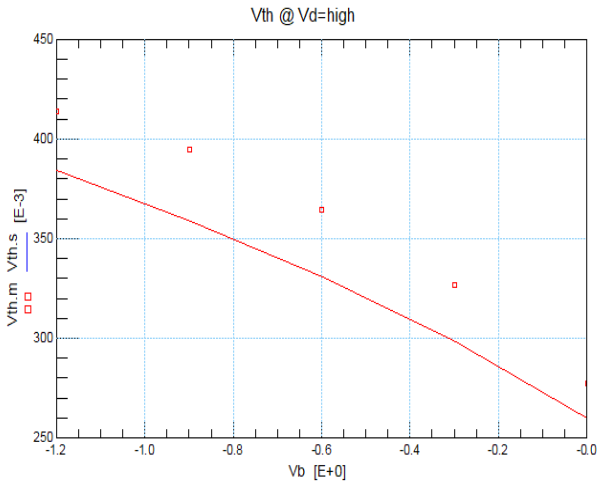
idvg, Vd = 1.2V, T = 70°C

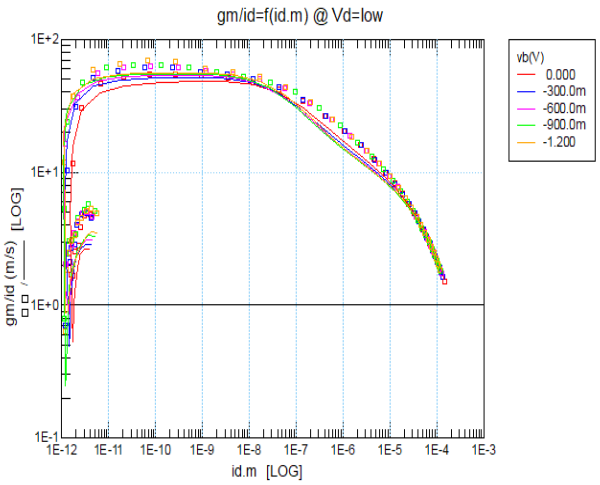


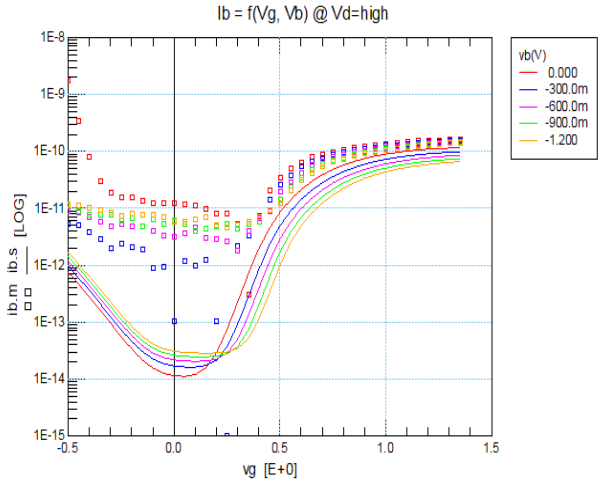


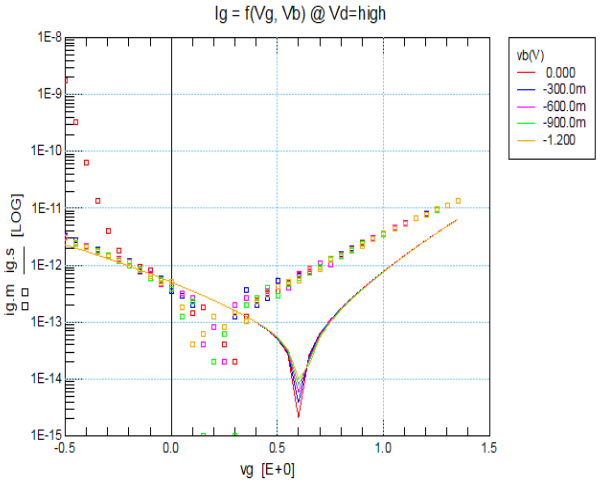




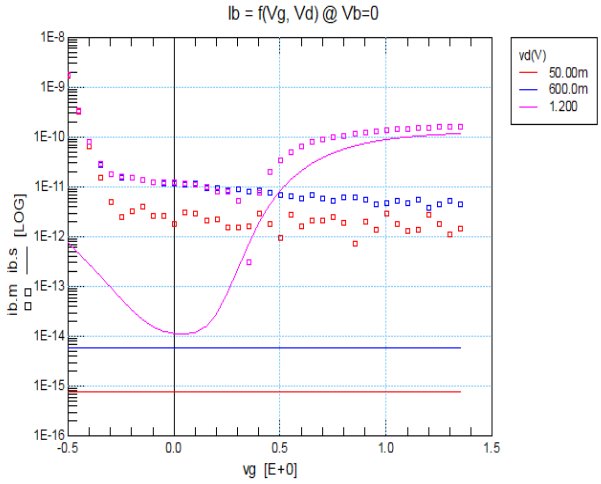


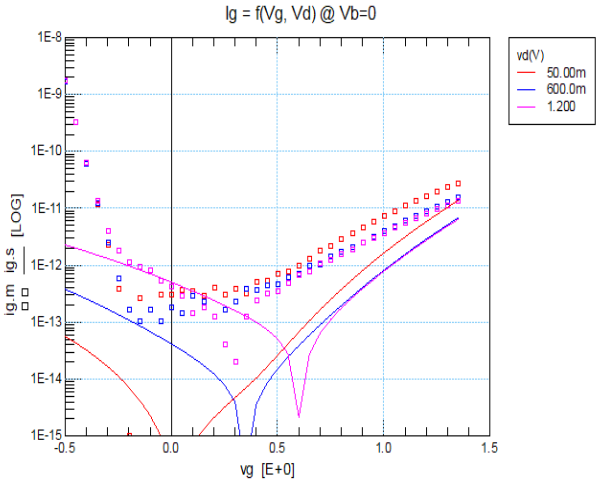




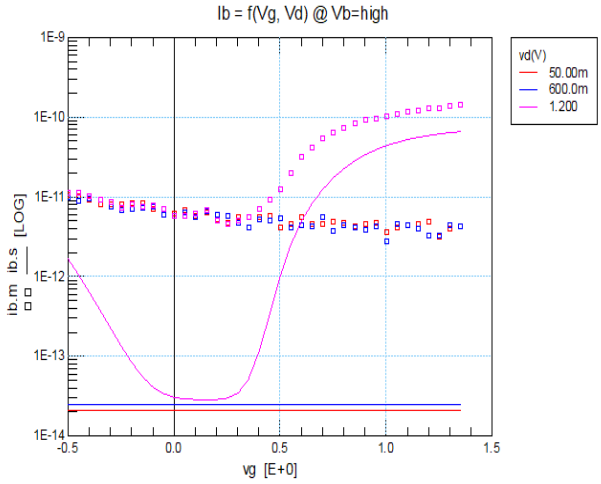


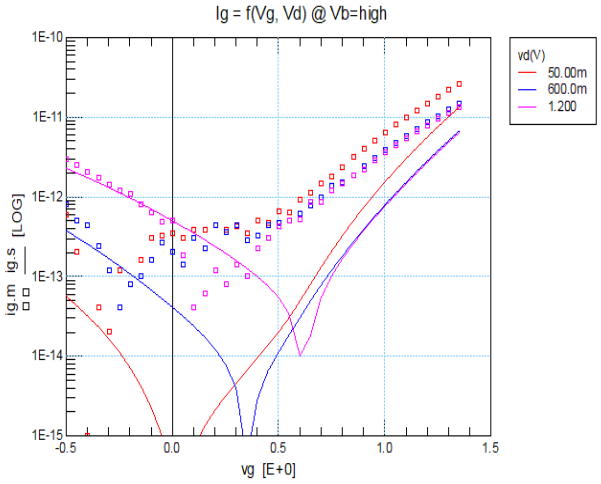
idvg, Ib,Ig, Vb = 0V, T = 70°C



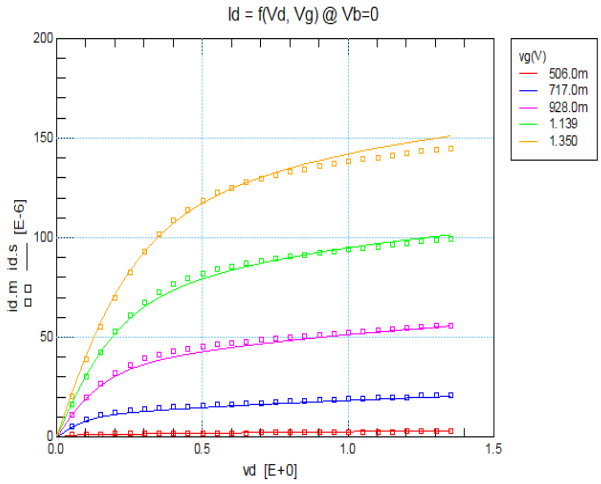


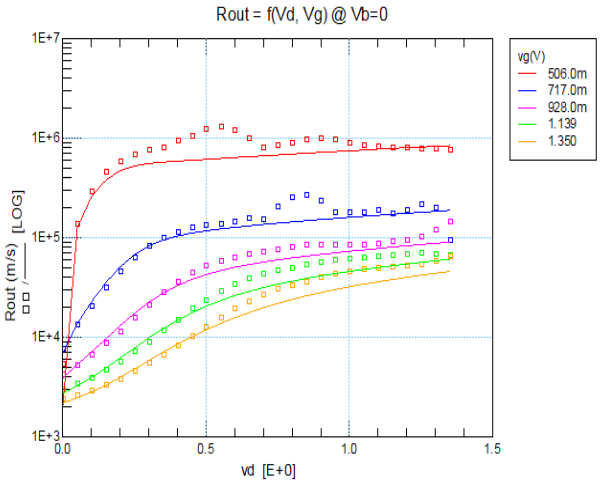
idvg, Ib,Ig, Vb = -1.2V, T = 70°C

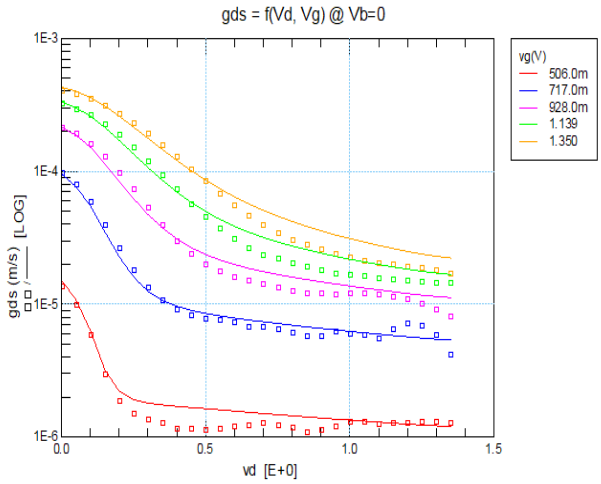




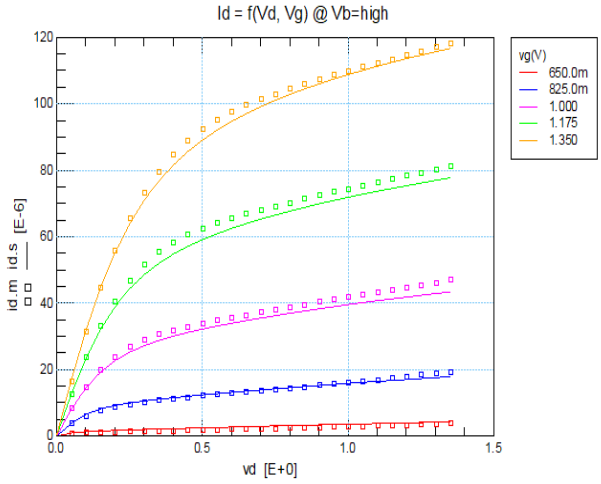
idvd, Vb = 0V, T = 70°C

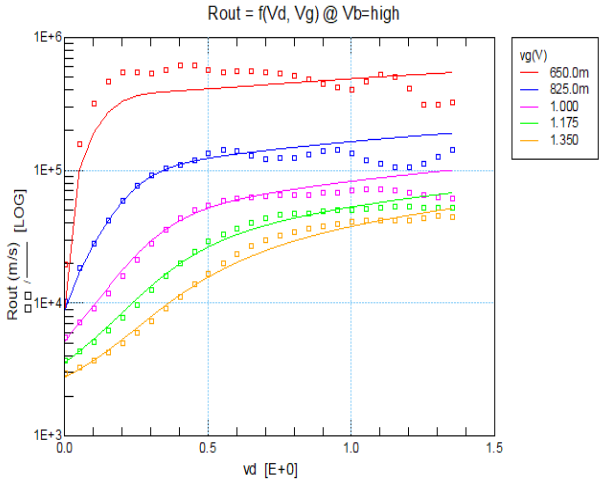


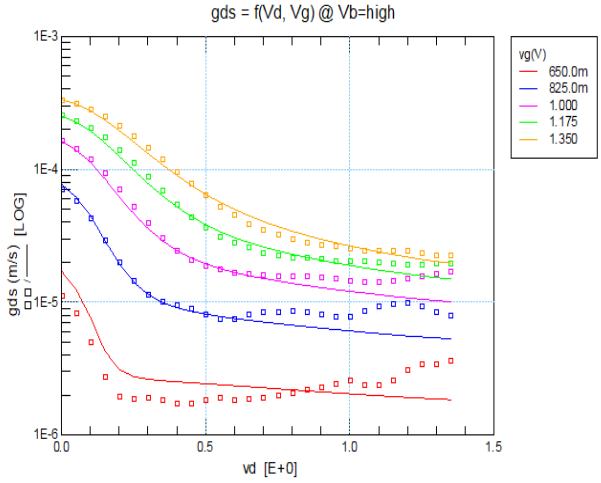




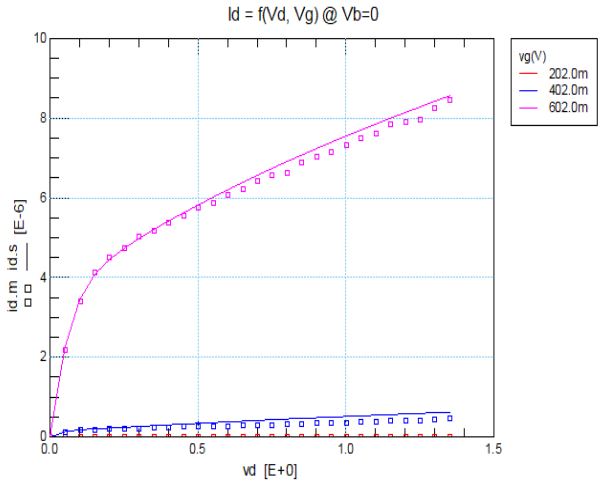
idvd_vbmin, Vb = -1.2V, T = 70°C

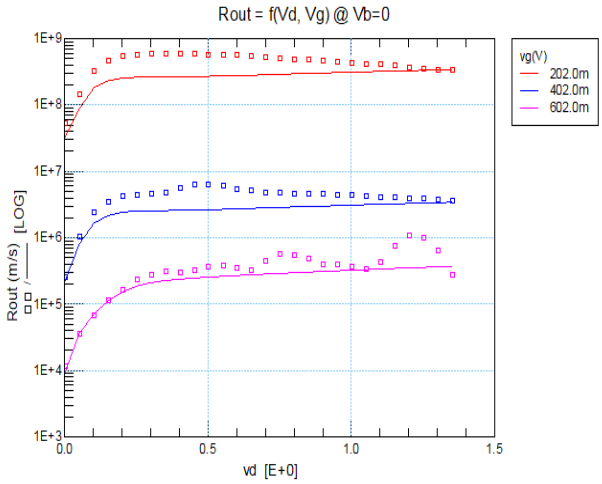


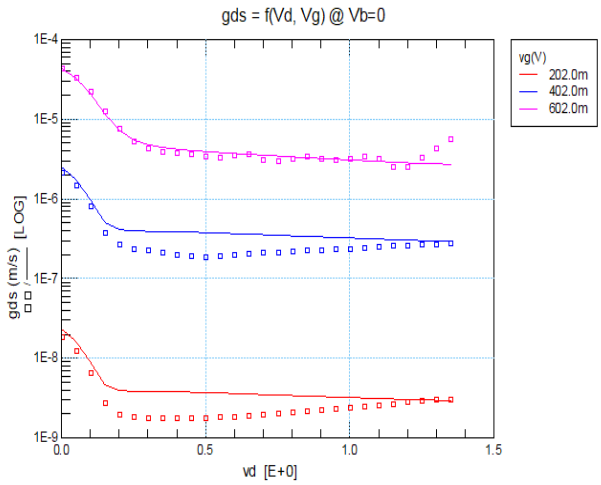




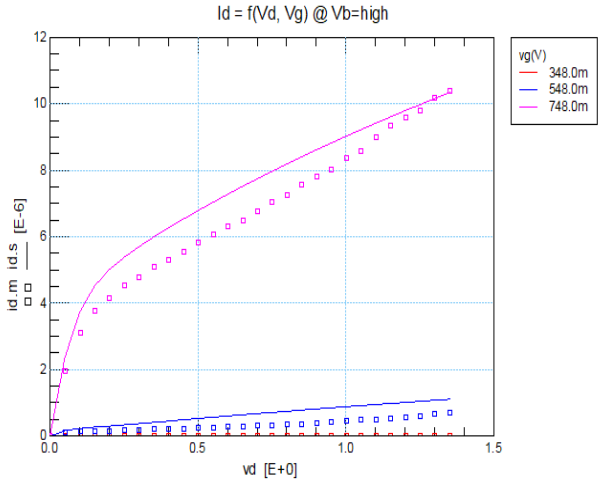
idvd_vth, Vb = 0V, T = 70°C

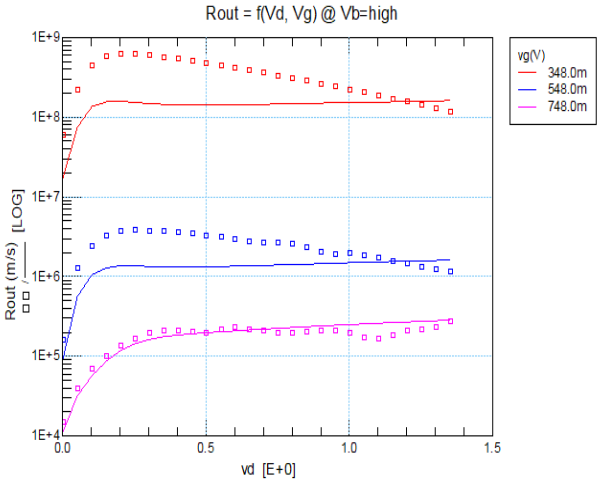


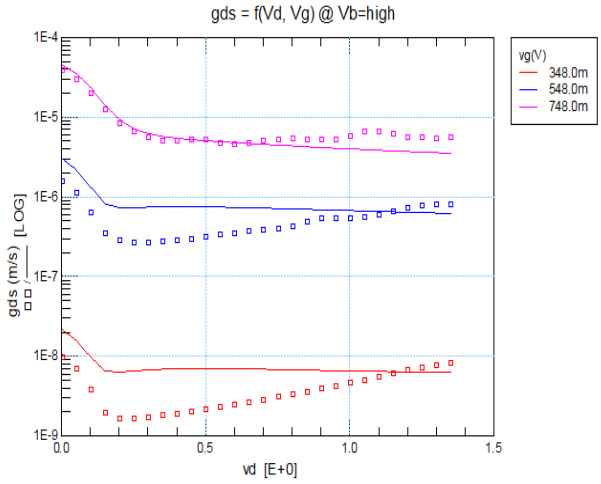




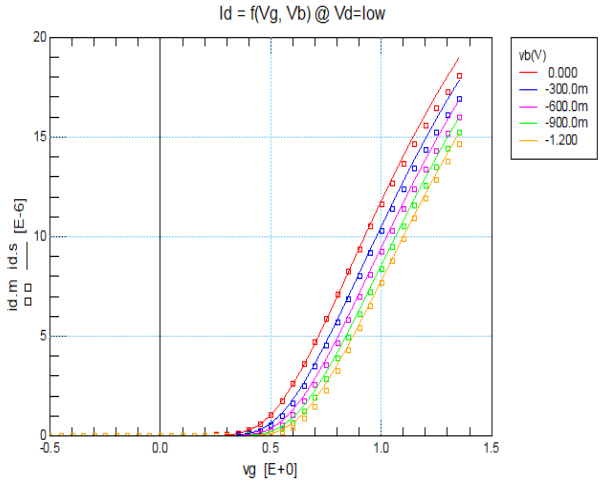
idvd_vth_vbmin, Vb = -1.2V, T = 70°C

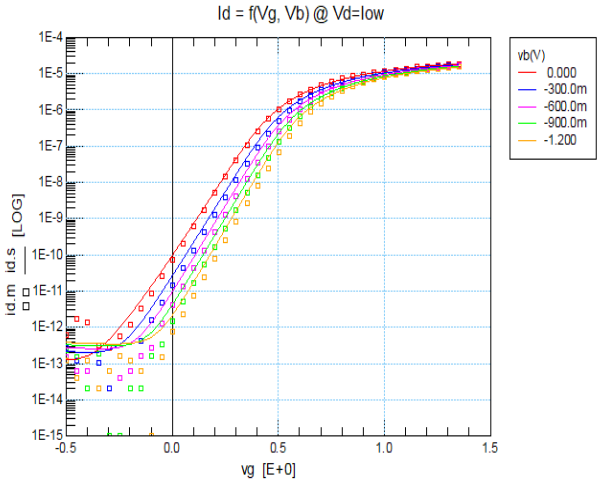


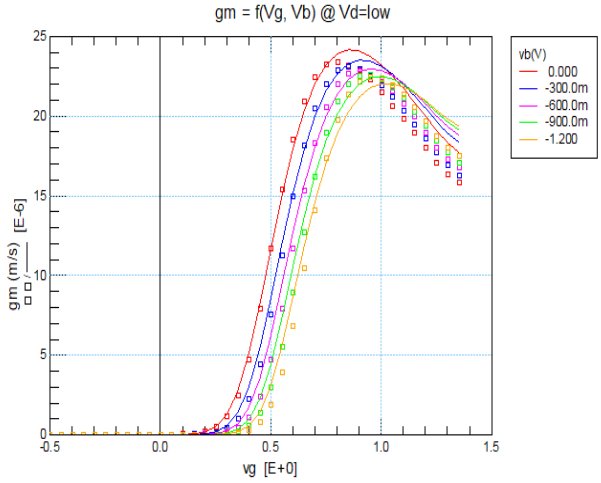


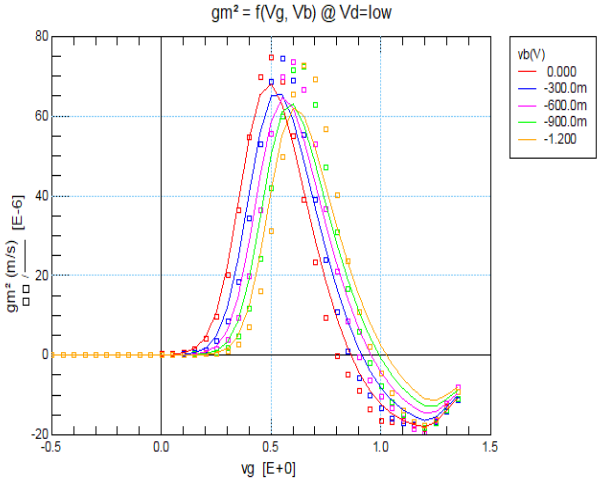


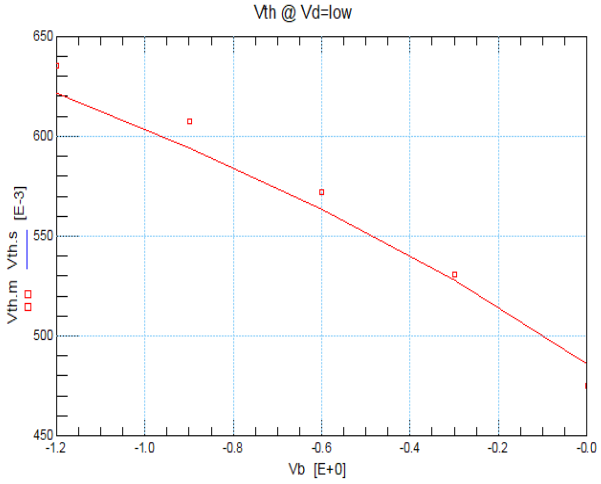
idvg, Vd = 0.05V, T = 125°C

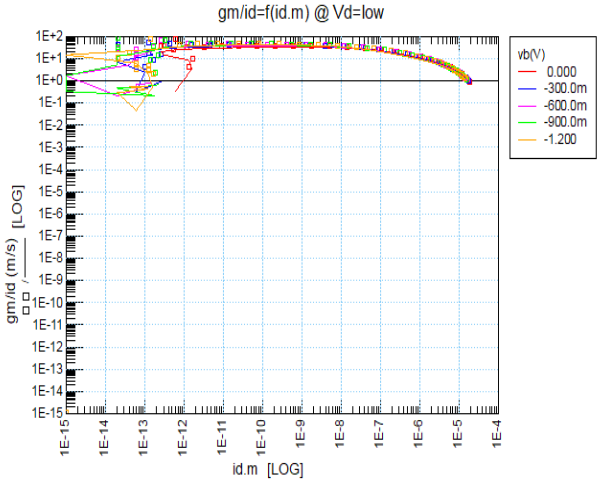


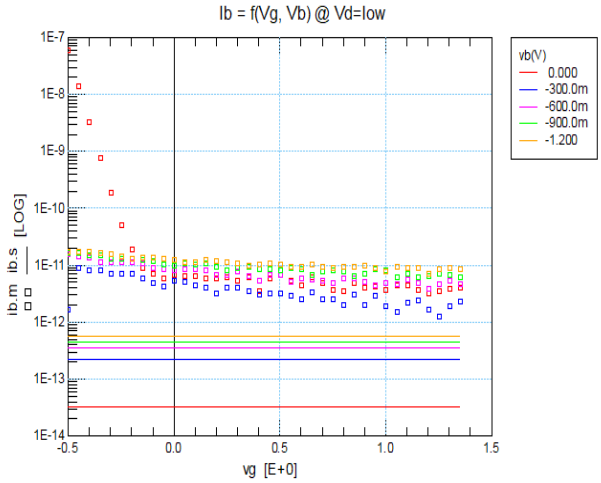


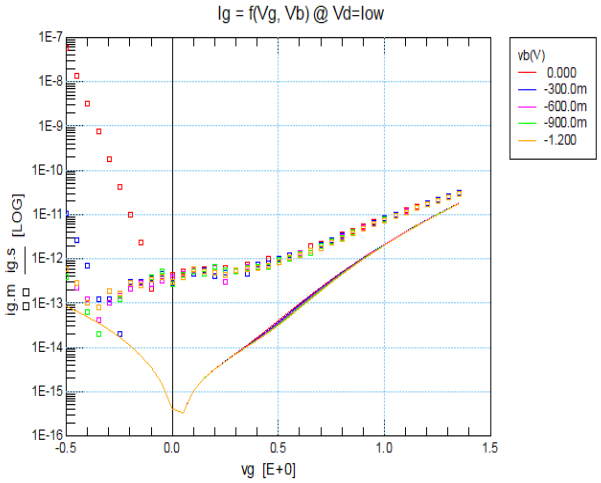




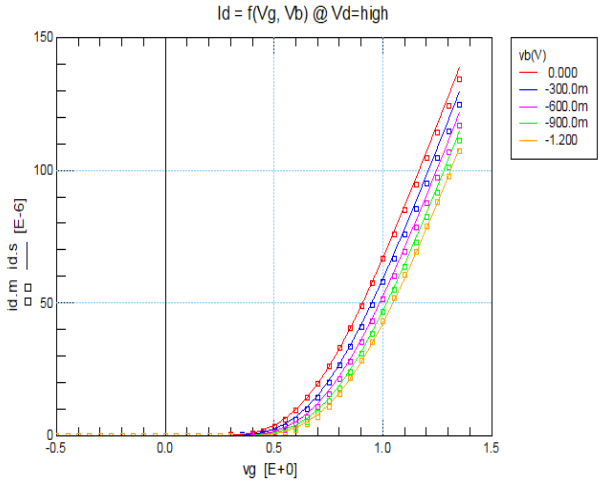


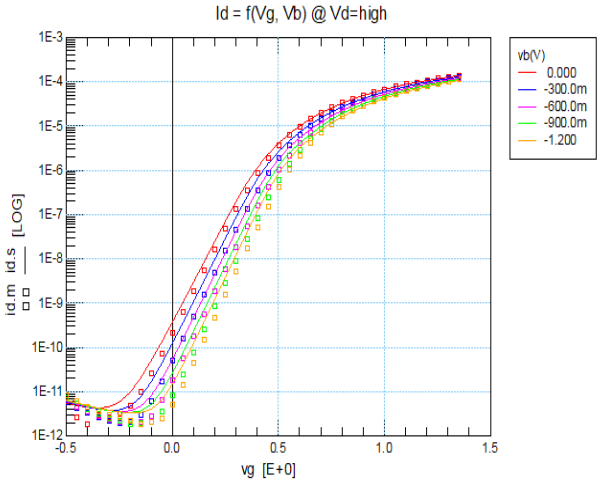


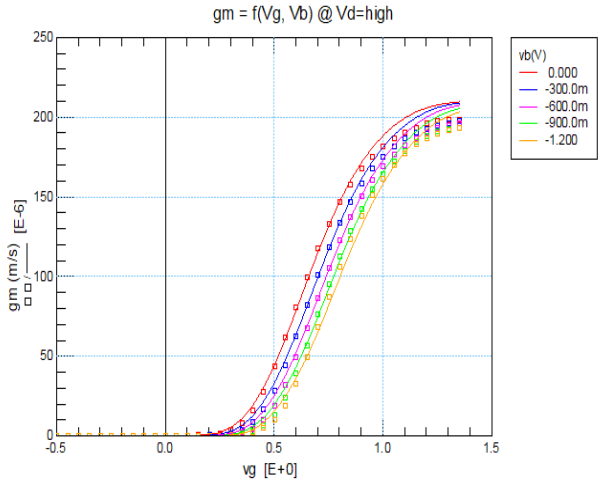


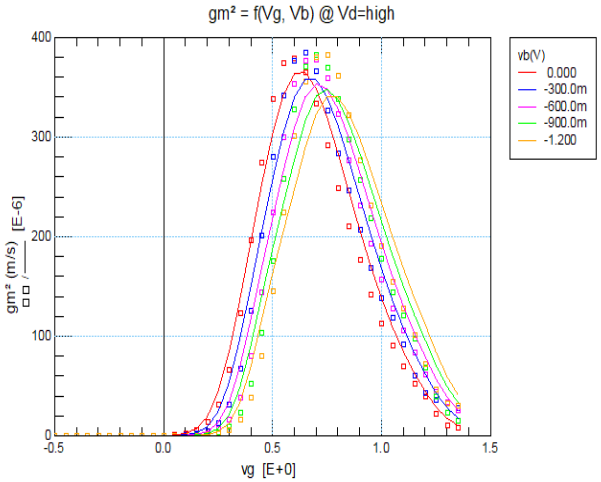


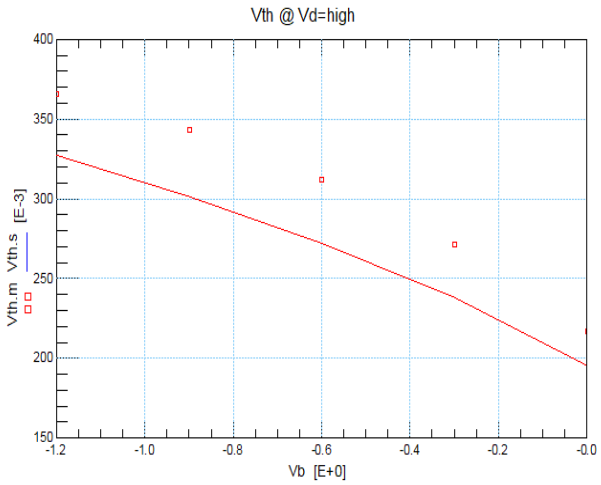
idvg, Vd = 1.2V, T = 125°C

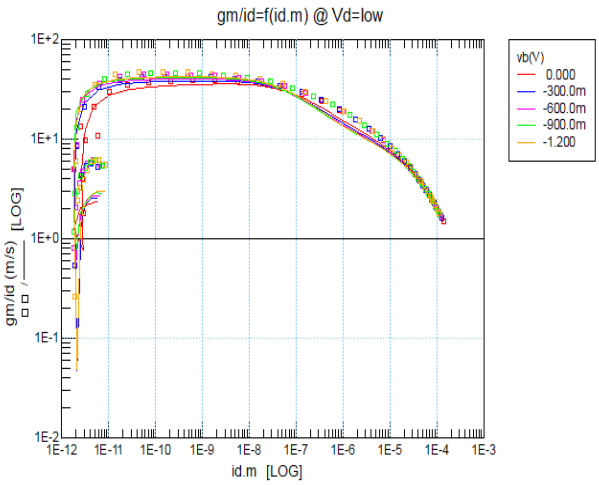


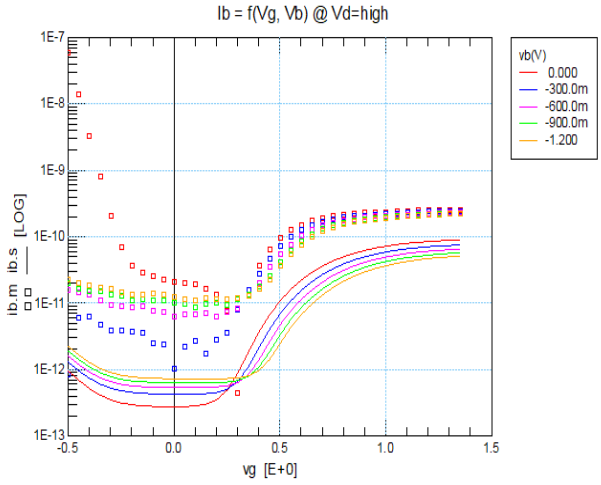


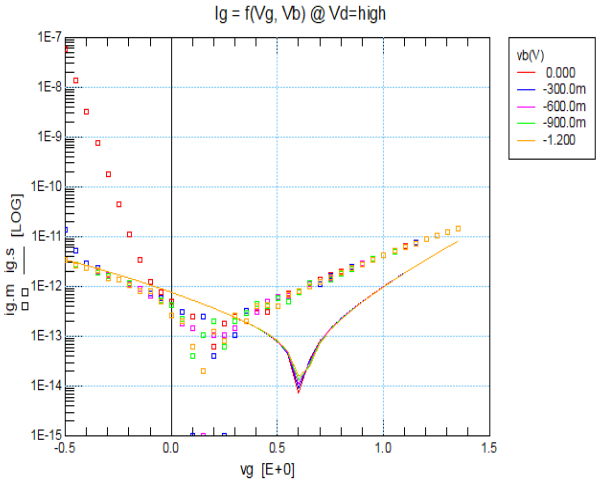




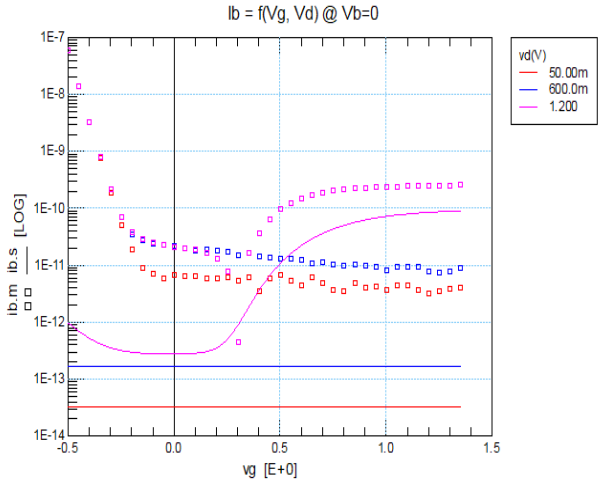


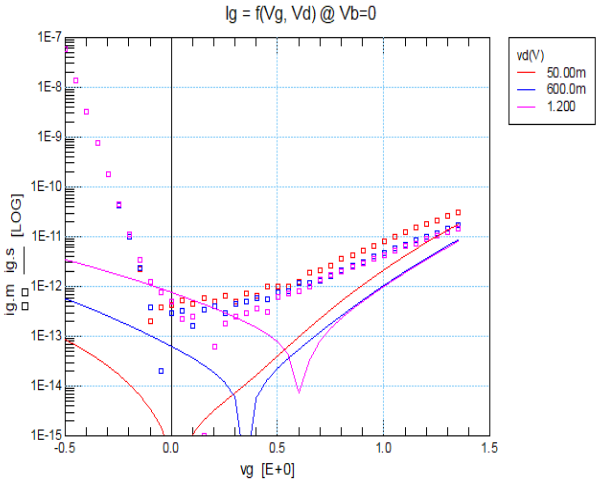




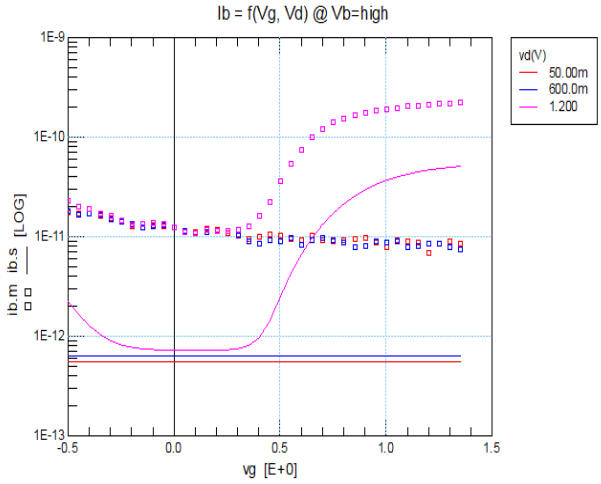


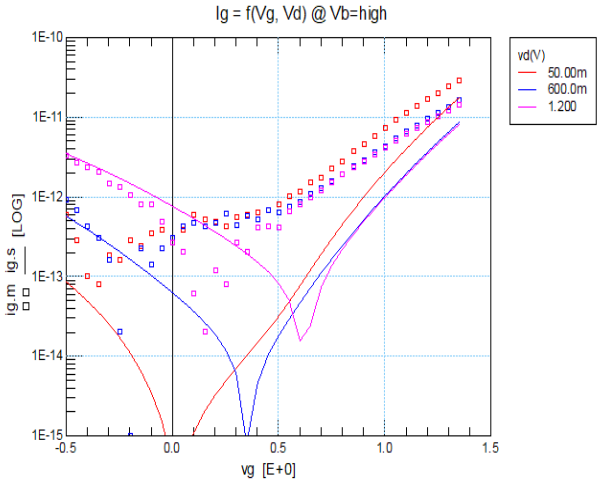
idvg, Ib,Ig, Vb = 0V, T = 125°C



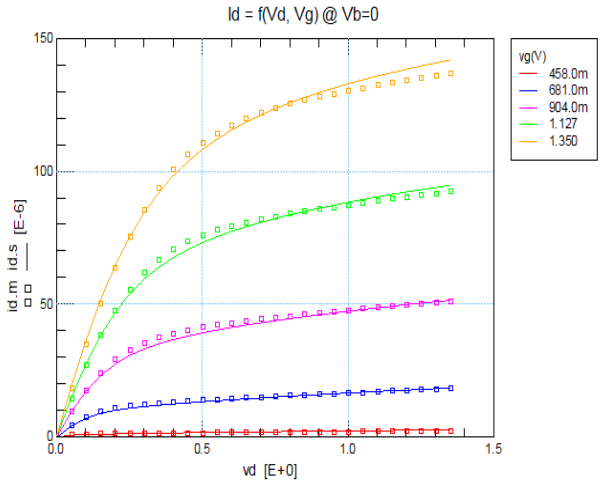


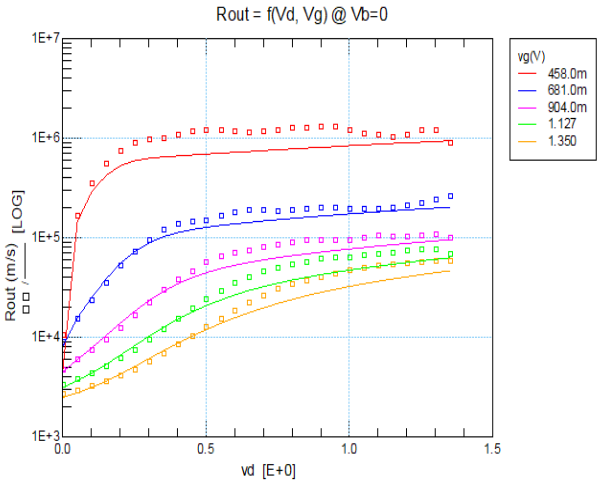
idvg, Ib,Ig, Vb = -1.2V, T = 125°C

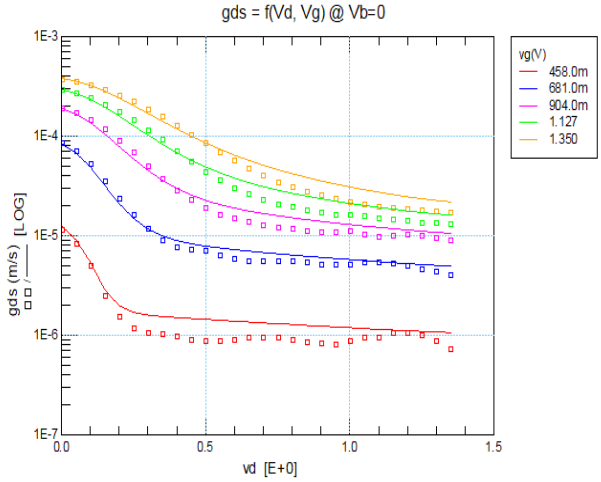




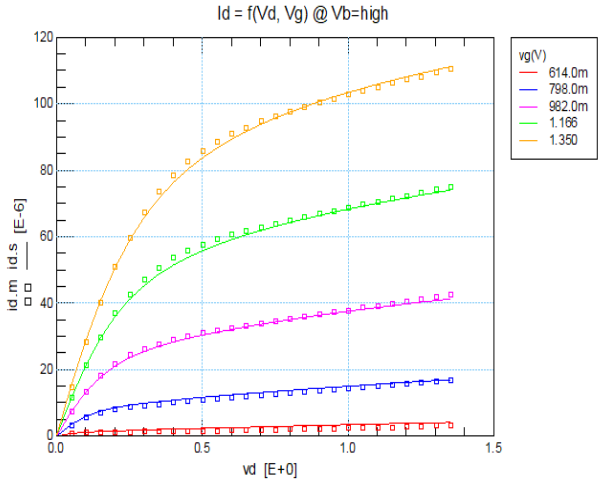
idvd, Vb = 0V, T = 125°C

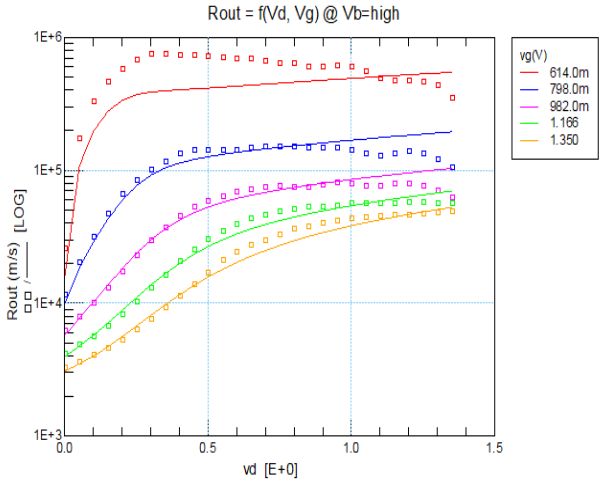


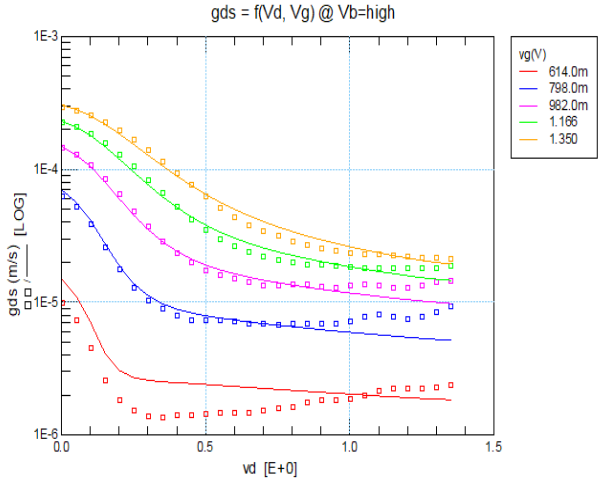




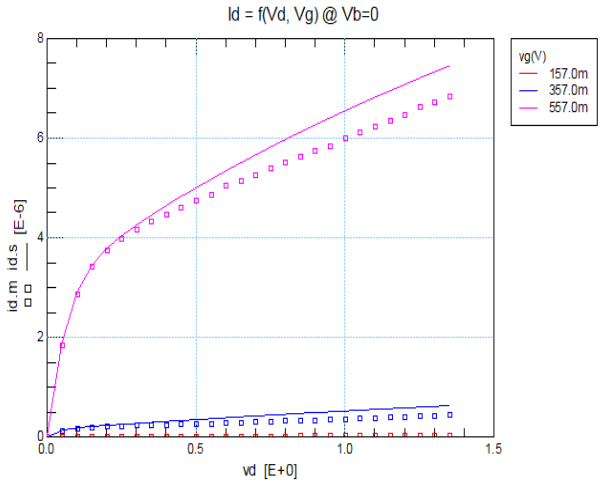
idvd_vbmin, Vb = -1.2V, T = 125°C

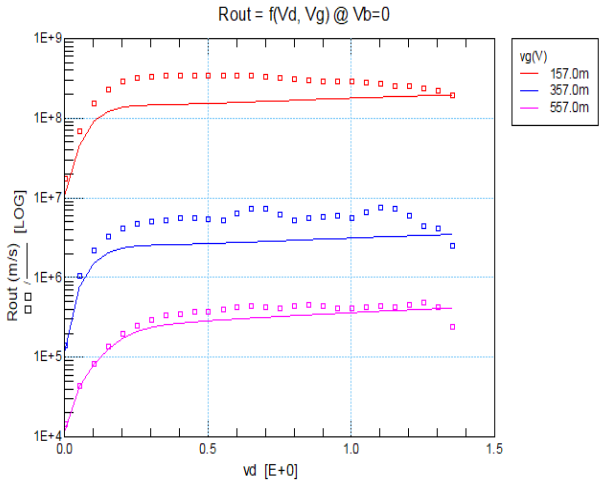


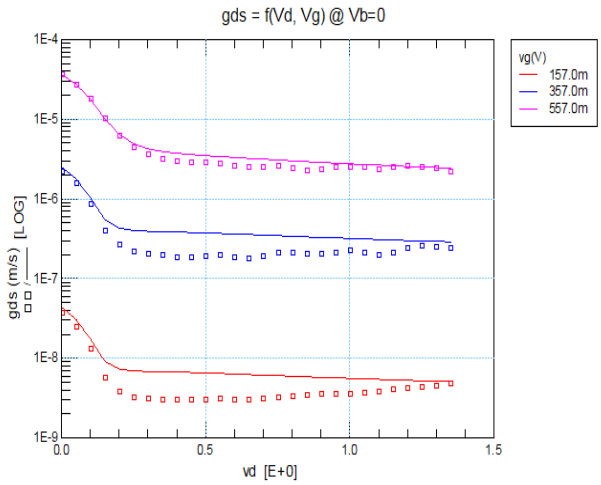




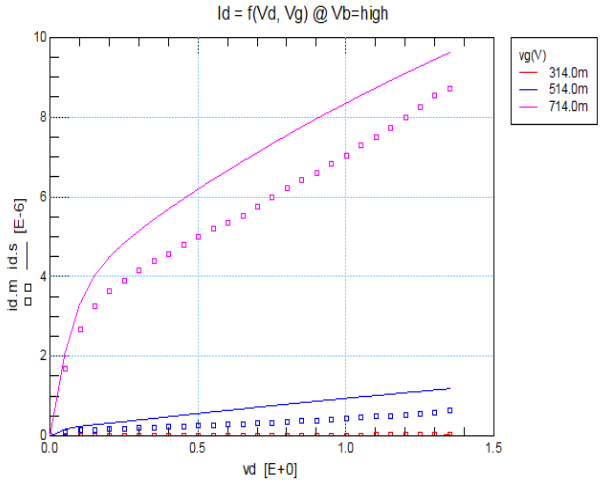
idvd_vth, Vb = 0V, T = 125°C

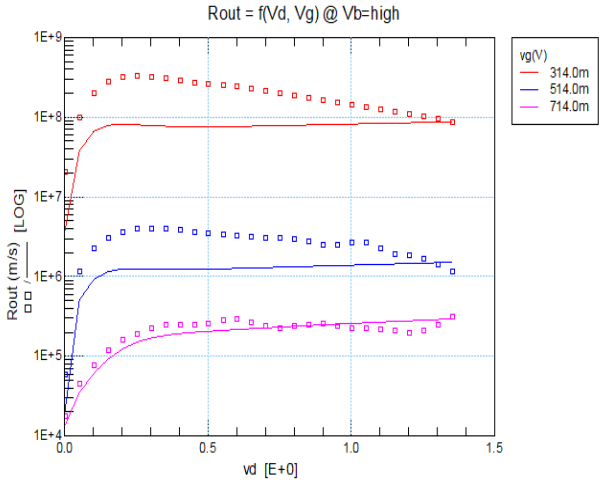


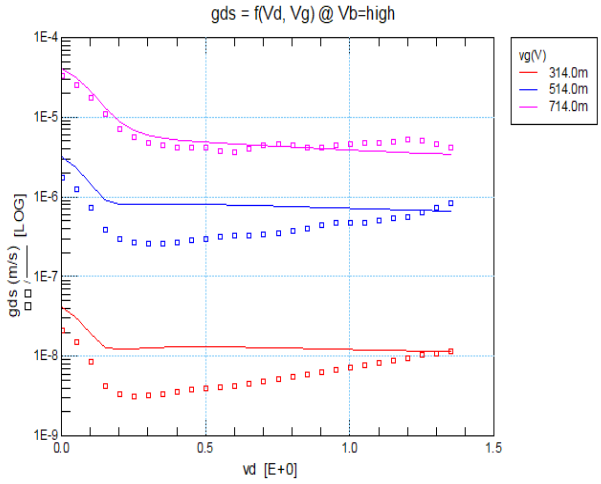




idvd_vth_vbmin, Vb = -1.2V, T = 125°C

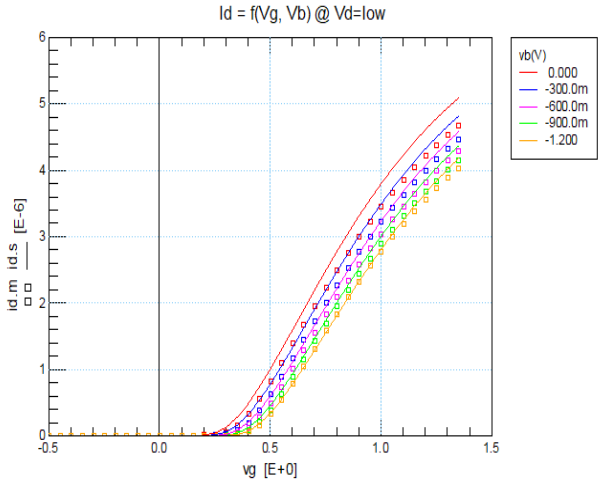


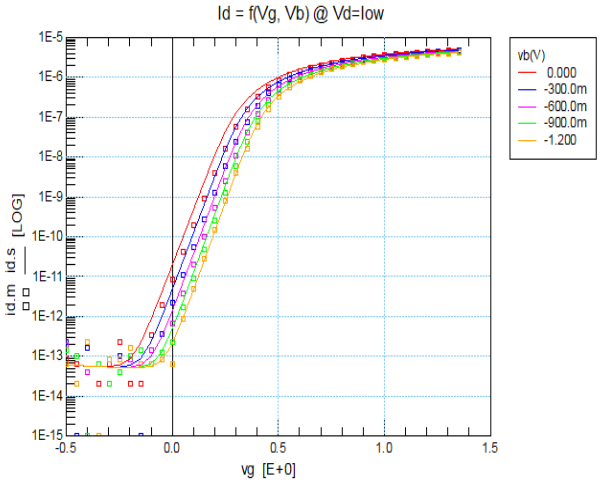


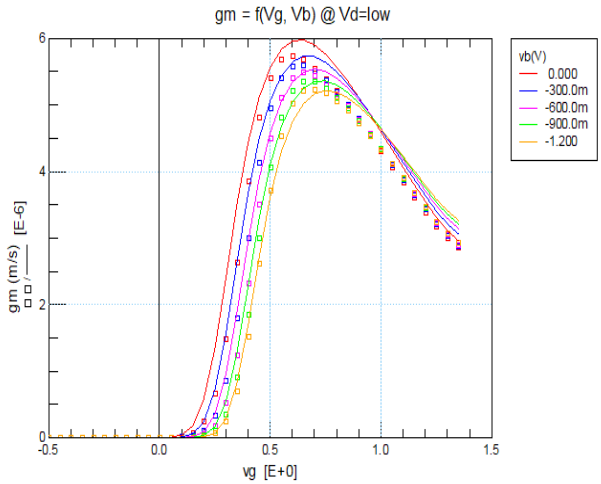


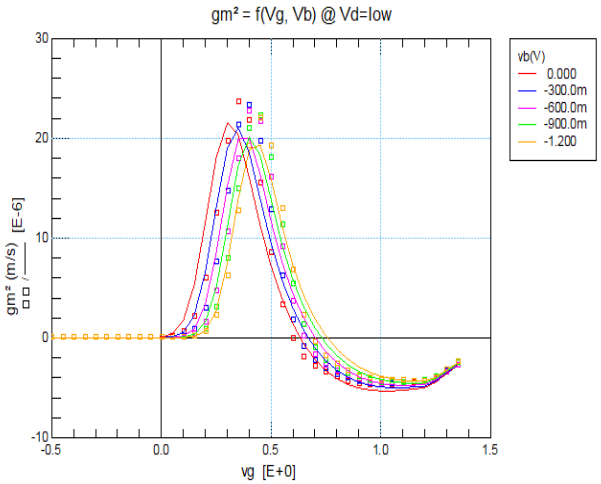
4.21 W0u15_L0u5_S547_2

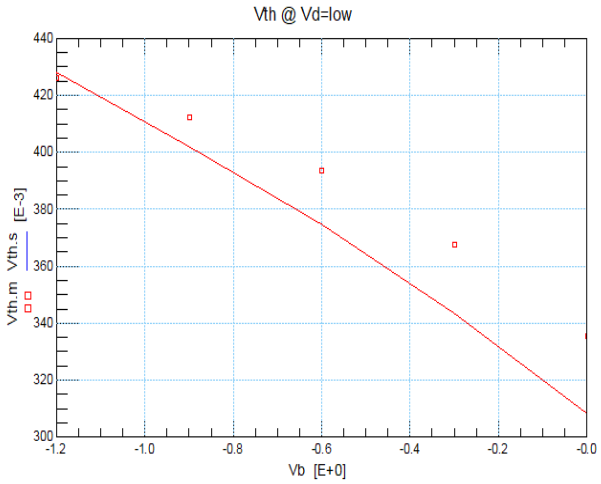
idvg, Vd = 0.05V, T = 27°C

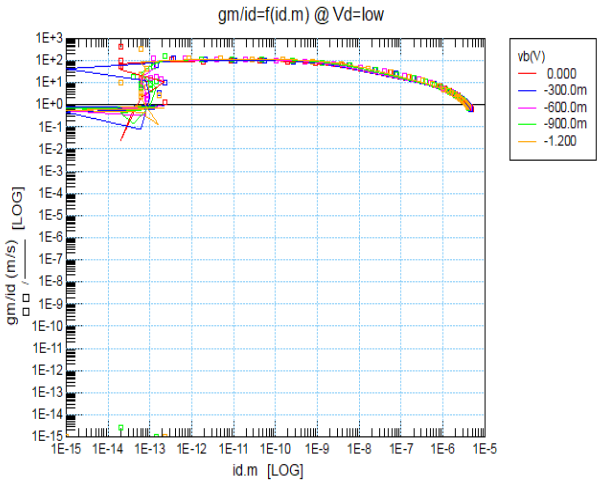


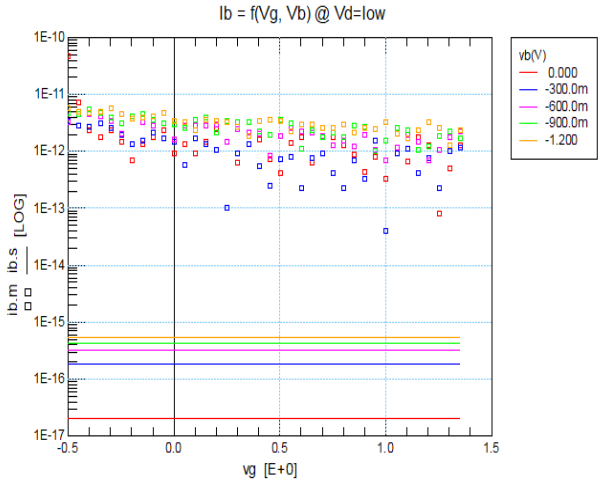


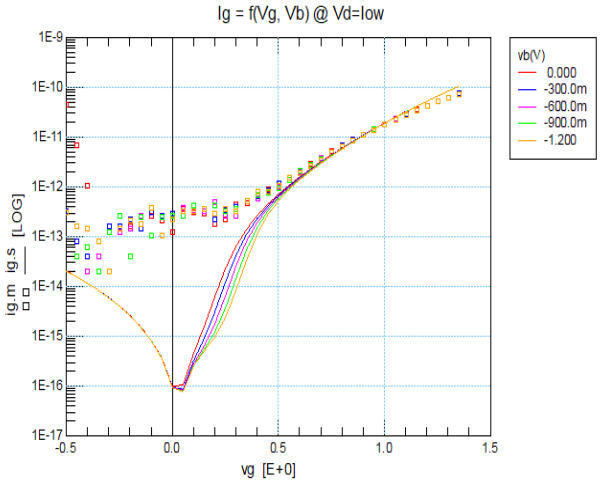




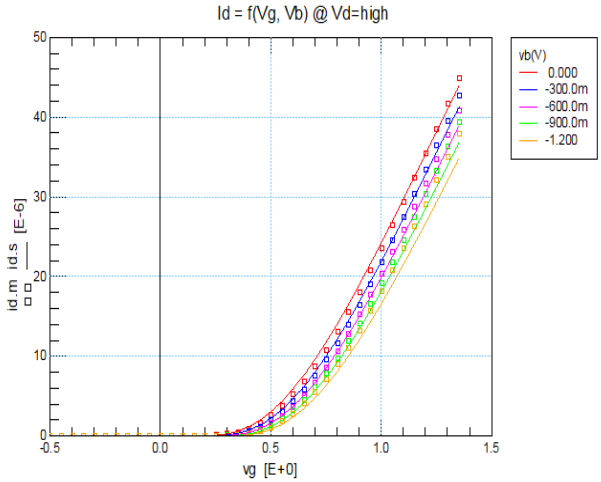


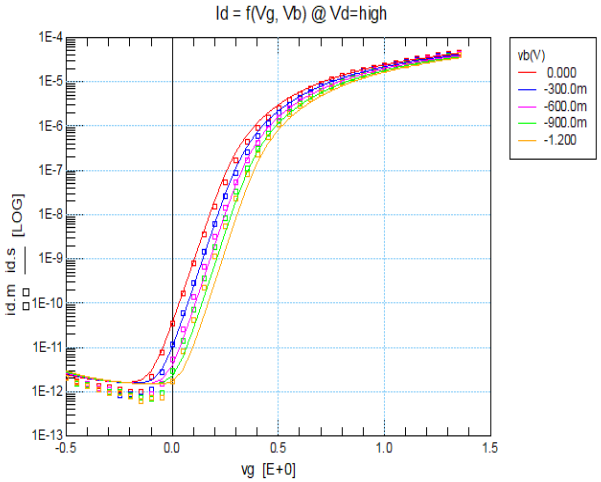


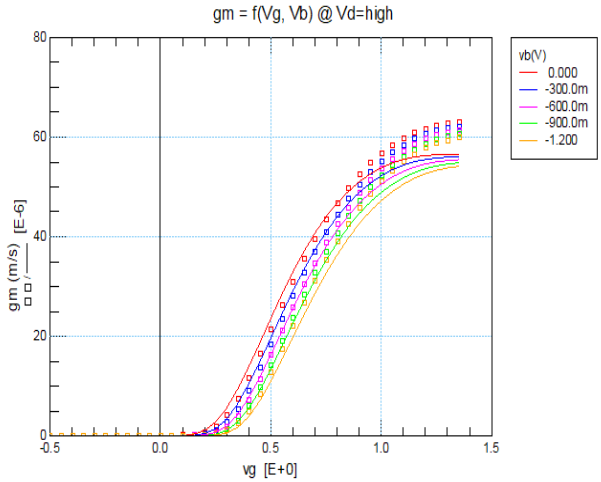


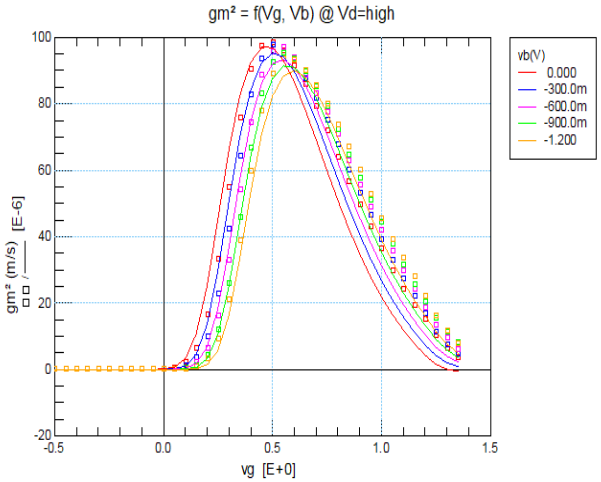


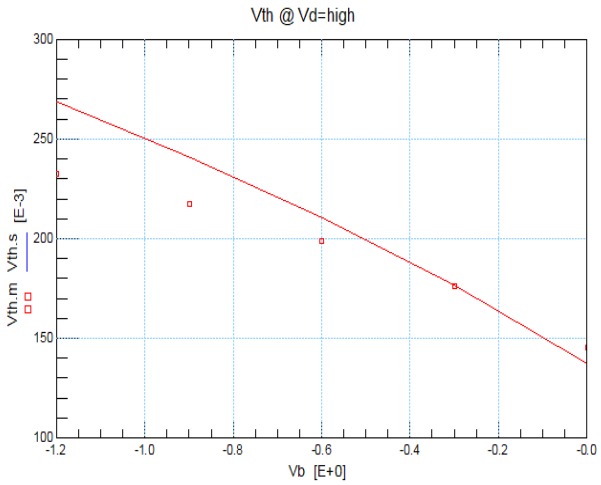
idvg, Vd = 1.2V, T = 27°C

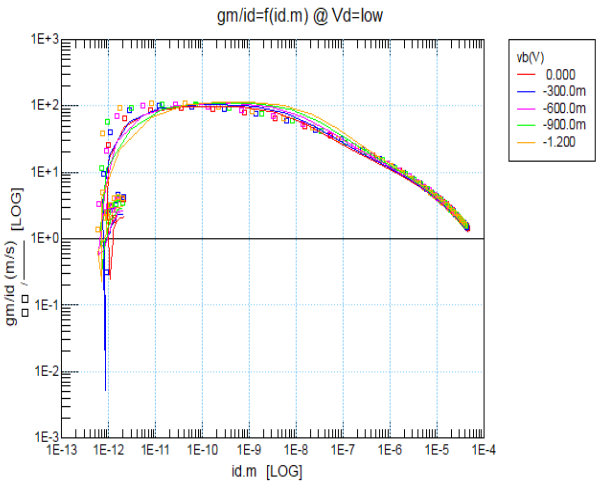


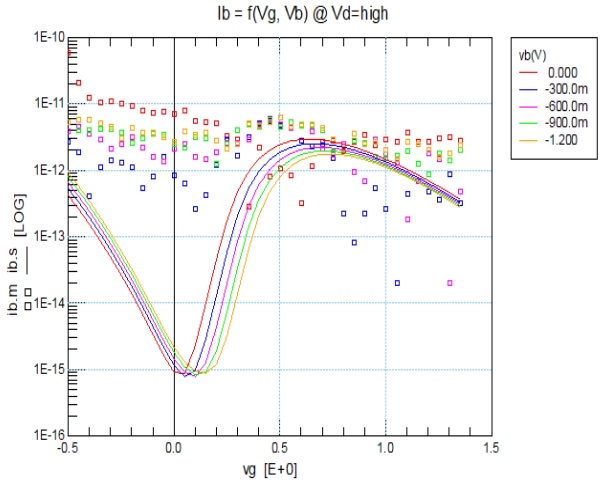


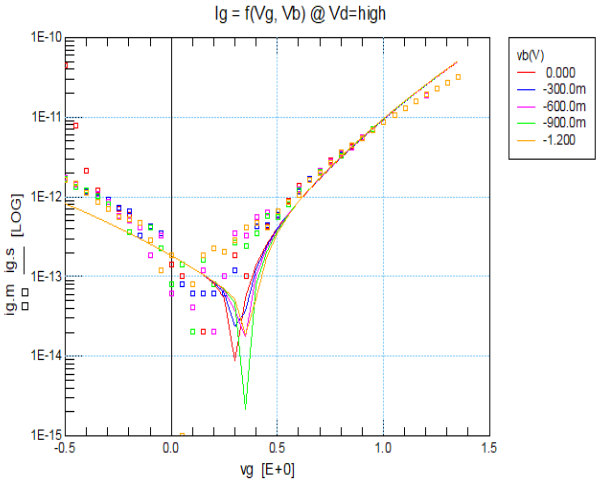




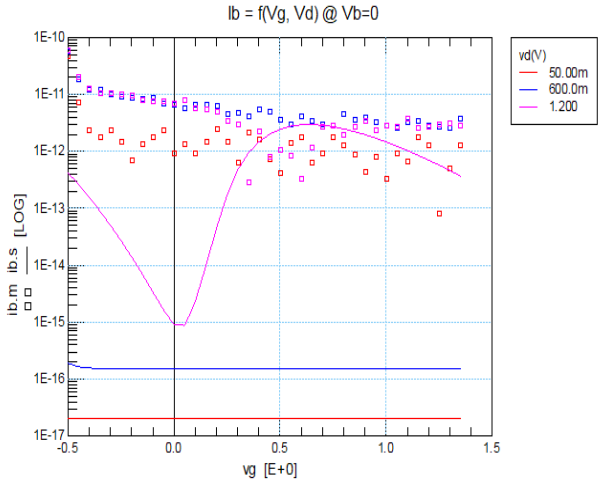


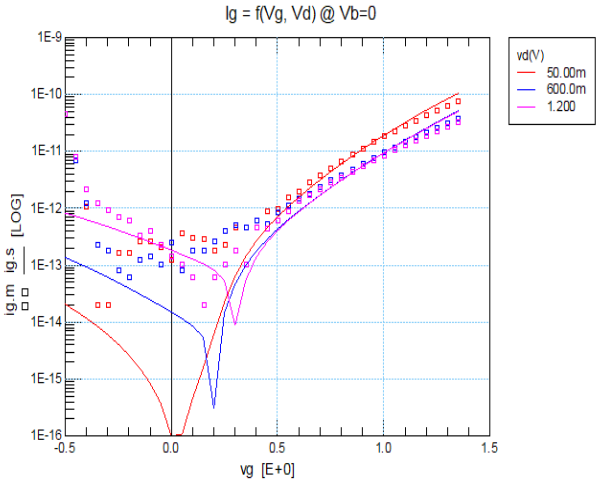




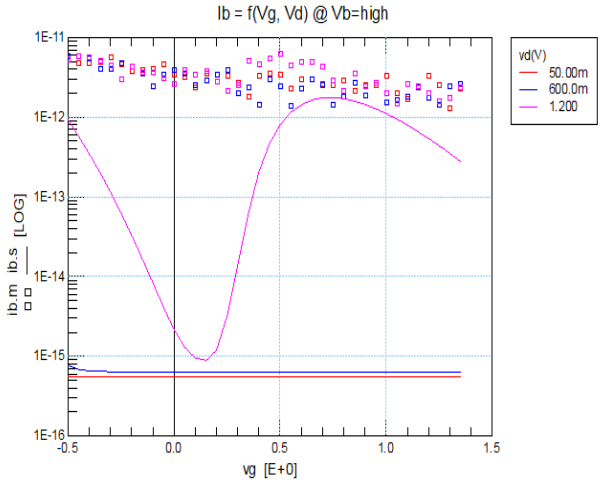


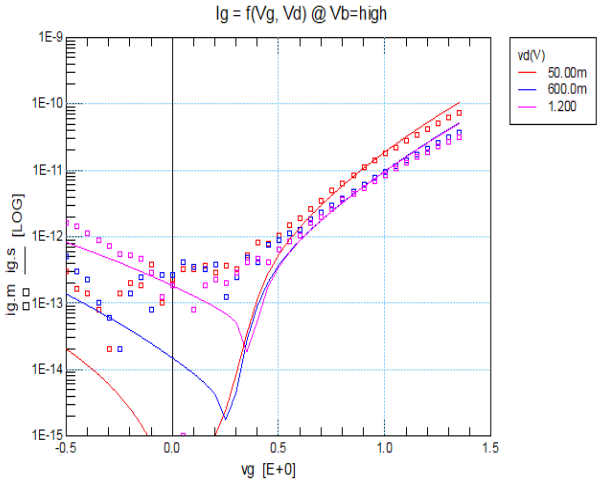
idvg, Ib,Ig, Vb = 0V, T = 27°C



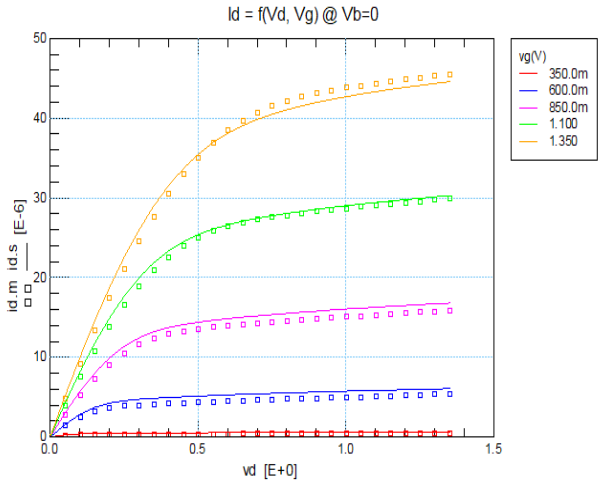


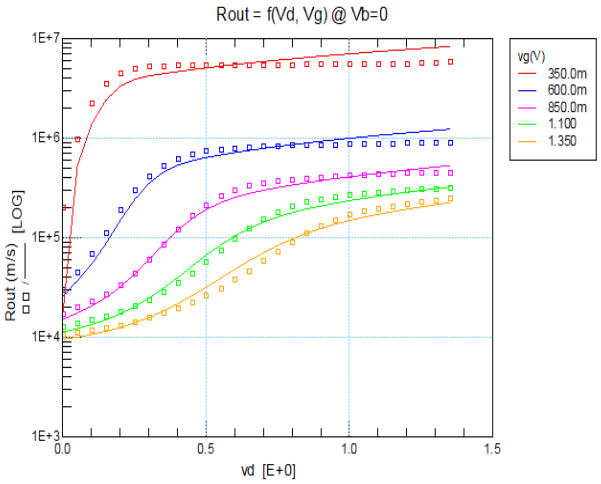
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

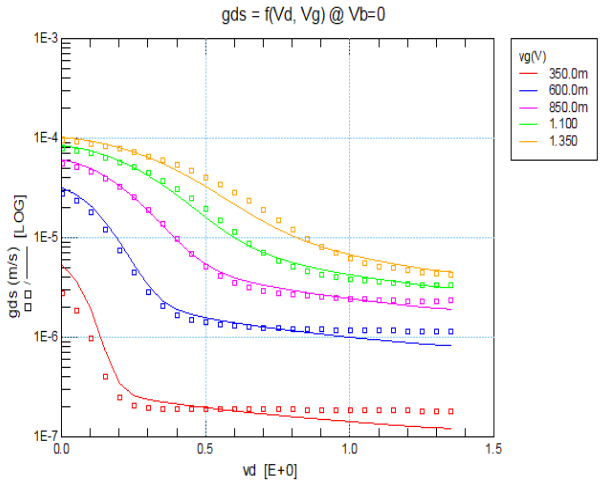




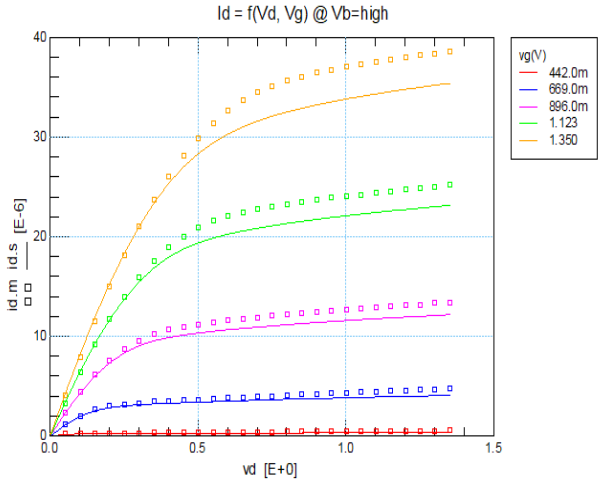
idvd, Vb = 0V, T = 27°C

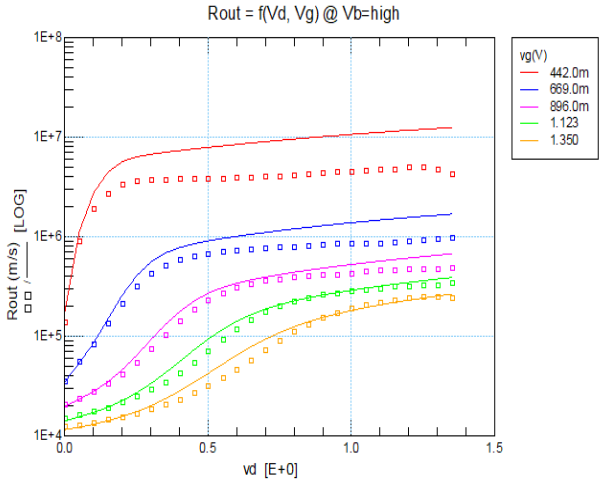


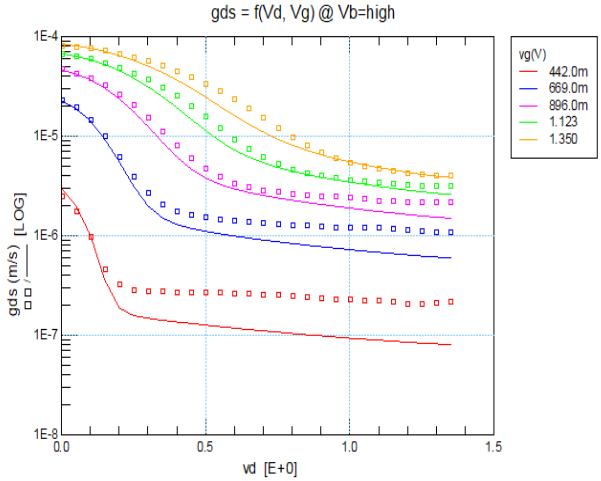




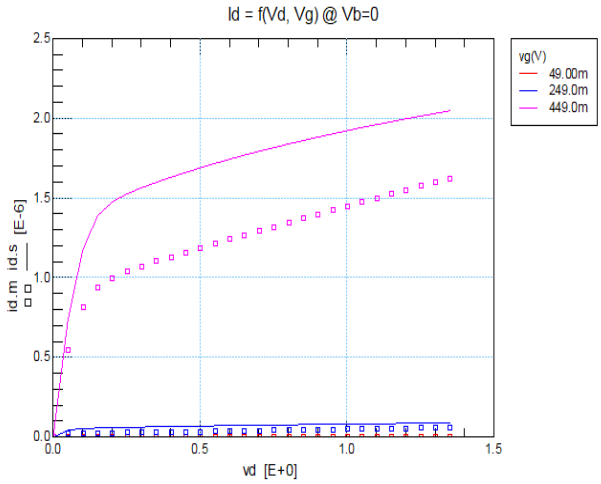
idvd_vbmin, Vb = -1.2V, T = 27°C

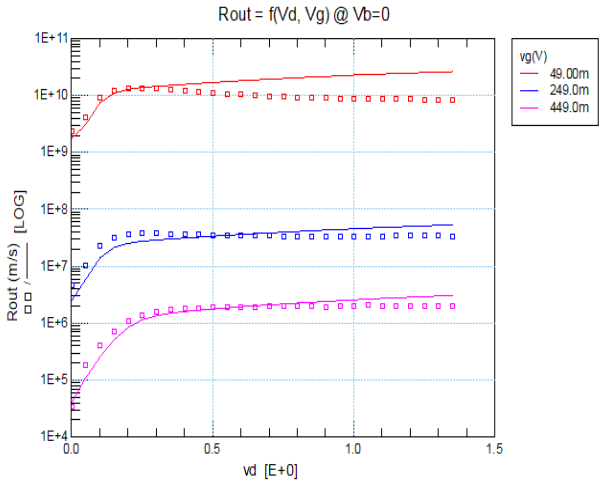


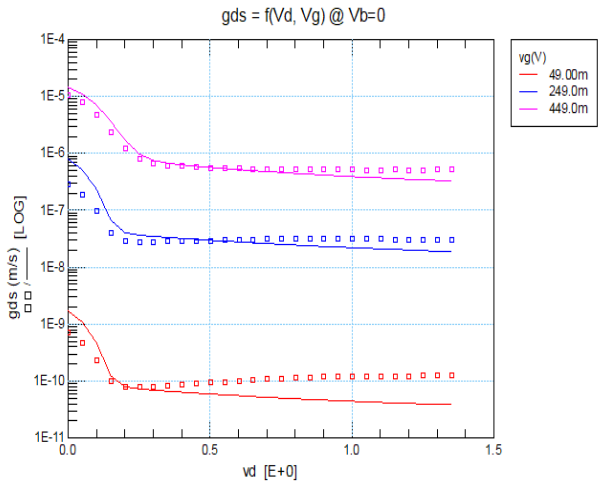




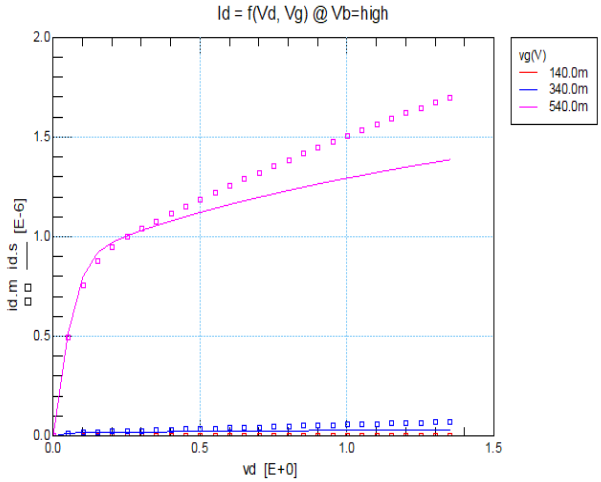
idvd_vth, Vb = 0V, T = 27°C

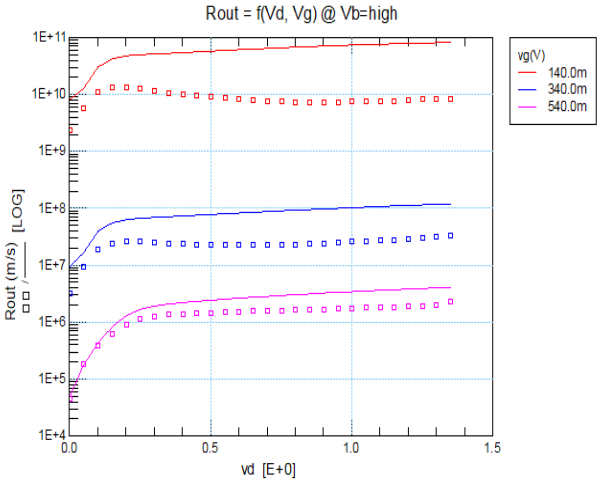


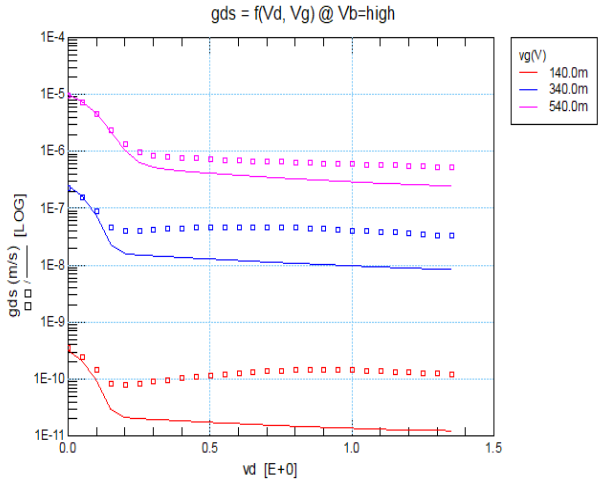




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

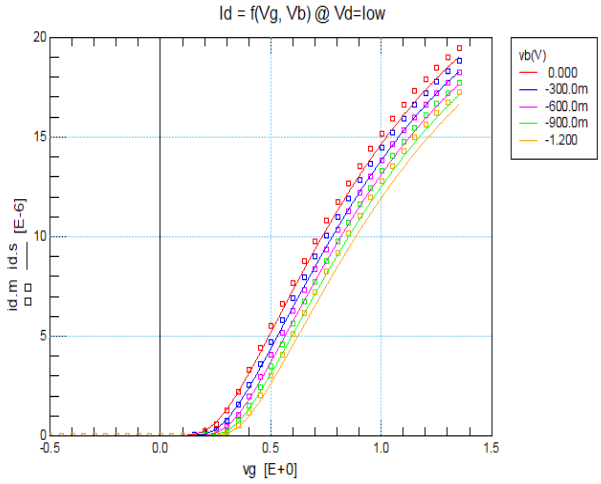


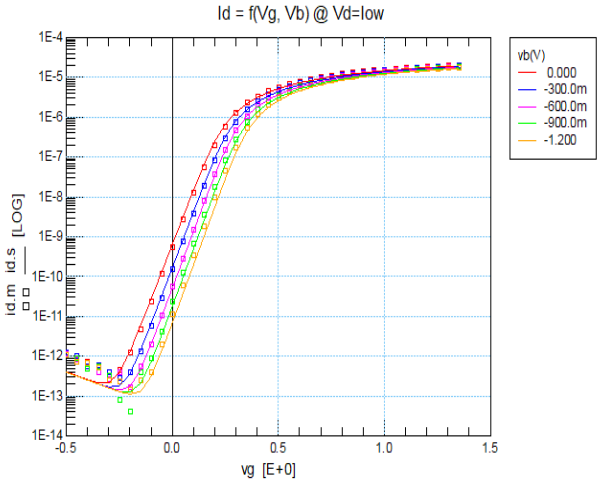


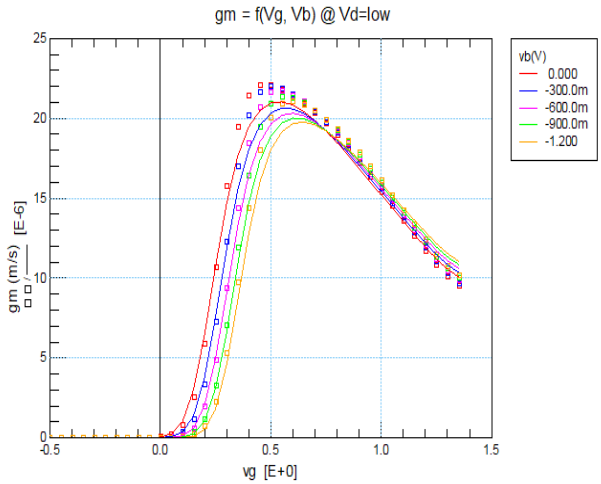


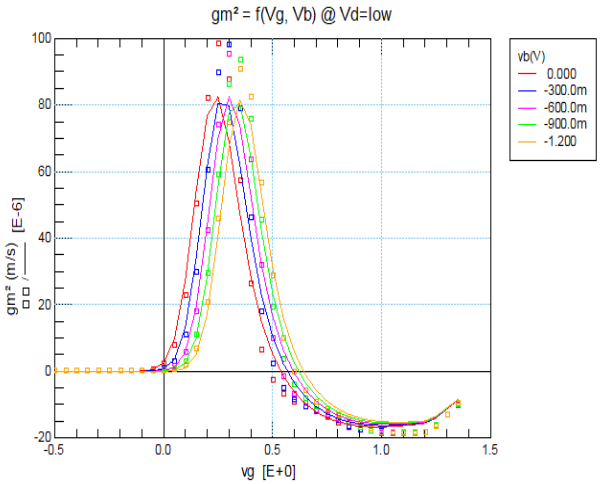
4.22 W05u0_L5u0_S542_2

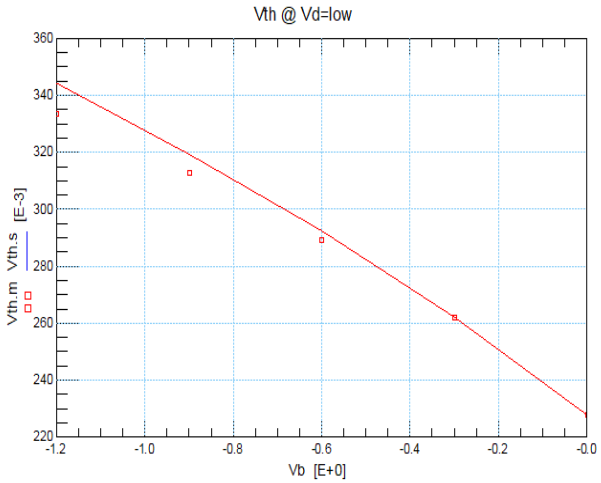
idvg, Vd = 0.05V, T = 27°C

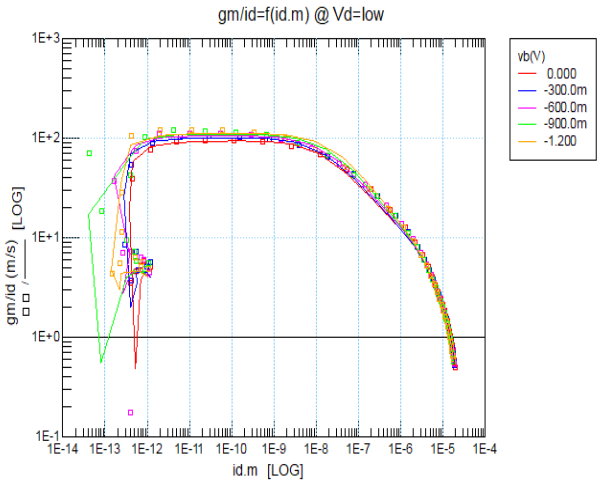


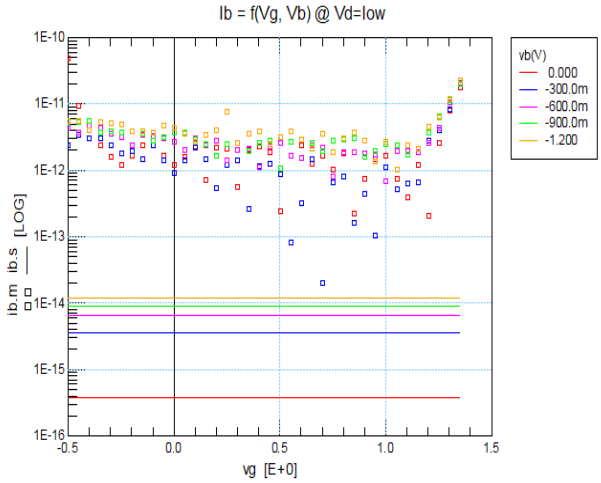


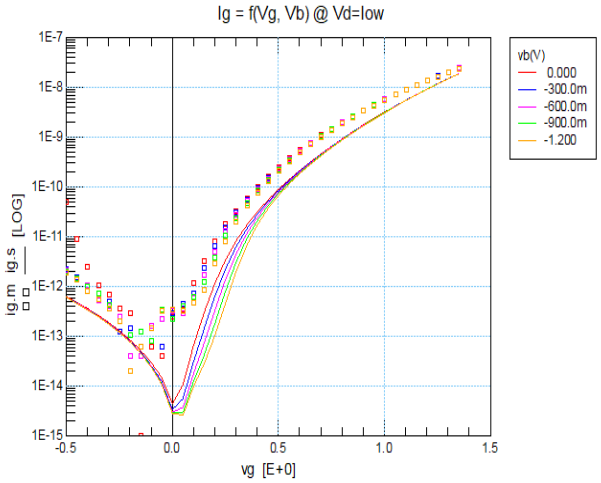




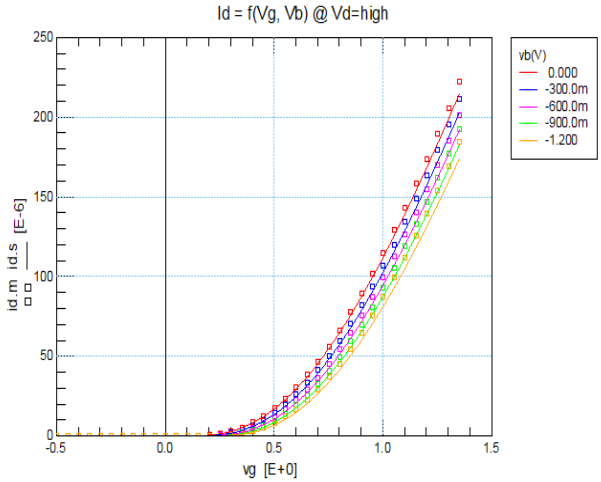


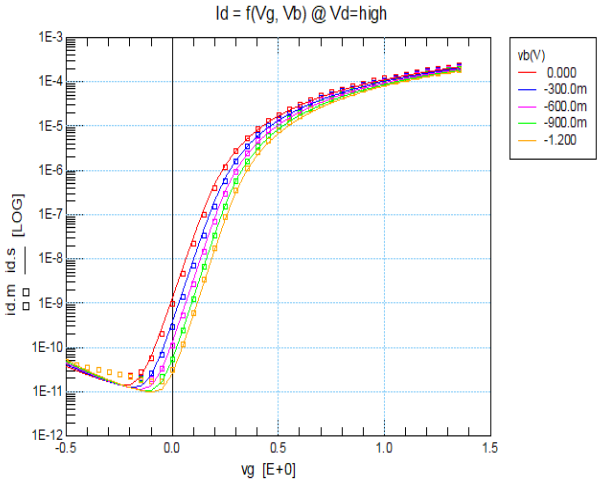


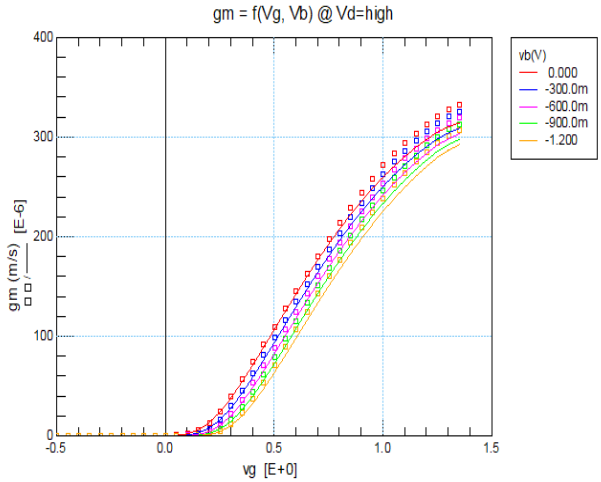


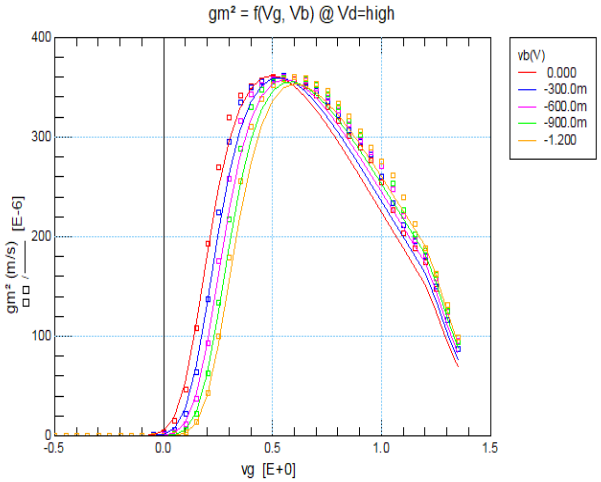


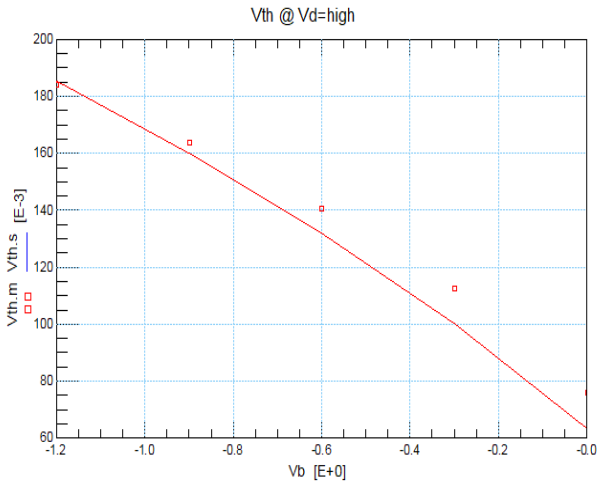
idvg, Vd = 1.2V, T = 27°C

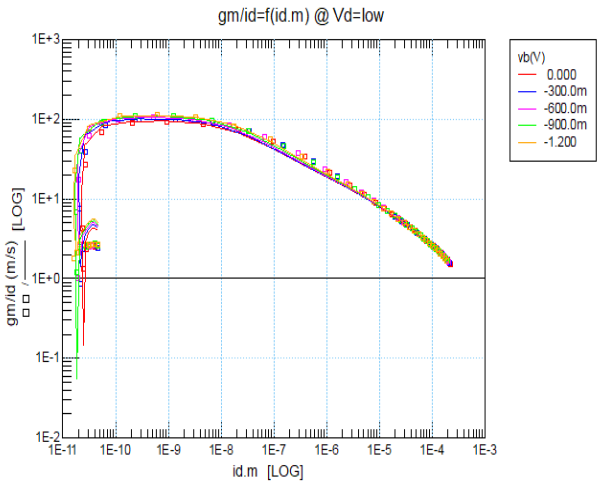


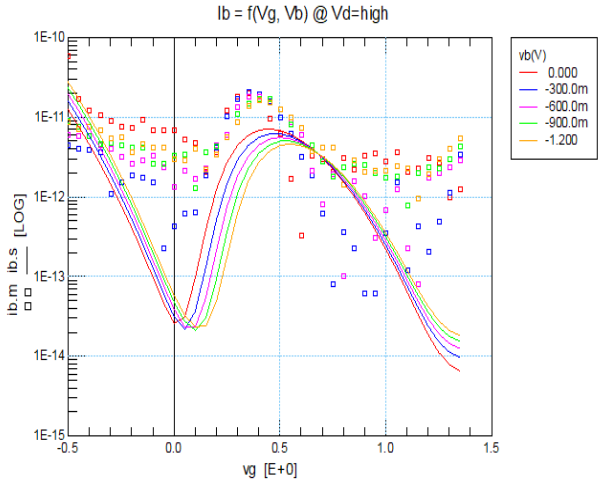


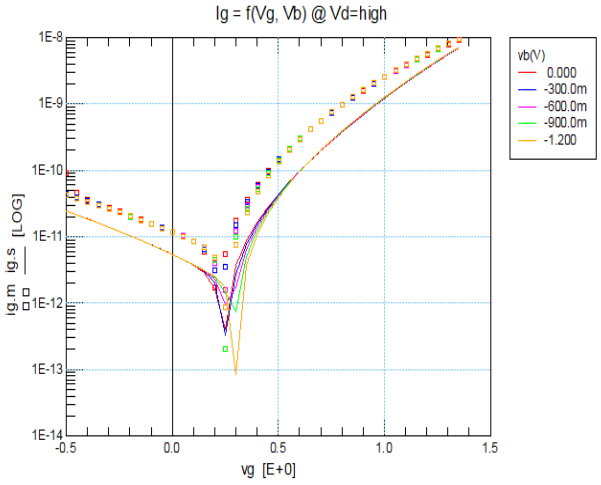




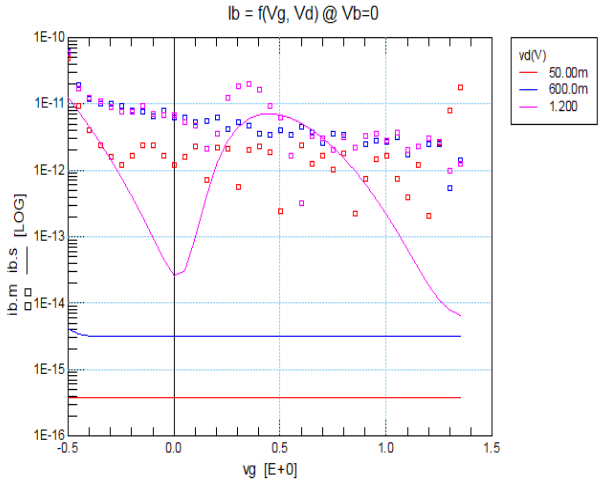


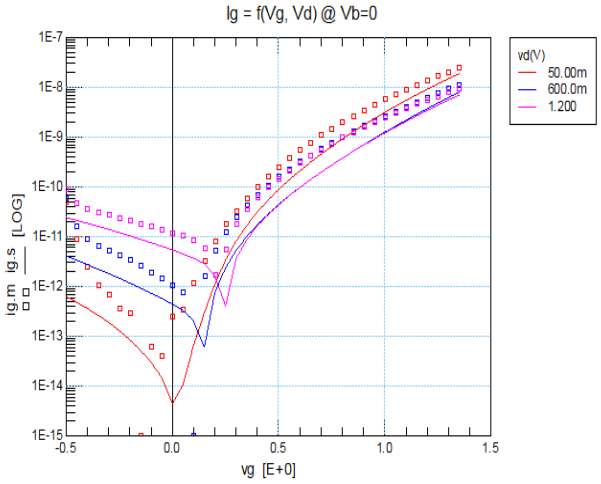




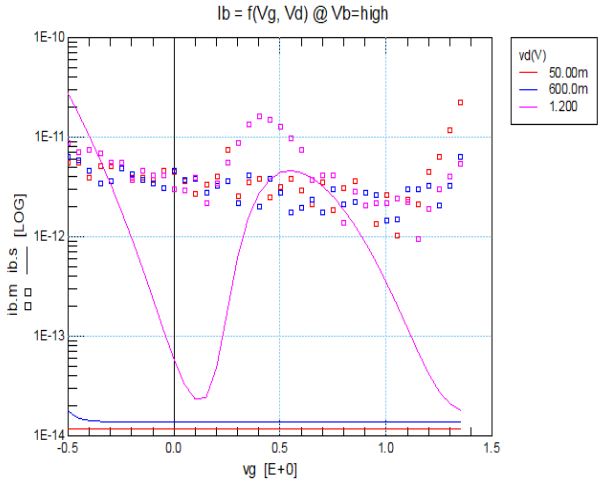


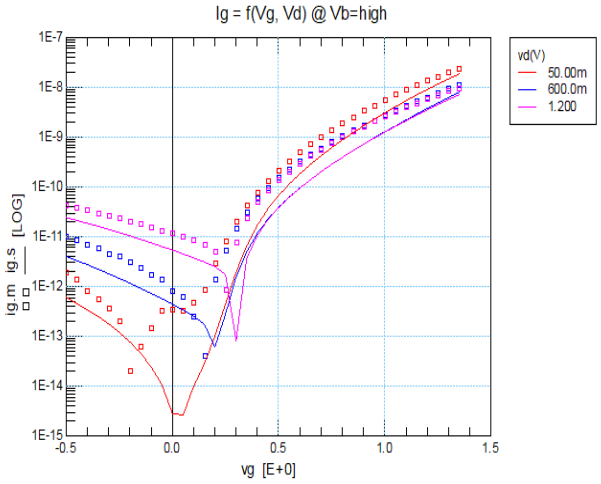
idvg, Ib,Ig, Vb = 0V, T = 27°C



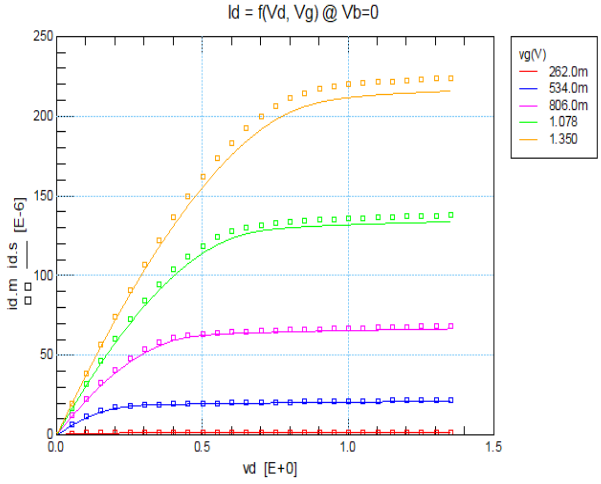


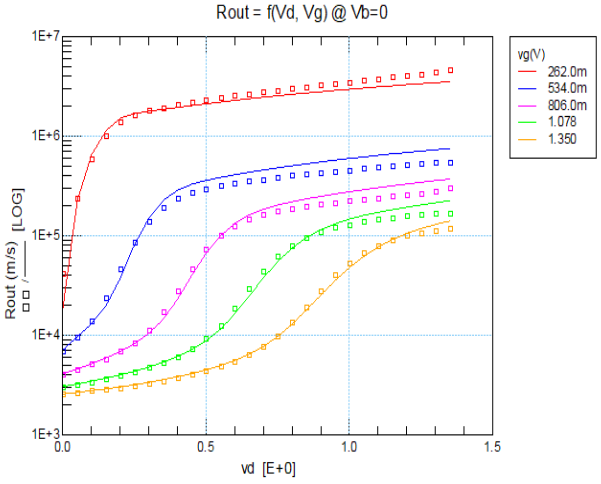
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

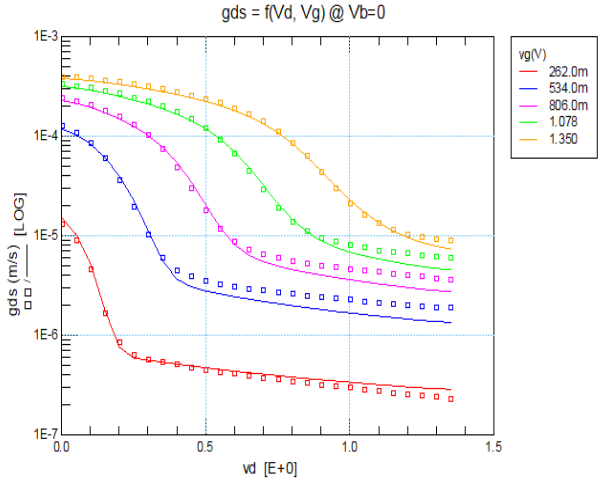




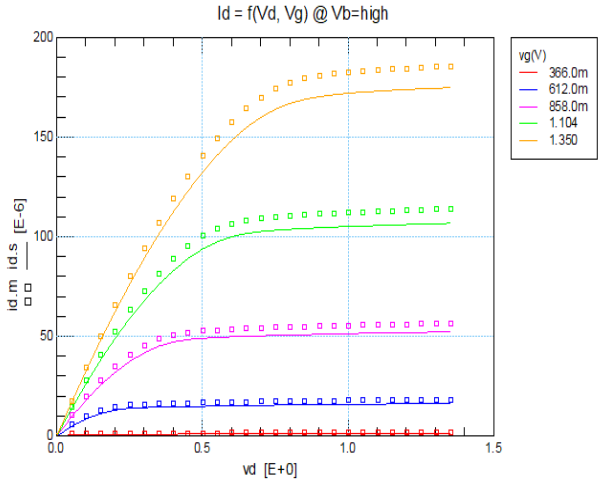
idvd, Vb = 0V, T = 27°C

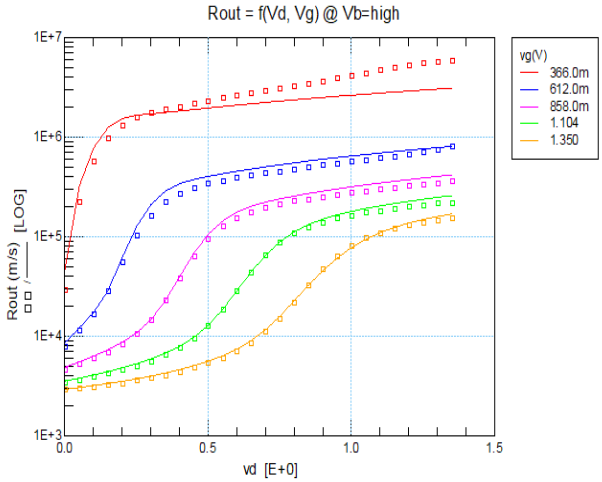


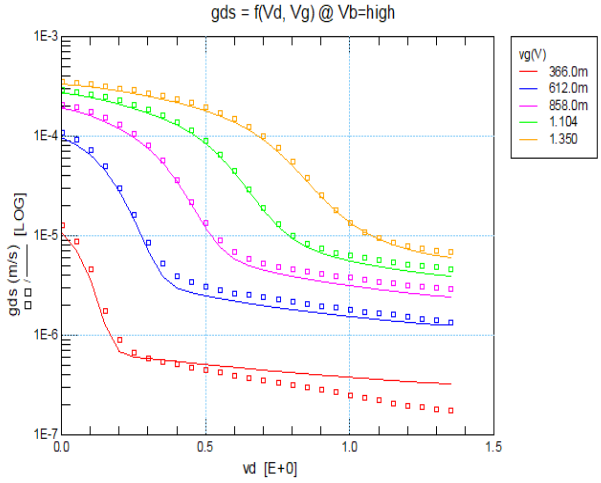




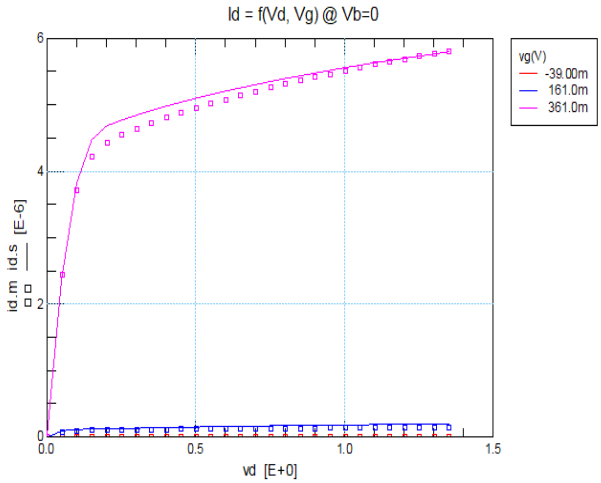
idvd_vbmin, Vb = -1.2V, T = 27°C

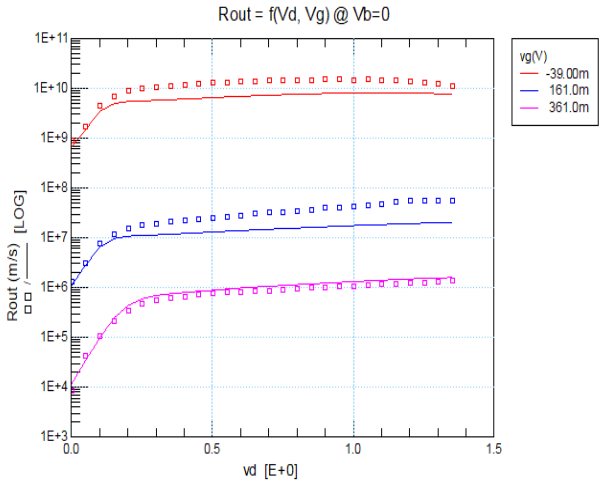


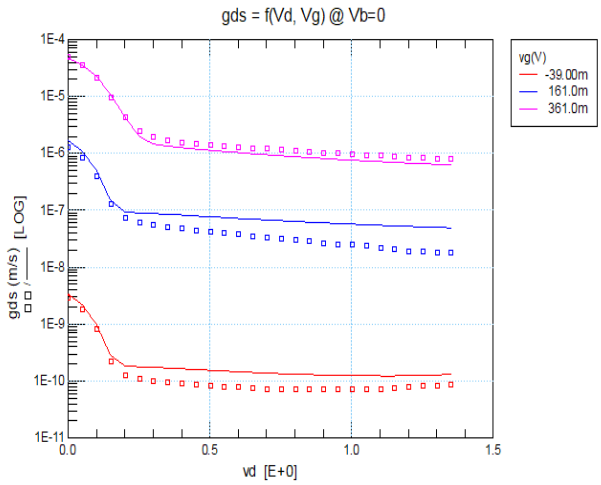




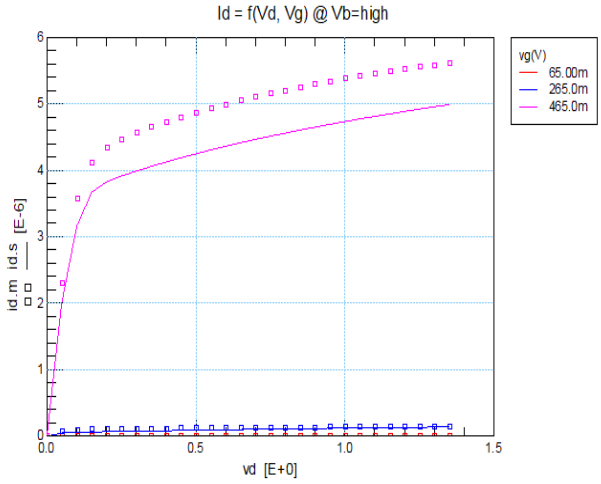
idvd_vth, Vb = 0V, T = 27°C

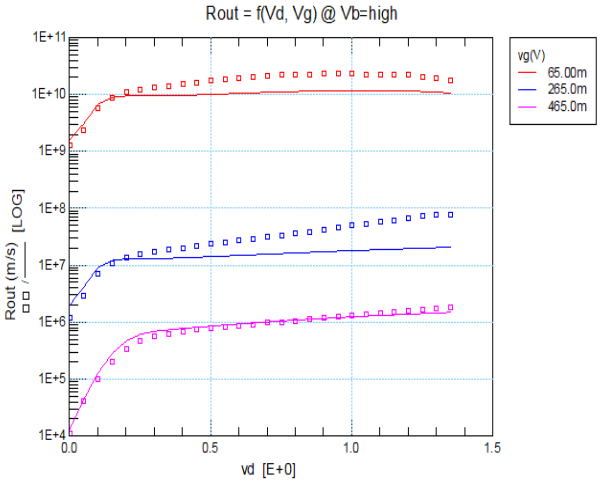


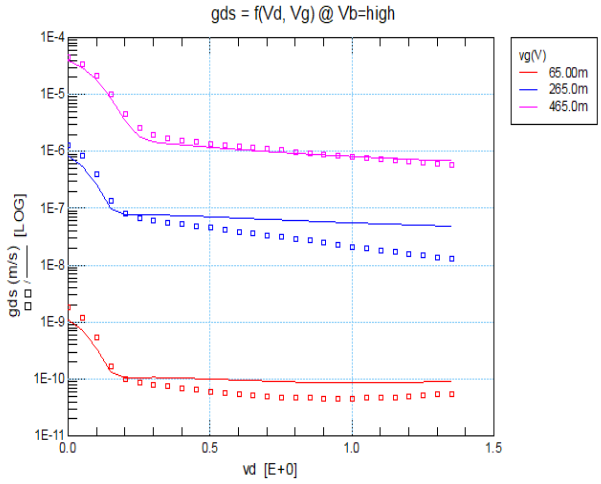




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

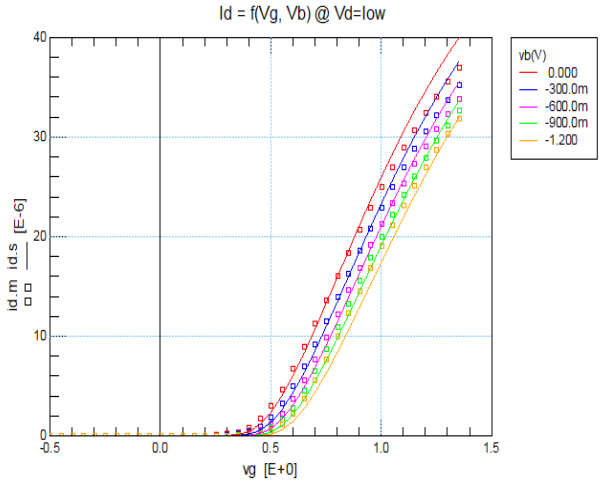


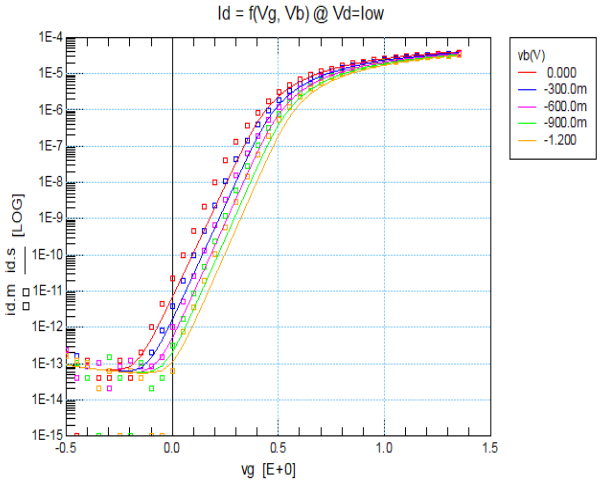


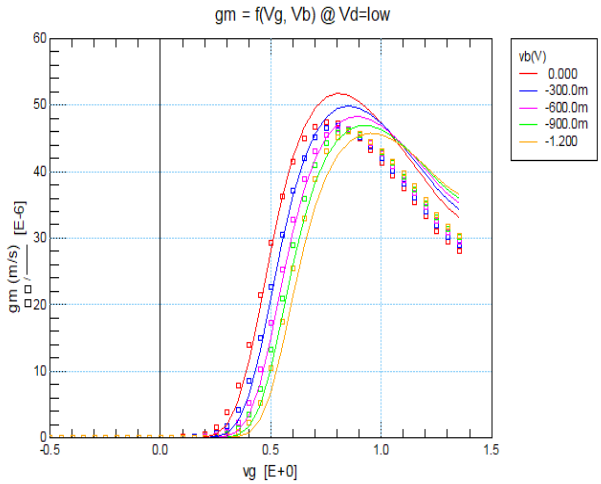


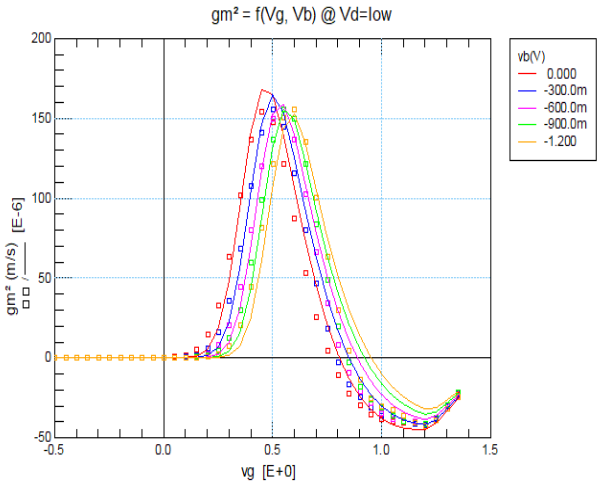
4.23 W0u6_L0u18_S543_3

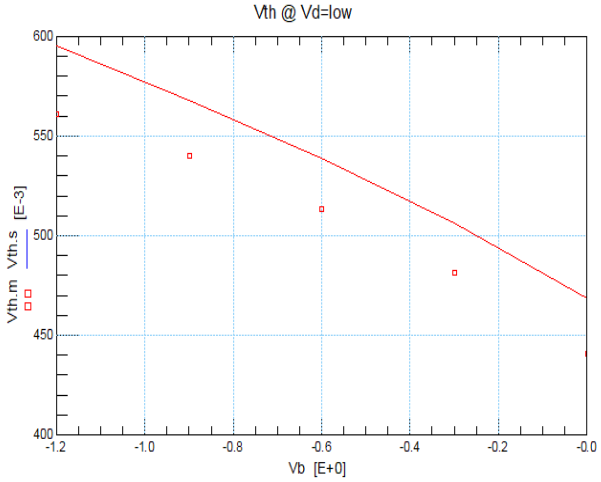
idvg, Vd = 0.05V, T = 27°C

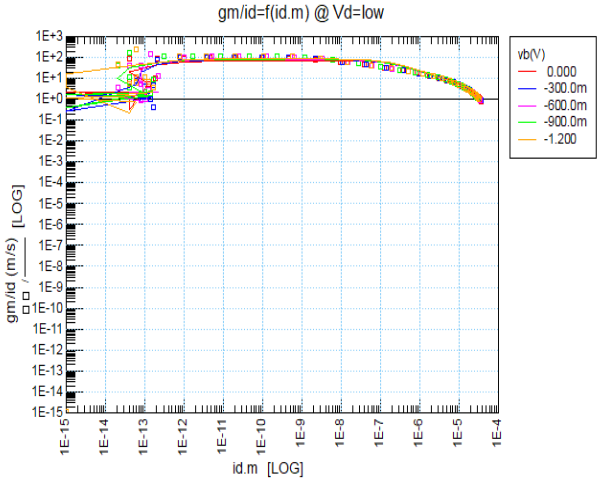


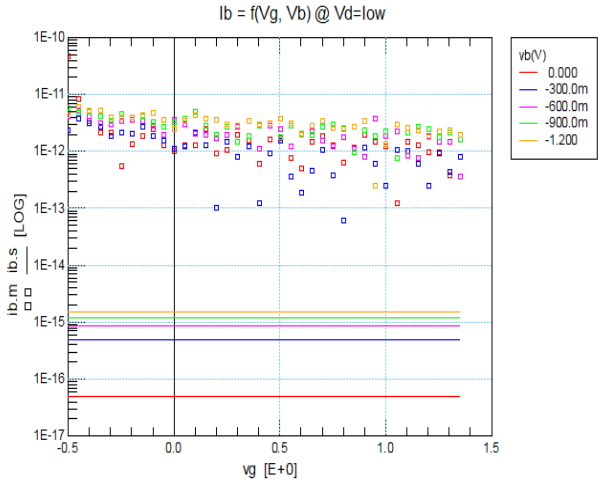


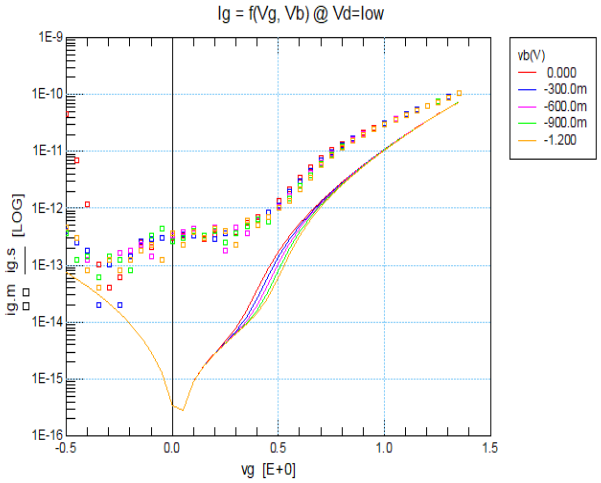




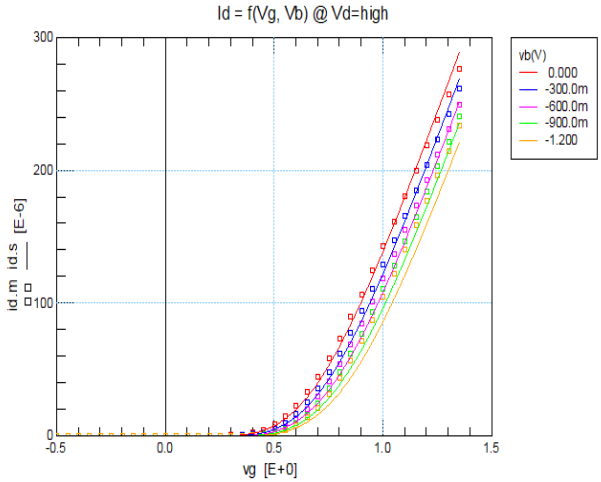


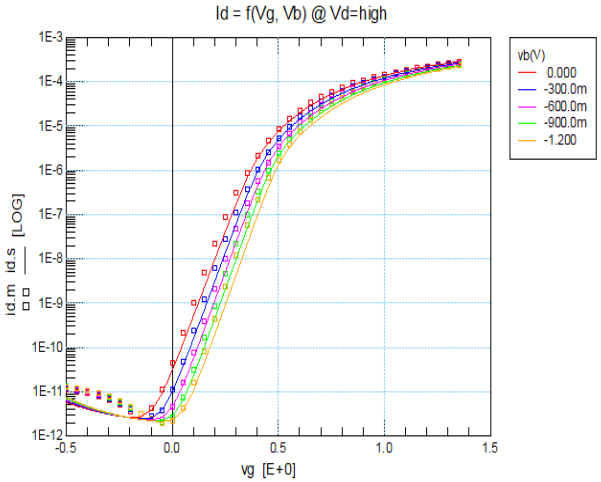


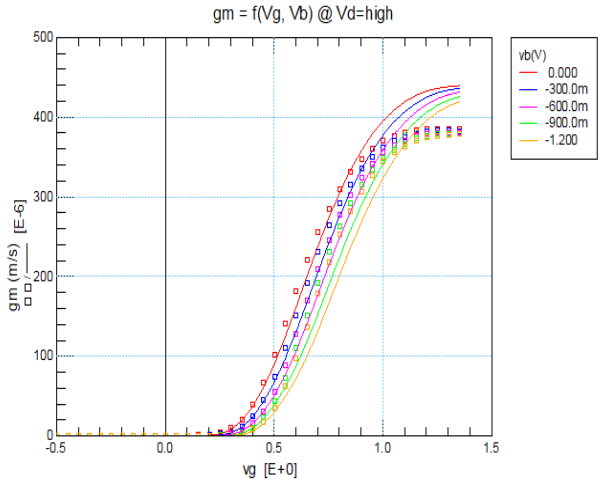


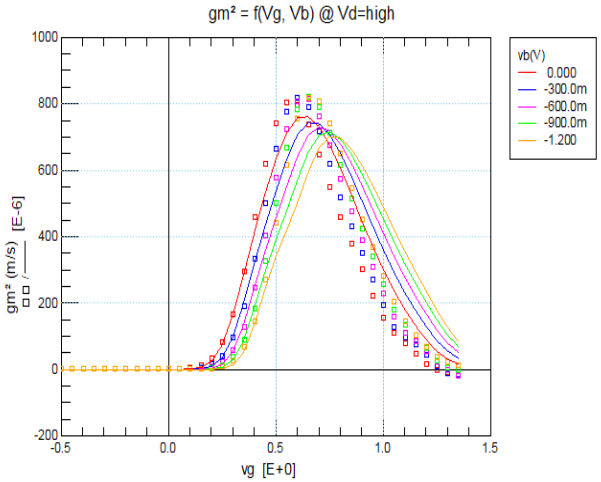


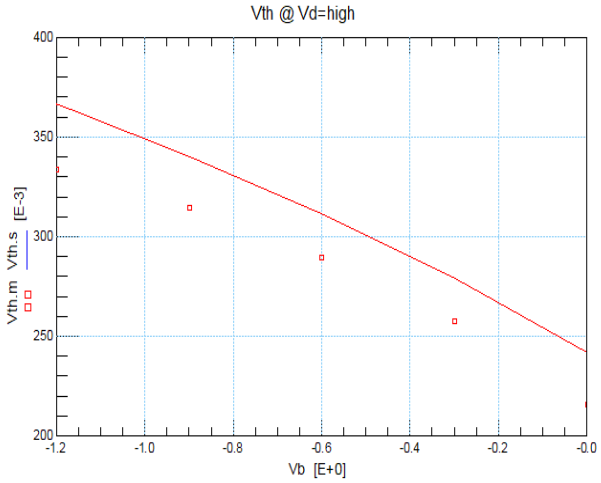
idvg, Vd = 1.2V, T = 27°C

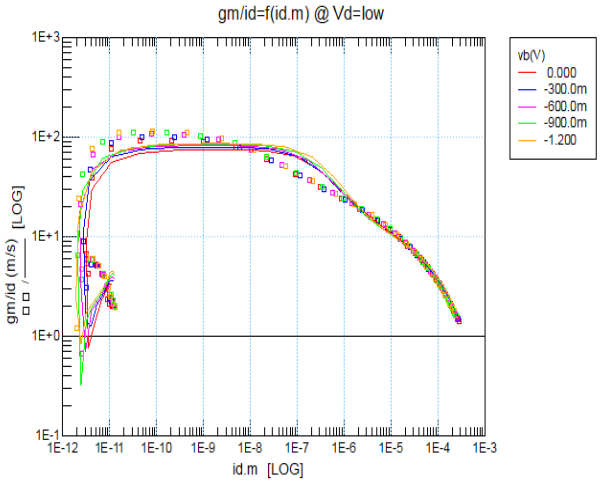


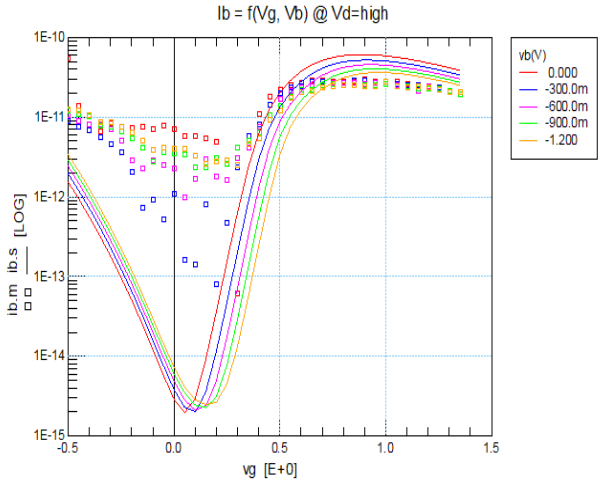


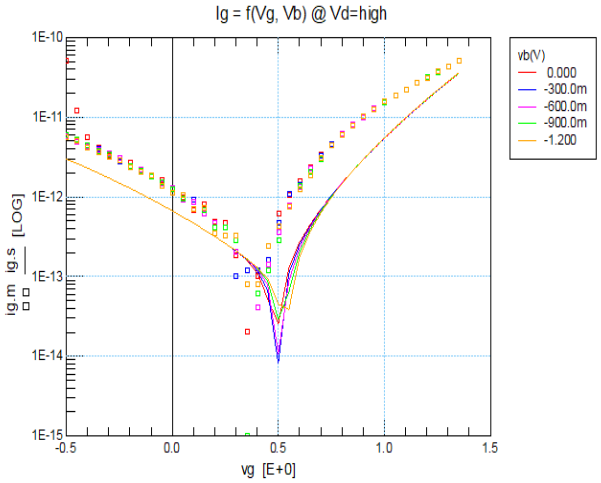




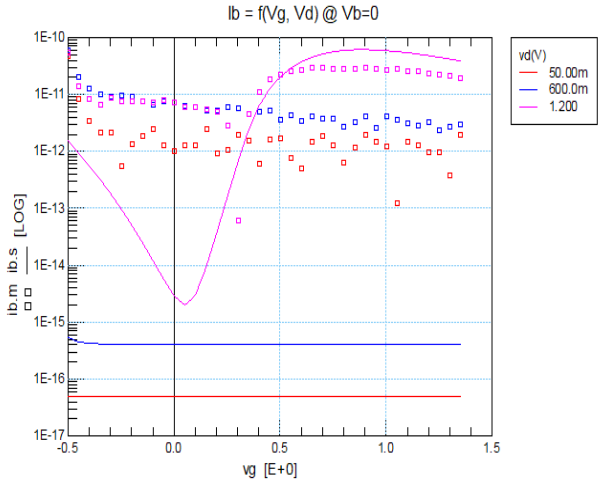


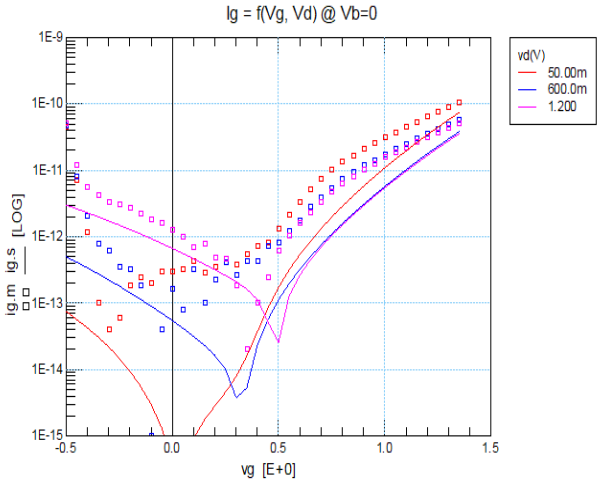




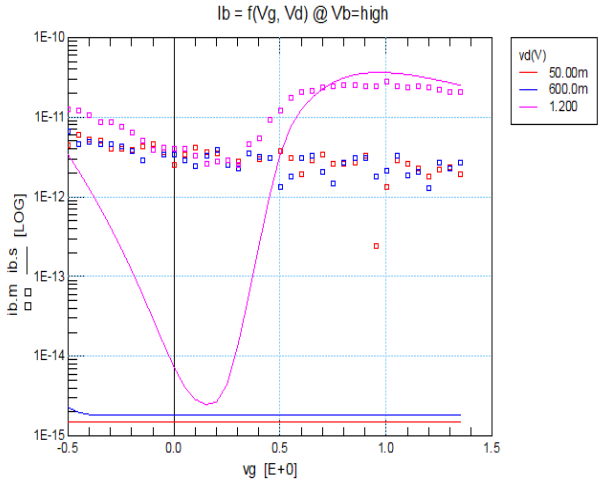


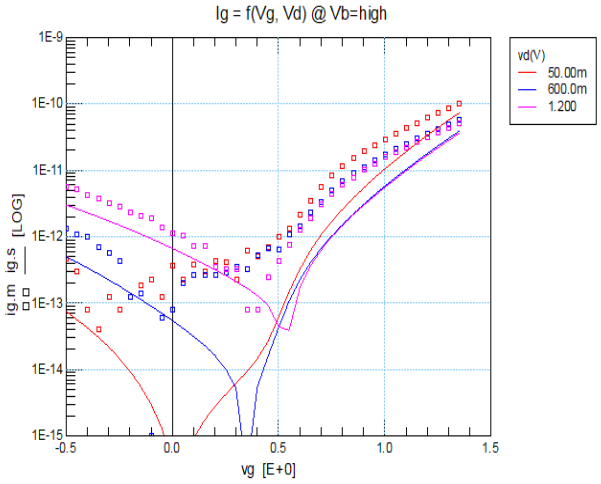
idvg, Ib,Ig, Vb = 0V, T = 27°C



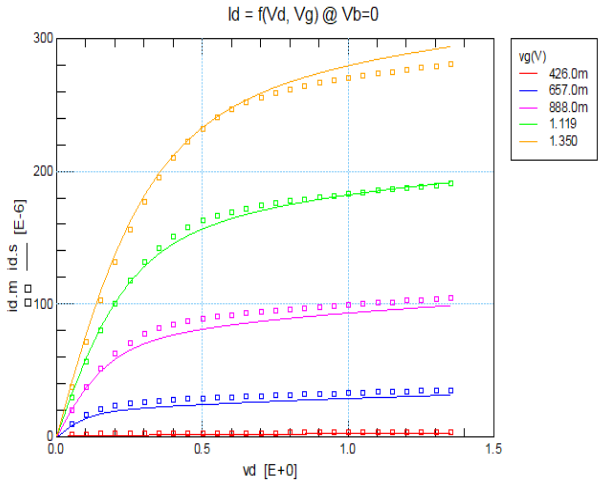


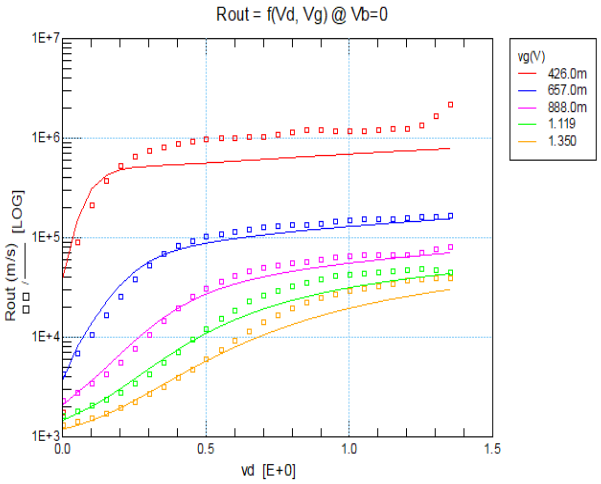
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

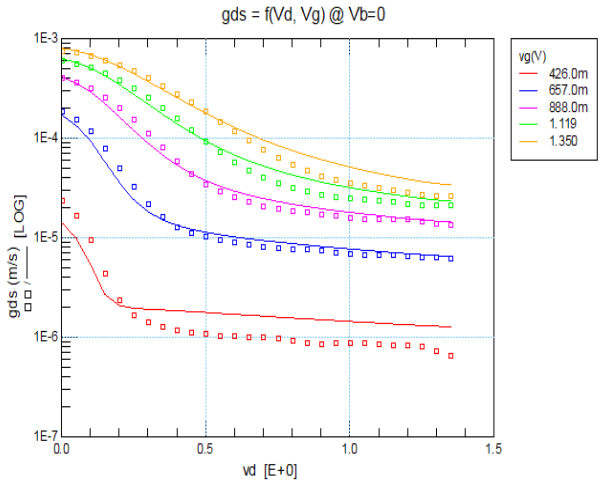




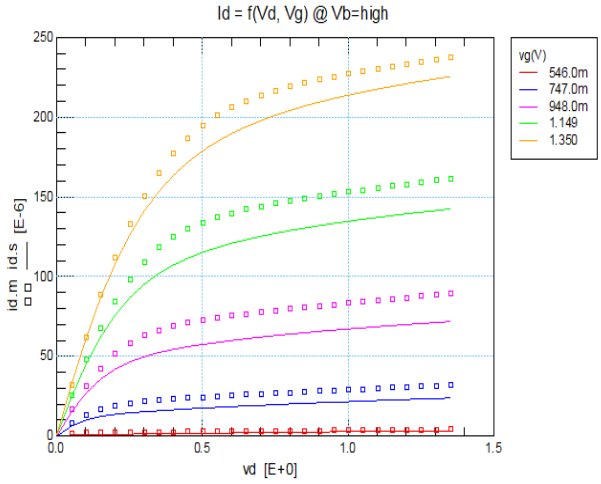
idvd, Vb = 0V, T = 27°C

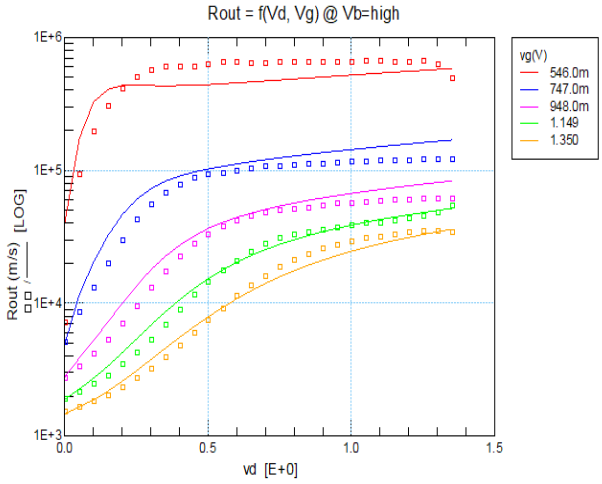


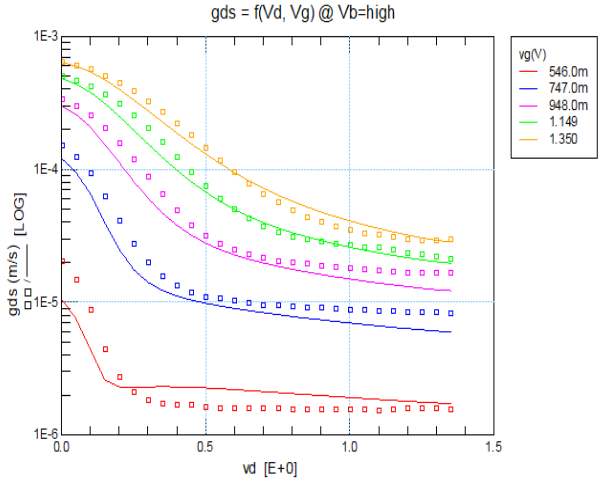




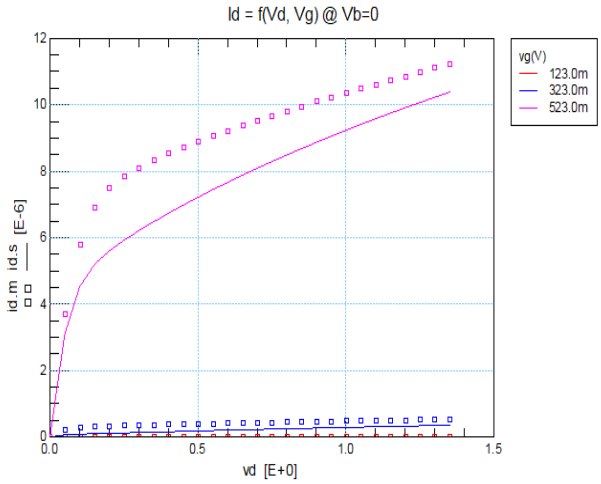
idvd_vbmin, Vb = -1.2V, T = 27°C

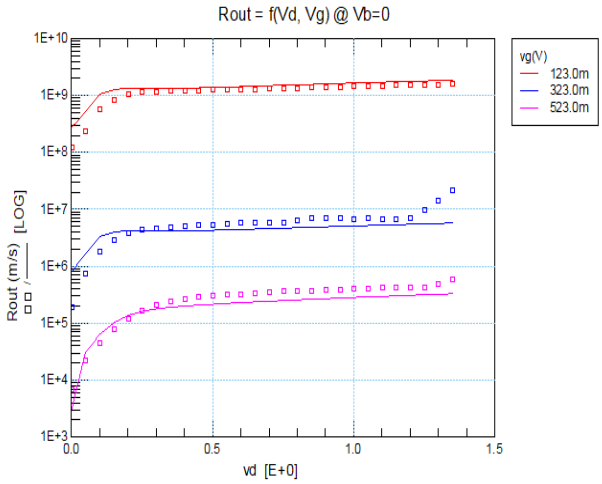


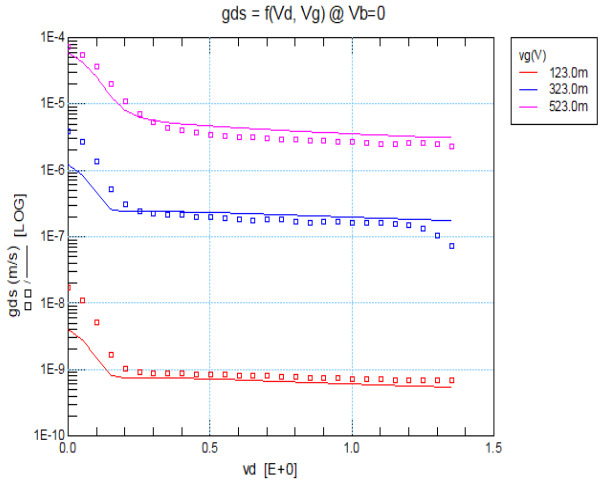




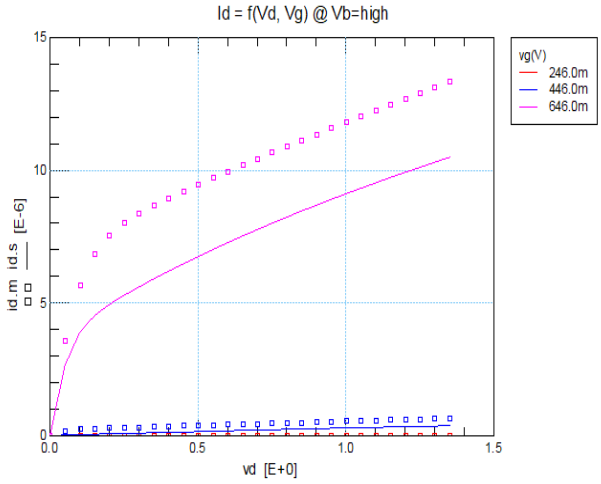
idvd_vth, Vb = 0V, T = 27°C

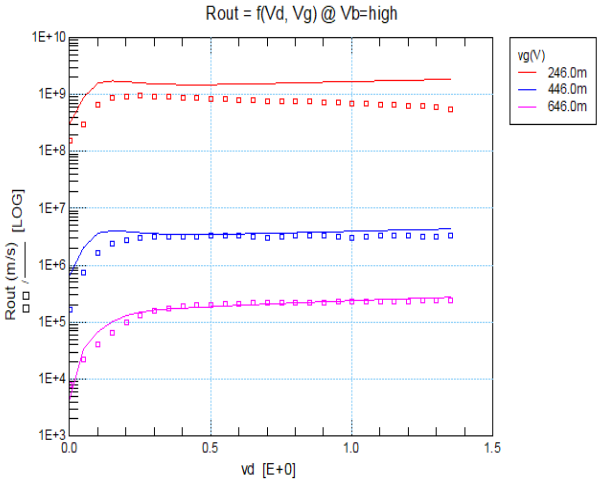


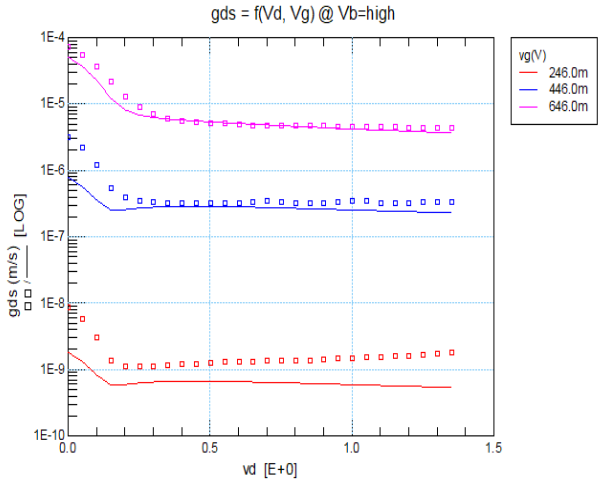




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

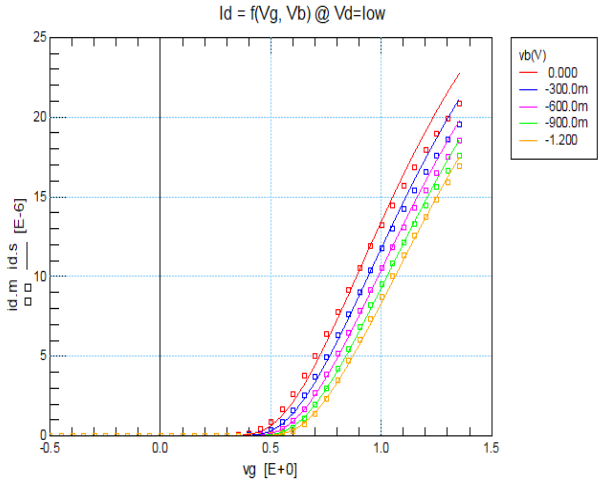


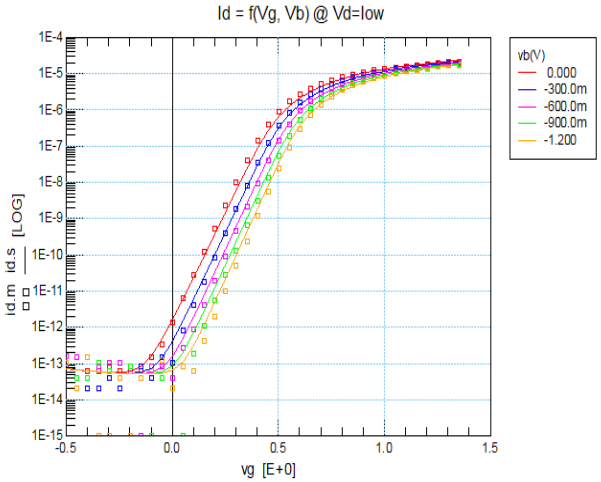


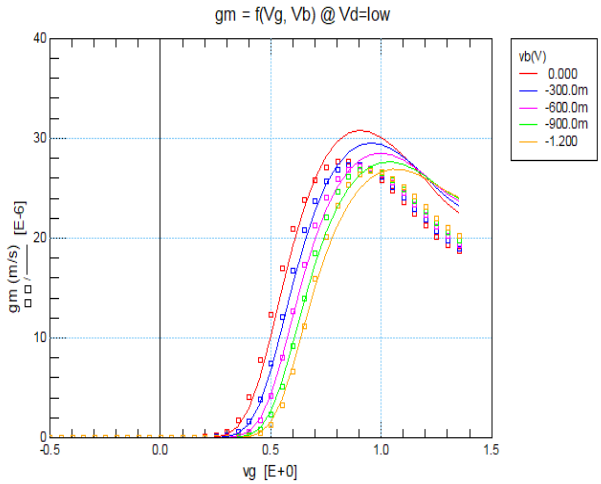


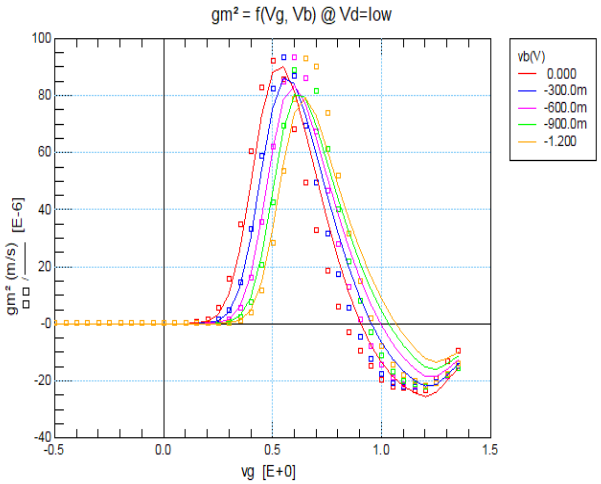
4.24 W0u3_L0u14_S544_3

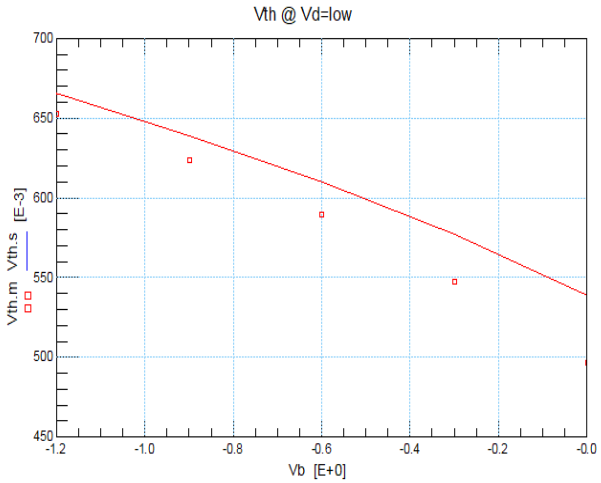
idvg, Vd = 0.05V, T = 27°C

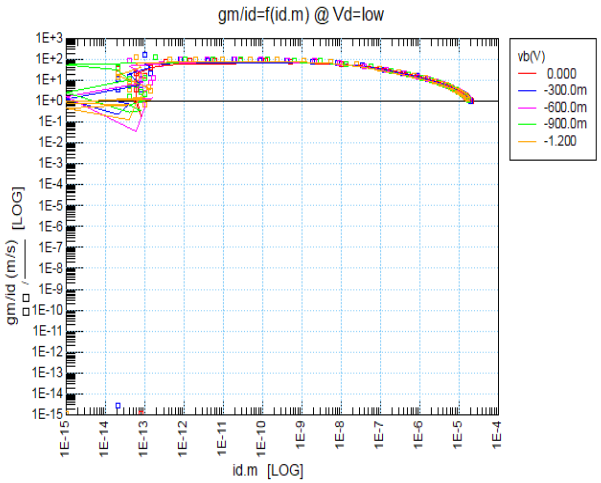


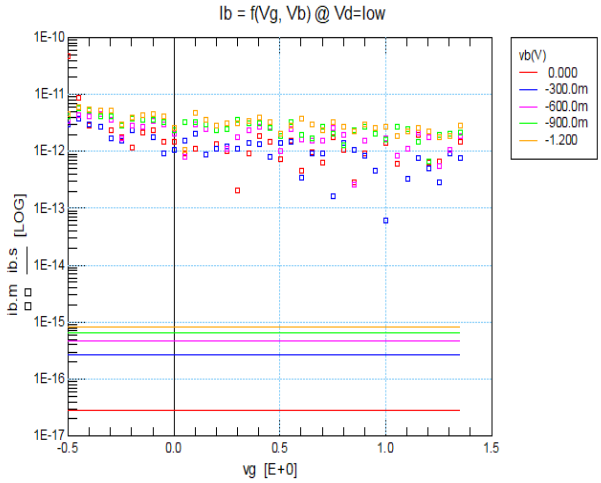


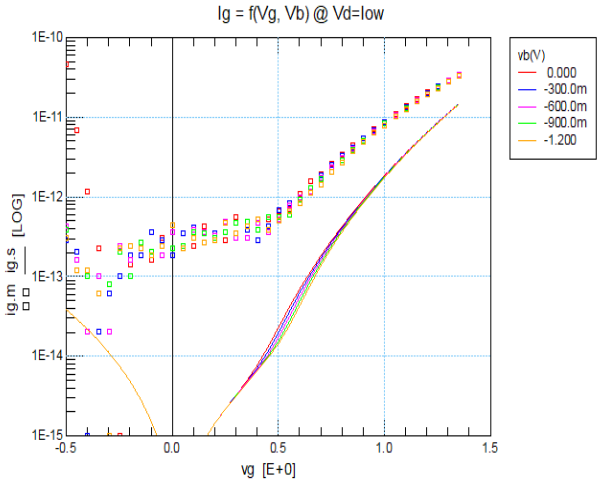




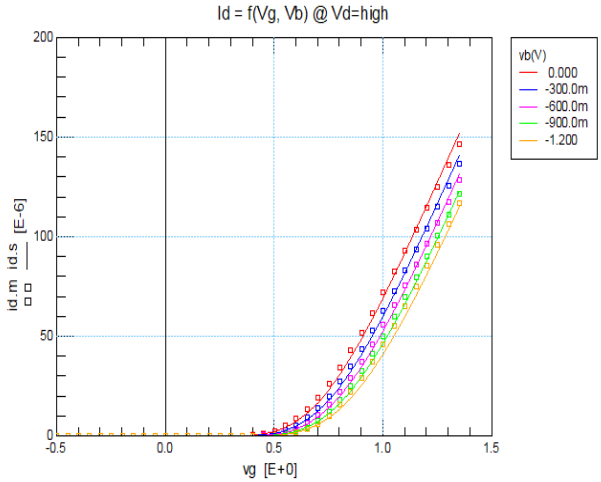


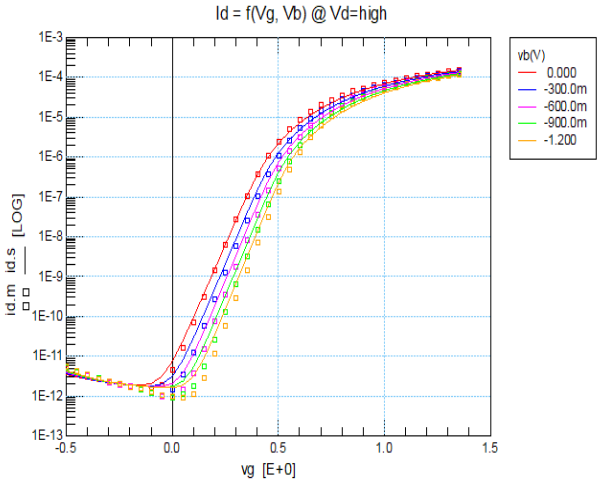


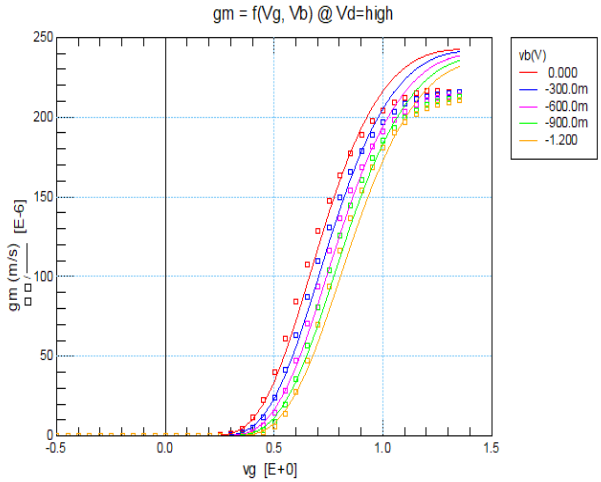


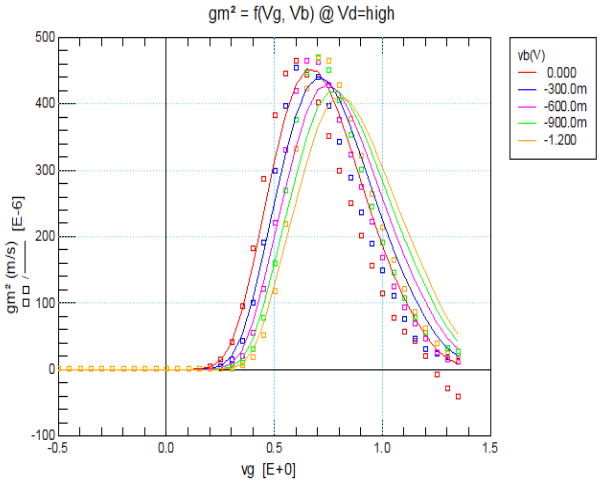


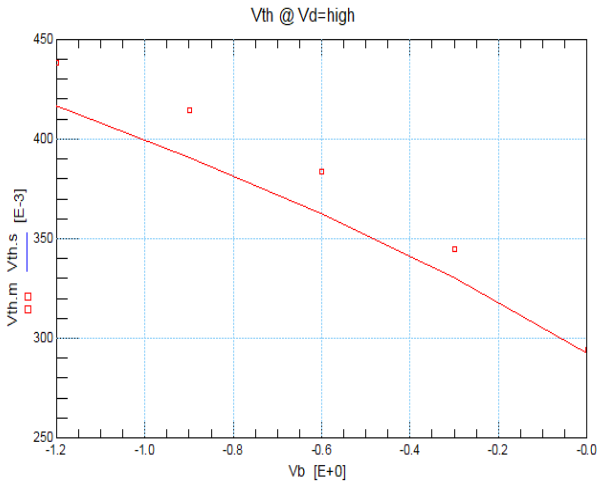
idvg, Vd = 1.2V, T = 27°C

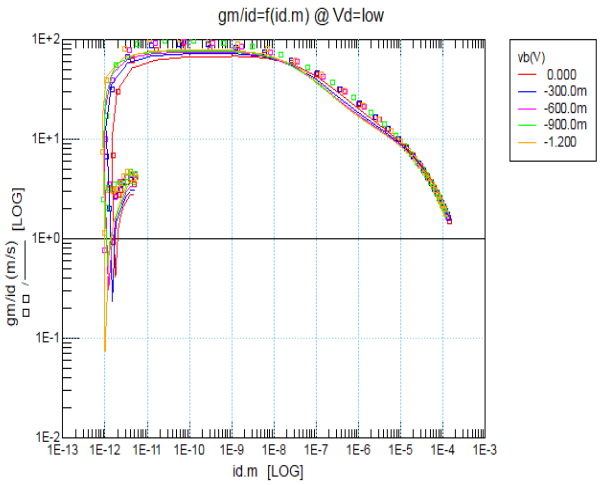


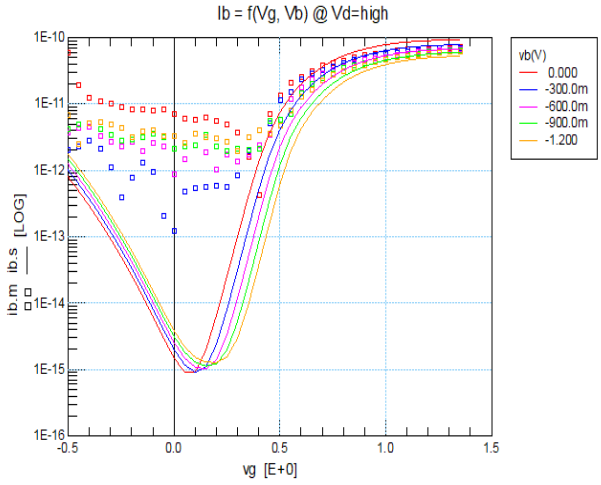


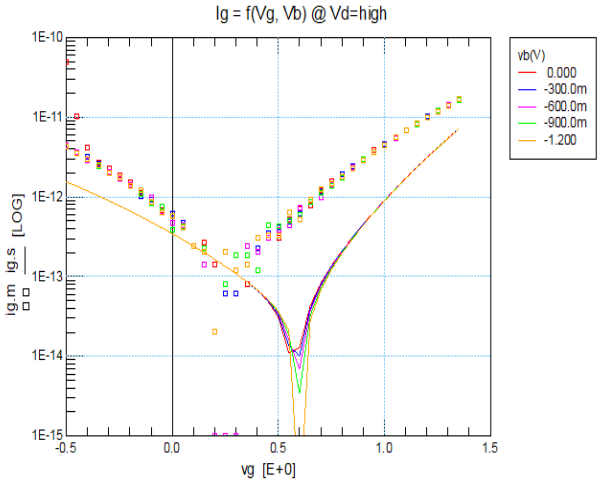




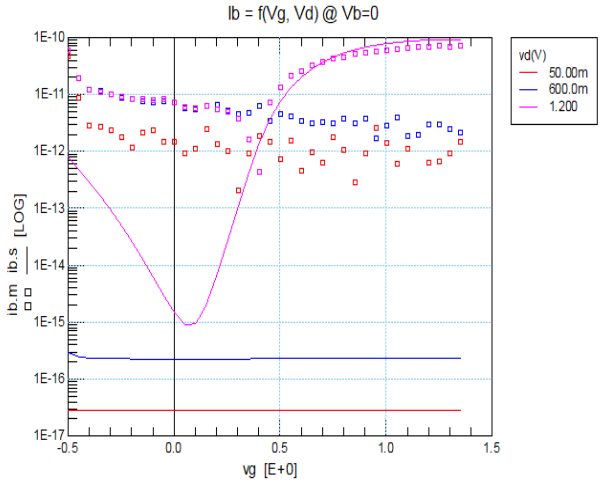


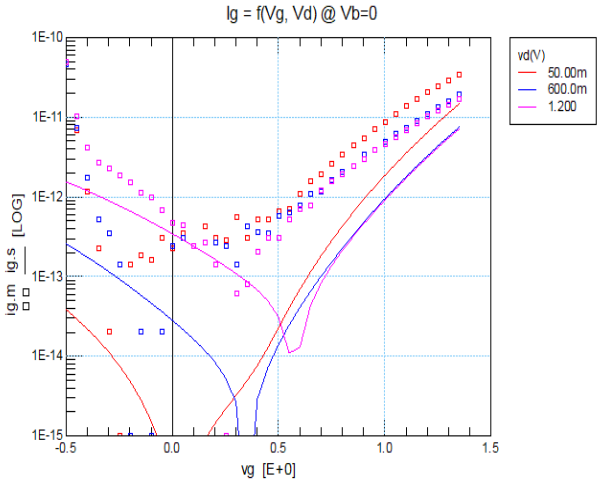




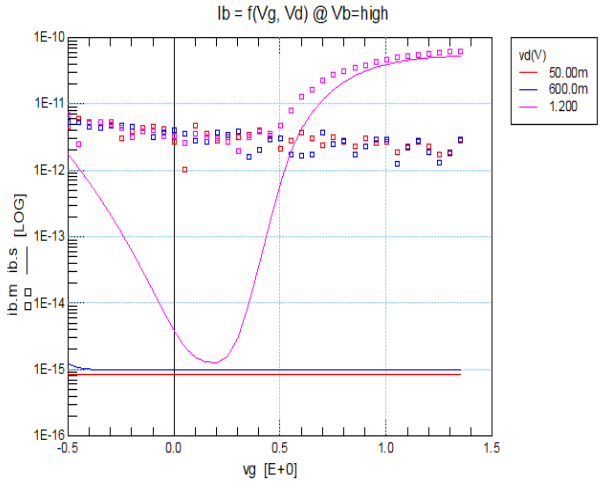


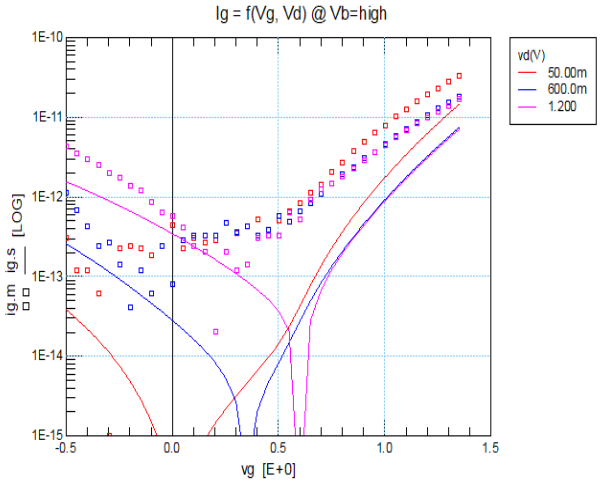
idvg, Ib,Ig, Vb = 0V, T = 27°C



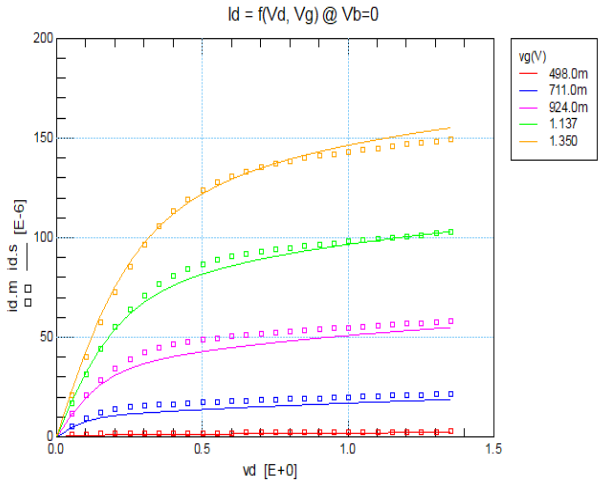


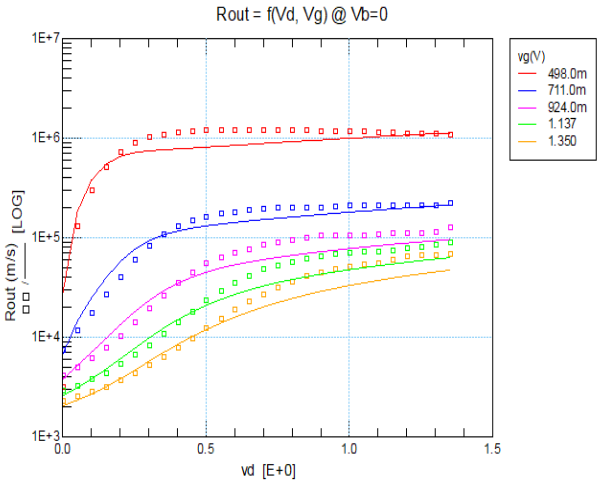
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

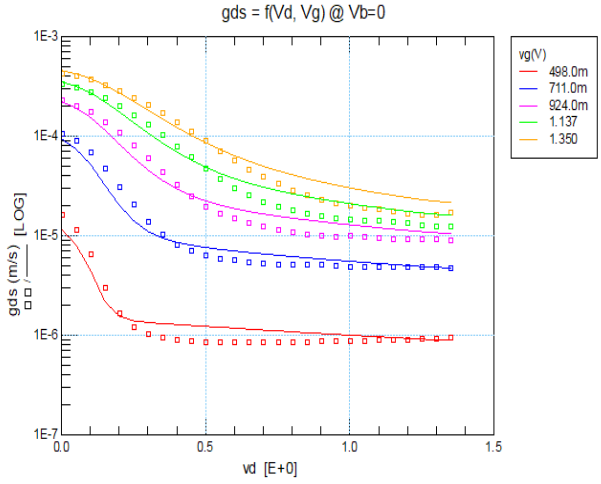




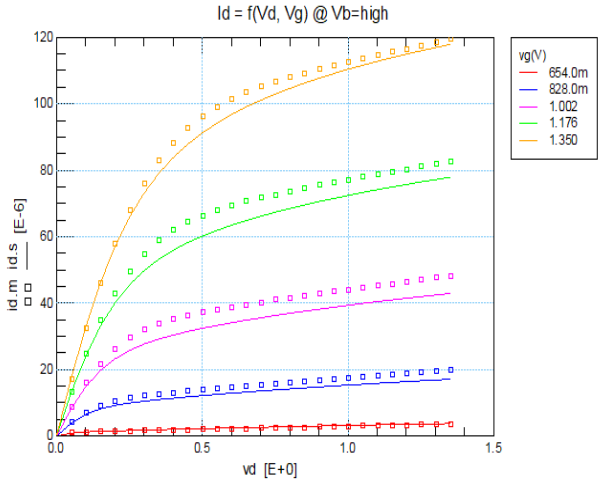
idvd, Vb = 0V, T = 27°C

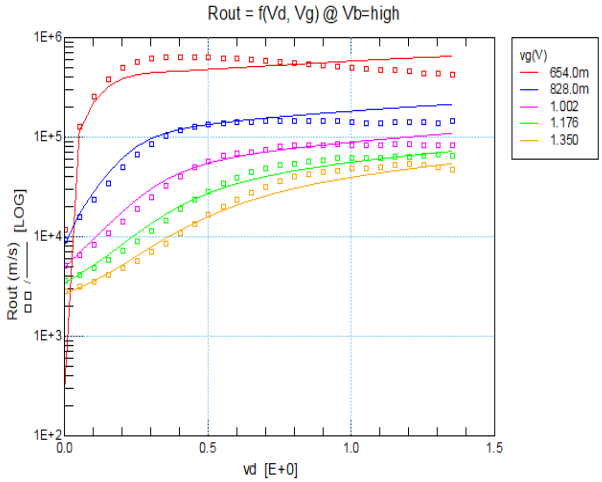


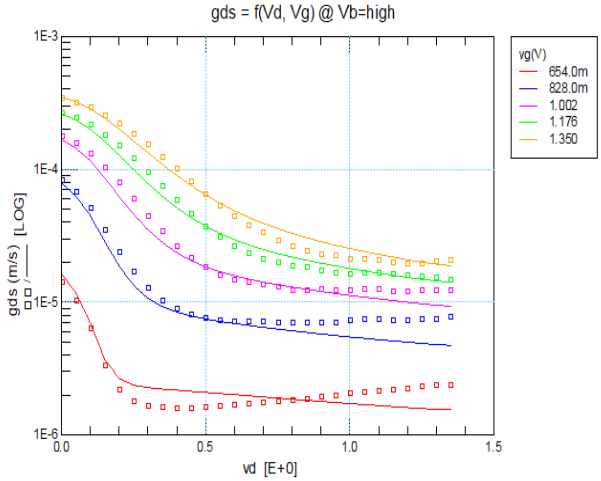




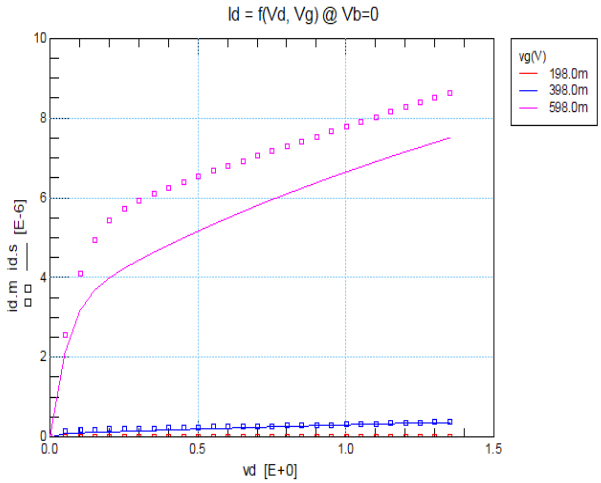
idvd_vbmin, Vb = -1.2V, T = 27°C

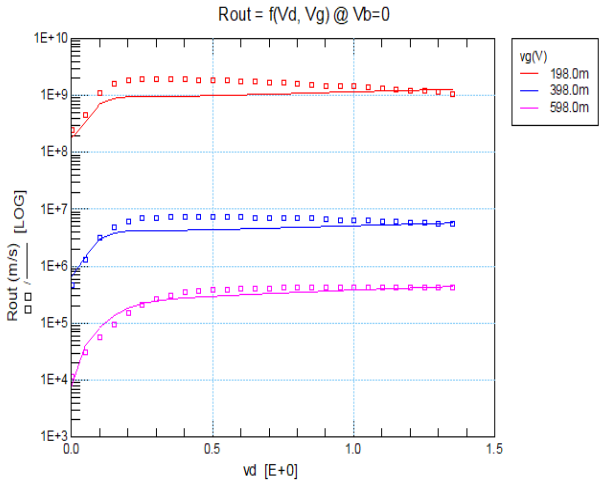


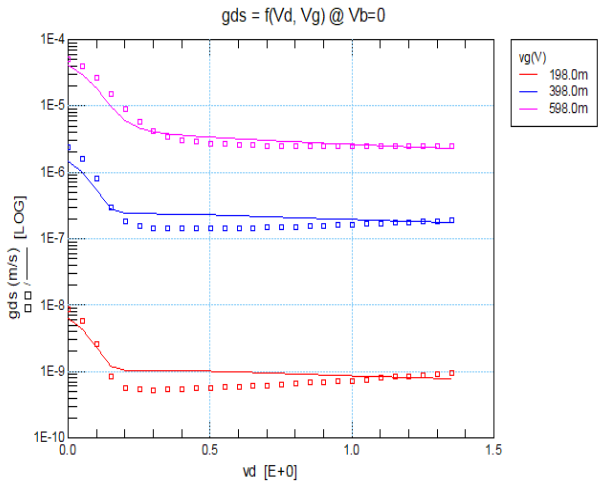




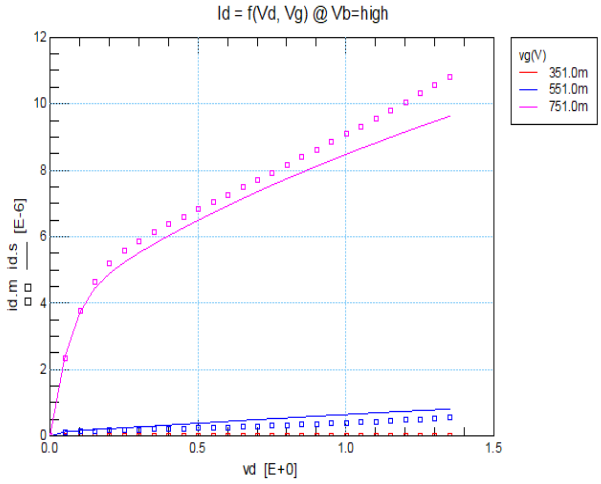
idvd_vth, Vb = 0V, T = 27°C

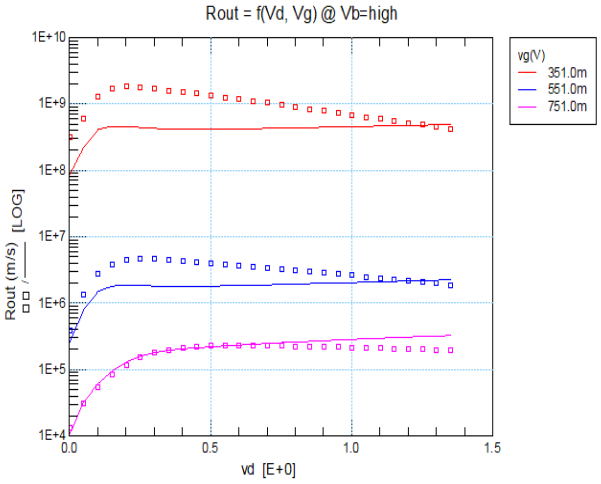


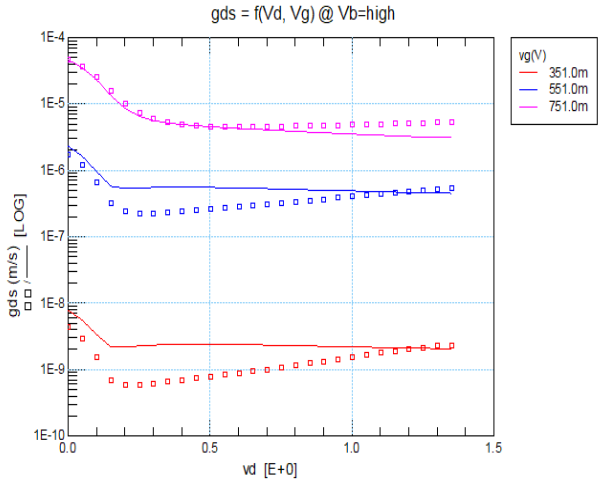




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

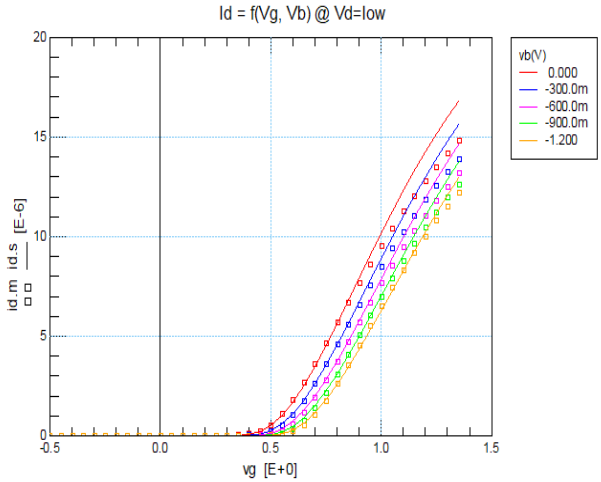


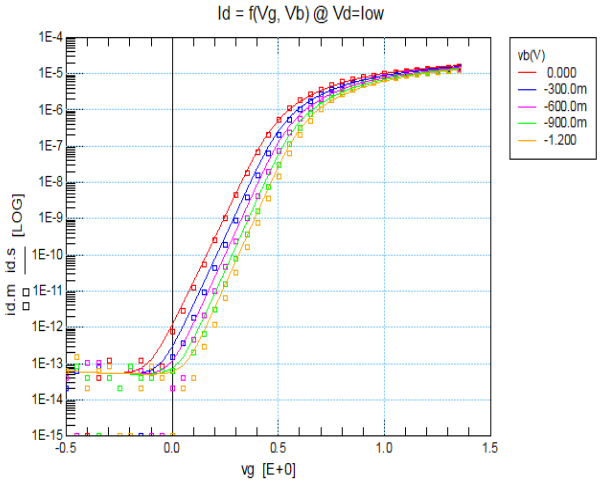


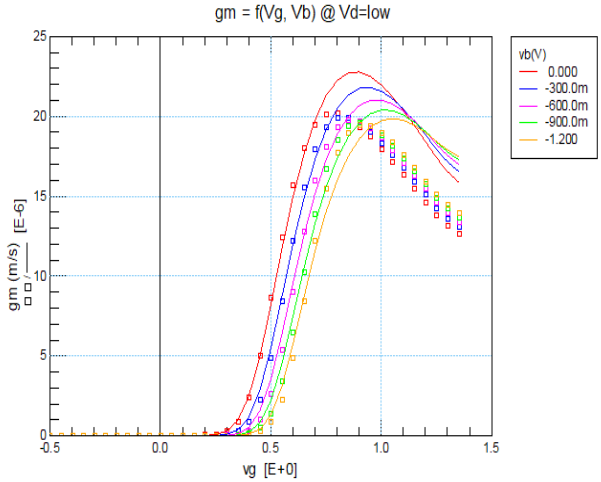


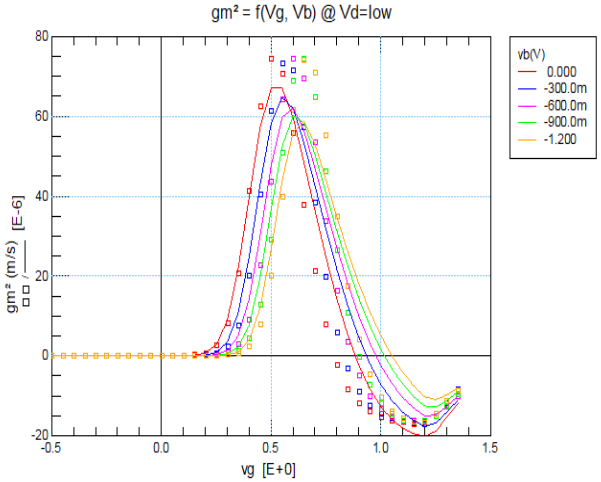
4.25W0u18_L0u14_S545_4

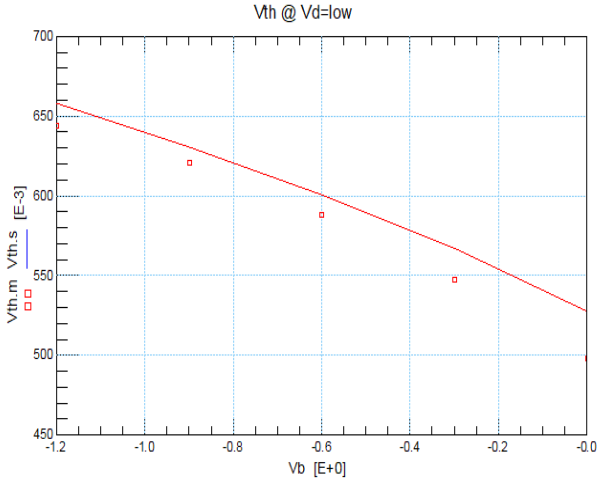
idvg, Vd = 0.05V, T = 27°C

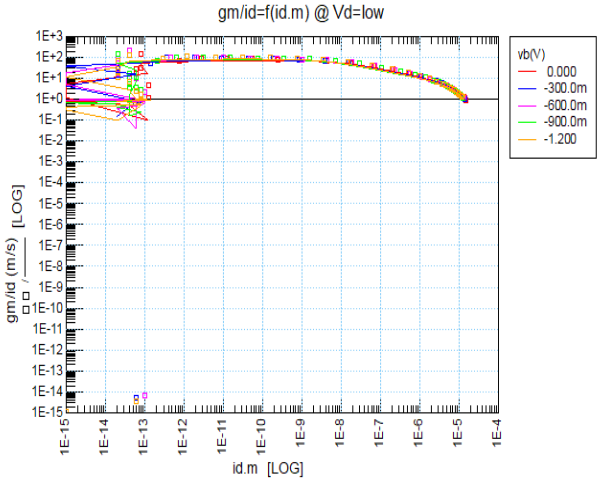


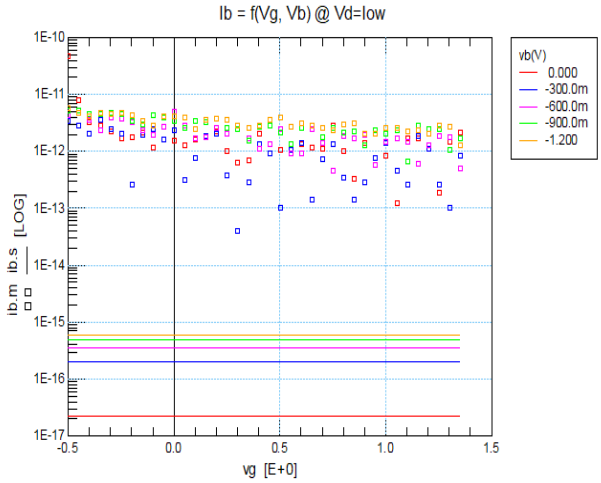


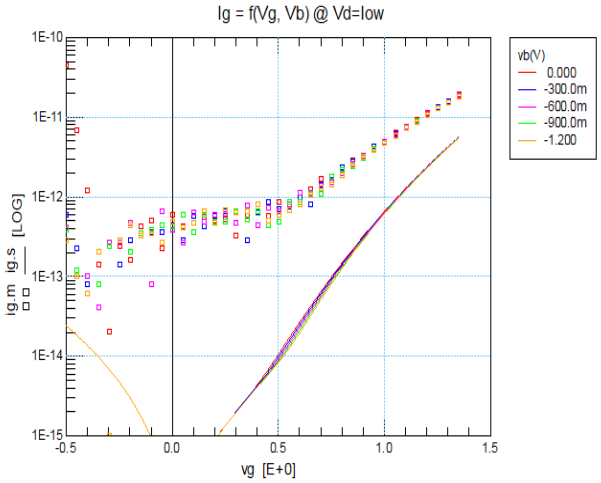




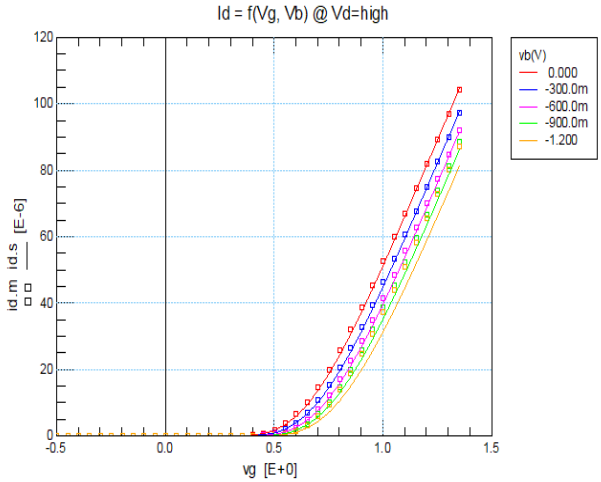


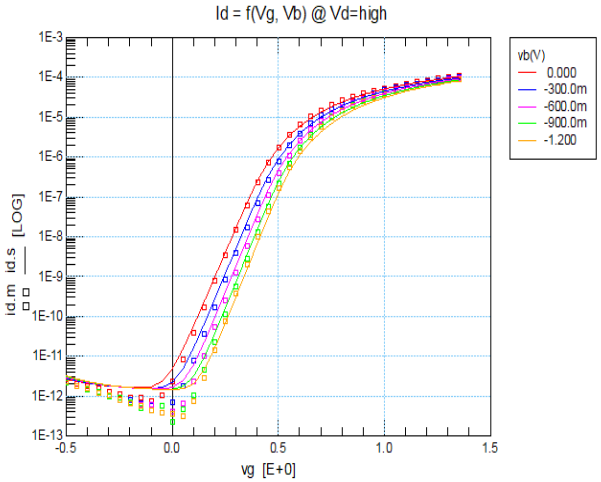


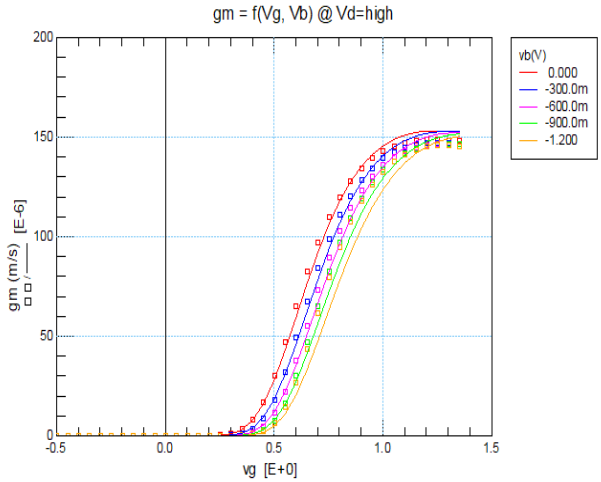


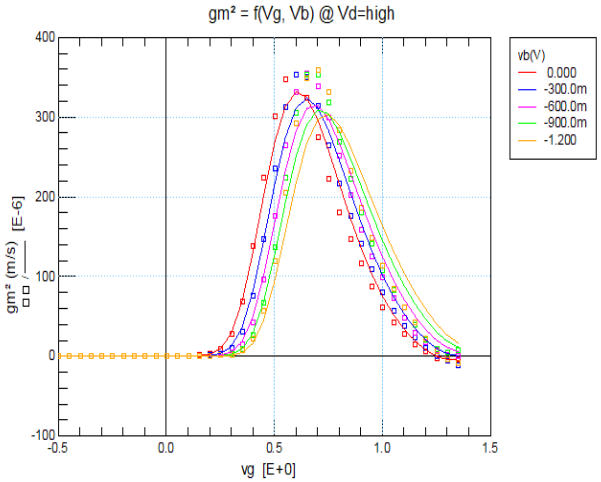


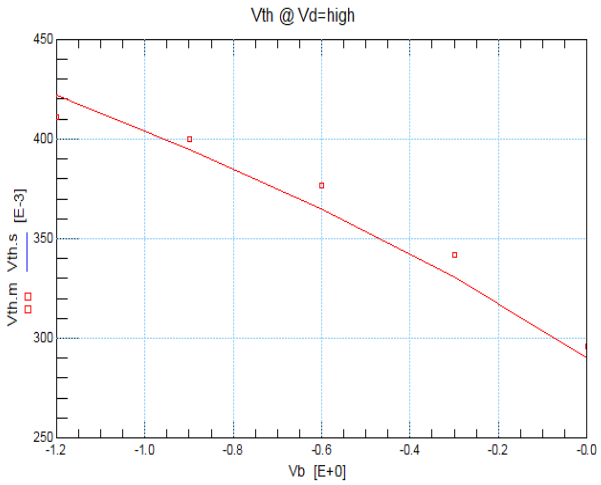
idvg, Vd = 1.2V, T = 27°C

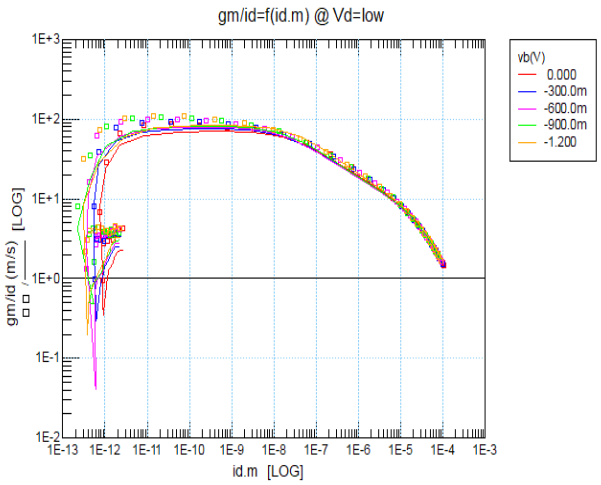


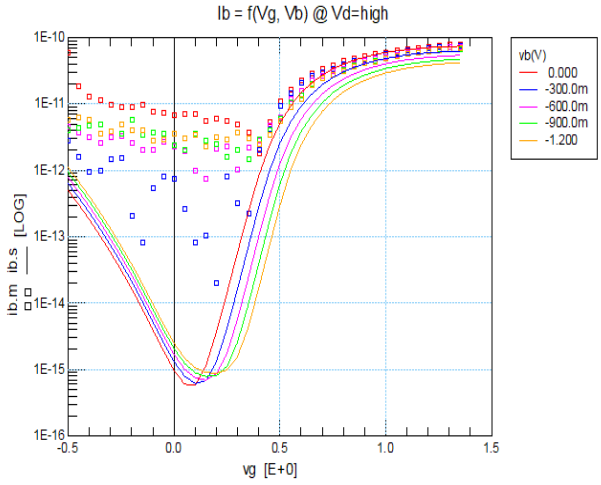


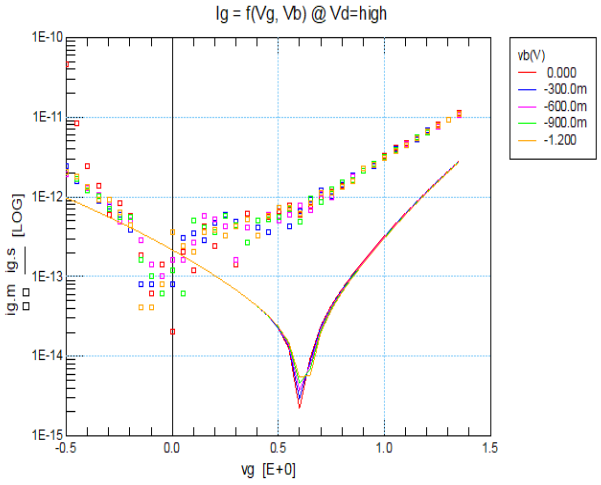




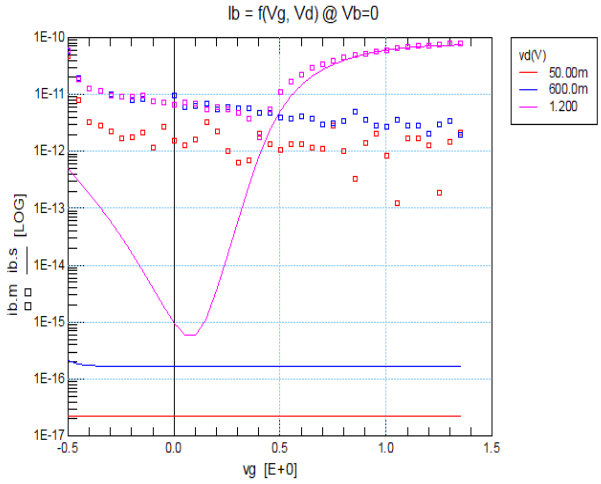


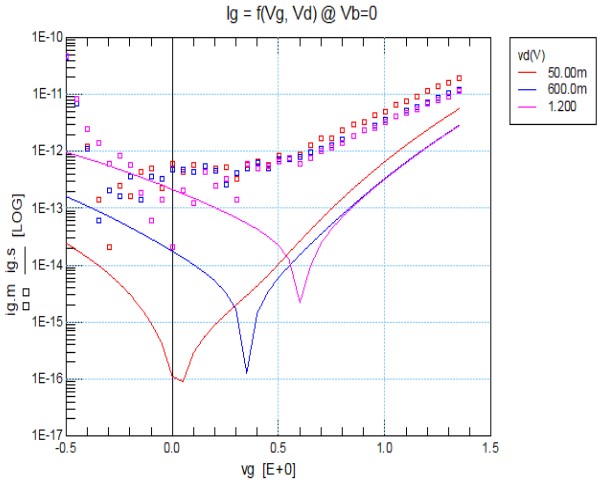




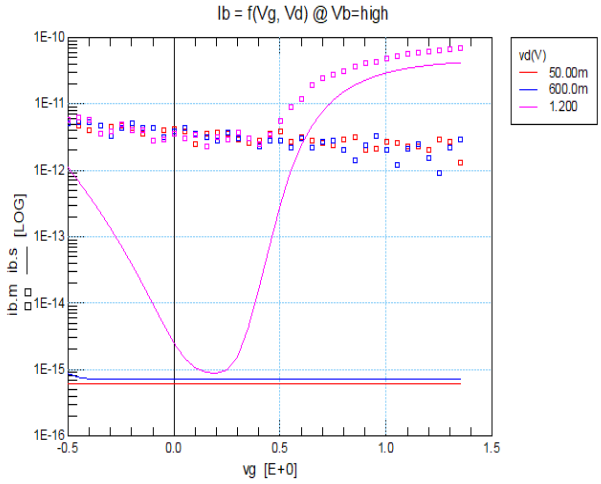


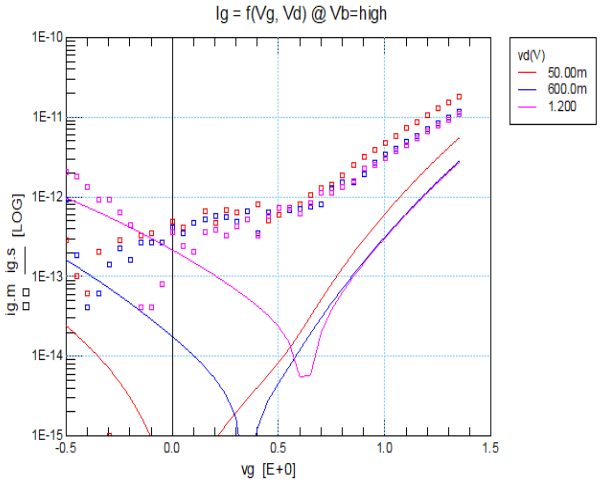
idvg, Ib,Ig, Vb = 0V, T = 27°C



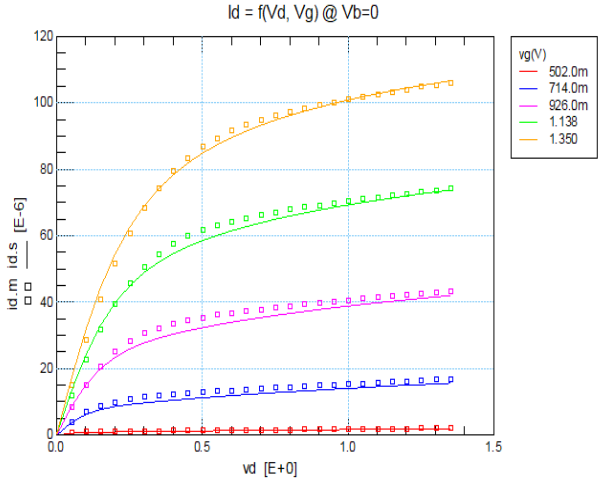


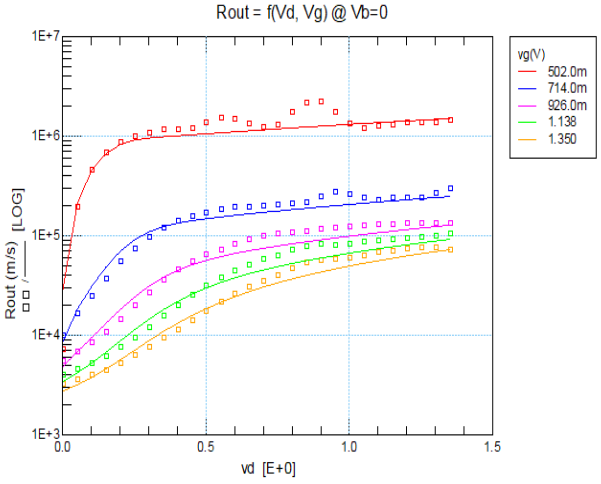
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

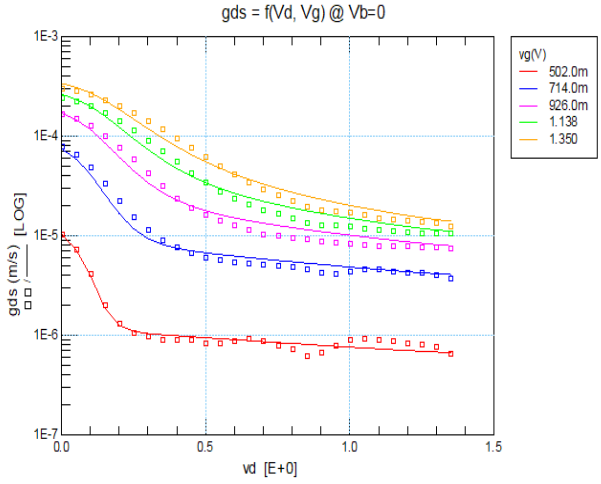




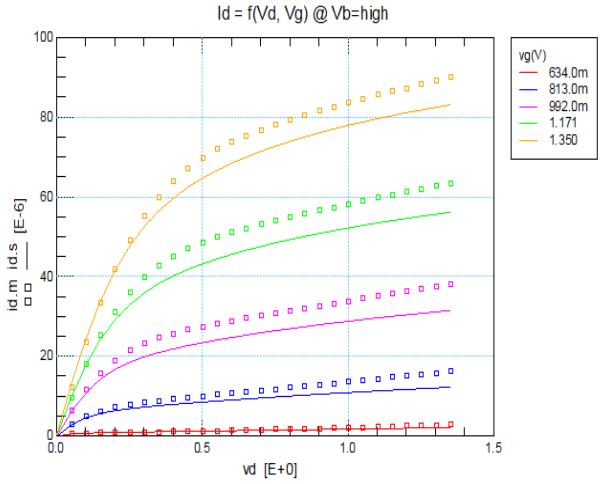
idvd, Vb = 0V, T = 27°C

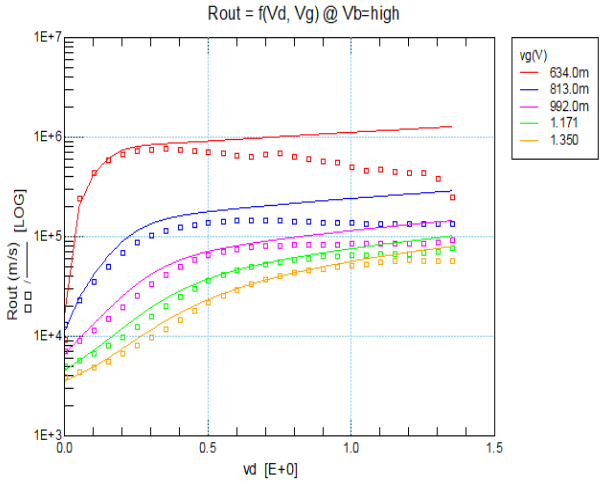


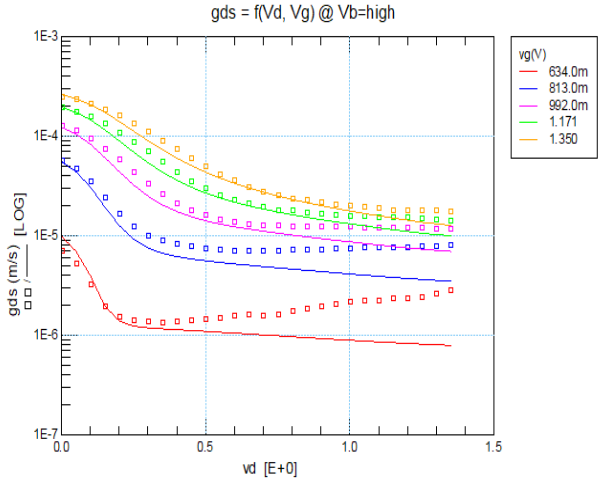




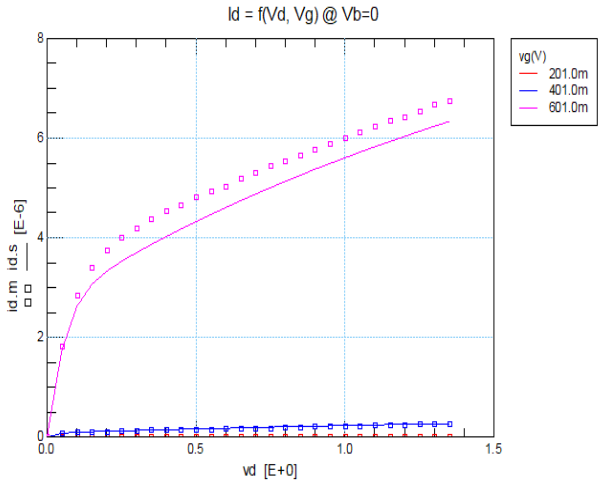
idvd_vbmin, Vb = -1.2V, T = 27°C

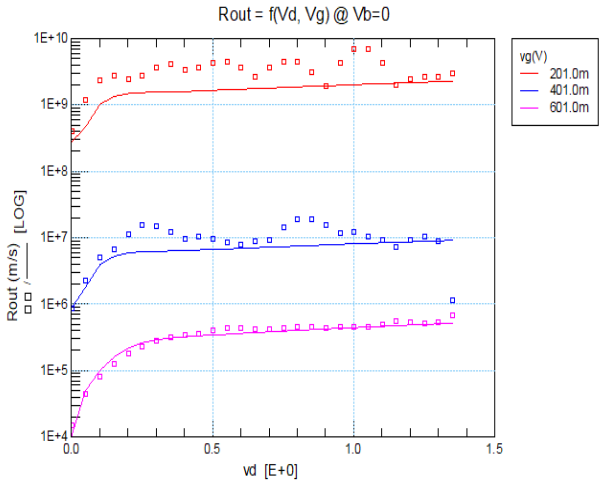


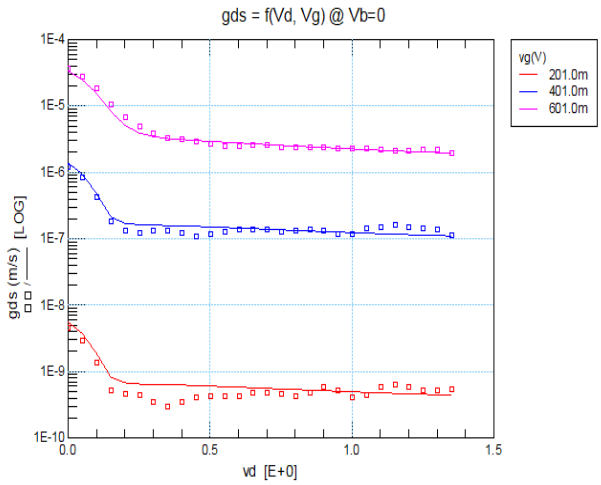




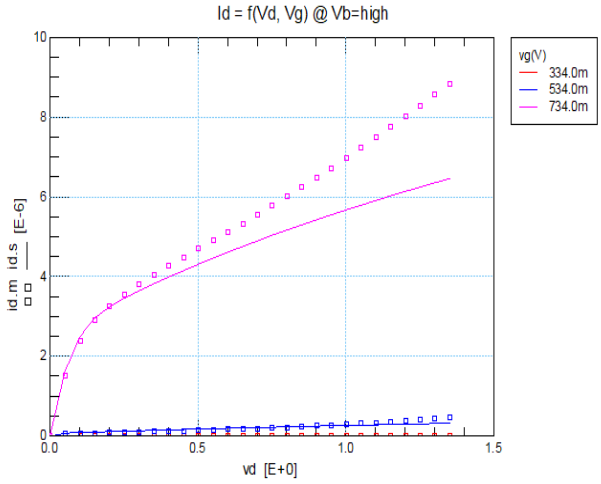
idvd_vth, Vb = 0V, T = 27°C

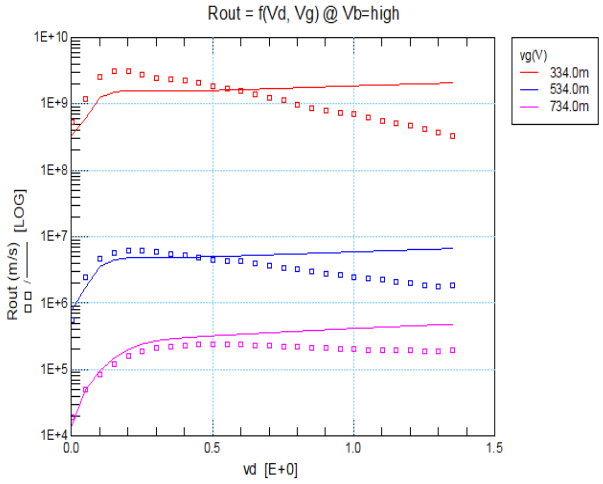


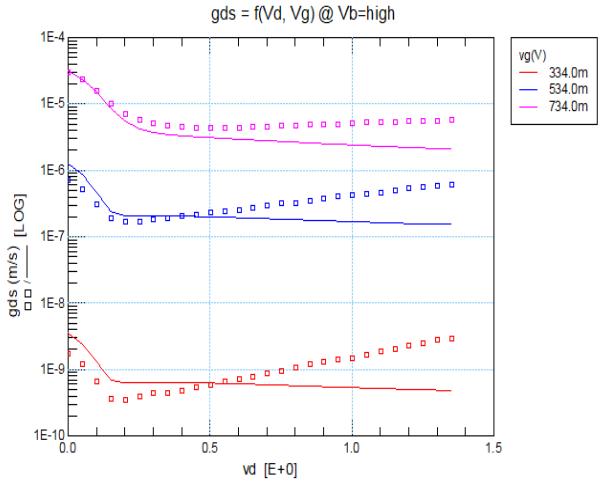




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

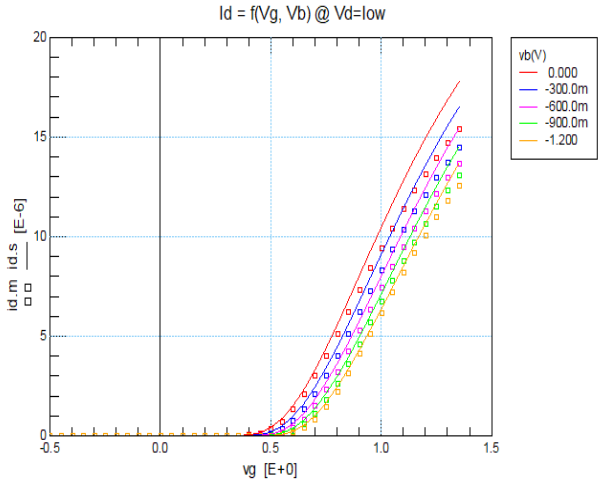


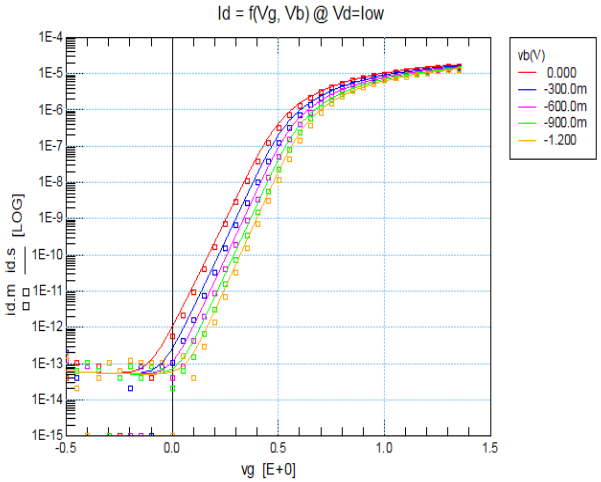


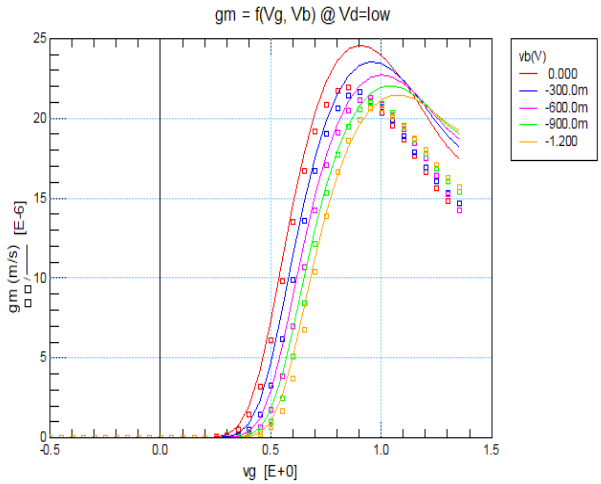


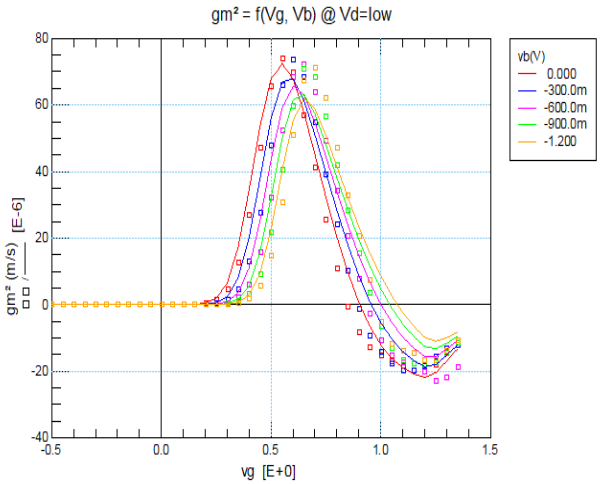
4.26W0u18_L0u13_S545_3

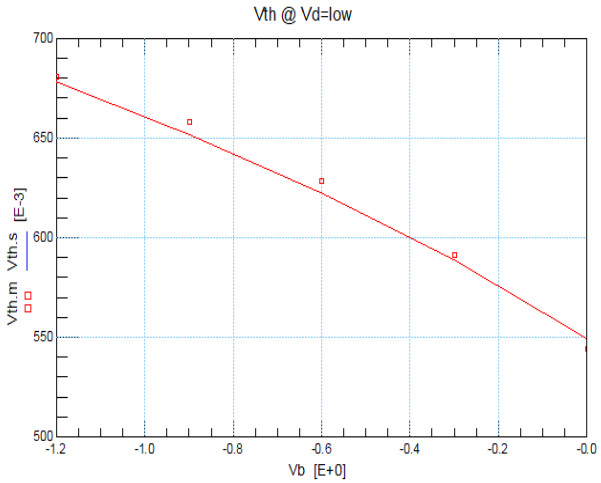
idvg, Vd = 0.05V, T = 27°C

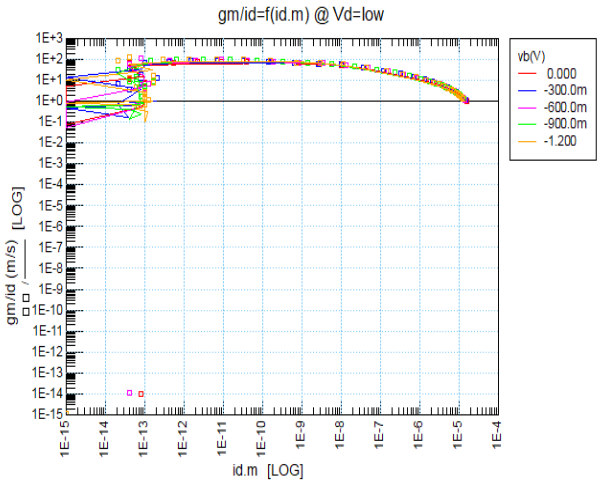


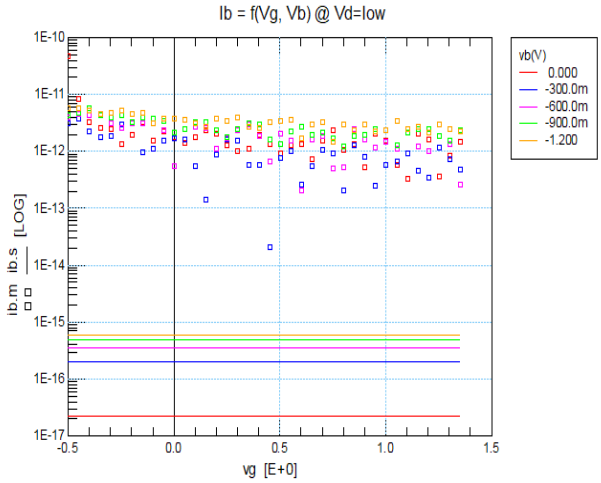


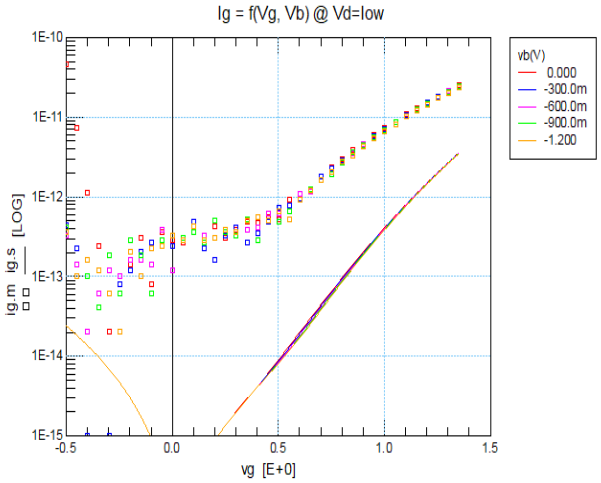




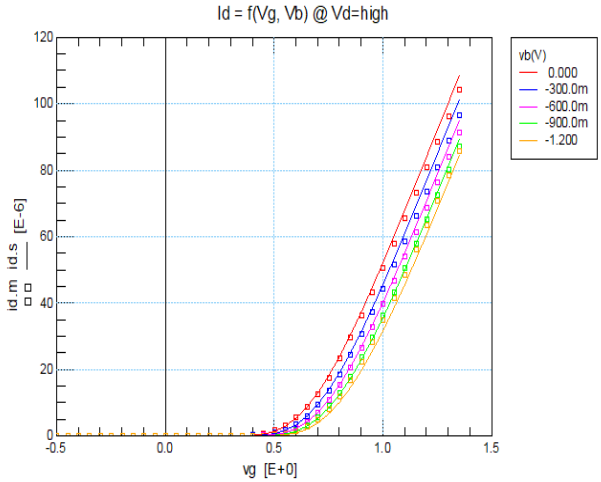


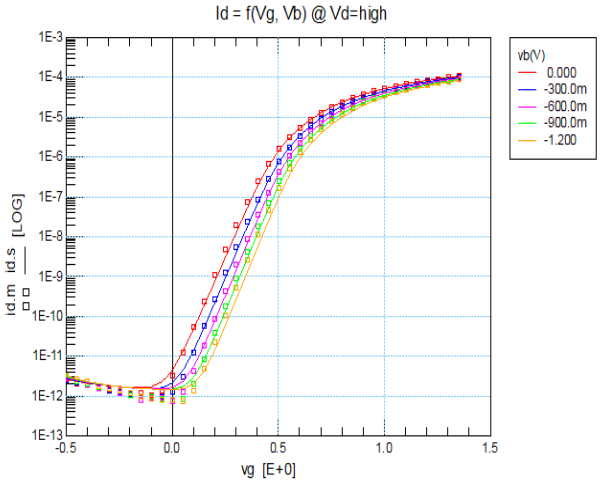


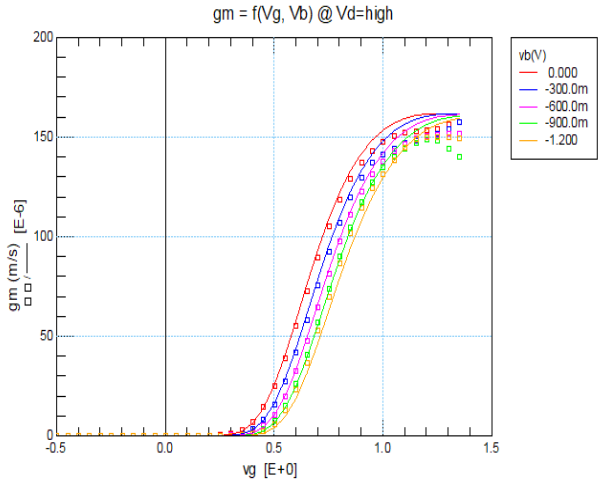


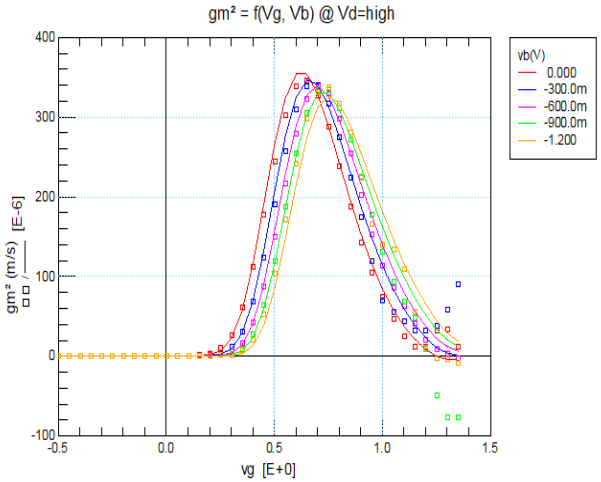


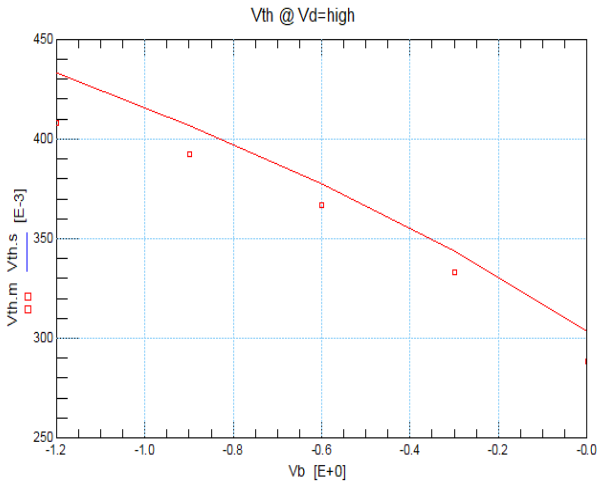
idvg, Vd = 1.2V, T = 27°C

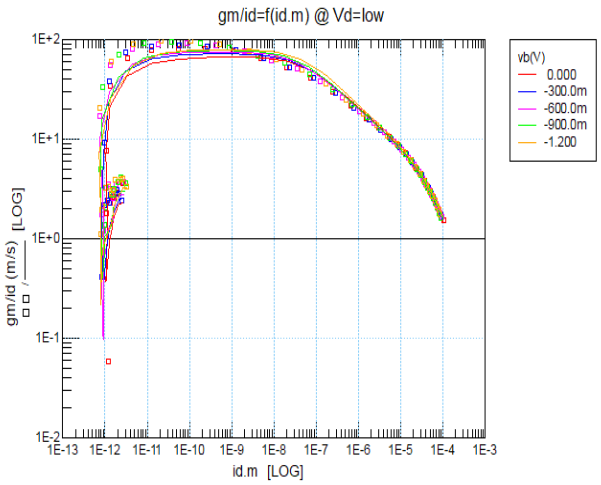


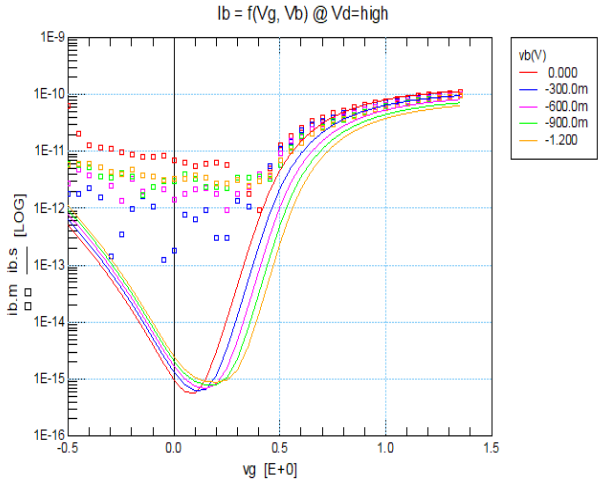


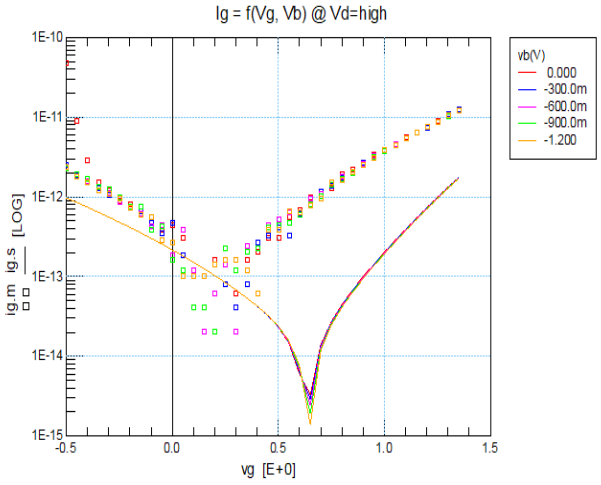




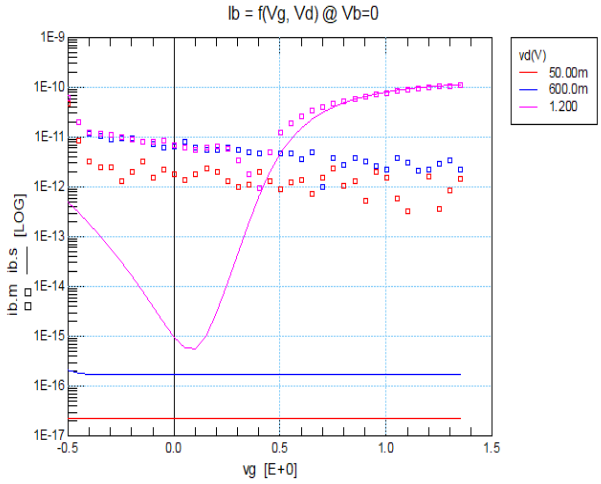


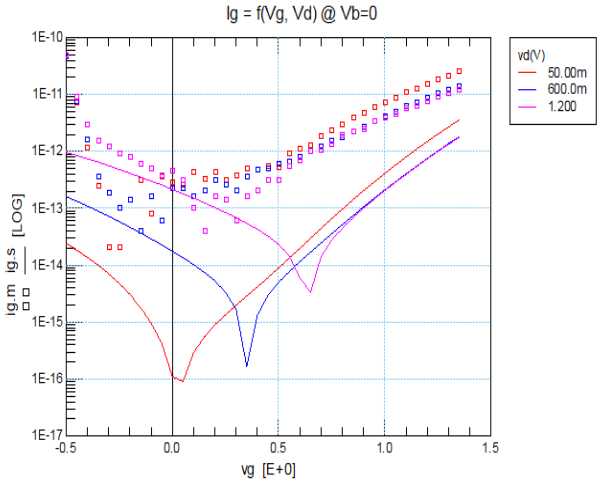




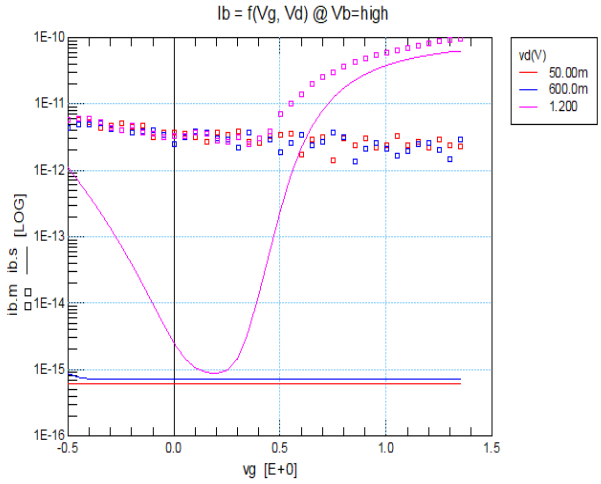


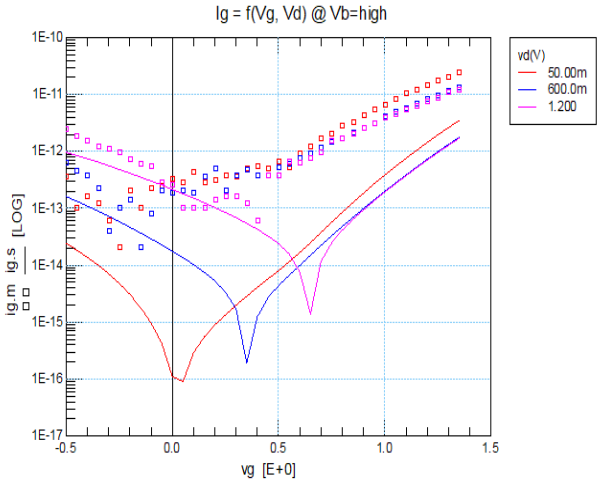
idvg, Ib,Ig, Vb = 0V, T = 27°C



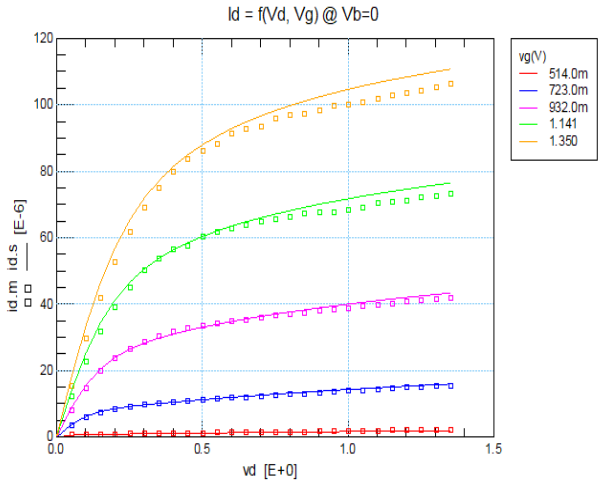


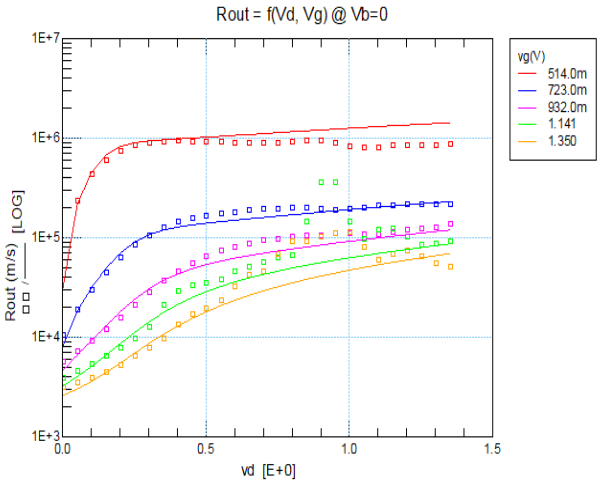
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

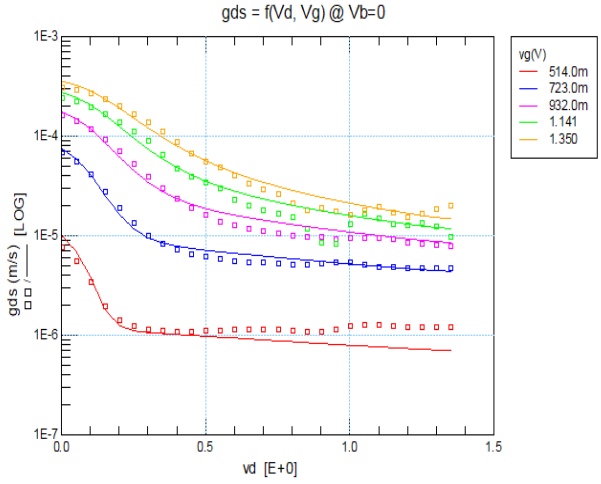




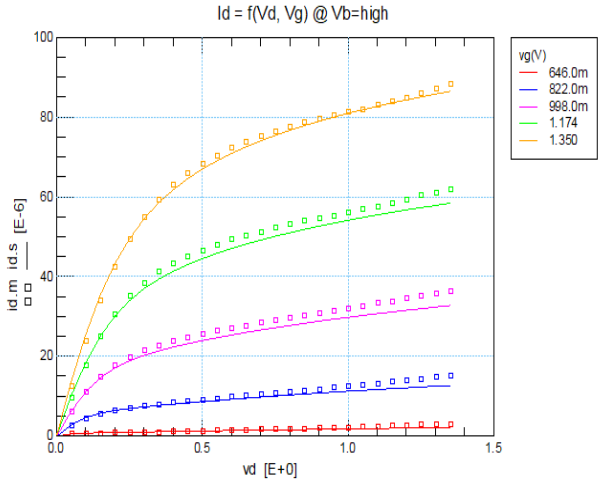
idvd, Vb = 0V, T = 27°C

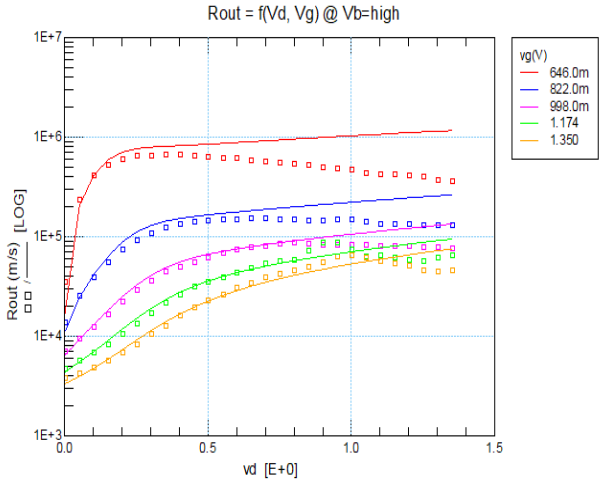


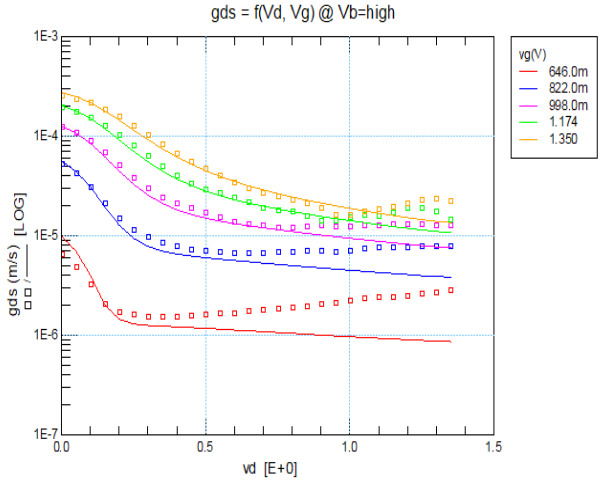




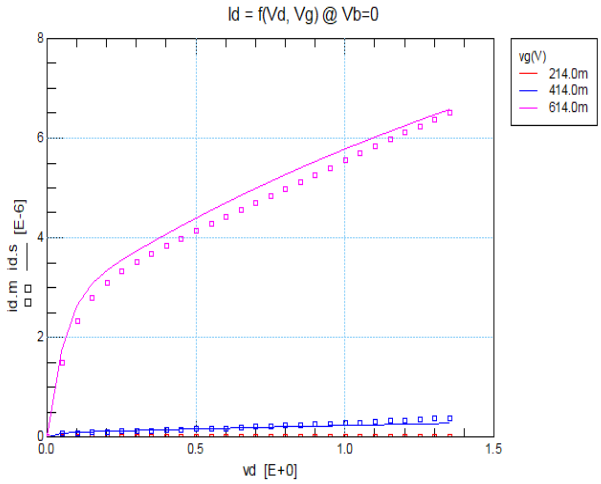
idvd_vbmin, Vb = -1.2V, T = 27°C

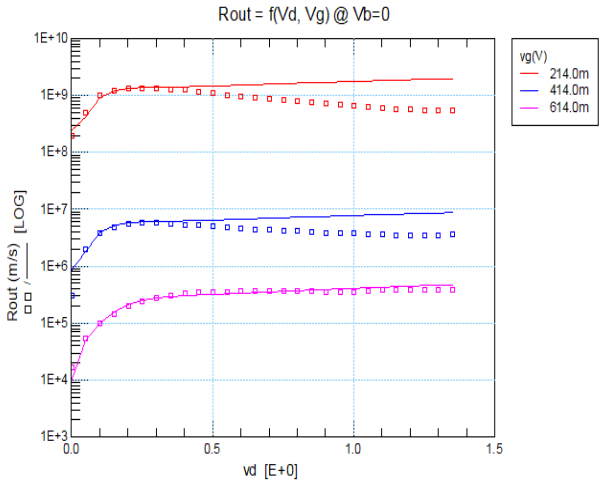


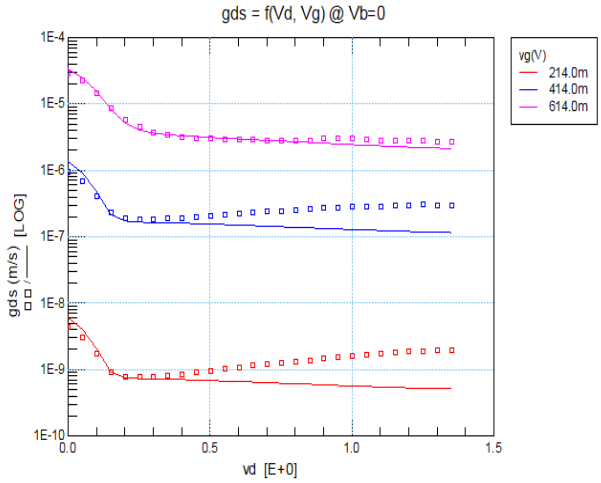




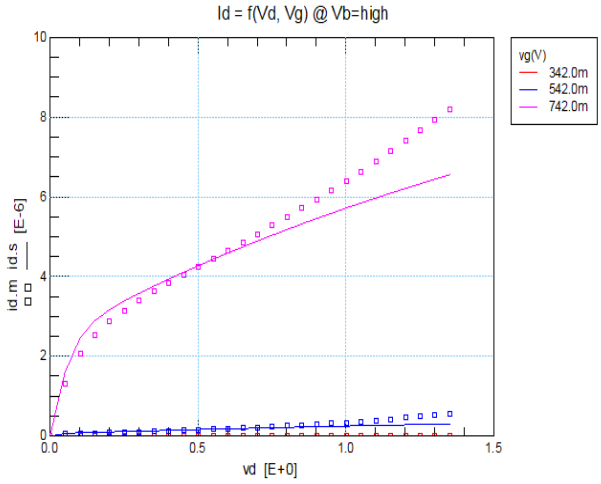
idvd_vth, Vb = 0V, T = 27°C

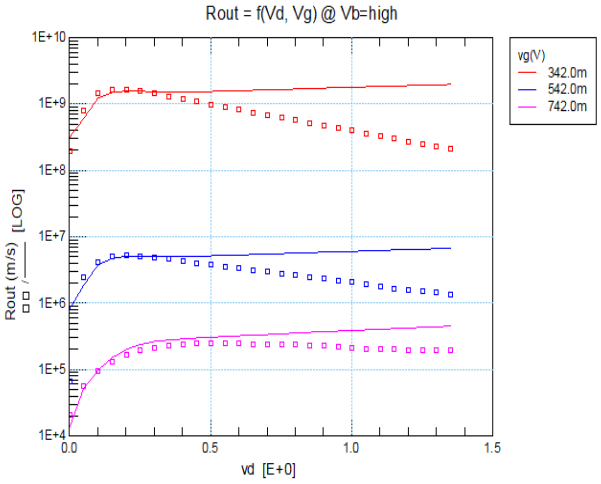


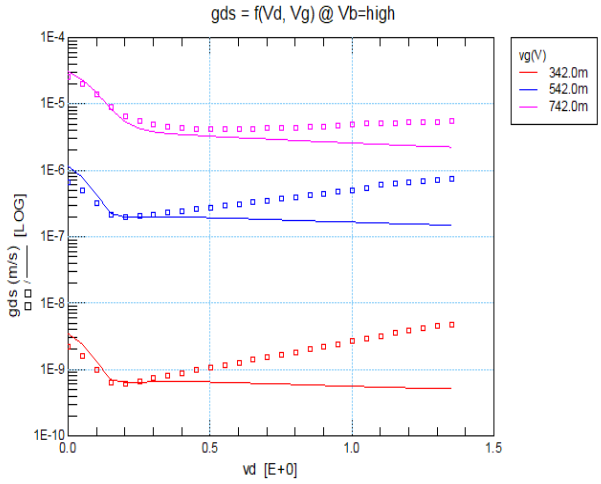




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

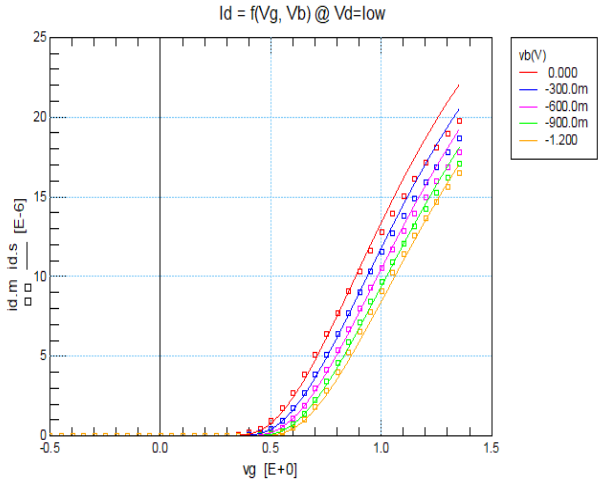


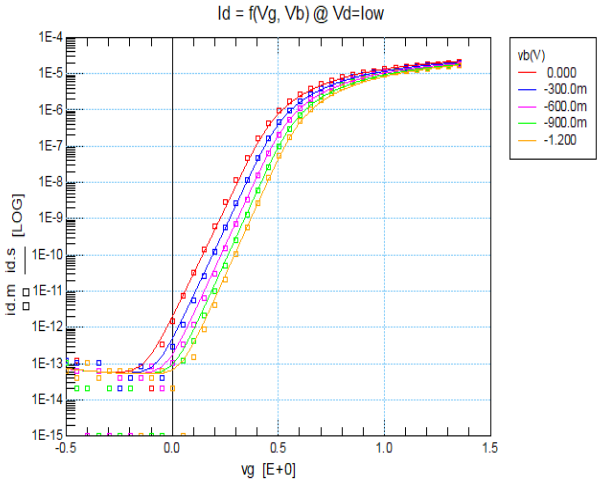


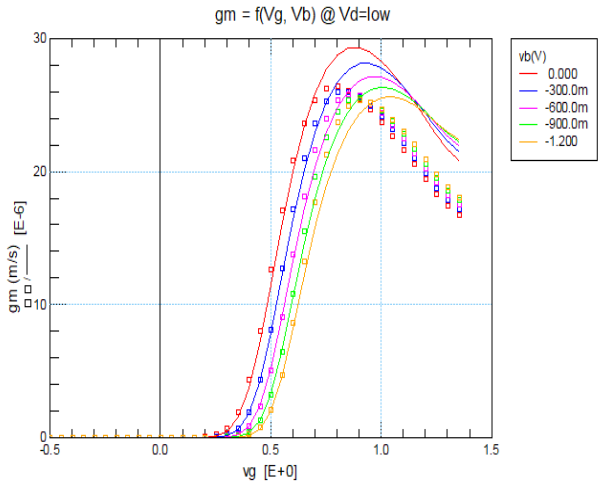


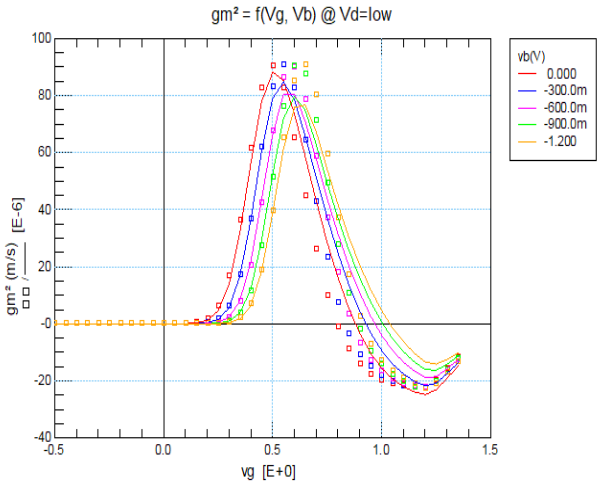
4.27W0u3_L0u15_S544_4

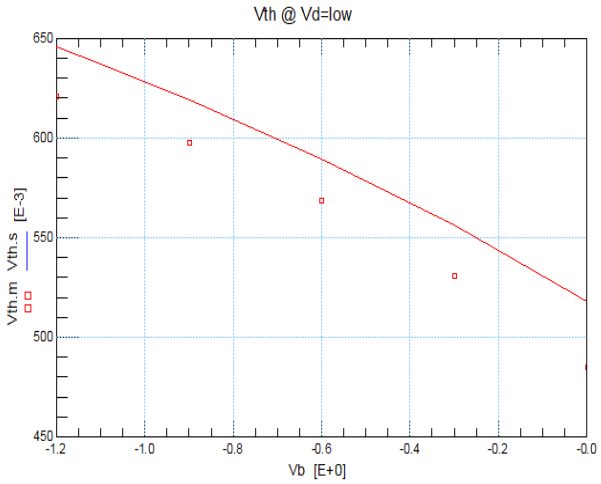
idvg, Vd = 0.05V, T = 27°C

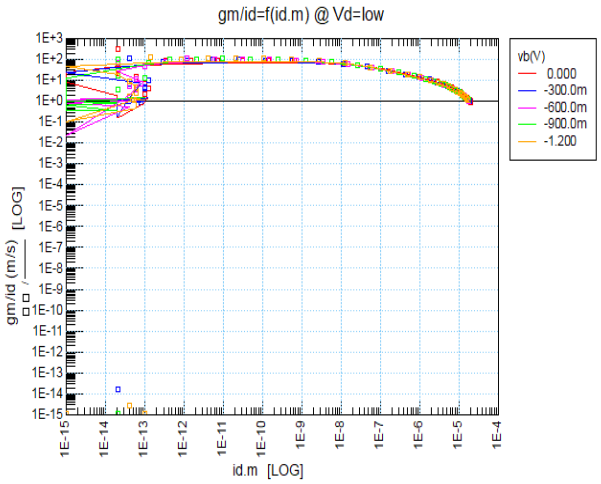


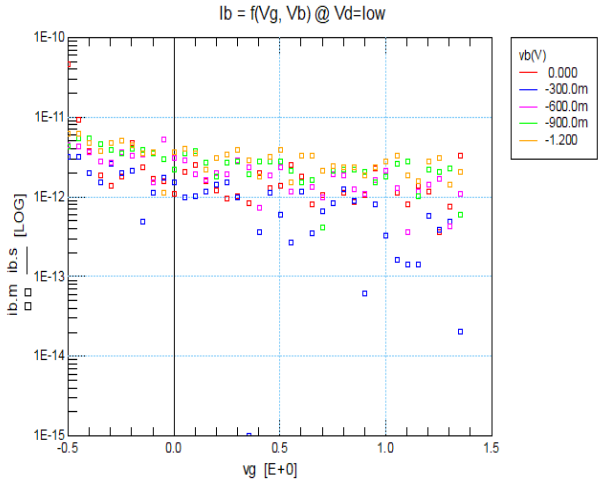


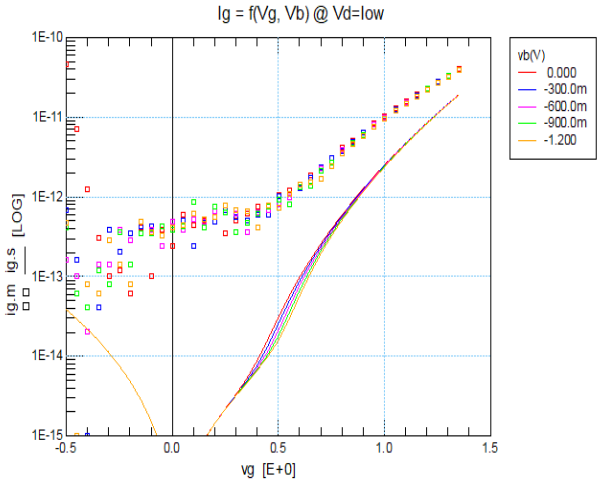




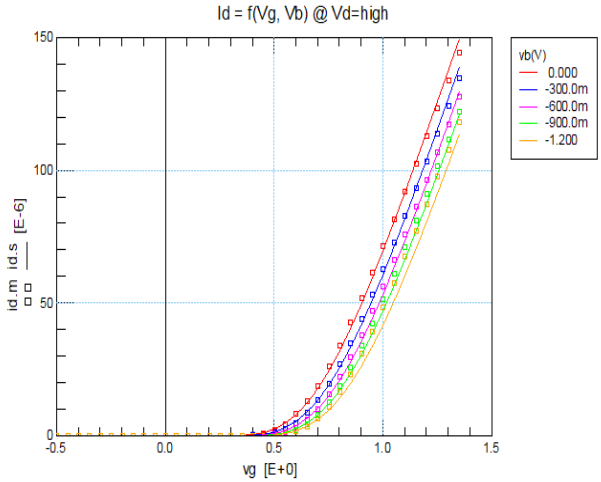


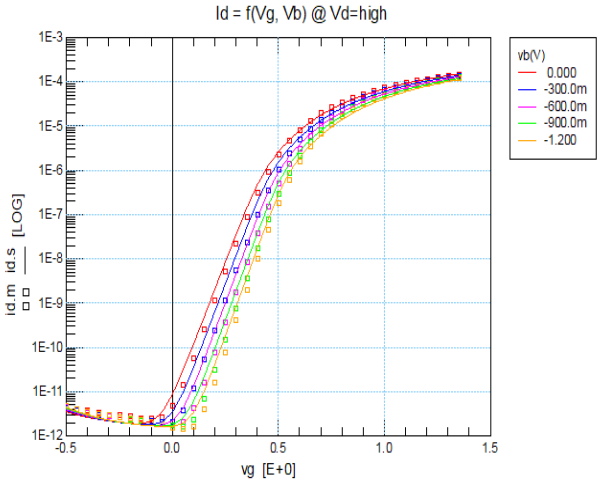


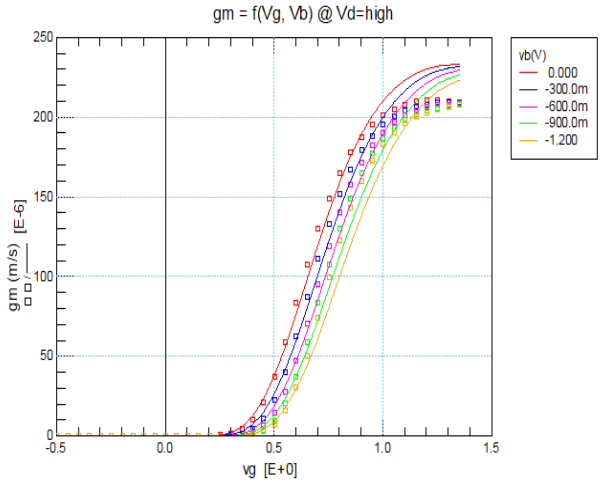


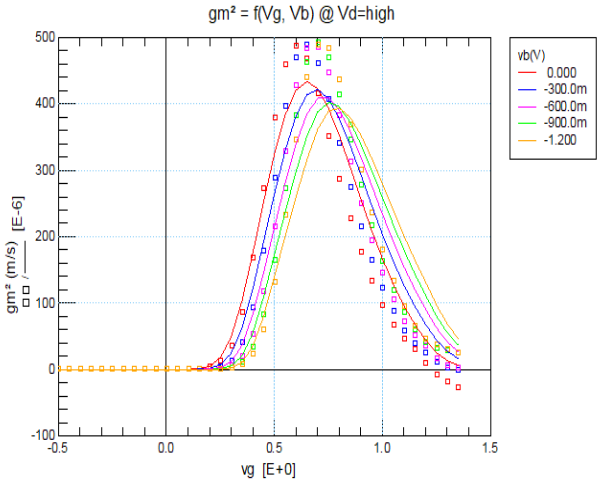


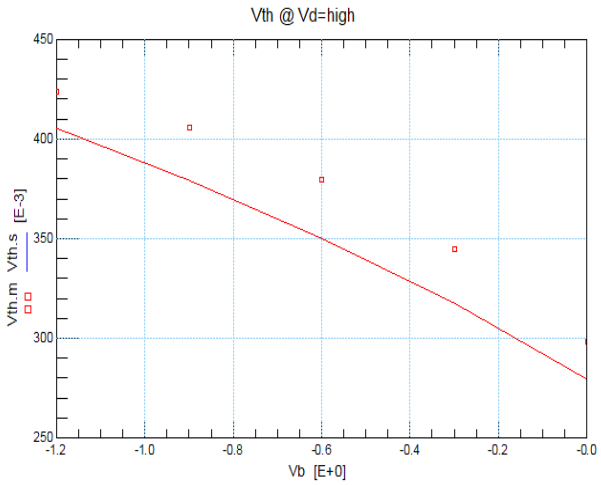
idvg, Vd = 1.2V, T = 27°C

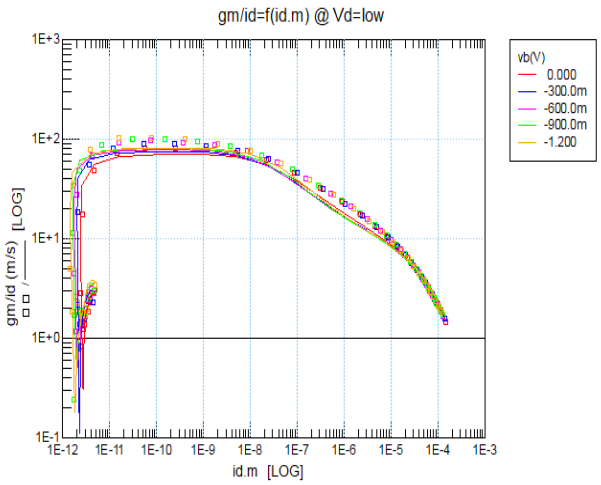


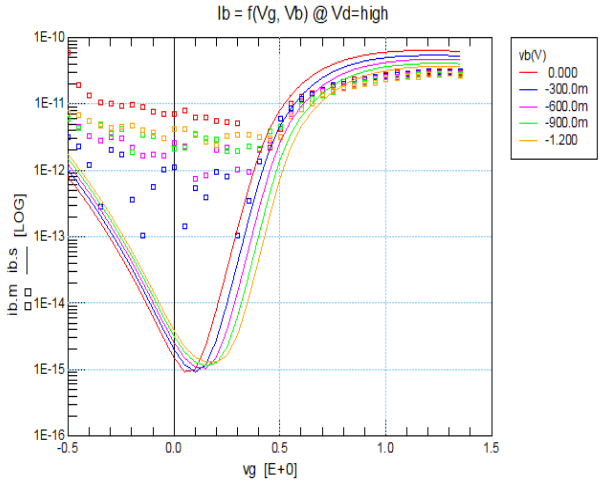


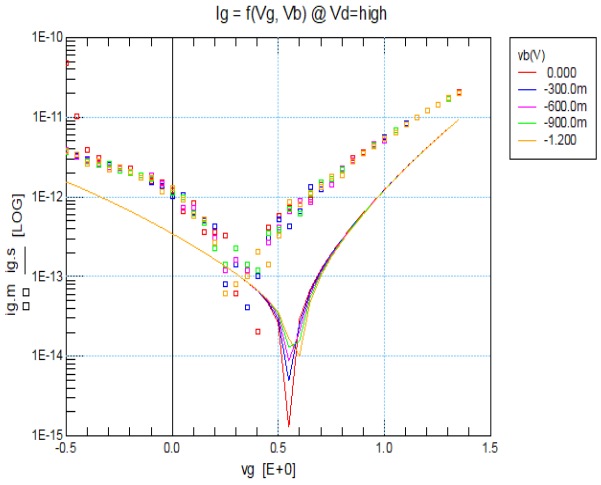




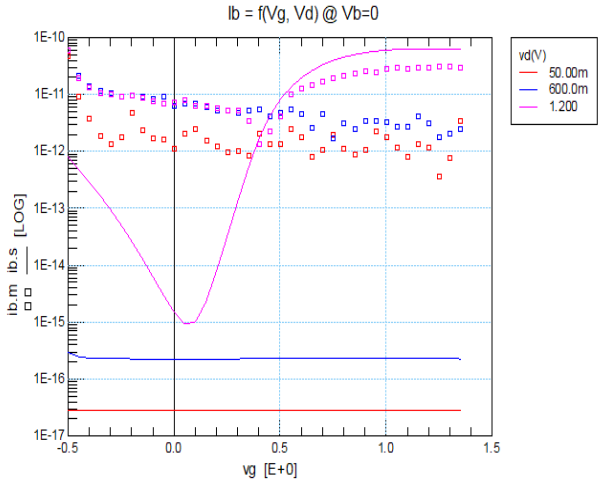


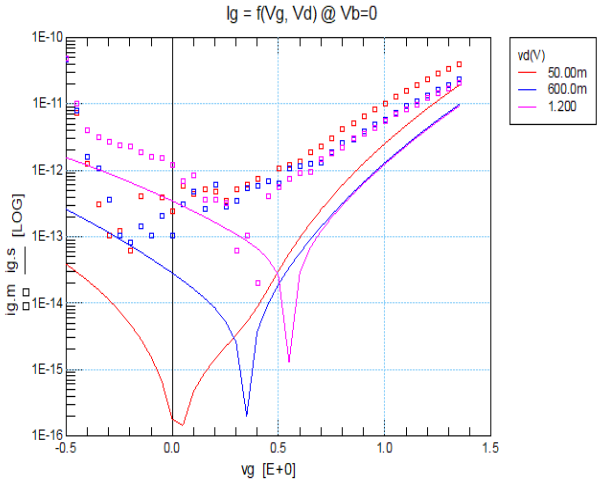




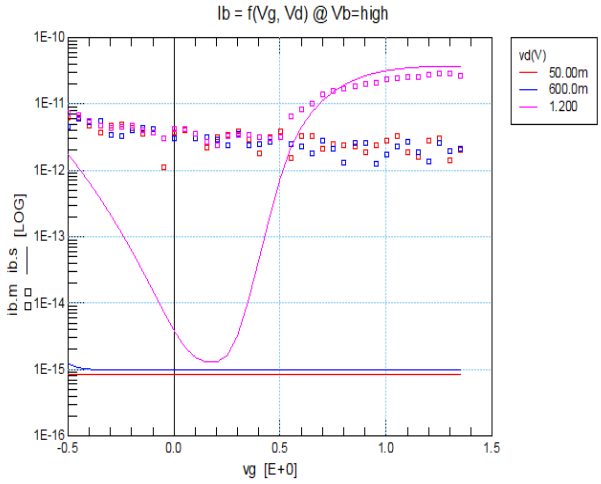


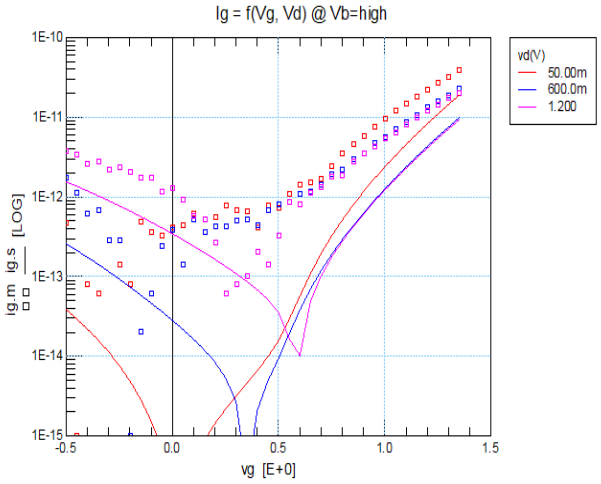
idvg, Ib,Ig, Vb = 0V, T = 27°C



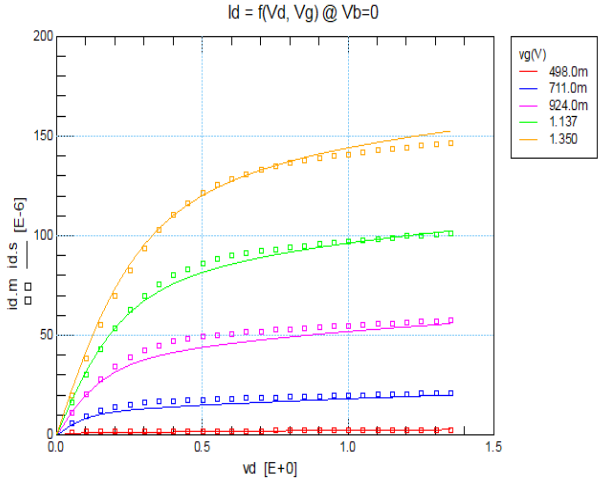


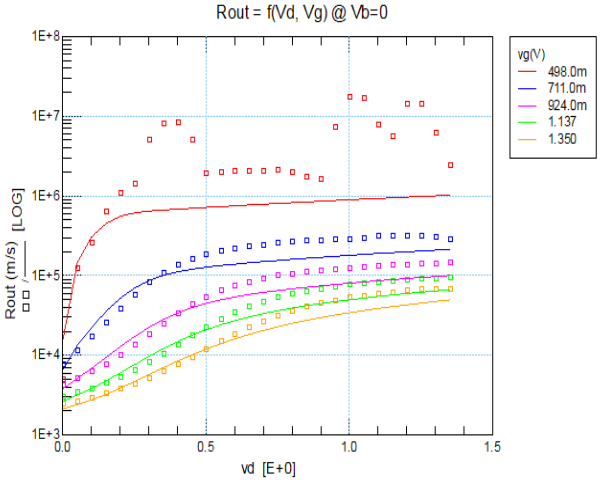
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

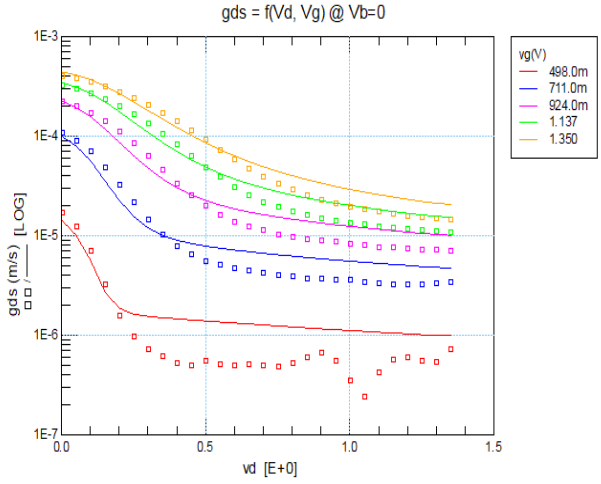




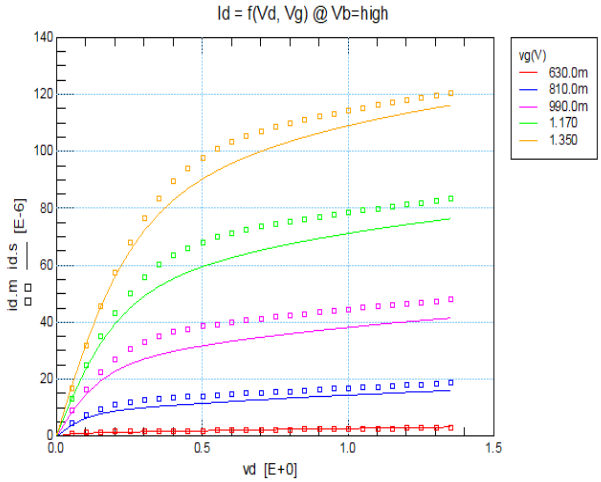
idvd, Vb = 0V, T = 27°C

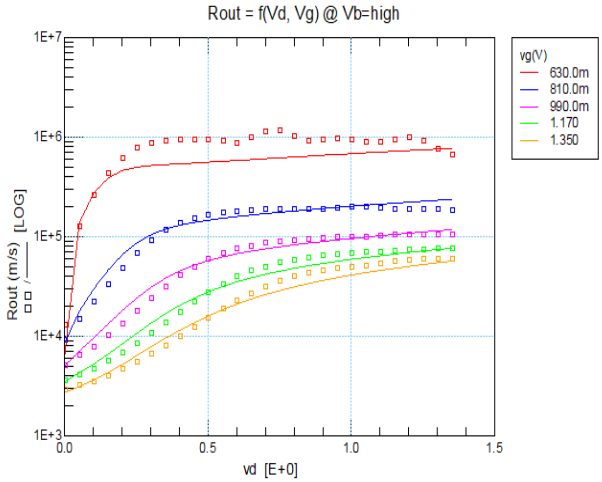


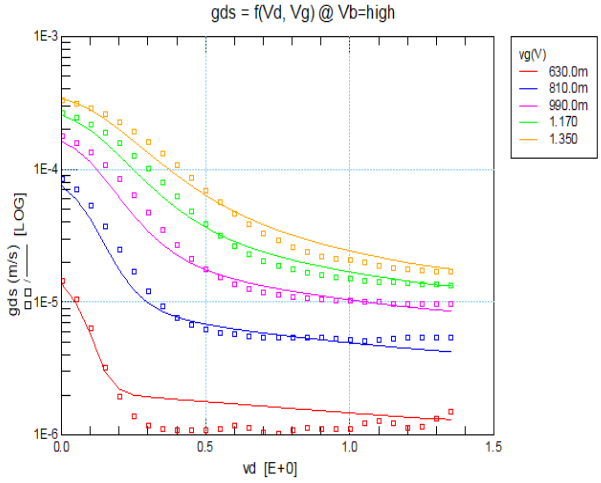




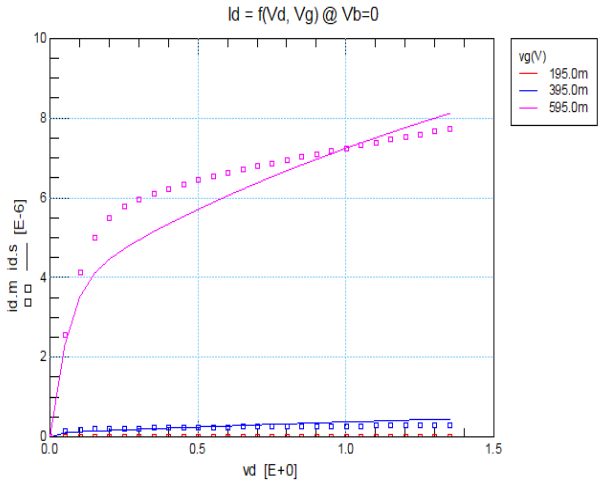
idvd_vbmin, Vb = -1.2V, T = 27°C

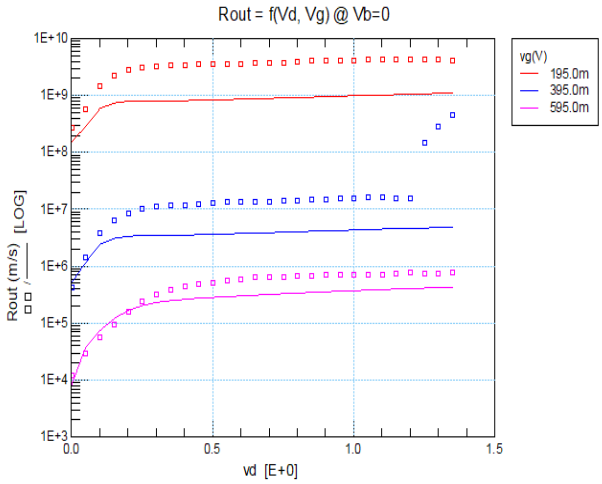


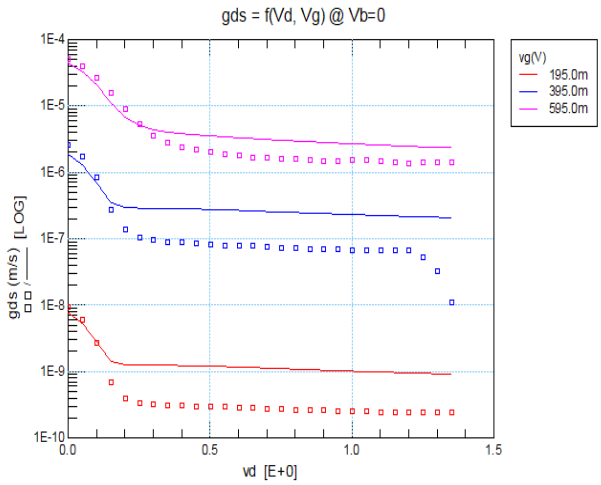




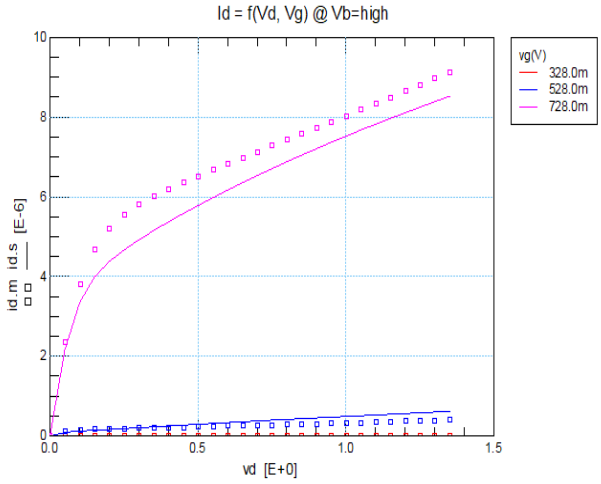
idvd_vth, Vb = 0V, T = 27°C

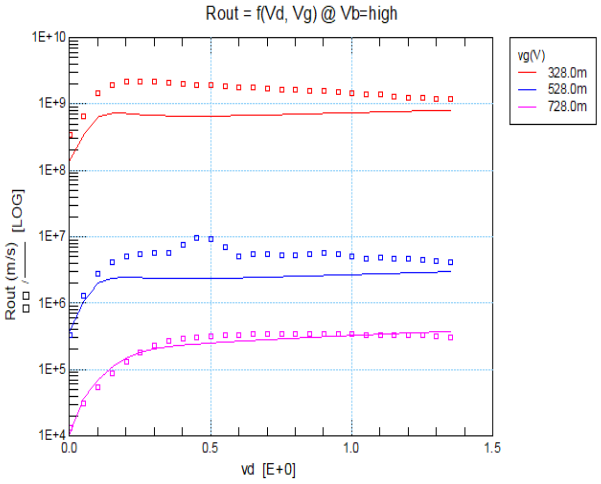


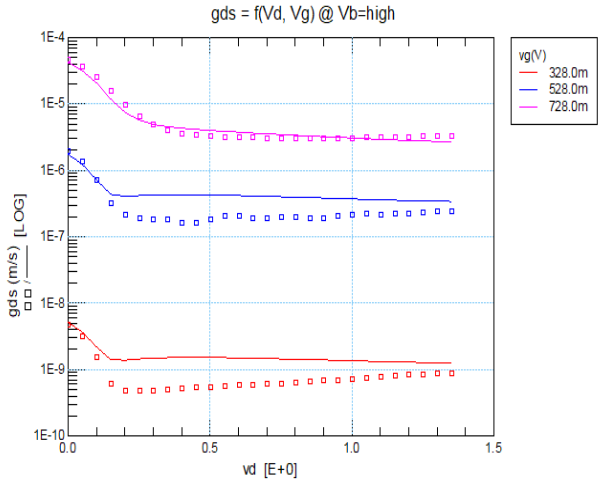




idvd_vth_vbmin, Vb = -1.2V, T = 27°C



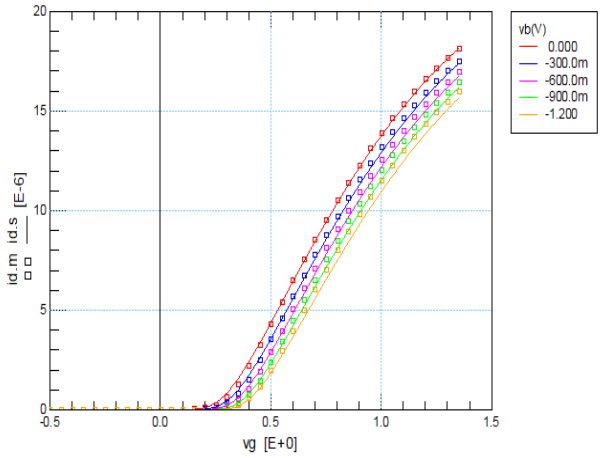




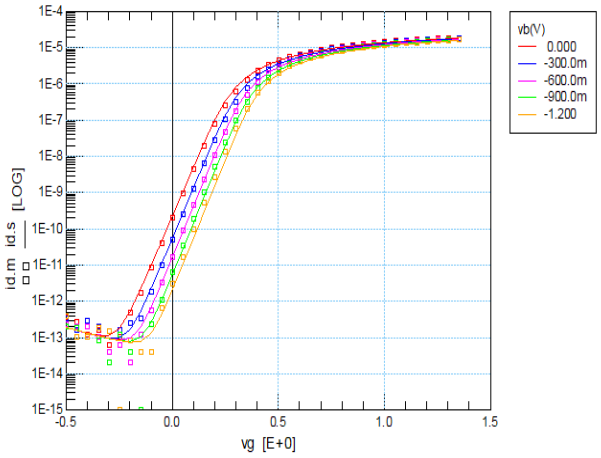
4.28W02u0_L2u0_S542_5

idvg, Vd = 0.05V, T = 27°C

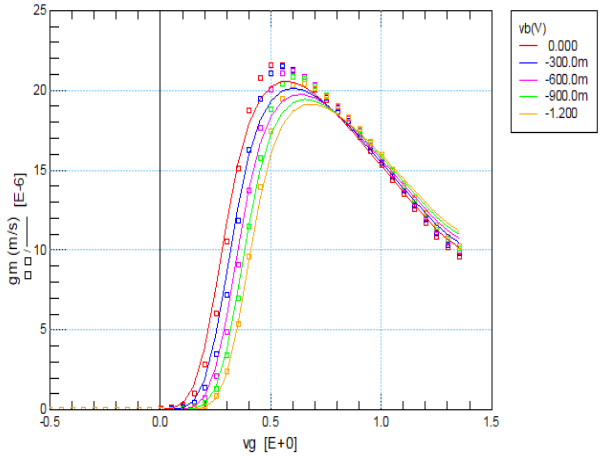
Id = f(Vg, Vb) @ Vd=low



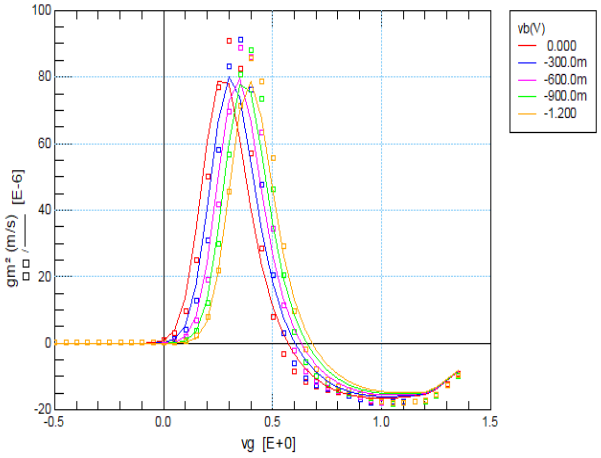
Id = f(Vg, Vb) @ Vd=low



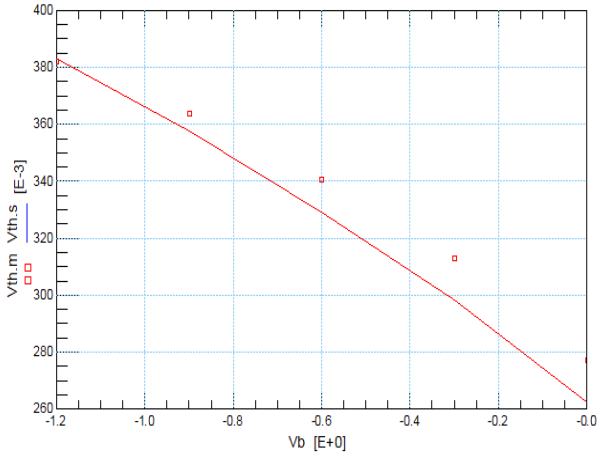
gm = f(Vg, Vb) @ Vd=low



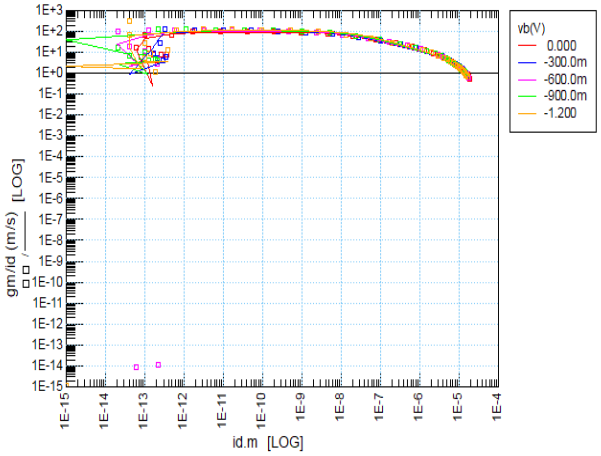
gm² = f(Vg, Vb) @ Vd=low

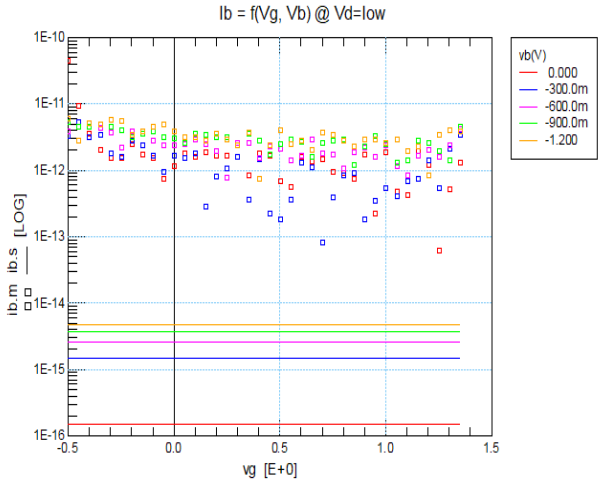


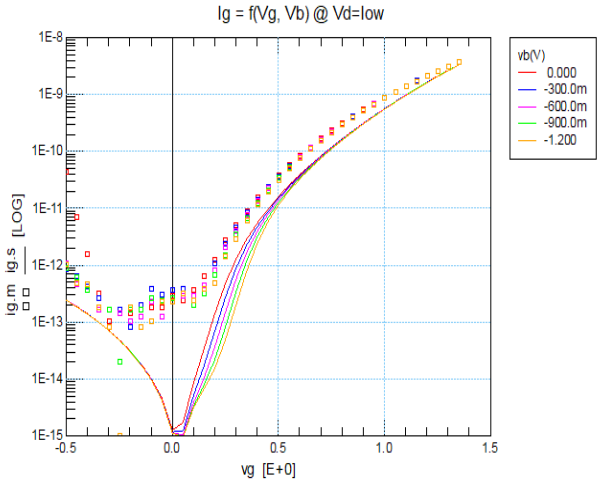
Vth @ Vd=low



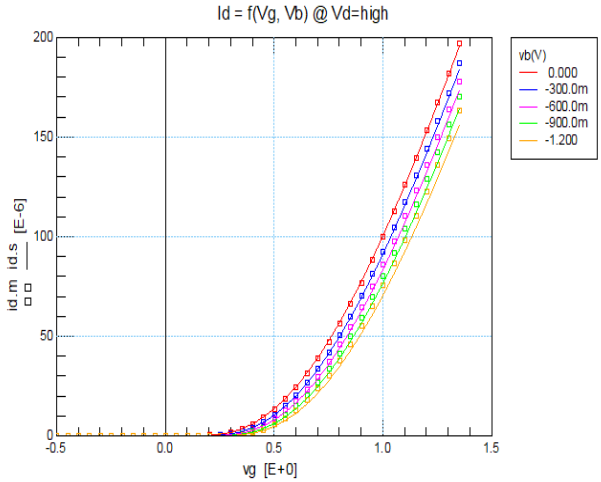
gm/id=f(id.m) @ Vd=low

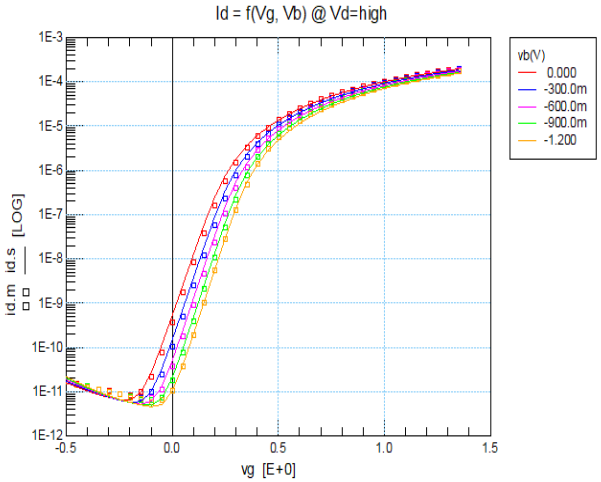


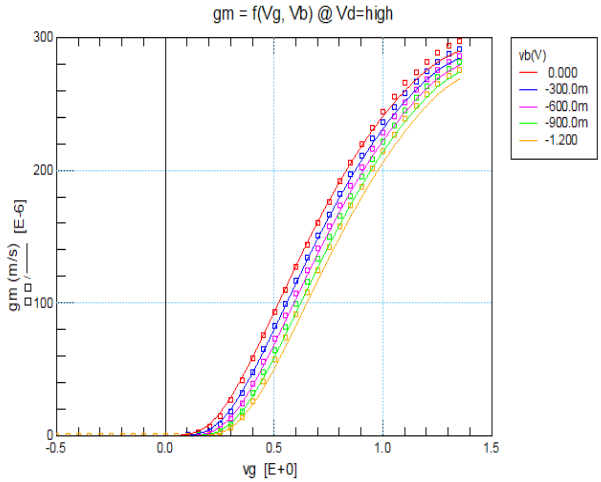


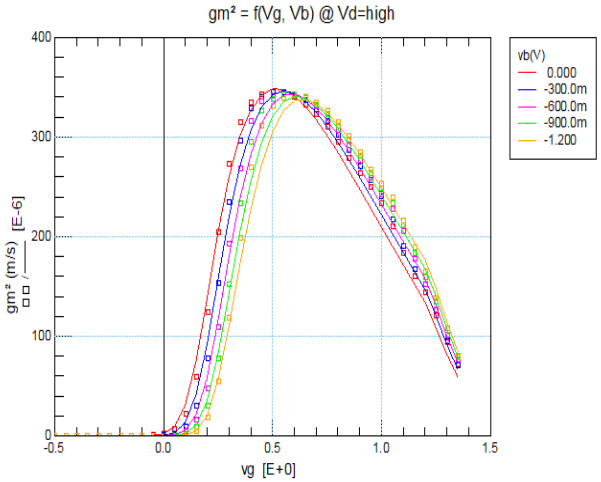


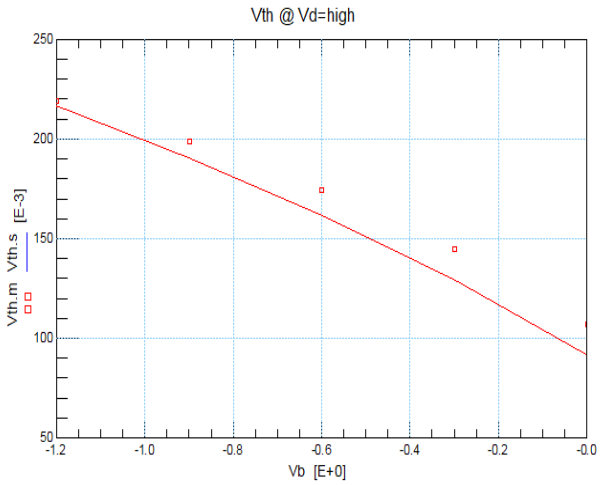
idvg, Vd = 1.2V, T = 27°C

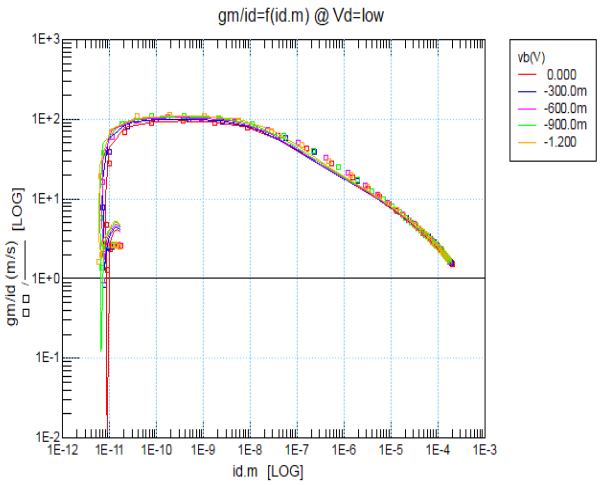


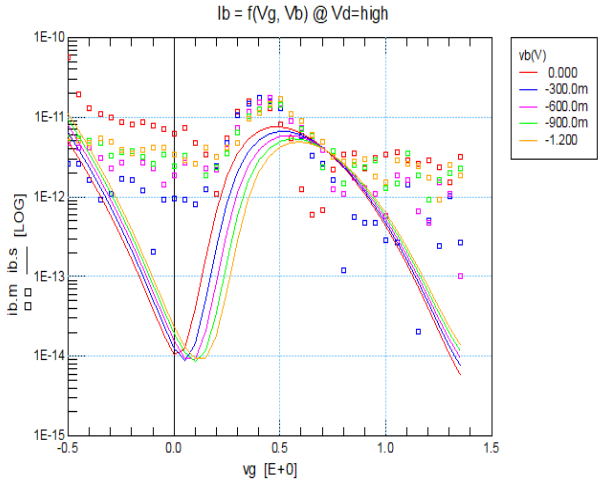


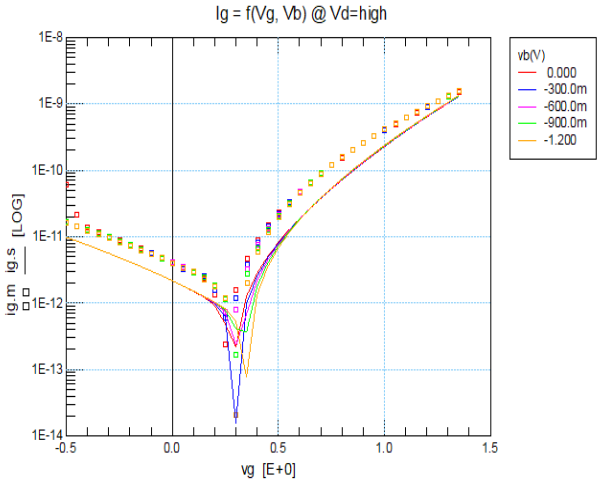




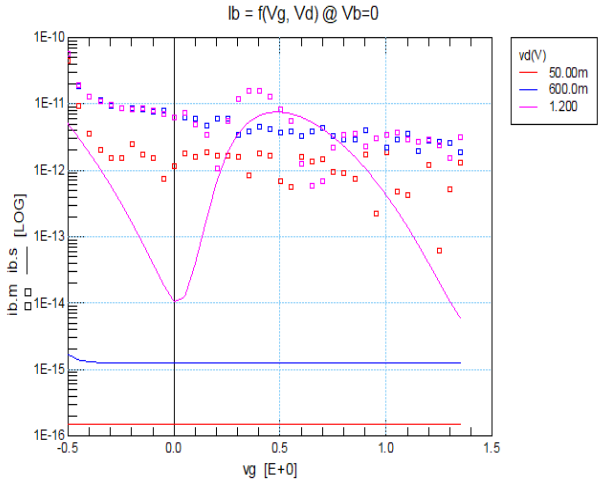


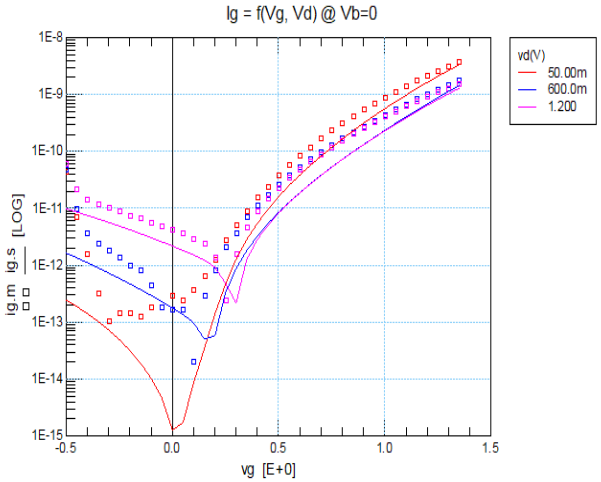




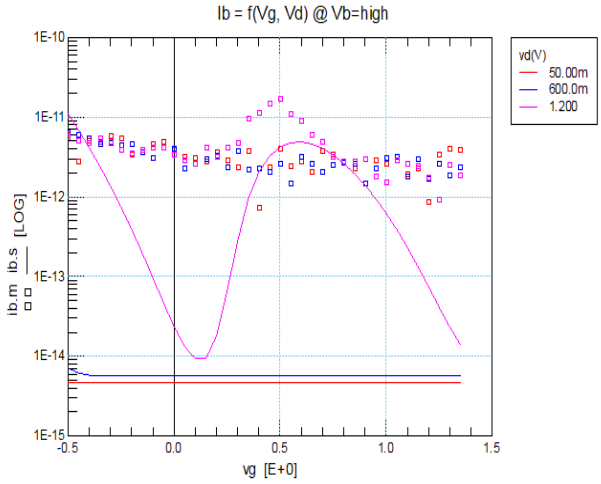


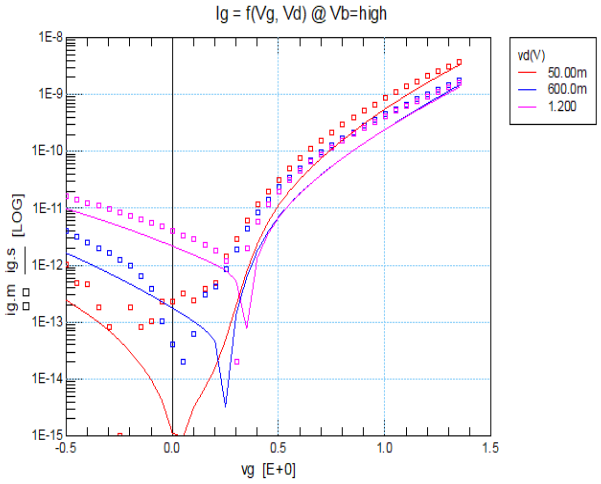
idvg, Ib,Ig, Vb = 0V, T = 27°C



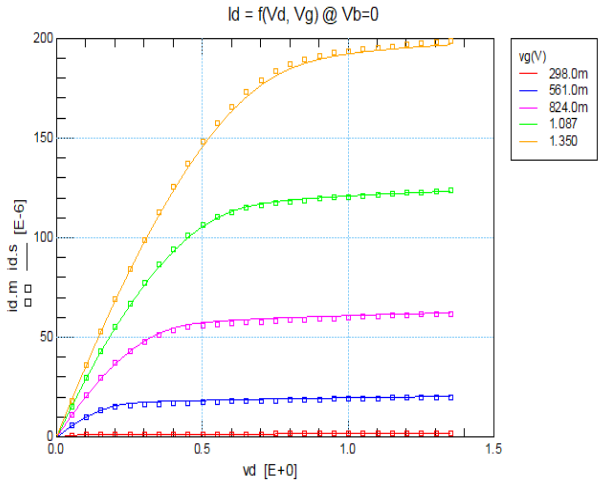


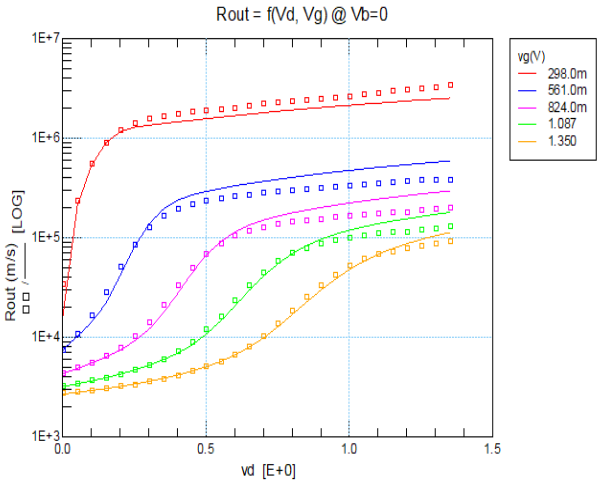
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

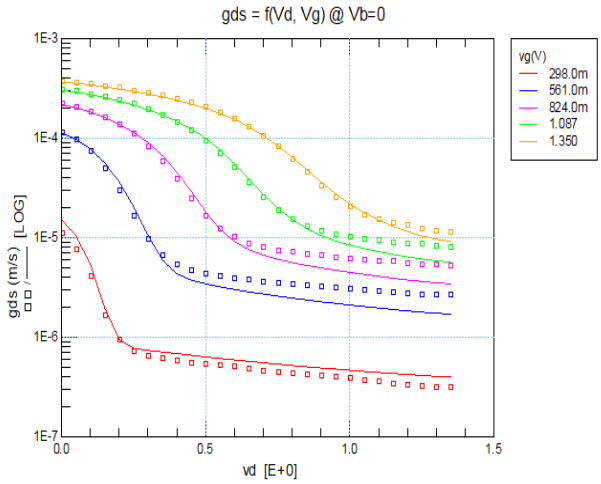




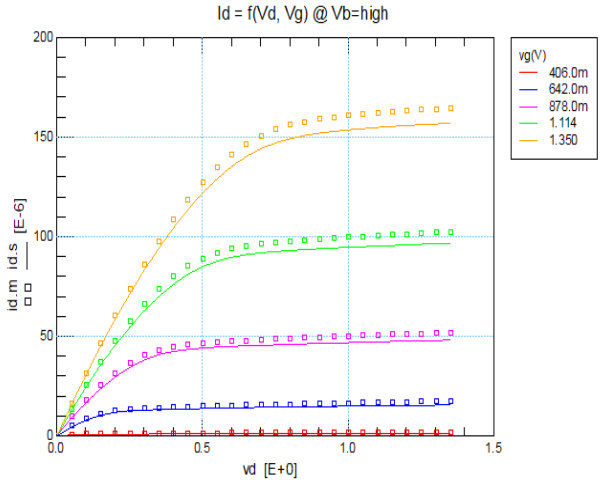
idvd, Vb = 0V, T = 27°C

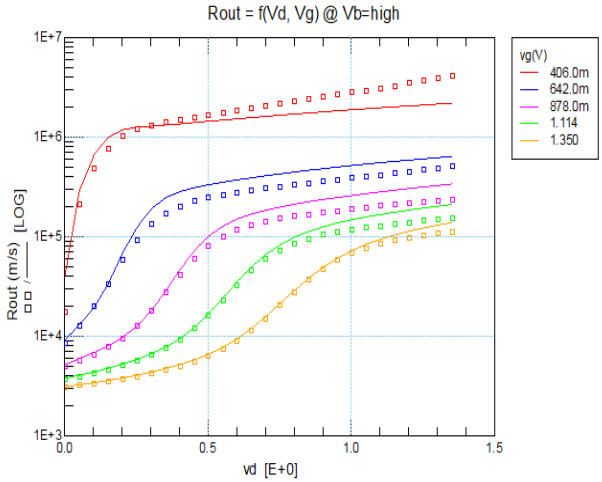


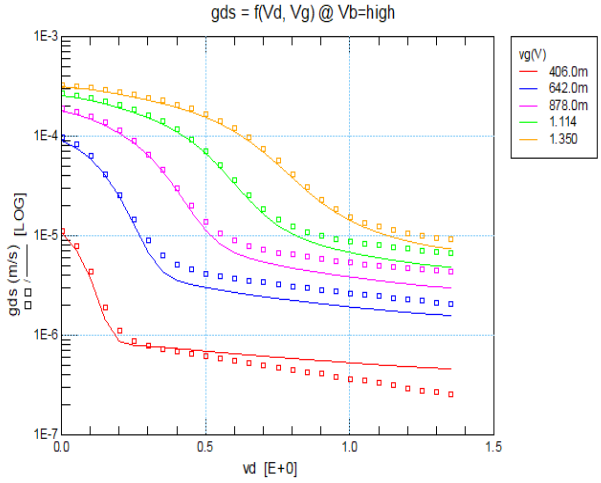




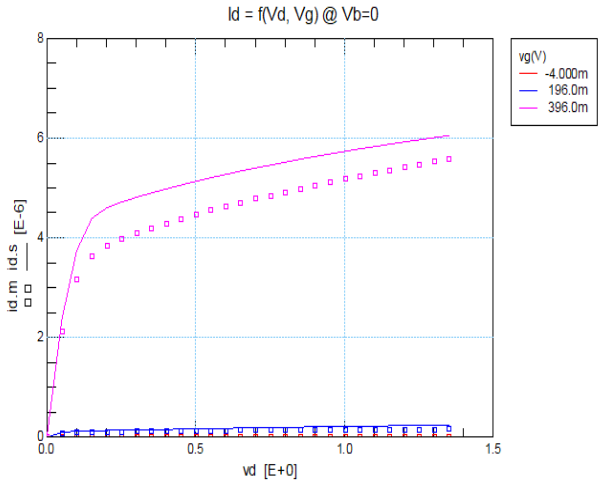
idvd_vbmin, Vb = -1.2V, T = 27°C

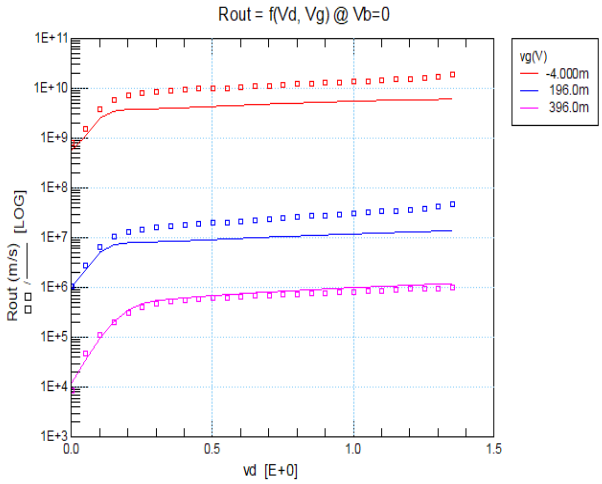


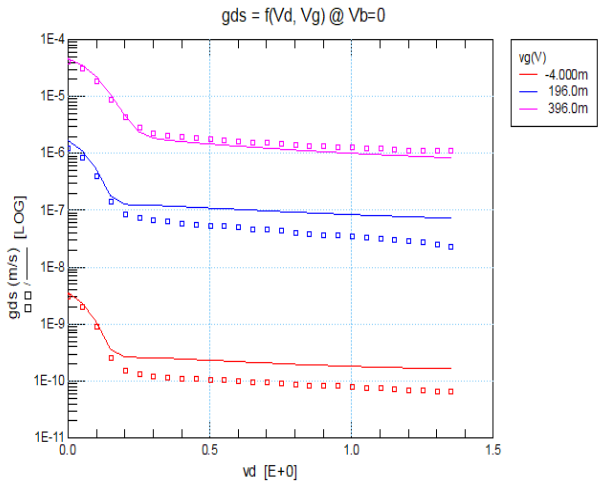




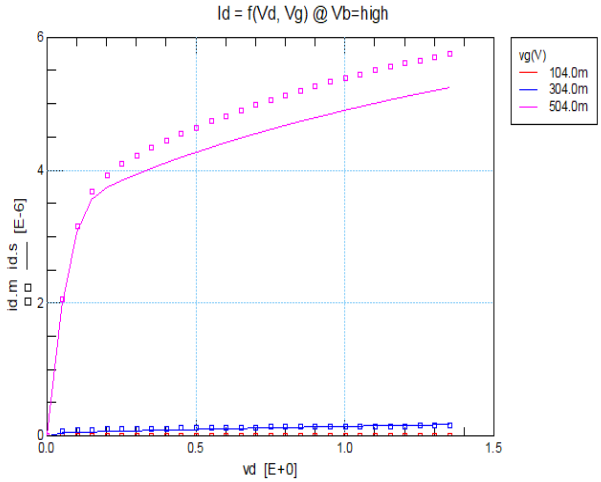
idvd_vth, Vb = 0V, T = 27°C

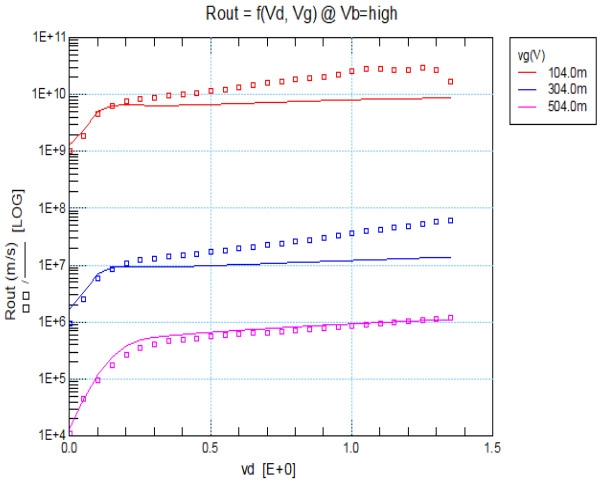


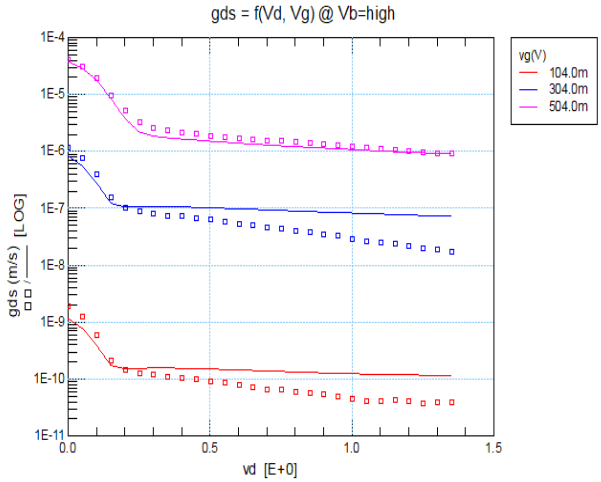




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

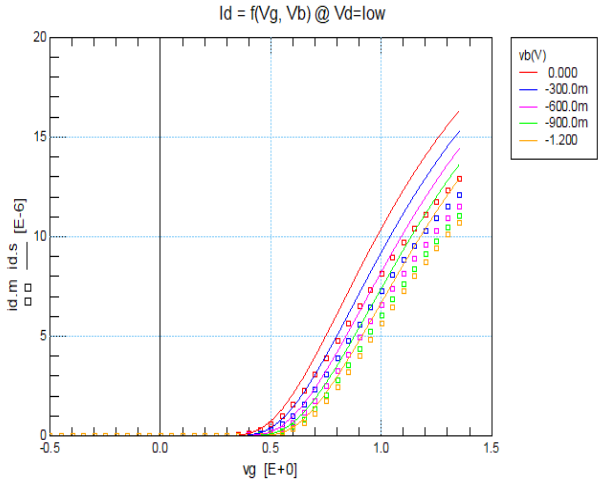


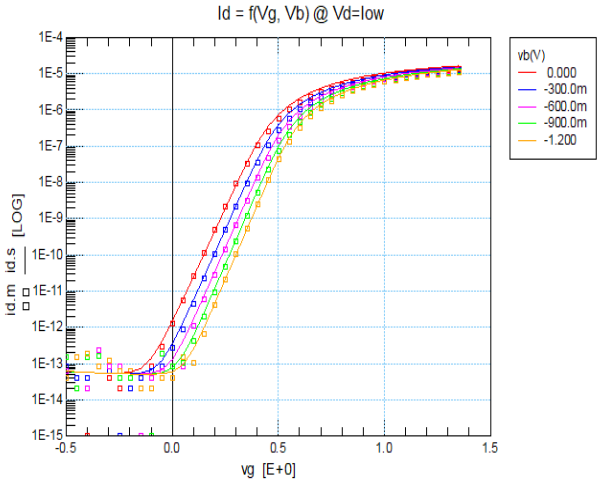


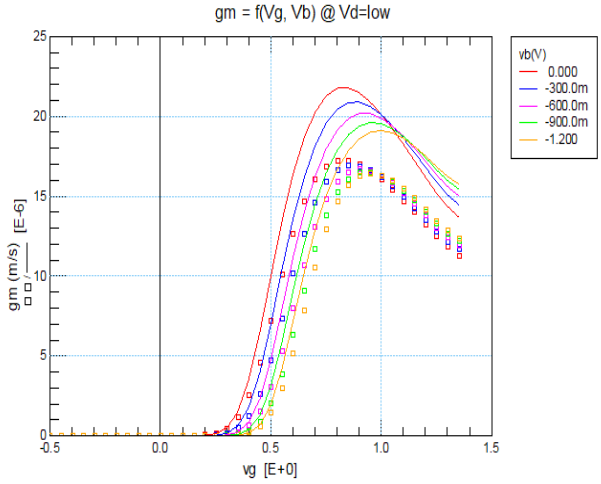


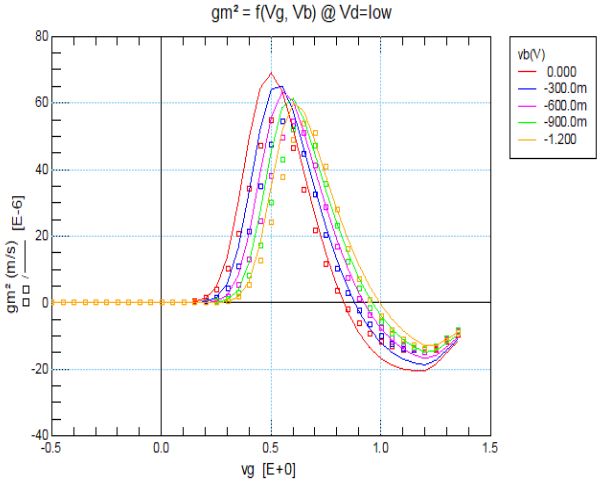
4.29W0u15_L0u15_S546_5

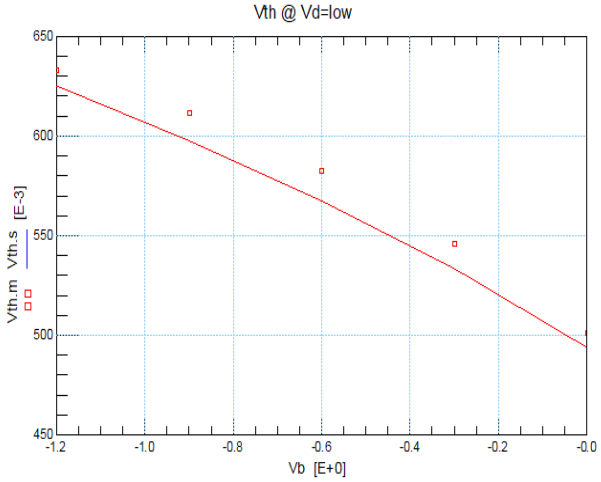
idvg, Vd = 0.05V, T = 27°C

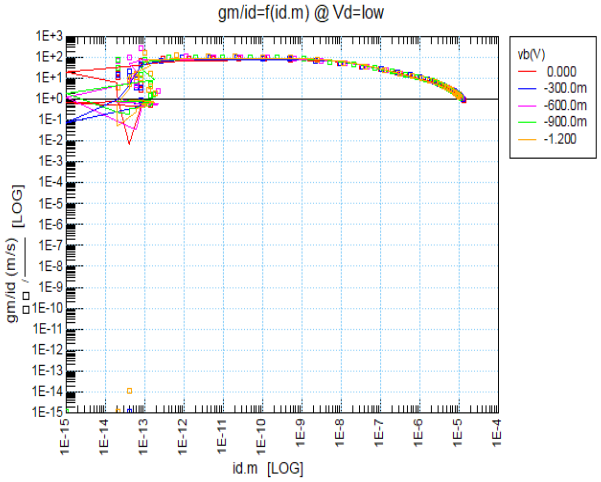


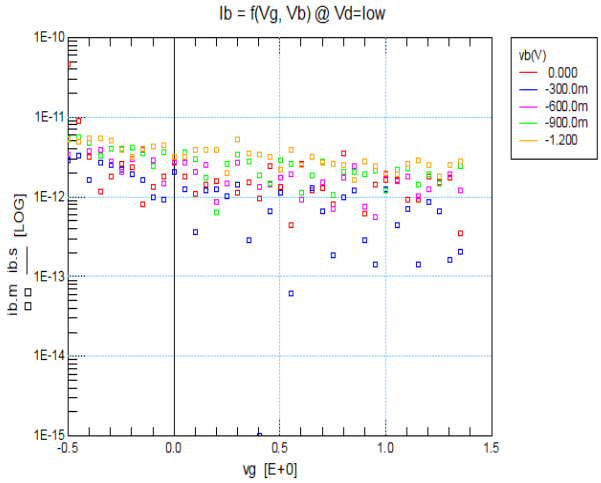


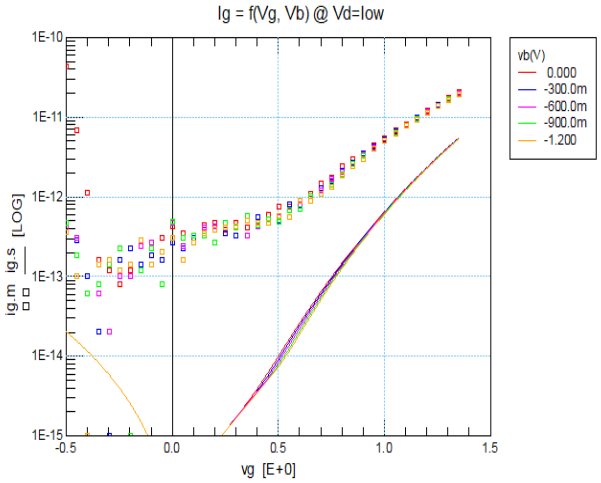




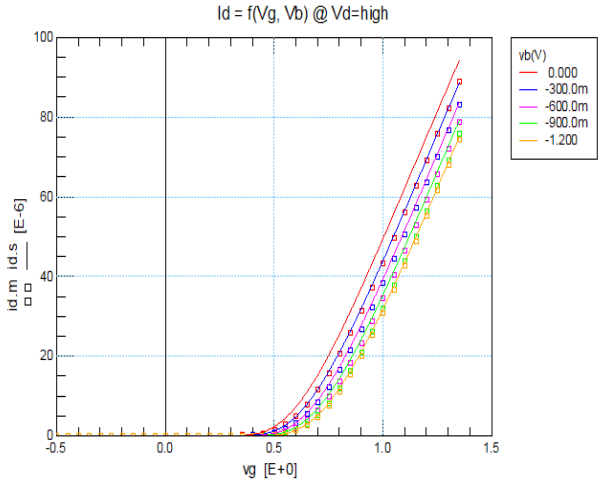


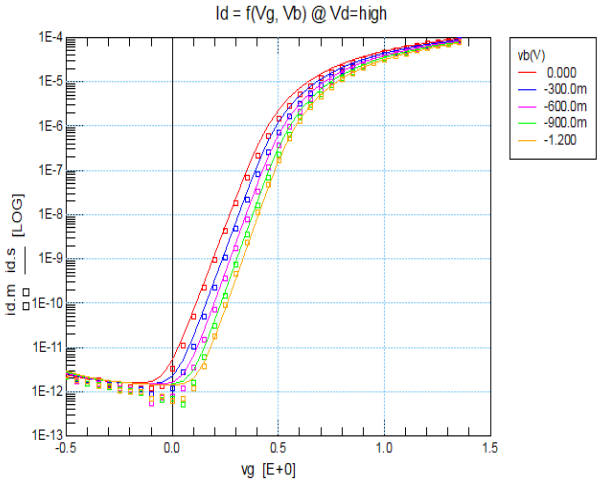


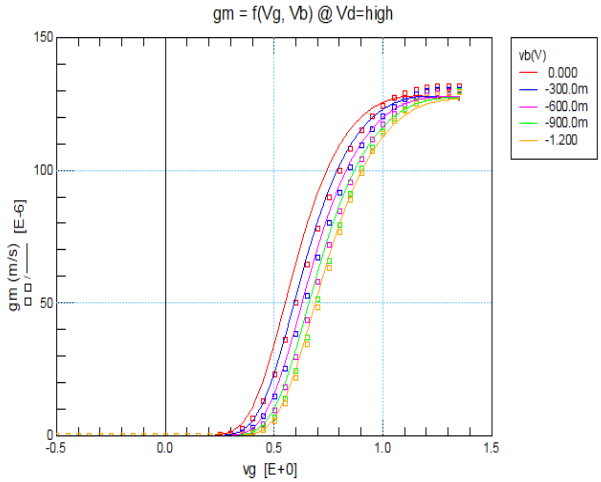


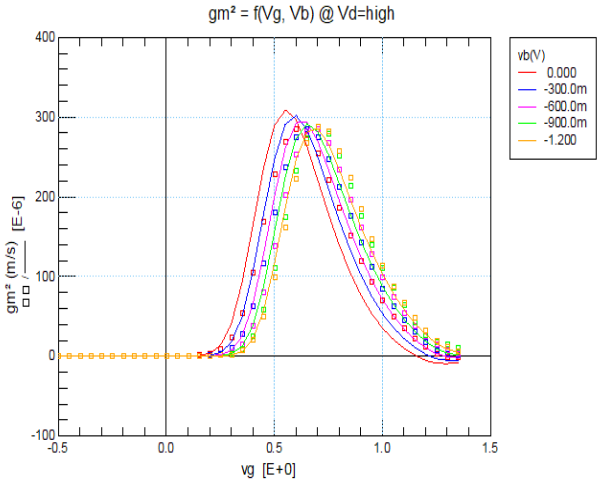


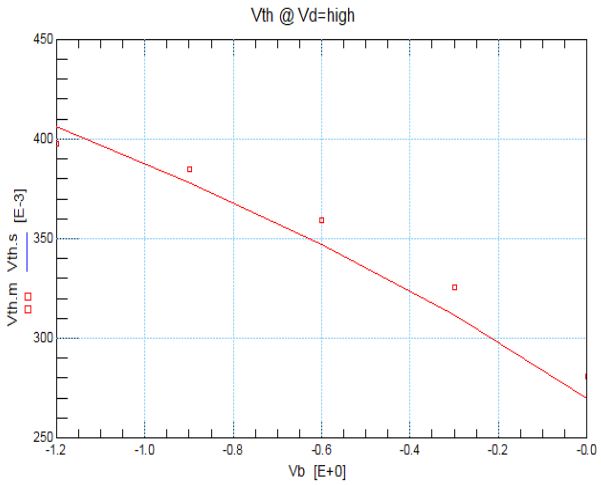
idvg, Vd = 1.2V, T = 27°C

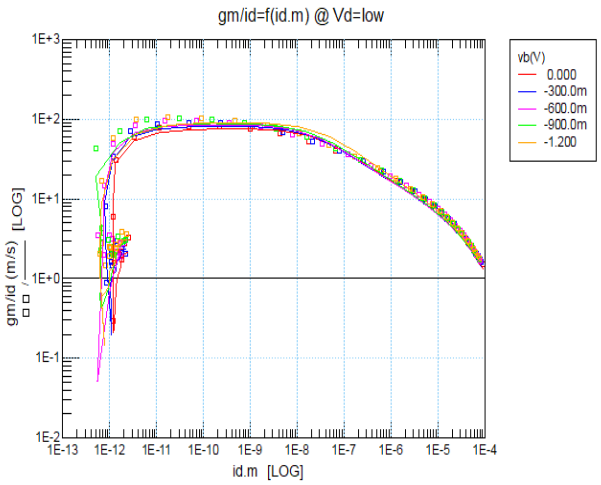


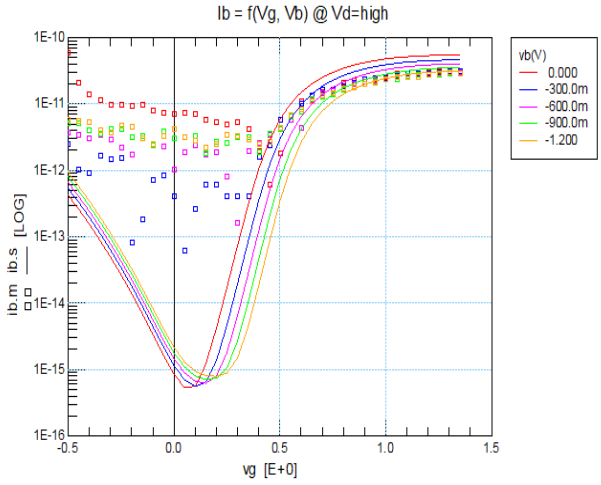


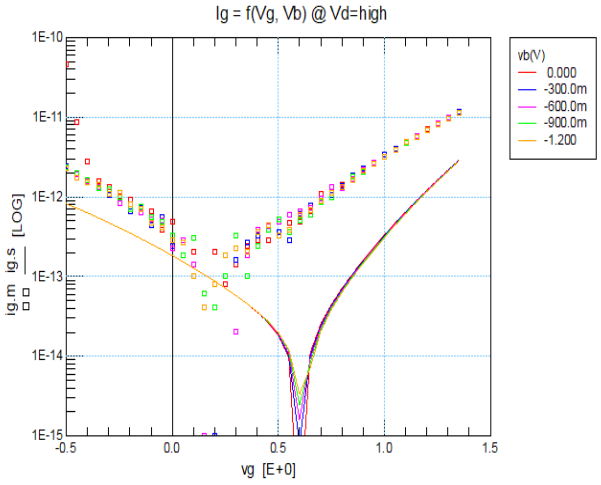




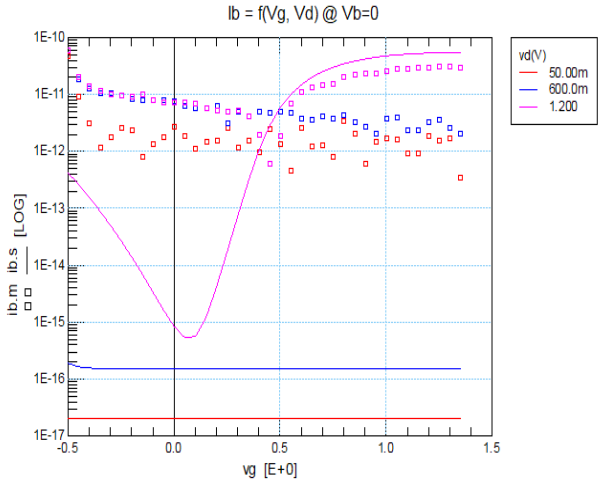


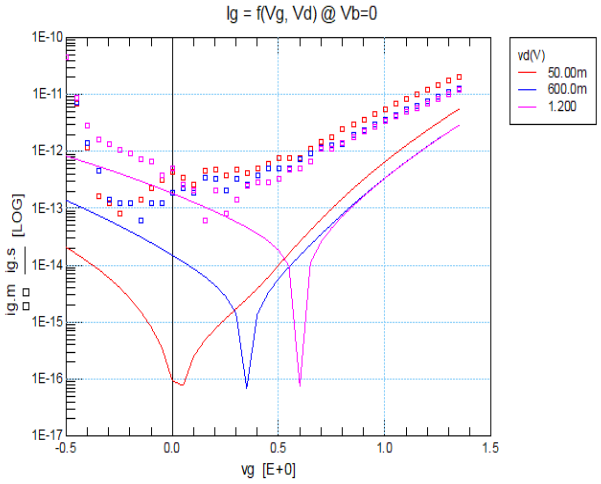




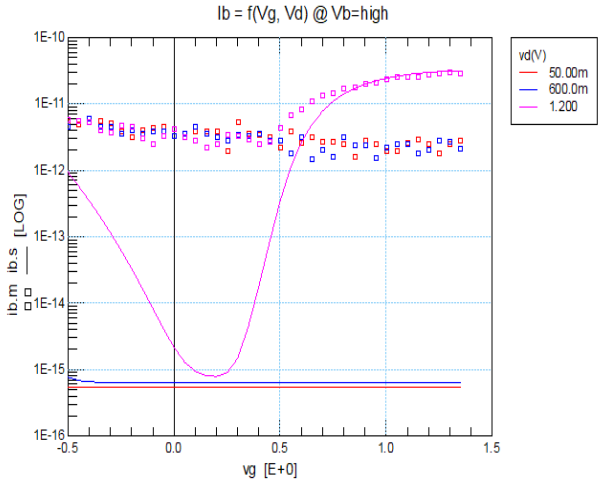


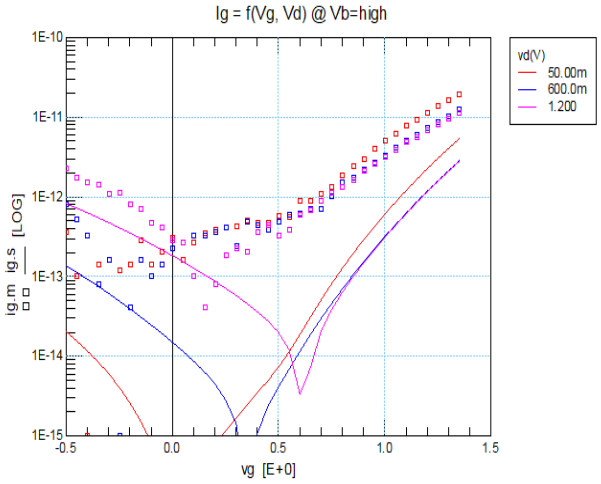
idvg, Ib,Ig, Vb = 0V, T = 27°C



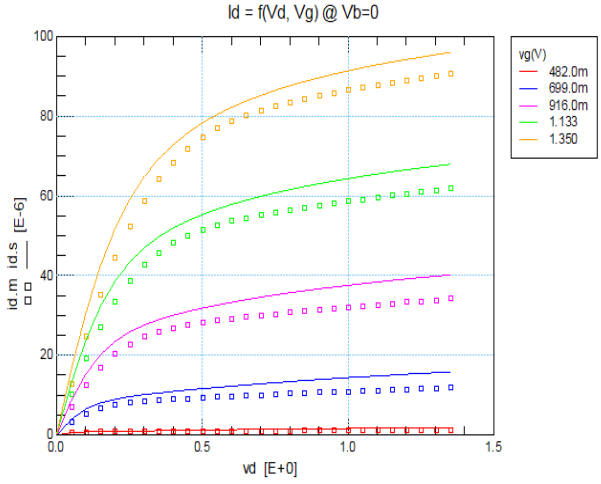


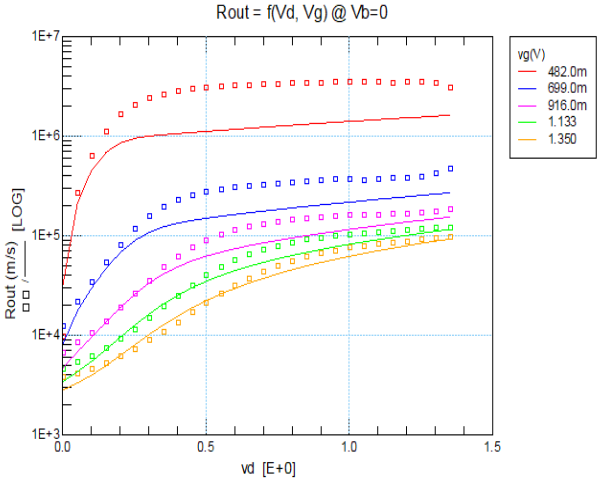
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

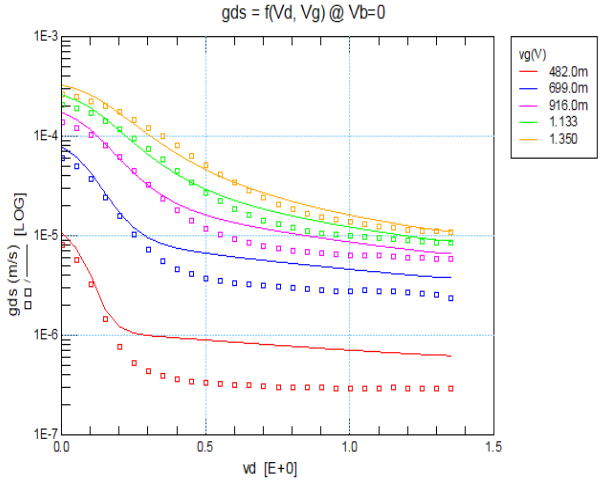




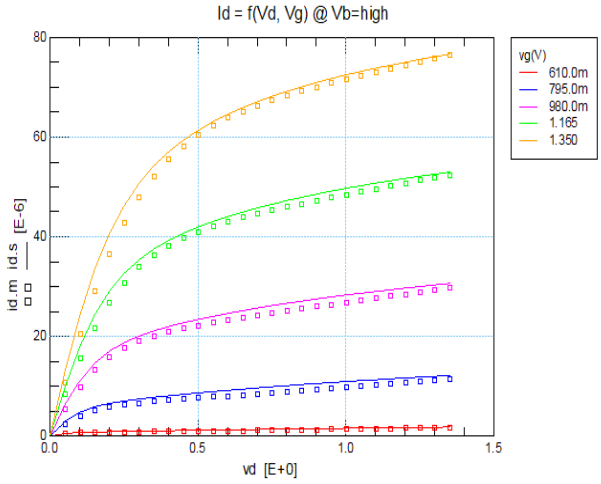
idvd, Vb = 0V, T = 27°C

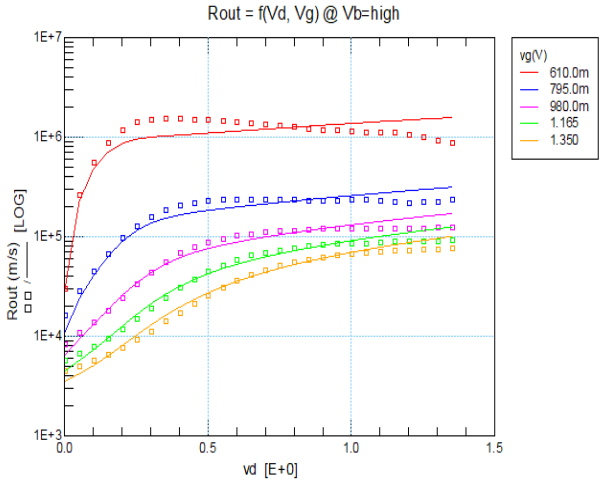


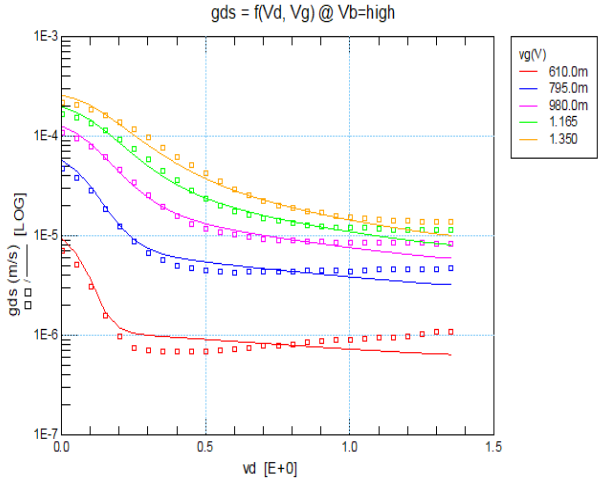




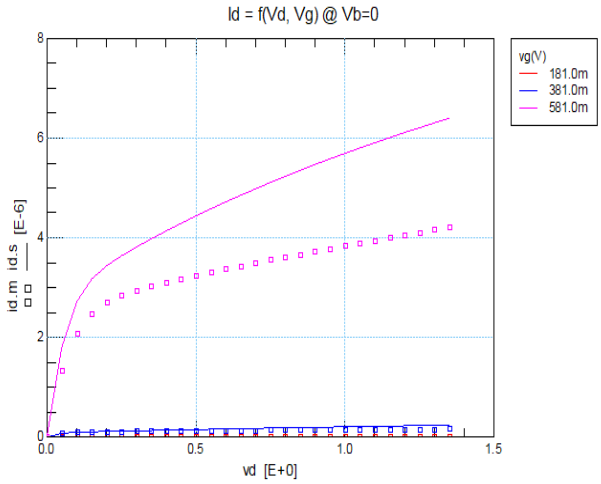
idvd_vbmin, Vb = -1.2V, T = 27°C

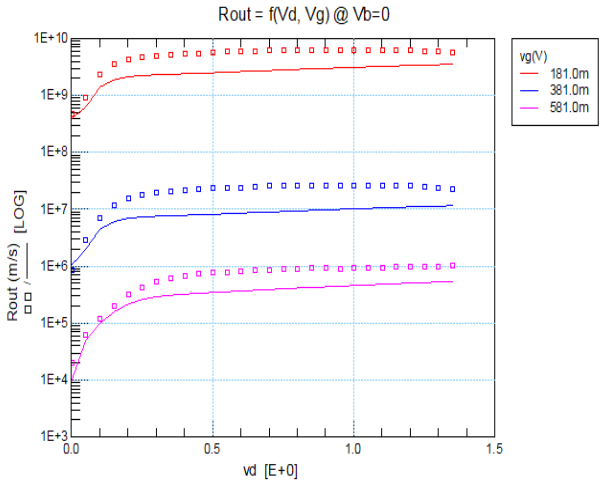


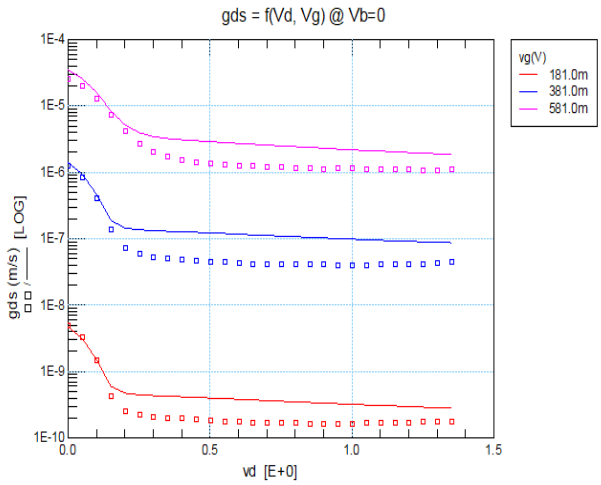




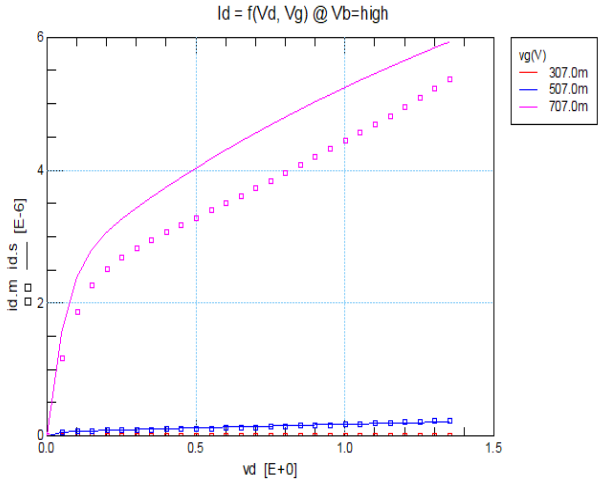
idvd_vth, Vb = 0V, T = 27°C

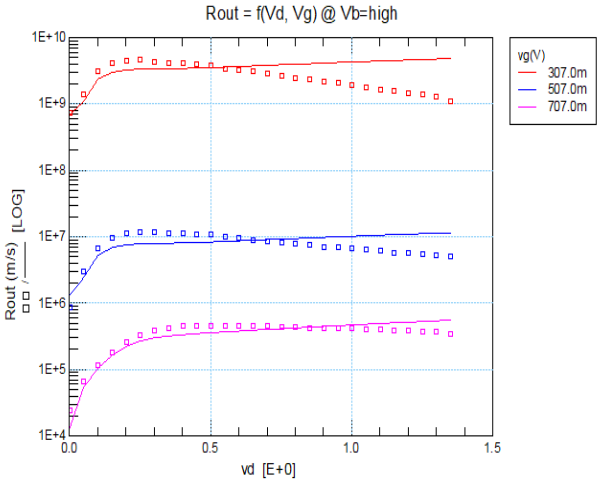


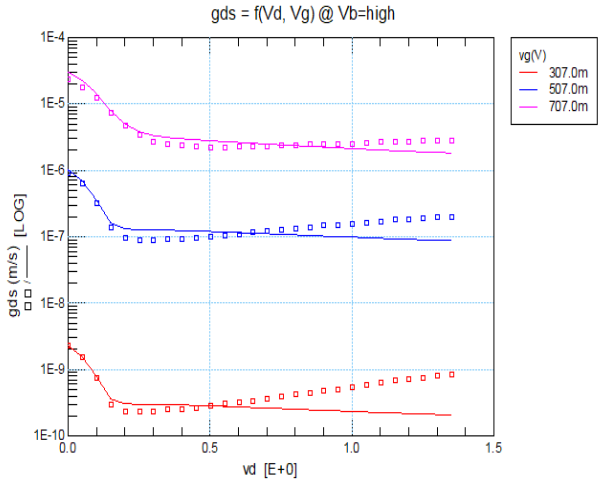




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

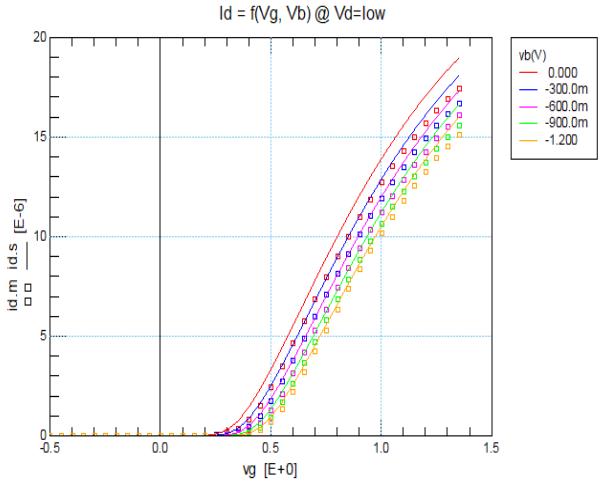


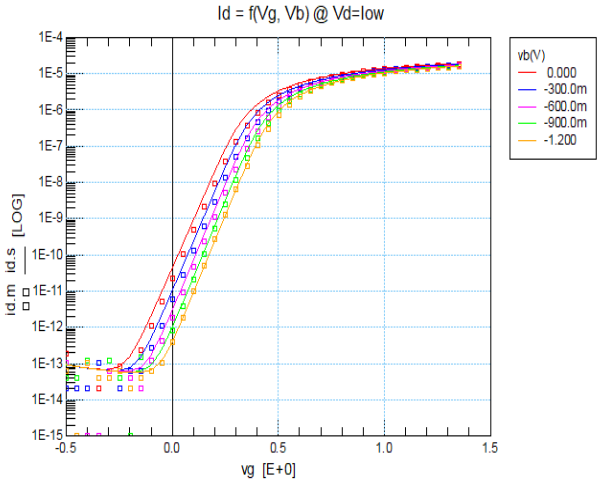


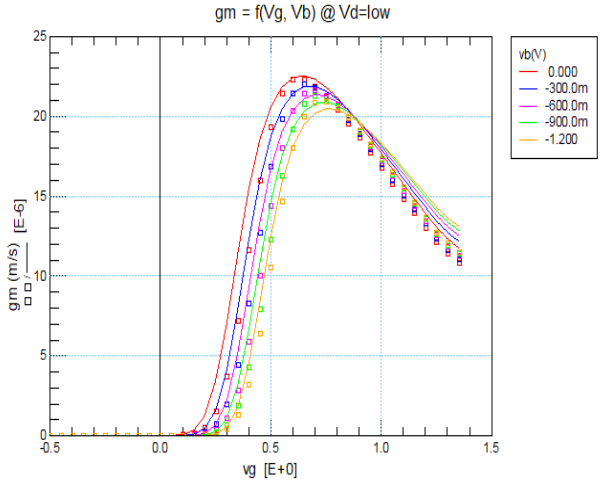


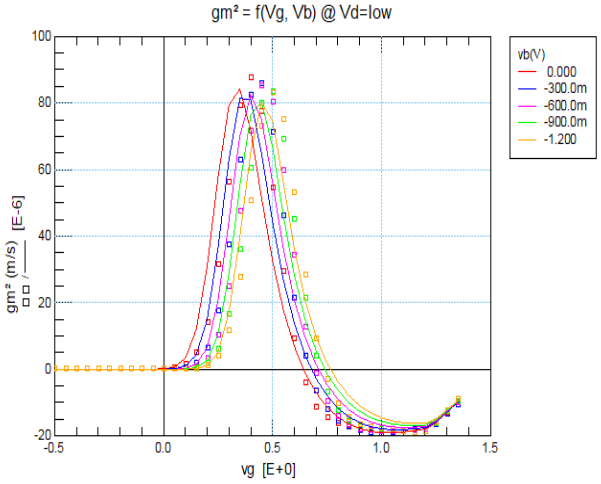
4.30 W0u6_L0u5_S543_4

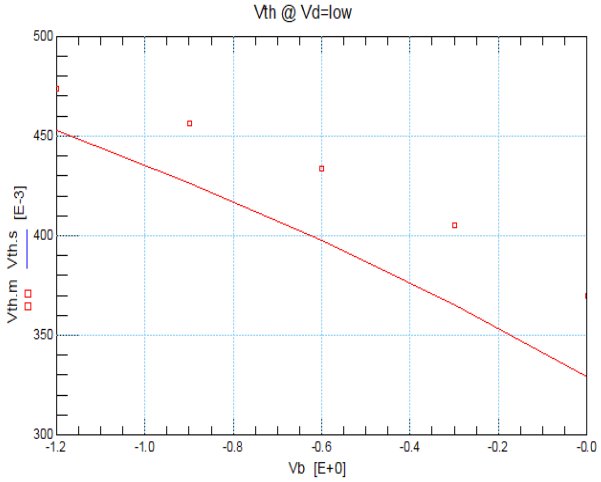
idvg, Vd = 0.05V, T = 27°C

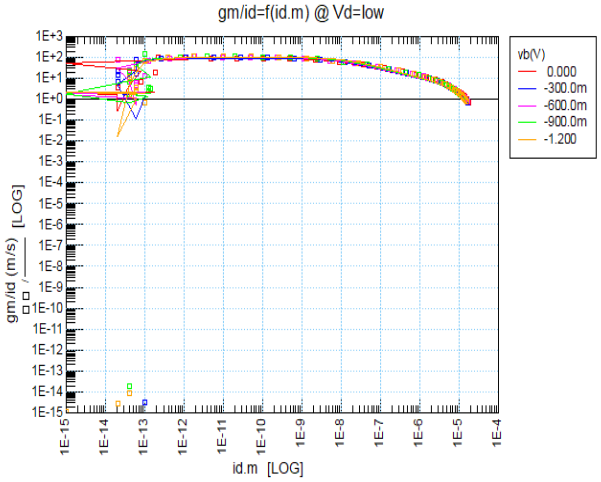


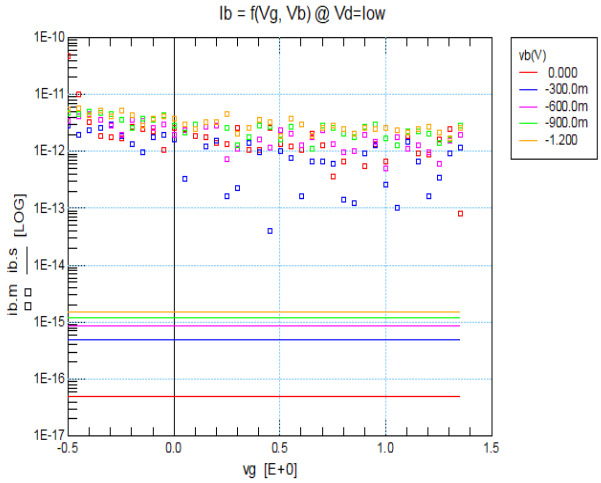


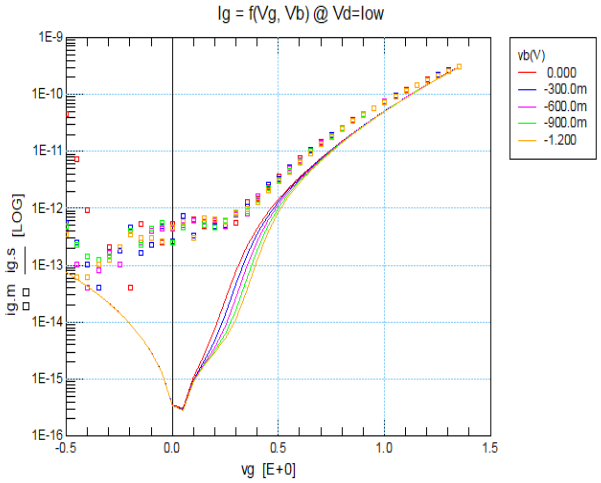




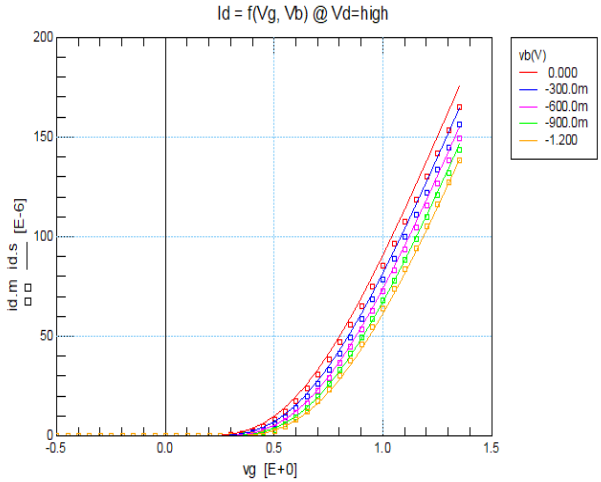


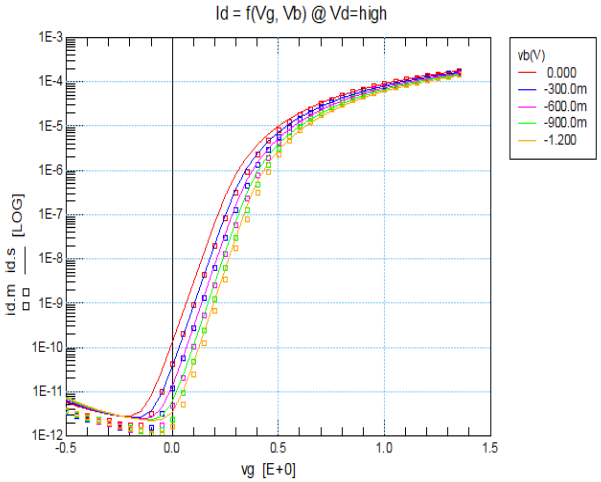


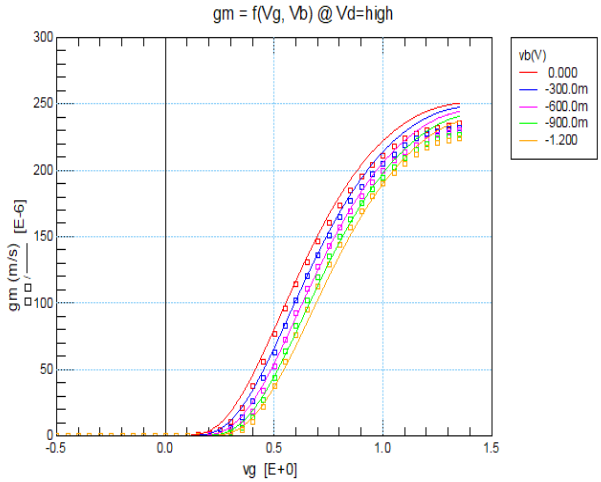


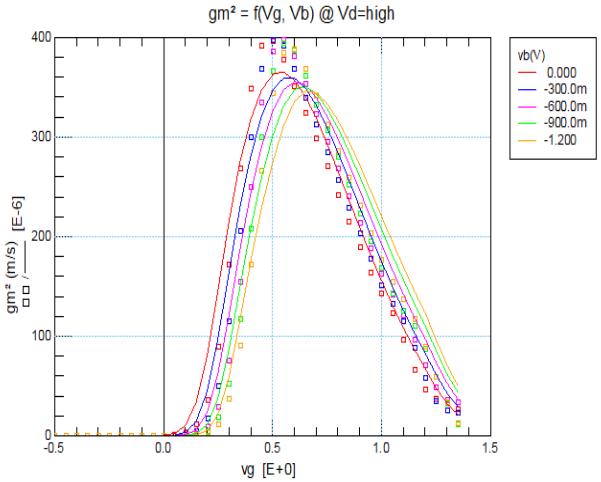


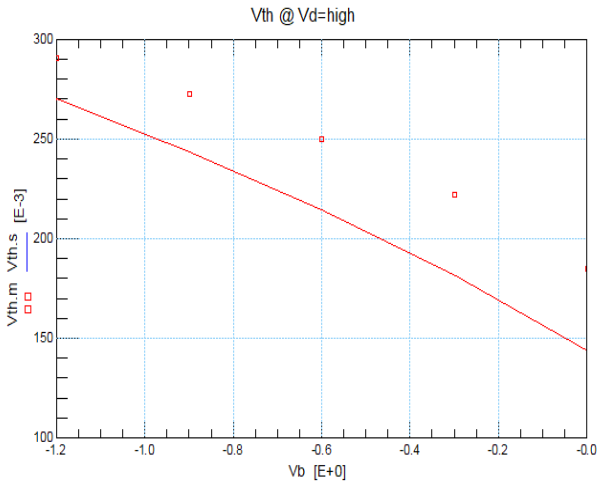
idvg, Vd = 1.2V, T = 27°C

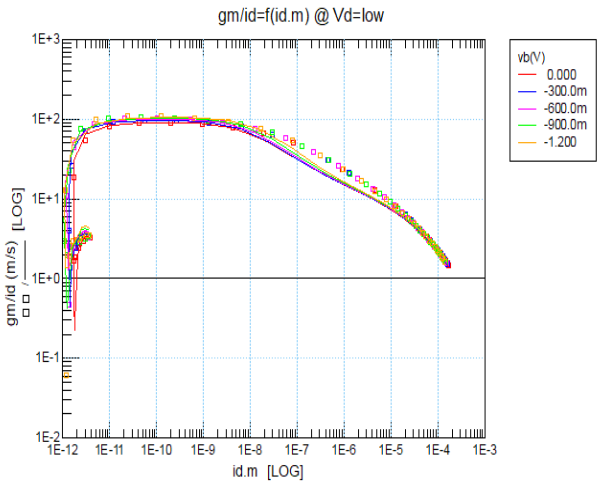


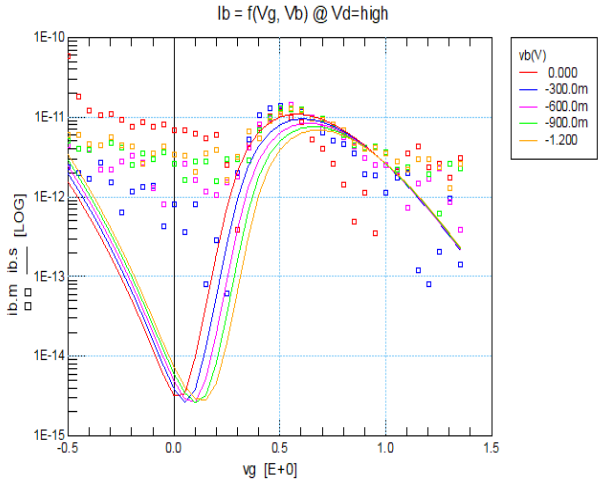


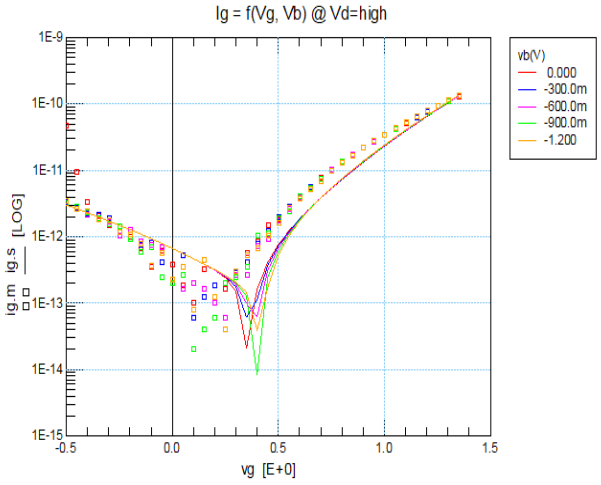




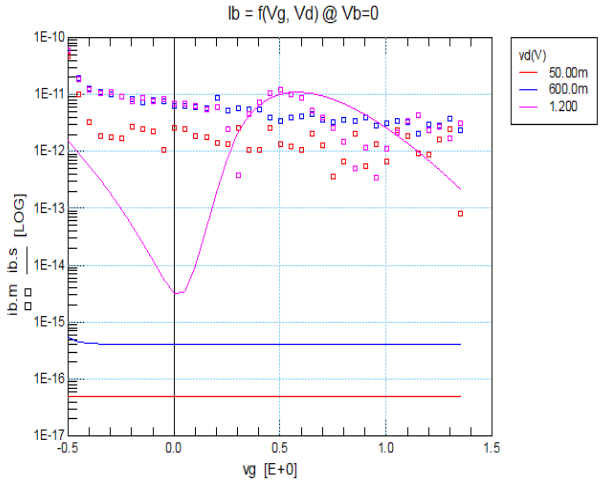


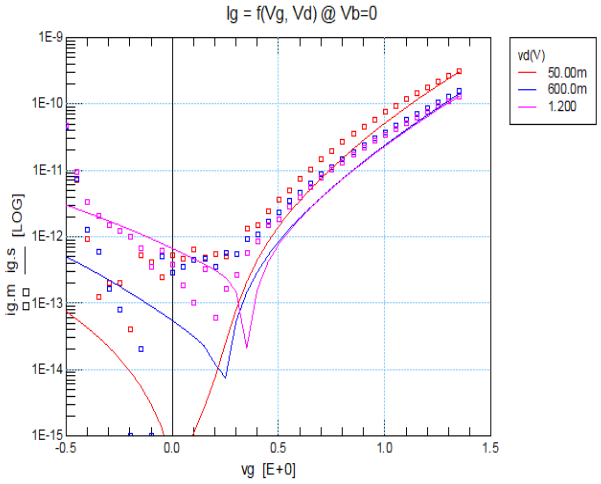




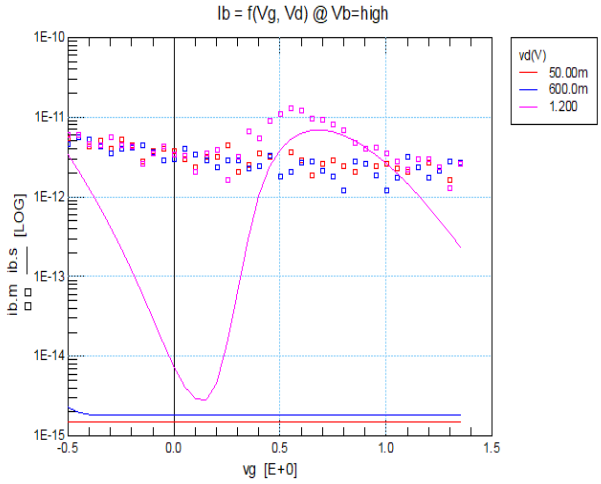


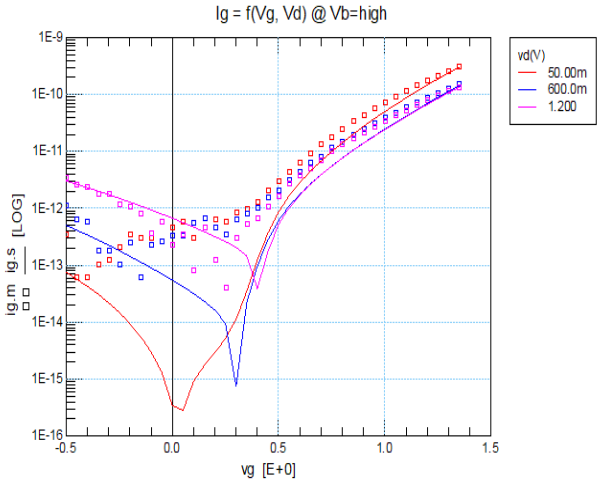
idvg, Ib,Ig, Vb = 0V, T = 27°C



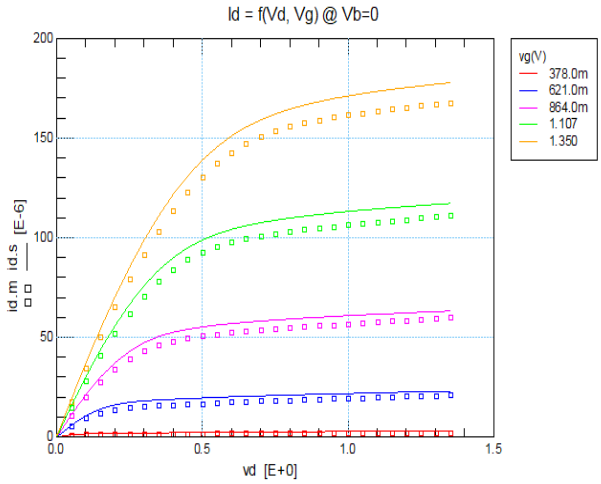


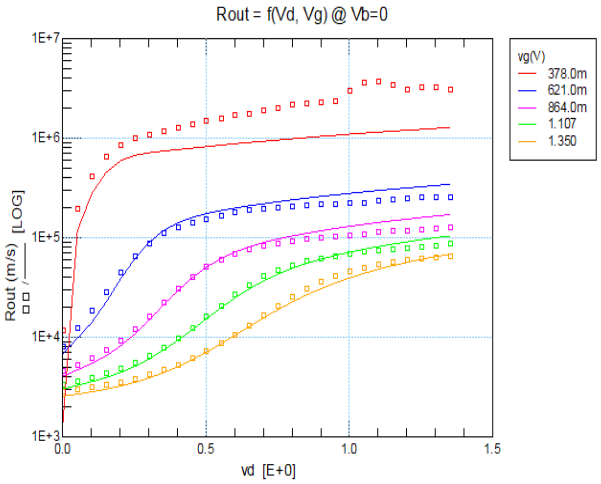
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

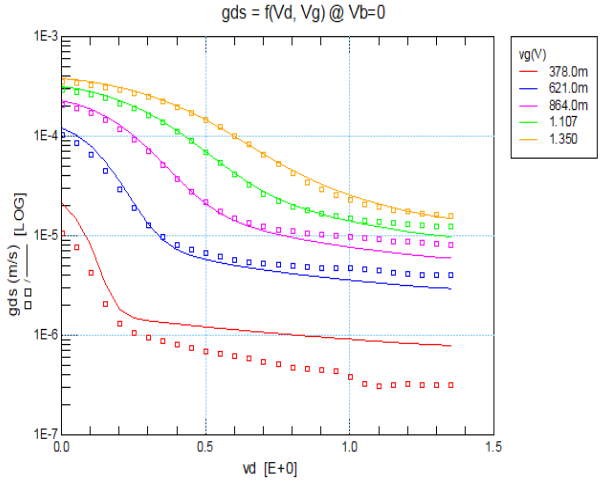




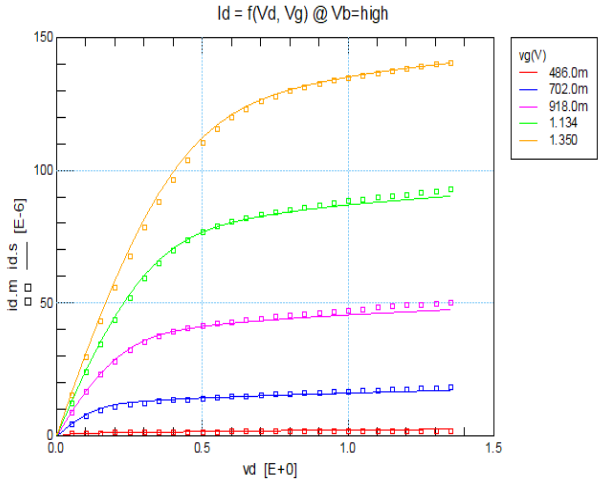
idvd, Vb = 0V, T = 27°C

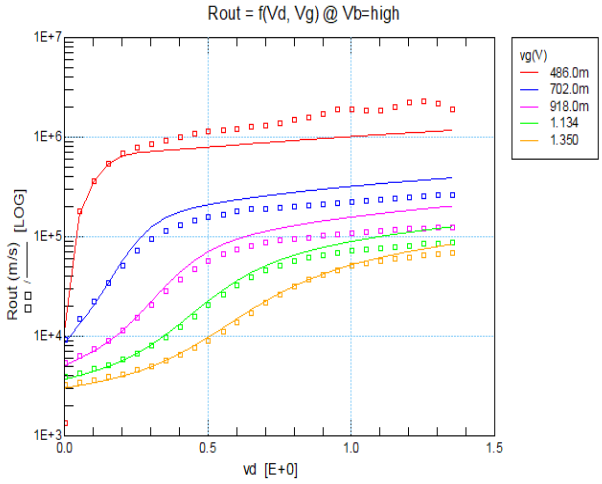


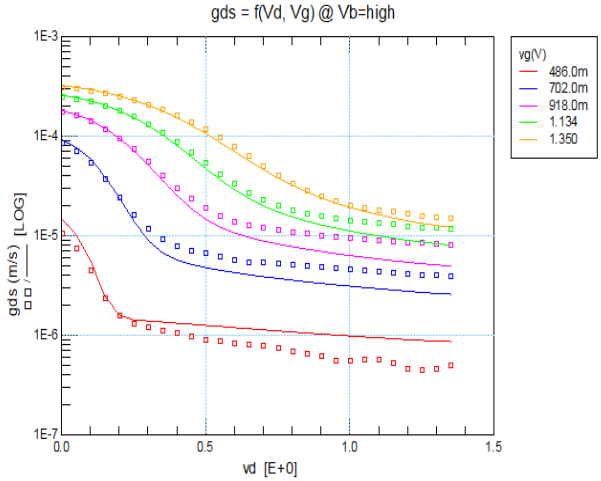




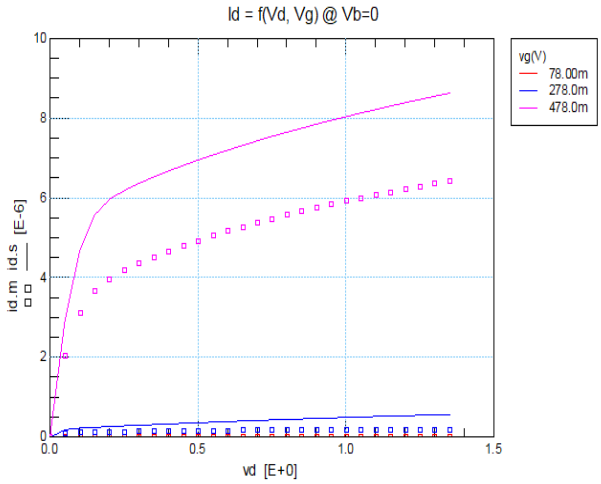
idvd_vbmin, Vb = -1.2V, T = 27°C

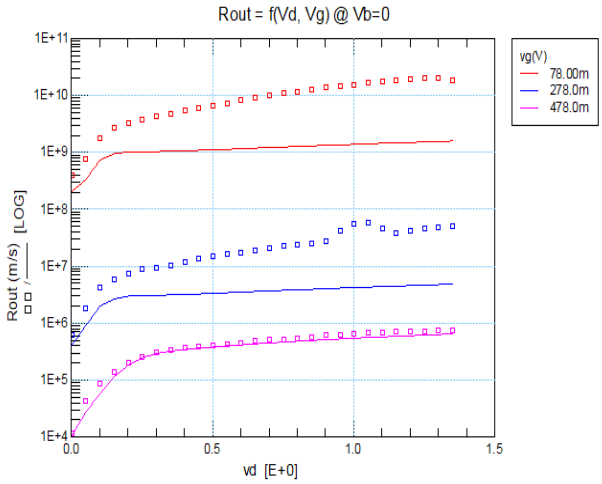


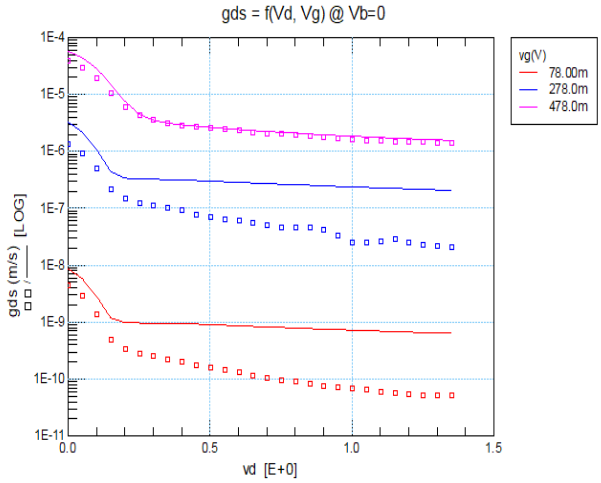




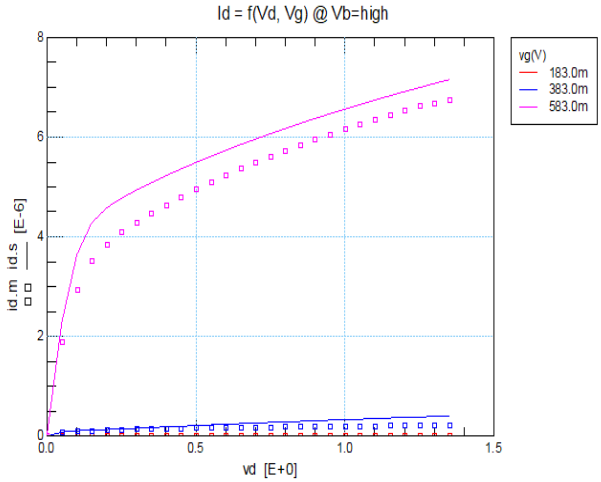
idvd_vth, Vb = 0V, T = 27°C

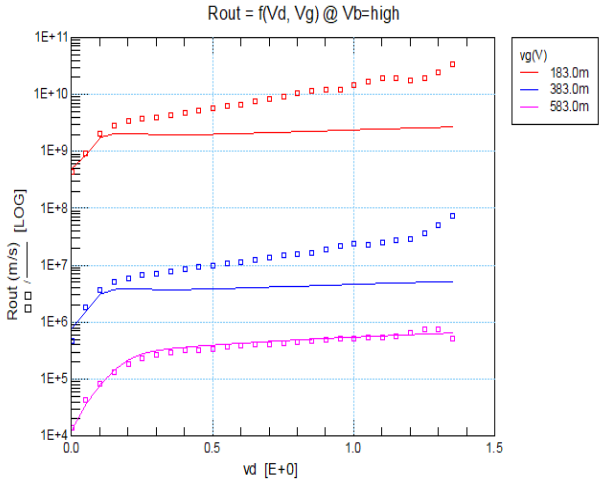


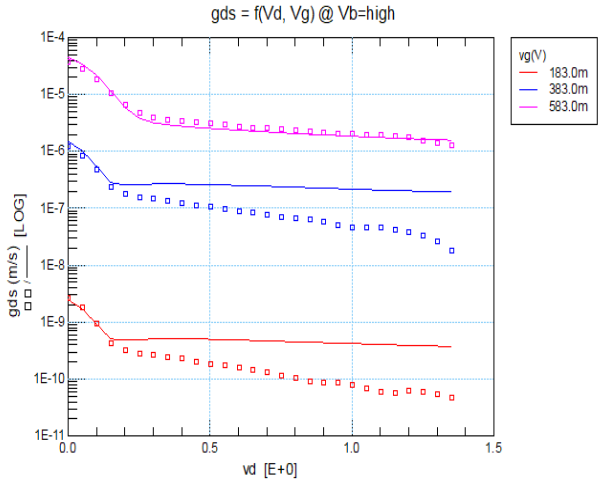




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

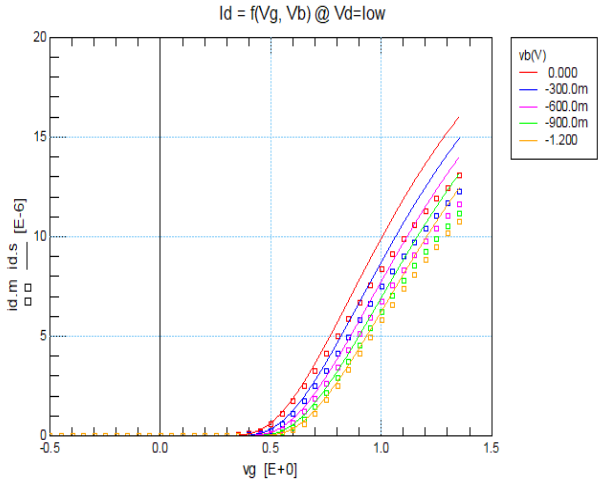


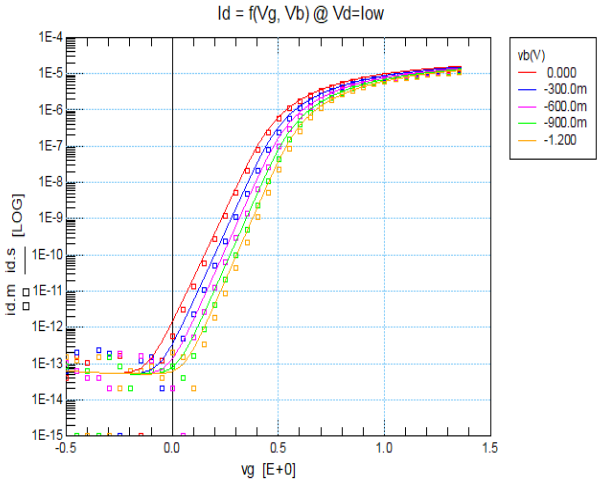


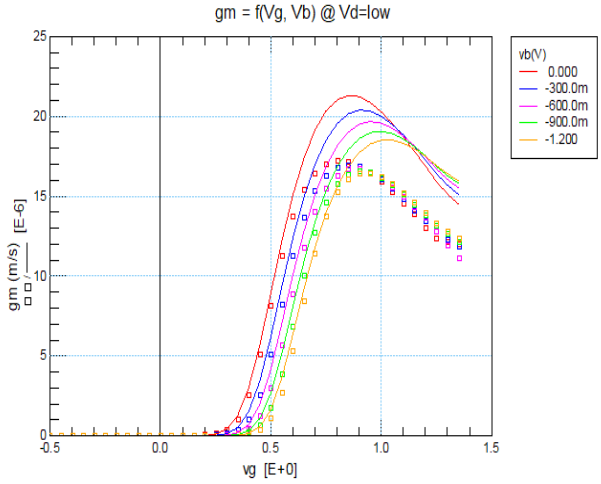


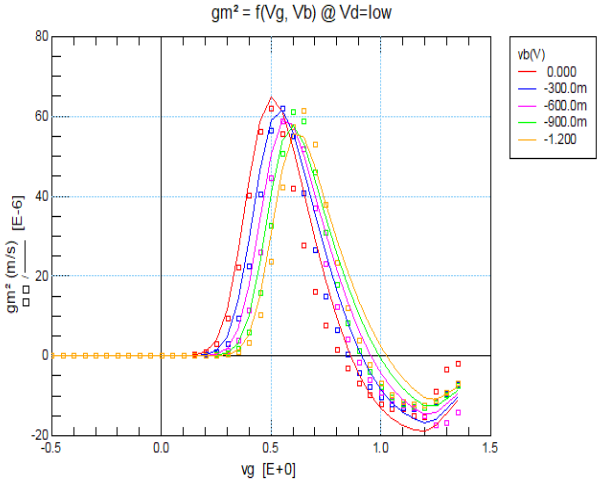
4.31 W0u18_L0u15_S545_5

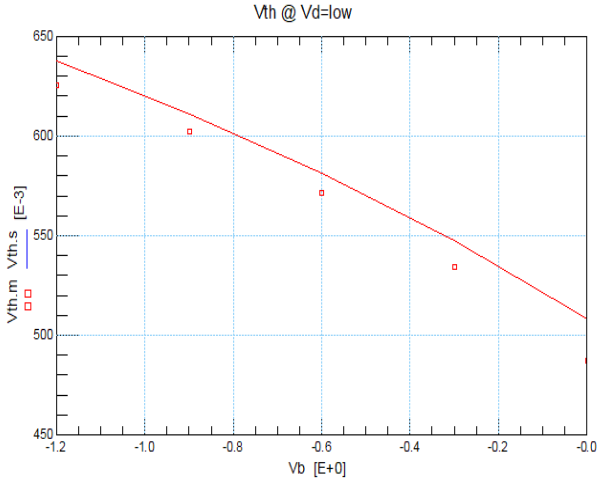
idvg, Vd = 0.05V, T = 27°C

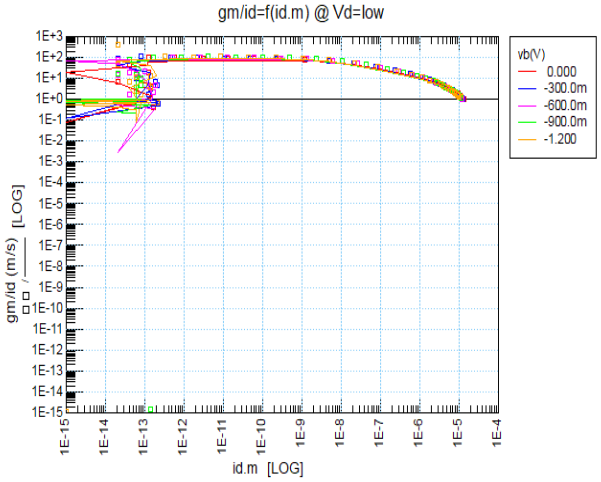


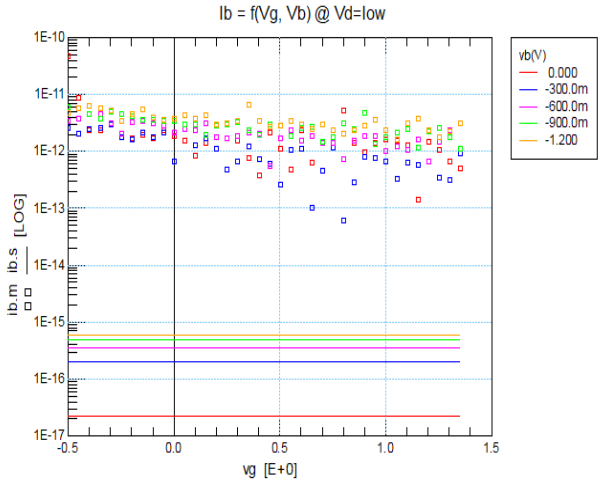


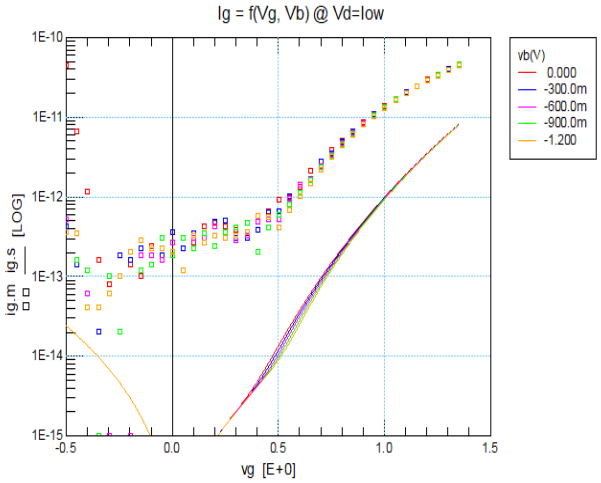




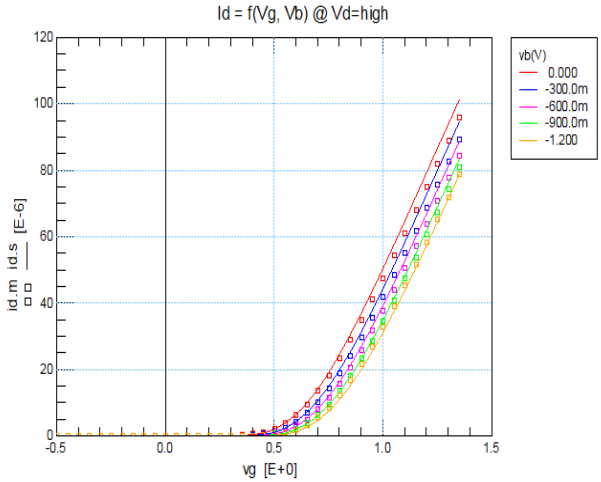


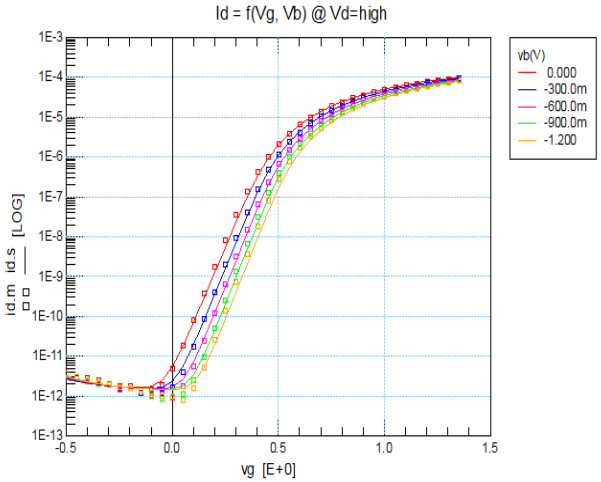


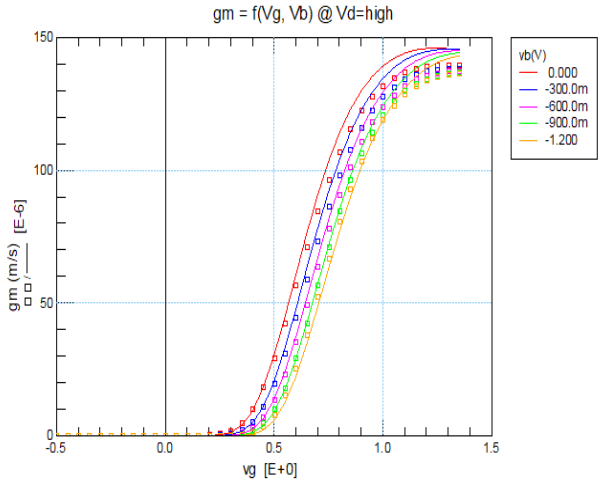


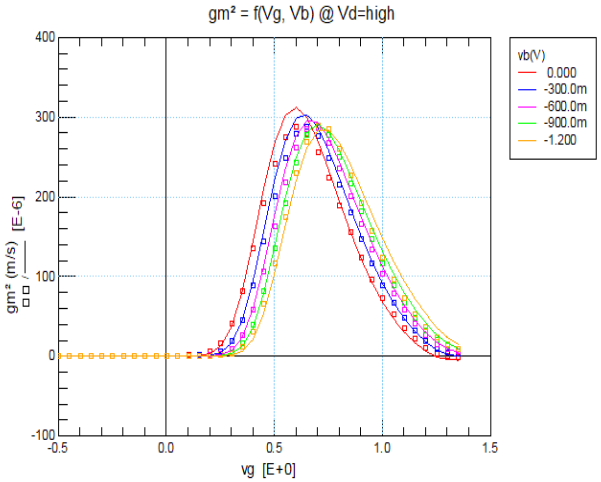


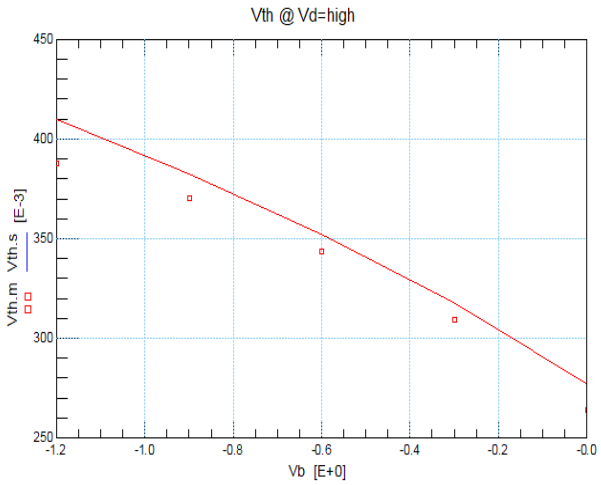
idvg, Vd = 1.2V, T = 27°C

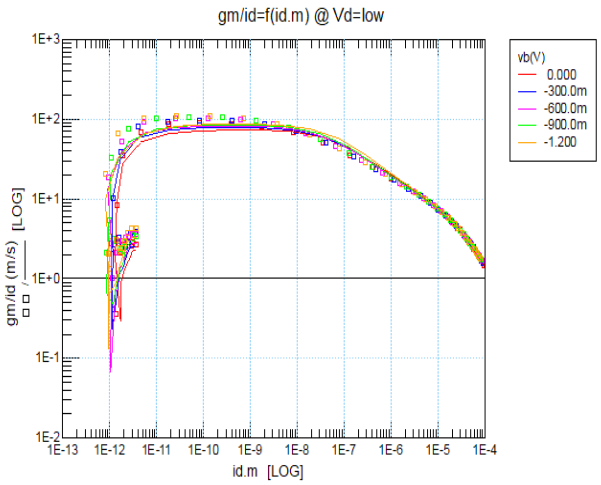


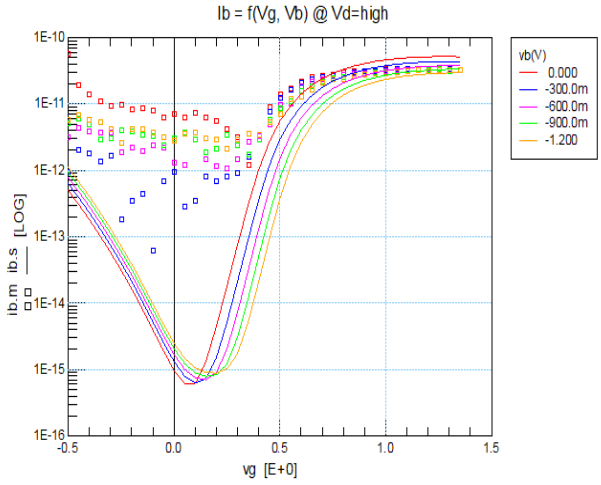


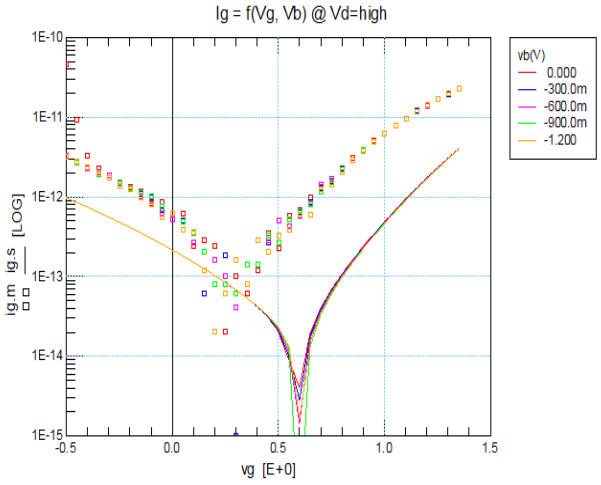




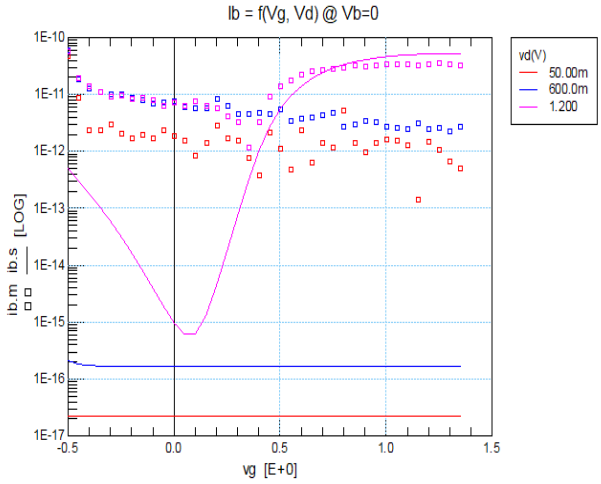


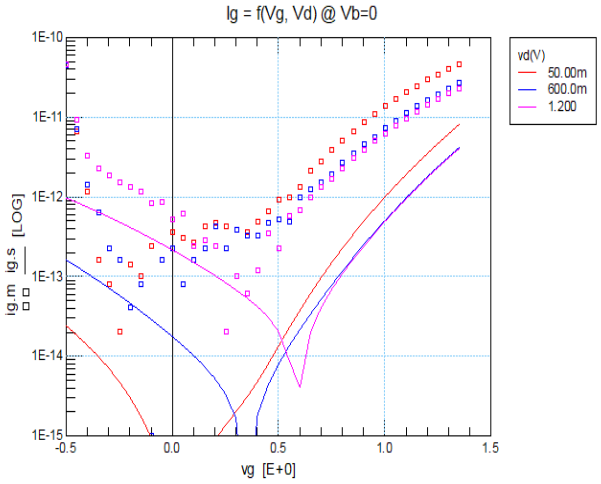




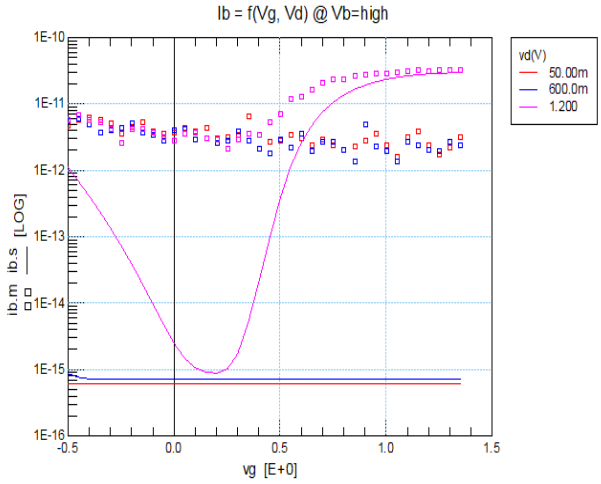


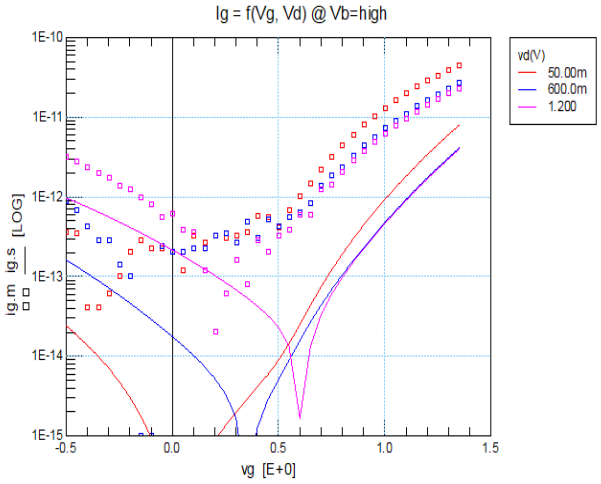
idvg, Ib,Ig, Vb = 0V, T = 27°C



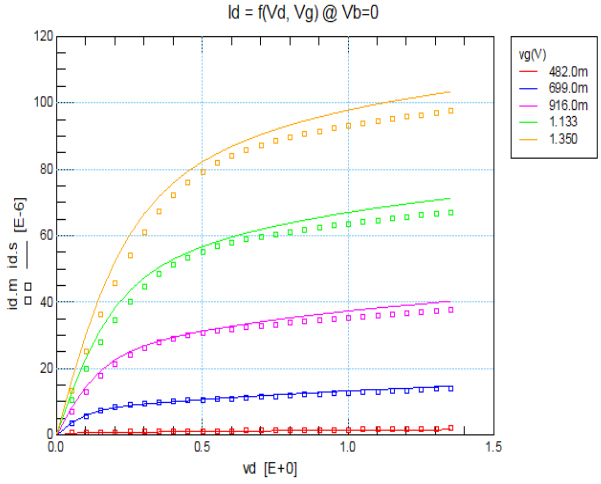


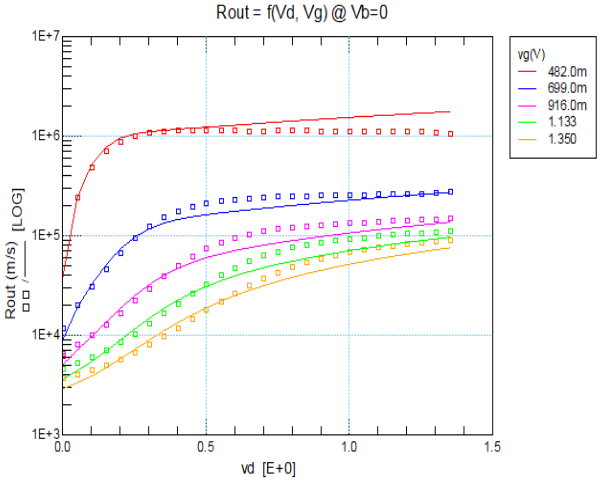
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

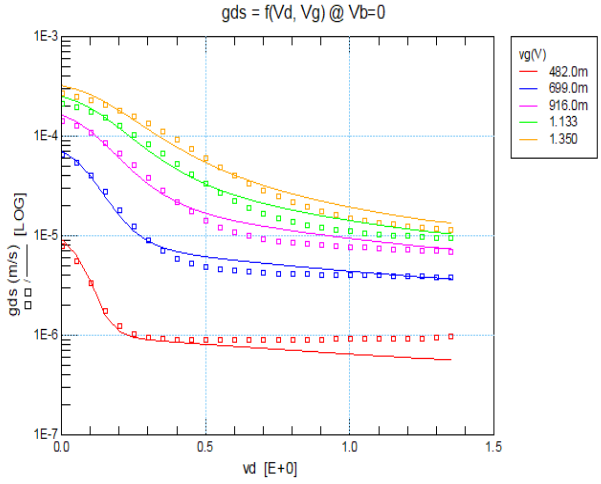




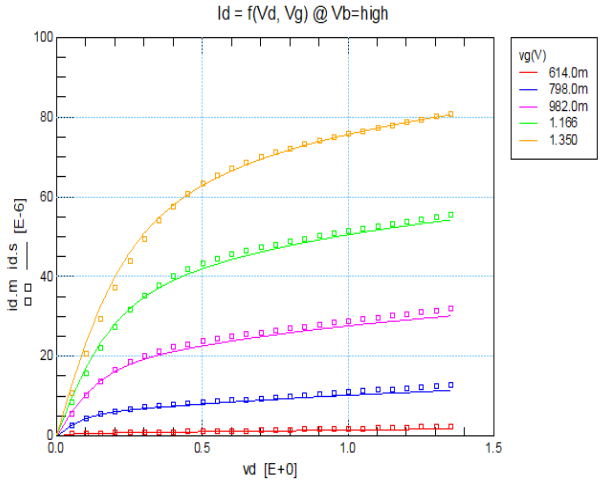
idvd, Vb = 0V, T = 27°C

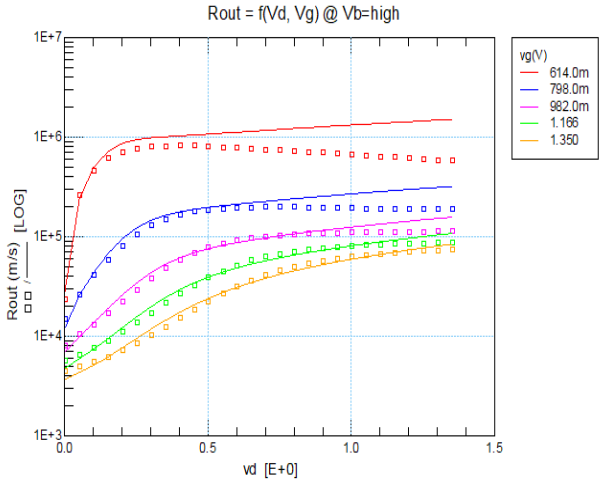


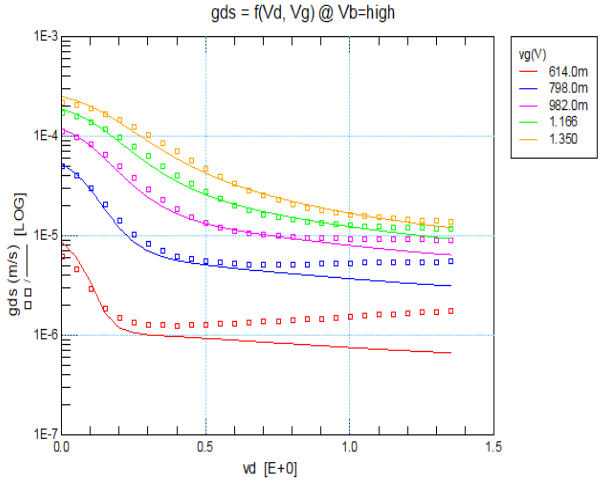




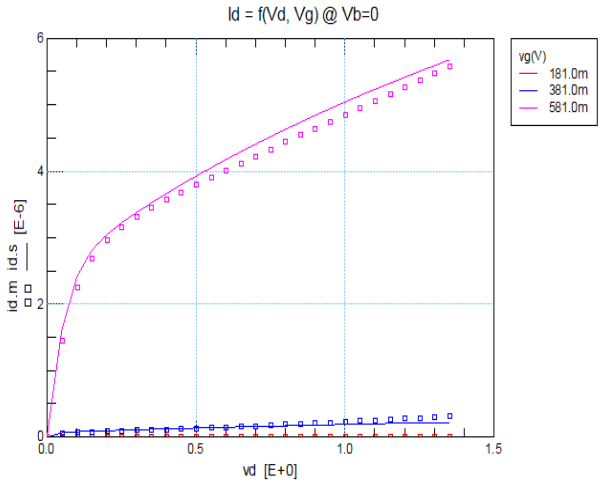
idvd_vbmin, Vb = -1.2V, T = 27°C

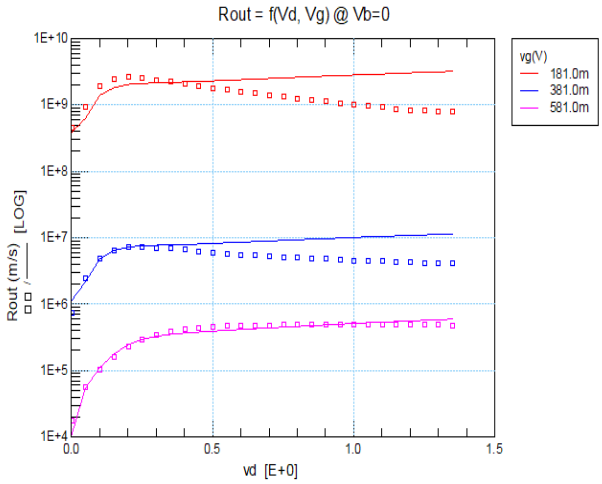


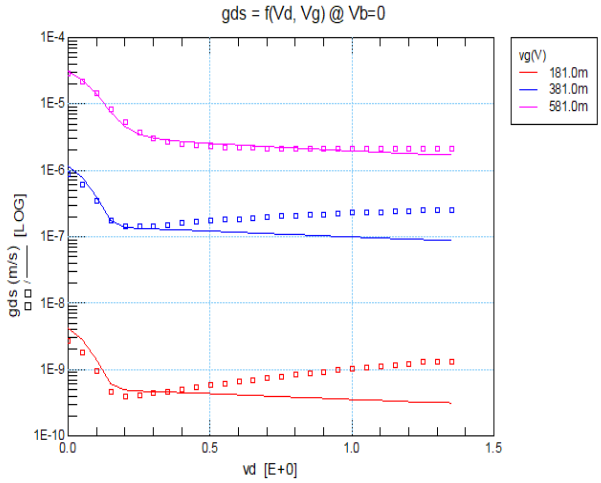




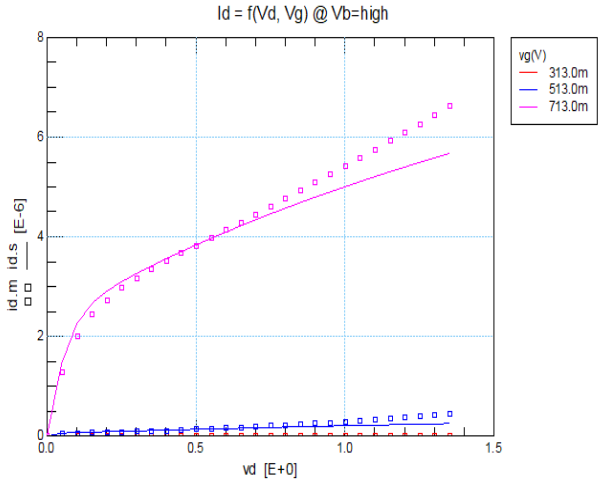
idvd_vth, Vb = 0V, T = 27°C

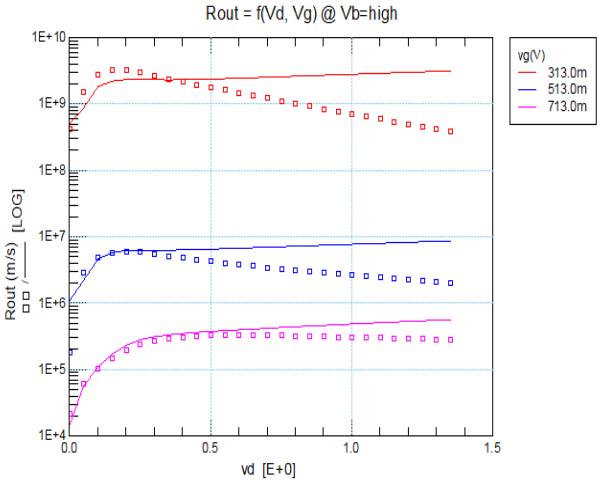


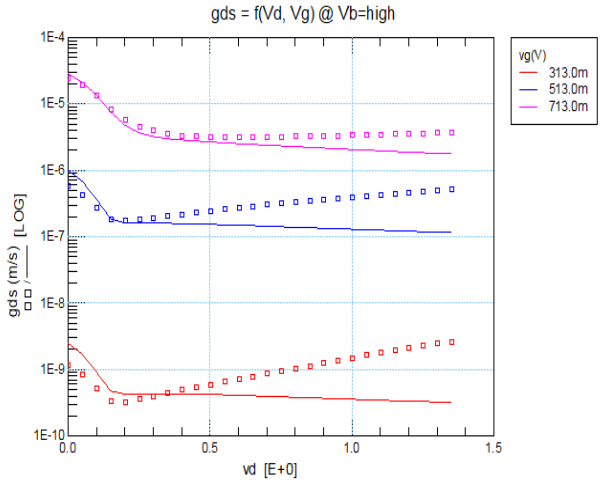




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

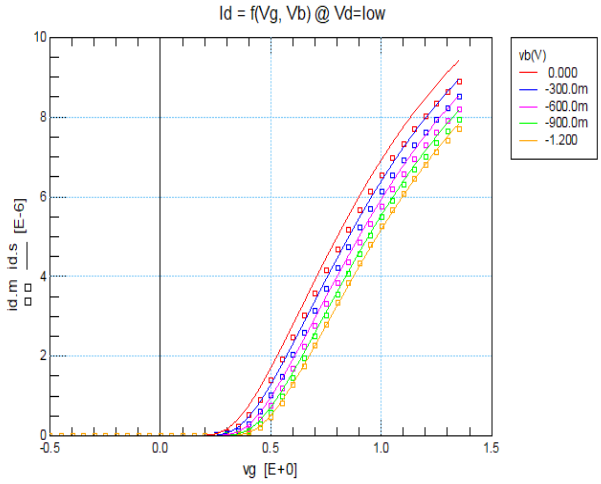


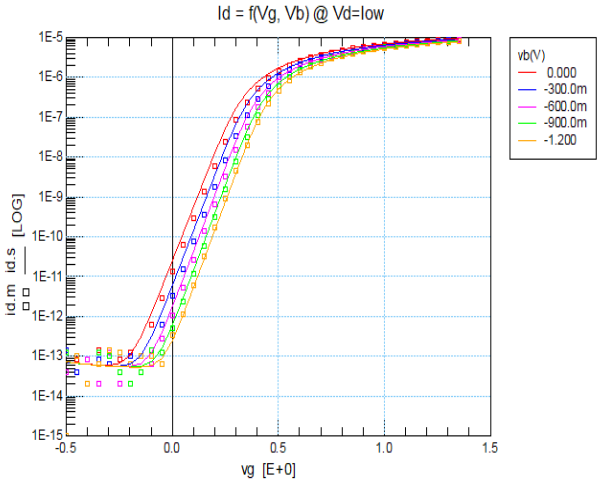


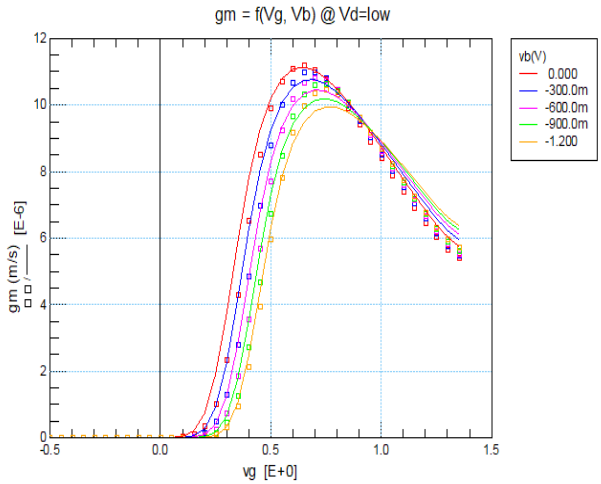


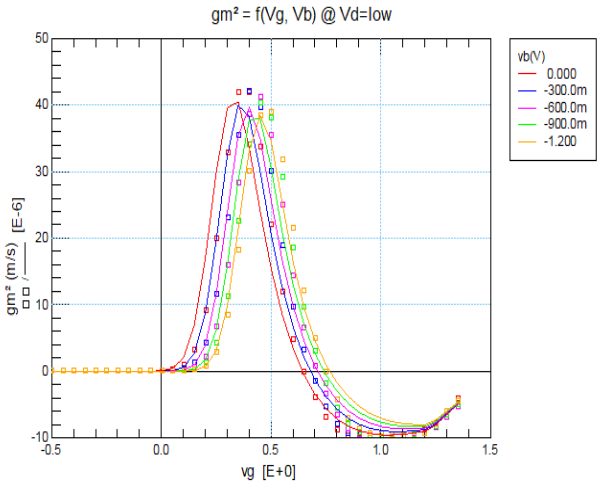
4.32 W0u3_L0u5_S545_1

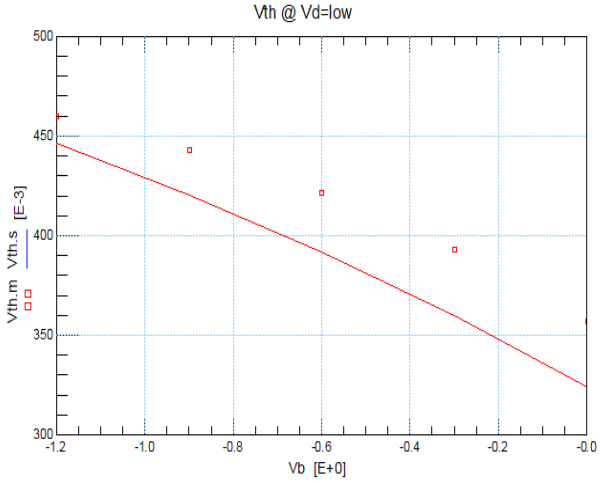
idvg, Vd = 0.05V, T = 27°C

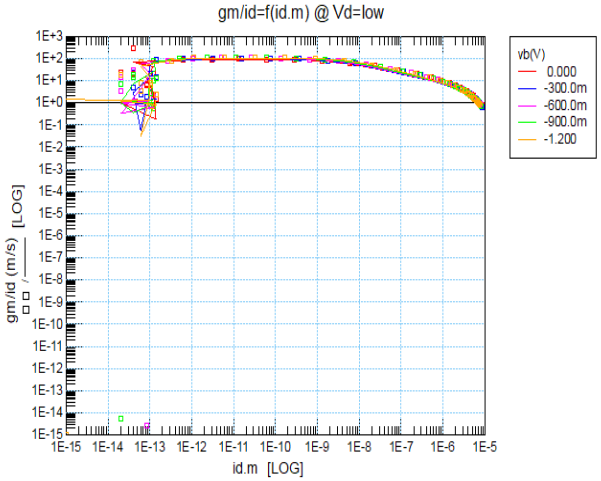


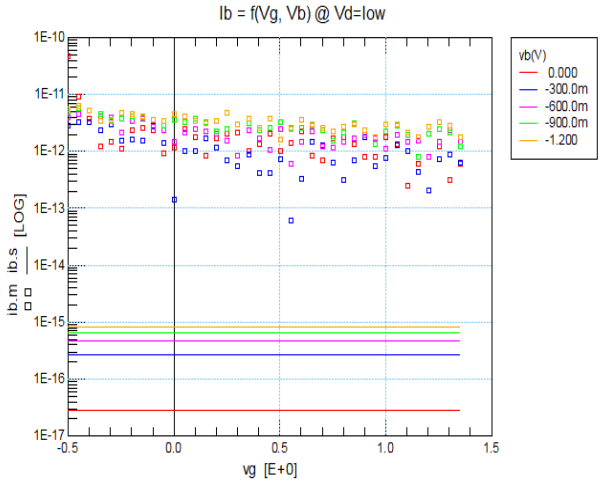


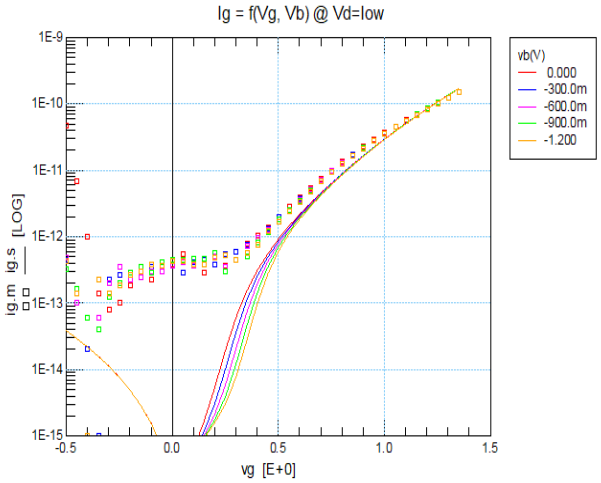




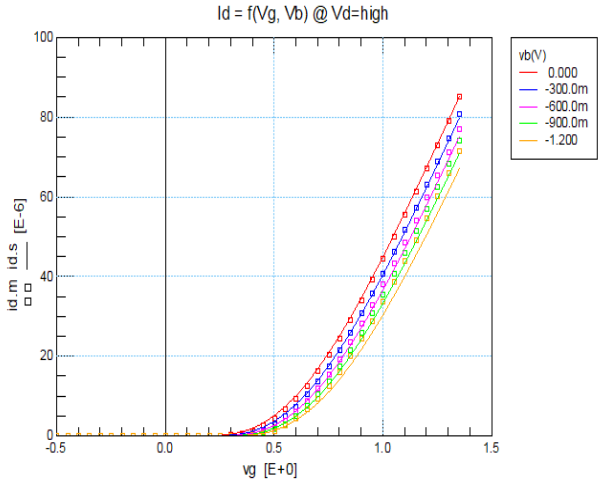


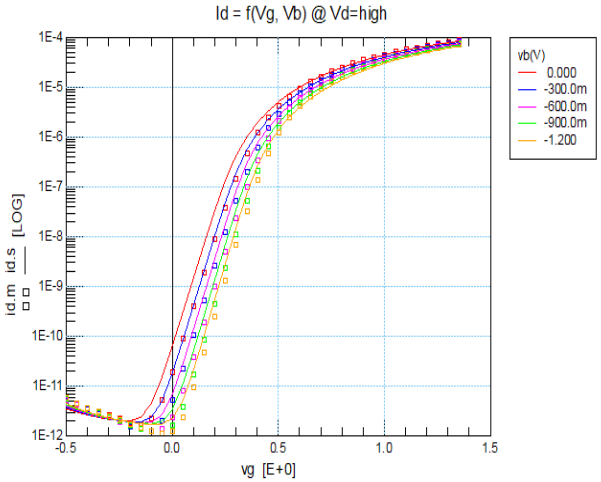


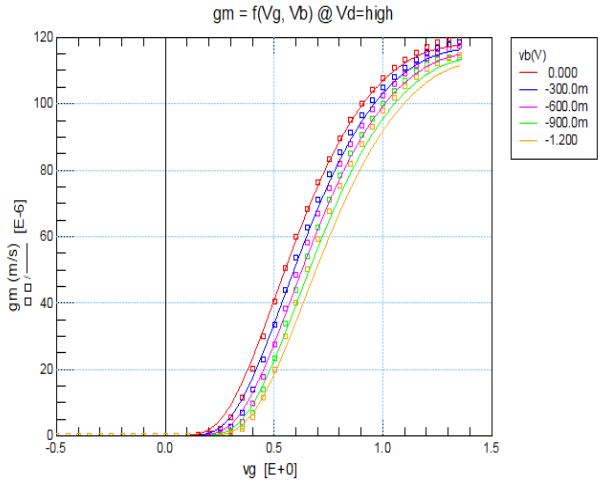


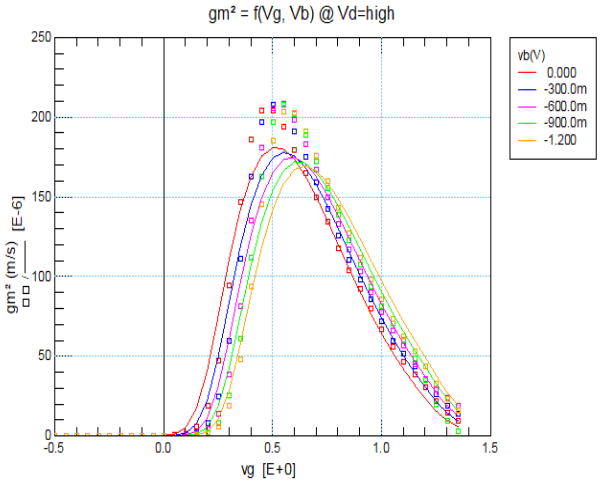


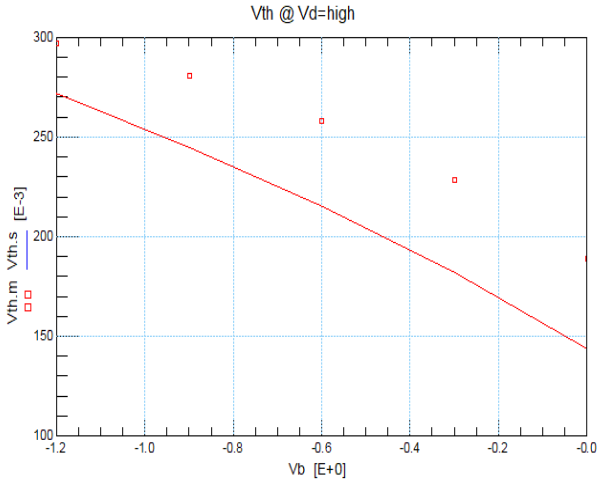
idvg, Vd = 1.2V, T = 27°C

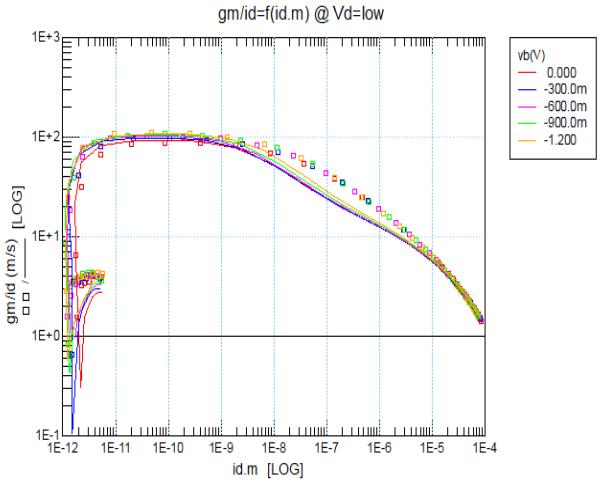


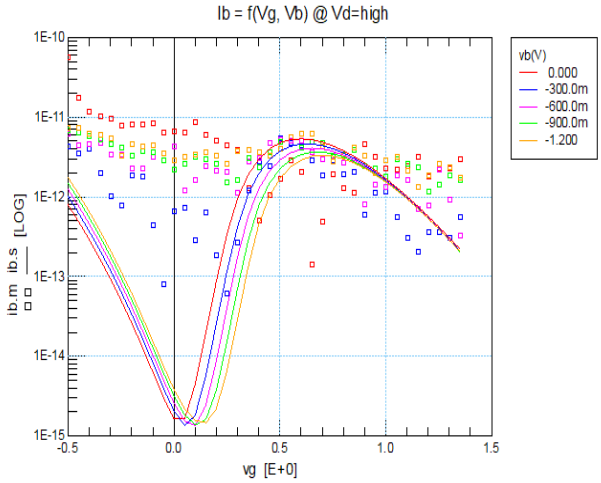


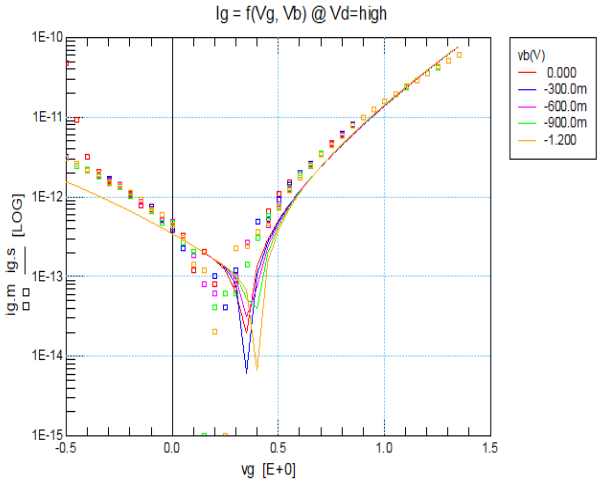




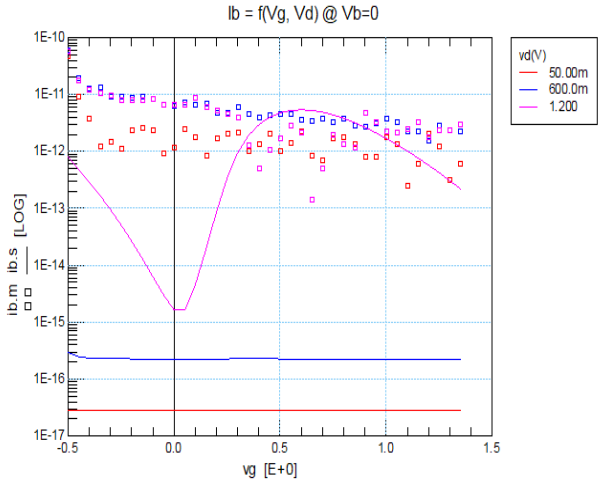


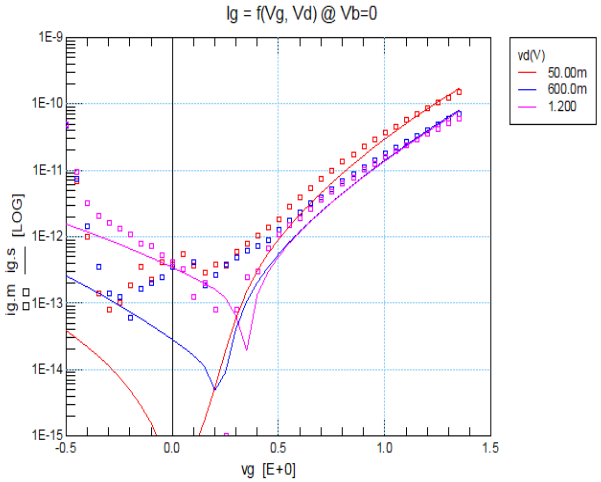




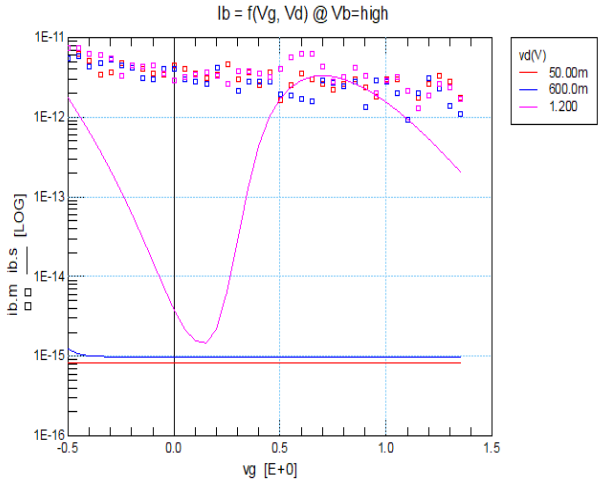


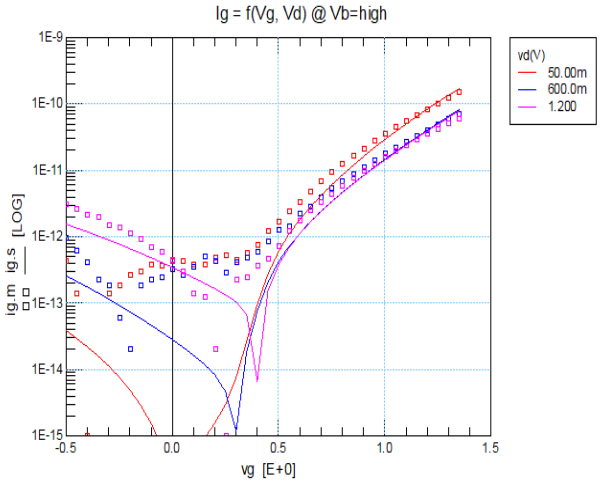
idvg, Ib,Ig, Vb = 0V, T = 27°C



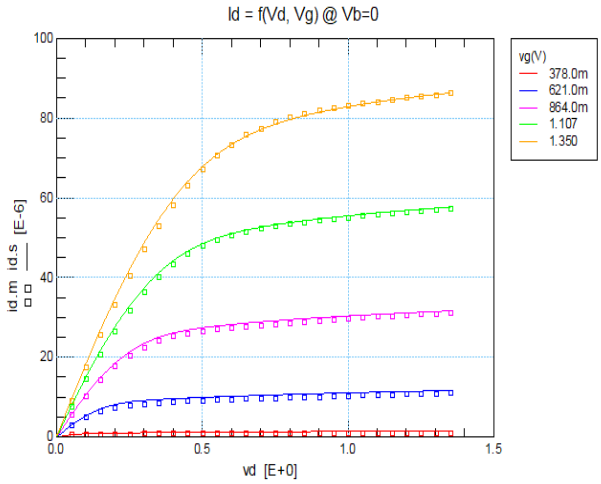


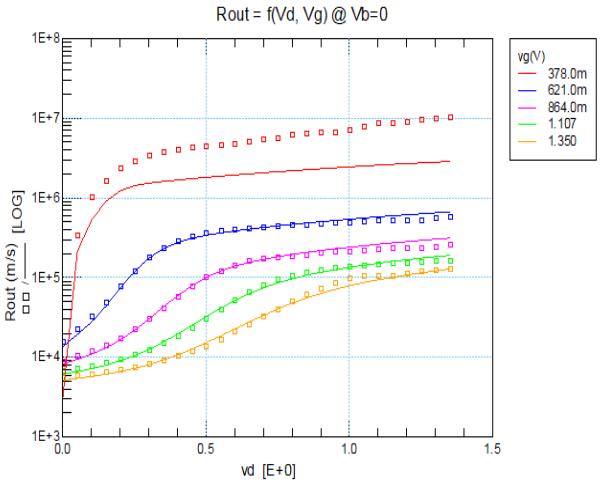
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

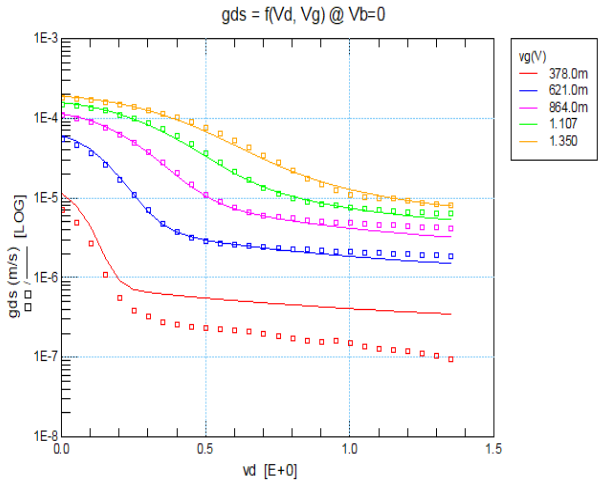




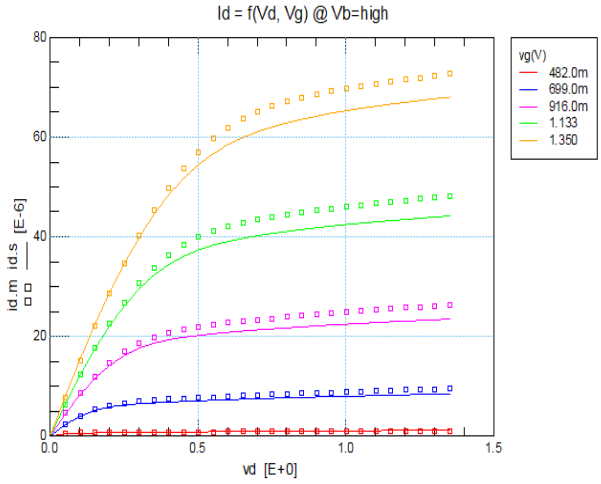
idvd, Vb = 0V, T = 27°C

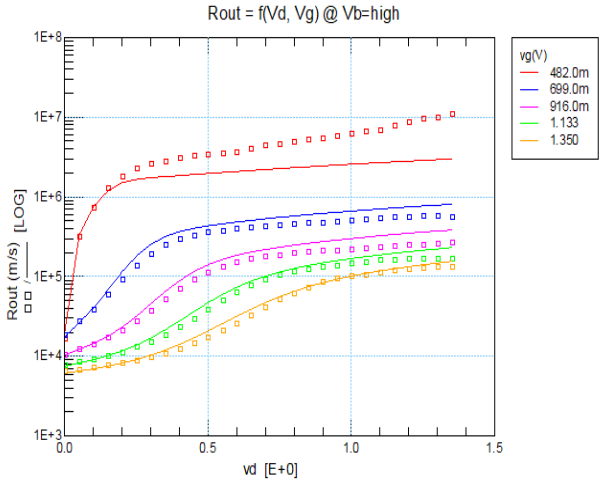


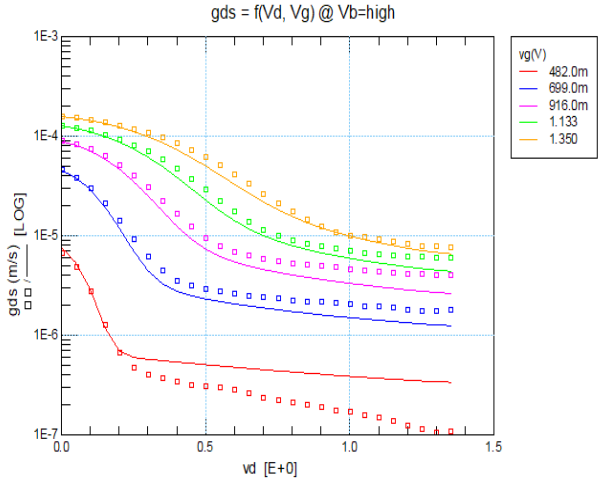




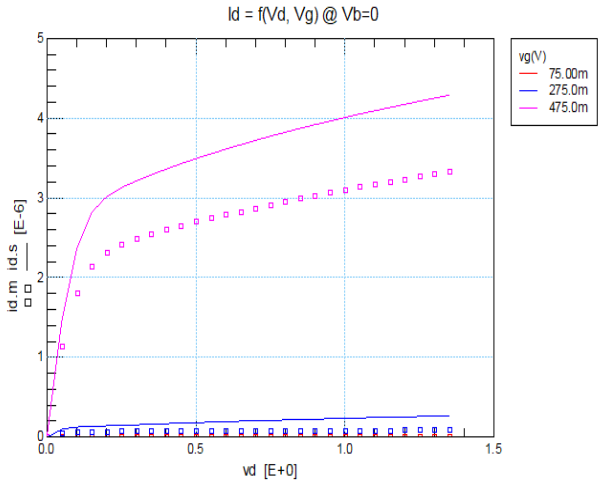
idvd_vbmin, Vb = -1.2V, T = 27°C

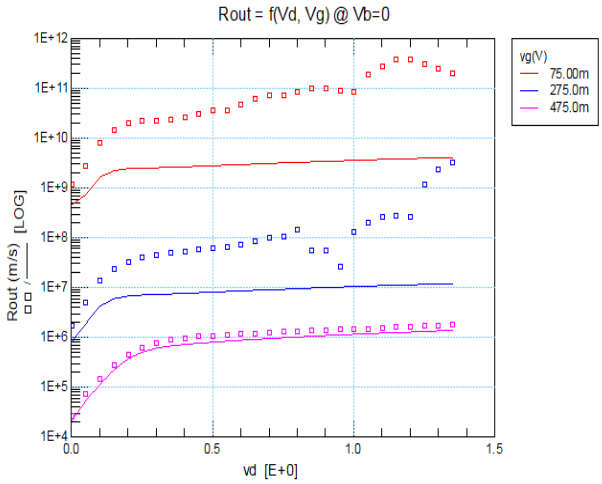


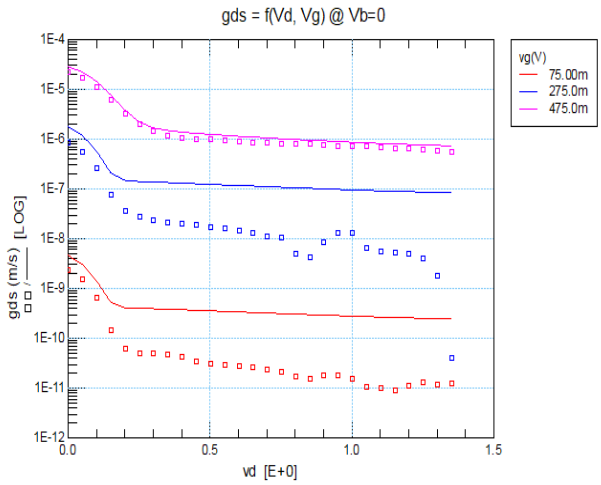




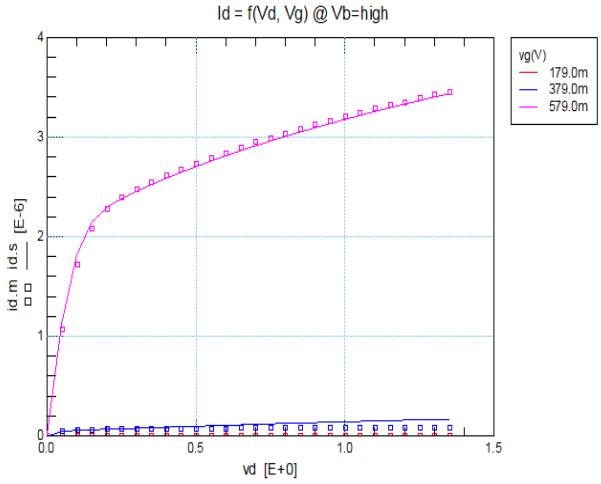
idvd_vth, Vb = 0V, T = 27°C

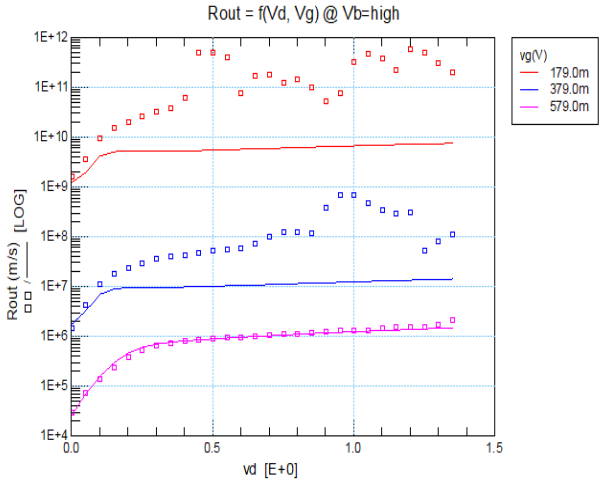


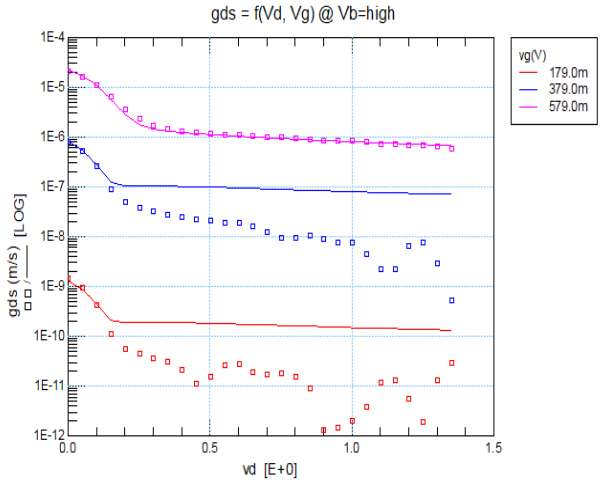




idvd_vth_vbmin, Vb = -1.2V, T = 27°C



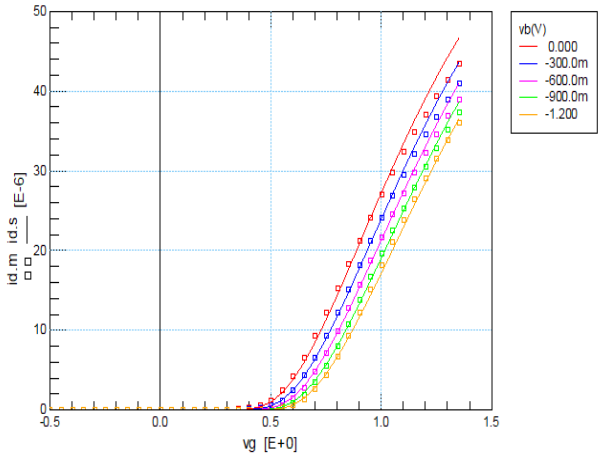




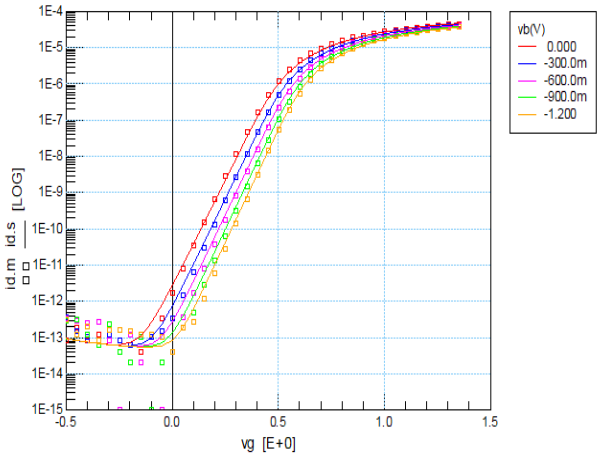
4.33 W0u6_L0u13_S543_2

idvg, Vd = 0.05V, T = 27°C

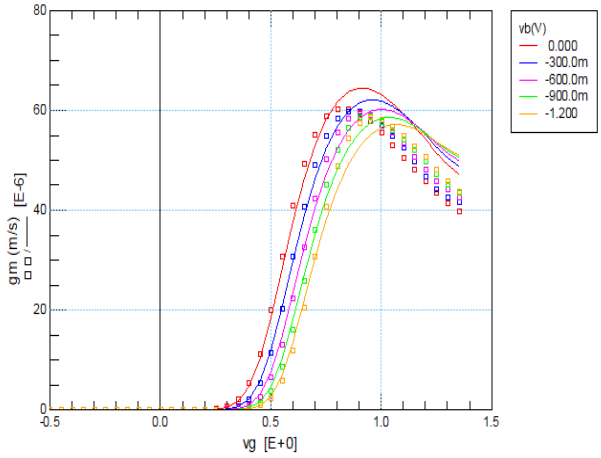
Id = f(Vg, Vb) @ Vd=low



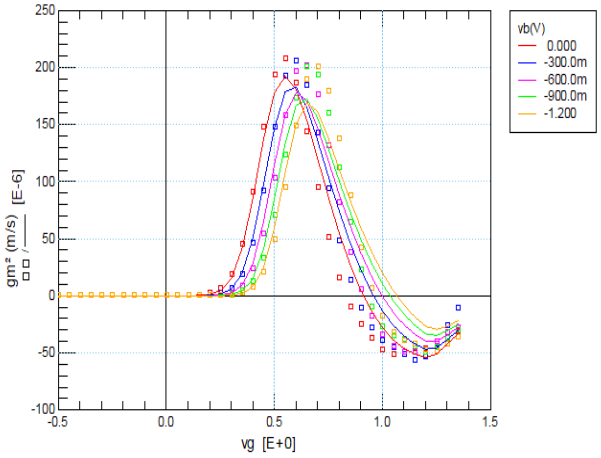
Id = f(Vg, Vb) @ Vd=low



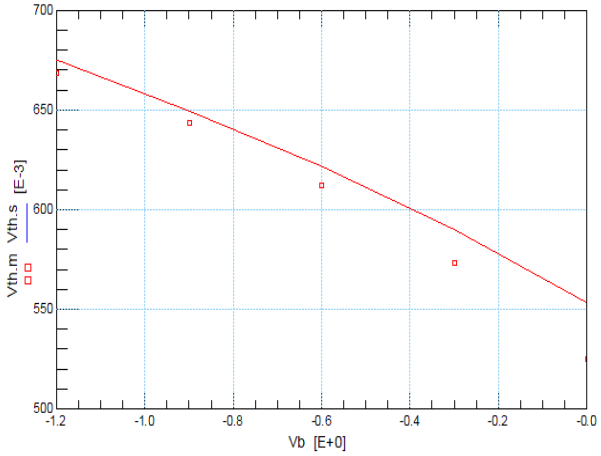
gm = f(Vg, Vb) @ Vd=low



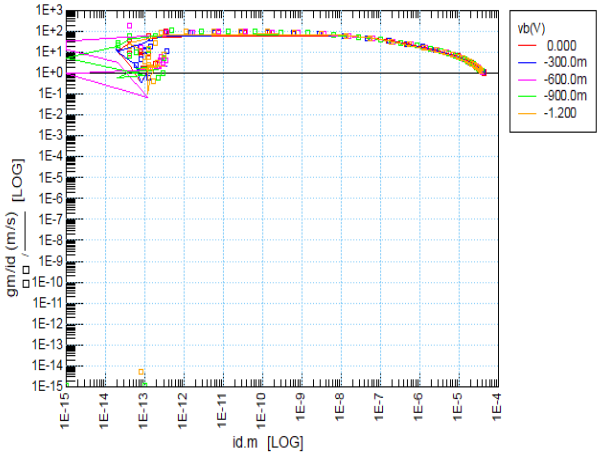
gm² = f(Vg, Vb) @ Vd=low

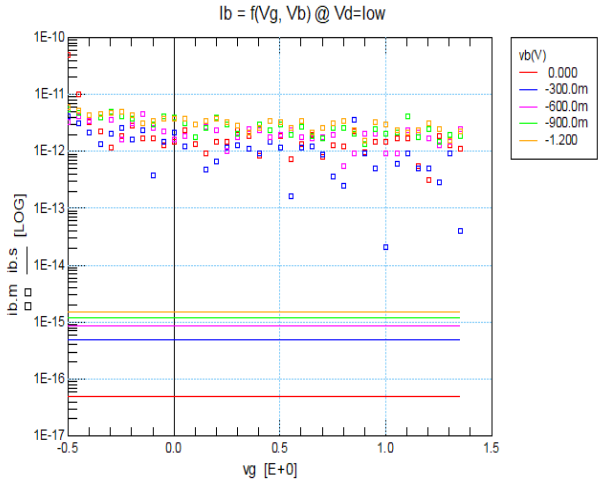


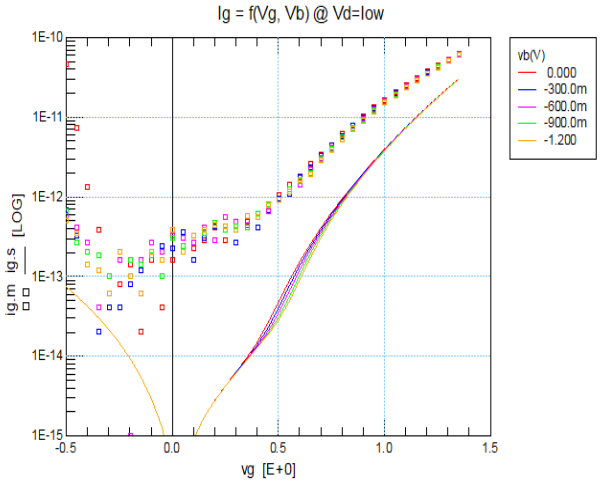
Vth @ Vd=low



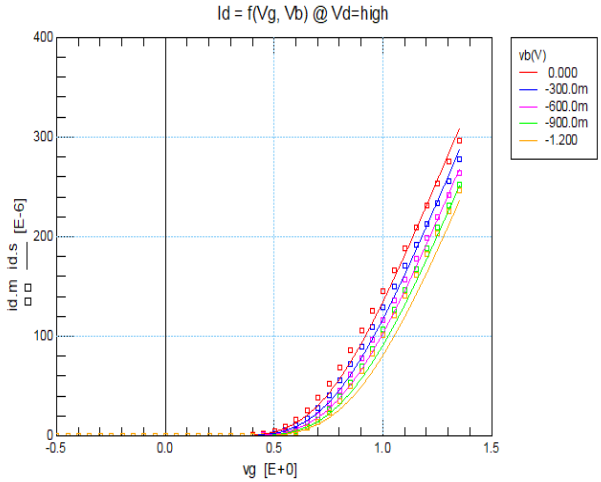
gm/id=f(id.m) @ Vd=low

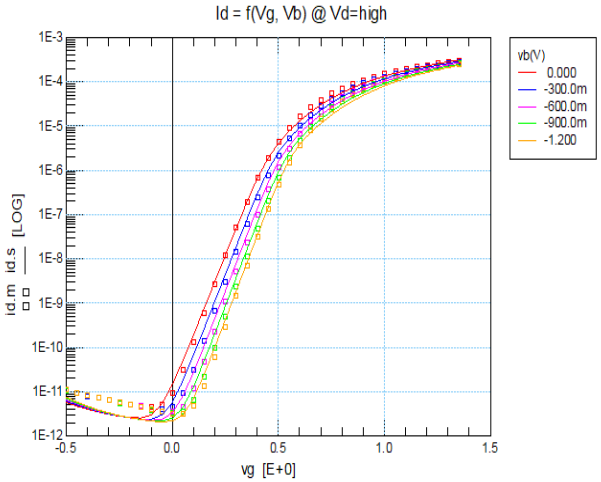


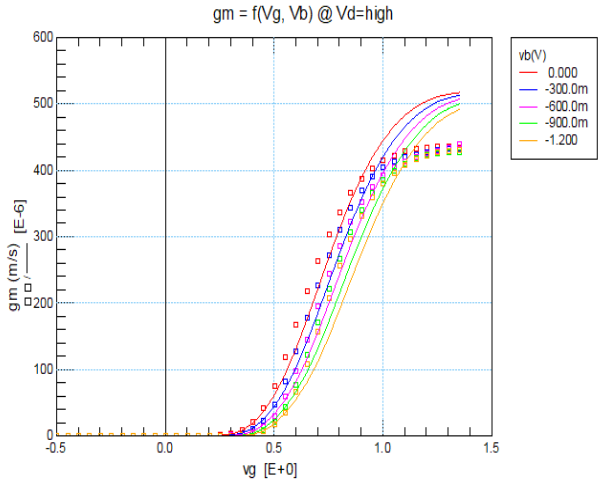


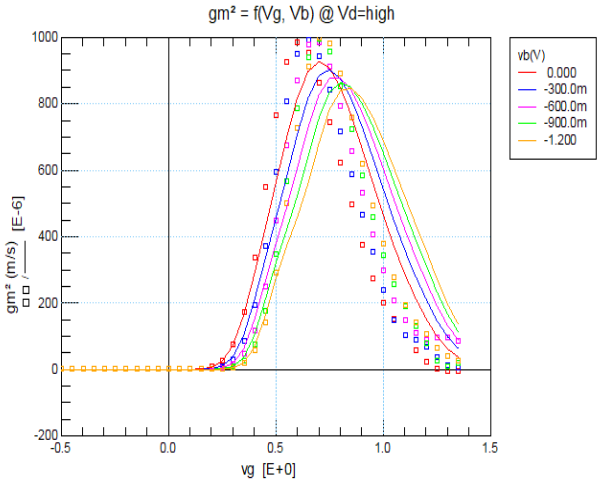


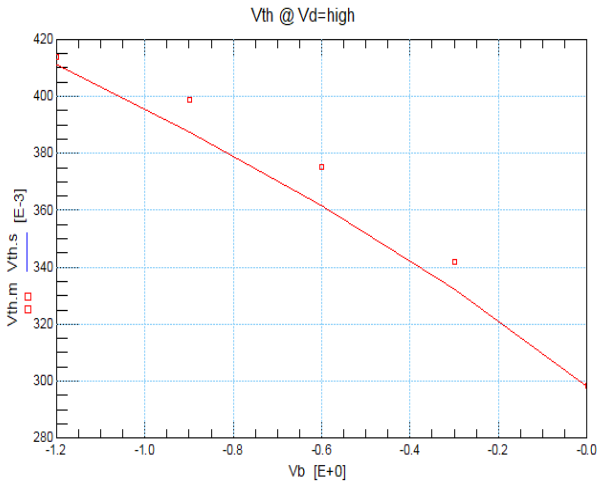
idvg, Vd = 1.2V, T = 27°C

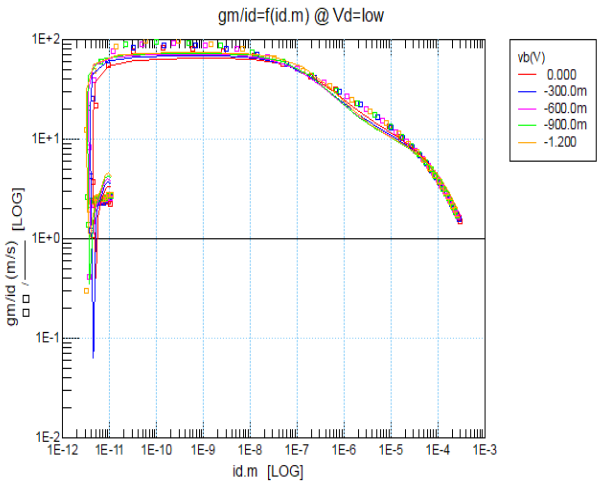


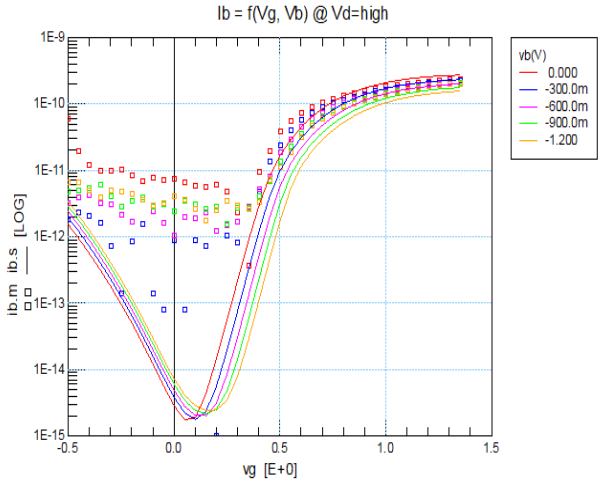


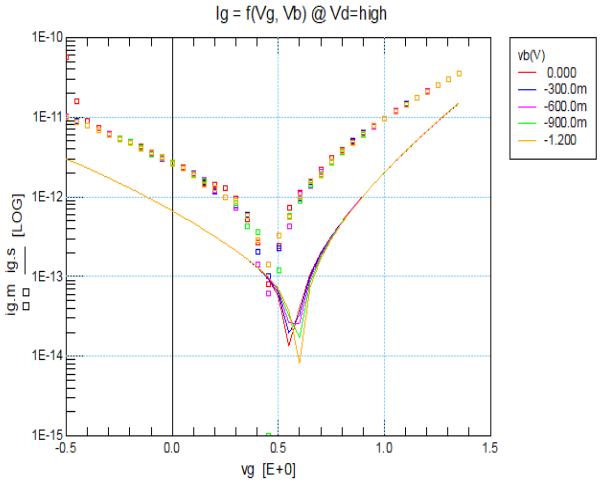




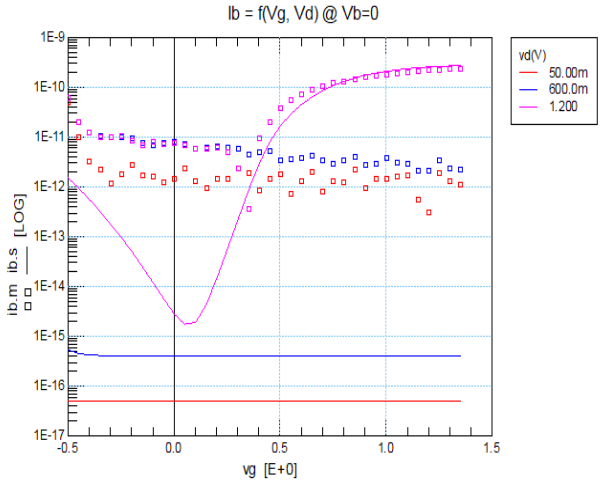


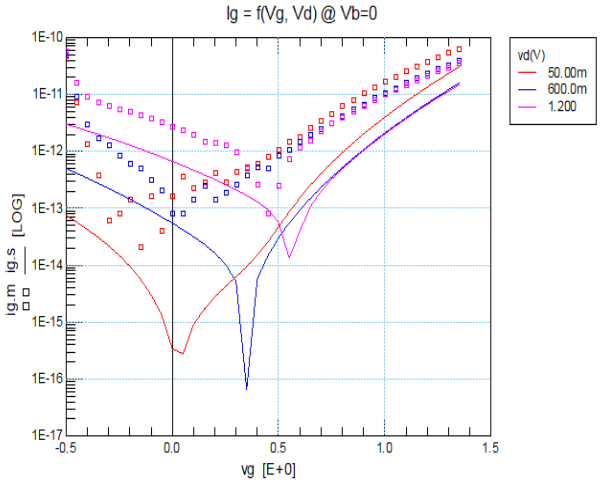




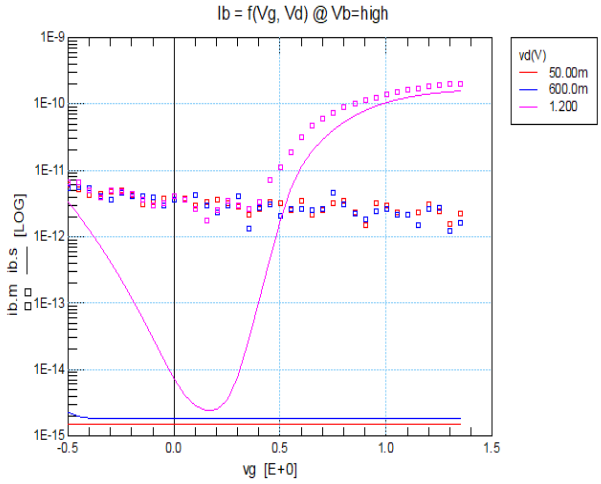


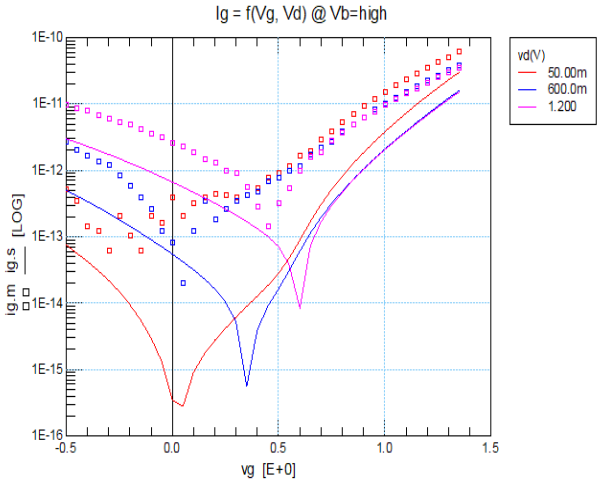
idvg, Ib,Ig, Vb = 0V, T = 27°C



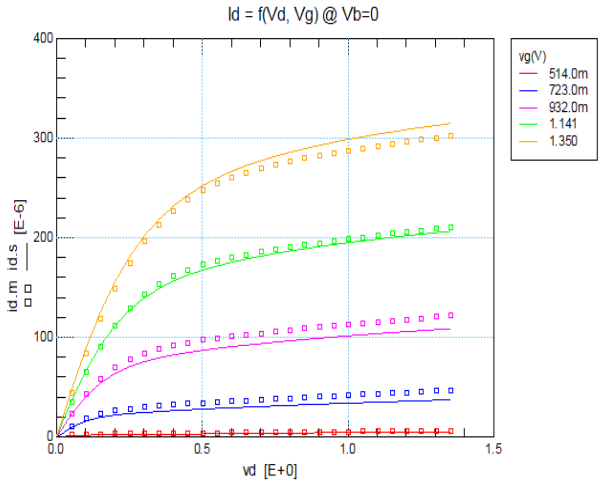


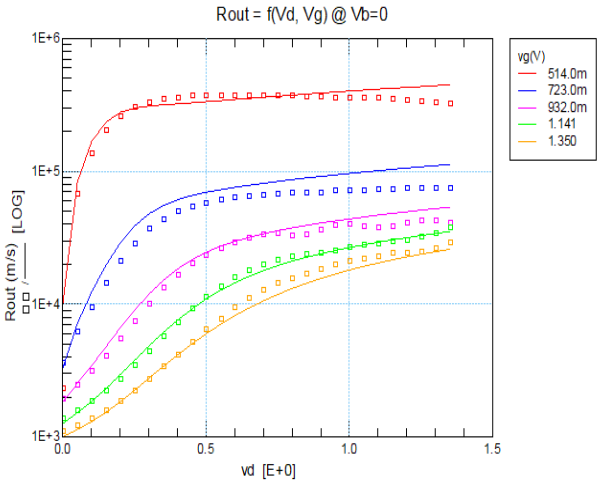
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

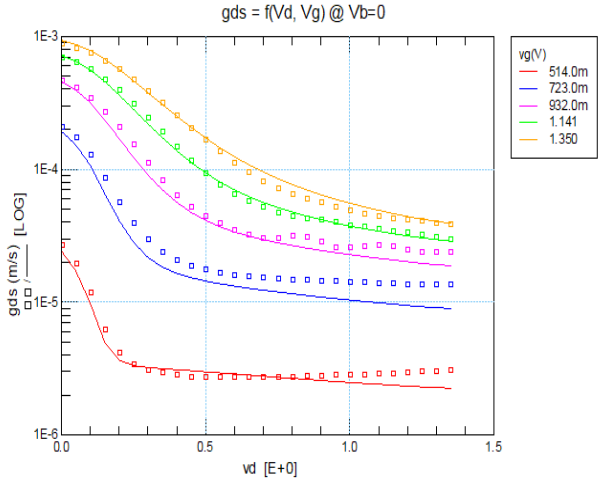




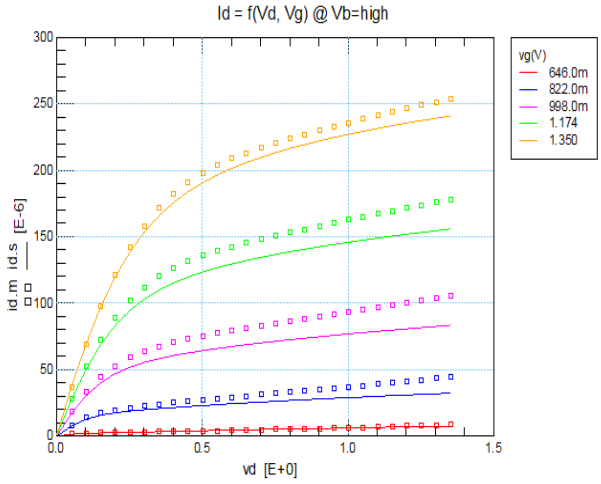
idvd, Vb = 0V, T = 27°C

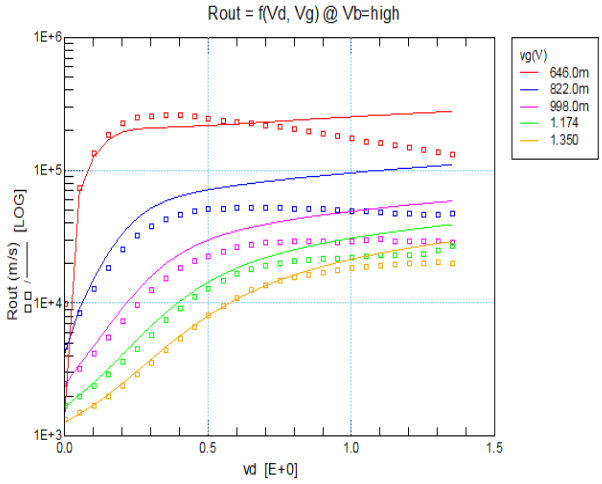


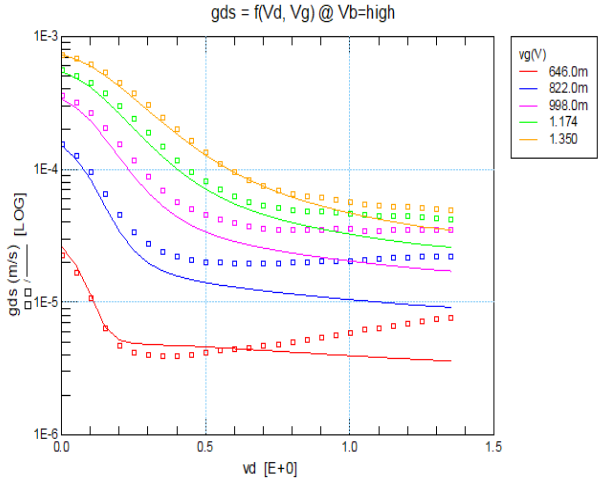




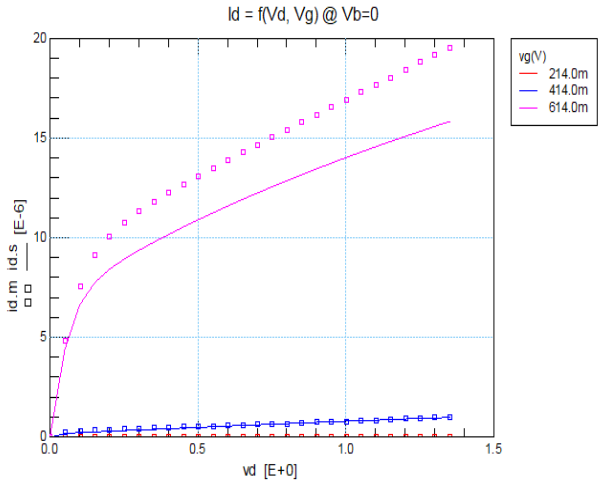
idvd_vbmin, Vb = -1.2V, T = 27°C

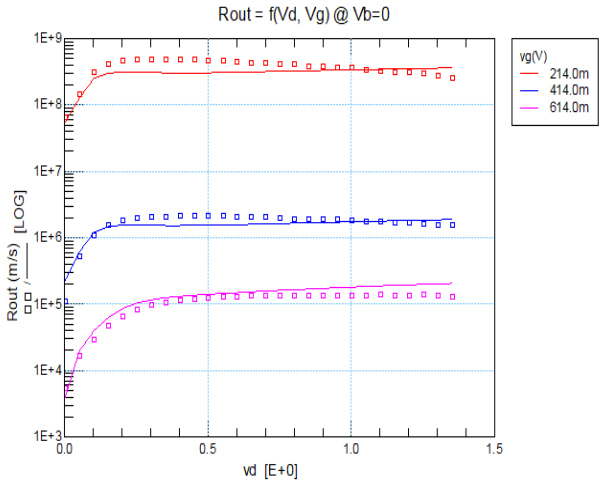


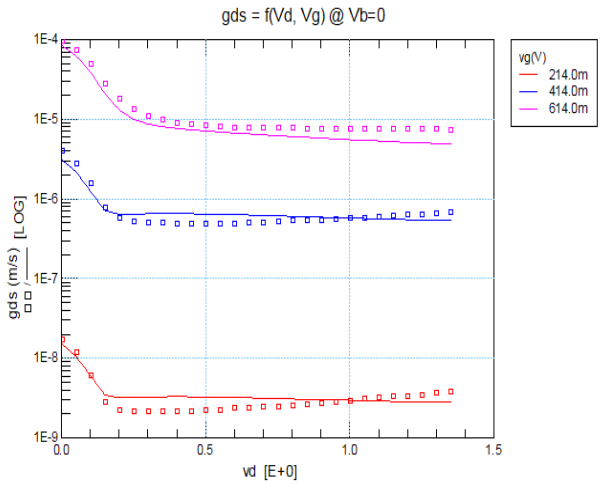




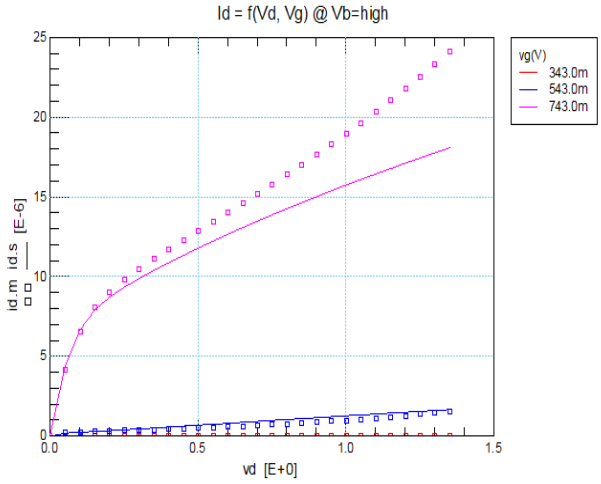
idvd_vth, Vb = 0V, T = 27°C

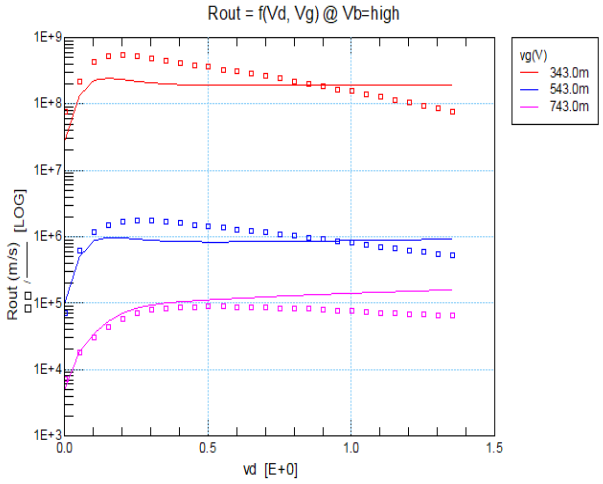


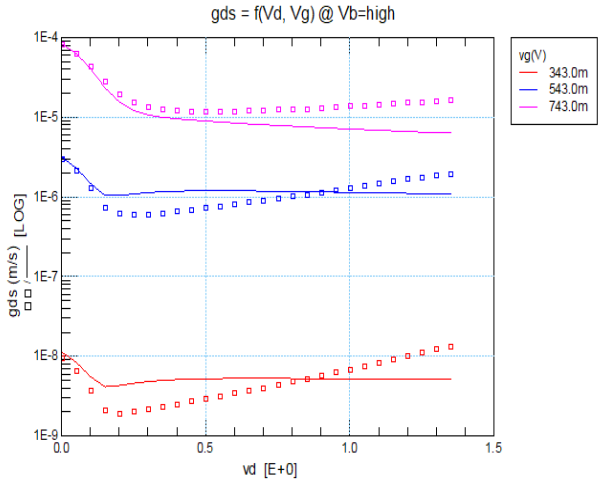




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

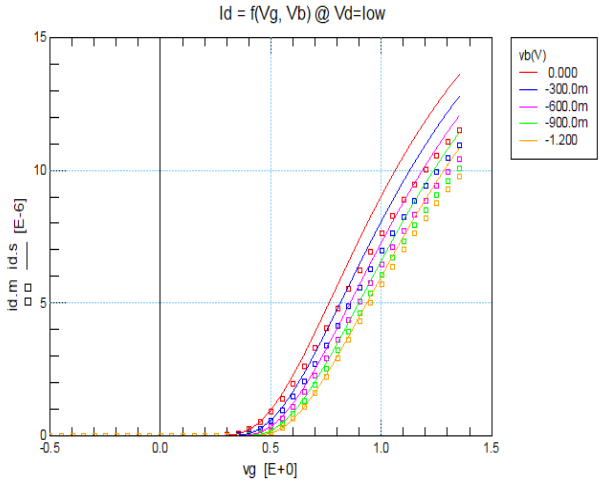


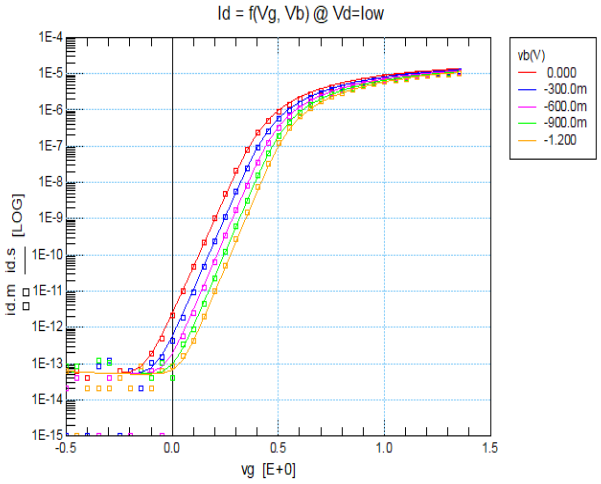


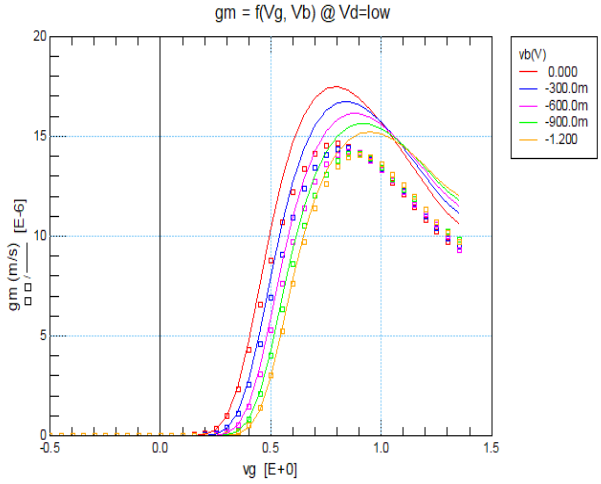


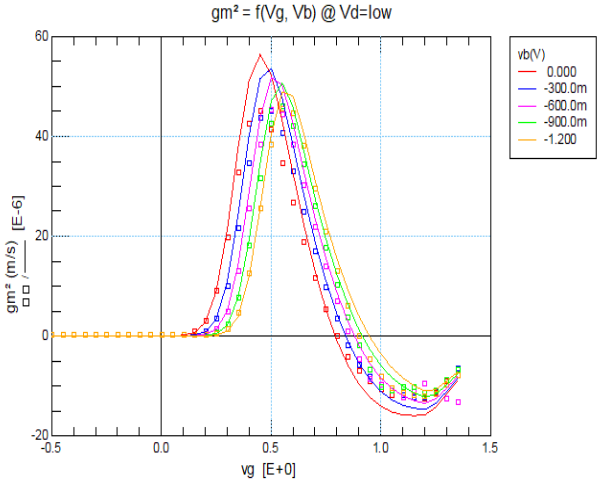
4.34 W0u15_L0u18_S547_1

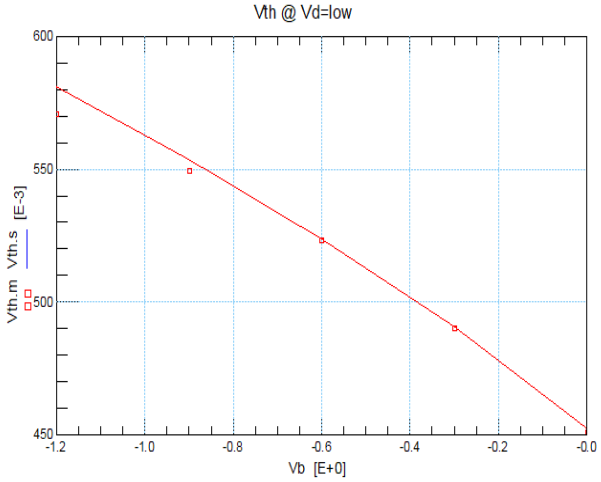
idvg, Vd = 0.05V, T = 27°C

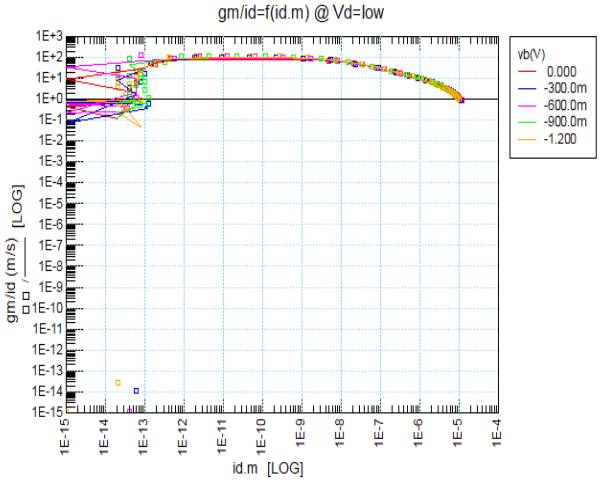


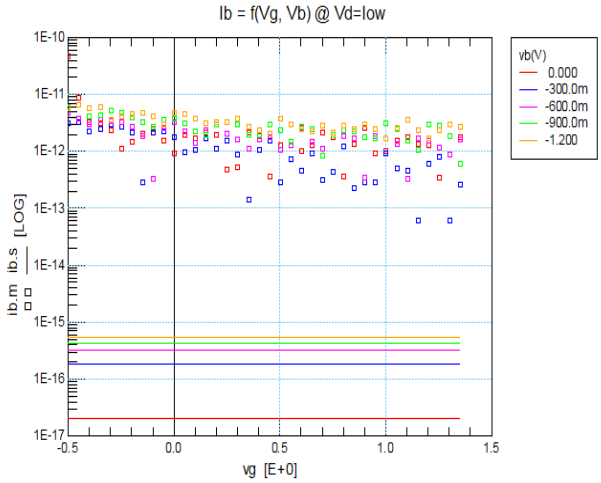


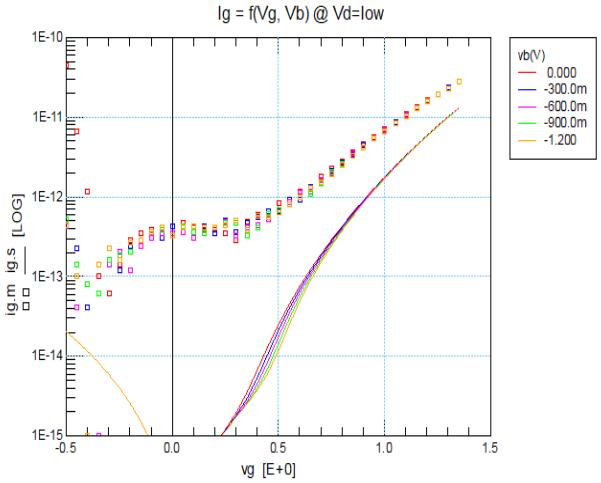




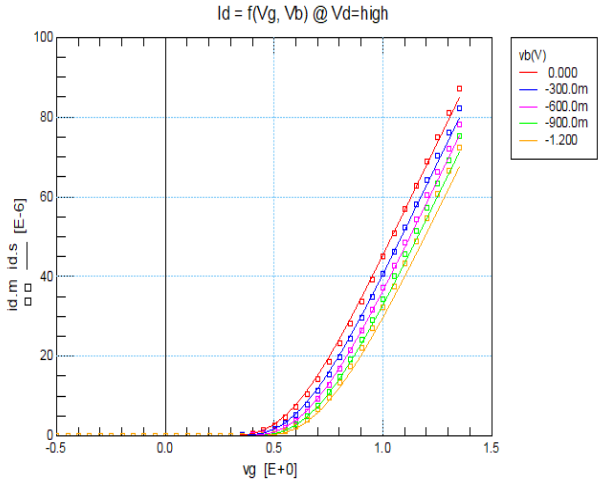


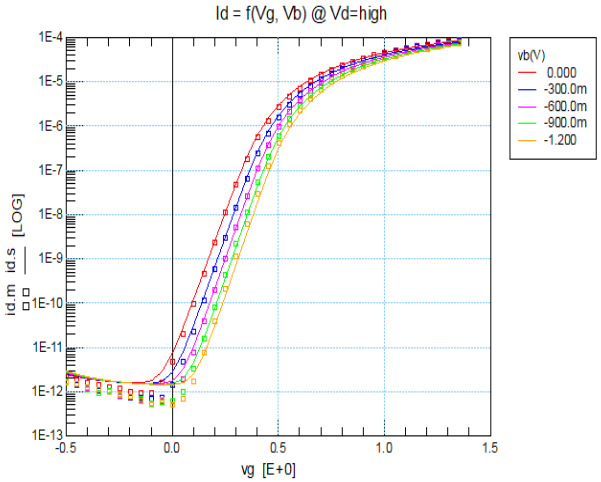


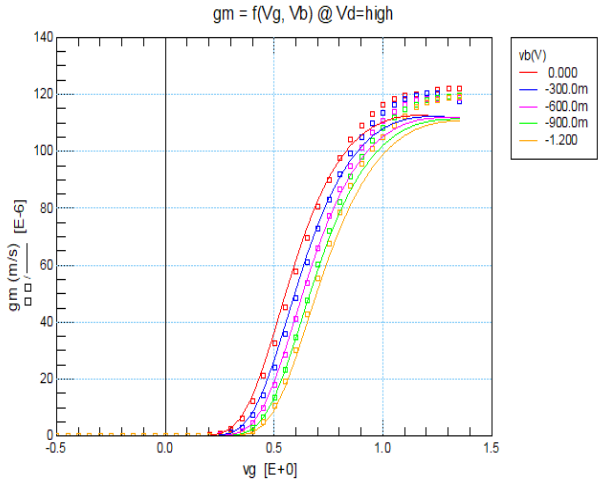


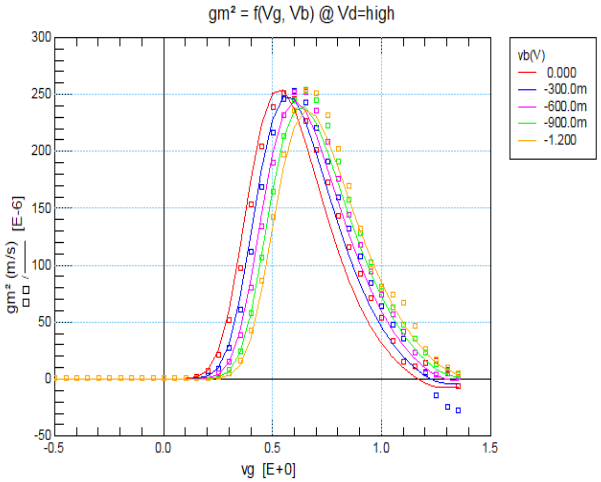


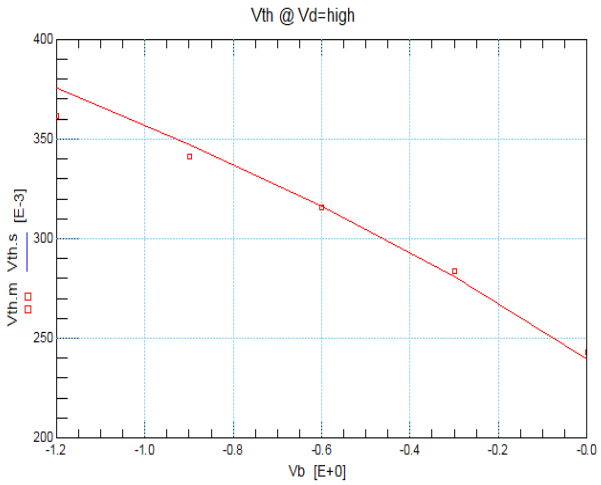
idvg, Vd = 1.2V, T = 27°C

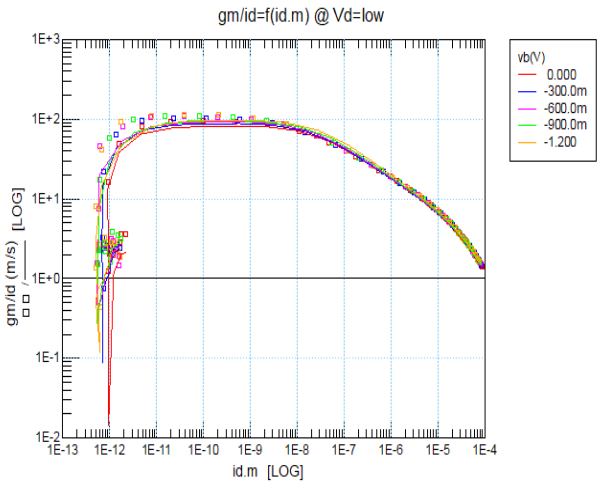


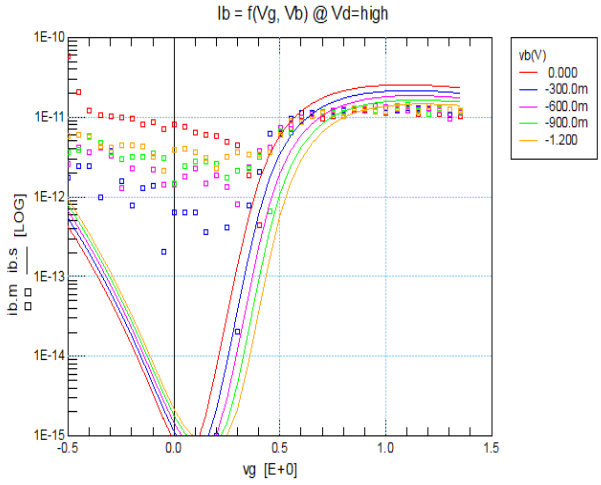


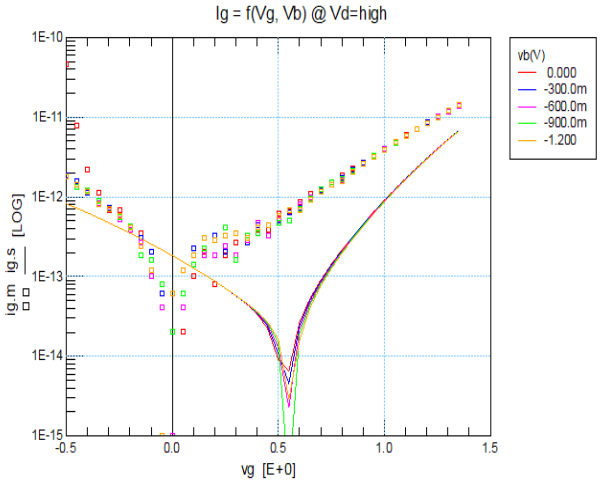




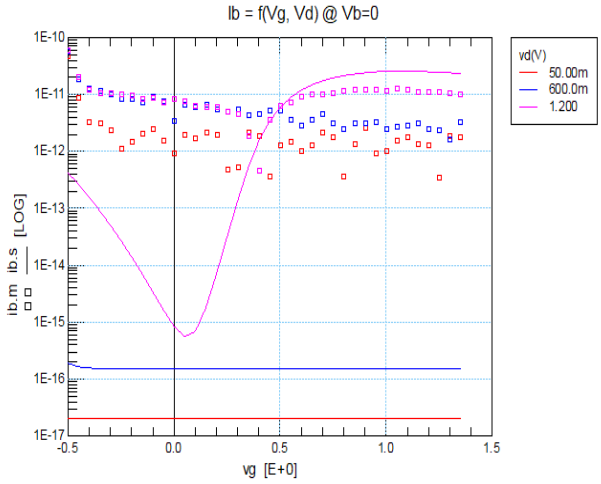


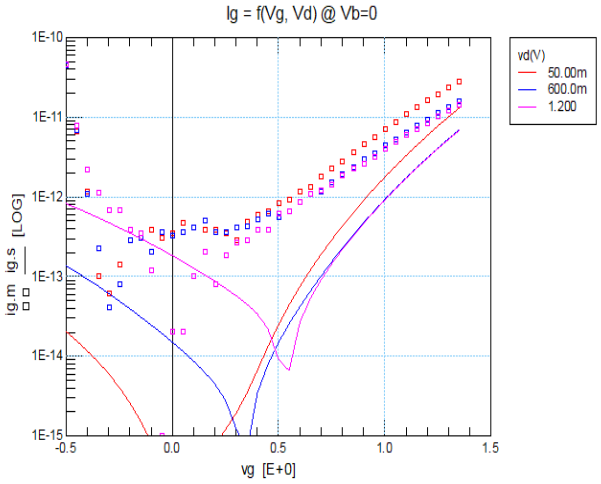




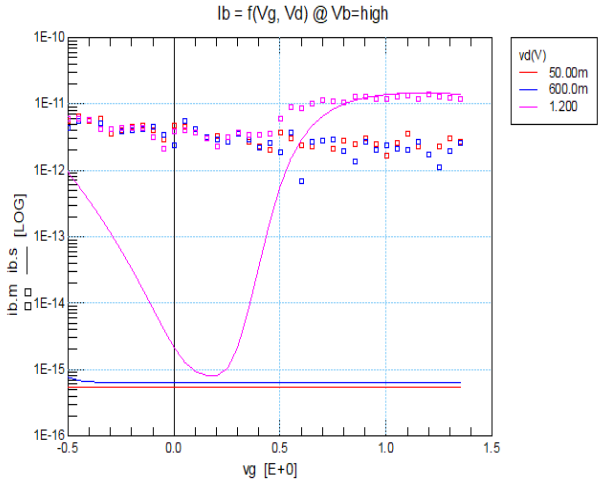


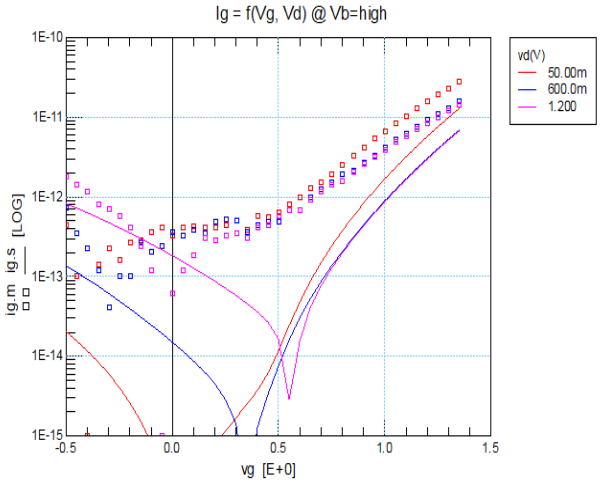
idvg, Ib,Ig, Vb = 0V, T = 27°C



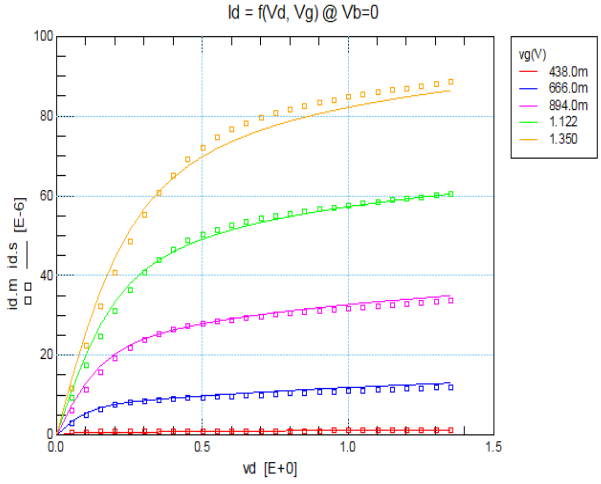


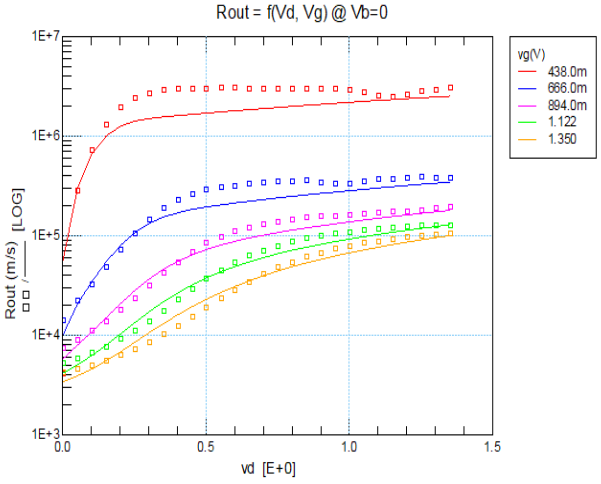
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

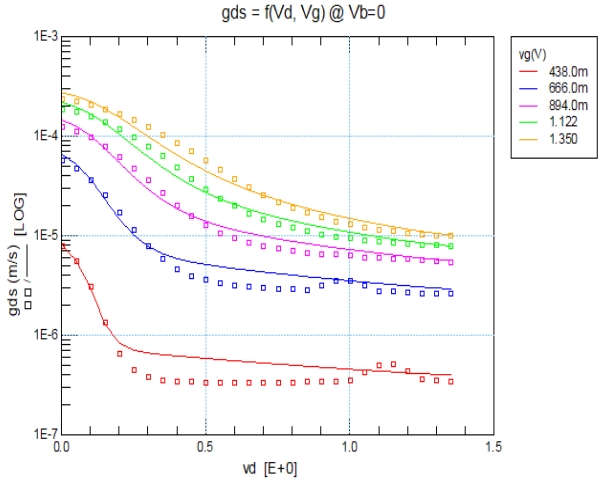




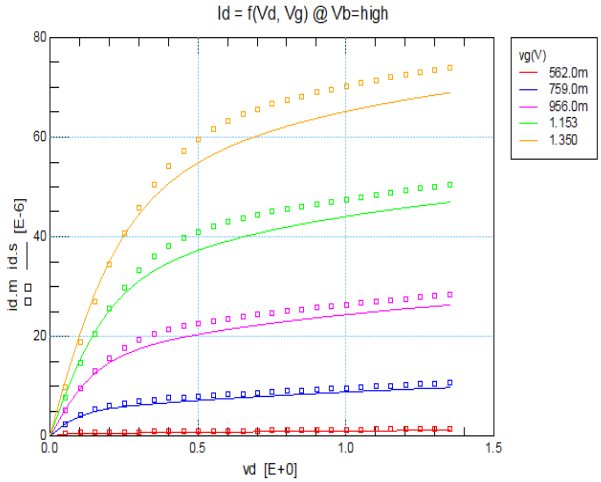
idvd, Vb = 0V, T = 27°C

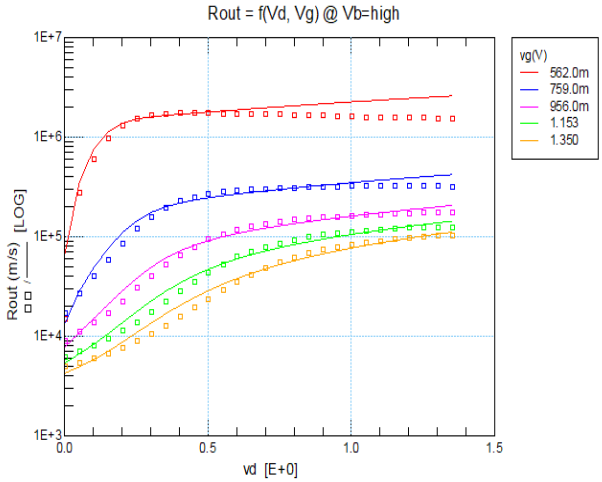


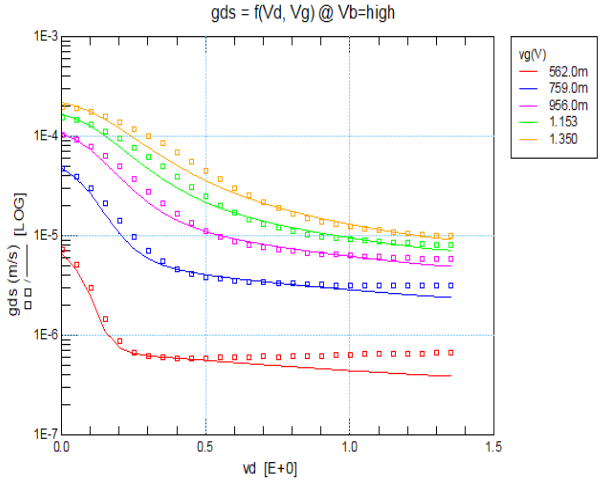




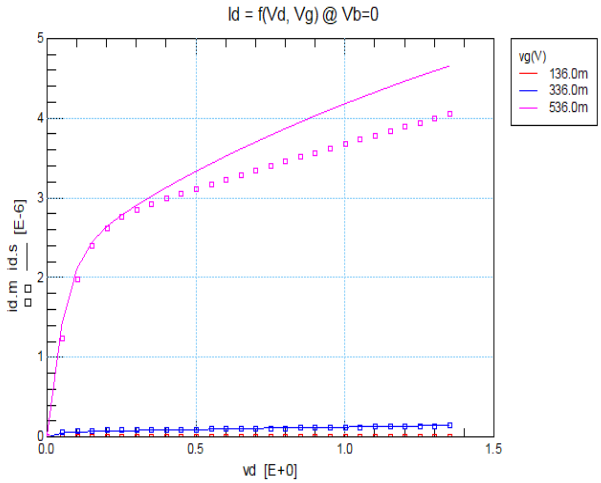
idvd_vbmin, Vb = -1.2V, T = 27°C

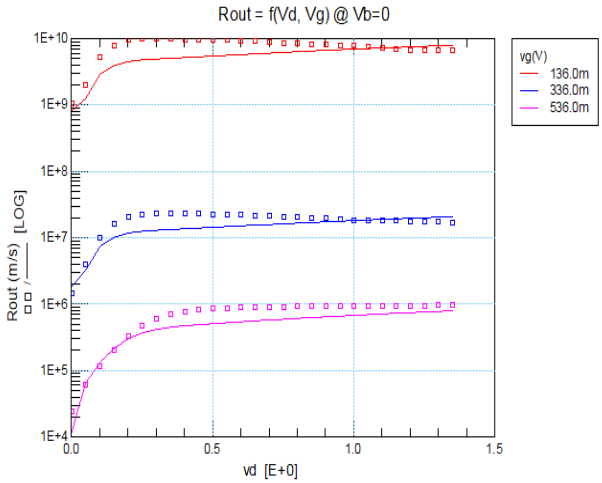


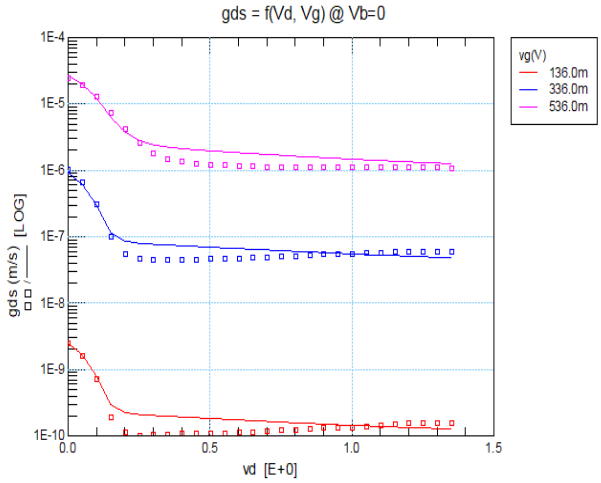




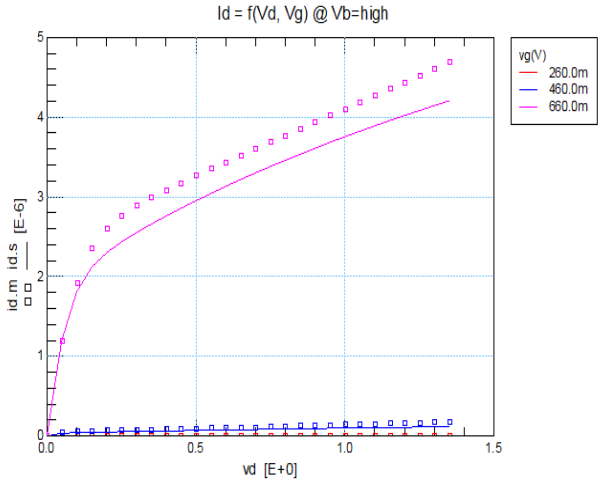
idvd_vth, Vb = 0V, T = 27°C

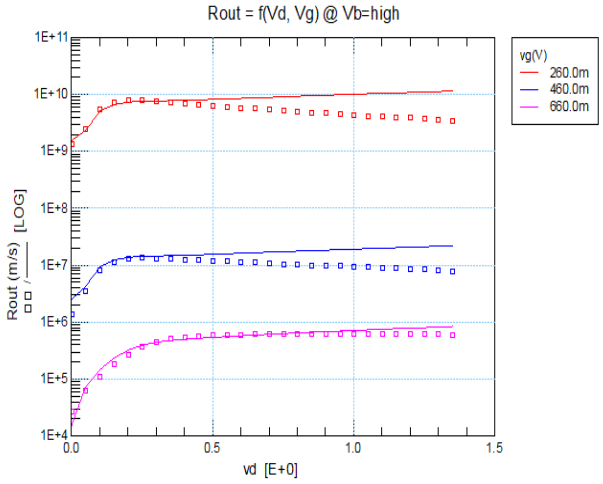


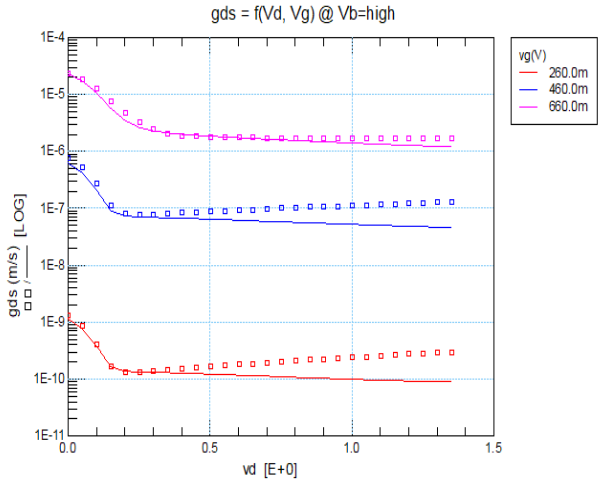




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

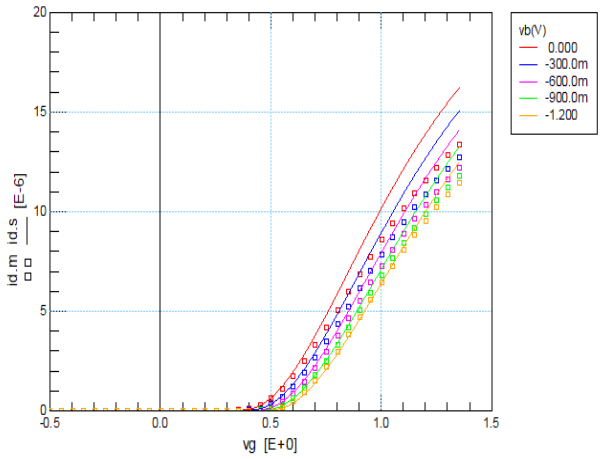




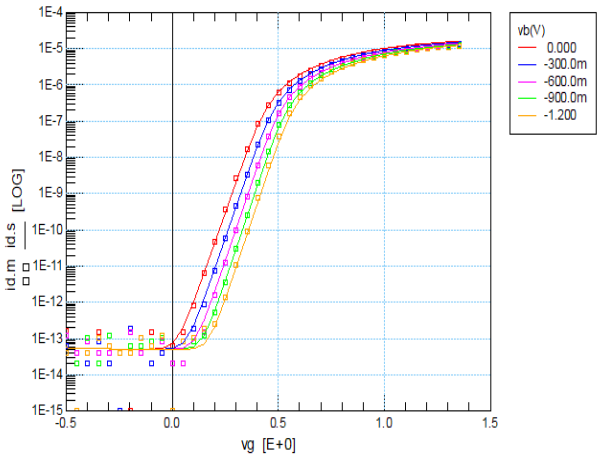


idvg, Vd = 0.05V, T = -40°C

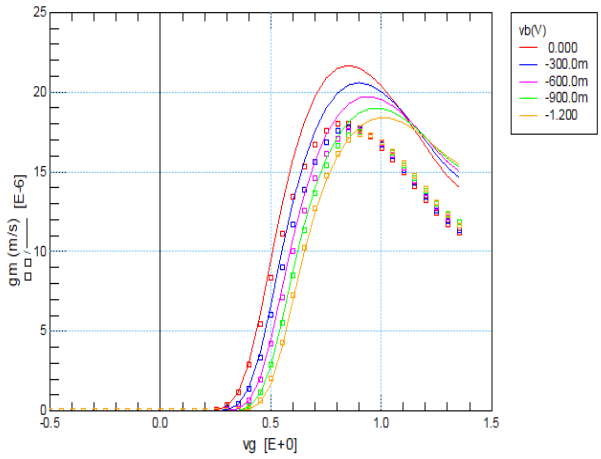
Id = f(Vg, Vb) @ Vd=low



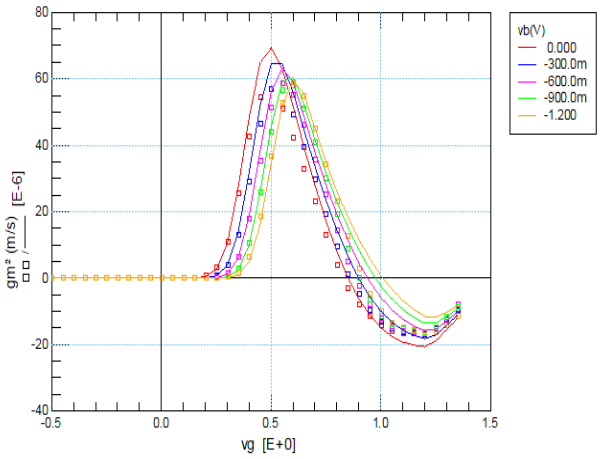
Id = f(Vg, Vb) @ Vd=low



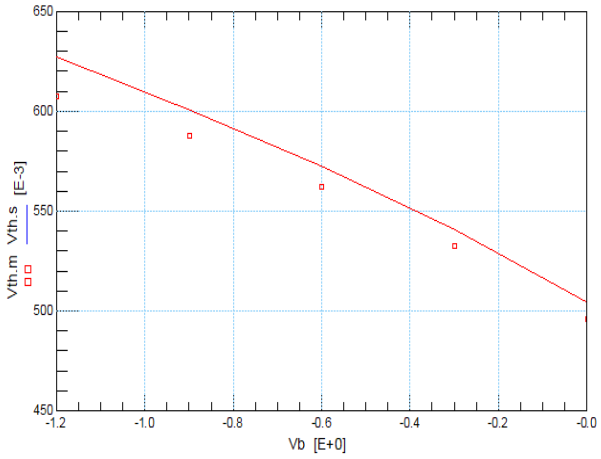
gm = f(Vg, Vb) @ Vd=low



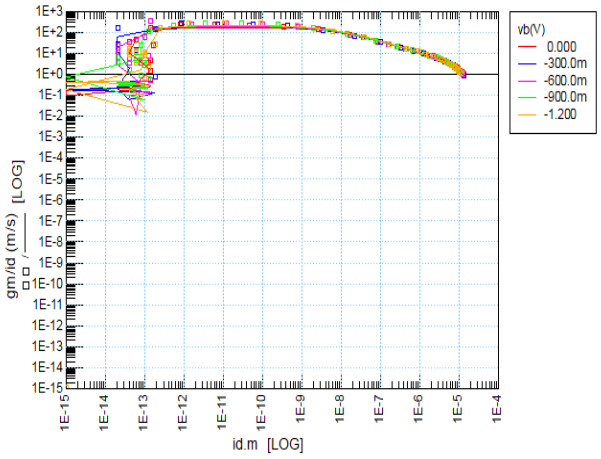
gm² = f(Vg, Vb) @ Vd=low

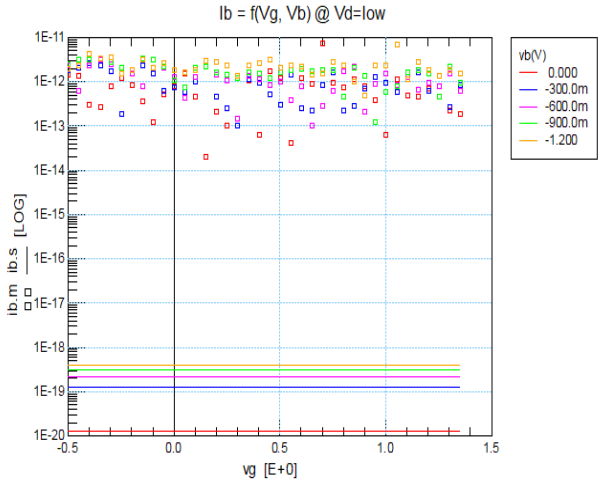


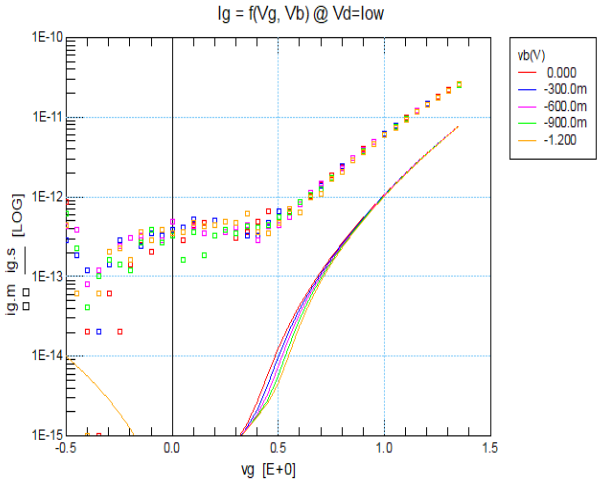
Vth @ Vd=low



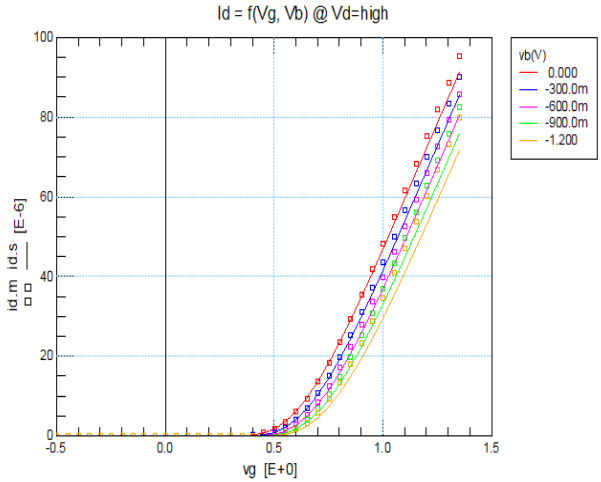
gm/id=f(id.m) @ Vd=low

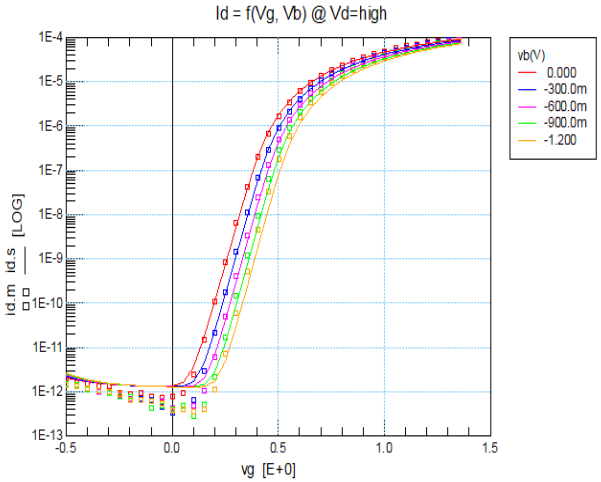


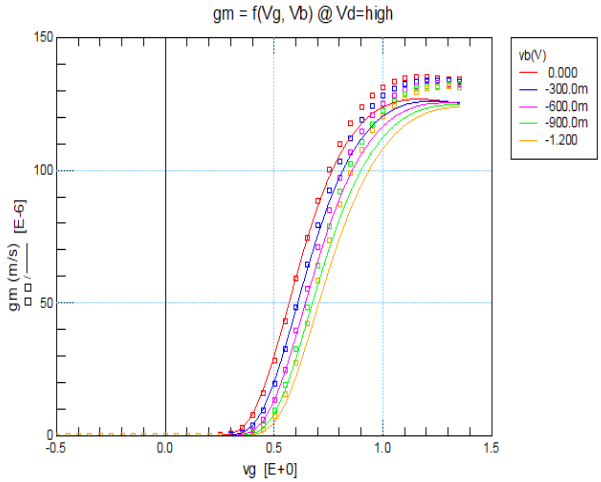


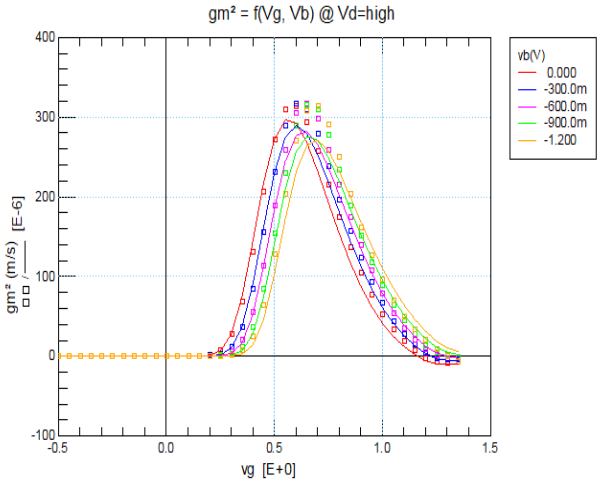


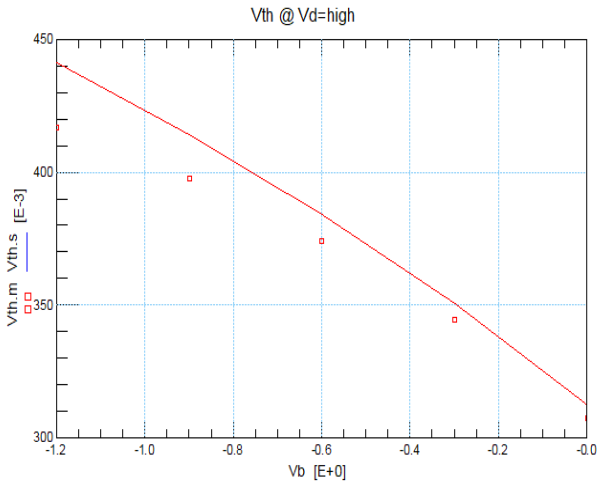
idvg, Vd = 1.2V, T = -40°C

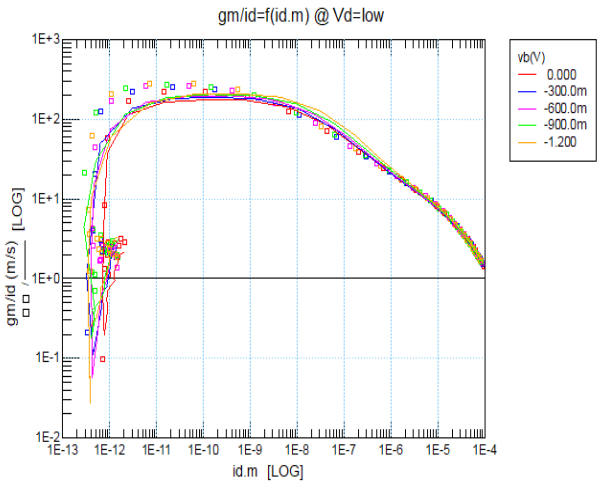


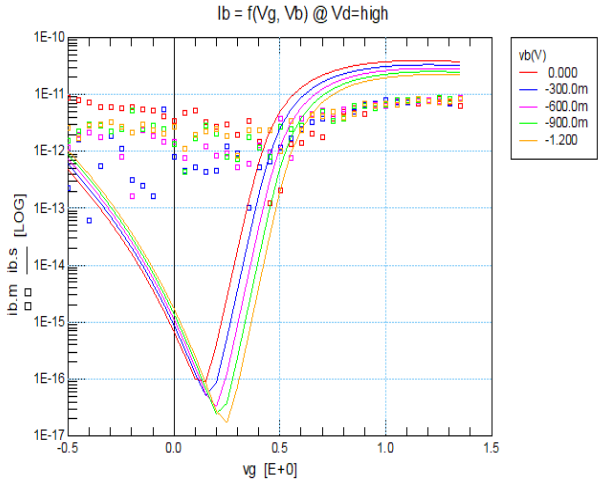


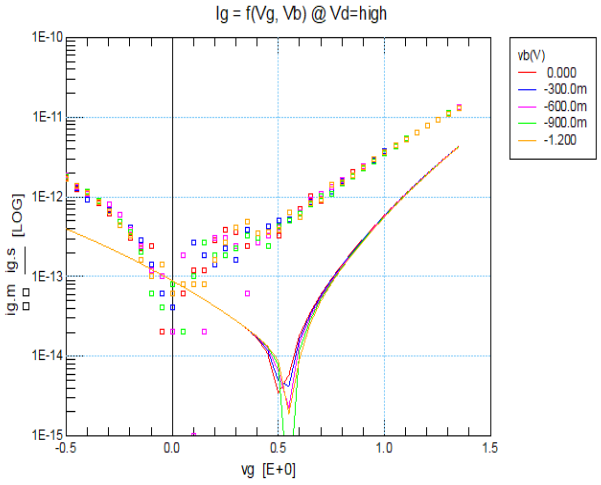




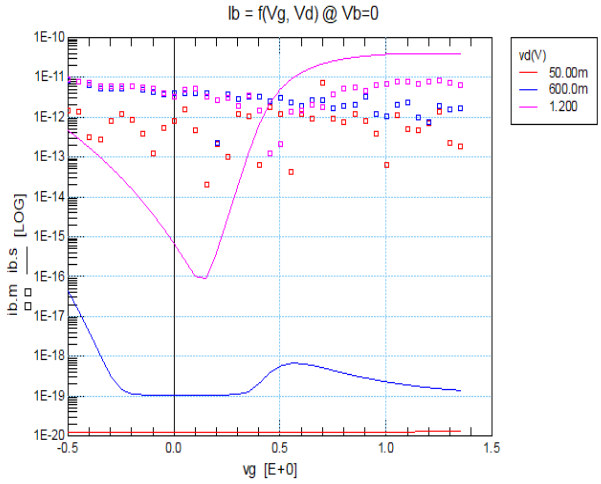


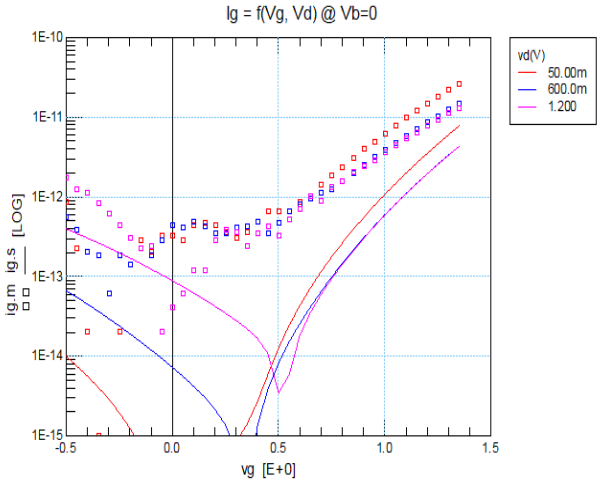




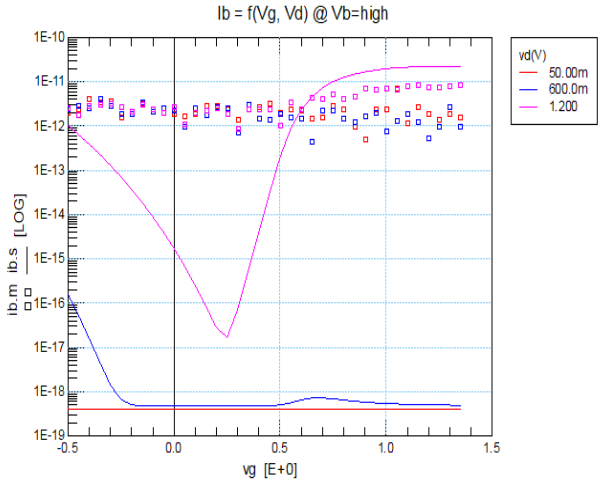


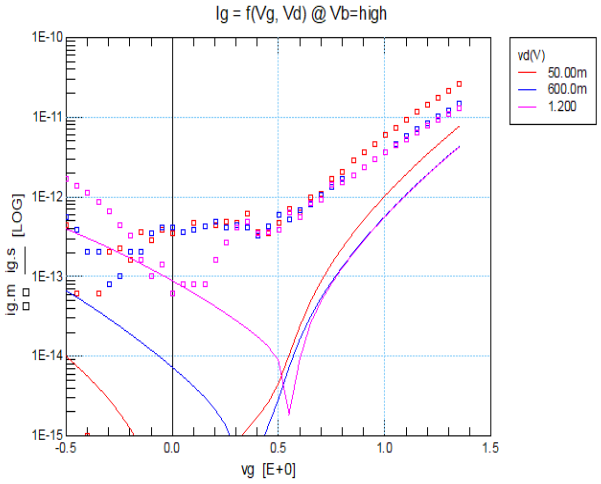
idvg, Ib,Ig, Vb = 0V, T = -40°C



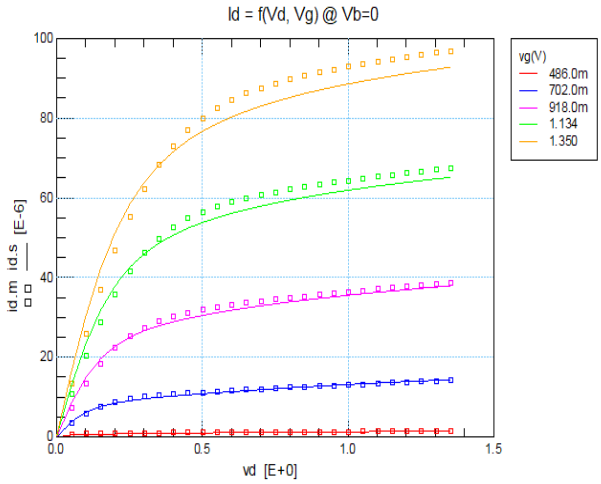


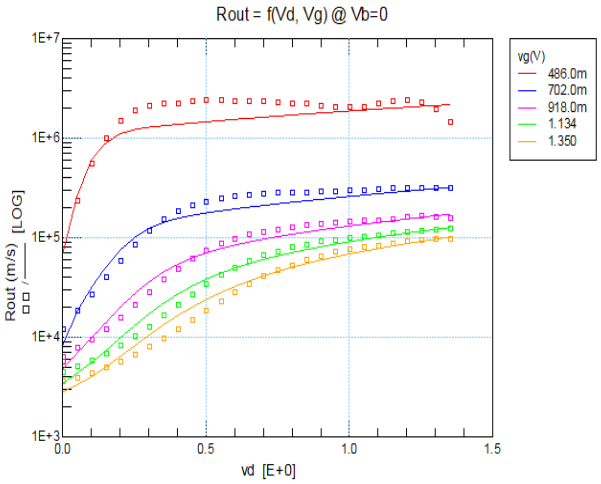
idvg, Ib,Ig, Vb = -1.2V, T = -40°C

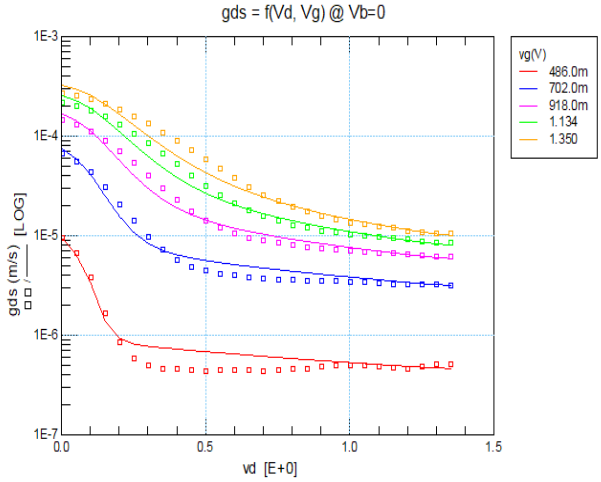




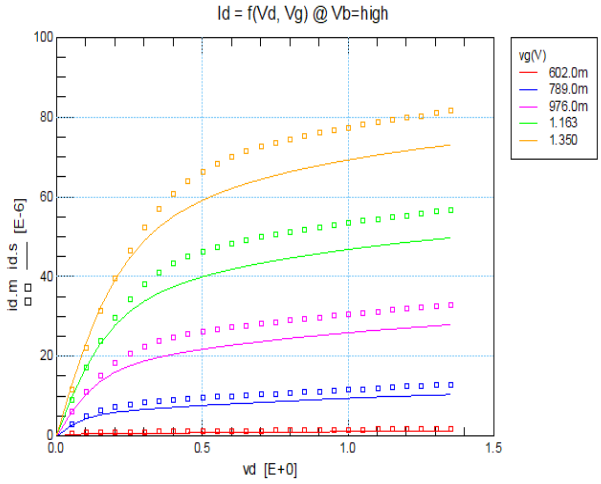
idvd, Vb = 0V, T = -40°C

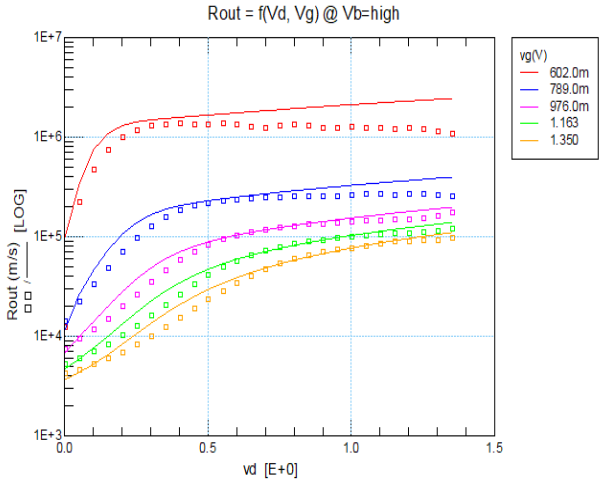


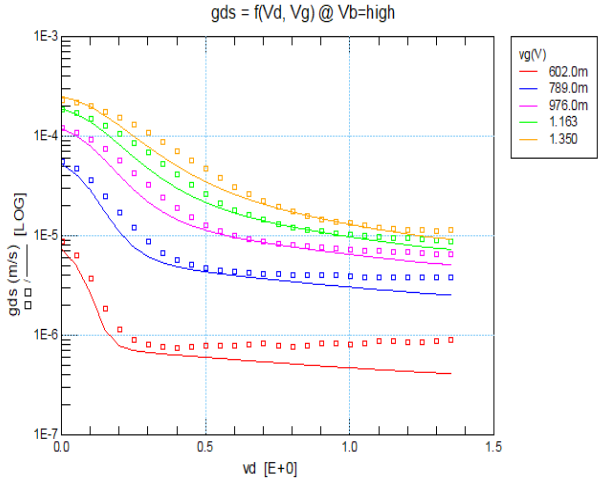




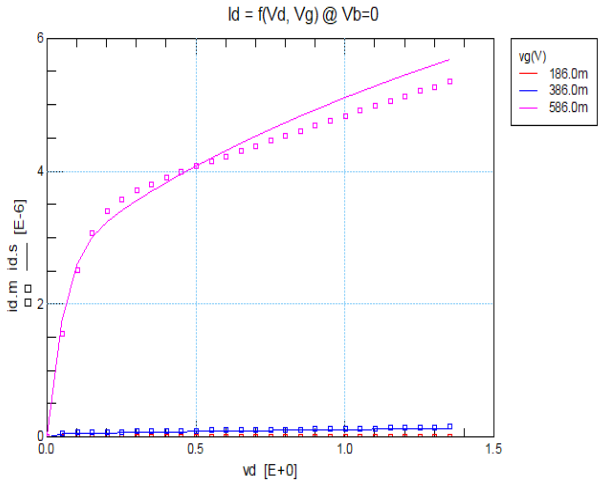
idvd_vbmin, Vb = -1.2V, T = -40°C

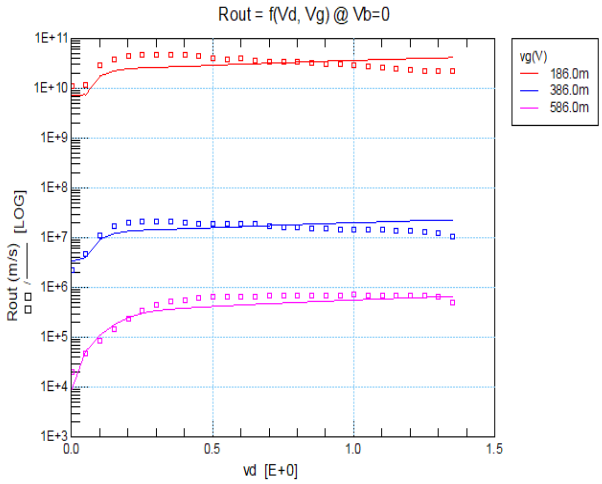


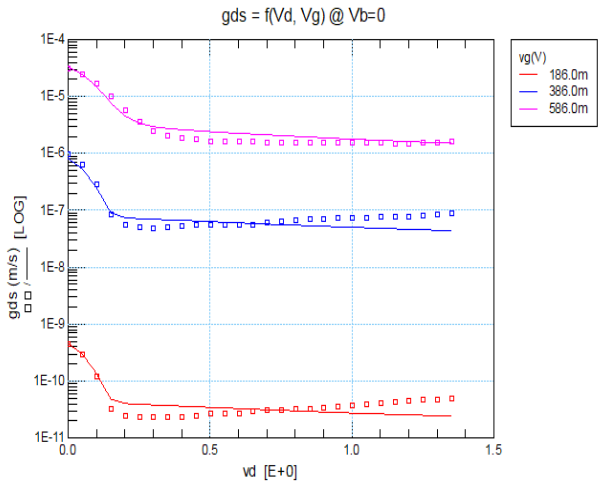




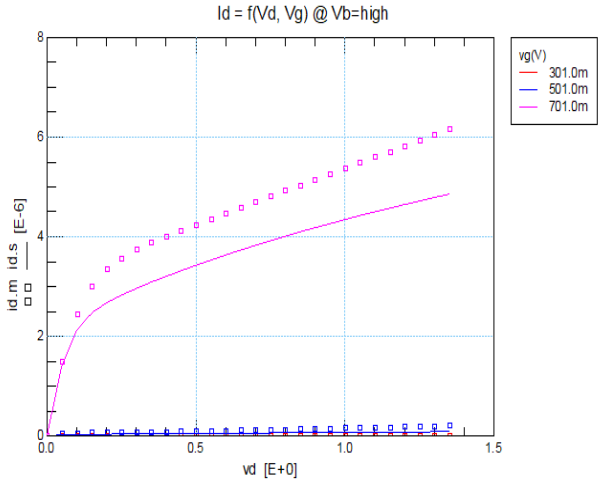
idvd_vth, Vb = 0V, T = -40°C

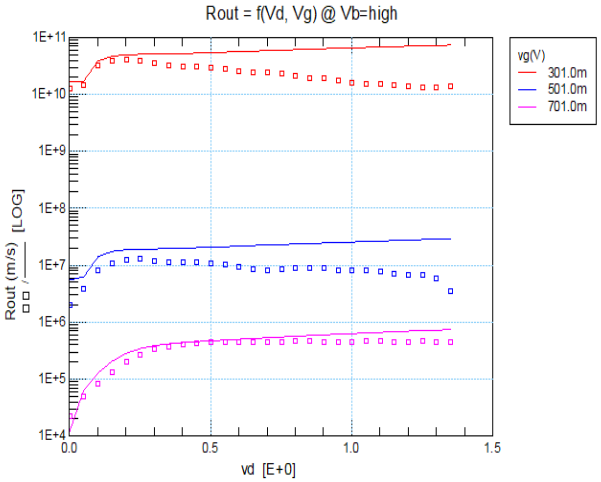


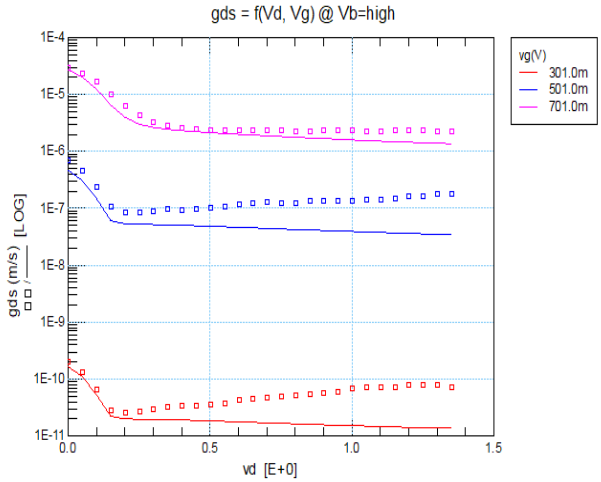




idvd_vth_vbmin, Vb = -1.2V, T = -40°C

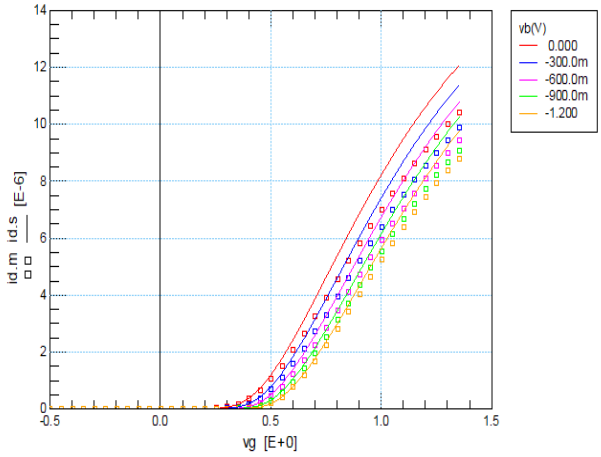




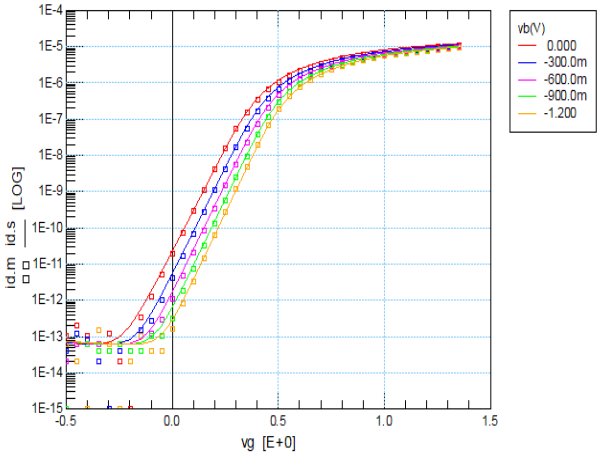


idvg, Vd = 0.05V, T = 70°C

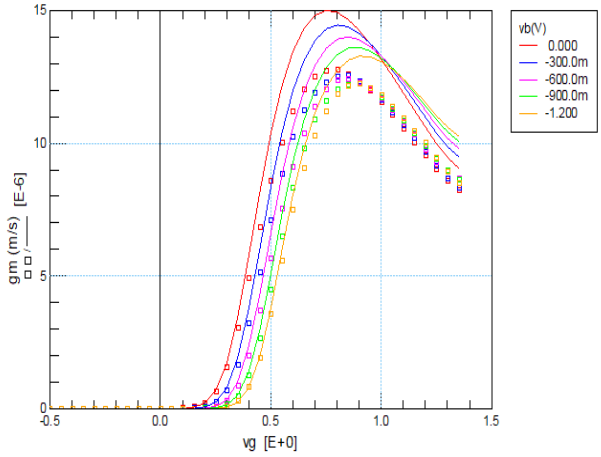
Id = f(Vg, Vb) @ Vd=low



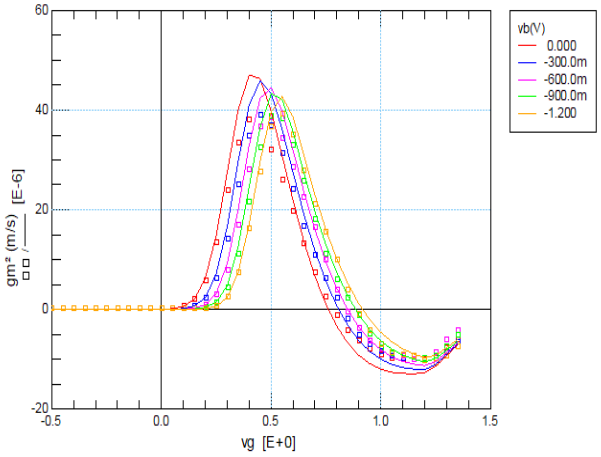
Id = f(Vg, Vb) @ Vd=low



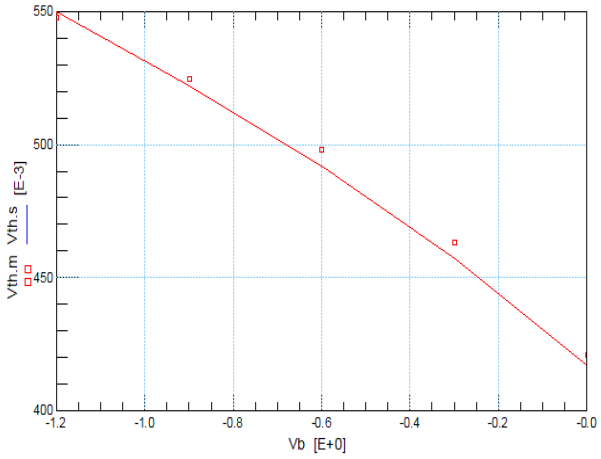
gm = f(Vg, Vb) @ Vd=low



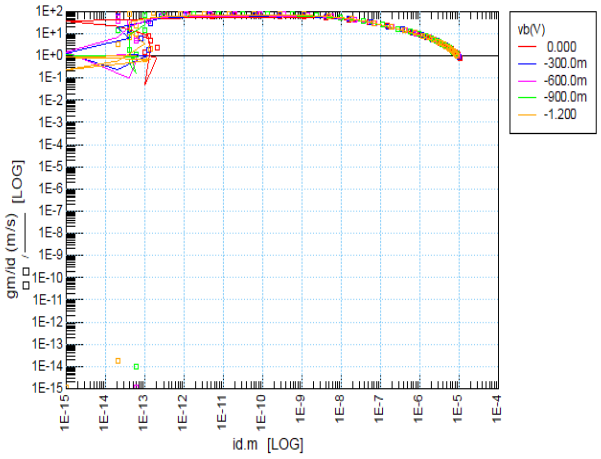
gm² = f(Vg, Vb) @ Vd=low

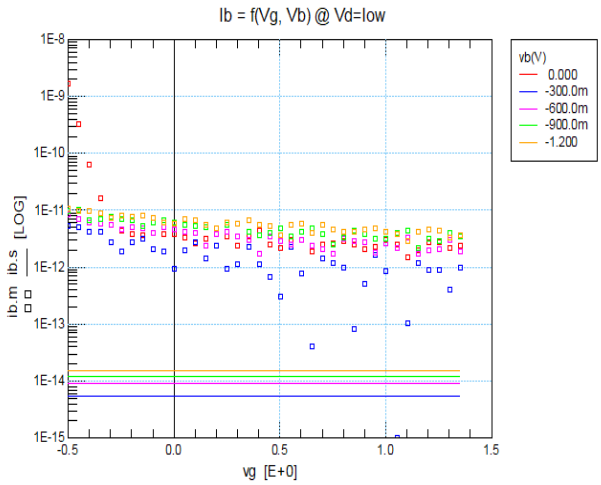


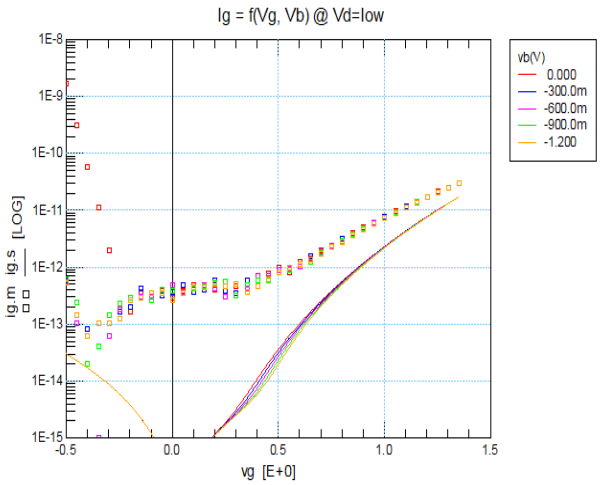
Vth @ Vd=low



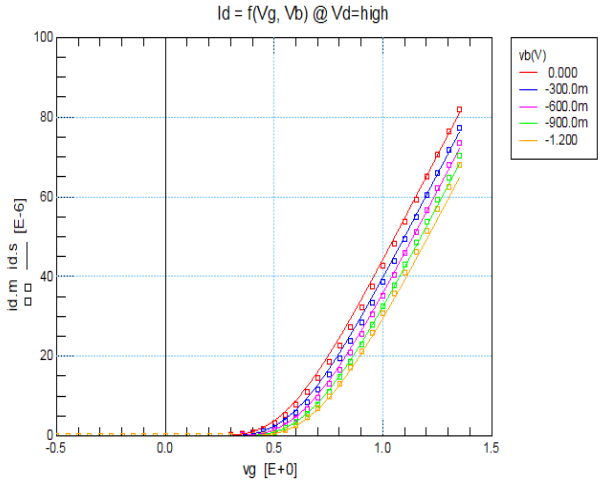
gm/id=f(id.m) @ Vd=low

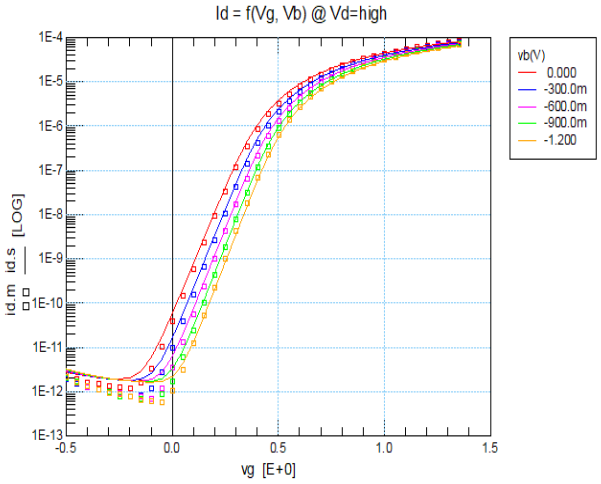


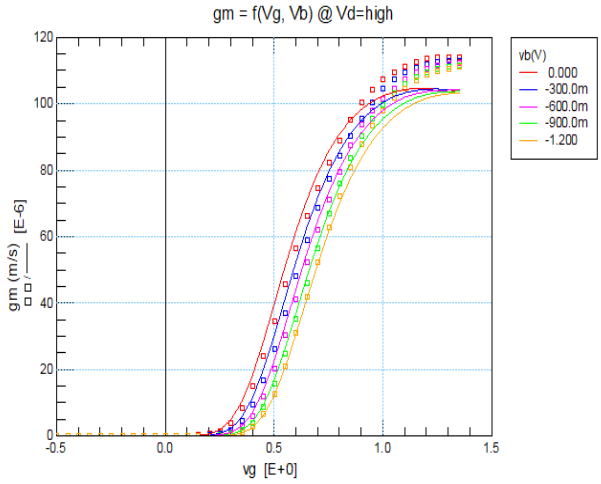


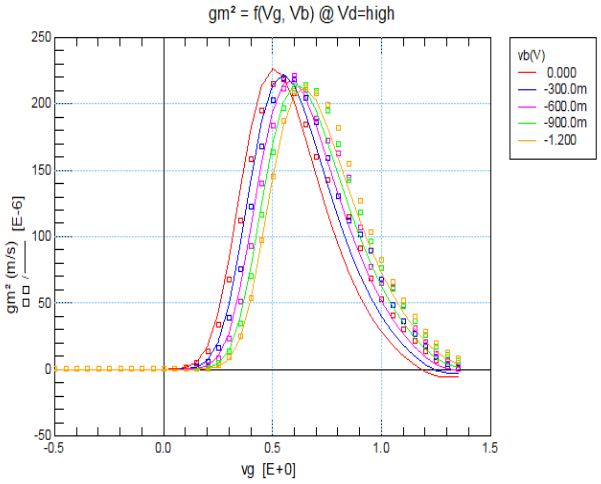


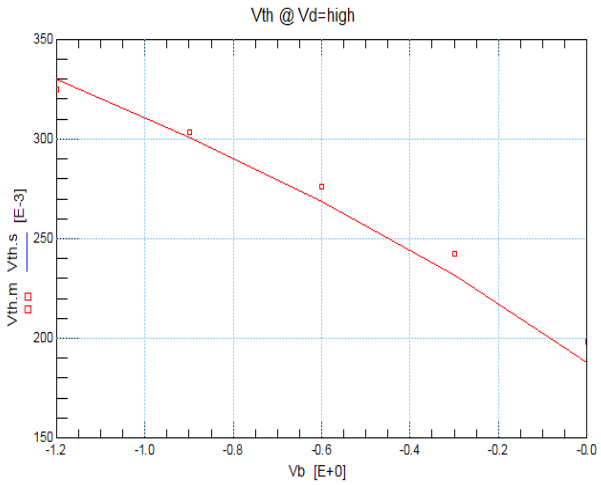
idvg, Vd = 1.2V, T = 70°C

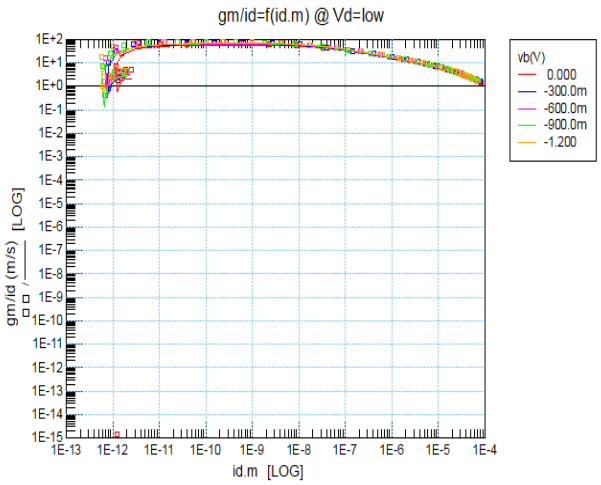


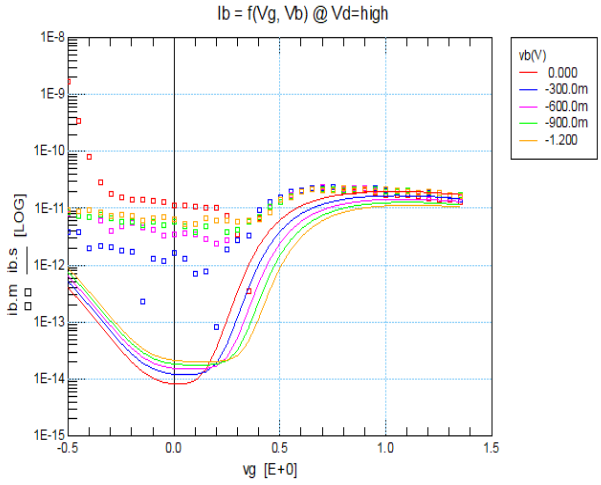


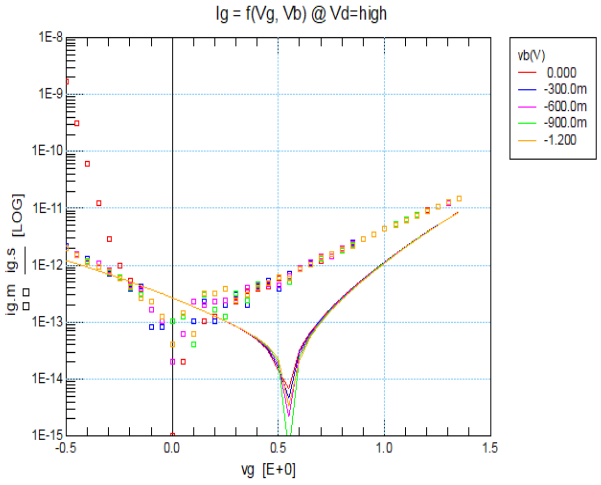




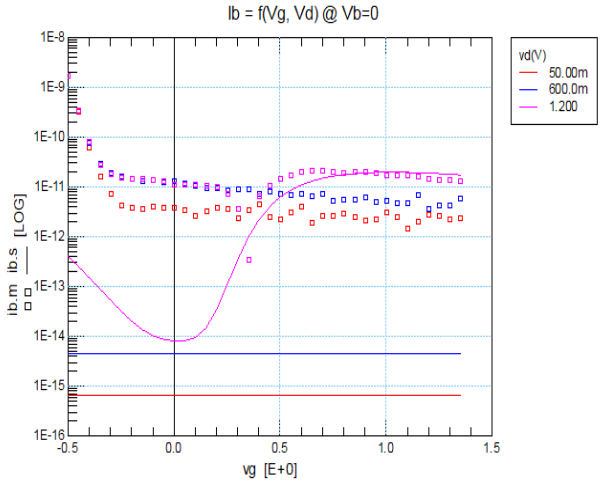


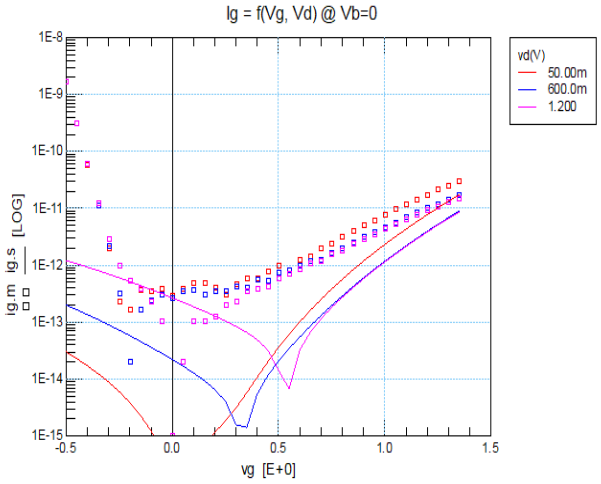




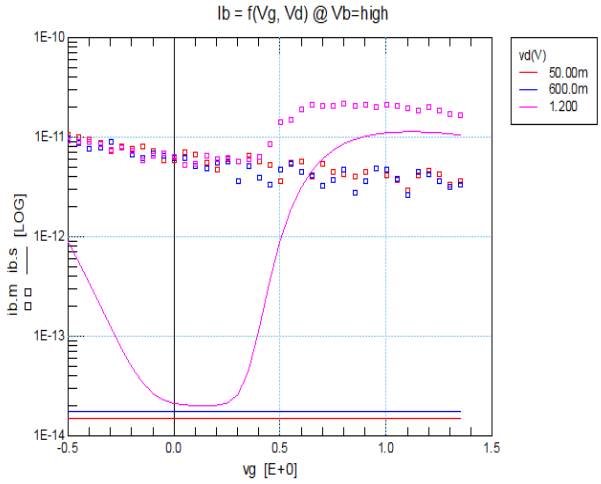


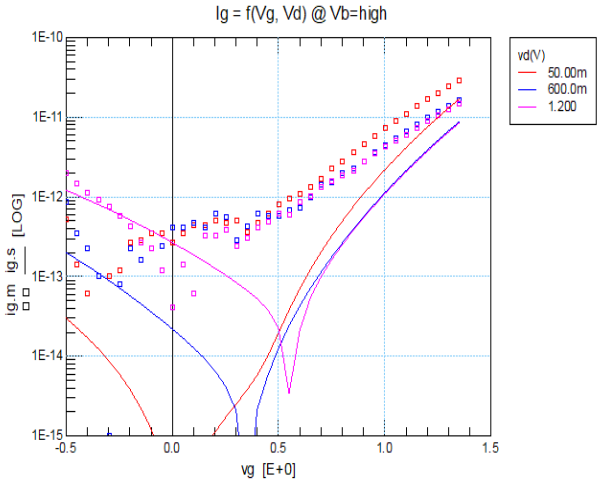
idvg, Ib,Ig, Vb = 0V, T = 70°C



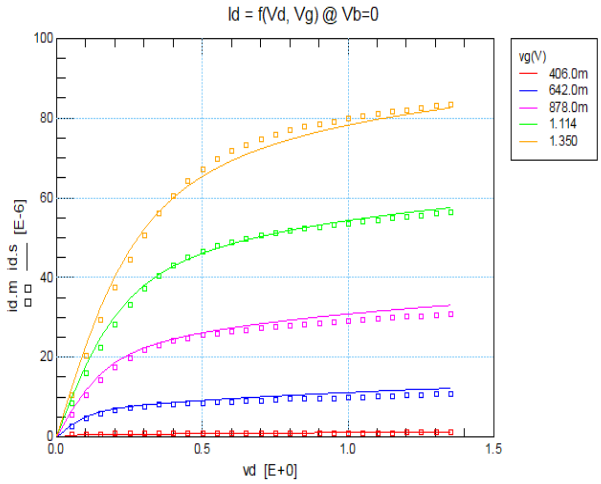


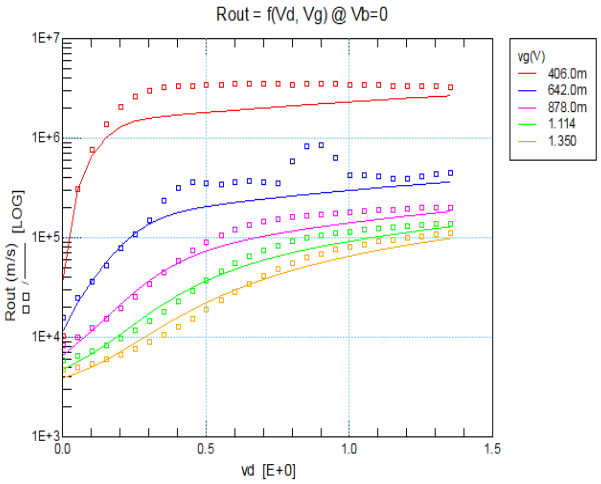
idvg, Ib,Ig, Vb = -1.2V, T = 70°C

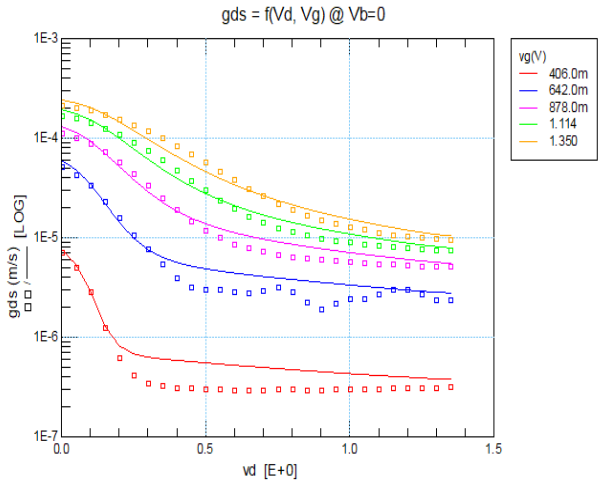




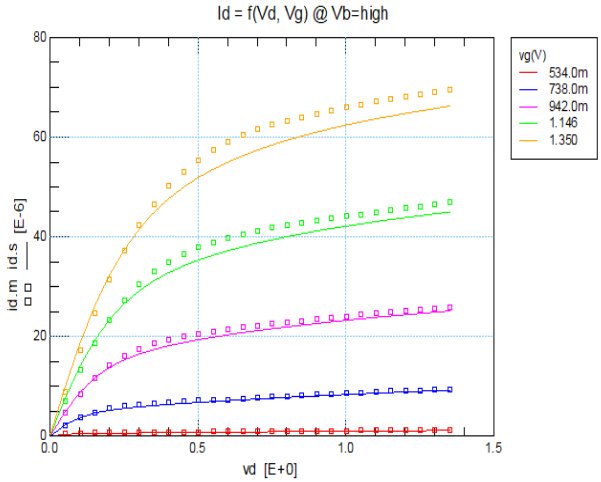
idvd, Vb = 0V, T = 70°C

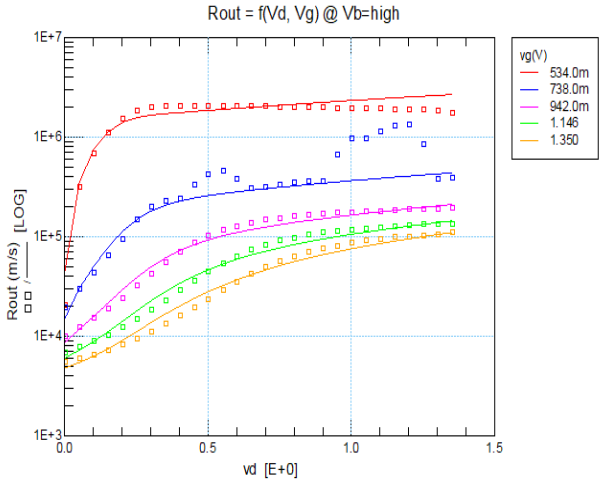


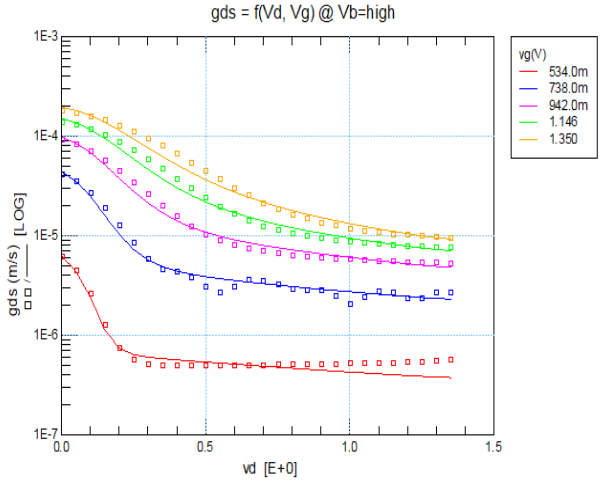




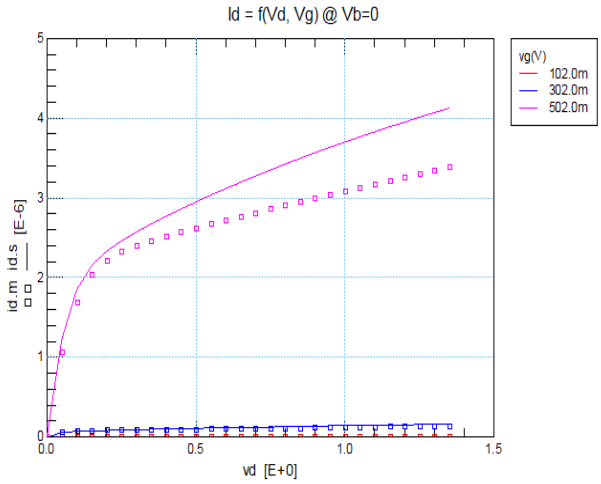
idvd_vbmin, Vb = -1.2V, T = 70°C

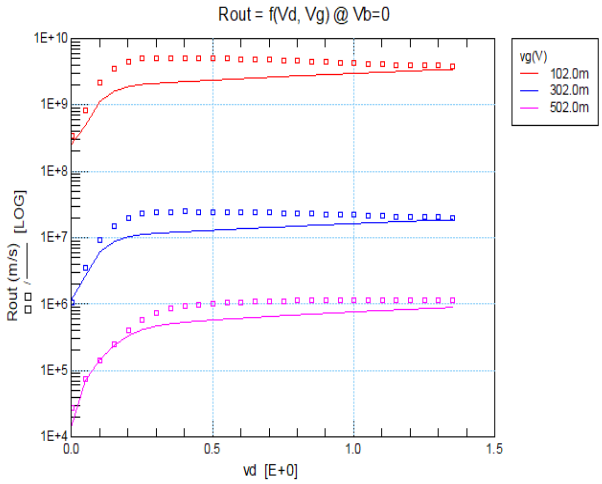


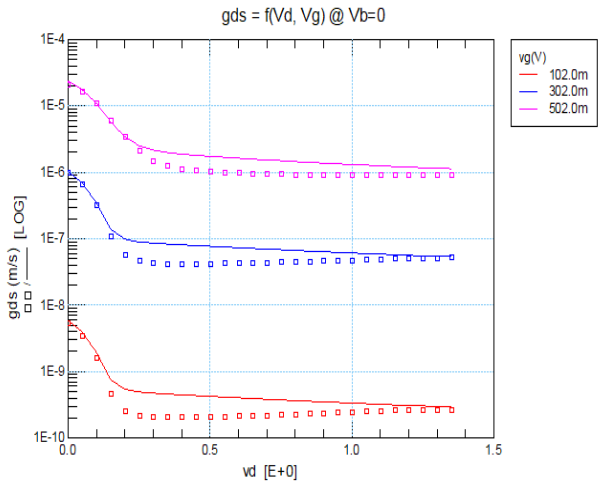




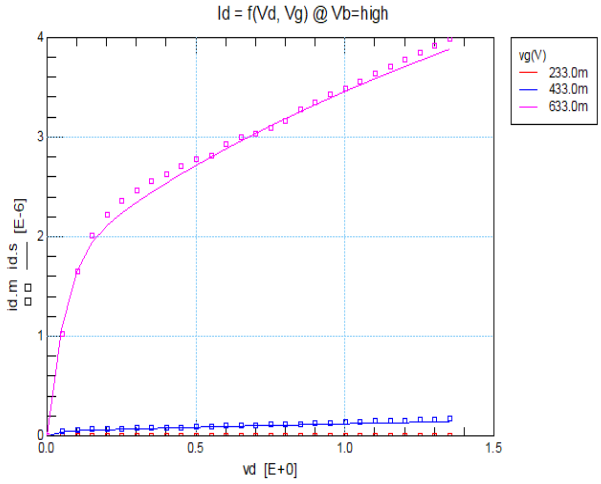
idvd_vth, Vb = 0V, T = 70°C

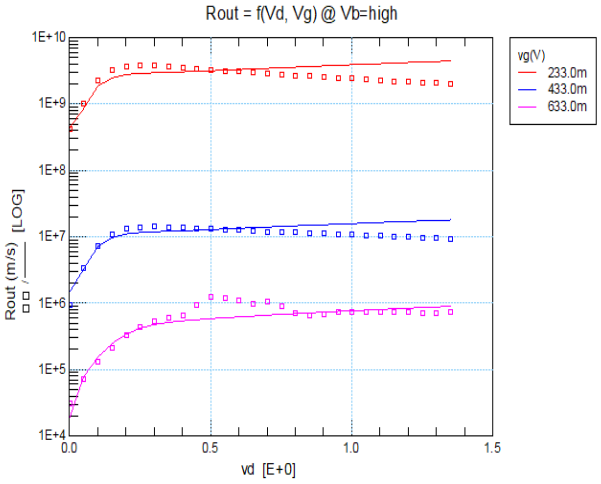


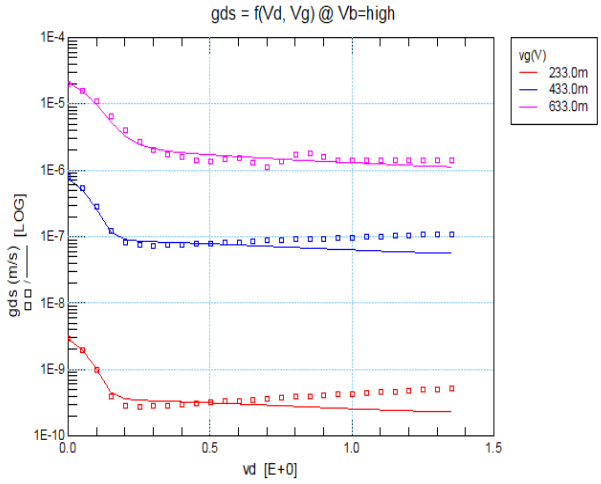




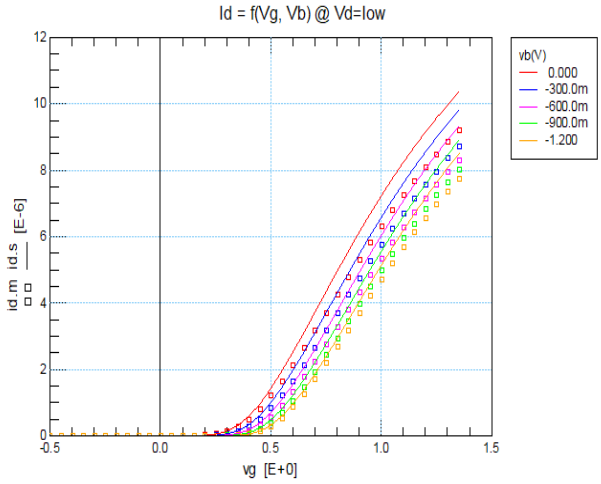
idvd_vth_vbmin, Vb = -1.2V, T = 70°C

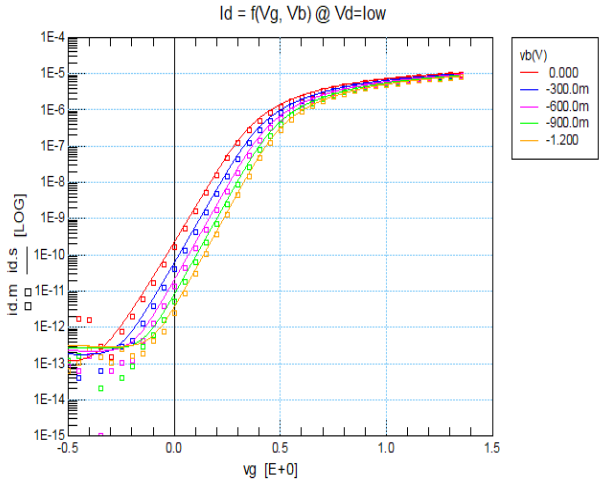


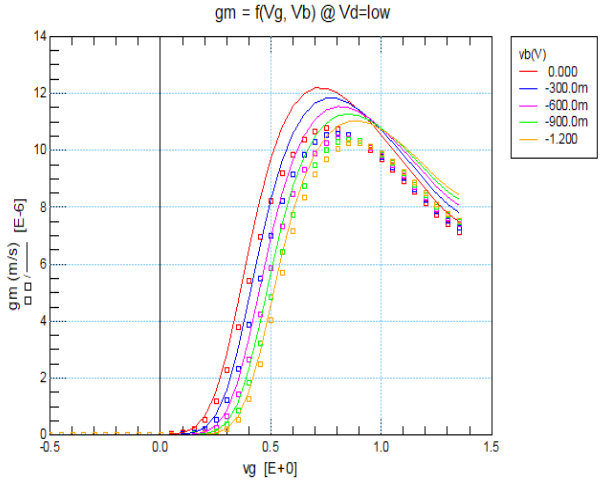


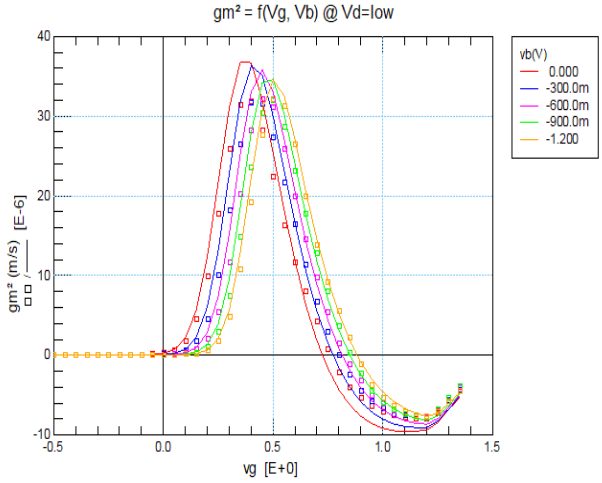


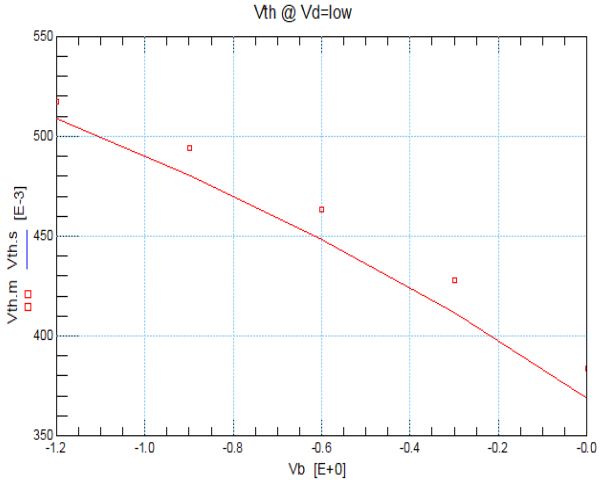
idvg, Vd = 0.05V, T = 125°C

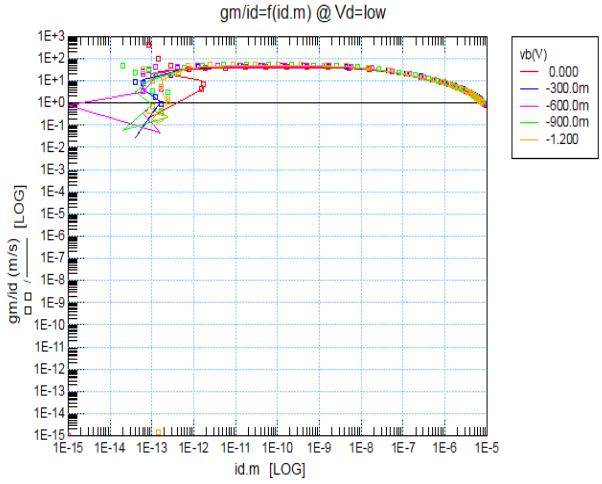


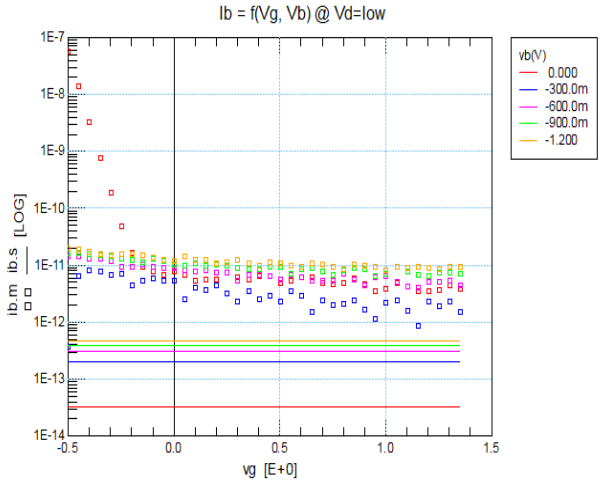


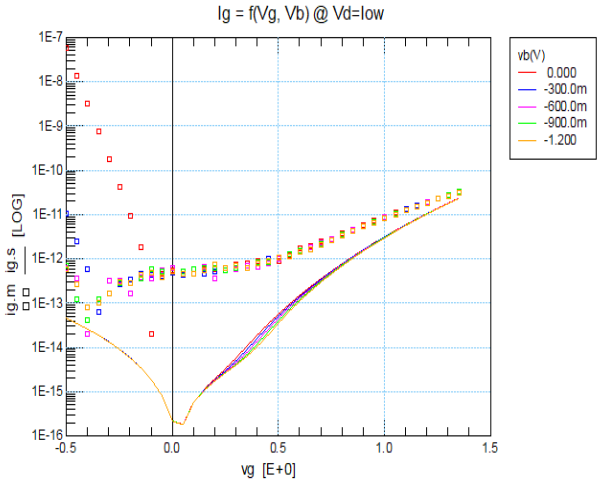




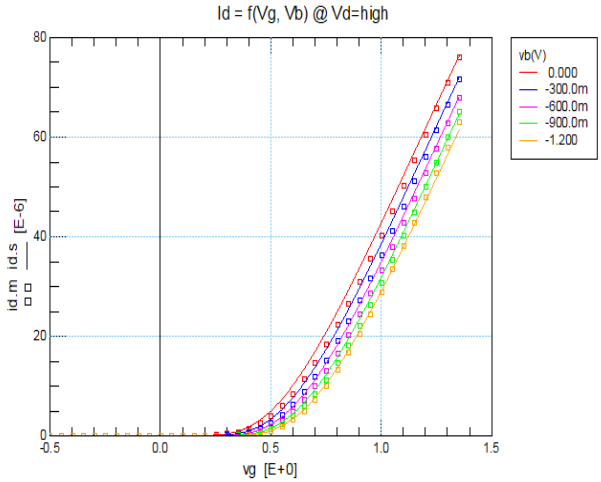


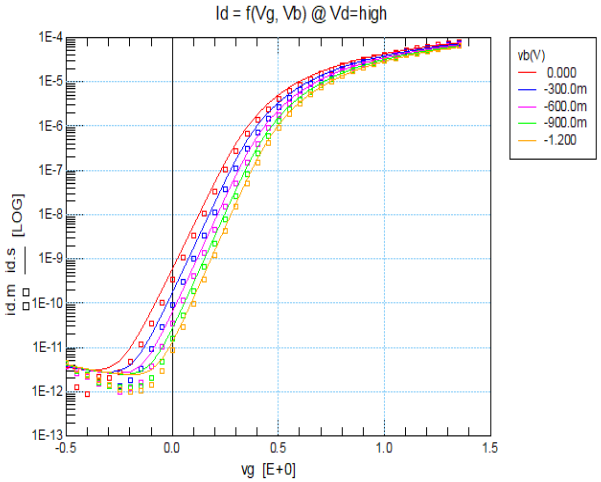


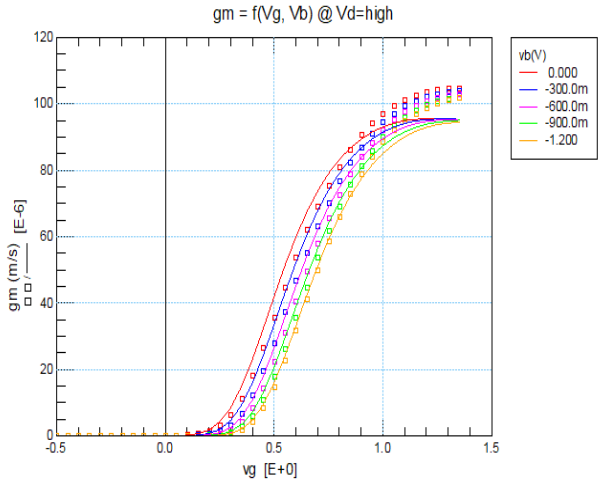


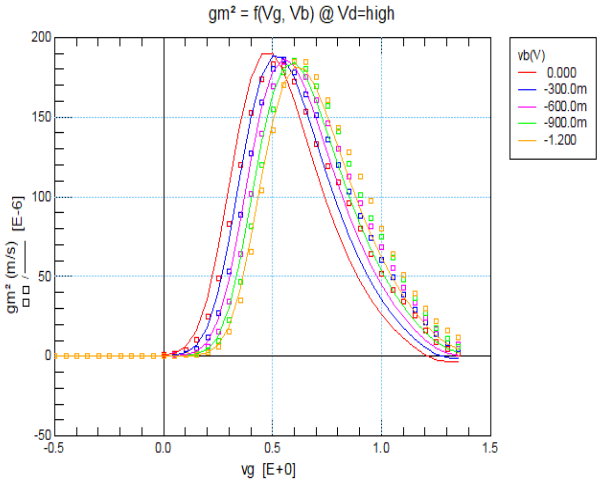


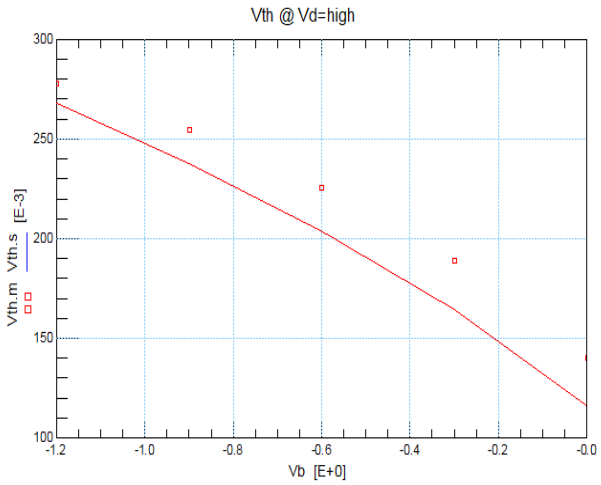
idvg, Vd = 1.2V, T = 125°C

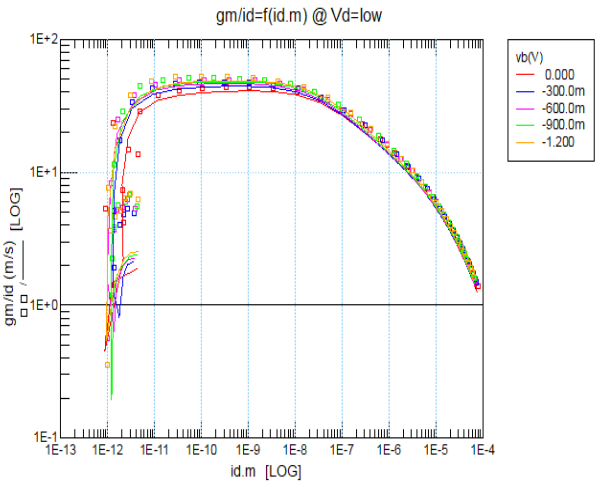


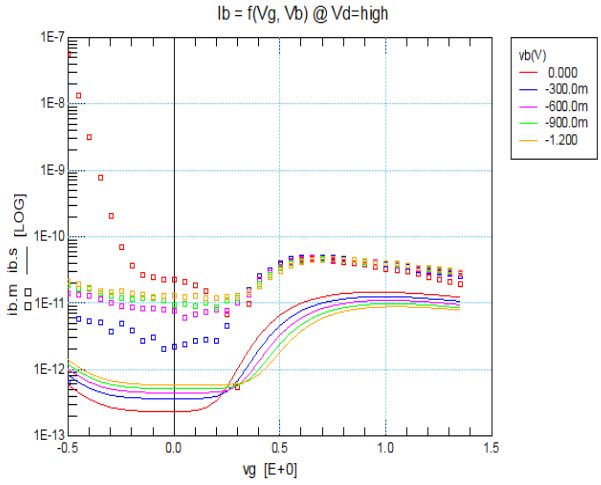


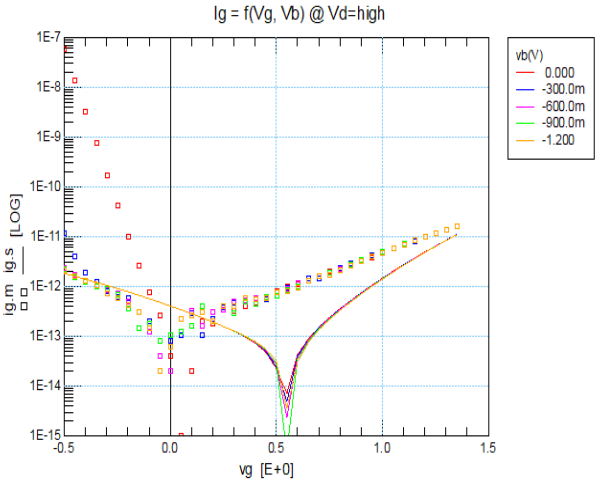




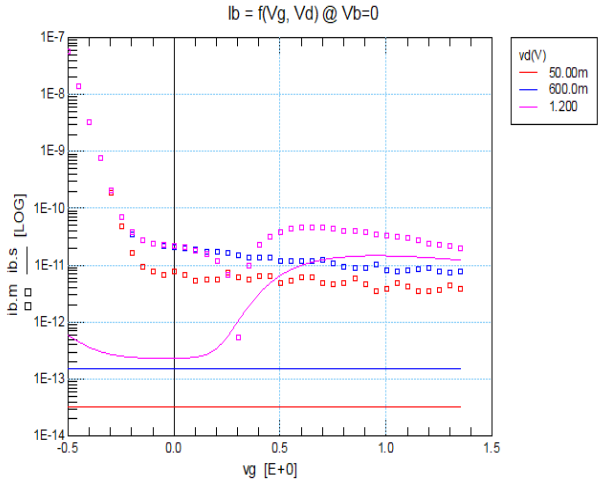


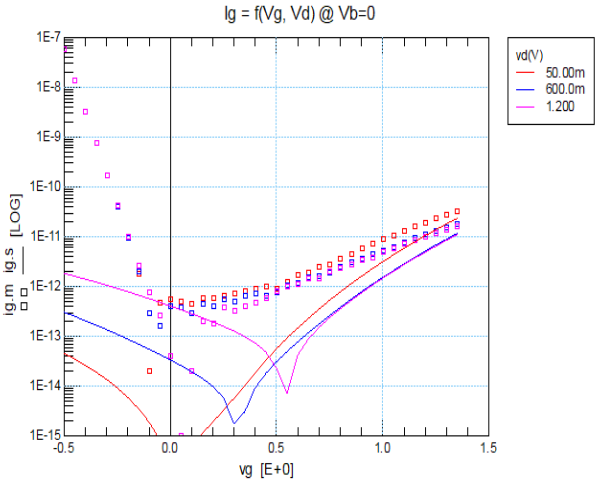




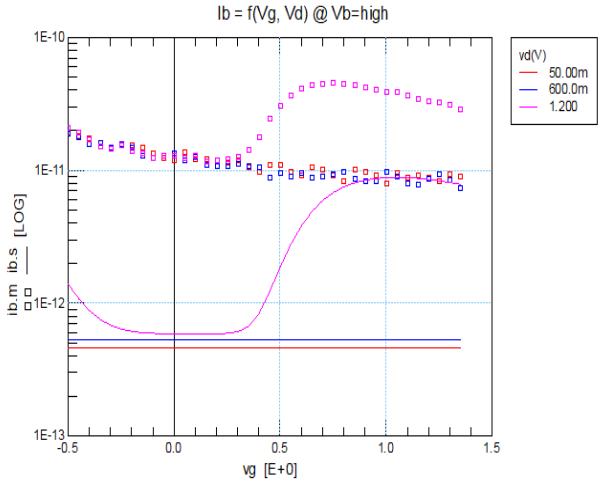


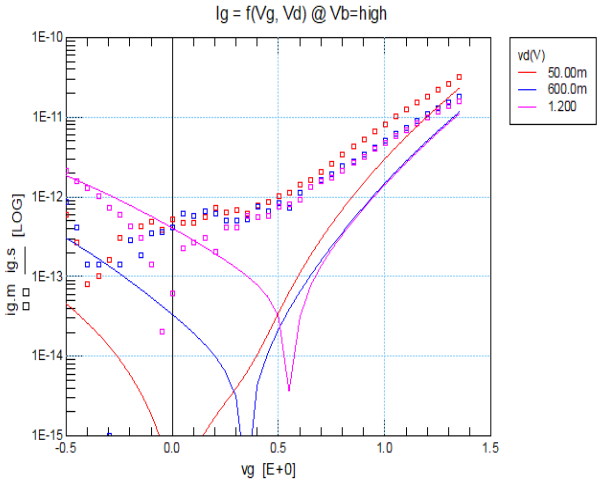
idvg, Ib,Ig, Vb = 0V, T = 125°C



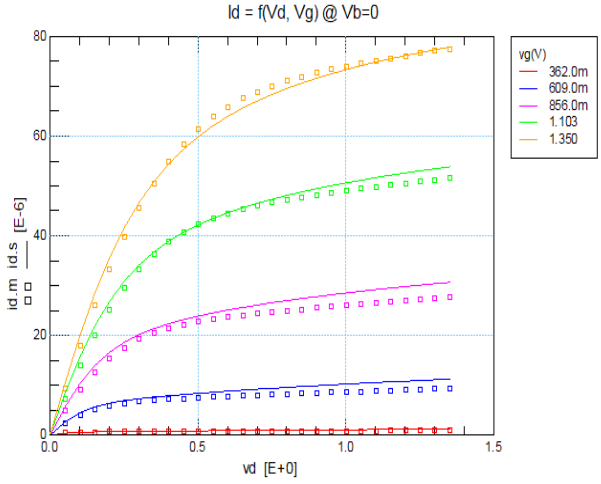


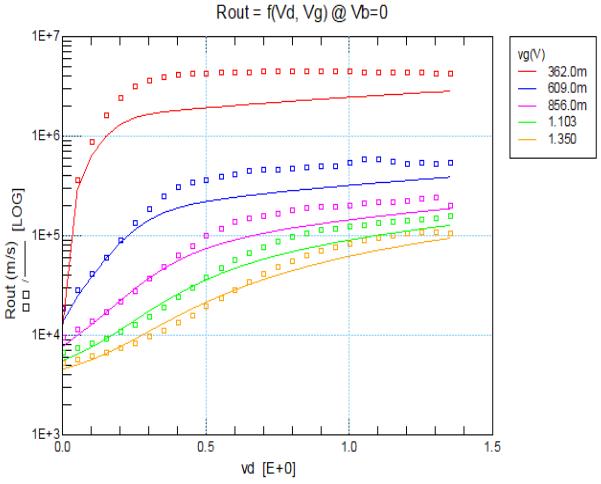
idvg, Ib,Ig, Vb = -1.2V, T = 125°C

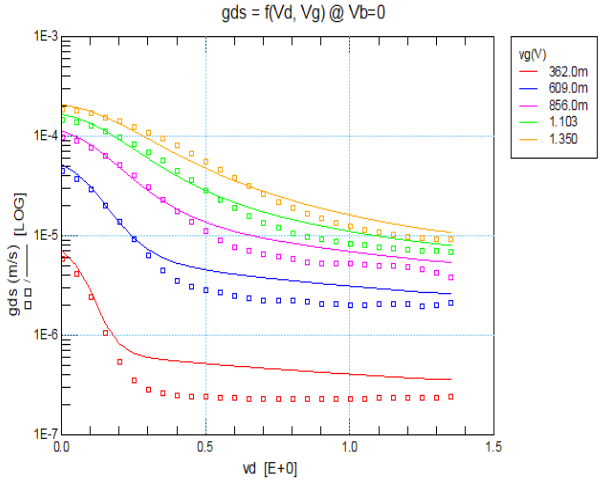




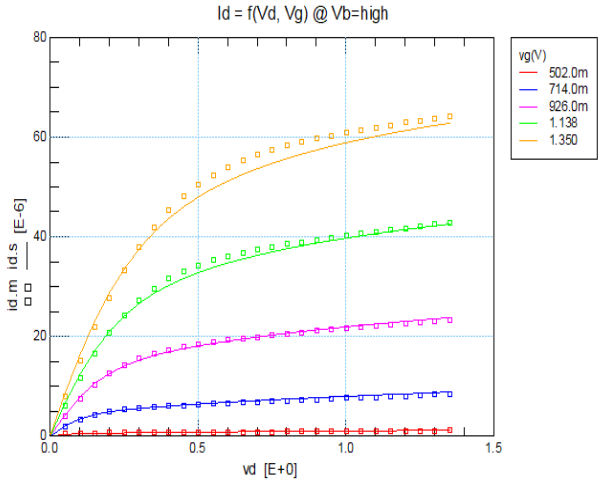
idvd, Vb = 0V, T = 125°C

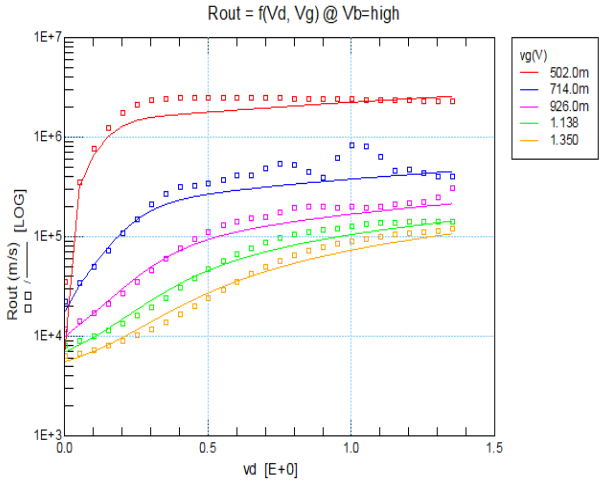


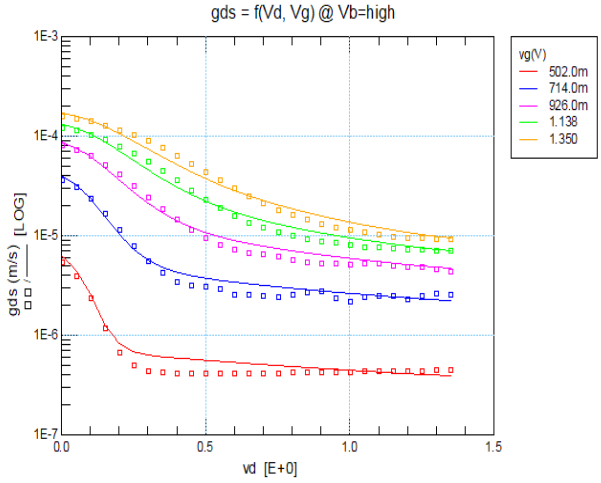




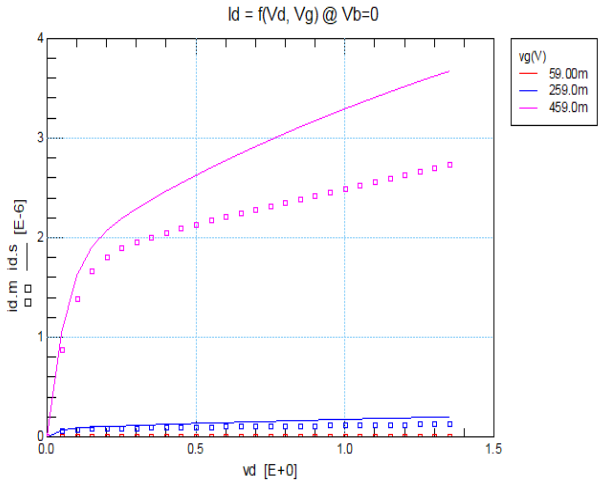
idvd_vbmin, Vb = -1.2V, T = 125°C

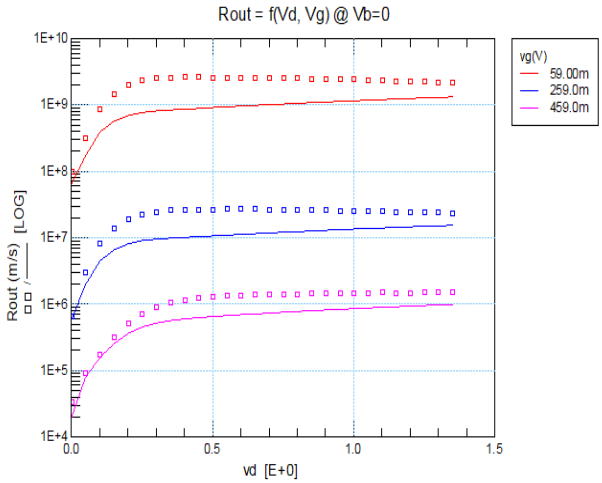


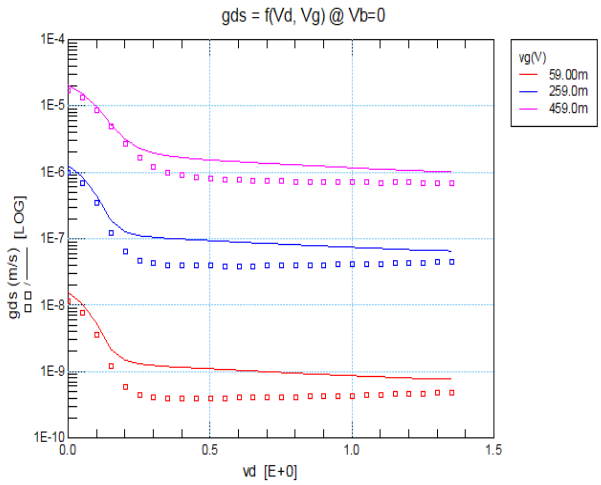




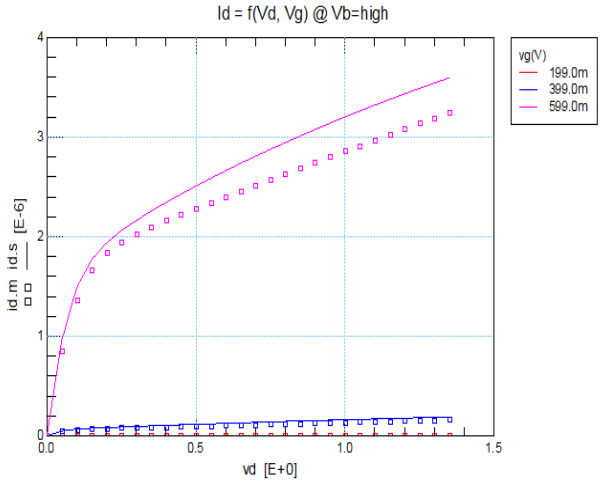
idvd_vth, Vb = 0V, T = 125°C

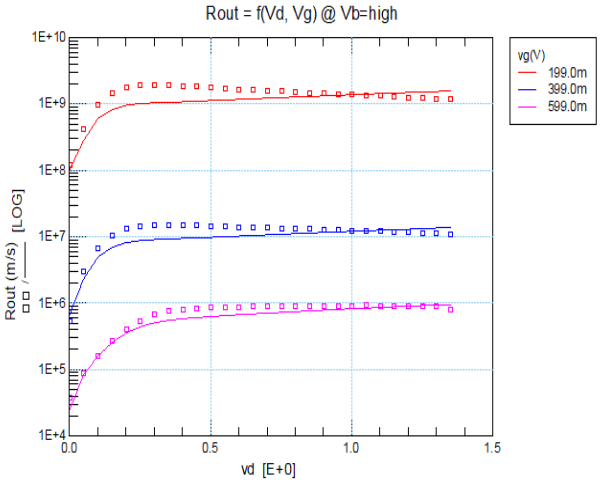


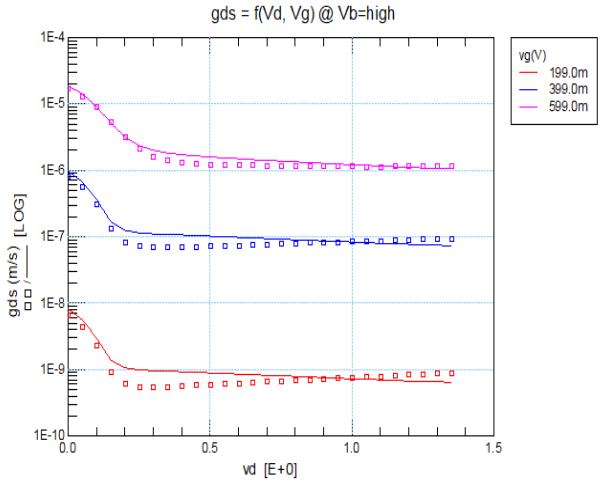




idvd_vth_vbmin, Vb = -1.2V, T = 125°C

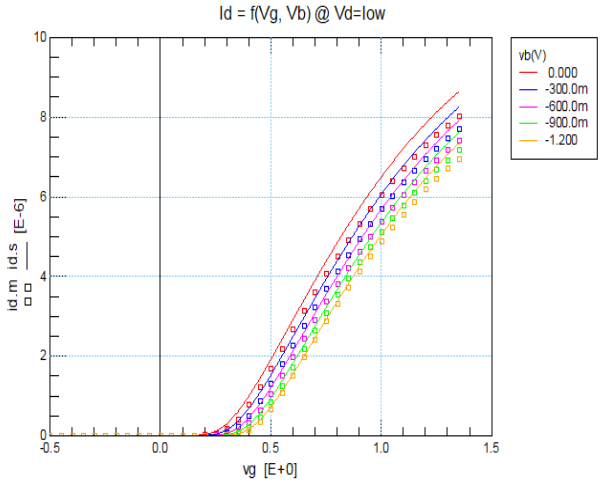


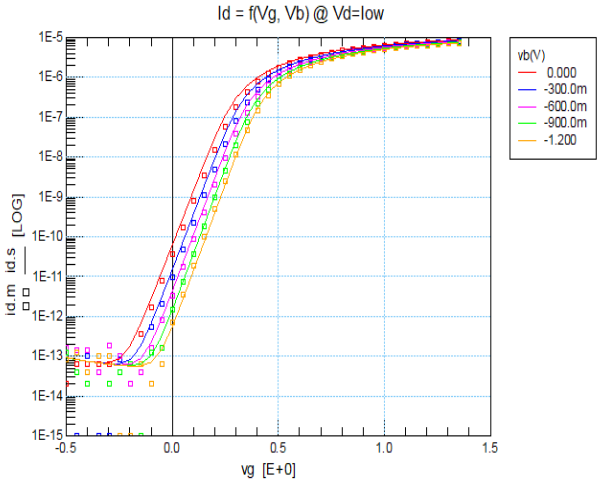


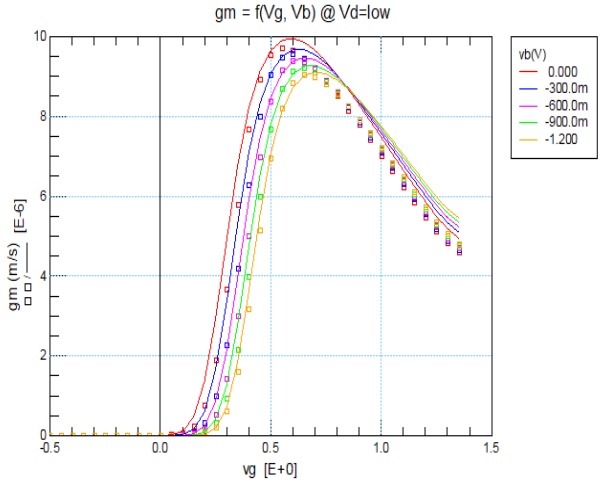


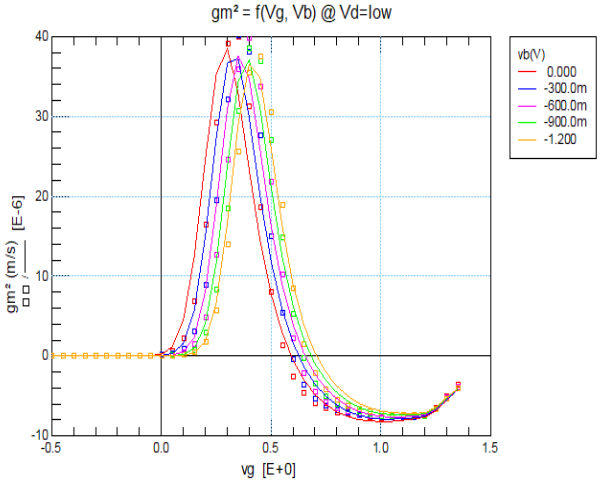
4.35W0u6_L1u2_S543_5

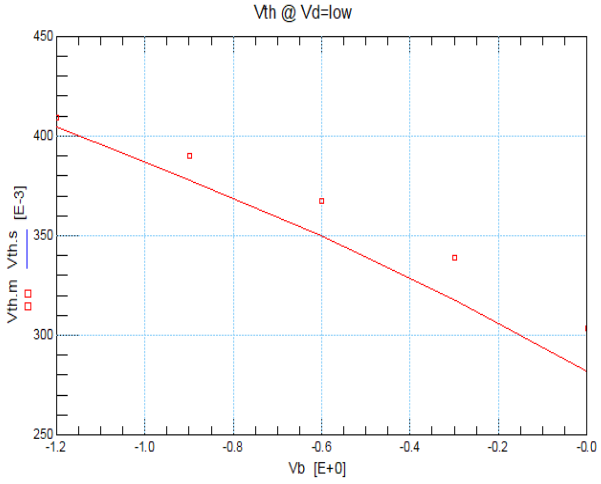
idvg, Vd = 0.05V, T = 27°C

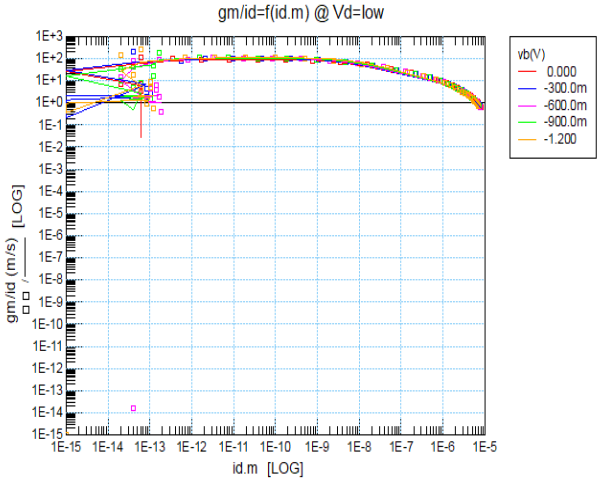


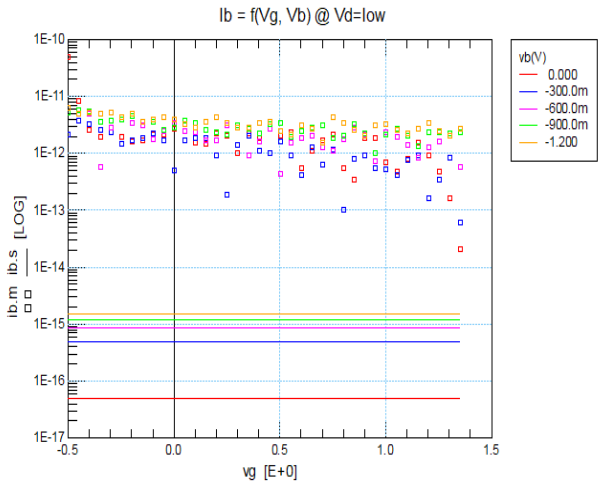


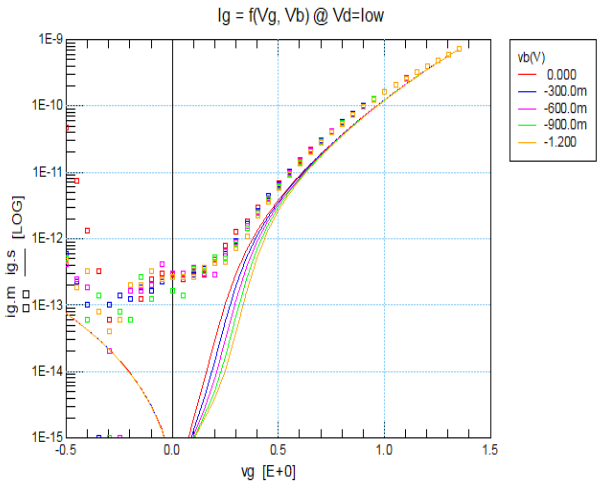




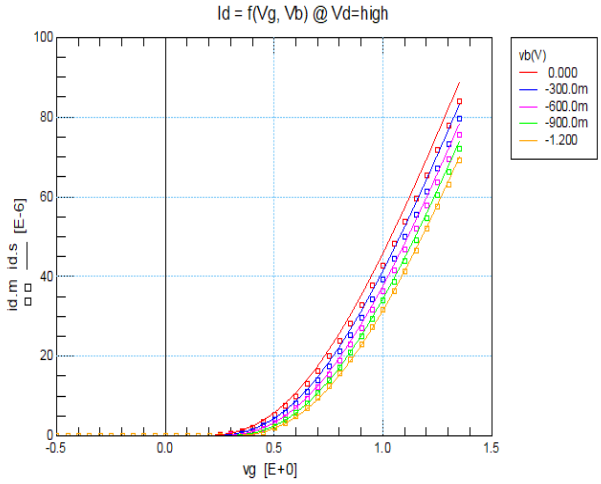


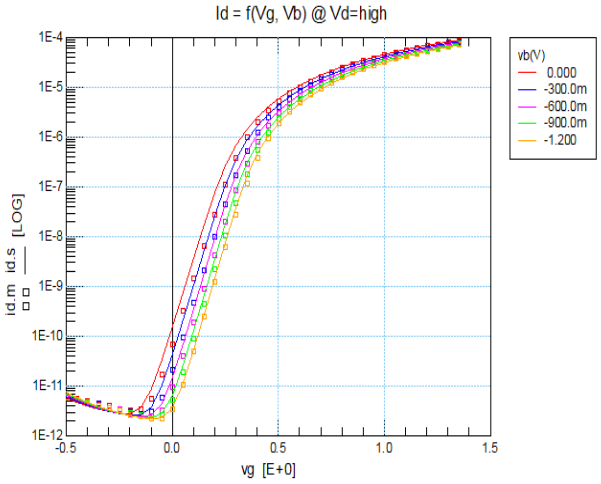


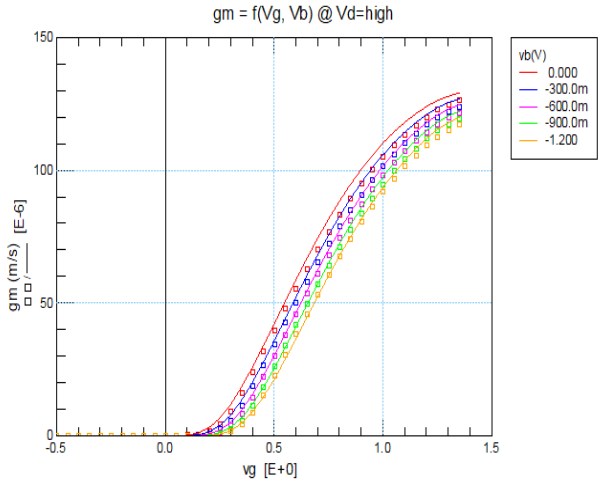


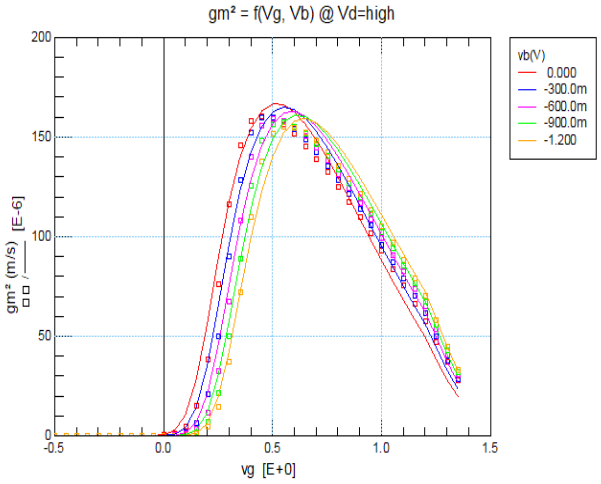


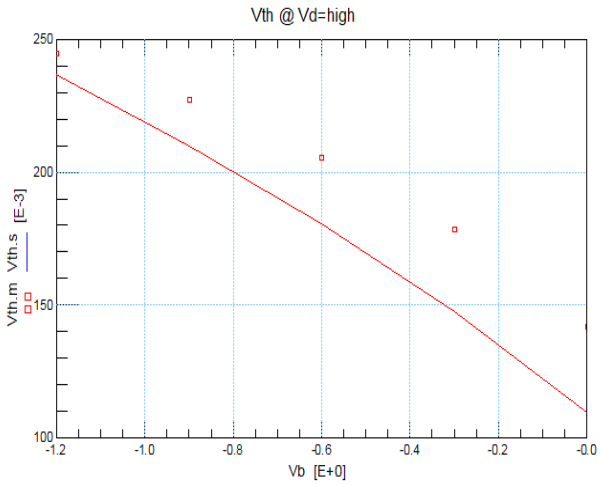
idvg, Vd = 1.2V, T = 27°C

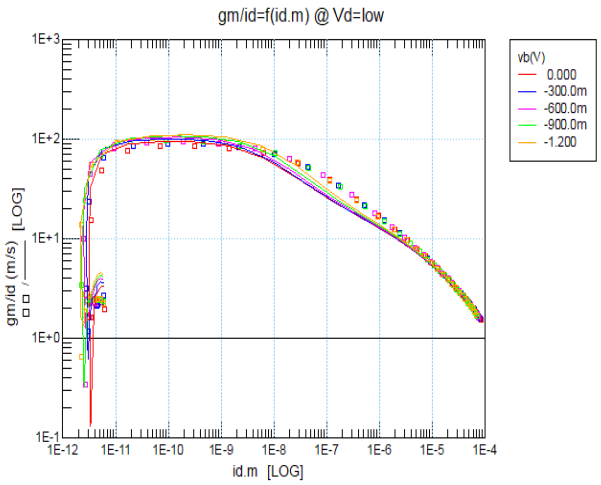


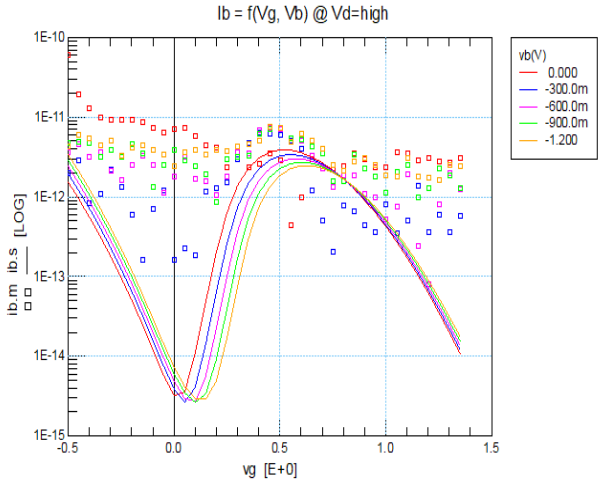


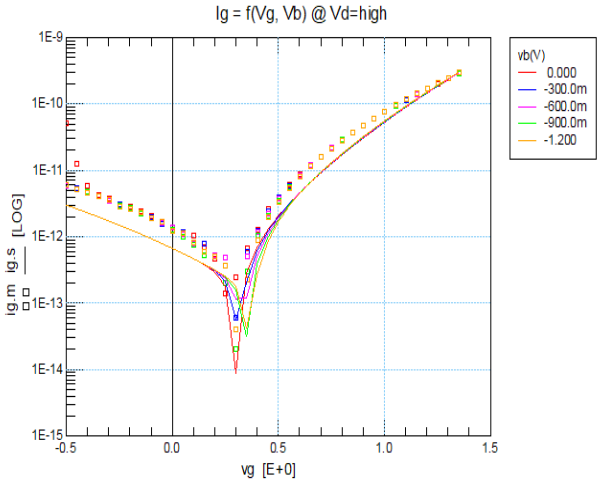




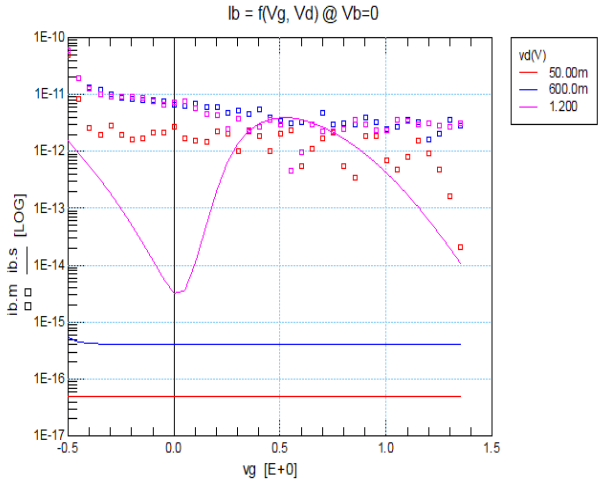


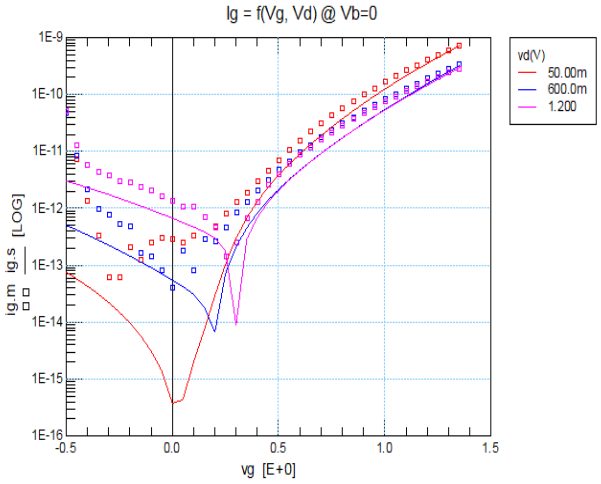




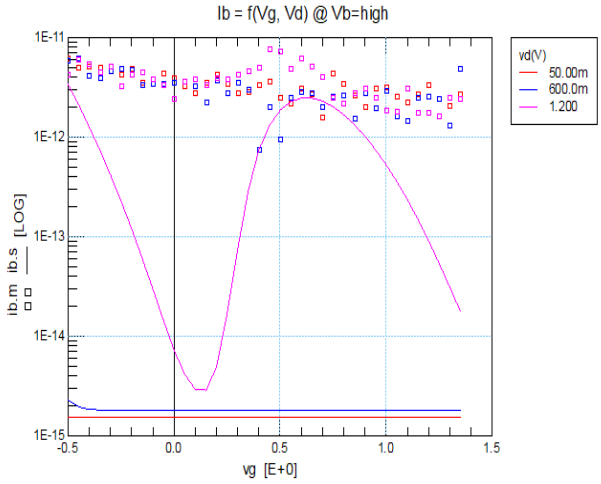


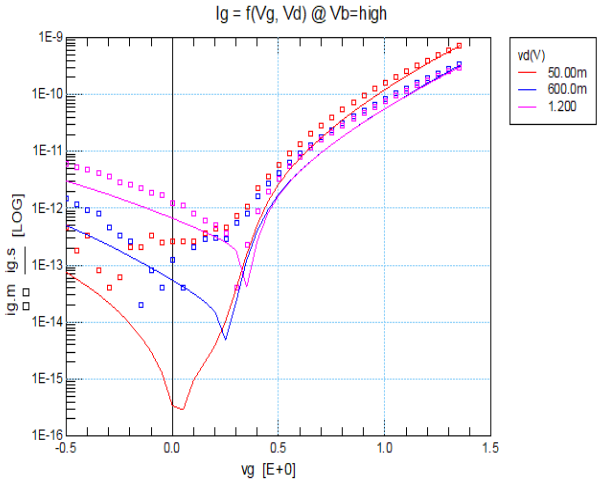
idvg, Ib,Ig, Vb = 0V, T = 27°C



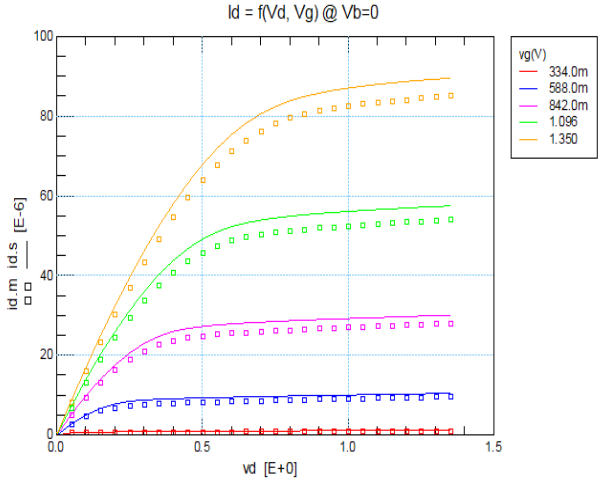


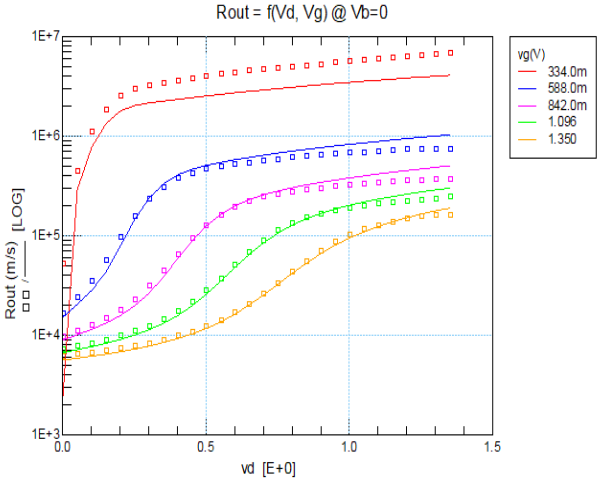
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

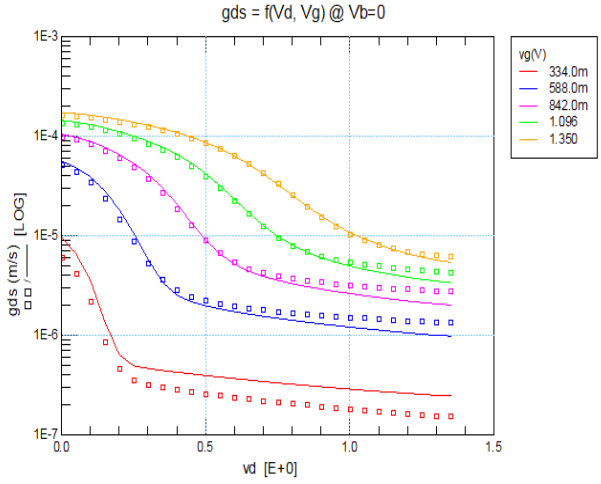




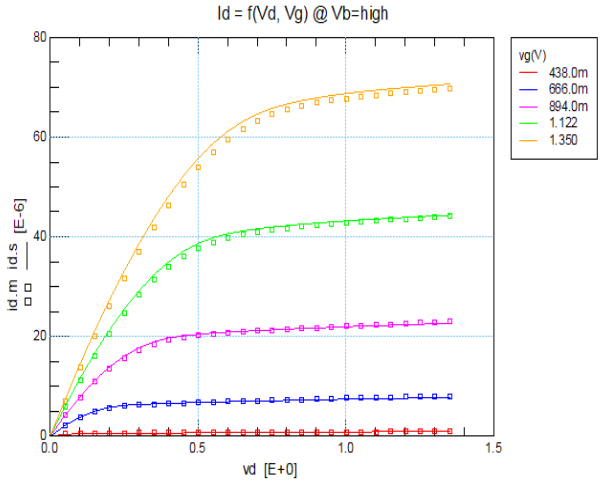
idvd, Vb = 0V, T = 27°C

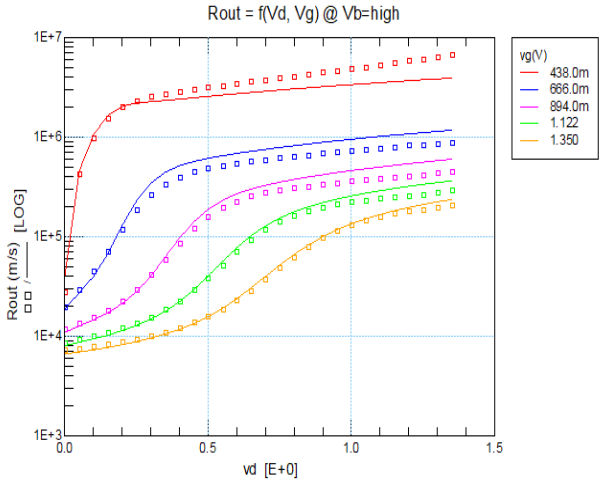


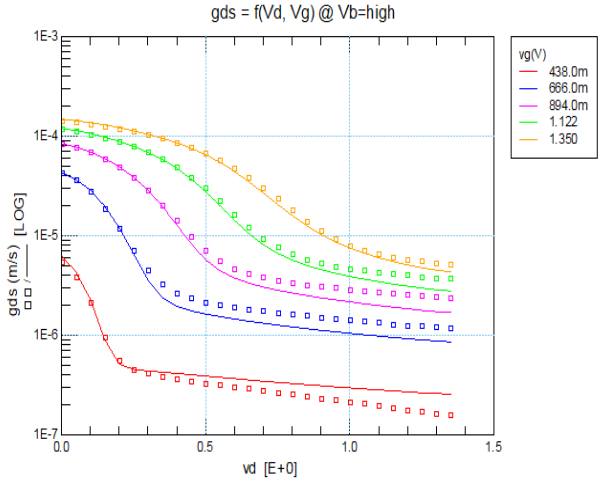




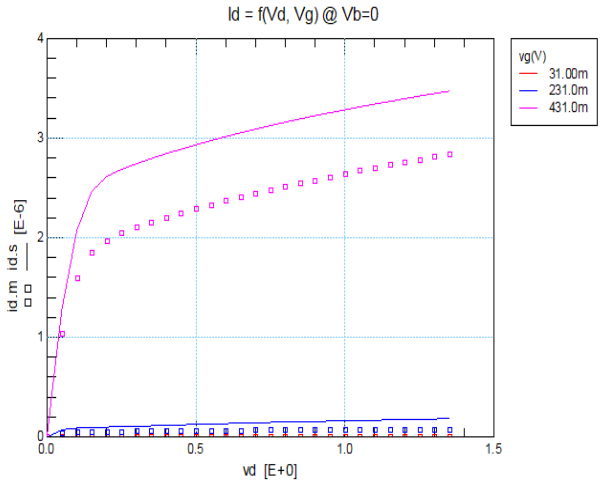
idvd_vbmin, Vb = -1.2V, T = 27°C

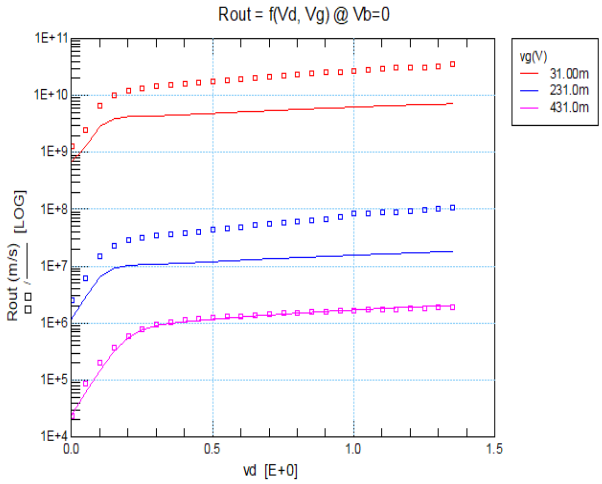


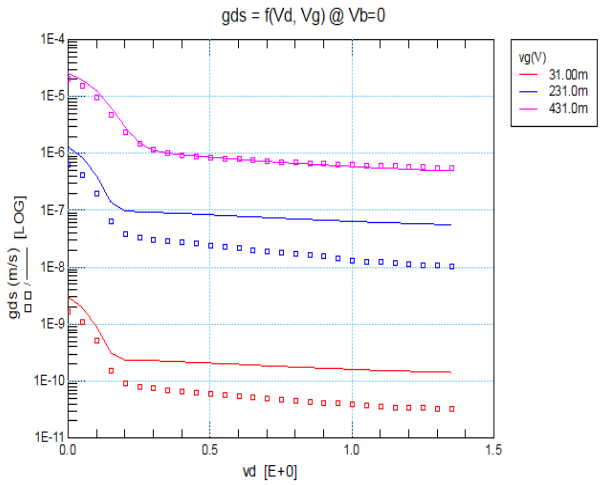




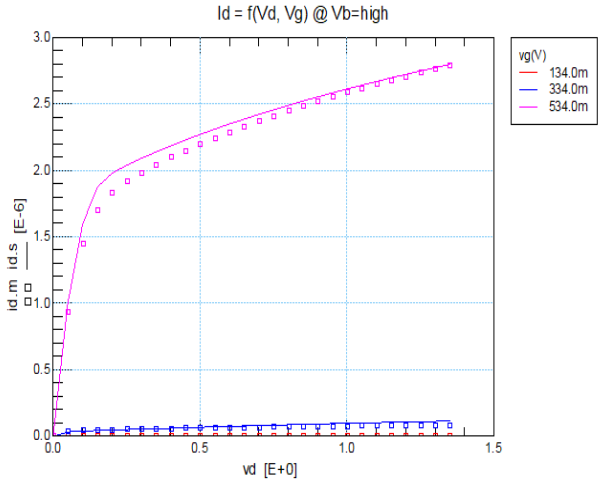
idvd_vth, Vb = 0V, T = 27°C

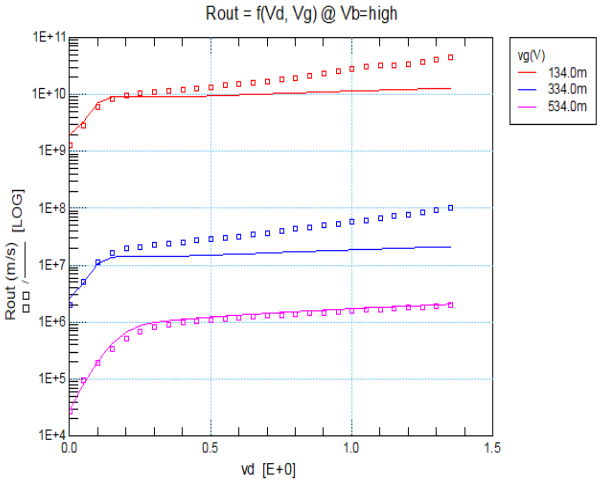


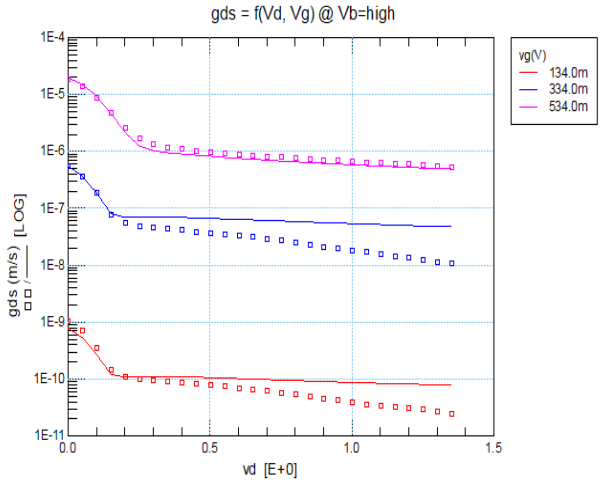




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

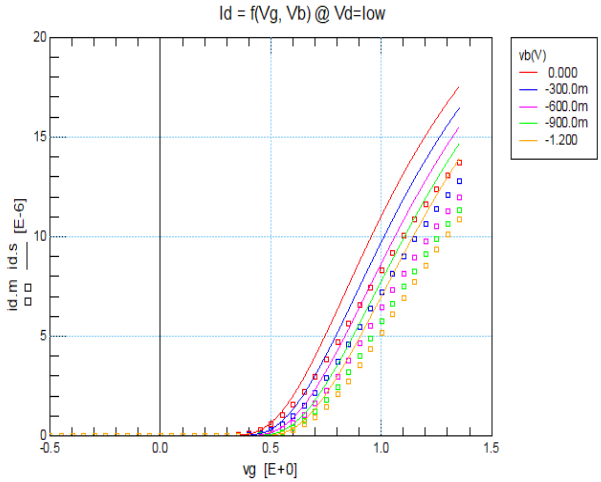


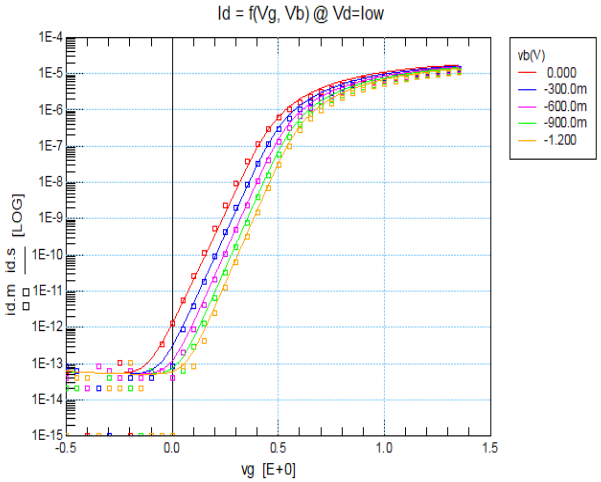


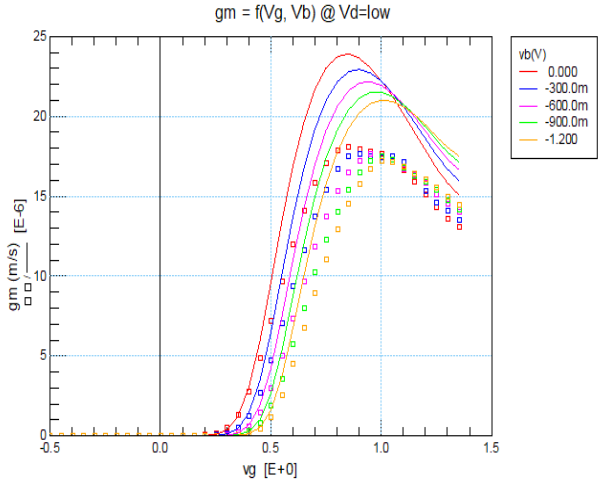


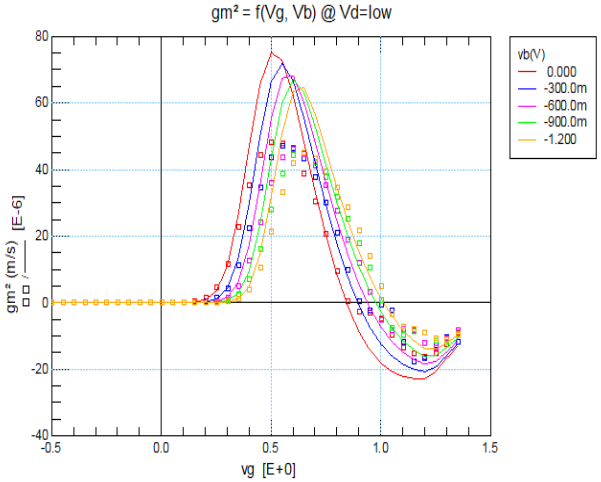
4.36 W0u15_L0u14_S546_4

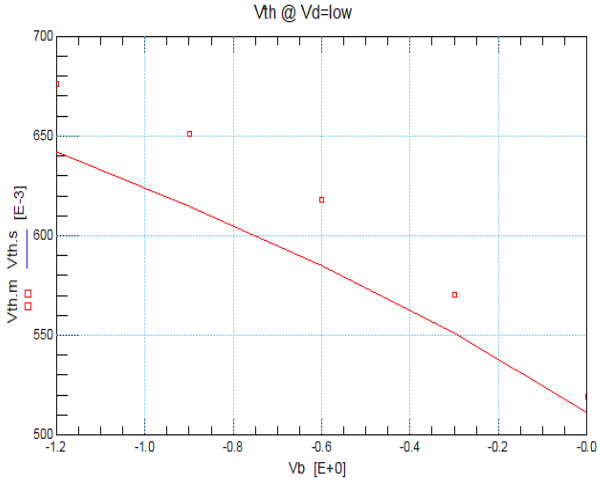
idvg, Vd = 0.05V, T = 27°C

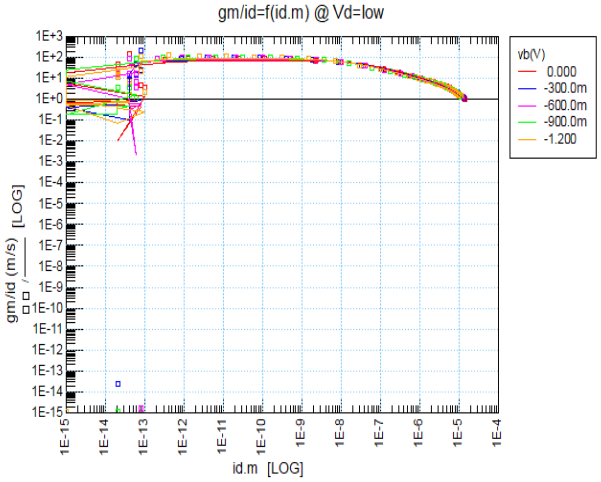


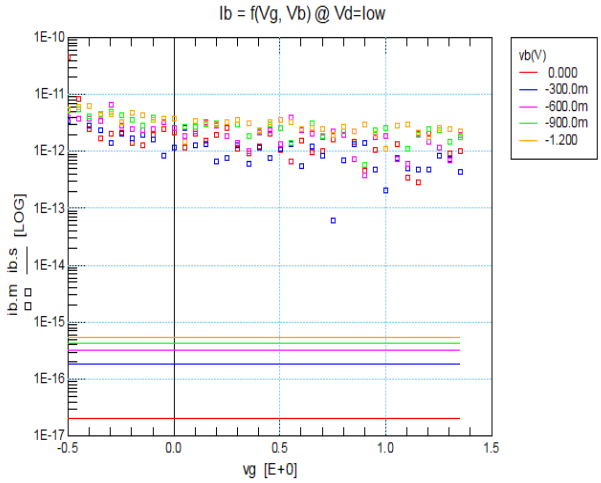


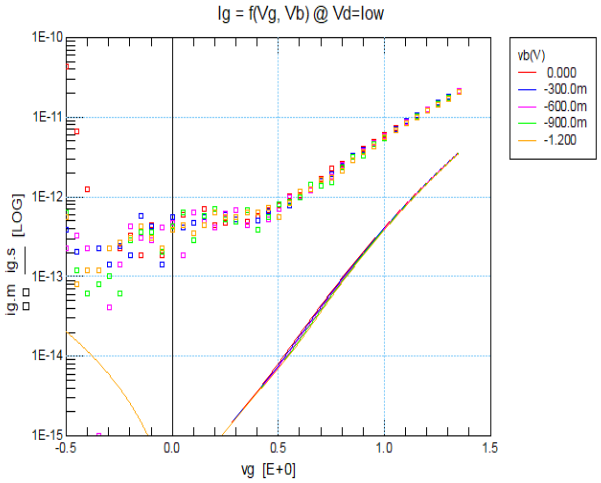




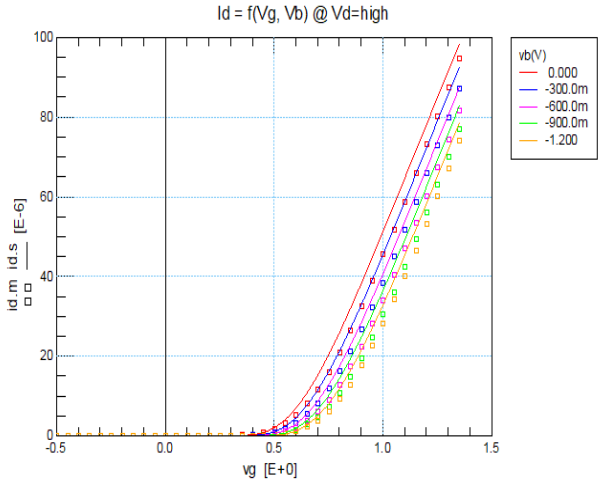


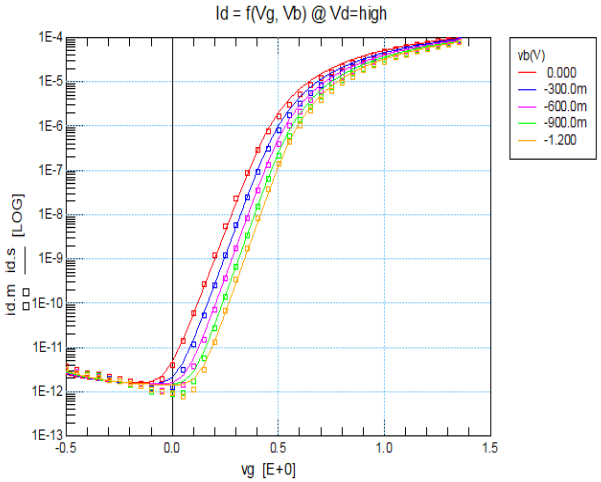


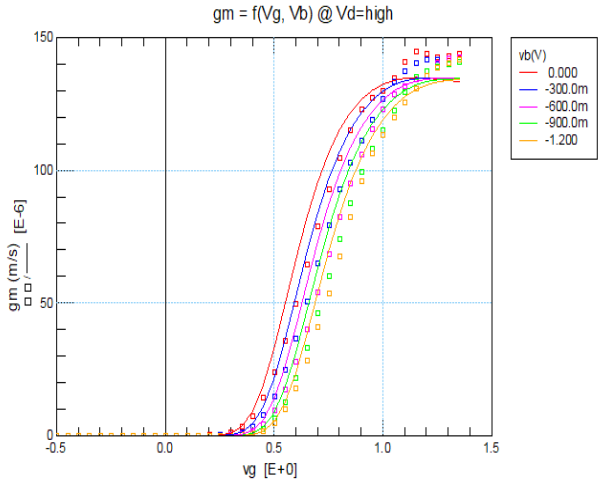


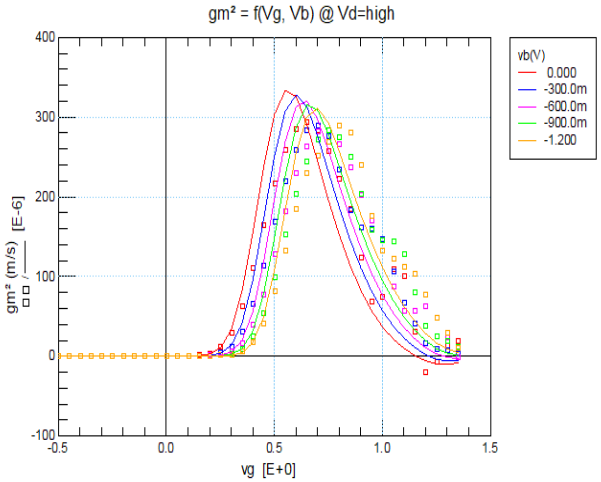


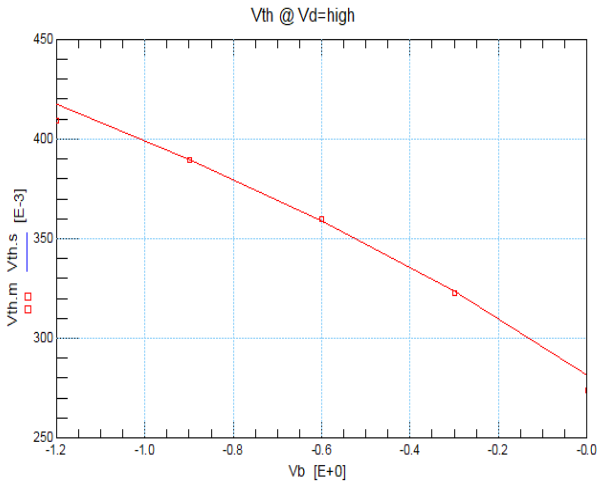
idvg, Vd = 1.2V, T = 27°C

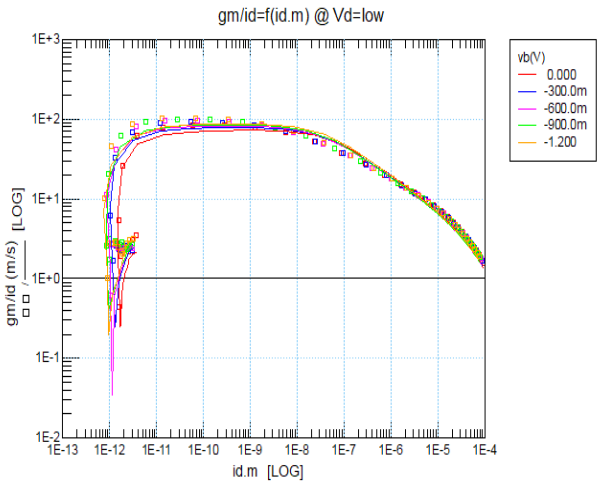


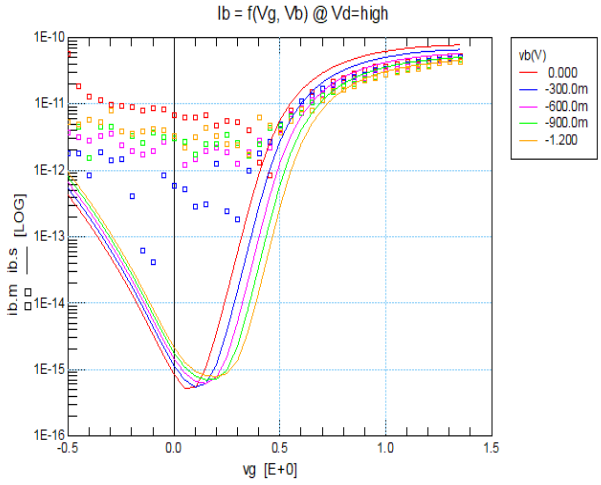


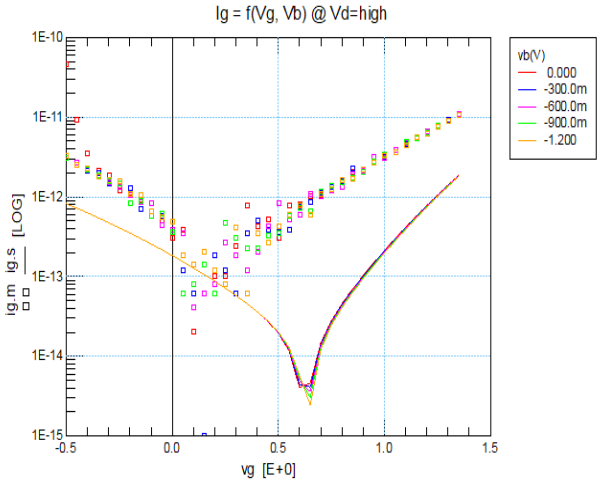




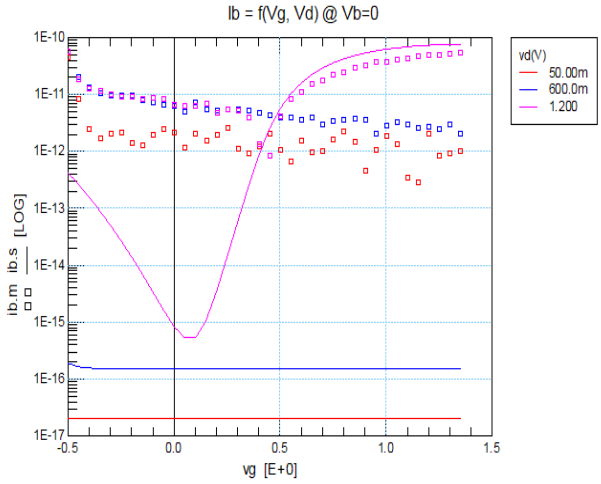


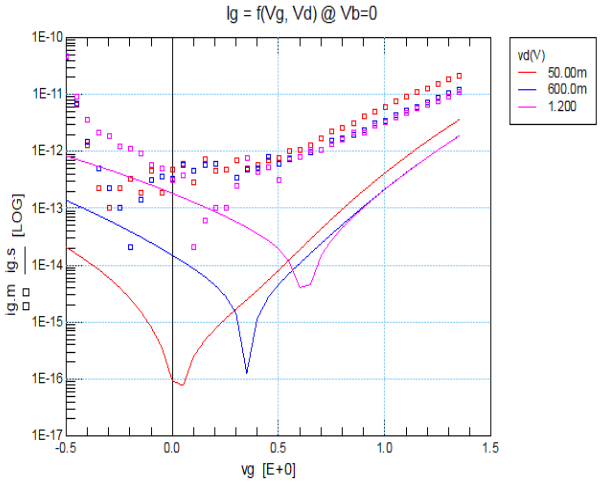




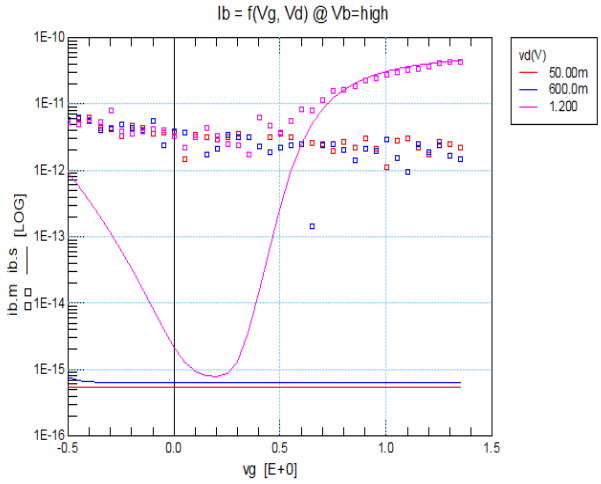


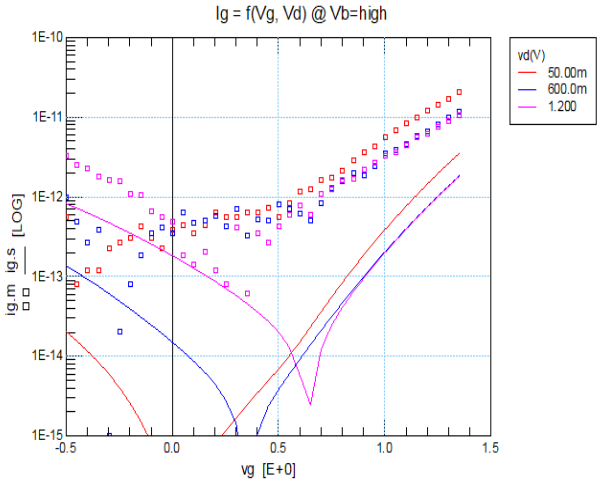
idvg, Ib,Ig, Vb = 0V, T = 27°C



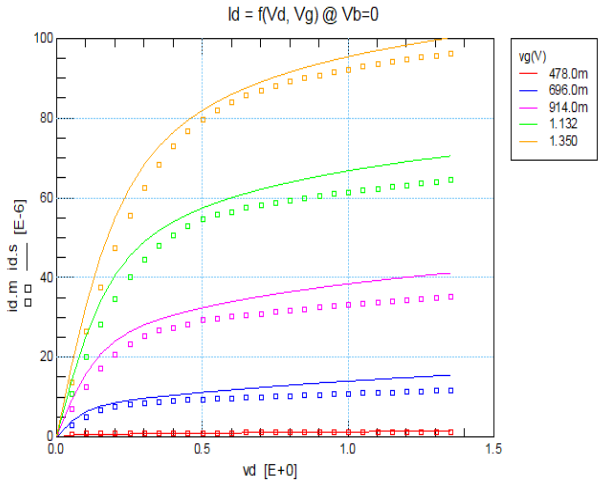


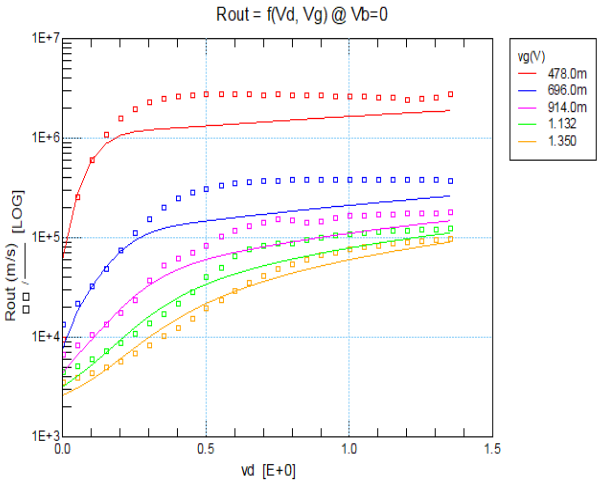
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

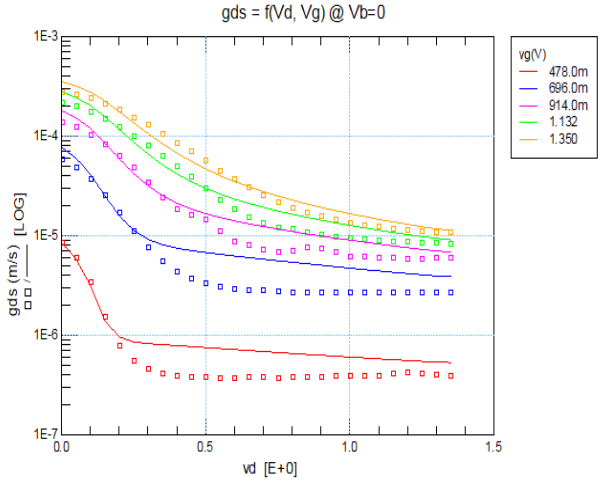




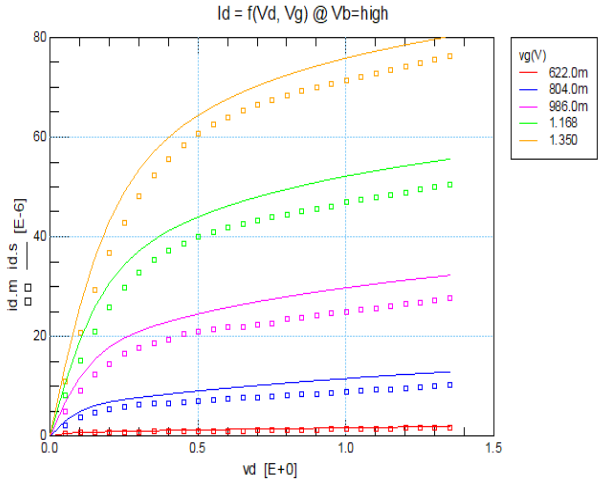
idvd, Vb = 0V, T = 27°C

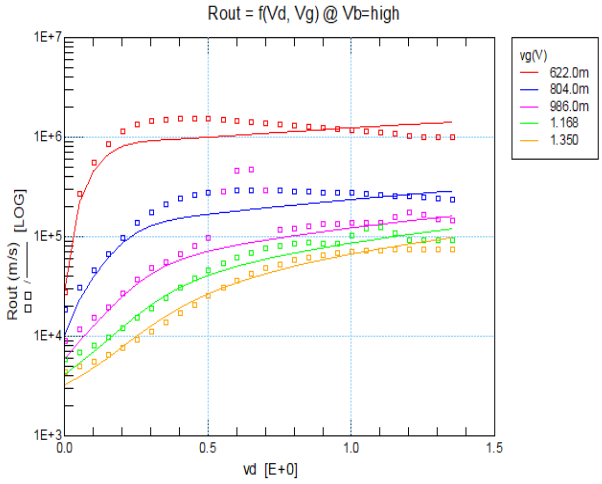


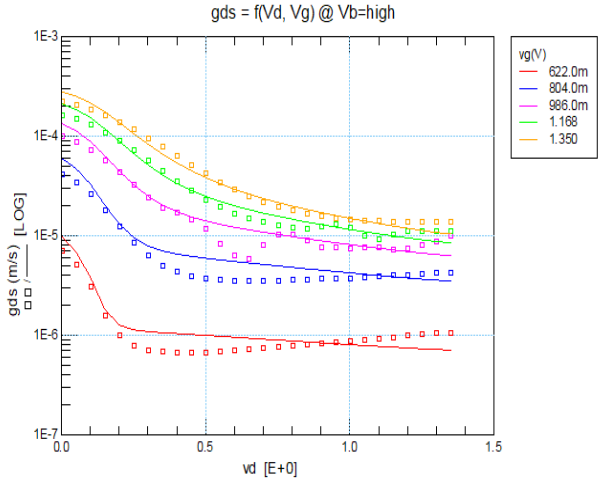




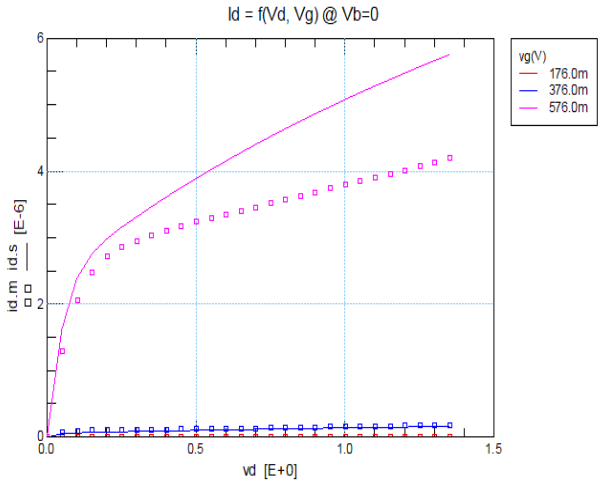
idvd_vbmin, Vb = -1.2V, T = 27°C

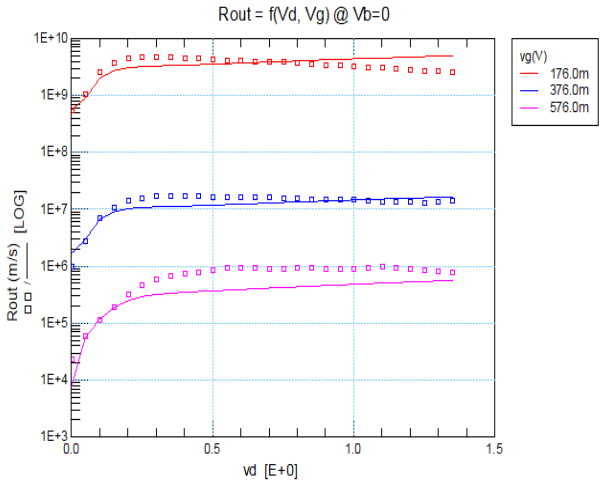


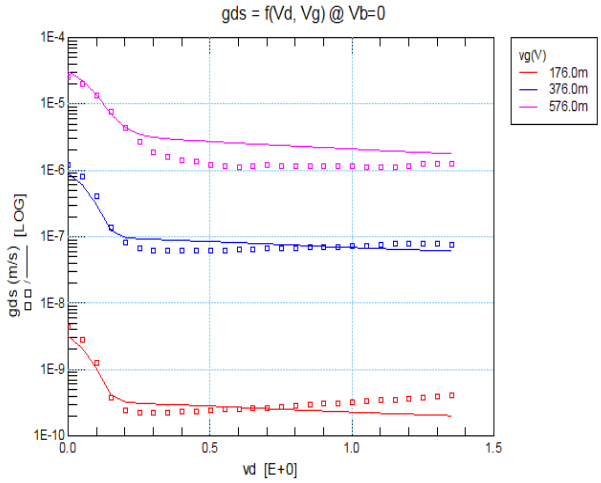




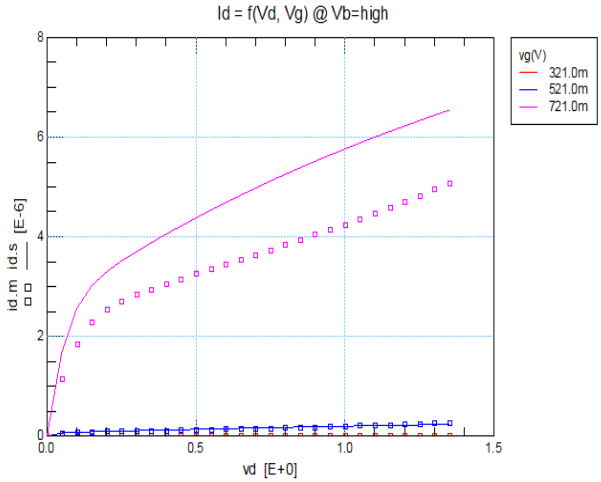
idvd_vth, Vb = 0V, T = 27°C

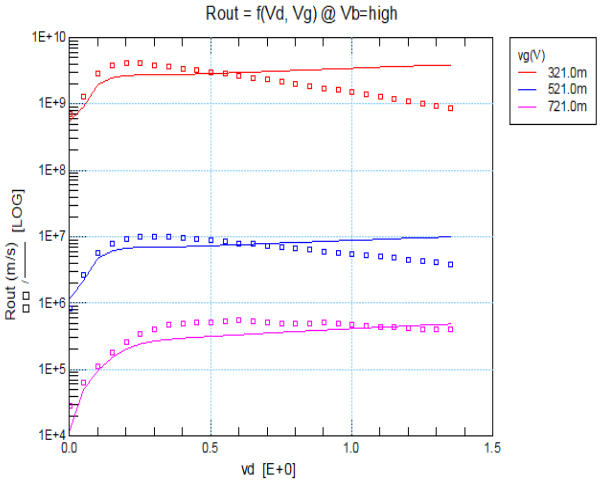


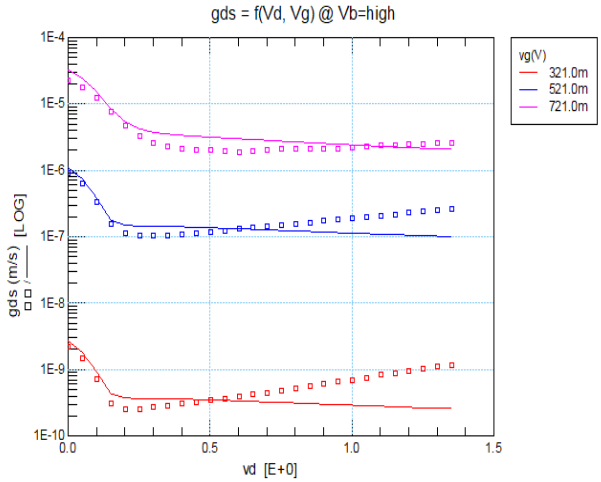




idvd_vth_vbmin, Vb = -1.2V, T = 27°C



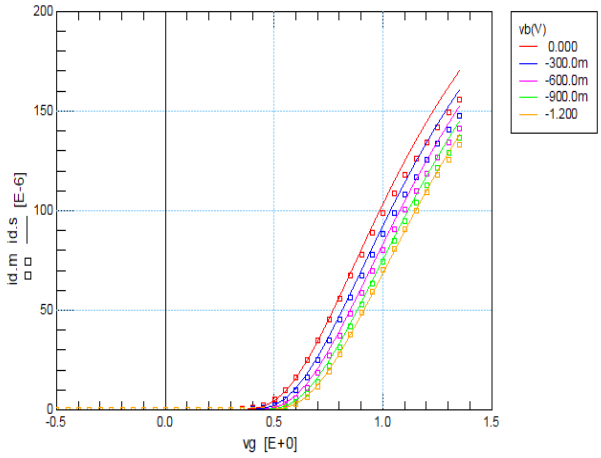




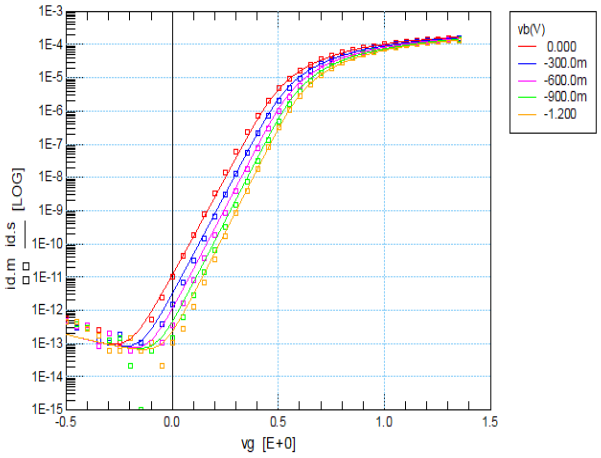
4.37W02u0_L0u13_S542_4

idvg, Vd = 0.05V, T = 27°C

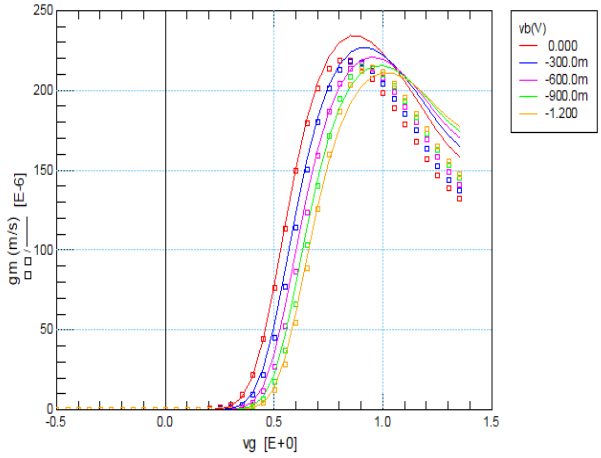
Id = f(Vg, Vb) @ Vd=low



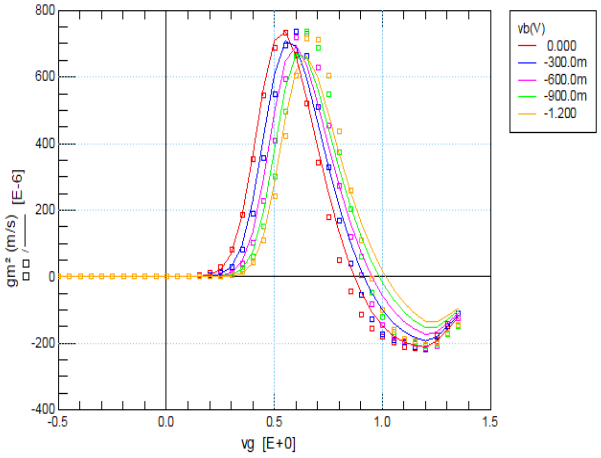
Id = f(Vg, Vb) @ Vd=low



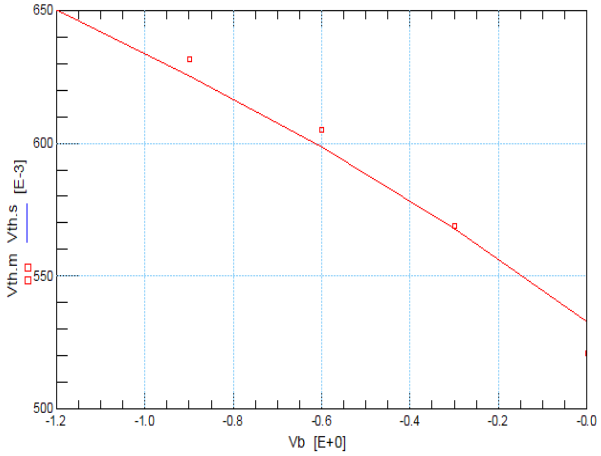
gm = f(Vg, Vb) @ Vd=low



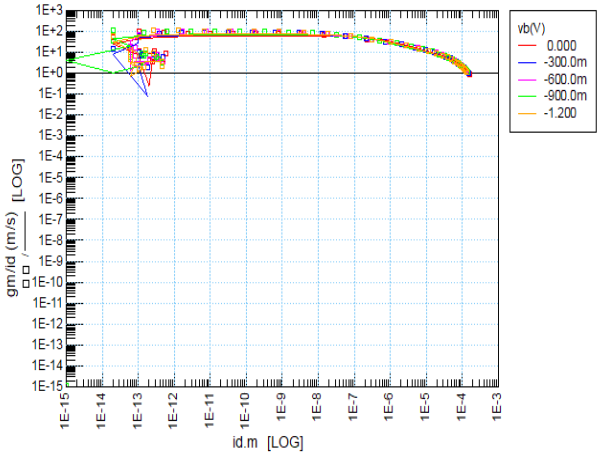
gm² = f(Vg, Vb) @ Vd=low

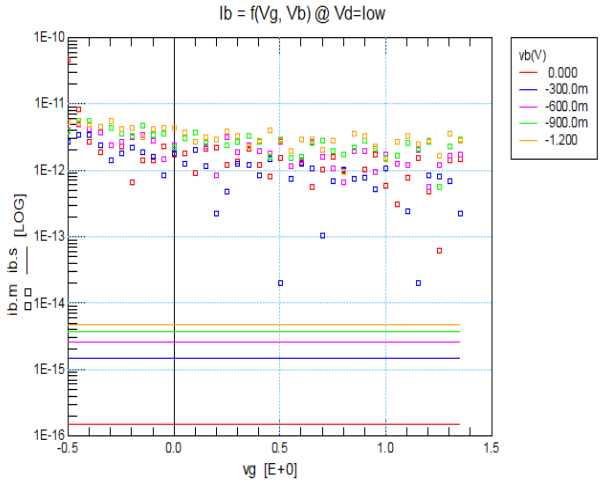


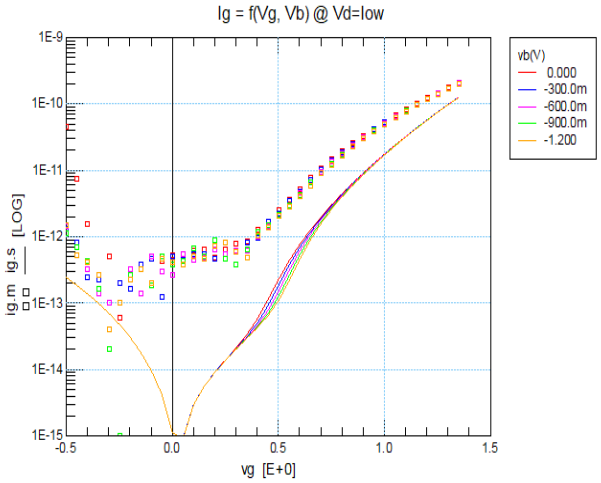
Vth @ Vd=low



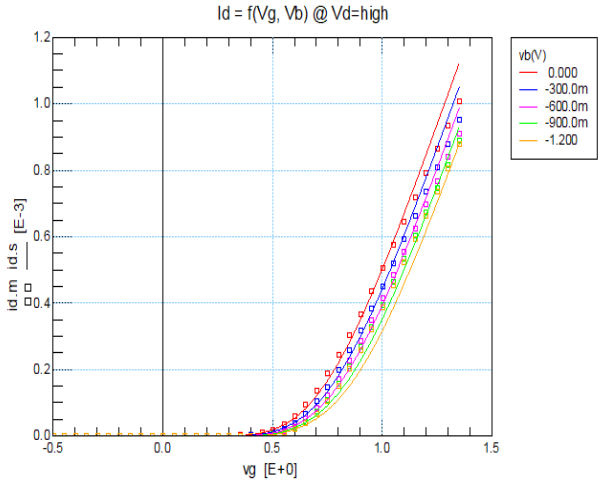
gm/id=f(id.m) @ Vd=low

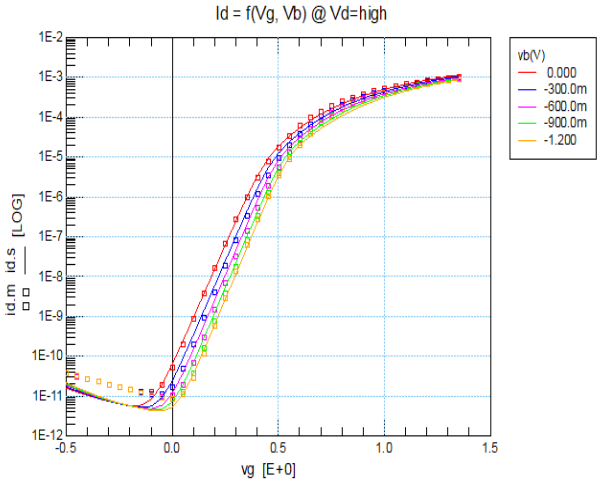


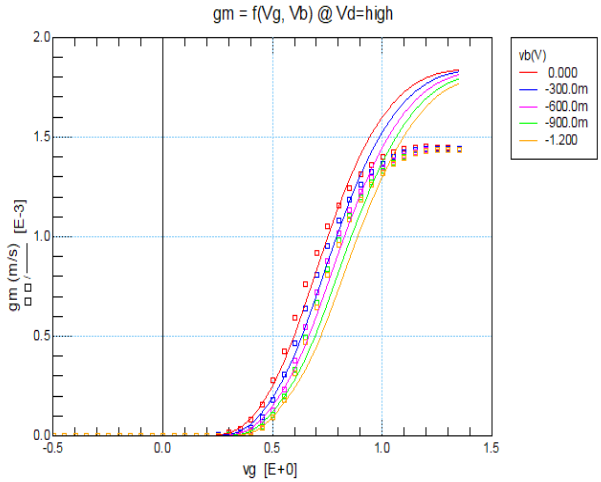


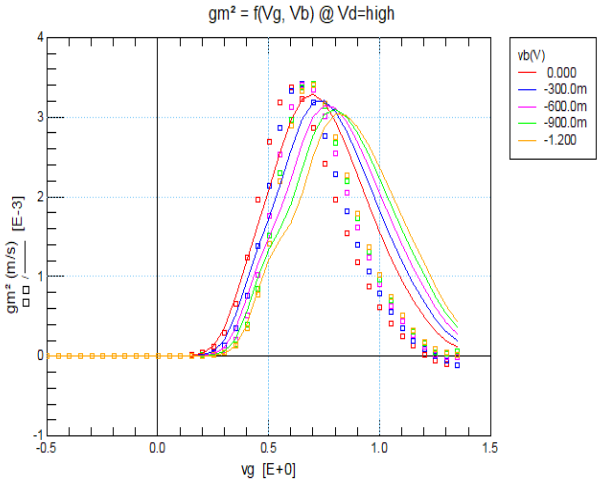


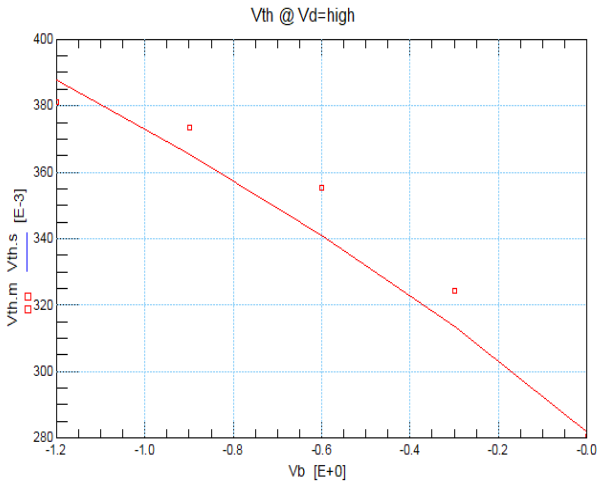
idvg, Vd = 1.2V, T = 27°C

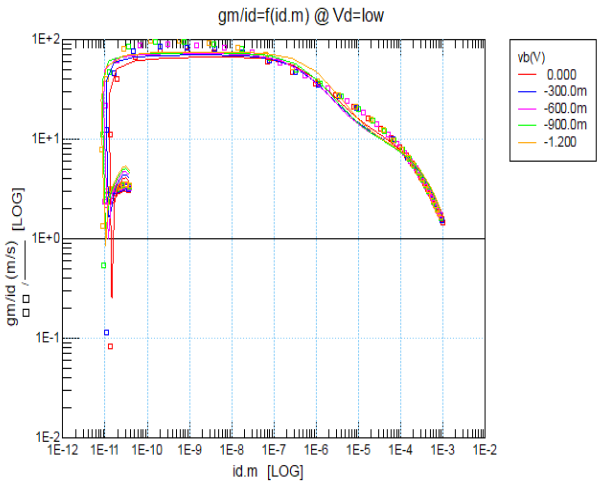


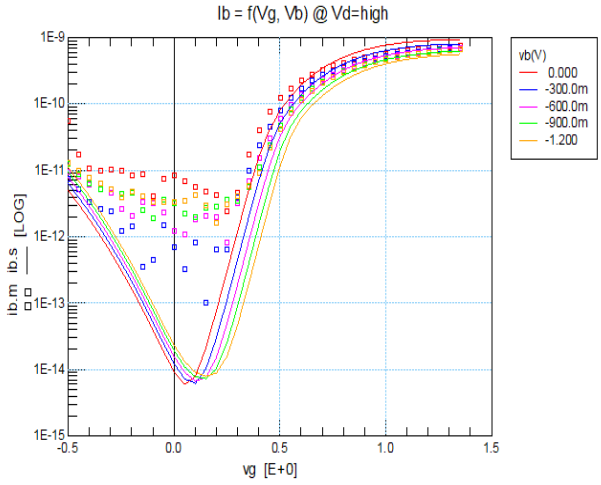


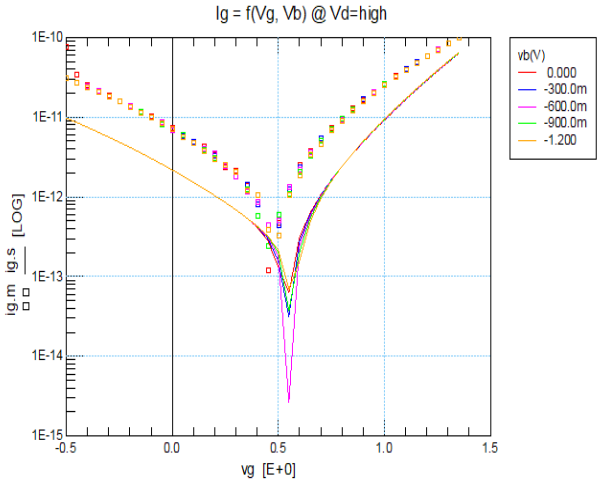




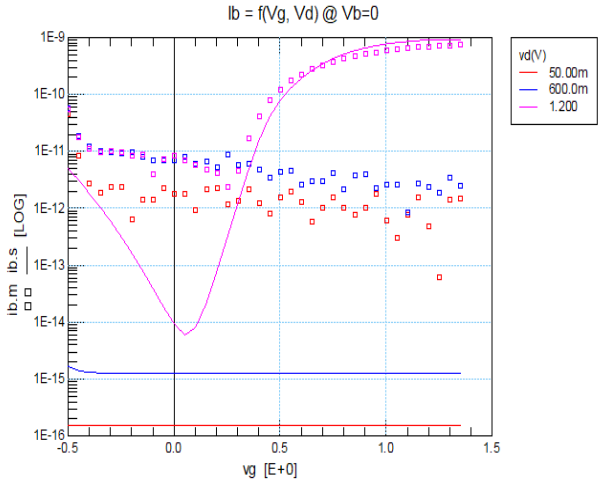


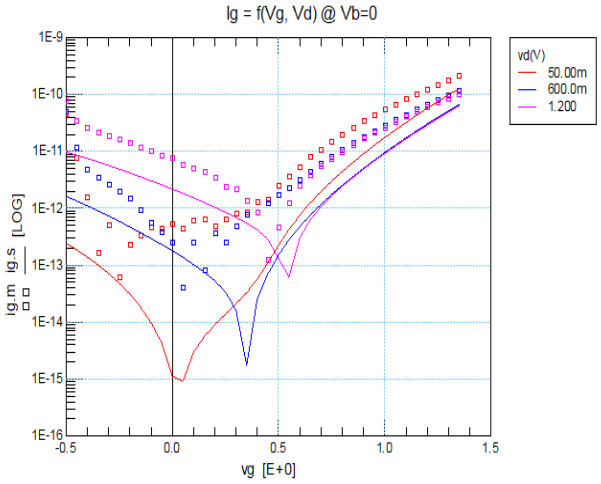




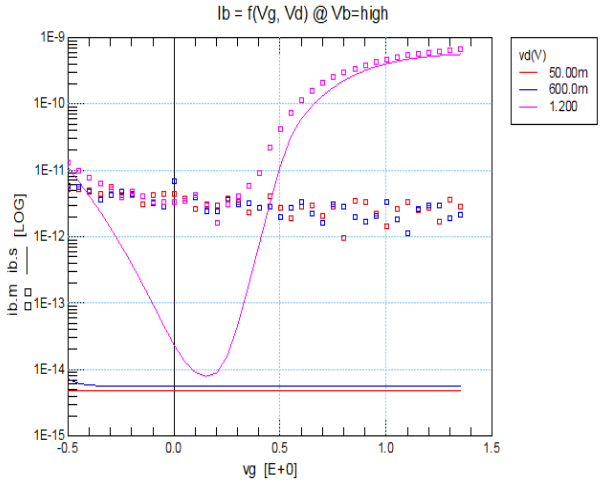


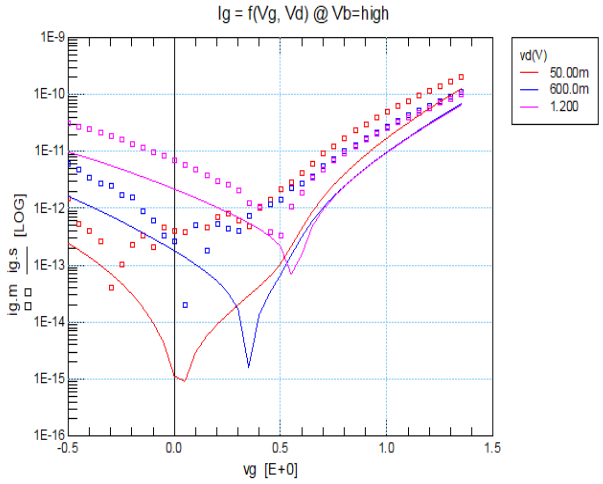
idvg, Ib,Ig, Vb = 0V, T = 27°C



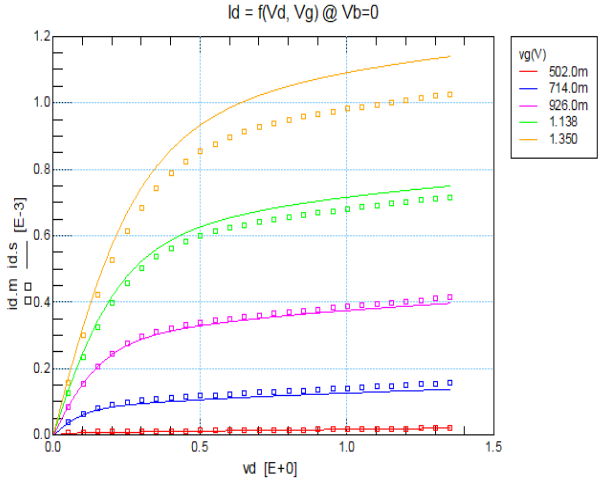


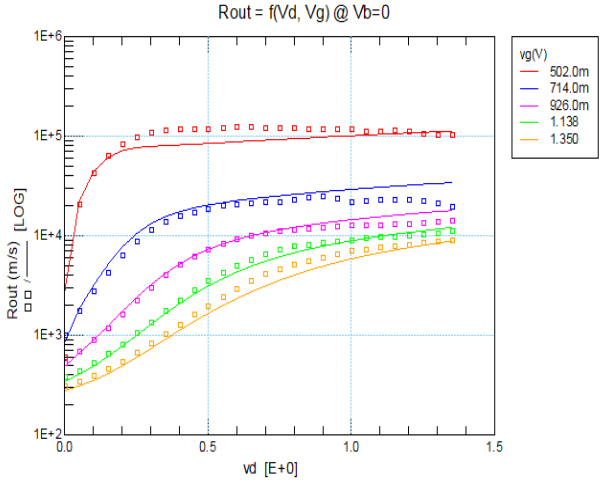
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

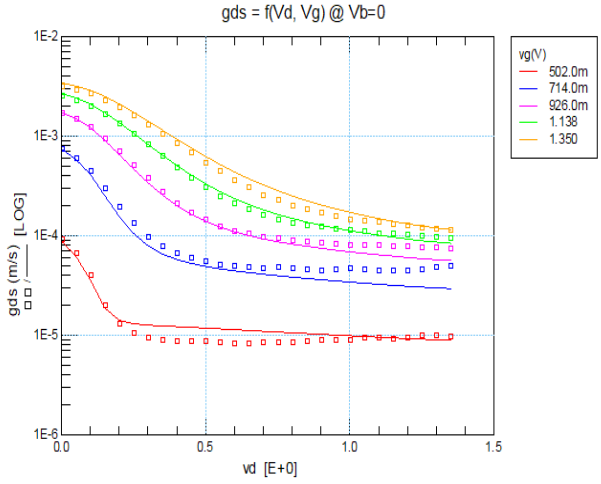




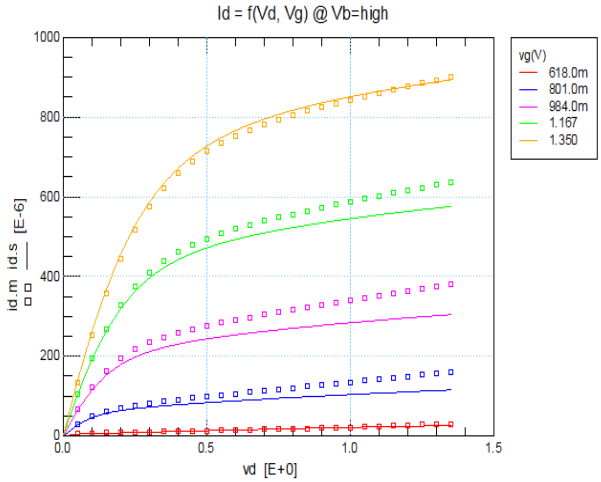
idvd, Vb = 0V, T = 27°C

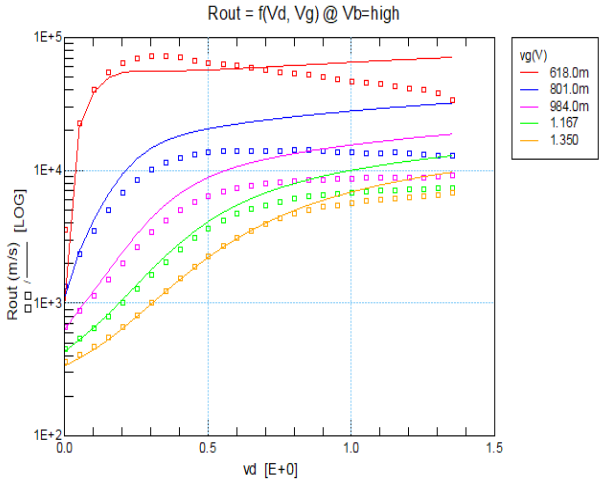


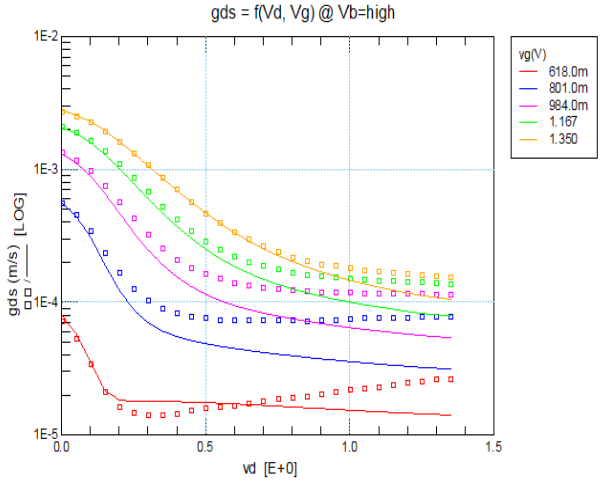




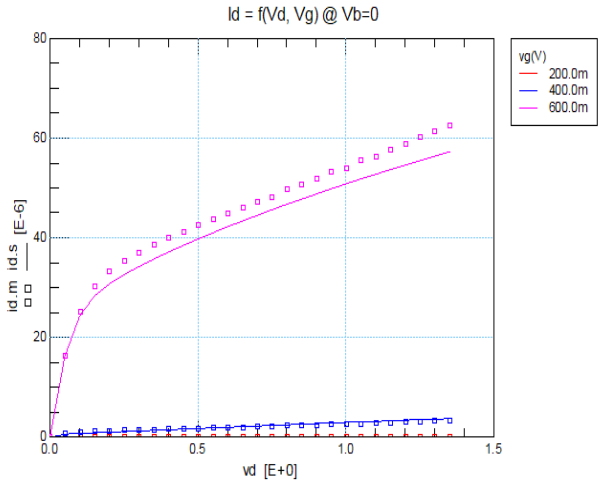
idvd_vbmin, Vb = -1.2V, T = 27°C

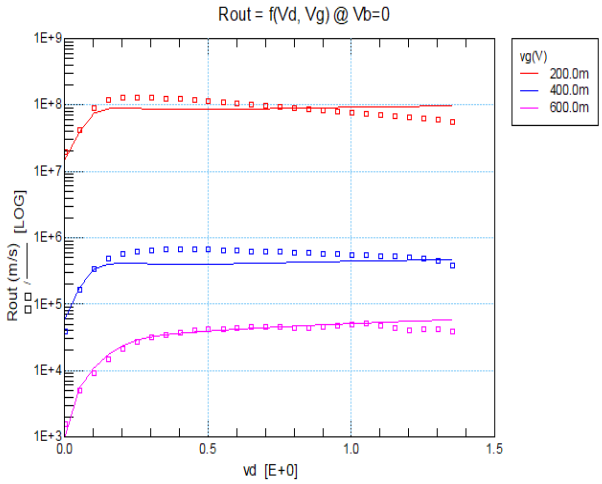


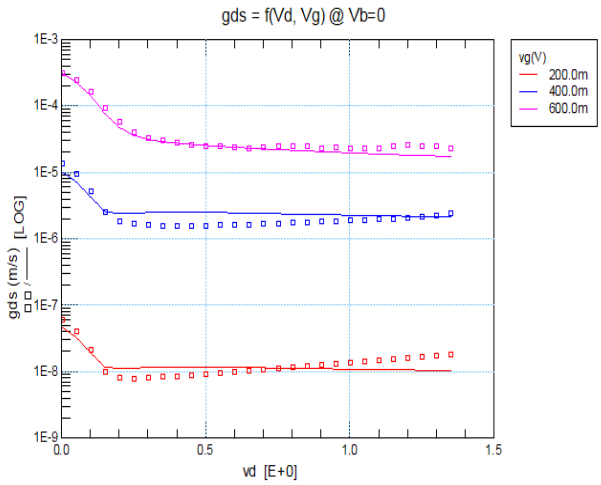




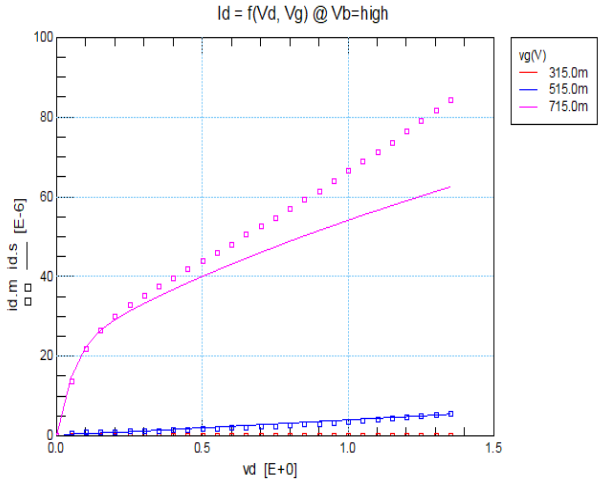
idvd_vth, Vb = 0V, T = 27°C

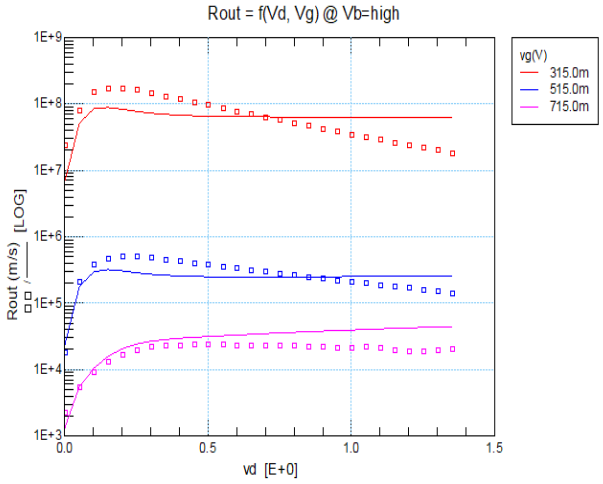


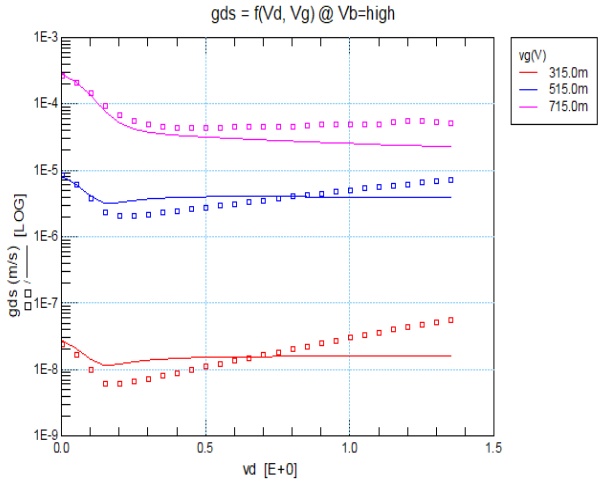




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

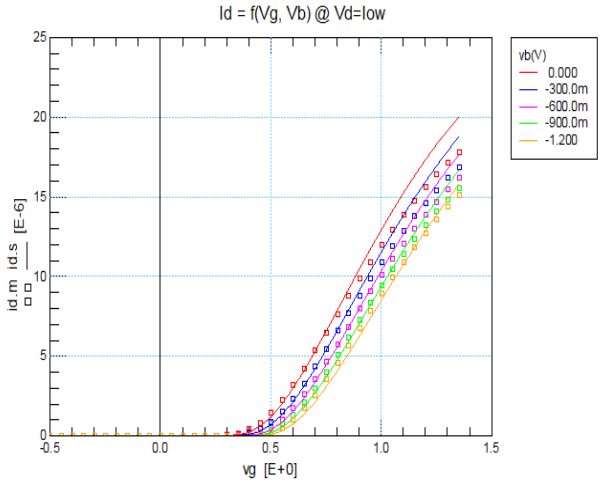


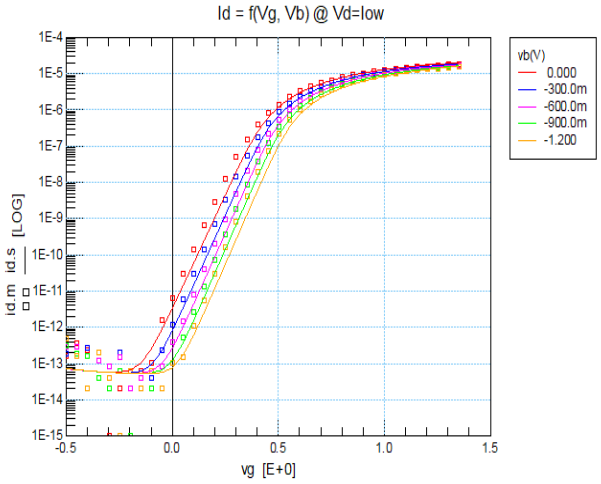


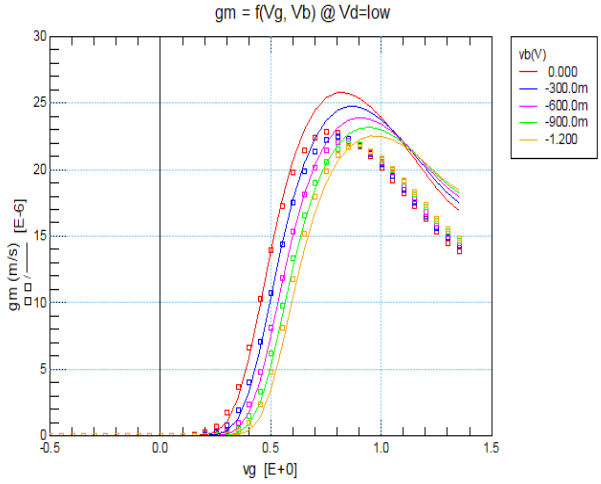


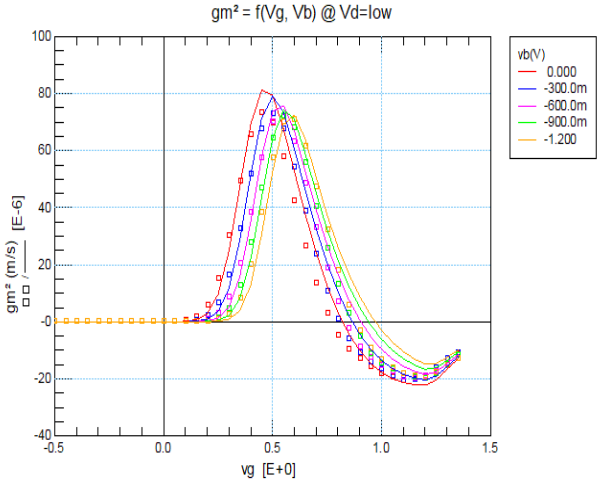
4.38 W0u3_L0u18_S544_5

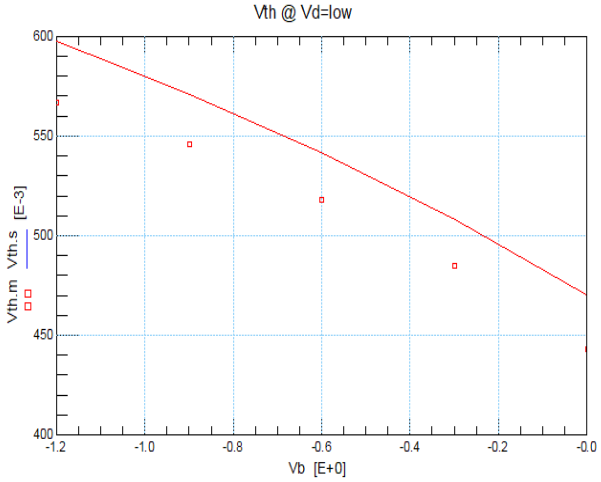
idvg, Vd = 0.05V, T = 27°C

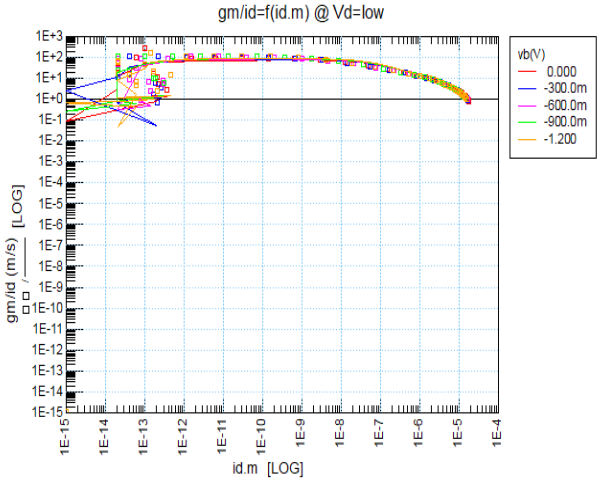


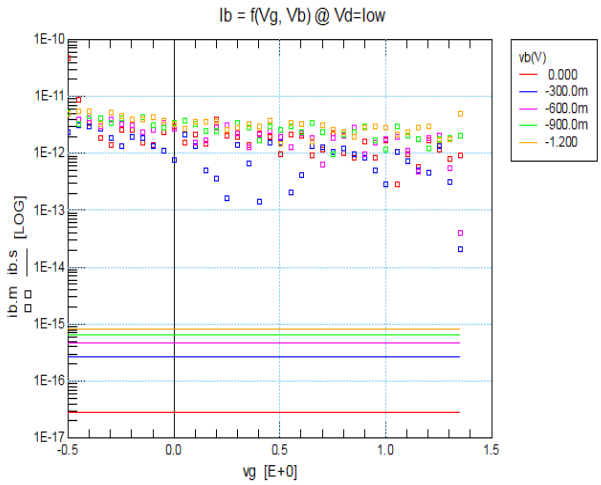


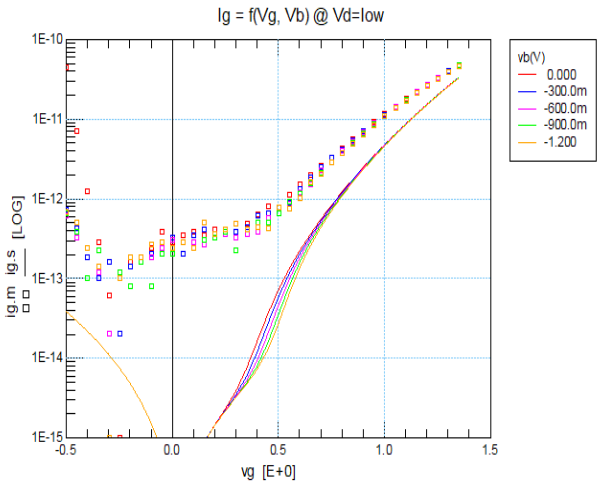




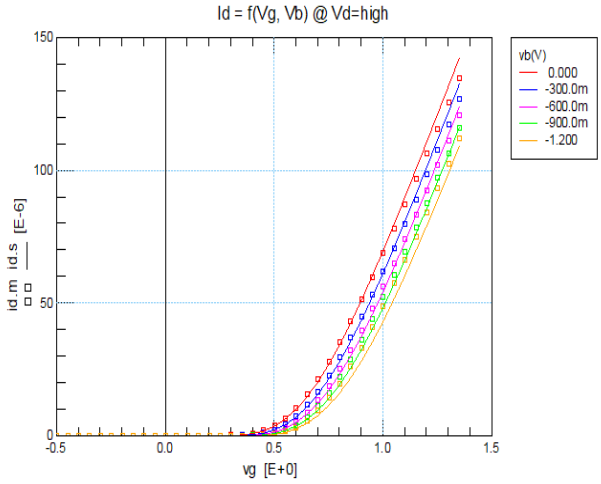


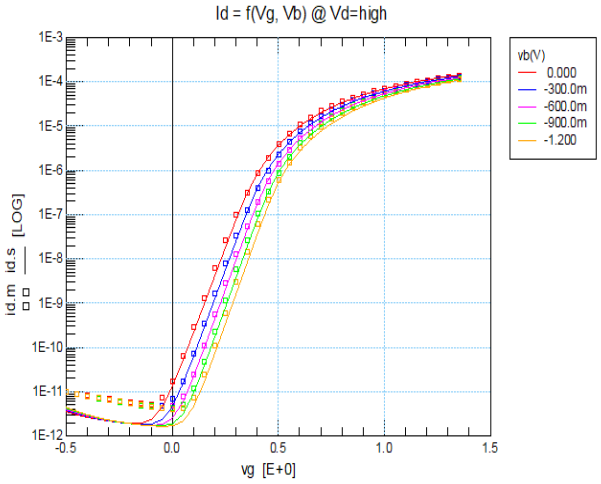


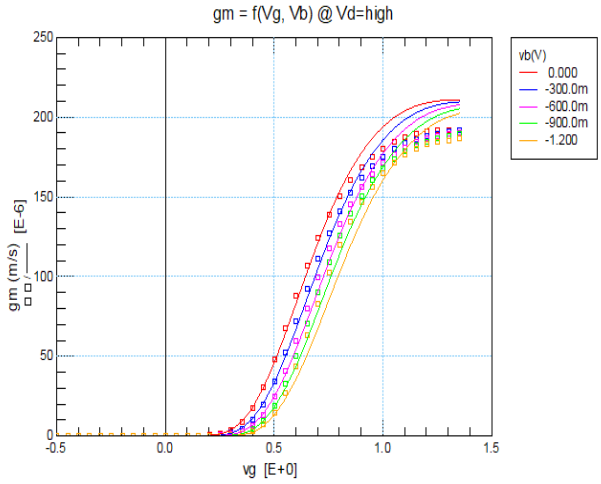


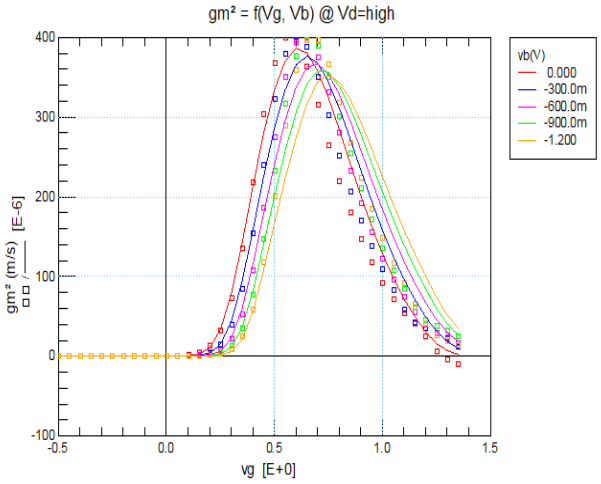


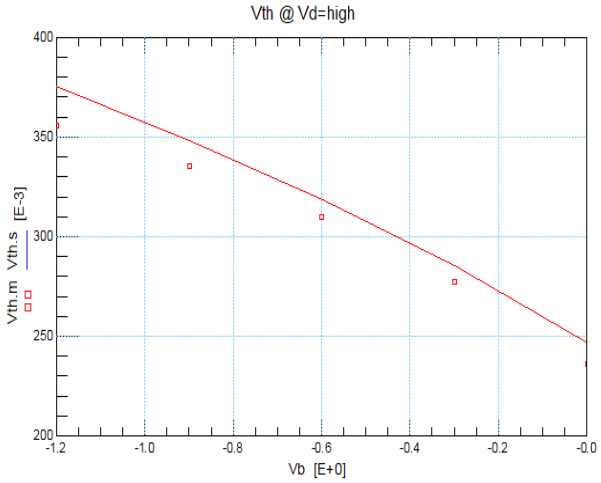
idvg, Vd = 1.2V, T = 27°C

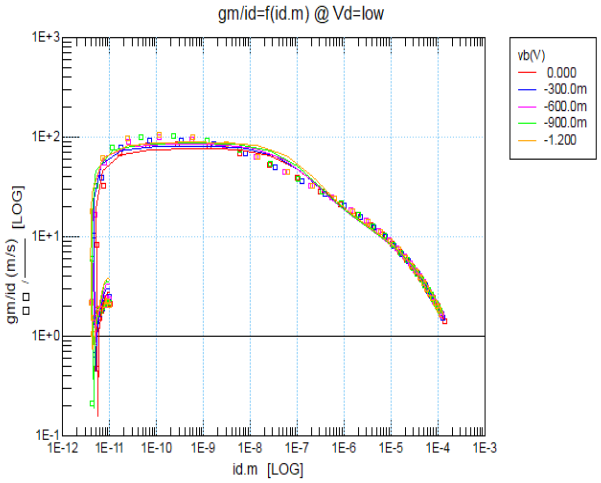


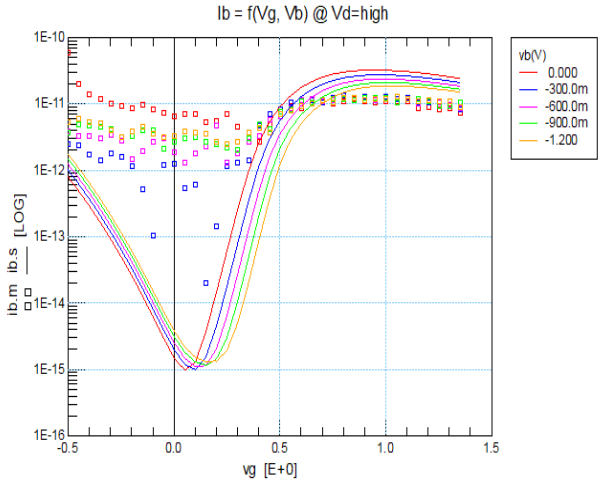


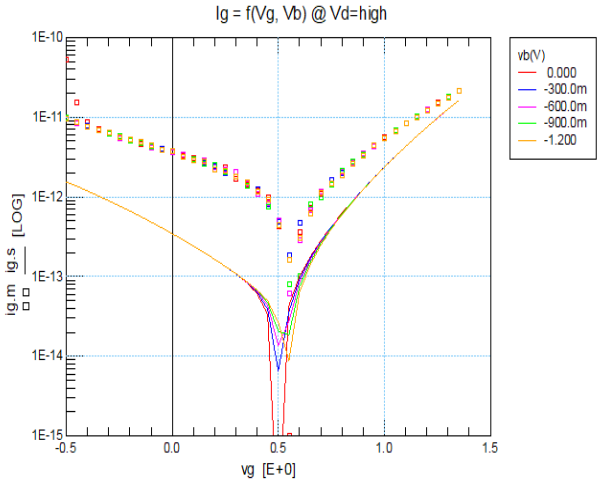




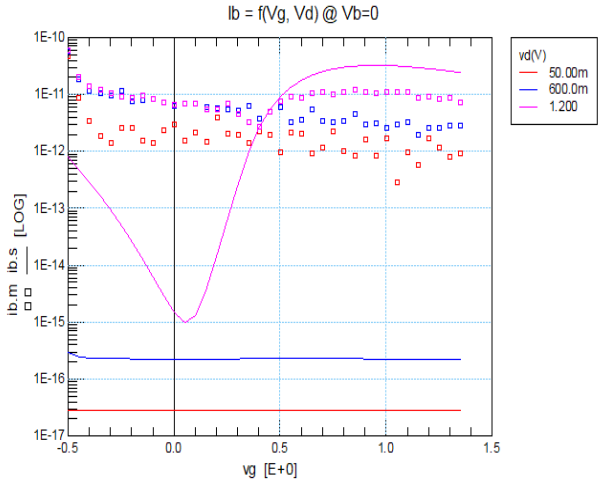


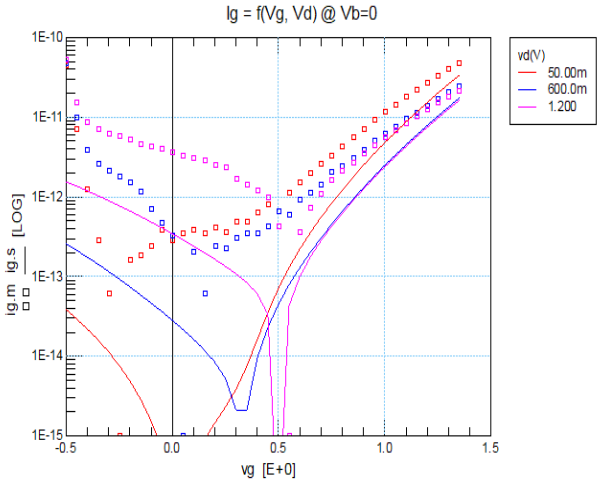




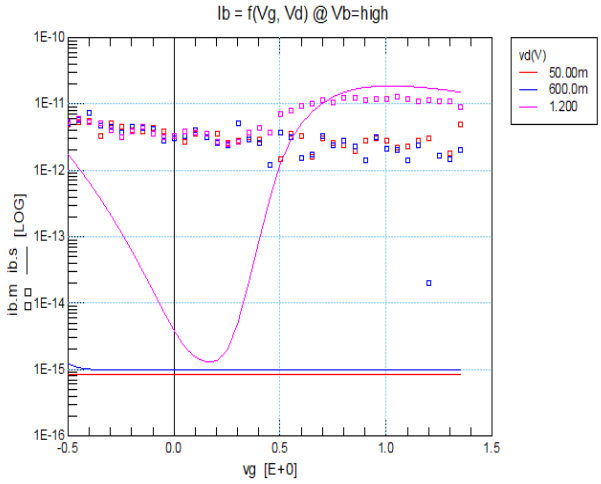


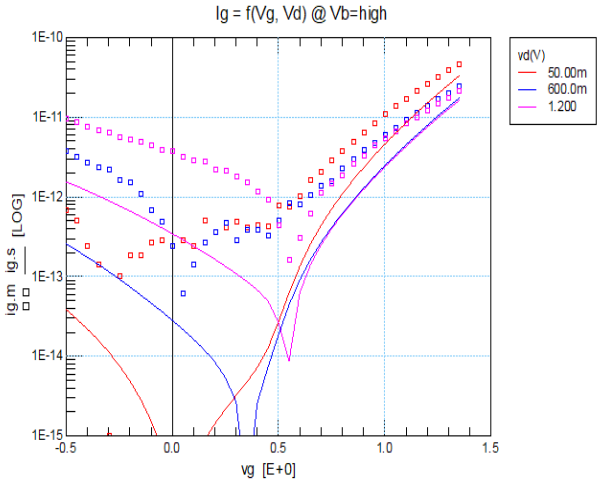
idvg, Ib,Ig, Vb = 0V, T = 27°C



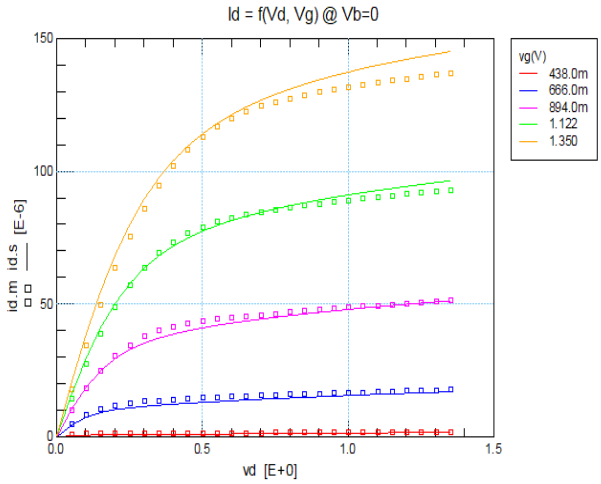


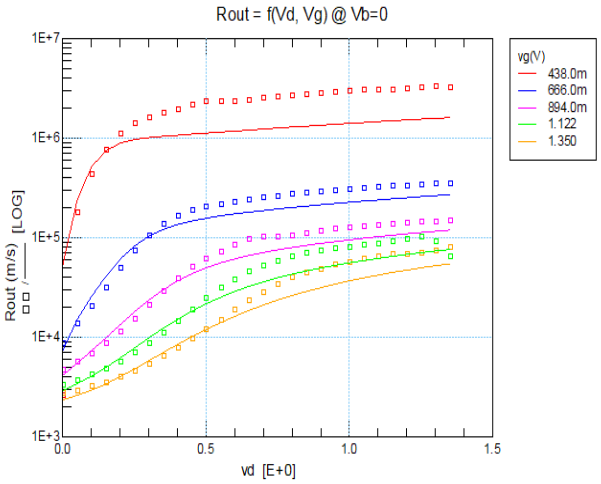
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

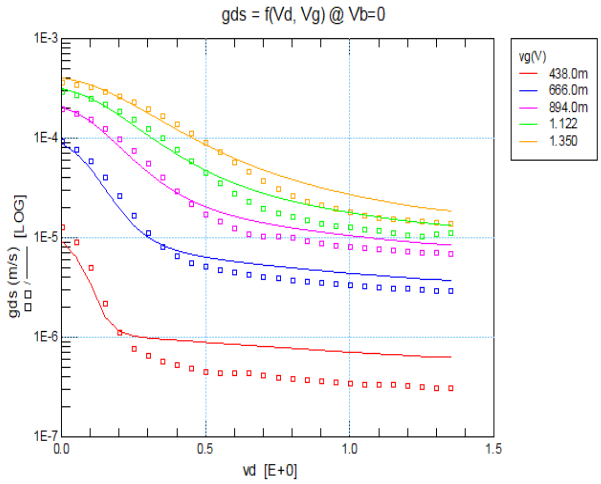




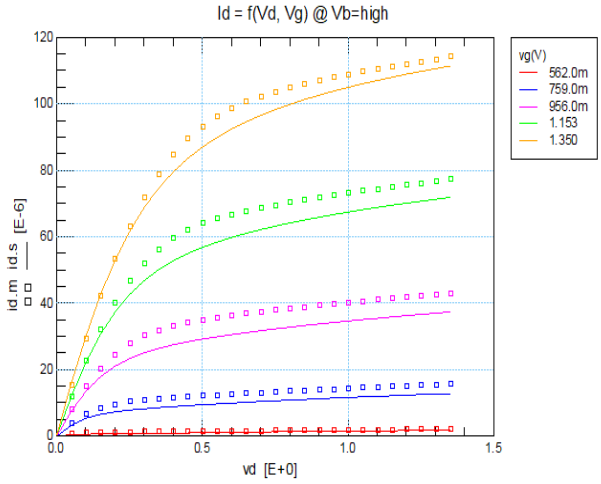
idvd, Vb = 0V, T = 27°C

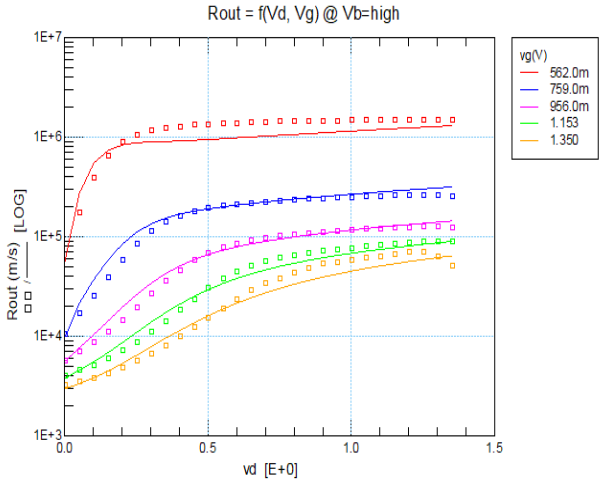


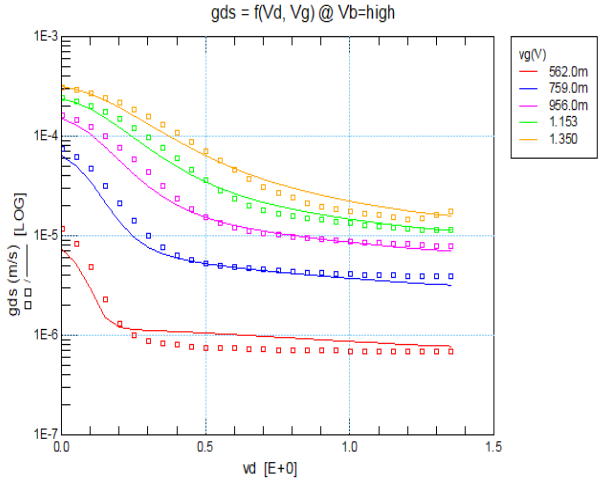




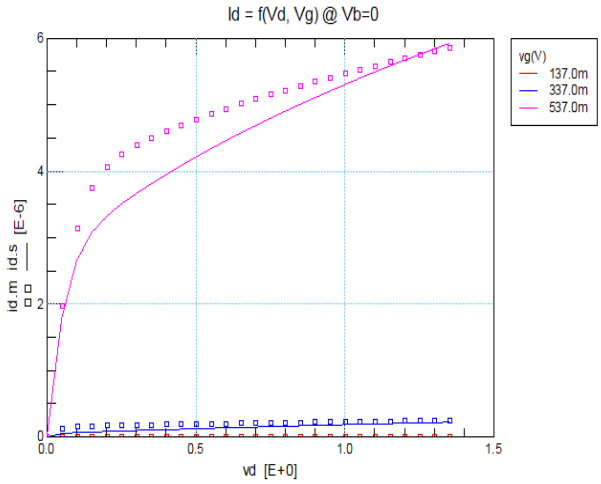
idvd_vbmin, Vb = -1.2V, T = 27°C

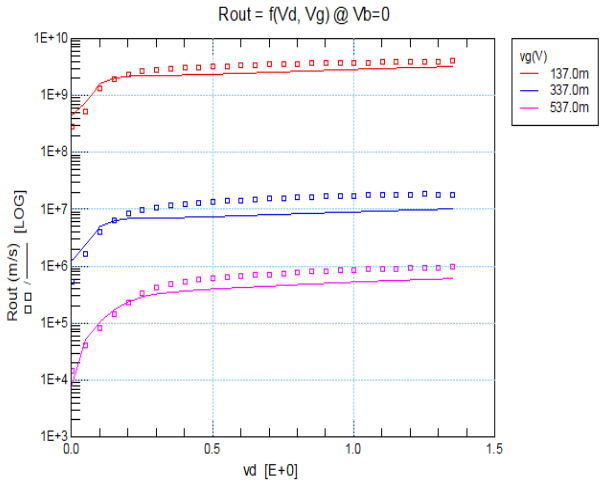


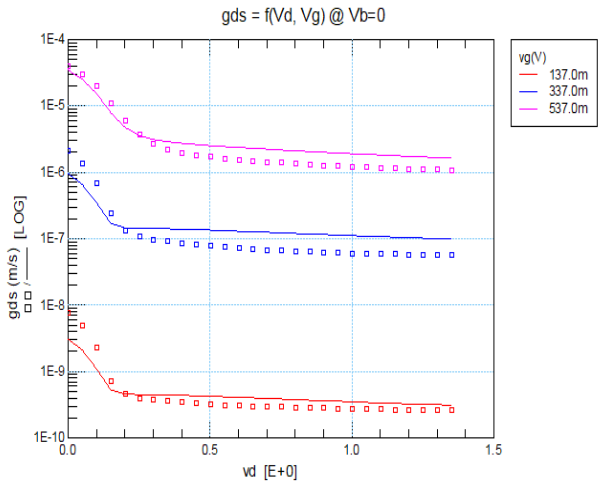




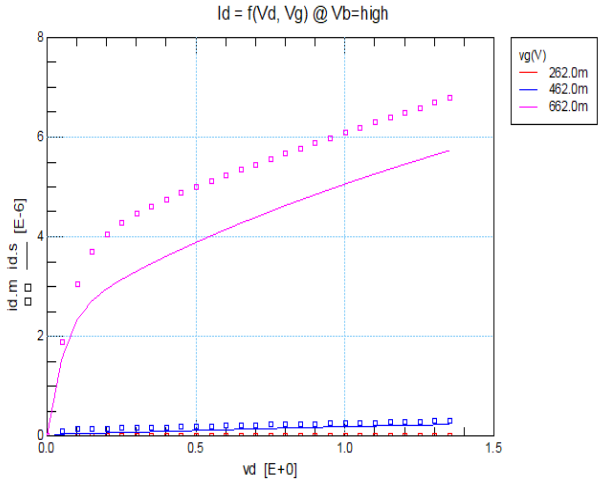
idvd_vth, Vb = 0V, T = 27°C

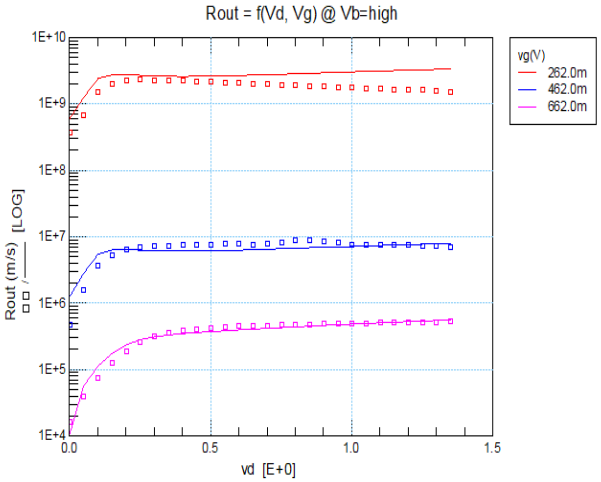


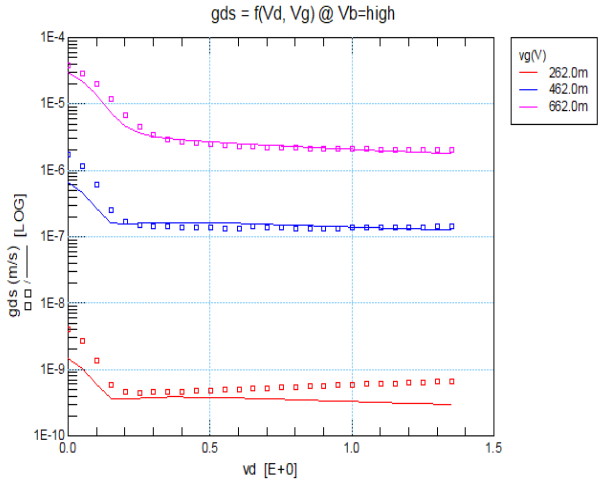




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

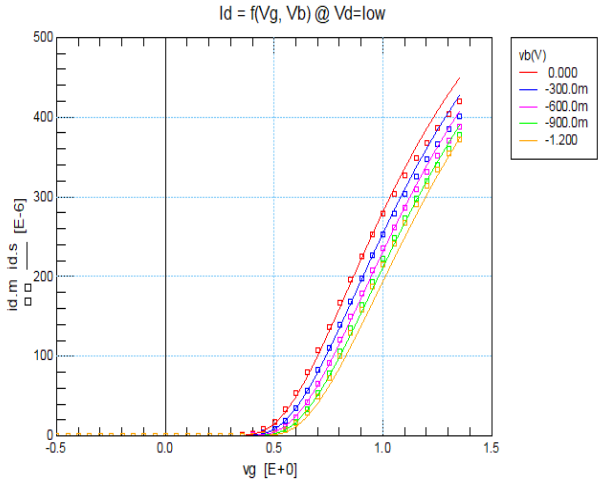


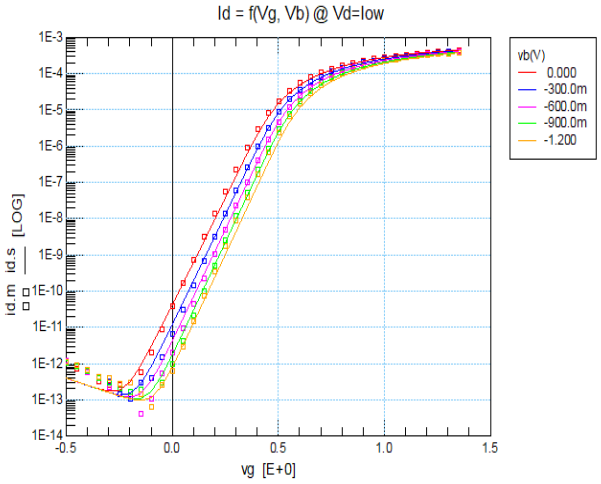


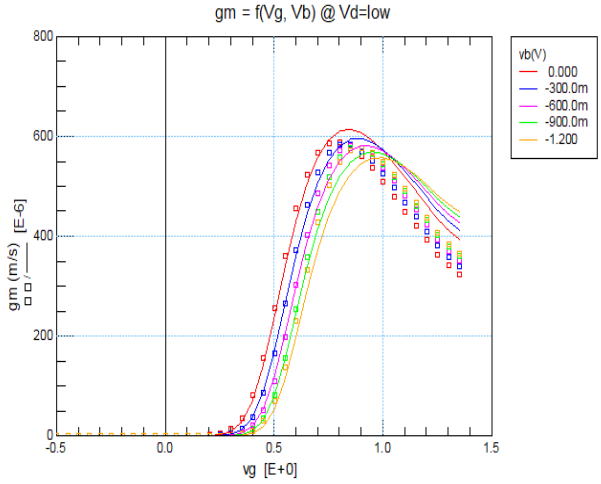


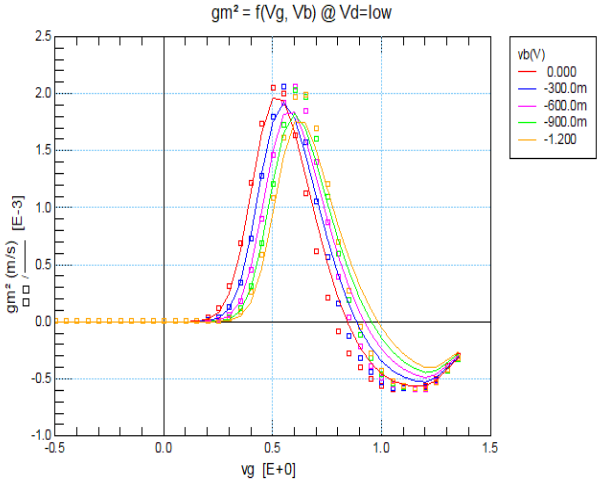
4.39W05u0_L0u13_S542_1

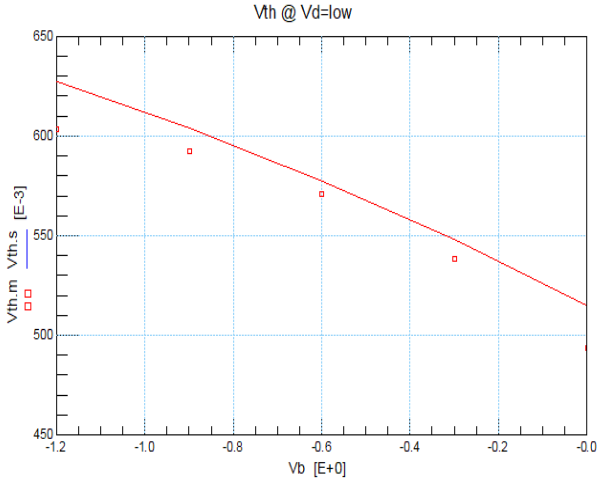
idvg, Vd = 0.05V, T = 27°C

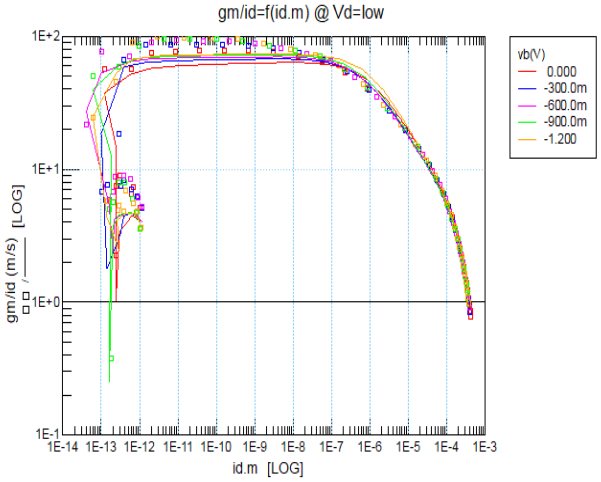


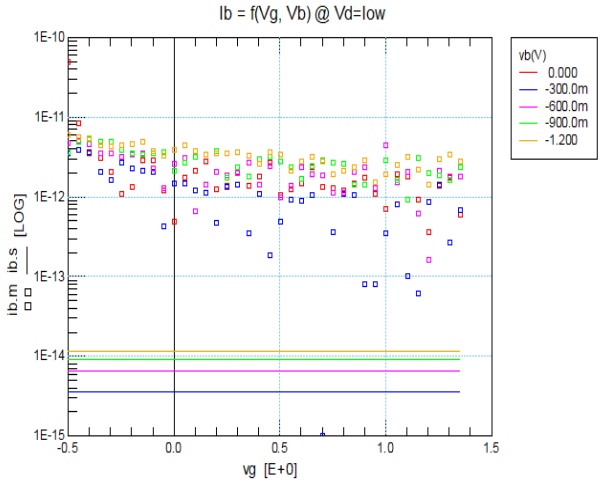


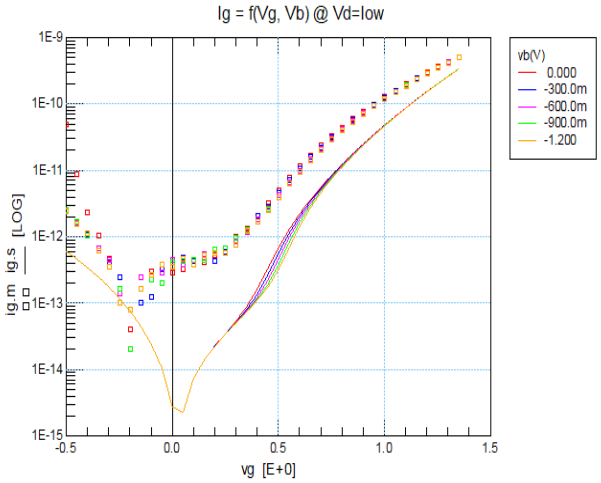




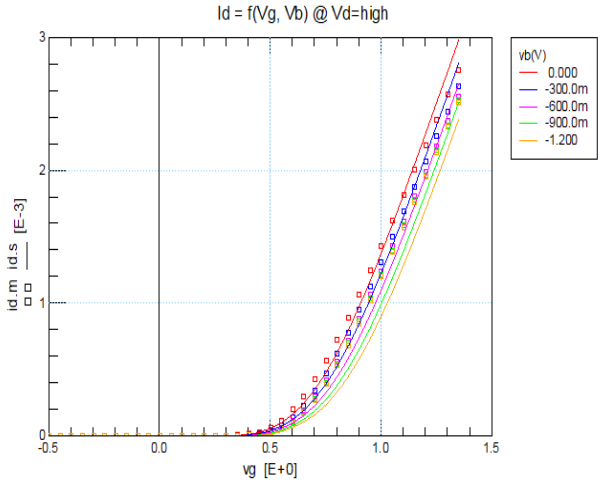


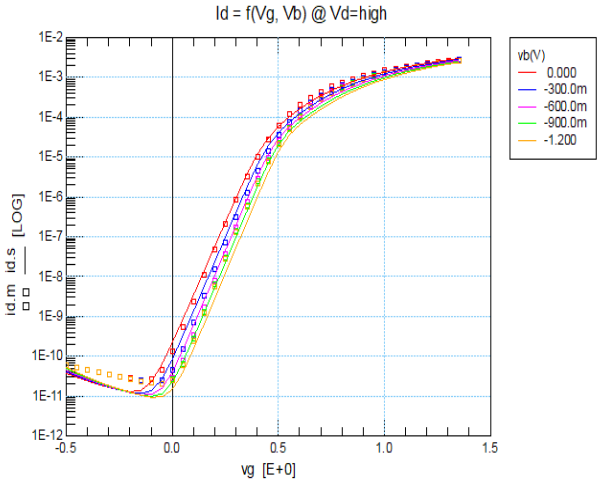


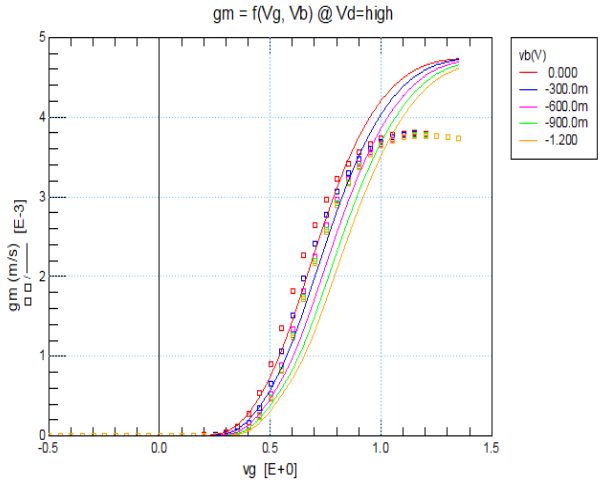


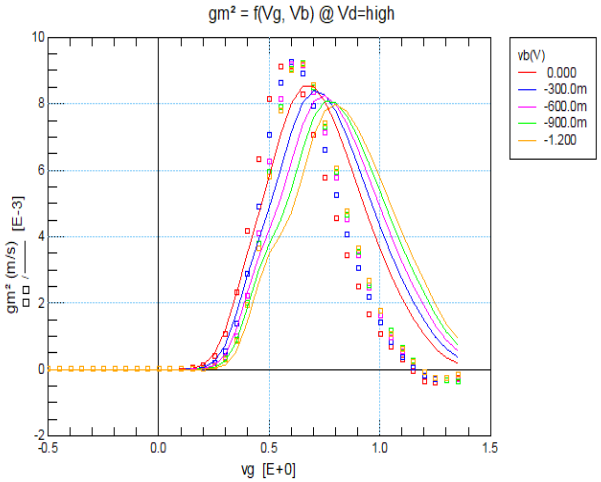


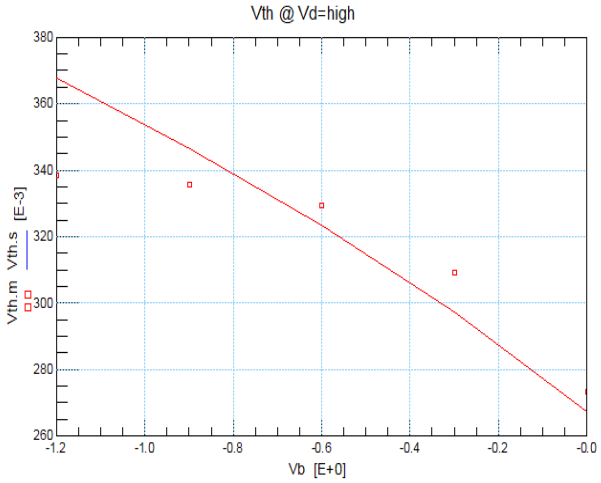
idvg, Vd = 1.2V, T = 27°C

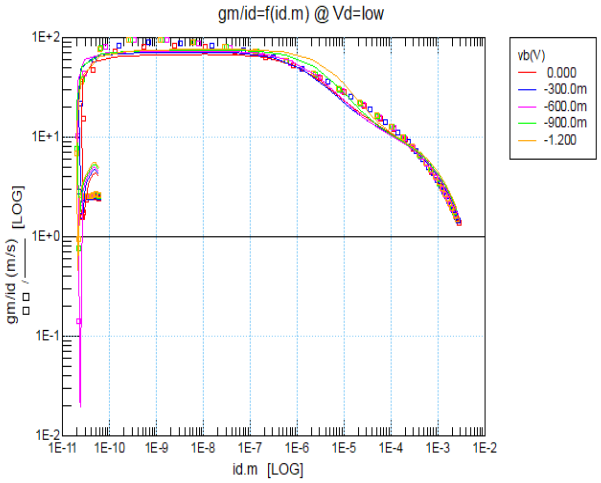


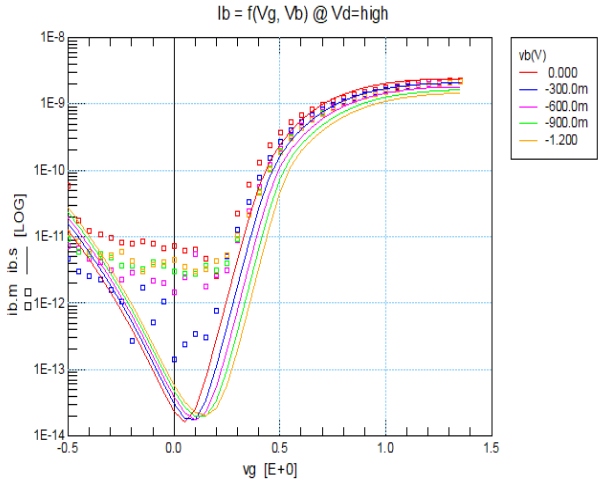


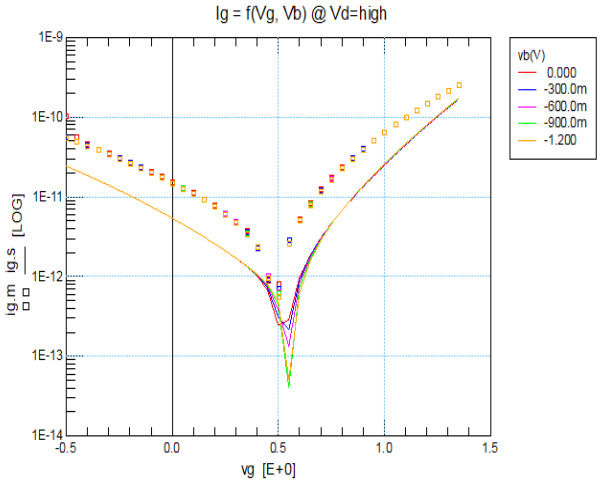




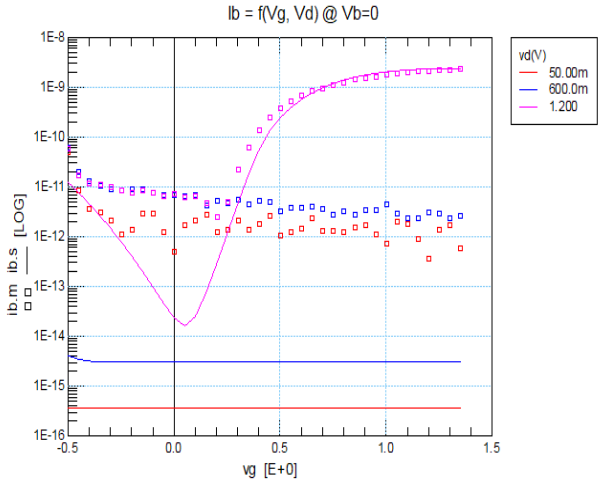


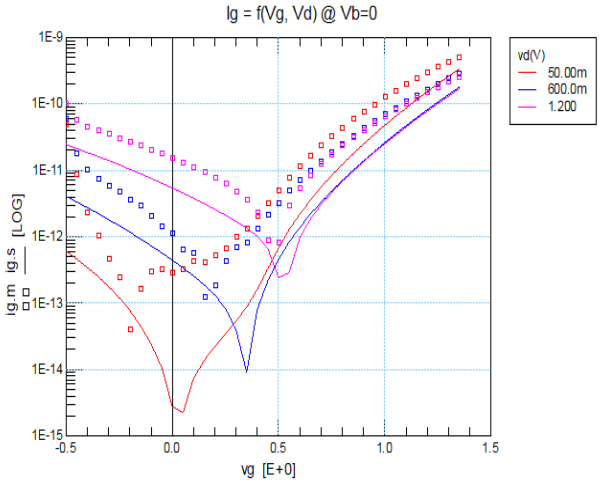




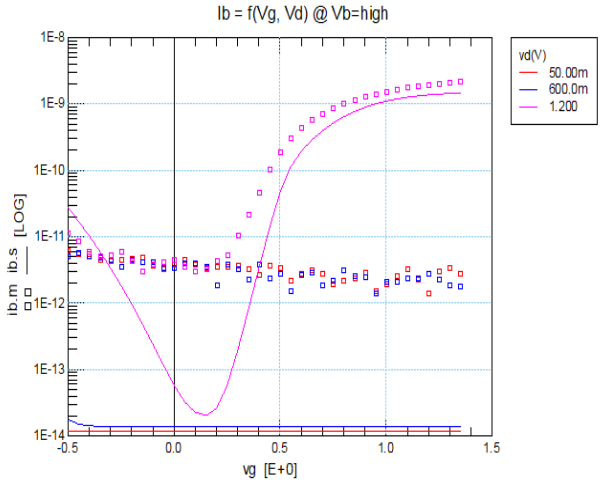


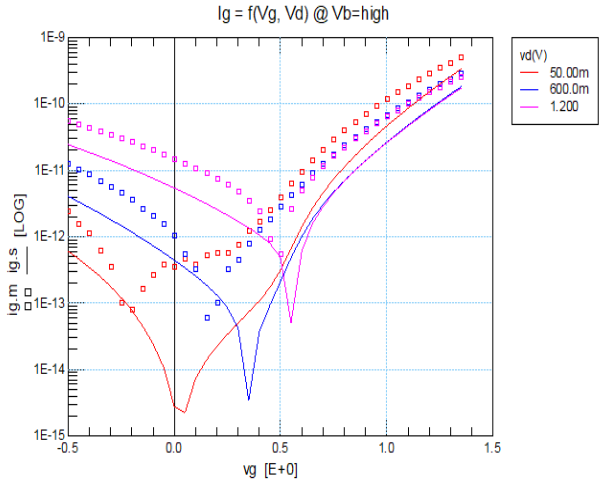
idvg, Ib,Ig, Vb = 0V, T = 27°C



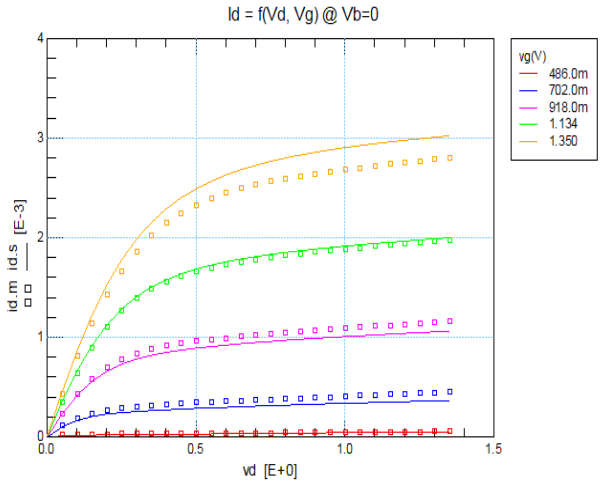


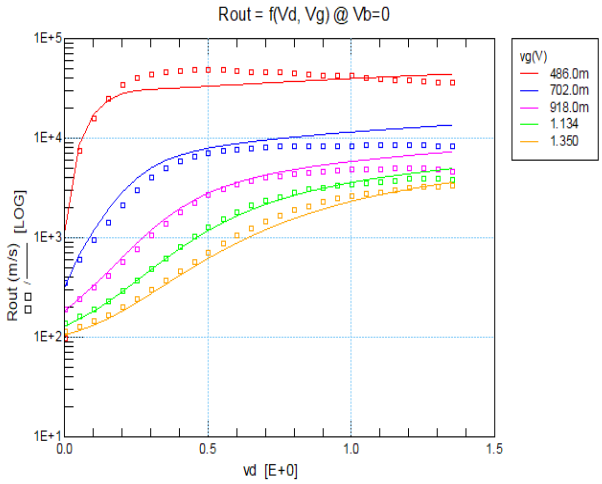
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

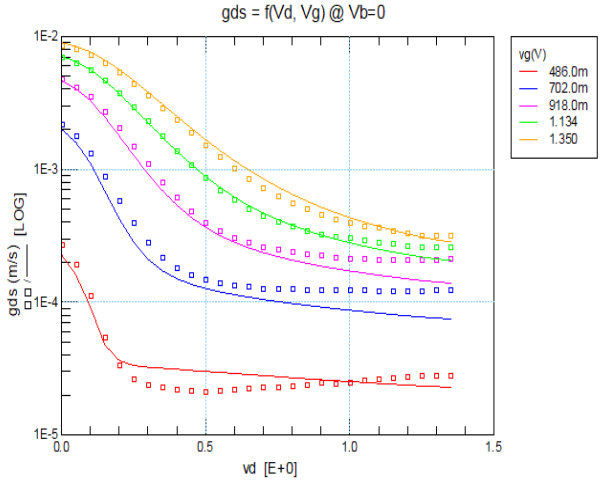




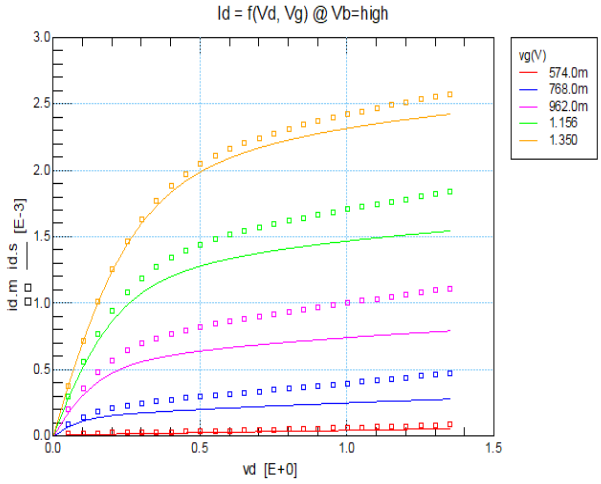
idvd, Vb = 0V, T = 27°C

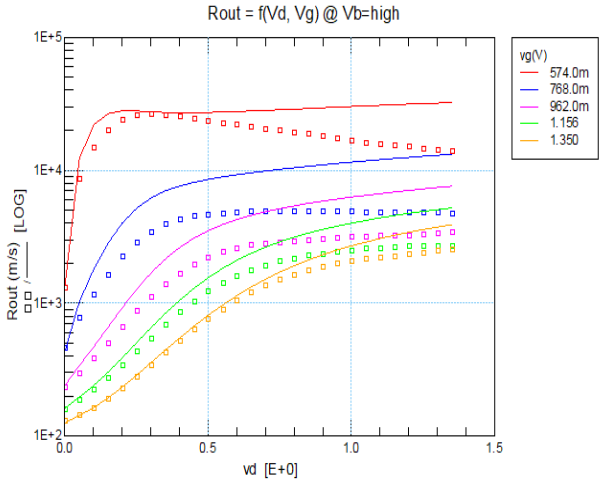


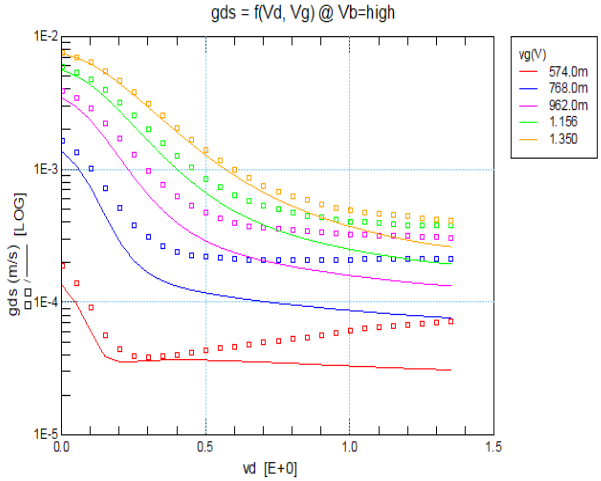




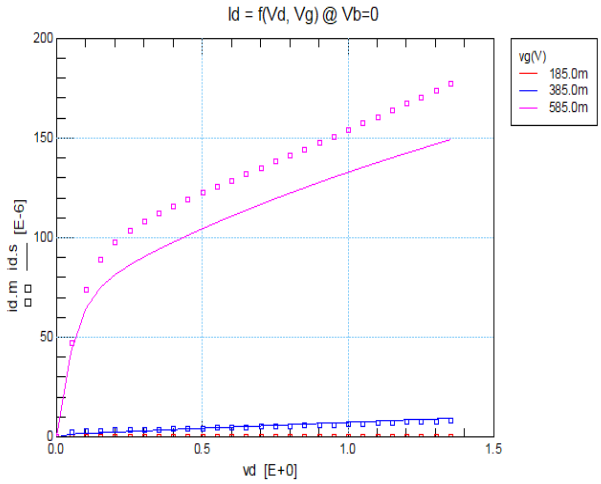
idvd_vbmin, Vb = -1.2V, T = 27°C

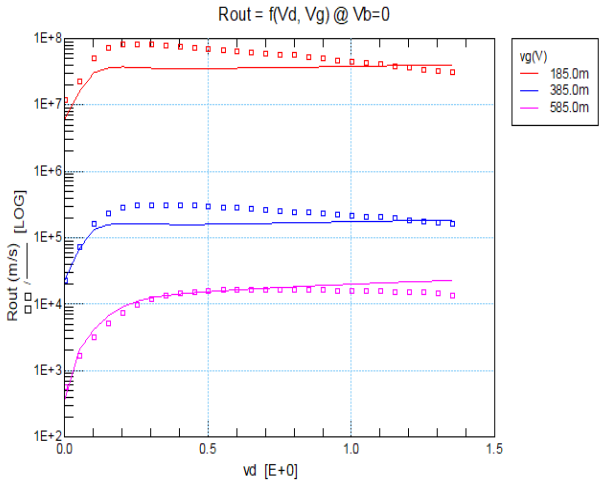


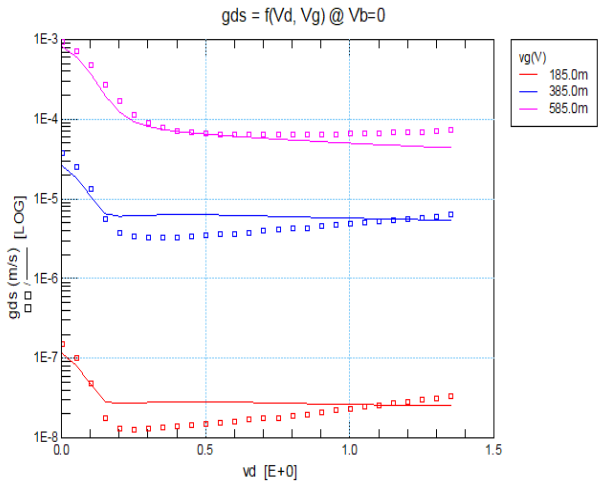




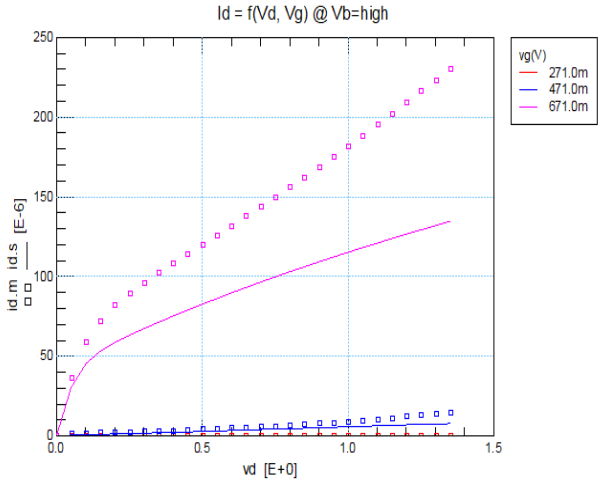
idvd_vth, Vb = 0V, T = 27°C

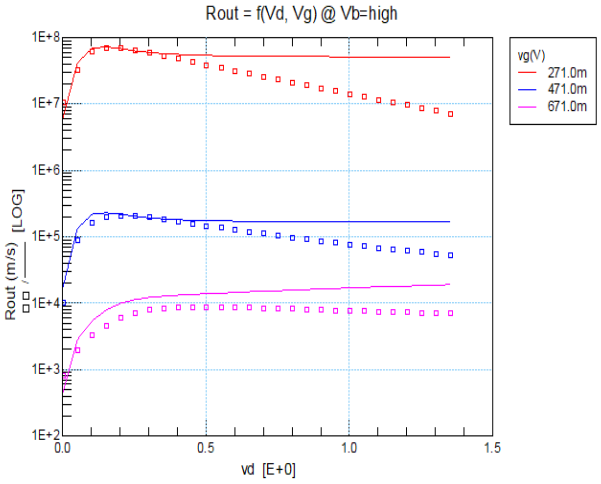


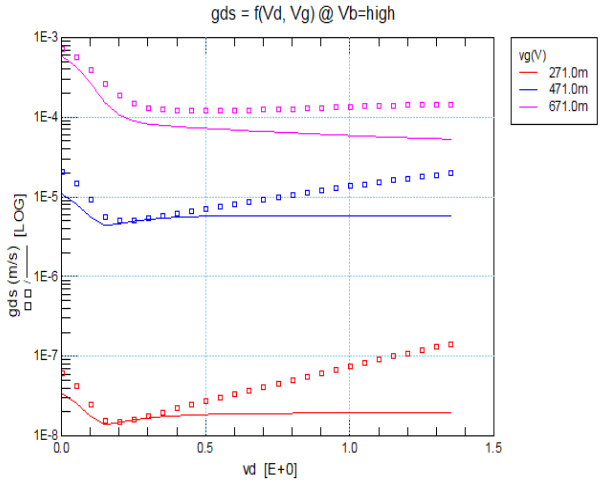




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

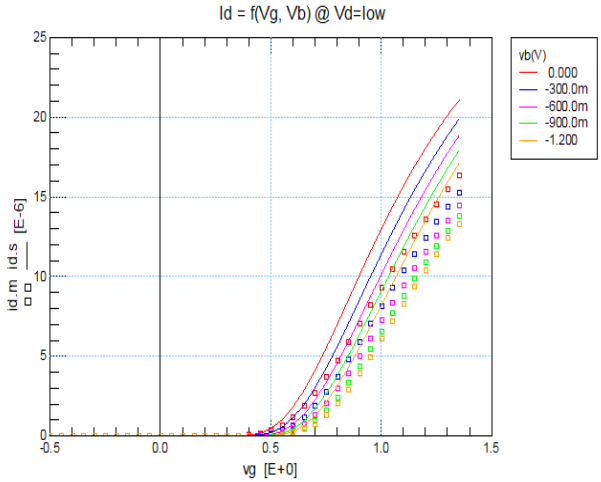


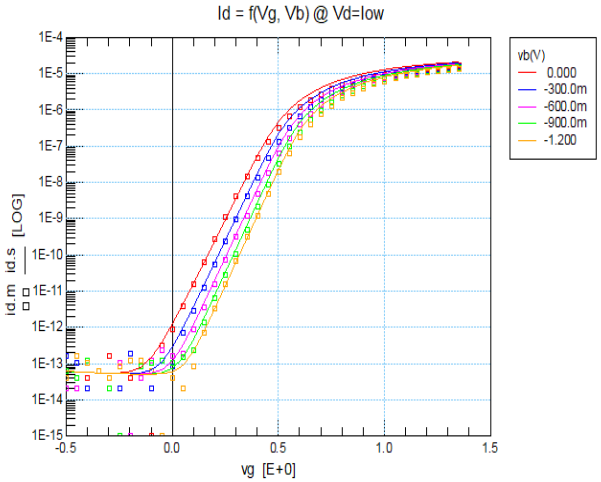


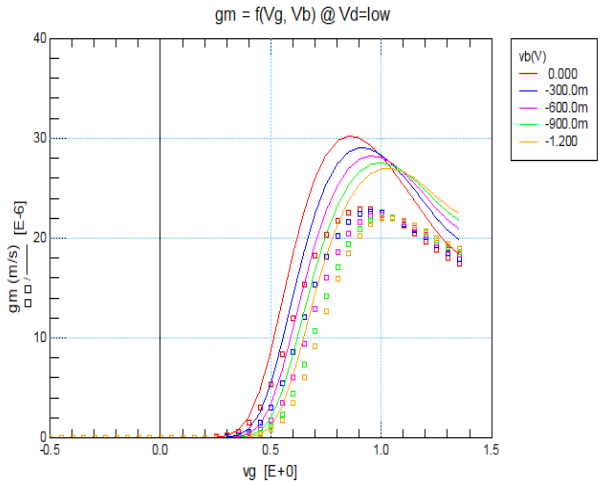


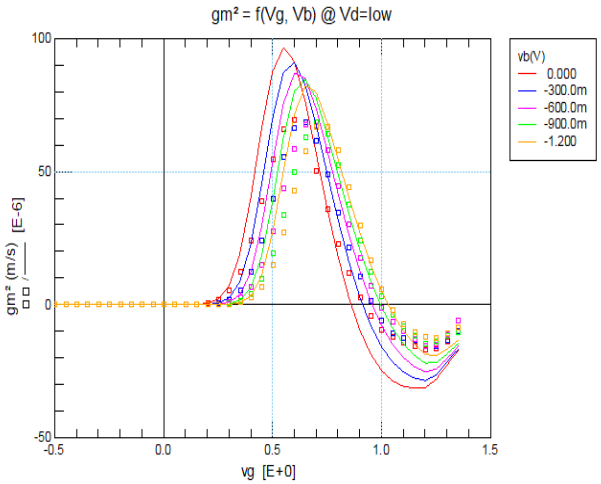
4.40 W0u15_L0u12_S546_2

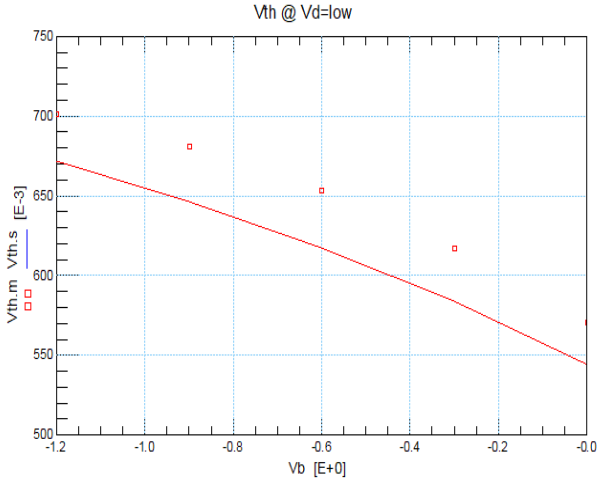
idvg, Vd = 0.05V, T = 27°C

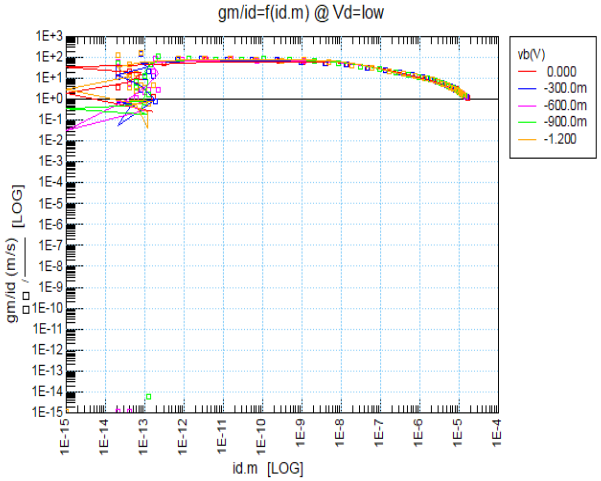


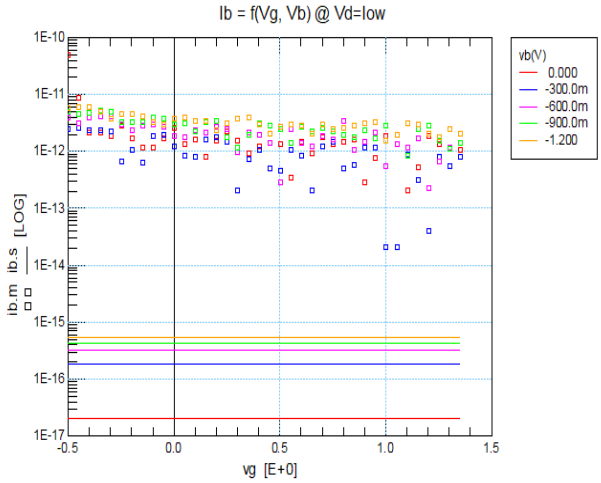


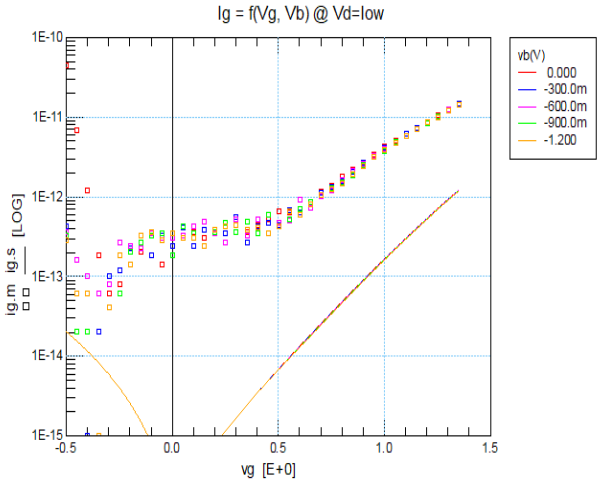




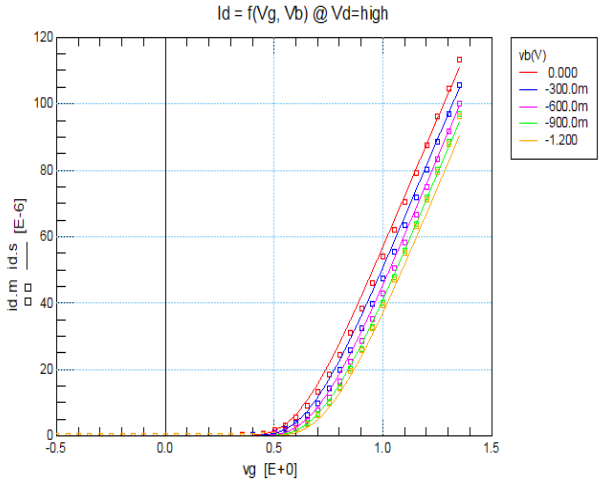


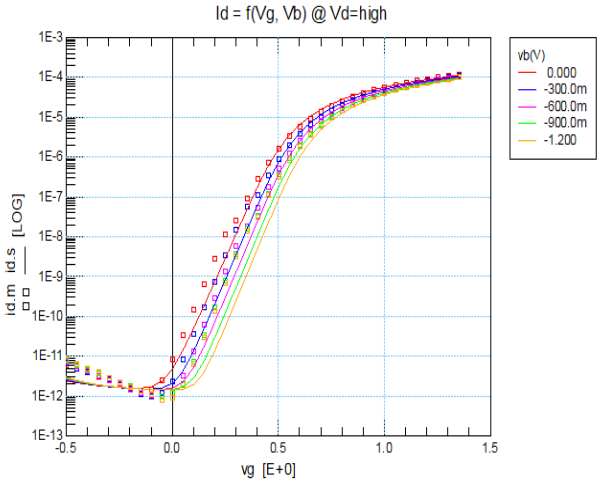


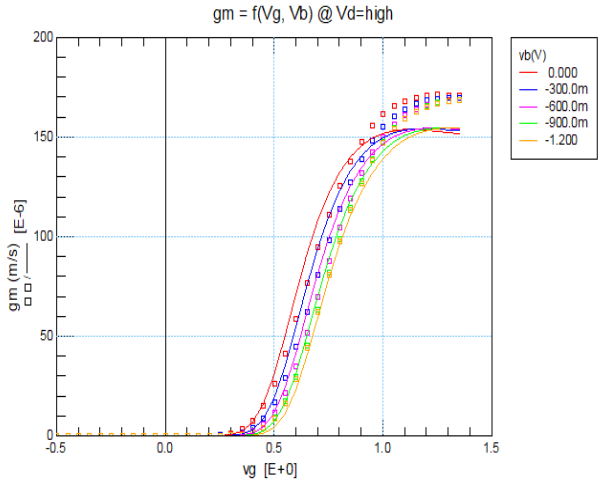


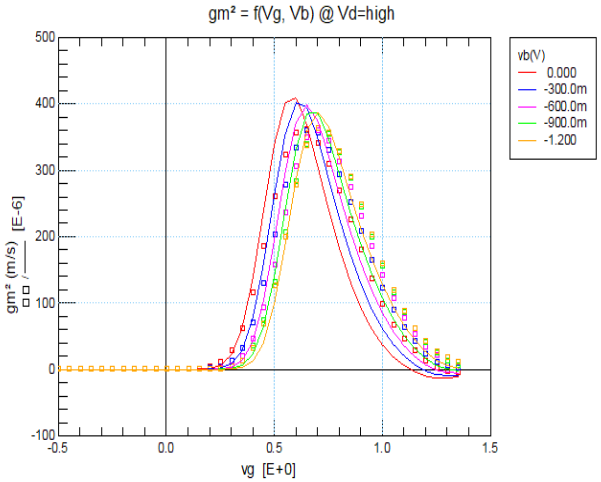


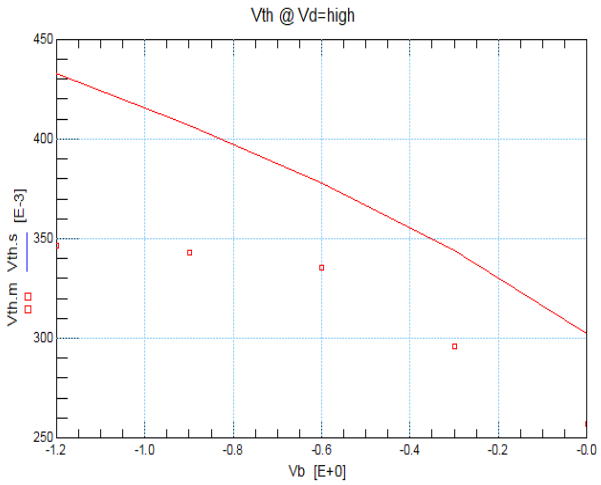
idvg, Vd = 1.2V, T = 27°C

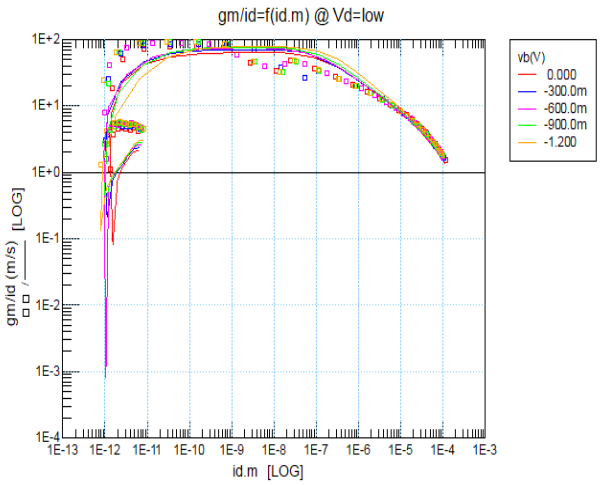


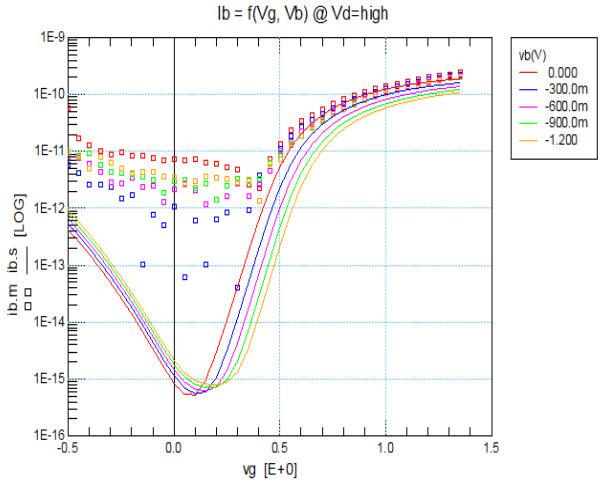


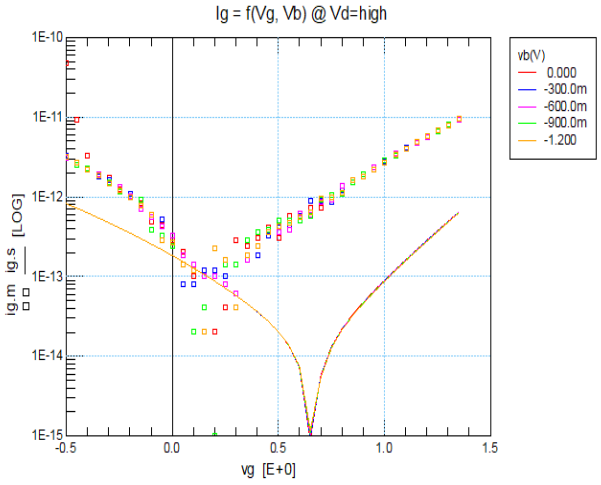




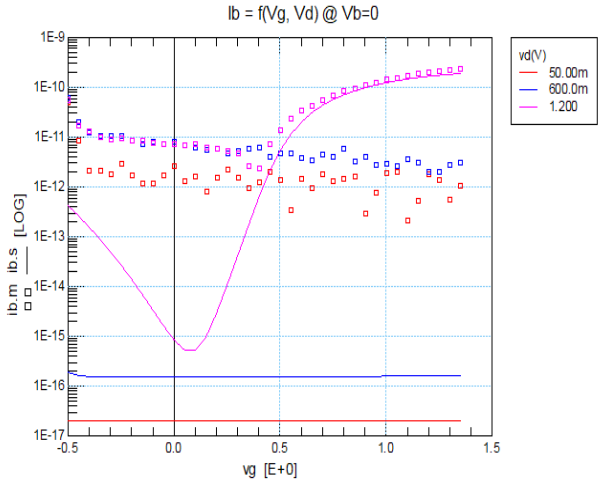


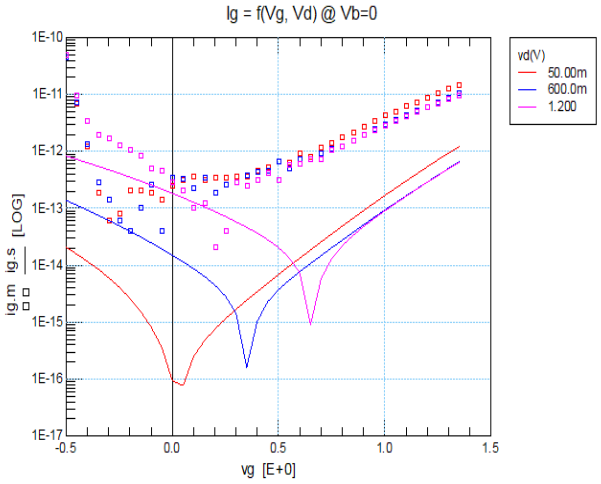




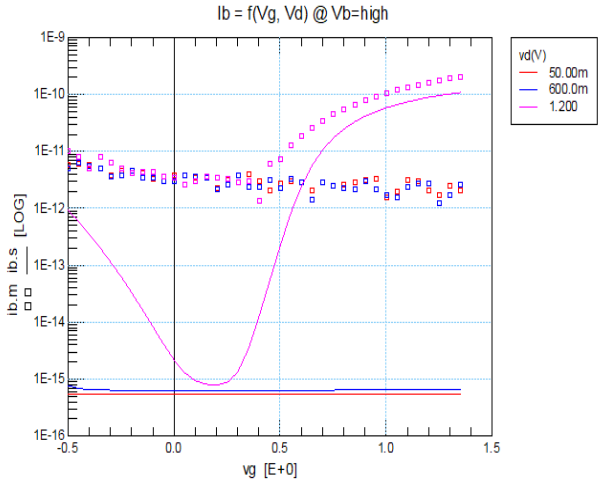


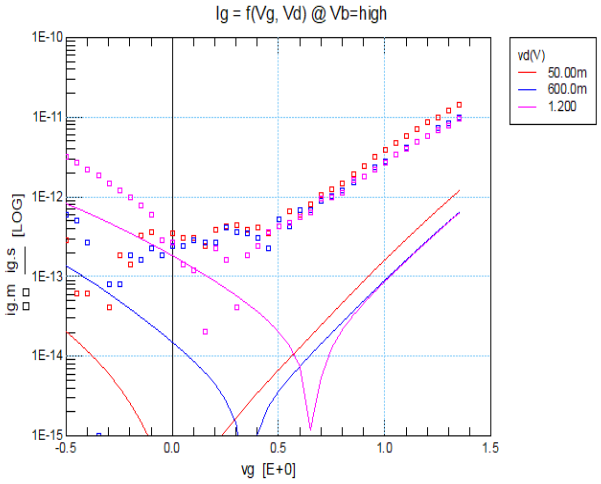
idvg, Ib,Ig, Vb = 0V, T = 27°C



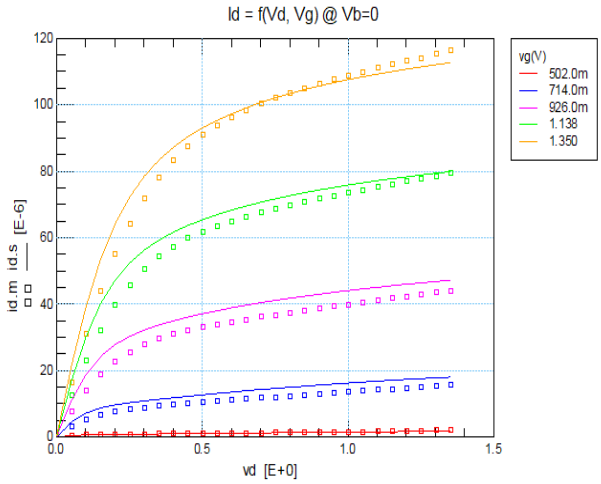


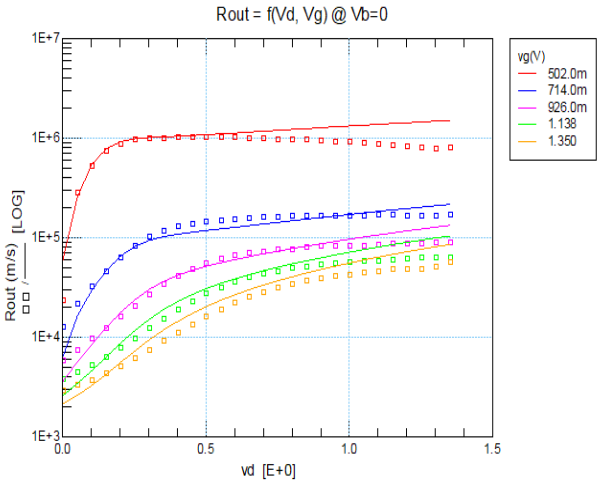
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

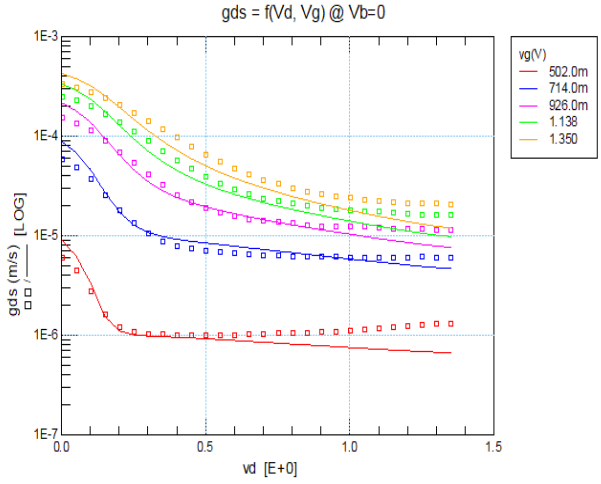




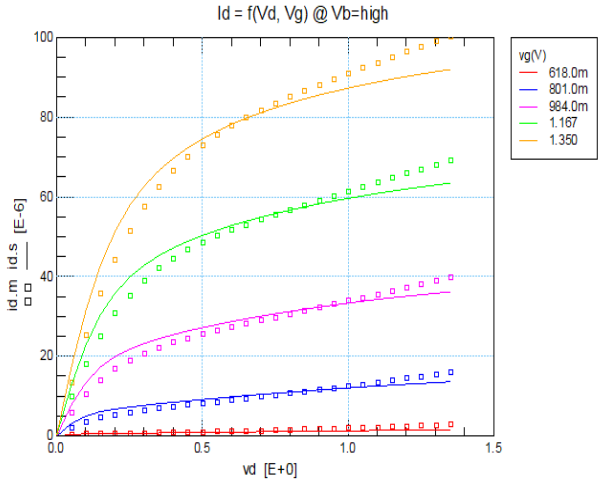
idvd, Vb = 0V, T = 27°C

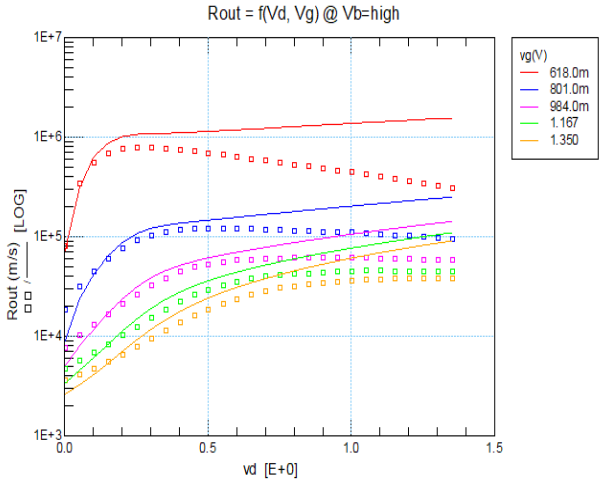


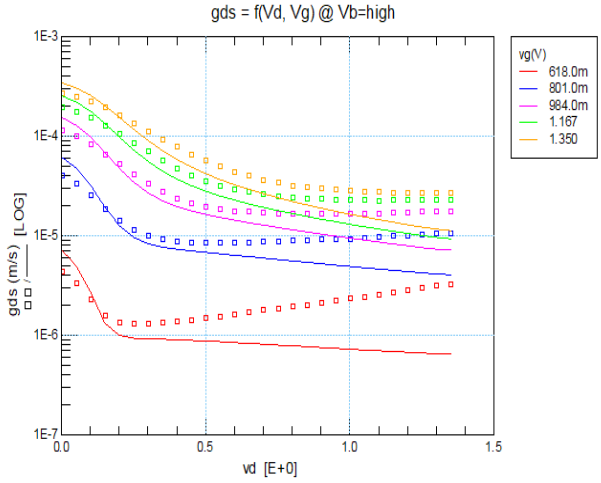




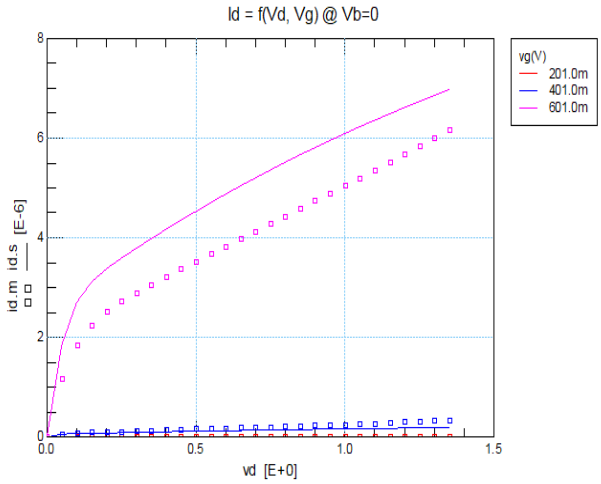
idvd_vbmin, Vb = -1.2V, T = 27°C

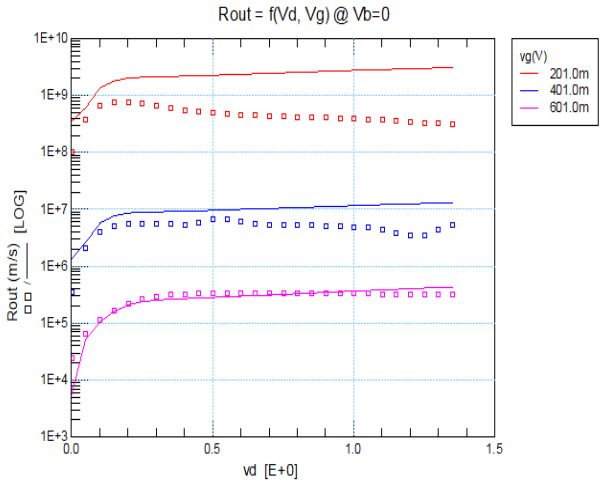


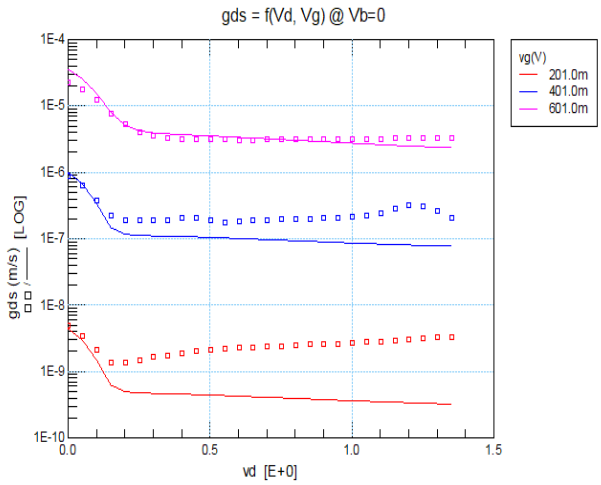




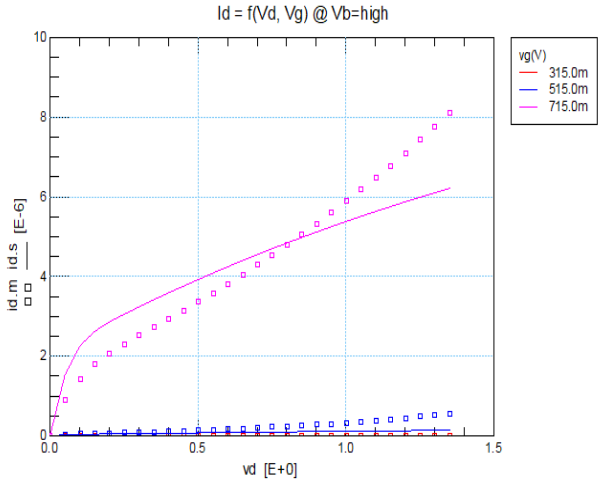
idvd_vth, Vb = 0V, T = 27°C

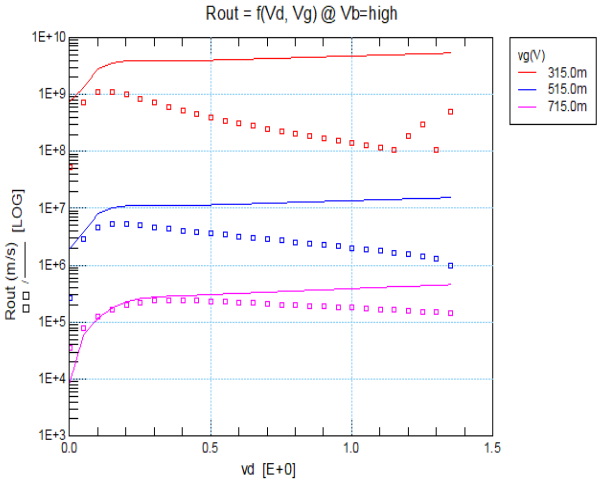


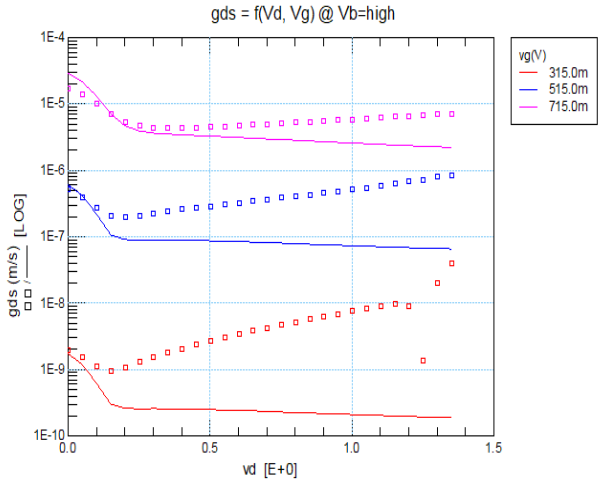




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

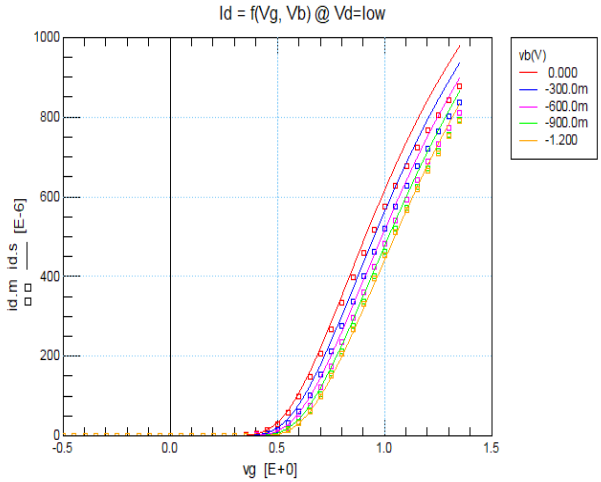


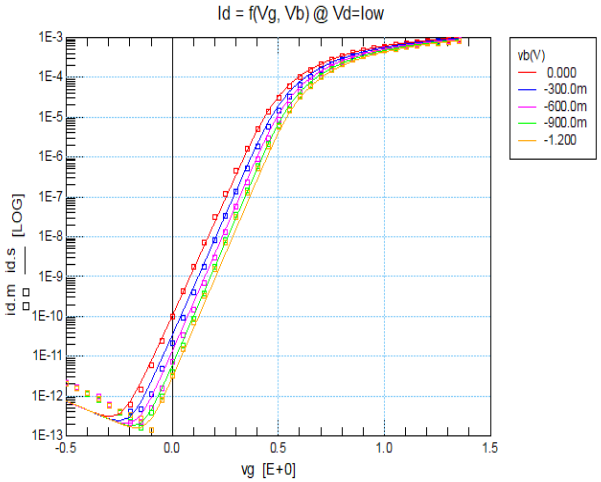


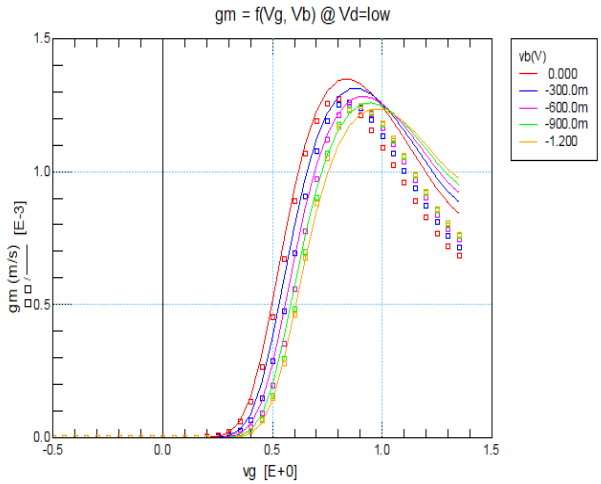


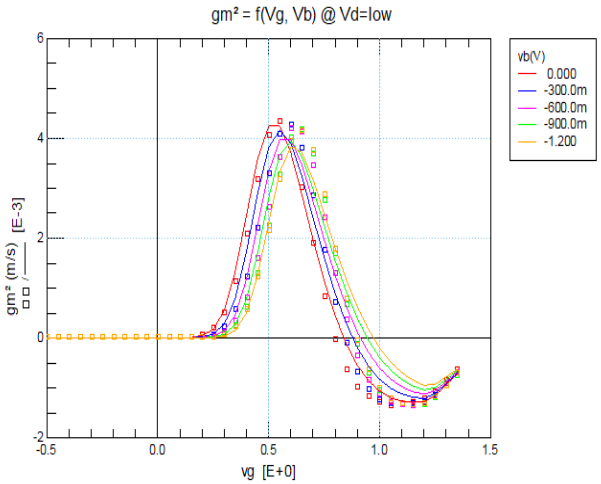
4.41 W10u0_L0u12_S540_1

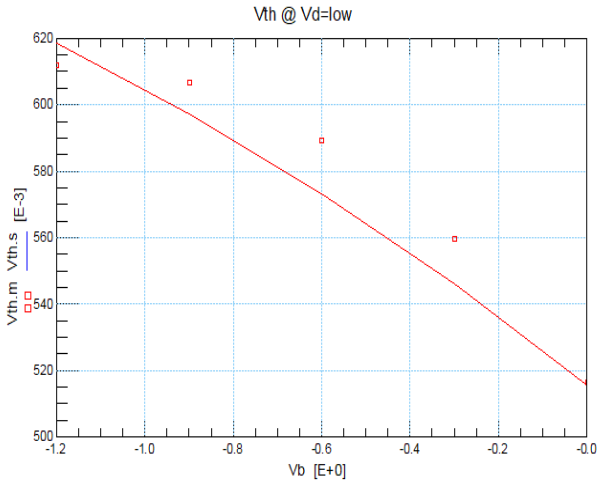
idvg, Vd = 0.05V, T = 27°C

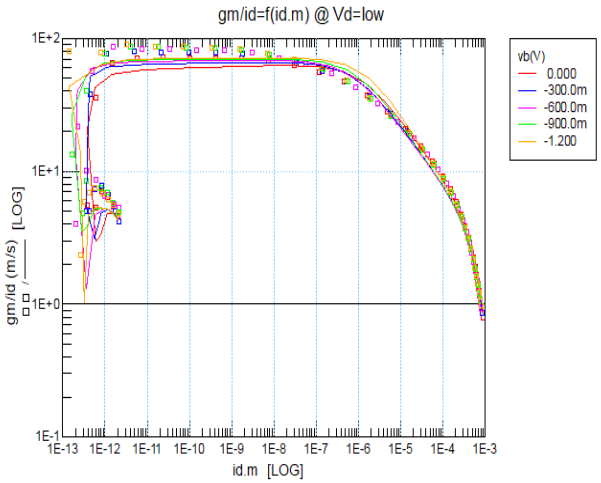


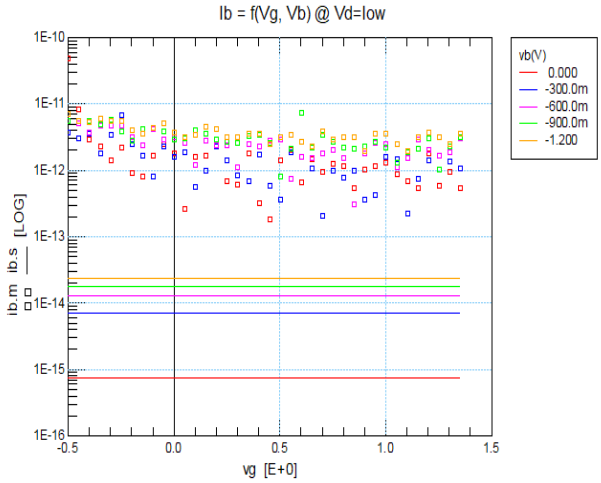


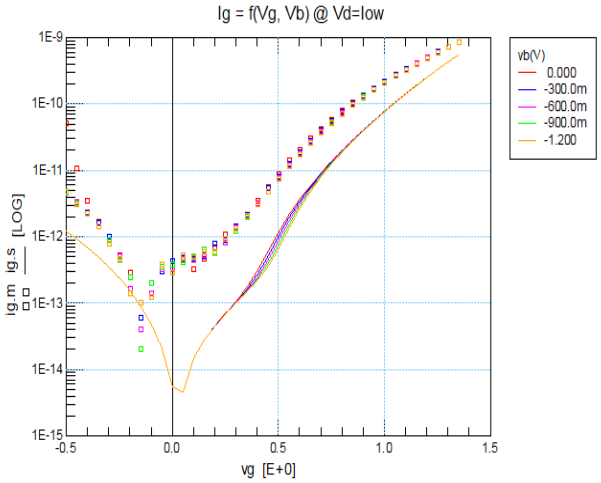




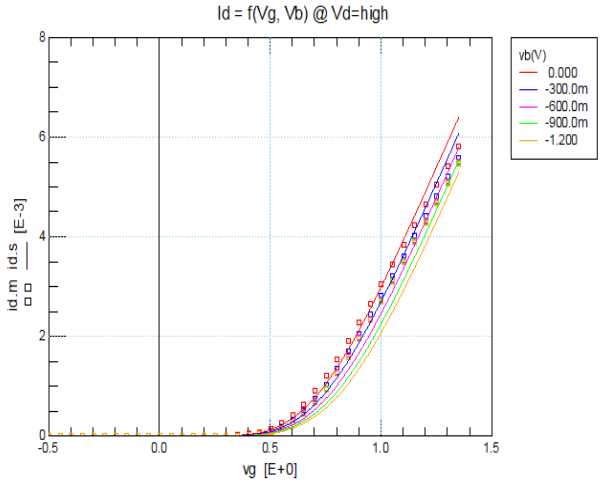


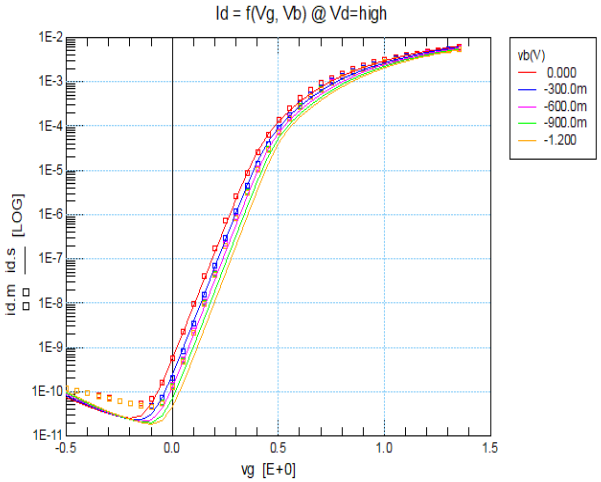


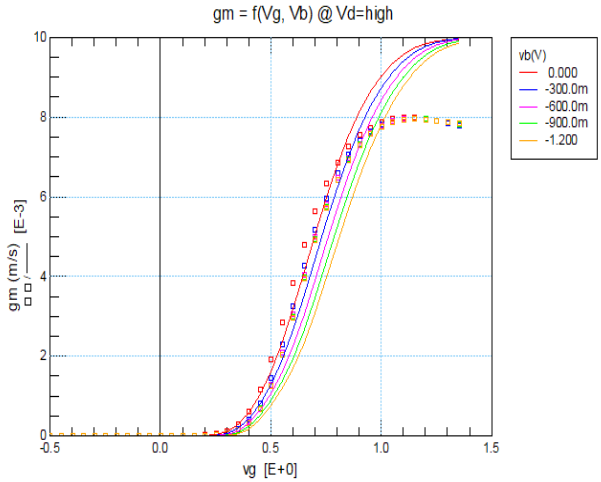


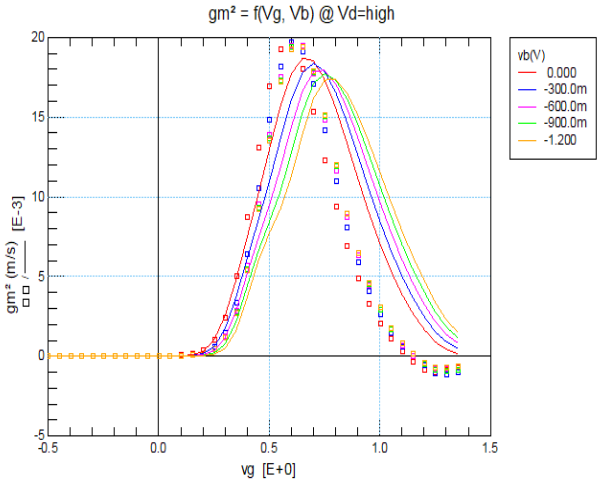


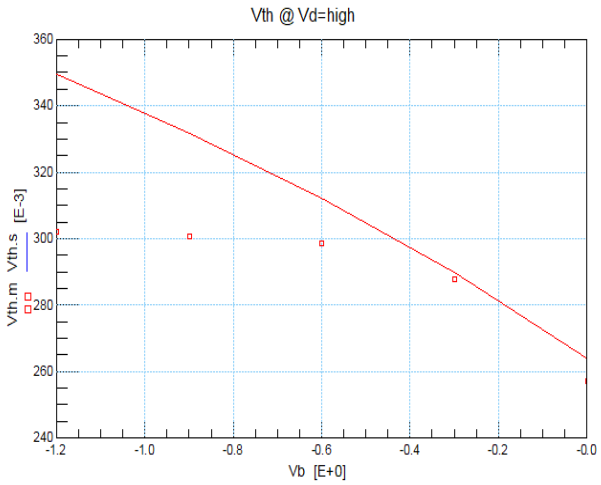
idvg, Vd = 1.2V, T = 27°C

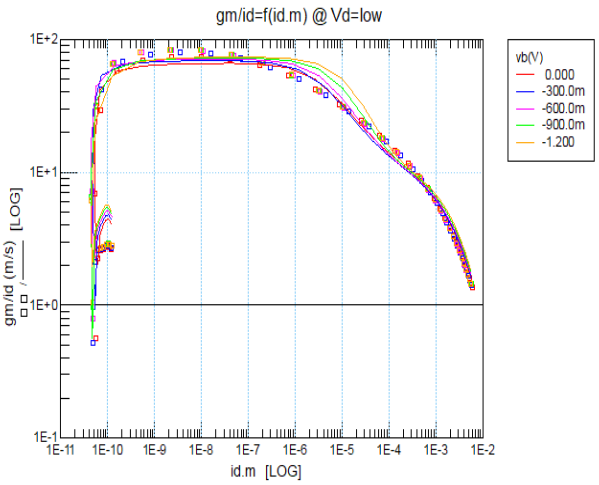


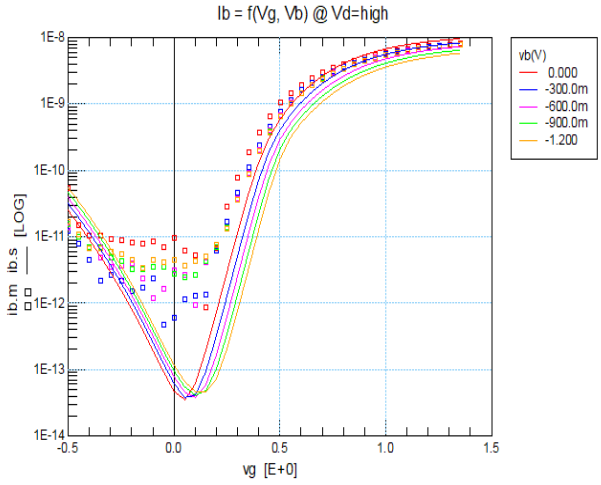


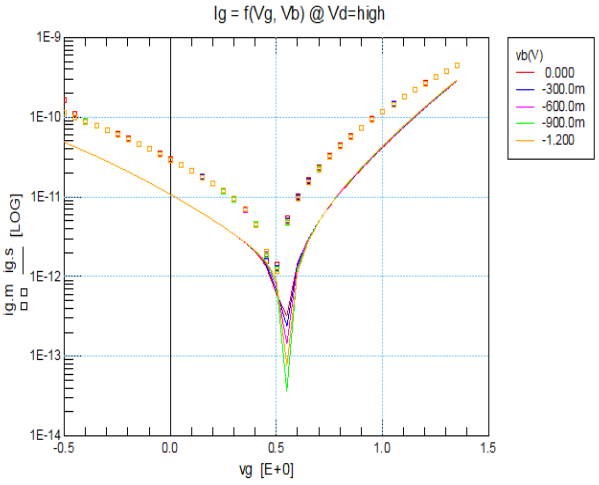




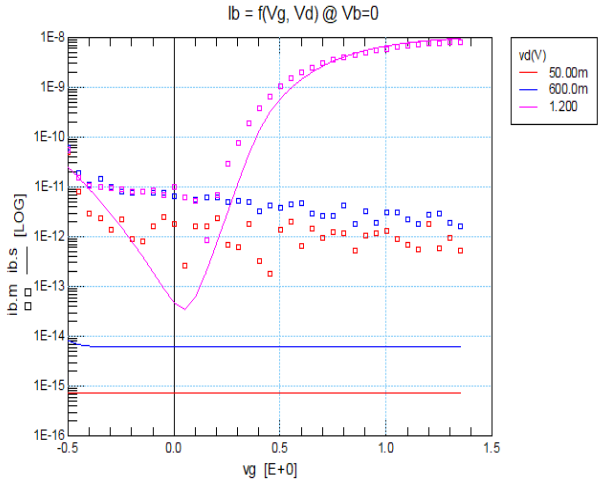


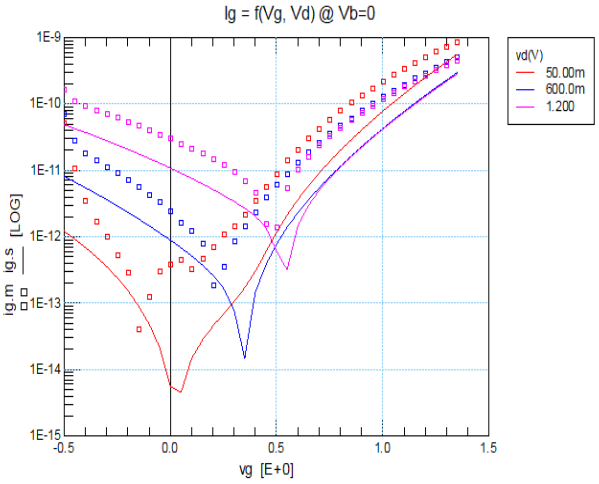




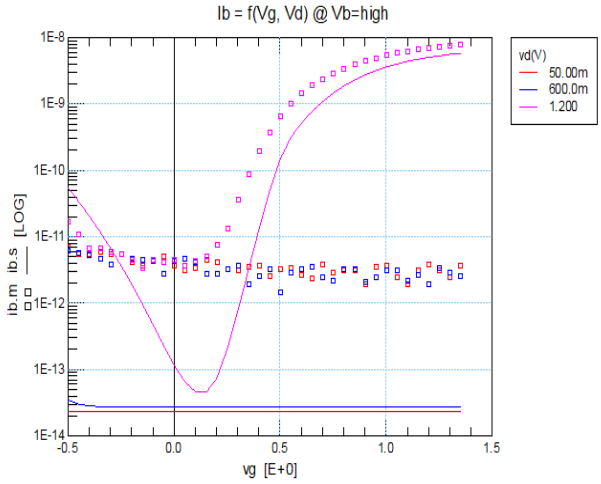


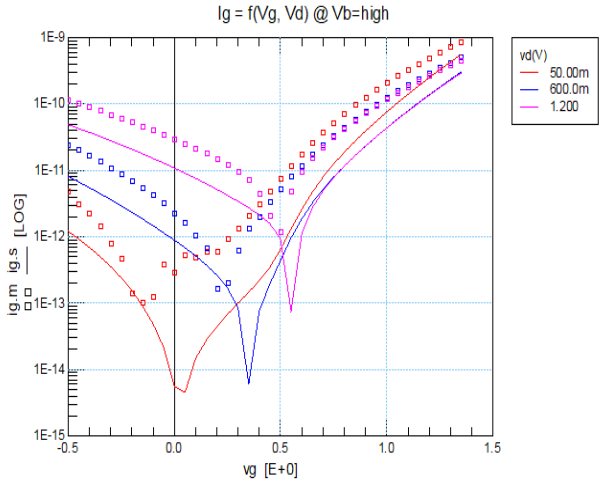
idvg, Ib,Ig, Vb = 0V, T = 27°C



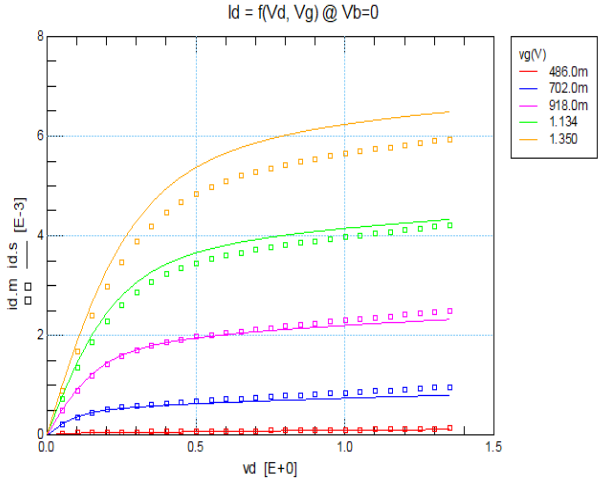


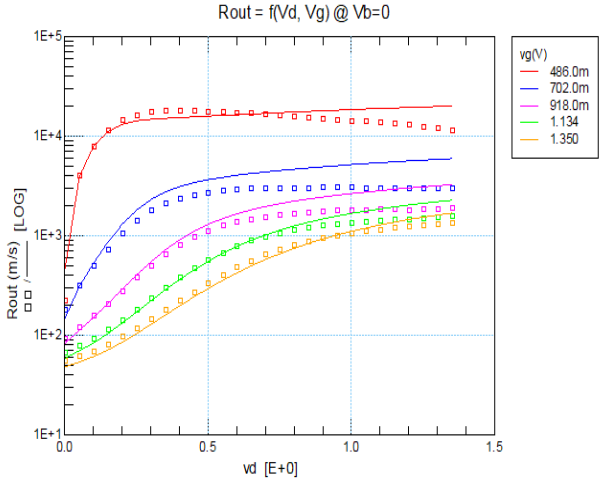
idvg, Ib,Ig, Vb = -1.2V, T = 27°C

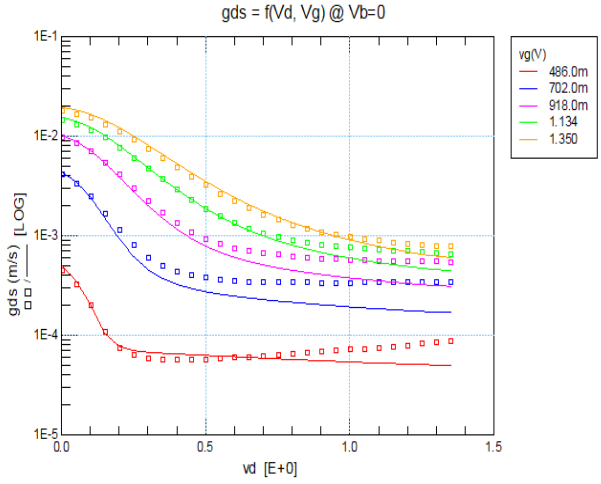




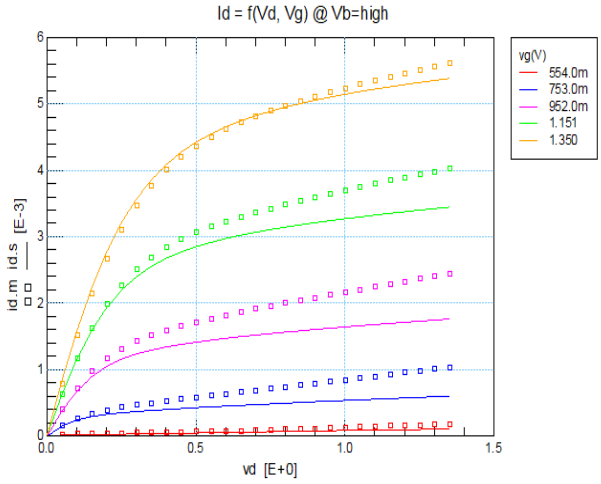
idvd, Vb = 0V, T = 27°C

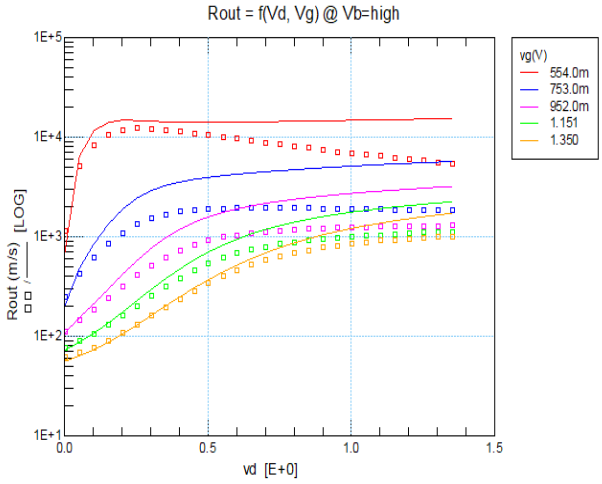


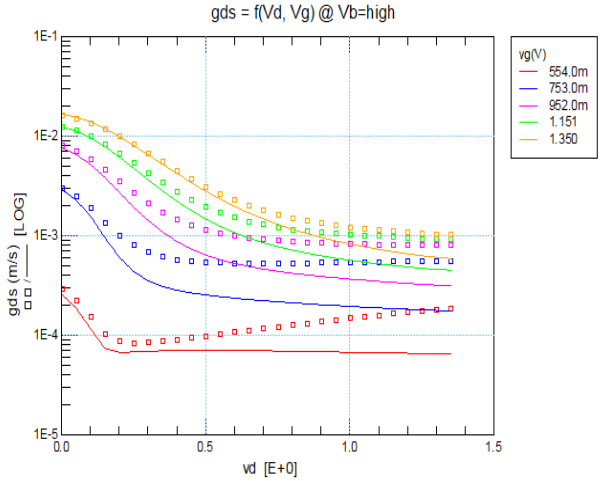




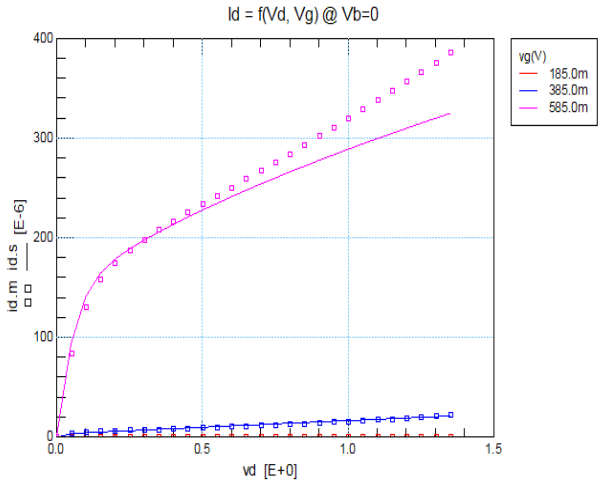
idvd_vbmin, Vb = -1.2V, T = 27°C

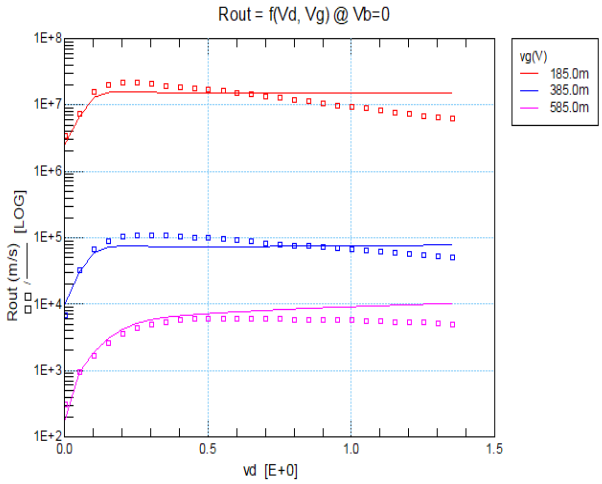


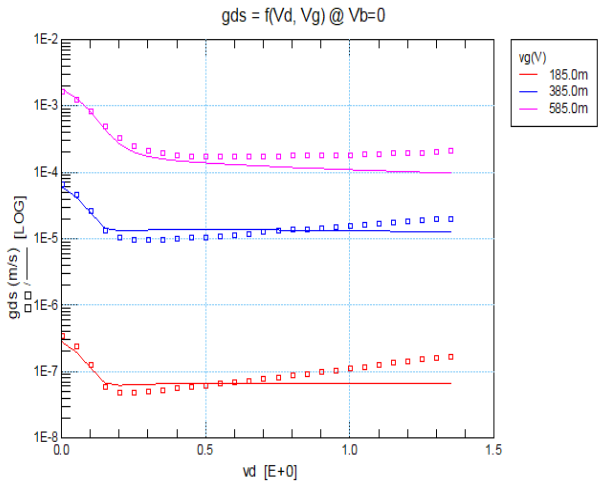




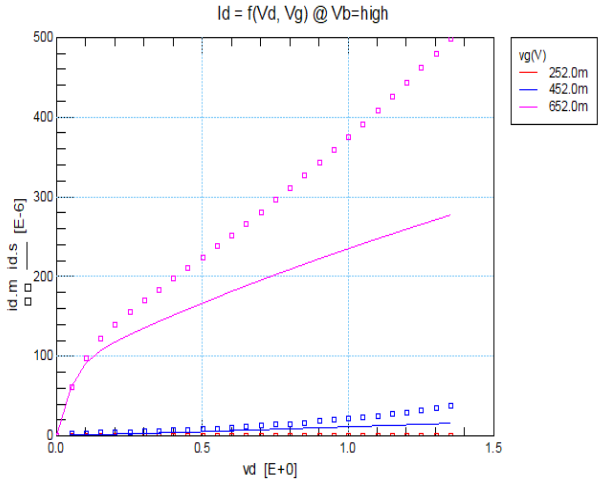
idvd_vth, Vb = 0V, T = 27°C

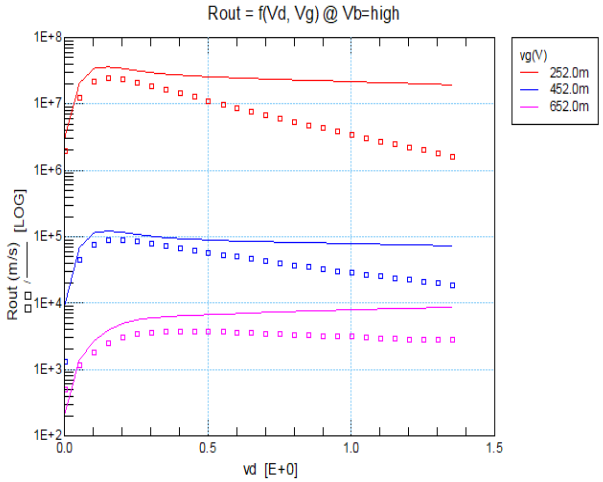


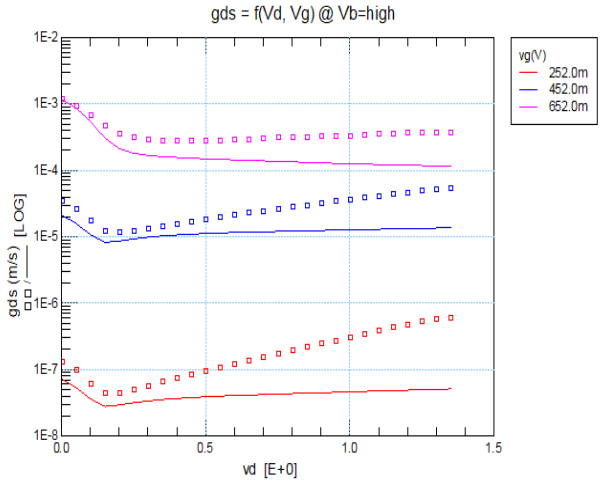




idvd_vth_vbmin, Vb = -1.2V, T = 27°C

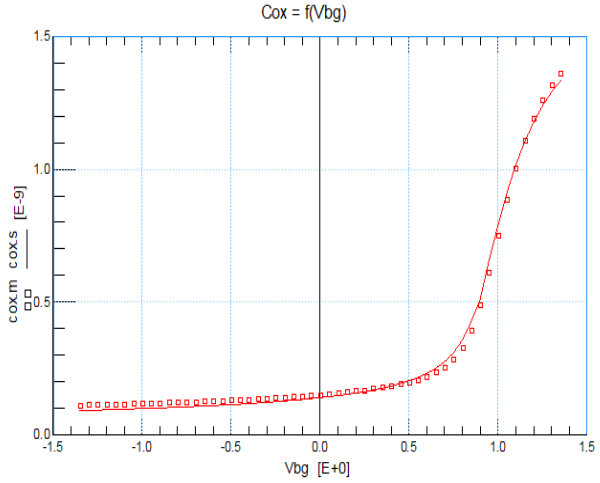




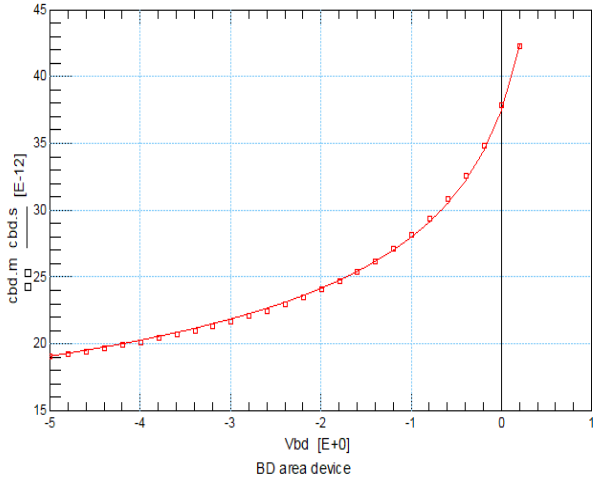


5 Capacitances

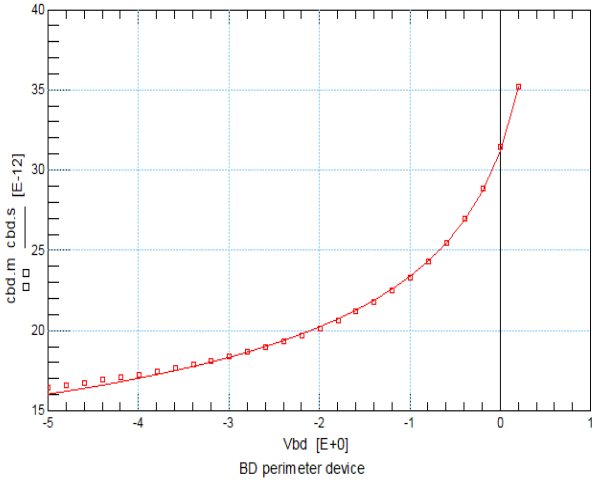
5.1 Gate Oxide Capacitance



5.2 Junction Capacitance Bulk - Drain

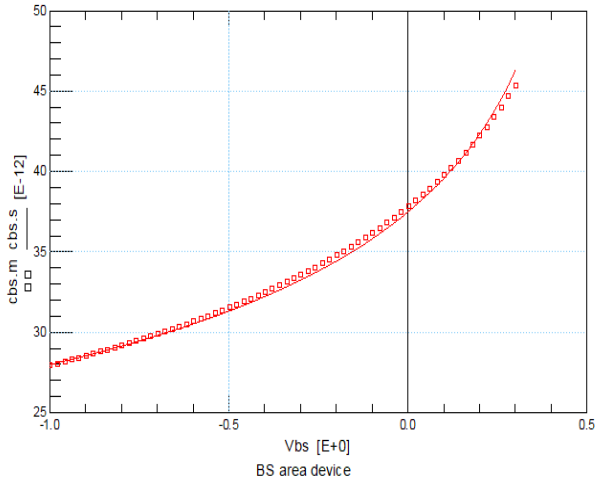


Area

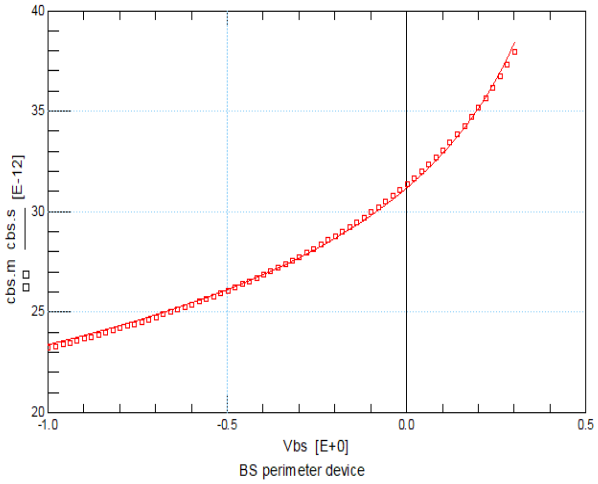


Perimeter

5.3 Junction Capacitance Bulk - Source

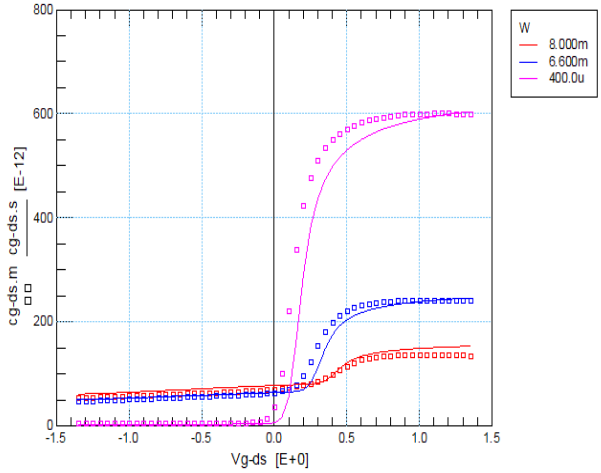


Area

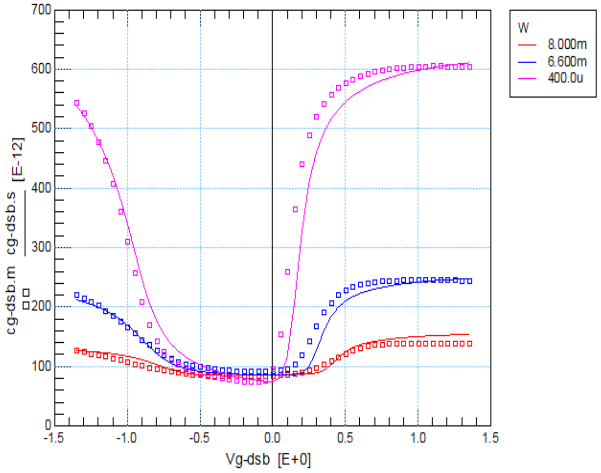


Perimeter

5.4 Overlap Capacitance



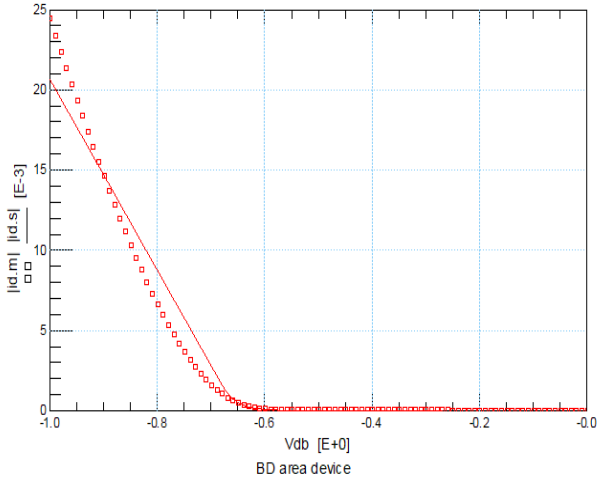
Gate - Drain,Source



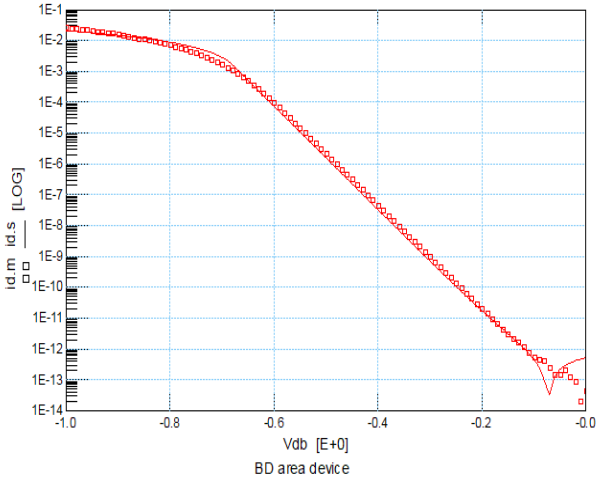
Gate - Drain,Source,Bulk

6 Diodes

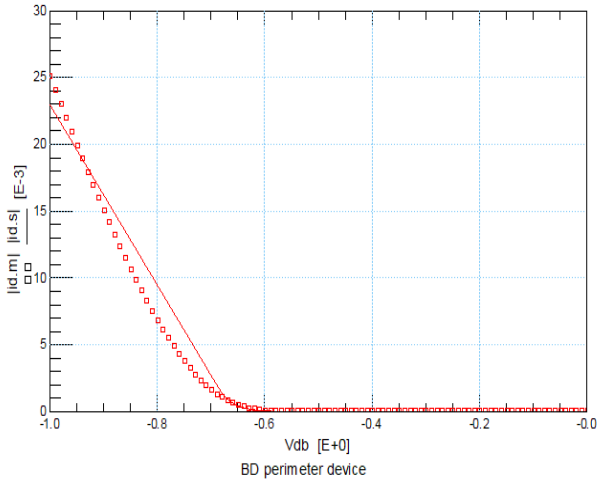
6.1 Diode Bulk - Drain



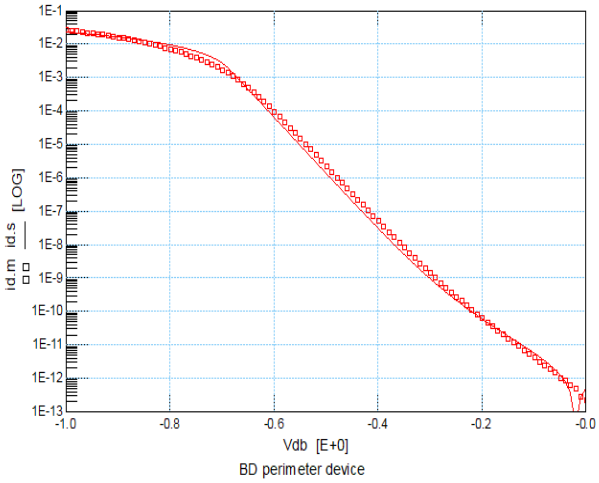
Area



Area

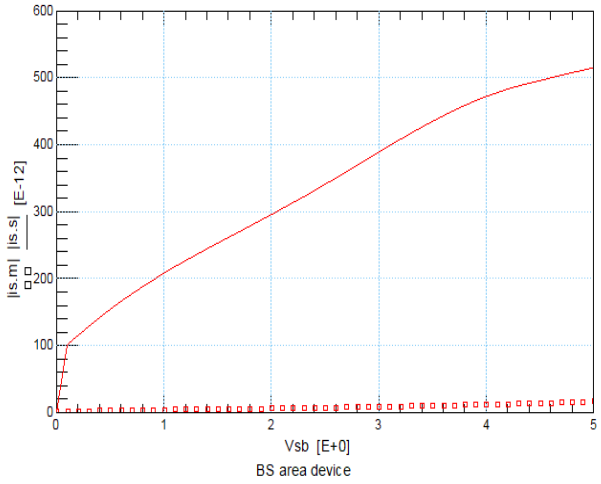


Perimeter

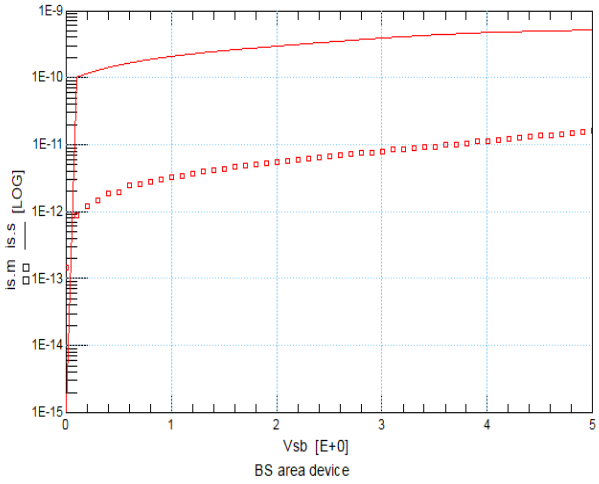


Perimeter

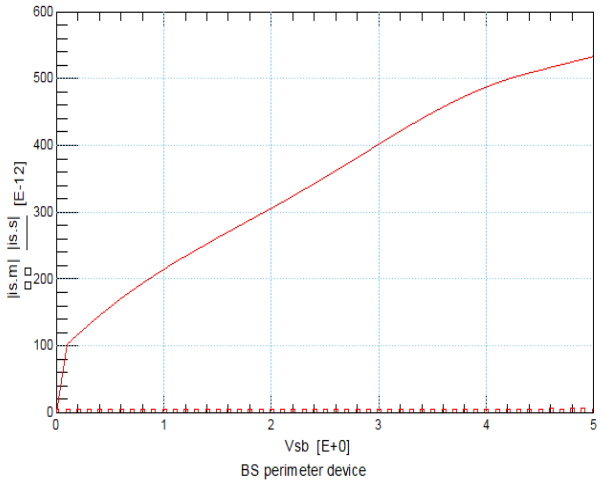
6.2 Diode Bulk - Source



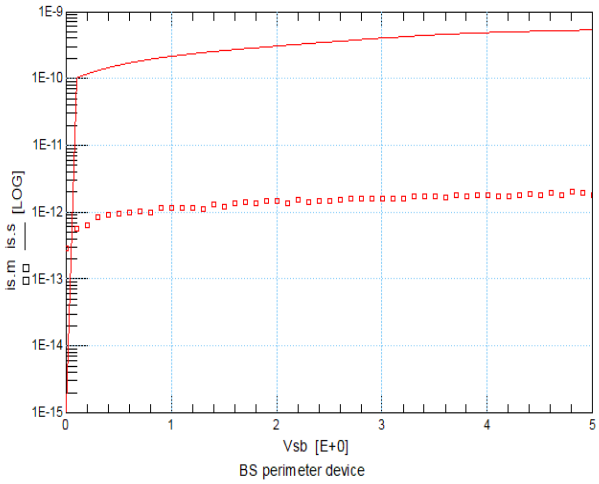
Area



Area



Perimeter

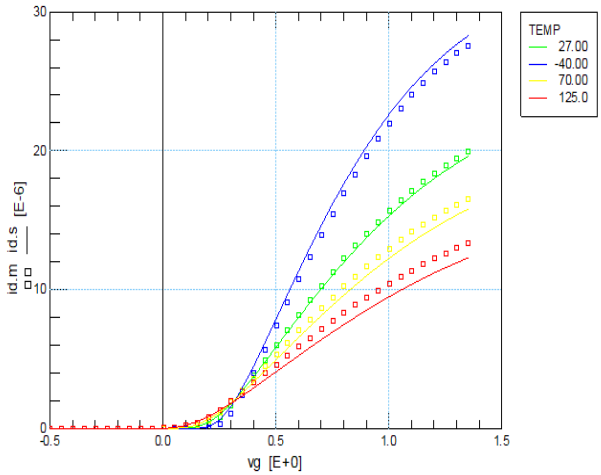


Perimeter

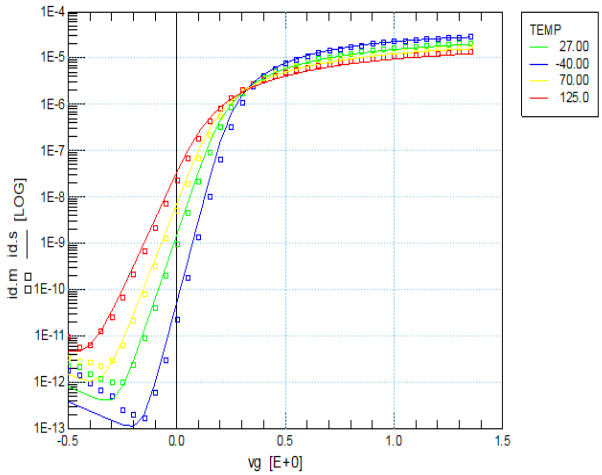
7 Temperature Measurement

7.1 Transistors

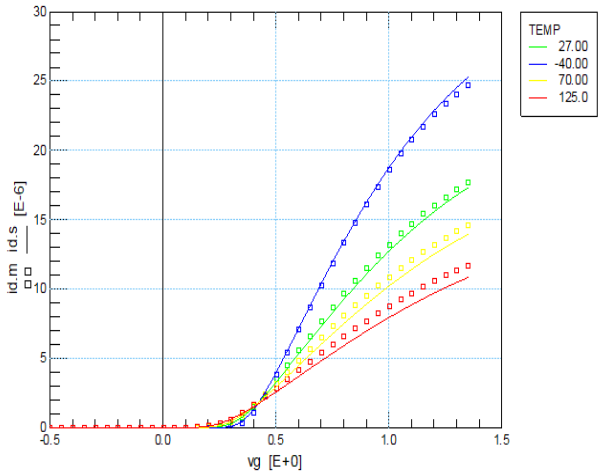
7.1.1 W10u0_L10u0_S541_5



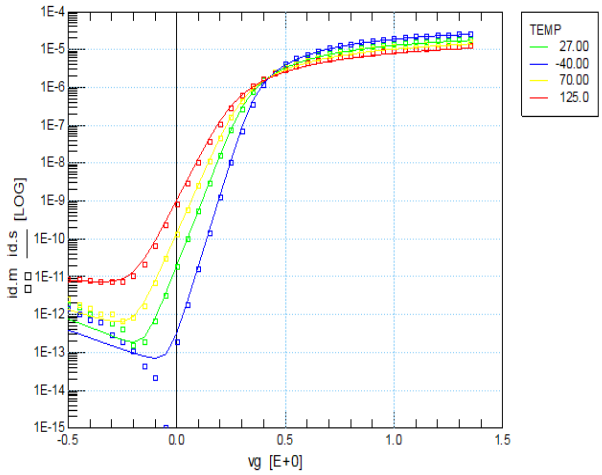
Id = f (vg, T) @ low Vds, Vbs = 0



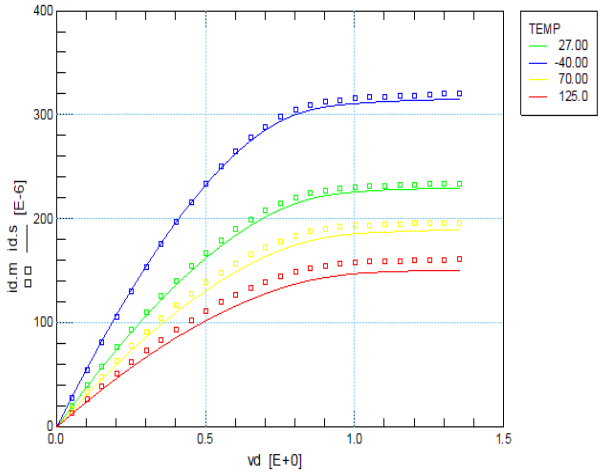
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

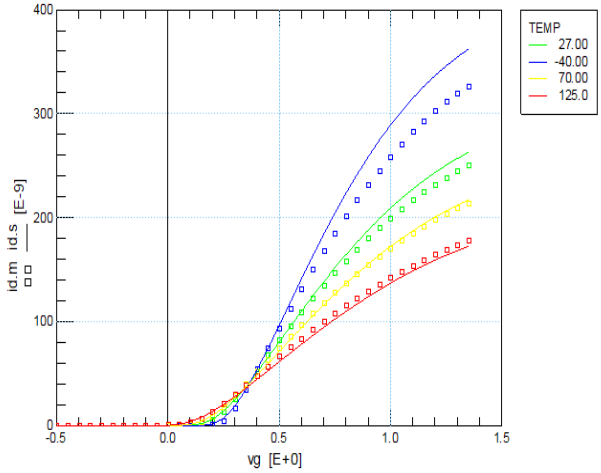


log(Id) = f (vg, T) @ low Vds, Vbs = max

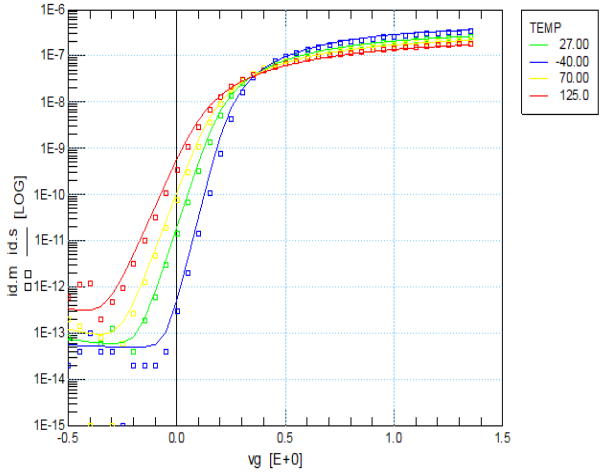


Id = f (Vd, Vg, T) @ Vbs = 0

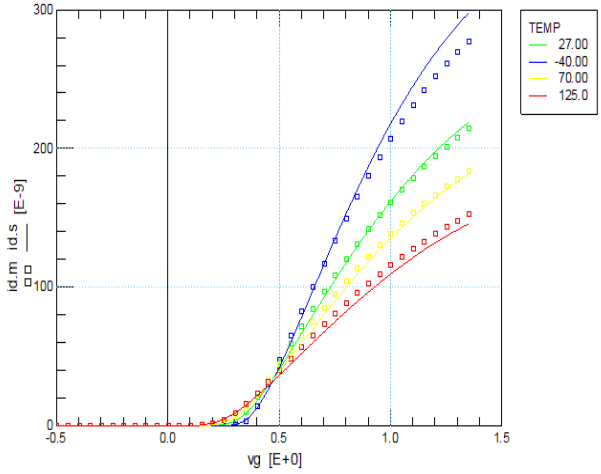
7.1.2 W0u15_L10u0_S547_3



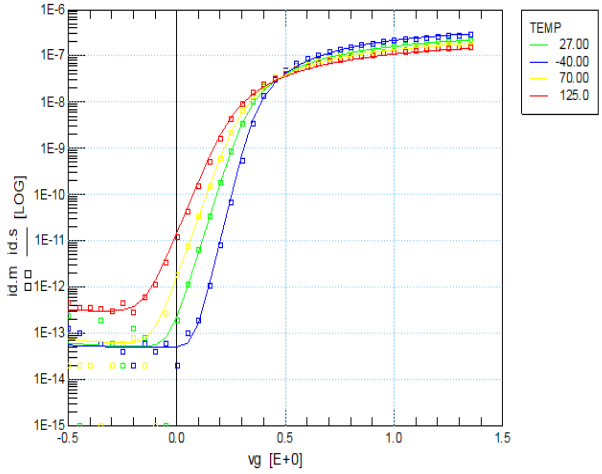
Id = f (vg, T) @ low Vds, Vbs = 0



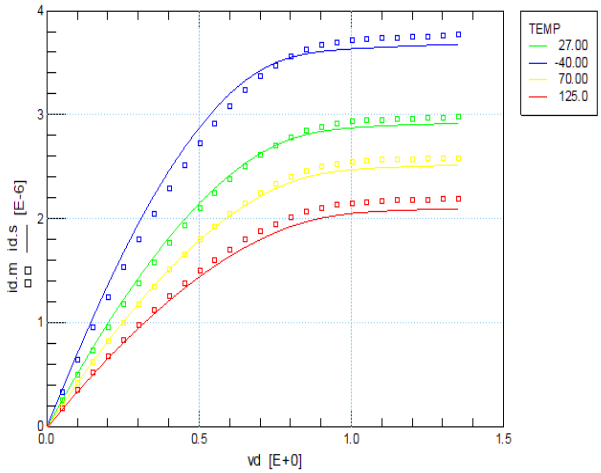
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

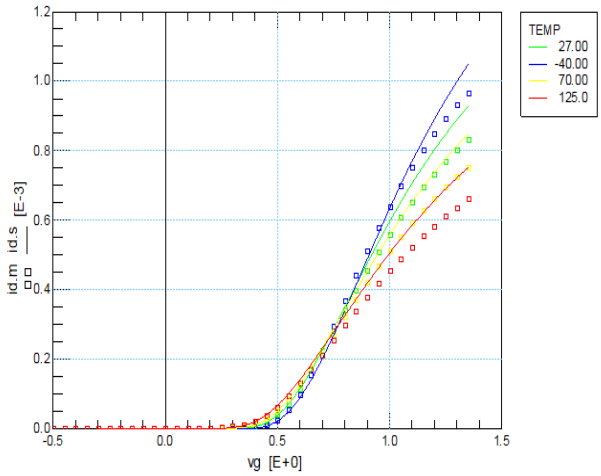


log(Id) = f (vg, T) @ low Vds, Vbs = max

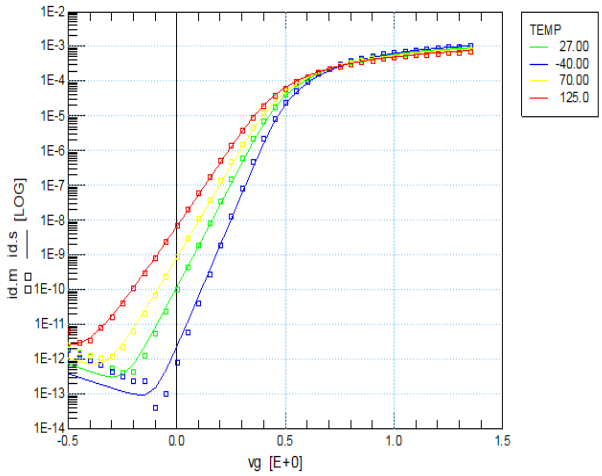


Id = f (Vd, Vg, T) @ Vbs = 0

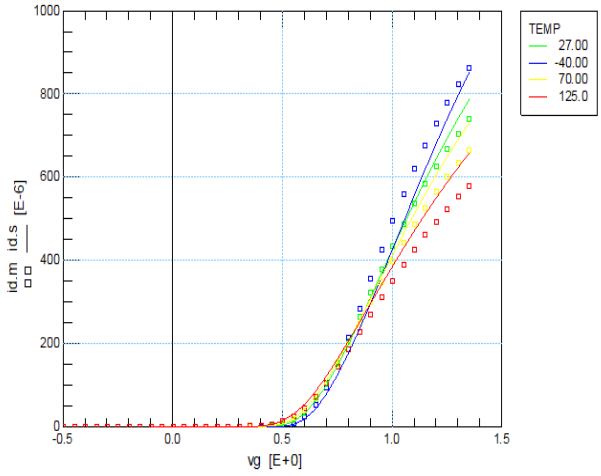
7.1.3 W10u0_L0u13_S540_2



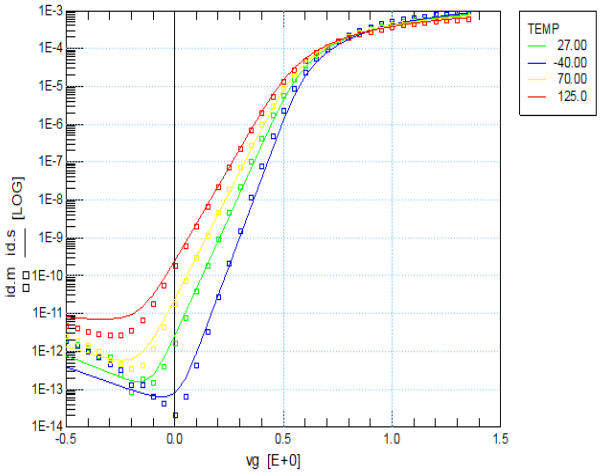
Id = f (vg, T) @ low Vds, Vbs = 0



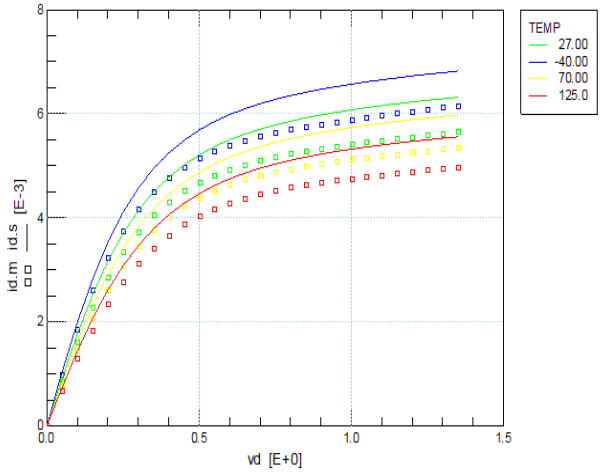
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

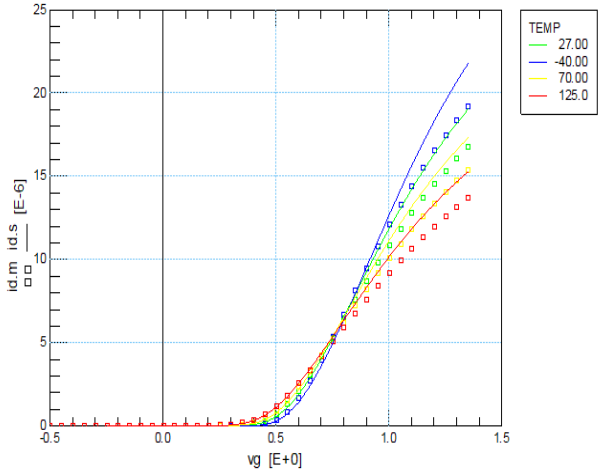


log(Id) = f (vg, T) @ low Vds, Vbs = max

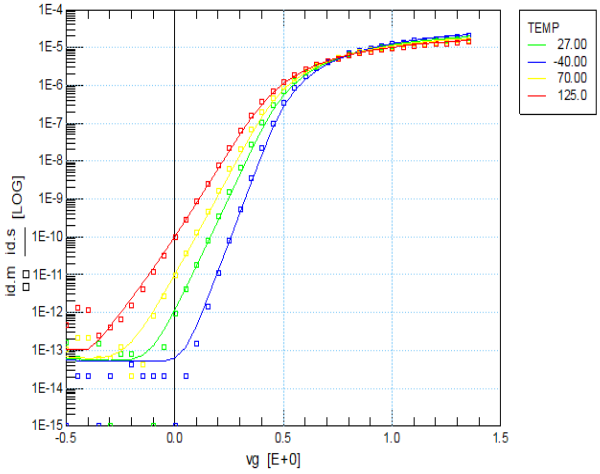


Id = f (Vd, Vg, T) @ Vbs = 0

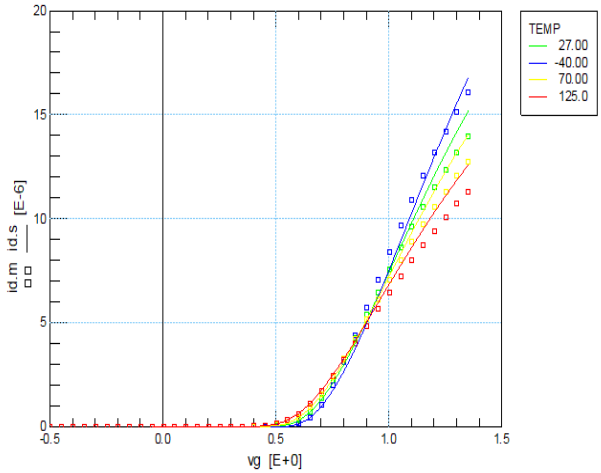
7.1.4 W0u15_L0u13_S546_3



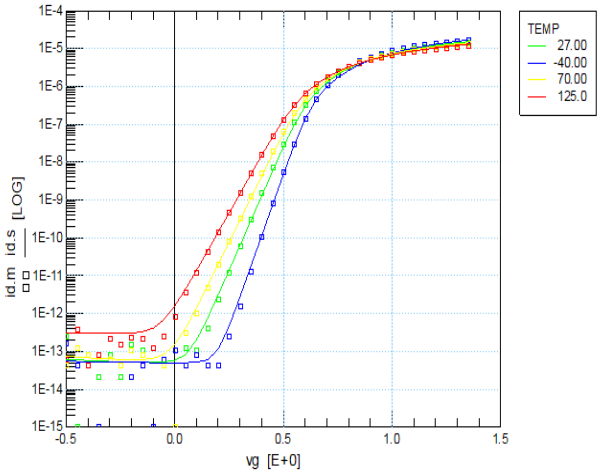
Id = f (vg, T) @ low Vds, Vbs = 0



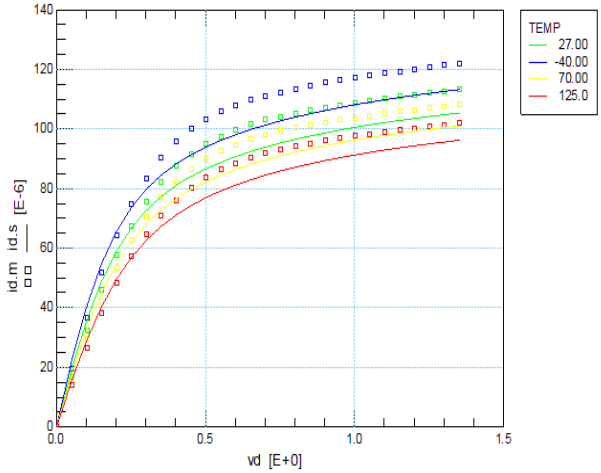
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

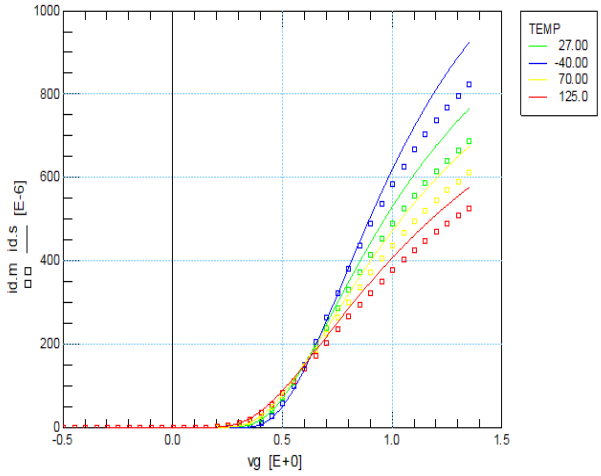


log(Id) = f (vg, T) @ low Vds, Vbs = max

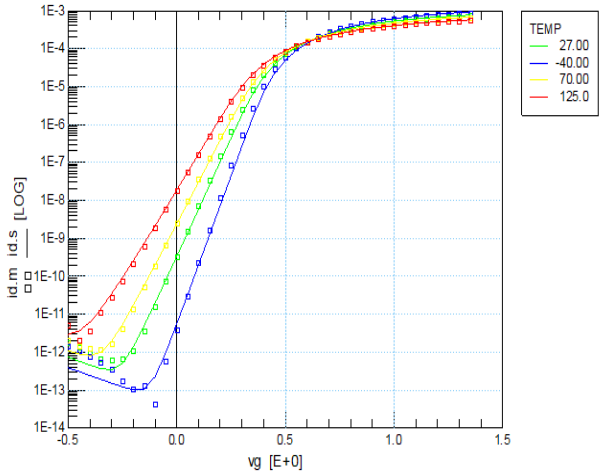


Id = f (Vd, Vg, T) @ Vbs = 0

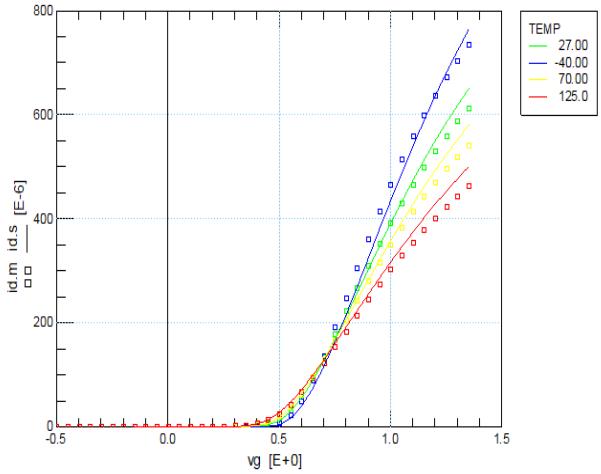
7.1.5 W10u0_L0u18_S540_5



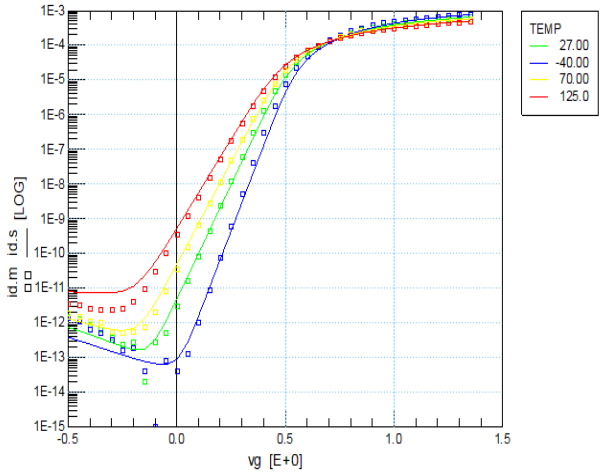
Id = f (vg, T) @ low Vds, Vbs = 0



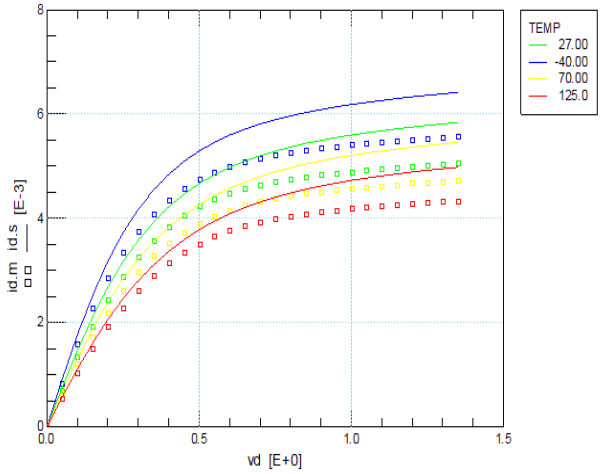
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

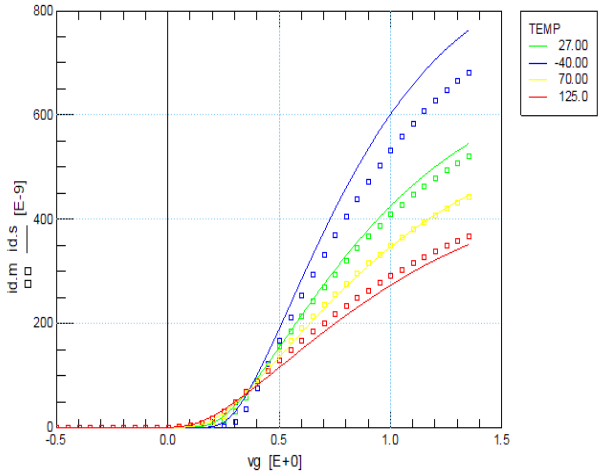


log(Id) = f (vg, T) @ low Vds, Vbs = max

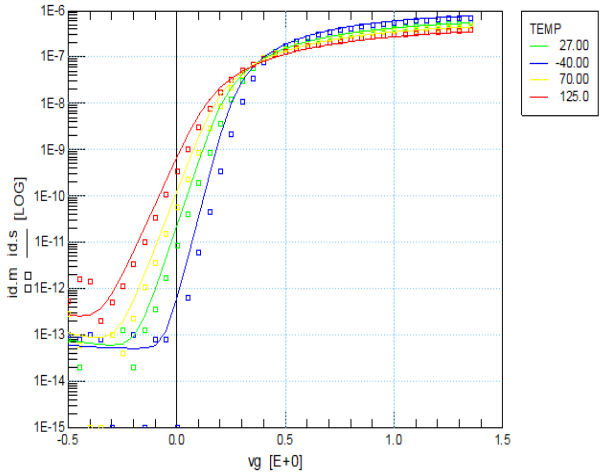


Id = f (Vd, Vg, T) @ Vbs = 0

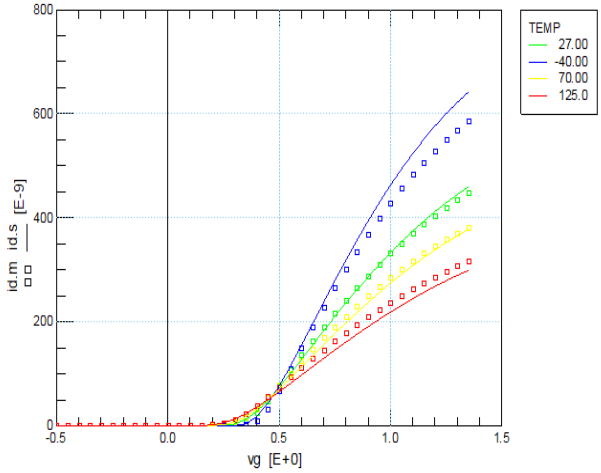
7.1.6 W0u3_L10u0_S545_2



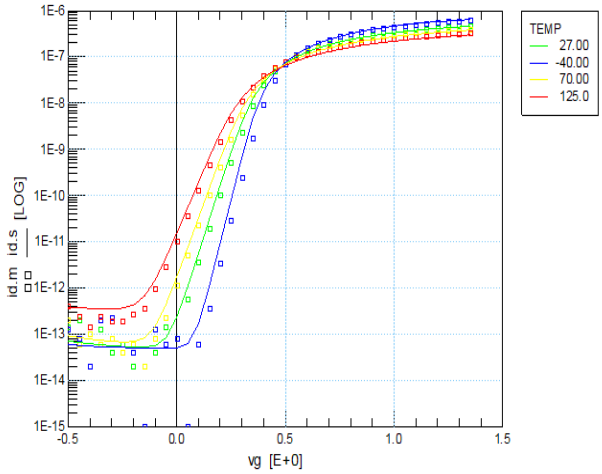
Id = f (vg, T) @ low Vds, Vbs = 0



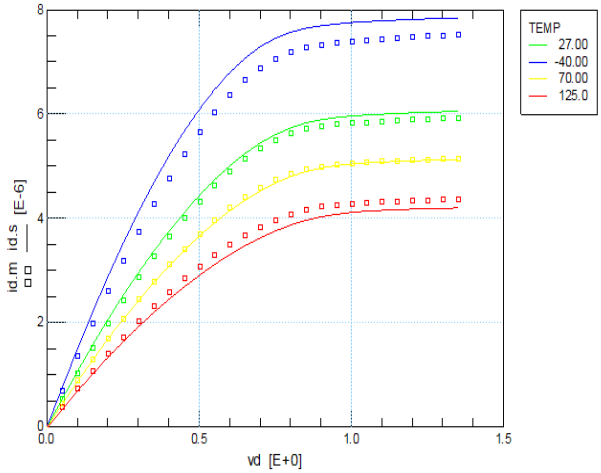
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

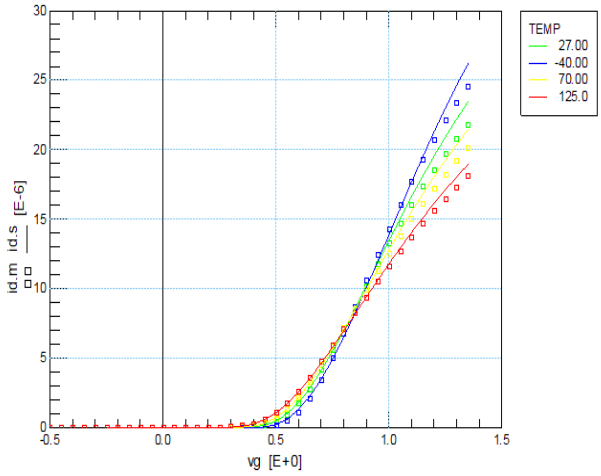


log(Id) = f (vg, T) @ low Vds, Vbs = max

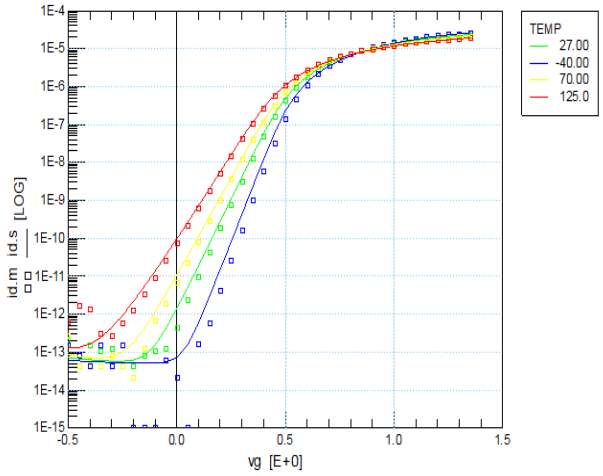


Id = f (Vd, Vg, T) @ Vbs = 0

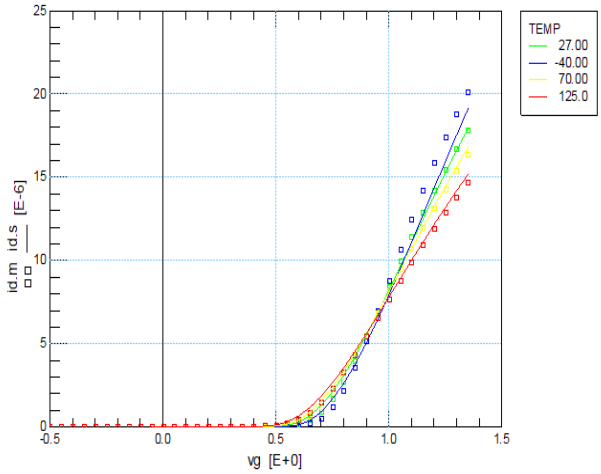
7.1.7 W0u3_L0u13_S544_2



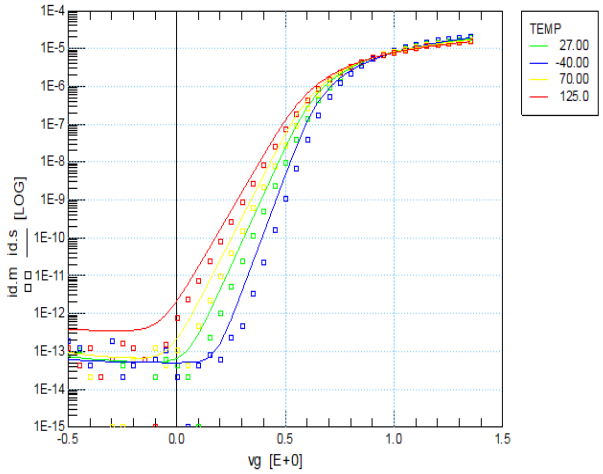
Id = f (vg, T) @ low Vds, Vbs = 0



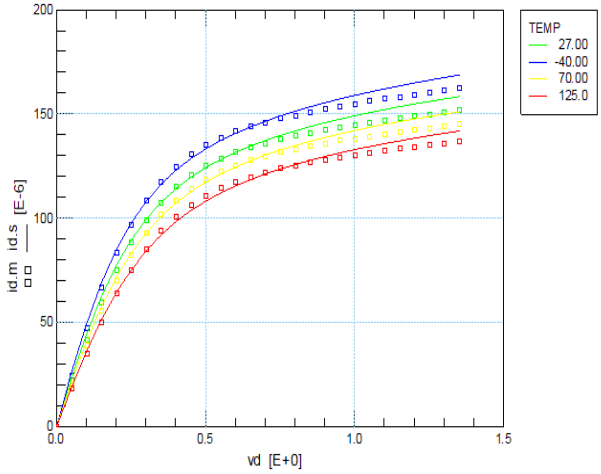
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

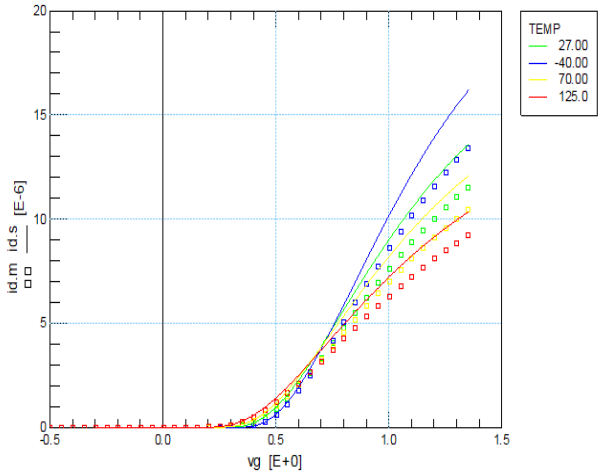


log(Id) = f (vg, T) @ low Vds, Vbs = max

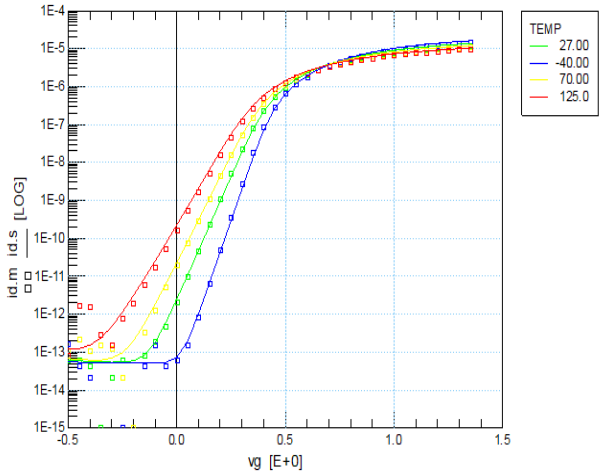


Id = f (Vd, Vg, T) @ Vbs = 0

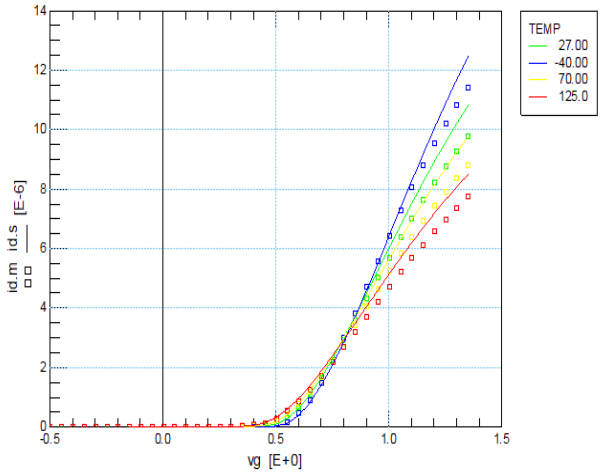
7.1.8 W0u15_L0u18_S547_1



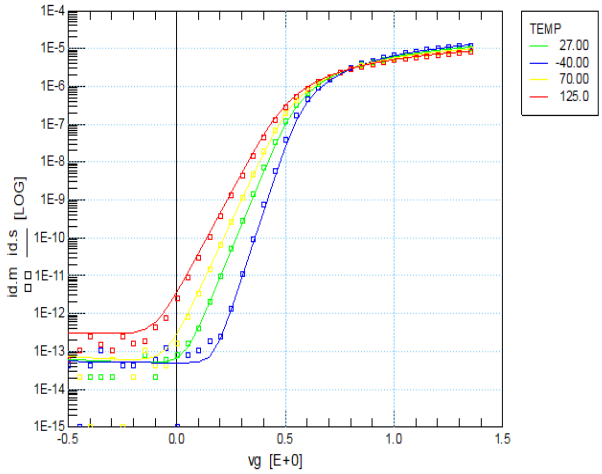
Id = f (vg, T) @ low Vds, Vbs = 0



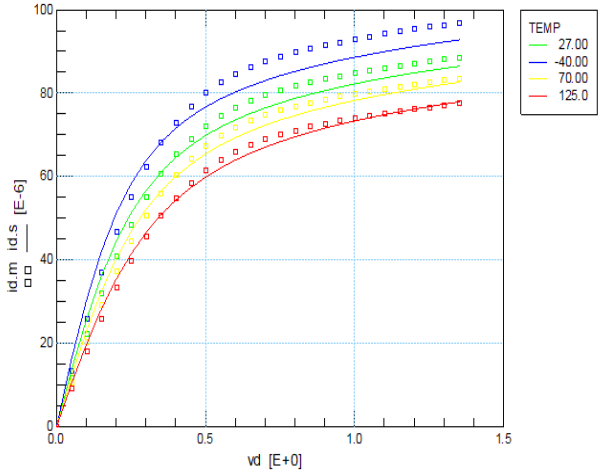
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max



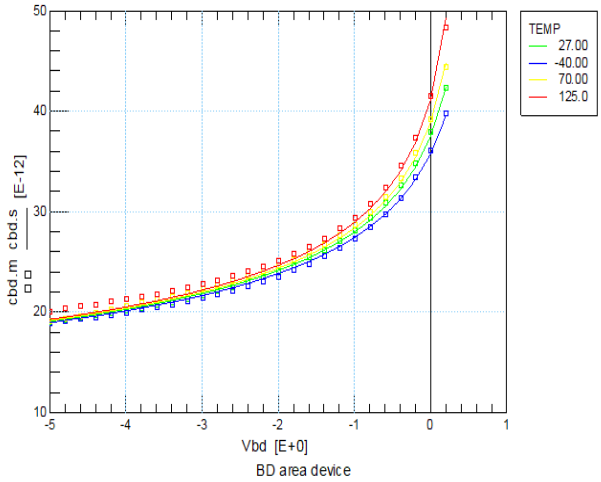
log(Id) = f (vg, T) @ low Vds, Vbs = max



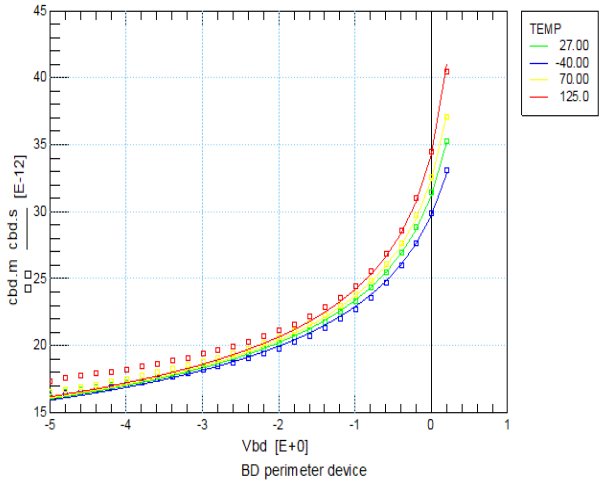
Id = f (Vd, Vg, T) @ Vbs = 0

7.2 Junction Capacitance

7.2.1 Bulk - Drain

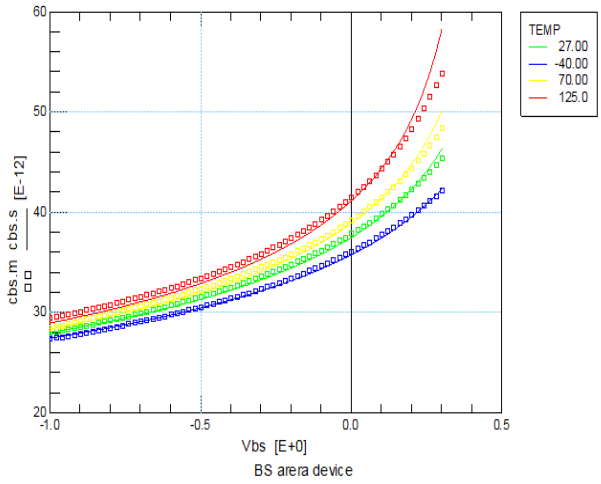


Area

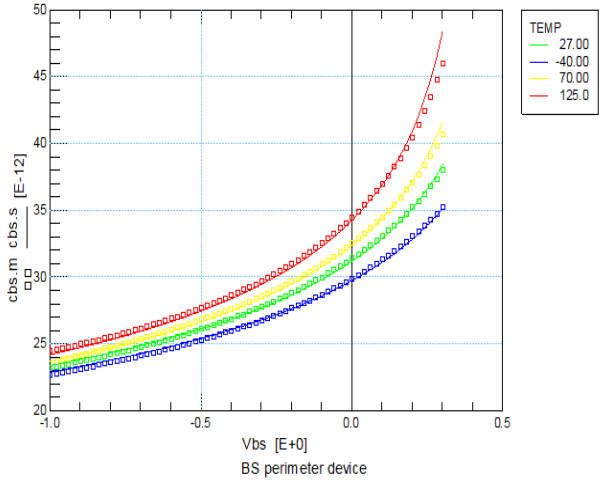


Perimeter

7.2.2 Bulk - Source



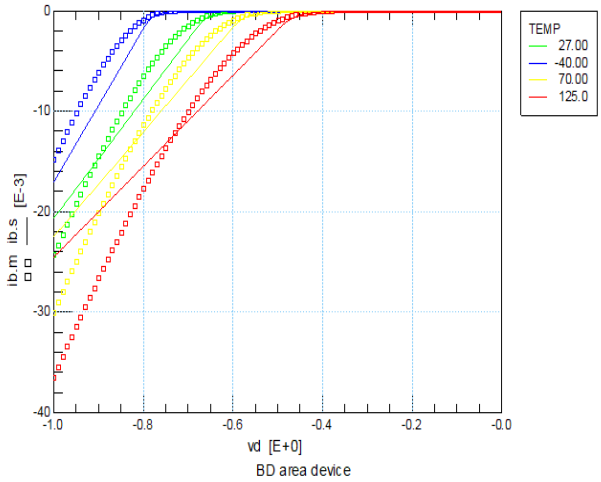
Area



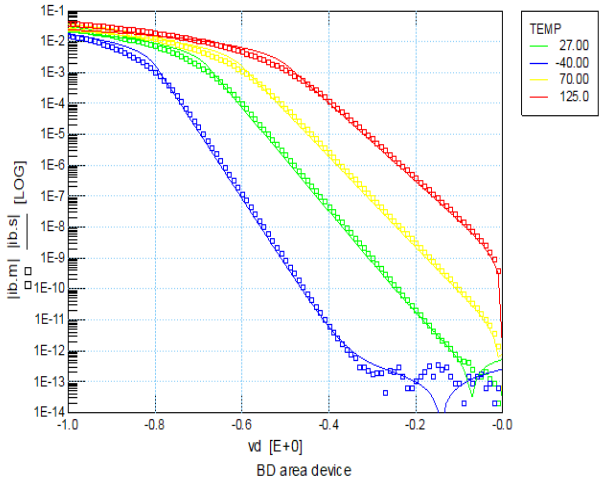
Perimeter

7.3 Parasitic Diodes

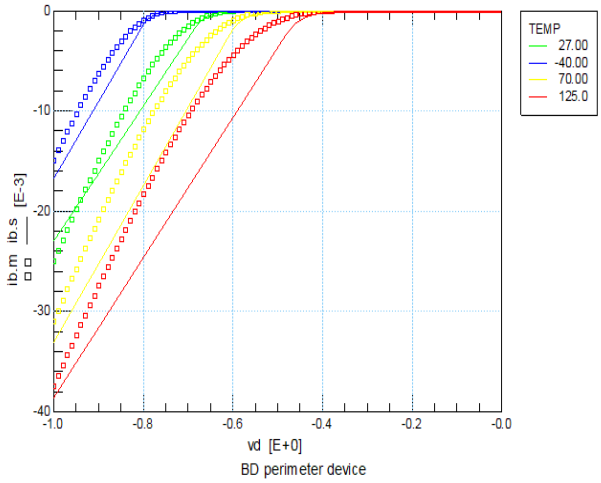
7.3.1 Bulk - Drain



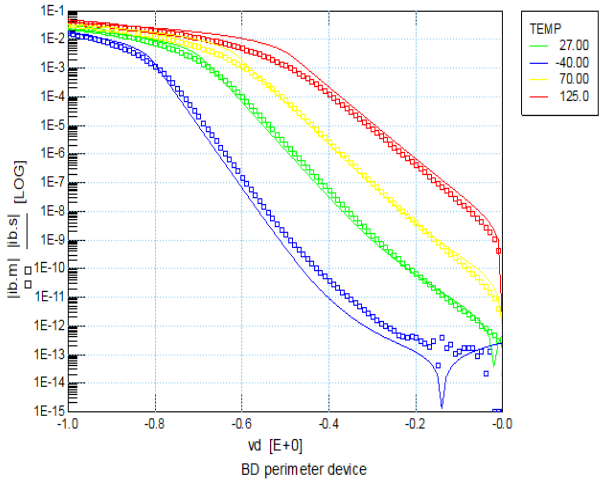
Area



Area

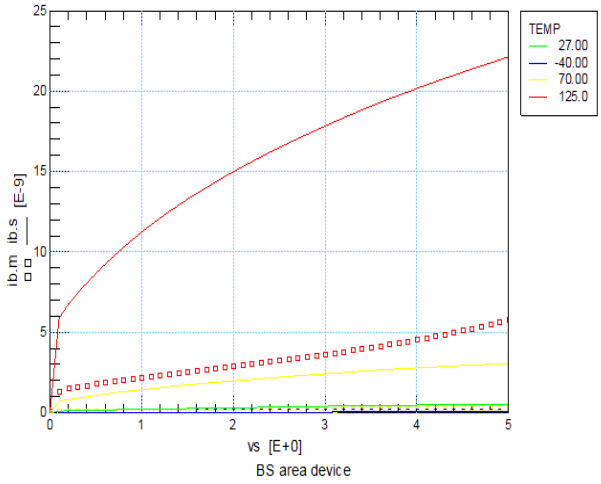


Perimeter

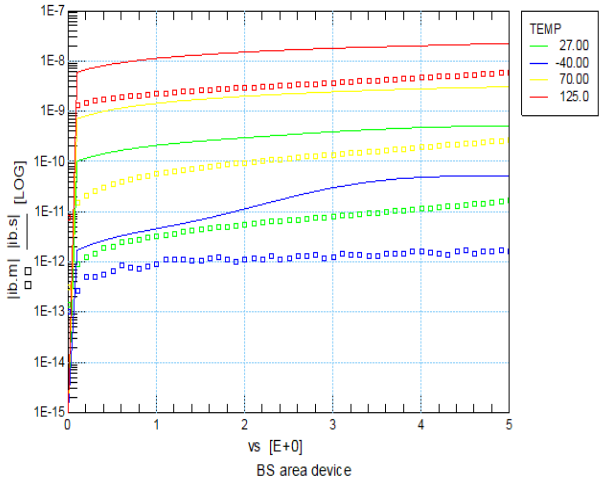


Perimeter

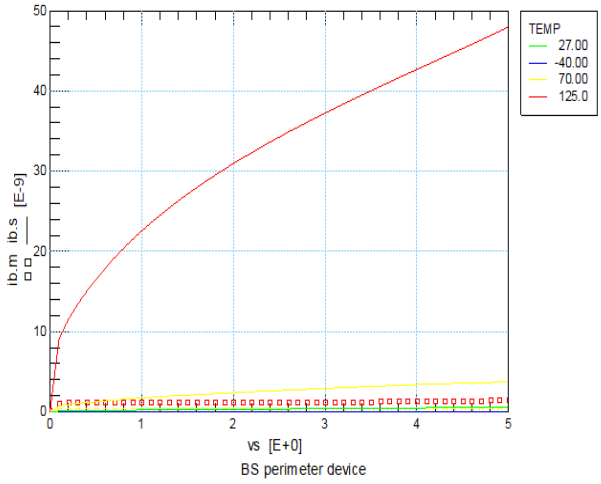
7.3.2 Bulk - Source



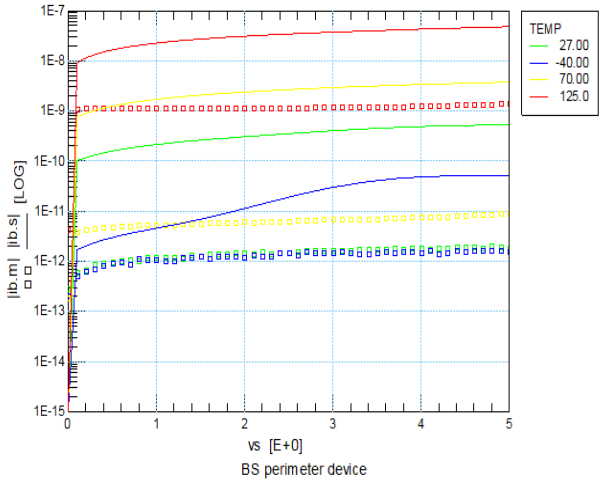
Area



Area



Perimeter



Perimeter

8 Flicker Noise

Flicker noise model parameters are taken from the SG13 simulation models.

8.1 Measurement Conditions

Not defined as no flicker noise measurements are yet done on SG13G2 devices.

8.2 Devices

Not defined as no flicker noise measurements are yet done on SG13G2 devices.

9 Change History

Revision	Device	Changes
190606	All	Initial version
190730	All	Adjust temperature and RF behavior
190924	Summarized Results	Legend added